

SMITHIAN FOLD THEORY - CHEMISTRY BRANCH PAPER 001

From Fold to Chemistry

An Exact, Parameter-Free and Machine-Closed Reconstruction of Chemical Science
from Smithian Fold Theory

Ernos Labs

Open Source Science Platform and Knowledge Tree

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Version 1.3 - complete to the registered current standard and open to lawful extension

272/272 obligations - 281 claims - 71,936 candidates - 1,124 controls

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Paper: CC BY 4.0 - Code: Apache-2.0

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Abstract

This paper reports the complete current Chemistry reconstruction in the third clean-room build of Smithian Fold Theory. All **272 of 272 registered discipline obligations** are separately receipt-backed and the branch remains open to lawful extension. The complete live Chemistry surface contains **281 admitted claims**, including nine supplemental lineage and Smithium consequence claims; **71,936** completely enumerated candidates; **281** unique survivors; **1,124** mandatory controls; **281** implementation-distinct reconstructions; **273** post-seal empirical packages; and **eight** explicitly formal-only boundaries. Closure is depth-independent for 276 claims and finite-complete for five. Every claim traces through named dependencies to the premise-free root theorem and was accepted by the same sealed fail-closed engine with zero imported axioms and zero free parameters.

The field reconstruction spans chemical identity and measurement; elements and periodicity; composition and stoichiometry; bonding and molecular structure; acid-base, redox and electrochemistry; thermodynamics, kinetics and transport; inorganic, organic, polymer, nuclear, analytical and computational chemistry; complete empirical vectors; and cross-branch ownership. Its headline exact results include orbit capacities 2, 6, 10, 14, 18; nuclear closures 2, 8, 20, 28, 50, 82, 126, 184; g-block opening at Z=121; Smithium at Z=126, N=184, A=310, [Og] 8s2 5g6; and the structural endpoint Z=137. These extrapolations are standing predictions, not falsely labelled observations. External records test consequences after sealing; they never choose the grammar or survivor.

Results first: a complete current Chemistry reconstruction

Headline	Exact result	Evidence and meaning
Complete branch	272/272 registered obligations; 281 admitted claims	Complete to Maria Smith's registered current standard, explicitly open to lawful extension rather than permanently locked.
Full machine surface	71,936 candidates; 281 survivors; 1,124 controls; 281 independent reconstructions	Every registered grammar was exhaustively enumerated and admitted separately through the untouched sealed engine.
Empirical surface	273 post-seal empirical packages; 8 explicit formal-only boundaries	Missing measurements are not invented; favorable, adverse, absent, unavailable and unresolved rows remain distinct.
Atomic orbit capacities	2, 6, 10, 14, 18	Forced from three-space, boundary rank two, two held Fold labels and exclusion; known periodic organization tests the sealed result.
Nuclear closure sequence	2, 8, 20, 28, 50, 82, 126, 184	Exact registered correspondence; 126/184 supplies the island-of-stability carrier used by Chemistry.
g-block and Smithium	opening Z=121; Smithium Z=126, N=184, A=310, [Og] 8s2 5g6	Standing unobserved predictions. The known table through 118 is retained separately and cannot masquerade as their observation.
Structural endpoint	Z=137	Standing prediction from the greatest-whole order below the exact Physics-owned inverse-alpha carrier.
Physics-owned prerequisite	$\alpha^{-1} = 503846395469/3676744786 = 137.035999177180855...$	Derived and measured in Physics; Chemistry consumes the admitted dependency without refitting or relocating it.
Molecular property unification	13 families; 9,025 source rows; 1,104 carriers; 676 overlap carriers; 6,676 overlap rows	One source-bound carrier serves every property family; separate fitted coefficients are a registered falsifier.
Entropy correspondence	13 water rows; phase jump 13637763/125000 J mol ⁻¹ K ⁻¹ ; independent ratio 4066845420/37275593	Exact distinction accounting is primary; the measured vector opens after sealing.
Colligative response	276 records: 144 boiling, 37 freezing, 95 osmotic	Complete solvent/solute custody with orientation held as a label and no fitted derivational constant.
Kinetic isotope effects	90 explicit rate ratios and 3 direct decay effects across 47 pages and 23 worksheets	Normal, inverse, near-equal and adverse rows remain in the same evidence surface.

Headline	Exact result	Evidence and meaning
Aromatic recurrence	supports 6, 10, 14; stability excess 150.39 kJ mol ⁻¹ ; uncertainty 6.60; lower envelope 143.79	Three outcome-unopened CCCBDB sources, 59 tables and 353 rows test the sealed recurrence law.
Quantitative electrochemistry	13/13 laws	Half-reactions, potentials, work, electrolysis, conductivity, kinetics, polarization, interfaces, corrosion and storage handoff are separately forced.
Nuclear and radiochemistry	12/12 laws	Carrier, transformation, activity, branching, equilibria, isotope effects, tracer custody, separation, yields and radiation networks remain distinct.
Analytical science	22/22 laws	Complete NMR, Raman, fluorescence, phosphorescence, diffraction, separation and multimodal records retain accuracy, uncertainty and adverse evidence.
Computational chemistry	14/14 laws	Exact molecular graphs, isomers, conformers, reactions, atom maps, mechanism proof traces, provenance, symbolic evaluation and classical-quantum correspondence.
Polymer chemistry	13/13 laws	Size, dispersity, growth, composition, architecture, gelation, conformation, phase, degradation and Materials handoff are independently closed.
Chemistry empirical Grand Lock	263 pre-VALID claims: 255 empirical and 8 formal-only; 26,486 evidence lines; 1,177 source occurrences	Every receipt and evidence class remains claim-specific; no aggregate success score hides a failed row.
One-owner cross-branch graph	1,484 frozen claims; 25,013 dependency edges; 18,553 cross-branch uses; all root-traced	Materials, Biology, Medicine, Earth Science and Astronomy consume Chemistry through directed dependencies and do not re-own its laws.

Public scientific mission and admission boundary

Ernos Labs is an open-source science movement, verification platform and public tree of knowledge founded by Maria Smith. Its purpose is not to replace one authority with another. It is to make scientific authority narrow, inspectable and revocable: a claim is admitted only through its complete derivation chain, generated alternatives, eliminations, unique survivor, adverse controls, measurement custody where applicable and unchanged-engine receipt. Open criticism is unrestricted and necessary; scientific admission is the separate machine-checked act of satisfying that public standard.

Maria Smith developed Smithian Fold Theory outside formal academic education, institutional research employment and conventional grant funding. That fact is not offered as evidence for a theorem; the derivations and observations carry the entire scientific burden. It is evidence about access. A credential-first system loses more than individual opportunity: it loses unknown questions, methods and discoveries from minds that capital and status never authorize. This work therefore treats Maria Smith's authorship neither as exceptionalism nor as a reason for dismissal, but as an indictment of every scientific contribution lost when financial gatekeeping is presented as rigor.

The institutional argument is empirical, not a claim that every institution or funded researcher acts in bad faith. Published studies document sponsor-linked differences in outcomes and conclusions, commercial influence over research agendas, limits and sensitivities in grant review, underpublication of null results, and inequalities created by paywalls and article-processing charges. Those findings establish that funding, prestige, publication and consensus are selection systems with incentives and failure modes. They cannot substitute for a public proof and evidence chain. Expertise, measurement and adversarial review remain indispensable; institutional permission does not select a fundamental law.

For Chemistry, a periodic-table entry, handbook definition, fitted Hamiltonian or opaque predictor may test or translate a sealed consequence but cannot choose the law. Measured agreement, disagreement, non-observation and standing prediction remain separate evidence classes. No institution, credential or funding body can substitute for the public derivation and evidence chain.

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describe itself as Ernos Labs only while it preserves the public empirical constitution, complete adverse evidence, unchanged admission route, critical review and community standards.

Independent replications, lawful extensions, corrections and attempted invalidations are invited. Credentials cannot rescue a failed gate and lack of credentials cannot prevent a reproducible result from being evaluated. Contact Maria.Smith.Sftoe@gmail.com, submit through <https://discord.gg/ucwGryVxGr>, or inspect the public project at <https://github.com/MettaMazza>.

1. Authorship, rights and publication boundary

Maria Smith is the author and founder of Ernos Labs. Contact: Maria.Smith.Sftoe@gmail.com. Submissions and reproducibility reports: <https://discord.gg/ucwGryVxGr>. GitHub: <https://github.com/MettaMazza>.

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2. Exact scope and meaning of completion

The registered discipline census contains 272 obligations in 23 dependency-ordered fields, all now receipt-backed. Nine additional claims retain the separately returned rovibronic and Smithium consequences, giving 281 live Chemistry claims. “Complete” means complete to this dated, registered current-knowledge and evidence standard. It does not mean immutable, permanently locked, or immune from a valid discovery. A lawful extension adds a versioned claim and successor census through the same engine; it never rewrites an old receipt.

Chemistry owns substances, species, composition, bonds, molecular structures and chemical transformations. Materials owns bulk response; Biology owns living organization and function; Medicine owns intervention and health outcome; Earth and Environmental Science owns system history, transport and planetary context; Astronomy owns source populations and cosmic context. Applications cannot select the laws.

3. Constitutional mathematics

Structural absence is represented by $\emptyset$ but does not denote conventional numerical zero. Scientific proof magnitudes are generated positive whole counts and exact positive rational parts or ratios. Direction, opposition, gain/loss and charge sense are held labels rather than negative magnitudes. Irrational, imaginary, floating, continuum, fitted or learned values are not admitted as Fold-native proof objects. Host counters may enumerate files and rows but cannot carry scientific authority.

Every law has zero imported axioms and zero free parameters. External values may test an already sealed result; they may not choose its form. A failed attempt retires nothing. It earns no completion credit, remains preserved, and is followed by a genuinely distinct lawful route unless a complete structural impossibility proof closes the registered grammar.

4. One sealed admission engine

Every claim is registered against the root theorem and already admitted dependencies. The engine requires complete candidate identity and cardinality, one decision per candidate, exactly one survivor, closure, minimality, named-shape uniqueness, four mandatory controls and an implementation-distinct reconstruction. Empirical claims additionally require capability isolation, target absence before the matching prediction seal, custody-checked release, all-row retention and an explicit falsifier. Any violated condition halts. Neither this paper builder nor any branch-specific tool can alter the protected engine or verification authority.

5. External evidence and observation

Observation is indispensable empirical science. IUPAC, NIST, IAEA, primary supporting information, complete databases and authoritative branch-specific sources test sealed consequences. Each measurement retains source, unit, condition, uncertainty and result class where supplied. The work does not use “empirical” to mean institutional approval; it means founded on observation or experiment. Blindness applies to target custody before prediction, not to pretending prior observations do not exist.

6. Preserved failures and lawful repair

Failed capture routes, incomplete tables, unavailable packages, source defects, adverse measurements and implementation-interface halts remain in the audit record. None was promoted into a model failure or used to manufacture performative balance. The HAND family itself preserved a zero-credit graph halt: the census carries owners but not dependencies, so the corrected route reconstructed all 25,013 edges from authoritative registrations before admission. The successful route did not alter a law, target, engine or verifier.

7. How to read the exhaustive ledger

The following 281 claim sections reproduce each statement, root-bound dependencies, generated grammar, complete candidate census, unique survivor, axis eliminations, controls, independent reconstruction, post-seal evidence, exclusions and immutable receipt. The claim packages and receipts remain authoritative; this manuscript is a human-readable projection of them.

8. Measurement, identity, nomenclature and traceability - 8 admitted claims

Chemical science begins with retained identity, declared observation class, exact amount carrier, reversible naming, uncertainty and source traceability. This entire registered subcategory is included; no proper subset is presented as field completion.

1. Chemical entity and retained identity

Claim identity: SFT-CHEM-MEAS-CHEMICAL-ENTITY-001

Question and theorem. A chemical entity is the least singular Fold carrier that remains separately distinguishable after constitution, isotopic identity and the observation context's required precision are retained.

singular-carrier__constitution-and-isotope-retained__separately-distinguishable__context-bounded-precision

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001

Generated grammar. Generate the literal product of the registered entity carrier, constitution, isotope, distinction, precision, record, provenance and extension choices.

Boundary: Every finite singular chemical carrier formed from admitted distinctions and any generated positive-finite extension of its constitution, isotope, state or contextual precision record.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *singular-carrier__constitution-retained__isotope-retained__separately-distinguishable__context-bounded-precision__held-observation-record__source-bound-proof-trace__no-extra-rule*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	singular-carrier	ensemble-without-member	An ensemble alone does not identify one chemical entity.
constitution	constitution-retained	constitution-erased	Erasing constitution merges chemically distinct carriers.
isotope	isotope-retained	isotope-erased	Erasing isotopic identity merges distinguishable entities.
distinction	separately-distinguishable	indistinguishable-answer-label	An answer label without a retained distinction cannot identify an entity.
precision	context-bounded-precision	unbounded-or-fixed-precision	A fixed universal precision either loses a required distinction or adds an unforced one.
record	held-observation-record	result-without-observation-record	Without an observation record the identity boundary cannot be audited.
provenance	source-bound-proof-trace	unbound-provenance	Unbound identity cannot be traced to the Fold dependency spine.
extension	no-extra-rule	free-extra-identity-rule	An extra identity rule can arbitrarily split or merge entities.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:5f110d5c9e6a40120b2d1a5cb5a6dfa2738fc19b611acdb41f21092208f2a84e.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:564377b083b5fd48f74bcd6ffe5e88fbeb00d970d9fb10253723fd39ff85ba41f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:5f6268c44c846a4e91dea61554c4b01f5c85b882c9e894009c0d04e325d2a0a2.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:eec730468fa661f596b4ac9b005665b9817aa9b7a63bb0878ca3cf0ec33b255d.

Independent reconstruction. Implementation hash

sha256:1e162cfb721018f6da4cb2c45eccff8324a1cba92e801169174696005e395938; independent certificate sha256:ab5e4ae45a37d0b2ac8f02f883990992b7c6492fcd59e809620be8a416d6fd86; external-validation hash sha256:3381b260e9fbfb038b71d3a7d4a1114354f6ec3a0127caba3285783315e11ac1. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-M03986-2026

Comparison records:

- chemical-entity-iupac-m03986: predicted
singular-carrier__constitution-and-isotope-retained__separately-distinguishable__context-bounded-precision; source-derived
singular-carrier__constitution-and-isotope-retained__separately-distinguishable__context-bounded-precision; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the current source-derived IUPAC entity record lacks singular distinguishability, constitution/isotope identity or contextual precision, or if any altered row is accepted.

Measurement receipt: sha256:a8cbdf1c9e73a3fac8c4a6da6bd1f3ba81500712a22914fff004ab8af9acafaa.

Isolation certificate: sha256:bebecdd2d7531e6cfd0c22336b0c47ee14ecf94e37d904322b3e2d7c74640.

Custody certificate: sha256:982fd03619647fc6b15956c26ca02bfd28cce9c149de8cf71c209158c7d93e2e.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/M03986/json> (sha256:6b0f6e183ef109d6f3483bcd1d70ca73a6d57e496f9ca32da23964bca652de8e)

Registered target identities:

- chemical-entity-iupac-m03986 from IUPAC-GOLD-BOOK-M03986-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every finite singular chemical carrier formed from admitted distinctions and any generated positive-finite extension of its constitution, isotope, state or contextual precision record.

Immutable identities. Source manifest

sha256:edd9a8adf1425f02310e8915ca2e9398c713acb20ff85f926a1e2bda5126b079; derivation seal sha256:e7ef6a425786f7cf1a84dfce7899765d52c3f1f124586871672fe53a70d79de4; engine receipt sha256:83fa734a2488422429d604cc55e330a76c5aa8de96029007640440ef5162223d at receipts/engine/model_admitted/SFT-CHEM-MEAS-CHEMICAL-ENTITY-001-83fa734a24884224.json; empirical hash sha256:d327ff6d2ffc65a0d6108c17605f093f35dde0e6178b9c6df8f7ab61074c06c6; measurement receipt sha256:a8cbdf1c9e73a3fac8c4a6da6bd1f3ba81500712a22914fff004ab8af9acafaa.

2. Chemical species and observation-equivalence class

Claim identity: SFT-CHEM-MEAS-CHEMICAL-SPECIES-001

Question and theorem. A chemical species is the least Fold quotient whose support is an ensemble of chemically identical entities sharing accessible state support at one declared experimental timescale.

ensemble-carrier__chemical-identity-equivalence__shared-energy-support__experiment-timescale-boundary

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001

Generated grammar. Generate the literal product of the registered species carrier, member identity, state support, timescale, solid support, record, provenance and extension choices.

Boundary: Every finite observation-equivalence class formed from admitted singular chemical entities under an exact declared experiment-timescale relation and any positive-finite successor extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `ensemble-carrier__chemical-identity-equivalence__shared-energy-support__experiment-timescale-boundary__molecular-or-solid-unit-support__held-member-and-class-record__source-bound-proof-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	ensemble-carrier	singular-only	One member alone is not a chemical species class.
member_identity	chemical-identity-equivalence	mixed-chemical-identities	Mixed chemical identities do not form one species.
support	shared-energy-support	energy-support-erased	Erasing accessible-state support merges experimentally distinct species.
timescale	experiment-timescale-boundary	timeless-equivalence	Interconversion can change whether two forms are experimentally distinct.
solid_support	molecular-or-solid-unit-support	isolated-molecule-only	A species law limited to isolable molecules loses solid structural units.
record	held-member-and-class-record	class-without-member-trace	A quotient without member trace cannot be reconstructed.
provenance	source-bound-proof-trace	unbound-provenance	An unbound class can silently change its equivalence rule.
extension	no-extra-rule	free-extra-equivalence	A discretionary equivalence can force any desired species partition.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: a7193b15135c4dbbd92429905bc94c36075dd22dcadc3a76f4198f85f360f0a4.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: e41ee4909562f76a1c4e66cce01a19dc0f86a6e9eb482555c81b63598cf9d228.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: ecff63e0a1fe45c9d7427360c1ff52460290dafcb6e8b33ace20a61491aa215e.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 4363a31941fcec233e36ff41c3cc5d6f7fdef7a860258c722f8ef96cae519289.

Independent reconstruction. Implementation hash

sha256: 0d4dfc16c46530663cc1eb405e4ca6b2ed0599e69792c046629a88fb4f98ff35; independent certificate sha256: cee56a524f22fcf251ad18689682ad1b6469ea97710e2683277a0350c05c517e; external-validation hash sha256: 8f10f11a5511b590d4cfebcf5fd9fc0f07822fd25c8b48d77f006d7f81f21824. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CT01038-2026

Comparison records:

- chemical-species-iupac-ct01038: predicted
ensemble-carrier__chemical-identity-equivalence__shared-energy-support__experiment-timescale-boundary; source-derived
ensemble-carrier__chemical-identity-equivalence__shared-energy-support__experiment-timescale-boundary; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC species record does not retain ensemble identity, common accessible support and experimental timescale, or if an altered row is accepted.

Measurement receipt: sha256:a70d26ed035f31edb2d80558885a6f96149a985aa3a55068af541dc4156f59c8.

Isolation certificate: sha256:1e49153472b2e59ad6a66b3b79ade990393e2b3840459cc9542a87a8ed0bdbff.

Custody certificate: sha256:9f38c749abc435e6773be4a20b482bf214ebe30f95e4787daf402569aa92dd5a.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/CT01038/json>
(sha256:538dd1087bc057eec8b68fdb713987cc8e1f0ff7548b4d03192d67b3e328eafe)

Registered target identities:

- chemical-species-iupac-ct01038 from IUPAC-GOLD-BOOK-CT01038-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every finite observation-equivalence class formed from admitted singular chemical entities under an exact declared experiment-timescale relation and any positive-finite successor extension.

Immutable identities. Source manifest

sha256:b3a32af04a4953c6b3a8db15997ac327b6e96c746dd69d0f8c5a8c2bc038dbec; derivation seal
sha256:d9f9531dc4380dd5d9381c531389925f561870b4af815a126ff215a26a1a2cc3; engine receipt
sha256:865ccab10650d67fedcb6a6f3c31d20887a08fff0ec3ce58de51d4e02a0e3ec4 at receipts/engine/model_admitted/SFT-CHEM-MEAS-CHEMICAL-SPECIES-001-865ccab10650d67f.json; empirical hash
sha256:54fe5ab2437f3f6d564bad61d5d59294faf4231f8f9a17616fbbf9943f6d4f87; measurement receipt
sha256:a70d26ed035f31edb2d80558885a6f96149a985aa3a55068af541dc4156f59c8.

3. Chemical substance and composition identity

Claim identity: SFT-CHEM-MEAS-SUBSTANCE-001

Question and theorem. A chemical substance is the least source-bounded Fold carrier that retains constant composition, its constituent entity support and the complete registered property characterization.

constant-composition__constituent-entity-support__property-characterization__source-bounded-substance-identity

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001

Generated grammar. Generate the literal product of the registered substance composition, constituents, properties, source, phase, record, provenance and extension choices.

Boundary: Every finite chemical substance carrier formed from one generated constant-composition support of chemical entities plus any positive-finite extension of its source-bound property record.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `constant-composition__constituent-entity-support__property-characterization__source-bounded-substance-identity__phase-record-retained__complete-property-record__source-bound-proof-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
composition	constant-composition	variable-unrecorded-composition	Unrecorded variable composition cannot identify one substance.
constituents	constituent-entity-support	anonymous-matter	Matter without retained constituent entities loses chemical identity.
properties	property-characterization	composition-only-answer	Composition alone can leave observation-equivalent candidates unresolved.
source	source-bounded-substance-identity	source-free-substance-label	A source-free label cannot distinguish sample and convention boundaries.
phase	phase-record-retained	phase-silently-collapsed	Silently collapsing phase can erase an observed substance distinction.
record	complete-property-record	property-selection-only	Selecting one favorable property is not a complete characterization record.
provenance	source-bound-proof-trace	unbound-provenance	Unbound composition and property claims cannot be audited.
extension	no-extra-rule	free-extra-substance-rule	A discretionary rule can split or merge substances after observation.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:ad2432d5ad2ab35e20e37628d4c7a34f5468b37f79470eecbe483f318bea4b2a`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:d3b75c1e2afca58bd1ace3b6c79c0db08b4790c8b5614a06f227e517436b2806`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b2afcb7ccd08f7de7686f06d0ee02426d69c0b76843fe38af1c13d393267e211.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:aa8c859776acb4b9507dadcd36885aa443478164dce3f33ad26e21c9f415a45.

Independent reconstruction. Implementation hash

sha256:c2a593300db78987c89c58b98d64c4f82f87cf79cbd340127f5f33ff93303b38; independent certificate
sha256:bb690d7cc17309542fe5180ead6830f124ab9fa1db52c15739778d23046a5210; external-validation
hash sha256:a9ec1291c4725fb003bee8f89dda03907ca047420f66793d66ffc487b68de5ea. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-C01039-2026

Comparison records:

- chemical-substance-iupac-c01039: predicted
constant-composition__constituent-entity-support__property-characterization__source-bounded-substance-identity;
source-derived
constant-composition__constituent-entity-support__property-characterization__source-bounded-substance-identity; exact match
True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC substance record lacks constant composition, constituent entities or property characterization, or if an altered row is accepted.

Measurement receipt: sha256:07dd2ce114199522ed77e99db1c1423e0121dc9169c788538115fa9e67ebe7c5.

Isolation certificate: sha256:5599868e1b678abd3e91192a679c33fae2dcb372d58c23307fdb7b72393d321.

Custody certificate: sha256:ea1357b9654a03e9543cac92aa5a6d39b0f21535a013f1949315d1fe4dedd2c5.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01039/json>
(sha256:97af068278c31568a03402f8dcb3e5de2327833ab9ac8da78c1322e730997498)

Registered target identities:

- chemical-substance-iupac-c01039 from IUPAC-GOLD-BOOK-C01039-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every finite chemical substance carrier formed from one generated constant-composition support of chemical entities plus any positive-finite extension of its source-bound property record.

Immutable identities. Source manifest

sha256:3a7356ac738c7d7e059d6f06ad84b424b3c1fece6b68e58acdbf1d5eeb047238; derivation seal

sha256:78832b5c8c2b0b089a986128bd0af1f14f56040a2ca7ff2fa67043afd2c813ff; engine receipt

sha256:c24408e53e46d5873b5c2f77d638045df99bf6250d5cbae3895cc9a092df15e7 at receipts/engine/model_admitted/SFT-CHEM-MEAS-SUBSTANCE-001-c24408e53e46d587.json; empirical hash sha256:18c5020cf21e14182317d1cefa0c0746f1e322a95e068f71704941d2cb758c2b; measurement receipt sha256:07dd2ce114199522ed77e99db1c1423e0121dc9169c788538115fa9e67ebe7c5.

4. Amount-of-substance count and reference carrier

Claim identity: SFT-CHEM-MEAS-AMOUNT-001

Question and theorem. Amount of substance is the exact Fold measure carried by a positive-finite generated count of one specified elementary-entity kind, with the entity reference and count trace retained.

amount-carrier__count-measure__specified-entity-reference__generated-entity-support

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-SUBSTANCE-001

Generated grammar. Generate the literal product of the registered amount carrier, entity reference, count, measure, support, record, unit and extension choices.

Boundary: Every positive-finite generated collection of one specified admitted chemical-entity kind, together with its exact member/reference trace and separately registered unit comparison.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *amount-carrier__specified-entity-reference__generated-cardinality__count-measure__generated-entity-support__member-and-reference-trace__separate-unit-comparison__no-extra-rule*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	amount-carrier	answer-only-amount	An answer label does not retain the entities being counted.
entity_reference	specified-entity-reference	unspecified-entities	A count without a specified elementary-entity kind changes meaning under substitution.
count	generated-cardinality	mass-proxy-count	A mass proxy can vary with isotope and cannot establish entity cardinality.
measure	count-measure	unrelated-count-and-quantity	An unrelated quantity does not measure how many referenced entities occur.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
support	generated-entity-support	single-presupposed-entity-kind	Presupposing one conventional entity kind excludes other generated elementary carriers.
record	member-and-reference-trace	total-without-members	A total without its entity reference cannot be independently reconstructed.
unit	separate-unit-comparison	unit-scale-as-proof	A conventional scale cannot select the amount law.
extension	no-extra-rule	free-amount-factor	An extra multiplier is a free answer-selecting parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:530a1ab514b800a0675a6f969439574a59117f367012e3ef8055b6cec3ebcc12.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:caafe4fbfd3413d2a4eeb2d4e1e893b744e6af78312e4a16209bccb295e91fef.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b4ba2f18586ba7dd2c311b1e9d1a1be3fdbd7f3980668604bd20becae6b4cd02.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:939ec2105ede1067163abcfe2685d946baa15236b910a9e6f925a5d31ef0d46f.

Independent reconstruction. Implementation hash

sha256:7876fb1d2c9a700c5019ad446b918591401f0b872b74241cd017f0208c2f1a22; independent certificate sha256:64d8d926722782a15875c9eaf0a522aae7879683644af93d38b2082865374622; external-validation hash sha256:8e7e27f1fcbafec108f14001399de4427ffa6d1b3760851975627c965368923d. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-A00297-2026

Comparison records:

- amount-of-substance-iupac-a00297: predicted
amount-carrier__count-measure__specified-entity-reference__generated-entity-support; source-derived
amount-carrier__count-measure__specified-entity-reference__generated-entity-support; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC amount record lacks number measure, specified entities or general elementary-entity support, or if an altered row is accepted.

Measurement receipt: sha256:217a7bfff7b4189a60c81a31ab1d17b5f886aaf6740a7a42638ce0ce65cc9372.

Isolation certificate: sha256:e1ebc507ece894d6014db01022e62223303f7673330476adeb4d362749f55019.

Custody certificate: sha256:e7526ad87c327a16c35fbaba8bb7226210bc07d0588ae94f76122ce8e94fe0b1.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/A00297/json>
(sha256:ddf59230caa12b510125795af6f624797e904a1724fe970b3c144af47167077e)

Registered target identities:

- amount-of-substance-iupac-a00297 from IUPAC-GOLD-BOOK-A00297-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence

boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every positive-finite generated collection of one specified admitted chemical-entity kind, together with its exact member/reference trace and separately registered unit comparison.

Immutable identities. Source manifest

sha256:1a1d716baf7c98101641cfc4441fe9d6cafa91853cd7c9a61592b2738143f94c; derivation seal sha256:9ca771cfc770aa83330108216d98647ad18621e7dd2fccfc935ba2df2e4e4b28; engine receipt sha256:4624037342fc423836b2ca0e1ffae15976d21b67b02a71cbf9e6202063bfd990 at receipts/engine/model_admitted/SFT-CHEM-MEAS-AMOUNT-001-4624037342fc4238.json; empirical hash sha256:6636cf5a5f91815ab2d0cd763bab1c1676f88b01e740d66f38dd55855df606d3; measurement receipt sha256:217a7bfff7b4189a60c81a31ab1d17b5f886aaf6740a7a42638ce0ce65cc9372.

5. Chemical formula as exact composition encoding

Claim identity: SFT-CHEM-MEAS-FORMULA-001

Question and theorem. A molecular formula is the least decodable Fold word that retains a discrete molecule's constituent entity labels, exact positive multiplicities and any structure coordinate required by the declared identity boundary.

discrete-molecule-domain__formula-encoding__composition-mass-correspondence__structure-correspondence

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-MEAS-AMOUNT-001

Generated grammar. Generate the literal product of the registered formula domain, carrier, constituents, multiplicity, mass, structure, decoding and extension choices.

Boundary: Every positive-finite discrete-molecule composition word generated from admitted entity labels, exact multiplicities and a held structural-order record.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `discrete-molecule-domain__formula-encoding__held-entity-symbols__exact-positive-multiplicity__composition-mass-correspondence__structure-correspondence__composition-decodable__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
domain	discrete-molecule-domain	unbounded-object-label	A formula without a declared chemical carrier domain can encode unrelated objects.
carrier	formula-encoding	name-only-string	A name alone does not retain exact constituent multiplicity.
constituents	held-entity-symbols	anonymous-symbols	Anonymous symbols erase chemical-entity identity.
multiplicity	exact-positive-multiplicity	unrecorded-repetition	Unrecorded repetition cannot reconstruct composition or relative mass.
mass	composition-mass-correspondence	mass-unrelated-to-composition	A mass label not reconstructed from the constituent record is not a formula consequence.
structure	structure-correspondence	order-erased	Equal counts can represent distinct molecular structures.
decode	composition-decodable	nondecodable-inscription	An inscription that cannot return its constituent record is not an exact encoding.
extension	no-extra-rule	free-formula-exception	A discretionary exception can manufacture a desired formula.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3845fb5f1a0403ba69cc287efe987ff69f8c3e9f1acf7a480f9aea44f03fdadd`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:2a15902eb8c0b79689cca8acb12b7c410d6e39222d9f24e4135db5e49740ba4f`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6d635c7023bcc25b4967611c84b32596233cfd7d5cf35afa38b87999e78d31a0`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:aa7e7ab21abb28659b496319db0ef2d0e3706b69d707305a9813d1f5ef243140`.

Independent reconstruction. Implementation hash

`sha256:16580eadc4660d4c0de70006350f24437e5dbafe427f71d0925f4cb62a62eb45`; independent certificate `sha256:cae7f79d06ad779f29123b54de65efd692475f4b1b10617a6ff41ce3049089a9`; external-validation hash `sha256:95d3cf4a9e1ee4a85722e5970080d238d504a95c6e10b1564efb0092af0df3fb`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-M03987-2026

Comparison records:

- **molecular-formula-iupac-m03987:** predicted `discrete-molecule-domain__formula-encoding__composition-mass-correspondence__structure-correspondence`; source-derived `discrete-molecule-domain__formula-encoding__composition-mass-correspondence__structure-correspondence`; exact match

True

- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC molecular-formula record lacks its discrete-molecule domain, formula carrier, relative-mass or structure correspondence, or if an altered row is accepted.

Measurement receipt: sha256:5dcb89990eec58be42b4396126b12f49c3d364dcf96b9d8e79a829b063c375f7.

Isolation certificate: sha256:35a01b248162986becb6c2371412a2d7839fabbf24af45dd08edfde9cc3c17e5.

Custody certificate: sha256:e91b20ddfb23218c110bb4441b26da56fff7e08db404b20268e892a5a79faa6c.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/M03987/json> (sha256:b33959596338e0e2797247c379e8214ecd6a58abd3e929a4dfad36f42b081704)

Registered target identities:

- molecular-formula-iupac-m03987 from IUPAC-GOLD-BOOK-M03987-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every positive-finite discrete-molecule composition word generated from admitted entity labels, exact multiplicities and a held structural-order record.

Immutable identities. Source manifest

sha256:b40c263d86bd9154d3ea4519bf72bfb0b0706aab90ec2dd8fc55b648b2ce67ea; derivation seal

sha256:3c9f9731257eee1f02754dd85a0dd9689e407718c97fb304d402498181ea8d0a; engine receipt

sha256:785d16f1c8f61f25624bfff347d919829dae332431e5a90d27446cc25db6dcb9c at

receipts/engine/model_admitted/SFT-CHEM-MEAS-FORMULA-001-785d16f1c8f61f25.json; empirical

hash sha256:8147c90803fce0cbc11682f6a5be2a14ce2176d6e78e8a1bac66ef329ca0bd8c; measurement

receipt sha256:5dcb89990eec58be42b4396126b12f49c3d364dcf96b9d8e79a829b063c375f7.

6. Reversible chemical nomenclature and identity boundary

Claim identity: SFT-CHEM-MEAS-NOMENCLATURE-001

Question and theorem. A systematic chemical name is a complete ordered Fold word assembled from registered name parts and optional count prefixes; exact reversibility is admitted only on the generated injective name-to-identity boundary.

complete-name-carrier__generated-name-parts__count-prefix-support

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001

- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-FORMULA-001

Generated grammar. Generate the literal product of the registered name carrier, parts, prefix, order, formula binding, decoding, record and extension choices.

Boundary: Every positive-finite ordered chemical-name word assembled wholly from registered syllable labels and optional generated count prefixes, with an explicit injective-decoding boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-name-carrier__generated-name-parts__count-prefix-support__held-name-order__formula-bound-name__injective-boundary-decoding__complete-name-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-name-carrier	partial-name-fragment	A fragment alone does not carry one complete registered chemical name.
parts	generated-name-parts	unregistered-free-text	Free text can import arbitrary answer-bearing distinctions.
prefix	count-prefix-support	implicit-count	An implicit multiplicity cannot be recovered from the name.
order	held-name-order	unordered-syllable-bag	Unordered parts can merge names with distinct composition.
binding	formula-bound-name	name-detached-from-formula	A detached name cannot preserve chemical identity.
decode	injective-boundary-decoding	unconditional-reversibility	Not every conventional name is uniquely decodable, so reversibility cannot be assumed.
record	complete-name-trace	surface-name-only	A surface name alone hides its assembly and decoding boundary.
extension	no-extra-rule	free-naming-exception	A discretionary naming exception can select any desired identity.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:363d36bc6f6549e4e0530b91da9cf081f8d8618f41d78e08b72d9307ca63e793`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:5eed2098f5a9a626e7aa35bb5ca2da5d3c9ae4c9a5bf310a635b7bcfc898655e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:213e9b19aa90d67471f010fc8b8919b0c63247f7c71195d457576cf1b0da6e6e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:19185c053f8a90242fa96295f5bcd2329eb4dee3c69ece49d5d51993607c075a.

Independent reconstruction. Implementation hash

sha256:35e60d416791137fa320e94b41a12eed0e18a1099c1b862537158c24d45f4fb9; independent certificate sha256:33534ecd55d79bd46e31fa8e85c98a78d1a0d48f961fa421774c1c17b1936a14; external-validation hash sha256:8fcaedfc67328dfafa8a26714a7ea5849996201f02186b63ef92bdf917422ed6. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-S06236-2026

Comparison records:

- systematic-name-iupac-s06236: predicted complete-name-carrier__generated-name-parts__count-prefix-support; source-derived complete-name-carrier__generated-name-parts__count-prefix-support; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC systematic-name record lacks complete name composition, selected syllable parts or optional numerical prefixes, or if reversibility is claimed outside the registered injective boundary.

Measurement receipt: sha256:aadb478faba130ffd564df0f9da1440f1ad9e2f798993fc19cbcc4ebf6db89ec.

Isolation certificate: sha256:d3fbc2a40a1fff419a86af2844cbfb552b8258da7a78a362028bb7c27349a113.

Custody certificate: sha256:a2e1c29a7b881e3e1877135ac45a0c563247d2e5aaaf92a9e04eb5a1439b0cb1.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/S06236/json> (sha256:a3163fe9e9694283c7280d87679fb49286dd44631d908c19fcc164ce8f2984ac)

Registered target identities:

- systematic-name-iupac-s06236 from IUPAC-GOLD-BOOK-S06236-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every positive-finite ordered chemical-name word assembled wholly from registered syllable labels and optional generated count prefixes, with an explicit injective-decoding boundary.

Immutable identities. Source manifest

sha256:fc36ae332375e584807b3698f64fe9ed2ce4635fbe869b7b76f87a670cb6c246; derivation seal

sha256:0a2b1ad4e8c2490a72b95966095911679522bb11742122ce511f6b9694be6756; engine receipt

sha256:12634338b801beb64d48817a76f405efc7a77593e5e74c62507fb0a458c63870 at

receipts/engine/model_admitted/SFT-CHEM-MEAS-NOMENCLATURE-001-12634338b801beb6.json;

empirical hash sha256:d4b533d7430144a24c17ecf3ab92668e137f236291d3cc2f112d64a8601edb0a;

measurement receipt sha256:aadb478faba130ffd564df0f9da1440f1ad9e2f798993fc19cbcc4ebf6db89ec.

7. Chemical measurement uncertainty and retained alternatives

Claim identity: SFT-CHEM-MEAS-UNCERTAINTY-001

Question and theorem. Chemical target uncertainty is a source-bound Fold record of unresolved measurement alternatives with a preregistered upper limit fixed by intended use and attached to the exact result identity.

`uncertainty-carrier__upper-bound-support__use-bounded-resolution__result-record-boundary`

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-AMOUNT-001

Generated grammar. Generate the literal product of the registered uncertainty carrier, alternatives, bound, use, result, source, record and extension choices.

Boundary: Every source-bounded finite chemical measurement record whose unresolved alternatives, exact upper bound, intended use and result identity are retained without entering the derivation as proof values.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `uncertainty-carrier__retained-alternative-class__upper-bound-support__use-bounded-resolution__result-record-boundary__source-bound-uncertainty__complete-uncertainty-record__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	uncertainty-carrier	single-answer-without-resolution	A lone answer erases the unresolved observation class.
alternatives	retained-alternative-class	discarded-alternatives	Discarding compatible alternatives falsely turns resolution into exact identity.
bound	upper-bound-support	unbounded-vagueness	Unbounded uncertainty cannot falsify a measured correspondence.
use	use-bounded-resolution	context-free-resolution	Required resolution depends on the declared purpose of the result.
result	result-record-boundary	bound-detached-from-result	A bound detached from its result cannot be applied or audited.
source	source-bound-uncertainty	source-free-bound	A source-free bound can be changed after comparison.
record	complete-uncertainty-record	favourable-row-only	Selected rows cannot test uncertainty transport.
extension	no-extra-rule	free-uncertainty-margin	An adjustable margin can force agreement with any target.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : f33978481bee18a785e521a82b83da05e66ad23051bf34af7e6ba8e6222893ac.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 2f46e0952e59ff9947a89d6942dd04a2175a7182e033d5c137c0d7d097b50848.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 259fc17f95d9dab542b48fca5a0d0362fabbb280b7315b7130540653701ef78d6.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : de5d0150aeecfffacf4ac229cb1552a22b70e6afa6c0f27ecb72cdd228a34927.

Independent reconstruction. Implementation hash

sha256 : 8b0ba2478d537cb6f9383f894be69dab1485aecb826ef31b9205e556bdc6fbaf; independent certificate sha256 : e9f1432b80fd5ed55c607e74ee387292fbb941e02ed52f76f923b53cca2f70dd; external-validation hash sha256 : fe29a6826bd4dc01efdd4d585e7752ed04037e736102d83e8d4842a797923614. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-08133-2026

Comparison records:

- target-uncertainty-iupac-08133: predicted
uncertainty-carrier__upper-bound-support__use-bounded-resolution__result-record-boundary; source-derived
uncertainty-carrier__upper-bound-support__use-bounded-resolution__result-record-boundary; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC target-uncertainty record lacks an upper limit, intended-use boundary or measurement-result attachment, or if an altered row is accepted.

Measurement receipt: sha256 : 6b359a0b5cbe12775b8a18e26925db947b9ca0599bc79d175809f4aed70747b.

Isolation certificate: sha256 : ec8fd145f113a00ab0d5297f1323dbef429d8e2e2ebc6c59ef95524c106f650f.

Custody certificate: sha256 : 55d5567e4a24c34b28add73194fb0f5e8d43a69d68c11e1495d1a4a0ed51b702.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/08133/json>
(sha256:014b6b034695f2b25c07cd4e5474b856aeecf8907e4022595cc93df2f6674dc1)

Registered target identities:

- target-uncertainty-iupac-08133 from IUPAC-GOLD-BOOK-08133-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody

- Every source-bounded finite chemical measurement record whose unresolved alternatives, exact upper bound, intended use and result identity are retained without entering the derivation as proof values.

Immutable identities. Source manifest

sha256:eb2f6af8596ba6e137596d6c6f7f23234bd6396ec3528c9934ff69425d695001; derivation seal
 sha256:b52e317a4560154f76b869d3146f52554d979c3fee511a0e49826478cda3e439; engine receipt
 sha256:50c9951b09bb54919aafde3a431ce4b294cf2f6c18f93e8abf52f5d5b78ba37a at
 receipts/engine/model_admitted/SFT-CHEM-MEAS-UNCERTAINTY-001-50c9951b09bb5491.json;
 empirical hash sha256:c7bde3a970ab4faf78e8ff61fd1e39f736aaa61c8ccd05bbd7df2fd68a127458;
 measurement receipt sha256:6b359a0b5cbec12775b8a18e26925db947b9ca0599bc79d175809f4aed70747b.

8. Chemical reference traceability and complete record

Claim identity: SFT-CHEM-MEAS-TRACEABILITY-001

Question and theorem. Chemical traceability is the complete Fold predecessor chain binding a property value to an identified reference material while retaining uncertainty, material identity and its certificate record at every link.

reference-material-property__uncertainty-retained__traceability-retained__certificate-bound-record

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-MEAS-REFERENCE-REALIZATION-001
- SFT-PHYS-MEAS-CALIBRATION-001
- SFT-CHEM-MEAS-UNCERTAINTY-001

Generated grammar. Generate the literal product of the registered property, reference, uncertainty, trace, certificate, identity, record and extension choices.

Boundary: Every positive-finite certified chemical-property chain whose reference material, property identity, uncertainty, predecessor links and certificate record are retained to an admitted reference realization.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *reference-material-property__identified-reference-realization__uncertainty-retained__traceability-retained__certificate-bound-record__material-identity-held__complete-chain-record__no-extra-rule*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
property	reference-material-property	value-without-property	A value without a property identity cannot be traced.
reference	identified-reference-realization	anonymous-reference	An anonymous comparison object cannot anchor a trace chain.
uncertainty	uncertainty-retained	uncertainty-erased	Erasing uncertainty overstates what each comparison preserves.
trace	traceability-retained	broken-predecessor-chain	A missing predecessor prevents reconstruction to the reference.
certificate	certificate-bound-record	uncertified-result-label	A result label alone does not bind property, reference and trace.
identity	material-identity-held	substitutable-sample	Silent sample substitution breaks traceability while preserving a surface value.
record	complete-chain-record	terminal-value-only	A terminal value cannot reproduce the chain.
extension	no-extra-rule	free-trace-link	An unregistered link can hide a discontinuity.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:97e310c97ca4153e8085aa29d2b588832b6425e4220e59493ed9a11b6d0a9c48.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:0a72df6d765e4516c4b2e09d9092b1bcef4c9665423eb32a35aa1eec79899488.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b1151e9de01fe193c6b7934ff2bca5000776b28dc01480a985bb08dc9de5e007.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:6743deb0003e20c5af2dac5d032ad5d67023c98dd7446991ec75a06c01abee11.

Independent reconstruction. Implementation hash

sha256:87aa49285efc58f739ef9b98990ae232ca5b846a3d052775904c353924df74bd; independent certificate sha256:1de8c820992757a3ec3499dda3159ad7f6e874af66e63b62d67c21a873a17a8c; external-validation hash sha256:57a15a3660637f886f514270ed86b157f1fa6a1610d48414ad536901900b234a. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-08013-2026

Comparison records:

- certified-property-iupac-08013: predicted
reference-material-property__uncertainty-retained__traceability-retained__certificate-bound-record; source-derived
reference-material-property__uncertainty-retained__traceability-retained__certificate-bound-record; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC certified-property record lacks reference material, associated uncertainty, traceability or certificate binding, or if an altered row is accepted.

Measurement receipt: sha256:c4214ad8a9353be59a92f1a00e7e2c340c41bdf9451ff71ef0cef1601c274ead.

Isolation certificate: sha256:d0bc0d3ed023ac1f2829441a7d6ec28ed35f35481e02f234fcc4d8595e4f05ad.

Custody certificate: sha256:491258a615c4f9185cf831ad849043a799c75552f20990e1d146e6efe793269d.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/08013/json> (sha256:34656c42ec7d741c00a0006b462487115d51dcd3d918d2fa64982ce100e24868)

Registered target identities:

- certified-property-iupac-08013 from IUPAC-GOLD-BOOK-08013-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every positive-finite certified chemical-property chain whose reference material, property identity, uncertainty, predecessor links and certificate record are retained to an admitted reference realization.

Immutable identities. Source manifest

sha256:78050b6e1018b12af790992ef4a12aa2de05345a1fbba245ddd446dbdcfbf86c1; derivation seal sha256:49df28197171b8dd875f208c53f01a010ee0263c4d69c15e72f666284b53189a; engine receipt sha256:74157ee587b7b0d43847e4e753c69637e4b0eecec84a443c2098d40078492269 at receipts/engine/model_admitted/SFT-CHEM-MEAS-TRACEABILITY-001-74157ee587b7b0d4.json; empirical hash sha256:fa39761239fa870491e1533365dc6a82d5792ed3a4521d6fbddd8ed3b2c30efb; measurement receipt sha256:c4214ad8a9353be59a92f1a00e7e2c340c41bdf9451fff71ef0cef1601c274ead.

9. Elements, isotopes, ions and periodic structure - 10 admitted claims

Elements, isotopes, ions and periodic recurrence retain exact proton, neutron, electronic and observation boundaries. This entire registered subcategory is included; no proper subset is presented as field completion.

9. Chemical element identity

Claim identity: SFT-CHEM-ELEM-ELEMENT-001

Question and theorem. A chemical element is the complete Fold class of generated atoms sharing one held nuclear proton count; isotope mass refinements preserve that identity, while pure-substance use remains a separate correspondence.

atom-species-carrier__common-proton-count__nuclear-identity__substance-carrier-correspondence

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001

- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001
- SFT-PHYS-NUCLEAR-LEVELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001

Generated grammar. Generate the literal product of the registered element carrier, identity, location, membership, isotope, substance, record and extension choices.

Boundary: Every positive-finite atom-species class generated from one held nuclear proton-count identity, including all mass-number refinements and its separately recorded pure-substance correspondence.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `atom-species-carrier__common-proton-count__nuclear-identity__all-generated-atoms-with-identity__mass-refinement-permitted__substance-carrier-correspondence__complete-nuclear-membership-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	atom-species-carrier	substance-name-only	A substance name does not define the atom class.
identity	common-proton-count	mass-selected-identity	Mass can vary between isotopes of one element.
location	nuclear-identity	protons-without-nuclear-carrier	An unlocated count does not preserve atomic constitution.
membership	all-generated-atoms-with-identity	sampled-atom-list	A sample cannot close the complete generated class.
isotope	mass-refinement-permitted	one-mass-number-only	Restricting an element to one mass number erases its isotopes.
substance	substance-carrier-correspondence	species-substance-conflation	An atom species and a pure substance are related but not identical carriers.
record	complete-nuclear-membership-trace	element-label-only	A label alone cannot reproduce class membership.
extension	no-extra-rule	free-element-exception	A discretionary exception can split or merge element classes.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:76dd7261cbd3b67042302e0d1db175f36c10d018ae297517613179d05638a9d5`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:13b7765dcc0f65b3252315344aad5c8a70c3fc576d445f435ec090dcd81e8e61`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:09f6d83fe72cd5c693cec7b72ad03d414521081d8b0934b0289a7580b0df232f`.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : f00796ad6f37a439619cea9a44469afcd69aa1d406af0949ab0b2172048d418a.

Independent reconstruction. Implementation hash

sha256 : 099700048127608e9d9b7d95a855a9c6cf5f79d2e8a37ebc5198fadbba1ca6b5; independent certificate
sha256 : be152c894873bde8bc0c4fa13caed3619dcde05f4a97b6d1351f0fcbf5bc3027; external-validation
hash sha256 : f16c090887352e1d067bf0b19f69696aaed1dde8636a2f8148d837c352fd0820. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-C01022-2026

Comparison records:

- chemical-element-iupac-c01022: predicted
atom-species-carrier__common-proton-count__nuclear-identity__substance-carrier-correspondence; source-derived
atom-species-carrier__common-proton-count__nuclear-identity__substance-carrier-correspondence; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC element record lacks an atom-species carrier, common proton count, nuclear identity or pure-substance correspondence, or if an altered row is accepted.

Measurement receipt: sha256 : c07fa3d86a90350abbc497d3fd63e8dd18b790d0e9b6382af5773ca4c4a29698.
Isolation certificate: sha256 : cf4c0c68a1bf4548322311658474ea6b7abd916450c6bc1bd6ecade0c82a7d5f.
Custody certificate: sha256 : ab17c654aad25499f3808ff68f2a8aa5a5fe132b5fab9482cf10f1bfcccb0d9f.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01022/json>
(sha256:6d75c5b2d5299402b7f35243e248cb60d4972553d3bb622490236e7b889d9575)

Registered target identities:

- chemical-element-iupac-c01022 from IUPAC-GOLD-BOOK-C01022-2026

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, periodic-table arrangement, measured element value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every positive-finite atom-species class generated from one held nuclear proton-count identity, including all mass-number refinements and its separately recorded pure-substance correspondence.

Immutable identities. Source manifest

sha256 : 8c0a23a9984440b8c5f8491be1c2a789fa5f391df305785b65f28055681f23e4; derivation seal
sha256 : f915e1cf6b8a26efdb39b22a7421cd1cccebf80f2467ecf9588f333f2c56f613; engine receipt
sha256 : be03c5e39699037ada9127372d6c8c02d713538c36de31904a5015100751f385 at
receipts/engine/model_admitted/SFT-CHEM-ELEM-ELEMENT-001-be03c5e39699037a.json; empirical
hash sha256 : 1ea5b00bec5fc468766bb583749e338ee865b5684f2ca421d071ac2e1afcc922; measurement
receipt sha256 : c07fa3d86a90350abbc497d3fd63e8dd18b790d0e9b6382af5773ca4c4a29698.

10. Atomic-number ordering and element distinction

Claim identity: SFT-CHEM-ELEM-ATOMIC-NUMBER-001

Question and theorem. Atomic number is the exact generated positive-finite count of protons in an atom's nucleus and therefore the unique order coordinate of its chemical-element class.

exact-proton-count__nuclear-count-carrier

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001
- SFT-PHYS-NUCLEAR-LEVELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-ELEM-ELEMENT-001

Generated grammar. Generate the literal product of the registered atomic-number carrier, location, order, element map, isotope, charge, record and extension choices.

Boundary: Every generated positive-finite nuclear proton collection and its one-to-one order coordinate over admitted chemical-element classes.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *exact-proton-count__nuclear-count-carrier__positive-count-order__one-count-one-element-class__isotope-invariant__charge-state-invariant__count-nucleus-element-trace__no-extra-rule*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	exact-proton-count	symbol-position-only	A table position cannot define the nuclear quantity.
location	nuclear-count-carrier	whole-atom-particle-count	Counting every atomic particle changes across isotopes and charge states.
order	positive-count-order	unordered-label	An unordered label cannot supply periodic succession.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
element_map	one-count-one-element-class	many-elements-one-count	Two element classes with one proton count violate their identity condition.
isotope	isotope-invariant	mass-dependent-number	Mass-number variation must not alter element order.
charge	charge-state-invariant	electron-count-dependent	Ion formation changes electrons without changing the element.
record	count-nucleus-element-trace	number-without-nucleus	A bare number loses the physical carrier and class mapping.
extension	no-extra-rule	free-order-offset	An offset can relabel the periodic order arbitrarily.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:4ddd54cd7947aa0e3b2aa3d03493a8c34c888edcfc7dea36e9282471cc6c308c.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:ed80251591ba927846d8a1247ed87eb592801a08102e5dbde5cac482563e7ef7.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:be2dd24a591be28884b941432fdafcdb712120c27fd0008b55795cad15fab594.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:db5a33c3d4c0529fb072f64aaf74b52b89281189de1ad94dc76b985ad0466b81.

Independent reconstruction. Implementation hash

sha256:a562cb4295bee04704e6d6f298f9400a671a00306a4c3e99f78b3192e9dec8c7; independent certificate sha256:50a96aa5a9fc736f9b208bcbbaa5ad98be89a75d160feb645174f208c0c28123d; external-validation hash sha256:8201d561d359f522fa8ab70184d0dae56a88d1ca12acc124793cf79f0aaa804e. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-A00499-2026

Comparison records:

- atomic-number-iupac-a00499: predicted exact-proton-count__nuclear-count-carrier; source-derived exact-proton-count__nuclear-count-carrier; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC atomic-number record is not the proton count in the atomic nucleus, or if an altered row is accepted.

Measurement receipt: sha256:d48f452685c24d17685eaaeeb8cfff838227cc5590cd9d33883a6cfa122e17b9b.

Isolation certificate: sha256:f9cc17100213391f51917c76c92638470637cea2c407723aed729b957ae7866b.

Custody certificate: sha256:d71f462ba49a7eeff4fba1ae47053a2182337b2684b7c5673dfb234a711367e9.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/A00499/json> (sha256:6ba53740329bcbdde38d8a8734282fb9414a1c7e62fad09743d0890890ff3e127)

Registered target identities:

- atomic-number-iupac-a00499 from IUPAC-GOLD-BOOK-A00499-2026

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered

grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, periodic-table arrangement, measured element value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every generated positive-finite nuclear proton collection and its one-to-one order coordinate over admitted chemical-element classes.

Immutable identities. Source manifest

sha256:c4f7a9d322a2a21a254af13f52d7339605fff891f0cf63cfa8311d78f9d72fab; derivation seal sha256:46fc75e6ffcf5e68259efbacb62e7114a8809dbad3c22b2403290d0053214908; engine receipt sha256:c63df67479dff616130c735b7e2f407ddba06d242ebdbe45067be655337a9cd9 at receipts/engine/model_admitted/SFT-CHEM-ELEM-ATOMIC-NUMBER-001-c63df67479dff616.json; empirical hash sha256:a7c0d7f6a745501691648cdfb492827fbb94d5bd6143d59d9358084589933d2f; measurement receipt sha256:d48f452685c24d17685eaaeeb8cff838227cc5590cd9d33883a6cfa122e17b9b.

11. Isotope identity within an element

Claim identity: SFT-CHEM-ELEM-ISOTOPE-001

Question and theorem. Isotopes are the generated Fold family of nuclide carriers sharing one atomic number while retaining different exact mass-number coordinates.

nuclide-carriers__element-identity-retained__mass-number-distinction

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001
- SFT-PHYS-NUCLEAR-LEVELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001

- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001

Generated grammar. Generate the literal product of the registered isotope carrier, element, mass, complementary nuclear count, chemistry, observation, record and extension choices.

Boundary: Every positive-finite family of admitted nuclides sharing one atomic-number identity while retaining each distinct generated mass-number coordinate and complete nuclear record.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `nuclide-carriers__element-identity-retained__mass-number-distinction__complementary-nuclear-count-held__common-element-chemistry-boundary__resolution-bounded-distinction__complete-nuclide-family-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	nuclide-carriers	atom-label-without-nucleus	An atom label alone does not establish nuclide identity.
element	element-identity-retained	mixed-atomic-numbers	Different proton counts are different elements, not isotopes.
mass	mass-number-distinction	mass-number-erased	Erasing mass number merges distinct nuclides.
neutron	complementary-nuclear-count-held	unrecorded-nuclear-difference	The source of mass-number distinction cannot remain anonymous.
chemistry	common-element-chemistry-boundary	different-element-chemistry	Changing element identity does not define an isotope relation.
observation	resolution-bounded-distinction	always-indistinguishable	Isotopic distinctions are resolvable in registered measurements.
record	complete-nuclide-family-trace	isotope-name-only	A name cannot reconstruct proton and mass coordinates.
extension	no-extra-rule	free-isotope-equivalence	An arbitrary equivalence can merge different elements or erase mass distinctions.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8b484361d78650d9a8282675607175a29a3f07500429fe30cdbc8be83fb806e8`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:d12deb5f406b68526427218796918b712da246387d65d777d60f2500d0d4f4a0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3b803889d00f08f745238ff5c3ba28066c73e87c0d9722104dc3c1b7ef514893`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:aca608666ff8ba3aa7e88511159717815c73d806611043bb4da1be7d86e7cc4b`.

Independent reconstruction. Implementation hash

`sha256:d5f0ab0003ff6099b3b93cd20d190c1ff1c2b231255d493a8e3275e5c580c8d8`; independent certificate `sha256:79130b42b41148a1bc0c2d74d50493a322b3eba6375c8a84a6f992c33a84db56`; external-validation hash `sha256:b4c81894550f33d0e71e526f5476b8343aa1540be378a25381e04693a1af9d49`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-I03331-2026

Comparison records:

- isotopes-iupac-i03331: predicted nuclide-carriers__element-identity-retained__mass-number-distinction; source-derived nuclide-carriers__element-identity-retained__mass-number-distinction; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC isotope record does not retain nuclides, common atomic number and distinct mass numbers, or if an altered row is accepted.

Measurement receipt: sha256:22f9282f80173d56aab5eb43ec05334720d329a7ccf94a22da57e2ffba960e39.

Isolation certificate: sha256:e57015b04a46d486eb229ba255bdb1db58a25fb4d84d2859326e6ef437e0fbd1.

Custody certificate: sha256:feaab618e9246f8c295cc7cdd957388e9eaac6b022c49c7b2fb74e32678a2ed2.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/I03331/json> (sha256:7b5ca7fe293902804ca76194a6ac35a57a35bfa489245c93979eb81370a6635c)

Registered target identities:

- isotopes-iupac-i03331 from IUPAC-GOLD-BOOK-I03331-2026

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, periodic-table arrangement, measured element value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every positive-finite family of admitted nuclides sharing one atomic-number identity while retaining each distinct generated mass-number coordinate and complete nuclear record.

Immutable identities. Source manifest

sha256:53ad765ceac8ab8e1e256f794b30d3f3aea5509cc9d8251335317491df2570ba; derivation seal

sha256:747be26bec8ce69d2d436a3bbcc70621cb9161794a79e576ebf1f5ba52c9c5ea; engine receipt

sha256:7cad83d03dcae4993df715fc0b9b92aecbd991c847b48e20d5825a5b12e70d31 at

receipts/engine/model_admitted/SFT-CHEM-ELEM-ISOTOPE-001-7cad83d03dcae499.json; empirical

hash sha256:956e53b8175ca37baa75f8ec307e05c01ff526e23e3d53cb6f3850b3b18c6ef4; measurement

receipt sha256:22f9282f80173d56aab5eb43ec05334720d329a7ccf94a22da57e2ffba960e39.

12. Atomic-weight record and isotopic composition boundary

Claim identity: SFT-CHEM-ELEM-ATOMIC-WEIGHT-001

Question and theorem. Relative atomic mass is the exact source-bounded Fold ratio of a finite atom population's retained average mass to a separately registered unified atomic-mass reference, with isotope support and measurement uncertainty preserved.

exact-ratio-carrier__average-atomic-mass__reference-unit-comparison

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001
- SFT-PHYS-NUCLEAR-LEVELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-MEAS-AMOUNT-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-ELEM-ISOTOPE-001

Generated grammar. Generate the literal product of the registered atomic-weight carrier, population, isotope support, mass pairs, reference, source, uncertainty and extension choices.

Boundary: Every finite source-bounded atom population with retained isotope membership, exact positive rational abundance parts, atomic-mass records and a separate unified-reference-unit comparison.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `exact-ratio-carrier__average-atomic-mass__isotope-support-retained__isotope-mass-pairs__reference-unit-comparison__source-bounded-population__measurement-interval-retained__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	exact-ratio-carrier	decimal-answer-only	A decimal answer hides population, isotope and reference provenance.
population	average-atomic-mass	single-atom-mass	Atomic weight concerns a population average rather than an arbitrary single atom.
isotopes	isotope-support-retained	isotope-mixture-erased	Erasing isotope support loses the source of population variation.
mass	isotope-mass-pairs	unbound-mass-labels	Unbound masses cannot be paired with their isotope parts.
reference	reference-unit-comparison	unit-as-derived-law	A conventional unit scale cannot select the structural ratio.
source	source-bounded-population	universal-fixed-composition	Natural isotope composition can vary by source.
uncertainty	measurement-interval-retained	exact-decimal-pretense	A reported decimal must not erase its measurement interval.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
extension	no-extra-rule	free-weight-adjustment	An adjustable correction can force a measured target.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:90d954fc2cd84571376d947dca60f0e84ee9fb1325ede38bfd707aa41ef2ea0f.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:2ff64c465529ab6a5c91e2ea3023965c9e499826385218803b3983644b4974ce.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:c341944a48f79104fa347f17ecb18ceeeddad0388d4e720e6ad69745b6e70ee8.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:fc1f589db5a76a8aadd4fd38cd7cce85db80eced58a1aa027d11df0c552742eb.

Independent reconstruction. Implementation hash

sha256:039223c857c830d8414ee0110d8b2216a5f2c4f22714d0c2242d71e34dbb29e2; independent certificate sha256:a0c1484aa9c368048506eda8a5164f4678cc44bcde54be62e38a94bbcb126bb9; external-validation hash sha256:4ff4fa952038a736f86f96e7505b82ddb63491a4f56d1a294e688aa00b3a9241. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-R05258-2026

Comparison records:

- relative-atomic-mass-iupac-r05258: predicted exact-ratio-carrier__average-atomic-mass__reference-unit-comparison; source-derived exact-ratio-carrier__average-atomic-mass__reference-unit-comparison; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC relative-atomic-mass record lacks a ratio, average atomic mass or unified atomic-mass reference, or if an altered row is accepted.

Measurement receipt: sha256:c977d6c137a1e6c7c1115ef597d8fbbe4ec15b6b6a57f0d5f472888bb178de1e.

Isolation certificate: sha256:a9e8ad43edada7f16cfb09e6f0a2198290274b66368fff3a45c8edbee73dce09.

Custody certificate: sha256:43618f5d83fbf96bf7156be5c34ab68e50ce8f709b8287f8c7aca82c406d238f.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/R05258/json>
(sha256:9a08b1dfe8699f2be546b1a0fcb9b359d49c78419099d853603c6c750a04f17a)

Registered target identities:

- relative-atomic-mass-iupac-r05258 from IUPAC-GOLD-BOOK-R05258-2026

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, periodic-table arrangement, measured element value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter

- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite source-bounded atom population with retained isotope membership, exact positive rational abundance parts, atomic-mass records and a separate unified-reference-unit comparison.

Immutable identities. Source manifest

sha256:b24a0cec4643f0e8f9eb1ae962a6ab365591c758b804091a359943d864e9ecef; derivation seal sha256:9c58baab3b565cedfd561c639d5a3cba034d56f4a8c04c076b990a78e9fc95b3; engine receipt sha256:56247bdb02421bb72d76605e00a82ac71708b08fb09f0cf34d242408e08ecfa9 at receipts/engine/model_admitted/SFT-CHEM-ELEM-ATOMIC-WEIGHT-001-56247bdb02421bb7.json; empirical hash sha256:626970a47503d41d13b9c63bdae377bcdd64c73a84cd3dfdb3c5cb5baed2741e; measurement receipt sha256:c977d6c137a1e6c7c1115ef597d8fbbe4ec15b6b6a57f0d5f472888bb178de1e.

13. Periodic ordering by retained nuclear identity

Claim identity: SFT-CHEM-ELEM-PERIODIC-ORDER-001

Question and theorem. Chemical elements possess one invariant exact total order: the generated positive proton-count order, with the complete element class retained at each coordinate.

complete-element-class-support__atomic-number-coordinate__positive-count-total-order__element-identity-retained

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001

Generated grammar. Generate the literal product of the registered periodic-order carrier, coordinate, order, identity, succession, observation, record and extension choices.

Boundary: Every positive-finite chemical-element class carrying its admitted atomic-number identity, ordered only by generated proton-count succession and retaining the complete element record at each coordinate.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-element-class-support__atomic-number-coordinate__positive-count-total-order__element-identity-retained__one-proton-successor__post-seal-known-table-correspondence__number-element-source-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	<code>complete-element-class-support</code>	<code>names-without-element-records</code>	Names alone do not preserve the generated element classes.
coordinate	<code>atomic-number-coordinate</code>	<code>layout-selected-coordinate</code>	A drawn location can be changed without changing nuclear identity.
order	<code>positive-count-total-order</code>	<code>unordered-element-set</code>	An unordered set cannot express successor structure.
identity	<code>element-identity-retained</code>	<code>order-erases-isotopes</code>	Ordering must not discard refinements of an element record.
succession	<code>one-proton-successor</code>	<code>free-table-rearrangement</code>	A discretionary rearrangement can select any pattern.
observation	<code>post-seal-known-table-correspondence</code>	<code>known-table-as-proof-premise</code>	The observed table cannot create the order law.
record	<code>number-element-source-trace</code>	<code>atomic-number-only-list</code>	A number list loses symbols, identities and source boundary.
extension	<code>no-extra-rule</code>	<code>free-order-exception</code>	An exception can silently permute successors.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: fb79be00e6697a390e2401a3f35d4a36e790dd0e25f17ce2d82cdf109459fdc8.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 2468f6a821c5f46777ffec9979273168f4fb4a44c9a3b07a459d3885cd591fe4.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: fcd4a72f1f0a76bcc6ab20b90e22587f439fd13e32a9a32b98ab2c2a256b2cf2.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 8f2e3cc1418f0d11a932467a232d2806e7e2ad150c76df6cbf655f22d58a3a91.

Independent reconstruction. Implementation hash

sha256: 1ab5b9c59e4bf934778ced8adb3532d3129d525e09be9ba7215d8b53249aee1d; independent certificate sha256: e3e8d00b305566a1ce3a029a1b2fe23d581ff06b4c904848b79cadf71121781e; external-validation hash sha256: 7280d26df6f345563945989327936d24fe5affc69b020ba97354973d4b9fe75a. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- `periodic-order-iupac-2022`: predicted `complete-known-element-support__atomic-number-coordinate-order`; source-derived `complete-known-element-support__atomic-number-coordinate-order`; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official source lacks the complete registered element support or atomic-number coordinate, or if a changed row is accepted.

Measurement receipt: sha256:f62e7c273172fb00b04a1e7e47c8ad197f25a50e3de2f3da28fd5469eb07dbde.

Isolation certificate: sha256:44a4433e02559e65a5749eae4fcbca9cc6c6bc305547ea09e56e0f37d3e02772.

Custody certificate: sha256:51eccf54ae60dc41a16cc61cf63f95c35bc64f6e8ce48489074b17f0fc81a3b7.

Registered source descriptions:

- International Union of Pure and Applied Chemistry:
https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf
 (sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- periodic-order-iupac-2022 from IUPAC-PERIODIC-TABLE-2022

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC table arrangement, term definition, measured value, electron-shell equation or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every positive-finite chemical-element class carrying its admitted atomic-number identity, ordered only by generated proton-count succession and retaining the complete element record at each coordinate.

Immutable identities. Source manifest

sha256:4412f4cd967861c6874dd9ce7a267d43888e9c1857ae135127bfeb9076e6aa30; derivation seal

sha256:23ffdc5ff63cd43950460419c0b66c4235ca4f6b34f6b7f64157ece2ecdc3ff4; engine receipt

sha256:b7092fdbc3d12a0149fcf2f857230bb022a7e0badda51d2b8fe530f864433d15 at

receipts/engine/model_admitted/SFT-CHEM-ELEM-PERIODIC-ORDER-001-b7092fdbc3d12a01.json;

empirical hash sha256:0287e745e229c2fc5b26527ffeb7a839f7a3473934e42bc8304be04b673f6154;

measurement receipt sha256:f62e7c273172fb00b04a1e7e47c8ad197f25a50e3de2f3da28fd5469eb07dbde.

14. Periodic recurrence of outer chemical organization

Claim identity: SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001

Question and theorem. Periodicity is the return, along atomic-number succession, of an earlier complete outer chemical observation class generated by finite bound-state recurrence and exclusion-preserving occupation.

boundary-closed-outer-support__atomic-number-successor-progression__complete-chemical-trace-equivalence__return-of-observation-class

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-LOGIC-PROOF-001

- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001

Generated grammar. Generate the literal product of the registered recurrence carrier, progression, equivalence, closure, occupation, periodicity, record and extension choices.

Boundary: Every generated positive-finite atomic-number successor family whose boundary-closed outer observation support is quotiented only by complete chemical transition equivalence.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `boundary-closed-outer-support__atomic-number-successor-progression__complete-chemical-trace-equivalence__finite-bound-state-recurrence__exclusion-preserving-occupation__return-of-observation-class__support-transition-recurrence-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	boundary-closed-outer-support	property-name-series	A list of property names contains no physical recurrence carrier.
progression	atomic-number-successor-progression	mass-selected-progression	Isotope mass does not determine element succession.
equivalence	complete-chemical-trace-equivalence	visual-column-equivalence	A drawing cannot establish common chemical behavior.
closure	finite-bound-state-recurrence	unbounded-orbital-continuum	An ungenerated continuum violates the finite exact domain.
occupation	exclusion-preserving-occupation	duplicate-state-without-trace	Unrecorded duplicate occupation violates exclusion.
periodicity	return-of-observation-class	assumed-fixed-period	A fixed imported period can manufacture recurrence.
record	support-transition-recurrence-trace	recurrence-label-only	A repeated label does not reproduce the equivalence test.
extension	no-extra-rule	free-shell-rule	A discretionary shell equation would import the desired organization.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:79f9f2d5c12e6b82db58e3e375f1eacc44ecb05fab2fa50516b603680f28412c.

- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:28b430eac9c449e5d0a8d0fb4bba092ed7abb057120feb4adf7acffbdcc2322d4.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:bbc1a7eb2265d08c410bfba1e8e59977b6d22242f4ecc31d573126547e3564e9.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:beaad842b2bffc5a9a6ff2d5afadcabaa2e10ef7901d916f73c931394e1d43f4.

Independent reconstruction. Implementation hash

sha256:f41c5f8015683765486787511a50e71e5470ce686a2d01b82e98842cc0aa0c64; independent certificate sha256:497e7cb1c9504707525c356e731610f93b86140f319380641e6854ed1e9e1d85; external-validation hash sha256:8f322e53a08f9d6eeb1c0ba8cce7f8c81d2e057b23a2ba71d1e3e457e378ac4a. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- periodic-recurrence-iupac-2022: predicted recurrent-element-organization__ordered-periodic-table-support; source-derived recurrent-element-organization__ordered-periodic-table-support; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official periodic source lacks recurrent ordered element organization at the registered boundary, or if a changed row is accepted.

Measurement receipt: sha256:45f84934b3e95e258a36ae9d76b15abe2ed25595beb90e2f323ef43c41b4a5ae.

Isolation certificate: sha256:67af8d3ba8605c019c1e3574382c19e00dff4a04df864e41d0c42e324992378c.

Custody certificate: sha256:e519a856fa0069f4891536f4ba89d6e7c4c77ca5001ebdb16def307518b769c3.

Registered source descriptions:

- International Union of Pure and Applied Chemistry:
https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf
(sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- periodic-recurrence-iupac-2022 from IUPAC-PERIODIC-TABLE-2022

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC table arrangement, term definition, measured value, electron-shell equation or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every generated positive-finite atomic-number successor family whose boundary-closed outer observation support is quotiented only by complete chemical transition equivalence.

Immutable identities. Source manifest

sha256:ad5eeb8e798d2a915db6c4554a449709745c56695903dcd715e6e635c8458c3a; derivation seal
 sha256:57ca4b7f552b99fa4e91df0bfff1cdbc8cc865d672ae76873ce26544b70c19eb6; engine receipt
 sha256:cc2d14cb33a302f8d6f329c59bfd6c013096e2824e9ebee89284ccfc01cd4042 at receipts/engine/model_admitted/SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001-cc2d14cb33a302f8.json; empirical hash
 sha256:fe695d81bd6fbfffd2aea34af8bad35a1b62f2fe04262a13fe510b0af977a70b2; measurement receipt
 sha256:45f84934b3e95e258a36ae9d76b15abe2ed25595beb90e2f323ef43c41b4a5ae.

15. Group and period compositional coordinates

Claim identity: SFT-CHEM-ELEM-GROUP-PERIOD-001

Question and theorem. A Fold group is an outer-chemical-trace equivalence class and a Fold period is its counted closure-to-reopening depth; their pair is independent of how a table is drawn.

periodic-element-carrier__outer-trace-equivalence-class__closure-reopening-depth__joint-group-period-coordinate

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001

Generated grammar. Generate the literal product of the registered coordinate carrier, group, period, pairing, ordering, equivalence, record and extension choices.

Boundary: Every finite periodic-order prefix partitioned by exact outer-trace recurrence, with each element assigned one recurrence-class coordinate and one counted closure-to-reopening depth.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *periodic-element-carrier__outer-trace-equivalence-class__closure-reopening-depth__joint-group-period-coordinate*

e__atomic-number-order-retained__complete-trace-quotient__element-order-recurrence-trace__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	periodic-element-carrier	printed-cell-coordinate	A page coordinate is representational rather than chemical.
group	outer-trace-equivalence-class	column-number-as-premise	A conventional column number cannot select equivalence.
period	closure-reopening-depth	row-number-as-premise	A conventional row number cannot select recurrence depth.
pairing	joint-group-period-coordinate	group-only-or-period-only	Either coordinate alone loses a required distinction.
ordering	atomic-number-order-retained	layout-dependent-order	Alternate drawings must not alter chemical coordinates.
equivalence	complete-trace-quotient	namesake-family-only	A family name cannot prove matching transition support.
record	element-order-recurrence-trace	coordinate-without-witness	A coordinate without a recurrence witness is not auditable.
extension	no-extra-rule	free-layout-exception	A layout exception can move an element without a chemical distinction.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:c3f412be7e6caf1ae399956b649c37ddf393710eddf9fdb12016eea631e0b019.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:98da74bd924801244f5e8caa8728810ec0af692816c78ff6a8d630faccfb69f0.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:60a6d8de4a10afab9cd43595398e1dead9b886863e69e3047bce476d9cfe5b9d.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:137fa1bbe50de52fb9ced8a59a7e7c9aaba62081905c588226eb4b55378df7bd.

Independent reconstruction. Implementation hash

sha256:70bed5c2690f2e67d9c06dae6a5d91346efc2ffdd3a831a7be754f51740875b3; independent certificate sha256:f4762e4f73f2b81be48db4aee3269731dd14ed8cf2cd2ee5f6e57bb13db1c7f2; external-validation hash sha256:ae629891bd6b882bf3cd651f18a59b75fb4b902af24aa6eaa9cde4a7c4c420a0. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- group-period-iupac-2022: predicted group-column-classes__period-row-progression__atomic-number-order-retained; source-derived group-column-classes__period-row-progression__atomic-number-order-retained; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official table does not retain column classes, row progression and atomic-number records together, or if a changed row is accepted.

Measurement receipt: sha256:5827bda9734827f55158b4f6bc4b50cec8a457f7393b21a7af4038661436e76e.

Isolation certificate: sha256:f588278083d2cb9b9fea952818f75e6fcd38336ef1f501ad5a354fd9cc7eba77.

Custody certificate: sha256:162cf8ef7110504bb0a34beadec75b71e9a651152cf1417e918530340cb8f237.

Registered source descriptions:

- International Union of Pure and Applied Chemistry:
https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf
 (sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- group-period-iupac-2022 from IUPAC-PERIODIC-TABLE-2022

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC table arrangement, term definition, measured value, electron-shell equation or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite periodic-order prefix partitioned by exact outer-trace recurrence, with each element assigned one recurrence-class coordinate and one counted closure-to-reopening depth.

Immutable identities. Source manifest

sha256:707c0e159d5fa1fed19a8a0a1466e1ad220aee208323f0ae54b8babf7f0a4d1b; derivation seal

sha256:95799048c0712a47fb38cdf1049f13a18924383a737f6a663533305af5b4bf9; engine receipt

sha256:b99160e0648fadc49668ba6f0f4d6c241115a7d5264c17e878175442e8fb8b62 at

receipts/engine/model_admitted/SFT-CHEM-ELEM-GROUP-PERIOD-001-b99160e0648fadc4.json;

empirical hash sha256:8bd849dfa80e00963624853aaa58c8dd836297d00f48688571c1348444d64c7a;

measurement receipt sha256:5827bda9734827f55158b4f6bc4b50cec8a457f7393b21a7af4038661436e76e.

16. Valence availability and chemical combination boundary

Claim identity: SFT-CHEM-ELEM-VALENCE-001

Question and theorem. Valence is the exact greatest realized univalent joining or substitution count in the complete generated combination support of a retained atom or fragment and chemical context.

atom-or-fragment-carrier__complete-generated-combination-support__univalent-partner-count__maximum-realized-count

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-LOGIC-PROOF-001

- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001
- SFT-CHEM-ELEM-GROUP-PERIOD-001

Generated grammar. Generate the literal product of the registered valence carrier, support, unit, selection, substitution, context, record and extension choices.

Boundary: Every finite atom or chemical fragment with complete generated combination and substitution traces, where the largest realized univalent joining count is retained with its chemical context.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `atom-or-fragment-carrier__complete-generated-combination-support__univalent-partner-count__maximum-realized-count__combination-or-substitution__chemical-context-retained__combination-maximum-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	atom-or-fragment-carrier	element-name-only	An element name contains no combination witness.
support	complete-generated-combination-support	assumed-valence-number	An assumed number imports the answer.
unit	univalent-partner-count	mixed-partner-count	Partners of unrecorded multiplicity cannot define one comparison.
selection	maximum-realized-count	typical-count	A typical value can omit a lawful larger combination.
substitution	combination-or-substitution	combination-only	Valence also constrains lawful substitution for the carrier.
context	chemical-context-retained	universal-context-free-number	Chemical environment may alter available combinations.
record	combination-maximum-trace	number-without-traces	A bare number cannot reproduce its maximum witness.
extension	no-extra-rule	free-valence-correction	An adjustment can force a conventional value.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256 : 5806447da7a755ea31ac1d6fd82ead9fe731b639e24caddb330920603419c201.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : c1fe82f9e1c25112465debbf7a0addc85fad42abe7789fe1bd67cb615eaf18d2.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256 : 091cb85be6fa237c253e1c4bc53ae5f474f1aba2031682b507ccab02ce6a5752.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256 : 549fd50e5b28eee9a33c5d54cf5a6e7e30bd1ea06f63e8cbcb36d869dd1c2b78.

Independent reconstruction. Implementation hash

sha256 : 65ca8bb5246cd20eb6c823f8ffd231729d9340a5330aca2a3c2b19751167f6d0; independent certificate sha256 : 40019dd01e7a18c83373b3ae05cfe65130dac33c549325a093ecdb169e7d389d; external-validation hash sha256 : 35902241a5172549e58be1c41e93ced958162497b3eadbbdf01326899ab0958. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-V06588-2026

Comparison records:

- valence-iupac-v06588: predicted maximum-univalent-combination-count__atom-or-fragment-carrier__substitution-boundary; source-derived maximum-univalent-combination-count__atom-or-fragment-carrier__substitution-boundary; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC record lacks a maximum univalent combination count, atom/fragment carrier or substitution boundary, or if a changed row is accepted.

Measurement receipt: sha256 : 17154a95f8d4c2b1a9ad2ed5e5211cbc482d3812d629e3341ea6969086dc7745.

Isolation certificate: sha256 : bd6301a1f56b1268b0a5f7c88d8958d3e458a6292eb0797f3baf1bfa8f85f694.

Custody certificate: sha256 : ce6d0c69d428bd9b0c31da10d17944a1e1b06a558188c6df00cbd87c84b16a00.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/V06588/json> (sha256:99ae406a9ac89a8f0a89bf3c775ff9ba7c5d9d0c96f89f83a57a0b35c76c4077)

Registered target identities:

- valence-iupac-v06588 from IUPAC-GOLD-BOOK-V06588-2026

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC table arrangement, term definition, measured value, electron-shell equation or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite atom or chemical fragment with complete generated combination and substitution traces, where the largest realized univalent joining count is retained with its chemical context.

Immutable identities. Source manifest

sha256:bde012bf288acde78fb9a213f27cb3193189e812170852fa2a62e15d817eccf2; derivation seal sha256:7669420367899f35589ba79654d46dd44a058d18bb84a3260eb783490eebb810; engine receipt sha256:033b82ac5d7e97cafd88c84ddf9204eb7ab6d4a4fe07d93f47ca6a7a2f2ee4 at receipts/engine/model_admitted/SFT-CHEM-ELEM-VALENCE-001-033b82ac5d7e97ca.json; empirical hash sha256:ba63c5bd46a5845aba5d05961482a5995fd0af5f8aee5908600a5ebe8a8d2659; measurement receipt sha256:17154a95f8d4c2b1a9ad2ed5e5211cbc482d3812d629e3341ea6969086dc7745.

17. Ion formation with retained elemental identity

Claim identity: SFT-CHEM-ELEM-ION-001

Question and theorem. An ion is an atomic or molecular Fold particle whose closed electric-label transfer leaves a net held charge orientation while retaining its nuclear and chemical identity.

atomic-or-molecular-particle__net-held-electric-charge__label-transfer-with-identity-retained__closed-charge-transfer

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-VALENCE-001

Generated grammar. Generate the literal product of the registered ion carrier, charge, formation, conservation, element, orientation, record and extension choices.

Boundary: Every finite atomic or molecular chemical carrier whose complete conserved electric-label accounting leaves one non-neutral held orientation while retaining its elemental and molecular identity records.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *atomic-or-molecular-particle__net-held-electric-charge__label-transfer-with-identity-retained__closed-charge-transfer__atomic-number-retained__cation-anion-held-fibres__carrier-charge-transfer-trac*

e__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	atomic-or-molecular-particle	charge-label-without-particle	A charge label detached from matter is not a chemical ion.
charge	net-held-electric-charge	signed-proof-number	A signed proof magnitude violates the positive exact domain.
formation	label-transfer-with-identity-retained	identity-destroying-change	Destroying the carrier does not create a charge state of it.
conservation	closed-charge-transfer	unpaired-charge-creation	Unpaired creation violates closed label flow.
element	atomic-number-retained	proton-count-change	Changing proton count changes element identity.
orientation	cation-anion-held-fibres	positive-negative-proof-scalars	Signed scalars are not SFT proof objects.
record	carrier-charge-transfer-trace	ion-name-only	A name cannot reconstruct carrier, charge and transfer.
extension	no-extra-rule	free-charge-exception	An exception can create or erase charge without transfer.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:2ae6650a4e3754c24fb9a4f8f09536c51e7b81ac2b96455c63b56e597cb94fe0.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:3b2456e07702e7ce9cbe5e02573a4d787dcd52859d789ab8bbc3c418bc1633ae.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:df200a2b7556795d20a2d92057fdcb90f979904158d594e9d314cd2223d281f1.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:75999b23650345a5d5e4effbad9ba3bbf7ec85d09e0d64c8e675cc381cb72a3c.

Independent reconstruction. Implementation hash

sha256:87f51143641461df3f93e8bc4ab0dc75357216b47b04bf5df604d469ee8f223f; independent certificate sha256:7f6e160af0765a688078df30a929390192b67361cb72598c39e46183ca9d4863; external-validation hash sha256:69f253a67622135339d64cbd638ed37837c5e2b66b8365bd7c9fc563c128923e. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-I03158-2026

Comparison records:

- ion-iupac-i03158: predicted atomic-or-molecular-particle__net-electric-charge; source-derived atomic-or-molecular-particle__net-electric-charge; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC ion record lacks an atomic-or-molecular particle and net electric charge, or if a changed row is accepted.

Measurement receipt: sha256:c0858d5bbb6e663ca112aed3cc8bdd4fa681cbbd33fb82f3bee61cb91683c055.
 Isolation certificate: sha256:8a2b926923b8510ba1b91e2654b2f496a3d0a74f87318e65fbd844032a38c86.
 Custody certificate: sha256:9490ca12922924907ad992aa86d3bbee8e16241cd072eef7b9988a168fb5fa7a.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/I03158/json> (sha256:6b54a0f6d9cf83f30251501b58b75764a39396e56d819b420b1556476f9f13f6)

Registered target identities:

- ion-iupac-i03158 from IUPAC-GOLD-BOOK-I03158-2026

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC table arrangement, term definition, measured value, electron-shell equation or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite atomic or molecular chemical carrier whose complete conserved electric-label accounting leaves one non-neutral held orientation while retaining its elemental and molecular identity records.

Immutable identities. Source manifest

sha256:c2e9f6d86e4cdd13f7238a71432514d988178d117fc769df8d002725b507402b; derivation seal
 sha256:c2e6df7e9cdb78b8dc0ffdf46523150e74254843d3dc5823b42c1dd0af9e8da8; engine receipt
 sha256:abc77e9274b595b56650212d3eb4f83323c92ffee85569fae93eab34491ca5a7 at
 receipts/engine/model_admitted/SFT-CHEM-ELEM-ION-001-abc77e9274b595b5.json; empirical hash
 sha256:53200de51747879d76501c996a8b7eb32132b2c0e9c0cdd46b014deb2efba9a3; measurement receipt
 sha256:c0858d5bbb6e663ca112aed3cc8bdd4fa681cbbd33fb82f3bee61cb91683c055.

18. Known periodic-table observation boundary

Claim identity: SFT-CHEM-ELEM-PERIODIC-BOUNDARY-001

Question and theorem. The known periodic boundary is the complete source-dated IUPAC element census through its greatest listed atomic-number coordinate; it is an observation boundary and not a theorem forbidding successors.

complete-source-bounded-census__greatest-listed-coordinate__element-record-retained__unobserved-successor-open

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-ELEM-GROUP-PERIOD-001
- SFT-CHEM-ELEM-ION-001

Generated grammar. Generate the literal product of the registered known-boundary carrier, extent, identity, time, completeness, extension, record and rule choices.

Boundary: Every official finite periodic-table release treated as a source-dated observation census, retaining all listed atomic-number/element records while leaving any unobserved successor scientifically open.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-source-bounded-census__greatest-listed-coordinate__element-record-retained__release-date-retained__all-listed-rows-retained__unobserved-successor-open__source-hash-census-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-source-bounded-census	largest-number-only	A largest number alone omits the observed element census.
extent	greatest-listed-coordinate	timeless-terminal-claim	A current release cannot prove that no successor can exist.
identity	element-record-retained	count-without-element	A number without its element record is not an observation boundary.
time	release-date-retained	undated-table	An undated table cannot define a reproducible empirical boundary.
completeness	all-listed-rows-retained	selected-terminal-row	Selecting only the last row cannot establish the registered census.
extension	unobserved-successor-open	observed-means-impossible-beyond	Absence from one release is not impossibility.
record	source-hash-census-trace	boundary-without-source	An unsourced endpoint cannot be reproduced.
rule	no-extra-rule	free-terminal-rule	A terminal rule chosen from the table would overfit observation.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:07957a12479b7fc21cb0f19bd56d222e157ebb86d5ecc6de767af082f09b4529`.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 9529beed9e6c984672c342003db5011e89be4e68d0e1591ab8de1c93c429482e.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: ef04b83aec104f822c81ac402cd320bdea3fd18aa95096286e409badaa6e4a4f.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: 2eec45d3cb99a30f59046b3098e2f4659fb5d313949bb9dd9f1ac6d56fffd2cf6.

Independent reconstruction. Implementation hash

sha256: cf9ea6d2271b92fe7b4dddb45245248dc45abe31f824d86bde302e418fe92c65; independent certificate sha256: f7c1c445a2435d2d0997ca0b49d94cc60bf174d5f973464f39c4f256593279be; external-validation hash sha256: c098a373db4e8b6730ee656cd4ab8e19b930a07470e18f22aaff1829dc1548d6. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- periodic-boundary-iupac-2022: predicted source-bounded-known-element-census__greatest-listed-atomic-number-118; source-derived source-bounded-known-element-census__greatest-listed-atomic-number-118; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the registered official release does not contain the complete stated current census through atomic number 118, or if a changed row is accepted.

Measurement receipt: sha256: 7745e73d227b50347c5123149adab8369df094ed775d20915c969faab26283ca.

Isolation certificate: sha256: 3d8148e921d8b6f02d545889d23e528e93aa4fc20aef9551928b693d83f5784a.

Custody certificate: sha256: 0de1af0a124c000f878f0e24d103f1dfd709e6b5674632a5aeab00d9cf8a1cf3.

Registered source descriptions:

- International Union of Pure and Applied Chemistry:
https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf
(sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- periodic-boundary-iupac-2022 from IUPAC-PERIODIC-TABLE-2022

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC table arrangement, term definition, measured value, electron-shell equation or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every official finite periodic-table release treated as a source-dated observation census, retaining all listed atomic-number/element records while leaving any unobserved successor scientifically open.

Immutable identities. Source manifest

sha256: c93fd28b3335f4f2deb91e5392c8c7481fedb7e4837d86c8f3a6e88ee2f2180c; derivation seal

sha256:eba95a589508601d55d41cf04256d862dfb55457bf325d835cec87be903f234a; engine receipt
 sha256:80fd05e50706039e8d8c1a0cca9dcd264c31b7ca0a393199df1de58d7ec320eb at receipts/engine/model_admitted/SFT-CHEM-ELEM-PERIODIC-BOUNDARY-001-80fd05e50706039e.json; empirical hash
 sha256:0bd74e7d079ec18f40a3b8970e85aa936af5c87deed188fe19759f4ae85c9db6; measurement receipt
 sha256:7745e73d227b50347c5123149adab8369df094ed775d20915c969faab26283ca.

10. Composition, mixtures, solutions and stoichiometry - 7 admitted claims

Every constituent remains in a positive exact ledger across mixtures, solutions and complete reaction maps. This entire registered subcategory is included; no proper subset is presented as field completion.

19. Exact chemical composition

Claim identity: SFT-CHEM-STOICH-COMPOSITION-001

Question and theorem. Chemical composition is the complete finite Fold carrier of distinct constituent identities paired with exact positive rational parts that exhaust the One.

complete-chemical-carrier__distinct-constituent-identities__exact-positive-rational-parts__parts-exhaust-the-One

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-MEAS-AMOUNT-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001

Generated grammar. Generate the literal product of the registered composition carrier, components, quantity, closure, absence, identity, record and extension choices.

Boundary: Every finite chemical carrier whose distinct constituent identities have exact positive rational parts that exhaust the One and preserve the measurement/source boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-chemical-carrier__distinct-constituent-identities__exact-positive-rational-parts__parts-exhaust-the-0`

ne__absent-component-omitted__constituent-identity-retained__identity-part-source-trace__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-chemical-carrier	bulk-label-only	A bulk label erases constituent support.
components	distinct-constituent-identities	anonymous-parts	Anonymous parts cannot preserve chemical identity.
quantity	exact-positive-rational-parts	floating-percentages	Floating percentages cannot certify exact closure.
closure	parts-exhaust-the-One	partial-selected-composition	Selected components need not exhaust the carrier.
absence	absent-component-omitted	zero-valued-components	Numerical zero is not an SFT composition object.
identity	constituent-identity-retained	interchangeable-constituents	Equal amounts do not make substances identical.
record	identity-part-source-trace	composition-answer-only	An answer loses part and source provenance.
extension	no-extra-rule	free-normalization-correction	A correction could force closure after the fact.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256 : a72500ce5db6e8ec585702f2e1a8dbfed043e7e52c2ee8304e5647b7e3abf5e7.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : c75c9b3e1be8fe6f88b5afb87d59e106cfc119080312c51c41d0e99922b5167a.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256 : fcecb84871bf9e043bc807dccc5e3c148e05b26ac43b05af6806cfd0025fc304.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256 : 6961ea1b574acf85e4ab8c4dc2c15a306a8853e50ddf3e7c30bb6e707858c87f.

Independent reconstruction. Implementation hash

sha256 : 17727dd7b5815338c9679ab3cc2682e0e423ae50b27cf1305f044b57369c4959; independent certificate sha256 : ad7e04bbe9d538d460664d08bba33e86cee6c30555ff31191f5fc4cbce190eb7; external-validation hash sha256 : f108a2e4fd4b1f8372ec551ff36ec3c111ce05926c0f0c0d1e63bfa8c419a8bc. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-M03722-2026

Comparison records:

- composition-iupac-m03722: predicted constituent-carrier__constituent-part-over-total-mixture; source-derived constituent-carrier__constituent-part-over-total-mixture; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the IUPAC record lacks constituent identity, a constituent-to-total relation or mixture support, or if a changed row is accepted.

Measurement receipt: sha256:cf4ca2d263e59e8c0fbc97e96f9d8d7c025c004498f42d6399e63df1dea5aca2.

Isolation certificate: sha256:a0061a81ed475bdad9bb550a451a26c790a4812b83e7bfcac8f6db21f4b62a32.

Custody certificate: sha256:7b41e60e267c0dac2d36effab242ab8dcc19d8311f7fecca693ed3b85b11a8fa.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/M03722/json> (sha256:cc715cdd306779e5032685105dacbbc7d1ef79ea8241269496befb1dc37a7f2f)

Registered target identities:

- composition-iupac-m03722 from IUPAC-GOLD-BOOK-M03722-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite chemical carrier whose distinct constituent identities have exact positive rational parts that exhaust the One and preserve the measurement/source boundary.

Immutable identities. Source manifest

sha256:9a0ee515e65d9f95c3cddb1c489b96a5c409fcdf0fcd935bf9d635e43b556acfa; derivation seal

sha256:86b3ee6ce9fe0b441d635e972e831128cb00bb35b9f63887847d5a4b32101f1c; engine receipt

sha256:074fbc50af44548032b31d2b871481f33477f8a1b1830164c34c9dd4a6a08dae at

receipts/engine/model_admitted/SFT-CHEM-STOICH-COMPOSITION-001-074fbc50af445480.json;

empirical hash sha256:615c18ca3f3ccd17003e92083761d6ba5dee8553b182a3340432a15dd37549ca;

measurement receipt sha256:cf4ca2d263e59e8c0fbc97e96f9d8d7c025c004498f42d6399e63df1dea5aca2.

20. Conservation of elemental identity through reaction

Claim identity: SFT-CHEM-STOICH-CONSERVATION-001

Question and theorem. A chemical reaction is closed only when complete held reactant/product words have exactly equal positive multiplicity totals for every retained elemental identity, including named boundary carriers.

complete-reaction-word__held-reactant-product-sides__elemental-identities-retained__all-element-totals-paired

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001

- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-MEAS-AMOUNT-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-STOICH-COMPOSITION-001

Generated grammar. Generate the literal product of the registered conservation carrier, orientation, identity, count, closure, boundary, record and extension choices.

Boundary: Every finite chemical transition word whose reactant and product sides retain complete species formulae, positive multiplicities and equal per-element totals.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-reaction-word__held-reactant-product-sides__elemental-identities-retained__exact-positive-multiplicity__all-element-totals-paired__named-boundary-carriers__species-formula-total-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-reaction-word	product-list-without-input	Products alone cannot demonstrate conservation.
orientation	held-reactant-product-sides	signed-stoichiometric-scalars	Signed proof quantities violate the positive domain.
identity	elemental-identities-retained	mass-only-balance	Equal bulk mass can hide changed element labels.
count	exact-positive-multiplicity	approximate-decimal-balance	Approximation cannot prove equality of discrete carriers.
closure	all-element-totals-paired	selected-elements-only	Omitting an element can manufacture balance.
boundary	named-boundary-carriers	unrecorded-open-system	Unrecorded boundary transfer can mimic loss or creation.
record	species-formula-total-trace	balanced-label-only	A label cannot reconstruct the equality.
extension	no-extra-rule	free-balancing-exception	An exception could excuse an unpaired carrier.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:5108a653085bf72b6824763bc3d1b16ed015738b65ff8dcd78f9367699bc2f27`.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:079af8a29365c118d92b3732ad4284de9697833b1c3698b2e3da9b48679a85fc.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:455afb7670db3dca84beea9f74cbfbc874dd8b6a3bbaa348262b0aafd2c7282d2.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:bf89eecbc5d713f6278f738bfc83f579a0c871960d060961c54922ca77d71ce0.

Independent reconstruction. Implementation hash

sha256:77092730858fcc8d021aca7483bbc7c25c64ef8b6cd071c34b0f409bb5e6901a; independent certificate sha256:85f9be325874c16051fc4b181a3689b3516089ad25438316ae14404254699402; external-validation hash sha256:260172ed629591fc01fc5d21450e038fffb47c16b2b197d5b5b3a5961f1b5224. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-S06026-2026

Comparison records:

- stoichiometry-iupac-s06026: predicted reactant-amount-support__product-amount-support__reaction-relation; source-derived reactant-amount-support__product-amount-support__reaction-relation; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The formal conservation theorem fails on any unequal element map; the categorical correspondence fails if IUPAC lacks related reactant and product amount supports or if a changed row is accepted.

Measurement receipt: sha256:c4993a6fe9f489b069c90674551d07e41a6c3865a91b60196cb1d53f6260d433. Isolation certificate: sha256:cdf2e0d2a2a2b763440bd34d98b56b6723e0a716c3be751c460ac112a075b01f. Custody certificate: sha256:8a8f99e69a13021f278ccc7a23c4235edffd05b5219678326bf4095b45bbdfb8.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/S06026/json> (sha256:0354d161fb1754d50dee2cfdc3bf051c6a005dbde765a112398f8e717e9952bc)

Registered target identities:

- stoichiometry-iupac-s06026 from IUPAC-GOLD-BOOK-S06026-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite chemical transition word whose reactant and product sides retain complete species formulae, positive multiplicities and equal per-element totals.

Immutable identities. Source manifest

sha256:b7543a8b7087b9af934a5a05cad493cf8a9018e547ba027ec5f48ccebfe02fdd; derivation seal

sha256:eab1a622948bc8c0e0702bbd0cf42f8310801573dd357b2d14aa7d10e9902af4; engine receipt
 sha256:4bbc2d9ff0b0af55f6764854012febfa99ee46bdea5ebde1fd66e7577692f6b9 at
 receipts/engine/model_admitted/SFT-CHEM-STOICH-CONSERVATION-001-4bbc2d9ff0b0af55.json;
 empirical hash sha256:61bed69744a041f4c6fa8aa2b587ea522c3acf4cc1628f6ec6b06c12364f0728;
 measurement receipt sha256:c4993a6fe9f489b069c90674551d07e41a6c3865a91b60196cb1d53f6260d433.

21. Positive stoichiometric coefficient and reaction balance

Claim identity: SFT-CHEM-STOICH-COEFFICIENT-001

Question and theorem. Stoichiometric coefficients are the unique primitive vector of positive generated species multiplicities balancing every element map, with reactant/product direction held structurally and nonparticipants omitted.

formula-bound-species-support__held-side-positive-count__element-balance-preserved__primitive-positive-vector

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-MEAS-AMOUNT-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

Generated grammar. Generate the literal product of the registered coefficient carrier, orientation, balance, scale, absence, uniqueness, record and extension choices.

Boundary: Every balanceable finite chemical reaction support with participating species only, represented by the least common-factor-free vector of positive generated multiplicities.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *formula-bound-species-support__held-side-positive-count__element-balance-preserved__primitive-positive-vector__nonparticipants-omitted__unique-up-to-common-multiple__formula-matrix-reduction-trace__no-extra-rule*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	formula-bound-species-support	coefficient-list-without-species	Counts detached from formulae have no chemical meaning.
orientation	held-side-positive-count	negative-reactant-count	Negative proof counts are inadmissible.
balance	element-balance-preserved	unbalanced-count-vector	An unbalanced vector violates element conservation.
scale	primitive-positive-vector	arbitrary-common-multiple	Common multiples are equivalent but not minimal.
absence	nonparticipants-omitted	zero-coefficient-species	A zero coefficient imports numerical zero.
uniqueness	unique-up-to-common-multiple	one-picked-balancing-vector	Picking one vector does not eliminate rescalings.
record	formula-matrix-reduction-trace	coefficient-answer-only	A count list cannot prove its balance.
extension	no-extra-rule	free-coefficient-adjustment	An adjustment can force a desired equation.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256 : c078d573e8a1e9e8a75c4bf44728a255e36bf2ce05c9bdb553302bc964024fb0.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 9a54b1e4fc558b7b721a27cfb8aef9fd9fd2b1b8a7f2a46eceed85bdba37fd4.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256 : 32577dd9fbd901b3ba894763049cc1ed4f6a95e6b0741fb56d9befed0823016f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256 : e7f45e303be1880d3bd54dde084ced16329f3cbfbcd507ceb19c40ddd1200de9.

Independent reconstruction. Implementation hash

sha256 : 11bea52dca022dcd31511c76455602348cffc106f1c11d30a6f48454a49b6f41; independent certificate sha256 : 864a5d480e06f6a3c429374da8dc9e45ef4c20957a0a4713cbf3f9f00d91f0ca; external-validation hash sha256 : 1a23c71ed66cb9b657e03b7417b277ba0d98c8f5766559fc270b3c8717775f6d. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-C01034-2026

Comparison records:

- reaction-equation-iupac-c01034: predicted reactant-product-held-sides__absolute-stoichiometric-counts__formula-bound-coefficients; source-derived reactant-product-held-sides__absolute-stoichiometric-counts__formula-bound-coefficients; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the IUPAC equation record lacks reactant/product sides, absolute stoichiometric counts or formula binding, or if a changed row is accepted.

Measurement receipt: sha256:b816fbd7ef147b25ef1ac144107058ef6a42653e84a738dea448f866668803fa.
 Isolation certificate: sha256:fb5c8473c29a74cfdc925c0608eb94e0a01acea1637cde0b72948e9e03d0cd0e.
 Custody certificate: sha256:68f499937db1429343ecf08d4fc8ebddd8702c063248704831d0516194aacad2.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01034/json>
 (sha256:a582f71bc6c555d5513f365d18d5d46a6a5da61e0c1668d231ad860b9a32dea3)

Registered target identities:

- reaction-equation-iupac-c01034 from IUPAC-GOLD-BOOK-C01034-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every balanceable finite chemical reaction support with participating species only, represented by the least common-factor-free vector of positive generated multiplicities.

Immutable identities. Source manifest

sha256:bf82c1309993ae04314219ea4a629d8d2c0f1ef6641bbb5d9c2d72f4ef6cc353; derivation seal
 sha256:7ecd910200bd8ce63e0ae9177159b3255a10ed88c6925c092b281214919aa837; engine receipt
 sha256:26cbd66986ff38ff66640b5a85290c9aa4fee32c7b728360e8a45057b89d109b at
 receipts/engine/model_admitted/SFT-CHEM-STOICH-COEFFICIENT-001-26cbd66986ff38ff.json;
 empirical hash sha256:96a0a50009b552c5dc8201fb549608169489593baaab06cece6862497bb72fa;
 measurement receipt sha256:b816fbd7ef147b25ef1ac144107058ef6a42653e84a738dea448f866668803fa.

22. Limiting-component and complete-consumption boundary

Claim identity: SFT-CHEM-STOICH-LIMITING-001

Question and theorem. The limiting support is the complete tied least set of exact available-amount to primitive-required-count ratios; it bounds reaction progress before any unavailable carrier would be required.

available-reactant-support__primitive-coefficient-requirement__exact-available-required-ratios__complete-least-ratio-support

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001

- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-MEAS-AMOUNT-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-STOICH-COEFFICIENT-001

Generated grammar. Generate the literal product of the registered limiting carrier, requirement, comparison, selection, consumption, excess, record and extension choices.

Boundary: Every finite reaction input with exact positive available amounts and primitive required coefficients, where complete reaction progress is bounded by every available-to-required ratio.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `available-reactant-support__primitive-coefficient-requirement__exact-available-required-ratios__complete-least-ratio-support__complete-consumption-boundary__retained-excess-support__amount-coefficient-ratio-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	available-reactant-support	named-reagent-only	A name without amount and coefficient cannot limit progress.
requirement	primitive-coefficient-requirement	unbalanced-requirement	Consumption requirements must come from the balanced reaction.
comparison	exact-available-required-ratios	floating-ratio-ranking	Floating comparison cannot certify ties or minima.
selection	complete-least-ratio-support	first-considered-reactant	Ordering of inspection cannot select the limiter.
consumption	complete-consumption-boundary	negative-remainder	Overrunning the least support would require unavailable matter.
excess	retained-excess-support	excess-erased	Nonlimiting remainder is part of the result.
record	amount-coefficient-ratio-trace	limiter-name-only	A name cannot reproduce the comparison.
extension	no-extra-rule	free-limiter-rule	A preference could choose a nonminimal reagent.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256: 6ebd09dcebbbc38118e45114f8cc922497e0587e6cdfe8cd4eabb024443338070.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 74667f499635c03544e37dbda2e53204e74a332e76fe799df64c6be0f44a9754.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:656b446ccde1d50e6e371ff7dc0cad490f962eef10fabddfd8a43fcbcc4bfd4f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:c0ba9f1cd9197180791356d6e661445fe7460b90b6b9ce6446d2f8f18bcf07bc.

Independent reconstruction. Implementation hash

sha256:8495ce255510f5059960137d6e1e9b3ee7214201e8d830661353e4c7d1274ecf; independent certificate sha256:1406ee0f53cea26f83d16e7cd23a3d49c5a694e77edd6020d3544febd31e8c34; external-validation hash sha256:6e1a2b3f2c2718edf9e790dabc64937c5e4f6e97d4d28b355ace42fc5d22ee2e. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-D01771-2026

Comparison records:

- limiting-reagent-iupac-d01771: predicted limiting-reagent-carrier__amount-boundary; source-derived limiting-reagent-carrier__amount-boundary; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the IUPAC record lacks a limiting reagent and its amount relation, an exact nonleast carrier is selected, or a changed row is accepted.

Measurement receipt: sha256:445f633709f19891fb999a3458646043f99a67e9032dc204adc8b3e1ade36ffa.

Isolation certificate: sha256:2d6e18e2904bf82f07b591395e3d0e41b51c43c3ac08f5985ef65bdf8c4507bc.

Custody certificate: sha256:e7d657db331b75ff429e02fd9da0d65fdc2cfe6f4cfe4d4e31d72b3a6aee4ab9.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/D01771/json> (sha256:352e753acf025b61ce54a5b00b61d9dd5b6d112d0d35133b48f2eba212c3b807)

Registered target identities:

- limiting-reagent-iupac-d01771 from IUPAC-GOLD-BOOK-D01771-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite reaction input with exact positive available amounts and primitive required coefficients, where complete reaction progress is bounded by every available-to-required ratio.

Immutable identities. Source manifest

sha256:a228daf8163ff0b0b5ce5da4e44137b16b77e35fcd49b1313cf15ec0e94ae752; derivation seal

sha256:698851631dedbf6671fdb06b809b2d7bb427ddebd7198bc1bd2cc9ca5ee35dd2; engine receipt

sha256:cc930de298924d974a42c9fd214aa2ee0f156f5dc5e2d2c5bbeed884292284e6 at

receipts/engine/model_admitted/SFT-CHEM-STOICH-LIMITING-001-cc930de298924d97.json;
 empirical hash sha256:de416cfb5d3280a636bd90229df732b868bd0b59ddc0afb26a5277c72ce4c6e;
 measurement receipt sha256:445f633709f19891fb999a3458646043f99a67e9032dc204adc8b3e1ade36ffa.

23. Reaction yield as retained product share

Claim identity: SFT-CHEM-STOICH-YIELD-001

Question and theorem. Chemical yield is the exact positive Fold part of a retained element-or-compound product amount relative to its registered expected support after a specified reaction or separation.

element-or-compound-product__specified-reaction-or-separation__complete-observed-product-amount__exact-positive-amount-fraction

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-MEAS-AMOUNT-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-STOICH-LIMITING-001

Generated grammar. Generate the literal product of the registered yield carrier, process, numerator, reference, quantity, absence, record and extension choices.

Boundary: Every specified reaction or separation with a retained expected product support and a positive observed product amount represented as an exact part of that reference support.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *element-or-compound-product__specified-reaction-or-separation__complete-observed-product-amount__registered-expected-support__exact-positive-amount-fraction__empty-product-outcome-form__product-process-reference-trace__no-extra-rule*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	element-or-compound-product	percentage-without-product	A percentage detached from a product is not chemical yield.
process	specified-reaction-or-separation	unspecified-process	Yield depends on the declared reaction or separation.
numerator	complete-observed-product-amount	selected-favorable-product	Selecting a favorable amount hides other outcomes.
reference	registered-expected-support	unbound-denominator	A fraction without its reference is uninterpretable.
quantity	exact-positive-amount-fraction	floating-percent	Floating output is not exact proof.
absence	empty-product-outcome-form	numerical-zero-yield	Zero is not an SFT proof value.
record	product-process-reference-trace	yield-answer-only	An answer loses product, process and reference provenance.
extension	no-extra-rule	free-yield-correction	A correction can inflate the observed result.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : d25409ab09b66f54e66658f55ef35bd4c7b994814725c3cecd49c5a18ffc6ede.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 9a81c40f9ff9716ea4d44351b4d379929c6a2e124154eaa6bb55b4e775388ee4.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : f7c821f8d7494f3c08d13fd43f4971cebcd6ea3d5f45d85f62b112202583abcb.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 2819fb08332303920b39a0fc659c93cea8485c85ea07e708d3be83a8055bf9d7.

Independent reconstruction. Implementation hash

sha256 : e84db5b8638298fffd8290fb69d355e310f64a6201fd700ce8d188b614972d04c; independent certificate sha256 : 27ce7f5b9a159971c4c513624866a8244cf854b3cd65b9bb3f465fd4e67f8dfb; external-validation hash sha256 : 9283bbbed6dde05037dd040c11b87af15843704ac3fc0e9d02f92a0d21dca87. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-C01041-2026

Comparison records:

- chemical-yield-iupac-c01041: predicted amount-fraction__element-or-compound-product__specified-reaction-or-separation; source-derived amount-fraction__element-or-compound-product__specified-reaction-or-separation; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the IUPAC record lacks an amount fraction, element/compound carrier or specified reaction/separation, or if a changed row is accepted.

Measurement receipt: sha256 : f27991d5a8976e80b78630dd4c222bd6f8c1a863a6e127b8c7c81e15cbbe7ab4.

Isolation certificate: sha256 : 5a5242bd4db2459bfb56e342b7c0a0d19a0eec8f08f914545f39645671fa3dda.

Custody certificate: sha256 : da05be045a8958c29e253c5a14ecd6fdcad51c5cc19f888d856fbd108771ac1.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01041/json> (sha256:47979dfe9e05be8a7c207779853ce25538a8753e05b648a8a43471ed040b732f)

Registered target identities:

- chemical-yield-iupac-c01041 from IUPAC-GOLD-BOOK-C01041-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every specified reaction or separation with a retained expected product support and a positive observed product amount represented as an exact part of that reference support.

Immutable identities. Source manifest

sha256:75507170b653f78cb85ecd27adb87b46d482252914f1848d5b2428bb51094074; derivation seal sha256:09b70d6abb774e86e912f9a39abf628e5c1a729751384ef106b741211e3cfee2; engine receipt sha256:86bc873a5b4b7126e96d289477d8e3b77de9ccb7f69749da0d1d22c14455c8db at receipts/engine/model_admitted/SFT-CHEM-STOICH-YIELD-001-86bc873a5b4b7126.json; empirical hash sha256:cf5336afbdc5d514eac9a90d66ebdcfae04f5f5be07241f4b677c8047ef8b2787; measurement receipt sha256:f27991d5a8976e80b78630dd4c222bd6f8c1a863a6e127b8c7c81e15cbbe7ab4.

24. Mixture components and composition support

Claim identity: SFT-CHEM-STOICH-MIXTURE-001

Question and theorem. A mixture is a finite portion-of-matter Fold carrier with multiple retained chemical-substance identities and exact positive composition parts, without assuming formation of a new substance.

portion-of-matter-carrier__multiple-constituent-support__chemical-substance-identities__exact-positive-composition-parts

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001

- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-MEAS-AMOUNT-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-YIELD-001

Generated grammar. Generate the literal product of the registered mixture carrier, multiplicity, identity, composition, bonding, phase, record and extension choices.

Boundary: Every finite portion of matter containing multiple retained chemical-substance identities and their exact positive composition parts without requiring a new bonded substance identity.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `portion-of-matter-carrier__multiple-constituent-support__chemical-substance-identities__exact-positive-composition-parts__constituent-identities-retained__phase-support-recorded-not-forced__component-part-phase-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	portion-of-matter-carrier	substance-name-only	One substance name does not define a mixture.
multiplicity	multiple-constituent-support	single-constituent	A single constituent is not a mixture.
identity	chemical-substance-identities	anonymous-components	Anonymous components erase chemical distinction.
composition	exact-positive-composition-parts	unmeasured-presence-list	Presence alone cannot reconstruct composition.
bonding	constituent-identities-retained	new-substance-conflation	Mixing does not by itself force a new chemical identity.
phase	phase-support-recorded-not-forced	single-phase-required	Mixtures may contain one or several phases.
record	component-part-phase-trace	mixture-label-only	A label cannot reconstruct components.
extension	no-extra-rule	free-mixture-equivalence	An arbitrary equivalence can merge distinct mixtures.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256: acb80f57db5c57ab1df7b007f1543f85ba3748e02f7d0e3d95d125a1cfb31630`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256: de36e78e252d653293e37ec113b0dc81ece173288e089f8d49fe23ac16133fdd`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:56afba8866e13bcfdc397b0f3eee00ada26e353e74d5095ac4466e87ba6615f3.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:97238131131cae2f7c598609c639c60476b18cb2e6290608d7b8f37dfbd1b0de.

Independent reconstruction. Implementation hash

sha256:85d3ff3531cced11d325c2211b6ea9680e0d6626a89522b8cd3e8872a322d103; independent certificate
sha256:96bb6f5fffb2813d6bd37c857ebcefbaea52fb4dbc0be4cf181c61e211b5ed55f; external-validation
hash sha256:34a836d4155c6adcc519db71df17e1e88b0bf8410580d5975add9911a705976a. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-M03949-2026

Comparison records:

- mixture-iupac-m03949: predicted matter-portion__two-or-more-substances__constituent-identities; source-derived matter-portion__two-or-more-substances__constituent-identities; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the IUPAC record lacks a matter portion, multiple chemical substances or constituent identities, or if a changed row is accepted.

Measurement receipt: sha256:4a9029755f68c1dcfee6328806f13fc53ef41c291ade2d9d411142716def89dc.

Isolation certificate: sha256:524b633738b03987b2829dfc549ebb26f223ca8145a7d6f19c2fcf3b52315744.

Custody certificate: sha256:a34c30b843d0157de7d0f213bb9ccf4c48fd63381f655000ecc2447725649dfc.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/M03949/json>
(sha256:71ba494096d995967227b9ef9fb4e6145b98c9cbcd46460bc63b94c06966f)

Registered target identities:

- mixture-iupac-m03949 from IUPAC-GOLD-BOOK-M03949-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite portion of matter containing multiple retained chemical-substance identities and their exact positive composition parts without requiring a new bonded substance identity.

Immutable identities. Source manifest

sha256:86bc2beefb095018a28b93ac40633cf7c5b82c148c38e5162ba8b84694446ee4; derivation seal

sha256:0d4bd1589336427e4ec4a3c2bbb4beeb2b3fec5d26122d080cd360dfbd92ca9c; engine receipt

sha256:85775961abbba9e37de249d62dd2a7653ff2fbf0f92949b00f105cbfbf6b8226 at

receipts/engine/model_admitted/SFT-CHEM-STOICH-MIXTURE-001-85775961abbba9e3.json; empirical

hash sha256:c5b6f171c506a272d5fbf676ddc210c0fefbc5c764cc11d6971c59a1c01ff8d8; measurement

receipt sha256:4a9029755f68c1dcfee6328806f13fc53ef41c291ade2d9d411142716def89dc.

25. Solution composition and solute-solvent distinction

Claim identity: SFT-CHEM-STOICH-SOLUTION-001

Question and theorem. A solution is a liquid or solid single-phase mixture containing multiple retained substances under an explicit contextual solvent/solute role partition.

liquid-or-solid-phase__single-phase-support__multiple-substance-support__solvent-solute-role-partition

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-MEAS-AMOUNT-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-STOICH-MIXTURE-001

Generated grammar. Generate the literal product of the registered solution carrier, phase, composition, roles, identity, context, record and extension choices.

Boundary: Every finite liquid or solid single-phase mixture with multiple retained substances and an explicit contextual partition between solvent carrier roles and solute carrier roles.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *liquid-or-solid-phase__single-phase-support__multiple-substance-support__solvent-solute-role-partition__substance-identities-retained__declared-context-role__phase-component-role-trace__no-extra-rule*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	liquid-or-solid-phase	gas-or-unclassified-matter	The registered solution boundary is a liquid or solid phase.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
phase	single-phase-support	multiple-unresolved-phases	Unresolved phase multiplicity does not define one solution carrier.
composition	multiple-substance-support	single-substance-phase	A pure phase lacks solute/solvent distinction.
roles	solvent-solute-role-partition	unpartitioned-components	A solution record distinguishes solvent from solutes.
identity	substance-identities-retained	roles-replace-identity	A role label must not erase substance identity.
context	declared-context-role	intrinsic-permanent-role	Solvent/solute assignment can depend on the description context.
record	phase-component-role-trace	solution-name-only	A name cannot reproduce phase and roles.
extension	no-extra-rule	free-solution-exception	An exception can relabel a multiphase mixture as one solution.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:4fe30a8a55d56b5ab42bf2b742148820ad68162cf9d5e70ee5a38004e0f457b9.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:4ea951e01a0a70099147774aea8b7a15b2ec3dc32a151b75faf81596cee969d7.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:24316754e898abeac2d19748aaf72de6a2f28f7cd99dd250b68a28cc3fa74def.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e1aad5725bea83acba570f25c4e0438cbff9c9620e5629755c76672e3f2b015a.

Independent reconstruction. Implementation hash

sha256:c0b00a98c9f8125cb8cf283d7fa1e8a33b328efeb815cb3e559cedc6740126bd; independent certificate sha256:105ee387d1fc186480013933b78a95f785fb02c759ac692c1cb36547a421997f; external-validation hash sha256:05aaaf5d349c42b312cec4e2819a004d12e4a9fb3ee07fa1ff110bec4626bdcf. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-S05746-2026

Comparison records:

- solution-iupac-s05746: predicted single-liquid-or-solid-phase__multiple-substance-support__solvent-solute-role-partition; source-derived single-liquid-or-solid-phase__multiple-substance-support__solvent-solute-role-partition; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the IUPAC solution record lacks liquid/solid phase, multiple substances or solvent/solute distinction, or if a changed row is accepted.

Measurement receipt: sha256:1f3495665eb39c150a4504ddc72d3dc9c85023807dbf005482a6721eca8ab780.

Isolation certificate: sha256:e03d3c3dfbbabc511092ea7cfa5fd61916d93d057ebb86c7bc237abd6382cd44.

Custody certificate: sha256:50e864e6293db75e8a1f402777dd89c133d9c9ba6fa73c5bfe2f28d127f1610a.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/S05746/json> (sha256:39f8916fdced6993ceb67e374641b95375570866a24123508ce26f9ec0049f28)

Registered target identities:

- solution-iupac-s05746 from IUPAC-GOLD-BOOK-S05746-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite liquid or solid single-phase mixture with multiple retained substances and an explicit contextual partition between solvent carrier roles and solute carrier roles.

Immutable identities. Source manifest

sha256: f5233dffc3c5df755d552eac064e4c4cb0418e20d604032c49c58745309f52cc; derivation seal
 sha256: b38c6b8072fe40cda7fd8d248c7bb8428e0a9704318d02eb0c53fc6689ea11e1; engine receipt
 sha256: 53c34d8473dac1d8e2567063e83f2d8ad1eab604fa4563b513b4084fa8bf934e at
 receipts/engine/model_admitted/SFT-CHEM-STOICH-SOLUTION-001-53c34d8473dac1d8.json;
 empirical hash sha256: b1253b86b3709b577b8e7516b1fde9cfe8bfc340ceb118eac9e1076558404a3a;
 measurement receipt sha256: 1f3495665eb39c150a4504ddc72d3dc9c85023807dbf005482a6721eca8ab780.

11. Bonding and molecular organization - 12 admitted claims

Bonds and molecules are generated joint-support and recurrence structures with held adjacency and orientation. This entire registered subcategory is included; no proper subset is presented as field completion.

26. Chemical bond as retained interaction closure

Claim identity: SFT-CHEM-BOND-CHEMICAL-BOND-001

Question and theorem. A chemical bond exists when complete source-bound interaction support between distinct atomic carrier occurrences or groups closes into stable joint recurrence distinguishable as an independent chemical carrier.

atoms-or-atomic-groups__source-bound-interaction-support__stable-joint-recurrence__independent-chemical-entity

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-CONDENSED-BAND-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-VALENCE-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001

Generated grammar. Generate the literal product of the registered bond carrier, interaction, closure, independence, energy, reversal, record and extension choices.

Boundary: Every finite pair or group of admitted atomic carriers and complete interaction support whose joint recurrence is stable as an independently distinguishable chemical carrier at the registered boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `atoms-or-atomic-groups__source-bound-interaction-support__stable-joint-recurrence__independent-chemical-entity__measured-stability-record-separated__opening-closes-to-separated-support__carrier-interaction-recurrence-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	atoms-or-atomic-groups	bond-name-without-atoms	A bond name detached from carriers has no chemical support.
interaction	source-bound-interaction-support	coincidental-proximity	Proximity alone need not produce a stable entity.
closure	stable-joint-recurrence	transient-contact-only	A contact without recurrence is not an independent bonded carrier.
independence	independent-chemical-entity	parts-only-observation	If only separated parts persist, no joined entity is formed.
energy	measured-stability-record-separated	imported-potential-function	A consensus potential cannot select closure.
reversal	opening-closes-to-separated-support	irreversible-by-definition	Bond opening may be lawful and must retain carriers.
record	carrier-interaction-recurrence-trace	bond-label-only	A label cannot reproduce interaction and stability.
extension	no-extra-rule	free-bond-exception	An exception can call any contact a bond.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:c79949b0bfe9498cf4d374c3041cad4bd3d2c76f13ee438fb68d1e4fed80a468.

- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:67d9bddfde9089152683c855abe345423e079bb9c69cbb4c2ec01eaea88c89ff.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:829ed6daf9d8afea57c415f332a212cf2dd68a7834c99709e75ed5ca411522b8.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:8647853beaa9c2398420258b61194b99ef938fb4a29bf75625cae36762f09e7b.

Independent reconstruction. Implementation hash

sha256:13efe64c7d43bb48869c408cbab8c44f2ccf1d7ed6e0e875e332e021c448ec6b; independent certificate sha256:13b94ba61ffe44d8d734add0368d83768a5e8e2384b92b4517df7bc2422418a6; external-validation hash sha256:98f7b13b41ab9ab294d4210af3adfbe746b7ca763a64adcf821b18763552b362. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CT07009-2026

Comparison records:

- chemical-bond-iupac-ct07009: predicted atoms-or-atomic-groups__stable-independent-molecular-entity__interaction-forces; source-derived atoms-or-atomic-groups__stable-independent-molecular-entity__interaction-forces; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if IUPAC lacks atomic/group carriers, stable independent molecular entity or interaction-force support, or if a changed row is accepted.

Measurement receipt: sha256:60ed33e3c2025cf530ee3b834573339b9c21731f0090c43ece4e1d13d34db4c8.

Isolation certificate: sha256:836e67cb59bba6af5d64f7f5cd231dd7e76af8613f6060bea06479eb7f7f7656.

Custody certificate: sha256:fc196685478ebe057a0b1ab61ba03dadf1b7c02fca6efd84b3281912baf86268.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/CT07009/json> (sha256:691e8522684b2e4290c207ae988a4689d1ab255cdd5fca4ed66a1bd557fa4207)

Registered target identities:

- chemical-bond-iupac-ct07009 from IUPAC-GOLD-BOOK-CT07009-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC bond definition, orbital equation, electronegativity scale, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- no universal bond-length/strength formula inferred from a categorical correspondence
- official target content opens only through post-seal custody
- Every finite pair or group of admitted atomic carriers and complete interaction support whose joint recurrence is stable as an independently distinguishable chemical carrier at the registered boundary.

Immutable identities. Source manifest

sha256: 45f6e312c4867083332ecfee9967a4efe1fcea77e04a309cf4f614ccbed25f5a; derivation seal
 sha256: 62af179236bb2dcc0b0ec73eafc94b7fabf03529b79701d7fcea3474144272ac; engine receipt
 sha256: 7c6c37891bbf8d54a3c8e41b6bafef24c29b35c7c380f1266cf62f8db2e28313e at
 receipts/engine/model_admitted/SFT-CHEM-BOND-CHEMICAL-BOND-001-7c6c37891bbf8d54.json;
 empirical hash sha256: 9b8cb421f61a5a59ec9c4387b443140d8e9d1524df6174ad4b3702f89b7c2f85;
 measurement receipt sha256: 60ed33e3c2025cf530ee3b834573339b9c21731f0090c43ece4e1d13d34db4c8.

27. Covalent shared-support bond

Claim identity: SFT-CHEM-BOND-COVALENT-001

Question and theorem. A covalent bond is stable joint recurrence in which one or more exclusion-preserving electron-labelled channels are jointly accessible between identified nuclear carriers without duplicating the conserved carrier.

identified-nuclear-pair__joint-electron-support__one-support-shared-access__internuclear-accessible-region

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-CONDENSED-BAND-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-VALENCE-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001

Generated grammar. Generate the literal product of the registered covalent carrier, electron, sharing, density, attraction, occupation, record and extension choices.

Boundary: Every finite bonded atomic support with one or more electron-labelled recurrence channels jointly accessible to both nuclear carriers and exclusion-preserving occupation.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `identified-nuclear-pair__joint-electron-support__one-support-shared-access__internuclear-accessible-region__stable-contractive-response__exclusion-preserving-channels__nuclei-channel-sharing-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	identified-nuclear-pair	bond-without-nuclei	Covalent support relates identified atomic centres.
electron	joint-electron-support	electron-assigned-to-one-centre	One-centre support alone is not shared bonding support.
sharing	one-support-shared-access	copied-electron-label	Copying one conserved carrier violates exclusion and conservation.
density	internuclear-accessible-region	empty-internuclear-path	No connecting support cannot join the nuclei.
attraction	stable-contractive-response	unbound-force-label	A force name without recurrence does not establish a bond.
occupation	exclusion-preserving-channels	duplicate-identical-cell	Duplicated identical occupation violates exclusion.
record	nuclei-channel-sharing-trace	covalent-label-only	A label cannot reconstruct sharing.
extension	no-extra-rule	free-orbital-rule	An imported orbital choice can select a desired bond.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f59fe4de319dd05bd3b6b6d48f1de3d15a39d2deee4bc65cbf5214affe909a3c`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:92ed58420e5d68434c41c74d2cb5eb78fb3782d79080388c4cf7c4efdbed92d0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ee7b74ae479c8c65c71393fac1927bf6dc7eb709d54e5f3a9f14167debde57f7`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a6c12d5e1a6339f3a4c4318629455f975ea16a517e11d0bb1cf857097839ea88`.

Independent reconstruction. Implementation hash

`sha256:4b68316cb8fb10965e70db523fc6335b89af5a692c612d7eaa9634157368aa0f`; independent certificate `sha256:3544a1af89d8576edf7c999b7ab8c70fada8cf2ad63bc3b0bef03541d0c4992c`; external-validation hash `sha256:5abb6ad5bcac23ef23e6129c1b10291d2a99be78918e51f1f4d8b8ef633e4221`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-C01384-2026

Comparison records:

- `covalent-bond-iupac-c01384`: predicted `internuclear-electron-density__electron-sharing__attractive-characteristic-distance`; source-derived `internuclear-electron-density__electron-sharing__attractive-characteristic-distance`; exact match `True`

- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if IUPAC lacks internuclear electron density, sharing or attractive characteristic distance, or if a changed row is accepted.

Measurement receipt: sha256:da76bd1b4898f3cfd5d9aaefedc5f5abec3bd76699f8e913a96537f01bb8777d.

Isolation certificate: sha256:58387cb8c623f636c8f05b24ea0d131f845798a3bca9bf827a889d810f38326b.

Custody certificate: sha256:19451c60c0ec98f763d3bba06a11548845a85342b23a08d86245b944a881df2c.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01384/json> (sha256:5106a40f5b1c372a570fc42f3cb1641b63a270454e4a7138699c584754294622)

Registered target identities:

- covalent-bond-iupac-c01384 from IUPAC-GOLD-BOOK-C01384-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC bond definition, orbital equation, electronegativity scale, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- no universal bond-length/strength formula inferred from a categorical correspondence
- official target content opens only through post-seal custody
- Every finite bonded atomic support with one or more electron-labelled recurrence channels jointly accessible to both nuclear carriers and exclusion-preserving occupation.

Immutable identities. Source manifest

sha256:83a8f02b154128b6ae7b30583d51ddd28d1e1a024468c757f7646c15e6536774; derivation seal

sha256:3ed8bc2b5e320ea65c0a63c5859311b9f7a234a224fd13db3d2574319b6fe26e; engine receipt

sha256:fd341d4ed9119d0e13bada61ffec798c5779cf4a8488bb8ea056f491732d0ef5 at

receipts/engine/model_admitted/SFT-CHEM-BOND-COVALENT-001-fd341d4ed9119d0e.json; empirical

hash sha256:56ae3c4b3309c5ac5695055d052750ad0815ec6f80fb474352c28aebb7114f08; measurement

receipt sha256:da76bd1b4898f3cfd5d9aaefedc5f5abec3bd76699f8e913a96537f01bb8777d.

28. Ionic transferred-label bond

Claim identity: SFT-CHEM-BOND-IONIC-001

Question and theorem. An ionic bond is stable joint recurrence of opposed cation/anion held charge fibres produced by closed charge transfer, with electrostatic attraction and both chemical identities retained.

cation-anion-carriers__closed-charge-transfer__opposed-held-charge-fibres__electrostatic-attraction-support

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-CONDENSED-BAND-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-VALENCE-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001

Generated grammar. Generate the literal product of the registered ionic carrier, formation, orientation, interaction, identity, character, record and extension choices.

Boundary: Every finite charge-transfer product containing opposed cation/anion held fibres whose closed electric interaction yields stable joint recurrence while both chemical identities remain retained.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `cation-anion-carriers__closed-charge-transfer__opposed-held-charge-fibres__electrostatic-attraction-support__chemical-identities-retained__exact-support-composition__ion-transfer-interaction-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	cation-anion-carriers	neutral-atom-pair-only	The ionic class requires retained ion carriers.
formation	closed-charge-transfer	unpaired-charge-creation	Charge cannot appear without conserved transfer.
orientation	opposed-held-charge-fibres	signed-charge-magnitudes	Negative proof magnitude is inadmissible.
interaction	electrostatic-attraction-support	generic-proximity	Proximity does not explain ionic stability.
identity	chemical-identities-retained	element-identity-erased	Ion formation preserves nuclear element identity.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
character	exact-support-comp osition	pure-binary-dogma	Observed bonds may retain both ionic and shared support.
record	ion-transfer-inter action-trace	ionic-label-only	A label cannot reconstruct transfer and attraction.
extension	no-extra-rule	free-electronegativity-cu toff	A fitted cutoff would import the classification.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:9cbd2a42338ef908d2f97d639e4e163a85f7c01e8f842a1027ea1bd7d82156f7.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:e01b7d4d4f8d957a884aca8560cad6aa53ce0add2f825323d75db608900ab55d.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:2259224cc1c5e1e8fe927ed597b35a75ef8ace609b8f28dab7e2a23e6537117b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:92462b6af0cedac2f75734bf5743e5454e360fc260e71256956609f5be56c39.

Independent reconstruction. Implementation hash

sha256:d343bcfb4e46576e97186ef06aba3aa0ae1e3d63bb00cafe938cc6172a3320bb; independent certificate sha256:3fb249ae42a0f7cfe14ee71adccafdc0dc851d4d7cc205a8b438663482a103ce; external-validation hash sha256:ddf0a613855feb1050fc67af920b451c93ff1900c43f2e74a389202aeaa66b16. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-IT07058-2026

Comparison records:

- ionic-bond-iupac-it07058: predicted cation-anion-carriers__electrostatic-attraction__ionic-character-boundary; source-derived cation-anion-carriers__electrostatic-attraction__ionic-character-boundary; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if IUPAC lacks cation/anion carriers, electrostatic attraction or the ionic-character boundary, or if a changed row is accepted.

Measurement receipt: sha256:78b00094cf52c285d2cd7a418074807d5123d92885b3c1659150870ad0ce617e.

Isolation certificate: sha256:8b5dc7db2c623bbd737cc505939ed5e10f93329f4f73a647d88cdd15e43674c3.

Custody certificate: sha256:18c409ddb69b5f3abedbb178e8f383908a9d3b8aa724ed1200f4fada92f21049.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/IT07058/json>
(sha256:85d726bab39480e944dfd95e31889931459f4436480d751d73df058c68ad232e)

Registered target identities:

- ionic-bond-iupac-it07058 from IUPAC-GOLD-BOOK-IT07058-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC bond definition, orbital equation, electronegativity scale, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- no universal bond-length/strength formula inferred from a categorical correspondence
- official target content opens only through post-seal custody
- Every finite charge-transfer product containing opposed cation/anion held fibres whose closed electric interaction yields stable joint recurrence while both chemical identities remain retained.

Immutable identities. Source manifest

sha256:3a04393c8396a53c4c35ca865a85b9787dbf894058cabd28ae3b9e1c8e837064; derivation seal
 sha256:7acbe40730e841eb98002bd110b3c308ad561bb1cd0376d510d099df3e8de583; engine receipt
 sha256:a77c3443d2e937680cf731cca982bb71b567d83f591baef101f04f8824be8b58 at
 receipts/engine/model_admitted/SFT-CHEM-BOND-IONIC-001-a77c3443d2e93768.json; empirical hash
 sha256:201790a5dc6966d75c17a0bad64a42ec3498fdd9ecd99fc304a2ae6b3d2cae31; measurement receipt
 sha256:78b00094cf52c285d2cd7a418074807d5123d92885b3c1659150870ad0ce617e.

29. Metallic collective-support bond

Claim identity: SFT-CHEM-BOND-METALLIC-001

Question and theorem. Metallic bonding is stable collective recurrence of a connected atomic network supported by electron-labelled paths delocalized across multiple cells of the complete finite network.

connected-atomic-network__delocalized-electron-support__finite-whole-network-extent__collective-shared-support

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-CONDENSED-BAND-001

- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-VALENCE-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001

Generated grammar. Generate the literal product of the registered metallic carrier, electron, extent, joining, recurrence, identity, record and extension choices.

Boundary: Every finite connected atomic network whose electron-labelled accessible support extends across multiple atomic cells and whose collective recurrence stabilizes the whole carrier.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `connected-atomic-network__delocalized-electron-support__finite-whole-network-extent__collective-shared-support__stable-collective-recurrence__atomic-and-network-identities__network-electron-recurrence-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	connected-atomic-network	isolated-atom-pair	Metallic support is collective rather than one fixed pair.
electron	delocalized-electron-support	localized-pair-only	A fixed pair cannot mediate the complete network.
extent	finite-whole-network-extent	unrecorded-infinite-lattice	A completed infinite lattice is not generated.
joining	collective-shared-support	independent-local-bonds-only	Independent local pairs omit collective redistribution.
recurrence	stable-collective-recurrence	transient-conduction-only	Transport alone does not establish stable bonding.
identity	atomic-and-network-identities	anonymous-metal-label	A metal label cannot create the law.
record	network-electron-recurrence-trace	metallic-label-only	A label cannot reproduce topology and support.
extension	no-extra-rule	free-electron-sea-premise	An imported metaphor cannot select the structure.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: f274ec7ff0584a9b1aa9ba3b9c7ecc7e0bc0a0c9e2a02f0280cc33fb7cf4b0d5.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 9e188caae50ec6514eba65a8d667abb1076a7ded34f9e0a45f6ec4ae466c1f27.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 6fe108c3748d6af8f85ab8790147389a9628a359676b39b909121c43441035ff.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:1801270185f1d5190300b4a9bd1f172f07b407b7f606e915414dc3da23914cd1.

Independent reconstruction. Implementation hash

sha256:f2495ddd58f17bc3d5859581fa2a310c528cd32e38b9434b51221ada9d9b22db; independent certificate sha256:32d15d7feb44f6a46c04372c2c1dd7eb026023f09f119126adbf729397408be7; external-validation hash sha256:6c0c29e63dde1b5df93f5a5279700b3a6410d5f326e5e4432274e1de5767b11e. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-08789-2026

Comparison records:

- metallic-delocalization-iupac-08789: predicted not-one-atom-or-bond__extended-whole-lattice-support__typical-of-metals; source-derived not-one-atom-or-bond__extended-whole-lattice-support__typical-of-metals; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if IUPAC lacks nonlocalized electrons, extended whole-lattice support or explicit metal correspondence, or if a changed row is accepted.

Measurement receipt: sha256:5cced161b50264b5bd4485c9c3245b46df80aa722f500ff3d46c02ff5dd12e34.

Isolation certificate: sha256:70ae5be489d99c31ca74b0621c9a33b31f427a739d26765af50db03eb32223cc.

Custody certificate: sha256:fec9c7af3d5d561ed1633eaf994975d29612be99eddf525e44d0a34c6d3b75c1.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/08789/json> (sha256:570755940f01bfa32741b03b6b2f22b02742101605a2263e57369966ea433abd)

Registered target identities:

- metallic-delocalization-iupac-08789 from IUPAC-GOLD-BOOK-08789-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC bond definition, orbital equation, electronegativity scale, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- no universal bond-length/strength formula inferred from a categorical correspondence
- official target content opens only through post-seal custody
- Every finite connected atomic network whose electron-labelled accessible support extends across multiple atomic cells and whose collective recurrence stabilizes the whole carrier.

Immutable identities. Source manifest

sha256:c369d1dc50eef809940247e481946b5e23801a11c936b59e6692ee57663c020d; derivation seal

sha256:58159c40dba161ffc72882d05d1ab1f820cac2ba2cdcdebc9131d4fb2ac3e14a; engine receipt

sha256:afd35cdaad243c1fa9f9988c0ce7d5a6ee199d081d687f2115d59329041399c7 at

receipts/engine/model_admitted/SFT-CHEM-BOND-METALLIC-001-afd35cdaad243c1f.json; empirical

hash sha256:d5b37c4f467daa5e6fc056a5dffaa74c0d7d696f2fd2689062d60c291a8168ae; measurement

receipt sha256:5cced161b50264b5bd4485c9c3245b46df80aa722f500ff3d46c02ff5dd12e34.

30. Bond order as retained joining multiplicity

Claim identity: SFT-CHEM-BOND-ORDER-001

Question and theorem. Bond order is the exact positive rational degree of complete joining-channel support for one identified bond relative to one localized electron-pair joining unit, with the observation partition retained.

identified-bond-carrier__complete-joining-channel-support__single-localized-pair-reference__exact-positive-rational-degree

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-CONDENSED-BAND-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-VALENCE-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001

Generated grammar. Generate the literal product of the registered bond-order carrier, support, reference, quantity, delocalization, method, record and extension choices.

Boundary: Every finite identified bond with complete joining-channel support, measured relative to one localized electron-pair joining unit using exact positive counts or rational support parts.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *identified-bond-carrier__complete-joining-channel-support__single-localized-pair-reference__exact-positive-rational-degree__rational-support-sharing-retained__partition-bound-index__bond-channel-re*

ference-trace__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	identified-bond-carrier	index-without-bond	A number detached from a bond has no meaning.
support	complete-joining-channel-support	selected-channel-count	Omitting channels can force the degree.
reference	single-localized-pair-reference	arbitrary-scale	A free scale can select any order.
quantity	exact-positive-rational-degree	floating-index	Floating arithmetic cannot certify the exact support ratio.
delocalization	rational-support-sharing-retained	integer-only-order	Delocalized support can distribute a joining unit across links.
method	partition-bound-index	one-partition-as-absolute	Different observational partitions can report different indices.
record	bond-channel-reference-trace	order-answer-only	An index alone cannot reproduce support.
extension	no-extra-rule	free-order-correction	A correction can force a conventional integer.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:e2db93d7acdd9cf1733eff52d8fd837ae587de58815c0f9e206eff130e6bfc39.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:0730cf4b586be0e8feb19dc34f293a1d2078cf398ce3ca253c97db2a0c352073.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:60a2093f539375e58bac443e95b50dbf98cd187090e8d91adb8100ffbc1a0d87.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:1f104e993203597e06f02f2080cf9f28583704674402ade6ceb7e601696867a7.

Independent reconstruction. Implementation hash

sha256:c65f2049ad12f2d618472090042a4f0a56dda8e31fabea284b65ecece6942c09; independent certificate sha256:cc744913cdce2a2bb18a73bddf76ea9c4e37f2ca0e0069b43d07769c5174fd49; external-validation hash sha256:259f33bfcfccfd6a94126ba047348947426754771d49fbe2dbbbfc6238f50b5. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-B00707-2026

Comparison records:

- bond-order-iupac-b00707: predicted bonding-degree-index__single-bond-reference__localized-electron-pair; source-derived bonding-degree-index__single-bond-reference__localized-electron-pair; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if IUPAC lacks a bonding-degree index, single-bond reference or localized electron-pair reference, or if a changed row is accepted.

Measurement receipt: sha256:d5dab602be33f40f67a9b153047af4184958f9e915f7096dc2e3bfdc240118e7.
 Isolation certificate: sha256:7d61aaaaa4ade59559d3376734eca9e824d1e990cf4a6f9b9328ed499bddb9a8.
 Custody certificate: sha256:672add9e9f7b59244ad3984493603821da4fcf479f61bdab8f639030c87fcbad.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/B00707/json> (sha256:f8056279c1ae14cc144d184f01797de9aa2d8fb69ed868b9753ba2e63255dccb)

Registered target identities:

- bond-order-iupac-b00707 from IUPAC-GOLD-BOOK-B00707-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC bond definition, orbital equation, electronegativity scale, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- no universal bond-length/strength formula inferred from a categorical correspondence
- official target content opens only through post-seal custody
- Every finite identified bond with complete joining-channel support, measured relative to one localized electron-pair joining unit using exact positive counts or rational support parts.

Immutable identities. Source manifest

sha256:4e7b9117280b9a849a93c95307dfffb1cd1511546b5013a7b43bfb1c0e770b44c; derivation seal
 sha256:711f82cfb57eb79402a0922ba6dd8de9282622cbd5efc762cda77c9860c2e284; engine receipt
 sha256:171120df1e712d3def9a761bca06f86050c596c0db773ff0d0f01b6265c84546 at
 receipts/engine/model_admitted/SFT-CHEM-BOND-ORDER-001-171120df1e712d3d.json; empirical hash
 sha256:a7d813b07ec1f077ebecf01ddba91c06b07ae27402fe8f433b8fd08d2de11778; measurement receipt
 sha256:d5dab602be33f40f67a9b153047af4184958f9e915f7096dc2e3bfdc240118e7.

31. Bond length, energy and identity correspondence

Claim identity: SFT-CHEM-BOND-LENGTH-STRENGTH-001

Question and theorem. Bond length and strength are separate source-bounded records paired by one specific bond identity: inter-centre distance and positive dissociation transfer under retained methods and conditions; no universal formula is imported.

specific-identified-bond__method-bounded-centre-distance__positive-dissociation-transfer__common-bond-identity-pairing

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-CONDENSED-BAND-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-VALENCE-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-ORDER-001

Generated grammar. Generate the literal product of the registered length/strength carrier, length, strength, conditions, correspondence, measurement, record and extension choices.

Boundary: Every identified bond with one source/method-bounded inter-centre distance record and one separately source/condition-bounded positive dissociation-transfer record, joined only by common bond identity.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `specific-identified-bond__method-bounded-centre-distance__positive-dissociation-transfer__measurement-conditions-retained__common-bond-identity-pairing__separate-source-bound-records__bond-length-strength-source-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	specific-identified-bond	generic-bond-type-only	A type average can erase the specific molecular carrier.
length	method-bounded-centre-distance	context-free-distance	Measured bond length depends on method and state.
strength	positive-dissociation-transfer	qualitative-strong-weak-label	A qualitative label cannot reproduce dissociation.
conditions	measurement-conditions-retained	conditions-erased	Phase, state and cleavage path affect measured records.
correspondence	common-bond-identity-pairing	universal-inverse-equation	A categorical relation does not force one universal formula.
measurement	separate-source-bound-records	decimal-as-proof-scalar	Measured decimals cannot enter derivation.
record	bond-length-strength-source-trace	paired-values-without-provenance	A pair without methods cannot be compared.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
extension	no-extra-rule	free-length-strength-fit	A fitted curve can force measured examples.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 7e1b3cf5cdf81e62b593729ae380c6b6151fbb768b02e8bf683fd10d67c2c8f.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 6d7e7989df91ca411877aaff4507ef5c4000a87e17468950721c429d0230f768.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: fd7a974bad4e601cb6c1b60bb8f0389b4d00b597b8529173ea28367aedbfff8f9.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 1026b70bc7c6d6d1fcaac20815b08bbbd0f66f656c3ee48a5cb01d4a3d22f86b.

Independent reconstruction. Implementation hash

sha256: 1ab7ee39098a0f17e5dc2489b48970f1f57d1444593a8701101beed99fc8fd04; independent certificate sha256: 93ff0d4b30b1dc3cadf664a6d1271083634771cc896a074b5af4ad60dc8dc0b2; external-validation hash sha256: c3a6c3d6319fdb6fb2facbc7c90850e7b228010c380c241d7eccdc99db75eae4. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-B00702-2026

Comparison records:

- bond-length-strength-iupac-b00702: predicted bond-length__bond-dissociation-energy__bond-order-empirical-relationship; source-derived bond-length__bond-dissociation-energy__bond-order-empirical-relationship; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if IUPAC lacks the registered empirical relationship among bond length, dissociation energy and order, or if a changed row is accepted; it does not fail merely because no universal monotone formula exists.

Measurement receipt: sha256: c851de6f97527329aa6b79dbe76bde3a19b51e5d7d1e7c9674758c1959a09f95.

Isolation certificate: sha256: 58a0db186b0ad2988bccc0665ab9ae99e97ae165babc7150f568a72c1859e910.

Custody certificate: sha256: a46586309a81a1ab8462dd750a326fc9dbe97830cf88e879fe58cbe715203ccd.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/B00702/json> (sha256:f53dadb6357406a52d41d9d4be1f7698d4db3b8954b04f798ba823b18c40c821)

Registered target identities:

- bond-length-strength-iupac-b00702 from IUPAC-GOLD-BOOK-B00702-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC bond definition, orbital equation, electronegativity scale, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter

- no application result or opaque predictor
- no universal bond-length/strength formula inferred from a categorical correspondence
- official target content opens only through post-seal custody
- Every identified bond with one source/method-bounded inter-centre distance record and one separately source/condition-bounded positive dissociation-transfer record, joined only by common bond identity.

Immutable identities. Source manifest

sha256:e981a081d6d424e39601ec42a83c921cb09a0847c1aa05225a8846aa6efdbb6d; derivation seal
 sha256:c6481ebbc263cd81703eb9f2e749c88a03efd6a41cb6f96c198e232a837fcf1; engine receipt
 sha256:12bf183bd0effffa233f2b3cffc4c423789a93eb997587d9e584c2bff96a6c09 at
 receipts/engine/model_admitted/SFT-CHEM-BOND-LENGTH-STRENGTH-001-12bf183bd0effffa.json;
 empirical hash sha256:e79a4d212af6c0a4ddc0b59e08cf39df3433ea735ba056cdd52b318677c20a32;
 measurement receipt sha256:c851de6f97527329aa6b79dbe76bde3a19b51e5d7d1e7c9674758c1959a09f95.

32. Molecular identity and complete bonded carrier

Claim identity: SFT-CHEM-MOL-MOLECULE-001

Question and theorem. A molecule is one finite neutral multi-atomic carrier whose complete bonded graph is connected and whose constituent occurrence and element identities remain retained.

multiple-atomic-occurrences__complete-bond-support__one-connected-component__closed-charge-boundary__constituent-identities-retained

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001

- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-BOND-LENGTH-STRENGTH-001

Generated grammar. Generate the literal product of the registered molecule cardinality, bonding, connectivity, charge, composition, identity, record and extension choices.

Boundary: Every finite generated multi-atomic carrier with complete connected bond support, retained constituent identity and no open net-charge fibre at the registered observation boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `multiple-atomic-occurrences__complete-bond-support__one-connected-component__closed-charge-boundary__constituent-identities-retained__complete-structure-identity__complete-carrier-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
cardinality	multiple-atomic-occurrences	single-atomic-occurrence	One occurrence has no internal molecular bond graph.
bonding	complete-bond-support	unbonded-collection	A collection without joining support is not one molecule.
connectivity	one-connected-component	disconnected-components	Disconnected components are separate carriers.
charge	closed-charge-boundary	open-net-charge-fibre	An open net fibre describes a molecular ion rather than the neutral molecular class.
composition	constituent-identities-retained	anonymous-atom-count	Counts without element identities cannot reproduce the carrier.
identity	complete-structure-identity	formula-alone	Equal composition need not imply one molecular structure.
record	complete-carrier-trace	molecule-label-only	A label cannot reconstruct atoms and bonds.
extension	no-extra-rule	free-molecule-exception	An exception can admit arbitrary aggregates.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: bbed887164a606acb5cca576ab0d7da79e9254ab958d14e368f2d1b686224b7c.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: d21385a0ac8db65b285a1e5cf9b46eb58fb7d306b196d5c08fd900001d7916b1.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 811a93ff2e246131afd67a2e53ecba9ba2d2c083c686188c278f8e537ee69785.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 809480cc84ffdf38484c8ef07fe549a9a9e44f9c25a0943073cc937513b52886d.

Independent reconstruction. Implementation hash

sha256: 22db6e9a5a35d9e670f83677a119a13cb1eb94927c93ed2e9e10636410408ce1; independent certificate sha256: 94b03b38ed013a00a089d3cb57cc45f51fba0e3cacc8ccc0b6697a98eb70c366; external-validation hash sha256: 732f4300695f8ac8579d44c5967e3a902163ad21dfdf34b3291f25d35bf392e. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-M04002-2026

Comparison records:

- molecule-iupac-m04002: predicted
multi-atomic-carrier__connected-complete-bond-support__closed-charge-boundary__constituent-identities-retained;
source-derived
multi-atomic-carrier__connected-complete-bond-support__closed-charge-boundary__constituent-identities-retained; exact
match `True`
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official molecular record lacks multiple atomic constituents, one bonded chemical entity or retained constituent identity, or if a changed row is accepted.

Measurement receipt: sha256:607feb81a94957c8b75533fc5d517f5b2ad77f390f8a85f9f33ea08151a68507.

Isolation certificate: sha256:e92b14a94255d887e93614fb008c647fd2906a4a0f9268996ac1618f946aa995.

Custody certificate: sha256:134fa9deb99dd268daab1dd248b72b1261b1ebe2ab4f13c6ab6b15bb914e5aac.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/M04002/json>
(sha256:a5058dad716a0233d1a49fb8869d30b581cff6f483317ffdaaf10a3293c52e36)

Registered target identities:

- molecule-iupac-m04002 from IUPAC-GOLD-BOOK-M04002-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no chemical dictionary, coordinate archive, fitted force field, measured geometry or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application or opaque prediction result
- no claim that topology alone predicts every measured coordinate or energy
- external content remains inaccessible until after the consequence is sealed
- Every finite generated multi-atomic carrier with complete connected bond support, retained constituent identity and no open net-charge fibre at the registered observation boundary.

Immutable identities. Source manifest

sha256:b70d0dee60058c1e0d457054b673ee81f37815149df754639349492185636944; derivation seal

sha256:940a580f5e80256f53d54e15ba2db51e36cc4888ddf579ac3ef9da0dee601fd; engine receipt

sha256:a075ff988b4db9d4c16a199fa6402277bb945aabfa38f32f9eebcdbd784f1fb1 at

receipts/engine/model_admitted/SFT-CHEM-MOL-MOLECULE-001-a075ff988b4db9d4.json; empirical

hash sha256:43e6fa2ff5fabef891fb70cdbb7b27312a6fc7cba4a89dda1ab01eff202b15e7; measurement

receipt sha256:607feb81a94957c8b75533fc5d517f5b2ad77f390f8a85f9f33ea08151a68507.

33. Molecular geometry from held adjacency and orientation

Claim identity: SFT-CHEM-MOL-GEOMETRY-001

Question and theorem. Molecular geometry is the complete relative adjacency-and-orientation form of one identified finite molecular carrier, invariant under a global rigid host-frame relabelling and separate from measured coordinate values.

identified-molecular-carrier__complete-atom-adjacency__held-relative-orientation__complete-pair-support__rigid-relabel-equivalence

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-BOND-LENGTH-STRENGTH-001
- SFT-CHEM-MOL-MOLECULE-001

Generated grammar. Generate the literal product of the registered geometry carrier, adjacency, orientation, completeness, equivalence, measurement, record and extension choices.

Boundary: Every finite admitted molecular carrier with a complete pairwise relative-orientation record, interpreted up to one global rigid relabelling and retained with its source and observation conditions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *identified-molecular-carrier__complete-atom-adjacency__held-relative-orientation__complete-pair-support__rigid-relabel-equivalence__source-bounded-geometry-record__adjacency-orientation-source-trace__no-extra-rule*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	identified-molecular-carrier	geometry-without-molecule	A shape detached from an identified carrier is not molecular geometry.
adjacency	complete-atom-adjacency	atom-list-only	An atom list omits which occurrences are joined.
orientation	held-relative-orientation	distance-only-record	Distances alone can omit handed or ordered orientation.
completeness	complete-pair-support	selected-coordinate-subset	A favorable subset cannot establish the whole geometry.
equivalence	rigid-relabel-equivalence	absolute-host-frame	A host coordinate frame cannot define chemical identity.
measurement	source-bounded-geometry-record	coordinate-as-proof-premise	A measured coordinate cannot select the law.
record	adjacency-orientation-source-trace	shape-name-only	A shape name cannot reproduce the relations.
extension	no-extra-rule	free-angle-fit	A fitted angle can force a target geometry.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 37ade72544361890f8b377d0be9fb8246aad5cccc01e85702f531dfea43806cf.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 2eef597bf64b1918b1dfd70e164ce86c8560f1d6cefe455e285fb5744b445b4a.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 42d88d1105e6572412e6725c928b48cf33f934003dcae249480f642068779791.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: b52da0b02b3272f33b9bbfb53e964d411909027aac6b11fac7589e1149d05c24.

Independent reconstruction. Implementation hash

sha256: 4f780754facfb7667f6db5ee9eedb8ae99aaf3ea8043383e94de88c4647caafc; independent certificate sha256: 8031cf31bbfbeb30b574d6ea6d4361d68c9fdb6cda3ce73f851af5f6bb1c512; external-validation hash sha256: 1b42100ba5378f5065e252bde551930747d9c7686fcddea86ae1136958564158. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-S06061-2026

Comparison records:

- molecular-geometry-iupac-s06061: predicted
identified-molecular-carrier__complete-atom-adjacency__held-relative-orientation__source-bounded-geometry-record;
source-derived
identified-molecular-carrier__complete-atom-adjacency__held-relative-orientation__source-bounded-geometry-record; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authoritative geometry record lacks identified atoms, relational spatial organization or source-bounded coordinates, or if a changed row is accepted.

Measurement receipt: sha256: ad21369442e5b2c57feb8b2e3369beea9d627af1b476f91db250fc066192b4a2.

Isolation certificate: sha256: cfc03e12f295b3928c662fac4754882f46fe4e9f43dcad8f168cb6e1f0dbbfe3.

Custody certificate: sha256: 870366e124bb052bee33bd51339a7fc3e14425b4ba1c3d54e4d6df2396dd8d5b.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/S06061/json> (sha256:65cc9feaea028f9ab7bd7ed34927652255ebcc66d2f079dc94d55fe9dc7230b)

Registered target identities:

- molecular-geometry-iupac-s06061 from IUPAC-GOLD-BOOK-S06061-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no chemical dictionary, coordinate archive, fitted force field, measured geometry or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application or opaque prediction result
- no claim that topology alone predicts every measured coordinate or energy
- external content remains inaccessible until after the consequence is sealed
- Every finite admitted molecular carrier with a complete pairwise relative-orientation record, interpreted up to one global rigid relabelling and retained with its source and observation conditions.

Immutable identities. Source manifest

sha256:5f1930bb586466cedb09433f2571b4601e39190cca6736c4cb6ee52ec97692f6; derivation seal sha256:d376c83016e57ce1a32db75872af7bc2c629a83cf16577a9ee79f06c14c0f41a; engine receipt sha256:ac60f99b83d623409f27d5997eae114ad5384e6fb2fdbcfc3c71b50d9a0aa054 at receipts/engine/model_admitted/SFT-CHEM-MOL-GEOMETRY-001-ac60f99b83d62340.json; empirical hash sha256:4d2df2952afb0c9c85752e96c6c4bfff3c5d0a8bde6d3856eb931bde635e0d650; measurement receipt sha256:ad21369442e5b2c57feb8b2e3369beea9d627af1b476f91db250fc066192b4a2.

34. Isomer distinction under equal composition

Claim identity: SFT-CHEM-MOL-ISOMER-001

Question and theorem. Isomers have exactly equal retained composition but inequivalent complete connectivity or orientation signatures after every identity-preserving finite relabelling.

two-identified-carriers__equal-exact-composition__complete-structure-signatures__all-identity-preserving-relabelings__connectivity-or-orientation-difference

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-BOND-LENGTH-STRENGTH-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-GEOMETRY-001

Generated grammar. Generate the literal product of the registered isomer pair, composition, structure, equivalence, distinction, identity, record and extension choices.

Boundary: Every finite pair of admitted chemical carriers with exactly equal composition signatures but unequal complete connectivity or held-orientation signatures after every identity-preserving relabelling.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `two-identified-carriers__equal-exact-composition__complete-structure-signatures__all-identity-preserving-relabelings__connectivity-or-orientation-difference__separate-chemical-identities__composition-and-structure-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
pair	two-identified-carriers	single-carrier-description	Isomerism compares at least two chemical carriers.
composition	equal-exact-composition	different-compositions	Different composition is not isomerism.
structure	complete-structure-signatures	formula-equated-identity	A formula does not exhaust bonded organization.
equivalence	all-identity-preserving-relabelings	name-string-inequality	Different names can denote the same structure.
distinction	connectivity-or-orientation-difference	equivalent-signatures	Equivalent complete signatures denote the same isomeric form.
identity	separate-chemical-identities	distinction-erased	Erasing the difference destroys the chemical question.
record	composition-and-structure-trace	isomer-label-only	A label cannot reproduce equality and difference.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
extension	no-extra-rule	free-isomer-class	A free class label can manufacture a distinction.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:cff91e3ddb362ea4243a3a751487636d8cbe5db830e215e91e1e8691584f0dcf.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:c2dbbbc0bbb52a72785bc5da5d04267b55b28aa26d80002fe05bd5ed7543bbcc.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:dc161f389ebcdec14659c79d57ce7f512f7f193d1ed845be8b7ae8da6361f5e3.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:ab3dfb81b50681603de1f62d122810cad4e7ad1bfa16ac3d1abc9d1abe830290.

Independent reconstruction. Implementation hash

sha256:7aa0f2f5ddb4f83e66d9ad7ac46a812691bd7707808f34962af813d170388c67; independent certificate
sha256:c465beb4071d7c4bbdf65b73e04f4e05d9f999142100dbbc350ac265c948d679; external-validation
hash sha256:9c6804453449153203b66533c8853b75b33f16769e337713d2d434075cc45d21. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-I03289-2026

Comparison records:

- isomer-iupac-i03289: predicted
equal-composition__distinct-structural-signature__connectivity-or-orientation-distinction__separate-chemical-identity;
source-derived
equal-composition__distinct-structural-signature__connectivity-or-orientation-distinction__separate-chemical-identity; exact
match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official isomer record does not require equal composition with distinct structure or spatial arrangement, or if a changed row is accepted.

Measurement receipt: sha256:83648160e3c07f72af7c7830dd773f77cce3547c9d65841eecf76cd4927647c68.

Isolation certificate: sha256:d72a2986f0f11e18dddc1e71eb5cb2e5f3274c43521a4cf0bb5146de65b892a6.

Custody certificate: sha256:025a6fe28402f9960fb04e581a376875e9e893bc46cd3c22ed8dfdbc113e00be.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/I03289/json>
(sha256:64fa2c9cc58b18b399d22481ee7f37692c7f9c904e4d5fff27c65dd3325393e7)

Registered target identities:

- isomer-iupac-i03289 from IUPAC-GOLD-BOOK-I03289-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no chemical dictionary, coordinate archive, fitted force field, measured geometry or V2 answer may select a candidate

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application or opaque prediction result
- no claim that topology alone predicts every measured coordinate or energy
- external content remains inaccessible until after the consequence is sealed
- Every finite pair of admitted chemical carriers with exactly equal composition signatures but unequal complete connectivity or held-orientation signatures after every identity-preserving relabelling.

Immutable identities. Source manifest

sha256:d416f25ba2e9a223bcf548f0f6804e5f2485ddf6f6fe2caf2f0640b8ad119bf6; derivation seal
 sha256:5ad6bf418c5bbb971dea651b7486e58e54491c8a2857f9ad5cfbe675e2e9bbcd; engine receipt
 sha256:93b049013df4a30b2fdd1be11c3b8fd05ade4c1126f22aa7527ae34bf54e335d at
 receipts/engine/model_admitted/SFT-CHEM-MOL-ISOMER-001-93b049013df4a30b.json; empirical hash
 sha256:d5d9bfe57b1c95e06e284442db738a9dfea4484893e04d17b59d4498823b2ca8; measurement receipt
 sha256:83648160e3c07f72afc7830dd773f77cce3547c9d65841eecf76cd4927647c68.

35. Intermolecular interaction and residual joining

Claim identity: SFT-CHEM-MOL-INTERMOLECULAR-001

Question and theorem. An intermolecular interaction is a registered response channel between already complete molecular carriers that retains each constituent molecular identity and supports collective recurrence.

distinct-molecular-carriers__between-whole-interaction-support__constituent-identities-retained__collective-recurrence__explicit-molecular-boundary

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001

- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-BOND-LENGTH-STRENGTH-001
- SFT-CHEM-MOL-MOLECULE-001

Generated grammar. Generate the literal product of the registered intermolecular carrier, support, identity, recurrence, boundary, strength, record and extension choices.

Boundary: Every finite interaction support joining two or more already complete molecular carriers while retaining each molecular identity and the boundary between inter- and intramolecular support.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `distinct-molecular-carriers__between-whole-interaction-support__constituent-identities-retained__collective-recurrence__explicit-molecular-boundary__source-bounded-interaction-record__carrier-channel-recurrence-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carriers	distinct-molecular-carriers	one-molecule-internal-edge	An internal edge is intramolecular support.
support	between-whole-interaction-support	proximity-without-channel	Proximity alone supplies no interaction trace.
identity	constituent-identities-retained	constituents-merged-erased	Erasing each molecule changes the class to a new bonded carrier.
recurrence	collective-recurrence	transient-coincidence	A coincidence without response cannot organize an assembly.
boundary	explicit-molecular-boundary	inter-intra-conflation	Conflation makes every covalent edge intermolecular.
strength	source-bounded-interaction-record	universal-weaker-law	Not every context forces one strength ordering.
record	carrier-channel-recurrence-trace	force-name-only	A force name cannot reconstruct endpoints and response.
extension	no-extra-rule	free-force-family	An imported family can select desired interactions.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:8c200aa3e761e0e8e583df86cea1e401ef6ac0666629e7e42bf01790e147cfaa.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:66170d1985dd0b0b805f940cb21d00d932c3bc1d354a0ee4387e3f3b857262c0.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:d5722e6b4ee1020dd3d52b8d3223fe482d257809ca683e78114888c2684038a1.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:fa996e1ad0ab9e877ebe04f8ca07a8deb28146632f4b5b5e8a2ee67cc8a7cb34.

Independent reconstruction. Implementation hash

sha256:d215c873502192ac1d7219b514b59406b547ce9b6be65c68c8902105a89cde0f; independent certificate sha256:1eb83e9304bfe476b0b2cd840e20e5867dd505fb5a069a8f8f95d35db08aeb49; external-validation hash sha256:3c2639fff93bd34d9eb1c13d952312c9f7778be6be8e60ffd1fa834635832379. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-I03098-2026

Comparison records:

- intermolecular-iupac-i03098: predicted
distinct-molecular-carriers__between-whole-interaction-support__constituent-identities-retained__collective-recurrence;
source-derived
distinct-molecular-carriers__between-whole-interaction-support__constituent-identities-retained__collective-recurrence; exact
match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official record lacks interaction between distinct molecules or preserved molecular endpoints, or if a changed row is accepted.

Measurement receipt: sha256:884b17739acc302f6797da5676051afc7d87e32cc5e3e3af09430bfe55407a73.

Isolation certificate: sha256:da24f0c43f2dcf3ea14583f76f94c98353d0ab425a2315c8f619f3a8430273a3.

Custody certificate: sha256:837e9e22a56215a045ba191a690a43bf841968f1b3b6f056aa52957452ac9918.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/I03098/json>
(sha256:ba7264c36447a463893ffdf66e0bce7478e764db59b928911a8f82b59af4b3e)

Registered target identities:

- intermolecular-iupac-i03098 from IUPAC-GOLD-BOOK-I03098-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no chemical dictionary, coordinate archive, fitted force field, measured geometry or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application or opaque prediction result
- no claim that topology alone predicts every measured coordinate or energy
- external content remains inaccessible until after the consequence is sealed
- Every finite interaction support joining two or more already complete molecular carriers while retaining each molecular identity and the boundary between inter- and intramolecular support.

Immutable identities. Source manifest

sha256:69507e148199a034bd6c20b2c664a9d4c1b7da25c6c7ecd17b9172b3f471b7b7; derivation seal

sha256:a3fd163cd110798bc13ae4bd246a201fe8e560938334c53391cb85b89aca2966; engine receipt

sha256:1ec50ca6ddec341fbce800beedbb12d473460da34391c9047b93c4c684750b40 at

receipts/engine/model_admitted/SFT-CHEM-MOL-INTERMOLECULAR-001-1ec50ca6ddec341f.json;

empirical hash sha256:fc95d25318b50430b9a06afef22c5e07f23ccc96d4ae7ceb5df32a5ff9e9a6d1;

measurement receipt sha256:884b17739acc302f6797da5676051afc7d87e32cc5e3e3af09430bfe55407a73.

36. Supramolecular organization by reversible recognition

Claim identity: SFT-CHEM-MOL-SUPRAMOLECULAR-001

Question and theorem. A supramolecular whole is a finite assembly of multiple intact molecular carriers joined by reversible complementary recognition support, with all components recoverable.

multiple-molecular-components__noncovalent-interaction-support__complementary-recognition__reversible-component-release__assembled-whole-identity

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-BOND-LENGTH-STRENGTH-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001

Generated grammar. Generate the literal product of the registered supramolecular components, joining, recognition, reversal, closure, scope, record and extension choices.

Boundary: Every finite assembly of multiple complete molecular carriers joined through reversible recognition support, with all components recoverable and the collective whole separately distinguishable.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `multiple-molecular-components__noncovalent-interaction-support__complementary-recognition__reversible-component-release__assembled-whole-identity__finite-declared-assembly__component-recognition-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
components	multiple-molecular-components	single-molecule-only	A supramolecular assembly contains multiple molecular components.
joining	noncovalent-interaction-support	new-internal-covalent-graph	Replacing components by one covalent graph erases the supramolecular boundary.
recognition	complementary-recognition	arbitrary-collision	A collision has no complementary selection trace.
reversal	reversible-component-release	components-unrecoverable	If components cannot be recovered, their identities were not retained.
closure	assembled-whole-identity	components-only	Separate components do not form an assembled whole.
scope	finite-declared-assembly	completed-infinite-assembly	A completed infinity is not generated.
record	component-recognition-trace	assembly-name-only	A name cannot reconstruct recognition and release.
extension	no-extra-rule	free-host-guest-rule	An exception can force any desired assembly.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 4624527ef8a225f350b93ce389704a0ca9f1c3f27a4fea4781d818ed9fd8b1d5.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 39aa04f9f35d247da547a24dd03d6d7994e5812a23cd1563bb5fd6a73ee2f846.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: af9101491f206ffe81a9ab14550635b603e45d1c4ee776f501cb4326b4556fdb.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: d14f7bef34eb9b65bd346b19ba3fc422b78107c054db310315f56e20fd7e11ec.

Independent reconstruction. Implementation hash

sha256: 0a59a72d662c2a0a8f04c59801c5cf4384a68ee42c609bc133443b8c3589b0e4; independent certificate sha256: 1d0d5cadf95fd4ea413ff2a3d3fcb215d96a64f1e82e125012659bd464ffa85c; external-validation hash sha256: 9bd0db47ea7cd392f932f378ab8b55075090158d890ed94430784c9202055757. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-13867-2026

Comparison records:

- `supramolecular-association-iupac-13867`: predicted
multiple-molecular-components__reversible-noncovalent-recognition__component-identities-retained__assembled-whole;
source-derived
multiple-molecular-components__reversible-noncovalent-recognition__component-identities-retained__assembled-whole; exact

match True

- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official supramolecular record lacks multiple molecular components, noncovalent organization or retained component identities, or if a changed row is accepted.

Measurement receipt: sha256:9bf7348107ea9a0e4500bf9295b7ad7e13a26872d48296341e0939db5cd3212c.

Isolation certificate: sha256:bb87893c9ebc198b556f03db3795b0f47554cc4b91e489f0fcbb6297182da282.

Custody certificate: sha256:c1e6416533d56a413a6c2d7b649f924edbc5557a92ed323d1920b5237a7633ff.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/13867/json> (sha256:34dd0e2a436c762eefbb6ddf8ab84b16b716b252632fc0835ec083d4d4a2551a)

Registered target identities:

- supramolecular-association-iupac-13867 from IUPAC-GOLD-BOOK-13867-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no chemical dictionary, coordinate archive, fitted force field, measured geometry or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application or opaque prediction result
- no claim that topology alone predicts every measured coordinate or energy
- external content remains inaccessible until after the consequence is sealed
- Every finite assembly of multiple complete molecular carriers joined through reversible recognition support, with all components recoverable and the collective whole separately distinguishable.

Immutable identities. Source manifest

sha256:f9022f39fda950e7d4966459bead83ac43b81a919b7dfed3601f8c49755ff2dc; derivation seal

sha256:01e849c6b20324f84b4dd3fdb8719329aef58f57df41baf145ba73d697c885c7; engine receipt

sha256:a38e3bdbb55e2cfe79d4491665da0f991ab463f874273e8e7b3c13adfa7538d1 at

receipts/engine/model_admitted/SFT-CHEM-MOL-SUPRAMOLECULAR-001-a38e3bdbb55e2cfe.json;

empirical hash sha256:eea420d0618c24416142c648cc8b1b3d17e86dc2a1c474cb17d9d31510dbd661;

measurement receipt sha256:9bf7348107ea9a0e4500bf9295b7ad7e13a26872d48296341e0939db5cd3212c.

37. Molecular network and connected chemical whole

Claim identity: SFT-CHEM-MOL-NETWORK-001

Question and theorem. A molecular network is one finite maximal connected component of named molecular nodes and registered interaction edges with complete reachability and provenance.

*finite-molecular-node-set__registered-interaction-edges__connected-maximal-component__complete-reachability-cens
us__finite-maximal-boundary*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-BOND-LENGTH-STRENGTH-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-MOL-SUPRAMOLECULAR-001

Generated grammar. Generate the literal product of the registered molecular-network nodes, edges, connectivity, paths, boundary, composition, record and extension choices.

Boundary: Every finite generated set of molecular nodes and registered interaction edges whose maximal declared component is connected, with complete node, edge, path and boundary provenance.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `finite-molecular-node-set__registered-interaction-edges__connected-maximal-component__complete-reachability-census__finite-maximal-boundary__node-identities-retained__complete-network-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
nodes	finite-molecular-node-set	anonymous-node-count	A count cannot recover molecular identities.
edges	registered-interaction-edges	implicit-relatedness	An implicit relation cannot be audited.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
connectivity	connected-maximal-component	disconnected-union-called-one	Disconnected components are not one connected whole.
paths	complete-reachability-census	selected-path-only	One favorable path cannot establish network connectivity.
boundary	finite-maximal-boundary	unbounded-network-name	An unbounded name does not enumerate a finite object.
composition	node-identities-retained	network-erases-molecules	Nodes remain chemical carriers, not anonymous points.
record	complete-network-trace	network-answer-only	A connectivity answer cannot reproduce the network.
extension	no-extra-rule	free-connectivity-edge	Adding an ungenerated edge can force connectedness.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256: aecb155adfee2fa5c2569fcce2754efa066481a4569263794de20d763d1eb430.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 88618a1b00901ee0c581709ea8f5ed231b4d9e37c51f0c9d5a4ec7d4bcb94730.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 0d5086ed6a62b7e51f3ae5bef5470bf27fc01435c48ec2fd806f412699448e52.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: 134c1b01b41fb77ae299a443f0fe0be5edc83e08dd803e6331c14bdef0cef7e.

Independent reconstruction. Implementation hash

sha256: 4803b87a071efe0f05f28748a53f932a80b12b5025d108a6bceb53444e66f1cf; independent certificate sha256: 1c37d683d2ac8ff6c01fb6d7e2afe60fa7ea9d0dc2d0ba54915c360000eaacdf; external-validation hash sha256: 9387d741c918a7c70b1789dd3464e50e4321a4eabc8cea21f5e7e279d2c3790b. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-N04112-2026

Comparison records:

- molecular-network-iupac-n04112: predicted
finite-molecular-node-set__registered-interaction-edges__connected-maximal-component__complete-network-trace;
source-derived
finite-molecular-node-set__registered-interaction-edges__connected-maximal-component__complete-network-trace; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official molecular-network record lacks molecular nodes, interaction links or connected organization, or if a changed row is accepted.

Measurement receipt: sha256: 47729bac3b3ad9a5cebd8e8713750b5bc9a4aed2f837c88db297524aa655a667.

Isolation certificate: sha256: 6ff1d93795f6d3d4abc0e06c50a75b73721854f5e3e1c6aa72803195becf5b81.

Custody certificate: sha256: 87bbe0a98c36fbc807fbb4e2ccd3f2eaa04bf80a81f8a89ac14176934f97c882.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/N04112/json>
(sha256:a146d63f8340f0cf5043e1f9fe2f1cfa8e5fa7c3bf27b8a409e665f40ae2c046)

Registered target identities:

- molecular-network-iupac-n04112 from IUPAC-GOLD-BOOK-N04112-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no chemical dictionary, coordinate archive, fitted force field, measured geometry or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application or opaque prediction result
- no claim that topology alone predicts every measured coordinate or energy
- external content remains inaccessible until after the consequence is sealed
- Every finite generated set of molecular nodes and registered interaction edges whose maximal declared component is connected, with complete node, edge, path and boundary provenance.

Immutable identities. Source manifest

sha256:7cec3f79ff2a7802fdafc7cfc58b48ab5008aa7ab2c54643e7038a116f5dd25f; derivation seal
 sha256:9a35148af2485a14fb3d8e42ef5e1ca3059744fe6551776eaf1ddaa79f47bb16; engine receipt
 sha256:7a59431e2c40b65e566213a7730b55112b2f1e80edaa9c23f9459bee06b4570c at
 receipts/engine/model_admitted/SFT-CHEM-MOL-NETWORK-001-7a59431e2c40b65e.json; empirical
 hash sha256:2e213116da07890897b06c941275b5c2ce29aeb338c5985ffe834990638aafdf; measurement
 receipt sha256:47729bac3b3ad9a5cebd8e8713750b5bc9a4aed2f837c88db297524aa655a667.

12. Acid-base, electronegativity, redox and electrochemistry foundations - 10 admitted claims

Transfer sense and charge orientation are held labels; no negative proof magnitude enters the derivation. This entire registered subcategory is included; no proper subset is presented as field completion.

38. Conjugate acid-base partition

Claim identity: SFT-CHEM-AB-ACID-BASE-001

Question and theorem. A conjugate acid/base relation is a pair of retained chemical identities differing by exactly one named proton carrier: the protonated member donates it and the deprotonated member accepts it.

two-conjugate-identities__one-held-proton-occurrence__proton-donor-role__proton-acceptor-role__differ-by-one-proton

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001

Generated grammar. Generate the literal product of the registered conjugate pair, carrier, acid role, base role, difference, orientation, record and extension choices.

Boundary: Every finite generated pair of chemical carriers related by exactly one retained proton occurrence, with the protonated member able to donate that occurrence and the deprotonated member able to accept it.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `two-conjugate-identities__one-held-proton-occurrence__proton-donor-role__proton-acceptor-role__differ-by-one-proton__held-donor-acceptor-orientation__pair-and-proton-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
pair	two-conjugate-identities	one-species-label	A conjugate relation compares two carrier identities.
carrier	one-held-proton-occurrence	anonymous-hydrogen-count	A count cannot trace the transferred occurrence.
acid-role	proton-donor-role	acid-by-name	A name cannot establish a transition.
base-role	proton-acceptor-role	base-by-name	A name cannot establish a transition.
difference	differ-by-one-proton	unbounded-composition-change	Additional changes destroy conjugacy.
orientation	held-donor-acceptor-orientation	signed-acidity-scalar	A signed scalar is not a carrier trace.
record	pair-and-proton-trace	acid-base-answer-only	An answer cannot reconstruct conjugacy.
extension	no-extra-rule	free-acid-base-exception	An exception can label arbitrary species.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c636c0c317cb207c4839f718a739e5a683b6040155125536f9280ca737ac0e4c`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:05b2c6729d885da3646df12d9ee9a76888e057a0da289866cd518789a4558085`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : f316b770b291518f418e84e5b494397d05ba8c8ec8525bd5a01ff8d30b72f537.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 408b29d1a7a54536d715a8da4f7f9947a771419db6481093b17a405f5ae716f4.

Independent reconstruction. Implementation hash

sha256 : 69b4921f1868f68bf465f91a281a6c5f67e8da2de9f06b518a49325116c84b52; independent certificate sha256 : 88152d90d6de4349d732338892b209665ac0f6dff91ab5a7099a95bf1bfd25c4; external-validation hash sha256 : 21f084d6cffa09dc98694f8761e171a79bb5d57e36db943af98a624720220a85. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ACID-BASE-COMPOSITE-2026

Comparison records:

- conjugate-acid-base-iupac-composite: predicted
proton-donor-acid__proton-acceptor-base__conjugates-differ-by-one-proton__paired-relation; source-derived
proton-donor-acid__proton-acceptor-base__conjugates-differ-by-one-proton__paired-relation; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authoritative conjugate acid/base record lacks proton donation, proton acceptance or a one-proton conjugate relation, or if a changed row is accepted.

Measurement receipt: sha256 : bb6a626c8a543157cd574cf93dca97d80f397f81b43e28bc3b72ef19e7260581.

Isolation certificate: sha256 : a44b297d6b52fd8be5d715124099da1e2985116cbe28bb2af11f040aa8201c0a.

Custody certificate: sha256 : e6ddaba184a7d905ea8c13e595611c5465952a12def55c61e78e7887e65e85fb.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01266>
(sha256:1ac0d6138e316d6230d10b2e7f99d4837175fcb640e824a8cb2249100221a504)

Registered target identities:

- conjugate-acid-base-iupac-composite from IUPAC-GOLD-BOOK-ACID-BASE-COMPOSITE-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no acid/base dictionary, pH scale, equilibrium equation, dissociation constant, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no unpaired creation, copying or loss of a proton or electron carrier
- no claim that structural closure alone fixes every solution-dependent strength or capacity value
- external target content remains inaccessible until the prediction is sealed
- Every finite generated pair of chemical carriers related by exactly one retained proton occurrence, with the protonated member able to donate that occurrence and the deprotonated member able to accept it.

Immutable identities. Source manifest

sha256 : 1ffd07cc47c7e9aa7f09612c400a17b98c414d2a809f9ce80d466bc14ac266f32; derivation seal

sha256:2f18d9aec07914d26e0486b4e43936e21c5b0e15be1bdf0c0b22a3eb3850600a; engine receipt sha256:318ea7b4fdc414c30a64a83df0ca5a0dfdcac348a4fc207743da2b9d6496827f at receipts/engine/model_admitted/SFT-CHEM-AB-ACID-BASE-001-318ea7b4fdc414c3.json; empirical hash sha256:46d787ebe2f6ed6c4f5dc4bc5c4f2f9bbdc8c147974df3b951bfd75f983604d8; measurement receipt sha256:bb6a626c8a543157cd574cf93dca97d80f397f81b43e28bc3b72ef19e7260581.

39. Proton-transfer acid-base relation

Claim identity: SFT-CHEM-AB-PROTON-TRANSFER-001

Question and theorem. Proton transfer moves one named proton carrier from one registered donor occurrence to one distinct acceptor occurrence while conserving carrier identity and both endpoint records.

one-named-proton-carrier__registered-donor__registered-acceptor__donor-to-acceptor-transfer__same-carrier-before-after

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001

Generated grammar. Generate the literal product of the registered proton carrier, source, destination, transition, identity, closure, record and extension choices.

Boundary: Every finite chemical transition in which one named proton carrier leaves one registered donor occurrence and enters one distinct registered acceptor occurrence with identical carrier identity before and after.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *one-named-proton-carrier__registered-donor__registered-acceptor__donor-to-acceptor-transfer__same-carrier-before-after__complete-endpoint-closure__carrier-endpoint-transition-trace__no-extra-rule*.

Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-named-proton-carrier	proton-count-only	A count cannot establish conservation of an occurrence.
source	registered-donor	source-erased	Without a donor, the carrier could be created freely.
destination	registered-acceptor	destination-erased	Without an acceptor, the carrier could be lost.
transition	donor-to-acceptor-transfer	copy-or-delete	Copying or deleting violates carrier conservation.
identity	same-carrier-before-after	before-after-carrier-change	A changed carrier is not transfer conservation.
closure	complete-endpoint-closure	unpaired-composition-change	Unpaired change leaves the reaction record open.
record	carrier-endpoint-transition-trace	product-label-only	A product label cannot reconstruct the path.
extension	no-extra-rule	free-proton-source	A free source can force any reaction.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 69777e7c94f864618f23bb8c316cb4755aa69932bbd5024824f7d651970eb1ed.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: c4d8f6608680b8ea69ff9a3d1aff3134aeb226b21fc7553992373580dbbb6b56.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 9e07bea854076486d00d939b92c32227e48d5576783837be7dc0119c913f1f5f.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 7183352ddde61f7554681ce6e9790f2a1b8e9c929bdf68cb50845f22b1c10ca3.

Independent reconstruction. Implementation hash

sha256: 4256a0b14274123ec7c79959d029bf6359f031402bce163da90a11f367076c4d; independent certificate sha256: 59456ef8b2f6dbb57ed62c61f0c0384384fd8aa61d956eb93ad1981bf8a4102a; external-validation hash sha256: 3dcea1b0798f0d4f9132607ef0305e25b905d822bb38a1f5869b5fefe5f30fed. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-P04915-2026

Comparison records:

- `proton-transfer-iupac-p04915`: predicted
one-proton-carrier__donor-to-acceptor-transfer__carrier-conserved__endpoint-identities-retained; source-derived
one-proton-carrier__donor-to-acceptor-transfer__carrier-conserved__endpoint-identities-retained; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authoritative proton-transfer record lacks a proton donor, proton acceptor or conserved transfer relation, or if a changed row is accepted.

Measurement receipt: sha256:eb40e28051009a16adec422f7c4babea938ca5cb5f48a5157486d26a572acb08.
 Isolation certificate: sha256:5e697a7892cb6a71f7f07976e00f2ba1c9a43ff8e9de5b3636b414ae66d94f35.
 Custody certificate: sha256:10da44114222589568fb99ae7b1ded1156580e5740b2ec058afa402691e9d3e1.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/P04915>
 (sha256:72d9b1431f3daa6b0190094d8400bc73b31e786b7d69b4052632297b4b709d33)

Registered target identities:

- proton-transfer-iupac-p04915 from IUPAC-GOLD-BOOK-P04915-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no acid/base dictionary, pH scale, equilibrium equation, dissociation constant, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no unpaired creation, copying or loss of a proton or electron carrier
- no claim that structural closure alone fixes every solution-dependent strength or capacity value
- external target content remains inaccessible until the prediction is sealed
- Every finite chemical transition in which one named proton carrier leaves one registered donor occurrence and enters one distinct registered acceptor occurrence with identical carrier identity before and after.

Immutable identities. Source manifest

sha256:70590ccb77ad3565b409c5508d23c998235e1246ee523f0acd7b8881759e1e26; derivation seal
 sha256:84ae485e19dae65aafeaaba17678180c50f27ce832b1c48dcd93aa9410d60223; engine receipt
 sha256:ddcb09522d19aa8840e8e57745b546ee62be1116bfd03a850643fe1d24ca6df3 at
 receipts/engine/model_admitted/SFT-CHEM-AB-PROTON-TRANSFER-001-ddcb09522d19aa88.json;
 empirical hash sha256:d30b62f5865b2596b87847452eb769a8912f7e497d7bc90e51f35169cd86346b;
 measurement receipt sha256:eb40e28051009a16adec422f7c4babea938ca5cb5f48a5157486d26a572acb08.

40. Electron-pair donor-acceptor relation

Claim identity: SFT-CHEM-AB-LEWIS-001

Question and theorem. A Lewis acid/base relation joins a distinct electron-pair donor and acceptor through the same two exclusion-preserving electron occurrences, with both identities retained.

*distinct-donor-acceptor__two-electron-occurrences__pair-donor-support__pair-acceptor-support__shared-or-transfer
 red-pair*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001

Generated grammar. Generate the literal product of the registered Lewis endpoints, support, donor, acceptor, joining, conservation, record and extension choices.

Boundary: Every finite donor/acceptor pair joined by one exclusion-preserving pair of named electron occurrences, with both electron identities conserved and the donor/acceptor orientation retained.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `distinct-donor-acceptor__two-electron-occurrences__pair-donor-support__pair-acceptor-support__shared-or-transferred-pair__both-electron-identities-retained__endpoint-pair-joining-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
endpoints	distinct-donor-acceptor	one-unoriented-species	The relation requires donor and acceptor roles.
support	two-electron-occurrences	single-electron-only	The registered relation concerns paired support.
donor	pair-donor-support	electron-pair-name-only	A name cannot demonstrate available support.
acceptor	pair-acceptor-support	unregistered-vacancy	An unregistered destination cannot close the interaction.
joining	shared-or-transferred-pair	copied-electron-pair	Copying violates exclusion and conservation.
conservation	both-electron-identities-retained	electron-identity-erased	Erasure cannot reverse the relation.
record	endpoint-pair-joining-trace	Lewis-label-only	A label cannot reconstruct pair support.
extension	no-extra-rule	free-orbital-premise	An imported orbital rule can select a desired classification.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : 2cc98aa40593481481ad39b0cb07b7ce428222ea723733f562c37b1394b2b6b6.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 95cc34d98ca5a838b98c1074a1db31ed70571fbb0c387b70c5e910d134f18cab.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : c1fa64533d715cd9e3911fa10b34f9b093332ff954c844029a150f2f90ab4fcf.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : abc13b877a1bebe4f04b54c9618bdc2137ed324f325e4d0f3499a6153c8680f6.

Independent reconstruction. Implementation hash

sha256 : 5e16152a66733442a6bde36d7de224f715f0f64f53c671e108eb36c57deef79f; independent certificate sha256 : 8bfe72a6239326e94c2ec7358fbe227e047702cb5ccf66ba4f41343be1deaa26; external-validation hash sha256 : a471d23d6e04a1b399b41fef7902f4aaf130a1ecda942e25f025a021a5a14570. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-LEWIS-COMPOSITE-2026

Comparison records:

- lewis-acid-base-iupac-composite: predicted
electron-pair-donor__electron-pair-acceptor__shared-or-transferred-pair__pair-conserved; source-derived
electron-pair-donor__electron-pair-acceptor__shared-or-transferred-pair__pair-conserved; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authoritative Lewis record lacks an electron-pair donor, electron-pair acceptor or conserved pair relation, or if a changed row is accepted.

Measurement receipt: sha256 : 5d0525fb63f86d96bd0477af62d7822ba68677bb7445231b0cbf9fc00e5f9776.

Isolation certificate: sha256 : 33f45a6afa4305b6a4136dae7b7f3d3275671bc48a4d99b5efe21d39d20d852d.

Custody certificate: sha256 : 9344b0290bb18ca2810c0d924e68efb637c9101bd005cd63b1a76a7c309b7fba.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/L03508>
(sha256:9fa013b12d6d0e3f305883a0db53dac7484e292e0288e2c3d0ba9625cfb2011c)

Registered target identities:

- lewis-acid-base-iupac-composite from IUPAC-GOLD-BOOK-LEWIS-COMPOSITE-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no acid/base dictionary, pH scale, equilibrium equation, dissociation constant, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no unpaired creation, copying or loss of a proton or electron carrier
- no claim that structural closure alone fixes every solution-dependent strength or capacity value

- external target content remains inaccessible until the prediction is sealed
- Every finite donor/acceptor pair joined by one exclusion-preserving pair of named electron occurrences, with both electron identities conserved and the donor/acceptor orientation retained.

Immutable identities. Source manifest

sha256:32ffebe25ae42141bcd183b6ab787d62ffa64cb0cf2b688a6699dc742683159d; derivation seal
 sha256:a55699c0fdb6f21bed8b854b4a27227c0700d1c8bed86a06d6c7e0d06ec57ab; engine receipt
 sha256:ef68becb2c22b1c5f781958280bd22f8abfad5a4cbb3c0dfe86cb56c7d16f598 at
 receipts/engine/model_admitted/SFT-CHEM-AB-LEWIS-001-ef68becb2c22b1c5.json; empirical hash
 sha256:49033e1397ba8e6f701acca93b7d469d00dbc145325850cc0fd899eda035806c; measurement receipt
 sha256:5d0525fb63f86d96bd0477af62d7822ba68677bb7445231b0cbf9fc00e5f9776.

41. Amphoteric dual-role relation

Claim identity: SFT-CHEM-AB-AMPHOTERIC-001

Question and theorem. An amphoteric species is the same retained chemical carrier appearing in two separately held contexts, once in a valid acid role and once in a valid base role.

same-species-identity__valid-donor-event__valid-acceptor-event__contexts-separately-held__both-roles-realized

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001

Generated grammar. Generate the literal product of the registered amphoteric species, acid context, base context, context separation, duality, identity, record and extension choices.

Boundary: Every finite chemical carrier for which two separately registered contexts contain a valid donor transition and a valid acceptor transition involving that same retained species identity.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `same-species-identity__valid-donor-event__valid-acceptor-event__contexts-separately-held__both-roles-realized__species-identity-retained__dual-event-context-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
species	same-species-identity	two-unrelated-species	Dual behavior must belong to one retained carrier identity.
acid-context	valid-donor-event	acid-label-without-event	A label does not establish donation.
base-context	valid-acceptor-event	base-label-without-event	A label does not establish acceptance.
context	contexts-separately-held	roles-conflated	Conflation can assert incompatible events simultaneously.
duality	both-roles-realized	one-role-only	One role does not close amphoteric behavior.
identity	species-identity-retained	species-changed-between-roles	A changed carrier does not establish dual capacity.
record	dual-event-context-trace	amphoteric-label-only	A label cannot reproduce both contexts.
extension	no-extra-rule	free-dual-role-exception	An exception can admit any species.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 5ade57913bbd3c0150acc9c9fb75df8d67ed170ae255e8b89209829a95841595.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: f922ab1bf0af3b619c30a84ce5dabf9646cfe4aeedce383f3029f901c58cac3d.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 1ad519e6b1b5619447797c273c1f06084e45dfa0283bef8c23618549db219f5e.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 8ea2d3cfbcf557a1aa8f9c6a6400a0ebca9d6921702a109600abd4a47e87c374.

Independent reconstruction. Implementation hash

sha256: ac8f026dbc2ed95ba69ffdfaacc29bd5e7de7d5699e902f37d1b09665cf4528db; independent certificate sha256: 48312bf23953149b2f7e9b86e4955f1fa4c4a2d0c1fc994bb1035905b0bdc8ac; external-validation hash sha256: 87aa0a9051da892a6671887802915210d3e2e47c9902b202f3379583b5e5c3e8. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-A00306-2026

Comparison records:

- amphoteric-iupac-a00306: predicted
same-species-acid-role__same-species-base-role__context-dependent-dual-response__both-traces-retained; source-derived
same-species-acid-role__same-species-base-role__context-dependent-dual-response__both-traces-retained; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authoritative amphoteric record lacks acid and base behavior by one species under retained contexts, or if a changed row is accepted.

Measurement receipt: sha256:1c4f68ff7623709e1b693029ad22ce5d388a5c675d5e2f02d4b9034ec4b04b1e.

Isolation certificate: sha256:1f4eeaa86e10ba54c8c741168fc7d8c5b54abcc13817030dada35d00c0bc9d63.

Custody certificate: sha256:194fb1bf95ecb5b82825b568a045eea0a8e7a6f5c3b944ed7d3c15ae13f38a2.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/A00306>
(sha256:70ab5021338400d9f02cbecc5c45b9f4271792e919d880aafff81684739022ab)

Registered target identities:

- amphoteric-iupac-a00306 from IUPAC-GOLD-BOOK-A00306-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no acid/base dictionary, pH scale, equilibrium equation, dissociation constant, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no unpaired creation, copying or loss of a proton or electron carrier
- no claim that structural closure alone fixes every solution-dependent strength or capacity value
- external target content remains inaccessible until the prediction is sealed
- Every finite chemical carrier for which two separately registered contexts contain a valid donor transition and a valid acceptor transition involving that same retained species identity.

Immutable identities. Source manifest

sha256:c53e755cd0c53593d6f2eff03f3fa98928a67ff43187d5032d2077ce1619da2c; derivation seal

sha256:521f0d5c4baf8f799ab2bd19a1e0542d279d815772f160144e15c99ea613c9aa; engine receipt

sha256:94649424e1a4c49ab8939c2f4eab025306b9262e766ec388c2d1fc97e99eb4ad at

receipts/engine/model_admitted/SFT-CHEM-AB-AMPHOTERIC-001-94649424e1a4c49a.json; empirical

hash sha256:6b063de4c54531311c1548ff91ff48221a914402d47a83f9385b94c9fafecebd; measurement

receipt sha256:1c4f68ff7623709e1b693029ad22ce5d388a5c675d5e2f02d4b9034ec4b04b1e.

42. Buffer response and conjugate-pair retention

Claim identity: SFT-CHEM-AB-BUFFER-001

Question and theorem. A buffer is a finite positive reservoir of both members of a conjugate pair whose complementary paths consume added proton support or supply proton support to added base while retaining pair identity and finite capacity.

conjugate-pair-reservoir__positive-support-of-both__base-member-accepts-proton__acid-member-supplies-proton__conjugate-pair-retained

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-AMPHOTERIC-001

Generated grammar. Generate the literal product of the registered buffer components, support, acid response, base response, retention, capacity, record and extension choices.

Boundary: Every finite reservoir containing positive support of both members of a conjugate pair and two complementary response paths that consume added proton support or supply proton support to an added base within capacity.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `conjugate-pair-reservoir__positive-support-of-both__base-member-accepts-proton__acid-member-supplies-proton__conjugate-pair-retained__finite-source-bounded-capacity__pair-count-response-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
components	conjugate-pair-reservoir	single-unpaired-species	One species cannot supply both complementary responses.
support	positive-support-of-both	one-component-absent	An absent member leaves one response unavailable.
acid-response	base-member-accepts-proton	added-proton-unanswered	No response permits the proton distinction to remain open.
base-response	acid-member-supplies-proton	added-base-unanswered	No response permits proton removal without paired change.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
retention	conjugate-pair-retained	reservoir-identity-erased	A response without conjugate tracking cannot be reversed.
capacity	finite-source-bounded-capacity	unlimited-buffer-claim	A finite reservoir cannot have unlimited capacity.
record	pair-count-response-trace	pH-answer-only	A pH value cannot reconstruct composition and response.
extension	no-extra-rule	free-equilibrium-fit	A fitted equation can force a measured response.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:15d9bacd7e2b1a04d502efb2b9edf19cc831eb2248f12f176669be2ed55b7727.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:7e69a9d59c2e2894ca3ea5c30cf7467cf5128645d30c60e368cc800a0eb067ed.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:1f4d2e4ef0767fc1a44344a407756603a283dad0ee6c5407c8604ac5338bc788.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:696c34f1a9d474e9875c4d03a58e6b5215831e6504ed23e52477dd7d20c84fe8.

Independent reconstruction. Implementation hash

sha256:8646664e19c6946f77ce8eeb44b31ed57ac3ef2d67c745f9aa666f3a3246d09; independent certificate sha256:393f76310d7a38322171e44145f2cfcc0652668cdd4f8bdca9cd8eeaeab112d6; external-validation hash sha256:2a3dff8f62a1772c817847893d6ebb523c90b9539ccaf79f12d8e6612719976d. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-DIDAC-E15-2026

Comparison records:

- buffer-iupac-didac-e15: predicted
weak-acid-conjugate-base-pair__responds-to-added-acid-or-base__composition-change-limited__source-bounded-capacity;
source-derived
weak-acid-conjugate-base-pair__responds-to-added-acid-or-base__composition-change-limited__source-bounded-capacity;
exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authoritative buffer record lacks a weak acid/conjugate base pair, response to added acid/base or finite composition-dependent capacity, or if a changed row is accepted.

Measurement receipt: sha256:a5e091e1bb008ff48586bd4ee78d47ea3acc735f794c524d2b82237566fd3fa7.

Isolation certificate: sha256:066d7ce3fd2a30b65c9b31dfb975abc51c269a7e7b1ee6bf30bc7237726db898.

Custody certificate: sha256:098033aca6e69192d1f3618c78b98ee4311e774a037c02c10fb00650e2948860.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://old.iupac.org/didac/Didac%20Eng/Didac02/Content/E15.htm> (sha256:570a91d1a7de35b22a8469c117f10af9e4bed3012ae61f54362fe45534b3d01e)

Registered target identities:

- buffer-iupac-didac-e15 from IUPAC-DIDAC-E15-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no acid/base dictionary, pH scale, equilibrium equation, dissociation constant, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no unpaired creation, copying or loss of a proton or electron carrier
- no claim that structural closure alone fixes every solution-dependent strength or capacity value
- external target content remains inaccessible until the prediction is sealed
- Every finite reservoir containing positive support of both members of a conjugate pair and two complementary response paths that consume added proton support or supply proton support to an added base within capacity.

Immutable identities. Source manifest

sha256:68ad8ec1d33309905d963b89f9d6c225584213690ea0d26b820b6f6fa5e7518a; derivation seal
 sha256:a6ea9892ec2a886e5ec451a939ef58eea7b5a2916c10346814533a831218701d; engine receipt
 sha256:b76553aa9e22f85717dd7cbda96a1e8ba880fb5396756d8049e79fac0f885ca1 at
 receipts/engine/model_admitted/SFT-CHEM-AB-BUFFER-001-b76553aa9e22f857.json; empirical hash
 sha256:f351067de0f9edac06934e0e244b44c2efe0e096a401d19f1be2209f0f8e2be0; measurement receipt
 sha256:a5e091e1bb008ff48586bd4ee78d47ea3acc735f794c524d2b82237566fd3fa7.

43. Electronegativity ordering

Claim identity: SFT-CHEM-ELECTRONEGATIVITY-001

Question and theorem. Electronegativity order is the transitive preorder of complete pairwise endpoint recurrence for identical shared electron support; equal traces remain an exact equivalence class rather than receiving an arbitrary rank.

*two-named-atomic-occurrences__one-shared-electron-support__complete-endpoint-recurrence-comparison__held-pref
 erred-endpoint__transitive-preorder-closure*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001

- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001

Generated grammar. Generate the literal product of carrier, support, comparison, orientation, tie, extension, record and extra-rule choices.

Boundary: Every finite pairwise comparison of two named atomic occurrences joined by one shared electron-labelled support, with the complete recurrence trace determining endpoint preference or an explicitly retained tie.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `two-named-atomic-occurrences__one-shared-electron-support__complete-endpoint-recurrence-comparison__held-preferred-endpoint__trace-equivalence-class__transitive-preorder-closure__complete-comparison-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carriers	two-named-atomic-occurrences	anonymous-element-values	Values erase endpoint identity.
support	one-shared-electron-support	free-affinity-scalar	A scalar imports the ordering.
comparison	complete-endpoint-recurrence-comparison	name-selected-order	Names cannot select preference.
orientation	held-preferred-endpoint	signed-difference	A signed magnitude violates the proof domain.
ties	trace-equivalence-class	forced-strict-order	Equal complete traces cannot be ordered uniquely.
extension	transitive-preorder-closure	pairwise-cycle-admitted	A cycle destroys lawful ordering.
record	complete-comparison-trace	rank-answer-only	A rank cannot reconstruct comparisons.
extra	no-extra-rule	free-scale-constant	A scale constant can tune any ordering.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:0b4cdc1f4aee52d241ee6c50286036754d52606fb34b3123efdf8788925569`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:75b4db050fe85f16506da2c636dce9c265cf2318fc57636085303aa2888a7b34`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : e328e529b6c738adc4b931a31da3aa43513455bfb2797dff915c4b9c5f39bc88.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : c1c883deac6b9242a292b10d7a1e05a83ca81c2276fad9e8e34732e35f6f959f.

Independent reconstruction. Implementation hash

sha256 : a07b6ba2f87119929254b9ab82389f69c9e26a2659a799af6cfd6fdee3b42622; independent certificate
sha256 : f09179fb7686c4719baaceff6b28302171cbb14392ab3eb079be703c9f8acc70; external-validation
hash sha256 : 97bec98206897408f69d9c87089c55b9eae34745d11cd50e0f49fdc7b573f814. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ELECTRONEGATIVITY-COMPOSITE-2026

Comparison records:

- electronegativity-iupac-composite: predicted
atomic-attraction-for-shared-electrons__pairwise-affinity-order__ties-retained__scale-values-source-bounded; source-derived
atomic-attraction-for-shared-electrons__pairwise-affinity-order__ties-retained__scale-values-source-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks atomic attraction for shared electrons or permits a scale value to replace the retained comparison relation, or if a changed row is accepted.

Measurement receipt: sha256 : 4d53587cb0293ecf34b026aaecb36674e672251726c183c58c8ac4a1e27c7a1e.

Isolation certificate: sha256 : 556d168b4fd5eb236d7c6b2bce3f77745effe24218ceb609d289f6869ea00af.

Custody certificate: sha256 : 4e4f633dd04ed8f3150540ae7be7818ad032b152aed784dbc73645aa657703ca.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/E01990>
(sha256:bc2888f84a2a3efc53f89738797c88528af75cd0564295e2a1d4138cd1c778e3)

Registered target identities:

- electronegativity-iupac-composite from IUPAC-GOLD-BOOK-ELECTRONEGATIVITY-COMPOSITE-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no electronegativity scale, oxidation-state table, electrode potential, external definition, measured value or V2 result may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no electron carrier may be created, copied, silently erased or assigned to two endpoints
- no qualitative structural law is represented as a parameter-free prediction of a measured decimal scale
- external target content remains inaccessible until the complete prediction is sealed
- Every finite pairwise comparison of two named atomic occurrences joined by one shared electron-labelled support, with the complete recurrence trace determining endpoint preference or an explicitly retained tie.

Immutable identities. Source manifest

sha256 : cda2f348483bf8c0edaec153c40d814f9d411a625225fc279621247eaa178e25; derivation seal

sha256:79310e0db9bb785e38e04c2da5f2af1c506ab396734dd8cd158932979a0a2457; engine receipt sha256:a22665b562002fc2eae1f487748405511ca02f6fb00cd071e660731a5dc0d10d at receipts/engine/model_admitted/SFT-CHEM-ELECTRONEGATIVITY-001-a22665b562002fc2.json; empirical hash sha256:9fad8587abb5430953a59bbdb7ffb3d2755241d92473e2203b90f69d6d08636a; measurement receipt sha256:4d53587cb0293ecf34b026aaecb36674e672251726c183c58c8ac4a1e27c7a1e.

44. Bond polarity from unequal held affinity

Claim identity: SFT-CHEM-BOND-POLARITY-001

Question and theorem. Unequal endpoint affinity for the identical shared support forces an oriented but still shared bond partition, with opposed partial-charge fibres held as labels rather than signed proof numbers.

registered-covalent-bond__unequal-complete-affinity__same-shared-electron-support__unequal-shared-partition__held-partial-charge-fibres

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001

Generated grammar. Generate the literal product of bond, affinity, support, partition, orientation, closure, record and extra-rule choices.

Boundary: Every finite covalent bond whose two named endpoints have unequal complete affinity traces for the same shared support, producing one held orientation of the support partition while retaining the bond whole.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `registered-covalent-bond__unequal-complete-affinity__same-shared-electron-support__unequal-shared-partition__held-partial-charge-fibres__paired-endpoint-response__bond-affinity-partition-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
bond	registered-covalent-bond	two-unjoined-atoms	Polarity belongs to a shared bond.
affinity	unequal-complete-affinity	equal-or-unknown-affinity	No direction follows from equality or missing evidence.
support	same-shared-electron-support	different-electrons-compare	Different support cannot orient one bond.
partition	unequal-shared-partition	full-transfer	Full transfer changes the bonding class.
orientation	held-partial-charge-fibres	negative-charge-number	A negative scalar is unnecessary.
closure	paired-endpoint-response	one-endpoint-only	One endpoint leaves the bond record open.
record	bond-affinity-partition-trace	polar-label-only	A label cannot reproduce the comparison.
extra	no-extra-rule	free-polarity-threshold	A threshold imports a parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:bc9919833ca65448defb66482960c2ac35503f2445d3cf46e5a2a8d745c1a6d6.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:c24cafbcb7496d10a8ec0cedbe90979502eb498af4dd5e8e8ee9dd9837b28933d.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:81b2675f657711b9a4a484f076fe8dfb4d12779fae46fbd14f41a9f3aa753391.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:496e405f3e97843fe901246d4f831feb336c6e91ebb49dc12bf17fa6ca42503c.

Independent reconstruction. Implementation hash

sha256:4c6ba5a086ff250256dee1c270231597286030abf637798a6138e036820db6fc; independent certificate sha256:97626f4d4ea6cc78c33c2a67460285297e027e25af4c7c7cec3f822cf909ff36; external-validation hash sha256:95de0dadbd2b1c1940a413e7e7fc8e0f16aa5c36f0cb4376d9fae1930b7555bed. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-BOND-POLARITY-COMPOSITE-2026

Comparison records:

- bond-polarity-iupac-composite: predicted
unequal-electron-attraction__polar-covalent-bond__partial-charge-separation__bond-support-remains-shared; source-derived
unequal-electron-attraction__polar-covalent-bond__partial-charge-separation__bond-support-remains-shared; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks unequal attraction, polar covalent sharing or partial charge separation, or if a changed row is accepted.

Measurement receipt: sha256:da3566fbefb3661e78cd21b52a9eac47ecb445d0a30aaf8b1f4ea1997c79e99c.

Isolation certificate: sha256:a70bf3c5b80edccc7befe973311d738156f7cdac1846b0e5cbec3d6bc7936a9b.

Custody certificate: sha256:a74af93fead51a0f6e790c92bce200aff1b4deab2b9aca67b1c17eda4bb0eaf7.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/08195>
(sha256:a907133200cc3296e7ecdb7c8970ae8ce35af7b6d3b44d6746f7d18ad5086d41)

Registered target identities:

- bond-polarity-iupac-composite from IUPAC-GOLD-BOOK-BOND-POLARITY-COMPOSITE-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no electronegativity scale, oxidation-state table, electrode potential, external definition, measured value or V2 result may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no electron carrier may be created, copied, silently erased or assigned to two endpoints
- no qualitative structural law is represented as a parameter-free prediction of a measured decimal scale
- external target content remains inaccessible until the complete prediction is sealed
- Every finite covalent bond whose two named endpoints have unequal complete affinity traces for the same shared support, producing one held orientation of the support partition while retaining the bond whole.

Immutable identities. Source manifest

sha256:7a9b118ebd7004dacce381daf30cf1caf54e4ce51aafc7dfa7d6fc9a47f7d86a; derivation seal

sha256:ee142200a3952b2c02392f6bddd5f5fb15b2ede578d3660deab2011c690b6a6f; engine receipt

sha256:0510531a9b1aa21d40a92effc5592d673d5d631fd052c4f1b6e3e434c66016b7 at

receipts/engine/model_admitted/SFT-CHEM-BOND-POLARITY-001-0510531a9b1aa21d.json; empirical

hash sha256:88def9aa9a23b62ba495958e29b14cadf7ed6b0e525d03e08a1089f35cc2c9c7; measurement

receipt sha256:da3566fbefb3661e78cd21b52a9eac47ecb445d0a30aaf8b1f4ea1997c79e99c.

45. Oxidation-state accounting

Claim identity: SFT-CHEM-REDOX-OXIDATION-STATE-001

Question and theorem. Oxidation state is a complete source-bound ionic-approximation accounting; each bond-electron occurrence is assigned exactly once by endpoint affinity, with magnitude positive and orientation separately held.

complete-bonded-chemical-whole__declared-ionic-approximation__assign-to-preferred-endpoint__equal-endpoint-division__each-electron-assigned-once

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001

Generated grammar. Generate the literal product of whole, partition, unequal-bond, equal-bond, count, conservation, record and extra-rule choices.

Boundary: Every finite bonded chemical whole under one declared ionic-approximation partition, assigning each bond-electron occurrence exactly once by complete affinity order (or equal division for tied endpoints), retaining atom identity, orientation and total carrier conservation.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-bonded-chemical-whole__declared-ionic-approximation__assign-to-preferred-endpoint__equal-endpoint-division__positive-count-plus-held-orientation__each-electron-assigned-once__atom-bond-assignment-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
whole	complete-bonded-chemical-whole	isolated-atom-answer	Oxidation accounting is composition-relative.
partition	declared-ionic-approximation	unstated-electron-rule	An unstated rule permits arbitrary answers.
unequal-bond	assign-to-preferred-endpoint	split-despite-preference	It contradicts complete affinity order.
equal-bond	equal-endpoint-division	arbitrary-endpoint	Equal traces cannot choose an endpoint.
count	positive-count-plus-held-orientation	signed-integer-primitive	A signed number hides orientation.
conservation	each-electron-assigned-once	electron-created-or-lost	Open assignment violates the whole.
record	atom-bond-assignment-trace	oxidation-number-only	A number cannot reproduce the assignment.
extra	no-extra-rule	free-state-table	A table can select desired values.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: eec9720b659074465f3bedf55d78324ab1d030d36810d96e0261e9ae158f95a9.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: b38971bae9d17c53c09af666594f47d248c37c82c5e5ac2f821cfea4eccf1a2b.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: b7ff129c257d37fbaeb72d84a6aca6cf5a240ecff39ca00dfee215ef9ba230ab.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: f9451d77ea5492dbc9190e7320c55355c41a38dfc67711cf3c06479836059344.

Independent reconstruction. Implementation hash

sha256: dde5adfebf754c437df1cfb00ac8e89e98023da3859ecc8e189227024cf5f072; independent certificate sha256: 4c303af97dc7aa5fa8ab7803de872f79e74602902c10343f340eccf5f0457caa; external-validation hash sha256: 5afddd6a10621d037d0cebc0268e855994c552b7e995ae963627c350e28bb59d. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-004365-2026

Comparison records:

- oxidation-state-iupac-o04365: predicted ionic-approximation__bond-electrons-assigned-by-electronegativity__per-atom-charge-accounting__complete-assignment-conserved; source-derived ionic-approximation__bond-electrons-assigned-by-electronegativity__per-atom-charge-accounting__complete-assignment-conserved; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks ionic approximation, electronegativity-based assignment or complete per-atom accounting, or if a changed row is accepted.

Measurement receipt: sha256: 4e94764c4f254466bc72537c36c107ff56a4433c5dbd3ac114d91bfdc252a464.

Isolation certificate: sha256: eec93931d1fa2574c19acd904b3254eabe55f1f610f90b4a5ca004fb7af92149.

Custody certificate: sha256: 839f88ab45b7d579621b23dd8cafca1e40bda11d070c4cb69d60a7042770b82.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/O04365> (sha256:5413171bfaa9152ff9a6a2dee0c79a2ec3e31a4d1efbfff0f8c76804c7fdb06e)

Registered target identities:

- oxidation-state-iupac-o04365 from IUPAC-GOLD-BOOK-O04365-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no electronegativity scale, oxidation-state table, electrode potential, external definition, measured value or V2 result may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no electron carrier may be created, copied, silently erased or assigned to two endpoints
- no qualitative structural law is represented as a parameter-free prediction of a measured decimal scale
- external target content remains inaccessible until the complete prediction is sealed
- Every finite bonded chemical whole under one declared ionic-approximation partition, assigning each bond-electron occurrence exactly once by complete affinity order (or equal division for tied endpoints), retaining atom identity, orientation and total carrier conservation.

Immutable identities. Source manifest

sha256:5dd34510642f10139fceb199c237b68968dbe166919b7ef99e8fec30ad064ac8; derivation seal
 sha256:a824c0190f355cf12ca89d387e1d6c1281055edb25d641c13599ba64188a5049; engine receipt
 sha256:dc3f55fe127993303d0bec15a2f6a53ee4c3ab349377f681bf8b596fa044d2d1 at receipts/engine/model_admitted/SFT-CHEM-REDOX-OXIDATION-STATE-001-dc3f55fe12799330.json; empirical hash
 sha256:6a588990aaa5ee34d89f18be4af3f24bb02b428ed7fc949ea9de90cf3f59561c; measurement receipt
 sha256:4e94764c4f254466bc72537c36c107ff56a4433c5dbd3ac114d91bfdc252a464.

46. Coupled oxidation and reduction

Claim identity: SFT-CHEM-REDOX-COUPPING-001

Question and theorem. Oxidation and reduction are the opposed endpoint records of one conserved electron transfer; neither half closes without the other.

named-electron-carrier__registered-oxidation-endpoint__registered-reduction-endpoint__one-paired-transfer__same-carrier-before-after

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001

Generated grammar. Generate the literal product of carrier, donor, acceptor, coupling, identity, accounting, record and extra-rule choices.

Boundary: Every finite closed electron-labelled transfer between distinct chemical endpoints, with identical carrier identity before and after and paired donor loss/acceptor gain or their complete oxidation-assignment equivalents.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `named-electron-carrier__registered-oxidation-endpoint__registered-reduction-endpoint__one-paired-transfer__same-carrier-before-after__opposed-assignment-updates__carrier-endpoint-coupling-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	named-electron-carrier	anonymous-charge-change	It cannot establish electron conservation.
donor	registered-oxidation-endpoint	donor-erased	The carrier could be freely created.
acceptor	registered-reduction-endpoint	acceptor-erased	The carrier could be lost.
coupling	one-paired-transfer	independent-half-changes	One open half violates conservation.
identity	same-carrier-before-after	different-carrier-after	That is replacement, not transfer.
accounting	opposed-assignment-updates	one-sided-state-change	One side cannot close the assignment.
record	carrier-endpoint-coupling-trace	redox-label-only	A label cannot reverse the event.
extra	no-extra-rule	free-electron-source	A free source breaks closure.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:93dc0f0dca3a8d0637be0f8366a9c9f7efbabe274caf377dc0d68648b26162ca.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:135e22bd09a5e264309f9102ca7412c0debb155083e86f4b50edca884843fd63.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:f7c8cad06259b068ad6ad36db72aebfbdde432e2cb83719d9e7cf26d4c680e0b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:fcdd87f9ce4203ac4f9b1284a0db8ea61f1f4ab2f08bbe6176dbaf3e6d2dcd12c.

Independent reconstruction. Implementation hash

sha256:a4955d74f704c817aac676ffc3af783df9b6bb7f328576e68920563fed210eab; independent certificate sha256:76ec599dda0c5859c14d2c306a87f6205b3a6c018234b8512be63c2f3d9c3db8; external-validation hash sha256:5bc8adc3aa6a4a9ec5f46723b60e70daf98cf6ff95fba9c9feab41dc99b85cc6. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-REDOX-COMPOSITE-2026

Comparison records:

- redox-coupling-iupac-composite: predicted
oxidation-electron-loss__reduction-electron-gain__paired-half-reactions__electron-transfer-conserved; source-derived
oxidation-electron-loss__reduction-electron-gain__paired-half-reactions__electron-transfer-conserved; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks paired oxidation/reduction, electron loss/gain or conserved transfer, or if a changed row is accepted.

Measurement receipt: sha256:0e526b5a363f71cc9197d97440720c42ed67a2e6d957838580a9b7294c44e4ab.

Isolation certificate: sha256:0e76be888cb4f92fa54a0c9af0c19ad7359f4686fcae082f491d469c2b57b996.

Custody certificate: sha256:ea8e2fb18aa224d07b45f653bf5d49418abea94fdfff25355877c9bb51033360.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/O04362>
(sha256:9e367d54ae7e7f9e5a40366f9298b1c8dffbacc4677847f60458b78e7e16fc68)

Registered target identities:

- redox-coupling-iupac-composite from IUPAC-GOLD-BOOK-REDOX-COMPOSITE-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no electronegativity scale, oxidation-state table, electrode potential, external definition, measured value or V2 result may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no electron carrier may be created, copied, silently erased or assigned to two endpoints

- no qualitative structural law is represented as a parameter-free prediction of a measured decimal scale
- external target content remains inaccessible until the complete prediction is sealed
- Every finite closed electron-labelled transfer between distinct chemical endpoints, with identical carrier identity before and after and paired donor loss/acceptor gain or their complete oxidation-assignment equivalents.

Immutable identities. Source manifest

sha256:4716ca0dc4e1b02be4550aed520ad9d6666f5b4c06ec6a4c179e40d6263609bd; derivation seal
 sha256:d1c62949f08368e109de85d3abbd09097e5c298b8eed8991415c5aa915fdf379; engine receipt
 sha256:8e1f6048b5dc8e6fb29718ca5b5075dd0fcdea5bbc4eed6e1d7d1e30b1ac8fb7 at
 receipts/engine/model_admitted/SFT-CHEM-REDOX-COUPLING-001-8e1f6048b5dc8e6f.json; empirical
 hash sha256:508d883df739899e26bd450ccaf6d537d4acb040cc9f0a9cdba0a078d7c185f7; measurement
 receipt sha256:0e526b5a363f71cc9197d97440720c42ed67a2e6d957838580a9b7294c44e4ab.

47. Electrochemical cell and separated redox closure

Claim identity: SFT-CHEM-ELECTROCHEM-CELL-001

Question and theorem. An electrochemical cell is a separated coupled-redox process with complete external electron and compensating ionic paths, retaining site identities and source-bounded work transfer.

*distinct-oxidation-reduction-sites__complete-external-electron-path__complete-compensating-ionic-path__one-redox-t
 transfer-cycle__source-bounded-work-transfer*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001

- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001

Generated grammar. Generate the literal product of site, electron-path, ionic-path, coupling, work, orientation, record and extra-rule choices.

Boundary: Every finite separated pair of coupled redox sites with a complete electron path from oxidation to reduction, a complete compensating ionic path, retained chemical boundaries and source-bounded work transfer.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `distinct-oxidation-reduction-sites__complete-external-electron-path__complete-compensating-ionic-path__one-redox-transfer-cycle__source-bounded-work-transfer__held-cell-orientation__sites-paths-transfer-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
sites	distinct-oxidation-reduction-sites	one-conflated-site	Separation is required for an external path.
electron-path	complete-external-electron-path	electron-teleportation	It omits the transfer trace.
ionic-path	complete-compensating-ionic-path	charge-accumulation-unclosed	Accumulation halts sustained transfer.
coupling	one-redox-transfer-cycle	independent-half-reactions	Unpaired halves violate conservation.
work	source-bounded-work-transfer	free-energy-output	Untraced work creates support.
orientation	held-cell-orientation	negative-potential-primitive	A negative scalar is not needed.
record	sites-paths-transfer-trace	cell-label-only	A label cannot reproduce either path.
extra	no-extra-rule	free-electrode-potential	A fitted potential can select behavior.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:63371e499dce6d50287fc484b0f0cbf70c3dfd532ed6b9703aa7bf01d4945db0`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8c3c9a1814cd2e0e60db3b4e95284e76c9a5eb2f3500e44588cc6f80b50c61cb`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:815ea97bdb3eea0fe85ee56fef6342a6e546c2c71f10bcfba66baee6eb6247ce`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:458f2a1f78edf5913a5538be6612662eb14b0c60edf88ce2f52524182f1a2383`.

Independent reconstruction. Implementation hash

`sha256:4e3a0f2270afaa2bf36345a962acbb96bc9e41cdf8f268cc01a75f3ff5854c3f`; independent certificate `sha256:ce4e3f99a3026cab0531efad71c9733fc92a01eab2650628c4ed9bcce6636279`; external-validation

hash sha256:1611191550a17c05b352ceacf7219d0acb19549098545655fd02d570684ce3c8. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ELECTROCHEMICAL-CELL-COMPOSITE-2026

Comparison records:

- electrochemical-cell-iupac-composite: predicted
separated-redox-half-cells__external-electron-path__ionic-circuit-closure__chemical-to-electrical-work-transfer; source-derived
separated-redox-half-cells__external-electron-path__ionic-circuit-closure__chemical-to-electrical-work-transfer; exact match
True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks separated redox sites, electron and ionic paths or chemical/electrical work conversion, or if a changed row is accepted.

Measurement receipt: sha256:ed2c18d4f111c4951e31882355b8174ee760e0b1bfaf84fc42854b464f4d4071.

Isolation certificate: sha256:ddd9642c0fe88df891ae4d9eba58fbed23c49da0b3e43f6ed363dd3023e339f8.

Custody certificate: sha256:108530f73810c1ffe6f221c638697434ba3ce1ca3eb28a62bce071456be4045c.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/09058>
(sha256:0016d0976a51c44b0208458810ad455a459f988c7573d186293edff0aea33d68)

Registered target identities:

- electrochemical-cell-iupac-composite from IUPAC-GOLD-BOOK-ELECTROCHEMICAL-CELL-COMPOSITE-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no electronegativity scale, oxidation-state table, electrode potential, external definition, measured value or V2 result may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no electron carrier may be created, copied, silently erased or assigned to two endpoints
- no qualitative structural law is represented as a parameter-free prediction of a measured decimal scale
- external target content remains inaccessible until the complete prediction is sealed
- Every finite separated pair of coupled redox sites with a complete electron path from oxidation to reduction, a complete compensating ionic path, retained chemical boundaries and source-bounded work transfer.

Immutable identities. Source manifest

sha256:dc04bfcbe7e646cbb9374cdefacc8427db47a2b020cfd54d7fc5829caad8d5e4; derivation seal

sha256:7875b7061b5c5442832e2b350da6f9c07b04f64c09ea896b3b80c32c1398fe67; engine receipt

sha256:bbb6c005a3ccd1c37e538fe895aace47663809fa4ccdaea21caa8183a652eaf4 at

receipts/engine/model_admitted/SFT-CHEM-ELECTROCHEM-CELL-001-bbb6c005a3ccd1c3.json;

empirical hash sha256:1617e1c9b28893ab028b44cd8e988a03e0f397e416cb38c537f7a31af7cfce7d;

measurement receipt sha256:ed2c18d4f111c4951e31882355b8174ee760e0b1bfaf84fc42854b464f4d4071.

13. Reaction, kinetics, equilibrium and thermochemistry foundations - 12 admitted claims

Reaction paths retain source, intermediate, condition, energy, time and terminal identity without importing a fitted law. This entire registered subcategory is included; no proper subset is presented as field completion.

48. Chemical reaction as source-bound identity transformation

Claim identity: SFT-CHEM-RXN-IDENTITY-001

Question and theorem. A chemical reaction is a complete source-bound reactant-to-product transformation conserving every elemental carrier while changing at least one chemical relation.

source-bound-transition__complete-reactant-organization__complete-product-organization__all-elemental-carriers-conserved__at-least-one-relation-changed

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001

• SFT-CHEM-ELECTROCHEM-CELL-001

Generated grammar. Generate the literal product of source, input, output, carrier, change, orientation, record and extension choices.

Boundary: Every finite source-bound chemical transition between complete named reactant and product organizations, conserving every elemental carrier while changing at least one registered chemical relation.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `source-bound-transition__complete-reactant-organization__complete-product-organization__all-elemental-carriers-conserved__at-least-one-relation-changed__held-reactant-product-orientation__complete-endpoint-carrier-relation-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
source	source-bound-transition	unregistered-transformation	No chemical provenance is retained.
input	complete-reactant-organization	reactants-erased	A product alone cannot define change.
output	complete-product-organization	products-erased	An input alone cannot define change.
carriers	all-elemental-carriers-conserved	element-created-or-lost	It violates chemical closure.
change	at-least-one-relation-changed	identity-transition	No changed relation means no reaction event.
orientation	held-reactant-product-orientation	unordered-endpoints	Direction cannot be audited.
record	complete-endpoint-carrier-relation-trace	reaction-name-only	A name cannot reproduce the transition.
extra	no-extra-rule	free-reaction-exception	An exception can admit arbitrary change.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:fb754a6029d272793d88a9743f1a5ca1c2e28481c465d2910713413cb2b900f3`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:959bb4b2950fa5b8ca9f5ac5db294ffbcf0d889230a15c3bfffaf0026b4ccc46`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:767e576739cb6dc5eaa7f0d4ae141f26bf618676e521133b66f4407821b1e7`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:15b247c519ea516c0b538768e82bebe17813c244712eb2495d2664222da5cd88`.

Independent reconstruction. Implementation hash

`sha256:15c43d7930b426b88179e32fba7fbf6289d9c438b043a1e571c14243f6f908a2`; independent certificate `sha256:f250baeb1f01ff51426dcf57f3edc484ff4bfa408801c5e5aa6dd631239b1d9d`; external-validation hash `sha256:7ab5daf4b2863630faf3c3bc95b6aa460a22f0c89b33f10844d7db84605edff5`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CHEMICAL-REACTION-COMPOSITE-2026

Comparison records:

- chemical-reaction-iupac-composite: predicted
reactants-transformed-to-products__chemical-identities-change__elemental-carriers-conserved__source-bound-reaction-record;
source-derived
reactants-transformed-to-products__chemical-identities-change__elemental-carriers-conserved__source-bound-reaction-record;
exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks reactant/product transformation, chemical change or carrier conservation, or if a changed row is accepted.

Measurement receipt: sha256:40ab81e82e83f0f753e15301f74cdb6cf444d81ba760015aa26e57634c6efb9d.

Isolation certificate: sha256:c01dce160f1fb326111a5aff3b6662145a41cb959d9ee100da2746f7e3322afc.

Custody certificate: sha256:f73f788180b3356810c7f01958ce03a8aab3b4c9c95f86493e1d01e05f3e7b5d.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01033>
(sha256:83d7891114ad65c012ce2e35f4021beaee6a37d4473c416170d6643db4dc7347)

Registered target identities:

- chemical-reaction-iupac-composite from IUPAC-GOLD-BOOK-CHEMICAL-REACTION-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite source-bound chemical transition between complete named reactant and product organizations, conserving every elemental carrier while changing at least one registered chemical relation.

Immutable identities. Source manifest

sha256:d2f0173a6d05e0cd204e80689cddb86dc5a7aef43c2001495f62703d67f9600b; derivation seal

sha256:cb6f88146f0eb8a99d8c775d2a7caf9989c8a401a13bc42ea9eafb0d27408622; engine receipt

sha256:ec1b06b42e323744b8166f5498671868cd4f9ce5ef68c1f43d718a894712e42d at

receipts/engine/model_admitted/SFT-CHEM-RXN-IDENTITY-001-ec1b06b42e323744.json; empirical

hash sha256:ee5a3c51d4778efda6f367cb7d00b50e29dc5385627ab7565b14d547b6126445; measurement

receipt sha256:40ab81e82e83f0f753e15301f74cdb6cf444d81ba760015aa26e57634c6efb9d.

49. Reaction mechanism as complete elementary-step trace

Claim identity: SFT-CHEM-RXN-MECHANISM-001

Question and theorem. A reaction mechanism is the complete ordered composition of elementary chemical transitions whose internal states join exactly and whose external endpoints equal the overall reaction.

ordered-elementary-step-word__adjacent-state-composition__declared-indivisible-steps__internal-states-held__same-overall-endpoints

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001

Generated grammar. Generate the literal product of step, adjacency, elementarity, internal-state, endpoint, conservation, record and extension choices.

Boundary: Every finite ordered sequence of elementary chemical transitions whose adjacent states compose exactly and whose endpoint composition equals one registered overall reaction after internal states cancel.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `ordered-elementary-step-word__adjacent-state-composition__declared-indivisible-steps__internal-states-held__sa`

me-overall-endpoints__carrier-conservation-each-step__complete-ordered-step-trace__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
steps	ordered-elementary-step-word	unordered-step-list	Unordered changes do not form a path.
adjacency	adjacent-state-composition	unmatched-step-endpoints	The proposed path is discontinuous.
elementarity	declared-indivisible-steps	hidden-substeps	The trace is incomplete at its declared boundary.
internal	internal-states-held	internal-species-erased	Cancellation cannot be audited.
endpoints	same-overall-endpoints	different-overall-reaction	The path does not explain the claim.
conservation	carrier-conservation-each-step	carrier-loss-between-steps	Local openness breaks global closure.
record	complete-ordered-step-trace	mechanism-label-only	A label cannot reproduce the path.
extra	no-extra-rule	free-hidden-path	An unseen path can explain any endpoint.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256: 2b08b15144e7448122e88a2a305aad7c8ac006ff06273c79a20de192a4534887.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 5a8ce953ed1770bff5b7b4cc8d8b8b44826a0eb87215bec2ebdbef36dc82ecdf.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 81abefbb5f378e8edd9b45f331410f29b0fc2b044732df627c2030f63501c9b9.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: cca8d7ea96b3dd5513a244338663507233b8fa000c8323f450f93ae9113130ea.

Independent reconstruction. Implementation hash

sha256: 52cf78c584d78eea0045c48685d0bca81995b54d84f0a39dc59fa91be71cec1b; independent certificate sha256: e015218ecd7e1982bb5a476c05f79d02779995e38b0a668982c9638121c14058; external-validation hash sha256: d9b7d4275d4318d9d3013d3e79cd57ed415f4c1aeb8c28924c1dabffd539d7c0. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-REACTION-MECHANISM-COMPOSITE-2026

Comparison records:

- reaction-mechanism-iupac-composite: predicted overall-reaction-decomposed__ordered-elementary-steps__intermediate-states-retained__step-composition-reproduces-endpoints; source-derived overall-reaction-decomposed__ordered-elementary-steps__intermediate-states-retained__step-composition-reproduces-endpoints; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks ordered elementary steps, retained intermediates or composition to the overall reaction, or if a changed row is accepted.

Measurement receipt: sha256:ac968be8a7826c61228a8e6759ac2b987a2bc82ba8bd9a25f31cc491f7dd3aa4.

Isolation certificate: sha256:b697c70f6a4341bbaadb6d3e93fb1cc9491becbee0c7ca375206220c27e99493.

Custody certificate: sha256:6bf41b2a3491641c01b94308a830208cdf93c216eeefe2bba423f378da7d96d8.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/M03804> (sha256:b84ddab1f13dbc71482ed3f7653eeab3e2cfe5b7ffba64c30ef9d08513d6be33)

Registered target identities:

- reaction-mechanism-iupac-composite from IUPAC-GOLD-BOOK-REACTION-MECHANISM-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite ordered sequence of elementary chemical transitions whose adjacent states compose exactly and whose endpoint composition equals one registered overall reaction after internal states cancel.

Immutable identities. Source manifest

sha256:f707c79fe4f531b5cfcfac5d8d8d6159a5ec730317f58730dd4d5fba9056889c; derivation seal

sha256:330276f336f931d377fa0bdcf6a796b667275b16cf1c4ba2421748cc9ac155e2; engine receipt

sha256:4045d3ae4048a774bb4c5bde502120261bccd2f991683cf8c8415b20ce32dc02 at

receipts/engine/model_admitted/SFT-CHEM-RXN-MECHANISM-001-4045d3ae4048a774.json; empirical

hash sha256:9b22daf1e91aba70756d0d0501e0ccb921663724650ddf3adc0fc05bfed70853; measurement

receipt sha256:ac968be8a7826c61228a8e6759ac2b987a2bc82ba8bd9a25f31cc491f7dd3aa4.

50. Reaction intermediate and retained path identity

Claim identity: SFT-CHEM-RXN-INTERMEDIATE-001

Question and theorem. A reaction intermediate is a named internal species produced by one mechanism step and consumed by a later step, canceling from net endpoints while remaining in the complete path trace.

*named-internal-species__produced-by-one-step__consumed-by-later-step__production-precedes-consumption__cancel
s-from-net-endpoints*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001

- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001

Generated grammar. Generate the literal product of identity, production, consumption, order, net, retention, record and extension choices.

Boundary: Every named chemical species produced by one internal elementary step and consumed by a later step of the same complete mechanism, absent from the net endpoint difference but retained in the path trace.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `named-internal-species__produced-by-one-step__consumed-by-later-step__production-precedes-consumption__cancels-from-net-endpoints__intermediate-path-identity-held__production-consumption-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
identity	named-internal-species	anonymous-transient	Its recurrence cannot be checked.
production	produced-by-one-step	not-produced	It is not generated by the mechanism.
consumption	consumed-by-later-step	not-consumed	It remains a product, not an intermediate.
order	production-precedes-consumption	consumed-before-produced	The trace is causally open.
net	Cancels-from-net-endpoints	present-in-net-products	It is then an endpoint product.
retention	intermediate-path-identity-held	intermediate-erased	Erasure destroys the mechanism proof.
record	production-consumption-trace	intermediate-name-only	A name lacks step witnesses.
extra	no-extra-rule	free-short-lived-premise	Lifetime alone cannot define path role.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : a2752d09725cb5de86a3b147b38a4775c0d0becc9b26791e17fd354ce5226e48.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : d64901b4b0f43026487180a4ab4442dca557b6ac66e399ebaa765e1d14e9028f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 2fdca90259d1e28cb742f7dc20ae1ba5f976a83c45c3f10a0c1e382fb943b0a7.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : f2b5675a4b4d7b517d7b472aec706efc56e4d23acf3987a148244f3fbccde8f9.

Independent reconstruction. Implementation hash

sha256 : c903ed412ee204f58fc66bb48dd011c89b5c4b3b5fe31abbd90c28a67b8d4473; independent certificate sha256 : 068785565e96bdc6f82dc2adf032fa8837690bfab3da2ea0a7b9ccbddb92901d; external-validation hash sha256 : c12427575af25883ab85ab321f4eb0e2a50091f5bf17b62a5c4093a63e9010e6. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-I03096-2026

Comparison records:

- reaction-intermediate-iupac-i03096: predicted
formed-in-one-elementary-step__consumed-in-later-step__absent-from-overall-reaction__retained-in-mechanism-trace;
source-derived
formed-in-one-elementary-step__consumed-in-later-step__absent-from-overall-reaction__retained-in-mechanism-trace; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks produced/consumed internal species behavior or net cancellation, or if a changed row is accepted.

Measurement receipt: sha256 : babad6b1f9ccfe3db110c7685b3c9b6bb12731c20de9bb37436c994bb072ad01.
Isolation certificate: sha256 : 9a56f10e63794dfcc9ec59149ea2c83ae166c2e25608e1eed83c9ba6dc34b23d.
Custody certificate: sha256 : 0a91bb23328b0c624ef16f4cd51e95225564ffec77801741c3393e42b35b8939.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/I03096> (sha256:9de39353e9af29cbe1b9dcb7f7972e2125378129e82924805f5d97696d93fd4a)

Registered target identities:

- reaction-intermediate-iupac-i03096 from IUPAC-GOLD-BOOK-I03096-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every named chemical species produced by one internal elementary step and consumed by a later step of the same complete mechanism, absent from the net endpoint difference but retained in the path trace.

Immutable identities. Source manifest

sha256:90b0a285f59d9a56f1059fe4a10878b2c3d57e7ac60cbd1cb71329447648cb4d; derivation seal sha256:e4e466d04a0430b215a03e537da73d80cba6a720277cccade70c5007db1837e6; engine receipt sha256:d53d9a103971aa8eee09162a1763b71e9572dbc40c1c546c5042d7f571dcf4af at receipts/engine/model_admitted/SFT-CHEM-RXN-INTERMEDIATE-001-d53d9a103971aa8e.json; empirical hash sha256:e22d8e90029ce389694d0ee25c61f0e155bb7b9aa7f7f68e59c978f017d31c26; measurement receipt sha256:babad6b1f9ccfe3db110c7685b3c9b6bb12731c20de9bb37436c994bb072ad01.

51. Activation barrier and transition boundary

Claim identity: SFT-CHEM-KIN-ACTIVATION-001

Question and theorem. Activation support is the least source-bound addition required for a generated reaction path to reach its highest transition boundary relative to the reactant boundary.

complete-generated-reaction-path__source-bound-support-account__highest-path-boundary__reactant-relative-support__least-crossing-support

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001

Generated grammar. Generate the literal product of path, source, transition, reference, minimality, orientation, record and extension choices.

Boundary: Every finite generated reaction path from a retained reactant organization to product organization, with the least additional source support needed to reach its unique highest transition boundary recorded relative to the reactant boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-generated-reaction-path__source-bound-support-account__highest-path-boundary__reactant-relative-support__least-crossing-support__held-reactant-to-transition-orientation__path-source-boundary-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
path	complete-generated-reaction-path	endpoint-only	No crossing requirement is represented.
source	source-bound-support-account	free-energy-number	An imported number selects the barrier.
transition	highest-path-boundary	arbitrary-high-state	It need not lie on the path.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
reference	reactant-relative-support	absolute-energy-origin	An arbitrary origin imports a parameter.
minimality	least-crossing-support	nonminimal-added-support	Extra support does not define activation.
orientation	held-reactant-to-transition-orientation	signed-barrier	A negative scalar is outside the proof domain.
record	path-source-boundary-trace	activation-value-only	A value cannot reproduce the path.
extra	no-extra-rule	free-temperature-fit	A fitted input can tune the result.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:660d0bba8291ba95de30f0439f20507f8a51eac2f38cbaf90287d61f61dd0645.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:b027b4fdc5b38f0c0ef3c9a3232b0eabfa547393ee6c41f228abbb05e86328ad.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:94281f85f5d6d9d09d06e622c59d30d07f5246431edbbba2ab67d03b2cc131be4.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:4784092078887c2ffacaa9f8423c814871eb6d69a86186db2664831d2e22cac0.

Independent reconstruction. Implementation hash

sha256:e383c27e7a4bb6acf54b280cfc8dbc77ee6df11b4dd68d4495893046a0b8c5d9; independent certificate sha256:2549feed7fb6748985ea7bfc80cd168b8d37152be45c05d441f5fc7a39972385; external-validation hash sha256:becc777742df17e5e3b6b8efc1ad00c55e7fc6c31edcce0abc702251f881bbbeb. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ACTIVATION-COMPOSITE-2026

Comparison records:

- activation-barrier-iupac-composite: predicted
reactants-to-transition-state__minimum-required-energy__barrier-relative-to-reactants__reaction-path-source-bounded;
source-derived
reactants-to-transition-state__minimum-required-energy__barrier-relative-to-reactants__reaction-path-source-bounded; exact
match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks a transition-state energy difference or activation barrier relative to reactants, or if a changed row is accepted.

Measurement receipt: sha256:aa37a56bf39414cba9dd60307bfcd06db1199d28c85e97a6f519abdd2e2802ee.

Isolation certificate: sha256:c8caa9b7387790f3f27f105079e211ef62dfebd7234c652a89f2008d637e6f1d.

Custody certificate: sha256:79796318f459bc0268440ab492f62f374d49eab32932062202feb4d719536a53.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/G02631>
(sha256:c51e2c7167603bd633c54e11d238728c7313aa8584662ea4bfc3c3a4bf97a04c)

Registered target identities:

- activation-barrier-iupac-composite from IUPAC-GOLD-BOOK-ACTIVATION-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite generated reaction path from a retained reactant organization to product organization, with the least additional source support needed to reach its unique highest transition boundary recorded relative to the reactant boundary.

Immutable identities. Source manifest

sha256:1e09085df6d33f783fa45df82d6e74160bb85e9f681a7b2762fa79bfffde1006d; derivation seal sha256:610e20b5fb74d6fd866b155b306605e690407a30d8e21bc58c43578832f38409; engine receipt sha256:bd0c43a9f211959e83b423962038626e7bc1190a33e82ca1baec27a89d492d8b at receipts/engine/model_admitted/SFT-CHEM-KIN-ACTIVATION-001-bd0c43a9f211959e.json; empirical hash sha256:94e9c9357c63b20faad55db948d89a2e2c47c21f4ab458d36bb68eda9041f010; measurement receipt sha256:aa37a56bf39414cba9dd60307bfc06db1199d28c85e97a6f519abdd2e2802ee.

52. Reaction rate from counted transitions

Claim identity: SFT-CHEM-KIN-RATE-001

Question and theorem. Reaction rate is the exact positive count of completed instances of one registered reaction per positive reference recurrence and retained observation boundary, with conditions source-bound.

one-registered-reaction-event__positive-completed-event-count__positive-reference-recurrence-count__registered-species-observation-boundary__event-count-per-recurrence

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001

- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001

Generated grammar. Generate the literal product of event, count, duration, boundary, relation, condition, record and extension choices.

Boundary: Every finite observation boundary with a positive counted set of completed instances of one reaction transition and a positive counted set of registered recurrence intervals, retaining species amount and conditions as source records.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `one-registered-reaction-event__positive-completed-event-count__positive-reference-recurrence-count__registered-species-observation-boundary__event-count-per-recurrence__source-bound-condition-record__event-duration-boundary-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
event	one-registered-reaction-event	unidentified-change	Events cannot be counted consistently.
count	positive-completed-event-count	continuous-rate-primitive	It imports a continuum quantity.
duration	positive-reference-recurrence-count	clock-free-count	No rate relation follows.
boundary	registered-species-observation-boundary	unbounded-observation	Amount change is undefined.
relation	event-count-per-recurrence	fitted-rate-value	A fitted value is target-derived.
conditions	source-bound-condition-record	conditions-erased	Rates cannot be compared reproducibly.
record	event-duration-boundary-trace	rate-answer-only	It cannot reconstruct events or time.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
extra	no-extra-rule	free-rate-constant	A constant can tune any target.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:2ca5149a0037bdc6c5af664a16fbe871bd044417c8461d3d1799bf414c4b85a.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:1ab78e2ba0ff39f6f97b554e3c8e84baeaf11592a3e925622243241c756f0513.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:a570e4838fecc768effd3446ebfe6dd2165ad3b10676081251be393901f4813f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:8c1067c6d6dd67c38da69b8631a81a9466230c9781a06506a5d3b2b95bcf7ab7.

Independent reconstruction. Implementation hash

sha256:e6152457338984a6274e730a62ca2fe591bc2e4483bd7e7da9993b3f7eebef18; independent certificate sha256:6f370ecdb33bbca8628cb1164fc257a347847ae3548c54d3b7cb185ac3d21225; external-validation hash sha256:c33a94fffb46668103de593e639b069fb4ab74873fc1e4d5a3a6a4a0b9685bd60. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-REACTION-RATE-COMPOSITE-2026

Comparison records:

- reaction-rate-iupac-composite: predicted
extent-of-reaction-change__per-time-interval__observation-boundary-retained__conditions-and-method-source-bounded;
source-derived
extent-of-reaction-change__per-time-interval__observation-boundary-retained__conditions-and-method-source-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks reaction change per time or an observation/condition boundary, or if a changed row is accepted.

Measurement receipt: sha256:2bc7a90bb3e425f4599a2ba358f4742d83bc2add536db517b65550ed4f014e0b.

Isolation certificate: sha256:2400a336b98423d17a076759ea2c6b5621e3984ea8ad733fc79537058eac66e5.

Custody certificate: sha256:22cf474da198613f76d09d1d596e77ffb08342c9597a41094857f468991ed8f6.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/R05147>
(sha256:4f61cb4a9d72b0266dd464f84a2349aee57aec161521fb2466bcf8febed361fe)

Registered target identities:

- reaction-rate-iupac-composite from IUPAC-GOLD-BOOK-REACTION-RATE-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate

- no numerical zero, negative, irrational, imaginary or floating point quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite observation boundary with a positive counted set of completed instances of one reaction transition and a positive counted set of registered recurrence intervals, retaining species amount and conditions as source records.

Immutable identities. Source manifest

sha256:5d48554eb1fd827972bd64f62413803926b25262cf378bbd9b49f7cca7461edd; derivation seal
 sha256:8d53fc1fad13fe8b90f880ea07260116fe57582d23368d0207e5707fc220c9b6; engine receipt
 sha256:c4c4e2acabcf3b44e5da03ed097bb132c729e017549fbc79dc93d8e64c0a2d1 at
 receipts/engine/model_admitted/SFT-CHEM-KIN-RATE-001-c4c4e2acabcf3b4.json; empirical hash
 sha256:c1e8cc44b39471323ebb4c083591f1c37939bde73e279f31a9530deaf3193925; measurement receipt
 sha256:2bc7a90bb3e425f4599a2ba358f4742d83bc2add536db517b65550ed4f014e0b.

53. Kinetic order and dependency multiplicity

Claim identity: SFT-CHEM-KIN-ORDER-001

Question and theorem. Kinetic order is the exact condition-bound multiplicity of a named reactant coordinate in the complete intervention-to-rate dependency trace; independence is Empty rather than numerical zero.

named-reactant-coordinate__source-bound-coordinate-intervention__complete-rate-response-census__forced-dependency-occurrence-count__empty-dependency-form

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001

- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-KIN-RATE-001

Generated grammar. Generate the literal product of reactant, intervention, response, multiplicity, independence, scope, record and extension choices.

Boundary: Every finite source-bound intervention census for one reactant coordinate, retaining the exact multiplicity with which distinguishable occurrences of that coordinate participate in the rate dependency; absence is represented structurally.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `named-reactant-coordinate__source-bound-coordinate-intervention__complete-rate-response-census__forced-dependency-occurrence-count__empty-dependency-form__condition-and-mechanism-bound-order__reactant-intervention-response-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
reactant	named-reactant-coordinate	anonymous-concentration	The dependency coordinate is lost.
intervention	source-bound-coordinate-intervention	passive-correlation	It cannot establish dependence.
response	complete-rate-response-census	single-rate-answer	Multiplicity cannot be recovered.
multiplicity	forced-dependency-occurrence-count	stoichiometric-coefficient-assumed	Kinetic and endpoint counts need not coincide.
independence	empty-dependency-form	numerical-zero-order	Zero is prohibited as proof quantity.
scope	condition-and-mechanism-bound-order	universal-condition-free-order	Observed order can be condition-specific.
record	reactant-intervention-response-trace	order-number-only	It cannot reproduce the intervention.
extra	no-extra-rule	free-power-law-fit	A fit can select the exponent.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:b2ec76b3e1f64ef6bbe60439d7c492f27614b93516d4c65092e90f2f94fc8985`.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 295e33bd061e7b035038ac143c6ff58b9c0d6055338ea690a7eac1f7ef087903.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: b1000a7635db0efabd0fbd160210eef648866a5cf90229cc0f5b0bdd55efeb31.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: dc3f77164ac93ae31817f766ae36505994210688eb8237c8a701a7f67084bc5e.

Independent reconstruction. Implementation hash

sha256: ef4271771caeb278a9789ab2a7d72ab743de0a0cae928d1bec475ef3bdb934a2; independent certificate sha256: 7d4b1f0011c6d5b0bce0edadeb270489f2a5924c3f7758d972ed5c640a79e26d; external-validation hash sha256: 6850cda3848dcd0f2e4e7f1bbb07df18ecd53e7a3e1ac4b02245186038a263a3. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-O04322-2026

Comparison records:

- reaction-order-iupac-o04322: predicted
rate-law-concentration-dependence__reactant-specific-exponent__experimentally-determined__not-forced-by-stoichiometry;
source-derived
rate-law-concentration-dependence__reactant-specific-exponent__experimentally-determined__not-forced-by-stoichiometry;
exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks reactant-specific rate dependence, experimental determination or separation from stoichiometry, or if a changed row is accepted.

Measurement receipt: sha256: 7eb797b25fdeb87c6f2e7c2bd3c6ff7b9930c5d39576362cbaf110044663d179.

Isolation certificate: sha256: fdfa233607d0e26b1df6df1db1494c2c8299ef48be9b653ac34233cb1248f47c.

Custody certificate: sha256: 1fff1b93452f2f09922adb516389b06e44d47a3530a29b9bfac617b3e01ef0d4.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/O04322>
(sha256: acb0c1bcc39fca71b9fad77cfbc8813c6ffdc1728be3943d56dcd7631c90b9f4)

Registered target identities:

- reaction-order-iupac-o04322 from IUPAC-GOLD-BOOK-O04322-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed

- Every finite source-bound intervention census for one reactant coordinate, retaining the exact multiplicity with which distinguishable occurrences of that coordinate participate in the rate dependency; absence is represented structurally.

Immutable identities. Source manifest

sha256:85001ab3be411c1a37ae654f04759245d4361e25edb10909f92a0b6ef645fecdd; derivation seal
 sha256:a07fbec88a101579fe82cf4cd643ff33b3d174a47b5adc69c1962d2d79cbdfc9; engine receipt
 sha256:37abcdd1d68e0f50d2df51585e9b700e106c363b97ff6bc36ef150b3dc6d3a1a at
 receipts/engine/model_admitted/SFT-CHEM-KIN-ORDER-001-37abcdd1d68e0f50.json; empirical hash
 sha256:9926a6aec59e0699faa6b193846a4ed8534f30aa894ed58f2beb408c5095d2ce; measurement receipt
 sha256:7eb797b25fdeb87c6f2e7c2bd3c6ff7b9930c5d39576362cbaf110044663d179.

54. Chemical equilibrium as balanced reversible support

Claim identity: SFT-CHEM-EQ-CHEMICAL-001

Question and theorem. Chemical equilibrium is persistent forward and reverse reaction recurrence on a closed boundary whose paired carrier changes preserve the macroscopic composition class.

forward-and-reverse-transitions__persistent-opposed-events__paired-forward-reverse-support__macroscopic-composition-class-recurs__closed-observation-boundary

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001

- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-KIN-RATE-001

Generated grammar. Generate the literal product of reversibility, activity, balance, composition, boundary, condition, record and extension choices.

Boundary: Every finite reversible chemical transition on one closed observation boundary for which forward and reverse event recurrences persist while their complete carrier changes pair and the retained macroscopic composition class recurs.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `forward-and-reverse-transitions__persistent-opposed-events__paired-forward-reverse-support__macroscopic-composition-class-recurs__closed-observation-boundary__source-bound-condition-record__forward-reverse-composition-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
reversibility	forward-and-reverse-transitions	one-way-reaction	No reverse support exists.
activity	persistent-opposed-events	static-no-events	Chemical equilibrium is dynamically supported.
balance	paired-forward-reverse-support	unpaired-event-counts	Composition would drift.
composition	macroscopic-composition-class-recurs	microscopic-state-fixed	Individual events need not stop.
boundary	closed-observation-boundary	open-material-flow	External flow can imitate balance.
conditions	source-bound-condition-record	condition-free-equilibrium	The class can change with conditions.
record	forward-reverse-composition-trace	equilibrium-constant-only	A number cannot reconstruct dynamics.
extra	no-extra-rule	free-equilibrium-fit	A fitted constant can force a target.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:efbdf446dafef76afa9d768e8ab86b366db99bb763a88014f1559f5279b7a246b.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:101db4c738b823c2aaa66bf158c163a9c9f0b6dff0ebcaa24e4f669f0e8b77b6.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:bef1a0371237a62a4e9f2ccebde1c97119c7797967f5081c7d6014b377fdd4fc.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:1d3e059d0e008a62401bd5313bb3466cef844adb001f118d0cdd58a821fa6dae.

Independent reconstruction. Implementation hash

sha256:9795f2eafcfce4afe0c107b84963aeab091d2859dec0229c033bea11e3d13b0c6; independent certificate sha256:83220324729c01a4a2bdb47ec45c4b6bfc93606c5266a3cdca91c5e3165d6c21; external-validation hash sha256:673eb045dfffa27eb408f7c31e43ae572ca26de3280b8c55c0b2ec5b64d110d33. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CHEMICAL-EQUILIBRIUM-COMPOSITE-2026

Comparison records:

- chemical-equilibrium-iupac-composite: predicted forward-and-reverse-reactions-continue__rates-equal-at-equilibrium__macroscopic-composition-constant__condition-bound-dynamic-state; source-derived forward-and-reverse-reactions-continue__rates-equal-at-equilibrium__macroscopic-composition-constant__condition-bound-dynamic-state; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks continuing opposed reactions, equal rates or constant macroscopic composition under fixed conditions, or if a changed row is accepted.

Measurement receipt: sha256:c784f640938b7ee31b298152ea612f30f6022459eaa06752018f4d4b8f16b6a1.

Isolation certificate: sha256:03a826df6d439759177d91f17085fee068cb7ce9b682c7a44746496c601b1e41.

Custody certificate: sha256:d15804f433bd45d106fa58418e2eebb201ef447cef42f83a700e563104fa76e0.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01023> (sha256:8720b60c684dd7869500da1577be0fa91f2362e02420d755fbbd1abbad6582b)

Registered target identities:

- chemical-equilibrium-iupac-composite from IUPAC-GOLD-BOOK-CHEMICAL-EQUILIBRIUM-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite reversible chemical transition on one closed observation boundary for which forward and reverse event recurrences persist while their complete carrier changes pair and the retained macroscopic composition class recurs.

Immutable identities. Source manifest

sha256:1bbd965823dc374c8a3cc74661e6b432c60ca3c5d6abb5c91549790f1cebe82c; derivation seal

sha256:bd7304163ebebe069824cacee79a015b56c6ea9fe18831cc1604171be324c068; engine receipt

sha256:8f7da55dbccf2e835b129adc43335e84c1ce3336ad696a5b5d7e58b4851c32b7 at

receipts/engine/model_admitted/SFT-CHEM-EQ-CHEMICAL-001-8f7da55dbccf2e83.json; empirical

hash sha256:76e8659e535b5cf8574d4c92c22b04712fefbc88d5a9ebaadd267a983887ab5e; measurement

receipt sha256:c784f640938b7ee31b298152ea612f30f6022459eaa06752018f4d4b8f16b6a1.

55. Reaction thermochemical accounting

Claim identity: SFT-CHEM-THERMO-REACTION-001

Question and theorem. Reaction thermochemistry is the complete source-bound energy ledger across one conserved chemical transformation, with transfer orientation, phase, conditions and reference state retained.

registered-reaction-trace__complete-reactant-energy-support__complete-product-energy-support__energy-transfer-balanced__held-absorbed-or-released-orientation

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001

- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-RXN-INTERMEDIATE-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-KIN-ORDER-001
- SFT-CHEM-EQ-CHEMICAL-001

Generated grammar. Generate the literal product of reaction, input, output, conservation, orientation, condition, record and extension choices.

Boundary: Every finite closed chemical reaction with complete source-bound energy carriers at reactant and product boundaries, held transfer orientation, phase, conditions and reference state.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `registered-reaction-trace__complete-reactant-energy-support__complete-product-energy-support__energy-transfer-balanced__held-absorbed-or-released-orientation__phase-condition-reference-held__reaction-energy-source-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
reaction	registered-reaction-trace	energy-value-without-reaction	No chemical identity is retained.
input	complete-reactant-energy-support	reactant-energy-erased	The ledger cannot close.
output	complete-product-energy-support	product-energy-erased	The ledger cannot close.
conservation	energy-transfer-balanced	free-energy-creation	It violates closure.
orientation	held-absorbed-or-released-orientation	signed-heat-number	A signed scalar is unnecessary.
conditions	phase-condition-reference-held	condition-free-value	Thermochemical quantities are state-bound.
record	reaction-energy-source-trace	enthalpy-answer-only	It cannot reproduce accounting.
extra	no-extra-rule	free-calibration-fit	A fit can select any value.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f0d09549fc7aba9e3e959cdcd632a04ccd046cf9645b3a55925f0779021368d1`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:57872b69512c1c9f80e2e7c241245dd8bd4bed71f6fdb2ff6b57b3ebc2f931a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:76d290aaa533e03e6abe301a16e9471b1e7120820c210f8187c6bef6d0a0a7d0`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:fdb60292a3bddf4eb85038e4d768a41a44dc18bdc79863a07913318ce58e8db3`.

Independent reconstruction. Implementation hash

sha256: f04a8631eb33adf1005011b19ed41ebf2b14be6ce7d9ebffcd79c8026cbd7fb; independent certificate sha256: 3701008c6ea7b1b78f1f9909e75c5d5edac9a3877f7772c9607399a0232932d9; external-validation hash sha256: 507b1e7d45383ad3a337630eebbd4ae76d736ff285c638b934a98e7c497e2319. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-REACTION-THERMOCHEMISTRY-COMPOSITE-2026

Comparison records:

- reaction-thermochemistry-iupac-composite: predicted
reaction-energy-change__products-minus-reactants__absorbed-or-released-transfer__state-and-condition-source-bounded;
source-derived
reaction-energy-change__products-minus-reactants__absorbed-or-released-transfer__state-and-condition-source-bounded; exact
match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks reaction energy difference, transfer direction or state/condition boundary, or if a changed row is accepted.

Measurement receipt: sha256: f8db2327ec43c473372c9e66a203203cd8ff6f5a947c296831cf4edcaa65b056.

Isolation certificate: sha256: e072269eb5feb103ca69602e8e8e308455df8da702c11bac7443f450ba09108b.

Custody certificate: sha256: 33a57286e90a272fa59bf94398ee21d19dcfc9746b796f1092c46ef323684c66.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/S05923>
(sha256:e976fc5090d3344af00af1a64fea3bc564f126f53d920a0387c1fbf56ba07bf6)

Registered target identities:

- reaction-thermochemistry-iupac-composite from
IUPAC-GOLD-BOOK-REACTION-THERMOCHEMISTRY-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no thermochemical table, spontaneity rule, phase diagram, solubility value, photochemical database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no energy, matter or photon carrier may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite closed chemical reaction with complete source-bound energy carriers at reactant and product boundaries, held transfer orientation, phase, conditions and reference state.

Immutable identities. Source manifest

sha256: 6d1fc40d1dc97bbf9de5b3e33865126fc1b421aa5c16953e04e75a0c45d3725a; derivation seal

sha256: eea73da477e06383a04703feb7b3ccdc90dcde9221e85b4e27bdd31a1e40122; engine receipt

sha256: ceebcccc68fd6438210ac1ee70f21b40819f37b683772597f563902cd17a6c44 at

receipts/engine/model_admitted/SFT-CHEM-THERMO-REACTION-001-ceedcccc68fd6438.json;

empirical hash sha256: 4eac278377c2f93b0dc6b3bb5d3161e4609816bae2290d38fb15e2dc322c616d;

measurement receipt sha256 : f8db2327ec43c473372c9e66a203203cd8ff6f5a947c296831cf4edcaa65b056.

56. Reaction direction under complete energy and distinction accounting

Claim identity: SFT-CHEM-THERMO-DIRECTION-001

Question and theorem. Reaction direction is the held orientation selected by strict complete support order between forward and reverse paths; equality yields no selected direction.

forward-and-reverse-paths__complete-energy-transfer-ledgers__complete-accessible-distinction-census__strict-support-order__held-favored-direction

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001

- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-RXN-INTERMEDIATE-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-KIN-ORDER-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001

Generated grammar. Generate the literal product of path, energy, distinction, comparison, orientation, equilibrium, record and extension choices.

Boundary: Every finite pair of forward and reverse reaction paths on one retained condition boundary, compared by complete source, energy-transfer and distinction-production support, selecting an orientation only when one path strictly closes the available transition support.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `forward-and-reverse-paths__complete-energy-transfer-ledgers__complete-accessible-distinction-census__strict-support-order__held-favored-direction__empty-direction-at-equality__path-energy-distinction-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
paths	forward-and-reverse-paths	forward-path-only	No directional comparison exists.
energy	complete-energy-transfer-ledgers	energy-change-erased	Available support is unknown.
distinctions	complete-accessible-distinction-census	entropy-label-only	A label cannot count support.
comparison	strict-support-order	imported-spontaneity-sign	A sign assumes the answer.
orientation	held-favored-direction	negative-Gibbs-primitive	A negative scalar is prohibited.
equilibrium	empty-direction-at-equality	forced-direction-at-tie	Equal support cannot select.
record	path-energy-distinction-trace	direction-answer-only	It cannot reproduce comparison.
extra	no-extra-rule	free-temperature-threshold	A parameter can force a target.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:daf33949c5cd054da35a9146169943e1dda63e5891e02fb4492534265b797eb8`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:bf934a0d381cf363f0d6e66be8f507d530b60942e2322d5230310374c465f779`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b87f78bc4cf618f78510d9a021bb6ce4de8289791c3af93cd66300979fbf8e65`.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:5343842832622e21645afc61d2c5a53702e7ee8f0ebee61619e03653bbb8d1a.

Independent reconstruction. Implementation hash

sha256:6576a2864e07da87d8d37adce128d68cdf2776c2fcfcc90f2b7e5deef9925c4e; independent certificate sha256:2189dca95b759411e57de2a862e881e2ded04255a51f0b74b44c3d7eb15631d0; external-validation hash sha256:8824b92dcfa50437fa7566aa2686ffd005649b386f7f6859dbda662624d5b60b. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-REACTION-DIRECTION-COMPOSITE-2026

Comparison records:

- reaction-direction-iupac-composite: predicted
forward-or-reverse-spontaneous-tendency__Gibbs-energy-condition__equilibrium-at-no-driving-force__direction-state-bounded;
source-derived forward-or-reverse-spontaneous-tendency__Gibbs-energy-condition__equilibrium-at-no-driving-force__direction-state-bounded; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks condition-bound forward/reverse tendency and equilibrium at no driving force, or if a changed row is accepted.

Measurement receipt: sha256:3abf199635514001604ff56ad1ef22dd1cd95f6e9158eaac7b68bf75a969abf7.
Isolation certificate: sha256: eaa33e005c03a61c1152be065773603398293927731a333d148a8571743b11e5.
Custody certificate: sha256: ea94faf446bed772393ea449aa9afa560a881bc04204215f8447e4bdab520052.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/A00178>
(sha256:29e64f1b53aba905340496145e5393f853b118f32684689de8e79c573e2642ef)

Registered target identities:

- reaction-direction-iupac-composite from IUPAC-GOLD-BOOK-REACTION-DIRECTION-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no thermochemical table, spontaneity rule, phase diagram, solubility value, photochemical database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no energy, matter or photon carrier may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite pair of forward and reverse reaction paths on one retained condition boundary, compared by complete source, energy-transfer and distinction-production support, selecting an orientation only when one path strictly closes the available transition support.

Immutable identities. Source manifest

sha256:b3816276c7b5c10b21dc5055166c5939af2bf8a859a046a44f158601cda95e1f; derivation seal sha256:9964663e20092e692ae09feceb4ef534c7bcb24ef6d081cc23b4582ab5d6ddb1; engine receipt sha256:78ce22d5ad1ff53bc945906184614c11901a1504bedb5950d5b7bf0ebad5e09a at

receipts/engine/model_admitted/SFT-CHEM-THERMO-DIRECTION-001-78ce22d5ad1ff53b.json;
 empirical hash sha256:b8fdd451da95737c6ddbfc240e46cb6ed43c01c0fd18ee7f528a3b774700065d;
 measurement receipt sha256:3abf199635514001604ff56ad1ef22dd1cd95f6e9158eaac7b68bf75a969abf7.

57. Chemical phase and coexistence identity

Claim identity: SFT-CHEM-PHASE-CHEMICAL-001

Question and theorem. A chemical phase is a maximal connected region of complete intensive-trace equivalence; coexistence retains two phase identities and their recurrent interface under fixed conditions.

maximal-connected-region__complete-intensive-trace-equivalence__extent-independent-class__retained-phase-interface__paired-interface-recurrence

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001

- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-RXN-INTERMEDIATE-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-KIN-ORDER-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001

Generated grammar. Generate the literal product of region, homogeneity, extent, boundary, coexistence, condition, record and extension choices.

Boundary: Every finite maximal connected chemical region whose complete intensive observation trace is homogeneous under the declared resolution, separated from a distinct region by a retained interface; coexistence is paired recurrence at that interface.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `maximal-connected-region__complete-intensive-trace-equivalence__extent-independent-class__retained-phase-interface__paired-interface-recurrence__temperature-pressure-composition-held__region-trace-interface-record__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
region	maximal-connected-region	arbitrary-sample-label	It does not define phase identity.
homogeneity	complete-intensive-trace-equivalence	single-point-observation	It cannot establish uniformity.
extent	extent-independent-class	size-dependent-phase	Phase identity is intensive.
boundary	retained-phase-interface	interface-erased	Distinct phases cannot be separated.
coexistence	paired-interface-recurrence	one-phase-only	No coexistence relation exists.
conditions	temperature-pressure-composition-held	condition-free-phase	Phase depends on source conditions.
record	region-trace-interface-record	phase-name-only	It cannot reproduce homogeneity.
extra	no-extra-rule	free-phase-diagram	An imported diagram selects the class.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256: eaad5bde6903bc5cb65e88d04f3ee3e8bd4a9421c6af7f9751047698beeb46b0`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256: 71c4e3eb21b71d48f16e94127cbf7c5b2bee89e466b76e2db5d2d8ee43e98db1`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:bf9dcdbde12718e36cc70d466631dfe841fa2855c168100de4cc9fc351f596967.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:ae5241a8b94116a4a4db862b70868b989231718eb7d846d5376fdb5cacee158c.

Independent reconstruction. Implementation hash

sha256:2b5773927b802554089e42a4a081ec5460410a5064e57aece41893d4a9408eab; independent certificate sha256:6c954b256c8c36582c6097aacb9c944371cfd17d01731f9fc6a336517e6cef23; external-validation hash sha256:3a55dc04800159d3c7c38981f1cda70677ccac200d0f9525f709c205c9b776d8. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CHEMICAL-PHASE-COMPOSITE-2026

Comparison records:

- chemical-phase-iupac-composite: predicted
homogeneous-material-region__uniform-intensive-properties__phase-boundary-retained__coexistence-condition-bounded;
source-derived
homogeneous-material-region__uniform-intensive-properties__phase-boundary-retained__coexistence-condition-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks homogeneous intensive properties, a phase boundary or condition-bound coexistence, or if a changed row is accepted.

Measurement receipt: sha256:a44611b1df8f365f5a2703d9b14484827969e22d43980f84119efa120e65de01.

Isolation certificate: sha256:49e37ed5ad397281704c8b3200dbdcb0cd31425ddc55b95d087efc2875dbd3b8.

Custody certificate: sha256:688518ffa1fb57594525c0b4b310c2db1b347630d869d0915a70a03246f92602.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/P04528>
(sha256:c903c27903b67f47a61403a4de408c23bb21c50100ee37e5d05e9a8aa168c4eb)

Registered target identities:

- chemical-phase-iupac-composite from IUPAC-GOLD-BOOK-CHEMICAL-PHASE-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no thermochemical table, spontaneity rule, phase diagram, solubility value, photochemical database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no energy, matter or photon carrier may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite maximal connected chemical region whose complete intensive observation trace is homogeneous under the declared resolution, separated from a distinct region by a retained interface; coexistence is paired recurrence at that interface.

Immutable identities. Source manifest

sha256:8ba6ae7ef167d5493c8caadb8973c3fc875766f9f0b9bac2d275755d13b8d215; derivation seal sha256:46f9d11a8a2bf7efc58734c22adb215013bd5788f42af68c386457459dc6a17c; engine receipt sha256:80720a90afc5c84c8c5aa4a6524da264e88628c125b004d9fb370fd81d9baeac at receipts/engine/model_admitted/SFT-CHEM-PHASE-CHEMICAL-001-80720a90afc5c84c.json; empirical hash sha256:39370c174e13e98e01f043f988967f21183e2f6cec27029f09c71eaec1a2cca9; measurement receipt sha256:a44611b1df8f365f5a2703d9b14484827969e22d43980f84119efa120e65de01.

58. Solution and solubility equilibrium

Claim identity: SFT-CHEM-SOLUTION-EQUILIBRIUM-001

Question and theorem. Solubility equilibrium is paired dissolution and return recurrence on a retained solution boundary, preserving a condition-bound macroscopic dissolved-composition class at finite saturation.

retained-solute-solvent-organization__dissolution-and-return-transitions__paired-opposed-event-support__source-bounded-capacity-boundary__macroscopic-dissolved-class-recurs

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001

- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-RXN-INTERMEDIATE-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-KIN-ORDER-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-PHASE-CHEMICAL-001

Generated grammar. Generate the literal product of solution, dissolution, balance, saturation, composition, condition, record and extension choices.

Boundary: Every finite solution boundary containing a retained solvent/solute organization and opposed dissolution/return transitions whose complete event supports pair at saturation while the macroscopic dissolved-composition class recurs.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: retained-solute-solvent-organization__dissolution-and-return-transitions__paired-opposed-event-support__source-bounded-capacity-boundary__macroscopic-dissolved-class-recurs__phase-temperature-pressure-held__solution-event-composition-trace__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
solution	retained-solute-solvent-organization	unidentified-mixture	Solute and solvent roles are lost.
dissolution	dissolution-and-return-transitions	one-way-dissolution-only	No equilibrium can close.
balance	paired-opposed-event-support	unpaired-event-support	Composition would drift.
saturation	source-bounded-capacity-boundary	free-solubility-number	A measured number imports the target.
composition	macroscopic-dissolved-class-recurs	microscopic-state-fixed	Individual events may continue.
conditions	phase-temperature-pressure-held	condition-free-solubility	Solubility changes with conditions.
record	solution-event-composition-trace	solubility-answer-only	It cannot reproduce the dynamics.
extra	no-extra-rule	free-equilibrium-fit	A fit can select any saturation.

Adverse controls. All mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:759effeb3a46056715c37ecb42ca70aab80ba89291af7b19bc00e39f9ff790ad.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:1444804b3af5b2334fab3e15a71486349ff605b48105b8b37a8e3f7916b0cbc.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:0794da732271434dc54f50e6cf7985ead0f54ea16e2550bd49c249b7947af18f.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:941cae2d85c4715f190d7e25bf31cc19ff04e0c64b3c5b8f0ec3d6218ced6651.

Independent reconstruction. Implementation hash

sha256:04f14dc7cddb5249ae6e469a5f9a7d95e1cdd028e8618ba000ecfe236c7bbda1; independent certificate sha256:7016e64075d4ce9a6ea43e2ca21334b72162eac1201bf3ca2634f086cff9ec6d; external-validation hash sha256:37ac7bd4173f9d2242dccbf5f55da1d0992b9076c50253ebbf331710d733f3fc. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-SOLUTION-EQUILIBRIUM-COMPOSITE-2026

Comparison records:

- solution-equilibrium-iupac-composite: predicted dissolution-and-crystallization-continue__rates-balanced-at-saturation__dissolved-composition-constant__solubility-condition-bounded; source-derived dissolution-and-crystallization-continue__rates-balanced-at-saturation__dissolved-composition-constant__solubility-condition-bounded; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks opposed dissolution/crystallization, saturation balance or condition-bounded solubility, or if a changed row is accepted.

Measurement receipt: sha256:c353e0ed8bafb35d69641666fce1fc3487dec610707067ae9c1a5c5dcb3673d0.

Isolation certificate: sha256:fe06c0c4ece7c5429790c9e6d8cc4880568bbbeaf3ca93d8eca92065adc1d054.

Custody certificate: sha256:ea84b7ee26f7ea451964e6f50578e3701624d797bf7dc5287b3007b49c08fe9d.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/14656> (sha256:bd07661ff6120f150e0a53e67a72b1e212293bf96bfad41628c1e8f7ec9ffdcdb)

Registered target identities:

- solution-equilibrium-iupac-composite from IUPAC-GOLD-BOOK-SOLUTION-EQUILIBRIUM-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no thermochemical table, spontaneity rule, phase diagram, solubility value, photochemical database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no energy, matter or photon carrier may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite solution boundary containing a retained solvent/solute organization and opposed dissolution/return transitions whose complete event supports pair at saturation while the macroscopic dissolved-composition class recurs.

Immutable identities. Source manifest

sha256:6119aed2f1d01cbb7bb9467ee7eff806296a2bcd27c70f842b4b76e878ec9d70; derivation seal
 sha256:413c129d443db2c75e3295c1cf4064d2b539b34a9c572953bb16bdfc84bd3f9c; engine receipt
 sha256:7e7c23b23d756d5bb9ec8d828b0920427adec44e83aefc24fcba1335b446d3 at
 receipts/engine/model_admitted/SFT-CHEM-SOLUTION-EQUILIBRIUM-001-7e7c23b23d756d5b.json;
 empirical hash sha256:4dac19d825e967b9b44ae90328fb9f50c0b5728747527f5ef51ef40c1458b87f;
 measurement receipt sha256:c353e0ed8bafb35d69641666fce1fc3487dec610707067ae9c1a5c5dcb3673d0.

59. Photochemical excitation and reaction channel

Claim identity: SFT-CHEM-PHOTOCHEM-001

Question and theorem. A photochemical event absorbs one retained photon carrier into a named molecular carrier, produces distinct excited support and closes through a generated reaction or relaxation channel with complete energy accounting.

one-retained-photon-carrier__carrier-absorbed-once__distinct-excited-state-support__source-bound-reaction-or-relaxation-channel__complete-photon-molecule-energy-ledger

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001

- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-RXN-INTERMEDIATE-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-KIN-ORDER-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001

Generated grammar. Generate the literal product of radiation, absorption, state, channel, conservation, selection, record and extension choices.

Boundary: Every finite chemical path initiated when one retained radiation carrier is absorbed by a named molecular carrier, producing a distinct excited-state support and a source-bound reaction or relaxation channel with complete energy accounting.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `one-retained-photon-carrier__carrier-absorbed-once__distinct-excited-state-support__source-bound-reaction-or-relaxation-channel__complete-photon-molecule-energy-ledger__generated-channel-census__photon-state-channel-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
radiation	one-retained-photon-carrier	anonymous-energy-input	It cannot establish carrier identity.
absorption	carrier-absorbed-once	photon-copied	Copying violates conservation.
state	distinct-excited-state-support	ground-state-unchanged	No excitation occurs.
channel	source-bound-reaction-or-relaxation-channel	unregistered-outcome	No chemical path is retained.
conservation	complete-photon-molecule-energy-ledger	energy-erased	The path cannot close.
selection	generated-channel-census	free-quantum-yield-fit	A fit can force outcomes.
record	photon-state-channel-trace	photochemical-label-only	It cannot reproduce excitation.
extra	no-extra-rule	free-spectral-rule	An imported rule selects a target.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:604c53c9f236ae0f1bbe2349ede60a046b166a3e5310ea1874f6118f092477a4.

- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:db644f9f56463d159d682018a30e2c3e1b261e3080722afb120458ce503a11ee.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:bbd4f835c874e51ff9de61a5de9b060ed22e3cc5400b77f7b6a1e19cda121d35.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:79ae19af5ff3de24c52ed6a09785a6d314a2b60b931535b8373fee3ff3743582.

Independent reconstruction. Implementation hash

sha256:15de61fa6b9a5576d3eef1e3535ff6b4376d8041b93d8757fda8751a81ad1abe; independent certificate sha256:ff70f8241cc8f564e7d87df9ebdc5621ebd377a57b73bda7386a5ffdd1749376; external-validation hash sha256:66bdcd058fe82b537d0ac42c74fa1a6a83cf971cbcefabc3a5dbe6209783984c. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-PHOTOCHEMISTRY-COMPOSITE-2026

Comparison records:

- photochemistry-iupac-composite: predicted
light-absorption-initiates-chemical-change__molecular-excited-state__reaction-or-relaxation-pathways__photon-energy-accounted;
source-derived light-absorption-initiates-chemical-change__molecular-excited-state__reaction-or-relaxation-pathways__photon-energy-accounted; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks light absorption, excited molecular state, reaction/relaxation pathways or photon-energy accounting, or if a changed row is accepted.

Measurement receipt: sha256:2aa621376fec1e3bc37a02ca5535208fdb60ae75e984ec4595754d06752f3f87.

Isolation certificate: sha256:ff87797b1631291791c7460956da60e760ff485757959b7191d7f5804e78f5d2.

Custody certificate: sha256:fec354f2ed73e4497292c5a76ab4cbab777a56f39114fc507cc23e15d7263602.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/P04585>
(sha256:5e052ba4dda35e0152ec2de28c4d90a748933bd99a82481d5b0b31d9f0c0ed3d)

Registered target identities:

- photochemistry-iupac-composite from IUPAC-GOLD-BOOK-PHOTOCHEMISTRY-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no thermochemical table, spontaneity rule, phase diagram, solubility value, photochemical database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no energy, matter or photon carrier may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed

- Every finite chemical path initiated when one retained radiation carrier is absorbed by a named molecular carrier, producing a distinct excited-state support and a source-bound reaction or relaxation channel with complete energy accounting.

Immutable identities. Source manifest

sha256:8a1043ca6825d9b7f456c7c415407c6f8495a402b6239a630013102c1996e06d; derivation seal
 sha256:1a32f871fde4c9c18af7afb465ba5e224f2d38d40ec35d985bb67c4a8a0d52bc; engine receipt
 sha256:0ec198ef64ee4470b66083ece9cce7ef8911bff02b330fd13cbb1a7e9bc6b0e9 at
 receipts/engine/model_admitted/SFT-CHEM-PHOTOCHEM-001-0ec198ef64ee4470.json; empirical hash
 sha256:fa5b244a681c11c0a150321cd9291ec1f4354eb03c4f1644ee0a5852a00b85b3; measurement receipt
 sha256:2aa621376fec1e3bc37a02ca5535208fdb60ae75e984ec4595754d06752f3f87.

14. Catalysis, networks, surfaces and interfaces - 8 admitted claims

Catalysts, networks and interfaces preserve alternative paths, returned carriers and explicit boundaries. This entire registered subcategory is included; no proper subset is presented as field completion.

60. Catalyst identity conserved through a reaction cycle

Claim identity: SFT-CHEM-CAT-CATALYST-001

Question and theorem. A catalyst is a retained chemical carrier that opens an alternative closed reaction path and returns with exact identity while the net reaction endpoints are unchanged.

*reaction-cycle-participant__exact-catalyst-identity-retained__alternative-step-path-retained__reaction-rate-increased__
 _net-reaction-unchanged*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001

- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite source-bound reaction cycle whose catalyst entry, step occurrences, return identity, net endpoints and recurrence interval are completely recorded.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `reaction-cycle-participant__exact-catalyst-identity-returned__alternative-step-path-retained__reaction-rate-increased__net-reaction-unchanged__both-directions-share-path__complete-cycle-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
role	reaction-cycle-participant	ordinary-reactant-only	The carrier is consumed.
identity	exact-catalyst-identity-returned	carrier-identity-changed	It is not recovered.
path	alternative-step-path-retained	overall-reaction-only	No catalytic path is distinguished.
rate	reaction-rate-increased	rate-unchanged-by-cycle	No catalytic action occurs.
net	net-reaction-unchanged	catalyst-in-net-products	The carrier was consumed or produced.
direction	both-directions-share-path	one-direction-rule	A reversible cycle is incomplete.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
record	complete-cycle-trace	catalyst-name-only	Identity return cannot be checked.
extra	no-extra-rule	free-catalytic-fit	A fit can select the result.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : a0ae5548ab64a5314989e01db584c23471703055d234de86e8ecf9285c346a9e.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : fc5ab89d3cfab7add282f8d230b65bbc434f7ef691e76e8ed714e1ca4f5f4c71.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 9c78d99f45777e5234b96e68742b6f38d3807dddab51f0149c8aa228825b70df.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 0f4686e4369b29d1c58e4bad23b4d7954f59c0dbfe99122ffaddef061229f026.

Independent reconstruction. Implementation hash

sha256 : 58ec12d7c67ad1fd917c9639d1ff8ced462e348d4c471e56192d4f5b8a7a314a; independent certificate sha256 : f573cafbd6724ed8c57b766c376536ba7562110459be982c9209dae132e385f8; external-validation hash sha256 : 2fa83fab5a2e35d78ded002a011c38a6fc7091bcb0d205d5dc5e11298c2a2c9c. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- CHEM-AUTHORITY-CATALYST-COMPOSITE-2026

Comparison records:

- catalyst-authority-composite: predicted
reaction-rate-increased__alternative-reaction-pathway__catalyst-regenerated__net-reaction-unchanged; source-derived
reaction-rate-increased__alternative-reaction-pathway__catalyst-regenerated__net-reaction-unchanged; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if an official authority does not retain changed rate, an alternative path, regenerated catalyst identity and unchanged net reaction, or if a tampered row is accepted.

Measurement receipt: sha256 : 434e14464ec053c73dd4f4773e7342ad7a5ddb3feaa4442447fab056e70e3c57.

Isolation certificate: sha256 : bf4a5b3532d448f7ed09c595fe4fd71a7e656f5b9d370cc959552cf93b901825.

Custody certificate: sha256 : 7fedb848ee7b9d79a713be5fa41f50e51b8ac0398bb61d0846527a87019809d2.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C00876>
(sha256:a960e78123592da7dd8ad054b65a590961706c25e22a578f9088f91fb3c89e5e)

Registered target identities:

- catalyst-authority-composite from CHEM-AUTHORITY-CATALYST-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate

- no numerical zero, negative, irrational, imaginary or floating point quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite source-bound reaction cycle whose catalyst entry, step occurrences, return identity, net endpoints and recurrence interval are completely recorded.

Immutable identities. Source manifest

sha256:b3bd8ab71faedc5d050d083193458e452c033585250011ba6266c2a7ae1f2d76; derivation seal
 sha256:02b882f2690edd1b84ca21c86f4b58ff3fb20df90af5e05951790bb9ba1cfe8a; engine receipt
 sha256:937aa5f579196ccd541a1fe09dfff3eb5cd095a64ccd378f29c566b97df3793e at
 receipts/engine/model_admitted/SFT-CHEM-CAT-CATALYST-001-937aa5f579196ccd.json; empirical
 hash sha256:6d488e0b27f55e5590338d7a3c80e053a3304bfff7acd1a1e8d06da7e3c3f3f3c; measurement
 receipt sha256:434e14464ec053c73dd4f4773e7342ad7a5ddb3feaa4442447fab056e70e3c57.

61. Catalytic alternative-path relation

Claim identity: SFT-CHEM-CAT-PATHWAY-001

Question and theorem. A catalytic pathway is the condition-bound complete step trace sharing overall reaction endpoints with an uncatalysed path, ordered by activation support and closing the catalyst cycle.

same-overall-reaction__alternative-reaction-mechanism__lower-activation-support__catalyst-cycle-retained__all-intermediates-balanced

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001

- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-CAT-CATALYST-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite pair of balanced reaction mechanisms with common endpoints, complete intermediate and activation records, and one exact catalyst-return cycle.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `same-overall-reaction__alternative-reaction-mechanism__lower-activation-support__catalyst-cycle-retained__all-intermediates-balanced__conditions-held__pathwise-energy-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
endpoints	same-overall-reaction	different-overall-products	The paths are not alternatives.
mechanism	alternative-reaction-mechanism	step-trace-erased	No pathway comparison exists.
barrier	lower-activation-support	imported-energy-number	A target value is not forced.
carrier	catalyst-cycle-retained	catalyst-consumed	The cycle does not close.
conservation	all-intermediates-balanced	free-intermediate	Matter is unaccounted.
conditions	conditions-held	condition-free-path	Path availability is unbounded.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
record	pathwise-energy-trace	endpoint-answer-only	The alternative path is unreproducible.
extra	no-extra-rule	free-mechanism-choice	A choice can force any path.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : d56c7c24dd710906db456ed73dff4648c5cbacb903334859dbc6eaedd87247d5.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : b1a3c0ce3bbe224fe6adfaca32428de47fcd2130bf09026d2d09617fb085506f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 62a01744198cbb8cb2e61f83caf3adc1b591dcfe2bfd037895bec0be9f32c9e.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : a54f92031e4559f61065b6a954f4941b2298198daabf36d8b3930933d0279811.

Independent reconstruction. Implementation hash

sha256 : 4d8c60ce565b3448c751cf5e52ccfefa1c9bcdd202ffd84f9e595d0d55365b4b; independent certificate sha256 : f6bb5877f01fc2c5ebd935d2d55a1fe52134d695628d90f9755638631f692193; external-validation hash sha256 : 991dd146d562180250760d92c5e4fc55d9cc7beb7ca388ab6e593b7b225bcd6f. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- CHEM-AUTHORITY-CATALYTIC-PATHWAY-COMPOSITE-2026

Comparison records:

- catalytic-pathway-authority-composite: predicted
alternative-reaction-mechanism__lower-activation-energy-path__same-overall-reaction__catalyst-cycle-retained; source-derived
alternative-reaction-mechanism__lower-activation-energy-path__same-overall-reaction__catalyst-cycle-retained; exact match
True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks an alternative lower-activation mechanism with unchanged overall reaction and retained catalyst cycle, or if a tampered row is accepted.

Measurement receipt: sha256 : 3785ad258985abdbb75884b68ab547efd24b02fa673bf22abddc77f1ef6652a1.

Isolation certificate: sha256 : 5fa7659d3d22d5e4e44730850eb15b4b04cd7b6183529e9a230b0fcb9be28871.

Custody certificate: sha256 : ff250f5e427a48dcfb8213e27e26a5e91c527ac5b7f5dfb216ca50e21be7ea79.

Registered source descriptions:

- International Union of Pure and Applied Chemistry and Royal Society of Chemistry:
<https://goldbook.iupac.org/terms/view/C00876>
(sha256:a960e78123592da7dd8ad054b65a590961706c25e22a578f9088f91fb3c89e5e)

Registered target identities:

- catalytic-pathway-authority-composite from CHEM-AUTHORITY-CATALYTIC-PATHWAY-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite pair of balanced reaction mechanisms with common endpoints, complete intermediate and activation records, and one exact catalyst-return cycle.

Immutable identities. Source manifest

sha256:d09c3c6348f988522f161c19e40cfe4f645dd492aae15344e73e637e25c0fade; derivation seal
 sha256:f5e48c1d871608b5d28690e5380958bfac51ca1927b2c65f4c8fd7ed1d32f041; engine receipt
 sha256:088dfaddc5b0a7b94907fd8bd92a967db8a27ad7c97fef56358dede752f45207 at
 receipts/engine/model_admitted/SFT-CHEM-CAT-PATHWAY-001-088dfaddc5b0a7b9.json; empirical
 hash sha256:a9b16c7347f8b2d13184fea569e6964459440b33dd8ebe64d2a62129b58dec1e; measurement
 receipt sha256:3785ad258985abdbb75884b68ab547efd24b02fa673bf22abddc77f1ef6652a1.

62. Catalytic selectivity among generated products

Claim identity: SFT-CHEM-CAT-SELECTIVITY-001

Question and theorem. Catalytic selectivity is the exact maximal product-recurrence class over all generated competing products at one reactant, catalyst, condition and observation boundary; ties remain a cofavored class.

all-accessible-products-generated__complete-product-recurrence-order__one-product-favored__cofavored-class-at-tie__reactant-consumption-held

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001

- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-CAT-CATALYST-001
- SFT-CHEM-CAT-PATHWAY-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite complete product census for one retained catalytic input boundary with exact positive recurrence counts and all ties retained.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `all-accessible-products-generated__complete-product-recurrence-order__one-product-favored__cofavored-class-at-tie__reactant-consumption-held__condition-and-method-bounded__complete-competing-product-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
products	all-accessible-products-generated	one-product-presupposed	Alternatives were not generated.
comparison	complete-product-recurrence-order	yield-number-imported	A measurement selects the answer.
selection	one-product-favored	all-products-equated	No preference is represented.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
ties	cofavored-class-at-tie	arbitrary-tie-break	Equal support cannot force identity.
reactant	reactant-consumption-held	consumption-erased	Selectivity lacks a denominator.
conditions	condition-and-method-bounded	universal-preference	Selectivity can change with context.
record	complete-competing-product-trace	selected-name-only	Competing products are hidden.
extra	no-extra-rule	free-selectivity-fit	A fit can choose any product.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:5a1a0226f6360ac2b96c67319fff4be09265d2181106cd3396fc7cd908d5542e.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:7a4e988ef2cfc8faa54e8492f557dc4e4d2b096938a9759bc4e90657f1ac86d0.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:694038c24110a7ede1b6ebe7a63ee0b88c201fe288c773816c65416663b5d448.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:2788954e8e9dccbff66dbe5c81d681fc83223ad6cd276ebe12ff20dc1c1e337.

Independent reconstruction. Implementation hash

sha256:754cf0c2b2c7594c8784cc47def21254bbe89b6db3682727307ffad8ada6be12; independent certificate sha256:c45ba906fb4d3c4aa0f4d308389ebb355098d0766df03a451d237d5d88866196; external-validation hash sha256:c93c4d37adbf1a2282f3a0235d82d8d36f73f44a201f3d24b8ab0d37871acd33. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CATALYTIC-SELECTIVITY-COMPOSITE-2026

Comparison records:

- catalytic-selectivity-iupac-composite: predicted one-product-favored-among-alternatives__specified-reactant-consumption__condition-and-method-bounded__competing-products-retained; source-derived one-product-favored-among-alternatives__specified-reactant-consumption__condition-and-method-bounded__competing-products-retained; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks product preference relative to alternatives, specified reactant consumption, a bounded method/condition or retained competitors, or if a tampered row is accepted.

Measurement receipt: sha256:656aad51fd3dc0347d99090d5f5c20c90ec57c3ed0e94c8f0a51a008b5376963.

Isolation certificate: sha256:f364b83a946eea1188b14800973c883657572a958329c9b4693859e2aa1ed74a.

Custody certificate: sha256:4f9978a765a4eed8bc273751de7b21fe9b118c5d506ae53084b53f64787f7562.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/S05563>
(sha256:72489fb0beb2453befdf2628aa48d7fe5a18f8a8ce4e09573c09b3bb464d7506)

Registered target identities:

- catalytic-selectivity-iupac-composite from IUPAC-GOLD-BOOK-CATALYTIC-SELECTIVITY-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite complete product census for one retained catalytic input boundary with exact positive recurrence counts and all ties retained.

Immutable identities. Source manifest

sha256:13606327bf7b4bc2d42179302ea7a202c0148cf3a0420d62be33644bb2a53732; derivation seal
 sha256:a9e6974897a6c1d74f9d9235346c286bd48d093a6f263ff171c85eb2b0754053; engine receipt
 sha256:0210a8962416f13c5e82b3c11bd91c562fc98a8ad7f0e8ecf7fd8cbd87edf3cf at
 receipts/engine/model_admitted/SFT-CHEM-CAT-SELECTIVITY-001-0210a8962416f13c.json;
 empirical hash sha256:bac8eb3c34b866bbb99cca1743281048c7d7e6724fe3d7d2a2663b17629c50c3;
 measurement receipt sha256:656aad51fd3dc0347d99090d5f5c20c90ec57c3ed0e94c8f0a51a008b5376963.

63. Chemical reaction-network composition

Claim identity: SFT-CHEM-NET-REACTION-001

Question and theorem. A chemical reaction network is the complete finite directed incidence organization generated by balanced reaction steps sharing retained chemical-species identities, including branches, merges, cycles and external boundaries.

chemical-species-nodes__reaction-steps-connect-species__shared-species-compose-paths__source-product-boundary-held__all-generated-paths-retained

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001

- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-CAT-PATHWAY-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite set of retained chemical-species carriers and balanced reaction-step incidences with complete external source/product boundaries.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `chemical-species-nodes__reaction-steps-connect-species__shared-species-compose-paths__source-product-boundary-held__all-generated-paths-retained__reaction-cycles-retained__complete-node-edge-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
species	chemical-species-nodes	anonymous-state-count	Chemical identities are lost.
reactions	reaction-steps-connect-species	unlabelled-adjacency	Transitions are chemically ambiguous.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
composition	shared-species-com pose-paths	isolated-reaction-list	Shared carriers are not composed.
boundary	source-product-bou ndary-held	open-unrecorded-flux	The network cannot balance.
branching	all-generated-path s-retained	single-path-only	Network alternatives are erased.
cycles	reaction-cycles-re tained	cycle-collapsed	Recurrence is lost.
record	complete-node-edge -trace	network-picture-only	The network cannot be replayed.
extra	no-extra-rule	free-network-topology	An imported graph selects the result.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:c57d530691d21b8498f915c7441c1c1c725a6c7fe1e30d386eb9c97135bb20d9.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:204274c29c30cf943174df0510debdc011859a5541c2f550be0a193709e490e2.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:9b7d1d4cc915cbac4b24e9145301723b46be11253e0968d7d57b88349708238e.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:5be5d2c477de356160384c12d2baf67ce3b355ccce7adb22dec3b542764681.

Independent reconstruction. Implementation hash

sha256:ffd05d6f83262ada8319d0cb30a8e8cdc9427acf976ee02b05d8020bb1e2584d; independent certificate
sha256:f848abc22cd52e77c1a07e21dc4db6dc7f9fa6577f918a27f4c4c7b1dbb82768; external-validation
hash sha256:df3acf244d844dbb22a99cf0230126ccdaa4f87e6be1c428fabb6356cb2f359a. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-REACTION-NETWORK-COMPOSITE-2026

Comparison records:

- reaction-network-iupac-composite: predicted
chemical-species-nodes__reaction-steps-connect-species__combined-network-paths__complete-source-product-boundary;
source-derived
chemical-species-nodes__reaction-steps-connect-species__combined-network-paths__complete-source-product-boundary; exact
match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks species, reaction connections, composed network paths or a complete boundary, or if a tampered row is accepted.

Measurement receipt: sha256:851d9975a3e1af769a9a6715b8804c181c22b0faff0228fad7fe775a34919838.

Isolation certificate: sha256:6634bea0480a1517d31db3afea552a02e8af83d78387e360b6e2aeec3c0a996b.

Custody certificate: sha256:6ae90eb78dc0b8cbb2d4e80982579d64a70b964f3817a9ff14c85da5acf4f57e.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01210>
(sha256:f7949843026a8fe090e71f1d952633b8c63eb8277ad6b067b4bcfa5babf3598d)

Registered target identities:

- reaction-network-iupac-composite from IUPAC-GOLD-BOOK-REACTION-NETWORK-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite set of retained chemical-species carriers and balanced reaction-step incidences with complete external source/product boundaries.

Immutable identities. Source manifest

sha256:d4655a22fad5c8bdf4832879ae94222bac941691dd53c7f3027abb070764f0f1; derivation seal
 sha256:238a79de26238ba854fd317ac120e53d4fe4aa4260b7518b9d47295ea01514c0; engine receipt
 sha256:e6af94d430f01d2e10ec7053d39b13705ed69dfb6510c8c0c4504d97721bbe2a at
 receipts/engine/model_admitted/SFT-CHEM-NET-REACTION-001-e6af94d430f01d2e.json; empirical
 hash sha256:220257d8131e33578dc66a242d1931359b4438b6e1653ab83958cd196d38e58a; measurement
 receipt sha256:851d9975a3e1af769a9a6715b8804c181c22b0faff0228fad7fe775a34919838.

64. Autocatalytic closure and self-amplifying path

Claim identity: SFT-CHEM-NET-AUTOCATALYSIS-001

Question and theorem. Autocatalysis is a finite resource-bounded reaction-network cycle in which a retained product carrier re-enters as catalyst for another production cycle of the same identity.

reaction-product-is-catalyst__product-accelerates-own-formation__same-identity-feedback__resource-consuming-amplification__positive-feedback-cycle

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001

- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-CAT-CATALYST-001
- SFT-CHEM-NET-REACTION-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite reaction network whose product/catalyst identity equality, feedback edge, consumed inputs and stopping boundary are completely recorded.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: reaction-product-is-catalyst__product-accelerates-own-formation__same-identity-feedback__resource-consuming-amplification__positive-feedback-cycle__finite-resource-bounded__complete-feedback-trace__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
product	reaction-product-is-catalyst	external-catalyst-only	Self-production is absent.
feedback	product-accelerates-own-formation	product-inert	Output cannot affect its formation.
identity	same-identity-feedback	product-role-erased	The feedback carrier is unknown.
growth	resource-consuming-amplification	unbounded-free-copy	Creation violates closure.
cycle	positive-feedback-cycle	linear-step-only	No feedback path closes.
limit	finite-resource-bounded	infinite-support-assumed	Resources are not counted.
record	complete-feedback-trace	growth-label-only	Carrier balance cannot be checked.
extra	no-extra-rule	free-growth-constant	A fitted rate can force behavior.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:0cee95067a1a5431ddc8780c8864ae551f96a26bb2c001743642747736cac51d.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:2b89cecb7863ccd95d6a1d95bf55533e2c1ad530a14eed634a88ca8534be5bac.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b95e37ad7b16c81b5d7997fd345b2151209a5f626569efa1032891f153b64d4f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:6f3686eed26f288c5bc85889d39921ff67795676ea2ba9075b06efdc5e039bfe.

Independent reconstruction. Implementation hash

sha256:fdd0d83368dd716f7302b78183a3d93e151ae640703569d3fc20ae81a2299ebc; independent certificate sha256:dc9fca524512a204e305f1f2262f8ffc25587cb8c349b09c19b137519e8a8b88; external-validation hash sha256:ccd5e5f1fa4997f35bb9d3d7cd2c56c2b39b6614d644dc83210cb641d1805a68. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-AUTOCATALYSIS-COMPOSITE-2026

Comparison records:

- autocatalysis-iupac-composite: predicted
reaction-product-is-catalyst__product-accelerates-own-formation__positive-feedback-cycle__finite-resource-bounded;
source-derived
reaction-product-is-catalyst__product-accelerates-own-formation__positive-feedback-cycle__finite-resource-bounded; exact
match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks a product acting catalytically in its own formation, feedback closure or the declared finite boundary, or if a tampered row is accepted.

Measurement receipt: sha256:2736f6aa7b994a0b108f31feeb7d9dba728970ce9f5fa6590a0f80279e794d34.

Isolation certificate: sha256:e50d57345851402d7866deb1bee0f1584639e838e9032c3a8fd1ed94f20568f9.

Custody certificate: sha256:c0e34bd0a1f8453e3d02c73cd210c3e6aeecd9eb54f300df1d14dc20847cea53.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/A00525> (sha256:cd6e2b68b8eefaa71fd7212cb577bd9996810a0d64771508748f7d3599469b06)

Registered target identities:

- autocatalysis-iupac-composite from IUPAC-GOLD-BOOK-AUTOCATALYSIS-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite reaction network whose product/catalyst identity equality, feedback edge, consumed inputs and stopping boundary are completely recorded.

Immutable identities. Source manifest

sha256:6d780bf7aa646faf5d18b1e2d6ee19a3412397b6c43d7a3cfc19e9b078d66496; derivation seal sha256:401c0cf9e3ac0f9051edb936434ee7394e502cd06c070596dfc8a244fa41bd2c; engine receipt sha256:8005e646115c8bab3c72478669c99e70f85419d388f4344bbc0e059eb8d175f2 at receipts/engine/model_admitted/SFT-CHEM-NET-AUTOCATALYSIS-001-8005e646115c8bab.json; empirical hash sha256:d5df643c7b1587c648bc3b5e5ec1cc18c6b7f5c26ecac1ed7d28724b3262ca68; measurement receipt sha256:2736f6aa7b994a0b108f31feeb7d9dba728970ce9f5fa6590a0f80279e794d34.

65. Adsorption at a retained interface

Claim identity: SFT-CHEM-SURFACE-ADSORPTION-001

Question and theorem. Adsorption is reversible localization of retained adsorbate carriers at a retained adsorbent interface, producing surface support distinct from bulk support under declared conditions.

retained-interface__adsorbate-and-adsorbent-retained__accumulation-at-interface__surface-concentration-differs__reversible-desorption-boundary

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite two-region chemical boundary with named adsorbate and adsorbent, counted arrivals/residence/departures and source-bound conditions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `retained-interface__adsorbate-and-adsorbent-retained__accumulation-at-interface__surface-concentration-differs__reversible-desorption-boundary__condition-and-surface-bounded__complete-interface-event-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
boundary	retained-interface	bulk-only	No interface is present.
roles	adsorbate-and-adsorbent-retained	carrier-identity-erased	Surface membership is ambiguous.
location	accumulation-at-interface	uniform-bulk-distribution	No surface accumulation occurs.
composition	surface-concentration-differs	surface-equals-bulk	The interface adds no distinction.
binding	reversible-desorption-boundary	irreversible-erasure	The adsorbate cannot be recovered.
conditions	condition-and-surface-bounded	condition-free-coverage	Surface support is context-bound.
record	complete-interface-event-trace	adsorption-name-only	Localization cannot be reproduced.
extra	no-extra-rule	free-isotherm-fit	A fit can select coverage.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:1222829e939050aadd0f8822802fd259e38a34efc4956031923d6901d9376f77.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:5e79aced536f06e06417b9bd97561298a94bae3a6ed69a8edfc2238aa255bd21.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:4208793f4c9b2043c17f721fc4db84dacd13580da5eb86b7480bbabfd24e615b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:59eae4ccdfd45006bede3138562ce34a170fd3ccea0c58c81440c9b92f26b70.

Independent reconstruction. Implementation hash

sha256:594768dbdeffe5e68cea02238fda64b8a1d762d3f6bf2af4706a08291bc6db7c; independent certificate sha256:083528ba959417a9c14e7a719aef5b1cc340b3bfbc296a7ef03ad2d78b7542f7; external-validation hash sha256:897d6e40788b121cbbf7cd6d591ef18248220fb5dd26382e130ce45bac9e5f8e. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ADSORPTION-COMPOSITE-2026

Comparison records:

- adsorption-iupac-composite: predicted accumulation-at-interface__adsorbate-and-adsorbent-retained__surface-concentration-differs-from-bulk__reversible-desorption-boundary; source-derived accumulation-at-interface__adsorbate-and-adsorbent-retained__surface-concentration-differs-from-bulk__reversible-desorption-boundary; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks interfacial accumulation, retained adsorbate/adsorbent roles, surface/bulk distinction or a desorption boundary, or if a tampered row is accepted.

Measurement receipt: sha256:046fa4e8964ffc029b96e04caa15111bf84afa346ffd5ba3e0d328eb8c6e58c3.

Isolation certificate: sha256:ff91e5cfe59e9a4f4593a364eb6931c89e5d28dcc63f7b15a1e6b4f15a8738ab.

Custody certificate: sha256:b8ba897f7d7acd2a538133120723ff158c7580354e240ee63c9d84ffe1b738a5.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/A00155> (sha256:8d8d32d4fad6f36f8be4aeed8056bac1efb3ef5f9d03c3ec26fddaade7cce582)

Registered target identities:

- adsorption-iupac-composite from IUPAC-GOLD-BOOK-ADSORPTION-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite two-region chemical boundary with named adsorbate and adsorbent, counted arrivals/residence/departures and source-bound conditions.

Immutable identities. Source manifest

sha256:e275ba820615ea693eddf065f12e1c7e9644427e2f7c23681b55cc0fb33b55d1; derivation seal sha256:e8c64e7beb58a1be407f04d7e5435a4002eb41fc1a3558c7509462d0e1270e87; engine receipt sha256:5d30fb2ced993666788efb16c24431b275bb5847ffdfaf187282ab4b12f97249 at receipts/engine/model_admitted/SFT-CHEM-SURFACE-ADSORPTION-001-5d30fb2ced993666.json; empirical hash sha256:7e8b153394bbfddc3019a0b03144ba7eb4de333e0aa75b4c90fdd018983dbe8a4; measurement receipt sha256:046fa4e8964fffc029b96e04caa15111bf84afa346ffd5ba3e0d328eb8c6e58c3.

66. Colloidal dispersion and phase-interface support

Claim identity: SFT-CHEM-COLLOID-DISPERSION-001

Question and theorem. A colloidal dispersion is a source-bounded population of finite dispersed-phase regions distributed through a retained continuous phase, each with a retained interface and observation-scale boundary.

dispersed-and-continuous-phases__finite-dispersed-particles__phase-interface-retained__particles-distributed-through-medium__particle-size-source-bounded

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001

- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite population of separated same-phase regions within a distinct continuous phase, with complete interfaces, sizes, positions, conditions and observation interval.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `dispersed-and-continuous-phases__finite-dispersed-particles__phase-interface-retained__particles-distributed-through-medium__particle-size-source-bounded__dispersion-condition-bounded__particle-interface-distribution-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
phases	dispersed-and-cont inuous-phases	single-uniform-phase	No dispersion exists.
extent	finite-dispersed-p articles	molecular-solution-only	No mesoscopic carrier is retained.
interface	phase-interface-re tained	phase-boundary-erased	Particle identity is lost.
distribution	particles-distribu ted-through-medium	one-settled-region	Dispersion is absent.
scale	particle-size-sour ce-bounded	size-free-label	Colloid identity is unbounded.
stability	dispersion-conditi on-bounded	eternal-stability-assumed	Persistence is condition-dependent.
record	particle-interface -distribution-trac e	colloid-name-only	Phase support cannot be checked.
extra	no-extra-rule	free-dispersion-fit	A fit can force stability.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : edd8f0455d2f299a3ad9630f6622b49d804706b79d355ae6319912a942000e0c.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 723ba74dce8f73823e5842be52b5b1dd8ecf3cde9c4f11da3b779a472b975234.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 083d54c83dfc2538aee457363b236556741bb91855fe1a35c5554550b3e837b5.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : f01f5eab860c62ac8679d4555e9eb9ce1c8f72eb35560ce05eff3a3858e0ae2a.

Independent reconstruction. Implementation hash

sha256 : 1733d436397c2812555cf24e9c2225cf28f9d2e77c81560dbd60b58f74ae99aa; independent certificate
sha256 : 455bdd9c0d4dc97eb1652ea2880c1a40fd3a2f25097e9e24371f13a601ed227a; external-validation
hash sha256 : e5f2eeb23c4d43bf72b40622c6f6ecfc693352190c993d2ce213ca3c76c466d4. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-COLLOIDAL-DISPERSION-COMPOSITE-2026

Comparison records:

- colloidal-dispersion-iupac-composite: predicted microscopic-phase-dispersed-in-continuous-phase__phase-interface-retained__p
article-size-boundary__dispersion-condition-bounded; source-derived microscopic-phase-dispersed-in-continuous-phase__phase
-interface-retained__particle-size-boundary__dispersion-condition-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks dispersed/continuous phases, retained interfaces, a particle-size boundary or condition-bounded dispersion, or if a tampered row is accepted.

Measurement receipt: sha256 : 756f04cc2dc5911953f4944a57d6cc5ca311e30a032189d21f03ca0da204aa0f.

Isolation certificate: sha256 : 2756f3764cebd728dff5b52f8b6da6d433261ee34a4d46bdecc73ca03ce90fae.

Custody certificate: sha256 : 52aec327fc4dee9c5644ab9dd553a4fe6343f81da4b5e7d251a846e96aa5d773.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01174> (sha256:a8261e1400d58c847f37722d7c1cb39f782a97844d847648156bf01dfad7a636)

Registered target identities:

- colloidal-dispersion-iupac-composite from IUPAC-GOLD-BOOK-COLLOIDAL-DISPERSION-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite population of separated same-phase regions within a distinct continuous phase, with complete interfaces, sizes, positions, conditions and observation interval.

Immutable identities. Source manifest

sha256:84f3f220ee578c83a09c6dec737ba0e22b9c824563af556b3ab9635ed2959bb5; derivation seal sha256:36d55422010e0cad0f8c4a9c43d686de7a284230748102246b68e1bd42b2e22b; engine receipt sha256:1de99723c818a575de1f05a5c8a230b16ebdc834102d39f9600fe6b89b0e5141 at receipts/engine/model_admitted/SFT-CHEM-COLL0ID-DISPERSION-001-1de99723c818a575.json; empirical hash sha256:7563aa767bcb90c2549ca96055c32c76715cf89d219acb97b8baaf92873114a8; measurement receipt sha256:756f04cc2dc5911953f4944a57d6cc5ca311e30a032189d21f03ca0da204aa0f.

67. Chemical transfer across an interface

Claim identity: SFT-CHEM-INTERFACE-TRANSFER-001

Question and theorem. Interfacial chemical transfer is the counted, oriented crossing of a retained chemical-species carrier from a named donor phase to a distinct receiver phase, pairing departure and arrival under declared conditions.

two-phase-boundary__chemical-species-retained__held-donor-to-receiver-orientation__mass-balance-preserved__boundary-crossing-transition

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-SURFACE-ADSORPTION-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite two-phase boundary with named donor, receiver and chemical carrier, paired crossing events, complete balance and source-bound rate conditions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: two-phase-boundary__chemical-species-retained__held-donor-to-receiver-orientation__mass-balance-preserved__boundary-crossing-transition__transfer-rate-condition-bounded__complete-donor-interface-receiver-trace__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
interface	two-phase-boundary	one-phase-only	No crossing boundary exists.
carrier	chemical-species-retained	anonymous-flux	Chemical identity is lost.
orientation	held-donor-to-receiver-orientation	unsigned-transfer-number	Direction is ambiguous.
balance	mass-balance-preserved	carrier-created-at-boundary	Conservation fails.
process	boundary-crossing-transition	instantaneous-relabel	No transfer event is represented.
rate	transfer-rate-condition-bounded	universal-transfer-value	Rate depends on context.
record	complete-donor-interface-receiver-trace	flux-answer-only	The crossing cannot be replayed.
extra	no-extra-rule	free-transfer-coefficient	A fitted coefficient can select any value.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : c930e6944f1a09f686a08d3c96eb6287aba3fb827d5bad341ba8d8f4b7548113.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : f24b244547dcfdb4e01c3b88b91ca4cb460bee0b24c241efe4f14e01f1406e96.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 72fe92a1dbf643723166962831da07bfc71712dda65fa11d6fd07ceb9cc88d78.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 3d25aea62ec6d8302fefb5738b84969989268a5917e5d8a22d080a65215cd0b3.

Independent reconstruction. Implementation hash

sha256 : f9ce306e7d93983fb7c043ef150e796b853c53283ed531a9c34d2332848d44a9; independent certificate sha256 : 85ceb7f7d1bf236cc83718fb44d1f95b727ee51b66f36d86ba5e4608540032f44; external-validation hash sha256 : 5d392dd5b5aebdf41f1e62f3ffff3753680574a964135886aca9580fb8fd3fe67. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-INTERFACE-TRANSFER-COMPOSITE-2026

Comparison records:

- interface-transfer-iupac-composite: predicted chemical-species-crosses-phase-boundary__donor-and-receiver-phases-retained__transfer-rate-condition-bounded__mass-balance-preserved; source-derived chemical-species-crosses-phase-boundary__donor-and-receiver-phases-retained__transfer-rate-condition-bounded__mass-balance-preserved; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks a species crossing, two retained phases, a condition-bounded rate or mass balance, or if a tampered row is accepted.

Measurement receipt: sha256 : 763a188c4228d2a783b959cf6c178551c0a35110311c5d939fb61ebad99d690a.

Isolation certificate: sha256 : 229292cf33dd67add4f32465bb6d011999cbd074a2c91933acf04ddf3111640c.

Custody certificate: sha256:542d8da90558da5042311bf0981de4cb7954793008716b382db1cdb64bbb33f3.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/P04536>
(sha256:b60eba64e32f00b5bcb7ecf818b25abc53208476546555ae05349a52cbd7e7b6)

Registered target identities:

- interface-transfer-iupac-composite from IUPAC-GOLD-BOOK-INTERFACE-TRANSFER-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite two-phase boundary with named donor, receiver and chemical carrier, paired crossing events, complete balance and source-bound rate conditions.

Immutable identities. Source manifest

sha256:08ad2eb0804d0dffa998faae7cc62d75910f0bd667dd5fe77b4222813df5053e; derivation seal
sha256:8ed9d246bcb25f4903f556fcab7bb54e0e6cf40f39f87233d0736661263fa073; engine receipt
sha256:ccc945354866ff49d1cb40dc69622afadce5be4e706bac640477352f6c4292a7 at
receipts/engine/model_admitted/SFT-CHEM-INTERFACE-TRANSFER-001-ccc945354866ff49.json;
empirical hash sha256:2d8cad096834aa47203a20ad369b1310ba68a24206aa7ce1570f7ffd64813713;
measurement receipt sha256:763a188c4228d2a783b959cf6c178551c0a35110311c5d939fb61ebad99d690a.

15. Stereochemistry, organic and polymer foundations - 8 admitted claims

Constitutional, orientation and chain distinctions arise from complete finite equivalence and connected support. This entire registered subcategory is included; no proper subset is presented as field completion.

68. Chirality and non-superposable held orientation

Claim identity: SFT-CHEM-STEREO-CHIRALITY-001

Question and theorem. Chirality is the retained distinction between a complete oriented molecular carrier and its exact reflection when exhaustive identity-preserving proper superposition admits no equivalence.

complete-three-dimensional-object__exact-mirror-image-generated__composition-preserved__adjacency-preserved__no-proper-superposition

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite complete three-dimensional chemical structure with named atoms, adjacency, orientation and exhaustive identity-preserving proper maps.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-three-dimensional-object__exact-mirror-image-generated__composition-preserved__adjacency-preserved__all-proper-finite-relabelings-enumerated__no-proper-superposition__orientation-and-map-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-three-dimensional-object	formula-only	A formula has no spatial orientation.
reflection	exact-mirror-image-generated	mirror-image-not-generated	Chirality cannot be decided.
composition	composition-preserved	mirror-composition-changed	The comparison is not reflective identity.
adjacency	adjacency-preserved	mirror-adjacency-changed	The comparison changes constitution.
superposition	all-proper-finite-relabelings-enumerated	visual-guess	A guess cannot establish equivalence.
result	no-proper-superposition	arbitrary-handedness-name	A name cannot force distinction.
record	orientation-and-map-trace	chirality-label-only	The result cannot be reproduced.
extra	no-extra-rule	free-chirality-rule	An extra rule can classify anything.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : a90fecae5d272bfff9642b6c1cbb2246da4a5e5ea36336d27de3e3f38c61b0842.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 2f4df44ccf7ea37f290000c67426ecfaea88c1d5dc75908b21ddeba196961327.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : e30626a49e357397a5b503d28c26a3b0fb8439bd2b6284caa64566cde654d7d2.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 2dc9b28c06452315912058f81e49812f2abb24e5d4a343fce7464529a67affe1.

Independent reconstruction. Implementation hash

sha256 : c58e29aeec8dd7045a2a6a0a35bac932bb4ebe42211a40aa3378176cb79256d6; independent certificate sha256 : c5dba95a5d23f251698226727ab9cad978861019221b4f9eeeff049ad2237d87; external-validation hash sha256 : f001aa2a71f5d5fc61afaacf3b8a343a5cb2a87129bb792418fb51040626d725. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CHIRALITY-COMPOSITE-2026

Comparison records:

- chirality-iupac-composite: predicted
object-not-superposable-on-mirror-image__chirality-retained__reflection-distinction__observation-boundary-held;
source-derived
object-not-superposable-on-mirror-image__chirality-retained__reflection-distinction__observation-boundary-held; exact match
True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks reflection, non-superposability, retained chirality or an observation boundary, or if a tampered row is accepted.

Measurement receipt: sha256:b7e2673434262f4fd4b324e08759553fdc24e9fcd51797a3ea30b2bd1b154b99.

Isolation certificate: sha256:0dfb4ec13345c293c10809ac8c921ef688e71bc36432025b0c195197838680ff.

Custody certificate: sha256:a79c30528e65690be2ae987e4daca452ad6516de57261bb8dba05543b9326984.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01058>
(sha256:a787109fa5a2c606b0628013669773016b0245479ed6387584ba027d43e5f01b)

Registered target identities:

- chirality-iupac-composite from IUPAC-GOLD-BOOK-CHIRALITY-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite complete three-dimensional chemical structure with named atoms, adjacency, orientation and exhaustive identity-preserving proper maps.

Immutable identities. Source manifest

sha256:746edf9303a440594e8430a4b65673c68165c1a91685b595b1a632a296a5dc60; derivation seal

sha256:3bf55b4df34bbd642d86d9348e1af0c459b3bf9312120c8f08d3f34fe90e7db8; engine receipt

sha256:2f29569bdc7dd166c19b48a69d3dfd95c3299fe9db0fd32f9b00449bf51bfc91 at

receipts/engine/model_admitted/SFT-CHEM-STEREO-CHIRALITY-001-2f29569bdc7dd166.json;

empirical hash sha256:ff8c4c94813ae88e2ffa135f9eb7f9a6f2c114c434b1a11d50c9bf5985cec965;

measurement receipt sha256:b7e2673434262f4fd4b324e08759553fdc24e9fcd51797a3ea30b2bd1b154b99.

69. Enantiomer pair and chiral observation boundary

Claim identity: SFT-CHEM-STEREO-ENANTIOMER-001

Question and theorem. Enantiomers are the paired complete stereoisomer carriers related by exact reflection and by no identity-preserving proper superposition, with opposed configurations retained.

paired-stereoisomers__same-composition-and-connectivity__exact-mirror-relation__non-superposable-pair__opposed-configurations-held

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001

- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-STEREO-CHIRALITY-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite pair of equal-composition, equal-connectivity oriented molecular carriers with exhaustive mirror and proper-superposition tests.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `paired-stereoisomers__same-composition-and-connectivity__exact-mirror-relation__non-superposable-pair__opposed-configurations-held__chiral-observation-boundary__paired-map-and-observation-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
members	paired-stereoisomers	one-structure-only	No pair exists.
constitution	same-composition-and-connectivity	different-connectivity	They are constitutional isomers.
mirror	exact-mirror-relation	unrelated-orientations	The pair is not enantiomeric.
superposition	non-superposable-pair	superposable-pair	The reflected structures are identical.
orientation	opposed-configurations-held	orientation-erased	Opposed configurations merge.
environment	chiral-observation-boundary	universal-property-difference	Response depends on observation context.
record	paired-map-and-observation-trace	pair-name-only	The mirror test cannot be checked.
extra	no-extra-rule	free-enantiomer-rule	A discretionary label can force the pair.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:78874adb9051f2cda593f82c2378adc94592159d0fce0287299f8bc1501a94dc`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:80898deff1518c63ea5d791407d62d7bbb29dc25b5cbfc6477a8240f1135088d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d892bdf9ea15e6523b592d1b7b9439d6d5022e55032d91d191b7bdc843c221d5`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7f8cd4ae6999f6643eb7752e0d261eb4389e0a29ea01b45e51c0da078e636c67`.

Independent reconstruction. Implementation hash

`sha256:b9e0407b93ac2315487b7c6dcda9bc34f65c4a439fa518cbdc0f817c39a0321d`; independent certificate `sha256:2c44dff49dba2fb9f7f9259ec34e3ed7e75956aab0a1073f1a5df99c26b52d23`; external-validation hash `sha256:6d1c147125d797b049a80a8e01a9cdeeb2a77ceefa8a4d05b5e9fd96552c3b3e`. The independent

program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ENANTIOMER-COMPOSITE-2026

Comparison records:

- enantiomer-iupac-composite: predicted
molecular-entities-mirror-images__non-superposable__opposed-configurations__chiral-observation-bounded; source-derived
molecular-entities-mirror-images__non-superposable__opposed-configurations__chiral-observation-bounded; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks molecular mirror relation, non-superposability, opposed configuration or chiral observation boundary, or if a tampered row is accepted.

Measurement receipt: sha256:9ebcbc56392162f13d95b18ecba6430dbcb2121a365a1aed92df11b78b5bb686.

Isolation certificate: sha256:338da9ac010e7ca6cecd35ddc8cc6a6d0bff06f1cfb26ba86513ba69b62eccd6.

Custody certificate: sha256:d42f56dcb8b00141ed11db99fa34c037bc8779efd944976f901cc7419726ab15.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/E02069>
(sha256:833912292d5eba7d3528d2a31a748a95f4cd4d28b92e896f6e1b5d9ef914935d)

Registered target identities:

- enantiomer-iupac-composite from IUPAC-GOLD-BOOK-ENANTIOMER-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite pair of equal-composition, equal-connectivity oriented molecular carriers with exhaustive mirror and proper-superposition tests.

Immutable identities. Source manifest

sha256:7248743a426c15db4c8ee1b7899a75bb6b573487f7b0561112e477e4f04a308a; derivation seal

sha256:4e61140242b466e2f4f87be1f590bc7db0b54bddb8daaf050001025ec15e6802; engine receipt

sha256:d9a1134bbb9af58c62e6fbbc65f72c47071183fa1ed0274c2d796392d210077d at

receipts/engine/model_admitted/SFT-CHEM-STEREO-ENANTIOMER-001-d9a1134bbb9af58c.json;

empirical hash sha256:259683e4df5735ee9acffdc7c83cd130ddcfb3e420b071a996a54e91e51174733;

measurement receipt sha256:9ebcbc56392162f13d95b18ecba6430dbcb2121a365a1aed92df11b78b5bb686.

70. Diastereomer distinction

Claim identity: SFT-CHEM-STEREO-DIASTEREOMER-001

Question and theorem. Diastereomers are paired non-superposable stereoisomers of equal composition and connectivity that are not related as exact mirror images.

paired-stereoisomers__same-composition-and-connectivity__non-superposable-pair__not-mirror-related__complete-relative-configurations-held

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001

- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-CHEM-STEREO-ENANTIOMER-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite equal-constitution oriented molecular pair with exhaustive identity, reflection and proper-superposition classifications.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `paired-stereoisomers__same-composition-and-connectivity__non-superposable-pair__not-mirror-related__complete-relative-configurations-held__property-distinction-open__complete-pair-classification-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
members	paired-stereoisomers	one-structure-only	No pair exists.
constitution	same-composition-and-connectivity	different-composition-or-connectivity	They are not stereoisomers.
superposition	non-superposable-pair	superposable-pair	No stereochemical distinction remains.
mirror	not-mirror-related	exact-mirror-pair	That pair is enantiomeric.
configuration	complete-relative-configurations-held	all-centres-erased	The difference is unreproducible.
properties	property-distinction-open	properties-forced-equal	Non-mirror structures need not share response.
record	complete-pair-classification-trace	diastereomer-label-only	The exclusions cannot be checked.
extra	no-extra-rule	free-diastereomer-rule	A taxonomy can select the answer.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 8682d8dcee95265f2e77c106b5f8b4cea6d08860c097fe761367e7042ae95966.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 3a570f720ae4e97c27103ded8bc92baa50d7d553202e4053e17e83a4c0f7aebd.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: d351449dc0617402486df5ae3d907c79a641ac271c17fb87bcb906f884a6a24.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:ad5e7ea3c443b07d374848fb6d542ad5d4a5351ccb2ed46fb83b0be1f3bb9318.

Independent reconstruction. Implementation hash

sha256:3936122354bc8e32351cbd45cb5e1b9835a7bb5b3a12546e2893154c9daf820c; independent certificate sha256:2c1b89fe37e89c64402624627281e4a28c22f9a9305246322c89e5ec3b433d5a; external-validation hash sha256:91f03f8244b102455caad2cd49b179f0980562de9363de5da9606c39f5531585. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-DIASTEREOMER-COMPOSITE-2026

Comparison records:

- diastereomer-iupac-composite: predicted
stereoisomers__not-mirror-images__distinct-configurations__property-distinction-possible; source-derived
stereoisomers__not-mirror-images__distinct-configurations__property-distinction-possible; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks stereoisomer identity, non-mirror relation, configurational distinction or possible property distinction, or if a tampered row is accepted.

Measurement receipt: sha256:d90d2f0193fb7f0ed5e98953f993c87200da07f101af9141d653cb30231bec09.

Isolation certificate: sha256:caf9a73d4d6e7d5b0b8b4d11fd0c8f5d7baf6a4fd903bcb25905ffb09cbde93a.

Custody certificate: sha256:7f9a03dca8cb9883fc46b9639c827eaf73a0621f2187ed429b2a5c19fe62d6fd.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/D01679>
(sha256:4fa09c83930088c3eaea3fd428d370918e6bf98c361673287c0fc8ef9d394078)

Registered target identities:

- diastereomer-iupac-composite from IUPAC-GOLD-BOOK-DIASTEREOMER-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite equal-constitution oriented molecular pair with exhaustive identity, reflection and proper-superposition classifications.

Immutable identities. Source manifest

sha256:412ed0eeb32a6a43956bfa96f82d1c9dd4cdc2a712b4bf302c3e0fbf2a2fb8a8; derivation seal

sha256:b5a8de446b152bcb33379bde9b3e2b6145cce569dafae3bb4c3c3f381c139a3; engine receipt

sha256:54222fa195d500b1528f8c93492bedd144c895243dba02c49ba8b189f8629bc9 at

receipts/engine/model_admitted/SFT-CHEM-STEREO-DIASTEREOMER-001-54222fa195d500b1.json;

empirical hash sha256:fb79f114e5344349495e488e580a4d6800bd07f4a52266059f63b1655dfd03d1;

measurement receipt sha256:d90d2f0193fb7f0ed5e98953f993c87200da07f101af9141d653cb30231bec09.

71. Functional-group recurrence and reaction role

Claim identity: SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001

Question and theorem. A functional group is a retained atom-or-group subcarrier whose complete local structure recurs across molecular embeddings and carries a characteristic generated reaction role within a declared context.

named-atom-or-group-support__molecular-embedding-held__recurs-across-molecules__characteristic-reaction-role__group-identity-retained

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001

- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite molecular carrier with complete named substructures, boundary bonds and generated source-bound reaction traces.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `named-atom-or-group-support__molecular-embedding-held__recurs-across-molecules__characteristic-reaction-role__group-identity-retained__molecular-context-bounded__substructure-and-reaction-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
support	named-atom-or-group-support	whole-molecule-only	No recurrent substructure is identified.
embedding	molecular-embedding-held	isolated-fragment	A functional group acts in a molecule.
recurrence	recurs-across-molecules	one-molecule-only	No group class is established.
reaction	characteristic-reaction-role	shape-without-transition-role	Function requires chemical behavior.
identity	group-identity-retained	context-erases-group	The group cannot be followed.
context	molecular-context-bounded	universal-context-free-response	Neighboring structure can modify response.
record	substructure-and-reaction-trace	group-name-only	Recurrence cannot be reproduced.
extra	no-extra-rule	free-functional-taxonomy	An imported list selects the group.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c188cf693c44cb41b1a98a4cac9a337f8c0045f92bc18d237416427b14b2f689`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3727abd174787d207fb36092b3e137e2f11ec5878725228dbc473d2ad541183b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:bd73c500dd3af7832d2bc4f118d46f20ceda9568f900c6b69d93ab2c7001b32b`.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 72208840975e03044131c459563f053a4d989014820fecebf1b28b8d6f418397.

Independent reconstruction. Implementation hash

sha256 : 600186025be20195d13b50b988a6c1454172503b1e3c9af181c2c3579850101b; independent certificate sha256 : 93cb8ec0892c4409a11bf050cc142e7c068898f82bfeb071fcc325d3df3b5f39; external-validation hash sha256 : 24a94854de814576b4427c1f35b1be4a8d09b569691ecef f3c1d6288bc7d5e41. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-FUNCTIONAL-GROUP-COMPOSITE-2026

Comparison records:

- functional-group-iupac-composite: predicted
atom-or-group-with-characteristic-properties__recurs-across-molecules__reaction-role-retained__molecular-context-bounded;
source-derived
atom-or-group-with-characteristic-properties__recurs-across-molecules__reaction-role-retained__molecular-context-bounded;
exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks an atom/group, characteristic behavior, recurrence, reaction role or context boundary, or if a tampered row is accepted.

Measurement receipt: sha256 : 170ee809d69a0850d62859ec4338762fc9fbf08bc3fe90e57ef5d5b37080110f.

Isolation certificate: sha256 : 98551e23514f264636497258172076ed0686621e8ffbe29d5a8de6e1ac17fd99.

Custody certificate: sha256 : d63627c0e31efea83ee1fb03cfe9a74ac768150c4d870e036ca0f2712eef f8bc.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/F02555>
(sha256:1fc0f254f488921a762c25454615ea9ec4f40d54736be524fa9f4ff8d79869d1)

Registered target identities:

- functional-group-iupac-composite from IUPAC-GOLD-BOOK-FUNCTIONAL-GROUP-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite molecular carrier with complete named substructures, boundary bonds and generated source-bound reaction traces.

Immutable identities. Source manifest

sha256 : 3ca722fa02847aa3c7c3e8067cfa328d1a4560fde9457703d6ca58390589f240; derivation seal

sha256 : 84059bdbbe6d9493bdf904ec062536a64a3fbc9dbaf89923d1607662a7a88cc1; engine receipt

sha256 : 84d6b1302565580828881323d45c81917e74c150b85ede407160bc90e33989b8 at receipts/engine/model_admitted/SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001-84d6b13025655808.json; empirical hash

sha256 : 5550eb0c76fe5930c73ef28bef2abf7b3644b7830e9a2bd4b1ed896b8d49afd1; measurement receipt

sha256:170ee809d69a0850d62859ec4338762fc9fbf08bc3fe90e57ef5d5b37080110f.

72. Organic reaction-family compositional law

Claim identity: SFT-CHEM-ORGANIC-REACTION-FAMILY-001

Question and theorem. An organic reaction family is the equivalence class of complete balanced reactant-product maps sharing one minimal bond-change pattern and role correspondence without assuming a common mechanism.

multiple-generated-reactions__complete-reactant-product-maps__shared-bond-change-pattern__reactant-product-role-correspondence__elemental-carriers-balanced

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001

- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite generated set of balanced organic reactant-product atom maps with complete bond changes, functional roles, conditions and separate mechanism records.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `multiple-generated-reactions__complete-reactant-product-maps__shared-bond-change-pattern__reactant-product-role-correspondence__elemental-carriers-balanced__mechanism-not-assumed__member-and-pattern-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
members	multiple-generated-reactions	one-reaction-only	No family is formed.
encoding	complete-reactant-product-maps	reaction-names-only	Names do not establish common structure.
pattern	shared-bond-change-pattern	unrelated-bond-changes	No shared family relation exists.
roles	reactant-product-role-correspondence	reactant-product-roles-erased	Pattern orientation is ambiguous.
composition	elemental-carriers-balanced	carrier-balance-erased	The transformations cannot close.
mechanism	mechanism-not-assumed	mechanism-assumed-from-pattern	Equal net changes need not share a mechanism.
record	member-and-pattern-trace	family-label-only	Membership cannot be audited.
extra	no-extra-rule	free-reaction-taxonomy	An imported list can force membership.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256: 2794e829a8c2cd1c9b2bf0a50c8294a70400ffe90be534b840b1363dec4b90c6`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256: 35977ac840a0e2d15e7593c90926ed8ba21498624ab5ca81adce3b34fbc24762`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 94e838e1e737d341840056cf2b4c397e9745eaed535e4a80dd9e9b7e65e12e93.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: db912a2311b33366ee25d520989015f8231a0f4959c831e03149d92dad4561c3.

Independent reconstruction. Implementation hash

sha256: 1af395b5f6f38579aaffb318538190387ebb8f470c3d46f7881ec9ad2c3a6cf6; independent certificate
sha256: ff5a8998035aab50f52a103e7a6886cfcf88a42d9f1049d282fd448408ba77c7; external-validation
hash sha256: 5b529be1dbb5e8f35ff13b00325228eaa410353002aae31956ad72a02f58ebda. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ORGANIC-REACTION-FAMILY-COMPOSITE-2026

Comparison records:

- organic-reaction-family-iupac-composite: predicted
shared-bond-change-pattern__reactant-product-role-correspondence__mechanism-not-assumed__conditions-source-bounded;
source-derived
shared-bond-change-pattern__reactant-product-role-correspondence__mechanism-not-assumed__conditions-source-bounded;
exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks shared reaction pattern, reactant/product roles, separation from mechanism or condition boundary, or if a tampered row is accepted.

Measurement receipt: sha256: 767bceec7ec74a76c6d65ee0faac453cbda4541783c13820f3a07e40b7c7097c.

Isolation certificate: sha256: acefbcf92192dff31c8df64f0aec6147c420c4590747b964240f2528d3e6c4a3.

Custody certificate: sha256: d4f6fcd8db12a5a7c07b2fdb9e9b35071df47f7b6c44673453c8e9fad90faa58.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/R05164>
(sha256:e7253285c30faeb217a73b62d4ca4cea7636a626c1306e2eebe8b71960abf6eb)

Registered target identities:

- organic-reaction-family-iupac-composite from IUPAC-GOLD-BOOK-ORGANIC-REACTION-FAMILY-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite generated set of balanced organic reactant-product atom maps with complete bond changes, functional roles, conditions and separate mechanism records.

Immutable identities. Source manifest

sha256:06d9e0c7eda713f35b939bbdcee35dd65832a6616d0dc4e1b87080f87a6cefacc; derivation seal
 sha256:3121b5fe6544a9301355d31d2c27ddf2f2c653a1dc117454286d94ce0da12221; engine receipt
 sha256:89120ae1d10844a8d907ab00b6db94bcb7dfccb75e9ccc93b072f84eef8f8526 at receipts/engine/model_admitted/SFT-CHEM-ORGANIC-REACTION-FAMILY-001-89120ae1d10844a8.json; empirical hash
 sha256:a5ce171523d3880a26ca25d1c6656bc4108b0b2b60653665cb7290f33be6ac44; measurement receipt
 sha256:767bceec7ec74a76c6d65ee0faac453cbda4541783c13820f3a07e40b7c7097c.

73. Polymer chain from repeated monomer identity

Claim identity: SFT-CHEM-POLYMER-CHAIN-001

Question and theorem. A polymer chain is one finite connected macromolecular carrier formed by covalently ordered occurrences of a retained repeat-unit identity with exact positive extent and explicit ends.

connected-macromolecular-carrier__repeated-low-mass-unit-identity__covalent-sequence-connections__ordered-chain-incidence__positive-finite-repeat-count

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001

- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite connected molecular incidence path containing repeated complete low-mass unit traces, covalent connections and explicit terminal boundaries.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `connected-macromolecular-carrier__repeated-low-mass-unit-identity__covalent-sequence-connections__ordered-chain-incidence__positive-finite-repeat-count__chain-end-boundary-retained__repeat-connection-end-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	connected-macromolecular-carrier	unconnected-monomer-mixture	No chain exists.
repeat	repeated-low-mass-unit-identity	anonymous-segments	Repeat identity is lost.
connection	covalent-sequence-connections	noncovalent-coincidence	The sequence is not a polymer chain.
order	ordered-chain-incidence	occurrence-order-erased	Chain topology is lost.
extent	positive-finite-repeat-count	fixed-universal-length	Length is population- and synthesis-bound.
ends	chain-end-boundary-retained	infinite-chain-assumed	A finite carrier needs a boundary.
record	repeat-connection-end-trace	polymer-name-only	The chain cannot be reconstructed.
extra	no-extra-rule	free-polymer-rule	A threshold can arbitrarily select chains.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:8726c5ae36f1d1dda15c6109697ba8f767017f4b5806a911dc62ff6d46f82db7.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:4527583bc01617f404e6e917d56a214b3f33732d24bb19ab270d9e50793a8f99.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:3154c241dd7a4f1d0d151c880c2e340905664003bc6411691de8ba3be063ae7d.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:7e6b7adcf2ff991f470a76322fad0d788027a04b2ee07e6087c6a331bf625206.

Independent reconstruction. Implementation hash

sha256:fc9ba42e65fefb02b61cbf5a7b32554e3c38cf5d592a5371b70f07f620fc7a16; independent certificate sha256:8e5fd2dd33f54399c8a6a3416e84050d860244c45d6b8b202d877bd8c1de9268; external-validation hash sha256:4b5480f5c2e2ccd4c30192c9555127c12f2997d393b93ce8ae4523162861ef28. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-POLYMER-CHAIN-COMPOSITE-2026

Comparison records:

- polymer-chain-iupac-composite: predicted
high-relative-molecular-mass-molecule__repeated-low-mass-units__covalent-sequence__chain-end-boundary; source-derived
high-relative-molecular-mass-molecule__repeated-low-mass-units__covalent-sequence__chain-end-boundary; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks a macromolecular carrier, repeated units, covalent sequence or chain boundary, or if a tampered row is accepted.

Measurement receipt: sha256:84eb46a6d0d2fc95e918a5b3ba66933306b1b4488874f38de6fefcb65fa52794.

Isolation certificate: sha256:e4186aa15c542d261f56c16ace8d95c53e0c0d7b3a542db14c748212a43d10a5.

Custody certificate: sha256:f7392b10e6c9f01779d59d366d423671a3a0a30e7a2088ede8aa37034dc859c1.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/M03667>
(sha256:b38c6c106e35763169204b065edaabf18df22bf97f4ea9094db3b8b72383e85a)

Registered target identities:

- polymer-chain-iupac-composite from IUPAC-GOLD-BOOK-POLYMER-CHAIN-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero

- external target content remains inaccessible until the prediction is sealed
- Every finite connected molecular incidence path containing repeated complete low-mass unit traces, covalent connections and explicit terminal boundaries.

Immutable identities. Source manifest

sha256:448eede8378e51d088184107602b54318de72be66428d29ae5ebe675eb85cbda; derivation seal sha256:f7751c8eed2343daec06913a3cba18250c606a753fc9d88a9e255bb141aa0aea; engine receipt sha256:03f536704d27b6e91a52c136572c1d7596bc5c5d95681b0d56292cbc8e472201 at receipts/engine/model_admitted/SFT-CHEM-POLYMER-CHAIN-001-03f536704d27b6e9.json; empirical hash sha256:d5bc02078bba4c54f6be98301828b5c08f34e4cd4ae21b42029cca7d519641c5; measurement receipt sha256:84eb46a6d0d2fc95e918a5b3ba66933306b1b4488874f38de6fefcb65fa52794.

74. Polymer population and retained chain-length distribution

Claim identity: SFT-CHEM-POLYMER-DISTRIBUTION-001

Question and theorem. A polymer distribution is the complete source-bound census of resolved chain carriers by repeat count and positive multiplicity, with number and mass weightings retained separately.

all-observed-chain-carriers__exact-chain-length-counts__positive-chain-multiplicities__number-and-mass-support-separated__complete-molar-mass-distribution

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001

- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-POLYMER-CHAIN-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite observed polymer sample with complete chain identities, repeat counts, positive multiplicities, weighting rule, sample and method boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `all-observed-chain-carriers__exact-chain-length-counts__positive-chain-multiplicities__number-and-mass-support-separated__complete-molar-mass-distribution__method-and-sample-bounded__chain-population-on-trace__no-extra-rule`. Closure is `depth-independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
population	all-observed-chain-carriers	single-representative-chain	A sample distribution is erased.
length	exact-chain-length-counts	mean-only	A mean loses alternatives.
multiplicity	positive-chain-multiplicities	duplicate-chains-collapse	Population frequency is lost.
measure	number-and-mass-support-separated	one-universal-average	Different summaries retain different information.
distribution	complete-molar-mass-distribution	answer-scalar-only	The population cannot be reconstructed.
method	method-and-sample-bounded	method-free-census	Resolution changes the observed population.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
record	chain-population-trace	distribution-name-only	The census cannot be audited.
extra	no-extra-rule	free-distribution-fit	A fitted shape can hide unfavorable chains.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : 88065ddd66a529ae65468a34e9385d4efa760e54ccd0c87359658a17f83ea231.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 3e9fe0202d7069000f6003a37591693635f37214510a56dd32ee5d33c3c7881c.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 90d31eee582fad2b4a6b2b14cc405a233e8a7c2c0a7db750908f4137db7b3800.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : ee81365154d6fc7ee014cdac864f9d47eb8e41328b830b95829704096f87ed57.

Independent reconstruction. Implementation hash

sha256 : 8628fe82a0a16f901776576b549a688e90d8c857ac9004825836c20252b39bd3; independent certificate sha256 : ffe61525dbbc336a4802001ad25570d65a61f25408c301d1484cf53898dda99d; external-validation hash sha256 : f5bb6567226f41b160ecfbc3ad40ec8f45f678b34b9ccaadc9f78efacb8b2b4c. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-POLYMER-DISTRIBUTION-COMPOSITE-2026

Comparison records:

- polymer-distribution-iupac-composite: predicted
polymer-sample-many-chain-lengths__molar-mass-distribution__number-and-mass-fractions-retained__method-bounded;
source-derived
polymer-sample-many-chain-lengths__molar-mass-distribution__number-and-mass-fractions-retained__method-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks chain-length diversity, molar-mass distribution, distinct weightings or method boundary, or if a tampered row is accepted.

Measurement receipt: sha256 : 919ae7f73bb04f65d9bb2b3c403fdb6eab2ce59d580149b77c78d3fc0b86802.

Isolation certificate: sha256 : 34eff9b560906f5615b4b3c1ada49c516533ed7e64b697b70d2e51120dc75b7c.

Custody certificate: sha256 : f373b6e869a72becff087cf48694c6fdd13552727592676af5d46789ba5d4d4a.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/12215>
(sha256:98c415c9e851b41083ffc7ee3b9e1ba2d2b9e81bc6e05bb0dce170dd9733e91a)

Registered target identities:

- polymer-distribution-iupac-composite from IUPAC-GOLD-BOOK-POLYMER-DISTRIBUTION-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite observed polymer sample with complete chain identities, repeat counts, positive multiplicities, weighting rule, sample and method boundary.

Immutable identities. Source manifest

sha256:ac686db673d7292add6ad2e9d0b51b2433e2f3fdd6991b137dd84e225a799ebf; derivation seal
 sha256:c7cb1ba5914c89f0e418171b4f1035544119f86b3160fd51010b02ff33bf1f00; engine receipt
 sha256:ad81bf1f69ee772b22896d80d25bcb312522f69a90c52418f6f8362a31b29a84 at
 receipts/engine/model_admitted/SFT-CHEM-POLYMER-DISTRIBUTION-001-ad81bf1f69ee772b.json;
 empirical hash sha256:953816317fdfb027e491091ebfc84f48230eccb9276b6c8e822b02ecfd84c7fa;
 measurement receipt sha256:919ae7f73bb04f65d9bb2b3c403fdb6eab2ce59d580149b77c78d3fc0b86802.

75. Chemistry-to-biology molecular handoff boundary

Claim identity: SFT-CHEM-BIOMOLECULAR-BOUNDARY-001

Question and theorem. The chemistry-to-biology boundary passes only a complete source-bound molecular identity, structure and chemical state; biological function is a separately retained downstream relation and cannot select the chemical law.

chemical-molecular-identity-retained__composition-preserved__molecular-structure-preserved__chemical-state-boundary-held__biological-function-separately-held

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001

- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite chemical molecular carrier used by a downstream biological process, with complete composition, structure, state, conditions and separately registered function relation.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `chemical-molecular-identity-retained__composition-preserved__molecular-structure-preserved__chemical-state-boundary-held__biological-function-separately-held__chemistry-to-biology-handoff__composition-structure-state-function-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	chemical-molecular-identity-retained	biological-function-without-molecule	Chemistry requires a molecular carrier.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
composition	composition-preserved	composition-erased-at-handoff	The object cannot be traced chemically.
structure	molecular-structure-preserved	structure-erased-at-handoff	Biological recognition loses its substrate.
state	chemical-state-boundary-held	timeless-molecule	Function acts on physical chemical states.
function	biological-function-separately-held	function-as-chemical-identity	Different functions need not change chemical identity.
boundary	chemistry-to-biology-handoff	biology-imported-into-chemistry	A downstream law cannot select chemistry.
record	composition-structure-state-function-trace	biomolecule-name-only	The handoff cannot be audited.
extra	no-extra-rule	free-life-criterion	A life label can select any molecule.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:d24fc84b3d2eb24db795ab0e5e24264436d5b446a3d0db66271770e515baf85f.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:d9532c7da21be65210d77703ceafb00a1d0ed36e06cb60b83d3ed2582d4e9a9f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:e3b5635d8c33cd0e99f9d01f3d9368a5206a4d053087c72ecc3fc2a813a78e46.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:7c47d008b17ce95a41abeb1071ec1c3281cfac6d126816e7529602799b36c302.

Independent reconstruction. Implementation hash

sha256:d831eace1b996519fb40bc5b13e8897cdf6c007f87d9159e66bf77edeef4130a; independent certificate
sha256:f4d1b90f1ef528e8bb267bdc296905e85dcbd67d0ea2eee3e84ff09c71e7561e; external-validation
hash sha256:db6a36654236f3fb8fcb1bd5a5c155d48c6e5a878b7aa5755117506a18e7f6ea. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- EMBL-EBI-CHEBI-BIOMOLECULAR-BOUNDARY-2026

Comparison records:

- biomolecular-boundary-authority-composite: predicted chemical-molecular-identity-retained__biological-function-not-chemical-identity__structure-composition-handoff__biology-boundary-explicit; source-derived chemical-molecular-identity-retained__biological-function-not-chemical-identity__structure-composition-handoff__biology-boundary-explicit; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence cannot separate chemical molecular identity from biological function at an explicit handoff boundary, or if a tampered row is accepted.

Measurement receipt: sha256:c7069c2f9bde225b62a7aa88c291736badf28ea1d7a36cd1e2c05335e6d245fe.

Isolation certificate: sha256:39d474ab831cde20b5801b5d28395ee2833614382c9bee25c2e6c381e4e116c6.

Custody certificate: sha256:f33454a495d91e6a738af4a65d06b6a27fb01c378e45c77f0446dc9394fd0de2.

Registered source descriptions:

- EMBL-European Bioinformatics Institute: [https://www.ebi.ac.uk/chebi/](https://www.ebi.ac.uk/chebi/(sha256:d862deab28de555e80751d9fce3fecf4ac4de970d17d9a4c1481dc6976731f24))
(sha256:d862deab28de555e80751d9fce3fecf4ac4de970d17d9a4c1481dc6976731f24)

Registered target identities:

- biomolecular-boundary-authority-composite from EMBL-EBI-CHEBI-BIOMOLECULAR-BOUNDARY-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite chemical molecular carrier used by a downstream biological process, with complete composition, structure, state, conditions and separately registered function relation.

Immutable identities. Source manifest

sha256:ca210a3eea83e6f933e8f17d4ad6d5fcd2dc406715f4ba931f280b617dd9b164; derivation seal
sha256:a165cad506a0cef78eacf0f67942163bb2fa09c5241af0271488da9a6a7aa12b; engine receipt
sha256:dfed7cd5d124ff390399d5adf8b06a01a7edcc102a9e6d0ff59beb201bfd547e at receipts/engine/model_admitted/SFT-CHEM-BIOMOLECULAR-BOUNDARY-001-dfed7cd5d124ff39.json; empirical hash
sha256:c4337b17744f0ac2e605e985ce04db0dae2448c48d038c916e9d1e9489837c55; measurement receipt
sha256:c7069c2f9bde225b62a7aa88c291736badf28ea1d7a36cd1e2c05335e6d245fe.

16. Analytical and spectroscopic foundations - 8 admitted claims

Measurement identity, selectivity and spectra remain downstream tests with complete uncertainty and adverse rows. This entire registered subcategory is included; no proper subset is presented as field completion.

76. Analytical sample, analyte and matrix distinction

Claim identity: SFT-CHEM-ANALYTICAL-SAMPLE-001

Question and theorem. An analytical sample is a declared material portion whose complete resolved component support is partitioned by held roles into the analyte and retained matrix, together with its source-to-sample map.

declared-material-portion__complete-resolved-component-support__analyte-role-held__matrix-support-held__source-to-sample-map-held

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-TRACEABILITY-001

• SFT-CHEM-BIOMOLECULAR-BOUNDARY-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite declared material portion with complete resolved components, one or more analyte roles and retained non-analyte support.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `declared-material-portion__complete-resolved-component-support__analyte-role-held__matrix-support-held__source-to-sample-map-held__component-response-map-held__sample-analyte-matrix-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
portion	declared-material-portion	unbounded-material-name	No sampled carrier is identified.
components	complete-resolved-component-support	component-support-erased	The portion cannot be compositionally audited.
analyte	analyte-role-held	analyte-not-designated	No measurand-bearing component is selected.
matrix	matrix-support-held	non-analyte-components-discarded	Interference support is hidden.
sampling	source-to-sample-map-held	sample-equals-whole-source	A portion need not reproduce the source whole.
observation	component-response-map-held	response-without-component-map	A response cannot be assigned.
record	sample-analyte-matrix-trace	result-only	The sample distinction cannot be reproduced.
extra	no-extra-rule	free-sample-rule	A discretionary rule can select any portion.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3efc490e1bb313eb3a34c6f86d105d7b0feb050e3dc07907cb22f9e557e3db2a`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f6628399c538353c7fb79523e499076b0985aa2d7845782e3bab7e269ba2402e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:54733cd21625ad9d2061bffcbbfc18170a824966c03c94ce4abd76a18e66cdfcb`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7c171463044bd6ddcb0a3699fa40084a8b5f356e72f0015643dd009118f5dab8`.

Independent reconstruction. Implementation hash

`sha256:2e248b730cee3eeb2aeb40d4018c0beafc12e8ac5692a7b8c6ed0b72657168bc`; independent certificate `sha256:29057279273fb54b1e5c5c6f56b99df6749458dbbb89aec2895105ee80e84ac8`; external-validation hash `sha256:65c970ece5b7e8a45d9071e3fc94c3e32ef74c97336f1cd2b10e6bdce9bd9c1d`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ANALYTICAL-SAMPLE-COMPOSITE-2026

Comparison records:

- analytical-sample-iupac-composite: predicted sample-is-declared-material-portion__analyte-is-component-under-determination__matrix-is-other-sample-components__sampling-boundary-held; source-derived sample-is-declared-material-portion__analyte-is-component-under-determination__matrix-is-other-sample-components__sampling-boundary-held; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks sample portion, analyte, matrix or sampling boundary, or if a tampered row is accepted.

Measurement receipt: sha256:341c9df03892ee0750b39506c3d228f62f0e25926a67d4183c81d1c31f263dcb.

Isolation certificate: sha256:cdbebc05824c89bc83a0c65aa42473fa114f801b17052666f86980c0a859fe9a.

Custody certificate: sha256:7b0c27394fc4a1069ec4bcd1096cf3a93b9f6ab94cca5998d3311bb25b6c6254.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/S05451>
(sha256:f2d630c6a9a8ec0ab1304a6e9e3acf1bc495b9b3f2d5f3958375aa0c3fafa184)

Registered target identities:

- analytical-sample-iupac-composite from IUPAC-GOLD-BOOK-ANALYTICAL-SAMPLE-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite declared material portion with complete resolved components, one or more analyte roles and retained non-analyte support.

Immutable identities. Source manifest

sha256:6ce4941faf17edb0c45fa8f17db0e8ad3550c1ca0343826628c1c6c2a348f0e6; derivation seal

sha256:84f0c7306151fb8c786915b99daf0eb91317c631b5b143d271f70a4db7a59dd9; engine receipt

sha256:1fd5c56d3970342f09bb0c871b8f2fb12f84d562e0f580299e89e91c624b00 at

receipts/engine/model_admitted/SFT-CHEM-ANALYTICAL-SAMPLE-001-1fd5c56d3970342f.json;

empirical hash sha256:726e111ea202dc58879d24d05d73b6c000a025d441b67bc5ba8b7bb3606b1f04;

measurement receipt sha256:341c9df03892ee0750b39506c3d228f62f0e25926a67d4183c81d1c31f263dcb.

77. Analytical calibration and traceable comparison

Claim identity: SFT-CHEM-ANALYTICAL-CALIBRATION-001

Question and theorem. Calibration is the complete traceable comparison of held reference input/response pairs with a sample response inside an explicit finite method domain, retaining uncertainty and every row.

traceable-reference-input__reference-response-pair-held__exact-generated-comparison-order__sample-compared-with-reference-map__declared-calibration-domain

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001

- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001
- SFT-CHEM-ANALYTICAL-SAMPLE-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite registered reference-response set and sample response sharing one declared observation organization.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `traceable-reference-input__reference-response-pair-held__exact-generated-comparison-order__sample-compared-with-reference-map__declared-calibration-domain__reference-and-sample-uncertainty-held__reference-map-and-result-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
reference	traceable-reference-input	unknown-reference-input	Comparison has no fixed support.
response	reference-response-pair-held	reference-response-erased	The comparison cannot be reconstructed.
ordering	exact-generated-comparison-order	unordered-fit	A fit can conceal incompatible rows.
sample	sample-compared-with-reference-map	sample-outside-reference-map	No calibrated comparison is possible.
range	declared-calibration-domain	universal-unbounded-map	Calibration support is finite and method-bound.
uncertainty	reference-and-sample-uncertainty-held	uncertainty-erased	Alternatives are falsely collapsed.
record	reference-map-and-result-trace	calibrated-answer-only	Traceability is lost.
extra	no-extra-rule	free-calibration-parameter	A fitted value can select the answer.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:032e873ba65355fe014ee4ea11bc71860ad6fe7164823becca09a813425962ad.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:ec401bf322205d24affee40106306cd23ff06e1726aaa862af3db593374ff9bb.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:454c698f8da30d9662a41b9526c30f56193e6a98cee4296f04011056542b30dc.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:f2d3b9679de44e3fec73bc19dec2b45f5bf4a9a76a71918982afc544d84b3980.

Independent reconstruction. Implementation hash

sha256:1dbf78592e06bd73224c7c6afeee79718cf93ce2936d633abbc26654f817b49b; independent certificate sha256:94a382fa81641e7ff83e6cef4cb4683a3db0e00ef1acce42986f34d6b71c5a72; external-validation hash sha256:337171a5b52b39aa087cd73ef361183371b30fc6b50402ffe584ba88f327cc03. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ANALYTICAL-CALIBRATION-COMPOSITE-2026

Comparison records:

- analytical-calibration-iupac-composite: predicted calibration-relates-response-to-reference-values__traceability-chain-retained__calibration-domain-explicit__uncertainty-method-bounded; source-derived calibration-relates-response-to-reference-values__traceability-chain-retained__calibration-domain-explicit__uncertainty-method-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks response/reference relation, traceability, domain or uncertainty/method boundary, or if a tampered row is accepted.

Measurement receipt: sha256:d4f1ee33ca16a454ce6f5b5fd392feafbd0b1af01b69f82ac02a5079e82b7cd0.

Isolation certificate: sha256:708632b4fb52d0ea69a009667e12e7952ddb808cdfc540dc12748e17f77e1abc.

Custody certificate: sha256:dd5e8bd8cb4699823086814e0c631ba9f387a76cdcb5ec3e3505fddbe7f37982.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C00775> (sha256:9a50ffb4e909ae78ac6ccf2cfa69db372a7ff659db4fb88b48e89e79322b69c0)

Registered target identities:

- analytical-calibration-iupac-composite from IUPAC-GOLD-BOOK-ANALYTICAL-CALIBRATION-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite registered reference-response set and sample response sharing one declared observation organization.

Immutable identities. Source manifest

sha256:7ab30b7fc98b9fcc2069b6b1feca6674670316a4243fc70d17c91da348a8c10b; derivation seal

sha256:b0f21e5c4549f4f80b6e3e4167295fffb06f3969b5f9567f20ba57dcc445429; engine receipt

sha256:94e195edc3e188f525232747b5c5ca4f893b490ecd46ef9d8bb377ef1d2d2a54 at receipts/engine/model_admitted/SFT-CHEM-ANALYTICAL-CALIBRATION-001-94e195edc3e188f5.json; empirical hash

sha256:a6f57a9ea8b23a843c92fbb5f9a04fe1abde9afac45aea22e6268fe7c98ef2e; measurement receipt

sha256:d4f1ee33ca16a454ce6f5b5fd392feafbd0b1af01b69f82ac02a5079e82b7cd0.

78. Analytical selectivity and interference boundary

Claim identity: SFT-CHEM-ANALYTICAL-SELECTIVITY-001

Question and theorem. Analytical selectivity is exact distinguishability of the analyte response from every declared interferent response within a retained method-and-condition boundary; merged responses are recorded as interference.

analyte-response-held__declared-interferent-support__all-analyte-interferent-responses-compared__distinguishable-analyte-response__interference-boundary-retained

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001

- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001
- SFT-CHEM-ANALYTICAL-SAMPLE-001
- SFT-CHEM-ANALYTICAL-CALIBRATION-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite analyte/interferent response support under one registered method and condition class.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `analyte-response-held__declared-interferent-support__all-analyte-interferent-responses-compared__distinguishable-analyte-response__interference-boundary-retained__method-and-condition-bounded__complete-response-comparison-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
target	analyte-response-held	response-without-analyte	Selectivity has no target.
alternatives	declared-interferent-support	other-components-erased	Competing support is hidden.
comparison	all-analyte-interferent-responses-compared	single-response-only	No discrimination is tested.
decision	distinguishable-analyte-response	name-based-assignment	A name cannot prove attribution.
interference	interference-boundary-retained	failed-alternatives-deleted	Unfavorable rows disappear.
conditions	method-and-condition-bounded	universal-selectivity	Response depends on method and conditions.
record	complete-response-comparison-trace	selectivity-scalar-only	The tested alternatives cannot be audited.
extra	no-extra-rule	free-selectivity-threshold	An arbitrary threshold can force success.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:3b6ea28b86431cb84b37ffd90b03311ea422cc2dc805d76224971189594641e8.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:a7d04dc9d2f7bc12b43d5e5e8df3b7a4bcf26911119584e555d5d74f01160b64.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:18a0bf0c39f89ee23bcf5b4c7a0092ac625f88fea4c3d15c8e860f04698bc715.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:98ad52c643c40c247122849ca5db493c0f3ab8e4cb17c141826be16d1d16b702.

Independent reconstruction. Implementation hash

sha256:62b133e342f7dd1974b887e976c0df6a472e824750afb8ec0872087d4d5427bf; independent certificate sha256:fe120fe14ad5e2b636c45d8014360b4858a8c3d87b8d79139fc67719289f4487; external-validation hash sha256:b0b97ae25e3375c62a629cee86325340ffed9e51fb59f2819ccb36bd53fa4344. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ANALYTICAL-SELECTIVITY-COMPOSITE-2026

Comparison records:

- analytical-selectivity-iupac-composite: predicted analyte-response-distinguished-amid-components__interference-retained__selectivity-not-universal__method-condition-boundary-held; source-derived analyte-response-distinguished-amid-components__interference-retained__selectivity-not-universal__method-condition-boundary-held; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks analyte discrimination, interference, non-universality or method/condition boundary, or if a tampered row is accepted.

Measurement receipt: sha256:75b883b0415a7c35a47e89715545fa1e9222aee2ab15657af21e429a2a7053f0.

Isolation certificate: sha256:94c85cc8784fcb3eeb209828d860c8dfe7f73e5843fbfa7bb9242841c4680520.

Custody certificate: sha256:b1b1c7ad359336ceeb31288e081ff7b1a6bd9afdd1db5a5352277cb6d7900765.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/S05564> (sha256:24cfb5654b5ba14eb99cc4dfe0c01efa7a251be91937a33000f362ba8a9730f6)

Registered target identities:

- analytical-selectivity-iupac-composite from IUPAC-GOLD-BOOK-ANALYTICAL-SELECTIVITY-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite analyte/interferent response support under one registered method and condition class.

Immutable identities. Source manifest

sha256:59a2f4741ab80c5463705fb7786c013d18124a08971905d75d1f6d263f0e6125; derivation seal
 sha256:bd2feda791f4b6838264f63ee0b0ee5a9ec214f8fa532f86f4a3692fb57000ec; engine receipt
 sha256:6e86b8d37deaba116872b23c046a6cabd8cc701713d7e1d36ffbc0d7478f6bca at receipts/engine/model_admitted/SFT-CHEM-ANALYTICAL-SELECTIVITY-001-6e86b8d37deaba11.json; empirical hash
 sha256:bfb25df7c49f3e6272ef130ec3e4c538e7debd8d81686fd57e5efd9037b251b7; measurement receipt
 sha256:75b883b0415a7c35a47e89715545fa1e9222aee2ab15657af21e429a2a7053f0.

79. Mass-spectral composition correspondence

Claim identity: SFT-CHEM-SPEC-MASS-001

Question and theorem. A mass-spectral record is the complete map from generated ionized carriers to retained mass/charge response classes and counted abundances, including precursor-fragment relations and observation conditions.

generated-ionized-carriers__composition-and-isotope-support-held__mass-charge-classes-held__class-position-and-abundance-held__precursor-fragment-map-held

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001

- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001
- SFT-CHEM-ANALYTICAL-SAMPLE-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite generated ionized-carrier support with complete composition, isotope, mass/charge, response and precursor-fragment records.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `generated-ionized-carriers__composition-and-isotope-support-held__mass-charge-classes-held__class-position-and-abundance-held__precursor-fragment-map-held__ionization-and-instrument-boundary__complete-carrier-response-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	generated-ionized-carriers	neutral-sample-label-only	No detected carrier class is generated.
identity	composition-and-isotope-support-held	carrier-composition-erased	Composition correspondence is impossible.
partition	mass-charge-classes-held	undifferentiated-response	Distinct carrier classes merge.
response	class-position-and-abundance-held	peak-position-only	Multiplicity support is lost.
fragments	precursor-fragment-map-held	fragment-origin-erased	Products cannot be related to precursors.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
conditions	ionization-and-instrument-boundary	universal-spectrum	Carrier production is method-bound.
record	complete-carrier-response-trace	spectrum-image-only	The composition map cannot be checked.
extra	no-extra-rule	free-peak-assignment	A library guess can select composition.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:b30aae3c4fac29492c023e1ddecc2f8fcb59015da79eeadf2fb7ea23baaca22.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:7023aee7d974957bfd7bd46f6c83b997fa078f56c68a69d46fb051980153b357.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:5fbc7a983050ed25c72382178f46875fdf3fbb11d6ec7c91dc77044761a41e49.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e39d6c8b12dfe47a2149f642070ebdeba1fad62702440d164fd805569d2c2f98.

Independent reconstruction. Implementation hash

sha256:63f8223999cb6f43867dfa34e17c98dc20e6a5a1c49bc7286e9867776ab6c544; independent certificate
sha256:68951ba69b1e384489a5e141bf0855dc916c1454d8548dcaab442b39f328392d; external-validation
hash sha256:575485ce21677ea5dc647e5503cd816bdbdfcad4275fdc85e33dcebbba0e488a6. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-MASS-SPECTROSCOPY-COMPOSITE-2026

Comparison records:

- mass-spectroscopy-iupac-composite: predicted ions-separated-by-mass-to-charge__mass-spectrum-retains-relative-abundance__composition-fragment-correspondence__ionization-method-bounded; source-derived ions-separated-by-mass-to-charge__mass-spectrum-retains-relative-abundance__composition-fragment-correspondence__ionization-method-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks ion mass/charge separation, abundance, composition/fragment correspondence or method boundary, or if a tampered row is accepted.

Measurement receipt: sha256:935aa89464eb739cee31ae0fcd0b615c9faa861c311c8ee912f1a4b0696e1671.

Isolation certificate: sha256:ed727405e4832ff2643ff0c878e6dabf4536202d84d17435ca117aecf5a345aa.

Custody certificate: sha256:2ecd0f934df3d18776416582e1446f05b008bed807ba5ee74f965d1d823e647d.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/M03749>
(sha256:bb7b4631fab65d61fd3c0f8b6257ae91bc87da91ceea23a961f4c824f20b4e56)

Registered target identities:

- mass-spectroscopy-iupac-composite from IUPAC-GOLD-BOOK-MASS-SPECTROSCOPY-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite generated ionized-carrier support with complete composition, isotope, mass/charge, response and precursor-fragment records.

Immutable identities. Source manifest

sha256:d6cf30eae512e29a505f73edde6ff30debe87500b0a93b1c4f52c6ea9cd106a4; derivation seal
 sha256:96e7de3bb4ce7877d499c14eb6f6c224c1e8ef78f3ef984975de0b5c004e6268; engine receipt
 sha256:dd28b321a35b48d8cb382cfa7b64b2c9ac917bce959c4b52ceeddb9fdcab7a1c at
 receipts/engine/model_admitted/SFT-CHEM-SPEC-MASS-001-dd28b321a35b48d8.json; empirical hash
 sha256:0a046bcbd9abc5345631c9047e60715eee70e6b6a322568ca914f282e434770b; measurement receipt
 sha256:935aa89464eb739cee31ae0fcd0b615c9faa861c311c8ee912f1a4b0696e1671.

80. Infrared recurrence and molecular distinction

Claim identity: SFT-CHEM-SPEC-INFRARED-001

Question and theorem. Infrared organization is the complete condition-bound pattern of incident recurrence classes coupled to retained molecular vibrational state transitions, with whole-pattern rather than isolated-line identity.

incident-infrared-support-held__complete-molecular-carrier-held__vibrational-state-transition-held__dipole-response-change-required__complete-frequency-response-pattern

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001

- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001
- SFT-CHEM-PHOTOCHEM-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite molecular carrier with generated internal recurrence states, paired transitions, interaction-change tests and a declared observation domain.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `incident-infrared-support-held__complete-molecular-carrier-held__vibrational-state-transition-held__dipole-response-change-required__complete-frequency-response-pattern__whole-pattern-molecular-distinction__state-method-condition-boundary__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
input	incident-infrared-support-held	unregistered-incident-recurrence	No excitation comparison is defined.
molecule	complete-molecular-carrier-held	molecular-identity-erased	Response cannot distinguish carriers.
transition	vibrational-state-transition-held	static-geometry-only	A spectrum requires a state change.
coupling	dipole-response-change-required	all-transitions-assumed-visible	Observation requires a changed interaction trace.
spectrum	complete-frequency-response-pattern	one-line-answer	Alternative recurrences are erased.
identity	whole-pattern-molecular-distinction	universal-one-line-identifier	Different carriers may share a line.
conditions	state-method-condition-boundary	condition-free-spectrum	State and method affect observation.
extra	no-extra-rule	free-infrared-line-rule	A lookup can force assignment.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:b2ffe399a008258b480ff5f4c6f502fe06f1554a8544521f5d1ece040955890a.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:a603e622aeaed9f12c66289143a2bd1b22cec3d5a1b7d996404b77fa9d144334.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:1d6b9d317916812bb704757723d4399c03e03f6d00a9e5e985f0bbc82060bbad.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:3d6cea107788fcc04f575ed6126f2ad493ed4dc3187751c7aa2e56e002b9db24.

Independent reconstruction. Implementation hash

sha256:9623a7a8ad4f940f91f869bb1b7224df95a3399ea5ee013763d34c92a621724b; independent certificate sha256:81480db63e0f925d2062ec153fa02b0bb1df1d38b15c2471b5794949c032bd86; external-validation hash sha256:71177e5710fecab7d0b2f8524b617e6e58168c96cea9b893cc451df2df313777. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-INFRARED-SPECTROSCOPY-COMPOSITE-2026

Comparison records:

- **infrared-spectroscopy-iupac-composite:** predicted infrared-absorption-corresponds-to-molecular-vibration__frequency-pattern-retained__molecular-distinction-uses-whole-spectrum__state-method-bounded; source-derived infrared-absorption-corresponds-to-molecular-vibration__frequency-pattern-retained__molecular-distinction-uses-whole-spectrum__state-method-bounded; exact match True
- **deliberately tampered unfavorable control rejected**

Falsification boundary: The claim fails if authority evidence lacks infrared/vibrational correspondence, a retained frequency pattern, whole-spectrum molecular distinction or state/method boundary, or if a tampered row is accepted.

Measurement receipt: sha256:d155039532903ffcc649b12162b8ddff3ab38ebb2fa396e9fae96df9f4366e10.

Isolation certificate: sha256:c3435bea650fe17928aec663c66db5d34ab8b9c826c50f5c47943d29ac792a10.

Custody certificate: sha256:f1d5307874731029ded147f3852ac958f311f3387fc8293d821ec321f6b277e1.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/08618> (sha256:0be3cc15940bfcf345e61af54ef378022de8f2c32b7e566b7d4ed25f69c553ab)

Registered target identities:

- infrared-spectroscopy-iupac-composite from IUPAC-GOLD-BOOK-INFRARED-SPECTROSCOPY-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite molecular carrier with generated internal recurrence states, paired transitions, interaction-change tests and a declared observation domain.

Immutable identities. Source manifest

sha256:20ab9e1522821fc629e431f3c874ddacc4e2126c312604efb5e1b78120013ee0; derivation seal sha256:2a4a79aca67992d5b049f068f9ed8774e39449f43b91521b15d66df98735bac0; engine receipt sha256:399a21477f3aedee80bdce487e17bbd4ff81213bcc79d35326f359dc5b2949df at receipts/engine/model_admitted/SFT-CHEM-SPEC-INFRARED-001-399a21477f3aedee.json; empirical hash sha256:422c785663107b7d09e8067f9be1ae47524536b14ddb00039a3902f03c705a5a; measurement receipt sha256:d155039532903ffcc649b12162b8ddff3ab38ebb2fa396e9fae96df9f4366e10.

81. Electronic absorption and molecular-state distinction

Claim identity: SFT-CHEM-SPEC-UVVIS-001

Question and theorem. Ultraviolet-visible organization is the complete condition-bound response pattern of molecular or ionic carriers under retained transitions between ordered electronic states.

ultraviolet-visible-support-held__molecular-or-ionic-carrier-held__paired-electronic-states__higher-state-transition-held__wavelength-response-pattern-held

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001
- SFT-CHEM-PHOTOCHEM-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite chemical carrier with paired generated electronic state classes, ordered transitions and a declared ultraviolet-visible observation domain.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `ultraviolet-visible-support-held__molecular-or-ionic-carrier-held__paired-electronic-states__higher-state-transition-held__wavelength-response-pattern-held__carrier-state-response-map__state-method-condition-boundary__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
input	ultraviolet-visible-support-held	unregistered-radiation	No excitation support is identified.
carrier	molecular-or-ionic-carrier-held	chemical-carrier-erased	Absorption cannot be assigned.
state	paired-electronic-states	single-state-only	No transition occurs.
transition	higher-state-transition-held	energy-order-erased	Absorption direction is ambiguous.
response	wavelength-response-pattern-held	absorption-number-only	Spectral support is lost.
composition	carrier-state-response-map	response-detached-from-carrier	Molecular distinction cannot be tested.
conditions	state-method-condition-boundary	universal-response	Environment and method affect observation.
extra	no-extra-rule	free-electronic-band-rule	A fitted band can select the answer.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 8f109438a2184ff5bd94ed5c3baa2779939b48fa5908c03a9494ead18ff4c43a.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: bdaad010665344c3bc6d4315e843eea1f36ae20f9d6016475c3f06bac89b5953.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 256566afdc578e393902758a27a005c03759954a45509d5445c7b8427a425f5a.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 4cf9ae33fa257656b34c7fe81af112869f07b904db81a4937acccd502270d63a.

Independent reconstruction. Implementation hash

sha256: 09021bd00f42035e6774890cefc66fd971761de04775ef636494b024c7a45c8; independent certificate sha256: cdb28a031fb92bd003449669ff7410e83e34c5809bdcd5cc065f98c9471e5c7c; external-validation hash sha256: ae502e9249d8a3567d62f58b549da13c3ff33f88454d3c7589344b0c10cd1b25. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-UVVISIBLE-SPECTROSCOPY-COMPOSITE-2026

Comparison records:

- uv-visible-spectroscopy-iupac-composite: predicted ultraviolet-visible-absorption-involves-electronic-transition__carrier-and-state-retained__response-pattern-recorded__state-method-bounded; source-derived ultraviolet-visible-absorption-involves-electronic-transition__carrier-and-state-retained__response-pattern-recorded__state-method-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks UV-visible absorption, electronic transition, retained carrier/state response or method boundary, or if a tampered row is accepted.

Measurement receipt: sha256:47cf49d63bc991b5d2fedb5f293c268b60d083c3f614e10c2818467452f65c18.

Isolation certificate: sha256:8fa79aca28ebf1ad65f4233ae1de11b95fa9488ec28706ebf1071b49b17270f6.

Custody certificate: sha256:491de535489e29b482bffee92445467ec4a05058c506e22dac0ef368820b5079.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/S05696> (sha256:bcf4022ed1743c6445d1c6a2dc88d616adad7f833c299734f1b3fc3b45652a7d)

Registered target identities:

- uv-visible-spectroscopy-iupac-composite from IUPAC-GOLD-BOOK-UVVISIBLE-SPECTROSCOPY-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite chemical carrier with paired generated electronic state classes, ordered transitions and a declared ultraviolet-visible observation domain.

Immutable identities. Source manifest

sha256:38c77692efe605cee2e938d02e28084972e3ee9d2151f54130e81248cf571532; derivation seal

sha256:18df34814176a7143ba2578488a5c75a0e5a5f9a7a41b50cd2f75b8242bddbc3; engine receipt

sha256:38f612b5b2fdc62dfa158d3c592c930c8611a00d0b1447d9ba59af313bc173f9 at

receipts/engine/model_admitted/SFT-CHEM-SPEC-UVVIS-001-38f612b5b2fdc62d.json; empirical hash

sha256:3c2471ff91e9c3df63cc154c980d744ffa71db30c3f8a29255f327785fc697e7; measurement receipt

sha256:47cf49d63bc991b5d2fedb5f293c268b60d083c3f614e10c2818467452f65c18.

82. Rotational-vibrational molecular spectra

Claim identity: SFT-CHEM-SPEC-ROT-VIB-001

Question and theorem. A rotational-vibrational spectrum is the complete condition-bound transition organization of retained molecular orientation-recurrence and internal relative-displacement recurrence classes.

complete-molecular-geometry-held__rotational-state-classes-held__vibrational-state-classes-held__rotational-vibrational-composition__allowed-state-transition-trace

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001

- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001
- SFT-CHEM-SPEC-INFRARED-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite molecular geometry with generated rotational and vibrational recurrence classes, paired transitions and declared observation support.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-molecular-geometry-held__rotational-state-classes-held__vibrational-state-classes-held__rotational-vibrational-composition__allowed-state-transition-trace__complete-transition-frequency-pattern__state-method-condition-boundary__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-molecular-geometry-held	point-label-only	Internal molecular motion is absent.
rotation	rotational-state-classes-held	orientation-history-erased	Rotational recurrence cannot be distinguished.
vibration	vibrational-state-classes-held	relative-displacement-erased	Vibrational recurrence cannot be distinguished.
composition	rotational-vibrational-composition	motions-collapsed-together	Joint spectral structure is lost.
transition	allowed-state-transition-trace	states-without-transition-map	No spectral event is generated.
spectrum	complete-transition-frequency-pattern	one-frequency-only	Resolved organization is erased.
conditions	state-method-condition-boundary	universal-spectrum	Population and resolution are bounded.
extra	no-extra-rule	free-rotor-oscillator-equation	An imported model can select the spectrum.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:4323d8285eafe3506586879204c1788f573d36d9a713b75d9b52ab8745e41bfa`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ca67c32795fdd77c1b6ba5a17ba0a5984f0c5433217db50572ebcc84a840f588`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:bef4f75c40ebc0d0e5933659a7ea33f77a3bad65f0ed79a5877261fbbf03c980`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:179ca61d9ff6baa1547972cfa7013cebb029156f005a5193a807e86ff46661d3.

Independent reconstruction. Implementation hash

sha256:833c5b45b93d721aa1ab38cc1261be7e6e312fd3be9721a6ba087236b26d7eb1; independent certificate sha256:416fbcd13691a5a9887624f22611dcd6b60a31f2f6e113167ace8efeb96a0183; external-validation hash sha256:dd26ee4f0c3b07a0ade826d75c164ed243d6e8a2e6aa81eabc2b6c4851f3295f. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ROTATION-VIBRATION-COMPOSITE-2026

Comparison records:

- rotation-vibration-iupac-composite: predicted molecular-spectra-retain-rotational-and-vibrational-transitions__state-composition-held__transition-pattern-complete__population-resolution-bounded; source-derived molecular-spectra-retain-rotational-and-vibrational-transitions__state-composition-held__transition-pattern-complete__population-resolution-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks rotational/vibrational transitions, state composition, transition pattern or population/resolution boundary, or if a tampered row is accepted.

Measurement receipt: sha256:59a8c57f7fb1a5825682542c80a5ca0f21dde5d8e1ac8596ab931b577c68988b.

Isolation certificate: sha256:edd6d79a3963e17fe45cd0fd3e110803c91f84edcc54f7e91acc8f93a7ea745d.

Custody certificate: sha256:43efccaaed17b5ee382443e7f09d4334a44635de5d3e5edfd2bce8a772e8a5dc.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/08676> (sha256:fa1bdd34ab2e93cce09dd22311bba6d9c1e3ee4221a285510eca2b7ce8e3e647)

Registered target identities:

- rotation-vibration-iupac-composite from IUPAC-GOLD-BOOK-ROTATION-VIBRATION-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite molecular geometry with generated rotational and vibrational recurrence classes, paired transitions and declared observation support.

Immutable identities. Source manifest

sha256:915570416fa41ea0818165beb4c6fd5e18035df45a95139b52a794f58bfab3c6; derivation seal

sha256:dcc35fda03d67c27218e9ea3cc7957c6fee858ca7fb07a5b8f46cf39a1d111fa; engine receipt

sha256:20367bb1fe68cb5795f63f6d62da369e7cfc5295098f621cc6736c7fd3075e3d at

receipts/engine/model_admitted/SFT-CHEM-SPEC-ROT-VIB-001-20367bb1fe68cb57.json; empirical

hash sha256:af9b74ac34ae3ee05c79d44d5eec848774da635587b0e326ad1cc2a383797638; measurement

receipt sha256:59a8c57f7fb1a5825682542c80a5ca0f21dde5d8e1ac8596ab931b577c68988b.

83. Complete analytical result and falsification record

Claim identity: SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001

Question and theorem. A complete analytical result retains the measurand, sample/analyte/matrix, method, conditions, reference chain, every raw and derived observation, uncertainty, failed and tampered controls, and an explicit falsification condition.

measurand-and-purpose-held__sample-analyte-matrix-record__method-and-condition-record__calibration-and-reference-chain__all-raw-and-derived-rows-held__uncertainty-and-resolution-held__failed-tampered-and-falsification-record

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001

- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001
- SFT-CHEM-ANALYTICAL-SAMPLE-001
- SFT-CHEM-ANALYTICAL-CALIBRATION-001
- SFT-CHEM-ANALYTICAL-SELECTIVITY-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite analytical execution with registered question, material, method, reference, observations, transformations, uncertainty and controls.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `measurand-and-purpose-held__sample-analyte-matrix-record__method-and-condition-record__calibration-and-reference-chain__all-raw-and-derived-rows-held__uncertainty-and-resolution-held__failed-tampered-and-falsification-record__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
question	measurand-and-purpose-held	unstated-measurand	The result has no declared target.
sample	sample-analyte-matrix-record	sample-identity-erased	The result cannot be located.
method	method-and-condition-record	method-erased	The transformation cannot be reproduced.
reference	calibration-and-reference-chain	reference-erased	Traceability is lost.
observation	all-raw-and-derived-rows-held	accepted-values-only	Unfavorable evidence disappears.
uncertainty	uncertainty-and-resolution-held	single-value-certainty	Admissible alternatives are hidden.
falsification	failed-tampered-and-falsification-record	success-only-report	The claim cannot fail publicly.
extra	no-extra-rule	free-reporting-omission	Discretion can hide a contradiction.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:0cab94f55d8f3ec6788fc21b068becad07822404dccecdfaa4aaef265212cad1.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:1ee4fcf2da1f0174c3b1e3c980363e6bca1731c16e17bea678247da58b52565d.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:93a5466a246fc746f2db197b298846a920a9544ce5df6ee0b6ebd4bbb02b6726.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:739798d50cb8adcec6ce6d95e38de2516a30dd5354ebca6d8c46646e40ec048e.

Independent reconstruction. Implementation hash

sha256:2ca15f1a1c1c4a464dbbe10d8860ab9a34743a8ac02dfd0ab04a032a5c999881; independent certificate sha256:a82ede737db1b63ea05e93848b8c61587572baf23629fcc9fd3127be01fc3249; external-validation hash sha256:e8642f27691fbfe5534fd622286bc0ec863dab486d1ad0e52a6ffd040c734507. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-NIST-COMPLETE-ANALYTICAL-RECORD-COMPOSITE-2026

Comparison records:

- complete-analytical-record-authority-composite: predicted analytical-result-identifies-measurand-sample-and-method__traceability-and-uncertainty-retained__all-results-and-controls-preserved__falsification-condition-explicit; source-derived analytical-result-identifies-measurand-sample-and-method__traceability-and-uncertainty-retained__all-results-and-controls-preserved__falsification-condition-explicit; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks result identity, traceability/uncertainty, retained controls or explicit falsification boundary, or if a tampered row is accepted.

Measurement receipt: sha256:56baf90991e88c03e7c2ba7742074100fa1b66c2ce13533a3632db8e2765c78b.

Isolation certificate: sha256:e7acb7809ec492088087665ab79c04112dd235d598217b8de002cd701b73fa5b.

Custody certificate: sha256:c5d98b8fb461d5f0092d168048e8d967682b22f6e07ca84586fffd2f2539bdc8a.

Registered source descriptions:

- International Union of Pure and Applied Chemistry and National Institute of Standards and Technology:
<https://goldbook.iupac.org/terms/view/M03796>
(sha256:f27104514cc2af6ad31064e74244121a076fc3c795f3f636e4f7f8face60e9a1)

Registered target identities:

- complete-analytical-record-authority-composite from
IUPAC-NIST-COMPLETE-ANALYTICAL-RECORD-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity

- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite analytical execution with registered question, material, method, reference, observations, transformations, uncertainty and controls.

Immutable identities. Source manifest

sha256:721214fc995c71f5ca0e7669d4a83d70e6e633f1f8ee48e394a6097e1555dc3c; derivation seal
 sha256:4648cc1a81f414ba0df403a374338344f542e6981df1709bef71068cc7f3bc99; engine receipt
 sha256:7787001c4d7b90235d1bf7f3e19482cf5f05cb5f37f98a980be1efc4de413669 at receipts/engine/model_admitted/SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001-7787001c4d7b9023.json; empirical
 hash sha256:909fa1807b7ed309219a6920df56d30c351aa7ae0256f8c60b439d7b351cb6bd; measurement
 receipt sha256:56baf90991e88c03e7c2ba7742074100fa1b66c2ce13533a3632db8e2765c78b.

17. g-block, Smithium and the periodic endpoint - 11 admitted claims

Known periodic support, exact structural extrapolation and standing unobserved prediction are separate evidence classes. This entire registered subcategory is included; no proper subset is presented as field completion.

84. Generated g-block structural prediction

Claim identity: SFT-CHEM-PRED-G-BLOCK-001

Question and theorem. The admitted Fold orbit capacities ordered by increasing joint principal/orbit cover, with principal rank breaking a cover tie, generate noble closures 2, 10, 18, 36, 54, 86 and 118; fill 8s at 119-120; and open the 5g block at 121. The known prefix is externally checked and the g-block placement remains a sealed unobserved prediction.

The exact walk has capacities 2,6,10,14,18; noble closures 2,10,18,36,54,86,118; 8s occupations at 119-120; and first 5g occupation at 121.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001
- SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001
- SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001
- SFT-CHEM-ELEM-PERIODIC-BOUNDARY-001

Generated grammar. Generate the complete eight-axis product of positive orbit carrier, capacity, joint-cover order, predicted coordinate, provenance, target custody, evidence status and extra-rule forms.

Boundary: All finite subshell walks built from the admitted positive-rank capacities, complete valid principal/orbit cells, increasing joint cover and source-rank tie order before the atomic endpoint.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `positive-principal-and-orbit-ranks__admitted-Fold-orbit-capacity__increasing-joint-cover-then-principal-rank__first-5g-occupation-at-121__complete-fill-and-dependency-trace__official-table-opened-after-seal__known-prefix-validated-future-coordinate-unobserved__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	positive-principal-and-orbit-ranks	borrowed-orbital-notation	A borrowed notation has no Fold support trace.
capacity	admitted-Fold-orbit-capacity	imported-subshell-widths	Imported widths would supply the answer.
order	increasing-joint-cover-then-principal-rank	memorized-aufbau-list	A memorized order is a target-bearing table.
coordinate	first-5g-occupation-at-121	selected-future-coordinate	A selected coordinate is not forced.
provenance	complete-fill-and-dependency-trace	answer-only-prediction	An answer-only prediction cannot be reproduced.
target	official-table-opened-after-seal	official-table-readable-before-seal	The known table could select the walk.
status	known-prefix-validated-future-coordinate-unobserved	future-prediction-called-measured	That conflates unobserved consequence with empirical confirmation.
extension	no-extra-rule	free-exception-or-fit	An exception can be tuned to a desired element.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:7029631ba2389d152c39777bc1b5fdc51d436ec265ac0641b9f3cef50e51fc70`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:250a87885493ce97efb4e3db74f1eb38976bc61c1cb2b6271fce279423030e19`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8e07b41044d8c88aea1e8e8bc250609590e03205172694eac518f05cc0012c3a`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7f0ea0abcc06647ebf2729218c6ad037a15810eced930bfe1b51e9914c4447ab`.

Independent reconstruction. Implementation hash

`sha256:e323670a7c5c5d5a5c69b3138b71f1339902d495c29f0e7a9569905a7ad707d3`; independent certificate `sha256:eec03859c29d9208b967fb97d8f354e997dcea1b0f8026466fb738a677ff1bfff`; external-validation hash `sha256:f23016c51b52d05ad316080e574e2346c4434683f435803c01748dec6811b75c`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- IUPAC-G-BLOCK-KNOWN-PREFIX: predicted
known-noble-closures-2-10-18-36-54-86-118__g-block-121-standing-unobserved-prediction; source-derived
known-noble-closures-2-10-18-36-54-86-118__g-block-121-standing-unobserved-prediction; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The generated known closure prefix differs from the official source, the 5g opening is not 121, a target enters before sealing, a future prediction is called observed, or a tampered row is accepted.

Measurement receipt: sha256:649af8445c652c96c1e858a2b9e1ac81e4b0c641592ea13434a5f19aeae88ede.

Isolation certificate: sha256:3a233dfa2e1799c10168e45381e28331dfc83e68e0b869f74465c0b9c9b413f2.

Custody certificate: sha256:dc41fd3dc57da343fccf363ecd9b4f336abb6e8c4b5460562caaa04716017987.

Registered source descriptions:

- International Union of Pure and Applied Chemistry:
https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf
(sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- IUPAC-G-BLOCK-KNOWN-PREFIX from IUPAC-PERIODIC-TABLE-2022

Meaning. The final subbranch applies separately admitted exact physical prerequisites to a complete positive-rank fill walk and distinguishes known-domain validation from unobserved prediction. This particular result is closed only at its exact registered grammar and evidence boundary. The official record checks only the known prefix; every coordinate beyond element 118 remains a standing unobserved prediction.

Explicit exclusions.

- no imported aufbau list or observed shell width
- no V1/V2 answer as a derivational input
- no semantic numerical zero, negative or nonexact proof value
- no claim that element 121 has been observed

Immutable identities. Source manifest

sha256:8473fe67b5792ece7552d6f8565103faab596941256b2d6ecd627ad8043d906e; derivation seal

sha256:63f231e18c8993ad0601d41f78a53b1bf433006328496c7f92ad28519b46201a; engine receipt

sha256:5e21d5cb1ad9758257dfa1fb732792adf4e9b7aa7ae0300fdaf21cb2426772bb at

receipts/engine/model_admitted/SFT-CHEM-PRED-G-BLOCK-001-5e21d5cb1ad97582.json; empirical

hash sha256:d8bbb85c1d1fd073660446af86ccb82bb4e4da6c6f25fab2a2730c598be2d8e3; measurement

receipt sha256:649af8445c652c96c1e858a2b9e1ac81e4b0c641592ea13434a5f19aeae88ede.

85. Smithium element-126 standing prediction

Claim identity: SFT-CHEM-PRED-SMITHIUM-001

Question and theorem. The independently sealed nuclear recurrence places the next proton/neutron double closure at 126/184. The generated electronic walk gives element 126 the active configuration 8s2 5g6, mass coordinate 310, eight valence carriers and structurally admissible positive oxidation counts +2 through +8.

Smithium is the standing prediction $Z=126$, $N=184$, $A=310$ with active configuration 8s2 5g6, valence count 8 and predicted positive oxidation counts +2 through +8.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001
- SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001
- SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-QUANTUM-EXCLUSION-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001
- SFT-CHEM-ELEM-PERIODIC-BOUNDARY-001
- SFT-PHYS-NUCLEAR-CLOSURE-SEQUENCE-001
- SFT-PHYS-VALIDATION-NUCLEAR-CLOSURES-001
- SFT-CHEM-PRED-G-BLOCK-001

Generated grammar. Generate the complete eight-axis product of positive orbit carrier, capacity, joint-cover order, nuclear/electronic coordinate, provenance, target custody, evidence status and extra-rule forms.

Boundary: All double-closure/electronic-fill consequences at the next generated proton closure after 82 using the admitted nuclear recurrence and complete subshell walk.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `positive-principal-and-orbit-ranks__admitted-Fold-orbit-capacity__sealed-joint-cover-fill-walk__next-double-closure-126-184-with-8s2-5g6__complete-fill-and-dependency-trace__official-table-opened-after-seal__known-prefix-validated-future-coordinate-unobserved__no-extra-rule`. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	positive-principal-and-orbit-ranks	borrowed-orbital-notation	A borrowed notation has no Fold support trace.
capacity	admitted-Fold-orbit-capacity	imported-subshell-widths	Imported widths would supply the answer.
order	sealed-joint-cover-fill-walk	memorized-aufbau-list	A memorized order is a target-bearing table.
coordinate	next-double-closure-126-184-with-8s2-5g6	selected-future-coordinate	A selected coordinate is not forced.
provenance	complete-fill-and-dependency-trace	answer-only-prediction	An answer-only prediction cannot be reproduced.
target	official-table-opened-after-seal	official-table-readable-before-seal	The known table could select the walk.
status	known-prefix-validated-future-coordinate-unobserved	future-prediction-called-measured	That conflates unobserved consequence with empirical confirmation.
extension	no-extra-rule	free-exception-or-fit	An exception can be tuned to a desired element.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2b9281d1a762cc7bdb1579504ba24551a5b444ea492bab0951a4287e69276fbc`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:fef001517e2f8109bea8ec2718b66087f655d84ce9032d5dcf4bc6f32e033f52`.

- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:3f0509e56d6e645f0f27d9d64972dc7be8ce260b8fc027edfd1ccec7e3b2e6d.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e3c97ef11cec09172621574fe62e1ceeea7995cc9ff6eaac97785d3456443933.

Independent reconstruction. Implementation hash

sha256:dbf2dee9cba1c6c7ccb65a677bb3214e5f4aeefb986c74c37f57d5305bf0a956; independent certificate
sha256:c46787ad2d1ee4ec941ade01601086040d8389a934fb9f2de55f817cf3467695; external-validation
hash sha256:4d09d460ee45542f9f59fde147d2151f996640b6f6f21298d8e2554b250587c9. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- IUPAC-SMITHIUM-OBSERVATION-BOUNDARY: predicted
official-table-through-118__smithium-126-standing-unobserved-prediction; source-derived
official-table-through-118__smithium-126-standing-unobserved-prediction; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: A future verified element 126 lacks the sealed nuclear/electronic coordinates, the official known boundary differs, target data enter before seal, or a tampered row is accepted.

Measurement receipt: sha256:093ffb2143b4051ffb48fbda2574f3161f93116bd2c313f283e619942988aae8.

Isolation certificate: sha256:29098725cff155020210fb3d3e82157f2f2f839c6618303885437bf0425ac3e2.

Custody certificate: sha256:43a3e5083ec72c77609455b3f6f3381891e9f04fe45617c9a43a6869c6705b40.

Registered source descriptions:

- International Union of Pure and Applied Chemistry:
https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf
(sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- IUPAC-SMITHIUM-OBSERVATION-BOUNDARY from IUPAC-PERIODIC-TABLE-2022

Meaning. The final subbranch applies separately admitted exact physical prerequisites to a complete positive-rank fill walk and distinguishes known-domain validation from unobserved prediction. This particular result is closed only at its exact registered grammar and evidence boundary. The official record checks only the known prefix; every coordinate beyond element 118 remains a standing unobserved prediction.

Explicit exclusions.

- no observed Smithium claim
- no imported island-of-stability coordinate
- no fitted nuclear coupling or chemical exception
- no semantic numerical zero, negative oxidation magnitude or nonexact proof value

Immutable identities. Source manifest

sha256:a7334009abed2d4a56a4a25413cef27be9a3c6f4c60031a61189f6f0a5885770; derivation seal

sha256:c8c868f83448a38c195fb4ccdb5029a9566441d936da00364eea4374ead92b65; engine receipt

sha256:8028271d2d1e27a0c1351974b39d09131916ed17cd349b36268eaa6b8b2b2613 at

receipts/engine/model_admitted/SFT-CHEM-PRED-SMITHIUM-001-8028271d2d1e27a0.json; empirical

hash sha256:d9f8d8930ef9da0f06f3a4937ce1f2236c4a6b419e9fc6381dbae28c30059cc2; measurement

receipt sha256:093ffb2143b4051ffb48fbda2574f3161f93116bd2c313f283e619942988aae8.

86. Generated periodic endpoint standing prediction

Claim identity: SFT-CHEM-PRED-PERIODIC-ENDPOINT-001

Question and theorem. The independently sealed exact inverse fine-structure ratio and One binding ceiling admit positive whole atomic coordinate 137 and reject its successor 138. Chemistry therefore records 137 as the model's structural endpoint while the official observed table remains source-bounded through 118.

The model's exact structural periodic endpoint is positive atomic number 137; the IUPAC 2022 observed boundary remains 118, so 119-137 including the endpoint are unobserved standing predictions.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001
- SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001
- SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001
- SFT-CHEM-ELEM-PERIODIC-BOUNDARY-001
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-CHEM-PRED-G-BLOCK-001
- SFT-CHEM-PRED-SMITHIUM-001

Generated grammar. Generate the complete eight-axis product of positive atomic carrier, exact binding boundary, greatest-whole construction, predicted coordinate, provenance, target custody, evidence status and extra-rule forms.

Boundary: All Chemistry endpoint consequences of the admitted exact atomic-existence boundary and positive atomic-number order, separated from the source-dated observed table boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `positive-principal-and-orbit-ranks__admitted-Fold-orbit-capacity__sealed-exact-binding-boundary__structural-endpoint-137-successor-138-excluded__complete-fill-and-dependency-trace__official-table-opened-after-seal__known-prefix-validated-future-coordinate-unobserved__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	positive-principal-and-orbit-ranks	borrowed-orbital-notation	A borrowed notation has no Fold support trace.
capacity	admitted-Fold-orbit-capacity	imported-subshell-widths	Imported widths would supply the answer.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
order	sealed-exact-binding-boundary	memorized-aufbau-list	A memorized order is a target-bearing table.
coordinate	structural-endpoint-137-successor-138-excluded	selected-future-coordinate	A selected coordinate is not forced.
provenance	complete-fill-and-dependency-trace	answer-only-prediction	An answer-only prediction cannot be reproduced.
target	official-table-opened-after-seal	official-table-readable-before-seal	The known table could select the walk.
status	known-prefix-validated-future-coordinate-unobserved	future-prediction-called-measured	That conflates unobserved consequence with empirical confirmation.
extension	no-extra-rule	free-exception-or-fit	An exception can be tuned to a desired element.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 9d93b8a29fb07ab3a7f8f5d4efe373c45864448cb4bc6b54ac7656b4f2ae0a7a.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 314897440a8fb21e21bbe6d1be41620f1cc2b1435261325748c8d401dbc6ab28.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 4c7ebcab86120fc5c11fb4089e51b6de3225b9f60c844de96937fe2e3d2370a8.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: a9260d97a3d1af2d03cce0d6cf6037fcfb8039e34a7f91e66000ce1a4e28ad78.

Independent reconstruction. Implementation hash

sha256: 419962bc01b6a1a8c4399dddc62b15cb7b621808d204ef66530146b89b8d85a6; independent certificate sha256: ee90ca1e5a00142feb28aacce9965f6f8aab8f8d9cdcec7a1ec2673ebc898507; external-validation hash sha256: ec27a31b318b41962a99c82c49af1efda936aaa5aaf3207b4362de52ef3576d1. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- IUPAC-ENDPOINT-OBSERVATION-BOUNDARY: predicted
official-table-through-118__endpoint-137-standing-unobserved-prediction; source-derived
official-table-through-118__endpoint-137-standing-unobserved-prediction; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: A verified neutral element beyond 137 satisfies the registered One-ceiling boundary, the exact prerequisite fails replay, the observed/source boundary is conflated with the structural law, or a tampered row is accepted.

Measurement receipt: sha256: 18644e5b05031b5cb0ba43dea63dec4de8cbc40f8807f871696329ffff7e7aea.

Isolation certificate: sha256: f7d2a51fa88e8cf36e00160185db83ae8cc8c70bec1f1c61122db418345f7c05.

Custody certificate: sha256: c0c8cbc503b581965072fd8312c8ee721a8fd02b2c57896ca012411d8cf9db5c.

Registered source descriptions:

- International Union of Pure and Applied Chemistry:
https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf

(sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- IUPAC-ENDPOINT-OBSERVATION-BOUNDARY from IUPAC-PERIODIC-TABLE-2022

Meaning. The final subbranch applies separately admitted exact physical prerequisites to a complete positive-rank fill walk and distinguishes known-domain validation from unobserved prediction. This particular result is closed only at its exact registered grammar and evidence boundary. The official record checks only the known prefix; every coordinate beyond element 118 remains a standing unobserved prediction.

Explicit exclusions.

- no observed absence promoted to impossibility
- no bounded endpoint scan
- no imported critical-charge value
- no claim that element 137 has been observed

Immutable identities. Source manifest

sha256:6d4c0670b937acc0b11801cf7e1104ba7fecced9c8ea9b43bbc1aa8c17206334; derivation seal
 sha256:7a7286410b88849023373a1509223c286928a20650e17510b3c64f01792552e6; engine receipt
 sha256:df9ea26853f6ac0678893d0e005f2a25a4d86ab3084ab3147353432050d1bf16 at receipts/engine/model_admitted/SFT-CHEM-PRED-PERIODIC-ENDPOINT-001-df9ea26853f6ac06.json; empirical hash
 sha256:bb394b25866a92779ebd18d42bf4f071a4bb250eb67198d7c24d9bb624f61772; measurement receipt
 sha256:18644e5b05031b5cb0ba43dea63dec4de8cbc40f8807f871696329ffff7e7aea.

87. Complete Smithium synthesis-conservation grammar

Claim identity: SFT-CHEM-SMITHIUM-SYNTHESIS-CONSERVATION-001

Question and theorem. The complete conservation grammar contains exactly 22,875 labelled positive two-body entrance partitions. Every row sums to $Z=126$, $N=184$ and $A=310$. No partition is promoted to a feasible reaction, unique synthesis route, beam energy, cross-section or yield without additional sealed law and measurement.

The complete conservation grammar contains exactly 22,875 labelled positive two-body entrance partitions. Every row sums to $Z=126$, $N=184$ and $A=310$. No partition is promoted to a feasible reaction, unique synthesis route, beam energy, cross-section or yield without additional sealed law and measurement.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-PRED-SMITHIUM-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-PHYS-NUCLEAR-FUSION-001

Generated grammar. Generate the complete eight-axis Smithium synthesis product and every positive labelled two-body Z/N entrance partition.

Boundary: All positive labelled pairs $(Z_1, N_1), (Z_2, N_2)$ with $Z_1 + Z_2 = 126$ and $N_1 + N_2 = 184$, plus all 256 methodological forms.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: sealed-126-184-310-coordinate__complete-positive-carrier-ledger__all-22875-conserving-labelled-entrance-partitions__complete-declared-product__formal-seal-before-target-access__favourable-adverse-absent-unresolved-held__structural-absence-and-explicit-halt__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
coordinate	sealed-126-184-310-coordinate	selected-element-name	A selected name does not reconstruct a nuclear/electronic coordinate.
conservation	complete-positive-carrier-ledger	outcome-only-transformation	An outcome without the complete carrier ledger can hide loss or invention.
relation	all-22875-conserving-labelled-entrance-partitions	free-or-imported-relation	A free or imported relation is not forced by the dependencies.
enumeration	complete-declared-product	selected-favourable-example	One favourable example cannot establish uniqueness.
measurement	formal-seal-before-target-access	target-before-seal	Target access before the derivation seal is fitting.
record	favourable-adverse-absent-unresolved-held	favourable-only-record	Selective retention destroys empirical falsifiability.
absence	structural-absence-and-explicit-halt	invented-numerical-placeholder	A placeholder turns absence into a false number.
extension	no-extra-rule	free-extra-rule	An exception is an unforced parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256 : 5439aebaf61b5e3e7438495082c4853f08fdd9a75c31e62da5c27b1da5547af7.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256 : f68cf521265b85d2d66a78b587fae6a3133426c9e68380e2c500b78e856a2186.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 380aba5e86c5a36057ed6e3bfed690ee860f004670114c132153dd2456f6ff4d.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256 : 6b923f79e135688dff8a19c5daf7c5b58574c6035e2382cc6770ba9e58333f42.

Independent reconstruction. Implementation hash

sha256 : 37f6ad9c651e645fd69788e591e4045ea372fd44d08036cdd7e79b02425bf371; independent certificate
sha256 : 2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dacb5d687df7fca82fa60; external-validation
hash sha256 : 585e23b86246987038790ed2c5090964daabacee96b54140ed3504f7e12d81fc. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. This is a sealed formal law with an implementation-distinct reconstruction and no separate external-observation package at this boundary. It receives no empirical-comparison credit that is not present in its claim package.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The final subbranch applies separately admitted exact physical prerequisites to a complete positive-rank fill walk and distinguishes known-domain validation from unobserved prediction. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no apparatus, projectile, target, beam energy, cross-section or production yield selected as a law
- no lifetime, branching ratio, transition energy, wavelength or line intensity invented without measurement

- no current non-observation relabelled as disproof and no standing prediction relabelled as discovery
- no numerical absence used as a proof value; absence is a held structural record
- no negative, irrational, imaginary, floating, fitted, free or target-selected proof magnitude
- no V1/V2 answer used to select a survivor; prior work supplies only the reconstruction question

Immutable identities. Source manifest

sha256:850dcc3e8b89f3657f6fdad6436d9320358fa2e54e46607df8022e7961268f6d; derivation seal sha256:b57f6796f4ef2f502e0d6796c28ac1100a24cd13d5d227d5ab9da5d096962933; engine receipt sha256:c26635123b4b59798b649c11d3980194cf216242ebbaceb7bec83bdbebcca12e at receipts/engine/model_admitted/SFT-CHEM-SMITHIUM-SYNTHESIS-CONSERVATION-001-c26635123b4b5979.json; empirical hash None; measurement receipt None.

88. Complete Smithium decay-channel coordinate ledger

Claim identity: SFT-CHEM-SMITHIUM-DECAY-CHANNEL-LEDGER-001

Question and theorem. Gamma retains (126,184,310); alpha yields daughter (124,182,306) plus (2,2,4); beta-minus yields (127,183,310); beta-plus or electron capture yields (125,185,310); and labelled fission exhausts the same 22,875 positive conservation partitions. No branch is asserted active and no branching ratio is invented.

Gamma retains (126,184,310); alpha yields daughter (124,182,306) plus (2,2,4); beta-minus yields (127,183,310); beta-plus or electron capture yields (125,185,310); and labelled fission exhausts the same 22,875 positive conservation partitions. No branch is asserted active and no branching ratio is invented.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-SMITHIUM-SYNTHESIS-CONSERVATION-001
- SFT-PHYS-NUCLEAR-RADIOACTIVE-DECAY-TERMINAL-005
- SFT-PHYS-NUCLEAR-FISSION-001
- SFT-CHEM-RADIOACTIVE-CHEMICAL-TRANSFORMATION-002

Generated grammar. Generate all 256 decay-ledger forms and the complete registered gamma, alpha, beta and labelled-fission coordinate classes.

Boundary: The sealed Smithium parent under gamma, alpha, beta-minus, beta-plus/electron-capture and every positive labelled two-fragment conservation partition.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: sealed-126-184-310-coordinate__complete-positive-carrier-ledger__complete-gamma-alpha-beta-and-labelled-fission-ledger__complete-declared-product__formal-seal-before-target-access__favourable-adverse-absent-unresolved-held__structural-absence-and-explicit-halt__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
coordinate	sealed-126-184-310-coordinate	selected-element-name	A selected name does not reconstruct a nuclear/electronic coordinate.
conservation	complete-positive-carrier-ledger	outcome-only-transformation	An outcome without the complete carrier ledger can hide loss or invention.
relation	complete-gamma-alpha-beta-and-labelled-fission-ledger	free-or-imported-relation	A free or imported relation is not forced by the dependencies.
enumeration	complete-declared-product	selected-favourable-example	One favourable example cannot establish uniqueness.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
measurement	formal-seal-before -target-access	target-before-seal	Target access before the derivation seal is fitting.
record	favourable-adverse -absent-unresolved -held	favourable-only-record	Selective retention destroys empirical falsifiability.
absence	structural-absence -and-explicit-halt	invented-numerical-placeholder	A placeholder turns absence into a false number.
extension	no-extra-rule	free-extra-rule	An exception is an unforced parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256: 35efd84bbcb6ef25e5daa342dc8b25e65933c38cbaf396b164d09e24da4c3842e.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256: 08e672c143b97b4d7e2844b718a5005eb59978e264ae784be46c501d7d7ce122.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 29a8caac7bcedf44b212dab07897245700e24adb4a0474b39745e882d5853305.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256: 78a2d0c348662bfd3ea0d2d15f5ca9e10866dbc33ea9f8c605b506e93e22e976.

Independent reconstruction. Implementation hash

sha256: 37f6ad9c651e645fd69788e591e4045ea372fd44d08036cdd7e79b02425bf371; independent certificate
sha256: 2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dacb5d687df7fca82fa60; external-validation
hash sha256: e5146ae47d59c2d61b4c1c8e8eab0e95e7b771eb029248b804130b2b77fc82fd. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. This is a sealed formal law with an implementation-distinct reconstruction and no separate
external-observation package at this boundary. It receives no empirical-comparison credit that is not present in its claim package.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The final subbranch applies separately admitted exact physical prerequisites to a complete positive-rank fill walk and
distinguishes known-domain validation from unobserved prediction. This particular result is closed only at its exact registered
grammar and evidence boundary.

Explicit exclusions.

- no apparatus, projectile, target, beam energy, cross-section or production yield selected as a law
- no lifetime, branching ratio, transition energy, wavelength or line intensity invented without measurement
- no current non-observation relabelled as disproof and no standing prediction relabelled as discovery
- no numerical absence used as a proof value; absence is a held structural record
- no negative, irrational, imaginary, floating, fitted, free or target-selected proof magnitude
- no V1/V2 answer used to select a survivor; prior work supplies only the reconstruction question

Immutable identities. Source manifest

sha256: 983cbef25792348b3a70d826b37386fa79234588850d30b23460d050fdf44306; derivation seal
sha256: a112e0e3234450cf8b434daf1bd1e47ce755cf74fe9a70ff3c19765c0694a6f6; engine receipt

sha256:b274737acdb5f638d7cecdff4f7f79c42b46a50a11d130ff40381d56586ac90a at receipts/engine/model_admitted/SFT-CHEM-SMITHIUM-DECAY-CHANNEL-LEDGER-001-b274737acdb5f638.json; empirical hash None; measurement receipt None.

89. Smithium lifetime identifiability and mandatory-halt boundary

Claim identity: SFT-CHEM-SMITHIUM-LIFETIME-BOUNDARY-001

Question and theorem. Smithium's 126/184 double closure forces a stability-relevant structural coordinate but not a numerical lifetime. Numerical evaluation must halt until a positive transition width and registered time unit are independently derived or measured. No half-life or branching value is claimed by V3 at this boundary.

Smithium's 126/184 double closure forces a stability-relevant structural coordinate but not a numerical lifetime. Numerical evaluation must halt until a positive transition width and registered time unit are independently derived or measured. No half-life or branching value is claimed by V3 at this boundary.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-SMITHIUM-DECAY-CHANNEL-LEDGER-001
- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001
- SFT-PHYS-ATOMIC-TRANSITION-RATE-TERMINAL-005

Generated grammar. Generate all 256 lifetime-identifiability forms from closure, channel, width, unit, record and target-custody alternatives.

Boundary: Every numerical-lifetime request using the sealed double closure and channel ledger, distinguished by presence or absence of a positive transition width and registered time unit.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: sealed-126-184-310-coordinate__complete-positive-carrier-ledger__numeric-lifetime-only-from-positive-width-and-unit__complete-declared-product__formal-seal-before-target-access__favourable-adverse-absent-unresolved-held__structural-absence-and-explicit-halt__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
coordinate	sealed-126-184-310-coordinate	selected-element-name	A selected name does not reconstruct a nuclear/electronic coordinate.
conservation	complete-positive-carrier-ledger	outcome-only-transformation	An outcome without the complete carrier ledger can hide loss or invention.
relation	numeric-lifetime-only-from-positive-width-and-unit	free-or-imported-relation	A free or imported relation is not forced by the dependencies.
enumeration	complete-declared-product	selected-favourable-example	One favourable example cannot establish uniqueness.
measurement	formal-seal-before-target-access	target-before-seal	Target access before the derivation seal is fitting.
record	favourable-adverse-absent-unresolved-held	favourable-only-record	Selective retention destroys empirical falsifiability.
absence	structural-absence-and-explicit-halt	invented-numerical-placeholder	A placeholder turns absence into a false number.
extension	no-extra-rule	free-extra-rule	An exception is an unforced parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256: 79625a5dedf221f93c340b875bc2b66b7bb172afa910a1f3900d8e005c6c90fe.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256: b386cb759025fe3616b6fc6ffd455cfd1587285049eff181aaec5e2092a6117.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: dd77994c7d11e80a7a8522ab26e41640d916edc33f1e854ea31b54c12723a2.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256: e9c7ddf4829eb7d5d45319b7f7aabb5fa74f5d1459c58ceb063ec42fcb35174b.

Independent reconstruction. Implementation hash

sha256: 37f6ad9c651e645fd69788e591e4045ea372fd44d08036cdd7e79b02425bf371; independent certificate sha256: 2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dadb5d687df7fca82fa60; external-validation hash sha256: 3fe98f86608985b4f92f868eb3c8c71e8064f25dd8dbdef2767b65b00b336ebe. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. This is a sealed formal law with an implementation-distinct reconstruction and no separate external-observation package at this boundary. It receives no empirical-comparison credit that is not present in its claim package.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The final subbranch applies separately admitted exact physical prerequisites to a complete positive-rank fill walk and distinguishes known-domain validation from unobserved prediction. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no apparatus, projectile, target, beam energy, cross-section or production yield selected as a law
- no lifetime, branching ratio, transition energy, wavelength or line intensity invented without measurement
- no current non-observation relabelled as disproof and no standing prediction relabelled as discovery
- no numerical absence used as a proof value; absence is a held structural record
- no negative, irrational, imaginary, floating, fitted, free or target-selected proof magnitude
- no V1/V2 answer used to select a survivor; prior work supplies only the reconstruction question

Immutable identities. Source manifest

sha256: 66381643c4efdf2cb4d76463f11a150e670137089882cc8a18346b18f0723f0; derivation seal sha256: 5439b4bb6ffe85070af1d3b32e60878017a218886dd65fbb373ea75d69541aae; engine receipt sha256: fe1d8b934a1edd046d578e632368981fabb92944aa4fa7bc48668190af091377 at receipts/engine/model_admitted/SFT-CHEM-SMITHIUM-LIFETIME-BOUNDARY-001-fe1d8b934a1edd04.json; empirical hash None; measurement receipt None.

90. Exact Smithium positive-ion and oxidation ladder

Claim identity: SFT-CHEM-SMITHIUM-ION-OXIDATION-LADDER-001

Question and theorem. The exact structural ladder has seven records: +2 leaves [Og]5g6 with 124 electrons; each successor removes one further 5g carrier; +8 leaves the [Og] core with 118 electrons. These are structural admissibility classes, not measured abundances or guaranteed isolable compounds.

The exact structural ladder has seven records: +2 leaves [Og]5g6 with 124 electrons; each successor removes one further 5g carrier; +8 leaves the [Og] core with 118 electrons. These are structural admissibility classes, not measured abundances or guaranteed isolable compounds.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-SMITHIUM-LIFETIME-BOUNDARY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-MOLECULAR-IONIZATION-ENERGY-007

Generated grammar. Generate all 256 ion-ladder forms and every positive removal count from the 8s pair through the six 5g carriers.

Boundary: The seven structurally admissible positive oxidation counts +2 through +8 of the sealed 8s2 5g6 Smithium active configuration.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: sealed-126-184-310-coordinate__complete-positive-carrier-ledger__ordered-8s-then-5g-removal-with-identity-retained__complete-declared-product__formal-seal-before-target-access__favourable-adverse-absent-unresolved-held__structural-absence-and-explicit-halt__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
coordinate	sealed-126-184-310-coordinate	selected-element-name	A selected name does not reconstruct a nuclear/electronic coordinate.
conservation	complete-positive-carrier-ledger	outcome-only-transformation	An outcome without the complete carrier ledger can hide loss or invention.
relation	ordered-8s-then-5g-removal-with-identity-retained	free-or-imported-relation	A free or imported relation is not forced by the dependencies.
enumeration	complete-declared-product	selected-favourable-example	One favourable example cannot establish uniqueness.
measurement	formal-seal-before-target-access	target-before-seal	Target access before the derivation seal is fitting.
record	favourable-adverse-absent-unresolved-held	favourable-only-record	Selective retention destroys empirical falsifiability.
absence	structural-absence-and-explicit-halt	invented-numerical-placeholder	A placeholder turns absence into a false number.
extension	no-extra-rule	free-extra-rule	An exception is an unforced parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:072f3d83f8f3e7388f79891cea9b9edbbe40f8b576ca3943f989ce1808977d6e.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:12b37cf92b19e1b31177bf7558229111195f16efba8972361381df7ec07f78cc.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:15d7a0ce0516aeeda907f72c1141c497653a4366134b5d1cb3f5d51c36ee9bb8.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt

sha256:060e26dbc9100f8df3bd03753f381c90db927672afb89531e8689f7142595a21.

Independent reconstruction. Implementation hash

sha256:37f6ad9c651e645fd69788e591e4045ea372fd44d08036cdd7e79b02425bf371; independent certificate sha256:2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dacb5d687df7fca82fa60; external-validation hash sha256:00535c0da977da5653f1158ac2d13af8222912343bbb68addb70a5ca29bedca9. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. This is a sealed formal law with an implementation-distinct reconstruction and no separate external-observation package at this boundary. It receives no empirical-comparison credit that is not present in its claim package.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The final subbranch applies separately admitted exact physical prerequisites to a complete positive-rank fill walk and distinguishes known-domain validation from unobserved prediction. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no apparatus, projectile, target, beam energy, cross-section or production yield selected as a law
- no lifetime, branching ratio, transition energy, wavelength or line intensity invented without measurement
- no current non-observation relabelled as disproof and no standing prediction relabelled as discovery
- no numerical absence used as a proof value; absence is a held structural record
- no negative, irrational, imaginary, floating, fitted, free or target-selected proof magnitude
- no V1/V2 answer used to select a survivor; prior work supplies only the reconstruction question

Immutable identities. Source manifest

sha256:4de18b7dc11c9b70215116c013397fa873f2242dd449fbb611e8261f2e4e83c3; derivation seal sha256:9a727c361ef8fb6cc863c3c058db9dc2aa8f668779ae174437fbd35f16ad4a30; engine receipt sha256:477fbfde3f3abfbc9e30d51fe49fef3e6ccd92e77b99a4bfbcb05bd27b023a61 at receipts/engine/model_admitted/SFT-CHEM-SMITHIUM-ION-OXIDATION-LADDER-001-477fbfde3f3abfbc.json; empirical hash None; measurement receipt None.

91. Smithium spectroscopic transition-class law

Claim identity: SFT-CHEM-SMITHIUM-SPECTROSCOPIC-CLASSES-001

Question and theorem. The conventional g orbital has $l=4$ and Fold orbit rank 5, exact capacity 18, occupation 6 and 12 holes. One-electron E1 orbital adjacency permits $g \leftrightarrow f$ and $g \leftrightarrow h$ classes. Numerical line energies, wavelengths, intensities and lifetimes remain standing measurements and cannot be fabricated from class identity.

The conventional g orbital has $l=4$ and Fold orbit rank 5, exact capacity 18, occupation 6 and 12 holes. One-electron E1 orbital adjacency permits $g \leftrightarrow f$ and $g \leftrightarrow h$ classes. Numerical line energies, wavelengths, intensities and lifetimes remain standing measurements and cannot be fabricated from class identity.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-SMITHIUM-ION-OXIDATION-LADDER-001
- SFT-CHEM-SELECTION-RULE-STRUCTURE-010
- SFT-PHYS-MOLECULAR-SPECTROSCOPY-TERMINAL-005

Generated grammar. Generate all 256 spectroscopy forms and the complete capacity, occupation, hole and E1-adjacent class record for the sealed 5g6 support.

Boundary: The Smithium 5g active carrier under exact Fold capacity and one-electron electric-dipole orbital-rank adjacency, separated from unmeasured line values.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: sealed-126-184-310-coordinate__complete-positive-carrier-ledger__5g6-capacity-occupation-holes-and-E1-adjacency__complete-declared-product__formal-seal-before-target-access__favourable-adverse-absent-unresolved-held__structural-absence-and-explicit-halt__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
coordinate	sealed-126-184-310-coordinate	selected-element-name	A selected name does not reconstruct a nuclear/electronic coordinate.
conservation	complete-positive-carrier-ledger	outcome-only-transformation	An outcome without the complete carrier ledger can hide loss or invention.
relation	5g6-capacity-occupation-holes-and-E1-adjacency	free-or-imported-relation	A free or imported relation is not forced by the dependencies.
enumeration	complete-declared-product	selected-favourable-example	One favourable example cannot establish uniqueness.
measurement	formal-seal-before-target-access	target-before-seal	Target access before the derivation seal is fitting.
record	favourable-adverse-absent-unresolved-held	favourable-only-record	Selective retention destroys empirical falsifiability.
absence	structural-absence-and-explicit-halt	invented-numerical-placeholder	A placeholder turns absence into a false number.
extension	no-extra-rule	free-extra-rule	An exception is an unforced parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256: eecbbe2969685e474b76d5d309dbb913fc7d232c950898f1a0cb6dc1c7ae1f5e.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256: 570b230267a84a5220ddc900a3dde8bfb1ee3d77d22c270a308ede0dba2a7d17.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: ff912c2b7dad0b4d201c1767745cb500290ced7bd764aedbedfe3cbdc9871611.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256: 97651cea51c89cc5c111cd1604c3250cf802feedeabdd80acf4cf145ba4bf9f6.

Independent reconstruction. Implementation hash

sha256: 37f6ad9c651e645fd69788e591e4045ea372fd44d08036cdd7e79b02425bf371; independent certificate sha256: 2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dacb5d687df7fca82fa60; external-validation hash sha256: 9e7d418632f0318b09e19a029fc9463cd93d14423dfa825d837fd2fc170ed953. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. This is a sealed formal law with an implementation-distinct reconstruction and no separate external-observation package at this boundary. It receives no empirical-comparison credit that is not present in its claim package.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The final subbranch applies separately admitted exact physical prerequisites to a complete positive-rank fill walk and distinguishes known-domain validation from unobserved prediction. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no apparatus, projectile, target, beam energy, cross-section or production yield selected as a law
- no lifetime, branching ratio, transition energy, wavelength or line intensity invented without measurement
- no current non-observation relabelled as disproof and no standing prediction relabelled as discovery
- no numerical absence used as a proof value; absence is a held structural record
- no negative, irrational, imaginary, floating, fitted, free or target-selected proof magnitude
- no V1/V2 answer used to select a survivor; prior work supplies only the reconstruction question

Immutable identities. Source manifest

sha256:9a31886c8c9bd6f907728a91c0dbedcfb0c91abf8af9526c889460061460bf5b; derivation seal
sha256:b3934b02e84e290ef0dc1e6799275af5b52d63e0677888110a284f1f8ce1f798; engine receipt
sha256:11ce87ed6b65a993ec7b1923871cca267490ef000025521848d382a2977ef9ab at receipts/engine/model_admitted/SFT-CHEM-SMITHIUM-SPECTROSCOPIC-CLASSES-001-11ce87ed6b65a993.json;
empirical hash None; measurement receipt None.

92. Complete Smithium chemical-state distinction and separation boundary

Claim identity: SFT-CHEM-SMITHIUM-CHEMICAL-SEPARATION-001

Question and theorem. The seven Smithium +2...+8 state identities generate exactly 21 unordered pairwise distinctions, each held with $Z=126$, $N=184$ and $A=310$. This closes the mathematical separation target grammar; it does not assert a separation medium, recovery, decontamination factor, efficiency or yield.

The seven Smithium +2...+8 state identities generate exactly 21 unordered pairwise distinctions, each held with $Z=126$, $N=184$ and $A=310$. This closes the mathematical separation target grammar; it does not assert a separation medium, recovery, decontamination factor, efficiency or yield.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-SMITHIUM-SPECTROSCOPIC-CLASSES-001
- SFT-CHEM-RADIOCHEMICAL-SEPARATION-DECONTAMINATION-010

Generated grammar. Generate all 256 separation forms and every unordered pair of the seven retained positive oxidation-state identities.

Boundary: All pairwise distinctions among Smithium's +2 through +8 structural states, conditioned on the sealed nuclear identity and without an apparatus model.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: sealed-126-184-310-coordinate__complete-positive-carrier-ledger__all-21-pairwise-oxidation-state-distinctions__complete-declared-product__formal-seal-before-target-access__favourable-adverse-absent-unresolved-held__structural-absence-and-explicit-halt__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
coordinate	sealed-126-184-310-coordinate	selected-element-name	A selected name does not reconstruct a nuclear/electronic coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
conservation	complete-positive-carrier-ledger	outcome-only-transformation	An outcome without the complete carrier ledger can hide loss or invention.
relation	all-21-pairwise-oxidation-state-distributions	free-or-imported-relation	A free or imported relation is not forced by the dependencies.
enumeration	complete-declared-product	selected-favourable-example	One favourable example cannot establish uniqueness.
measurement	formal-seal-before-target-access	target-before-seal	Target access before the derivation seal is fitting.
record	favourable-adverse-absent-unresolved-held	favourable-only-record	Selective retention destroys empirical falsifiability.
absence	structural-absence-and-explicit-halt	invented-numerical-placeholder	A placeholder turns absence into a false number.
extension	no-extra-rule	free-extra-rule	An exception is an unforced parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:57be8b275da76c12b6590c42a242812f334925c84ca7feca2c91f010134a4fee.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:c6d27a375c45a54a0c35f5ceb226ec16438bfc674c61eb3aa20c21ad0041c2f5.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:e9ddae0aa96e9795a815f7ea3e03fc977c2567e970b5ec81c5108c3bb2fd0874.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:cfd963cd0fca6d8c602305956389d319aa4a799be550c3f48fa8e6b69e86d5ca.

Independent reconstruction. Implementation hash

sha256:37f6ad9c651e645fd69788e591e4045ea372fd44d08036cdd7e79b02425bf371; independent certificate
sha256:2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dadb5d687df7fca82fa60; external-validation
hash sha256:2c2b2536e350b862e4198bd9cd556871a737be59a26800f4467b0dabf318423c. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. This is a sealed formal law with an implementation-distinct reconstruction and no separate external-observation package at this boundary. It receives no empirical-comparison credit that is not present in its claim package.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The final subbranch applies separately admitted exact physical prerequisites to a complete positive-rank fill walk and distinguishes known-domain validation from unobserved prediction. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no apparatus, projectile, target, beam energy, cross-section or production yield selected as a law
- no lifetime, branching ratio, transition energy, wavelength or line intensity invented without measurement
- no current non-observation relabelled as disproof and no standing prediction relabelled as discovery

- no numerical absence used as a proof value; absence is a held structural record
- no negative, irrational, imaginary, floating, fitted, free or target-selected proof magnitude
- no V1/V2 answer used to select a survivor; prior work supplies only the reconstruction question

Immutable identities. Source manifest

sha256:b196b6b3df7e79536f6a98bfff456b0e8db2966c5514d4b96e09090fc5a767363; derivation seal sha256:93ce7ba8ceed5b9e55db91bc77371e8725626eab0261f01a390a2511d1006ea0; engine receipt sha256:5f1e5ccee4cc5dbf7fc3a11c37abc9979770ffc441ee45b4e766ed60f3b16a4c at receipts/engine/model_admitted/SFT-CHEM-SMITHIUM-CHEMICAL-SEPARATION-001-5f1e5ccee4cc5dbf.json; empirical hash None; measurement receipt None.

93. Complete Smithium joint-identification protocol

Claim identity: SFT-CHEM-SMITHIUM-JOINT-DETECTION-001

Question and theorem. Complete SFT identification of Smithium requires the conjunction of (i) Z/N, (ii) A, (iii) genetically linked decay, (iv) ion/oxidation identity and (v) spectroscopic class, with source custody for every record. This is a deliberately stronger complete-project protocol, not a claim that IUPAC requires all five for priority recognition. Current status remains an unobserved standing prediction.

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Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-SMITHIUM-CHEMICAL-SEPARATION-001
- SFT-CHEM-RADIOTRACER-CUSTODY-INFERENCE-009
- SFT-CHEM-MULTIMODAL-MOLECULAR-IDENTITY-021

Generated grammar. Generate all 256 identification forms and the complete five-record conjunction for the Smithium standing prediction.

Boundary: Every finite evidence record assembled from nuclear coordinate, mass, genetic decay, ion/oxidation and spectroscopic classes, with complete provenance and explicit official-scope separation.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: sealed-126-184-310-coordinate__complete-positive-carrier-ledger__five-record-complete-SFT-identification-conjunction__complete-declared-product__formal-seal-before-target-access__favourable-adverse-absent-unresolved-held__structural-absence-and-explicit-halt__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
coordinate	sealed-126-184-310-coordinate	selected-element-name	A selected name does not reconstruct a nuclear/electronic coordinate.
conservation	complete-positive-carrier-ledger	outcome-only-transformation	An outcome without the complete carrier ledger can hide loss or invention.
relation	five-record-complete-SFT-identification-conjunction	free-or-imported-relation	A free or imported relation is not forced by the dependencies.
enumeration	complete-declared-product	selected-favourable-example	One favourable example cannot establish uniqueness.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
measurement	formal-seal-before -target-access	target-before-seal	Target access before the derivation seal is fitting.
record	favourable-adverse -absent-unresolved -held	favourable-only-record	Selective retention destroys empirical falsifiability.
absence	structural-absence -and-explicit-halt	invented-numerical-placeholder	A placeholder turns absence into a false number.
extension	no-extra-rule	free-extra-rule	An exception is an unforced parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:76dfd5fd278f80df827f765d17f1de610c483867e58c2a5f25ade7a32300ebeb.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:8b6fcb7731bf1e2018a4200d3366116306ef9fc9dc94747ae0b708c34a5a112e.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:74494f499aed0918c2bbd70ca42b4736317f54c00c80d7dc3eea874090b48913.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:4eaeac4e82e85cc884fec471e96c2de2b4288cc12186f9cb617a503e897bea1d.

Independent reconstruction. Implementation hash

sha256:37f6ad9c651e645fd69788e591e4045ea372fd44d08036cdd7e79b02425bf371; independent certificate sha256:2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dacb5d687df7fca82fa60; external-validation hash sha256:5fb5d53aafa2ed212cf7d7cd7717cbb855ca05db4f7a7d47995ae5ca74c48120. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. This is a sealed formal law with an implementation-distinct reconstruction and no separate external-observation package at this boundary. It receives no empirical-comparison credit that is not present in its claim package.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The final subbranch applies separately admitted exact physical prerequisites to a complete positive-rank fill walk and distinguishes known-domain validation from unobserved prediction. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no apparatus, projectile, target, beam energy, cross-section or production yield selected as a law
- no lifetime, branching ratio, transition energy, wavelength or line intensity invented without measurement
- no current non-observation relabelled as disproof and no standing prediction relabelled as discovery
- no numerical absence used as a proof value; absence is a held structural record
- no negative, irrational, imaginary, floating, fitted, free or target-selected proof magnitude
- no V1/V2 answer used to select a survivor; prior work supplies only the reconstruction question

Immutable identities. Source manifest

sha256:b7bb8281b3e183d2c869188dd5b4700385189a3fa0bc801984df7a5dddb6ce48; derivation seal sha256:0252acad647ea7081c443c604f8ddc36bb4151b097af6244479d9523380f76af; engine receipt

sha256:4c17d9f41d1cb558e31419f8d036a8fcf05ee7f87e06372958d5b341ed78487b at receipts/engine/model_admitted/SFT-CHEM-SMITHIUM-JOINT-DETECTION-001-4c17d9f41d1cb558.json; empirical hash None; measurement receipt None.

94. Complete post-seal Smithium external-status record

Claim identity: SFT-CHEM-VALIDATION-SMITHIUM-COMPLETE-FAMILY-001

Question and theorem. IUPAC's registered record places the current known element boundary at oganesson, $Z=118$; reports superheavy $Z \geq 104$ production by heavy-ion fusion and identification through implanted-residue decay chains with alpha decay and/or spontaneous fission; and NIST records one-electron $E1$ orbital change by one rank. These externally correspond to the frozen synthesis, decay and spectroscopy classes but do not observe Smithium, select a unique production route, or supply its lifetime, branch fractions or wavelengths. Smithium remains a precise standing prediction.

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Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-SMITHIUM-JOINT-DETECTION-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

Generated grammar. Generate all 256 complete-family comparison forms before opening the registered IUPAC and NIST snapshots.

Boundary: All seven formal Smithium successor receipts, the official current element boundary, superheavy production/identification practice, $E1$ orbital adjacency and favorable/adverse/absent/unresolved status classes.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: sealed-126-184-310-coordinate__complete-positive-carrier-ledger__sealed-seven-law-family-versus-complete-official-source-record__complete-declared-product__formal-seal-before-target-access__favourable-adverse-absent-unresolved-held__structural-absence-and-explicit-halt__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
coordinate	sealed-126-184-310-coordinate	selected-element-name	A selected name does not reconstruct a nuclear/electronic coordinate.
conservation	complete-positive-carrier-ledger	outcome-only-transformation	An outcome without the complete carrier ledger can hide loss or invention.
relation	sealed-seven-law-family-versus-complete-official-source-record	free-or-imported-relation	A free or imported relation is not forced by the dependencies.
enumeration	complete-declared-product	selected-favourable-example	One favourable example cannot establish uniqueness.
measurement	formal-seal-before-target-access	target-before-seal	Target access before the derivation seal is fitting.
record	favourable-adverse-absent-unresolved-held	favourable-only-record	Selective retention destroys empirical falsifiability.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
absence	structural-absence -and-explicit-halt	invented-numerical-placeholder	A placeholder turns absence into a false number.
extension	no-extra-rule	free-extra-rule	An exception is an unforced parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256 : 4b5d413cfc510a2689477684e05e7e31d5023b6d56d78664643e7b2963657e8d.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256 : 7a53dfa32e19db93299294b49f28a0a0bb193186cf34399d7972223b14d85cb8.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 8dc220e998e4d7079fc369611d80e53c0e93380a2ecb3e959c33b9a2dadf8e61.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256 : f25b33a40b36f2a35116b024a081a0aabb2a82238b9396b87e2166075d23080.

Independent reconstruction. Implementation hash

sha256 : 37f6ad9c651e645fd69788e591e4045ea372fd44d08036cdd7e79b02425bf371; independent certificate
sha256 : 2a4267b65de2ef6619b79023712e50bc8282cdcdfd7dacb5d687df7fca82fa60; external-validation
hash sha256 : da675ed86fe3547c6727de485fa1927f37d79df9c4014f86f4ae0fbdc9a0e5db. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-SUPERHEAVY-2018
- IUPAC-PERIODIC-TABLE-2022
- NIST-ATOMIC-SPECTROSCOPY

Comparison records:

- SMITHIUM-COMPLETE-OFFICIAL-STATUS-2026-07-28: exact source-derived categorical match **True**
- favorable correspondence retained: production, decay identification and E1 class
- absent retained: no recognized element 126 within the current boundary ending at 118
- unresolved retained: route, cross-section, lifetime, branches, lines and separation remain unmeasured
- adverse retained: none in the registered official record
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any formal receipt or official byte identity changes; if any required fragment or result class is removed; if source content opens before seal; if Smithium is called observed; or if absence is called retirement.

Measurement receipt: sha256 : 8c61077bf38cabdf648b1c596288d263f03a31502ac1fca0be1850e8a108e092.

Isolation certificate: sha256 : 54175e4b9a9f008eca4e2b76590c2ef5f7226864f5bcf967e382ffe5c22161b9.

Custody certificate: sha256 : ae78f75f4a45f4bd70c8195683bfe5f0bcaf2aea332ec0c58240ec826de440ba.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The final subbranch applies separately admitted exact physical prerequisites to a complete positive-rank fill walk and distinguishes known-domain validation from unobserved prediction. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no apparatus, projectile, target, beam energy, cross-section or production yield selected as a law
- no lifetime, branching ratio, transition energy, wavelength or line intensity invented without measurement
- no current non-observation relabelled as disproof and no standing prediction relabelled as discovery
- no numerical absence used as a proof value; absence is a held structural record
- no negative, irrational, imaginary, floating, fitted, free or target-selected proof magnitude
- no V1/V2 answer used to select a survivor; prior work supplies only the reconstruction question

Immutable identities. Source manifest

sha256:d0d4659d1f8ac2d59896917134355e53e4cb9e44ed5f8c22a459dd005dba9808; derivation seal
 sha256:2b923e659d0bb60ca582d41713d9a2ae91ae8e0c2203f587bc8abc915d05cce6; engine receipt
 sha256:b5e611832ae9f3fe36316d0168851e5783e504dd3bf4d42c36b399bb9ba1dcb5 at receipts/engine/model_admitted/SFT-CHEM-VALIDATION-SMITHIUM-COMPLETE-FAMILY-001-b5e611832ae9f3fe.json
 ; empirical hash sha256:3abd872a48da67afb9d7bf7a7722769bca8446ad0ae43432e6b9fe2b260ed19b;
 measurement receipt sha256:8c61077bf38cabdf648b1c596288d263f03a31502ac1fca0be1850e8a108e092.

18. Molecular electronic and quantum-chemical correspondence - 16 admitted claims

One exact state carrier supports occupancy, symmetry, transitions, measurement and classical-quantum correspondence. This entire registered subcategory is included; no proper subset is presented as field completion.

95. Molecular electronic carrier and retained state identity

Claim identity: SFT-CHEM-ELECTRONIC-STATE-IDENTITY-001

Question and theorem. A molecular electronic state is the complete finite arrangement of every retained electron occurrence, held spin label and joint electronic support inside one identified molecular carrier that preserves the admitted quantum constraints and remains distinguishable under a retained chemical observation record.

identified-molecular-carrier__electron-occurrences-retained__complete-joint-arrangement__admitted-quantum-constraints__molecular-composition-retained__observation-distinguishable-state__held-state-record__no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001

- SFT-PHYS-QUANTUM-PHYSICAL-STATE-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001

Generated grammar. Generate the literal Cartesian product of carrier, occurrence, arrangement, admissibility, composition, distinction, record and extension choices; decide every product by preservation of admitted molecular, quantum, information and observation dependencies.

Boundary: Every exact positive-finite molecular carrier and its generated finite electron-occurrence support at a declared chemical observation resolution. The certificate is depth-independent because adding one electron occurrence repeats the same required occurrence, support, held-label and record checks.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `identified-molecular-carrier__electron-occurrences-retained__complete-joint-arrangement__admitted-quantum-constraints__molecular-composition-retained__observation-distinguishable-state__held-state-record__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	identified-molecular-carrier	state-without-molecular-carrier	A free state label cannot identify which chemical whole carries it.
occurrences	electron-occurrences-retained	anonymous-electron-number	A number alone erases the separately retained electron occurrences.
arrangement	complete-joint-arrangement	energy-name-only	An energy or state name cannot reconstruct the electronic organization.
admissibility	admitted-quantum-constraints	arbitrary-electron-arrangement	An arbitrary arrangement can violate admitted spin, indistinguishability and exclusion laws.
composition	molecular-composition-retained	independent-electron-list	Independent labels omit the composed molecular state.
distinction	observation-distinguishable-state	state-name-without-observation	A name without a retained observation class supplies no state distinction.
record	held-state-record	electronic-result-without-trace	A result without carrier, occurrences and support cannot be reproduced.
extension	no-extra-rule	free-electronic-exception	A free exception can manufacture a desired state or target match.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d66cac9b49987f51c50d9e51790f2bfbdeb6882eadac40923519503e9dc9c6329`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f5857a62a57aa328b5709abed3e8976f8ab58835d4f3b512d42b0dab964a6421`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3a47ed1e33a3ae8a639cae25103402e6ed2c0d2f4efd96b0bf1ef71e89a4a08f`.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:04bb141f95fb7c366db831077615d730fa36c6167c7b16c1b412534f0be61265.

Independent reconstruction. Implementation hash

sha256:292c5198b60b28bf7c13a8a9e695dd51ae496fa37e27a25688deb6052b257c4f; independent certificate sha256:46780748e9b8dbabd10868095161e880de8960b91ebb2deb9de0b7ad824c3e2f; external-validation hash sha256:ea06b2820bfd447edf8f751d24cc11bf3e246af3ee826df5cffa7ae81650b7e6. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ET07026-2026

Comparison records:

- electronic-state-iupac-et07026: predicted
allowed-electronic-arrangement__electron-support-retained__molecular-carrier-bounded; source-derived
allowed-electronic-arrangement__electron-support-retained__molecular-carrier-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if any generated finite electron occurrence can be omitted, duplicated or detached from the molecular carrier while preserving electronic-state identity; if an arrangement violating an admitted quantum constraint survives; if the official source-derived target differs from the sealed consequence; or if any target, source, candidate, receipt or unfavorable control is omitted or altered.

Measurement receipt: sha256:1a8b013b1472fd8604b4a6b64acab25deefe142f44ee355e6420f22a84f43bdd.

Isolation certificate: sha256:291f721a8b099e47c6c9debebbe331145e7eac03a92870fbf1183970a521e120.

Custody certificate: sha256:b2f38140118e5b4991e4c3b236780d4468f0b2e63f39a03ffdeda2b76c913644.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/ET07026/json>
(sha256:c22eb092dceb58f901486e494d81b2528a74a60cdd6caa39d9890f69975bef13)

Registered target identities:

- electronic-state-iupac-et07026 from IUPAC-GOLD-BOOK-ET07026-2026

Meaning. The field-wide extension derives molecular electronic identity, state composition, occupancy, ordering, symmetry, correlation, transitions, measurement and operational classical-quantum correspondence from the same exact carrier laws. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording or target label in candidate generation or survivor selection
- no imported molecular Hamiltonian, wavefunction, orbital table or continuum state space
- no numerical zero, negative, irrational, imaginary or floating proof value
- no fitted, learned, target-derived or species-specific free parameter
- no application output or opaque predictor as chemical law
- external target opens only after the derivation and prediction seal

Immutable identities. Source manifest

sha256:a7bd04660f65f83d6703deced158f0c9a65ce208c6ede4ee891ee27ef4385d49; derivation seal

sha256:50fe212212c4e255dc54d467cc7681c48840669349b105d072251e13517821bb; engine receipt

sha256:330b9bbf58741bbca7648d7bd1aadede56d70fbfdd5e427f0232d80f870d7a41 at receipts/engine/model_admitted/SFT-CHEM-ELECTRONIC-STATE-IDENTITY-001-330b9bbf58741bbc.json; empirical

hash sha256:5a1ef1d47ecf58bf759f7659cd138ec26b33577c8ef5c049f5c761bae1021a00; measurement

receipt sha256:1a8b013b1472fd8604b4a6b64acab25deefe142f44ee355e6420f22a84f43bdd.

96. Exact nuclear-electronic, vibronic and rovibronic state composition

Claim identity: SFT-CHEM-ROVIBRONIC-COMPOSITION-001

Question and theorem. An exact molecular rovibronic state is the complete finite joint carrier of one admitted molecular electronic state, every retained nuclear occurrence and isotope label, positive vibrational and rotational recurrence coordinates, one held spin state and one declared observation record. The admitted electronic-over-vibrational-over-rotational hierarchy distinguishes the coordinates without importing a separation approximation; the H2/D2 isotope transport and resolved rotational/vibrational relations remain the exact measured consequence supplied by the admitted Physics dependency.

identified-molecular-state-carrier__nuclear-electronic-joint-support__admitted-electronic-vibrational-rotational-order__positive-odd-vibrational-recurrence__positive-counted-rotational-recurrence__held-spin-state__retained-rovibronic-observation__no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-MOLECULAR-SPECTRUM-HIERARCHY-004
- SFT-PHYS-MOLECULAR-SPECTROSCOPY-TERMINAL-005
- SFT-CHEM-ELEM-ISOTOPE-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-ELECTRONIC-STATE-IDENTITY-001

Generated grammar. Generate the literal eight-axis product of carrier, nuclear-electronic composition, scale order, vibrational recurrence, rotational recurrence, held spin, observation and extension forms; decide every form by preservation of the admitted dependency chain.

Boundary: Every positive-finite molecular electronic carrier with positive-finite retained nuclei, isotope labels, positive vibration and rotation ordinals, held spin and a declared finite observation resolution.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *identified-molecular-state-carrier__nuclear-electronic-joint-support__admitted-electronic-vibrational-rotational-order__positive-odd-vibrational-recurrence__positive-counted-rotational-recurrence__held-spin-state__retained-rovibronic-observation__no-extra-rule*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	identified-molecular-state-carrier	detached-state-name	A detached name cannot identify the molecular whole carrying the state.
composition	nuclear-electronic-joint-support	nuclei-or-electrons-erased	Erasing either support merges distinguishable isotopologues or electronic states.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
scale	admitted-electronic-vibrational-rotational-order	imported-separation-approximation	An imported approximation cannot select a fundamental state organization.
vibration	positive-odd-vibrational-recurrence	continuum-vibrational-coordinate	A continuum coordinate is outside the exact generated grammar.
rotation	positive-counted-rotational-recurrence	signed-or-continuum-rotation	A signed continuum angle is not an exact proof carrier.
spin	held-spin-state	spin-erased-state	Erasing spin merges spectroscopically distinguishable molecular states.
observation	retained-rovibronic-observation	result-without-readout-record	A state without a readout boundary cannot be reproduced or distinguished.
extension	no-extra-rule	free-species-exception	A species exception can fit any desired spectrum.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : 43920d03d40e96ffa3e5717d0954eb056c14c944a35f3e6af44a666f580b0721.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 1f856682ffe39c1d10ee4fe918e73426aba8835d3de76abb025721b31d70b822.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : abbb86f1565dfe4535a2b859c0453816831830da88f9cc4c5867b5aaadd704dc.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : adac792031608424057db06c9c6ec79a7969472d9cd29f57c16160d24e421c8f.

Independent reconstruction. Implementation hash

sha256 : 69309f4f2aa26433914908b249cea965e887837e29fb892552be4eeadc7bbf2a; independent certificate sha256 : 3acea3fb46d72a4670a82d29695a94dd8fd49cf9efed7a692628643f79914d1b; external-validation hash sha256 : 5c8e7662c3007d0f8c07a4c13ba1d70a67c0dade3b0f69aa1f2f63545798ae4f. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-MOLECULAR-SPECTROSCOPY-H2-D2-SUCCESSOR

Comparison records:

- **hydrogen_rotation_over_vibration:** exact
891050294212130200059015688492600075210824140/64445503903521854869538077484881292205289061521 contained in source-derived interval [1217059/88024270, 1217061/88024250]: **True**
- **hydrogen_anharmonic_over_vibration:** exact
177668288370797223397270039535383633647719432/64445503903521854869538077484881292205289061521 contained in source-derived interval [242671/8802427, 242673/8802425]: **True**
- **deuterium_rotation_over_hydrogen_rotation:** exact
127978935375939764924042592947263805/255814091290792903110270905899893418 contained in source-derived interval [202957/405687, 608873/1217059]: **True**
- **deuterium_vibration_over_hydrogen_vibration_squared:** exact
128184365564433438104117938993725957/255814091290792903110270905899893418 contained in source-derived interval [38825236380100/77482721090329, 1553019424804/3099307435225]: **True**

- NIST-MOLECULAR-H2-D2-COMPLETE-VECTOR: predicted terminal-molecular-JJplusOne-anharmonic-and-isotope-relation s-contained-in-complete-NIST-H2-D2-displayed-resolution-vector__observational-prediction-protocol-passed; source-derived terminal-molecular-JJplusOne-anharmonic-and-isotope-relations-contained-in-complete-NIST-H2-D2-displayed-resolution-vector__observational-prediction-protocol-passed; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if any retained nucleus, isotope, electronic, vibrational, rotational, spin or observation coordinate can be erased without merging a registered state; if the exact admitted H2/D2 ladder or isotope relations leave the complete NIST displayed-resolution vector; if a source is opened before the prediction seal; or if any favorable, unfavorable, predecessor or tampered row is omitted.

Measurement receipt: sha256:f44faf3133a9d2e7dd64e0fd4d6e96844caf5cc9eb358165952a1cf12426a117.
Isolation certificate: sha256:97d2bb3f62cc2d5e2b5ed84398ac37a27a424d7d34834989bd700f05cd2b264e.
Custody certificate: sha256:15eb1a56334502803d7c3d1dbb4bc54421ac4715520c63838dfa14fa5cedd485.

Registered source descriptions:

- National Institute of Standards and Technology, Chemistry WebBook SRD 69:
<https://webbook.nist.gov/cgi/cbook.cgi?ID=C1333740&Mask=1828>
(sha256:18036a188088f880122249544ceb6b384fabfba93b300b4f9f0fa01aa0ed9b24)
- National Institute of Standards and Technology, Chemistry WebBook SRD 69:
<https://webbook.nist.gov/cgi/cbook.cgi?ID=C7782390&Mask=1000>
(sha256:790276cd493dbede03fc9e83db95bc0808a36838eb31473aaf923493b4749936)

Registered target identities:

- NIST-MOLECULAR-H2-D2-COMPLETE-VECTOR from NIST-MOLECULAR-SPECTROSCOPY-H2-D2-SUCCESSOR

Meaning. The field-wide extension derives molecular electronic identity, state composition, occupancy, ordering, symmetry, correlation, transitions, measurement and operational classical-quantum correspondence from the same exact carrier laws. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no source value, NIST inscription or comparison interval in candidate generation or survivor selection
- no imported wavefunction, Hamiltonian, Born-Oppenheimer approximation, continuum surface or rigid-rotor premise
- no numerical-zero state, negative, irrational, imaginary or floating proof value
- no fitted molecular constant, isotope factor, species exception or target-derived parameter
- no absolute bond length, frequency or rotational constant claimed by this compositional prerequisite
- all earlier Physics and Chemistry receipts remain immutable and separately owned

Immutable identities. Source manifest

sha256:0f13089e8a8795eebf452ad66d2ce80131b27de3ee4fc6e9be530433b52a7000; derivation seal
sha256:0358f73d4a6c8b63dbf2da56a28cc21496286c0edec1b7ed85cfe90ecda0bca1; engine receipt
sha256:fea9ffff1a16343aa9480d62cc7642e47dc55b5394a362845401e9ace942a1fbf at receipts/engine/model_admitted/SFT-CHEM-ROVIBRONIC-COMPOSITION-001-fea9ffff1a16343aa.json; empirical hash
sha256:17953edfff387505be78d9d0e321b48096fc3d43b22eb75e811afecac60caa1bd; measurement receipt
sha256:f44faf3133a9d2e7dd64e0fd4d6e96844caf5cc9eb358165952a1cf12426a117.

97. Exact molecular electron count and held-spin organization

Claim identity: SFT-CHEM-ELECTRON-COUNT-SPIN-002

Question and theorem. For every positive finite molecular nuclear support, the complete electron occurrence count is forced by atomic-number occurrences and a held adjoin/remove transfer; every occurrence then has exactly one of two held spin fibres, same-fibre same-cell doubling is excluded, and the support decomposes uniquely at a declared spin width into complementary pairs plus unmatched held fibres.

atomic-number-occurrence-support__held-directed-electron-transfer__complete-electron-occurrence-census__two-held-spin-fibres__complementary-cell-occupation__complete-pairs-plus-held-surplus__support-compatible-spin-width__no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-ELECTRONIC-STATE-IDENTITY-001

Generated grammar. Generate the literal Cartesian product of nuclear support, charge representation, electron census, spin carrier, cell occupation, pair/surplus decomposition, observable width and extension choices. Decide every one of the 256 forms solely by preservation of admitted exact arithmetic, atomic-number, ion, spin, indistinguishability, exclusion, information and molecular-state laws.

Boundary: Every positive finite molecular multiset of admitted atomic-number occurrences, structural empty-One or a positive held electron-transfer count that leaves positive support, and every positive spin width. Closure is depth-independent: adjoining one nucleus, electron occurrence, complementary pair or unmatched fibre repeats the same local census, two-fibre, exclusion and complete-support checks.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *atomic-number-occurrence-support__held-directed-electron-transfer__complete-electron-occurrence-census__two-held-spin-fibres__complementary-cell-occupation__complete-pairs-plus-held-surplus__support-compatible-spin-width__no-extra-rule*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
nuclear_support	atomic-number-occurrence-support	formula-name-only	A formula name does not enumerate the positive nuclear support that forces electron count.
charge	held-directed-electron-transfer	signed-charge-scalar	A signed scalar imports negative quantity and conflates direction with magnitude.
electron_count	complete-electron-occurrence-census	asserted-electron-number	An asserted number is neither generated nor reconstructible.
spin	two-held-spin-fibres	signed-spin-magnitude	A signed magnitude imports a forbidden negative coordinate and loses fibre identity.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
occupation	complementary-cell-occupation	same-fibre-cell-duplication	Doubled same-fibre occupation violates admitted indistinguishability and exclusion.
decomposition	complete-pairs-plus-held-surplus	unstructured-spin-list	An unstructured list cannot reconstruct pairs, surplus or width.
observation	support-compatible-spin-width	state-width-detached-from-support	A detached multiplicity can contradict the complete electron support.
extension	no-extra-rule	species-specific-exception	An exception or fitted table can select a measured state.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256: 3bdaaa578e56bbafea5b3ac7b95c2cdbfe3cf0b8be3252a57b2532a6f4eda844.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: fefdb22a0f6e993555176009ac2ab8209b027d5ded1666578bc947f2b0ab5b0.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 702741be8f4d03ca2f2de9666b9fa6dae216ea63ec0220145a8de474acb06770.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: b7618d96dd0e90723590860f0ae12b53b8798b64b75a8fc2ae450250f5538221.

Independent reconstruction. Implementation hash

sha256: 74613b9e9711f7aaf25ed21fe7cf8994a5f3707853da552b0f39dfab92d49ade; independent certificate sha256: dc74f648ad0fd6b0e08b11113e3717ecd38b2517eab8ce773634c4dfdeecd91e; external-validation hash sha256: f278caa6d9f1f8137c7201c3110953ed9c8a2f20f724f02a3560d6b0aabbba77b. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025

Comparison records:

- hydrogen-neutral (H2, empty-One): forced electron count 2; forced odd-positive-width; NIST X-state $X^{1\Sigma_g^+ 1s\sigma^2}$ gives multiplicity 1; pairs 1; unmatched fibres empty-One; pass True
- nitrogen-neutral (N2, empty-One): forced electron count 14; forced odd-positive-width; NIST X-state $X^{1\Sigma_g^+}$ gives multiplicity 1; pairs 7; unmatched fibres empty-One; pass True
- oxygen-neutral (O2, empty-One): forced electron count 16; forced odd-positive-width; NIST X-state $X^{3\Sigma_g^-}$ gives multiplicity 3; pairs 7; unmatched fibres 2; pass True
- fluorine-neutral (F2, empty-One): forced electron count 18; forced odd-positive-width; NIST X-state $X^{1\Sigma_g^+}$ gives multiplicity 1; pairs 9; unmatched fibres empty-One; pass True
- carbon-monoxide-neutral (CO, empty-One): forced electron count 14; forced odd-positive-width; NIST X-state $X^{1\Sigma^+}$ gives multiplicity 1; pairs 7; unmatched fibres empty-One; pass True
- nitric-oxide-neutral (NO, empty-One): forced electron count 15; forced even-positive-width; NIST X-state $X^{2\Pi_r}$ gives multiplicity 2; pairs 7; unmatched fibres 1; pass True
- hydroxyl-neutral (HO, empty-One): forced electron count 9; forced even-positive-width; NIST X-state $X^{2\Pi_i}$ gives multiplicity 2; pairs 4; unmatched fibres 1; pass True
- imidogen-neutral (HN, empty-One): forced electron count 8; forced odd-positive-width; NIST X-state $X^{3\Sigma^-}$ gives multiplicity 3; pairs 3; unmatched fibres 2; pass True

- methylidyne-neutral (CH, empty-One): forced electron count 7; forced even-positive-width; NIST X-state $X^{\wedge}2\Pi_r$ gives multiplicity 2; pairs 3; unmatched fibres 1; pass True
- chlorine-neutral (Cl₂, empty-One): forced electron count 34; forced odd-positive-width; NIST X-state $X^{\wedge}1\Sigma_g^{+}$ gives multiplicity 1; pairs 17; unmatched fibres empty-One; pass True
- silicon-monoxide-neutral (OSi, empty-One): forced electron count 22; forced odd-positive-width; NIST X-state $X^{\wedge}1\Sigma^{+}$ gives multiplicity 1; pairs 11; unmatched fibres empty-One; pass True
- sulfur-monoxide-neutral (OS, empty-One): forced electron count 24; forced odd-positive-width; NIST X-state $X^{\wedge}3\Sigma^{-}$ gives multiplicity 3; pairs 11; unmatched fibres 2; pass True
- hydrogen-fluoride-neutral (FH, empty-One): forced electron count 10; forced odd-positive-width; NIST X-state $X^{\wedge}1\Sigma^{+}$ gives multiplicity 1; pairs 5; unmatched fibres empty-One; pass True
- hydrogen-chloride-neutral (ClH, empty-One): forced electron count 18; forced odd-positive-width; NIST X-state $X^{\wedge}1\Sigma^{+}$ gives multiplicity 1; pairs 9; unmatched fibres empty-One; pass True
- hydrogen-bromide-neutral (BrH, empty-One): forced electron count 36; forced odd-positive-width; NIST X-state $X^{\wedge}1\Sigma^{+}$ gives multiplicity 1; pairs 18; unmatched fibres empty-One; pass True
- iodine-neutral (I₂, empty-One): forced electron count 106; forced odd-positive-width; NIST X-state $X^{\wedge}1\Sigma_g^{+}$ gives multiplicity 1; pairs 53; unmatched fibres empty-One; pass True
- hydrogen-cation (H₂⁺, remove-electron): forced electron count 1; forced even-positive-width; NIST X-state $X^{\wedge}2\Sigma_g^{+} 1s\sigma$ gives multiplicity 2; pairs empty-One; unmatched fibres 1; pass True
- nitrogen-cation (N₂⁺, remove-electron): forced electron count 13; forced even-positive-width; NIST X-state $X^{\wedge}2\Sigma_g^{+}$ gives multiplicity 2; pairs 6; unmatched fibres 1; pass True
- oxygen-cation (O₂⁺, remove-electron): forced electron count 15; forced even-positive-width; NIST X-state $X^{\wedge}2\Pi_g$ gives multiplicity 2; pairs 7; unmatched fibres 1; pass True
- nitric-oxide-cation (NO⁺, remove-electron): forced electron count 14; forced odd-positive-width; NIST X-state $X^{\wedge}1\Sigma^{+}$ gives multiplicity 1; pairs 7; unmatched fibres empty-One; pass True
- oxygen-anion (O₂⁻, adjoin-electron): forced electron count 17; forced even-positive-width; NIST X-state $X^{\wedge}2\Pi_{gi}$ gives multiplicity 2; pairs 8; unmatched fibres 1; pass True
- nitric-oxide-anion (NO⁻, adjoin-electron): forced electron count 16; forced odd-positive-width; NIST X-state $X^{\wedge}3\Sigma^{-}$ gives multiplicity 3; pairs 7; unmatched fibres 2; pass True
- adverse tampered_charge_rejected: True
- adverse tampered_multiplicity_parity_rejected: True
- adverse omitted_row_rejected: True
- adverse tampered_snapshot_hash_rejected: True
- adverse all_registered_rows_retained: True

Falsification boundary: The claim fails if electron occurrence support differs from the complete atomic-number/held-transfer census; if a same-fibre duplicate occupies one cell; if complete support cannot be decomposed into complementary pairs and unmatched held fibres; if any NIST X-state multiplicity violates the sealed count/width parity; if any registered neutral, cation or anion row is omitted; or if a changed charge, state width, source byte, target row, prediction trace or adverse control is accepted.

Measurement receipt: sha256:8d415225af0e245d65b14efe3f2cda48c679e8368fb54d0eeddd32347a04b0cb.

Isolation certificate: sha256:042b7390c708c388cd8d1238c232cecc9cb53a81eb2baa0233988fa5ec329b9d.

Custody certificate: sha256:48de6aa51fbc5a7ddcdb06cf28311ed0e80eab1e3f05f04930e2b84ee59772a5.

Registered source descriptions:

- National Institute of Standards and Technology: <https://webbook.nist.gov/chemistry/> ()

Registered target identities:

- hydrogen-neutral from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025

- nitrogen-neutral from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- oxygen-neutral from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- fluorine-neutral from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- carbon-monoxide-neutral from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- nitric-oxide-neutral from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- hydroxyl-neutral from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- imidogen-neutral from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- methylidyne-neutral from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- chlorine-neutral from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- silicon-monoxide-neutral from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- sulfur-monoxide-neutral from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- hydrogen-fluoride-neutral from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- hydrogen-chloride-neutral from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- hydrogen-bromide-neutral from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- iodine-neutral from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- hydrogen-cation from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- nitrogen-cation from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- oxygen-cation from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- nitric-oxide-cation from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- oxygen-anion from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- nitric-oxide-anion from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025

Meaning. The field-wide extension derives molecular electronic identity, state composition, occupancy, ordering, symmetry, correlation, transitions, measurement and operational classical-quantum correspondence from the same exact carrier laws. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical-zero, negative, irrational, imaginary or floating proof value
- no signed charge or signed-spin proof coordinate
- no imported orbital occupation table, Hamiltonian, wavefunction or ground-state solver
- no measured state term or multiplicity in the derivation, candidate generator or prediction inputs
- no fitted, learned, species-specific or target-derived parameter
- no omitted, selected or silently corrected external row
- external state terms open only after the complete prediction vector is sealed

Immutable identities. Source manifest

sha256:0bf1abbd48c1f025356ce5e50154b89641f6d57e1842389b112fcbe706ffa24c; derivation seal
 sha256:ee43dfffc72d929269f5da9edba48c512f3cc4a10468ca9ebe1d4983b76ecfb62; engine receipt
 sha256:f9f7da16969c8222cb4a799c8e87a963f2b07d826d552567e35d15f0ceb17 at
 receipts/engine/model_admitted/SFT-CHEM-ELECTRON-COUNT-SPIN-002-f9f7da16969c8222.json;
 empirical hash sha256:fd6a8c9cdc75b3528bc0ca5180766e5af11dc05698d3616cf148045d8045e2bd;
 measurement receipt sha256:8d415225af0e245d65b14efe3f2cda48c679e8368fb54d0eeddd32347a04b0cb.

98. Fold-native molecular support composition and fermionic occupancy

Claim identity: SFT-CHEM-ORBITAL-SUPPORT-OCCUPANCY-003

Question and theorem. A molecular electronic support cell is a molecule-bound composition of positive radial recurrence, structural empty-One or positive axis recurrence, one of two held joining phases and applicable held symmetry observations. Its fermionic occupancy is structural empty One, one held electron occurrence or one complementary-spin pair, and the complete occupied support partitions every molecular electron occurrence exactly once.

molecule-bound-support-cell__two-held-joining-phases__empty-One-or-positive-axis-recurrence__held-exchange-and-reflection-labels__empty-One-single-or-complementary-pair__complementary-spin-double-occupation__complete-electron-support-partition__no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-ELECTRONIC-STATE-IDENTITY-001
- SFT-CHEM-ELECTRON-COUNT-SPIN-002

Generated grammar. Generate the literal Cartesian product of carrier, joining, axis organization, held symmetry, occupancy, spin, completeness and extension forms. Decide all 256 forms only by admitted molecular composition, exact recurrence, two-fibre, spin, indistinguishability, exclusion, electron-count and information-retention laws.

Boundary: Every positive finite molecular carrier; every positive radial recurrence; the structural axis-invariant boundary and every positive axis recurrence; both held joining phases; applicable held exchange/reflection labels; and every positive finite electron support. Closure is depth-independent because each newly generated cell and occurrence repeats the same coordinate-identity, occupancy, complementary-spin and complete-partition checks.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *molecule-bound-support-cell__two-held-joining-phases__empty-One-or-positive-axis-recurrence__held-exchange-and-reflection-labels__empty-One-single-or-complementary-pair__complementary-spin-double-occupation__complete-electron-support-partition__no-extra-rule*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	molecule-bound-sup port-cell	free-orbital-name	A free orbital name is detached from the molecule and its derivation.
joining	two-held-joining-p hases	unjoined-atomic-support-l ist	An unjoined list cannot express the two generated molecular phase relations.
axis	empty-One-or-posit ive-axis-recurrenc e	continuum-angular-coordin ate	A continuum angular coordinate imports an ungenerated state space.
symmetry	held-exchange-and- reflection-labels	unsigned-erased-symmetry	Erasing exchange and reflection distinctions merges spectroscopically distinct support.
occupancy	empty-One-single-o r-complementary-pa ir	signed-or-unbounded-occup ation	Signed, numerical-zero or unbounded fermion occupation violates the exact domain and exclusion.
spin	complementary-spin -double-occupation	same-fibre-double-occupat ion	Same-fibre doubling has no distinct admissible preimage under exclusion.
completeness	complete-electron- support-partition	selected-support-fragment	A selected fragment cannot reconstruct the molecular electronic state or electron census.
extension	no-extra-rule	species-or-symbol-excepti on	A species lookup or conventional symbol exception lets the target select the law.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:e8ba89ba1a9577ddb6acd17ce87b95e5bd21a06dfb2a5b07e4813ddd3b72c2b8.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:b4b83c12637be6aed01942e75976359984db76d6efbfff22c8e032561993c8b4f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:f6f323656d9d8931b839ad79f3531cd625174354f6b7edf605783e433a0494b3.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:6b3e314b412df590e768623c31d8725f7966bb589034cb295f470089819a649b.

Independent reconstruction. Implementation hash

sha256:dda118d238b6be7a4ab96ff2e7eb868c49fa8e86044cf957b4f2cc0547839186; independent certificate
sha256:272f70f6445fe141a5a4e8b49dd9583683c821a140a272bdf7964fba410ff414; external-validation
hash sha256:8ee186dbbd9e130470e8f30866e7a49fea92220347dcce4b4af65fb243493469. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025

Comparison records:

- C1333740-state-001 (hydrogen-neutral): NIST state $u \ ^3\Pi_u \ 6p\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-002 (hydrogen-neutral): NIST state $t \ ^3\Sigma_u^+ \ 5f\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-003 (hydrogen-neutral): NIST state $q \ (^3\Sigma_g^+) \ 5d\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-004 (hydrogen-neutral): NIST state $n \ ^3\Pi_u \ 5p\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True

- C1333740-state-005 (hydrogen-neutral): NIST state $m \ ^3\Sigma_u^+ 4f\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-006 (hydrogen-neutral): NIST state $s \ ^3\Delta_g 4d\delta$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-007 (hydrogen-neutral): NIST state $r \ ^3\Pi_g 4d\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-008 (hydrogen-neutral): NIST state $p \ ^3\Sigma_g^+ 4d\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-009 (hydrogen-neutral): NIST state $v \ (^3\Pi_g)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C1333740-state-010 (hydrogen-neutral): NIST state $k \ ^3\Pi_u 4p\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-011 (hydrogen-neutral): NIST state $f \ ^3\Sigma_u^+ 4p\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-012 (hydrogen-neutral): NIST state $o \ ^3\Sigma_u^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C1333740-state-013 (hydrogen-neutral): NIST state $l \ ^3\Pi_u$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C1333740-state-014 (hydrogen-neutral): NIST state $j \ ^3\Delta_g 3d\delta$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-015 (hydrogen-neutral): NIST state $i \ ^3\Pi_g 3d\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-016 (hydrogen-neutral): NIST state $h \ ^3\Sigma_g^+ 3s\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-017 (hydrogen-neutral): NIST state $g \ ^3\Sigma_g^+ 3d\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-018 (hydrogen-neutral): NIST state $d \ ^3\Pi_u 3p\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-019 (hydrogen-neutral): NIST state $e \ ^3\Sigma_u^+ 3p\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-020 (hydrogen-neutral): NIST state $a \ ^3\Sigma_g^+ 2s\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-021 (hydrogen-neutral): NIST state $c \ ^3\Pi_u 3p\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-022 (hydrogen-neutral): NIST state $D'' \ ^1\Pi_u 5p\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-023 (hydrogen-neutral): NIST state $D' \ ^1\Pi_u 4p\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-024 (hydrogen-neutral): NIST state $S \ ^1\Delta_g 4d\delta$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-025 (hydrogen-neutral): NIST state $O \ ^1\Sigma^+ 4s\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-026 (hydrogen-neutral): NIST state $R \ ^1\Pi_g 4d\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True

- C1333740-state-027 (hydrogen-neutral): NIST state $P \ ^1\Sigma_g^+ 4d\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-028 (hydrogen-neutral): NIST state $T \ ^1\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C1333740-state-029 (hydrogen-neutral): NIST state $B'' \ ^1\Sigma_u^+ 4p\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-030 (hydrogen-neutral): NIST state $N \ ^1\Sigma_g^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C1333740-state-031 (hydrogen-neutral): NIST state $U \ (^1\Sigma_g^+)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C1333740-state-032 (hydrogen-neutral): NIST state $M \ ^1\Sigma_g^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C1333740-state-033 (hydrogen-neutral): NIST state $L \ ^1\Sigma_g^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C1333740-state-034 (hydrogen-neutral): NIST state $H \ ^1\Sigma_g^+ 3s\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-035 (hydrogen-neutral): NIST state $D \ ^1\Pi_u 3p\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-036 (hydrogen-neutral): NIST state $J \ ^1\Delta_g 3d\delta$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-037 (hydrogen-neutral): NIST state $I \ ^1\Pi_g 3d\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-038 (hydrogen-neutral): NIST state $G \ ^1\Sigma_g^+ 3d\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-039 (hydrogen-neutral): NIST state $K \ (^1\Sigma_g^+)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C1333740-state-040 (hydrogen-neutral): NIST state $Q \ (^1\Pi_g)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C1333740-state-041 (hydrogen-neutral): NIST state $B' \ ^1\Sigma_u^+ 3p\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-042 (hydrogen-neutral): NIST state $F \ ^1\Sigma_g^+ 2p\sigma^2$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-043 (hydrogen-neutral): NIST state $E \ ^1\Sigma_g^+ 2s\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-044 (hydrogen-neutral): NIST state $C \ ^1\Pi_u 2p\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-045 (hydrogen-neutral): NIST state $B \ ^1\Sigma_u^+ 2p\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C1333740-state-046 (hydrogen-neutral): NIST state $X \ ^1\Sigma_g^+ 1s\sigma^2$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C7727379-state-001 (nitrogen-neutral): NIST state $x'' \ (^1\Sigma_u^+)$ 1; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-002 (nitrogen-neutral): NIST state $x' \ (^1\Pi_u)$ 2; term assignments 1 pass True; configuration assignments 0 pass True; row pass True

- C7727379-state-003 (nitrogen-neutral): NIST state $v \ ^1\Pi_g$ 4; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-004 (nitrogen-neutral): NIST state $o_5 \ ^1\Pi_u$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-005 (nitrogen-neutral): NIST state $O_5 \ (^3\Pi_u)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-006 (nitrogen-neutral): NIST state $c'_n \ ^1\Sigma_u^+ 21$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-007 (nitrogen-neutral): NIST state $c_n \ ^1\Pi_u$ 22; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-008 (nitrogen-neutral): NIST state $c_n \ ^1\Pi_u$ 26; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-009 (nitrogen-neutral): NIST state $o_4 \ ^1\Pi_u$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-010 (nitrogen-neutral): NIST state $O_4 \ (^3\Pi_u)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-011 (nitrogen-neutral): NIST state $c'_5 \ ^1\Sigma_u^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-012 (nitrogen-neutral): NIST state $c_4 \ ^1\Pi_u$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-013 (nitrogen-neutral): NIST state $c_4 \ ^1\Pi_u$ 31; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-014 (nitrogen-neutral): NIST state $z \ ^1\Delta_g$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-015 (nitrogen-neutral): NIST state $y \ ^1\Pi_g$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-016 (nitrogen-neutral): NIST state $k \ ^1\Pi_g$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-017 (nitrogen-neutral): NIST state $x \ ^1\Sigma_g^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-018 (nitrogen-neutral): NIST state $d' \ ^1\Sigma_u^-$ or $^1\Delta_u$; term assignments 2 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-019 (nitrogen-neutral): NIST state $o_3 \ ^1\Pi_u$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-020 (nitrogen-neutral): NIST state $H \ ^3\Phi_u$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-021 (nitrogen-neutral): NIST state $c'_4 \ ^1\Sigma_u^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-022 (nitrogen-neutral): NIST state $c_3 \ ^1\Pi_u$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-023 (nitrogen-neutral): NIST state $b' \ ^1\Sigma_u^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-024 (nitrogen-neutral): NIST state $D \ ^3\Sigma_u^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True

- C7727379-state-025 (nitrogen-neutral): NIST state $b^1\Pi_u$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-026 (nitrogen-neutral): NIST state $a^1\Sigma_g^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-027 (nitrogen-neutral): NIST state $C^3\Pi_u$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-028 (nitrogen-neutral): NIST state $E^3\Sigma_g^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-029 (nitrogen-neutral): NIST state $C^5\Pi_u$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-030 (nitrogen-neutral): NIST state $C^3\Pi_u$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-031 (nitrogen-neutral): NIST state $G^3\Delta_g$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-032 (nitrogen-neutral): NIST state $A^5\Sigma_g^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-033 (nitrogen-neutral): NIST state $w^1\Delta_u$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-034 (nitrogen-neutral): NIST state $a^1\Pi_g$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-035 (nitrogen-neutral): NIST state $a^1\Sigma_u^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-036 (nitrogen-neutral): NIST state $B^3\Sigma_u^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-037 (nitrogen-neutral): NIST state $W^3\Delta_u$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-038 (nitrogen-neutral): NIST state $B^3\Pi_g$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-039 (nitrogen-neutral): NIST state $A^3\Sigma_u^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7727379-state-040 (nitrogen-neutral): NIST state $X^1\Sigma_g^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-001 (oxygen-neutral): NIST state $Z(^3\Pi_u)_2$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-002 (oxygen-neutral): NIST state $W(^3\Sigma_u^-)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-003 (oxygen-neutral): NIST state $p^1\Phi_u$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-004 (oxygen-neutral): NIST state $H(^3\Pi_u)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-005 (oxygen-neutral): NIST state $L(^3\Pi_u)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-006 (oxygen-neutral): NIST state $k(^1\Delta_u)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True

- C7782447-state-007 (oxygen-neutral): NIST state j ($^1\Sigma_u^+$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-008 (oxygen-neutral): NIST state G ($^3\Sigma_u^+$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-009 (oxygen-neutral): NIST state i ($^1\Delta_u$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-010 (oxygen-neutral): NIST state i' ($^3\Delta_u$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-011 (oxygen-neutral): NIST state h ($^1\Pi_u$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-012 (oxygen-neutral): NIST state g ($^1\Pi_u$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-013 (oxygen-neutral): NIST state F ($^3\Pi_u$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-014 (oxygen-neutral): NIST state E ($^3\Sigma_u^-$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-015 (oxygen-neutral): NIST state f ($^1\Sigma_u^+$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-016 (oxygen-neutral): NIST state D ($^3\Sigma_u^+$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-017 (oxygen-neutral): NIST state e ($^1\Delta_u$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-018 (oxygen-neutral): NIST state e' ($^3\Delta_u$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-019 (oxygen-neutral): NIST state d ($^1\Pi_g$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-020 (oxygen-neutral): NIST state C ($^3\Pi_g$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-021 (oxygen-neutral): NIST state B ($^3\Sigma_u^-$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-022 (oxygen-neutral): NIST state A ($^3\Sigma_u^+$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-023 (oxygen-neutral): NIST state A' ($^3\Delta_u$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-024 (oxygen-neutral): NIST state c ($^1\Sigma_u^-$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-025 (oxygen-neutral): NIST state b ($^1\Sigma_g^+$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-026 (oxygen-neutral): NIST state a ($^1\Delta_g$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782447-state-027 (oxygen-neutral): NIST state X ($^3\Sigma_g^-$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782414-state-001 (fluorine-neutral): NIST state K ($^1\Pi_u$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True

- C7782414-state-002 (fluorine-neutral): NIST state $J \ ^1\Pi_u \ (^4p \ \sigma)$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C7782414-state-003 (fluorine-neutral): NIST state $I \ ^1\Sigma_u^+ \ (^4p \ \pi)$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C7782414-state-004 (fluorine-neutral): NIST state $H \ ^1\Pi_u \ (^3p \ \sigma)$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C7782414-state-005 (fluorine-neutral): NIST state $h \ ^3\Pi_{1u} \ (3p \ \sigma)$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C7782414-state-006 (fluorine-neutral): NIST state $G \ ^1\Sigma_u^+ \ (^3p \ \pi)$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C7782414-state-007 (fluorine-neutral): NIST state $E \ (^1\Sigma_u^+)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782414-state-008 (fluorine-neutral): NIST state $D \ (^1\Sigma_u^+)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782414-state-009 (fluorine-neutral): NIST state $f \ (^3\Pi_{1g}) \ (3s \ \sigma)$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C7782414-state-010 (fluorine-neutral): NIST state $C \ ^1\Sigma_u^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782414-state-011 (fluorine-neutral): NIST state $F \ ^1\Pi_g \ (3s \ \sigma)$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C7782414-state-012 (fluorine-neutral): NIST state $A \ ^1\Pi_u \ 17$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782414-state-013 (fluorine-neutral): NIST state $a \ ^3\Pi_{0+u} \ 19$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782414-state-014 (fluorine-neutral): NIST state $X \ ^1\Sigma_g^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C630080-state-001 (carbon-monoxide-neutral): NIST state $O_2 \ (^1\Pi)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C630080-state-002 (carbon-monoxide-neutral): NIST state $O_1 \ (^1\Pi)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C630080-state-003 (carbon-monoxide-neutral): NIST state $Z \ ^1\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C630080-state-004 (carbon-monoxide-neutral): NIST state $H15 \ (^1\Pi)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C630080-state-005 (carbon-monoxide-neutral): NIST state $H' \ ^1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C630080-state-006 (carbon-monoxide-neutral): NIST state $L \ ^1\Pi \ 4p \ \pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C630080-state-007 (carbon-monoxide-neutral): NIST state $L' \ (^1\Pi)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C630080-state-008 (carbon-monoxide-neutral): NIST state $K \ ^1\Sigma^+ \ 4p \ \sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C630080-state-009 (carbon-monoxide-neutral): NIST state $W \ ^1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True

- C630080-state-010 (carbon-monoxide-neutral): NIST state $W' \ ^1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C630080-state-011 (carbon-monoxide-neutral): NIST state $J \ ^1\Sigma^+ 4s\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C630080-state-012 (carbon-monoxide-neutral): NIST state $G \ ^1\Pi(3d\pi)$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C630080-state-013 (carbon-monoxide-neutral): NIST state $G' \ ^1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C630080-state-014 (carbon-monoxide-neutral): NIST state $h \ (^3\Sigma^+, 4s\sigma)$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C630080-state-015 (carbon-monoxide-neutral): NIST state $Y \ ^1\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C630080-state-016 (carbon-monoxide-neutral): NIST state $F \ ^1\Sigma^+(3d\sigma)$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C630080-state-017 (carbon-monoxide-neutral): NIST state $V \ ^1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C630080-state-018 (carbon-monoxide-neutral): NIST state $g \ ^3\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C630080-state-019 (carbon-monoxide-neutral): NIST state $E \ ^1\Pi 3p\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C630080-state-020 (carbon-monoxide-neutral): NIST state $c \ ^3\Pi 3p\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C630080-state-021 (carbon-monoxide-neutral): NIST state $C \ ^1\Sigma^+ 3p\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C630080-state-022 (carbon-monoxide-neutral): NIST state $j \ (^3\Sigma^+ 3p\sigma)$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C630080-state-023 (carbon-monoxide-neutral): NIST state $B \ ^1\Sigma^+ 3s\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C630080-state-024 (carbon-monoxide-neutral): NIST state $b \ ^3\Sigma^+ 3s\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C630080-state-025 (carbon-monoxide-neutral): NIST state $D \ ^1\Delta$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C630080-state-026 (carbon-monoxide-neutral): NIST state $I \ ^1\Sigma^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C630080-state-027 (carbon-monoxide-neutral): NIST state $A \ ^1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C630080-state-028 (carbon-monoxide-neutral): NIST state $e \ ^3\Sigma^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C630080-state-029 (carbon-monoxide-neutral): NIST state $d \ ^3\Delta_i$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C630080-state-030 (carbon-monoxide-neutral): NIST state $a' \ ^3\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C630080-state-031 (carbon-monoxide-neutral): NIST state $a \ ^3\Pi_r$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True

- C630080-state-032 (carbon-monoxide-neutral): NIST state $X \ ^2\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10102439-state-001 (nitric-oxide-neutral): NIST state $Z \ ^2\Sigma^+ \ 7s\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-002 (nitric-oxide-neutral): NIST state $Y \ ^2\Sigma^+ \ 6p\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-003 (nitric-oxide-neutral): NIST state $W \ ^2\Pi \ 6p\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-004 (nitric-oxide-neutral): NIST state $U \ ^2\Delta \ 5d\delta$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-005 (nitric-oxide-neutral): NIST state $T \ ^2\Sigma^+ \ 6s\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-006 (nitric-oxide-neutral): NIST state $R \ ^2\Sigma^+ \ 5p\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-007 (nitric-oxide-neutral): NIST state $Q \ ^2\Pi \ 5p\pi \ 26$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-008 (nitric-oxide-neutral): NIST state $Q \ ^2\Pi \ 5p\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-009 (nitric-oxide-neutral): NIST state $O' \ ^2\Pi \ 4d\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-010 (nitric-oxide-neutral): NIST state $O \ ^2\Sigma^+ \ 4d\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-011 (nitric-oxide-neutral): NIST state $N \ ^2\Delta \ 4d\delta$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-012 (nitric-oxide-neutral): NIST state $S \ ^2\Sigma^+ \ 5s\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-013 (nitric-oxide-neutral): NIST state $M \ ^2\Sigma^+ \ 4p\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-014 (nitric-oxide-neutral): NIST state $K \ ^2\Pi \ 4p\pi \ 34$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-015 (nitric-oxide-neutral): NIST state $K \ ^2\Pi \ 4p\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-016 (nitric-oxide-neutral): NIST state $I \ ^2\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10102439-state-017 (nitric-oxide-neutral): NIST state $G \ ^2\Sigma^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10102439-state-018 (nitric-oxide-neutral): NIST state $L \ ^2\Pi_i$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10102439-state-019 (nitric-oxide-neutral): NIST state $H' \ ^2\Pi \ 3d\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-020 (nitric-oxide-neutral): NIST state $H \ ^2\Sigma^+ \ 3d\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-021 (nitric-oxide-neutral): NIST state $F \ ^2\Delta \ 3d\delta$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True

- C10102439-state-022 (nitric-oxide-neutral): NIST state $E \ ^2\Sigma^+4s\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-023 (nitric-oxide-neutral): NIST state $B' \ ^2\Delta_i$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10102439-state-024 (nitric-oxide-neutral): NIST state $D \ ^2\Sigma^+ 3p\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-025 (nitric-oxide-neutral): NIST state $C \ ^2\Pi_r 3p\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-026 (nitric-oxide-neutral): NIST state $b \ (^4\Sigma^-)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10102439-state-027 (nitric-oxide-neutral): NIST state $B \ ^2\Pi_r$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10102439-state-028 (nitric-oxide-neutral): NIST state $A \ ^2\Sigma^+ 3s\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10102439-state-029 (nitric-oxide-neutral): NIST state $a \ (^4\Pi_i)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10102439-state-030 (nitric-oxide-neutral): NIST state $X \ ^2\Pi_r$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C3352576-state-001 (hydroxyl-neutral): NIST state $C \ ^2\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C3352576-state-002 (hydroxyl-neutral): NIST state $C \ ^2\Sigma^+ 5$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C3352576-state-003 (hydroxyl-neutral): NIST state $D \ ^2\Sigma^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C3352576-state-004 (hydroxyl-neutral): NIST state $B \ ^2\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C3352576-state-005 (hydroxyl-neutral): NIST state $A \ ^2\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C3352576-state-006 (hydroxyl-neutral): NIST state $X \ ^2\Pi_i$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C13774920-state-001 (imidogen-neutral): NIST state $d \ ^1\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C13774920-state-002 (imidogen-neutral): NIST state $c \ ^1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C13774920-state-003 (imidogen-neutral): NIST state $A \ ^3\Pi_i$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C13774920-state-004 (imidogen-neutral): NIST state $b \ ^1\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C13774920-state-005 (imidogen-neutral): NIST state $a \ ^1\Delta$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C13774920-state-006 (imidogen-neutral): NIST state $X \ ^3\Sigma^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C3315375-state-001 (methylidyne-neutral): NIST state $F \ ^2\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True

- C3315375-state-002 (methylidyne-neutral): NIST state $E \ ^2\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C3315375-state-003 (methylidyne-neutral): NIST state $D \ ^2\Pi_i$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C3315375-state-004 (methylidyne-neutral): NIST state $C \ ^2\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C3315375-state-005 (methylidyne-neutral): NIST state $B \ ^2\Sigma^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C3315375-state-006 (methylidyne-neutral): NIST state $A \ ^2\Delta$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C3315375-state-007 (methylidyne-neutral): NIST state $a \ ^4\Sigma^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C3315375-state-008 (methylidyne-neutral): NIST state $X \ ^2\Pi_r$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782505-state-001 (chlorine-neutral): NIST state $C \ ^1\Pi_u$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782505-state-002 (chlorine-neutral): NIST state $B \ ^3\Pi_0^+_{u_1}$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782505-state-003 (chlorine-neutral): NIST state $A \ (^3\Pi_1)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782505-state-004 (chlorine-neutral): NIST state $A' \ (^3\Pi_2)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7782505-state-005 (chlorine-neutral): NIST state $X \ ^1\Sigma_g^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10097286-state-001 (silicon-monoxide-neutral): NIST state $P \ ^1\Sigma^+ (5p\sigma)$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10097286-state-002 (silicon-monoxide-neutral): NIST state $O \ ^1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10097286-state-003 (silicon-monoxide-neutral): NIST state $N \ ^1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10097286-state-004 (silicon-monoxide-neutral): NIST state $M \ ^1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10097286-state-005 (silicon-monoxide-neutral): NIST state $L \ ^1\Sigma^+ (5s\sigma)$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10097286-state-006 (silicon-monoxide-neutral): NIST state $K \ ^1\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10097286-state-007 (silicon-monoxide-neutral): NIST state $J \ ^1\Pi (4p\pi)$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10097286-state-008 (silicon-monoxide-neutral): NIST state $h \ ^3\Pi_i$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10097286-state-009 (silicon-monoxide-neutral): NIST state $I \ ^1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10097286-state-010 (silicon-monoxide-neutral): NIST state $H \ ^1\Sigma^+ (4p\sigma)$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True

- C10097286-state-011 (silicon-monoxide-neutral): NIST state $G^1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10097286-state-012 (silicon-monoxide-neutral): NIST state $F^1\Sigma^+ (4s\sigma)$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10097286-state-013 (silicon-monoxide-neutral): NIST state $g^3\Sigma^+ (4s\sigma)$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C10097286-state-014 (silicon-monoxide-neutral): NIST state $f^3\Pi_i$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10097286-state-015 (silicon-monoxide-neutral): NIST state $c^3\Sigma^+ (+)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10097286-state-016 (silicon-monoxide-neutral): NIST state $E^1\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10097286-state-017 (silicon-monoxide-neutral): NIST state $A^1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10097286-state-018 (silicon-monoxide-neutral): NIST state $D^1\Delta$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10097286-state-019 (silicon-monoxide-neutral): NIST state $C^1\Sigma^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10097286-state-020 (silicon-monoxide-neutral): NIST state $e^3\Sigma^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10097286-state-021 (silicon-monoxide-neutral): NIST state $d^3\Delta_r$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10097286-state-022 (silicon-monoxide-neutral): NIST state $b^3\Pi_2$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10097286-state-023 (silicon-monoxide-neutral): NIST state $b^3\Pi_1$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10097286-state-024 (silicon-monoxide-neutral): NIST state $b^3\Pi_0$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10097286-state-025 (silicon-monoxide-neutral): NIST state $a^3\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10097286-state-026 (silicon-monoxide-neutral): NIST state $X^1\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C13827322-state-001 (sulfur-monoxide-neutral): NIST state $E (^3\Pi)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C13827322-state-002 (sulfur-monoxide-neutral): NIST state $D (^3\Pi)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C13827322-state-003 (sulfur-monoxide-neutral): NIST state $B^3\Sigma^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C13827322-state-004 (sulfur-monoxide-neutral): NIST state $A^3\Pi_2$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C13827322-state-005 (sulfur-monoxide-neutral): NIST state $A^3\Pi_1$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C13827322-state-006 (sulfur-monoxide-neutral): NIST state $A^3\Pi_0$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True

- C13827322-state-007 (sulfur-monoxide-neutral): NIST state $b \ ^1\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C13827322-state-008 (sulfur-monoxide-neutral): NIST state $a \ ^1\Delta$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C13827322-state-009 (sulfur-monoxide-neutral): NIST state $X \ ^3\Sigma^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7664393-state-001 (hydrogen-fluoride-neutral): NIST state $D \ ^1\Sigma^+ \ 1$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7664393-state-002 (hydrogen-fluoride-neutral): NIST state $C \ ^1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7664393-state-003 (hydrogen-fluoride-neutral): NIST state $b \ ^3\Pi \ 2$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7664393-state-004 (hydrogen-fluoride-neutral): NIST state $B \ ^1\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7664393-state-005 (hydrogen-fluoride-neutral): NIST state $X \ ^1\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7647010-state-001 (hydrogen-chloride-neutral): NIST state $M \ (^1\Sigma^+)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7647010-state-002 (hydrogen-chloride-neutral): NIST state $L \ (^1\Sigma^+, ^1\Pi)$; term assignments 2 pass True; configuration assignments 0 pass True; row pass True
- C7647010-state-003 (hydrogen-chloride-neutral): NIST state $K \ ^1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7647010-state-004 (hydrogen-chloride-neutral): NIST state $H \ ^1\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7647010-state-005 (hydrogen-chloride-neutral): NIST state $E \ ^1\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7647010-state-006 (hydrogen-chloride-neutral): NIST state $g \ (^3\Sigma^-)1$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7647010-state-007 (hydrogen-chloride-neutral): NIST state $f_1 \ ^3\Delta_1$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7647010-state-008 (hydrogen-chloride-neutral): NIST state $D \ ^1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7647010-state-009 (hydrogen-chloride-neutral): NIST state $d_0 \ ^3\Pi_0$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7647010-state-010 (hydrogen-chloride-neutral): NIST state $f_2 \ ^3\Delta_2$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7647010-state-011 (hydrogen-chloride-neutral): NIST state $f_3 \ ^3\Delta_3$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7647010-state-012 (hydrogen-chloride-neutral): NIST state $d_1 \ ^3\Pi_1$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7647010-state-013 (hydrogen-chloride-neutral): NIST state $d_2 \ ^3\Pi_2$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7647010-state-014 (hydrogen-chloride-neutral): NIST state $C \ ^1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True

- C7647010-state-015 (hydrogen-chloride-neutral): NIST state $V \ ^1\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7647010-state-016 (hydrogen-chloride-neutral): NIST state $V \ ^1\Sigma^+$ 21; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7647010-state-017 (hydrogen-chloride-neutral): NIST state $b_0 \ ^3\Pi_0$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7647010-state-018 (hydrogen-chloride-neutral): NIST state $b_1 \ ^3\Pi_1$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7647010-state-019 (hydrogen-chloride-neutral): NIST state $b_2 \ ^3\Pi_2$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7647010-state-020 (hydrogen-chloride-neutral): NIST state $A \ (^1\Pi) \ 24 \ 25$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7647010-state-021 (hydrogen-chloride-neutral): NIST state $X \ ^1\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10035106-state-001 (hydrogen-bromide-neutral): NIST state $M \ (^1\Sigma^+)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10035106-state-002 (hydrogen-bromide-neutral): NIST state $L \ (^1\Sigma^+, \ ^1\Pi)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10035106-state-003 (hydrogen-bromide-neutral): NIST state $g \ (^3\Sigma^-)0^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10035106-state-004 (hydrogen-bromide-neutral): NIST state $F \ ^1\Delta$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10035106-state-005 (hydrogen-bromide-neutral): NIST state $f_1 \ ^3\Delta_1$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10035106-state-006 (hydrogen-bromide-neutral): NIST state $D \ ^1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10035106-state-007 (hydrogen-bromide-neutral): NIST state $d_0 \ ^3\Pi_0$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10035106-state-008 (hydrogen-bromide-neutral): NIST state $E \ (^3\Sigma^+)0^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10035106-state-009 (hydrogen-bromide-neutral): NIST state $V \ ^1\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10035106-state-010 (hydrogen-bromide-neutral): NIST state $f_2 \ ^3\Delta_2$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10035106-state-011 (hydrogen-bromide-neutral): NIST state $f_3 \ ^3\Delta_3$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10035106-state-012 (hydrogen-bromide-neutral): NIST state $e \ ^3\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10035106-state-013 (hydrogen-bromide-neutral): NIST state $d_1 \ ^3\Pi_1$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10035106-state-014 (hydrogen-bromide-neutral): NIST state $d_2 \ ^3\Pi_2$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10035106-state-015 (hydrogen-bromide-neutral): NIST state $C \ ^1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True

- C10035106-state-016 (hydrogen-bromide-neutral): NIST state $b_0 \ ^3\Pi_0 \ 0^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10035106-state-017 (hydrogen-bromide-neutral): NIST state $b_0 \ ^3\Pi_0 \ 0^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10035106-state-018 (hydrogen-bromide-neutral): NIST state $b_1 \ ^3\Pi_1$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10035106-state-019 (hydrogen-bromide-neutral): NIST state $b_2 \ ^3\Pi_2$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10035106-state-020 (hydrogen-bromide-neutral): NIST state A ($^1\Pi$) 28; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C10035106-state-021 (hydrogen-bromide-neutral): NIST state X $^1\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7553562-state-001 (iodine-neutral): NIST state F ($^1\Sigma_u^+$) 6; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7553562-state-002 (iodine-neutral): NIST state G' ($^3\Pi_1g$) 9; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7553562-state-003 (iodine-neutral): NIST state E $^3\Pi_o+g$ 9; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7553562-state-004 (iodine-neutral): NIST state D $^1\Sigma_u^+$ 13; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7553562-state-005 (iodine-neutral): NIST state G $^3\Pi_2g$ 9; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7553562-state-006 (iodine-neutral): NIST state C $^3\Sigma_1u^+$ 13; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7553562-state-007 (iodine-neutral): NIST state B'' $^1\Pi_1u$ 19; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7553562-state-008 (iodine-neutral): NIST state B' $^3\Pi_0-u$ 19; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7553562-state-009 (iodine-neutral): NIST state B $^3\Pi_0+u$ 19; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7553562-state-010 (iodine-neutral): NIST state A $^3\Pi_1u$ 19; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7553562-state-011 (iodine-neutral): NIST state A' $^3\Pi_2u$ 19; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C7553562-state-012 (iodine-neutral): NIST state X $^1\Sigma_g^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C12184906-state-001 (hydrogen-cation): NIST state C $^2\Pi_u \ 2p\pi$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C12184906-state-002 (hydrogen-cation): NIST state B $^2\Sigma_g^+ \ 3d\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C12184906-state-003 (hydrogen-cation): NIST state A $^2\Sigma_u^+ \ 2p\sigma \ 3$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True
- C12184906-state-004 (hydrogen-cation): NIST state X $^2\Sigma_g^+ \ 1s\sigma$; term assignments 1 pass True; configuration assignments 1 pass True; row pass True

- C13966046-state-001 (nitrogen-cation): NIST state $G \ ^2\Sigma_g^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C13966046-state-002 (nitrogen-cation): NIST state $C \ ^2\Sigma_u^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C13966046-state-003 (nitrogen-cation): NIST state $D \ ^2\Pi_{gi}$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C13966046-state-004 (nitrogen-cation): NIST state $(a \ ^4\Sigma_u^+)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C13966046-state-005 (nitrogen-cation): NIST state $B \ ^2\Sigma_u^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C13966046-state-006 (nitrogen-cation): NIST state $A \ ^2\Pi_{ui}$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C13966046-state-007 (nitrogen-cation): NIST state $X \ ^2\Sigma_g^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C12185078-state-001 (oxygen-cation): NIST state $x_2 \ ^2\Sigma^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C12185078-state-002 (oxygen-cation): NIST state $x_1 \ ^4\Sigma^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C12185078-state-003 (oxygen-cation): NIST state $(^2\Sigma_g^-)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C12185078-state-004 (oxygen-cation): NIST state $(^4\Sigma_g^-)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C12185078-state-005 (oxygen-cation): NIST state $(^2\Sigma_u^-)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C12185078-state-006 (oxygen-cation): NIST state $c \ ^4\Sigma_u^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C12185078-state-007 (oxygen-cation): NIST state $(^2\Pi_u)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C12185078-state-008 (oxygen-cation): NIST state $B \ ^2\Sigma_g^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C12185078-state-009 (oxygen-cation): NIST state $D \ (^2\Delta_g)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C12185078-state-010 (oxygen-cation): NIST state $C \ (^2\Phi_u)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C12185078-state-011 (oxygen-cation): NIST state $b \ ^4\Sigma_g^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C12185078-state-012 (oxygen-cation): NIST state $b' \ (^4\Pi_g)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C12185078-state-013 (oxygen-cation): NIST state $A \ ^2\Pi_u$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C12185078-state-014 (oxygen-cation): NIST state $a \ ^4\Pi_{ui}$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C12185078-state-015 (oxygen-cation): NIST state $X \ ^2\Pi_g$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True

- C14452938-state-001 (nitric-oxide-cation): NIST state $\wedge 1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14452938-state-002 (nitric-oxide-cation): NIST state $\wedge 3\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14452938-state-003 (nitric-oxide-cation): NIST state $\wedge 1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14452938-state-004 (nitric-oxide-cation): NIST state $\wedge 3\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14452938-state-005 (nitric-oxide-cation): NIST state $\wedge 1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14452938-state-006 (nitric-oxide-cation): NIST state $\wedge 3\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14452938-state-007 (nitric-oxide-cation): NIST state $(B' \wedge 1\Sigma^+)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14452938-state-008 (nitric-oxide-cation): NIST state $c \wedge 3\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14452938-state-009 (nitric-oxide-cation): NIST state $A \wedge 1\Pi$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14452938-state-010 (nitric-oxide-cation): NIST state $W \wedge 1\Delta$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14452938-state-011 (nitric-oxide-cation): NIST state $A' \wedge 1\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14452938-state-012 (nitric-oxide-cation): NIST state $b' \wedge 3\Sigma^-$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14452938-state-013 (nitric-oxide-cation): NIST state $w \wedge 3\Delta$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14452938-state-014 (nitric-oxide-cation): NIST state $b \wedge 3\Pi_p$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14452938-state-015 (nitric-oxide-cation): NIST state $a \wedge 3\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14452938-state-016 (nitric-oxide-cation): NIST state $X \wedge 1\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C11062774-state-001 (oxygen-anion): NIST state $a (\wedge 4\Pi_u)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C11062774-state-002 (oxygen-anion): NIST state $A (\wedge 2\Pi_u)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C11062774-state-003 (oxygen-anion): NIST state $X \wedge 2\Pi_{gi}$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14967783-state-001 (nitric-oxide-anion): NIST state $(\wedge 3\Sigma^-)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14967783-state-002 (nitric-oxide-anion): NIST state $(\wedge 3\Sigma^+)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14967783-state-003 (nitric-oxide-anion): NIST state $(\wedge 3\Pi)$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True

- C14967783-state-004 (nitric-oxide-anion): NIST state ($\wedge^1\Sigma^+$); term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14967783-state-005 (nitric-oxide-anion): NIST state b $\wedge^1\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14967783-state-006 (nitric-oxide-anion): NIST state a $\wedge^1\Delta$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- C14967783-state-007 (nitric-oxide-anion): NIST state X $\wedge^3\Sigma^+$; term assignments 1 pass True; configuration assignments 0 pass True; row pass True
- adverse unknown_support_symbol_rejected: True
- adverse triple_occupancy_rejected: True
- adverse wrong_multiplicity_parity_rejected: True
- adverse omitted_row_rejected: True
- adverse tampered_snapshot_rejected: True
- adverse complete_assignment_counts_retained: True

Falsification boundary: The claim fails if a joined molecular support lacks one of the two phase fibres; if a required axis class cannot be represented by structural empty One or positive recurrence; if a spatial cell admits a third electron or same-spin pair; if complete occupied support loses or duplicates an electron occurrence; if any of the 360 registered NIST state rows or 362 term assignments leaves the generated axis-support grammar; if any of the 87 explicit spectroscopic configuration assignments leaves the occupancy grammar; if a measured state multiplicity violates the ELEC-002 electron-count parity law; or if any source, row or adverse control is altered.

Measurement receipt: sha256:7d9b1e2a3f6679ddc372e01550e1074e835466cc4334e15ee58098e267c55620.

Isolation certificate: sha256:6787fc1d4bf180e27931f59a3a28059a779493afb2b96f764575fa19d4ee3392.

Custody certificate: sha256:3b01b21145fba21a0dbb4b23c7079348cbfadc1a9c542e90a121fcea2f289943.

Registered source descriptions:

- National Institute of Standards and Technology: <https://webbook.nist.gov/chemistry/> ()

Registered target identities:

- C1333740-state-001 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C1333740-state-002 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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Meaning. The field-wide extension derives molecular electronic identity, state composition, occupancy, ordering, symmetry, correlation, transitions, measurement and operational classical-quantum correspondence from the same exact carrier laws. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical-zero, negative, irrational, imaginary or floating proof value
- no continuum angular coordinate or imported molecular wavefunction
- no conventional sigma/pi/delta/phi symbol in candidate generation or survivor selection
- no imported molecular-orbital energy ordering or species configuration table
- no occupancy above one complementary spin pair in a spatial support cell
- no measured NIST state or configuration assignment before prediction seal
- no fitted, learned, species-specific or target-derived parameter
- no selected, omitted or silently normalized spectroscopic assignment row

Immutable identities. Source manifest

sha256:39f060ce18c09376a4114b37703b0c5aa7f3408f376818a5959898ce6a1ce0c9; derivation seal
 sha256:13948e01e0975e0def8c7bb556fc5b9bb31a79d9547aba8412638f61ab1d0f6b; engine receipt
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99. Exact finite ground and excited molecular-state ordering

Claim identity: SFT-CHEM-STATE-ENERGY-ORDER-004

Question and theorem. At one declared molecular composition and geometry, every finite distinguishable electronic-state support has exactly one structural least state and every excited state occupies one exact positive successor position; all distinct state pairs are comparable, every higher/lower separation remains a retained positive gap, and the complete state/support identity record is preserved.

one-molecule-state-support__state-and-support-identity__structural-empty-One-ground__positive-successor-excitation__complete-finite-total-order__retained-positive-order-gap__complete-state-order-record__no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001

- SFT-CHEM-ELECTRONIC-STATE-IDENTITY-001
- SFT-CHEM-ELECTRON-COUNT-SPIN-002
- SFT-CHEM-ORBITAL-SUPPORT-OCCUPANCY-003

Generated grammar. Generate the literal product of carrier, identity, least-state, excitation, order, gap, record and extension forms. Decide all 256 forms only by admitted exact order, distinguishability, discrete spectrum, electronic state, electron-count and molecular-support laws.

Boundary: Every positive finite set of distinguishable molecular electronic states at one declared composition and geometry. The least element is structural empty One and each remaining state is one positive successor. Closure is depth-independent because adjoining one distinguishable state forces exactly the next positive position and its comparisons with every retained predecessor.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `one-molecule-state-support__state-and-support-identity__structural-empty-One-ground__positive-successor-excitation__complete-finite-total-order__retained-positive-order-gap__complete-state-order-record__no-extra-rule`. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-molecule-state-support	cross-molecule-state-list	States from different carriers have no common declared molecular order.
identity	state-and-support-identity	energy-only-anonymous-row	An anonymous energy cannot preserve which electronic support carries it.
least	structural-empty-One-ground	numerical-zero-ground	Numerical zero imports a forbidden proof magnitude and does not establish structural leastness.
excitation	positive-successor-excitation	signed-energy-displacement	A signed displacement imports negative proof quantities.
order	complete-finite-total-order	partial-selected-comparison	Selected comparisons cannot establish a complete finite state order.
gap	retained-positive-order-gap	unretained-transition-gap	Erasing the positive separation loses transition and reconstruction information.
record	complete-state-order-record	ordered-labels-without-trace	A label list without support and observation trace cannot reproduce the order.
extension	no-extra-rule	species-energy-lookup-rule	A species lookup or measured energy in generation lets the target select the law.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : b90d354668dadd63e80d745c1737ae1e0a7310d67e1a4d8b094598327bad4ecd.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 6fde0193bd34962b2b91d6578dbdcd5155b72ac61698da77f66bab80a0eada7f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : fbef2494f3668e38ca47f1ae3296b9b60b1558e5797a5166ecde36d43b87ef36.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : e76ea7268e17372df950c926adc608773769814a1ee91b95fb5736f0de82a40c.

Independent reconstruction. Implementation hash

sha256 : c1d3db3e78712da963746f27c97a9d77db6f6ea4aae01bb05c41cd008112ea47; independent certificate
sha256 : d3e65245c5421d9d03e3546c82fc36055dff8dff043d7108a57da5bb6dde8dec; external-validation

hash sha256:0670cb93bb8ff692cbdc551f752b5870f4748680096dae59e8a4ac995837978a. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025

Comparison records:

- C1333740-energy-045 (hydrogen-neutral): NIST X $^1\Sigma_g^+ 1s\sigma^2$; Te 0 cm⁻¹ = 0/1; ground True; exact gap 0/1; order structural-empty-One; pass True
- C1333740-energy-044 (hydrogen-neutral): NIST B $^1\Sigma_u^+ 2p\sigma$; Te 91700.0 cm⁻¹ = 91700/1; ground False; exact gap 91700/1; order 1; pass True
- C1333740-energy-021 (hydrogen-neutral): NIST c $^3\Pi_u 3p\pi$; Te 95838.5 cm⁻¹ = 191677/2; ground False; exact gap 191677/2; order 2; pass True
- C1333740-energy-020 (hydrogen-neutral): NIST a $^3\Sigma_g^+ 2s\sigma$; Te 95936.1 cm⁻¹ = 959361/10; ground False; exact gap 959361/10; order 3; pass True
- C1333740-energy-042 (hydrogen-neutral): NIST E $^1\Sigma_g^+ 2s\sigma$; Te 100082.3 cm⁻¹ = 1000823/10; ground False; exact gap 1000823/10; order 4; pass True
- C1333740-energy-043 (hydrogen-neutral): NIST C $^1\Pi_u 2p\pi$; Te 100089.8 cm⁻¹ = 500449/5; ground False; exact gap 500449/5; order 5; pass True
- C1333740-energy-041 (hydrogen-neutral): NIST F $^1\Sigma_g^+ 2p\sigma^2$; Te 100911 cm⁻¹ = 100911/1; ground False; exact gap 100911/1; order 6; pass True
- C1333740-energy-019 (hydrogen-neutral): NIST e $^3\Sigma_u^+ 3p\sigma$; Te 107774.7 cm⁻¹ = 1077747/10; ground False; exact gap 1077747/10; order 7; pass True
- C1333740-energy-040 (hydrogen-neutral): NIST B' $^1\Sigma_u^+ 3p\sigma$; Te 111642.8 cm⁻¹ = 558214/5; ground False; exact gap 558214/5; order 8; pass True
- C1333740-energy-038 (hydrogen-neutral): NIST K ($^1\Sigma_g^+$); Te (112669) cm⁻¹ = 112669/1; ground False; exact gap 112669/1; order 9; pass True
- C1333740-energy-018 (hydrogen-neutral): NIST d $^3\Pi_u 3p\pi$; Te 112700.3 cm⁻¹ = 1127003/10; ground False; exact gap 1127003/10; order 10; pass True
- C1333740-energy-037 (hydrogen-neutral): NIST G $^1\Sigma_g^+ 3d\sigma$; Te 112834 cm⁻¹ = 112834/1; ground False; exact gap 112834/1; order 11; pass True
- C1333740-energy-017 (hydrogen-neutral): NIST g $^3\Sigma_g^+ 3d\sigma$; Te 112854.4 cm⁻¹ = 564272/5; ground False; exact gap 564272/5; order 12; pass True
- C1333740-energy-016 (hydrogen-neutral): NIST h $^3\Sigma_g^+ 3s\sigma$; Te (112913) cm⁻¹ = 112913/1; ground False; exact gap 112913/1; order 13; pass True
- C1333740-energy-015 (hydrogen-neutral): NIST i $^3\Pi_g 3d\pi$; Te (113132) cm⁻¹ = 113132/1; ground False; exact gap 113132/1; order 14; pass True
- C1333740-energy-036 (hydrogen-neutral): NIST I $^1\Pi_g 3d\pi$; Te (113142) cm⁻¹ = 113142/1; ground False; exact gap 113142/1; order 15; pass True
- C1333740-energy-039 (hydrogen-neutral): NIST Q ($^1\Pi_g$); Te (113163) cm⁻¹ = 113163/1; ground False; exact gap 113163/1; order 16; pass True
- C1333740-energy-014 (hydrogen-neutral): NIST j $^3\Delta_g 3d\delta$; Te (113533) cm⁻¹ = 113533/1; ground False; exact gap 113533/1; order 17; pass True
- C1333740-energy-035 (hydrogen-neutral): NIST J $^1\Delta_g 3d\delta$; Te (113550) cm⁻¹ = 113550/1; ground False; exact gap 113550/1; order 18; pass True

- C1333740-energy-013 (hydrogen-neutral): NIST I $\Lambda^3\Pi_u$; Te 113825 cm^{-1} = 113825/1; ground False; exact gap 113825/1; order 19; pass True
- C1333740-energy-034 (hydrogen-neutral): NIST D $\Lambda^1\Pi_u$ $3p\pi$; Te 113888.7 cm^{-1} = 1138887/10; ground False; exact gap 1138887/10; order 20; pass True
- C1333740-energy-033 (hydrogen-neutral): NIST H $\Lambda^1\Sigma_g^+$ $3s\sigma$; Te 113899 cm^{-1} = 113899/1; ground False; exact gap 113899/1; order 21; pass True
- C1333740-energy-012 (hydrogen-neutral): NIST o $\Lambda^3\Sigma_u^+$; Te (114234) cm^{-1} = 114234/1; ground False; exact gap 114234/1; order 22; pass True
- C1333740-energy-031 (hydrogen-neutral): NIST M $\Lambda^1\Sigma_g^+$; Te (114485) cm^{-1} = 114485/1; ground False; exact gap 114485/1; order 23; pass True
- C1333740-energy-032 (hydrogen-neutral): NIST L $\Lambda^1\Sigma_g^+$; Te (114520) cm^{-1} = 114520/1; ground False; exact gap 114520/1; order 24; pass True
- C1333740-energy-029 (hydrogen-neutral): NIST N $\Lambda^1\Sigma_g^+$; Te (116287) cm^{-1} = 116287/1; ground False; exact gap 116287/1; order 25; pass True
- C1333740-energy-011 (hydrogen-neutral): NIST f $\Lambda^3\Sigma_u^+$ $4p\sigma$; Te (116705) cm^{-1} = 116705/1; ground False; exact gap 116705/1; order 26; pass True
- C1333740-energy-030 (hydrogen-neutral): NIST U ($\Lambda^1\Sigma_g^+$); Te [116707.7] cm^{-1} = 1167077/10; ground False; exact gap 1167077/10; order 27; pass True
- C1333740-energy-028 (hydrogen-neutral): NIST B' $\Lambda^1\Sigma_u^+$ $4p\sigma$; Te 117984.5 cm^{-1} = 235969/2; ground False; exact gap 235969/2; order 28; pass True
- C1333740-energy-009 (hydrogen-neutral): NIST v ($\Lambda^3\Pi_g$); Te (118330) cm^{-1} = 118330/1; ground False; exact gap 118330/1; order 29; pass True
- C1333740-energy-010 (hydrogen-neutral): NIST k $\Lambda^3\Pi_u$ $4p\pi$; Te 118366.2 cm^{-1} = 591831/5; ground False; exact gap 591831/5; order 30; pass True
- C1333740-energy-008 (hydrogen-neutral): NIST p $\Lambda^3\Sigma_g^+$ $4d\sigma$; Te 118509.8 cm^{-1} = 592549/5; ground False; exact gap 592549/5; order 31; pass True
- C1333740-energy-007 (hydrogen-neutral): NIST r $\Lambda^3\Pi_g$ $4d\pi$; Te 118613.7 cm^{-1} = 1186137/10; ground False; exact gap 1186137/10; order 32; pass True
- C1333740-energy-025 (hydrogen-neutral): NIST R $\Lambda^1\Pi_g$ $4d\pi$; Te (118688) cm^{-1} = 118688/1; ground False; exact gap 118688/1; order 33; pass True
- C1333740-energy-023 (hydrogen-neutral): NIST D' $\Lambda^1\Pi_u$ $4p\pi$; Te 118865.3 cm^{-1} = 1188653/10; ground False; exact gap 1188653/10; order 34; pass True
- C1333740-energy-006 (hydrogen-neutral): NIST s $\Lambda^3\Delta_g$ $4d\delta$; Te 118875.2 cm^{-1} = 594376/5; ground False; exact gap 594376/5; order 35; pass True
- C1333740-energy-005 (hydrogen-neutral): NIST m $\Lambda^3\Sigma_u^+$ $4f\sigma$; Te (119317) cm^{-1} = 119317/1; ground False; exact gap 119317/1; order 36; pass True
- C1333740-energy-027 (hydrogen-neutral): NIST T $\Lambda^1\Sigma^+$; Te [119512.6] cm^{-1} = 597563/5; ground False; exact gap 597563/5; order 37; pass True
- C1333740-energy-026 (hydrogen-neutral): NIST P $\Lambda^1\Sigma_g^+$ $4d\sigma$; Te [119531] cm^{-1} = 119531/1; ground False; exact gap 119531/1; order 38; pass True
- C1333740-energy-024 (hydrogen-neutral): NIST S $\Lambda^1\Delta_g$ $4d\delta$; Te [119893] cm^{-1} = 119893/1; ground False; exact gap 119893/1; order 39; pass True
- C1333740-energy-004 (hydrogen-neutral): NIST n $\Lambda^3\Pi_u$ $5p\pi$; Te 120952.9 cm^{-1} = 1209529/10; ground False; exact gap 1209529/10; order 40; pass True

- C1333740-energy-022 (hydrogen-neutral): NIST D" $\Lambda^1\Pi_u$ 5p π ; Te 121211.0 cm⁻¹ = 121211/1; ground False; exact gap 121211/1; order 41; pass True
- C1333740-energy-002 (hydrogen-neutral): NIST t $\Lambda^3\Sigma_u^+$ 5f σ ; Te (121292) cm⁻¹ = 121292/1; ground False; exact gap 121292/1; order 42; pass True
- C1333740-energy-003 (hydrogen-neutral): NIST q ($\Lambda^3\Sigma_g^+$) 5d σ ; Te (121295) cm⁻¹ = 121295/1; ground False; exact gap 121295/1; order 43; pass True
- C1333740-energy-001 (hydrogen-neutral): NIST u $\Lambda^3\Pi_u$ 6p π ; Te [123488._0] cm⁻¹ = 123488/1; ground False; exact gap 123488/1; order 44; pass True
- C7727379-energy-033 (nitrogen-neutral): NIST X $\Lambda^1\Sigma_g^+$; Te 0 cm⁻¹ = 0/1; ground True; exact gap 0/1; order structural-empty-One; pass True
- C7727379-energy-032 (nitrogen-neutral): NIST A $\Lambda^3\Sigma_u^+$; Te 50203.6_3 cm⁻¹ = 5020363/100; ground False; exact gap 5020363/100; order 1; pass True
- C7727379-energy-031 (nitrogen-neutral): NIST B $\Lambda^3\Pi_g$; Te 59619.3_5 cm⁻¹ = 1192387/20; ground False; exact gap 1192387/20; order 2; pass True
- C7727379-energy-030 (nitrogen-neutral): NIST W $\Lambda^3\Delta_u$; Te 59808 cm⁻¹ = 59808/1; ground False; exact gap 59808/1; order 3; pass True
- C7727379-energy-029 (nitrogen-neutral): NIST B' $\Lambda^3\Sigma_u^-$; Te 66272.4_7 cm⁻¹ = 6627247/100; ground False; exact gap 6627247/100; order 4; pass True
- C7727379-energy-028 (nitrogen-neutral): NIST a' $\Lambda^1\Sigma_u^-$; Te 68152.66 cm⁻¹ = 3407633/50; ground False; exact gap 3407633/50; order 5; pass True
- C7727379-energy-027 (nitrogen-neutral): NIST a $\Lambda^1\Pi_g$; Te 69283.06 cm⁻¹ = 3464153/50; ground False; exact gap 3464153/50; order 6; pass True
- C7727379-energy-026 (nitrogen-neutral): NIST w $\Lambda^1\Delta_u$; Te 72097.4 cm⁻¹ = 360487/5; ground False; exact gap 360487/5; order 7; pass True
- C7727379-energy-025 (nitrogen-neutral): NIST A' $\Lambda^5\Sigma_g^+$; Te (78800) cm⁻¹ = 78800/1; ground False; exact gap 78800/1; order 8; pass True
- C7727379-energy-024 (nitrogen-neutral): NIST G $\Lambda^3\Delta_g$; Te (87900) cm⁻¹ = 87900/1; ground False; exact gap 87900/1; order 9; pass True
- C7727379-energy-023 (nitrogen-neutral): NIST C $\Lambda^3\Pi_u$; Te 89136.88 cm⁻¹ = 2228422/25; ground False; exact gap 2228422/25; order 10; pass True
- C7727379-energy-022 (nitrogen-neutral): NIST C" $\Lambda^5\Pi_u$; Te (93500) cm⁻¹ = 93500/1; ground False; exact gap 93500/1; order 11; pass True
- C7727379-energy-021 (nitrogen-neutral): NIST E $\Lambda^3\Sigma_g^+$; Te (95858) cm⁻¹ = 95858/1; ground False; exact gap 95858/1; order 12; pass True
- C7727379-energy-020 (nitrogen-neutral): NIST C' $\Lambda^3\Pi_u$; Te 98351 cm⁻¹ = 98351/1; ground False; exact gap 98351/1; order 13; pass True
- C7727379-energy-019 (nitrogen-neutral): NIST a" $\Lambda^1\Sigma_g^+$; Te [100016._0] cm⁻¹ = 100016/1; ground False; exact gap 100016/1; order 14; pass True
- C7727379-energy-018 (nitrogen-neutral): NIST b $\Lambda^1\Pi_u$; Te (101675) cm⁻¹ = 101675/1; ground False; exact gap 101675/1; order 15; pass True
- C7727379-energy-015 (nitrogen-neutral): NIST c_3 $\Lambda^1\Pi_u$; Te 104476 cm⁻¹ = 104476/1; ground False; exact gap 104476/1; order 16; pass True
- C7727379-energy-016 (nitrogen-neutral): NIST b' $\Lambda^1\Sigma_u^+$; Te 104498 cm⁻¹ = 104498/1; ground False; exact gap 104498/1; order 17; pass True

- C7727379-energy-014 (nitrogen-neutral): NIST $c'_4 \wedge 1\Sigma_u^+$; Te 104519 cm^{-1} = 104519/1; ground False; exact gap 104519/1; order 18; pass True
- C7727379-energy-017 (nitrogen-neutral): NIST $D \wedge 3\Sigma_u^+$; Te [104746.6] cm^{-1} = 523733/5; ground False; exact gap 523733/5; order 19; pass True
- C7727379-energy-013 (nitrogen-neutral): NIST $H \wedge 3\Phi_u$; Te (105720) cm^{-1} = 105720/1; ground False; exact gap 105720/1; order 20; pass True
- C7727379-energy-012 (nitrogen-neutral): NIST $o_3 \wedge 1\Pi_u$; Te 105869 cm^{-1} = 105869/1; ground False; exact gap 105869/1; order 21; pass True
- C7727379-energy-011 (nitrogen-neutral): NIST $d' \wedge 1\Sigma_u^-$ or $\wedge 1\Delta_u$; Te [112500] cm^{-1} = 112500/1; ground False; exact gap 112500/1; order 22; pass True
- C7727379-energy-010 (nitrogen-neutral): NIST $x \wedge 1\Sigma_g^-$; Te 113438.0 cm^{-1} = 113438/1; ground False; exact gap 113438/1; order 23; pass True
- C7727379-energy-009 (nitrogen-neutral): NIST $k \wedge 1\Pi_g$; Te (113808) cm^{-1} = 113808/1; ground False; exact gap 113808/1; order 24; pass True
- C7727379-energy-008 (nitrogen-neutral): NIST $y \wedge 1\Pi_g$; Te 114305.2 cm^{-1} = 571526/5; ground False; exact gap 571526/5; order 25; pass True
- C7727379-energy-007 (nitrogen-neutral): NIST $z \wedge 1\Delta_g$; Te (115435) cm^{-1} = 115435/1; ground False; exact gap 115435/1; order 26; pass True
- C7727379-energy-006 (nitrogen-neutral): NIST $c_4 \wedge 1\Pi_u$; Te 115635.9 cm^{-1} = 1156359/10; ground False; exact gap 1156359/10; order 27; pass True
- C7727379-energy-005 (nitrogen-neutral): NIST $c'_5 \wedge 1\Sigma_u^+$; Te (115876) cm^{-1} = 115876/1; ground False; exact gap 115876/1; order 28; pass True
- C7727379-energy-004 (nitrogen-neutral): NIST $O_4 (\wedge 3\Pi_u)$; Te 121263 cm^{-1} = 121263/1; ground False; exact gap 121263/1; order 29; pass True
- C7727379-energy-003 (nitrogen-neutral): NIST $o_4 \wedge 1\Pi_u$; Te 122419 cm^{-1} = 122419/1; ground False; exact gap 122419/1; order 30; pass True
- C7727379-energy-002 (nitrogen-neutral): NIST $O_5 (\wedge 3\Pi_u)$; Te 127445 cm^{-1} = 127445/1; ground False; exact gap 127445/1; order 31; pass True
- C7727379-energy-001 (nitrogen-neutral): NIST $o_5 \wedge 1\Pi_u$; Te (127868) cm^{-1} = 127868/1; ground False; exact gap 127868/1; order 32; pass True
- C7782447-energy-026 (oxygen-neutral): NIST $X \wedge 3\Sigma_g^-$; Te 0 cm^{-1} = 0/1; ground True; exact gap 0/1; order structural-empty-One; pass True
- C7782447-energy-025 (oxygen-neutral): NIST $a \wedge 1\Delta_g$; Te 7918.1 cm^{-1} = 79181/10; ground False; exact gap 79181/10; order 1; pass True
- C7782447-energy-024 (oxygen-neutral): NIST $b \wedge 1\Sigma_g^+$; Te 13195.1 cm^{-1} = 131951/10; ground False; exact gap 131951/10; order 2; pass True
- C7782447-energy-023 (oxygen-neutral): NIST $c \wedge 1\Sigma_u^-$; Te 33057.3 cm^{-1} = 330573/10; ground False; exact gap 330573/10; order 3; pass True
- C7782447-energy-022 (oxygen-neutral): NIST $A' \wedge 3\Delta_u$; Te (34690) cm^{-1} = 34690/1; ground False; exact gap 34690/1; order 4; pass True
- C7782447-energy-021 (oxygen-neutral): NIST $A \wedge 3\Sigma_u^+$; Te 35397.8 cm^{-1} = 176989/5; ground False; exact gap 176989/5; order 5; pass True
- C7782447-energy-020 (oxygen-neutral): NIST $B \wedge 3\Sigma_u^-$; Te 49793.28 cm^{-1} = 1244832/25; ground False; exact gap 1244832/25; order 6; pass True

- C7782447-energy-019 (oxygen-neutral): NIST C ($\wedge^3\Pi_g$); Te (65530) $\text{cm}^{-1} = 65530/1$; ground False; exact gap 65530/1; order 7; pass True
- C7782447-energy-018 (oxygen-neutral): NIST d ($\wedge^1\Pi_g$); Te (69180) $\text{cm}^{-1} = 69180/1$; ground False; exact gap 69180/1; order 8; pass True
- C7782447-energy-017 (oxygen-neutral): NIST e' ($\wedge^3\Delta_{2u}$); Te (74915) $\text{cm}^{-1} = 74915/1$; ground False; exact gap 74915/1; order 9; pass True
- C7782447-energy-016 (oxygen-neutral): NIST e ($\wedge^1\Delta_{2u}$); Te (75254) $\text{cm}^{-1} = 75254/1$; ground False; exact gap 75254/1; order 10; pass True
- C7782447-energy-015 (oxygen-neutral): NIST D ($\wedge^3\Sigma_u^+$); Te (75260) $\text{cm}^{-1} = 75260/1$; ground False; exact gap 75260/1; order 11; pass True
- C7782447-energy-014 (oxygen-neutral): NIST f ($\wedge^1\Sigma_u^+$); Te 76091 $\text{cm}^{-1} = 76091/1$; ground False; exact gap 76091/1; order 12; pass True
- C7782447-energy-013 (oxygen-neutral): NIST E ($\wedge^3\Sigma_u^-$); Te (79883) $\text{cm}^{-1} = 79883/1$; ground False; exact gap 79883/1; order 13; pass True
- C7782447-energy-012 (oxygen-neutral): NIST F ($\wedge^3\Pi_u$); Te (85868) $\text{cm}^{-1} = 85868/1$; ground False; exact gap 85868/1; order 14; pass True
- C7782447-energy-011 (oxygen-neutral): NIST g ($\wedge^1\Pi_u$); Te (86604) $\text{cm}^{-1} = 86604/1$; ground False; exact gap 86604/1; order 15; pass True
- C7782447-energy-010 (oxygen-neutral): NIST h ($\wedge^1\Pi_u$); Te (86750) $\text{cm}^{-1} = 86750/1$; ground False; exact gap 86750/1; order 16; pass True
- C7782447-energy-009 (oxygen-neutral): NIST i' ($\wedge^3\Delta_{2u}$); Te (86843) $\text{cm}^{-1} = 86843/1$; ground False; exact gap 86843/1; order 17; pass True
- C7782447-energy-008 (oxygen-neutral): NIST i ($\wedge^1\Delta_{2u}$); Te (86846) $\text{cm}^{-1} = 86846/1$; ground False; exact gap 86846/1; order 18; pass True
- C7782447-energy-007 (oxygen-neutral): NIST G ($\wedge^3\Sigma_u^+$); Te (86998) $\text{cm}^{-1} = 86998/1$; ground False; exact gap 86998/1; order 19; pass True
- C7782447-energy-006 (oxygen-neutral): NIST j ($\wedge^1\Sigma_u^+$); Te (87209) $\text{cm}^{-1} = 87209/1$; ground False; exact gap 87209/1; order 20; pass True
- C7782447-energy-005 (oxygen-neutral): NIST k ($\wedge^1\Delta_u$); Te [89066] $\text{cm}^{-1} = 89066/1$; ground False; exact gap 89066/1; order 21; pass True
- C7782447-energy-004 (oxygen-neutral): NIST L ($\wedge^3\Pi_u$); Te [90044] $\text{cm}^{-1} = 90044/1$; ground False; exact gap 90044/1; order 22; pass True
- C7782447-energy-003 (oxygen-neutral): NIST H ($\wedge^3\Pi_u$); Te 99880 $\text{cm}^{-1} = 99880/1$; ground False; exact gap 99880/1; order 23; pass True
- C7782447-energy-002 (oxygen-neutral): NIST p ($\wedge^1\Phi_u$); Te 118951 $\text{cm}^{-1} = 118951/1$; ground False; exact gap 118951/1; order 24; pass True
- C7782447-energy-001 (oxygen-neutral): NIST W ($\wedge^3\Sigma_u^-$); Te (168290) $\text{cm}^{-1} = 168290/1$; ground False; exact gap 168290/1; order 25; pass True
- C7782414-energy-008 (fluorine-neutral): NIST X ($\wedge^1\Sigma_g^+$); Te 0 $\text{cm}^{-1} = 0/1$; ground True; exact gap 0/1; order structural-empty-One; pass True
- C7782414-energy-007 (fluorine-neutral): NIST F ($\wedge^1\Pi_g$ (3s σ)); Te 93099 $\text{cm}^{-1} = 93099/1$; ground False; exact gap 93099/1; order 1; pass True
- C7782414-energy-006 (fluorine-neutral): NIST f ($\wedge^3\Pi_{1g}$) (3s σ); Te [97314] $\text{cm}^{-1} = 97314/1$; ground False; exact gap 97314/1; order 2; pass True

- C7782414-energy-005 (fluorine-neutral): NIST G $^1\Sigma_u^+$ ($^3p\pi$); Te (104300) cm^{-1} = 104300/1; ground False; exact gap 104300/1; order 3; pass True
- C7782414-energy-004 (fluorine-neutral): NIST h $^3\Pi_u$ ($3p\sigma$); Te (104904) cm^{-1} = 104904/1; ground False; exact gap 104904/1; order 4; pass True
- C7782414-energy-003 (fluorine-neutral): NIST H $^1\Pi_u$ ($^3p\sigma$); Te 105520.14 cm^{-1} = 5276007/50; ground False; exact gap 5276007/50; order 5; pass True
- C7782414-energy-002 (fluorine-neutral): NIST I $^1\Sigma_u^+$ ($^4p\pi$); Te 113841 cm^{-1} = 113841/1; ground False; exact gap 113841/1; order 6; pass True
- C7782414-energy-001 (fluorine-neutral): NIST J $^1\Pi_u$ ($^4p\sigma$); Te 116409 cm^{-1} = 116409/1; ground False; exact gap 116409/1; order 7; pass True
- C630080-energy-020 (carbon-monoxide-neutral): NIST X $^1\Sigma^+$; Te 0 cm^{-1} = 0/1; ground True; exact gap 0/1; order structural-empty-One; pass True
- C630080-energy-019 (carbon-monoxide-neutral): NIST a $^3\Pi_r$; Te 48686.7 cm^{-1} = 486867/10; ground False; exact gap 486867/10; order 1; pass True
- C630080-energy-018 (carbon-monoxide-neutral): NIST a' $^3\Sigma^+$; Te 55825.49 cm^{-1} = 5582549/100; ground False; exact gap 5582549/100; order 2; pass True
- C630080-energy-017 (carbon-monoxide-neutral): NIST d $^3\Delta_i$; Te 61120 cm^{-1} = 61120/1; ground False; exact gap 61120/1; order 3; pass True
- C630080-energy-016 (carbon-monoxide-neutral): NIST e $^3\Sigma^-$; Te 64230.2_4 cm^{-1} = 1605756/25; ground False; exact gap 1605756/25; order 4; pass True
- C630080-energy-015 (carbon-monoxide-neutral): NIST A $^1\Pi$; Te 65075.7_7 cm^{-1} = 6507577/100; ground False; exact gap 6507577/100; order 5; pass True
- C630080-energy-014 (carbon-monoxide-neutral): NIST I $^1\Sigma^-$; Te 65084.4_0 cm^{-1} = 325422/5; ground False; exact gap 325422/5; order 6; pass True
- C630080-energy-013 (carbon-monoxide-neutral): NIST D $^1\Delta$; Te 65928 cm^{-1} = 65928/1; ground False; exact gap 65928/1; order 7; pass True
- C630080-energy-012 (carbon-monoxide-neutral): NIST b $^3\Sigma^+$ $3s\sigma$; Te (83814) cm^{-1} = 83814/1; ground False; exact gap 83814/1; order 8; pass True
- C630080-energy-011 (carbon-monoxide-neutral): NIST B $^1\Sigma^+$ $3s\sigma$; Te 86945.2 cm^{-1} = 434726/5; ground False; exact gap 434726/5; order 9; pass True
- C630080-energy-010 (carbon-monoxide-neutral): NIST j ($^3\Sigma^+$ $3p\sigma$); Te 90975 cm^{-1} = 90975/1; ground False; exact gap 90975/1; order 10; pass True
- C630080-energy-009 (carbon-monoxide-neutral): NIST C $^1\Sigma^+$ $3p\sigma$; Te 91916.5 cm^{-1} = 183833/2; ground False; exact gap 183833/2; order 11; pass True
- C630080-energy-007 (carbon-monoxide-neutral): NIST E $^1\Pi$ $3p\pi$; Te (92903) cm^{-1} = 92903/1; ground False; exact gap 92903/1; order 12; pass True
- C630080-energy-008 (carbon-monoxide-neutral): NIST c $^3\Pi$ $3p\pi$; Te [93158.5] cm^{-1} = 186317/2; ground False; exact gap 186317/2; order 13; pass True
- C630080-energy-006 (carbon-monoxide-neutral): NIST F $^1\Sigma^+$ ($3d\sigma$); Te (99803) cm^{-1} = 99803/1; ground False; exact gap 99803/1; order 14; pass True
- C630080-energy-005 (carbon-monoxide-neutral): NIST J $^1\Sigma^+$ $4s\sigma$; Te (101409) cm^{-1} = 101409/1; ground False; exact gap 101409/1; order 15; pass True
- C630080-energy-004 (carbon-monoxide-neutral): NIST L $^1\Pi$ $4p\pi$; Te 103251 cm^{-1} = 103251/1; ground False; exact gap 103251/1; order 16; pass True

- C630080-energy-003 (carbon-monoxide-neutral): NIST H ($\Lambda^1\Pi$); Te (105811) cm^{-1} = 105811/1; ground False; exact gap 105811/1; order 17; pass True
- C630080-energy-002 (carbon-monoxide-neutral): NIST O₁ ($\Lambda^1\Pi$); Te 121137 cm^{-1} = 121137/1; ground False; exact gap 121137/1; order 18; pass True
- C630080-energy-001 (carbon-monoxide-neutral): NIST O₂ ($\Lambda^1\Pi$); Te 126729 cm^{-1} = 126729/1; ground False; exact gap 126729/1; order 19; pass True
- C10102439-energy-026 (nitric-oxide-neutral): NIST X $\Lambda^2\Pi_r$; Te 119.82 cm^{-1} = 5991/50; ground True; exact gap 0/1; order structural-empty-One; pass True
- C10102439-energy-025 (nitric-oxide-neutral): NIST a ($\Lambda^4\Pi_i$); Te (38440) cm^{-1} = 38440/1; ground False; exact gap 1916009/50; order 1; pass True
- C10102439-energy-024 (nitric-oxide-neutral): NIST A $\Lambda^2\Sigma^+ 3s\sigma$; Te 43965.7 cm^{-1} = 439657/10; ground False; exact gap 1096147/25; order 2; pass True
- C10102439-energy-023 (nitric-oxide-neutral): NIST B $\Lambda^2\Pi_r$; Te 45942.6 cm^{-1} = 229713/5; ground False; exact gap 2291139/50; order 3; pass True
- C10102439-energy-022 (nitric-oxide-neutral): NIST b ($\Lambda^4\Sigma^-$); Te (48680) cm^{-1} = 48680/1; ground False; exact gap 2428009/50; order 4; pass True
- C10102439-energy-021 (nitric-oxide-neutral): NIST C $\Lambda^2\Pi_r 3p\pi$; Te 52126 cm^{-1} = 52126/1; ground False; exact gap 2600309/50; order 5; pass True
- C10102439-energy-020 (nitric-oxide-neutral): NIST D $\Lambda^2\Sigma^+ 3p\sigma$; Te 53084.7 cm^{-1} = 530847/10; ground False; exact gap 1324122/25; order 6; pass True
- C10102439-energy-019 (nitric-oxide-neutral): NIST B' $\Lambda^2\Delta_i$; Te 60364.2 cm^{-1} = 301821/5; ground False; exact gap 3012219/50; order 7; pass True
- C10102439-energy-018 (nitric-oxide-neutral): NIST E $\Lambda^2\Sigma^+ 4s\sigma$; Te 60628.8 cm^{-1} = 303144/5; ground False; exact gap 3025449/50; order 8; pass True
- C10102439-energy-017 (nitric-oxide-neutral): NIST F $\Lambda^2\Delta 3d\delta$; Te 61800 cm^{-1} = 61800/1; ground False; exact gap 3084009/50; order 9; pass True
- C10102439-energy-016 (nitric-oxide-neutral): NIST H $\Lambda^2\Sigma^+ 3d\sigma$; Te 62473.4 cm^{-1} = 312367/5; ground False; exact gap 3117679/50; order 10; pass True
- C10102439-energy-015 (nitric-oxide-neutral): NIST H' $\Lambda^2\Pi 3d\pi$; Te 62485.4 cm^{-1} = 312427/5; ground False; exact gap 3118279/50; order 11; pass True
- C10102439-energy-014 (nitric-oxide-neutral): NIST L $\Lambda^2\Pi_i$; Te (62500) cm^{-1} = 62500/1; ground False; exact gap 3119009/50; order 12; pass True
- C10102439-energy-013 (nitric-oxide-neutral): NIST G $\Lambda^2\Sigma^-$; Te 62913.0 cm^{-1} = 62913/1; ground False; exact gap 3139659/50; order 13; pass True
- C10102439-energy-012 (nitric-oxide-neutral): NIST I $\Lambda^2\Sigma^+$; Te (63500) cm^{-1} = 63500/1; ground False; exact gap 3169009/50; order 14; pass True
- C10102439-energy-011 (nitric-oxide-neutral): NIST M $\Lambda^2\Sigma^+ 4p\sigma$; Te 64437 cm^{-1} = 64437/1; ground False; exact gap 3215859/50; order 15; pass True
- C10102439-energy-010 (nitric-oxide-neutral): NIST S $\Lambda^2\Sigma^+ 5s\sigma$; Te 66900 cm^{-1} = 66900/1; ground False; exact gap 3339009/50; order 16; pass True
- C10102439-energy-009 (nitric-oxide-neutral): NIST N $\Lambda^2\Delta 4d\delta$; Te 67374 cm^{-1} = 67374/1; ground False; exact gap 3362709/50; order 17; pass True
- C10102439-energy-008 (nitric-oxide-neutral): NIST O $\Lambda^2\Sigma^+ 4d\sigma$; Te (67757) cm^{-1} = 67757/1; ground False; exact gap 3381859/50; order 18; pass True

- C10102439-energy-007 (nitric-oxide-neutral): NIST O' $\wedge 2\Pi$ 4d π ; Te (67762) cm⁻¹ = 67762/1; ground False; exact gap 3382109/50; order 19; pass True
- C10102439-energy-006 (nitric-oxide-neutral): NIST R $\wedge 2\Sigma^+$ 5p σ ; Te (68598) cm⁻¹ = 68598/1; ground False; exact gap 3423909/50; order 20; pass True
- C10102439-energy-005 (nitric-oxide-neutral): NIST T $\wedge 2\Sigma^+$ 6s σ ; Te (69728) cm⁻¹ = 69728/1; ground False; exact gap 3480409/50; order 21; pass True
- C10102439-energy-004 (nitric-oxide-neutral): NIST U $\wedge 2\Delta$ 5d δ ; Te (69977) cm⁻¹ = 69977/1; ground False; exact gap 3492859/50; order 22; pass True
- C10102439-energy-003 (nitric-oxide-neutral): NIST W $\wedge 2\Pi$ 6p π ; Te 70512 cm⁻¹ = 70512/1; ground False; exact gap 3519609/50; order 23; pass True
- C10102439-energy-002 (nitric-oxide-neutral): NIST Y $\wedge 2\Sigma^+$ 6p σ ; Te 70614 cm⁻¹ = 70614/1; ground False; exact gap 3524709/50; order 24; pass True
- C10102439-energy-001 (nitric-oxide-neutral): NIST Z $\wedge 2\Sigma^+$ 7s σ ; Te 71224 cm⁻¹ = 71224/1; ground False; exact gap 3555209/50; order 25; pass True
- C3352576-energy-005 (hydroxyl-neutral): NIST X $\wedge 2\Pi_i$; Te 0 cm⁻¹ = 0/1; ground True; exact gap 0/1; order structural-empty-One; pass True
- C3352576-energy-004 (hydroxyl-neutral): NIST A $\wedge 2\Sigma^+$; Te 32684.1 cm⁻¹ = 326841/10; ground False; exact gap 326841/10; order 1; pass True
- C3352576-energy-003 (hydroxyl-neutral): NIST B $\wedge 2\Sigma^+$; Te 69774 cm⁻¹ = 69774/1; ground False; exact gap 69774/1; order 2; pass True
- C3352576-energy-002 (hydroxyl-neutral): NIST D $\wedge 2\Sigma^+$; Te (82130) cm⁻¹ = 82130/1; ground False; exact gap 82130/1; order 3; pass True
- C3352576-energy-001 (hydroxyl-neutral): NIST C $\wedge 2\Sigma^+$; Te 89459.1 cm⁻¹ = 894591/10; ground False; exact gap 894591/10; order 4; pass True
- C13774920-energy-006 (imidogen-neutral): NIST X $\wedge 3\Sigma^-$; Te 0 cm⁻¹ = 0/1; ground True; exact gap 0/1; order structural-empty-One; pass True
- C13774920-energy-005 (imidogen-neutral): NIST a $\wedge 1\Delta$; Te (12566) cm⁻¹ = 12566/1; ground False; exact gap 12566/1; order 1; pass True
- C13774920-energy-004 (imidogen-neutral): NIST b $\wedge 1\Sigma^+$; Te 21202 cm⁻¹ = 21202/1; ground False; exact gap 21202/1; order 2; pass True
- C13774920-energy-003 (imidogen-neutral): NIST A $\wedge 3\Pi_i$; Te 29807.4 cm⁻¹ = 149037/5; ground False; exact gap 149037/5; order 3; pass True
- C13774920-energy-002 (imidogen-neutral): NIST c $\wedge 1\Pi$; Te (43744) cm⁻¹ = 43744/1; ground False; exact gap 43744/1; order 4; pass True
- C13774920-energy-001 (imidogen-neutral): NIST d $\wedge 1\Sigma^+$; Te 83160 cm⁻¹ = 83160/1; ground False; exact gap 83160/1; order 5; pass True
- C3315375-energy-008 (methylidyne-neutral): NIST X $\wedge 2\Pi_r$; Te 0 cm⁻¹ = 0/1; ground True; exact gap 0/1; order structural-empty-One; pass True
- C3315375-energy-007 (methylidyne-neutral): NIST a $\wedge 4\Sigma^-$; Te (5844) cm⁻¹ = 5844/1; ground False; exact gap 5844/1; order 1; pass True
- C3315375-energy-006 (methylidyne-neutral): NIST A $\wedge 2\Delta$; Te 23189.8 cm⁻¹ = 115949/5; ground False; exact gap 115949/5; order 2; pass True
- C3315375-energy-005 (methylidyne-neutral): NIST B $\wedge 2\Sigma^-$; Te (26044) cm⁻¹ = 26044/1; ground False; exact gap 26044/1; order 3; pass True

- C3315375-energy-004 (methylidyne-neutral): NIST C $\wedge^2\Sigma^+$; Te 31801.5 cm^{-1} = 63603/2; ground False; exact gap 63603/2; order 4; pass True
- C3315375-energy-003 (methylidyne-neutral): NIST D $\wedge^2\Pi_i$; Te [60394] cm^{-1} = 60394/1; ground False; exact gap 60394/1; order 5; pass True
- C3315375-energy-002 (methylidyne-neutral): NIST E $\wedge^2\Pi$; Te [65625] cm^{-1} = 65625/1; ground False; exact gap 65625/1; order 6; pass True
- C3315375-energy-001 (methylidyne-neutral): NIST F $\wedge^2\Sigma^+$; Te [65945] cm^{-1} = 65945/1; ground False; exact gap 65945/1; order 7; pass True
- C7782505-energy-004 (chlorine-neutral): NIST X $\wedge^1\Sigma_g^+$; Te 0 cm^{-1} = 0/1; ground True; exact gap 0/1; order structural-empty-One; pass True
- C7782505-energy-003 (chlorine-neutral): NIST A' ($\wedge^3\Pi_{2u}$); Te (17160) cm^{-1} = 17160/1; ground False; exact gap 17160/1; order 1; pass True
- C7782505-energy-002 (chlorine-neutral): NIST A ($\wedge^3\Pi_{1u}$); Te (17440) cm^{-1} = 17440/1; ground False; exact gap 17440/1; order 2; pass True
- C7782505-energy-001 (chlorine-neutral): NIST B $\wedge^3\Pi_{0^+u}$; Te 17809 cm^{-1} = 17809/1; ground False; exact gap 17809/1; order 3; pass True
- C10097286-energy-026 (silicon-monoxide-neutral): NIST X $\wedge^1\Sigma^+$; Te 0 cm^{-1} = 0/1; ground True; exact gap 0/1; order structural-empty-One; pass True
- C10097286-energy-025 (silicon-monoxide-neutral): NIST a $\wedge^3\Sigma^+$; Te (33630) cm^{-1} = 33630/1; ground False; exact gap 33630/1; order 1; pass True
- C10097286-energy-024 (silicon-monoxide-neutral): NIST b $\wedge^3\Pi_0$; Te 33874.8 cm^{-1} = 169374/5; ground False; exact gap 169374/5; order 2; pass True
- C10097286-energy-023 (silicon-monoxide-neutral): NIST b $\wedge^3\Pi_1$; Te 33947.4 cm^{-1} = 169737/5; ground False; exact gap 169737/5; order 3; pass True
- C10097286-energy-022 (silicon-monoxide-neutral): NIST b $\wedge^3\Pi_2$; Te 34018.5 cm^{-1} = 68037/2; ground False; exact gap 68037/2; order 4; pass True
- C10097286-energy-021 (silicon-monoxide-neutral): NIST d $\wedge^3\Delta_r$; Te 36487 cm^{-1} = 36487/1; ground False; exact gap 36487/1; order 5; pass True
- C10097286-energy-020 (silicon-monoxide-neutral): NIST e $\wedge^3\Sigma^-$; Te 38309 cm^{-1} = 38309/1; ground False; exact gap 38309/1; order 6; pass True
- C10097286-energy-019 (silicon-monoxide-neutral): NIST C $\wedge^1\Sigma^-$; Te 38624 cm^{-1} = 38624/1; ground False; exact gap 38624/1; order 7; pass True
- C10097286-energy-018 (silicon-monoxide-neutral): NIST D $\wedge^1\Delta$; Te 38823 cm^{-1} = 38823/1; ground False; exact gap 38823/1; order 8; pass True
- C10097286-energy-017 (silicon-monoxide-neutral): NIST A $\wedge^1\Pi$; Te 42835.4 cm^{-1} = 214177/5; ground False; exact gap 214177/5; order 9; pass True
- C10097286-energy-016 (silicon-monoxide-neutral): NIST E $\wedge^1\Sigma^+$; Te 52860.9 cm^{-1} = 528609/10; ground False; exact gap 528609/10; order 10; pass True
- C10097286-energy-015 (silicon-monoxide-neutral): NIST c $\wedge^3\Sigma^+(+)$; Te 57551.3 cm^{-1} = 575513/10; ground False; exact gap 575513/10; order 11; pass True
- C10097286-energy-014 (silicon-monoxide-neutral): NIST f $\wedge^3\Pi_i$; Te 59283.1 cm^{-1} = 592831/10; ground False; exact gap 592831/10; order 12; pass True
- C10097286-energy-013 (silicon-monoxide-neutral): NIST g $\wedge^3\Sigma^+(4s\sigma)$; Te [68091.4] cm^{-1} = 340457/5; ground False; exact gap 340457/5; order 13; pass True

- C10097286-energy-012 (silicon-monoxide-neutral): NIST F $\wedge 1\Sigma^+$ (4s σ); Te 68532.0 cm $^{-1}$ = 68532/1; ground False; exact gap 68532/1; order 14; pass True
- C10097286-energy-011 (silicon-monoxide-neutral): NIST G $\wedge 1\Pi$; Te (69670) cm $^{-1}$ = 69670/1; ground False; exact gap 69670/1; order 15; pass True
- C10097286-energy-010 (silicon-monoxide-neutral): NIST H $\wedge 1\Sigma^+$ (4p σ); Te 69727 cm $^{-1}$ = 69727/1; ground False; exact gap 69727/1; order 16; pass True
- C10097286-energy-009 (silicon-monoxide-neutral): NIST I $\wedge 1\Pi$; Te (70510) cm $^{-1}$ = 70510/1; ground False; exact gap 70510/1; order 17; pass True
- C10097286-energy-008 (silicon-monoxide-neutral): NIST h $\wedge 3\Pi_i$; Te (70790) cm $^{-1}$ = 70790/1; ground False; exact gap 70790/1; order 18; pass True
- C10097286-energy-007 (silicon-monoxide-neutral): NIST J $\wedge 1\Pi$ (4p π); Te 76381 cm $^{-1}$ = 76381/1; ground False; exact gap 76381/1; order 19; pass True
- C10097286-energy-006 (silicon-monoxide-neutral): NIST K $\wedge 1\Sigma^+$; Te 78369 cm $^{-1}$ = 78369/1; ground False; exact gap 78369/1; order 20; pass True
- C10097286-energy-005 (silicon-monoxide-neutral): NIST L $\wedge 1\Sigma^+$ (5s σ); Te 80783 cm $^{-1}$ = 80783/1; ground False; exact gap 80783/1; order 21; pass True
- C10097286-energy-004 (silicon-monoxide-neutral): NIST M $\wedge 1\Pi$; Te 81203 cm $^{-1}$ = 81203/1; ground False; exact gap 81203/1; order 22; pass True
- C10097286-energy-003 (silicon-monoxide-neutral): NIST N $\wedge 1\Pi$; Te (81232) cm $^{-1}$ = 81232/1; ground False; exact gap 81232/1; order 23; pass True
- C10097286-energy-001 (silicon-monoxide-neutral): NIST P $\wedge 1\Sigma^+$ (5p σ); Te 82019 cm $^{-1}$ = 82019/1; ground False; exact gap 82019/1; order 24; pass True
- C10097286-energy-002 (silicon-monoxide-neutral): NIST O $\wedge 1\Pi$; Te [82208.1] cm $^{-1}$ = 822081/10; ground False; exact gap 822081/10; order 25; pass True
- C13827322-energy-009 (sulfur-monoxide-neutral): NIST X $\wedge 3\Sigma^-$; Te 0 cm $^{-1}$ = 0/1; ground True; exact gap 0/1; order structural-empty-One; pass True
- C13827322-energy-008 (sulfur-monoxide-neutral): NIST a $\wedge 1\Delta$; Te (6350) cm $^{-1}$ = 6350/1; ground False; exact gap 6350/1; order 1; pass True
- C13827322-energy-007 (sulfur-monoxide-neutral): NIST b $\wedge 1\Sigma^+$; Te 10510.0 cm $^{-1}$ = 10510/1; ground False; exact gap 10510/1; order 2; pass True
- C13827322-energy-006 (sulfur-monoxide-neutral): NIST A $\wedge 3\Pi_0$; Te 38306 cm $^{-1}$ = 38306/1; ground False; exact gap 38306/1; order 3; pass True
- C13827322-energy-005 (sulfur-monoxide-neutral): NIST A $\wedge 3\Pi_1$; Te 38462 cm $^{-1}$ = 38462/1; ground False; exact gap 38462/1; order 4; pass True
- C13827322-energy-004 (sulfur-monoxide-neutral): NIST A $\wedge 3\Pi_2$; Te 38622 cm $^{-1}$ = 38622/1; ground False; exact gap 38622/1; order 5; pass True
- C13827322-energy-003 (sulfur-monoxide-neutral): NIST B $\wedge 3\Sigma^-$; Te 41629 cm $^{-1}$ = 41629/1; ground False; exact gap 41629/1; order 6; pass True
- C13827322-energy-002 (sulfur-monoxide-neutral): NIST D ($\wedge 3\Pi$); Te (54340) cm $^{-1}$ = 54340/1; ground False; exact gap 54340/1; order 7; pass True
- C13827322-energy-001 (sulfur-monoxide-neutral): NIST E ($\wedge 3\Pi$); Te (67884) cm $^{-1}$ = 67884/1; ground False; exact gap 67884/1; order 8; pass True
- C7664393-energy-003 (hydrogen-fluoride-neutral): NIST X $\wedge 1\Sigma^+$; Te 0 cm $^{-1}$ = 0/1; ground True; exact gap 0/1; order structural-empty-One; pass True

- C7664393-energy-002 (hydrogen-fluoride-neutral): NIST B $\wedge 1\Sigma^+$; Te 84776.65 cm^{-1} = 1695533/20; ground False; exact gap 1695533/20; order 1; pass True
- C7664393-energy-001 (hydrogen-fluoride-neutral): NIST C $\wedge 1\Pi$; Te (105820) cm^{-1} = 105820/1; ground False; exact gap 105820/1; order 2; pass True
- C7647010-energy-019 (hydrogen-chloride-neutral): NIST X $\wedge 1\Sigma^+$; Te 0 cm^{-1} = 0/1; ground True; exact gap 0/1; order structural-empty-One; pass True
- C7647010-energy-017 (hydrogen-chloride-neutral): NIST b_1 $\wedge 3\Pi_1$; Te (75195) cm^{-1} = 75195/1; ground False; exact gap 75195/1; order 1; pass True
- C7647010-energy-016 (hydrogen-chloride-neutral): NIST b_0 $\wedge 3\Pi_0$; Te (75617) cm^{-1} = 75617/1; ground False; exact gap 75617/1; order 2; pass True
- C7647010-energy-018 (hydrogen-chloride-neutral): NIST b_2 $\wedge 3\Pi_2$; Te [76322._2] cm^{-1} = 381611/5; ground False; exact gap 381611/5; order 3; pass True
- C7647010-energy-015 (hydrogen-chloride-neutral): NIST V $\wedge 1\Sigma^+$; Te 77293._0 cm^{-1} = 77293/1; ground False; exact gap 77293/1; order 4; pass True
- C7647010-energy-014 (hydrogen-chloride-neutral): NIST C $\wedge 1\Pi$; Te 77575 cm^{-1} = 77575/1; ground False; exact gap 77575/1; order 5; pass True
- C7647010-energy-013 (hydrogen-chloride-neutral): NIST d_2 $\wedge 3\Pi_2$; Te [83083] cm^{-1} = 83083/1; ground False; exact gap 83083/1; order 6; pass True
- C7647010-energy-012 (hydrogen-chloride-neutral): NIST d_1 $\wedge 3\Pi_1$; Te [83255.6] cm^{-1} = 416278/5; ground False; exact gap 416278/5; order 7; pass True
- C7647010-energy-011 (hydrogen-chloride-neutral): NIST f_3 $\wedge 3\Delta_3$; Te [83308.2] cm^{-1} = 416541/5; ground False; exact gap 416541/5; order 8; pass True
- C7647010-energy-010 (hydrogen-chloride-neutral): NIST f_2 $\wedge 3\Delta_2$; Te [83497.7] cm^{-1} = 834977/10; ground False; exact gap 834977/10; order 9; pass True
- C7647010-energy-009 (hydrogen-chloride-neutral): NIST d_0 $\wedge 3\Pi_0$; Te [83753.6] cm^{-1} = 418768/5; ground False; exact gap 418768/5; order 10; pass True
- C7647010-energy-008 (hydrogen-chloride-neutral): NIST D $\wedge 1\Pi$; Te [83972.0] cm^{-1} = 83972/1; ground False; exact gap 83972/1; order 11; pass True
- C7647010-energy-007 (hydrogen-chloride-neutral): NIST f_1 $\wedge 3\Delta_1$; Te [84006.1] cm^{-1} = 840061/10; ground False; exact gap 840061/10; order 12; pass True
- C7647010-energy-005 (hydrogen-chloride-neutral): NIST E $\wedge 1\Sigma^+$; Te (84193) cm^{-1} = 84193/1; ground False; exact gap 84193/1; order 13; pass True
- C7647010-energy-006 (hydrogen-chloride-neutral): NIST g ($\wedge 3\Sigma^-$)₁; Te [84329.7] cm^{-1} = 843297/10; ground False; exact gap 843297/10; order 14; pass True
- C7647010-energy-004 (hydrogen-chloride-neutral): NIST H $\wedge 1\Sigma^+$; Te (89120) cm^{-1} = 89120/1; ground False; exact gap 89120/1; order 15; pass True
- C7647010-energy-003 (hydrogen-chloride-neutral): NIST K $\wedge 1\Pi$; Te (89861) cm^{-1} = 89861/1; ground False; exact gap 89861/1; order 16; pass True
- C7647010-energy-002 (hydrogen-chloride-neutral): NIST L ($\wedge 1\Sigma^+$, $\wedge 1\Pi$); Te 111280 cm^{-1} = 111280/1; ground False; exact gap 111280/1; order 17; pass True
- C7647010-energy-001 (hydrogen-chloride-neutral): NIST M ($\wedge 1\Sigma^+$); Te (117811) cm^{-1} = 117811/1; ground False; exact gap 117811/1; order 18; pass True
- C10035106-energy-019 (hydrogen-bromide-neutral): NIST X $\wedge 1\Sigma^+$; Te 0 cm^{-1} = 0/1; ground True; exact gap 0/1; order structural-empty-One; pass True

- C10035106-energy-017 (hydrogen-bromide-neutral): NIST $b_1 \text{ } ^3\Pi_1$; Te (67180) $\text{cm}^{-1} = 67180/1$; ground False; exact gap 67180/1; order 1; pass True
- C10035106-energy-018 (hydrogen-bromide-neutral): NIST $b_2 \text{ } ^3\Pi_2$; Te [67663.0] $\text{cm}^{-1} = 67663/1$; ground False; exact gap 67663/1; order 2; pass True
- C10035106-energy-016 (hydrogen-bromide-neutral): NIST $b_0 \text{ } ^3\Pi_0 0^+$; Te (68998) $\text{cm}^{-1} = 68998/1$; ground False; exact gap 68998/1; order 3; pass True
- C10035106-energy-015 (hydrogen-bromide-neutral): NIST $C \text{ } ^1\Pi$; Te 70578 $\text{cm}^{-1} = 70578/1$; ground False; exact gap 70578/1; order 4; pass True
- C10035106-energy-014 (hydrogen-bromide-neutral): NIST $d_2 \text{ } ^3\Pi_2$; Te [74753] $\text{cm}^{-1} = 74753/1$; ground False; exact gap 74753/1; order 5; pass True
- C10035106-energy-013 (hydrogen-bromide-neutral): NIST $d_1 \text{ } ^3\Pi_1$; Te [74855] $\text{cm}^{-1} = 74855/1$; ground False; exact gap 74855/1; order 6; pass True
- C10035106-energy-012 (hydrogen-bromide-neutral): NIST $e \text{ } ^3\Sigma^+$; Te [75053] $\text{cm}^{-1} = 75053/1$; ground False; exact gap 75053/1; order 7; pass True
- C10035106-energy-011 (hydrogen-bromide-neutral): NIST $f_3 \text{ } ^3\Delta_3$; Te [75403.1] $\text{cm}^{-1} = 754031/10$; ground False; exact gap 754031/10; order 8; pass True
- C10035106-energy-010 (hydrogen-bromide-neutral): NIST $f_2 \text{ } ^3\Delta_2$; Te [75533.8] $\text{cm}^{-1} = 377669/5$; ground False; exact gap 377669/5; order 9; pass True
- C10035106-energy-009 (hydrogen-bromide-neutral): NIST $V \text{ } ^1\Sigma^+$; Te (75800) $\text{cm}^{-1} = 75800/1$; ground False; exact gap 75800/1; order 10; pass True
- C10035106-energy-007 (hydrogen-bromide-neutral): NIST $d_0 \text{ } ^3\Pi_0$; Te (76193) $\text{cm}^{-1} = 76193/1$; ground False; exact gap 76193/1; order 11; pass True
- C10035106-energy-006 (hydrogen-bromide-neutral): NIST $D \text{ } ^1\Pi$; Te (76310) $\text{cm}^{-1} = 76310/1$; ground False; exact gap 76310/1; order 12; pass True
- C10035106-energy-008 (hydrogen-bromide-neutral): NIST $E \text{ } (^3\Sigma^+)0^+$; Te (76691) $\text{cm}^{-1} = 76691/1$; ground False; exact gap 76691/1; order 13; pass True
- C10035106-energy-005 (hydrogen-bromide-neutral): NIST $f_1 \text{ } ^3\Delta_1$; Te (76814) $\text{cm}^{-1} = 76814/1$; ground False; exact gap 76814/1; order 14; pass True
- C10035106-energy-004 (hydrogen-bromide-neutral): NIST $F \text{ } ^1\Delta$; Te (78322.3) $\text{cm}^{-1} = 783223/10$; ground False; exact gap 783223/10; order 15; pass True
- C10035106-energy-003 (hydrogen-bromide-neutral): NIST $g \text{ } (^3\Sigma^-)0^+$; Te (79253.2) $\text{cm}^{-1} = 396266/5$; ground False; exact gap 396266/5; order 16; pass True
- C10035106-energy-002 (hydrogen-bromide-neutral): NIST $L \text{ } (^1\Sigma^+, 1\Pi)$; Te (104201) $\text{cm}^{-1} = 104201/1$; ground False; exact gap 104201/1; order 17; pass True
- C10035106-energy-001 (hydrogen-bromide-neutral): NIST $M \text{ } (^1\Sigma^+)$; Te (109473) $\text{cm}^{-1} = 109473/1$; ground False; exact gap 109473/1; order 18; pass True
- C7553562-energy-009 (iodine-neutral): NIST $X \text{ } ^1\Sigma_g^+$; Te 0 $\text{cm}^{-1} = 0/1$; ground True; exact gap 0/1; order structural-empty-One; pass True
- C7553562-energy-008 (iodine-neutral): NIST $A' \text{ } ^3\Pi_{2u}$; Te (10100) $\text{cm}^{-1} = 10100/1$; ground False; exact gap 10100/1; order 1; pass True
- C7553562-energy-007 (iodine-neutral): NIST $A \text{ } ^3\Pi_{1u}$; Te (11888) $\text{cm}^{-1} = 11888/1$; ground False; exact gap 11888/1; order 2; pass True
- C7553562-energy-006 (iodine-neutral): NIST $B \text{ } ^3\Pi_{0+u}$; Te 15769.01 $\text{cm}^{-1} = 1576901/100$; ground False; exact gap 1576901/100; order 3; pass True

- C7553562-energy-005 (iodine-neutral): NIST G $^3\Pi_{2g}$; Te (40300) cm^{-1} = 40300/1; ground False; exact gap 40300/1; order 4; pass True
- C7553562-energy-004 (iodine-neutral): NIST D $^1\Sigma_u^+$; Te (40679) cm^{-1} = 40679/1; ground False; exact gap 40679/1; order 5; pass True
- C7553562-energy-003 (iodine-neutral): NIST E $^3\Pi_{o+g}$; Te 41411.4 cm^{-1} = 207057/5; ground False; exact gap 207057/5; order 6; pass True
- C7553562-energy-002 (iodine-neutral): NIST G' ($^3\Pi_{1g}$); Te (42300) cm^{-1} = 42300/1; ground False; exact gap 42300/1; order 7; pass True
- C7553562-energy-001 (iodine-neutral): NIST F ($^1\Sigma_u^+$); Te 47217.8 cm^{-1} = 236089/5; ground False; exact gap 236089/5; order 8; pass True
- C12184906-energy-003 (hydrogen-cation): NIST X $^2\Sigma_g^+ 1s\sigma$; Te 0 cm^{-1} = 0/1; ground True; exact gap 0/1; order structural-empty-One; pass True
- C12184906-energy-002 (hydrogen-cation): NIST B $^2\Sigma_g^+ 3d\sigma$; Te (93804._5) cm^{-1} = 187609/2; ground False; exact gap 187609/2; order 1; pass True
- C12184906-energy-001 (hydrogen-cation): NIST C $^2\Pi_u 2p\pi$; Te (102696._7) cm^{-1} = 1026967/10; ground False; exact gap 1026967/10; order 2; pass True
- C13966046-energy-007 (nitrogen-cation): NIST X $^2\Sigma_g^+$; Te 0 cm^{-1} = 0/1; ground True; exact gap 0/1; order structural-empty-One; pass True
- C13966046-energy-006 (nitrogen-cation): NIST A $^2\Pi_{ui}$; Te 9166.9_5 cm^{-1} = 183339/20; ground False; exact gap 183339/20; order 1; pass True
- C13966046-energy-005 (nitrogen-cation): NIST B $^2\Sigma_u^+$; Te 25461.4_6 cm^{-1} = 1273073/50; ground False; exact gap 1273073/50; order 2; pass True
- C13966046-energy-004 (nitrogen-cation): NIST (a $^4\Sigma_u^+$); Te (25467) cm^{-1} = 25467/1; ground False; exact gap 25467/1; order 3; pass True
- C13966046-energy-003 (nitrogen-cation): NIST D $^2\Pi_{gi}$; Te 52318.2 cm^{-1} = 261591/5; ground False; exact gap 261591/5; order 4; pass True
- C13966046-energy-002 (nitrogen-cation): NIST C $^2\Sigma_u^+$; Te 64608._1 cm^{-1} = 646081/10; ground False; exact gap 646081/10; order 5; pass True
- C13966046-energy-001 (nitrogen-cation): NIST G $^2\Sigma_g^+$; Te (195000) cm^{-1} = 195000/1; ground False; exact gap 195000/1; order 6; pass True
- C12185078-energy-010 (oxygen-cation): NIST X $^2\Pi_g$; Te 197.3 cm^{-1} = 1973/10; ground True; exact gap 0/1; order structural-empty-One; pass True
- C12185078-energy-009 (oxygen-cation): NIST a $^4\Pi_{ui}$; Te 32964 cm^{-1} = 32964/1; ground False; exact gap 327667/10; order 1; pass True
- C12185078-energy-008 (oxygen-cation): NIST A $^2\Pi_u$; Te 40669._3 cm^{-1} = 406693/10; ground False; exact gap 40472/1; order 2; pass True
- C12185078-energy-007 (oxygen-cation): NIST b' ($^4\Pi_g$); Te (48000) cm^{-1} = 48000/1; ground False; exact gap 478027/10; order 3; pass True
- C12185078-energy-006 (oxygen-cation): NIST b $^4\Sigma_g^-$; Te 49552 cm^{-1} = 49552/1; ground False; exact gap 493547/10; order 4; pass True
- C12185078-energy-005 (oxygen-cation): NIST C ($^2\Phi_u$); Te (53620) cm^{-1} = 53620/1; ground False; exact gap 534227/10; order 5; pass True
- C12185078-energy-004 (oxygen-cation): NIST D ($^2\Delta_g$); Te 62730 cm^{-1} = 62730/1; ground False; exact gap 625327/10; order 6; pass True

- C12185078-energy-003 (oxygen-cation): NIST B $\wedge^2\Sigma_g^-$; Te 66719 cm^{-1} = 66719/1; ground False; exact gap 665217/10; order 7; pass True
- C12185078-energy-002 (oxygen-cation): NIST $(\wedge^2\Pi_u)$; Te (90000) cm^{-1} = 90000/1; ground False; exact gap 898027/10; order 8; pass True
- C12185078-energy-001 (oxygen-cation): NIST c $\wedge^4\Sigma_u^-$; Te (100914) cm^{-1} = 100914/1; ground False; exact gap 1007167/10; order 9; pass True
- C14452938-energy-010 (nitric-oxide-cation): NIST X $\wedge^1\Sigma^+$; Te 0 cm^{-1} = 0/1; ground True; exact gap 0/1; order structural-empty-One; pass True
- C14452938-energy-009 (nitric-oxide-cation): NIST a $\wedge^3\Sigma^+$; Te 52190 cm^{-1} = 52190/1; ground False; exact gap 52190/1; order 1; pass True
- C14452938-energy-008 (nitric-oxide-cation): NIST b $\wedge^3\Pi$; Te 59240 cm^{-1} = 59240/1; ground False; exact gap 59240/1; order 2; pass True
- C14452938-energy-007 (nitric-oxide-cation): NIST w $\wedge^3\Delta$; Te 61880 cm^{-1} = 61880/1; ground False; exact gap 61880/1; order 3; pass True
- C14452938-energy-006 (nitric-oxide-cation): NIST b' $\wedge^3\Sigma^-$; Te 67720 cm^{-1} = 67720/1; ground False; exact gap 67720/1; order 4; pass True
- C14452938-energy-005 (nitric-oxide-cation): NIST A' $\wedge^1\Sigma^+$; Te 69540 cm^{-1} = 69540/1; ground False; exact gap 69540/1; order 5; pass True
- C14452938-energy-004 (nitric-oxide-cation): NIST W $\wedge^1\Delta$; Te 71450 cm^{-1} = 71450/1; ground False; exact gap 71450/1; order 6; pass True
- C14452938-energy-003 (nitric-oxide-cation): NIST A $\wedge^1\Pi$; Te 73471.72 cm^{-1} = 1836793/25; ground False; exact gap 1836793/25; order 7; pass True
- C14452938-energy-002 (nitric-oxide-cation): NIST c $\wedge^3\Pi$; Te (100780) cm^{-1} = 100780/1; ground False; exact gap 100780/1; order 8; pass True
- C14452938-energy-001 (nitric-oxide-cation): NIST (B' $\wedge^1\Sigma^+$); Te (109400) cm^{-1} = 109400/1; ground False; exact gap 109400/1; order 9; pass True
- C11062774-energy-003 (oxygen-anion): NIST X $\wedge^2\Pi_g$; Te 0 cm^{-1} = 0/1; ground True; exact gap 0/1; order structural-empty-One; pass True
- C11062774-energy-002 (oxygen-anion): NIST A $(\wedge^2\Pi_u)$; Te (25300) cm^{-1} = 25300/1; ground False; exact gap 25300/1; order 1; pass True
- C11062774-energy-001 (oxygen-anion): NIST a $(\wedge^4\Pi_u)$; Te 97800 cm^{-1} = 97800/1; ground False; exact gap 97800/1; order 2; pass True
- C14967783-energy-007 (nitric-oxide-anion): NIST X $\wedge^3\Sigma^-$; Te 0 cm^{-1} = 0/1; ground True; exact gap 0/1; order structural-empty-One; pass True
- C14967783-energy-006 (nitric-oxide-anion): NIST a $\wedge^1\Delta$; Te 6050 cm^{-1} = 6050/1; ground False; exact gap 6050/1; order 1; pass True
- C14967783-energy-005 (nitric-oxide-anion): NIST b $\wedge^1\Sigma^+$; Te (9300) cm^{-1} = 9300/1; ground False; exact gap 9300/1; order 2; pass True
- C14967783-energy-004 (nitric-oxide-anion): NIST $(\wedge^1\Sigma^+)$; Te 40400 cm^{-1} = 40400/1; ground False; exact gap 40400/1; order 3; pass True
- C14967783-energy-003 (nitric-oxide-anion): NIST $(\wedge^3\Pi)$; Te 43400 cm^{-1} = 43400/1; ground False; exact gap 43400/1; order 4; pass True
- C14967783-energy-002 (nitric-oxide-anion): NIST $(\wedge^3\Sigma^+)$; Te 43800 cm^{-1} = 43800/1; ground False; exact gap 43800/1; order 5; pass True

- C14967783-energy-001 (nitric-oxide-anion): NIST ($\Lambda\Sigma^+$); Te 51700 cm⁻¹ = 51700/1; ground False; exact gap 51700/1; order 6; pass True
- complete strict measured state-pair comparisons: 3325
- adverse tied_ground_rejected: True
- adverse negative_gap_rejected: True
- adverse omitted_row_rejected: True
- adverse unit_confounded_row_rejected: True
- adverse tampered_source_rejected: True
- adverse complete_vector_retained: True

Falsification boundary: The claim fails if any declared finite molecular state support lacks one unique least state; if an excited state lacks a positive successor or retained positive gap; if any pair is incomparable or contradictory; if the source-designated X state is not the unique least orderable NIST state for any registered species; if any of the 284 excited-state values does not lie above its species ground record; if any of the 306 exact measured energy inscriptions, uncertainty-status labels or source identities is omitted; or if a tied ground, negative gap, omitted row, unit-confounded row or tampered source is accepted.

Measurement receipt: sha256:602f80b4fcfda500d4809d144812dceec3a45e15976a58fecff27dfc9760d80c.

Isolation certificate: sha256:d41d8515846cab140857e45ecd774f9008fd45f28686206c57a1a433fe5216c6.

Custody certificate: sha256:4577085b3b0e7ee903d21ddb02313a8a6de11483dcd5b67fc2160d9f0cb7661c.

Registered source descriptions:

- National Institute of Standards and Technology: <https://webbook.nist.gov/chemistry/> ()

Registered target identities:

- C1333740-energy-001 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C1333740-energy-002 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C7782447-energy-025 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7782447-energy-026 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7782414-energy-001 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7782414-energy-002 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7782414-energy-003 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7782414-energy-004 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7782414-energy-005 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7782414-energy-006 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7782414-energy-007 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7782414-energy-008 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C630080-energy-001 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C630080-energy-002 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C630080-energy-003 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C630080-energy-004 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C630080-energy-005 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C630080-energy-008 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C630080-energy-014 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C630080-energy-015 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C630080-energy-016 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C630080-energy-017 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C630080-energy-018 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C630080-energy-019 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C630080-energy-020 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10102439-energy-001 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10102439-energy-002 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025

- C10102439-energy-003 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10102439-energy-004 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10102439-energy-005 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C10102439-energy-011 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10102439-energy-012 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C3352576-energy-004 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C3352576-energy-005 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C13774920-energy-006 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C3315375-energy-002 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C3315375-energy-003 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025

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- C3315375-energy-008 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C10097286-energy-017 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C13827322-energy-003 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025

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- C13827322-energy-005 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13827322-energy-006 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13827322-energy-007 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13827322-energy-008 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C7664393-energy-002 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7664393-energy-003 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C7647010-energy-005 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7647010-energy-006 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7647010-energy-007 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7647010-energy-008 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C7647010-energy-014 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C7647010-energy-017 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C7647010-energy-019 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C10035106-energy-003 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C10035106-energy-005 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-energy-006 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-energy-007 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-energy-008 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-energy-009 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-energy-010 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025

- C10035106-energy-011 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-energy-012 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-energy-013 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-energy-014 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-energy-015 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-energy-016 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-energy-017 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-energy-018 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-energy-019 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C7553562-energy-003 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7553562-energy-004 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7553562-energy-005 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7553562-energy-006 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7553562-energy-007 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7553562-energy-008 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7553562-energy-009 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C12184906-energy-002 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12184906-energy-003 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C13966046-energy-002 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13966046-energy-003 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13966046-energy-004 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13966046-energy-005 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13966046-energy-006 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C12185078-energy-006 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12185078-energy-007 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12185078-energy-008 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12185078-energy-009 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12185078-energy-010 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025

- C14452938-energy-001 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C14452938-energy-002 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C11062774-energy-002 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C11062774-energy-003 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C14967783-energy-005 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C14967783-energy-006 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C14967783-energy-007 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025

Meaning. The field-wide extension derives molecular electronic identity, state composition, occupancy, ordering, symmetry, correlation, transitions, measurement and operational classical-quantum correspondence from the same exact carrier laws. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical-zero, negative, irrational, imaginary or floating proof value
- no measured NIST energy or state label in candidate generation or survivor selection
- no imported Hamiltonian, wavefunction, molecular-orbital energy formula or fitted scale
- no selected state pair or omitted unfavorable state-energy row
- no species-specific ground-state lookup rule
- external decimal and source-zero inscriptions remain post-seal measurement records
- no fitted, learned or target-derived parameter

Immutable identities. Source manifest

sha256:1e1aa0ca9f03c47d05853a02d2c97bf09354ecb3539efc4e8e5f07fff4b608fa; derivation seal
 sha256:9e1837f1591f48a25568c99f4ccc5efafb85a81236694bf9b18852d60ff13f2c; engine receipt
 sha256:7231405d08d5f63184bbc074f15578cdfef505e2051d8ed82077f5b4e893c690 at
 receipts/engine/model_admitted/SFT-CHEM-STATE-ENERGY-ORDER-004-7231405d08d5f631.json;
 empirical hash sha256:1d1575e9b208613e7956ca036de9ae7287131468bddae6a90fea2bc6a0c1d1d1;
 measurement receipt sha256:602f80b4fcfda500d4809d144812dceec3a45e15976a58fecff27dfc9760d80c.

100. Finite electronic-state equivalence, degeneracy and retained symmetry

Claim identity: SFT-CHEM-STATE-SYMMETRY-DEGENERACY-005

Question and theorem. At one molecular carrier, electronic states are equivalent exactly when their complete positive-spin, axis-support, inversion, reflection and component signatures agree. Axis-invariant support retains one orientation; every positive axis recurrence retains the complementary orientation pair; the finite equivalence class contains exactly the positive spin multiplicity times the positive orientation count, with every applicable symmetry distinction held and every absent distinction represented only by structural EmptyOne.

one-molecule-equivalence__complete-symmetry-signature__positive-spin-multiplicity__empty-One-or-positive-axis-rank__one-or-complementary-orientation-support__held-symmetry-label-composition__complete-positive-component-classes__no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-CHEM-ELECTRON-COUNT-SPIN-002
- SFT-CHEM-ORBITAL-SUPPORT-OCCUPANCY-003
- SFT-CHEM-STATE-ENERGY-ORDER-004

Generated grammar. Generate the literal product of carrier, identity, spin, axis, orientation, symmetry, class and extension forms. Decide all 256 forms solely by admitted exact equivalence, held-label, spin, exclusion, support and finite-order laws.

Boundary: Every positive finite molecular electronic-state support; every positive spin multiplicity; structural axis invariance and every positive axis recurrence; both held inversion/reflection fibres where applicable; and structural EmptyOne wherever a symmetry coordinate is absent. Closure is depth-independent because each added component, multiplicity successor or axis recurrence repeats the same exact signature and complete-component checks.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *one-molecule-equivalence__complete-symmetry-signature__positive-spin-multiplicity__empty-One-or-positive-axis-rank__one-or-complementary-orientation-support__held-symmetry-label-composition__complete-positive-component-class__no-extra-rule*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-molecule-equivalence	cross-carrier-equivalence	States on different molecules cannot share one declared chemical equivalence class.
identity	complete-symmetry-signature	energy-only-equivalence	Equal or nearby energy alone erases support and symmetry distinctions.
spin	positive-spin-multiplicity	unsigned-or-absent-multiplicity	Erasing positive spin width loses generated component support.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
axis	empty-One-or-positive-axis-rank	continuum-or-zero-projection	A continuum coordinate or numerical zero violates the generated axis grammar.
orientation	one-or-complementary-orientation-support	free-degeneracy-number	An asserted degeneracy count is an unforced parameter.
symmetry	held-symmetry-label-composition	erased-sign-labels	Erasing inversion, reflection and component labels merges distinguishable terms.
class	complete-positive-component-class	unbounded-anonymous-members	Anonymous or unbounded members cannot certify degeneracy.
extension	no-extra-rule	term-table-exception	A species or term lookup lets source assignments select the law.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:edcfc53a893102c7e786b91b214350befe8a5d2dac498d508024bfb407912070.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:77bd3b1ab3efdefe6e62a8f83bc941d433b55cb3dab25cf59be8342f870dbbe6.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ed2e1d44ee0458e12b7bf4a0a151ce64eae3eb0e1d25011c3344ea944fda6b24.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:904ffe3d6ddc370e3ad5f88265da1864d73d752766a1a12538effa22fe404d21.

Independent reconstruction. Implementation hash

sha256:208c3f64d75f44b4c25af0f7231115c899c3584edaaa422203549cb6c8bd5a1c; independent certificate sha256:2eee45796f701101b275bdd03ce9f2aace6daba31d4ddd608b5ce7e2d5a690cf; external-validation hash sha256:5d73a8e0c08758f6993adec96eec5f9ccaba7f34034768581b7cf094fee8beb6. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025

Comparison records:

- C1333740-symmetry-001-01 (hydrogen-neutral): NIST $u \wedge 3\Pi_u 6p\pi$; term $\wedge 3\Pi_u$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-002-01 (hydrogen-neutral): NIST $t \wedge 3\Sigma_u^+ 5f\sigma$; term $\wedge 3\Sigma_u^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-003-01 (hydrogen-neutral): NIST $q \wedge 3\Sigma_g^+ 5d\sigma$; term $\wedge 3\Sigma_g^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-004-01 (hydrogen-neutral): NIST $n \wedge 3\Pi_u 5p\pi$; term $\wedge 3\Pi_u$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-005-01 (hydrogen-neutral): NIST $m \wedge 3\Sigma_u^+ 4f\sigma$; term $\wedge 3\Sigma_u^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True

- C1333740-symmetry-006-01 (hydrogen-neutral): NIST s $\wedge^3\Delta_g 4d\delta$; term $\wedge^3\Delta_g$; spin multiplicity 3; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-007-01 (hydrogen-neutral): NIST r $\wedge^3\Pi_g 4d\pi$; term $\wedge^3\Pi_g$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-008-01 (hydrogen-neutral): NIST p $\wedge^3\Sigma_g^+ 4d\sigma$; term $\wedge^3\Sigma_g^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-009-01 (hydrogen-neutral): NIST v $(\wedge^3\Pi_g)$; term $\wedge^3\Pi_g$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-010-01 (hydrogen-neutral): NIST k $\wedge^3\Pi_u 4p\pi$; term $\wedge^3\Pi_u$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-011-01 (hydrogen-neutral): NIST f $\wedge^3\Sigma_u^+ 4p\sigma$; term $\wedge^3\Sigma_u^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-012-01 (hydrogen-neutral): NIST o $\wedge^3\Sigma_u^+$; term $\wedge^3\Sigma_u^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-013-01 (hydrogen-neutral): NIST l $\wedge^3\Pi_u$; term $\wedge^3\Pi_u$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-014-01 (hydrogen-neutral): NIST j $\wedge^3\Delta_g 3d\delta$; term $\wedge^3\Delta_g$; spin multiplicity 3; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-015-01 (hydrogen-neutral): NIST i $\wedge^3\Pi_g 3d\pi$; term $\wedge^3\Pi_g$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-016-01 (hydrogen-neutral): NIST h $\wedge^3\Sigma_g^+ 3s\sigma$; term $\wedge^3\Sigma_g^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-017-01 (hydrogen-neutral): NIST g $\wedge^3\Sigma_g^+ 3d\sigma$; term $\wedge^3\Sigma_g^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-018-01 (hydrogen-neutral): NIST d $\wedge^3\Pi_u 3p\pi$; term $\wedge^3\Pi_u$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-019-01 (hydrogen-neutral): NIST e $\wedge^3\Sigma_u^+ 3p\sigma$; term $\wedge^3\Sigma_u^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-020-01 (hydrogen-neutral): NIST a $\wedge^3\Sigma_g^+ 2s\sigma$; term $\wedge^3\Sigma_g^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-021-01 (hydrogen-neutral): NIST c $\wedge^3\Pi_u 3p\pi$; term $\wedge^3\Pi_u$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True

- C1333740-symmetry-022-01 (hydrogen-neutral): NIST D" $\wedge^1\Pi_u 5p\pi$; term $\wedge^1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-023-01 (hydrogen-neutral): NIST D' $\wedge^1\Pi_u 4p\pi$; term $\wedge^1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-024-01 (hydrogen-neutral): NIST S $\wedge^1\Delta_g 4d\delta$; term $\wedge^1\Delta_g$; spin multiplicity 1; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-025-01 (hydrogen-neutral): NIST O $\wedge^1\Sigma^+ 4s\sigma$; term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-026-01 (hydrogen-neutral): NIST R $\wedge^1\Pi_g 4d\pi$; term $\wedge^1\Pi_g$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-027-01 (hydrogen-neutral): NIST P $\wedge^1\Sigma_g^+ 4d\sigma$; term $\wedge^1\Sigma_g^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-028-01 (hydrogen-neutral): NIST T $\wedge^1\Sigma^+$; term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-029-01 (hydrogen-neutral): NIST B" $\wedge^1\Sigma_u^+ 4p\sigma$; term $\wedge^1\Sigma_u^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-030-01 (hydrogen-neutral): NIST N $\wedge^1\Sigma_g^+$; term $\wedge^1\Sigma_g^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-031-01 (hydrogen-neutral): NIST U ($\wedge^1\Sigma_g^+$); term $\wedge^1\Sigma_g^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-032-01 (hydrogen-neutral): NIST M $\wedge^1\Sigma_g^+$; term $\wedge^1\Sigma_g^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-033-01 (hydrogen-neutral): NIST L $\wedge^1\Sigma_g^+$; term $\wedge^1\Sigma_g^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-034-01 (hydrogen-neutral): NIST H $\wedge^1\Sigma_g^+ 3s\sigma$; term $\wedge^1\Sigma_g^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-035-01 (hydrogen-neutral): NIST D $\wedge^1\Pi_u 3p\pi$; term $\wedge^1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-036-01 (hydrogen-neutral): NIST J $\wedge^1\Delta_g 3d\delta$; term $\wedge^1\Delta_g$; spin multiplicity 1; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-037-01 (hydrogen-neutral): NIST I $\wedge^1\Pi_g 3d\pi$; term $\wedge^1\Pi_g$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True

- C1333740-symmetry-038-01 (hydrogen-neutral): NIST G $^1\Sigma_g^+ 3d\sigma$; term $^1\Sigma_g^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-039-01 (hydrogen-neutral): NIST K ($^1\Sigma_g^+$); term $^1\Sigma_g^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-040-01 (hydrogen-neutral): NIST Q ($^1\Pi_g$); term $^1\Pi_g$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-041-01 (hydrogen-neutral): NIST B' $^1\Sigma_u^+ 3p\sigma$; term $^1\Sigma_u^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-042-01 (hydrogen-neutral): NIST F $^1\Sigma_g^+ 2p\sigma^2$; term $^1\Sigma_g^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-043-01 (hydrogen-neutral): NIST E $^1\Sigma_g^+ 2s\sigma$; term $^1\Sigma_g^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-044-01 (hydrogen-neutral): NIST C $^1\Pi_u 2p\pi$; term $^1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-045-01 (hydrogen-neutral): NIST B $^1\Sigma_u^+ 2p\sigma$; term $^1\Sigma_u^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C1333740-symmetry-046-01 (hydrogen-neutral): NIST X $^1\Sigma_g^+ 1s\sigma^2$; term $^1\Sigma_g^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-001-01 (nitrogen-neutral): NIST x" ($^1\Sigma_u^+$) 1; term $^1\Sigma_u^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-002-01 (nitrogen-neutral): NIST x' ($^1\Pi_u$) 2; term $^1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-003-01 (nitrogen-neutral): NIST v $^1\Pi_g 4$; term $^1\Pi_g$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-004-01 (nitrogen-neutral): NIST o_5 $^1\Pi_u$; term $^1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-005-01 (nitrogen-neutral): NIST O_5 ($^3\Pi_u$); term $^3\Pi_u$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-006-01 (nitrogen-neutral): NIST c'_n $^1\Sigma_u^+ 21$; term $^1\Sigma_u^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-007-01 (nitrogen-neutral): NIST c_n $^1\Pi_u 22$; term $^1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True

- C7727379-symmetry-008-01 (nitrogen-neutral): NIST $c_n \wedge 1\Pi_u$ 26; term $\wedge 1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-009-01 (nitrogen-neutral): NIST $o_4 \wedge 1\Pi_u$; term $\wedge 1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-010-01 (nitrogen-neutral): NIST $O_4 (\wedge 3\Pi_u)$; term $\wedge 3\Pi_u$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-011-01 (nitrogen-neutral): NIST $c'_5 \wedge 1\Sigma_u^+$; term $\wedge 1\Sigma_u^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-012-01 (nitrogen-neutral): NIST $c_4 \wedge 1\Pi_u$; term $\wedge 1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-013-01 (nitrogen-neutral): NIST $c_4 \wedge 1\Pi_u$ 31; term $\wedge 1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-014-01 (nitrogen-neutral): NIST $z \wedge 1\Delta_g$; term $\wedge 1\Delta_g$; spin multiplicity 1; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-015-01 (nitrogen-neutral): NIST $y \wedge 1\Pi_g$; term $\wedge 1\Pi_g$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-016-01 (nitrogen-neutral): NIST $k \wedge 1\Pi_g$; term $\wedge 1\Pi_g$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-017-01 (nitrogen-neutral): NIST $x \wedge 1\Sigma_g^-$; term $\wedge 1\Sigma_g^-$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion g; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-018-01 (nitrogen-neutral): NIST $d' \wedge 1\Sigma_u^-$ or $\wedge 1\Delta_u$; term $\wedge 1\Sigma_u^-$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion u; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-018-02 (nitrogen-neutral): NIST $d' \wedge 1\Sigma_u^-$ or $\wedge 1\Delta_u$; term $\wedge 1\Delta_u$; spin multiplicity 1; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-019-01 (nitrogen-neutral): NIST $o_3 \wedge 1\Pi_u$; term $\wedge 1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-020-01 (nitrogen-neutral): NIST $H \wedge 3\Phi_u$; term $\wedge 3\Phi_u$; spin multiplicity 3; axis Φ (third-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-021-01 (nitrogen-neutral): NIST $c'_4 \wedge 1\Sigma_u^+$; term $\wedge 1\Sigma_u^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-022-01 (nitrogen-neutral): NIST $c_3 \wedge 1\Pi_u$; term $\wedge 1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True

- C7727379-symmetry-023-01 (nitrogen-neutral): NIST b' $\wedge^1\Sigma_u^+$; term $\wedge^1\Sigma_u^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-024-01 (nitrogen-neutral): NIST D $\wedge^3\Sigma_u^+$; term $\wedge^3\Sigma_u^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-025-01 (nitrogen-neutral): NIST b $\wedge^1\Pi_u$; term $\wedge^1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-026-01 (nitrogen-neutral): NIST a" $\wedge^1\Sigma_g^+$; term $\wedge^1\Sigma_g^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-027-01 (nitrogen-neutral): NIST C' $\wedge^3\Pi_u$; term $\wedge^3\Pi_u$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-028-01 (nitrogen-neutral): NIST E $\wedge^3\Sigma_g^+$; term $\wedge^3\Sigma_g^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-029-01 (nitrogen-neutral): NIST C" $\wedge^5\Pi_u$; term $\wedge^5\Pi_u$; spin multiplicity 5; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 10; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-030-01 (nitrogen-neutral): NIST C $\wedge^3\Pi_u$; term $\wedge^3\Pi_u$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-031-01 (nitrogen-neutral): NIST G $\wedge^3\Delta_g$; term $\wedge^3\Delta_g$; spin multiplicity 3; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-032-01 (nitrogen-neutral): NIST A' $\wedge^5\Sigma_g^+$; term $\wedge^5\Sigma_g^+$; spin multiplicity 5; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 5; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-033-01 (nitrogen-neutral): NIST w $\wedge^1\Delta_u$; term $\wedge^1\Delta_u$; spin multiplicity 1; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-034-01 (nitrogen-neutral): NIST a $\wedge^1\Pi_g$; term $\wedge^1\Pi_g$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-035-01 (nitrogen-neutral): NIST a' $\wedge^1\Sigma_u^-$; term $\wedge^1\Sigma_u^-$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion u; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-036-01 (nitrogen-neutral): NIST B' $\wedge^3\Sigma_u^-$; term $\wedge^3\Sigma_u^-$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion u; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-037-01 (nitrogen-neutral): NIST W $\wedge^3\Delta_u$; term $\wedge^3\Delta_u$; spin multiplicity 3; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-038-01 (nitrogen-neutral): NIST B $\wedge^3\Pi_g$; term $\wedge^3\Pi_g$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True

- C7727379-symmetry-039-01 (nitrogen-neutral): NIST A $\wedge^3\Sigma_u^+$; term $\wedge^3\Sigma_u^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7727379-symmetry-040-01 (nitrogen-neutral): NIST X $\wedge^1\Sigma_g^+$; term $\wedge^1\Sigma_g^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-001-01 (oxygen-neutral): NIST Z ($\wedge^3\Pi_u$) 2; term $\wedge^3\Pi_u$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-002-01 (oxygen-neutral): NIST W ($\wedge^3\Sigma_u^-$); term $\wedge^3\Sigma_u^-$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion u; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-003-01 (oxygen-neutral): NIST p $\wedge^1\Phi_u$; term $\wedge^1\Phi_u$; spin multiplicity 1; axis Φ (third-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-004-01 (oxygen-neutral): NIST H ($\wedge^3\Pi_u$); term $\wedge^3\Pi_u$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-005-01 (oxygen-neutral): NIST L ($\wedge^3\Pi_u$); term $\wedge^3\Pi_u$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-006-01 (oxygen-neutral): NIST k ($\wedge^1\Delta_u$); term $\wedge^1\Delta_u$; spin multiplicity 1; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-007-01 (oxygen-neutral): NIST j ($\wedge^1\Sigma_u^+$); term $\wedge^1\Sigma_u^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-008-01 (oxygen-neutral): NIST G ($\wedge^3\Sigma_u^+$); term $\wedge^3\Sigma_u^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-009-01 (oxygen-neutral): NIST i ($\wedge^1\Delta_{2u}$); term $\wedge^1\Delta_{2u}$; spin multiplicity 1; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component positive-component-2; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-010-01 (oxygen-neutral): NIST i' ($\wedge^3\Delta_{2u}$); term $\wedge^3\Delta_{2u}$; spin multiplicity 3; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component positive-component-2; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-011-01 (oxygen-neutral): NIST h ($\wedge^1\Pi_u$); term $\wedge^1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-012-01 (oxygen-neutral): NIST g ($\wedge^1\Pi_u$); term $\wedge^1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-013-01 (oxygen-neutral): NIST F $\wedge^3\Pi_u$; term $\wedge^3\Pi_u$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-014-01 (oxygen-neutral): NIST E $\wedge^3\Sigma_u^-$; term $\wedge^3\Sigma_u^-$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion u; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True

- C7782447-symmetry-015-01 (oxygen-neutral): NIST $f \Lambda^1\Sigma_u^+$; term $\Lambda^1\Sigma_u^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-016-01 (oxygen-neutral): NIST $D \Lambda^3\Sigma_u^+$; term $\Lambda^3\Sigma_u^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-017-01 (oxygen-neutral): NIST $e \Lambda^1\Delta_{2u}$; term $\Lambda^1\Delta_{2u}$; spin multiplicity 1; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component positive-component-2; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-018-01 (oxygen-neutral): NIST $e' \Lambda^3\Delta_{2u}$; term $\Lambda^3\Delta_{2u}$; spin multiplicity 3; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component positive-component-2; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-019-01 (oxygen-neutral): NIST $d \Lambda^1\Pi_g$; term $\Lambda^1\Pi_g$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-020-01 (oxygen-neutral): NIST $C \Lambda^3\Pi_g$; term $\Lambda^3\Pi_g$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-021-01 (oxygen-neutral): NIST $B \Lambda^3\Sigma_u^-$; term $\Lambda^3\Sigma_u^-$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion u; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-022-01 (oxygen-neutral): NIST $A \Lambda^3\Sigma_u^+$; term $\Lambda^3\Sigma_u^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-023-01 (oxygen-neutral): NIST $A' \Lambda^3\Delta_u$; term $\Lambda^3\Delta_u$; spin multiplicity 3; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-024-01 (oxygen-neutral): NIST $c \Lambda^1\Sigma_u^-$; term $\Lambda^1\Sigma_u^-$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion u; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-025-01 (oxygen-neutral): NIST $b \Lambda^1\Sigma_g^+$; term $\Lambda^1\Sigma_g^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-026-01 (oxygen-neutral): NIST $a \Lambda^1\Delta_g$; term $\Lambda^1\Delta_g$; spin multiplicity 1; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782447-symmetry-027-01 (oxygen-neutral): NIST $X \Lambda^3\Sigma_g^-$; term $\Lambda^3\Sigma_g^-$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion g; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782414-symmetry-001-01 (fluorine-neutral): NIST $K \Lambda^1\Pi_u$; term $\Lambda^1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782414-symmetry-002-01 (fluorine-neutral): NIST $J \Lambda^1\Pi_u (\Lambda^4p \sigma)$; term $\Lambda^1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782414-symmetry-003-01 (fluorine-neutral): NIST $I \Lambda^1\Sigma_u^+ (\Lambda^4p \pi)$; term $\Lambda^1\Sigma_u^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True

- C7782414-symmetry-004-01 (fluorine-neutral): NIST $H \ ^1\Pi_u \ (^3p\sigma)$; term $^1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782414-symmetry-005-01 (fluorine-neutral): NIST $h \ ^3\Pi_{1u} \ (3p\sigma)$; term $^3\Pi_{1u}$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component positive-component-1; component kind absence; source glyph 0 denotes absence False; pass True
- C7782414-symmetry-006-01 (fluorine-neutral): NIST $G \ ^1\Sigma_u^+ \ (^3p\pi)$; term $^1\Sigma_u^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782414-symmetry-007-01 (fluorine-neutral): NIST $E \ (^1\Sigma_u^+)$; term $^1\Sigma_u^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782414-symmetry-008-01 (fluorine-neutral): NIST $D \ (^1\Sigma_u^+)$; term $^1\Sigma_u^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782414-symmetry-009-01 (fluorine-neutral): NIST $f \ (^3\Pi_{1g}) \ (3s\sigma)$; term $^3\Pi_{1g}$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion g; reflection absence; axis component positive-component-1; component kind absence; source glyph 0 denotes absence False; pass True
- C7782414-symmetry-010-01 (fluorine-neutral): NIST $C \ ^1\Sigma_u^+$; term $^1\Sigma_u^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782414-symmetry-011-01 (fluorine-neutral): NIST $F \ ^1\Pi_g \ (3s\sigma)$; term $^1\Pi_g$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782414-symmetry-012-01 (fluorine-neutral): NIST $A \ ^1\Pi_u \ 17$; term $^1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782414-symmetry-013-01 (fluorine-neutral): NIST $a \ ^3\Pi_{0+u} \ 19$; term $^3\Pi_{0+u}$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence True; pass True
- C7782414-symmetry-014-01 (fluorine-neutral): NIST $X \ ^1\Sigma_g^+$; term $^1\Sigma_g^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-001-01 (carbon-monoxide-neutral): NIST $O_2 \ (^1\Pi)$; term $^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-002-01 (carbon-monoxide-neutral): NIST $O_1 \ (^1\Pi)$; term $^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-003-01 (carbon-monoxide-neutral): NIST $Z \ ^1\Sigma^+$; term $^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-004-01 (carbon-monoxide-neutral): NIST $H15 \ (^1\Pi)$; term $^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-005-01 (carbon-monoxide-neutral): NIST $H' \ ^1\Pi$; term $^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True

- C630080-symmetry-006-01 (carbon-monoxide-neutral): NIST L $\wedge 1\Pi\ 4p\pi$; term $\wedge 1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-007-01 (carbon-monoxide-neutral): NIST L' ($\wedge 1\Pi$); term $\wedge 1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-008-01 (carbon-monoxide-neutral): NIST K $\wedge 1\Sigma^+ 4p\sigma$; term $\wedge 1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-009-01 (carbon-monoxide-neutral): NIST W $\wedge 1\Pi$; term $\wedge 1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-010-01 (carbon-monoxide-neutral): NIST W' $\wedge 1\Pi$; term $\wedge 1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-011-01 (carbon-monoxide-neutral): NIST J $\wedge 1\Sigma^+ 4s\sigma$; term $\wedge 1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-012-01 (carbon-monoxide-neutral): NIST G $\wedge 1\Pi(3d\pi)$; term $\wedge 1\Pi(3d\pi)$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-013-01 (carbon-monoxide-neutral): NIST G' $\wedge 1\Pi$; term $\wedge 1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-014-01 (carbon-monoxide-neutral): NIST h ($\wedge 3\Sigma^+, 4s\sigma$); term $\wedge 3\Sigma^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-015-01 (carbon-monoxide-neutral): NIST Y $\wedge 1\Sigma^+$; term $\wedge 1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-016-01 (carbon-monoxide-neutral): NIST F $\wedge 1\Sigma^+(3d\sigma)$; term $\wedge 1\Sigma^+(3d\sigma)$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-017-01 (carbon-monoxide-neutral): NIST V $\wedge 1\Pi$; term $\wedge 1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-018-01 (carbon-monoxide-neutral): NIST g $\wedge 3\Sigma^+$; term $\wedge 3\Sigma^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-019-01 (carbon-monoxide-neutral): NIST E $\wedge 1\Pi\ 3p\pi$; term $\wedge 1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-020-01 (carbon-monoxide-neutral): NIST c $\wedge 3\Pi\ 3p\pi$; term $\wedge 3\Pi$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-021-01 (carbon-monoxide-neutral): NIST C $\wedge 1\Sigma^+ 3p\sigma$; term $\wedge 1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True

- C630080-symmetry-022-01 (carbon-monoxide-neutral): NIST j $^{\wedge}3\Sigma^{\wedge}+3p\sigma$; term $^{\wedge}3\Sigma^{\wedge}+3p\sigma$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-023-01 (carbon-monoxide-neutral): NIST B $^{\wedge}1\Sigma^{\wedge}+3s\sigma$; term $^{\wedge}1\Sigma^{\wedge}+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-024-01 (carbon-monoxide-neutral): NIST b $^{\wedge}3\Sigma^{\wedge}+3s\sigma$; term $^{\wedge}3\Sigma^{\wedge}+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-025-01 (carbon-monoxide-neutral): NIST D $^{\wedge}1\Delta$; term $^{\wedge}1\Delta$; spin multiplicity 1; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-026-01 (carbon-monoxide-neutral): NIST I $^{\wedge}1\Sigma^{\wedge}-$; term $^{\wedge}1\Sigma^{\wedge}-$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-027-01 (carbon-monoxide-neutral): NIST A $^{\wedge}1\Pi$; term $^{\wedge}1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-028-01 (carbon-monoxide-neutral): NIST e $^{\wedge}3\Sigma^{\wedge}-$; term $^{\wedge}3\Sigma^{\wedge}-$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-029-01 (carbon-monoxide-neutral): NIST d $^{\wedge}3\Delta_i$; term $^{\wedge}3\Delta_i$; spin multiplicity 3; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind i; source glyph 0 denotes absence False; pass True
- C630080-symmetry-030-01 (carbon-monoxide-neutral): NIST a' $^{\wedge}3\Sigma^{\wedge}+$; term $^{\wedge}3\Sigma^{\wedge}+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C630080-symmetry-031-01 (carbon-monoxide-neutral): NIST a $^{\wedge}3\Pi_r$; term $^{\wedge}3\Pi_r$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind r; source glyph 0 denotes absence False; pass True
- C630080-symmetry-032-01 (carbon-monoxide-neutral): NIST X $^{\wedge}1\Sigma^{\wedge}+$; term $^{\wedge}1\Sigma^{\wedge}+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-001-01 (nitric-oxide-neutral): NIST Z $^{\wedge}2\Sigma^{\wedge}+7s\sigma$; term $^{\wedge}2\Sigma^{\wedge}+$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-002-01 (nitric-oxide-neutral): NIST Y $^{\wedge}2\Sigma^{\wedge}+6p\sigma$; term $^{\wedge}2\Sigma^{\wedge}+$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-003-01 (nitric-oxide-neutral): NIST W $^{\wedge}2\Pi$ $6p\pi$; term $^{\wedge}2\Pi$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-004-01 (nitric-oxide-neutral): NIST U $^{\wedge}2\Delta$ $5d\delta$; term $^{\wedge}2\Delta$ $5d\delta$; spin multiplicity 2; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-005-01 (nitric-oxide-neutral): NIST T $^{\wedge}2\Sigma^{\wedge}+6s\sigma$; term $^{\wedge}2\Sigma^{\wedge}+6s\sigma$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True

- C10102439-symmetry-006-01 (nitric-oxide-neutral): NIST R $\wedge 2\Sigma^+5p\sigma$; term $\wedge 2\Sigma^+5p\sigma$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-007-01 (nitric-oxide-neutral): NIST Q $\wedge 2\Pi 5p\pi$ 26; term $\wedge 2\Pi 5p\pi$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-008-01 (nitric-oxide-neutral): NIST Q $\wedge 2\Pi 5p\pi$; term $\wedge 2\Pi 5p\pi$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-009-01 (nitric-oxide-neutral): NIST O' $\wedge 2\Pi 4d\pi$; term $\wedge 2\Pi$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-010-01 (nitric-oxide-neutral): NIST O $\wedge 2\Sigma^+ 4d\sigma$; term $\wedge 2\Sigma^+$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-011-01 (nitric-oxide-neutral): NIST N $\wedge 2\Delta 4d\delta$; term $\wedge 2\Delta 4d\delta$; spin multiplicity 2; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-012-01 (nitric-oxide-neutral): NIST S $\wedge 2\Sigma^+5s\sigma$; term $\wedge 2\Sigma^+5s\sigma$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-013-01 (nitric-oxide-neutral): NIST M $\wedge 2\Sigma^+4p\sigma$; term $\wedge 2\Sigma^+4p\sigma$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-014-01 (nitric-oxide-neutral): NIST K $\wedge 2\Pi 4p\pi$ 34; term $\wedge 2\Pi 4p\pi$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-015-01 (nitric-oxide-neutral): NIST K $\wedge 2\Pi 4p\pi$; term $\wedge 2\Pi 4p\pi$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-016-01 (nitric-oxide-neutral): NIST I $\wedge 2\Sigma^+$; term $\wedge 2\Sigma^+$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-017-01 (nitric-oxide-neutral): NIST G $\wedge 2\Sigma^-$; term $\wedge 2\Sigma^-$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-018-01 (nitric-oxide-neutral): NIST L $\wedge 2\Pi_i$; term $\wedge 2\Pi_i$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion absence; reflection absence; axis component absence; component kind i; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-019-01 (nitric-oxide-neutral): NIST H' $\wedge 2\Pi 3d\pi$; term $\wedge 2\Pi$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-020-01 (nitric-oxide-neutral): NIST H $\wedge 2\Sigma^+ 3d\sigma$; term $\wedge 2\Sigma^+$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-021-01 (nitric-oxide-neutral): NIST F $\wedge 2\Delta 3d\delta$; term $\wedge 2\Delta 3d\delta$; spin multiplicity 2; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True

- C10102439-symmetry-022-01 (nitric-oxide-neutral): NIST E $^2\Sigma^+4s\sigma$; term $^2\Sigma^+4s\sigma$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-023-01 (nitric-oxide-neutral): NIST B' $^2\Delta_i$; term $^2\Delta_i$; spin multiplicity 2; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion absence; reflection absence; axis component absence; component kind i; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-024-01 (nitric-oxide-neutral): NIST D $^2\Sigma^+3p\sigma$; term $^2\Sigma^+$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-025-01 (nitric-oxide-neutral): NIST C $^2\Pi_r3p\pi$; term $^2\Pi_r$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion absence; reflection absence; axis component absence; component kind r; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-026-01 (nitric-oxide-neutral): NIST b ($^4\Sigma^-$); term $^4\Sigma^-$; spin multiplicity 4; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 4; inversion absence; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-027-01 (nitric-oxide-neutral): NIST B $^2\Pi_r$; term $^2\Pi_r$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion absence; reflection absence; axis component absence; component kind r; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-028-01 (nitric-oxide-neutral): NIST A $^2\Sigma^+3s\sigma$; term $^2\Sigma^+$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-029-01 (nitric-oxide-neutral): NIST a ($^4\Pi_i$); term $^4\Pi_i$; spin multiplicity 4; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 8; inversion absence; reflection absence; axis component absence; component kind i; source glyph 0 denotes absence False; pass True
- C10102439-symmetry-030-01 (nitric-oxide-neutral): NIST X $^2\Pi_r$; term $^2\Pi_r$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion absence; reflection absence; axis component absence; component kind r; source glyph 0 denotes absence False; pass True
- C3352576-symmetry-001-01 (hydroxyl-neutral): NIST C $^2\Sigma^+$; term $^2\Sigma^+$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C3352576-symmetry-002-01 (hydroxyl-neutral): NIST C $^2\Sigma^+$ 5; term $^2\Sigma^+$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C3352576-symmetry-003-01 (hydroxyl-neutral): NIST D $^2\Sigma^-$; term $^2\Sigma^-$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C3352576-symmetry-004-01 (hydroxyl-neutral): NIST B $^2\Sigma^+$; term $^2\Sigma^+$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C3352576-symmetry-005-01 (hydroxyl-neutral): NIST A $^2\Sigma^+$; term $^2\Sigma^+$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C3352576-symmetry-006-01 (hydroxyl-neutral): NIST X $^2\Pi_i$; term $^2\Pi_i$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion absence; reflection absence; axis component absence; component kind i; source glyph 0 denotes absence False; pass True
- C13774920-symmetry-001-01 (imidogen-neutral): NIST d $^1\Sigma^+$; term $^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True

- C13774920-symmetry-002-01 (imidogen-neutral): NIST $c \wedge 1\Pi$; term $\wedge 1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C13774920-symmetry-003-01 (imidogen-neutral): NIST $A \wedge 3\Pi_i$; term $\wedge 3\Pi_i$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind i; source glyph 0 denotes absence False; pass True
- C13774920-symmetry-004-01 (imidogen-neutral): NIST $b \wedge 1\Sigma^+$; term $\wedge 1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C13774920-symmetry-005-01 (imidogen-neutral): NIST $a \wedge 1\Delta$; term $\wedge 1\Delta$; spin multiplicity 1; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C13774920-symmetry-006-01 (imidogen-neutral): NIST $X \wedge 3\Sigma^-$; term $\wedge 3\Sigma^-$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C3315375-symmetry-001-01 (methylidyne-neutral): NIST $F \wedge 2\Sigma^+$; term $\wedge 2\Sigma^+$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C3315375-symmetry-002-01 (methylidyne-neutral): NIST $E \wedge 2\Pi$; term $\wedge 2\Pi$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C3315375-symmetry-003-01 (methylidyne-neutral): NIST $D \wedge 2\Pi_i$; term $\wedge 2\Pi_i$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion absence; reflection absence; axis component absence; component kind i; source glyph 0 denotes absence False; pass True
- C3315375-symmetry-004-01 (methylidyne-neutral): NIST $C \wedge 2\Sigma^+$; term $\wedge 2\Sigma^+$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C3315375-symmetry-005-01 (methylidyne-neutral): NIST $B \wedge 2\Sigma^-$; term $\wedge 2\Sigma^-$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C3315375-symmetry-006-01 (methylidyne-neutral): NIST $A \wedge 2\Delta$; term $\wedge 2\Delta$; spin multiplicity 2; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C3315375-symmetry-007-01 (methylidyne-neutral): NIST $a \wedge 4\Sigma^-$; term $\wedge 4\Sigma^-$; spin multiplicity 4; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 4; inversion absence; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C3315375-symmetry-008-01 (methylidyne-neutral): NIST $X \wedge 2\Pi_r$; term $\wedge 2\Pi_r$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion absence; reflection absence; axis component absence; component kind r; source glyph 0 denotes absence False; pass True
- C7782505-symmetry-001-01 (chlorine-neutral): NIST $C \wedge 1\Pi_u$; term $\wedge 1\Pi_u$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7782505-symmetry-002-01 (chlorine-neutral): NIST $B \wedge 3\Pi_0^{+u}$; term $\wedge 3\Pi_0^{+u}$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence True; pass True
- C7782505-symmetry-003-01 (chlorine-neutral): NIST $A (\wedge 3\Pi_1u)$; term $\wedge 3\Pi_1u$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component positive-component-1; component kind absence; source glyph 0 denotes absence False; pass True

- C7782505-symmetry-004-01 (chlorine-neutral): NIST A' ($\wedge^3\Pi_{2u}$); term $\wedge^3\Pi_{2u}$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component positive-component-2; component kind absence; source glyph 0 denotes absence False; pass True
- C7782505-symmetry-005-01 (chlorine-neutral): NIST X $\wedge^1\Sigma_g^+$; term $\wedge^1\Sigma_g^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-001-01 (silicon-monoxide-neutral): NIST P $\wedge^1\Sigma^+$ ($5p\sigma$); term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-002-01 (silicon-monoxide-neutral): NIST O $\wedge^1\Pi$; term $\wedge^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-003-01 (silicon-monoxide-neutral): NIST N $\wedge^1\Pi$; term $\wedge^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-004-01 (silicon-monoxide-neutral): NIST M $\wedge^1\Pi$; term $\wedge^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-005-01 (silicon-monoxide-neutral): NIST L $\wedge^1\Sigma^+$ ($5s\sigma$); term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-006-01 (silicon-monoxide-neutral): NIST K $\wedge^1\Sigma^+$; term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-007-01 (silicon-monoxide-neutral): NIST J $\wedge^1\Pi$ ($4p\pi$); term $\wedge^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-008-01 (silicon-monoxide-neutral): NIST h $\wedge^3\Pi_i$; term $\wedge^3\Pi_i$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind i; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-009-01 (silicon-monoxide-neutral): NIST I $\wedge^1\Pi$; term $\wedge^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-010-01 (silicon-monoxide-neutral): NIST H $\wedge^1\Sigma^+$ ($4p\sigma$); term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-011-01 (silicon-monoxide-neutral): NIST G $\wedge^1\Pi$; term $\wedge^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-012-01 (silicon-monoxide-neutral): NIST F $\wedge^1\Sigma^+$ ($4s\sigma$); term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-013-01 (silicon-monoxide-neutral): NIST g $\wedge^3\Sigma^+$ ($4s\sigma$); term $\wedge^3\Sigma^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-014-01 (silicon-monoxide-neutral): NIST f $\wedge^3\Pi_i$; term $\wedge^3\Pi_i$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind i; source glyph 0 denotes absence False; pass True

- C10097286-symmetry-015-01 (silicon-monoxide-neutral): NIST c $\wedge^3\Sigma^+$; term $\wedge^3\Sigma^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-016-01 (silicon-monoxide-neutral): NIST E $\wedge^1\Sigma^+$; term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-017-01 (silicon-monoxide-neutral): NIST A $\wedge^1\Pi$; term $\wedge^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-018-01 (silicon-monoxide-neutral): NIST D $\wedge^1\Delta$; term $\wedge^1\Delta$; spin multiplicity 1; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-019-01 (silicon-monoxide-neutral): NIST C $\wedge^1\Sigma^-$; term $\wedge^1\Sigma^-$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-020-01 (silicon-monoxide-neutral): NIST e $\wedge^3\Sigma^-$; term $\wedge^3\Sigma^-$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-021-01 (silicon-monoxide-neutral): NIST d $\wedge^3\Delta_r$; term $\wedge^3\Delta_r$; spin multiplicity 3; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind r; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-022-01 (silicon-monoxide-neutral): NIST b $\wedge^3\Pi_2$; term $\wedge^3\Pi_2$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component positive-component-2; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-023-01 (silicon-monoxide-neutral): NIST b $\wedge^3\Pi_1$; term $\wedge^3\Pi_1$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component positive-component-1; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-024-01 (silicon-monoxide-neutral): NIST b $\wedge^3\Pi_0$; term $\wedge^3\Pi_0$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence True; pass True
- C10097286-symmetry-025-01 (silicon-monoxide-neutral): NIST a $\wedge^3\Sigma^+$; term $\wedge^3\Sigma^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10097286-symmetry-026-01 (silicon-monoxide-neutral): NIST X $\wedge^1\Sigma^+$; term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C13827322-symmetry-001-01 (sulfur-monoxide-neutral): NIST E ($\wedge^3\Pi$); term $\wedge^3\Pi$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C13827322-symmetry-002-01 (sulfur-monoxide-neutral): NIST D ($\wedge^3\Pi$); term $\wedge^3\Pi$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C13827322-symmetry-003-01 (sulfur-monoxide-neutral): NIST B $\wedge^3\Sigma^-$; term $\wedge^3\Sigma^-$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C13827322-symmetry-004-01 (sulfur-monoxide-neutral): NIST A $\wedge^3\Pi_2$; term $\wedge^3\Pi_2$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component positive-component-2; component kind absence; source glyph 0 denotes absence False; pass True

- C13827322-symmetry-005-01 (sulfur-monoxide-neutral): NIST A $\wedge^3\Pi_1$; term $\wedge^3\Pi_1$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component positive-component-1; component kind absence; source glyph 0 denotes absence False; pass True
- C13827322-symmetry-006-01 (sulfur-monoxide-neutral): NIST A $\wedge^3\Pi_0$; term $\wedge^3\Pi_0$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence True; pass True
- C13827322-symmetry-007-01 (sulfur-monoxide-neutral): NIST b $\wedge^1\Sigma^+$; term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C13827322-symmetry-008-01 (sulfur-monoxide-neutral): NIST a $\wedge^1\Delta$; term $\wedge^1\Delta$; spin multiplicity 1; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C13827322-symmetry-009-01 (sulfur-monoxide-neutral): NIST X $\wedge^3\Sigma^-$; term $\wedge^3\Sigma^-$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7664393-symmetry-001-01 (hydrogen-fluoride-neutral): NIST D $\wedge^1\Sigma^+ 1$; term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7664393-symmetry-002-01 (hydrogen-fluoride-neutral): NIST C $\wedge^1\Pi$; term $\wedge^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7664393-symmetry-003-01 (hydrogen-fluoride-neutral): NIST b $\wedge^3\Pi 2$; term $\wedge^3\Pi$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7664393-symmetry-004-01 (hydrogen-fluoride-neutral): NIST B $\wedge^1\Sigma^+$; term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7664393-symmetry-005-01 (hydrogen-fluoride-neutral): NIST X $\wedge^1\Sigma^+$; term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7647010-symmetry-001-01 (hydrogen-chloride-neutral): NIST M ($\wedge^1\Sigma^+$); term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7647010-symmetry-002-01 (hydrogen-chloride-neutral): NIST L ($\wedge^1\Sigma^+, \wedge^1\Pi$); term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7647010-symmetry-002-02 (hydrogen-chloride-neutral): NIST L ($\wedge^1\Sigma^+, \wedge^1\Pi$); term $\wedge^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7647010-symmetry-003-01 (hydrogen-chloride-neutral): NIST K $\wedge^1\Pi$; term $\wedge^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7647010-symmetry-004-01 (hydrogen-chloride-neutral): NIST H $\wedge^1\Sigma^+$; term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7647010-symmetry-005-01 (hydrogen-chloride-neutral): NIST E $\wedge^1\Sigma^+$; term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True

- C7647010-symmetry-006-01 (hydrogen-chloride-neutral): NIST $g (^3\Sigma^-)1$; term $^3\Sigma^-$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7647010-symmetry-007-01 (hydrogen-chloride-neutral): NIST $f_1 ^3\Delta_1$; term $^3\Delta_1$; spin multiplicity 3; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component positive-component-1; component kind absence; source glyph 0 denotes absence False; pass True
- C7647010-symmetry-008-01 (hydrogen-chloride-neutral): NIST $D ^1\Pi$; term $^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7647010-symmetry-009-01 (hydrogen-chloride-neutral): NIST $d_0 ^3\Pi_0$; term $^3\Pi_0$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence True; pass True
- C7647010-symmetry-010-01 (hydrogen-chloride-neutral): NIST $f_2 ^3\Delta_2$; term $^3\Delta_2$; spin multiplicity 3; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component positive-component-2; component kind absence; source glyph 0 denotes absence False; pass True
- C7647010-symmetry-011-01 (hydrogen-chloride-neutral): NIST $f_3 ^3\Delta_3$; term $^3\Delta_3$; spin multiplicity 3; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component positive-component-3; component kind absence; source glyph 0 denotes absence False; pass True
- C7647010-symmetry-012-01 (hydrogen-chloride-neutral): NIST $d_1 ^3\Pi_1$; term $^3\Pi_1$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component positive-component-1; component kind absence; source glyph 0 denotes absence False; pass True
- C7647010-symmetry-013-01 (hydrogen-chloride-neutral): NIST $d_2 ^3\Pi_2$; term $^3\Pi_2$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component positive-component-2; component kind absence; source glyph 0 denotes absence False; pass True
- C7647010-symmetry-014-01 (hydrogen-chloride-neutral): NIST $C ^1\Pi$; term $^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7647010-symmetry-015-01 (hydrogen-chloride-neutral): NIST $V ^1\Sigma^+$; term $^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7647010-symmetry-016-01 (hydrogen-chloride-neutral): NIST $V ^1\Sigma^+ 21$; term $^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7647010-symmetry-017-01 (hydrogen-chloride-neutral): NIST $b_0 ^3\Pi_0$; term $^3\Pi_0$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence True; pass True
- C7647010-symmetry-018-01 (hydrogen-chloride-neutral): NIST $b_1 ^3\Pi_1$; term $^3\Pi_1$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component positive-component-1; component kind absence; source glyph 0 denotes absence False; pass True
- C7647010-symmetry-019-01 (hydrogen-chloride-neutral): NIST $b_2 ^3\Pi_2$; term $^3\Pi_2$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component positive-component-2; component kind absence; source glyph 0 denotes absence False; pass True
- C7647010-symmetry-020-01 (hydrogen-chloride-neutral): NIST $A (^1\Pi) 24 25$; term $^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7647010-symmetry-021-01 (hydrogen-chloride-neutral): NIST $X ^1\Sigma^+$; term $^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True

- C10035106-symmetry-001-01 (hydrogen-bromide-neutral): NIST M ($\wedge^1\Sigma^+$); term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10035106-symmetry-002-01 (hydrogen-bromide-neutral): NIST L ($\wedge^1\Sigma^+$, 1Π); term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10035106-symmetry-003-01 (hydrogen-bromide-neutral): NIST g ($\wedge^3\Sigma^-$) 0^+ ; term $\wedge^3\Sigma^-$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10035106-symmetry-004-01 (hydrogen-bromide-neutral): NIST F $\wedge^1\Delta$; term $\wedge^1\Delta$; spin multiplicity 1; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10035106-symmetry-005-01 (hydrogen-bromide-neutral): NIST f_1 $\wedge^3\Delta_1$; term $\wedge^3\Delta_1$; spin multiplicity 3; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component positive-component-1; component kind absence; source glyph 0 denotes absence False; pass True
- C10035106-symmetry-006-01 (hydrogen-bromide-neutral): NIST D $\wedge^1\Pi$; term $\wedge^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10035106-symmetry-007-01 (hydrogen-bromide-neutral): NIST d_0 $\wedge^3\Pi_0$; term $\wedge^3\Pi_0$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence True; pass True
- C10035106-symmetry-008-01 (hydrogen-bromide-neutral): NIST E ($\wedge^3\Sigma^+$) 0^+ ; term $\wedge^3\Sigma^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10035106-symmetry-009-01 (hydrogen-bromide-neutral): NIST V $\wedge^1\Sigma^+$; term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10035106-symmetry-010-01 (hydrogen-bromide-neutral): NIST f_2 $\wedge^3\Delta_2$; term $\wedge^3\Delta_2$; spin multiplicity 3; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component positive-component-2; component kind absence; source glyph 0 denotes absence False; pass True
- C10035106-symmetry-011-01 (hydrogen-bromide-neutral): NIST f_3 $\wedge^3\Delta_3$; term $\wedge^3\Delta_3$; spin multiplicity 3; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component positive-component-3; component kind absence; source glyph 0 denotes absence False; pass True
- C10035106-symmetry-012-01 (hydrogen-bromide-neutral): NIST e $\wedge^3\Sigma^+$; term $\wedge^3\Sigma^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10035106-symmetry-013-01 (hydrogen-bromide-neutral): NIST d_1 $\wedge^3\Pi_1$; term $\wedge^3\Pi_1$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component positive-component-1; component kind absence; source glyph 0 denotes absence False; pass True
- C10035106-symmetry-014-01 (hydrogen-bromide-neutral): NIST d_2 $\wedge^3\Pi_2$; term $\wedge^3\Pi_2$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component positive-component-2; component kind absence; source glyph 0 denotes absence False; pass True
- C10035106-symmetry-015-01 (hydrogen-bromide-neutral): NIST C $\wedge^1\Pi$; term $\wedge^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10035106-symmetry-016-01 (hydrogen-bromide-neutral): NIST b_0 $\wedge^3\Pi_0$ 0^+ ; term $\wedge^3\Pi_0$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence True; pass True

- C10035106-symmetry-017-01 (hydrogen-bromide-neutral): NIST $b_0 \wedge 3\Pi_0 0^-$; term $\wedge 3\Pi_0$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence True; pass True
- C10035106-symmetry-018-01 (hydrogen-bromide-neutral): NIST $b_1 \wedge 3\Pi_1$; term $\wedge 3\Pi_1$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component positive-component-1; component kind absence; source glyph 0 denotes absence False; pass True
- C10035106-symmetry-019-01 (hydrogen-bromide-neutral): NIST $b_2 \wedge 3\Pi_2$; term $\wedge 3\Pi_2$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component positive-component-2; component kind absence; source glyph 0 denotes absence False; pass True
- C10035106-symmetry-020-01 (hydrogen-bromide-neutral): NIST A ($\wedge 1\Pi$) 28; term $\wedge 1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C10035106-symmetry-021-01 (hydrogen-bromide-neutral): NIST X ($\wedge 1\Sigma^+$); term $\wedge 1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7553562-symmetry-001-01 (iodine-neutral): NIST F ($\wedge 1\Sigma_u^+$) 6; term $\wedge 1\Sigma_u^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7553562-symmetry-002-01 (iodine-neutral): NIST G' ($\wedge 3\Pi_{1g}$) 9; term $\wedge 3\Pi_{1g}$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion g; reflection absence; axis component positive-component-1; component kind absence; source glyph 0 denotes absence False; pass True
- C7553562-symmetry-003-01 (iodine-neutral): NIST E ($\wedge 3\Pi_o + g$) 9; term $\wedge 3\Pi_o + g$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion g; reflection plus-fibre; axis component absence; component kind o; source glyph 0 denotes absence False; pass True
- C7553562-symmetry-004-01 (iodine-neutral): NIST D ($\wedge 1\Sigma_u^+$) 13; term $\wedge 1\Sigma_u^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C7553562-symmetry-005-01 (iodine-neutral): NIST G ($\wedge 3\Pi_{2g}$) 9; term $\wedge 3\Pi_{2g}$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion g; reflection absence; axis component positive-component-2; component kind absence; source glyph 0 denotes absence False; pass True
- C7553562-symmetry-006-01 (iodine-neutral): NIST C ($\wedge 3\Sigma_{1u}^+$) 13; term $\wedge 3\Sigma_{1u}^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion u; reflection plus-fibre; axis component positive-component-1; component kind absence; source glyph 0 denotes absence False; pass True
- C7553562-symmetry-007-01 (iodine-neutral): NIST B'' ($\wedge 1\Pi_{1u}$) 19; term $\wedge 1\Pi_{1u}$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion u; reflection absence; axis component positive-component-1; component kind absence; source glyph 0 denotes absence False; pass True
- C7553562-symmetry-008-01 (iodine-neutral): NIST B' ($\wedge 3\Pi_{0-u}$) 19; term $\wedge 3\Pi_{0-u}$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence True; pass True
- C7553562-symmetry-009-01 (iodine-neutral): NIST B ($\wedge 3\Pi_{0+u}$) 19; term $\wedge 3\Pi_{0+u}$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence True; pass True
- C7553562-symmetry-010-01 (iodine-neutral): NIST A ($\wedge 3\Pi_{1u}$) 19; term $\wedge 3\Pi_{1u}$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component positive-component-1; component kind absence; source glyph 0 denotes absence False; pass True
- C7553562-symmetry-011-01 (iodine-neutral): NIST A' ($\wedge 3\Pi_{2u}$) 19; term $\wedge 3\Pi_{2u}$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion u; reflection absence; axis component positive-component-2; component kind absence; source glyph 0 denotes absence False; pass True

- C7553562-symmetry-012-01 (iodine-neutral): NIST X $^1\Sigma_g^+$; term $^1\Sigma_g^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C12184906-symmetry-001-01 (hydrogen-cation): NIST C $^2\Pi_u$ $2p\pi$; term $^2\Pi_u$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C12184906-symmetry-002-01 (hydrogen-cation): NIST B $^2\Sigma_g^+ 3d\sigma$; term $^2\Sigma_g^+$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C12184906-symmetry-003-01 (hydrogen-cation): NIST A $^2\Sigma_u^+ 2p\sigma$ 3; term $^2\Sigma_u^+$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C12184906-symmetry-004-01 (hydrogen-cation): NIST X $^2\Sigma_g^+ 1s\sigma$; term $^2\Sigma_g^+$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C13966046-symmetry-001-01 (nitrogen-cation): NIST G $^2\Sigma_g^+$; term $^2\Sigma_g^+$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C13966046-symmetry-002-01 (nitrogen-cation): NIST C $^2\Sigma_u^+$; term $^2\Sigma_u^+$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C13966046-symmetry-003-01 (nitrogen-cation): NIST D $^2\Pi_g$; term $^2\Pi_g$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion g; reflection absence; axis component absence; component kind i; source glyph 0 denotes absence False; pass True
- C13966046-symmetry-004-01 (nitrogen-cation): NIST (a $^4\Sigma_u^+$); term $^4\Sigma_u^+$; spin multiplicity 4; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 4; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C13966046-symmetry-005-01 (nitrogen-cation): NIST B $^2\Sigma_u^+$; term $^2\Sigma_u^+$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion u; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C13966046-symmetry-006-01 (nitrogen-cation): NIST A $^2\Pi_{ui}$; term $^2\Pi_{ui}$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion u; reflection absence; axis component absence; component kind i; source glyph 0 denotes absence False; pass True
- C13966046-symmetry-007-01 (nitrogen-cation): NIST X $^2\Sigma_g^+$; term $^2\Sigma_g^+$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion g; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C12185078-symmetry-001-01 (oxygen-cation): NIST x_2 $^2\Sigma^-$; term $^2\Sigma^-$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion absence; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C12185078-symmetry-002-01 (oxygen-cation): NIST x_1 $^4\Sigma^-$; term $^4\Sigma^-$; spin multiplicity 4; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 4; inversion absence; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C12185078-symmetry-003-01 (oxygen-cation): NIST ($^2\Sigma_g^-$); term $^2\Sigma_g^-$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion g; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C12185078-symmetry-004-01 (oxygen-cation): NIST ($^4\Sigma_g^-$); term $^4\Sigma_g^-$; spin multiplicity 4; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 4; inversion g; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True

- C12185078-symmetry-005-01 (oxygen-cation): NIST ($\wedge^2\Sigma_u^-$); term $\wedge^2\Sigma_u^-$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion u; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C12185078-symmetry-006-01 (oxygen-cation): NIST c $\wedge^4\Sigma_u^-$; term $\wedge^4\Sigma_u^-$; spin multiplicity 4; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 4; inversion u; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C12185078-symmetry-007-01 (oxygen-cation): NIST ($\wedge^2\Pi_u$); term $\wedge^2\Pi_u$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C12185078-symmetry-008-01 (oxygen-cation): NIST B $\wedge^2\Sigma_g^-$; term $\wedge^2\Sigma_g^-$; spin multiplicity 2; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 2; inversion g; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C12185078-symmetry-009-01 (oxygen-cation): NIST D ($\wedge^2\Delta_g$); term $\wedge^2\Delta_g$; spin multiplicity 2; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C12185078-symmetry-010-01 (oxygen-cation): NIST C ($\wedge^2\Phi_u$); term $\wedge^2\Phi_u$; spin multiplicity 2; axis Φ (third-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C12185078-symmetry-011-01 (oxygen-cation): NIST b $\wedge^4\Sigma_g^-$; term $\wedge^4\Sigma_g^-$; spin multiplicity 4; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 4; inversion g; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C12185078-symmetry-012-01 (oxygen-cation): NIST b' ($\wedge^4\Pi_g$); term $\wedge^4\Pi_g$; spin multiplicity 4; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 8; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C12185078-symmetry-013-01 (oxygen-cation): NIST A $\wedge^2\Pi_u$; term $\wedge^2\Pi_u$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C12185078-symmetry-014-01 (oxygen-cation): NIST a $\wedge^4\Pi_{ui}$; term $\wedge^4\Pi_{ui}$; spin multiplicity 4; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 8; inversion u; reflection absence; axis component absence; component kind i; source glyph 0 denotes absence False; pass True
- C12185078-symmetry-015-01 (oxygen-cation): NIST X $\wedge^2\Pi_g$; term $\wedge^2\Pi_g$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion g; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C14452938-symmetry-001-01 (nitric-oxide-cation): NIST $\wedge^1\Pi$; term $\wedge^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C14452938-symmetry-002-01 (nitric-oxide-cation): NIST $\wedge^3\Pi$; term $\wedge^3\Pi$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C14452938-symmetry-003-01 (nitric-oxide-cation): NIST $\wedge^1\Pi$; term $\wedge^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C14452938-symmetry-004-01 (nitric-oxide-cation): NIST $\wedge^3\Pi$; term $\wedge^3\Pi$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C14452938-symmetry-005-01 (nitric-oxide-cation): NIST $\wedge^1\Pi$; term $\wedge^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True

- C14452938-symmetry-006-01 (nitric-oxide-cation): NIST $\wedge^3\Pi$; term $\wedge^3\Pi$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C14452938-symmetry-007-01 (nitric-oxide-cation): NIST (B' $\wedge^1\Sigma^+$); term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C14452938-symmetry-008-01 (nitric-oxide-cation): NIST c $\wedge^3\Pi$; term $\wedge^3\Pi$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C14452938-symmetry-009-01 (nitric-oxide-cation): NIST A $\wedge^1\Pi$; term $\wedge^1\Pi$; spin multiplicity 1; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C14452938-symmetry-010-01 (nitric-oxide-cation): NIST W $\wedge^1\Delta$; term $\wedge^1\Delta$; spin multiplicity 1; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C14452938-symmetry-011-01 (nitric-oxide-cation): NIST A' $\wedge^1\Sigma^+$; term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C14452938-symmetry-012-01 (nitric-oxide-cation): NIST b' $\wedge^3\Sigma^-$; term $\wedge^3\Sigma^-$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C14452938-symmetry-013-01 (nitric-oxide-cation): NIST w $\wedge^3\Delta$; term $\wedge^3\Delta$; spin multiplicity 3; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C14452938-symmetry-014-01 (nitric-oxide-cation): NIST b $\wedge^3\Pi_p$; term $\wedge^3\Pi_p$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind p; source glyph 0 denotes absence False; pass True
- C14452938-symmetry-015-01 (nitric-oxide-cation): NIST a $\wedge^3\Sigma^+$; term $\wedge^3\Sigma^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C14452938-symmetry-016-01 (nitric-oxide-cation): NIST X $\wedge^1\Sigma^+$; term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C11062774-symmetry-001-01 (oxygen-anion): NIST a ($\wedge^4\Pi_u$); term $\wedge^4\Pi_u$; spin multiplicity 4; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 8; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C11062774-symmetry-002-01 (oxygen-anion): NIST A ($\wedge^2\Pi_u$); term $\wedge^2\Pi_u$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion u; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C11062774-symmetry-003-01 (oxygen-anion): NIST X $\wedge^2\Pi_{gi}$; term $\wedge^2\Pi_{gi}$; spin multiplicity 2; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 4; inversion g; reflection absence; axis component absence; component kind i; source glyph 0 denotes absence False; pass True
- C14967783-symmetry-001-01 (nitric-oxide-anion): NIST ($\wedge^3\Sigma^-$); term $\wedge^3\Sigma^-$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C14967783-symmetry-002-01 (nitric-oxide-anion): NIST ($\wedge^3\Sigma^+$); term $\wedge^3\Sigma^+$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True

- C14967783-symmetry-003-01 (nitric-oxide-anion): NIST ($\wedge^3\Pi$); term $\wedge^3\Pi$; spin multiplicity 3; axis Π (first-recurrence); positive orientation count 2; forced combined degeneracy 6; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C14967783-symmetry-004-01 (nitric-oxide-anion): NIST ($\wedge^1\Sigma^+$); term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C14967783-symmetry-005-01 (nitric-oxide-anion): NIST b $\wedge^1\Sigma^+$; term $\wedge^1\Sigma^+$; spin multiplicity 1; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 1; inversion absence; reflection plus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C14967783-symmetry-006-01 (nitric-oxide-anion): NIST a $\wedge^1\Delta$; term $\wedge^1\Delta$; spin multiplicity 1; axis Δ (second-recurrence); positive orientation count 2; forced combined degeneracy 2; inversion absence; reflection absence; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- C14967783-symmetry-007-01 (nitric-oxide-anion): NIST X $\wedge^3\Sigma^-$; term $\wedge^3\Sigma^-$; spin multiplicity 3; axis Σ (structural-empty-One); positive orientation count 1; forced combined degeneracy 3; inversion absence; reflection minus-fibre; axis component absence; component kind absence; source glyph 0 denotes absence False; pass True
- complete count term_assignments: 362
- complete count inversion_labels: 170
- complete count reflection_labels: 167
- complete count positive_axis_components: 32
- complete count absent_axis_components: 330
- complete count source_zero_component_glyphs: 11
- complete count species: 22
- complete count registered_state_rows: 360
- adverse unknown_support_symbol_rejected: True
- adverse axis_invariant_wrong_orientation_rejected: True
- adverse positive_axis_without_complementary_pair_rejected: True
- adverse free_degeneracy_count_rejected: True
- adverse incomplete_equivalence_class_rejected: True
- adverse source_absence_glyph_as_numerical_value_rejected: True
- adverse omitted_assignment_rejected: True
- adverse tampered_snapshot_rejected: True
- adverse complete_external_vector_retained: True

Falsification boundary: The claim fails if equivalence merges different carriers or retained symmetry signatures; if axis-invariant support has more than one orientation; if a positive axis recurrence lacks the complementary pair; if the component census differs from positive spin multiplicity times positive orientation count; if any source absence glyph is treated as a number rather than EmptyOne; if any of 362 NIST term assignments, 170 inversion labels, 167 reflection labels, 32 positive axis-component labels, 330 absent axis-component coordinates, 11 source component-absence glyphs or their exact state inscriptions is omitted or mismatched; or if a changed label, free degeneracy, incomplete class, omitted row or tampered source is accepted.

Measurement receipt: sha256:1fa3fc6f9666f8ff71385c823ee531a193f9694a557202174918ea99eb97ac7b.
 Isolation certificate: sha256:6eeca1c0a4a521e3782ec215c1ecbe48aaea57a552c16d4096e1a377b6fe649e.
 Custody certificate: sha256:d11688a045de6de1f8b99acc8d373f266087ade2700f556e586ff8c43a23d582.

Registered source descriptions:

- National Institute of Standards and Technology: <https://webbook.nist.gov/chemistry/> ()

Registered target identities:

- [illegible]

- C1333740-symmetry-038-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C1333740-symmetry-039-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C1333740-symmetry-040-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C1333740-symmetry-041-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C1333740-symmetry-042-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C7727379-symmetry-028-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025

- [illegible]

- C7782447-symmetry-027-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7782414-symmetry-001-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C10102439-symmetry-022-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10102439-symmetry-023-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10102439-symmetry-024-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10102439-symmetry-025-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10102439-symmetry-026-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10102439-symmetry-027-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10102439-symmetry-028-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10102439-symmetry-029-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025

- C10102439-symmetry-030-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C3352576-symmetry-001-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C3352576-symmetry-002-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C3352576-symmetry-003-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C3352576-symmetry-004-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C3352576-symmetry-005-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C3352576-symmetry-006-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13774920-symmetry-001-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13774920-symmetry-002-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13774920-symmetry-003-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13774920-symmetry-004-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13774920-symmetry-005-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13774920-symmetry-006-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C3315375-symmetry-001-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C3315375-symmetry-002-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C3315375-symmetry-003-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C3315375-symmetry-004-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C3315375-symmetry-005-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C3315375-symmetry-006-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C3315375-symmetry-007-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C3315375-symmetry-008-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7782505-symmetry-001-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7782505-symmetry-002-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7782505-symmetry-003-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7782505-symmetry-004-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7782505-symmetry-005-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-001-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-002-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-003-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-004-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-005-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-006-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-007-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-008-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-009-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-010-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-011-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-012-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025

- C10097286-symmetry-013-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-014-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-015-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-016-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-017-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-018-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-019-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-020-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-021-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-022-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-023-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-024-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-025-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10097286-symmetry-026-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13827322-symmetry-001-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13827322-symmetry-002-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13827322-symmetry-003-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13827322-symmetry-004-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13827322-symmetry-005-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13827322-symmetry-006-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13827322-symmetry-007-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13827322-symmetry-008-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13827322-symmetry-009-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7664393-symmetry-001-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7664393-symmetry-002-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7664393-symmetry-003-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7664393-symmetry-004-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7664393-symmetry-005-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7647010-symmetry-001-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7647010-symmetry-002-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7647010-symmetry-002-02 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7647010-symmetry-003-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7647010-symmetry-004-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7647010-symmetry-005-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7647010-symmetry-006-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7647010-symmetry-007-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7647010-symmetry-008-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7647010-symmetry-009-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025

- C7647010-symmetry-010-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7647010-symmetry-011-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C7647010-symmetry-013-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7647010-symmetry-014-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C7647010-symmetry-016-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7647010-symmetry-017-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7647010-symmetry-018-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7647010-symmetry-019-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7647010-symmetry-020-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7647010-symmetry-021-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-001-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-002-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-003-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-004-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-005-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-006-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-007-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-008-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-009-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-010-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-011-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-012-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-013-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-014-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-015-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-016-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-017-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-018-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-019-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-020-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C10035106-symmetry-021-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7553562-symmetry-001-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7553562-symmetry-002-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7553562-symmetry-003-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7553562-symmetry-004-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7553562-symmetry-005-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025

- C7553562-symmetry-006-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7553562-symmetry-007-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7553562-symmetry-008-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7553562-symmetry-009-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7553562-symmetry-010-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7553562-symmetry-011-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C7553562-symmetry-012-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12184906-symmetry-001-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12184906-symmetry-002-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12184906-symmetry-003-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12184906-symmetry-004-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13966046-symmetry-001-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13966046-symmetry-002-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13966046-symmetry-003-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13966046-symmetry-004-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13966046-symmetry-005-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13966046-symmetry-006-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C13966046-symmetry-007-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12185078-symmetry-001-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12185078-symmetry-002-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12185078-symmetry-003-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12185078-symmetry-004-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12185078-symmetry-005-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12185078-symmetry-006-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12185078-symmetry-007-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12185078-symmetry-008-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12185078-symmetry-009-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12185078-symmetry-010-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12185078-symmetry-011-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12185078-symmetry-012-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12185078-symmetry-013-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12185078-symmetry-014-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C12185078-symmetry-015-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C14452938-symmetry-001-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C14452938-symmetry-002-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C14452938-symmetry-003-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C14452938-symmetry-004-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C14452938-symmetry-005-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025

- C14452938-symmetry-006-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C14452938-symmetry-007-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C14452938-symmetry-008-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C14452938-symmetry-013-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C14452938-symmetry-014-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C14452938-symmetry-015-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C14452938-symmetry-016-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C11062774-symmetry-001-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
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- C11062774-symmetry-003-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C14967783-symmetry-001-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C14967783-symmetry-002-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C14967783-symmetry-003-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C14967783-symmetry-004-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C14967783-symmetry-005-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C14967783-symmetry-006-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- C14967783-symmetry-007-01 from NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025

Meaning. The field-wide extension derives molecular electronic identity, state composition, occupancy, ordering, symmetry, correlation, transitions, measurement and operational classical-quantum correspondence from the same exact carrier laws. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; glyph 0 is source/interface notation for absence only
- no negative, irrational, imaginary, floating or continuum proof value
- no imported point group, character table or conventional degeneracy formula in candidate generation
- no measured NIST term or symmetry label before prediction seal
- no asserted free degeneracy count or species exception
- no omitted or normalized-away term assignment
- source component glyph 0 normalizes only to structural EmptyOne

Immutable identities. Source manifest

sha256:4714bd76347a8394545609cfff2a714a068235ba76e763365f8de3603f26e3047; derivation seal sha256:0c5cc5322c05ee6a565ce23261ae41a63d0ba0d4bee59807f3aa0f31f35270ec; engine receipt sha256:3fbf5236f491ba585d6d0baf70801d1d92db65a7b258abb3c0d1b525da3e3fd8 at receipts/engine/model_admitted/SFT-CHEM-STATE-SYMMETRY-DEGENERACY-005-3fbf5236f491ba58.json; empirical hash sha256:08a4ca47c7fdc250d7909810fff64489cbd507d473eb1ea3c7697dedacd589cb5; measurement receipt sha256:1fa3fc6f9666f8ff71385c823ee531a193f9694a557202174918ea99eb97ac7b.

101. Molecular exclusion and complementary electron-pair exchange organization

Claim identity: SFT-CHEM-MOLECULAR-EXCLUSION-EXCHANGE-006

Question and theorem. For every identical two-electron molecular pair, the total exchange word is alternating. The complete two-label spin census forces a positive One-width alternating spin sector with preserving spatial support and a positive three-width preserving spin sector with alternating spatial support. Same-cell pairing is admitted only in the preserving spatial sector; the complementary sectors remain distinct on the same orbital support. No signed, negative, imaginary, continuum or numerical-zero proof value is used.

one-molecule-electron-pair__exchange-equivalent-constituents__alternating-total-word__complete-one-and-three-spin-sectors__complementary-spin-spatial-fibres__spatial-exchange-controlled-occupation__complete-state-and-exchange-pair-record__pairwise-successor-with-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-SPIN-STATISTICS-CONDENSATION-TERMINAL-045
- SFT-CHEM-ELECTRON-COUNT-SPIN-002
- SFT-CHEM-ORBITAL-SUPPORT-OCCUPANCY-003
- SFT-CHEM-STATE-SYMMETRY-DEGENERACY-005

Generated grammar. Generate the literal product of carrier, constituent identity, total exchange, spin sector, complementary composition, same-cell support, complete record and extension forms. Decide all 256 forms solely by the admitted Fold indistinguishability, exclusion, spin-census, orbital-support and state-symmetry laws.

Boundary: Every molecular carrier containing an identical electron pair; both complete two-label spin-composition sectors; every retained spatial support and pairwise successor in a larger molecule. External closure uses the complete 46-state NIST molecular-hydrogen term census, both explicit same-cell pair records and all 14 same-configuration singlet/triplet observation pairs.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *one-molecule-electron-pair__exchange-equivalent-constituents__alternating-total-word__complete-one-and-three-spin-sectors__complementary-spin-spatial-fibres__spatial-exchange-controlled-occupation__complete-state-and-exchange-pair-record__pairwise-successor-with-no-extra-rule*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-molecule-electron-pair	cross-molecule-pair	A cross-molecule word does not define one molecular exchange organization.
identity	exchange-equivalent-constituents	named-distinguishable-electrons	Naming identical electrons imports a distinction absent from every observable exchange trace.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
total	alternating-total-word	free-total-exchange-label	A freely chosen total exchange class violates the admitted identical-fermion law.
spin	complete-one-and-three-spin-sectors	imported-spin-sign-function	A signed or continuum wavefunction is outside the exact Fold proof domain.
composition	complementary-spin-spatial-fibres	independent-spin-and-space-labels	Independent labels can compose to a preserving total word and violate fermionic exchange.
same_cell	spatial-exchange-controlled-occupation	selected-occupancy-cap	A selected cap is a free rule rather than an exclusion consequence.
record	complete-state-and-exchange-pair-record	selected-favourable-spin-states	Selected states cannot establish the complete molecular pair organization.
extension	pairwise-successor-with-no-extra-rule	species-or-energy-order-exception	A species lookup or fixed energy-order sign would add a target-selected rule.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : f6bb898cd13e86681765e46e9af5d1accaa8dbe8e02226b6caf60dc49a428122.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 5ca063a6907de48de36507af69d5bd86fb7c660cda6feb9f1416102deb823017.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : f152055bb51e9b41358618053d3e381e96957fc09b0cba5072dd46899aa3fa52.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : c9a8e8856169492a3a6a3540fa889572011cadba81872ccb445049c98bc6c4d1.

Independent reconstruction. Implementation hash

sha256 : 92c4fd12b26da46abb9a57a1d26649dd096e5a637d10b1209e1a840d9470daff; independent certificate sha256 : 214033b03081c081587204f552c19ba73abec2963966062f624b34358772d4e7; external-validation hash sha256 : 3527ca7a4deb118d17a7bc2fc64aa98776c64a8e7b9b8926517108c9592d4756. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025

Comparison records:

- H2-exchange-state-001: NIST $u \wedge 3\Pi_u 6\pi$; term $\wedge 3\Pi_u$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 6, 'source_family_label': 'p', 'axis_support_symbol': ' π ', 'positive_occupancy_count': 1}; energy inscription [123488._0] 3; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-002: NIST $t \wedge 3\Sigma_u^+ 5\sigma$; term $\wedge 3\Sigma_u^+$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 5, 'source_family_label': 'f', 'axis_support_symbol': ' σ ', 'positive_occupancy_count': 1}; energy inscription (121292) 5; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-003: NIST $q \wedge 3\Sigma_g^+ 5d\sigma$; term $\wedge 3\Sigma_g^+$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 5, 'source_family_label': 'd', 'axis_support_symbol': ' σ ', 'positive_occupancy_count': 1}; energy inscription (121295) 5; source glyph 0 is absence baseline False; pass True

- H2-exchange-state-004: NIST $n\ ^3\Pi_u\ 5p\pi$; term $^3\Pi_u$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 5, 'source_family_label': 'p', 'axis_support_symbol': ' π ', 'positive_occupancy_count': 1}; energy inscription 120952._9; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-005: NIST $m\ ^3\Sigma_u^+ 4f\sigma$; term $^3\Sigma_u^+$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 4, 'source_family_label': 'f', 'axis_support_symbol': ' σ ', 'positive_occupancy_count': 1}; energy inscription (119317) 8; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-006: NIST $s\ ^3\Delta_g 4d\delta$; term $^3\Delta_g$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 4, 'source_family_label': 'd', 'axis_support_symbol': ' δ ', 'positive_occupancy_count': 1}; energy inscription 118875._2; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-007: NIST $r\ ^3\Pi_g 4d\pi$; term $^3\Pi_g$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 4, 'source_family_label': 'd', 'axis_support_symbol': ' π ', 'positive_occupancy_count': 1}; energy inscription 118613._7; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-008: NIST $p\ ^3\Sigma_g^+ 4d\sigma$; term $^3\Sigma_g^+$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 4, 'source_family_label': 'd', 'axis_support_symbol': ' σ ', 'positive_occupancy_count': 1}; energy inscription 118509._8; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-009: NIST $v\ (^3\Pi_g)$; term $^3\Pi_g$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration absence; energy inscription (118330) 15; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-010: NIST $k\ ^3\Pi_u 4p\pi$; term $^3\Pi_u$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 4, 'source_family_label': 'p', 'axis_support_symbol': ' π ', 'positive_occupancy_count': 1}; energy inscription 118366._2 16; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-011: NIST $f\ ^3\Sigma_u^+ 4p\sigma$; term $^3\Sigma_u^+$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 4, 'source_family_label': 'p', 'axis_support_symbol': ' σ ', 'positive_occupancy_count': 1}; energy inscription (116705); source glyph 0 is absence baseline False; pass True
- H2-exchange-state-012: NIST $o\ ^3\Sigma_u^+$; term $^3\Sigma_u^+$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration absence; energy inscription (114234) 20; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-013: NIST $l\ ^3\Pi_u$; term $^3\Pi_u$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration absence; energy inscription 113825 21; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-014: NIST $j\ ^3\Delta_g 3d\delta$; term $^3\Delta_g$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 3, 'source_family_label': 'd', 'axis_support_symbol': ' δ ', 'positive_occupancy_count': 1}; energy inscription (113533); source glyph 0 is absence baseline False; pass True
- H2-exchange-state-015: NIST $i\ ^3\Pi_g 3d\pi$; term $^3\Pi_g$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 3, 'source_family_label': 'd', 'axis_support_symbol': ' π ', 'positive_occupancy_count': 1}; energy inscription (113132); source glyph 0 is absence baseline False; pass True
- H2-exchange-state-016: NIST $h\ ^3\Sigma_g^+ 3s\sigma$; term $^3\Sigma_g^+$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 3, 'source_family_label': 's', 'axis_support_symbol': ' σ ', 'positive_occupancy_count': 1}; energy inscription (112913); source glyph 0 is absence baseline False; pass True

- H2-exchange-state-017: NIST $g \ ^3\Sigma_g^+ 3d\sigma$; term $^3\Sigma_g^+$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 3, 'source_family_label': 'd', 'axis_support_symbol': ' σ ', 'positive_occupancy_count': 1}; energy inscription 112854._4; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-018: NIST $d \ ^3\Pi_u 3p\pi$; term $^3\Pi_u$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 3, 'source_family_label': 'p', 'axis_support_symbol': ' π ', 'positive_occupancy_count': 1}; energy inscription 112700._3 32; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-019: NIST $e \ ^3\Sigma_u^+ 3p\sigma$; term $^3\Sigma_u^+$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 3, 'source_family_label': 'p', 'axis_support_symbol': ' σ ', 'positive_occupancy_count': 1}; energy inscription 107774._7; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-020: NIST $a \ ^3\Sigma_g^+ 2s\sigma$; term $^3\Sigma_g^+$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 2, 'source_family_label': 's', 'axis_support_symbol': ' σ ', 'positive_occupancy_count': 1}; energy inscription 95936._1 36; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-021: NIST $c \ ^3\Pi_u 3p\pi$; term $^3\Pi_u$; positive spin multiplicity 3; spin exchange preserving-exchange; spatial exchange alternating-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 3, 'source_family_label': 'p', 'axis_support_symbol': ' π ', 'positive_occupancy_count': 1}; energy inscription 95838._5 40; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-022: NIST $D'' \ ^1\Pi_u 5p\pi$; term $^1\Pi_u$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 5, 'source_family_label': 'p', 'axis_support_symbol': ' π ', 'positive_occupancy_count': 1}; energy inscription 121211.0 49; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-023: NIST $D' \ ^1\Pi_u 4p\pi$; term $^1\Pi_u$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 4, 'source_family_label': 'p', 'axis_support_symbol': ' π ', 'positive_occupancy_count': 1}; energy inscription 118865.3 49; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-024: NIST $S \ ^1\Delta_g 4d\delta$; term $^1\Delta_g$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 4, 'source_family_label': 'd', 'axis_support_symbol': ' δ ', 'positive_occupancy_count': 1}; energy inscription [119893] 53 3; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-025: NIST $O \ ^1\Sigma^+ 4s\sigma$; term $^1\Sigma^+$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 4, 'source_family_label': 's', 'axis_support_symbol': ' σ ', 'positive_occupancy_count': 1}; energy inscription [(119870)] 55 3; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-026: NIST $R \ ^1\Pi_g 4d\pi$; term $^1\Pi_g$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 4, 'source_family_label': 'd', 'axis_support_symbol': ' π ', 'positive_occupancy_count': 1}; energy inscription (118688) 57; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-027: NIST $P \ ^1\Sigma_g^+ 4d\sigma$; term $^1\Sigma_g^+$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 4, 'source_family_label': 'd', 'axis_support_symbol': ' σ ', 'positive_occupancy_count': 1}; energy inscription [119531] 59 3; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-028: NIST $T \ ^1\Sigma^+$; term $^1\Sigma^+$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration absence; energy inscription [119512.6] 61 3; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-029: NIST $B'' \ ^1\Sigma_u^+ 4p\sigma$; term $^1\Sigma_u^+$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 4, 'source_family_label': 'p', 'axis_support_symbol': ' σ ', 'positive_occupancy_count': 1}; energy inscription [119512.6] 61 3; source glyph 0 is absence baseline False; pass True

- 1}); energy inscription 117984.5; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-030: NIST N $\Lambda^1\Sigma_g^+$; term $\Lambda^1\Sigma_g^+$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration absence; energy inscription (116287) 65; source glyph 0 is absence baseline False; pass True
 - H2-exchange-state-031: NIST U ($\Lambda^1\Sigma_g^+$); term $\Lambda^1\Sigma_g^+$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration absence; energy inscription [116707.7] 65 66; source glyph 0 is absence baseline False; pass True
 - H2-exchange-state-032: NIST M $\Lambda^1\Sigma_g^+$; term $\Lambda^1\Sigma_g^+$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration absence; energy inscription (114485) 65; source glyph 0 is absence baseline False; pass True
 - H2-exchange-state-033: NIST L $\Lambda^1\Sigma_g^+$; term $\Lambda^1\Sigma_g^+$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration absence; energy inscription (114520) 65; source glyph 0 is absence baseline False; pass True
 - H2-exchange-state-034: NIST H $\Lambda^1\Sigma_g^+ 3s\sigma$; term $\Lambda^1\Sigma_g^+$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 3, 'source_family_label': 's', 'axis_support_symbol': ' σ ', 'positive_occupancy_count': 1}; energy inscription 113899 69; source glyph 0 is absence baseline False; pass True
 - H2-exchange-state-035: NIST D $\Lambda^1\Pi_u 3p\pi$; term $\Lambda^1\Pi_u$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 3, 'source_family_label': 'p', 'axis_support_symbol': ' π ', 'positive_occupancy_count': 1}; energy inscription 113888.7; source glyph 0 is absence baseline False; pass True
 - H2-exchange-state-036: NIST J $\Lambda^1\Delta_g 3d\delta$; term $\Lambda^1\Delta_g$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 3, 'source_family_label': 'd', 'axis_support_symbol': ' δ ', 'positive_occupancy_count': 1}; energy inscription (113550); source glyph 0 is absence baseline False; pass True
 - H2-exchange-state-037: NIST I $\Lambda^1\Pi_g 3d\pi$; term $\Lambda^1\Pi_g$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 3, 'source_family_label': 'd', 'axis_support_symbol': ' π ', 'positive_occupancy_count': 1}; energy inscription (113142) 80; source glyph 0 is absence baseline False; pass True
 - H2-exchange-state-038: NIST G $\Lambda^1\Sigma_g^+ 3d\sigma$; term $\Lambda^1\Sigma_g^+$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 3, 'source_family_label': 'd', 'axis_support_symbol': ' σ ', 'positive_occupancy_count': 1}; energy inscription 112834 83; source glyph 0 is absence baseline False; pass True
 - H2-exchange-state-039: NIST K ($\Lambda^1\Sigma_g^+$); term $\Lambda^1\Sigma_g^+$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration absence; energy inscription (112669) 88; source glyph 0 is absence baseline False; pass True
 - H2-exchange-state-040: NIST Q ($\Lambda^1\Pi_g$); term $\Lambda^1\Pi_g$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration absence; energy inscription (113163) 89; source glyph 0 is absence baseline False; pass True
 - H2-exchange-state-041: NIST B' $\Lambda^1\Sigma_u^+ 3p\sigma$; term $\Lambda^1\Sigma_u^+$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 3, 'source_family_label': 'p', 'axis_support_symbol': ' σ ', 'positive_occupancy_count': 1}; energy inscription 111642_8 90; source glyph 0 is absence baseline False; pass True
 - H2-exchange-state-042: NIST F $\Lambda^1\Sigma_g^+ 2p\sigma^2$; term $\Lambda^1\Sigma_g^+$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation same-cell-pair-recorded; configuration {'positive_radial_recurrence': 2, 'source_family_label': 'p', 'axis_support_symbol': ' σ ', 'positive_occupancy_count': 2}; energy inscription 100911 98 99; source glyph 0 is absence baseline False; pass True
 - H2-exchange-state-043: NIST E $\Lambda^1\Sigma_g^+ 2s\sigma$; term $\Lambda^1\Sigma_g^+$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair;

- configuration {'positive_radial_recurrence': 2, 'source_family_label': 's', 'axis_support_symbol': 'σ', 'positive_occupancy_count': 1}; energy inscription 100082.3 98 104; source glyph 0 is absence baseline False; pass True
- H2-exchange-state-044: NIST C $^1\Pi_u\ 2p\pi$; term $^1\Pi_u$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 2, 'source_family_label': 'p', 'axis_support_symbol': 'π', 'positive_occupancy_count': 1}; energy inscription 100089.8 95; source glyph 0 is absence baseline False; pass True
 - H2-exchange-state-045: NIST B $^1\Sigma_u^+ 2p\sigma$; term $^1\Sigma_u^+$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation no-explicit-same-cell-pair; configuration {'positive_radial_recurrence': 2, 'source_family_label': 'p', 'axis_support_symbol': 'σ', 'positive_occupancy_count': 1}; energy inscription 91700.0 110; source glyph 0 is absence baseline False; pass True
 - H2-exchange-state-046: NIST X $^1\Sigma_g^+ 1s\sigma^2$; term $^1\Sigma_g^+$; positive spin multiplicity 1; spin exchange alternating-exchange; spatial exchange preserving-exchange; total alternating-exchange; same-cell observation same-cell-pair-recorded; configuration {'positive_radial_recurrence': 1, 'source_family_label': 's', 'axis_support_symbol': 'σ', 'positive_occupancy_count': 2}; energy inscription 0; source glyph 0 is absence baseline True; pass True
 - H2-exchange-pair-001: support {'positive_radial_recurrence': 2, 'source_family_label': 's', 'axis_support_symbol': 'σ', 'positive_occupancy_count': 1}; singlet E $^1\Sigma_g^+ 2s\sigma$ at 100082.3 98 104; triplet a $^3\Sigma_g^+ 2s\sigma$ at 95936.1 36; positive measured separation 20731/5 inverse-centimetre; held order triplet-below-singlet; pass True
 - H2-exchange-pair-002: support {'positive_radial_recurrence': 3, 'source_family_label': 'd', 'axis_support_symbol': 'δ', 'positive_occupancy_count': 1}; singlet J $^1\Delta_g\ 3d\delta$ at (113550); triplet j $^3\Delta_g\ 3d\delta$ at (113533); positive measured separation 17/1 inverse-centimetre; held order triplet-below-singlet; pass True
 - H2-exchange-pair-003: support {'positive_radial_recurrence': 3, 'source_family_label': 'd', 'axis_support_symbol': 'π', 'positive_occupancy_count': 1}; singlet I $^1\Pi_g\ 3d\pi$ at (113142) 80; triplet i $^3\Pi_g\ 3d\pi$ at (113132); positive measured separation 10/1 inverse-centimetre; held order triplet-below-singlet; pass True
 - H2-exchange-pair-004: support {'positive_radial_recurrence': 3, 'source_family_label': 'd', 'axis_support_symbol': 'σ', 'positive_occupancy_count': 1}; singlet G $^1\Sigma_g^+ 3d\sigma$ at 112834 83; triplet g $^3\Sigma_g^+ 3d\sigma$ at 112854.4; positive measured separation 102/5 inverse-centimetre; held order singlet-below-triplet; pass True
 - H2-exchange-pair-005: support {'positive_radial_recurrence': 3, 'source_family_label': 'p', 'axis_support_symbol': 'π', 'positive_occupancy_count': 1}; singlet D $^1\Pi_u\ 3p\pi$ at 113888.7; triplet d $^3\Pi_u\ 3p\pi$ at 112700.3 32; positive measured separation 5942/5 inverse-centimetre; held order triplet-below-singlet; pass True
 - H2-exchange-pair-006: support {'positive_radial_recurrence': 3, 'source_family_label': 'p', 'axis_support_symbol': 'π', 'positive_occupancy_count': 1}; singlet D $^1\Pi_u\ 3p\pi$ at 113888.7; triplet c $^3\Pi_u\ 3p\pi$ at 95838.5 40; positive measured separation 90251/5 inverse-centimetre; held order triplet-below-singlet; pass True
 - H2-exchange-pair-007: support {'positive_radial_recurrence': 3, 'source_family_label': 'p', 'axis_support_symbol': 'σ', 'positive_occupancy_count': 1}; singlet B' $^1\Sigma_u^+ 3p\sigma$ at 111642.8 90; triplet e $^3\Sigma_u^+ 3p\sigma$ at 107774.7; positive measured separation 38681/10 inverse-centimetre; held order triplet-below-singlet; pass True
 - H2-exchange-pair-008: support {'positive_radial_recurrence': 3, 'source_family_label': 's', 'axis_support_symbol': 'σ', 'positive_occupancy_count': 1}; singlet H $^1\Sigma_g^+ 3s\sigma$ at 113899 69; triplet h $^3\Sigma_g^+ 3s\sigma$ at (112913); positive measured separation 986/1 inverse-centimetre; held order triplet-below-singlet; pass True
 - H2-exchange-pair-009: support {'positive_radial_recurrence': 4, 'source_family_label': 'd', 'axis_support_symbol': 'δ', 'positive_occupancy_count': 1}; singlet S $^1\Delta_g\ 4d\delta$ at [119893] 53 3; triplet s $^3\Delta_g\ 4d\delta$ at 118875.2; positive measured separation 5089/5 inverse-centimetre; held order triplet-below-singlet; pass True
 - H2-exchange-pair-010: support {'positive_radial_recurrence': 4, 'source_family_label': 'd', 'axis_support_symbol': 'π', 'positive_occupancy_count': 1}; singlet R $^1\Pi_g\ 4d\pi$ at (118688) 57; triplet r $^3\Pi_g\ 4d\pi$ at 118613.7; positive measured separation 743/10 inverse-centimetre; held order triplet-below-singlet; pass True
 - H2-exchange-pair-011: support {'positive_radial_recurrence': 4, 'source_family_label': 'd', 'axis_support_symbol': 'σ', 'positive_occupancy_count': 1}; singlet P $^1\Sigma_g^+ 4d\sigma$ at [119531] 59 3; triplet p $^3\Sigma_g^+ 4d\sigma$ at 118509.8; positive measured separation 5106/5 inverse-centimetre; held order triplet-below-singlet; pass True
 - H2-exchange-pair-012: support {'positive_radial_recurrence': 4, 'source_family_label': 'p', 'axis_support_symbol': 'π', 'positive_occupancy_count': 1}; singlet D' $^1\Pi_u\ 4p\pi$ at 118865.3 49; triplet k $^3\Pi_u\ 4p\pi$ at 118366.2 16; positive measured

separation 4991/10 inverse-centimetre; held order triplet-below-singlet; pass True

- H2-exchange-pair-013: support {'positive_radial_recurrence': 4, 'source_family_label': 'p', 'axis_support_symbol': 'σ', 'positive_occupancy_count': 1}; singlet B" $\wedge 1\Sigma_u^+ 4p\sigma$ at 117984.5; triplet f $\wedge 3\Sigma_u^+ 4p\sigma$ at (116705); positive measured separation 2559/2 inverse-centimetre; held order triplet-below-singlet; pass True
- H2-exchange-pair-014: support {'positive_radial_recurrence': 5, 'source_family_label': 'p', 'axis_support_symbol': 'π', 'positive_occupancy_count': 1}; singlet D" $\wedge 1\Pi_u 5p\pi$ at 121211.0 49; triplet n $\wedge 3\Pi_u 5p\pi$ at 120952._9; positive measured separation 2581/10 inverse-centimetre; held order triplet-below-singlet; pass True
- complete count state_rows: 46
- complete count singlet_states: 25
- complete count triplet_states: 21
- complete count explicit_same_cell_singlets: 2
- complete count same_configuration_exchange_pairs: 14
- complete count triplet_below_singlet: 13
- complete count singlet_below_triplet: 1
- complete count source_ground_absence_glyphs: 1
- adverse invalid_two-electron_multiplicity_rejected: True
- adverse same_cell_triplet_rejected: True
- adverse third_same_support_occurrence_rejected: True
- adverse nonalternating_total_rejected: True
- adverse same_fibre_product_retained_as_preserving_not_fermionic_total: True
- adverse source_absence_glyph_as_numerical_value_rejected: True
- adverse omitted_state_rejected: True
- adverse omitted_exchange_pair_rejected: True
- adverse selected_favourable_energy_order_rejected: True
- adverse changed_measured_separation_rejected: True
- adverse tampered_snapshot_rejected: True
- adverse complete_external_vector_retained: True

Falsification boundary: The claim fails if a molecular electron pair is not exchange-equivalent; if its total exchange is not alternating; if a positive One-width spin sector is not paired with preserving spatial exchange; if a positive three-width spin sector is not paired with alternating spatial exchange; if a same-cell pair is accepted in the alternating spatial sector; if any of 46 NIST H2 states, 25 singlets, 21 triplets, two explicit same-cell singlet records, 14 same-configuration exchange pairs, 13 triplet-below-singlet records or the one retained opposite-order record is omitted or altered; if any positive measured separation is replaced by a fitted value; or if source glyph 0 is treated as a number rather than absence.

Measurement receipt: sha256:a41e15bed4038ffcb075f9106a6c3b9ba4261f75e8a1c5a008e7dcbb5cb89f36.

Isolation certificate: sha256:0c3324447b0a65df99b75e4d1e7411c0e883233af1d4da471143e53dc86685a0.

Custody certificate: sha256:6ade4372ac9646035674ad9e5da94f059a0d86ab913082e043b7478f2336c5c5.

Registered source descriptions:

- National Institute of Standards and Technology: <https://webbook.nist.gov/cgi/cbook.cgi?ID=C1333740&Mask=1000> ()

Registered target identities:

- H2-exchange-state-001 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- H2-exchange-state-002 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025

- [illegible]

- H2-exchange-state-041 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- H2-exchange-state-042 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- H2-exchange-state-043 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- H2-exchange-state-044 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- H2-exchange-state-045 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- H2-exchange-state-046 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- H2-exchange-pair-001 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- H2-exchange-pair-002 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- H2-exchange-pair-003 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- H2-exchange-pair-004 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- H2-exchange-pair-005 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- H2-exchange-pair-006 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- H2-exchange-pair-007 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- H2-exchange-pair-008 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- H2-exchange-pair-009 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- H2-exchange-pair-010 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- H2-exchange-pair-011 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- H2-exchange-pair-012 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- H2-exchange-pair-013 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- H2-exchange-pair-014 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025

Meaning. The field-wide extension derives molecular electronic identity, state composition, occupancy, ordering, symmetry, correlation, transitions, measurement and operational classical-quantum correspondence from the same exact carrier laws. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; glyph 0 denotes source/interface absence only
- no negative, irrational, imaginary, floating, signed-wavefunction or continuum proof magnitude
- no imported Pauli sign rule, antisymmetrized orbital equation, exchange integral or Hund ordering premise
- no measured NIST multiplicity, occupancy, energy order or separation before prediction seal
- no asserted energy-order sign or fitted exchange splitting
- no named-electron distinction, species exception, selected state or omitted unfavorable pair
- source energy inscription 0 remains an absence-baseline record and never becomes an SFT number

Immutable identities. Source manifest

sha256:5ccfa035375635fc3975292245a688b6aa6c9be19fb4418a50f748403aaa4535; derivation seal
 sha256:133de6732b6cd4c7cb9d86d1d01983741ead212cf1bb505bee9ee97c063b0fe9; engine receipt
 sha256:6b56289c48e5b57443ae836a0b0dffad769c1a717a7d9d14e3a31e43112cb18c at receipts/engine/model_admitted/SFT-CHEM-MOLECULAR-EXCLUSION-EXCHANGE-006-6b56289c48e5b574.json; empirical
 hash sha256:4680b6cff1daf49b35007eb18060054da34c2c8abc09f4f0cb72cff6ddf4e4e3; measurement
 receipt sha256:a41e15bed4038ffcb075f9106a6c3b9ba4261f75e8a1c5a008e7dcbb5cb89f36.

102. Nonfactorizable molecular joint support and dissociation correspondence

Claim identity: SFT-CHEM-JOINT-CORRELATION-DISSOCIATION-007

Question and theorem. Electron correlation is the retained exact joint-support distinction not reconstructible from independent one-carrier marginals. At a two-centre dissociation boundary, the complete indistinguishable-pair support contains both complementary cross-centre assignments, excludes both same-centre Cartesian words, and retains the bound carrier, separated product identities and transition trace. No fitted correction coefficient, signed amplitude, numerical zero or imported correlation functional enters the law.

bound-carrier-and-product-support__complete-complementary-joint-support__nonfactorizable-held-joint-relation__held-fibre-exchange-pair__bound-to-separated-support-transition__positive-post-seal-separation-record__complete-measurement-and-provenance-vector__pairwise-successor-with-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-CHEM-ORBITAL-SUPPORT-OCCUPANCY-003
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-STATE-SYMMETRY-DEGENERACY-005
- SFT-CHEM-MOLECULAR-EXCLUSION-EXCHANGE-006

Generated grammar. Generate the literal product of carrier, support, relation, exchange, transition, energy, record and extension forms. Decide all 256 forms solely by admitted indistinguishability, exclusion, joint-state, molecular-support, state-order and exchange laws.

Boundary: Every identical molecular electron pair at a two-centre separated-product boundary and every pairwise successor in a larger finite support graph. External closure uses all nine pre-registered APS and NIST hydrogen/deuterium neutral, ionic, ground-state, excited-state and threshold dissociation records.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *bound-carrier-and-product-support__complete-complementary-joint-support__nonfactorizable-held-joint-relation__held-fibre-exchange-pair__bound-to-separated-support-transition__positive-post-seal-separation-record__complete-measurement-and-provenance-vector__pairwise-successor-with-no-extra-rule*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	bound-carrier-and-product-support	unbound-energy-only-record	An energy alone erases the bound carrier and separated product identities.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
support	complete-complementary-joint-support	independent-single-product-word	One product word omits the complementary assignment required by indistinguishability and exchange.
relation	nonfactorizable-held-joint-relation	marginals-declared-complete	The two one-carrier marginals generate four Cartesian words and cannot identify the exact two-word joint support.
exchange	held-fibre-exchange-pair	named-electron-product-assignment	Named electrons import an unobservable constituent distinction.
transition	bound-to-separated-support-transition	dissociation-as-disconnected-lookup	A lookup does not preserve the molecular-to-product transition trace.
energy	positive-post-seal-separation-record	fitted-correlation-correction	A correction coefficient selected from dissociation data is a forbidden free parameter.
record	complete-measurement-and-provenance-vector	selected-single-isotopologue	Selecting one favourable molecule cannot test the complete registered dissociation vector.
extension	pairwise-successor-with-no-extra-rule	species-specific-correlation-rule	A species exception or imported correlation functional adds an unforced model.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 6b107e26cb0a8a2105d2092418d29d5324a54f33ab63a4ffe4ff82e63fb728eb.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: bd3a3167e886996b8d05bfff74e79a0a563e369d09e804f20f01da081a13a8bcc.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 1482c618ce8dc9d84f34eae13f14abfc0010279e80b9cb1ab98f51fd5c0e9704.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 933722dd9c22deebdc46413b2e53efff1dc5bc0d9ed052731da117e38a31c88b.

Independent reconstruction. Implementation hash

sha256: 43ec2936456e900eab2b2a6a689b619bca14bda3a6559d523875f3ab40421774; independent certificate sha256: c895e72ec5834d6cc95d64f3ce6835757cac0d3b5f6a49209e2e15028db78e81; external-validation hash sha256: d2b6435182a8dfdd6ae68e998fab2ee32b9608ff1bd39102989dc3b1b0df1172. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- APS-PRA-49-2460-1994
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-DISSOCIATION

Comparison records:

- APS-HYDROGEN-DISSOCIATION-001 (H2 B-prime-to-H-1s-plus-H-2s): 118377.06 inverse-centimetre; uncertainty 0.04; kind measured_dissociation_threshold; joint 2/Cartesian 4; pass **True**
- APS-HYDROGEN-DISSOCIATION-002 (D2 B-prime-to-D-1s-plus-D-2s): 119029.72 inverse-centimetre; uncertainty 0.04; kind measured_dissociation_threshold; joint 2/Cartesian 4; pass **True**
- APS-HYDROGEN-DISSOCIATION-003 (H2 X-ground): 36118.11 inverse-centimetre; uncertainty 0.08; kind measured_ground_state_dissociation_energy; joint 2/Cartesian 4; pass **True**
- APS-HYDROGEN-DISSOCIATION-004 (D2 X-ground): 36748.38 inverse-centimetre; uncertainty 0.07; kind measured_ground_state_dissociation_energy; joint 2/Cartesian 4; pass **True**

- APS-HYDROGEN-DISSOCIATION-005 (H2-positive-ion ground): 21379.37 inverse-centimetre; uncertainty 0.08; kind derived_from_measured-neutral-and-ionization-intervals; joint 2/Cartesian 4; pass True
- APS-HYDROGEN-DISSOCIATION-006 (D2-positive-ion ground): 21711.64 inverse-centimetre; uncertainty 0.07; kind derived_from_measured-neutral-and-ionization-intervals; joint 2/Cartesian 4; pass True
- NIST-H2-DISSOCIATION-001 (H2 B-state-dissociation-limit): 118377._6 inverse-centimetre; uncertainty absence; kind compiled_observed_dissociation_limit; joint 2/Cartesian 4; pass True
- NIST-H2-DISSOCIATION-002 (H2 B-state): 28174.2 inverse-centimetre; uncertainty absence; kind compiled_dissociation_energy; joint 2/Cartesian 4; pass True
- NIST-H2-DISSOCIATION-003 (H2 X-ground): 36118.3 inverse-centimetre; uncertainty 0.5; kind measured_upper_limit; joint 2/Cartesian 4; pass True
- count records: 9
- count APS_records: 6
- count NIST_records: 3
- count direct_or_compiled: 7
- count derived_ionic: 2
- count positive_uncertainties: 7
- count absent_uncertainties: 2
- count species: 4
- adverse incomplete_one_word_support_rejected: True
- adverse factorized_four_word_support_rejected: True
- adverse same_centre_support_rejected: True
- adverse numerical_zero_rejected: True
- adverse nonpositive_energy_rejected: True
- adverse omitted_record_rejected: True
- adverse omitted_derived_ion_rejected: True
- adverse selected_newer_only_rejected: True
- adverse selected_NIST_only_rejected: True
- adverse changed_value_rejected: True
- adverse changed_uncertainty_rejected: True
- adverse tampered_snapshot_rejected: True
- adverse complete_provenance_vector_retained: True

Falsification boundary: The claim fails if independent marginals reconstruct the exact joint support; if either complementary cross-centre word is missing; if either same-centre Cartesian word is admitted; if constituent names, fitted coefficients or imported correlation models select the support; if the bound-to-separated trace loses carrier or product identity; if any of nine APS/NIST records, six APS values, three NIST values, seven direct measured/compiled records, two explicitly derived ionic records, every uncertainty or exact positive inscription is omitted or changed; or if any source absence glyph is treated as an SFT number.

Measurement receipt: sha256:ec10f5a0192c637465f037efe99c6625ae7eea755982ca1f1a3ee321754529bc.
Isolation certificate: sha256:4bcb33d3c4811ccc732140ef2a3a4bdb323b2231b6ea0c4eed601781d8363f46.
Custody certificate: sha256:c17ef9d948a90f78d87e15e45b445f69fd8a7a280ff766fde91d6d6ec6fdc9ca.

Registered source descriptions:

- American Physical Society: <https://journals.aps.org/pr/abstract/10.1103/PhysRevA.49.2460> ()

- National Institute of Standards and Technology: <https://webbook.nist.gov/cgi/cbook.cgi?ID=C1333740&Mask=1000> ()

Registered target identities:

- APS-HYDROGEN-DISSOCIATION-001 from APS-PRA-49-2460-1994
- APS-HYDROGEN-DISSOCIATION-002 from APS-PRA-49-2460-1994
- APS-HYDROGEN-DISSOCIATION-003 from APS-PRA-49-2460-1994
- APS-HYDROGEN-DISSOCIATION-004 from APS-PRA-49-2460-1994
- APS-HYDROGEN-DISSOCIATION-005 from APS-PRA-49-2460-1994
- APS-HYDROGEN-DISSOCIATION-006 from APS-PRA-49-2460-1994
- NIST-H2-DISSOCIATION-001 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-DISSOCIATION
- NIST-H2-DISSOCIATION-002 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-DISSOCIATION
- NIST-H2-DISSOCIATION-003 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-DISSOCIATION

Meaning. The field-wide extension derives molecular electronic identity, state composition, occupancy, ordering, symmetry, correlation, transitions, measurement and operational classical-quantum correspondence from the same exact carrier laws. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; glyph 0 denotes source/interface absence only
- no negative, irrational, imaginary, floating, signed-amplitude or continuum proof magnitude
- no Hartree-Fock, configuration-interaction, coupled-cluster, density-functional or correlation-energy formula imported as a premise
- no fitted correlation coefficient, empirical functional, basis correction or dissociation-energy parameter
- no APS or NIST target value, uncertainty, isotope trend or ionic value before prediction seal
- no single-product factorization presented as complete joint support
- no omission of direct, compiled, derived-ion, older-reference or uncertainty records

Immutable identities. Source manifest

sha256:589cc8f15b334d9549bef506cc8754d53810dcca3f9be8a6c152d3fe9779175; derivation seal
 sha256:af62e27a7fa2c4d238a3fd4a5eae6be9ba88806b43f2730ed321fd3811e7386f; engine receipt
 sha256:f911d86a7563eee7f6589450425a7bcdd5d19cb8e1defb782fc4ffbdaf4d0924 at receipts/engine/model_admitted/SFT-CHEM-JOINT-CORRELATION-DISSOCIATION-007-f911d86a7563eee7.json;
 empirical hash sha256:e54ade4544579311933dc1088dded50557042c0cb3438c06daa82e57338ab116;
 measurement receipt sha256:ec10f5a0192c637465f037efe99c6625ae7eea755982ca1f1a3ee321754529bc.

103. Connected multicentre and delocalized molecular support

Claim identity: SFT-CHEM-MULTICENTRE-DELOCALIZED-SUPPORT-008

Question and theorem. A delocalized molecular electron support is one retained Fold word spanning every centre of a connected generated graph. At three or more centres it is not reducible to one localized two-centre support without losing the complete support identity. Path, cycle and polyhedral incidence force ribbon, surface and volume forms without an imported bonding model.

*complete-molecular-carrier__three-or-more-generated-centres__one-complete-extended-support__connected-generate
 d-graph__ribbon-surface-volume-census__multicentre-irreducible-support__complete-authority-and-geometry-vector__
 _connected-successor-with-no-extra-rule*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-CHEM-ORBITAL-SUPPORT-OCCUPANCY-003
- SFT-CHEM-MOLECULAR-EXCLUSION-EXCHANGE-006
- SFT-CHEM-JOINT-CORRELATION-DISSOCIATION-007

Generated grammar. Generate the literal product of molecular carrier, centre count, electron support, connection, topology, reduction, external record and extension forms. Decide all 256 forms solely by admitted Fold graph, joint-support, exchange and exclusion laws.

Boundary: Every positive finite connected molecular support containing at least three retained centres, with post-seal testing against all four IUPAC delocalization records, all nine NIST neutral-diborane geometry/bond-count records and all seven NIST benzene geometry/bond-count records.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-molecular-carrier__three-or-more-generated-centres__one-complete-extended-support__connected-generated-graph__ribbon-surface-volume-census__multicentre-irreducible-support__complete-authority-and-geometry-vector__connected-successor-with-no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-molecular-carrier	isolated-bond-list	A list erases the single molecular carrier joining the support.
centres	three-or-more-generated-centres	two-centre-only-premise	A two-centre restriction excludes generated supports before enumeration.
support	one-complete-extended-support	localized-edge-identities	Separate edge identities cannot preserve one electron-support identity spanning the graph.
connection	connected-generated-graph	disconnected-centre-union	A disconnected union is not one molecular support.
topology	ribbon-surface-volume-census	imported-or-single-topology	Selecting one familiar shape omits lawful connected alternatives.
reduction	multicentre-irreducible-support	pairwise-model-declared-complete	Independent localized pairs erase the one support identity extending beyond either endpoint.
record	complete-authority-and-geometry-vector	selected-example-only	One favourable example cannot test the registered topology and geometry surface.
extension	connected-successor-with-no-extra-rule	species-specific-extra-rule	A species exception or empirical bonding model adds an unforced premise.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:0303169402091a84f1524aad24ca3dfe9b62545b190cff6ca13937ed795d7bf0`.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 987e1af098eb3115a18151a0efa89b220d009b6d95fde509b3565ab4cadb524f.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 950d1433bd761bad834e9e2399cf1dac1d7670b2f5fa441313ce0fb94f099306.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: 86b4ca67c175a3aa62169169472af5d2dcfec8cf194147ea052afddc75a9aef.

Independent reconstruction. Implementation hash

sha256: 1478fce6c7dfddc9e2f781192be965021df9534b7bc57b2225efa025d4da71ec; independent certificate sha256: fa796bc55380822e42b518cfd12897155bee8b0977a1dbf5dfc4a42af32d7f9; external-validation hash sha256: cffd7c0e46f62f1688ea6129111b57e8fc92667b629413e96ad829ff17b0bf67. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-08789-2026
- NIST-CCCBDB-SRD101-DIBORANE
- NIST-CCCBDB-SRD101-BENZENE

Comparison records:

- **IUPAC-DELOCALIZATION-DEFINITION** definition: Redistribution of electron density in a molecular entity as compared with a model assuming individual atoms in the same valence states connected by localized bonds.; exact structural-absence; structural True; pass True
- **IUPAC-DELOCALIZATION-RIBBON** ribbon-topology: Topological modes of the electron delocalization include: ribbon (one-dimensional) delocalization of $\langle \psi | \mathbf{m} \rangle$ or $\langle \psi | \mathbf{\sigma} \rangle$ or both, which is typical of linear conjugated polymers that are therefore termed one-dimensional conductors; surface delocalization of $\langle \psi | \mathbf{m} \rangle$ or $\langle \psi | \mathbf{\sigma} \rangle$ or both through an overlap of radially oriented orbitals of a cyclic molecule such as a molecule of cyclopropane or benzene; volume delocalization of $\langle \psi | \mathbf{\sigma} \rangle$ through an overlap of $\langle \psi | \mathbf{\sigma} \rangle$ directed inside a molecular polyhedron, e.g., in tetrahedrane or carboranes.; exact structural-absence; structural True; pass True
- **IUPAC-DELOCALIZATION-SURFACE** surface-topology: Topological modes of the electron delocalization include: ribbon (one-dimensional) delocalization of $\langle \psi | \mathbf{m} \rangle$ or $\langle \psi | \mathbf{\sigma} \rangle$ or both, which is typical of linear conjugated polymers that are therefore termed one-dimensional conductors; surface delocalization of $\langle \psi | \mathbf{m} \rangle$ or $\langle \psi | \mathbf{\sigma} \rangle$ or both through an overlap of radially oriented orbitals of a cyclic molecule such as a molecule of cyclopropane or benzene; volume delocalization of $\langle \psi | \mathbf{\sigma} \rangle$ through an overlap of $\langle \psi | \mathbf{\sigma} \rangle$ directed inside a molecular polyhedron, e.g., in tetrahedrane or carboranes.; exact structural-absence; structural True; pass True
- **IUPAC-DELOCALIZATION-VOLUME** volume-topology: Topological modes of the electron delocalization include: ribbon (one-dimensional) delocalization of $\langle \psi | \mathbf{m} \rangle$ or $\langle \psi | \mathbf{\sigma} \rangle$ or both, which is typical of linear conjugated polymers that are therefore termed one-dimensional conductors; surface delocalization of $\langle \psi | \mathbf{m} \rangle$ or $\langle \psi | \mathbf{\sigma} \rangle$ or both through an overlap of radially oriented orbitals of a cyclic molecule such as a molecule of cyclopropane or benzene; volume delocalization of $\langle \psi | \mathbf{\sigma} \rangle$ through an overlap of $\langle \psi | \mathbf{\sigma} \rangle$ directed inside a molecular polyhedron, e.g., in tetrahedrane or carboranes.; exact structural-absence; structural True; pass True
- **NIST-DIBORANE-POINT-GROUP** point-group: D2h; exact structural-absence; structural True; pass True
- **NIST-DIBORANE-rBB** rBB: 1.763; exact 1763/1000; structural True; pass True
- **NIST-DIBORANE-rBH-OUTER** rBH-outer: 1.200; exact 1200/1000; structural True; pass True
- **NIST-DIBORANE-rBH-BRIDGING** rBH-bridging: 1.320; exact 1320/1000; structural True; pass True

- NIST-DIBORANE-aHBH-OUTER aHBH-outer: 121; exact 121/1; structural True; pass True
- NIST-DIBORANE-aHBH-BRIDGING aHBH-bridging: 96.2; exact 962/10; structural True; pass True
- NIST-DIBORANE-aBHB aBHB: 83.8; exact 838/10; structural True; pass True
- NIST-DIBORANE-aHBH-SYMMETRY aHBH-symmetry: 109.2; exact 1092/10; structural True; pass True
- NIST-DIBORANE-HB-LINK-COUNT H-B-count: 8; exact 8/1; structural True; pass True
- NIST-BENZENE-POINT-GROUP point-group: D6h; exact structural-absence; structural True; pass True
- NIST-BENZENE-rCC rCC: 1.397; exact 1397/1000; structural True; pass True
- NIST-BENZENE-rCH rCH: 1.084; exact 1084/1000; structural True; pass True
- NIST-BENZENE-aCCC aCCC: 120; exact 120/1; structural True; pass True
- NIST-BENZENE-aHCC aHCC: 120; exact 120/1; structural True; pass True
- NIST-BENZENE-AROMATIC-LINK-COUNT C:C-count: 6; exact 6/1; structural True; pass True
- NIST-BENZENE-HC-LINK-COUNT H-C-count: 6; exact 6/1; structural True; pass True
- count records: 20
- count IUPAC: 4
- count diborane: 9
- count benzene: 7
- count positive_numeric: 14
- count categorical_or_text: 6
- adverse two_centre_rejected: True
- adverse disconnected_support_rejected: True
- adverse incomplete_support_word_rejected: True
- adverse topology_mismatch_rejected: True
- adverse numerical_zero_rejected: True
- adverse measured_bridge_distinction_retained: True
- adverse diborane_BHB_connectivity_retained: True
- adverse benzene_six_equal_aromatic_links_retained: True
- adverse omitted_record_rejected: True
- adverse selected_IUPAC_only_rejected: True
- adverse selected_NIST_only_rejected: True
- adverse changed_value_rejected: True
- adverse tampered_snapshot_rejected: True
- adverse complete_vector_retained: True
- adverse all_generated_topologies_operational: True

Falsification boundary: The claim fails if the complete support is disconnected; if fewer than three centres are relabelled multicentre; if a one-pair support retains the same complete identity as a support spanning three or more centres; if path, cycle or polyhedral incidence requires an imported chemical model; if the measured B-H-B bridge or equal six-link benzene cycle is absent; if any of four IUPAC, nine diborane or seven benzene registered rows is omitted or changed; or if an absence glyph is treated as an SFT number.

Measurement receipt: sha256:9ee41c416b6af232c0fefec4f43a22d605d03220748eb68c521f2114e2325ba1.
Isolation certificate: sha256:562bdb499409f753a97b0f150dd66fdd96b950289a6a2a8a818c325496b1fa04.

Custody certificate: sha256 : f6780262516409e7dff6f96a25f689914154c6b3e21e70a3ca3602e4024055a3.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: ()
- National Institute of Standards and Technology: <https://cccbdb.nist.gov/expgeom2x.asp?casno=19287457> ()
- National Institute of Standards and Technology: <https://cccbdb.nist.gov/expgeom2x.asp?casno=71432> ()

Registered target identities:

- IUPAC-DELOCALIZATION-DEFINITION from IUPAC-GOLD-BOOK-08789-2026
- IUPAC-DELOCALIZATION-RIBBON from IUPAC-GOLD-BOOK-08789-2026
- IUPAC-DELOCALIZATION-SURFACE from IUPAC-GOLD-BOOK-08789-2026
- IUPAC-DELOCALIZATION-VOLUME from IUPAC-GOLD-BOOK-08789-2026
- NIST-DIBORANE-POINT-GROUP from NIST-CCCBDB-SRD101-DIBORANE
- NIST-DIBORANE-rBB from NIST-CCCBDB-SRD101-DIBORANE
- NIST-DIBORANE-rBH-OUTER from NIST-CCCBDB-SRD101-DIBORANE
- NIST-DIBORANE-rBH-BRIDGING from NIST-CCCBDB-SRD101-DIBORANE
- NIST-DIBORANE-aHBH-OUTER from NIST-CCCBDB-SRD101-DIBORANE
- NIST-DIBORANE-aHBH-BRIDGING from NIST-CCCBDB-SRD101-DIBORANE
- NIST-DIBORANE-aBHB from NIST-CCCBDB-SRD101-DIBORANE
- NIST-DIBORANE-aHBH-SYMMETRY from NIST-CCCBDB-SRD101-DIBORANE
- NIST-DIBORANE-HB-LINK-COUNT from NIST-CCCBDB-SRD101-DIBORANE
- NIST-BENZENE-POINT-GROUP from NIST-CCCBDB-SRD101-BENZENE
- NIST-BENZENE-rCC from NIST-CCCBDB-SRD101-BENZENE
- NIST-BENZENE-rCH from NIST-CCCBDB-SRD101-BENZENE
- NIST-BENZENE-aCCC from NIST-CCCBDB-SRD101-BENZENE
- NIST-BENZENE-aHCC from NIST-CCCBDB-SRD101-BENZENE
- NIST-BENZENE-AROMATIC-LINK-COUNT from NIST-CCCBDB-SRD101-BENZENE
- NIST-BENZENE-HC-LINK-COUNT from NIST-CCCBDB-SRD101-BENZENE

Meaning. The field-wide extension derives molecular electronic identity, state composition, occupancy, ordering, symmetry, correlation, transitions, measurement and operational classical-quantum correspondence from the same exact carrier laws. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; glyph 0 denotes source/interface absence only
- no negative, irrational, imaginary, floating, signed-amplitude or continuum proof magnitude
- no imported valence-bond, molecular-orbital, aromaticity, resonance, Wade, Hückel or two-centre bonding model
- no measured NIST geometry, IUPAC example or species name before prediction seal
- no fitted bond length, angle, order, topology or species correction
- no selected-source or selected-row validation

Immutable identities. Source manifest

sha256 : 49b2ddfc2cec71319cc9058b8fb3778e677328d757cfd5c3e1f6c902a19df07b; derivation seal
sha256 : d4ae5d6ec21dabf9bc575d594b7fe4ed3cacb43c2d85e190c0fea7757eb68915; engine receipt

sha256:0bbf92200254061e7a22f38a66abec14bf65c28f09f4d9fe38f38d2117994444 at receipts/engine/model_admitted/SFT-CHEM-MULTICENTRE-DELOCALIZED-SUPPORT-008-0bbf92200254061e.json;
 empirical hash sha256:0349a415ecbca4766cd33bc48fb72fcc99910fca44cffcb544fa4fee2391c780;
 measurement receipt sha256:9ee41c416b6af232c0fefec4f43a22d605d03220748eb68c521f2114e2325ba1.

104. Exact molecular-state transformation and transition law

Claim identity: SFT-CHEM-MOLECULAR-STATE-TRANSITION-009

Question and theorem. A molecular-state transformation retains one carrier, distinct state endpoints, a held orientation or coupling class, and its exact observation trace. Matching endpoints compose into finite transition paths. An absent transition coordinate is structural EmptyOne, not numerical zero. Selection criteria are not imported before their separate derivation.

retained-molecular-carrier__distinct-initial-and-terminal-states__held-forward-reverse-bidirectional-or-coupled__closed-terminal-EmptyOne__complete-state-transition-trace__matching-endpoint-composition__complete-presence-coupling-absence-vector__finite-path-successor-with-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-STATE-SYMMETRY-DEGENERACY-005
- SFT-CHEM-MULTICENTRE-DELOCALIZED-SUPPORT-008

Generated grammar. Generate the literal product of carrier, endpoints, orientation, absence, trace, composition, record and extension forms. Decide all 256 forms only from admitted state-transition, observation, molecular-state and exact-composition laws.

Boundary: Every positive finite molecular-state transition record and every matching finite path successor, externally tested against all 46 NIST H2 primary state rows and all 14 continuation transition rows.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *retained-molecular-carrier__distinct-initial-and-terminal-states__held-forward-reverse-bidirectional-or-coupled__closed-terminal-EmptyOne__complete-state-transition-trace__matching-endpoint-composition__complete-presence-coupling-absence-vector__finite-path-successor-with-no-extra-rule*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	retained-molecular-carrier	transition-label-alone	A label alone erases the transforming molecular carrier.
endpoints	distinct-initial-and-terminal-states	unbound-final-answer	An answer without endpoints cannot identify a state transformation.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
orientation	held-forward-reverse-bidirectional-or-coupled	signed-or-unoriented-change	A sign or erased orientation loses the observed direction class.
absence	closed-terminal-EmptyOne	absence-treated-as-number	Numerical absence violates the Fold domain.
trace	complete-state-transition-trace	endpoint-only-lookup	Endpoints alone erase the transition record and provenance.
composition	matching-endpoint-composition	nonmatching-path-concatenation	Transitions with unequal adjacent endpoints cannot form one path.
record	complete-presence-coupling-absence-vector	selected-present-transition	Selecting only observed rows cannot test absence or coupling records.
extension	finite-path-successor-with-no-extra-rule	species-selection-rule-premise	A selection rule at this stage imports the next obligation.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 923977513dab09c297c2ea579b147c490b62ed45af08c3c8997693b0e9510ecd.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 785314864a944b6bb0b18104b2eddc6a5908488aaffa817227fc9a8d4dfd4299.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 1fa5b9169db911ff7b57070a89f490ced1c8c5df848f946dba65ea3430cc94b7.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: c24e4f9b0e53ec04923fc8571f864bfe404e4a0dcc2f4b3dcc1f1ec6496257fa.

Independent reconstruction. Implementation hash

sha256: 73f8355a3901319d4e2f4c2c615138cf1252bc545bfe812cff05c901bb43cf5d; independent certificate sha256: 109a9b8aa658c3258cc9d965f5c7ff12d487d6ef0f6065e0ad0781704a1599e6; external-validation hash sha256: 57f689941e92e706f4eaa5afc789b6e34a02c060067f9b17acc57b42d76a2073. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS

Comparison records:

- H2-transition-state-001 $u \wedge 3\Pi_u 6p\pi: u \rightarrow a$; class observed-directional-transition; band $26232.3 \cdot 4 = 262323/10$; pass True
- H2-transition-state-002 $t \wedge 3\Sigma_u^+ 5f\sigma: t \rightarrow a$; class observed-directional-transition; band $(25342) = 25342/1$; pass True
- H2-transition-state-003 $q (\wedge 3\Sigma_g^+) 5d\sigma: q \rightarrow c$; class observed-directional-transition; band $(25325) \cdot 4 = 25325/1$; pass True
- H2-transition-state-004 $n \wedge 3\Pi_u 5p\pi: n \rightarrow a$; class observed-directional-transition; band $24847.3 \cdot 4 = 248473/10$; pass True
- H2-transition-state-005 $m \wedge 3\Sigma_u^+ 4f\sigma: m \rightarrow a$; class observed-directional-transition; band $23295.1 \cdot 9 = 232951/10$; pass True
- H2-transition-state-006 $s \wedge 3\Delta_g 4d\delta: s \rightarrow c$; class observed-directional-transition; band $22949.3 \cdot 12 = 229493/10$; pass True
- H2-transition-state-007 $r \wedge 3\Pi_g 4d\pi: r \rightarrow c$; class observed-directional-transition; band $22683.2 \cdot 13 = 113416/5$; pass True
- H2-transition-state-008 $p \wedge 3\Sigma_g^+ 4d\sigma: p \rightarrow k$ 14; class observed-coupled-state-relation; band absence = structural-absence; pass True

- H2-transition-continuation-001 $p \wedge 3\Sigma_g^+ 4d\sigma: p \rightarrow c$; class observed-directional-transition; band 22586.0 4 = 22586/1; pass True
- H2-transition-state-009 $v (\wedge 3\Pi_g): v \rightarrow c$; class observed-directional-transition; band (22430) = 22430/1; pass True
- H2-transition-state-010 $k \wedge 3\Pi_u 4p\pi: k \rightarrow a$; class observed-directional-transition; band 22271.0 4 = 22271/1; pass True
- H2-transition-state-011 $f \wedge 3\Sigma_u^+ 4p\sigma: f \rightarrow a$; class observed-directional-transition; band 20526.0 19 = 20526/1; pass True
- H2-transition-state-012 $o \wedge 3\Sigma_u^+: o \rightarrow a$; class observed-directional-transition; band (18160) = 18160/1; pass True
- H2-transition-state-013 $l \wedge 3\Pi_u: l \rightarrow a$; class observed-directional-transition; band 17846 4 = 17846/1; pass True
- H2-transition-state-014 $j \wedge 3\Delta_g 3d\delta: j \leftrightarrow c$ 23 R; class observed-directional-transition; band 17633.0 24 = 17633/1; pass True
- H2-transition-state-015 $i \wedge 3\Pi_g 3d\pi: i-d$ 26; class observed-coupled-state-relation; band absence = structural-absence; pass True
- H2-transition-continuation-002 $i \wedge 3\Pi_g 3d\pi: i \rightarrow e$ R; class observed-directional-transition; band 5384.81 27 = 538481/100; pass True
- H2-transition-continuation-003 $i \wedge 3\Pi_g 3d\pi: i \leftrightarrow c$ 23 R; class observed-directional-transition; band 17185.8 24 = 85929/5; pass True
- H2-transition-state-016 $h \wedge 3\Sigma_g^+ 3s\sigma: h \rightarrow c$; class observed-directional-transition; band 16990.8 29 = 84954/5; pass True
- H2-transition-state-017 $g \wedge 3\Sigma_g^+ 3d\sigma: g \rightarrow e$ R; class observed-directional-transition; band 5116.6 = 25583/5; pass True
- H2-transition-continuation-004 $g \wedge 3\Sigma_g^+ 3d\sigma: g \leftrightarrow c$ 23; class observed-directional-transition; band 16917.6 29 = 84588/5; pass True
- H2-transition-state-018 $d \wedge 3\Pi_u 3p\pi: d \rightarrow a$ 35 R; class observed-directional-transition; band 16619.0 29 = 16619/1; pass True
- H2-transition-state-019 $e \wedge 3\Sigma_u^+ 3p\sigma: e \rightarrow a$ R; class observed-directional-transition; band 11605.6 = 58028/5; pass True
- H2-transition-state-020 $a \wedge 3\Sigma_g^+ 2s\sigma: a \rightarrow b$ 38 39; class observed-directional-transition; band absence = structural-absence; pass True
- H2-transition-continuation-005 $a \wedge 3\Sigma_g^+ 2s\sigma: (a-X)$; class observed-coupled-state-relation; band 95076.4 36 = 475382/5; pass True
- H2-transition-state-021 $c \wedge 3\Pi_u 3p\pi: (c-X)$; class observed-coupled-state-relation; band 94881.0 43 = 94881/1; pass True
- H2-transition-state-022 $D'' \wedge 1\Pi_u 5p\pi: D'' \leftarrow X$ R; class observed-directional-transition; band 120176.0 49 = 120176/1; pass True
- H2-transition-state-023 $D' \wedge 1\Pi_u 4p\pi: D' \leftarrow X$ R; class observed-directional-transition; band 117835.2 49 = 589176/5; pass True
- H2-transition-state-024 $S \wedge 1\Delta_g 4d\delta: S \rightarrow B$ V; class observed-directional-transition; band 27510 = 27510/1; pass True
- H2-transition-state-025 $O \wedge 1\Sigma^+ 4s\sigma: O \rightarrow B$ V; class observed-directional-transition; band (27487) 56 = 27487/1; pass True
- H2-transition-state-026 $R \wedge 1\Pi_g 4d\pi: (R \rightarrow C)$; class observed-directional-transition; band (18488) = 18488/1; pass True
- H2-transition-continuation-006 $R \wedge 1\Pi_g 4d\pi: R \rightarrow B$ V; class observed-directional-transition; band 27376 45 = 27376/1; pass True
- H2-transition-state-027 $P \wedge 1\Sigma_g^+ 4d\sigma: (P \rightarrow C)$; class observed-directional-transition; band 18260 49 = 18260/1; pass True
- H2-transition-continuation-007 $P \wedge 1\Sigma_g^+ 4d\sigma: P \rightarrow B$ V; class observed-directional-transition; band 27148 60 = 27148/1; pass True
- H2-transition-state-028 $T \wedge 1\Sigma^+: T \rightarrow B$ V; class observed-directional-transition; band 27130.1 = 271301/10; pass True
- H2-transition-state-029 $B'' \wedge 1\Sigma_u^+ 4p\sigma: B'' \leftarrow X$ R; class observed-directional-transition; band 116886.9 64 = 1168869/10; pass True
- H2-transition-state-030 $N \wedge 1\Sigma_g^+: N \rightarrow B$ R; class observed-directional-transition; band 24896.4 = 124482/5; pass True
- H2-transition-state-031 $U (\wedge 1\Sigma_g^+): U \rightarrow B$ 67 R; class observed-directional-transition; band 24325.1 = 243251/10; pass True
- H2-transition-state-032 $M \wedge 1\Sigma_g^+: M \rightarrow B$ R; class observed-directional-transition; band 23190.0 68 = 23190/1; pass True

- H2-transition-state-033 $L \wedge 1\Sigma_g^+$: $L \rightarrow B$ R; class observed-directional-transition; band 23054.8 = 115274/5; pass True
- H2-transition-state-034 $H \wedge 1\Sigma_g^+ 3s\sigma$: $H \rightarrow C$ R; class observed-directional-transition; band 13866.6 = 69333/5; pass True
- H2-transition-continuation-008 $H \wedge 1\Sigma_g^+ 3s\sigma$: $H \rightarrow B$ V; class observed-directional-transition; band 22754.1 70 = 227541/10; pass True
- H2-transition-state-035 $D \wedge 1\Pi_u 3p\pi$: $D \rightarrow E$ R; class observed-directional-transition; band 13709._7 = 137097/10; pass True
- H2-transition-continuation-009 $D \wedge 1\Pi_u 3p\pi$: $D \leftrightarrow X$ 75 R; class observed-directional-transition; band 112872.3 76 = 1128723/10; pass True
- H2-transition-state-036 $J \wedge 1\Delta_g 3d\delta$: $J \rightarrow C$ 78 R; class observed-directional-transition; band 13435.6 79 = 67178/5; pass True
- H2-transition-continuation-010 $J \wedge 1\Delta_g 3d\delta$: $J \rightarrow B$ 78 V; class observed-directional-transition; band 22322.5 79 = 44645/2; pass True
- H2-transition-state-037 $I \wedge 1\Pi_g 3d\pi$: $I \rightarrow C$ 81 R; class observed-directional-transition; band 12982.5 82 = 25965/2; pass True
- H2-transition-continuation-011 $I \wedge 1\Pi_g 3d\pi$: $I \rightarrow B$ V; class observed-directional-transition; band 21869.5 82 = 43739/2; pass True
- H2-transition-state-038 $G \wedge 1\Sigma_g^+ 3d\sigma$: $G \rightarrow C$ 86 85 R; class observed-directional-transition; band 12722.2 87 = 63611/5; pass True
- H2-transition-continuation-012 $G \wedge 1\Sigma_g^+ 3d\sigma$: $G \rightarrow B$ 86 V; class observed-directional-transition; band 21609.2 87 = 108046/5; pass True
- H2-transition-state-039 $K (\wedge 1\Sigma_g^+)$: $K \rightarrow C$ R; class observed-directional-transition; band 12538.6 = 62693/5; pass True
- H2-transition-continuation-013 $K (\wedge 1\Sigma_g^+)$: $K \rightarrow B$ R; class observed-directional-transition; band 21425.4 = 107127/5; pass True
- H2-transition-state-040 $Q (\wedge 1\Pi_g)$: $Q \rightarrow B$ R; class observed-directional-transition; band 21151._1 = 211511/10; pass True
- H2-transition-state-041 $B' \wedge 1\Sigma_u^+ 3p\sigma$: $B' \rightarrow E, F$ 95; class observed-directional-transition; band 11311.5 96 = 22623/2; pass True
- H2-transition-continuation-014 $B' \wedge 1\Sigma_u^+ 3p\sigma$: $B' \leftarrow X$ 97 R; class observed-directional-transition; band 110478.2 = 552391/5; pass True
- H2-transition-state-042 $F \wedge 1\Sigma_g^+ 2p\sigma^2$: $F \rightarrow B$ 102 R 103; class observed-directional-transition; band absence = structural-absence; pass True
- H2-transition-state-043 $E \wedge 1\Sigma_g^+ 2s\sigma$: $E \rightarrow B$ 102 V; class observed-directional-transition; band 8961.23 = 896123/100; pass True
- H2-transition-state-044 $C \wedge 1\Pi_u 2p\pi$: $C \leftrightarrow X$ 108 109 R; class observed-directional-transition; band 99120.1_7 95 = 9912017/100; pass True
- H2-transition-state-045 $B \wedge 1\Sigma_u^+ 2p\sigma$: $B \leftrightarrow X$ 115 116 117 R; class observed-directional-transition; band 90203.3_5 = 1804067/20; pass True
- H2-transition-state-046 $X \wedge 1\Sigma_g^+ 1s\sigma^2$: absence; class absent-transition-coordinate; band absence = structural-absence; pass True
- count records: 60
- count primary: 46
- count continuation: 14
- count directional: 55
- count coupled: 4
- count absent_transition: 1
- count positive_band: 55
- count absent_band: 5

- adverse matching_path_composes: True
- adverse mismatched_endpoints_rejected: True
- adverse self_transition_rejected: True
- adverse absence_composition_rejected: True
- adverse numerical_zero_rejected: True
- adverse omitted_primary_rejected: True
- adverse omitted_continuation_rejected: True
- adverse present_only_selection_rejected: True
- adverse coupling_rows_retained: True
- adverse changed_band_rejected: True
- adverse tampered_source_rejected: True
- adverse complete_vector_retained: True

Falsification boundary: The claim fails if a transition loses its carrier, either endpoint, orientation or record; if unequal endpoints compose; if absence is treated numerically; if a selection rule is imported to define transformation before ELEC-010; if any of 46 primary rows, 14 continuation rows, 55 directional records, four coupled-state records, one absent transition coordinate, 55 positive band inscriptions or five absent band coordinates is omitted, changed or reclassified.

Measurement receipt: sha256:6eb4d49b27a619d57bc4ae3c34e6ba0b364153d749a7a01616910900d376976f.

Isolation certificate: sha256:3deadf438c912b9f4996fc1dd7dd171ecd16b4e66cb2a7284f6eeaa66cb422f1.

Custody certificate: sha256:0fdfe7eb81c234f6bb06975f5c17aa116608504f56520a3020230b3e5c7d617a.

Registered source descriptions:

- National Institute of Standards and Technology: <https://webbook.nist.gov/cgi/cbook.cgi?ID=C1333740&Mask=1000> ()

Registered target identities:

- H2-transition-state-001 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-002 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-003 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-004 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-005 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-006 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-007 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-008 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-continuation-001 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-009 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-010 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-011 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-012 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-013 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-014 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-015 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-continuation-002 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS

- H2-transition-continuation-003 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-016 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-017 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-continuation-004 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-018 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-019 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-020 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-continuation-005 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-021 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-022 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-023 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-024 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-025 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-026 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-continuation-006 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-027 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-continuation-007 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-028 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-029 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-030 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-031 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-032 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-033 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-034 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-continuation-008 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-035 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-continuation-009 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-036 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-continuation-010 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-037 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-continuation-011 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-038 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-continuation-012 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-039 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-continuation-013 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-040 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-041 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-continuation-014 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS

- H2-transition-state-042 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-043 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-044 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-045 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- H2-transition-state-046 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS

Meaning. The field-wide extension derives molecular electronic identity, state composition, occupancy, ordering, symmetry, correlation, transitions, measurement and operational classical-quantum correspondence from the same exact carrier laws. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; glyph 0 denotes source/interface absence only
- no negative, irrational, imaginary, floating, signed-amplitude or continuum proof magnitude
- no imported dipole, spin, parity, angular-momentum or spectroscopic selection rule
- no NIST state, transition, band or absence outcome before prediction seal
- no selected present-only row surface
- no species-specific exception

Immutable identities. Source manifest

sha256:708fe545889339a0e571c397cda8e548c6294c1c0ae6574717f5883fb34ce143; derivation seal
 sha256:39a5664e8fc84dd46a54489e6ab725efc44e79dc569b6d6292d216b37f2fffb40; engine receipt
 sha256:d239abbc191bc662020f3315df0dd8b74f95ae196158f4907a47206b531f0e89 at receipts/engin
 e/model_admitted/SFT-CHEM-MOLECULAR-STATE-TRANSITION-009-d239abbc191bc662.json; empirical
 hash sha256:2788e0cf53e75b2f9798b2c3a4ba02859b5ef874f1b281dabe50bceec6acc49f; measurement
 receipt sha256:6eb4d49b27a619d57bc4ae3c34e6ba0b364153d749a7a01616910900d376976f.

105. Exact chemical observation-class and selection-rule structure

Claim identity: SFT-CHEM-SELECTION-RULE-STRUCTURE-010

Question and theorem. A direct molecular observation retains spin-support multiplicity, changes a known inversion fibre, and changes axis support by no more than one generated recurrence. An observed relation outside that direct class requires a retained mediator or alternate observation channel; it does not erase the rule. Couplings, unresolved endpoints and channel absences remain separate exact classes, and absence is structural EmptyOne rather than numerical zero.

*retained-endpoint-signatures__declared-observation-class__direct-multiplicity-retention__known-inversion-fibre-chan
 ge__same-or-neighbour-axis-support__non-direct-requires-retained-mediator__channel-closed-EmptyOne__complete-
 adverse-inclusive-vector*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-CHEM-STATE-SYMMETRY-DEGENERACY-005

• SFT-CHEM-MOLECULAR-STATE-TRANSITION-009

Generated grammar. Generate the literal product of endpoint, observation-class, multiplicity, inversion, axis, mediation, absence and complete-record forms. Decide all 256 forms only from admitted Fold observation, information-retention, state-symmetry and molecular-transition laws.

Boundary: Every positive finite Fold axis recurrence under one direct observation action, together with mediated finite compositions and the complete registered NIST H2 transition, coupling, absence and adverse-note surface.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `retained-endpoint-signatures__declared-observation-class__direct-multiplicity-retention__known-inversion-fibre-change__same-or-neighbour-axis-support__non-direct-requires-retained-mediator__channel-closed-EmptyOne__complete-adverse-inclusive-vector`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	retained-endpoint-signatures	answer-only-rule	A selection answer without state endpoints erases the chemical carrier.
observation	declared-observation-class	channel-independent-permission	One universal yes/no label erases which observation operation was applied.
multiplicity	direct-multiplicity-retention	untracked-spin-support	Changing a retained support count in one direct Fold action erases a distinction.
inversion	known-inversion-fibre-change	same-fibre-direct-image	When both inversion fibres are known, an unchanged fibre supplies no direct observation distinction.
axis	same-or-neighbour-axis-support	arbitrary-axis-jump	One observation action cannot cross more than one generated axis recurrence.
mediation	non-direct-requires-retained-mediator	exception-erases-law	Calling every observed exception direct destroys the closed distinction.
absence	channel-closed-EmptyOne	numerical-zero-or-universal-forbidden	An empty source cell is neither a number nor a universal prohibition.
record	complete-adverse-inclusive-vector	selected-success-only	Keeping only ordinary transitions hides couplings, absences and mediated cases.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:47114ea62d0391a1b52d22c34af88baf89733999d9005f9bd5e02ef147f2ba55`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:6846bd783a74e3d329f9dd5e34281fd8329fc7f36cbd476d8171c7ecc9a87248`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:102ce1511196b1969c4d8f48e548fbfb4687b96c33b5167ddfe1c109f28b3346`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8fe0407f6bf7dbc1728798cb7b0b004bc9c315276edfe0631529a0b28256f2eb`.

Independent reconstruction. Implementation hash

`sha256:dc0309dce401244048a5969231768e2b2e35f32b9282af6c8b6c4c826ff08369`; independent certificate `sha256:9e7fdded67513d9582eb1ea04f8ea478af649b4f5542d2f5e10caa78159c9439`; external-validation hash `sha256:f9a72e1b857e1a533341d6a6179bef6a7e3210545465f5c8b10ab4b99decfacf`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE

Comparison records:

- selection-H2-transition-state-001: class direct-one-fold-observation; exact source hash sha256:82b2eed57d3d3ebe2a7aebdb3ea1a3574ac376670eb4e77af51e2760f01d78d8; pass True
- selection-H2-transition-state-002: class direct-one-fold-observation; exact source hash sha256:782d9d6dd8c6f22ac20babd340710df4c06db4b5c9dc5a687d42fc1d104e7773; pass True
- selection-H2-transition-state-003: class direct-one-fold-observation; exact source hash sha256:0c9fb42e63fea4a68e7b8cfe4982ffd2fa6444b3b5525636d479428a341cb316; pass True
- selection-H2-transition-state-004: class direct-one-fold-observation; exact source hash sha256:afd39e991cd9aa4b5d305a86ecc5a8a4395a30e20e6e84734c44a14c49aa7aa7; pass True
- selection-H2-transition-state-005: class direct-one-fold-observation; exact source hash sha256:d81dbf72856de47ac1033e0469876dd4df033de0061f669951c17a3d670873c7; pass True
- selection-H2-transition-state-006: class direct-one-fold-observation; exact source hash sha256:46e5b5225d243ace40b290dbb65fd37c7d6071358a508011ec3b20f8322b4358; pass True
- selection-H2-transition-state-007: class direct-one-fold-observation; exact source hash sha256:74738ba0c5dea360b975a255e9f4bbe1d5c7262cb242337f569530064d1bb464; pass True
- selection-H2-transition-state-008: class coupled-without-direction; exact source hash sha256:abe490730c03d3618648aff2cfc17197db4063e94e389d45c0da0d8d1102727d; pass True
- selection-H2-transition-continuation-001: class direct-one-fold-observation; exact source hash sha256:9faf2d684641813a1afa654090d12d951dfec2ee29782e0c7a9fa57619ef8897; pass True
- selection-H2-transition-state-009: class direct-one-fold-observation; exact source hash sha256:adaf30342bd5c155d22fcd0e965001e26dd93a88564d9fa509fb1daed698be16; pass True
- selection-H2-transition-state-010: class direct-one-fold-observation; exact source hash sha256:c733e15d0e022c65d31d738297ee16c3ef3d4d8a35f234196039550b484bc0ba; pass True
- selection-H2-transition-state-011: class direct-one-fold-observation; exact source hash sha256:76d5ee0c8d31648aa685624fbd5ad7267812f3d0156b19793dee3dc0962b8dce; pass True
- selection-H2-transition-state-012: class direct-one-fold-observation; exact source hash sha256:19af08b6ca22a6f0bcef445be90c928abb7485e69006d0836fb778cc9c262f4a; pass True
- selection-H2-transition-state-013: class direct-one-fold-observation; exact source hash sha256:28908cb4cfba5ee666434718627edab5dfdd21925a724c50ae70a47a621739d9; pass True
- selection-H2-transition-state-014: class direct-one-fold-observation; exact source hash sha256:a5d1b13a32fa8f23513728410df5e546ba867869439c6bd5f12a7324373799f5; pass True
- selection-H2-transition-state-015: class coupled-without-direction; exact source hash sha256:6292fcd9bdc24e95ab10a46f339d9d000732ca6a8d90d1972fef9d48b42244d25; pass True
- selection-H2-transition-continuation-002: class direct-one-fold-observation; exact source hash sha256:d822081a9a7cc4c31d5e75ff1dcb2ddad8131c33ca270aa608e301c6ada9a9b4; pass True
- selection-H2-transition-continuation-003: class direct-one-fold-observation; exact source hash sha256:65467b5113f8597ebac590483dc1edaf4c79dff3b47aae67f4e905a9b95d0d34; pass True
- selection-H2-transition-state-016: class direct-one-fold-observation; exact source hash sha256:758332fa6a063088e7a0b0486210e9fd7963bc13914ba286ee21ef47ce98c5d4; pass True
- selection-H2-transition-state-017: class direct-one-fold-observation; exact source hash sha256:d7c90ba3e4dec235cacfc63254138268ee0752525b9fe461d1382480880d31f9; pass True

- selection-H2-transition-continuation-004: class direct-one-fold-observation; exact source hash sha256:a159cfbc088669e06b4f2322055af15dac4e56e5cc49ded85e83604279062c2f; pass True
- selection-H2-transition-state-018: class direct-one-fold-observation; exact source hash sha256:ea8744a2ae62114ffb44dc028d7d985f17e9875cdebd0b7d09d86ad1a864756; pass True
- selection-H2-transition-state-019: class direct-one-fold-observation; exact source hash sha256:408930e8bd15caf2550432f00235fd4584d35ce2d8ccc0420af4ea789b3e91a3; pass True
- selection-H2-transition-state-020: class endpoint-signature-not-source-resolved; exact source hash sha256:c6fe39c07359997e9bfff457d856217b795d1ad41eaca75a3e0dd73a70512b45a; pass True
- selection-H2-transition-continuation-005: class coupled-without-direction; exact source hash sha256:89791ed615da7970599706a5e3f5fa0460f78305451ddbc330b1ccce04d8cfad; pass True
- selection-H2-transition-state-021: class coupled-without-direction; exact source hash sha256:b8083d371213573143e73772ab961fb839bfbf8ee4a9d43ba3b81ba695275a91; pass True
- selection-H2-transition-state-022: class direct-one-fold-observation; exact source hash sha256:8a0e55f573de72f1e5b9c973f490304fcd3a6bc9627cdda4a022fd2b9da8497d; pass True
- selection-H2-transition-state-023: class direct-one-fold-observation; exact source hash sha256:8aea49e36615e3c6304ea4eedd6d14926f308021d7c35b9ec2642f23d1238e2e; pass True
- selection-H2-transition-state-024: class mediated-multi-fold-observation; exact source hash sha256:12d1af9ddf9024773b506900677a8a8ca8732efe47b4fb24f1050f6d79e2af3c; pass True
- selection-H2-transition-state-025: class direct-one-fold-observation; exact source hash sha256:e143aa500d20bbb5935e8404e8fefff4dff42a08ffd56061f2a2828c8f3f993c; pass True
- selection-H2-transition-state-026: class direct-one-fold-observation; exact source hash sha256:224c0f8e7fa2fb6f63e0233d6f8f8e7ed8dddf3e54e184acc73f2e2740d483e; pass True
- selection-H2-transition-continuation-006: class direct-one-fold-observation; exact source hash sha256:078cbe9c89e47ab3ff5861608c28fae9c16fa420d49706af2166407b939ed22e; pass True
- selection-H2-transition-state-027: class direct-one-fold-observation; exact source hash sha256:afb4226800a383e7540afefec80a771020ef5a836bd30eb742cb31d3ced6e3ae; pass True
- selection-H2-transition-continuation-007: class direct-one-fold-observation; exact source hash sha256:ed342695dbe6b7350342dcfb396d409cde64de914f5804612428e75929368f65; pass True
- selection-H2-transition-state-028: class direct-one-fold-observation; exact source hash sha256:422debc236c1e2d5ebf63d2b13d15fb3b7cd379e6725c51b5c74ec82b1af04f8; pass True
- selection-H2-transition-state-029: class direct-one-fold-observation; exact source hash sha256:58640776eefd19844121c74469e534cd9e4710a634faba998c3aad5557b923fb; pass True
- selection-H2-transition-state-030: class direct-one-fold-observation; exact source hash sha256:1791d44ee3eff522007c5afebca65787f3c938bc2e79a250ccb013d7716a3870; pass True
- selection-H2-transition-state-031: class direct-one-fold-observation; exact source hash sha256:3a56614727320a97013b417a2ee2b5777e6b5d2899eb89911d8546547a465461; pass True
- selection-H2-transition-state-032: class direct-one-fold-observation; exact source hash sha256:54d7ebc54277fb2922ec23b0a4d4fca954314aed846aa00a7b40a5262e204f3; pass True
- selection-H2-transition-state-033: class direct-one-fold-observation; exact source hash sha256:7438e9ddff84dd2cad654ebb43c216e2472b351328cc93d9e9833c2cb2f752f5; pass True
- selection-H2-transition-state-034: class direct-one-fold-observation; exact source hash sha256:66db8359f1203f4fa720b2686ecdad993378dd7702ac0a665bad69782c604b12; pass True
- selection-H2-transition-continuation-008: class direct-one-fold-observation; exact source hash sha256:5e65d794f9b409c2ba26b0a1ae50973956f9df2ef8a82a779fbd34fdc42aa1d8; pass True

- selection-H2-transition-state-035: class direct-one-fold-observation; exact source hash sha256:932d6dc259966eff1211ab26fb627d00d99b613e16073c0c06dc8f53a2d61c07; pass True
- selection-H2-transition-continuation-009: class direct-one-fold-observation; exact source hash sha256:0f422bc418e933ca830287e61145d104ca6cc3a51b98170e50081ea5efc31366; pass True
- selection-H2-transition-state-036: class direct-one-fold-observation; exact source hash sha256:6a7cff6d8209a6ec9883d0c520b318f554e1d73d9933a549df8fa425f61f2a2a; pass True
- selection-H2-transition-continuation-010: class mediated-multi-fold-observation; exact source hash sha256:1a0cd37c557999b8a7ab3d97824e6ba35185126558955b8b4a7650931044f903; pass True
- selection-H2-transition-state-037: class direct-one-fold-observation; exact source hash sha256:302a36489e535e1b76e64d362dc7f8768e26e7a61547b3a1fbddc5814872c46c; pass True
- selection-H2-transition-continuation-011: class direct-one-fold-observation; exact source hash sha256:d46419e2e43b1cb8674b2175bae0305938a39d688fdd49d081baadef1a26a6c4; pass True
- selection-H2-transition-state-038: class direct-one-fold-observation; exact source hash sha256:436b15a1459b22aefe619d97d5e45513ed4f050c8cb880c7f39cee6f2fd94d2c; pass True
- selection-H2-transition-continuation-012: class direct-one-fold-observation; exact source hash sha256:8f32e48057ee9460b046c11c5f3fdf99a70d990ad2d87fd7f09e9d7be00bfa8b; pass True
- selection-H2-transition-state-039: class direct-one-fold-observation; exact source hash sha256:3b1068022f12a7bf2e35a74725dfd3176f9dfbc5541ceab37233926bc156d57f; pass True
- selection-H2-transition-continuation-013: class direct-one-fold-observation; exact source hash sha256:7f3c00b8f7d3b9e3890392f93711ebff76bed57652040edc15c688eb1e646630; pass True
- selection-H2-transition-state-040: class direct-one-fold-observation; exact source hash sha256:9d09ffee4e6b54650daa6454dc7748938c16f3d4d3805c5b138c121a9703b10a; pass True
- selection-H2-transition-state-041: class direct-one-fold-observation; exact source hash sha256:0e0e9a74936ed460ef9d29bc8a2eb3827ffb6577b4a598050eb75248acd21a6b; pass True
- selection-H2-transition-continuation-014: class direct-one-fold-observation; exact source hash sha256:c6668871881bd7828aebc6d53a5302643a027a5abd1536300c716884a88d646a; pass True
- selection-H2-transition-state-042: class direct-one-fold-observation; exact source hash sha256:c6e5bda6c983e81e5e34c44673118a894f60e7f032fc7b5b69933049ae6d260e; pass True
- selection-H2-transition-state-043: class direct-one-fold-observation; exact source hash sha256:ee0cddb97d38e0359f0fbc8027224e831ddcb0bbf631d40038c8999de9f18da5; pass True
- selection-H2-transition-state-044: class direct-one-fold-observation; exact source hash sha256:c4414d1287ae39ebac47e1c1a0fd93b6b52d836016ed184b2cc8e4d635e87f7; pass True
- selection-H2-transition-state-045: class direct-one-fold-observation; exact source hash sha256:857ebb92e0eef32b0c774d7ecdb12505b471b9dc46180c8269ff73db1c576922; pass True
- selection-H2-transition-state-046: class closed-observation-coordinate; exact source hash sha256:1dcca5e36ebbbdd72fc9ceb96c5bf6bb16c5e30e3f78c4cbd68511600bad6795; pass True
- selection-adverse-note-42: class adverse-note; exact source hash sha256:458ab3b954519d0bf85f3371b10d996bb4ffb061d7b6c3c9851a2ba7b1cd6195; pass True
- selection-adverse-note-73: class adverse-note; exact source hash sha256:9a6a70ac6e7b8cb9090e3d7264d3888abc711672717d65d230ee09ce41433cd; pass True
- selection-adverse-note-78: class adverse-note; exact source hash sha256:5ae6a301e0adc931fe7bf6a4be50643e02461b2d0e42e840938c768fe5c795cf; pass True
- count direct-one-fold-observation: 52
- count mediated-multi-fold-observation: 2
- count endpoint-signature-not-source-resolved: 1

- count coupled-without-direction: 4
- count closed-observation-coordinate: 1
- count adverse-note: 3
- adverse complete_counts: True
- adverse note_42_alternate_forbidden_channel_retained: True
- adverse note_73_emission_absence_and_absorption_presence_retained: True
- adverse note_78_forbidden_transition_requires_uncoupling: True
- adverse same_inversion_direct_control_rejected: True
- adverse multiplicity_change_direct_control_rejected: True

Falsification boundary: The claim fails if a direct observation changes multiplicity, preserves a known g/u fibre, jumps more than one axis recurrence, or loses either endpoint; if an observed non-direct case is admitted without a retained mediation requirement; if channel absence is treated as numerical zero or universal nonexistence; or if any of 60 NIST transition rows, three adverse notes, 52 direct one-step records, two mediated two-step records, one unresolved endpoint, four couplings or one closed coordinate is omitted or changed.

Measurement receipt: sha256:9a9d60bc391841b76a4f330d7ab5782cb98ec53cffd931e56f2dc924f242b7ab.

Isolation certificate: sha256:89aef930f898a3f2c515e2894a190f9dd845f38591658850647c53bbd532e79e.

Custody certificate: sha256:f5741d4162d31abf524770fa9f80d06815143f791aae96e08fc6eb6c84657c47.

Registered source descriptions:

- National Institute of Standards and Technology: <https://webbook.nist.gov/cgi/cbook.cgi?ID=C1333740&Mask=1000> ()

Registered target identities:

- selection-H2-transition-state-001 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-002 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-003 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-004 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-005 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-006 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-007 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-008 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-continuation-001 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-009 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-010 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-011 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-012 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-013 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-014 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-015 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-continuation-002 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-continuation-003 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-016 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-017 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE

- selection-H2-transition-continuation-004 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-018 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-019 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-020 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-continuation-005 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-021 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-022 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-023 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-024 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-025 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-026 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-continuation-006 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-027 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-continuation-007 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-028 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-029 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-030 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-031 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-032 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-033 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-034 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-continuation-008 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-035 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-continuation-009 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-036 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-continuation-010 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-037 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-continuation-011 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-038 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-continuation-012 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-039 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-continuation-013 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-040 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-041 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-continuation-014 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-042 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-043 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-044 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE

- selection-H2-transition-state-045 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-H2-transition-state-046 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-adverse-note-42 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-adverse-note-73 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- selection-adverse-note-78 from NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE

Meaning. The field-wide extension derives molecular electronic identity, state composition, occupancy, ordering, symmetry, correlation, transitions, measurement and operational classical-quantum correspondence from the same exact carrier laws. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; glyph 0 denotes source/interface absence only
- no negative, irrational, imaginary, floating, signed-amplitude or continuum proof magnitude
- no imported spectroscopic selection-rule formula or probability threshold
- no NIST endpoint, transition, coupling, absence or adverse-note outcome before prediction seal
- no observed exception relabelled as an ordinary direct transition
- no selected success-only surface and no erased channel distinction

Immutable identities. Source manifest

sha256:dced7d48559ba5be1a509bbb91c95f063c80df490541e608c4d36e3f769b6091; derivation seal
 sha256:1250444c044ddda7e9026bf18cb010b0c328b83585cc0a0471715e2b1e74fab2; engine receipt
 sha256:eca9cf4ab69aeb92660506cf11b87e1f4bb7aa0e3a1fa27f05c61680b6a8725f at receipts/engine/
 model_admitted/SFT-CHEM-SELECTION-RULE-STRUCTURE-010-eca9cf4ab69aeb92.json; empirical hash
 sha256:6b84b72f3546151e9a9b8f02f054ea160f79eb26a7c26d8c484d4686378dc247; measurement receipt
 sha256:9a9d60bc391841b76a4f330d7ab5782cb98ec53cffd931e56f2dc924f242b7ab.

106. Exact molecular configuration order, basin, barrier and path law

Claim identity: SFT-CHEM-CONFIGURATION-ORDER-PATH-011

Question and theorem. A molecular configuration surface is an exact generated configuration graph, not a continuum premise. Each node retains its carrier, coordinate, exact positive height above structural EmptyOne and record; complete neighbour order forces stable basins and barriers, connected adjacency forces reaction paths, and periodic recurrence identifies its terminal configuration with its initial class.

retained-molecular-carrier__generated-configuration-nodes__exact-positive-order-or-EmptyOne__local-minimum-by-complete-neighbours__local-barrier-by-complete-neighbours__connected-complete-configuration-path__exact-periodic-endpoint-identity__complete-favourable-and-adverse-vector

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-SELECTION-RULE-STRUCTURE-010

Generated grammar. Generate the literal product of carrier, configuration, order, stability, barrier, path, recurrence and record forms. Decide all 256 only from admitted exact order, graph, state-transition and molecular observation laws.

Boundary: Every positive finite generated configuration graph and periodic successor, tested against both complete 25-coordinate NIST experimental ethanol internal-rotation paths.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `retained-molecular-carrier__generated-configuration-nodes__exact-positive-order-or-EmptyOne__local-minimum-by-complete-neighbours__local-barrier-by-complete-neighbours__connected-complete-configuration-path__exact-periodic-endpoint-identity__complete-favourable-and-adverse-vector`.

Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	retained-molecular-carrier	coordinate-only-surface	Coordinates without the molecular carrier do not identify one chemical configuration space.
configuration	generated-configuration-nodes	continuum-point-premise	A continuum potential coordinate imports an ungenerated domain.
order	exact-positive-order-or-EmptyOne	signed-or-floating-energy	Signed or floating magnitudes violate the exact Fold domain.
stability	local-minimum-by-complete-neighbours	named-stable-structure	A stability name alone does not decide neighbouring alternatives.
barrier	local-barrier-by-complete-neighbours	imported-saddle-formula	A continuum saddle formula is unnecessary and ungenerated.
path	connected-complete-configuration-path	endpoint-only-reaction	Endpoints erase intermediates and barriers.
recurrence	exact-periodic-endpoint-identity	open-curve-assumption	An internal rotation that erases closure duplicates rather than identifies its terminal configuration.
record	complete-favourable-and-adverse-vector	selected-minima-and-maxima	Keeping extrema alone hides the empirical path.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:179b95273b3e79490c0ab7edbc4046a6c0906b3661b159bab0e653498bf859c4`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e1a1a63fe75d1ab1741227e7ab06341418d071bd77a0f76b337d6db0378a1107`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:2f8e7b74c5c55be37a5b8cc052b48cd9fdb91d48283386e778a3db5d6a7d89f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:765a0149a1a02af251f7f390e946f5c3c3b88d02de94163caa43eb10ead27989`.

Independent reconstruction. Implementation hash

`sha256:39b953ccd7c8d4cadd4279eb2bb3a92a840774415ef65f35bb1f0a130d5e3ce0`; independent certificate `sha256:cafe7f97078f7bfe6a8aedc229111a95f1125f4c47439f6d22e9223db8568d61`; external-validation hash `sha256:78c0a7ef9757d33c828f25153c7f7bbe42faa381e7584171ad47ba07b17743fe`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

• NIST-CCCBDB-SRD101-ETHANOL-EXPERIMENTAL-ROTATIONAL-BARRIER

Comparison records:

- ethanol-torsion-1-coordinate-01: kJ PositiveRatio(numerator=PositiveCount(value=541), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=452), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-02: kJ PositiveRatio(numerator=PositiveCount(value=47), denominator=PositiveCount(value=10)); cm PositiveRatio(numerator=PositiveCount(value=393), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-03: kJ PositiveRatio(numerator=PositiveCount(value=3), denominator=PositiveCount(value=1)); cm PositiveRatio(numerator=PositiveCount(value=251), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-04: kJ PositiveRatio(numerator=PositiveCount(value=32), denominator=PositiveCount(value=25)); cm PositiveRatio(numerator=PositiveCount(value=107), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-05: kJ PositiveRatio(numerator=PositiveCount(value=13), denominator=PositiveCount(value=25)); cm PositiveRatio(numerator=PositiveCount(value=43), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-06: kJ PositiveRatio(numerator=PositiveCount(value=113), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=94), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-07: kJ PositiveRatio(numerator=PositiveCount(value=271), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=227), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-08: kJ PositiveRatio(numerator=PositiveCount(value=429), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=359), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-09: kJ PositiveRatio(numerator=PositiveCount(value=49), denominator=PositiveCount(value=10)); cm PositiveRatio(numerator=PositiveCount(value=410), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-10: kJ PositiveRatio(numerator=PositiveCount(value=207), denominator=PositiveCount(value=50)); cm PositiveRatio(numerator=PositiveCount(value=346), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-11: kJ PositiveRatio(numerator=PositiveCount(value=241), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=202), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-12: kJ PositiveRatio(numerator=PositiveCount(value=7), denominator=PositiveCount(value=10)); cm PositiveRatio(numerator=PositiveCount(value=59), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-13: kJ EmptyOne(identity='empty-One'); cm EmptyOne(identity='empty-One'); pass True
- ethanol-torsion-1-coordinate-14: kJ PositiveRatio(numerator=PositiveCount(value=7), denominator=PositiveCount(value=10)); cm PositiveRatio(numerator=PositiveCount(value=59), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-15: kJ PositiveRatio(numerator=PositiveCount(value=241), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=202), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-16: kJ PositiveRatio(numerator=PositiveCount(value=207), denominator=PositiveCount(value=50)); cm PositiveRatio(numerator=PositiveCount(value=346), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-17: kJ PositiveRatio(numerator=PositiveCount(value=49), denominator=PositiveCount(value=10)); cm PositiveRatio(numerator=PositiveCount(value=410), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-18: kJ PositiveRatio(numerator=PositiveCount(value=429), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=359), denominator=PositiveCount(value=1)); pass True

- ethanol-torsion-1-coordinate-19: kJ PositiveRatio(numerator=PositiveCount(value=271), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=227), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-20: kJ PositiveRatio(numerator=PositiveCount(value=113), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=94), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-21: kJ PositiveRatio(numerator=PositiveCount(value=13), denominator=PositiveCount(value=25)); cm PositiveRatio(numerator=PositiveCount(value=43), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-22: kJ PositiveRatio(numerator=PositiveCount(value=32), denominator=PositiveCount(value=25)); cm PositiveRatio(numerator=PositiveCount(value=107), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-23: kJ PositiveRatio(numerator=PositiveCount(value=3), denominator=PositiveCount(value=1)); cm PositiveRatio(numerator=PositiveCount(value=251), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-24: kJ PositiveRatio(numerator=PositiveCount(value=47), denominator=PositiveCount(value=10)); cm PositiveRatio(numerator=PositiveCount(value=393), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-1-coordinate-25: kJ PositiveRatio(numerator=PositiveCount(value=541), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=452), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-26: kJ PositiveRatio(numerator=PositiveCount(value=398), denominator=PositiveCount(value=25)); cm PositiveRatio(numerator=PositiveCount(value=1331), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-27: kJ PositiveRatio(numerator=PositiveCount(value=1359), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=1136), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-28: kJ PositiveRatio(numerator=PositiveCount(value=199), denominator=PositiveCount(value=25)); cm PositiveRatio(numerator=PositiveCount(value=665), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-29: kJ PositiveRatio(numerator=PositiveCount(value=233), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=195), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-30: kJ EmptyOne(identity='empty-One'); cm EmptyOne(identity='empty-One'); pass True
- ethanol-torsion-2-coordinate-31: kJ PositiveRatio(numerator=PositiveCount(value=233), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=195), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-32: kJ PositiveRatio(numerator=PositiveCount(value=199), denominator=PositiveCount(value=25)); cm PositiveRatio(numerator=PositiveCount(value=665), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-33: kJ PositiveRatio(numerator=PositiveCount(value=1359), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=1136), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-34: kJ PositiveRatio(numerator=PositiveCount(value=398), denominator=PositiveCount(value=25)); cm PositiveRatio(numerator=PositiveCount(value=1331), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-35: kJ PositiveRatio(numerator=PositiveCount(value=1359), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=1136), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-36: kJ PositiveRatio(numerator=PositiveCount(value=199), denominator=PositiveCount(value=25)); cm PositiveRatio(numerator=PositiveCount(value=665), denominator=PositiveCount(value=1)); pass True

- ethanol-torsion-2-coordinate-37: kJ PositiveRatio(numerator=PositiveCount(value=233), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=195), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-38: kJ EmptyOne(identity='empty-One'); cm EmptyOne(identity='empty-One'); pass True
- ethanol-torsion-2-coordinate-39: kJ PositiveRatio(numerator=PositiveCount(value=233), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=195), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-40: kJ PositiveRatio(numerator=PositiveCount(value=199), denominator=PositiveCount(value=25)); cm PositiveRatio(numerator=PositiveCount(value=665), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-41: kJ PositiveRatio(numerator=PositiveCount(value=1359), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=1136), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-42: kJ PositiveRatio(numerator=PositiveCount(value=398), denominator=PositiveCount(value=25)); cm PositiveRatio(numerator=PositiveCount(value=1331), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-43: kJ PositiveRatio(numerator=PositiveCount(value=1359), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=1136), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-44: kJ PositiveRatio(numerator=PositiveCount(value=199), denominator=PositiveCount(value=25)); cm PositiveRatio(numerator=PositiveCount(value=665), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-45: kJ PositiveRatio(numerator=PositiveCount(value=233), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=195), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-46: kJ EmptyOne(identity='empty-One'); cm EmptyOne(identity='empty-One'); pass True
- ethanol-torsion-2-coordinate-47: kJ PositiveRatio(numerator=PositiveCount(value=233), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=195), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-48: kJ PositiveRatio(numerator=PositiveCount(value=199), denominator=PositiveCount(value=25)); cm PositiveRatio(numerator=PositiveCount(value=665), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-49: kJ PositiveRatio(numerator=PositiveCount(value=1359), denominator=PositiveCount(value=100)); cm PositiveRatio(numerator=PositiveCount(value=1136), denominator=PositiveCount(value=1)); pass True
- ethanol-torsion-2-coordinate-50: kJ PositiveRatio(numerator=PositiveCount(value=398), denominator=PositiveCount(value=25)); cm PositiveRatio(numerator=PositiveCount(value=1331), denominator=PositiveCount(value=1)); pass True
- count positive: 46
- count absence: 4
- count basin: 6
- count barrier: 6
- count ordinary: 36
- count recurrence_duplicate: 2

Falsification boundary: The claim fails if a configuration loses its carrier, coordinate, exact order or record; if a claimed basin or barrier fails complete-neighbour comparison; if a path skips a registered adjacent configuration; if a periodic endpoint fails identity

and height recurrence; if source absence is numerical; or if any of 50 NIST rows, 46 positive energy inscriptions, four structural least-energy coordinates, six basins, six barriers, 36 ordinary path nodes or two recurrence duplicates is omitted or changed.

Measurement receipt: sha256:f55272bca94866a3b19320b5c01d8e0cc9d2f37aaf6b6c591380fd44ebc87238.

Isolation certificate: sha256:afffd19b1ea529bbab78ab05cc7c33d9b2a3e7663cf723a0f9dd848d240dbb60.

Custody certificate: sha256:1daaed945c4bf7a76ed9ff9a51dfcc591f75525249d244797919c57a8ab23f42.

Registered source descriptions:

- National Institute of Standards and Technology: <https://cccbdb.nist.gov/exprotbar2x.asp?casno=64175&ti=1> ()

Registered target identities:

- ethanol-torsion-1-coordinate-01 from None
- ethanol-torsion-1-coordinate-02 from None
- ethanol-torsion-1-coordinate-03 from None
- ethanol-torsion-1-coordinate-04 from None
- ethanol-torsion-1-coordinate-05 from None
- ethanol-torsion-1-coordinate-06 from None
- ethanol-torsion-1-coordinate-07 from None
- ethanol-torsion-1-coordinate-08 from None
- ethanol-torsion-1-coordinate-09 from None
- ethanol-torsion-1-coordinate-10 from None
- ethanol-torsion-1-coordinate-11 from None
- ethanol-torsion-1-coordinate-12 from None
- ethanol-torsion-1-coordinate-13 from None
- ethanol-torsion-1-coordinate-14 from None
- ethanol-torsion-1-coordinate-15 from None
- ethanol-torsion-1-coordinate-16 from None
- ethanol-torsion-1-coordinate-17 from None
- ethanol-torsion-1-coordinate-18 from None
- ethanol-torsion-1-coordinate-19 from None
- ethanol-torsion-1-coordinate-20 from None
- ethanol-torsion-1-coordinate-21 from None
- ethanol-torsion-1-coordinate-22 from None
- ethanol-torsion-1-coordinate-23 from None
- ethanol-torsion-1-coordinate-24 from None
- ethanol-torsion-1-coordinate-25 from None
- ethanol-torsion-2-coordinate-26 from None
- ethanol-torsion-2-coordinate-27 from None
- ethanol-torsion-2-coordinate-28 from None
- ethanol-torsion-2-coordinate-29 from None
- ethanol-torsion-2-coordinate-30 from None
- ethanol-torsion-2-coordinate-31 from None

- ethanol-torsion-2-coordinate-32 from None
- ethanol-torsion-2-coordinate-33 from None
- ethanol-torsion-2-coordinate-34 from None
- ethanol-torsion-2-coordinate-35 from None
- ethanol-torsion-2-coordinate-36 from None
- ethanol-torsion-2-coordinate-37 from None
- ethanol-torsion-2-coordinate-38 from None
- ethanol-torsion-2-coordinate-39 from None
- ethanol-torsion-2-coordinate-40 from None
- ethanol-torsion-2-coordinate-41 from None
- ethanol-torsion-2-coordinate-42 from None
- ethanol-torsion-2-coordinate-43 from None
- ethanol-torsion-2-coordinate-44 from None
- ethanol-torsion-2-coordinate-45 from None
- ethanol-torsion-2-coordinate-46 from None
- ethanol-torsion-2-coordinate-47 from None
- ethanol-torsion-2-coordinate-48 from None
- ethanol-torsion-2-coordinate-49 from None
- ethanol-torsion-2-coordinate-50 from None

Meaning. The field-wide extension derives molecular electronic identity, state composition, occupancy, ordering, symmetry, correlation, transitions, measurement and operational classical-quantum correspondence from the same exact carrier laws. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; source glyph 0 denotes absence only
- no negative, irrational, imaginary, floating, signed or continuum proof magnitude
- no imported potential-energy function, differential equation, saddle formula or fitted coefficient
- no NIST angle, energy, minimum, barrier or path outcome before prediction seal
- no selected extrema-only vector

Immutable identities. Source manifest

sha256:088e5eccd1854614019537c1c05eb9ba9dacb25e2c8512e0751d77ca0ec6bdb2; derivation seal
 sha256:ef8184f7540681a83c9cb8efa4bcdd58e36f722e2dbad448d8d5b4a2f1d2ea99; engine receipt
 sha256:3860aa331987bef80cfe7be42265d5bb5eece24486e9e927f855e8b6fdf2bfaa at receipts/engine/model_admitted/SFT-CHEM-CONFIGURATION-ORDER-PATH-011-3860aa331987bef8.json; empirical hash
 sha256:38169a9422b0bec6cab1cb9fad3b9a724d71bb903f1cc338f282f947d4261cc3; measurement receipt
 sha256:f55272bca94866a3b19320b5c01d8e0cc9d2f37aaf6b6c591380fd44ebc87238.

107. Exact nuclear-electronic joint composition and isotopologue scale-separation law

Claim identity: SFT-CHEM-NUCLEAR-ELECTRONIC-COMPOSITION-012

Question and theorem. A molecular isotopologue is one retained joint carrier whose nuclear occurrences and isotope labels compose with, but do not erase, its electronic support and state. Their distinct exact coordinate families force lawful scale separation without a continuum approximation or fitted separation parameter; equal-electronic-support isotopologues transport distinct exact vibronic records while every absence remains structural EmptyOne and every external signed inscription remains

held provenance.

retained-joint-molecular-carrier__retained-occurrence-and-isotope-support__retained-electronic-support-and-state__finite-nuclear-electronic-joint-product__exact-coordinate-partition-without-separation-parameter__exact-isotope-labelled-coordinate-transport__EmptyOne-or-held-external-inscription__complete-isotopologue-and-vibronic-record

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-PHYS-NUCLEAR-DEUTERON-DINUCLEON-TERMINAL-006
- SFT-CHEM-ELEM-ISOTOPE-001
- SFT-CHEM-ELECTRONIC-STATE-IDENTITY-001
- SFT-CHEM-ORBITAL-SUPPORT-OCCUPANCY-003
- SFT-CHEM-CONFIGURATION-ORDER-PATH-011

Generated grammar. Generate the literal product of carrier, nuclear, electronic, composition, scale, transport, absence and record forms. Decide all 256 only from admitted exact composition, isotope, electronic-state and configuration laws.

Boundary: Every positive finite nuclear occurrence word jointly composed with one admitted molecular electronic state; externally tested against every state and cell in the complete NIST H2, HD and D2 diatomic tables.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *retained-joint-molecular-carrier__retained-occurrence-and-isotope-support__retained-electronic-support-and-state__finite-nuclear-electronic-joint-product__exact-coordinate-partition-without-separation-parameter__exact-isotope-labelled-coordinate-transport__EmptyOne-or-held-external-inscription__complete-isotopologue-and-vibronic-record*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	retained-joint-molecular-carrier	detached-nuclear-and-electronic-records	Detached records do not identify one molecular state.
nuclear	retained-occurrence-and-isotope-support	mass-number-without-occurrence-identity	A magnitude alone erases which nuclear occurrence carries an isotope.
electronic	retained-electronic-support-and-state	isotope-replacement-erases-electronic-identity	Erasing electronic identity prevents lawful equal-support comparison.
composition	finite-nuclear-electronic-joint-product	nuclear-or-electronic-factor-erased	Either erasure merges empirically distinguishable states.
scale	exact-coordinate-partition-without-separation-parameter	imported-continuum-separation	A continuum scale approximation is not generated by Fold structure.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
transport	exact-isotope-labeled-coordinate-transport	isotopologue-coordinates-declared-invariant	Invariant coordinates erase observed isotope transport.
absence	EmptyOne-or-held-external-inscription	numerical-zero-or-signed-proof-value	Numerical zero and signed magnitudes are outside the SFT proof domain.
record	complete-isotopologue-and-vibronic-record	selected-common-state-highlights	A selected intersection hides unmatched and absent source rows.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:919e831fff7676b717c45a987feeed4dadf22cb59dc60848530c525a6fb9c1554.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:0878c06a361ce16c22bd38163ae8b3453b97c134c11fcc8b2a0154038ee5f13c.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:f42bd540d9bea78dd53663bd96b14494b3d45b7bed57673ba455983257e9ead4.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:26bd7ae24a2794f73cbeba5352ff81fdfae64975119bded7d5c936690128e0cd.

Independent reconstruction. Implementation hash

sha256:c6d15d2a40a1b3b44c05db318557c6e7025cf317e4db5e71b53e851c46d54ed4; independent certificate sha256:2f9606890328c5d9f9a188818c7684eea6eb78003ddd080e6a4910bbd1f7adf3; external-validation hash sha256:6214b48844fc4c8d410cedfaea41440d886da14f762f4009a299461248ae64c4. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-HD-D2-DIATOMIC-CONSTANTS-2026

Comparison records:

- C1333740-nuclear-electronic-001: 13 complete cells; pass True
- C1333740-nuclear-electronic-002: 13 complete cells; pass True
- C1333740-nuclear-electronic-003: 13 complete cells; pass True
- C1333740-nuclear-electronic-004: 13 complete cells; pass True
- C1333740-nuclear-electronic-005: 13 complete cells; pass True
- C1333740-nuclear-electronic-006: 13 complete cells; pass True
- C1333740-nuclear-electronic-007: 13 complete cells; pass True
- C1333740-nuclear-electronic-008: 13 complete cells; pass True
- C1333740-nuclear-electronic-009: 13 complete cells; pass True
- C1333740-nuclear-electronic-010: 13 complete cells; pass True
- C1333740-nuclear-electronic-011: 13 complete cells; pass True
- C1333740-nuclear-electronic-012: 13 complete cells; pass True
- C1333740-nuclear-electronic-013: 13 complete cells; pass True
- C1333740-nuclear-electronic-014: 13 complete cells; pass True

- C1333740-nuclear-electronic-015: 13 complete cells; pass True
- C1333740-nuclear-electronic-016: 13 complete cells; pass True
- C1333740-nuclear-electronic-017: 13 complete cells; pass True
- C1333740-nuclear-electronic-018: 13 complete cells; pass True
- C1333740-nuclear-electronic-019: 13 complete cells; pass True
- C1333740-nuclear-electronic-020: 13 complete cells; pass True
- C1333740-nuclear-electronic-021: 13 complete cells; pass True
- C1333740-nuclear-electronic-022: 13 complete cells; pass True
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- C1333740-nuclear-electronic-025: 13 complete cells; pass True
- C1333740-nuclear-electronic-026: 13 complete cells; pass True
- C1333740-nuclear-electronic-027: 13 complete cells; pass True
- C1333740-nuclear-electronic-028: 13 complete cells; pass True
- C1333740-nuclear-electronic-029: 13 complete cells; pass True
- C1333740-nuclear-electronic-030: 13 complete cells; pass True
- C1333740-nuclear-electronic-031: 13 complete cells; pass True
- C1333740-nuclear-electronic-032: 13 complete cells; pass True
- C1333740-nuclear-electronic-033: 13 complete cells; pass True
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- C1333740-nuclear-electronic-035: 13 complete cells; pass True
- C1333740-nuclear-electronic-036: 13 complete cells; pass True
- C1333740-nuclear-electronic-037: 13 complete cells; pass True
- C1333740-nuclear-electronic-038: 13 complete cells; pass True
- C1333740-nuclear-electronic-039: 13 complete cells; pass True
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- C1333740-nuclear-electronic-044: 13 complete cells; pass True
- C1333740-nuclear-electronic-045: 13 complete cells; pass True
- C1333740-nuclear-electronic-046: 13 complete cells; pass True
- C13983205-nuclear-electronic-001: 13 complete cells; pass True
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- C13983205-nuclear-electronic-006: 13 complete cells; pass True

- C13983205-nuclear-electronic-007: 13 complete cells; pass True
- C13983205-nuclear-electronic-008: 13 complete cells; pass True
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- C7782390-nuclear-electronic-004: 13 complete cells; pass True
- C7782390-nuclear-electronic-005: 13 complete cells; pass True
- C7782390-nuclear-electronic-006: 13 complete cells; pass True
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- C7782390-nuclear-electronic-008: 13 complete cells; pass True
- C7782390-nuclear-electronic-009: 13 complete cells; pass True
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- C7782390-nuclear-electronic-013: 13 complete cells; pass True
- C7782390-nuclear-electronic-014: 13 complete cells; pass True
- C7782390-nuclear-electronic-015: 13 complete cells; pass True
- C7782390-nuclear-electronic-016: 13 complete cells; pass True
- C7782390-nuclear-electronic-017: 13 complete cells; pass True
- C7782390-nuclear-electronic-018: 13 complete cells; pass True
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- C7782390-nuclear-electronic-020: 13 complete cells; pass True
- C7782390-nuclear-electronic-021: 13 complete cells; pass True
- C7782390-nuclear-electronic-022: 13 complete cells; pass True
- C7782390-nuclear-electronic-023: 13 complete cells; pass True
- C7782390-nuclear-electronic-024: 13 complete cells; pass True

- C7782390-nuclear-electronic-025: 13 complete cells; pass True
- C7782390-nuclear-electronic-026: 13 complete cells; pass True
- C7782390-nuclear-electronic-027: 13 complete cells; pass True
- C7782390-nuclear-electronic-028: 13 complete cells; pass True
- C7782390-nuclear-electronic-029: 13 complete cells; pass True
- u H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=154360), denominator=PositiveCount(value=152957)); distinct True
- u H2/D2 column 5: PositiveRatio(numerator=PositiveCount(value=7325), denominator=PositiveCount(value=3759)); distinct True
- u H2/D2 column 8: PositiveRatio(numerator=PositiveCount(value=230), denominator=PositiveCount(value=53)); distinct True
- u H2/D2 column 10: PositiveRatio(numerator=PositiveCount(value=10690), denominator=PositiveCount(value=10551)); distinct True
- n H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=1209529), denominator=PositiveCount(value=1209769)); distinct True
- n H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=77380), denominator=PositiveCount(value=55091)); distinct True
- n H2/D2 column 3: PositiveRatio(numerator=PositiveCount(value=6286), denominator=PositiveCount(value=3425)); distinct True
- n H2/D2 column 5: PositiveRatio(numerator=PositiveCount(value=2995), denominator=PositiveCount(value=1504)); distinct True
- n H2/D2 column 6: PositiveRatio(numerator=PositiveCount(value=31), denominator=PositiveCount(value=14)); distinct True
- n H2/D2 column 8: PositiveRatio(numerator=PositiveCount(value=230), denominator=PositiveCount(value=53)); distinct True
- n H2/D2 column 10: PositiveRatio(numerator=PositiveCount(value=1057), denominator=PositiveCount(value=1055)); distinct True
- p H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=592549), denominator=PositiveCount(value=596210)); distinct True
- k H2/HD column 1: PositiveRatio(numerator=PositiveCount(value=65759), denominator=PositiveCount(value=65769)); distinct True
- k H2/HD column 2: PositiveRatio(numerator=PositiveCount(value=33491), denominator=PositiveCount(value=29008)); distinct True
- k H2/HD column 3: PositiveRatio(numerator=PositiveCount(value=6729), denominator=PositiveCount(value=5036)); distinct True
- k H2/HD column 5: PositiveRatio(numerator=PositiveCount(value=1367), denominator=PositiveCount(value=934)); distinct True
- k H2/HD column 6: PositiveRatio(numerator=PositiveCount(value=1462), denominator=PositiveCount(value=951)); distinct True
- k H2/HD column 10: PositiveRatio(numerator=PositiveCount(value=10547), denominator=PositiveCount(value=10550)); distinct True
- k H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=1183662), denominator=PositiveCount(value=1183967)); distinct True
- k H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=234437), denominator=PositiveCount(value=165885)); distinct True
- k H2/D2 column 3: PositiveRatio(numerator=PositiveCount(value=6729), denominator=PositiveCount(value=3388)); distinct True

- k H2/D2 column 4: PositiveRatio(numerator=PositiveCount(value=495), denominator=PositiveCount(value=254)); distinct True
- k H2/D2 column 5: PositiveRatio(numerator=PositiveCount(value=30074), denominator=PositiveCount(value=15075)); distinct True
- k H2/D2 column 6: PositiveRatio(numerator=PositiveCount(value=731), denominator=PositiveCount(value=283)); distinct True
- k H2/D2 column 8: PositiveRatio(numerator=PositiveCount(value=185), denominator=PositiveCount(value=46)); distinct True
- k H2/D2 column 10: PositiveRatio(numerator=PositiveCount(value=10547), denominator=PositiveCount(value=10538)); distinct True
- k HD/D2 column 1: PositiveRatio(numerator=PositiveCount(value=1183842), denominator=PositiveCount(value=1183967)); distinct True
- k HD/D2 column 2: PositiveRatio(numerator=PositiveCount(value=203056), denominator=PositiveCount(value=165885)); distinct True
- k HD/D2 column 3: PositiveRatio(numerator=PositiveCount(value=1259), denominator=PositiveCount(value=847)); distinct True
- k HD/D2 column 5: PositiveRatio(numerator=PositiveCount(value=20548), denominator=PositiveCount(value=15075)); distinct True
- k HD/D2 column 6: PositiveRatio(numerator=PositiveCount(value=951), denominator=PositiveCount(value=566)); distinct True
- k HD/D2 column 10: PositiveRatio(numerator=PositiveCount(value=5275), denominator=PositiveCount(value=5269)); distinct True
- f H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=23341), denominator=PositiveCount(value=23328)); distinct True
- f H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=5359), denominator=PositiveCount(value=4045)); distinct True
- f H2/D2 column 5: PositiveRatio(numerator=PositiveCount(value=1350), denominator=PositiveCount(value=733)); distinct True
- f H2/D2 column 10: PositiveRatio(numerator=PositiveCount(value=1110), denominator=PositiveCount(value=1069)); distinct True
- j H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=1135330), denominator=PositiveCount(value=1141941)); distinct True
- i H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=113132), denominator=PositiveCount(value=113093)); distinct True
- i H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=45071), denominator=PositiveCount(value=30838)); distinct True
- g H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=70534), denominator=PositiveCount(value=70535)); distinct True
- g H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=114543), denominator=PositiveCount(value=75565)); distinct True
- d H2/HD column 1: PositiveRatio(numerator=PositiveCount(value=1127003), denominator=PositiveCount(value=1127174)); distinct True
- d H2/HD column 2: PositiveRatio(numerator=PositiveCount(value=237158), denominator=PositiveCount(value=205459)); distinct True
- d H2/HD column 3: PositiveRatio(numerator=PositiveCount(value=2209), denominator=PositiveCount(value=1658)); distinct True
- d H2/HD column 5: PositiveRatio(numerator=PositiveCount(value=15182), denominator=PositiveCount(value=11405)); distinct True
- d H2/HD column 6: PositiveRatio(numerator=PositiveCount(value=103), denominator=PositiveCount(value=68)); distinct True

- d H2/HD column 10: PositiveRatio(numerator=PositiveCount(value=10496), denominator=PositiveCount(value=10489)); distinct True
- d H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=1127003), denominator=PositiveCount(value=1127298)); distinct True
- d H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=118579), denominator=PositiveCount(value=83911)); distinct True
- d H2/D2 column 3: PositiveRatio(numerator=PositiveCount(value=2209), denominator=PositiveCount(value=1098)); distinct True
- d H2/D2 column 4: PositiveRatio(numerator=PositiveCount(value=11), denominator=PositiveCount(value=3)); distinct True
- d H2/D2 column 5: PositiveRatio(numerator=PositiveCount(value=7591), denominator=PositiveCount(value=3800)); distinct True
- d H2/D2 column 6: PositiveRatio(numerator=PositiveCount(value=515), denominator=PositiveCount(value=184)); distinct True
- d H2/D2 column 8: PositiveRatio(numerator=PositiveCount(value=19100), denominator=PositiveCount(value=49)); distinct True
- d H2/D2 column 10: PositiveRatio(numerator=PositiveCount(value=5248), denominator=PositiveCount(value=5247)); distinct True
- d HD/D2 column 1: PositiveRatio(numerator=PositiveCount(value=563587), denominator=PositiveCount(value=563649)); distinct True
- d HD/D2 column 2: PositiveRatio(numerator=PositiveCount(value=205459), denominator=PositiveCount(value=167822)); distinct True
- d HD/D2 column 3: PositiveRatio(numerator=PositiveCount(value=829), denominator=PositiveCount(value=549)); distinct True
- d HD/D2 column 5: PositiveRatio(numerator=PositiveCount(value=2281), denominator=PositiveCount(value=1520)); distinct True
- d HD/D2 column 6: PositiveRatio(numerator=PositiveCount(value=85), denominator=PositiveCount(value=46)); distinct True
- d HD/D2 column 10: PositiveRatio(numerator=PositiveCount(value=10489), denominator=PositiveCount(value=10494)); distinct True
- e H2/HD column 1: PositiveRatio(numerator=PositiveCount(value=1077747), denominator=PositiveCount(value=1077766)); distinct True
- e H2/HD column 2: PositiveRatio(numerator=PositiveCount(value=219613), denominator=PositiveCount(value=190517)); distinct True
- e H2/HD column 3: PositiveRatio(numerator=PositiveCount(value=14), denominator=PositiveCount(value=11)); distinct True
- e H2/HD column 5: PositiveRatio(numerator=PositiveCount(value=4550), denominator=PositiveCount(value=3461)); distinct True
- e H2/HD column 6: PositiveRatio(numerator=PositiveCount(value=3), denominator=PositiveCount(value=2)); distinct True
- e H2/HD column 10: PositiveRatio(numerator=PositiveCount(value=11070), denominator=PositiveCount(value=10993)); distinct True
- e H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=1077747), denominator=PositiveCount(value=1077740)); distinct True
- e H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=219613), denominator=PositiveCount(value=155664)); distinct True
- e H2/D2 column 3: PositiveRatio(numerator=PositiveCount(value=940), denominator=PositiveCount(value=493)); distinct True
- e H2/D2 column 5: PositiveRatio(numerator=PositiveCount(value=6825), denominator=PositiveCount(value=3464)); distinct True

- e H2/D2 column 6: PositiveRatio(numerator=PositiveCount(value=1515), denominator=PositiveCount(value=451)); distinct True
- e H2/D2 column 10: PositiveRatio(numerator=PositiveCount(value=11070), denominator=PositiveCount(value=10991)); distinct True
- e HD/D2 column 1: PositiveRatio(numerator=PositiveCount(value=538883), denominator=PositiveCount(value=538870)); distinct True
- e HD/D2 column 2: PositiveRatio(numerator=PositiveCount(value=190517), denominator=PositiveCount(value=155664)); distinct True
- e HD/D2 column 3: PositiveRatio(numerator=PositiveCount(value=5170), denominator=PositiveCount(value=3451)); distinct True
- e HD/D2 column 5: PositiveRatio(numerator=PositiveCount(value=10383), denominator=PositiveCount(value=6928)); distinct True
- e HD/D2 column 6: PositiveRatio(numerator=PositiveCount(value=1010), denominator=PositiveCount(value=451)); distinct True
- e HD/D2 column 10: PositiveRatio(numerator=PositiveCount(value=10993), denominator=PositiveCount(value=10991)); distinct True
- a H2/HD column 1: PositiveRatio(numerator=PositiveCount(value=959361), denominator=PositiveCount(value=959471)); distinct True
- a H2/HD column 2: PositiveRatio(numerator=PositiveCount(value=266483), denominator=PositiveCount(value=230844)); distinct True
- a H2/HD column 3: PositiveRatio(numerator=PositiveCount(value=7165), denominator=PositiveCount(value=5377)); distinct True
- a H2/HD column 5: PositiveRatio(numerator=PositiveCount(value=34216), denominator=PositiveCount(value=25685)); distinct True
- a H2/HD column 6: PositiveRatio(numerator=PositiveCount(value=1671), denominator=PositiveCount(value=1099)); distinct True
- a H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=4796805), denominator=PositiveCount(value=4797904)); distinct True
- a H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=266483), denominator=PositiveCount(value=188584)); distinct True
- a H2/D2 column 3: PositiveRatio(numerator=PositiveCount(value=7165), denominator=PositiveCount(value=3596)); distinct True
- a H2/D2 column 4: PositiveRatio(numerator=PositiveCount(value=46), denominator=PositiveCount(value=17)); distinct True
- a H2/D2 column 5: PositiveRatio(numerator=PositiveCount(value=34216), denominator=PositiveCount(value=17109)); distinct True
- a H2/D2 column 6: PositiveRatio(numerator=PositiveCount(value=557), denominator=PositiveCount(value=202)); distinct True
- a H2/D2 column 8: PositiveRatio(numerator=PositiveCount(value=216), denominator=PositiveCount(value=55)); distinct True
- a H2/D2 column 10: PositiveRatio(numerator=PositiveCount(value=98879), denominator=PositiveCount(value=98910)); distinct True
- a HD/D2 column 1: PositiveRatio(numerator=PositiveCount(value=4797355), denominator=PositiveCount(value=4797904)); distinct True
- a HD/D2 column 2: PositiveRatio(numerator=PositiveCount(value=57711), denominator=PositiveCount(value=47146)); distinct True
- a HD/D2 column 3: PositiveRatio(numerator=PositiveCount(value=5377), denominator=PositiveCount(value=3596)); distinct True

- a HD/D2 column 5: PositiveRatio(numerator=PositiveCount(value=25685), denominator=PositiveCount(value=17109)); distinct True
- a HD/D2 column 6: PositiveRatio(numerator=PositiveCount(value=1099), denominator=PositiveCount(value=606)); distinct True
- c H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=958385), denominator=PositiveCount(value=967318)); distinct True
- c H2/D2 column 5: PositiveRatio(numerator=PositiveCount(value=6214), denominator=PositiveCount(value=3061)); distinct True
- c H2/D2 column 8: PositiveRatio(numerator=PositiveCount(value=975), denominator=PositiveCount(value=257)); distinct True
- c H2/D2 column 10: PositiveRatio(numerator=PositiveCount(value=5188), denominator=PositiveCount(value=5229)); distinct True
- D" H2/HD column 1: PositiveRatio(numerator=PositiveCount(value=606055), denominator=PositiveCount(value=606156)); distinct True
- D" H2/HD column 2: PositiveRatio(numerator=PositiveCount(value=231992), denominator=PositiveCount(value=200617)); distinct True
- D" H2/HD column 3: PositiveRatio(numerator=PositiveCount(value=63041), denominator=PositiveCount(value=45801)); distinct True
- D" H2/HD column 5: PositiveRatio(numerator=PositiveCount(value=6152), denominator=PositiveCount(value=4573)); distinct True
- D" H2/HD column 8: PositiveRatio(numerator=PositiveCount(value=10), denominator=PositiveCount(value=7)); distinct True
- D" H2/HD column 10: PositiveRatio(numerator=PositiveCount(value=10430), denominator=PositiveCount(value=10477)); distinct True
- D" H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=242422), denominator=PositiveCount(value=242455)); distinct True
- D" H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=57998), denominator=PositiveCount(value=41217)); distinct True
- D" H2/D2 column 3: PositiveRatio(numerator=PositiveCount(value=521), denominator=PositiveCount(value=278)); distinct True
- D" H2/D2 column 5: PositiveRatio(numerator=PositiveCount(value=30760), denominator=PositiveCount(value=15133)); distinct True
- D" H2/D2 column 6: PositiveRatio(numerator=PositiveCount(value=14500), denominator=PositiveCount(value=6521)); distinct True
- D" H2/D2 column 8: PositiveRatio(numerator=PositiveCount(value=375), denominator=PositiveCount(value=88)); distinct True
- D" H2/D2 column 10: PositiveRatio(numerator=PositiveCount(value=10430), denominator=PositiveCount(value=10517)); distinct True
- D" HD/D2 column 1: PositiveRatio(numerator=PositiveCount(value=1212312), denominator=PositiveCount(value=1212275)); distinct True
- D" HD/D2 column 2: PositiveRatio(numerator=PositiveCount(value=200617), denominator=PositiveCount(value=164868)); distinct True
- D" HD/D2 column 3: PositiveRatio(numerator=PositiveCount(value=45801), denominator=PositiveCount(value=33638)); distinct True
- D" HD/D2 column 5: PositiveRatio(numerator=PositiveCount(value=22865), denominator=PositiveCount(value=15133)); distinct True
- D" HD/D2 column 8: PositiveRatio(numerator=PositiveCount(value=525), denominator=PositiveCount(value=176)); distinct True

- D" HD/D2 column 10: PositiveRatio(numerator=PositiveCount(value=10477), denominator=PositiveCount(value=10517)); distinct True
- D' H2/HD column 1: PositiveRatio(numerator=PositiveCount(value=1188653), denominator=PositiveCount(value=1188792)); distinct True
- D' H2/HD column 2: PositiveRatio(numerator=PositiveCount(value=232997), denominator=PositiveCount(value=201491)); distinct True
- D' H2/HD column 3: PositiveRatio(numerator=PositiveCount(value=31570), denominator=PositiveCount(value=23509)); distinct True
- D' H2/HD column 5: PositiveRatio(numerator=PositiveCount(value=2989), denominator=PositiveCount(value=2235)); distinct True
- D' H2/HD column 6: PositiveRatio(numerator=PositiveCount(value=111), denominator=PositiveCount(value=125)); distinct True
- D' H2/HD column 8: PositiveRatio(numerator=PositiveCount(value=25), denominator=PositiveCount(value=22)); distinct True
- D' H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=1188653), denominator=PositiveCount(value=1188879)); distinct True
- D' H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=232997), denominator=PositiveCount(value=165315)); distinct True
- D' H2/D2 column 3: PositiveRatio(numerator=PositiveCount(value=6314), denominator=PositiveCount(value=3335)); distinct True
- D' H2/D2 column 5: PositiveRatio(numerator=PositiveCount(value=29890), denominator=PositiveCount(value=15041)); distinct True
- D' H2/D2 column 6: PositiveRatio(numerator=PositiveCount(value=925), denominator=PositiveCount(value=459)); distinct True
- D' H2/D2 column 8: PositiveRatio(numerator=PositiveCount(value=2500), denominator=PositiveCount(value=323)); distinct True
- D' HD/D2 column 1: PositiveRatio(numerator=PositiveCount(value=396264), denominator=PositiveCount(value=396293)); distinct True
- D' HD/D2 column 2: PositiveRatio(numerator=PositiveCount(value=201491), denominator=PositiveCount(value=165315)); distinct True
- D' HD/D2 column 3: PositiveRatio(numerator=PositiveCount(value=23509), denominator=PositiveCount(value=16675)); distinct True
- D' HD/D2 column 4: PositiveRatio(numerator=PositiveCount(value=633), denominator=PositiveCount(value=1130)); distinct True
- D' HD/D2 column 5: PositiveRatio(numerator=PositiveCount(value=22350), denominator=PositiveCount(value=15041)); distinct True
- D' HD/D2 column 6: PositiveRatio(numerator=PositiveCount(value=3125), denominator=PositiveCount(value=1377)); distinct True
- D' HD/D2 column 8: PositiveRatio(numerator=PositiveCount(value=2200), denominator=PositiveCount(value=323)); distinct True
- D' HD/D2 column 10: PositiveRatio(numerator=PositiveCount(value=212), denominator=PositiveCount(value=211)); distinct True
- B" H2/HD column 1: PositiveRatio(numerator=PositiveCount(value=1179845), denominator=PositiveCount(value=1179804)); distinct True
- B" H2/HD column 2: PositiveRatio(numerator=PositiveCount(value=7325), denominator=PositiveCount(value=6322)); distinct True

- B" H2/HD column 3: PositiveRatio(numerator=PositiveCount(value=5678), denominator=PositiveCount(value=4077)); distinct True
- B" H2/HD column 5: PositiveRatio(numerator=PositiveCount(value=1334), denominator=PositiveCount(value=1017)); distinct True
- B" H2/HD column 6: PositiveRatio(numerator=PositiveCount(value=595), denominator=PositiveCount(value=199)); distinct True
- B" H2/HD column 8: PositiveRatio(numerator=PositiveCount(value=34), denominator=PositiveCount(value=25)); distinct True
- B" H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=1179845), denominator=PositiveCount(value=1179707)); distinct True
- B" H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=109875), denominator=PositiveCount(value=78151)); distinct True
- B" H2/D2 column 3: PositiveRatio(numerator=PositiveCount(value=8517), denominator=PositiveCount(value=4427)); distinct True
- B" H2/D2 column 5: PositiveRatio(numerator=PositiveCount(value=232), denominator=PositiveCount(value=119)); distinct True
- B" H2/D2 column 6: PositiveRatio(numerator=PositiveCount(value=350), denominator=PositiveCount(value=113)); distinct True
- B" H2/D2 column 8: PositiveRatio(numerator=PositiveCount(value=425), denominator=PositiveCount(value=3)); distinct True
- B" HD/D2 column 1: PositiveRatio(numerator=PositiveCount(value=1179804), denominator=PositiveCount(value=1179707)); distinct True
- B" HD/D2 column 2: PositiveRatio(numerator=PositiveCount(value=94830), denominator=PositiveCount(value=78151)); distinct True
- B" HD/D2 column 3: PositiveRatio(numerator=PositiveCount(value=12231), denominator=PositiveCount(value=8854)); distinct True
- B" HD/D2 column 5: PositiveRatio(numerator=PositiveCount(value=4068), denominator=PositiveCount(value=2737)); distinct True
- B" HD/D2 column 6: PositiveRatio(numerator=PositiveCount(value=1990), denominator=PositiveCount(value=1921)); distinct True
- B" HD/D2 column 8: PositiveRatio(numerator=PositiveCount(value=625), denominator=PositiveCount(value=6)); distinct True
- B" HD/D2 column 10: PositiveRatio(numerator=PositiveCount(value=1111), denominator=PositiveCount(value=1106)); distinct True
- M H2/HD column 1: PositiveRatio(numerator=PositiveCount(value=16355), denominator=PositiveCount(value=16439)); distinct True
- M H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=228970), denominator=PositiveCount(value=229009)); distinct True
- M HD/D2 column 1: PositiveRatio(numerator=PositiveCount(value=230146), denominator=PositiveCount(value=229009)); distinct True
- M HD/D2 column 5: PositiveRatio(numerator=PositiveCount(value=13), denominator=PositiveCount(value=5)); distinct True
- M HD/D2 column 10: PositiveRatio(numerator=PositiveCount(value=155), denominator=PositiveCount(value=206)); distinct True
- D H2/HD column 1: PositiveRatio(numerator=PositiveCount(value=1138887), denominator=PositiveCount(value=1139017)); distinct True
- D H2/HD column 2: PositiveRatio(numerator=PositiveCount(value=235991), denominator=PositiveCount(value=203913)); distinct True

- D H2/HD column 3: PositiveRatio(numerator=PositiveCount(value=6256), denominator=PositiveCount(value=4447)); distinct True
- D H2/HD column 5: PositiveRatio(numerator=PositiveCount(value=15148), denominator=PositiveCount(value=11455)); distinct True
- D H2/HD column 6: PositiveRatio(numerator=PositiveCount(value=142), denominator=PositiveCount(value=97)); distinct True
- D H2/HD column 8: PositiveRatio(numerator=PositiveCount(value=67), denominator=PositiveCount(value=40)); distinct True
- D H2/HD column 10: PositiveRatio(numerator=PositiveCount(value=5254), denominator=PositiveCount(value=5235)); distinct True
- D H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=1138887), denominator=PositiveCount(value=1139140)); distinct True
- D H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=235991), denominator=PositiveCount(value=166760)); distinct True
- D H2/D2 column 3: PositiveRatio(numerator=PositiveCount(value=68816), denominator=PositiveCount(value=33343)); distinct True
- D H2/D2 column 5: PositiveRatio(numerator=PositiveCount(value=15148), denominator=PositiveCount(value=7555)); distinct True
- D H2/D2 column 6: PositiveRatio(numerator=PositiveCount(value=71), denominator=PositiveCount(value=27)); distinct True
- D H2/D2 column 8: PositiveRatio(numerator=PositiveCount(value=201), denominator=PositiveCount(value=50)); distinct True
- D H2/D2 column 10: PositiveRatio(numerator=PositiveCount(value=5254), denominator=PositiveCount(value=5265)); distinct True
- D HD/D2 column 1: PositiveRatio(numerator=PositiveCount(value=1139017), denominator=PositiveCount(value=1139140)); distinct True
- D HD/D2 column 2: PositiveRatio(numerator=PositiveCount(value=203913), denominator=PositiveCount(value=166760)); distinct True
- D HD/D2 column 3: PositiveRatio(numerator=PositiveCount(value=48917), denominator=PositiveCount(value=33343)); distinct True
- D HD/D2 column 5: PositiveRatio(numerator=PositiveCount(value=2291), denominator=PositiveCount(value=1511)); distinct True
- D HD/D2 column 6: PositiveRatio(numerator=PositiveCount(value=97), denominator=PositiveCount(value=54)); distinct True
- D HD/D2 column 8: PositiveRatio(numerator=PositiveCount(value=12), denominator=PositiveCount(value=5)); distinct True
- D HD/D2 column 10: PositiveRatio(numerator=PositiveCount(value=349), denominator=PositiveCount(value=351)); distinct True
- J H2/HD column 1: PositiveRatio(numerator=PositiveCount(value=56775), denominator=PositiveCount(value=56768)); distinct True
- J H2/HD column 2: PositiveRatio(numerator=PositiveCount(value=46823), denominator=PositiveCount(value=36656)); distinct True
- I H2/HD column 1: PositiveRatio(numerator=PositiveCount(value=56571), denominator=PositiveCount(value=56555)); distinct True
- I H2/HD column 2: PositiveRatio(numerator=PositiveCount(value=225915), denominator=PositiveCount(value=196214)); distinct True
- I H2/HD column 3: PositiveRatio(numerator=PositiveCount(value=7841), denominator=PositiveCount(value=5821)); distinct True
- I H2/HD column 5: PositiveRatio(numerator=PositiveCount(value=29259), denominator=PositiveCount(value=22360)); distinct True

- I H2/HD column 6: PositiveRatio(numerator=PositiveCount(value=72), denominator=PositiveCount(value=55)); distinct True
- I H2/HD column 8: PositiveRatio(numerator=PositiveCount(value=18), denominator=PositiveCount(value=7)); distinct True
- I H2/HD column 10: PositiveRatio(numerator=PositiveCount(value=10693), denominator=PositiveCount(value=10590)); distinct True
- I H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=226284), denominator=PositiveCount(value=226163)); distinct True
- I H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=75305), denominator=PositiveCount(value=53338)); distinct True
- I H2/D2 column 3: PositiveRatio(numerator=PositiveCount(value=7841), denominator=PositiveCount(value=3942)); distinct True
- I H2/D2 column 5: PositiveRatio(numerator=PositiveCount(value=9753), denominator=PositiveCount(value=4913)); distinct True
- I H2/D2 column 6: PositiveRatio(numerator=PositiveCount(value=792), denominator=PositiveCount(value=263)); distinct True
- I H2/D2 column 8: PositiveRatio(numerator=PositiveCount(value=36), denominator=PositiveCount(value=5)); distinct True
- I H2/D2 column 10: PositiveRatio(numerator=PositiveCount(value=10693), denominator=PositiveCount(value=10657)); distinct True
- I HD/D2 column 1: PositiveRatio(numerator=PositiveCount(value=226220), denominator=PositiveCount(value=226163)); distinct True
- I HD/D2 column 2: PositiveRatio(numerator=PositiveCount(value=98107), denominator=PositiveCount(value=80007)); distinct True
- I HD/D2 column 3: PositiveRatio(numerator=PositiveCount(value=5821), denominator=PositiveCount(value=3942)); distinct True
- I HD/D2 column 5: PositiveRatio(numerator=PositiveCount(value=22360), denominator=PositiveCount(value=14739)); distinct True
- I HD/D2 column 6: PositiveRatio(numerator=PositiveCount(value=605), denominator=PositiveCount(value=263)); distinct True
- I HD/D2 column 8: PositiveRatio(numerator=PositiveCount(value=14), denominator=PositiveCount(value=5)); distinct True
- I HD/D2 column 10: PositiveRatio(numerator=PositiveCount(value=10590), denominator=PositiveCount(value=10657)); distinct True
- G H2/HD column 1: PositiveRatio(numerator=PositiveCount(value=112834), denominator=PositiveCount(value=112843)); distinct True
- G H2/HD column 2: PositiveRatio(numerator=PositiveCount(value=23439), denominator=PositiveCount(value=18799)); distinct True
- G H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=112834), denominator=PositiveCount(value=112893)); distinct True
- G H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=23439), denominator=PositiveCount(value=14408)); distinct True
- G HD/D2 column 1: PositiveRatio(numerator=PositiveCount(value=112843), denominator=PositiveCount(value=112893)); distinct True
- G HD/D2 column 2: PositiveRatio(numerator=PositiveCount(value=18799), denominator=PositiveCount(value=14408)); distinct True
- K H2/HD column 1: PositiveRatio(numerator=PositiveCount(value=112669), denominator=PositiveCount(value=112663)); distinct True
- K H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=112669), denominator=PositiveCount(value=112610)); distinct True

- K H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=223259), denominator=PositiveCount(value=166000)); distinct True
- K H2/D2 column 5: PositiveRatio(numerator=PositiveCount(value=18), denominator=PositiveCount(value=11)); distinct True
- K H2/D2 column 10: PositiveRatio(numerator=PositiveCount(value=176), denominator=PositiveCount(value=159)); distinct True
- K HD/D2 column 1: PositiveRatio(numerator=PositiveCount(value=112663), denominator=PositiveCount(value=112610)); distinct True
- B' H2/HD column 1: PositiveRatio(numerator=PositiveCount(value=1116428), denominator=PositiveCount(value=1116497)); distinct True
- B' H2/HD column 2: PositiveRatio(numerator=PositiveCount(value=1821), denominator=PositiveCount(value=1585)); distinct True
- B' H2/HD column 3: PositiveRatio(numerator=PositiveCount(value=41703), denominator=PositiveCount(value=33830)); distinct True
- B' H2/HD column 5: PositiveRatio(numerator=PositiveCount(value=5341), denominator=PositiveCount(value=4000)); distinct True
- B' H2/HD column 6: PositiveRatio(numerator=PositiveCount(value=2781), denominator=PositiveCount(value=1280)); distinct True
- B' H2/HD column 8: PositiveRatio(numerator=PositiveCount(value=15), denominator=PositiveCount(value=2)); distinct True
- B' H2/HD column 10: PositiveRatio(numerator=PositiveCount(value=1399), denominator=PositiveCount(value=1400)); distinct True
- B' H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=558214), denominator=PositiveCount(value=558211)); distinct True
- B' H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=101976), denominator=PositiveCount(value=72599)); distinct True
- B' H2/D2 column 3: PositiveRatio(numerator=PositiveCount(value=83406), denominator=PositiveCount(value=45679)); distinct True
- B' H2/D2 column 5: PositiveRatio(numerator=PositiveCount(value=5341), denominator=PositiveCount(value=2721)); distinct True
- B' H2/D2 column 6: PositiveRatio(numerator=PositiveCount(value=2781), denominator=PositiveCount(value=920)); distinct True
- B' H2/D2 column 8: PositiveRatio(numerator=PositiveCount(value=240), denominator=PositiveCount(value=83)); distinct True
- B' H2/D2 column 10: PositiveRatio(numerator=PositiveCount(value=2798), denominator=PositiveCount(value=2773)); distinct True
- B' HD/D2 column 1: PositiveRatio(numerator=PositiveCount(value=1116497), denominator=PositiveCount(value=1116422)); distinct True
- B' HD/D2 column 2: PositiveRatio(numerator=PositiveCount(value=88760), denominator=PositiveCount(value=72599)); distinct True
- B' HD/D2 column 3: PositiveRatio(numerator=PositiveCount(value=3980), denominator=PositiveCount(value=2687)); distinct True
- B' HD/D2 column 4: PositiveRatio(numerator=PositiveCount(value=915), denominator=PositiveCount(value=524)); distinct True
- B' HD/D2 column 5: PositiveRatio(numerator=PositiveCount(value=4000), denominator=PositiveCount(value=2721)); distinct True
- B' HD/D2 column 6: PositiveRatio(numerator=PositiveCount(value=32), denominator=PositiveCount(value=23)); distinct True

- B' HD/D2 column 8: PositiveRatio(numerator=PositiveCount(value=32), denominator=PositiveCount(value=83)); distinct True
- B' HD/D2 column 10: PositiveRatio(numerator=PositiveCount(value=2800), denominator=PositiveCount(value=2773)); distinct True
- F H2/HD column 1: PositiveRatio(numerator=PositiveCount(value=67274), denominator=PositiveCount(value=67285)); distinct True
- F H2/HD column 2: PositiveRatio(numerator=PositiveCount(value=1090), denominator=PositiveCount(value=989)); distinct True
- F H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=504555), denominator=PositiveCount(value=504656)); distinct True
- F H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=1090), denominator=PositiveCount(value=781)); distinct True
- F HD/D2 column 1: PositiveRatio(numerator=PositiveCount(value=1009275), denominator=PositiveCount(value=1009312)); distinct True
- F HD/D2 column 2: PositiveRatio(numerator=PositiveCount(value=989), denominator=PositiveCount(value=781)); distinct True
- E H2/HD column 1: PositiveRatio(numerator=PositiveCount(value=1000823), denominator=PositiveCount(value=1001204)); distinct True
- E H2/HD column 2: PositiveRatio(numerator=PositiveCount(value=25889), denominator=PositiveCount(value=22044)); distinct True
- E H2/HD column 3: PositiveRatio(numerator=PositiveCount(value=435), denominator=PositiveCount(value=272)); distinct True
- E H2/HD column 5: PositiveRatio(numerator=PositiveCount(value=4085), denominator=PositiveCount(value=3071)); distinct True
- E H2/HD column 6: PositiveRatio(numerator=PositiveCount(value=909), denominator=PositiveCount(value=644)); distinct True
- E H2/HD column 8: PositiveRatio(numerator=PositiveCount(value=76), denominator=PositiveCount(value=41)); distinct True
- E H2/HD column 10: PositiveRatio(numerator=PositiveCount(value=10118), denominator=PositiveCount(value=10107)); distinct True
- E H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=1000823), denominator=PositiveCount(value=1001281)); distinct True
- E H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=129445), denominator=PositiveCount(value=89221)); distinct True
- E H2/D2 column 3: PositiveRatio(numerator=PositiveCount(value=2900), denominator=PositiveCount(value=1069)); distinct True
- E H2/D2 column 5: PositiveRatio(numerator=PositiveCount(value=20425), denominator=PositiveCount(value=10231)); distinct True
- E H2/D2 column 6: PositiveRatio(numerator=PositiveCount(value=4545), denominator=PositiveCount(value=1691)); distinct True
- E H2/D2 column 8: PositiveRatio(numerator=PositiveCount(value=38), denominator=PositiveCount(value=9)); distinct True
- E H2/D2 column 10: PositiveRatio(numerator=PositiveCount(value=25295), denominator=PositiveCount(value=25281)); distinct True
- E HD/D2 column 1: PositiveRatio(numerator=PositiveCount(value=1001204), denominator=PositiveCount(value=1001281)); distinct True
- E HD/D2 column 2: PositiveRatio(numerator=PositiveCount(value=10020), denominator=PositiveCount(value=8111)); distinct True
- E HD/D2 column 3: PositiveRatio(numerator=PositiveCount(value=5440), denominator=PositiveCount(value=3207)); distinct True

- E HD/D2 column 5: PositiveRatio(numerator=PositiveCount(value=15355), denominator=PositiveCount(value=10231)); distinct True
- E HD/D2 column 6: PositiveRatio(numerator=PositiveCount(value=3220), denominator=PositiveCount(value=1691)); distinct True
- E HD/D2 column 8: PositiveRatio(numerator=PositiveCount(value=41), denominator=PositiveCount(value=18)); distinct True
- E HD/D2 column 10: PositiveRatio(numerator=PositiveCount(value=5615), denominator=PositiveCount(value=5618)); distinct True
- C H2/HD column 1: PositiveRatio(numerator=PositiveCount(value=1000898), denominator=PositiveCount(value=1000929)); distinct True
- C H2/HD column 2: PositiveRatio(numerator=PositiveCount(value=81459), denominator=PositiveCount(value=70655)); distinct True
- C H2/HD column 3: PositiveRatio(numerator=PositiveCount(value=34762), denominator=PositiveCount(value=26655)); distinct True
- C H2/HD column 5: PositiveRatio(numerator=PositiveCount(value=313629), denominator=PositiveCount(value=235220)); distinct True
- C H2/HD column 6: PositiveRatio(numerator=PositiveCount(value=16647), denominator=PositiveCount(value=10960)); distinct True
- C H2/HD column 8: PositiveRatio(numerator=PositiveCount(value=223), denominator=PositiveCount(value=149)); distinct True
- C H2/HD column 10: PositiveRatio(numerator=PositiveCount(value=9389), denominator=PositiveCount(value=9390)); distinct True
- C H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=500449), denominator=PositiveCount(value=500486)); distinct True
- C H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=81459), denominator=PositiveCount(value=57664)); distinct True
- C H2/D2 column 3: PositiveRatio(numerator=PositiveCount(value=69524), denominator=PositiveCount(value=34917)); distinct True
- C H2/D2 column 5: PositiveRatio(numerator=PositiveCount(value=313629), denominator=PositiveCount(value=156731)); distinct True
- C H2/D2 column 6: PositiveRatio(numerator=PositiveCount(value=5549), denominator=PositiveCount(value=1893)); distinct True
- C H2/D2 column 8: PositiveRatio(numerator=PositiveCount(value=1115), denominator=PositiveCount(value=266)); distinct True
- C H2/D2 column 10: PositiveRatio(numerator=PositiveCount(value=103279), denominator=PositiveCount(value=103346)); distinct True
- C HD/D2 column 1: PositiveRatio(numerator=PositiveCount(value=1000929), denominator=PositiveCount(value=1000972)); distinct True
- C HD/D2 column 2: PositiveRatio(numerator=PositiveCount(value=70655), denominator=PositiveCount(value=57664)); distinct True
- C HD/D2 column 3: PositiveRatio(numerator=PositiveCount(value=17770), denominator=PositiveCount(value=11639)); distinct True
- C HD/D2 column 4: PositiveRatio(numerator=PositiveCount(value=1640), denominator=PositiveCount(value=653)); distinct True
- C HD/D2 column 5: PositiveRatio(numerator=PositiveCount(value=12380), denominator=PositiveCount(value=8249)); distinct True

- C HD/D2 column 6: PositiveRatio(numerator=PositiveCount(value=10960), denominator=PositiveCount(value=5679)); distinct True
- C HD/D2 column 8: PositiveRatio(numerator=PositiveCount(value=745), denominator=PositiveCount(value=266)); distinct True
- C HD/D2 column 10: PositiveRatio(numerator=PositiveCount(value=51645), denominator=PositiveCount(value=51673)); distinct True
- B H2/HD column 1: PositiveRatio(numerator=PositiveCount(value=917000), denominator=PositiveCount(value=916983)); distinct True
- B H2/HD column 2: PositiveRatio(numerator=PositiveCount(value=135809), denominator=PositiveCount(value=117716)); distinct True
- B H2/HD column 3: PositiveRatio(numerator=PositiveCount(value=10444), denominator=PositiveCount(value=7795)); distinct True
- B H2/HD column 5: PositiveRatio(numerator=PositiveCount(value=100077), denominator=PositiveCount(value=75355)); distinct True
- B H2/HD column 6: PositiveRatio(numerator=PositiveCount(value=2369), denominator=PositiveCount(value=1640)); distinct True
- B H2/HD column 8: PositiveRatio(numerator=PositiveCount(value=1625), denominator=PositiveCount(value=882)); distinct True
- B H2/HD column 10: PositiveRatio(numerator=PositiveCount(value=64641), denominator=PositiveCount(value=64520)); distinct True
- B H2/D2 column 1: PositiveRatio(numerator=PositiveCount(value=32750), denominator=PositiveCount(value=32749)); distinct True
- B H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=135809), denominator=PositiveCount(value=96308)); distinct True
- B H2/D2 column 3: PositiveRatio(numerator=PositiveCount(value=10444), denominator=PositiveCount(value=5519)); distinct True
- B H2/D2 column 5: PositiveRatio(numerator=PositiveCount(value=33359), denominator=PositiveCount(value=16780)); distinct True
- B H2/D2 column 6: PositiveRatio(numerator=PositiveCount(value=11845), denominator=PositiveCount(value=4198)); distinct True
- B H2/D2 column 8: PositiveRatio(numerator=PositiveCount(value=125), denominator=PositiveCount(value=31)); distinct True
- B H2/D2 column 10: PositiveRatio(numerator=PositiveCount(value=64641), denominator=PositiveCount(value=64472)); distinct True
- B HD/D2 column 1: PositiveRatio(numerator=PositiveCount(value=916983), denominator=PositiveCount(value=916972)); distinct True
- B HD/D2 column 2: PositiveRatio(numerator=PositiveCount(value=29429), denominator=PositiveCount(value=24077)); distinct True
- B HD/D2 column 3: PositiveRatio(numerator=PositiveCount(value=7795), denominator=PositiveCount(value=5519)); distinct True
- B HD/D2 column 5: PositiveRatio(numerator=PositiveCount(value=15071), denominator=PositiveCount(value=10068)); distinct True
- B HD/D2 column 6: PositiveRatio(numerator=PositiveCount(value=4100), denominator=PositiveCount(value=2099)); distinct True
- B HD/D2 column 8: PositiveRatio(numerator=PositiveCount(value=882), denominator=PositiveCount(value=403)); distinct True
- B HD/D2 column 10: PositiveRatio(numerator=PositiveCount(value=8065), denominator=PositiveCount(value=8059)); distinct True

- X H2/HD column 2: PositiveRatio(numerator=PositiveCount(value=1467071), denominator=PositiveCount(value=1271050)); distinct True
- X H2/HD column 3: PositiveRatio(numerator=PositiveCount(value=60668), denominator=PositiveCount(value=45825)); distinct True
- X H2/HD column 5: PositiveRatio(numerator=PositiveCount(value=60853), denominator=PositiveCount(value=45655)); distinct True
- X H2/HD column 6: PositiveRatio(numerator=PositiveCount(value=15311), denominator=PositiveCount(value=9930)); distinct True
- X H2/HD column 8: PositiveRatio(numerator=PositiveCount(value=942), denominator=PositiveCount(value=521)); distinct True
- X H2/HD column 10: PositiveRatio(numerator=PositiveCount(value=37072), denominator=PositiveCount(value=37071)); distinct True
- X H2/D2 column 2: PositiveRatio(numerator=PositiveCount(value=1467071), denominator=PositiveCount(value=1038500)); distinct True
- X H2/D2 column 3: PositiveRatio(numerator=PositiveCount(value=30334), denominator=PositiveCount(value=15455)); distinct True
- X H2/D2 column 5: PositiveRatio(numerator=PositiveCount(value=304265), denominator=PositiveCount(value=152218)); distinct True
- X H2/D2 column 6: PositiveRatio(numerator=PositiveCount(value=15311), denominator=PositiveCount(value=5393)); distinct True
- X H2/D2 column 8: PositiveRatio(numerator=PositiveCount(value=4710), denominator=PositiveCount(value=1141)); distinct True
- X H2/D2 column 10: PositiveRatio(numerator=PositiveCount(value=9268), denominator=PositiveCount(value=9269)); distinct True
- X HD/D2 column 2: PositiveRatio(numerator=PositiveCount(value=25421), denominator=PositiveCount(value=20770)); distinct True
- X HD/D2 column 3: PositiveRatio(numerator=PositiveCount(value=9165), denominator=PositiveCount(value=6182)); distinct True
- X HD/D2 column 5: PositiveRatio(numerator=PositiveCount(value=228275), denominator=PositiveCount(value=152218)); distinct True
- X HD/D2 column 6: PositiveRatio(numerator=PositiveCount(value=9930), denominator=PositiveCount(value=5393)); distinct True
- X HD/D2 column 8: PositiveRatio(numerator=PositiveCount(value=2605), denominator=PositiveCount(value=1141)); distinct True
- X HD/D2 column 10: PositiveRatio(numerator=PositiveCount(value=37071), denominator=PositiveCount(value=37076)); distinct True
- count positive-exact: 551
- count source-zero-EmptyOne: 3
- count held-negative-inscription: 8
- count held-text: 223
- count blank-EmptyOne: 450
- molecular weight H2: 50397/25000
- molecular weight HD: 75551/25000
- molecular weight D2: 10070508889/2500000000

Falsification boundary: Reject if any molecular carrier, nuclear occurrence, isotope label, electronic support, electronic state or source record is erased; if a continuum or fitted scale selects the law; if any of 95 NIST state rows or 1,235 cells is omitted or changed; if H2, HD and D2 fail distinct exact nuclear support and ordered positive molecular-weight inscriptions; if any of 330 jointly reported positive vibronic coordinate pairs fails isotope-labelled distinction; if any of 450 blank cells or three source-zero inscriptions becomes a number; if any of eight external negative inscriptions is consumed as an SFT proof value; or if target content enters before sealing.

Measurement receipt: sha256:6f38ce80e651e2e112448e38d4cbd18d1af4d561e9356d66e4c8418c2a2bc91d.

Isolation certificate: sha256:4861eb36b015cf3532dcf85fd7308796f7a0f2deaf2b92cac59e6260a036afcb.

Custody certificate: sha256:1a7688ccd744c9cc086ad5d37b5fad1ff9f4c02459d69ab905bffcbbc19a41651.

Registered source descriptions:

- National Institute of Standards and Technology: <https://webbook.nist.gov/chemistry/> ()

Registered target identities:

- C1333740-nuclear-electronic-001 from None
- C1333740-nuclear-electronic-002 from None
- C1333740-nuclear-electronic-003 from None
- C1333740-nuclear-electronic-004 from None
- C1333740-nuclear-electronic-005 from None
- C1333740-nuclear-electronic-006 from None
- C1333740-nuclear-electronic-007 from None
- C1333740-nuclear-electronic-008 from None
- C1333740-nuclear-electronic-009 from None
- C1333740-nuclear-electronic-010 from None
- C1333740-nuclear-electronic-011 from None
- C1333740-nuclear-electronic-012 from None
- C1333740-nuclear-electronic-013 from None
- C1333740-nuclear-electronic-014 from None
- C1333740-nuclear-electronic-015 from None
- C1333740-nuclear-electronic-016 from None
- C1333740-nuclear-electronic-017 from None
- C1333740-nuclear-electronic-018 from None
- C1333740-nuclear-electronic-019 from None
- C1333740-nuclear-electronic-020 from None
- C1333740-nuclear-electronic-021 from None
- C1333740-nuclear-electronic-022 from None
- C1333740-nuclear-electronic-023 from None
- C1333740-nuclear-electronic-024 from None
- C1333740-nuclear-electronic-025 from None
- C1333740-nuclear-electronic-026 from None
- C1333740-nuclear-electronic-027 from None
- C1333740-nuclear-electronic-028 from None

- C1333740-nuclear-electronic-029 from None
- C1333740-nuclear-electronic-030 from None
- C1333740-nuclear-electronic-031 from None
- C1333740-nuclear-electronic-032 from None
- C1333740-nuclear-electronic-033 from None
- C1333740-nuclear-electronic-034 from None
- C1333740-nuclear-electronic-035 from None
- C1333740-nuclear-electronic-036 from None
- C1333740-nuclear-electronic-037 from None
- C1333740-nuclear-electronic-038 from None
- C1333740-nuclear-electronic-039 from None
- C1333740-nuclear-electronic-040 from None
- C1333740-nuclear-electronic-041 from None
- C1333740-nuclear-electronic-042 from None
- C1333740-nuclear-electronic-043 from None
- C1333740-nuclear-electronic-044 from None
- C1333740-nuclear-electronic-045 from None
- C1333740-nuclear-electronic-046 from None
- C13983205-nuclear-electronic-001 from None
- C13983205-nuclear-electronic-002 from None
- C13983205-nuclear-electronic-003 from None
- C13983205-nuclear-electronic-004 from None
- C13983205-nuclear-electronic-005 from None
- C13983205-nuclear-electronic-006 from None
- C13983205-nuclear-electronic-007 from None
- C13983205-nuclear-electronic-008 from None
- C13983205-nuclear-electronic-009 from None
- C13983205-nuclear-electronic-010 from None
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- C13983205-nuclear-electronic-013 from None
- C13983205-nuclear-electronic-014 from None
- C13983205-nuclear-electronic-015 from None
- C13983205-nuclear-electronic-016 from None
- C13983205-nuclear-electronic-017 from None
- C13983205-nuclear-electronic-018 from None
- C13983205-nuclear-electronic-019 from None
- C13983205-nuclear-electronic-020 from None

- C7782390-nuclear-electronic-001 from None
- C7782390-nuclear-electronic-002 from None
- C7782390-nuclear-electronic-003 from None
- C7782390-nuclear-electronic-004 from None
- C7782390-nuclear-electronic-005 from None
- C7782390-nuclear-electronic-006 from None
- C7782390-nuclear-electronic-007 from None
- C7782390-nuclear-electronic-008 from None
- C7782390-nuclear-electronic-009 from None
- C7782390-nuclear-electronic-010 from None
- C7782390-nuclear-electronic-011 from None
- C7782390-nuclear-electronic-012 from None
- C7782390-nuclear-electronic-013 from None
- C7782390-nuclear-electronic-014 from None
- C7782390-nuclear-electronic-015 from None
- C7782390-nuclear-electronic-016 from None
- C7782390-nuclear-electronic-017 from None
- C7782390-nuclear-electronic-018 from None
- C7782390-nuclear-electronic-019 from None
- C7782390-nuclear-electronic-020 from None
- C7782390-nuclear-electronic-021 from None
- C7782390-nuclear-electronic-022 from None
- C7782390-nuclear-electronic-023 from None
- C7782390-nuclear-electronic-024 from None
- C7782390-nuclear-electronic-025 from None
- C7782390-nuclear-electronic-026 from None
- C7782390-nuclear-electronic-027 from None
- C7782390-nuclear-electronic-028 from None
- C7782390-nuclear-electronic-029 from None

Meaning. The field-wide extension derives molecular electronic identity, state composition, occupancy, ordering, symmetry, correlation, transitions, measurement and operational classical-quantum correspondence from the same exact carrier laws. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; source glyph 0 denotes structural EmptyOne only
- no negative, irrational, imaginary, floating, signed or continuum proof magnitude
- no Born-Oppenheimer, continuum potential, differential equation, fitted mass, fitted scale or isotope coefficient
- no NIST molecular weights, state cells or isotopologue outcomes before prediction seal
- no common-state-only or favorable-value-only subset

Immutable identities. Source manifest

sha256:3a9bde4ec0e7a261d79ccf376209cc59cf7690680c2452e719fc4bb4512d64ee; derivation seal sha256:2e5400156a37e5034ed35c651e16d968e5268a7c82fc8a38a006c78cfa57b1bc; engine receipt sha256:f970c3ae1bbacf791154156bb1fba1dc1a743ad0c2742ebea4739badab2fe16b at receipts/engine/model_admitted/SFT-CHEM-NUCLEAR-ELECTRONIC-COMPOSITION-012-f970c3ae1bbacf79.json; empirical hash sha256:a5f6b7b2c801d78a7799c487bb862253b4518c1b4421544d6b8ae1f656b6e970; measurement receipt sha256:6f38ce80e651e2e112448e38d4cbd18d1af4d561e9356d66e4c8418c2a2bc91d.

108. Exact resolved vibronic, rovibronic and spin-state joint-composition law

Claim identity: SFT-CHEM-RESOLVED-ROVIBRONIC-SPIN-COMPOSITION-013

Question and theorem. A resolved molecular spectral state is one finite joint product retaining its isotopologue carrier, electronic designation, positive whole spin multiplicity, exact positive vibrational and rotational coordinates or structural EmptyOne, transition/band record and complete source record. No coordinate may be erased, recombined independently or replaced by a continuum premise.

retained-molecular-and-isotope-carrier__retained-electronic-state-designation__positive-vibrational-coordinate-or-EmptyOne__positive-rotational-coordinate-or-EmptyOne__retained-positive-spin-multiplicity__finite-vibronic-rovibronic-spin-product__retained-transition-and-band-record__complete-resolved-spectrum-surface

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-PHYS-MOLECULAR-SPECTROSCOPY-TERMINAL-005
- SFT-CHEM-ROVIBRONIC-COMPOSITION-001
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-SELECTION-RULE-STRUCTURE-010
- SFT-CHEM-NUCLEAR-ELECTRONIC-COMPOSITION-012

Generated grammar. Generate the literal product of carrier, electronic, vibration, rotation, spin, composition, observation and record forms; decide all 256 only from admitted composition, transition, selection and exact spectroscopy dependencies.

Boundary: Every positive finite resolved molecular state word and every source absence, tested against all 95 state rows and all 1,235 cells in the complete NIST H2, HD and D2 diatomic surface.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *retained-molecular-and-isotope-carrier__retained-electronic-state-designation__positive-vibrational-coordinate-or-EmptyOne__positive-rotational-coordinate-or-EmptyOne__retained-positive-spin-multiplicity__finite-vibronic-rovibronic-spin-product__retained-transition-and-band-record__complete-resolved-spectrum-surface*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	retained-molecular-and-isotope-carrier	detached-spectral-row	A row detached from its molecule is not a chemical state.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
electronic	retained-electronic-state-designation	transition-name-without-electronic-state	A transition name alone erases its endpoint state.
vibration	positive-vibrational-coordinate-or-EmptyOne	continuum-vibration-or-erasure	A continuum coordinate or erasure is not an exact generated state.
rotation	positive-rotational-coordinate-or-EmptyOne	signed-rotation-or-erasure	Signed continuum rotation or erasure loses resolved structure.
spin	retained-positive-spin-multiplicity	spin-erased-molecular-state	Erasing multiplicity merges singlet and triplet records.
composition	finite-vibronic-or-vibronic-spin-product	independent-coordinate-list	A list without joint identity permits unlawful recombination.
observation	retained-transition-and-band-record	endpoint-only-transition	Endpoints alone erase the measured transition record and unresolved boundary.
record	complete-resolved-spectrum-surface	selected-resolved-lines	Selected lines cannot establish complete resolved composition.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:e464604ff9553577bc5ec347ee2fa94fc553c78bd3f60ac5b3d6a840e6971088.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:64fc4c4f2ab8e9ee57efa728ec33787a883fa5da6e4ea6492762af4f62b9be01.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:3ad68c8f7a793ec21bd6d3f43da078b2a470e25972d95fbbab5ff66aa644c67.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:36f04628b4465f75564db07276f093890db3edc831a39ead2c245dae4b50371d.

Independent reconstruction. Implementation hash

sha256:0c2ddf9d78c3f7a3c1cd9aab7cd6fc4f2ed587bb18ce3decae7c0798cb27a203; independent certificate sha256:ced76690f303a3fdd8dcc7234607225289bb88803123d10fc04223fb189fe37; external-validation hash sha256:8c19418b949ce72df9ad7889724258b4815240e4916c140f65c45fe57bb60d63. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-HD-D2-RESOLVED-DIATOMIC-SURFACE-2026

Comparison records:

- C1333740-nuclear-electronic-001: multiplicity 3; vibration blank-EmptyOne; rotation positive-exact; band positive-exact; pass True
- C1333740-nuclear-electronic-002: multiplicity 3; vibration positive-exact; rotation blank-EmptyOne; band positive-exact; pass True
- C1333740-nuclear-electronic-003: multiplicity 3; vibration positive-exact; rotation blank-EmptyOne; band positive-exact; pass True
- C1333740-nuclear-electronic-004: multiplicity 3; vibration positive-exact; rotation positive-exact; band positive-exact; pass True
- C1333740-nuclear-electronic-005: multiplicity 3; vibration positive-exact; rotation blank-EmptyOne; band positive-exact; pass True

- [illegible]

- C1333740-nuclear-electronic-040: multiplicity 1; vibration positive-exact; rotation held-text; band positive-exact; pass True
- C1333740-nuclear-electronic-041: multiplicity 1; vibration positive-exact; rotation positive-exact; band positive-exact; pass True
- C1333740-nuclear-electronic-042: multiplicity 1; vibration positive-exact; rotation blank-EmptyOne; band blank-EmptyOne; pass True
- C1333740-nuclear-electronic-043: multiplicity 1; vibration positive-exact; rotation positive-exact; band positive-exact; pass True
- C1333740-nuclear-electronic-044: multiplicity 1; vibration positive-exact; rotation positive-exact; band positive-exact; pass True
- C1333740-nuclear-electronic-045: multiplicity 1; vibration positive-exact; rotation positive-exact; band positive-exact; pass True
- C1333740-nuclear-electronic-046: multiplicity 1; vibration positive-exact; rotation positive-exact; band blank-EmptyOne; pass True
- C13983205-nuclear-electronic-001: multiplicity 3; vibration positive-exact; rotation positive-exact; band positive-exact; pass True
- C13983205-nuclear-electronic-002: multiplicity 3; vibration positive-exact; rotation positive-exact; band positive-exact; pass True
- C13983205-nuclear-electronic-003: multiplicity 3; vibration positive-exact; rotation positive-exact; band positive-exact; pass True
- C13983205-nuclear-electronic-004: multiplicity 3; vibration positive-exact; rotation positive-exact; band positive-exact; pass True
- C13983205-nuclear-electronic-005: multiplicity 3; vibration blank-EmptyOne; rotation blank-EmptyOne; band blank-EmptyOne; pass True
- C13983205-nuclear-electronic-006: multiplicity 1; vibration positive-exact; rotation positive-exact; band positive-exact; pass True
- C13983205-nuclear-electronic-007: multiplicity 1; vibration positive-exact; rotation positive-exact; band positive-exact; pass True
- C13983205-nuclear-electronic-008: multiplicity 1; vibration positive-exact; rotation positive-exact; band positive-exact; pass True
- C13983205-nuclear-electronic-009: multiplicity 1; vibration blank-EmptyOne; rotation positive-exact; band positive-exact; pass True
- C13983205-nuclear-electronic-010: multiplicity 1; vibration positive-exact; rotation positive-exact; band positive-exact; pass True
- C13983205-nuclear-electronic-011: multiplicity 1; vibration positive-exact; rotation blank-EmptyOne; band positive-exact; pass True
- C13983205-nuclear-electronic-012: multiplicity 1; vibration positive-exact; rotation positive-exact; band positive-exact; pass True
- C13983205-nuclear-electronic-013: multiplicity 1; vibration positive-exact; rotation blank-EmptyOne; band positive-exact; pass True
- C13983205-nuclear-electronic-014: multiplicity 1; vibration held-text; rotation blank-EmptyOne; band positive-exact; pass True
- C13983205-nuclear-electronic-015: multiplicity 1; vibration positive-exact; rotation positive-exact; band positive-exact; pass True
- C13983205-nuclear-electronic-016: multiplicity 1; vibration positive-exact; rotation blank-EmptyOne; band positive-exact; pass True
- C13983205-nuclear-electronic-017: multiplicity 1; vibration positive-exact; rotation positive-exact; band positive-exact; pass True
- C13983205-nuclear-electronic-018: multiplicity 1; vibration positive-exact; rotation positive-exact; band positive-exact; pass True

- [illegible]

- C7782390-nuclear-electronic-028: multiplicity 1; vibration positive-exact; rotation positive-exact; band positive-exact; pass True
- C7782390-nuclear-electronic-029: multiplicity 1; vibration positive-exact; rotation positive-exact; band blank-EmptyOne; pass True
- count rows: 95
- count cells: 1235
- count singlet: 54
- count triplet: 41
- count positive_vibration: 78
- count positive_rotation: 60
- count joint_vibration_rotation: 56
- count transition_designation: 92
- count positive_band_origin: 83
- count source_zero_electronic_origin: 3
- count absent_electronic_origin: 3
- count held_signed_inscriptions: 8

Falsification boundary: Reject if any carrier, isotope, electronic designation, multiplicity, vibrational coordinate, rotational coordinate, transition, band or source record is erased or recombined; if any of 95 rows or 1,235 cells changes; if the complete surface does not retain 54 singlet and 41 triplet rows, 78 positive vibrational coordinates, 60 positive rotational coordinates, 56 jointly resolved vibronic/rotational rows, 92 transition designations, 83 positive band origins, three source-zero and three absent electronic origins; if a signed inscription becomes an SFT number; or if targets enter before sealing.

Measurement receipt: sha256:6b954d40e0a50bd8a369d615ec50f09235789f5998199084c4d4a5394cfa2fbb.

Isolation certificate: sha256:872507adb2cabf24798d4a7905bb908aec3d30331ad780bc8bc4527a6ddbc17f.

Custody certificate: sha256:fcfae81f39ac9f8c8c895f43ccfe888f7b6c30bf71cb80b5d1b3d9846a6aa942.

Registered source descriptions:

- National Institute of Standards and Technology: ()

Registered target identities:

- C1333740-nuclear-electronic-001 from None
- C1333740-nuclear-electronic-002 from None
- C1333740-nuclear-electronic-003 from None
- C1333740-nuclear-electronic-004 from None
- C1333740-nuclear-electronic-005 from None
- C1333740-nuclear-electronic-006 from None
- C1333740-nuclear-electronic-007 from None
- C1333740-nuclear-electronic-008 from None
- C1333740-nuclear-electronic-009 from None
- C1333740-nuclear-electronic-010 from None
- C1333740-nuclear-electronic-011 from None
- C1333740-nuclear-electronic-012 from None
- C1333740-nuclear-electronic-013 from None
- C1333740-nuclear-electronic-014 from None

- C1333740-nuclear-electronic-015 from None
- C1333740-nuclear-electronic-016 from None
- C1333740-nuclear-electronic-017 from None
- C1333740-nuclear-electronic-018 from None
- C1333740-nuclear-electronic-019 from None
- C1333740-nuclear-electronic-020 from None
- C1333740-nuclear-electronic-021 from None
- C1333740-nuclear-electronic-022 from None
- C1333740-nuclear-electronic-023 from None
- C1333740-nuclear-electronic-024 from None
- C1333740-nuclear-electronic-025 from None
- C1333740-nuclear-electronic-026 from None
- C1333740-nuclear-electronic-027 from None
- C1333740-nuclear-electronic-028 from None
- C1333740-nuclear-electronic-029 from None
- C1333740-nuclear-electronic-030 from None
- C1333740-nuclear-electronic-031 from None
- C1333740-nuclear-electronic-032 from None
- C1333740-nuclear-electronic-033 from None
- C1333740-nuclear-electronic-034 from None
- C1333740-nuclear-electronic-035 from None
- C1333740-nuclear-electronic-036 from None
- C1333740-nuclear-electronic-037 from None
- C1333740-nuclear-electronic-038 from None
- C1333740-nuclear-electronic-039 from None
- C1333740-nuclear-electronic-040 from None
- C1333740-nuclear-electronic-041 from None
- C1333740-nuclear-electronic-042 from None
- C1333740-nuclear-electronic-043 from None
- C1333740-nuclear-electronic-044 from None
- C1333740-nuclear-electronic-045 from None
- C1333740-nuclear-electronic-046 from None
- C13983205-nuclear-electronic-001 from None
- C13983205-nuclear-electronic-002 from None
- C13983205-nuclear-electronic-003 from None
- C13983205-nuclear-electronic-004 from None
- C13983205-nuclear-electronic-005 from None
- C13983205-nuclear-electronic-006 from None

- C13983205-nuclear-electronic-007 from None
- C13983205-nuclear-electronic-008 from None
- C13983205-nuclear-electronic-009 from None
- C13983205-nuclear-electronic-010 from None
- C13983205-nuclear-electronic-011 from None
- C13983205-nuclear-electronic-012 from None
- C13983205-nuclear-electronic-013 from None
- C13983205-nuclear-electronic-014 from None
- C13983205-nuclear-electronic-015 from None
- C13983205-nuclear-electronic-016 from None
- C13983205-nuclear-electronic-017 from None
- C13983205-nuclear-electronic-018 from None
- C13983205-nuclear-electronic-019 from None
- C13983205-nuclear-electronic-020 from None
- C7782390-nuclear-electronic-001 from None
- C7782390-nuclear-electronic-002 from None
- C7782390-nuclear-electronic-003 from None
- C7782390-nuclear-electronic-004 from None
- C7782390-nuclear-electronic-005 from None
- C7782390-nuclear-electronic-006 from None
- C7782390-nuclear-electronic-007 from None
- C7782390-nuclear-electronic-008 from None
- C7782390-nuclear-electronic-009 from None
- C7782390-nuclear-electronic-010 from None
- C7782390-nuclear-electronic-011 from None
- C7782390-nuclear-electronic-012 from None
- C7782390-nuclear-electronic-013 from None
- C7782390-nuclear-electronic-014 from None
- C7782390-nuclear-electronic-015 from None
- C7782390-nuclear-electronic-016 from None
- C7782390-nuclear-electronic-017 from None
- C7782390-nuclear-electronic-018 from None
- C7782390-nuclear-electronic-019 from None
- C7782390-nuclear-electronic-020 from None
- C7782390-nuclear-electronic-021 from None
- C7782390-nuclear-electronic-022 from None
- C7782390-nuclear-electronic-023 from None
- C7782390-nuclear-electronic-024 from None

- C7782390-nuclear-electronic-025 from None
- C7782390-nuclear-electronic-026 from None
- C7782390-nuclear-electronic-027 from None
- C7782390-nuclear-electronic-028 from None
- C7782390-nuclear-electronic-029 from None

Meaning. The field-wide extension derives molecular electronic identity, state composition, occupancy, ordering, symmetry, correlation, transitions, measurement and operational classical-quantum correspondence from the same exact carrier laws. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; external zero is structural EmptyOne
- no negative, irrational, imaginary, floating, signed or continuum proof value
- no imported wavefunction, Hamiltonian, rigid rotor, harmonic oscillator, differential equation or fitted spectroscopic constant
- no NIST state, multiplicity, constant, transition or band outcome before prediction seal
- no singlet-only, triplet-only, common-state-only or resolved-value-only subset

Immutable identities. Source manifest

sha256:362a83d29f0478e383e8bc37e0df8f4e2e324f113f2f0ac5a4f12c6e67b4fcb9; derivation seal
 sha256:2b7d00dd5ac482575d6769799dc221d2eb0aeb50fcf95383b9356ee4dce86b74; engine receipt
 sha256:ed3419de39187595fd053d5f6e1e897f1994ebc8c4e562b322a5f5bd21977a1f at receipts/engine/
 model_admitted/SFT-CHEM-RESOLVED-ROVIBRONIC-SPIN-COMPOSITION-013-ed3419de39187595.json;
 empirical hash sha256:d01fbe58cd16d9bfd3337c3931bae35b867c2415d60178b4d36c42ffc9468df9;
 measurement receipt sha256:6b954d40e0a50bd8a369d615ec50f09235789f5998199084c4d4a5394cfa2fbb.

109. Exact chemical measurement-reduction law for molecular quantum states

Claim identity: SFT-CHEM-MOLECULAR-QUANTUM-MEASUREMENT-REDUCTION-014

Question and theorem. A molecular quantum-state observation is an exact process from one retained prepared state through one generated probe class to a retained finite readout and declared post-observation state. The record explicitly partitions retained from closed coordinates; only retained distinctions reconstruct, while every repetition, recovery, failed projection and boundary event remains evidence. No stochastic collapse or numerical probability is a premise.

retained-pre-observation-state__exact-generated-observation-class__explicit-retained-coordinate-set__explicit-closed-coordinate-set__retained-finite-readout-record__retained-post-observation-state__record-bounded-reconstruction__complete-repeat-and-adverse-record

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-RESOLVED-ROVIBRONIC-SPIN-COMPOSITION-013

Generated grammar. Generate the literal product of preparation, probe, retention, closure, record, poststate, reconstruction and replication forms; decide all 256 only from admitted observation, information-loss, state-transition and resolved molecular-state

laws.

Boundary: Every positive finite molecular preparation/readout sequence and its retained instrument/condition boundary, tested against the complete nine-file NIST MDS2-3389 CaH⁺ package containing 330 data rows and 796 cells.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `retained-pre-observation-state__exact-generated-observation-class__explicit-retained-coordinate-set__explicit-closed-coordinate-set__retained-finite-readout-record__retained-post-observation-state__record-bounded-reconstruction__complete-repeat-and-adverse-record`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
preparation	retained-pre-observation-state	outcome-without-prepared-state	An outcome alone cannot identify the state presented to the probe.
probe	exact-generated-observation-class	stochastic-collapse-premise	A stochastic collapse postulate imports an ungenerated law and numerical probability.
retention	explicit-retained-coordinate-set	all-or-nothing-observation	All-or-nothing readout hides which distinctions survive.
closure	explicit-closed-coordinate-set	silent-predecessor-erasure	Silent erasure makes later reconstruction claims untestable.
record	retained-finite-readout-record	unrecorded-measurement-result	A result without a retained record cannot be repeated or audited.
poststate	retained-post-observation-state	undefined-or-destroyed-poststate	An undefined post-state breaks process composition.
reconstruction	record-bounded-reconstruction	recover-unrecorded-predecessor	A merged predecessor cannot be recovered without a retained distinction.
replication	complete-repeat-and-adverse-record	selected-successful-readouts	Successful readouts alone erase failures and state changes.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 18ee5b35a1b4c672b08946df0f41efc2565a853eff8e47e62b46b3f2e0849725.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 53ad0d4bf427a81224ff7cd600e92e98d96f28a1c5725e41c578845e3d832740.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 18e479acdeb56dfb54d19d5b1aa0eedf1c9b18b7ad1805fd271315bf4560b0df.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 4b9d6a614e418bcc01be76879cb79a97397756555c55ab9a3b1c7e0da39d3d5.

Independent reconstruction. Implementation hash

sha256: f2d5e65b45846e89785751b48234fcd96f18e2dff13dbb20b12d9d538cd7b018; independent certificate sha256: cd76076e75d6309fda058427fdbb4915148e0e6e779e9d9e889f00754add1a85; external-validation hash sha256: ea176e4848f38c184b0af52fa4c4e9e69c6171269d268e7b177a0b73726c78f5. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- NIST-MDS2-3389-CAH-QUANTUM-STATE-TRACKING-CONTROL-V1.0.1

Comparison records:

- MDS2-3389-appearance-time-histogram-J-0-to-J1-0001: file appearance_time_histogram_J_0_to_J1.csv row 1 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0002: file appearance_time_histogram_J_0_to_J1.csv row 2 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0003: file appearance_time_histogram_J_0_to_J1.csv row 3 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0004: file appearance_time_histogram_J_0_to_J1.csv row 4 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0005: file appearance_time_histogram_J_0_to_J1.csv row 5 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0006: file appearance_time_histogram_J_0_to_J1.csv row 6 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0007: file appearance_time_histogram_J_0_to_J1.csv row 7 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0008: file appearance_time_histogram_J_0_to_J1.csv row 8 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0009: file appearance_time_histogram_J_0_to_J1.csv row 9 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0010: file appearance_time_histogram_J_0_to_J1.csv row 10 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0011: file appearance_time_histogram_J_0_to_J1.csv row 11 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0012: file appearance_time_histogram_J_0_to_J1.csv row 12 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0013: file appearance_time_histogram_J_0_to_J1.csv row 13 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0014: file appearance_time_histogram_J_0_to_J1.csv row 14 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0015: file appearance_time_histogram_J_0_to_J1.csv row 15 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0016: file appearance_time_histogram_J_0_to_J1.csv row 16 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0017: file appearance_time_histogram_J_0_to_J1.csv row 17 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0018: file appearance_time_histogram_J_0_to_J1.csv row 18 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0019: file appearance_time_histogram_J_0_to_J1.csv row 19 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0020: file appearance_time_histogram_J_0_to_J1.csv row 20 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0021: file appearance_time_histogram_J_0_to_J1.csv row 21 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0022: file appearance_time_histogram_J_0_to_J1.csv row 22 cells 3 pass True
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0023: file appearance_time_histogram_J_0_to_J1.csv row 23 cells 3 pass True
- MDS2-3389-entry-and-exit-times-J-1-0001: file entry_and_exit_times_J_1.csv row 1 cells 2 pass True
- MDS2-3389-entry-and-exit-times-J-1-0002: file entry_and_exit_times_J_1.csv row 2 cells 2 pass True
- MDS2-3389-entry-and-exit-times-J-1-0003: file entry_and_exit_times_J_1.csv row 3 cells 2 pass True
- MDS2-3389-entry-and-exit-times-J-1-0004: file entry_and_exit_times_J_1.csv row 4 cells 2 pass True
- MDS2-3389-entry-and-exit-times-J-1-0005: file entry_and_exit_times_J_1.csv row 5 cells 2 pass True
- MDS2-3389-entry-and-exit-times-J-1-0006: file entry_and_exit_times_J_1.csv row 6 cells 2 pass True
- MDS2-3389-entry-and-exit-times-J-1-0007: file entry_and_exit_times_J_1.csv row 7 cells 2 pass True
- MDS2-3389-entry-and-exit-times-J-1-0008: file entry_and_exit_times_J_1.csv row 8 cells 2 pass True
- MDS2-3389-entry-and-exit-times-J-1-0009: file entry_and_exit_times_J_1.csv row 9 cells 2 pass True
- MDS2-3389-entry-and-exit-times-J-1-0010: file entry_and_exit_times_J_1.csv row 10 cells 2 pass True
- MDS2-3389-entry-and-exit-times-J-1-0011: file entry_and_exit_times_J_1.csv row 11 cells 2 pass True
- MDS2-3389-entry-and-exit-times-J-1-0012: file entry_and_exit_times_J_1.csv row 12 cells 2 pass True
- MDS2-3389-entry-and-exit-times-J-1-0013: file entry_and_exit_times_J_1.csv row 13 cells 2 pass True
- MDS2-3389-entry-and-exit-times-J-1-0014: file entry_and_exit_times_J_1.csv row 14 cells 2 pass True
- MDS2-3389-entry-and-exit-times-J-1-0015: file entry_and_exit_times_J_1.csv row 15 cells 2 pass True

- [illegible]

- [illegible]

- [illegible]

- MDS2-3389-entry-and-exit-times-J-2-0113: file entry_and_exit_times_J_2.csv row 113 cells 2 pass True
- MDS2-3389-entry-and-exit-times-J-2-0114: file entry_and_exit_times_J_2.csv row 114 cells 2 pass True
- MDS2-3389-entry-and-exit-times-J-2-0115: file entry_and_exit_times_J_2.csv row 115 cells 2 pass True
- MDS2-3389-entry-and-exit-times-J-2-0116: file entry_and_exit_times_J_2.csv row 116 cells 2 pass True
- MDS2-3389-entry-and-exit-times-J-2-0117: file entry_and_exit_times_J_2.csv row 117 cells 2 pass True
- MDS2-3389-histogram-of-recoveries-from-J-0-and-J-2-0001: file histogram_of_recoveries_from_J_0_and_J_2.csv row 1 cells 2 pass True
- MDS2-3389-histogram-of-recoveries-from-J-0-and-J-2-0002: file histogram_of_recoveries_from_J_0_and_J_2.csv row 2 cells 2 pass True
- MDS2-3389-histogram-of-recoveries-from-J-0-and-J-2-0003: file histogram_of_recoveries_from_J_0_and_J_2.csv row 3 cells 2 pass True
- MDS2-3389-histogram-of-recoveries-from-J-0-and-J-2-0004: file histogram_of_recoveries_from_J_0_and_J_2.csv row 4 cells 2 pass True
- MDS2-3389-pressure-dependence-of-J-0-to-J-1-transitions-0001: file pressure_dependence_of_J_0_to_J_1_transitions.csv row 1 cells 5 pass True
- MDS2-3389-pressure-dependence-of-J-0-to-J-1-transitions-0002: file pressure_dependence_of_J_0_to_J_1_transitions.csv row 2 cells 5 pass True
- MDS2-3389-pressure-dependence-of-J-0-to-J-1-transitions-0003: file pressure_dependence_of_J_0_to_J_1_transitions.csv row 3 cells 5 pass True
- MDS2-3389-pressure-dependence-of-J-0-to-J-1-transitions-0004: file pressure_dependence_of_J_0_to_J_1_transitions.csv row 4 cells 5 pass True
- MDS2-3389-real-time-state-tracking-data-0001: file real_time_state_tracking_data.csv row 1 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0002: file real_time_state_tracking_data.csv row 2 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0003: file real_time_state_tracking_data.csv row 3 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0004: file real_time_state_tracking_data.csv row 4 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0005: file real_time_state_tracking_data.csv row 5 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0006: file real_time_state_tracking_data.csv row 6 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0007: file real_time_state_tracking_data.csv row 7 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0008: file real_time_state_tracking_data.csv row 8 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0009: file real_time_state_tracking_data.csv row 9 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0010: file real_time_state_tracking_data.csv row 10 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0011: file real_time_state_tracking_data.csv row 11 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0012: file real_time_state_tracking_data.csv row 12 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0013: file real_time_state_tracking_data.csv row 13 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0014: file real_time_state_tracking_data.csv row 14 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0015: file real_time_state_tracking_data.csv row 15 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0016: file real_time_state_tracking_data.csv row 16 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0017: file real_time_state_tracking_data.csv row 17 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0018: file real_time_state_tracking_data.csv row 18 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0019: file real_time_state_tracking_data.csv row 19 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0020: file real_time_state_tracking_data.csv row 20 cells 2 pass True

- MDS2-3389-real-time-state-tracking-data-0021: file real_time_state_tracking_data.csv row 21 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0022: file real_time_state_tracking_data.csv row 22 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0023: file real_time_state_tracking_data.csv row 23 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0024: file real_time_state_tracking_data.csv row 24 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0025: file real_time_state_tracking_data.csv row 25 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0026: file real_time_state_tracking_data.csv row 26 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0027: file real_time_state_tracking_data.csv row 27 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0028: file real_time_state_tracking_data.csv row 28 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0029: file real_time_state_tracking_data.csv row 29 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0030: file real_time_state_tracking_data.csv row 30 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0031: file real_time_state_tracking_data.csv row 31 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0032: file real_time_state_tracking_data.csv row 32 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0033: file real_time_state_tracking_data.csv row 33 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0034: file real_time_state_tracking_data.csv row 34 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0035: file real_time_state_tracking_data.csv row 35 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0036: file real_time_state_tracking_data.csv row 36 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0037: file real_time_state_tracking_data.csv row 37 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0038: file real_time_state_tracking_data.csv row 38 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0039: file real_time_state_tracking_data.csv row 39 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0040: file real_time_state_tracking_data.csv row 40 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0041: file real_time_state_tracking_data.csv row 41 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0042: file real_time_state_tracking_data.csv row 42 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0043: file real_time_state_tracking_data.csv row 43 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0044: file real_time_state_tracking_data.csv row 44 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0045: file real_time_state_tracking_data.csv row 45 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0046: file real_time_state_tracking_data.csv row 46 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0047: file real_time_state_tracking_data.csv row 47 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0048: file real_time_state_tracking_data.csv row 48 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0049: file real_time_state_tracking_data.csv row 49 cells 2 pass True
- MDS2-3389-real-time-state-tracking-data-0050: file real_time_state_tracking_data.csv row 50 cells 2 pass True
- MDS2-3389-sublevel-distribution-J0-to-J1-0001: file sublevel_distribution_J0_to_J1.csv row 1 cells 7 pass True
- MDS2-3389-sublevel-distribution-J0-to-J1-0002: file sublevel_distribution_J0_to_J1.csv row 2 cells 7 pass True
- MDS2-3389-sublevel-distribution-J0-to-J1-0003: file sublevel_distribution_J0_to_J1.csv row 3 cells 7 pass True
- MDS2-3389-sublevel-distribution-J0-to-J1-0004: file sublevel_distribution_J0_to_J1.csv row 4 cells 7 pass True
- MDS2-3389-sublevel-distribution-J0-to-J1-0005: file sublevel_distribution_J0_to_J1.csv row 5 cells 7 pass True
- MDS2-3389-sublevel-distribution-J0-to-J1-0006: file sublevel_distribution_J0_to_J1.csv row 6 cells 7 pass True
- MDS2-3389-tracking-time-series-0001: file tracking_time_series.csv row 1 cells 3 pass True
- MDS2-3389-tracking-time-series-0002: file tracking_time_series.csv row 2 cells 3 pass True

- [illegible]

- MDS2-3389-tracking-time-series-0041: file tracking_time_series.csv row 41 cells 3 pass True
- MDS2-3389-tracking-time-series-0042: file tracking_time_series.csv row 42 cells 3 pass True
- MDS2-3389-tracking-time-series-0043: file tracking_time_series.csv row 43 cells 3 pass True
- MDS2-3389-tracking-time-series-0044: file tracking_time_series.csv row 44 cells 3 pass True
- MDS2-3389-tracking-time-series-0045: file tracking_time_series.csv row 45 cells 3 pass True
- MDS2-3389-tracking-time-series-0046: file tracking_time_series.csv row 46 cells 3 pass True
- MDS2-3389-tracking-time-series-0047: file tracking_time_series.csv row 47 cells 3 pass True
- MDS2-3389-tracking-time-series-0048: file tracking_time_series.csv row 48 cells 3 pass True
- MDS2-3389-tracking-time-series-0049: file tracking_time_series.csv row 49 cells 3 pass True
- MDS2-3389-tracking-time-series-0050: file tracking_time_series.csv row 50 cells 3 pass True
- MDS2-3389-tracking-time-series-0051: file tracking_time_series.csv row 51 cells 3 pass True
- MDS2-3389-tracking-time-series-0052: file tracking_time_series.csv row 52 cells 3 pass True
- MDS2-3389-tracking-time-series-0053: file tracking_time_series.csv row 53 cells 3 pass True
- MDS2-3389-tracking-time-series-0054: file tracking_time_series.csv row 54 cells 3 pass True
- MDS2-3389-tracking-time-series-0055: file tracking_time_series.csv row 55 cells 3 pass True
- MDS2-3389-tracking-time-series-0056: file tracking_time_series.csv row 56 cells 3 pass True
- MDS2-3389-tracking-time-series-0057: file tracking_time_series.csv row 57 cells 3 pass True
- MDS2-3389-tracking-time-series-0058: file tracking_time_series.csv row 58 cells 3 pass True
- MDS2-3389-tracking-time-series-0059: file tracking_time_series.csv row 59 cells 3 pass True
- MDS2-3389-tracking-time-series-0060: file tracking_time_series.csv row 60 cells 3 pass True
- MDS2-3389-tracking-time-series-0061: file tracking_time_series.csv row 61 cells 3 pass True
- MDS2-3389-tracking-time-series-0062: file tracking_time_series.csv row 62 cells 3 pass True
- MDS2-3389-tracking-time-series-0063: file tracking_time_series.csv row 63 cells 3 pass True
- MDS2-3389-tracking-time-series-0064: file tracking_time_series.csv row 64 cells 3 pass True
- MDS2-3389-tracking-time-series-0065: file tracking_time_series.csv row 65 cells 3 pass True
- MDS2-3389-tracking-time-series-0066: file tracking_time_series.csv row 66 cells 3 pass True
- MDS2-3389-tracking-time-series-0067: file tracking_time_series.csv row 67 cells 3 pass True
- MDS2-3389-tracking-time-series-0068: file tracking_time_series.csv row 68 cells 3 pass True
- MDS2-3389-tracking-time-series-0069: file tracking_time_series.csv row 69 cells 3 pass True
- MDS2-3389-tracking-time-series-0070: file tracking_time_series.csv row 70 cells 3 pass True
- MDS2-3389-tracking-time-series-0071: file tracking_time_series.csv row 71 cells 3 pass True
- count package_components: 9
- count csv_files: 8
- count rows: 330
- count cells: 796
- count real_time_readouts: 50
- count J2_entry_exit_intervals: 117
- count J1_entry_exit_intervals: 55

- count prepared_sublevel_rows: 6
- count recovery_classes: 4
- count tracking_intervals: 71
- count pressure_control_rows: 4
- count appearance_time_rows: 23
- count positive_entry_exit_intervals: 172
- count positive_tracking_intervals: 70
- count EmptyOne_tracking_boundary: 1

Falsification boundary: Reject if preparation, probe, retained/closed partition, readout, post-state, instrument or condition is erased; if an unrecorded predecessor is claimed reconstructible; if any of nine NIST package files, 330 data rows or 796 cells changes; if the complete event surface does not retain 50 real-time readouts, 117 J=2 intervals, 55 J=1 intervals, six prepared-sublevel rows, four recovery classes, 71 tracking intervals, four pressure-control rows and 23 appearance-time rows; if the one reported zero-duration tracking boundary is discarded or consumed as a number; if any target enters before sealing; or if a stochastic/fitted model selects the law.

Measurement receipt: sha256:cbc7e53d5ebc601742deed1d2355c1aff7811cc414f1f857a9992494a850d582.

Isolation certificate: sha256:af81ae50d6c17180685a5836c7057bfc4737ec8f3e499c6fe2ab2e7ef079b130.

Custody certificate: sha256:563bd51acccab3c71fad1951fde8f589105abfb37cf4a477573f7602caea280f.

Registered source descriptions:

- National Institute of Standards and Technology: ()

Registered target identities:

- MDS2-3389-appearance-time-histogram-J-0-to-J1-0001 from None
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0002 from None
- MDS2-3389-appearance-time-histogram-J-0-to-J1-0003 from None
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- MDS2-3389-tracking-time-series-0001 from None
- MDS2-3389-tracking-time-series-0002 from None
- MDS2-3389-tracking-time-series-0003 from None
- MDS2-3389-tracking-time-series-0004 from None
- MDS2-3389-tracking-time-series-0005 from None
- MDS2-3389-tracking-time-series-0006 from None
- MDS2-3389-tracking-time-series-0007 from None
- MDS2-3389-tracking-time-series-0008 from None
- MDS2-3389-tracking-time-series-0009 from None
- MDS2-3389-tracking-time-series-0010 from None
- MDS2-3389-tracking-time-series-0011 from None
- MDS2-3389-tracking-time-series-0012 from None
- MDS2-3389-tracking-time-series-0013 from None
- MDS2-3389-tracking-time-series-0014 from None
- MDS2-3389-tracking-time-series-0015 from None
- MDS2-3389-tracking-time-series-0016 from None
- MDS2-3389-tracking-time-series-0017 from None
- MDS2-3389-tracking-time-series-0018 from None
- MDS2-3389-tracking-time-series-0019 from None
- MDS2-3389-tracking-time-series-0020 from None
- MDS2-3389-tracking-time-series-0021 from None
- MDS2-3389-tracking-time-series-0022 from None
- MDS2-3389-tracking-time-series-0023 from None
- MDS2-3389-tracking-time-series-0024 from None
- MDS2-3389-tracking-time-series-0025 from None
- MDS2-3389-tracking-time-series-0026 from None
- MDS2-3389-tracking-time-series-0027 from None

- MDS2-3389-tracking-time-series-0028 from None
- MDS2-3389-tracking-time-series-0029 from None
- MDS2-3389-tracking-time-series-0030 from None
- MDS2-3389-tracking-time-series-0031 from None
- MDS2-3389-tracking-time-series-0032 from None
- MDS2-3389-tracking-time-series-0033 from None
- MDS2-3389-tracking-time-series-0034 from None
- MDS2-3389-tracking-time-series-0035 from None
- MDS2-3389-tracking-time-series-0036 from None
- MDS2-3389-tracking-time-series-0037 from None
- MDS2-3389-tracking-time-series-0038 from None
- MDS2-3389-tracking-time-series-0039 from None
- MDS2-3389-tracking-time-series-0040 from None
- MDS2-3389-tracking-time-series-0041 from None
- MDS2-3389-tracking-time-series-0042 from None
- MDS2-3389-tracking-time-series-0043 from None
- MDS2-3389-tracking-time-series-0044 from None
- MDS2-3389-tracking-time-series-0045 from None
- MDS2-3389-tracking-time-series-0046 from None
- MDS2-3389-tracking-time-series-0047 from None
- MDS2-3389-tracking-time-series-0048 from None
- MDS2-3389-tracking-time-series-0049 from None
- MDS2-3389-tracking-time-series-0050 from None
- MDS2-3389-tracking-time-series-0051 from None
- MDS2-3389-tracking-time-series-0052 from None
- MDS2-3389-tracking-time-series-0053 from None
- MDS2-3389-tracking-time-series-0054 from None
- MDS2-3389-tracking-time-series-0055 from None
- MDS2-3389-tracking-time-series-0056 from None
- MDS2-3389-tracking-time-series-0057 from None
- MDS2-3389-tracking-time-series-0058 from None
- MDS2-3389-tracking-time-series-0059 from None
- MDS2-3389-tracking-time-series-0060 from None
- MDS2-3389-tracking-time-series-0061 from None
- MDS2-3389-tracking-time-series-0062 from None
- MDS2-3389-tracking-time-series-0063 from None
- MDS2-3389-tracking-time-series-0064 from None
- MDS2-3389-tracking-time-series-0065 from None

- MDS2-3389-tracking-time-series-0066 from None
- MDS2-3389-tracking-time-series-0067 from None
- MDS2-3389-tracking-time-series-0068 from None
- MDS2-3389-tracking-time-series-0069 from None
- MDS2-3389-tracking-time-series-0070 from None
- MDS2-3389-tracking-time-series-0071 from None

Meaning. The field-wide extension derives molecular electronic identity, state composition, occupancy, ordering, symmetry, correlation, transitions, measurement and operational classical-quantum correspondence from the same exact carrier laws. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero or probability parameter; external 0 remains a held source glyph or structural EmptyOne at its typed absence boundary
- no negative, irrational, imaginary, floating, signed or continuum proof value
- no imported stochastic collapse, Born rule, wavefunction, density matrix, Bayesian update or fitted detector model
- no NIST event, fidelity, transition, duration, occurrence or outcome before prediction seal
- no successful-projection-only, tracked-only or positive-duration-only subset

Immutable identities. Source manifest

sha256:b9f4fefe97fe981604ab82795966133b3e41a4fbfd8af33506ec0261acb8d3d1; derivation seal
 sha256:8a51faaaf1193cff3404bb69daa0964ca565dac52a2a75ec6e2d4ee9299c0ce6; engine receipt
 sha256:5fe377a187e37a0fe76eddfc954d2540d35223170b6e73ad7b99da813f481322 at receipts/engine/model_admitted/SFT-CHEM-MOLECULAR-QUANTUM-MEASUREMENT-REDUCTION-014-5fe377a187e37a0f.
 json; empirical hash sha256:46dd29ad29399004ed04b9f4fcfe47d9da154c72338ec344a719b47eefdb39a0;
 measurement receipt sha256:cbc7e53d5ebc601742deed1d2355c1aff7811cc414f1f857a9992494a850d582.

110. Exact operational classical--quantum chemical correspondence

Claim identity: SFT-CHEM-OPERATIONAL-CLASSICAL-QUANTUM-CORRESPONDENCE-015

Question and theorem. One admitted molecular process description embeds every classical chemical state into the reversible Fold-quantum submodel. Complete branchwise quantum execution, shared observation decoding, exact inverse restoration and a positive resource ledger preserve every classical result bidirectionally. Phase-labelled quantum traces remain explicit and are not collapsed into the classical trace.

common-generated-molecular-description__singleton-branch-classical-embedding__reversible-classical-transition-submodel__complete-branchwise-quantum-execution__one-admitted-chemical-transition-law__shared-chemical-decoder-and-record__bidirectional-result-and-inverse-preservation__exact-positive-overhead-ledger

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-QUANTUM-REVERSIBLE-MODEL-001
- SFT-QUANTUM-CIRCUIT-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009

• SFT-CHEM-MOLECULAR-QUANTUM-MEASUREMENT-REDUCTION-014

Generated grammar. Generate the literal product of description, embedding, reversibility, quantum execution, Chemistry ownership, observation, result and resource forms; decide all 256 only from admitted molecular transition, measurement and classical--quantum computation laws.

Boundary: Every positive finite reversible molecular transition table under one admitted Chemistry law, with complete classical rows, quantum branches, observation records, inverse trace and exact positive resource ledger.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `common-generated-molecular-description__singleton-branch-classical-embedding__reversible-classical-transition-submodel__complete-branchwise-quantum-execution__one-admitted-chemical-transition-law__shared-chemical-decoder-and-record__bidirectional-result-and-inverse-preservation__exact-positive-overhead-ledger`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
description	common-generated-molecular-description	different-classical-and-quantum-chemistries	Different chemical descriptions make result agreement accidental.
embedding	singleton-branch-classical-embedding	lossy-classical-reencoding	Lossy embedding cannot preserve a chemical state.
reversibility	reversible-classical-transition-submodel	hidden-irreversible-predecessor	A hidden predecessor prevents inverse execution.
quantum	complete-branchwise-quantum-execution	selected-branch-execution	Selected execution does not establish branchwise equivalence.
chemistry	one-admitted-chemical-transition-law	mode-specific-transition-law	Mode-specific laws allow the machine to select Chemistry.
observation	shared-chemical-decoder-and-record	result-without-shared-decoder	Unrelated outputs cannot be compared.
result	bidirectional-result-and-inverse-preservation	one-way-result-simulation	One-way simulation is not full operational correspondence.
resource	exact-positive-overhead-ledger	uncounted-correspondence-overhead	Uncounted overhead hides the operational distinction.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: f4b78ae6f44d91240edecca8243a12d53e8c88d021474c4e51e59dc9fa9d26f4.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: b1f3431ff932e789ba973cca82d8bbd758fb3f2c7ab12d266ac0fe3af4cde0ce.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 5c73fab8718fc933aa70250a3f044bb47b85c887ade7c50905a61075470e52bf.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 25c3cab70b0774c2828a41ce1f95615e322358c64bfe532efeb90c565a60aa1.

Independent reconstruction. Implementation hash

sha256: 67f88d00821d14eec67b1a7177c06500b732733b807add32aa17d92f4a078999; independent certificate
sha256: 45af5774ec4fe9ced10c2468a4534e50a40787c6c9e31a46e730305e0e75e476; external-validation

hash sha256:376a6ff88ba7b3741dce5e48bfff083b0b5d8e6df831e6232bc56827140fb510. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. This is a sealed formal law with an implementation-distinct reconstruction and no separate external-observation package at this boundary. It receives no empirical-comparison credit that is not present in its claim package.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The field-wide extension derives molecular electronic identity, state composition, occupancy, ordering, symmetry, correlation, transitions, measurement and operational classical-quantum correspondence from the same exact carrier laws. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero, negative, irrational, imaginary, floating, signed or continuum proof value
- no imported bit, qubit, amplitude, Hilbert space, stochastic collapse or conventional circuit premise
- no fitted, learned, hardware or application parameter
- no mode-specific chemical law or target-selected molecular process
- no claim that phase-sensitive or nonfactorable quantum traces are identical to classical traces

Immutable identities. Source manifest

sha256:8233084c39e929d170035855358b535520de98fbc339445a9455aa01e9ed268f; derivation seal sha256:4148fff19f7dc66b1c17ecf3fb3cfc74f3213326b869d11ee8eaa9698401583d; engine receipt sha256:5a7dcf58b16c374f3d0ef8f30cdfb7a592a63dc15b370818ab8857f2a79eadca at receipts/engine/model_admitted/SFT-CHEM-OPERATIONAL-CLASSICAL-QUANTUM-CORRESPONDENCE-015-5a7dcf58b16c374f.json; empirical hash None; measurement receipt None.

19. Quantitative molecular structure and properties - 14 admitted claims

Thirteen property families share a single carrier; no property-specific fitted parameter is admitted. This entire registered subcategory is included; no proper subset is presented as field completion.

111. Exact H2/D2 equilibrium bond-length relation

Claim identity: SFT-CHEM-EQUILIBRIUM-BOND-LENGTH-001

Question and theorem. For the complete registered gas-phase H2/D2 X-state boundary, the common Fold electronic base is seven-fifths of the held atomic-length carrier. The binary-order alpha return adds twenty-one alpha squared for H2 and twenty-four alpha squared for D2. Thus $r_e(\text{H}_2)/a_0 = 7/5 + 21 \alpha^2$ and $r_e(\text{D}_2)/a_0 = 7/5 + 24 \alpha^2$ before either NIST distance target is released.

named-isotopologue-state-and-condition__configuration-minimum-distance__held-atomic-length-reference__up-support-over-down-support__binary-order-alpha-return__distinct-light-and-heavy-typed-routes__sealed-exact-interval-trans port__finite-typed-vector-exhaustion

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-MOLECULAR-SPECTROSCOPY-TERMINAL-005
- SFT-PHYS-MEAS-UNIT-COMPARISON-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-CONFIGURATION-ORDER-PATH-011
- SFT-CHEM-NUCLEAR-ELECTRONIC-COMPOSITION-012
- SFT-CHEM-RESOLVED-ROVIBRONIC-SPIN-COMPOSITION-013

Generated grammar. Generate the literal product of molecular identity, equilibrium ownership, scale, common base, correction order, isotopologue route, measurement custody and extension forms; decide all 256 from the admitted Fold support and spectroscopy carriers.

Boundary: The complete registered H2 and D2 homonuclear-hydrogen X 1Sigma_g⁺ gas-phase equilibrium-distance vector, using one held atomic-length reference, the exact alpha carrier and each generated up, down, generator and terminal support only in its typed slot.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `named-isotopologue-state-and-condition__configuration-minimum-distance__held-atomic-length-reference__up-support-over-down-support__binary-order-alpha-return__distinct-light-and-heavy-typed-routes__sealed-exact-interval-transport__finite-typed-vector-exhaustion`. Closure is `finite_complete`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
identity	named-isotopologue-state-and-condition	generic-diatomic-with-era sed-state	A generic diatomic label cannot own a quantitative molecular value.
equilibrium	configuration-minimum-distance	arbitrary-distance-sample	A sampled distance is not the generated equilibrium fixed point.
scale	held-atomic-length-reference	free-angstrom-scale	A freely chosen physical scale is a fitted parameter.
base	up-support-over-down-support	free-leading-ratio	A chosen leading ratio can place the prediction near any target.
correction	binary-order-alpha-return	linear-or-open-correction-series	A linear or open series has no typed closure and can be extended to fit.
isotope	distinct-light-and-heavy-typed-routes	one-universal-isotope-correction	One correction erases the admitted nuclear-support distinction.
measurement	sealed-exact-interval-transport	target-readable-or-selected-value	Target-readable execution cannot provide an empirical prediction seal.
extension	finite-typed-vector-exhaustion	free-species-term-or-fit	A species correction not generated by the typed boundary is a free parameter.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:e02f662e7c0aa692f67d0e8ed4f95ff1659257b5f822d8b1edd4d124578eda34`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b6058901f49f5d30bdd0b9f772615573c5a67f1443350cdbf6e668e0e6ea09cb`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 2262681aa14d89d7542bb2d5e13de2e19cb730e877a496a3a01c2e929cf3d3ab.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 7668d5103f2085ff8a84019c7ee04624068999d18dde947556032b18b10fcfcfb.

Independent reconstruction. Implementation hash

sha256: 31b735b90f8658eb40104ba4aa65c64a3bcce0cf86aa168886bae2fa197e12e6; independent certificate sha256: 6e979ce5baea342a21add1959cd08de631bba709e3d28a2fa390d019d9dd088f; external-validation hash sha256: dd1b9eb2118fae157a6d8b284865bba37ccffd9a61d6d3afe4b89ff8b2403941. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS
- NIST-WEBBOOK-SRD69-H2-DIATOMIC-CONSTANTS
- NIST-WEBBOOK-SRD69-D2-DIATOMIC-CONSTANTS

Comparison records:

- H2: exact predicted interval [470557014695201322848924166499563917/634652975567759859324902500000000000, 470557014841034039912907896380265091/634652975567759859324902500000000000] angstrom; NIST 0.74144 within [148287/200000, 148289/200000]; overlap True
- D2: exact predicted interval [470610667121473368366229158732047057/634652975567759859324902500000000000, 470610667267322713126445183781277311/634652975567759859324902500000000000] angstrom; NIST 0.74152 within [148303/200000, 29661/40000]; overlap True
- adverse displaced H2 interval rejected: True
- adverse displaced D2 interval rejected: True

Falsification boundary: The claim fails if either exact sealed physical interval does not overlap its complete NIST displayed equilibrium-distance interval; if species, isotopologue, X state, gas-phase condition, CODATA scale provenance or source uncertainty is erased; if either target is readable before sealing; if any extra coefficient or species rule is appended; or if a deliberately displaced interval is accepted.

Measurement receipt: sha256: a4cc02639a1ada2e70cafe7ea3a71e4ba564f3f8af0584be7d57b8dfbef2ff27.
Isolation certificate: sha256: 06ab29061b07bd42b4c19d1569506b7feb6e0feb832ce402119aa41fa607e266.
Custody certificate: sha256: 05ecee08c201b2692c07c82d29fa82ce8215ed5ba52a1bf958eff741ff0bd621.

Registered source descriptions:

- National Institute of Standards and Technology: ()

Registered target identities:

- None.

Meaning. The quantitative property family derives bond length, dissociation, angle, torsion, dipole, polarizability, ionization, affinity, vibration, rotation, binding, magnetic response, formation energy and their common-carrier vector without per-property fitting. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero, negative, irrational, imaginary, floating, signed or continuum proof value
- no NIST distance, CODATA scale inscription or angstrom conversion in the executable law
- no imported wavefunction, potential, rigid-rotor length equation or conventional molecular model
- no fitted bond parameter, selected decimal coefficient, species lookup or free correction term

- no target access before the exact interval vector is sealed
- no claim beyond the registered H2/D2 X-state gas-phase boundary

Immutable identities. Source manifest

sha256:6553eca8df1bcc61702241dee2f1c9250020c1331bedb000ee621b86e2e0509b; derivation seal
 sha256:12c2b2c9cd881d17ce8a4e0b99556e8988d0eec2af32cd9eb3e1cdf9da5969df; engine receipt
 sha256:4a9b7cb81b8698229d13832eb27a5a17cdab9fb53f5fb1e244deefab5ec3ae5a at receipts/engine/model_admitted/SFT-CHEM-EQUILIBRIUM-BOND-LENGTH-001-4a9b7cb81b869822.json; empirical hash
 sha256:ccf5b0e09d7fa468f50769503014385f39fd9c8acef2467f43f70a63b9a67f81; measurement receipt
 sha256:a4cc02639a1ada2e70cafe7ea3a71e4ba564f3f8af0584be7d57b8dfbef2ff27.

112. Exact H2/D2 ground-state bond-dissociation path relation

Claim identity: SFT-CHEM-BOND-DISSOCIATION-ENERGY-002

Question and theorem. For each registered H2 or D2 channel, the B-prime threshold terminates in one ground-state atom and one 2s atom, while X-state ground dissociation terminates in two ground-state atoms. Their shared ground-atom prefix cancels structurally, so D0(M2) is exactly the B-prime threshold after the held atomic 1S-2S separation is Taken. This relation is sealed without measured values; only afterward does the complete threshold, atomic and historical/later ground-dissociation vector test it exactly.

named-isotopologue-and-X-state__both-product-channels-held__shared-prefix-path-composition__ordered-positive-held-Take__all-measurements-opened-post-seal__target-inaccessible-until-prediction-seal__complete-historical-and-current-vector__typed-two-row-exhaustion-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-ATOMIC-HYDROGEN-RYDBERG-TERMINAL-005
- SFT-CHEM-JOINT-CORRELATION-DISSOCIATION-007
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-CONFIGURATION-ORDER-PATH-011
- SFT-CHEM-EQUILIBRIUM-BOND-LENGTH-001

Generated grammar. Generate the literal product of identity, channel, path, operation, public-input, target-custody, record-completeness and extension forms; decide all 256 using only admitted transition/conservation laws and the registered source identities.

Boundary: The complete eight-row H2/D2 path vector: two B-prime thresholds, two atomic 1S-2S segments, two APS-1994 ground-state rows, and the later high-resolution H2-2009 and D2-2022 rows, with species, state, product channel, condition and uncertainty held and every number withheld until after sealing.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *named-isotopologue-and-X-state__both-product-channels-held__shared-prefix-path-composition__ordered-positive-held-Take__all-measurements-opened-post-seal__target-inaccessible-until-prediction-seal__complete-historical-and-current-vector__typed-two-row-exhaustion-no-extra-rule*. Closure is *finite_complete*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
identity	named-isotopologue-and-X-state	generic-molecule-with-era sed-isotope	A generic molecule cannot own an isotopologue-specific quantitative path.
channel	both-product-channels-held	dissociation-products-era sed	Erasing the atomic product states destroys the path whose segment is Taken.
path	shared-prefix-path-composition	answer-only-energy-lookup	An answer-only energy does not derive the relation between the two state paths.
operation	ordered-positive-held-Take	signed-or-floating-subtraction	A signed scalar or floating subtraction is outside the Fold exact domain.
input	all-measurements-opened-post-seal	measured-value-used-as-prediction-input	A measured threshold or atomic interval inside prediction would make observation select the number.
measurement	target-inaccessible-until-prediction-seal	dissociation-target-readable-before-seal	A readable target can select or alter the predicted interval.
record	complete-historical-and-current-vector	selected-historical-or-current-row	Selecting one resolution or era can conceal an unfavorable comparison.
extension	typed-two-row-exhaustion-no-extra-rule	species-coefficient-or-correction	A species coefficient or residual correction would be a free parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:32e6e6bf6e9fc516efebbfef3b2f1df1ba6cf64e71806b817b8d0133387351ff.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:24fc7ad592b645e88ccad2ed86ab9834c411690b5bf2167822535e889b4435e.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:921d014eebde1652e4313b89832b21e73eac61b4c00698c275a5b3449ab8eb78.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:2ca7b50cc21321d768cd774f1760d8722097bffc216482fe378d4189002c25d6.

Independent reconstruction. Implementation hash

sha256:197c36c493be53c9160e9c89cd3d01f18965a080d1a1575f35d8a0888256aa31; independent certificate sha256:2e70c75b5ea57a65ba0688cfd293bed9275ecdf158dfde656799936d6c6e507a; external-validation hash sha256:56b1636fe9c0e840fa8893ebfd24e8d6e04d5deba9cc7d93f86bcb1046100ad6. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- APS-PRA-49-2460-1994
- PRL-UDEM-H-1S2S-1997
- PRL-PARTHEY-HD-1S2S-SHIFT-2010
- SI-DEFINING-LIGHT-SPEED
- JCP-LIU-H2-DISSOCIATION-2009
- PRA-HUSSELS-D2-DISSOCIATION-2022

Comparison records:

- APS-1994-H2-X-GROUND-D0: post-seal exact Take interval [135349045807917/3747405725, 27069869120117/749481145] inverse-centimetre; source 36118.11 within [3611803/100, 3611819/100]; overlap True
- CURRENT-H2-X-GROUND-D0: post-seal exact Take interval [135349045807917/3747405725, 27069869120117/749481145] inverse-centimetre; source 36118.06962 within [144472277/4000, 3611806999/100000]; overlap True
- APS-1994-D2-X-GROUND-D0: post-seal exact Take interval [1101687626692543/29979245800, 1101690025033917/29979245800] inverse-centimetre; source 36748.38 within [3674831/100, 734969/20]; overlap True
- CURRENT-D2-X-GROUND-D0: post-seal exact Take interval [1101687626692543/29979245800, 1101690025033917/29979245800] inverse-centimetre; source 36748.362282 within [2296772641/62500, 9187090577/250000]; overlap True
- adverse complete_eight_row_vector: True
- adverse all_measurements_absent_from_prediction: True
- adverse states_and_channels_held: True
- adverse reversed_or_nonpositive_Take_halts: True
- adverse every_displaced_interval_rejected: True
- adverse historical_and_later_rows_both_tested: True
- adverse tampered_registry_identity_rejected: True

Falsification boundary: The claim fails if either post-seal threshold Take atomic interval does not overlap every registered historical and later ground-dissociation interval for its isotopologue; if any of eight measurement rows, a state, product channel, condition, source or uncertainty is erased; if any measured number is readable before the relation seal; if a deliberately displaced interval is accepted; or if a fitted coefficient or correction is introduced.

Measurement receipt: sha256: 7ea52e3943c49ad36c07db9fa165563c3172acd066fa17e009fb4da18f631b68.
Isolation certificate: sha256: 544a98513fe117df865e69afc64a21cde541bf9eeac577b00eb334c1e14e714f.
Custody certificate: sha256: 80316e4452c8105ccfeec85b9afc3323f3222835c7060142d66c5c0da5e052bf.

Registered source descriptions:

- American Physical Society: ()
- Bureau International des Poids et Mesures: ()
- American Institute of Physics: ()

Registered target identities:

- None.

Meaning. The quantitative property family derives bond length, dissociation, angle, torsion, dipole, polarizability, ionization, affinity, vibration, rotation, binding, magnetic response, formation energy and their common-carrier vector without per-property fitting. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero, negative, irrational, imaginary, floating, signed or continuum proof value
- no conventional molecular potential, wavefunction, ab-initio energy or fitted bond coefficient
- no threshold, atomic-transition or dissociation value in the executable law or prediction
- no target-derived atomic transition, selected residual correction or species exception
- no erased X state, rotational state, B-prime channel, atomic product state or measurement condition
- no omission of a historical or current registered target row
- no claim beyond the registered H2/D2 homolytic ground-state boundary

Immutable identities. Source manifest

sha256: b86f385526a9cd90bd3a104566cc4806238b4282ba1723ad5474baf65059a842; derivation seal

sha256:768bad055854acda5a9baf4d7973469636522893cb4974c3aafe5594040c3f8b; engine receipt sha256:23224e2d802e317a0c49a3d2ae914fab62454843a2d660e0b027afb7f73793b92 at receipts/engine/model_admitted/SFT-CHEM-BOND-DISSOCIATION-ENERGY-002-23224e2d802e317a.json; empirical hash sha256:b528f5d188c869f84a2c08bddd753673eb65ee63cd8b67302508df0d756ac29c; measurement receipt sha256:7ea52e3943c49ad36c07db9fa165563c3172acd066fa17e009fb4da18f631b68.

113. Exact symmetry-forced molecular bond-angle relation

Claim identity: SFT-CHEM-MOLECULAR-BOND-ANGLE-003

Question and theorem. A closed molecular turn divided into n symmetry-indistinguishable positive ligand sectors has no lawful unequal-sector form: unequal parts would introduce an ungenerated distinction. Each sector is therefore exactly $1/n$ turn and a retained k -sector separation is exactly k/n turn. The structural fractions seal without degrees; only afterward are all four NIST CCCBDB BF3, XeF2 and XeF4 angle rows opened.

named-state-geometry-sector-carrier__exact-generated-cyclic-order__symmetry-forced-equal-sectors__positive-sector-separation-over-count__value-free-exact-turn-fraction__all-angle-values-open-post-seal__complete-three-species-four-angle-vector__one-law-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-STATE-SYMMETRY-DEGENERACY-005
- SFT-CHEM-CONFIGURATION-ORDER-PATH-011
- SFT-CHEM-EQUILIBRIUM-BOND-LENGTH-001
- SFT-CHEM-BOND-DISSOCIATION-ENERGY-002

Generated grammar. Generate the literal product of carrier, cyclic order, sector partition, angle relation, prediction custody, measurement custody, record completeness and extension forms; decide all 256 using only admitted exact structure and the value-free registered target identities.

Boundary: The complete registered symmetry-forced molecular-angle vector at the official source display resolution: BF3 trigonal-planar adjacent aFBF, XeF2 linear opposite aFXeF, and XeF4 square-planar adjacent and opposite aFXeF, retaining species, neutral state, geometry, angle role, coordinate, method/condition, source identity and source comment.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *named-state-geometry-sector-carrier__exact-generated-cyclic-order__symmetry-forced-equal-sectors__positive-se-*

tor-separation-over-count__value-free-exact-turn-fraction__all-angle-values-open-post-seal__complete-three-species-four-angle-vector__one-law-no-extra-rule. Closure is finite_complete; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	named-state-geometry-sector-carrier	answer-only-angle-scalar	An answer-only scalar erases the molecular state, geometry and sector carrier.
order	exact-generated-cyclic-order	unordered-ligand-cloud	An unordered cloud cannot distinguish adjacent from opposite ligand paths.
partition	symmetry-forced-equal-sectors	unequal-or-fitted-sectors	Unequal sectors introduce an ungenerated distinction or fitted geometric parameter.
relation	positive-sector-separation-over-count	continuum-trigonometric-angle	A continuum or irrational trigonometric construction is outside the exact Fold domain.
prediction	value-free-exact-turn-fraction	degree-target-in-prediction	A degree value readable before sealing could select the predicted result.
measurement	all-angle-values-open-post-seal	target-open-before-seal	An open target before sealing destroys empirical custody.
record	complete-three-species-four-angle-vector	selected-species-or-angle-role	Selecting one favorable geometry or omitting an opposite row cannot close the registered vector.
extension	one-law-no-extra-rule	species-correction-or-angle-exception	A species correction or angle exception is a free rule.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:36f4d1704cd3fa66d89d5df1caf3df58f9b4ec9dacd6677c18ec94226af6c979.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:38aae32eda9b5b0a587a1b39917fed2c26bb73d79e98a25ab18bc99fdaef718.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:7f2eea905a508270aa07733dca6ff7a6b0e6b63c9f9a93459c7436dceccc39bb.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:7d1c6815bfded1fc3cbf815eeb7819198a5754a83cbaaaa7f7db5a165f8c8822.

Independent reconstruction. Implementation hash

sha256:12539b1d44ec6992a809f70c1bdc036048a7893c29aa9dd4b5bdb00bb726399a; independent certificate sha256:d3f7a6b3d4448c62f85bee2652cbefc288496d3f5b3c558cb28a279f12123be0; external-validation hash sha256:a98a20028a411a27babec561e7cbfa33584d495c6cf1f2791e56cb37206d5e84. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-CCCBDB-SRD101-BF3-EXPERIMENTAL-GEOMETRY
- NIST-CCCBDB-SRD101-XEF2-EXPERIMENTAL-GEOMETRY
- NIST-CCCBDB-SRD101-XEF4-EXPERIMENTAL-GEOMETRY

Comparison records:

- NIST-CCCBDB-BF3-FBF-ADJACENT: exact 1/3 turn -> 120 degree; source 120 degree; exact match True

- NIST-CCCBDB-XEF2-FXEF-OPPOSITE: exact 1/2 turn -> 180 degree; source 180 degree; exact match True
- NIST-CCCBDB-XEF4-FXEF-ADJACENT: exact 1/4 turn -> 90 degree; source 90 degree; exact match True
- NIST-CCCBDB-XEF4-FXEF-OPPOSITE: exact 1/2 turn -> 180 degree; source 180 degree; exact match True
- adverse complete_four_angle_vector: True
- adverse all_degree_values_absent_from_prediction: True
- adverse all_identity_and_condition_fields_held: True
- adverse every_displaced_value_rejected: True
- adverse adjacent_opposite_swap_rejected: True
- adverse wrong_geometry_count_rejected: True
- adverse unsupported_tetrahedral_continuum_form_rejected: True
- adverse tampered_snapshot_identity_rejected: True

Falsification boundary: The claim fails if any sealed turn fraction translated after release does not equal its source-bound degree inscription at the registered observation resolution; if any of the four rows or its species, state, geometry, angle definition, method/condition or comment is erased; if a degree value is readable before prediction sealing; if a displaced, swapped, wrong-count or unsupported-geometry control is accepted; or if a fitted correction or continuum construction is introduced.

Measurement receipt: sha256:8cba26d29c398990114e3803676c6ccb709e1b24e59b77fc645b4a8ee360c25e.

Isolation certificate: sha256:601d3bde31ecaa087e9c21aef509d651c8a5827fb1a52c53c0c22af9460e85f1.

Custody certificate: sha256:8405745660a2a175cc1f79823219f19c32a36e2895797ca9f1baaf78279369fd.

Registered source descriptions:

- NIST: ()

Registered target identities:

- None.

Meaning. The quantitative property family derives bond length, dissociation, angle, torsion, dipole, polarizability, ionization, affinity, vibration, rotation, binding, magnetic response, formation energy and their common-carrier vector without per-property fitting. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero, negative, irrational, imaginary, floating, signed or continuum proof value
- no continuum trigonometry, Cartesian coordinate fit, wavefunction, hybridization model or imported angle equation
- no measured degree value or degree-scale conversion in the law, candidate forcing or prediction
- no unequal sector, fitted angular correction, species coefficient or exception
- no erased species, state, point group, geometry, coordinate, angle role, method, condition or source comment
- no omission of either XeF4 row or selection of only one favorable molecule
- no claim for tetrahedral or other ungenerated non-cyclic-equal-sector geometry in this receipt

Immutable identities. Source manifest

sha256:8bf1fa46cac2e394ab2c419bbba9c27dd2d9a7c7c12c3dbc5454aea1e73fd2da; derivation seal

sha256:56f271157d431cd0997e00143caef84578eef540da0d7884cb41e280b71e1ebb; engine receipt

sha256:744ccd57c01a23f29ae6f4785b6e30d4d1b685e2cc6868674a4bc9acdfff11eef at

receipts/engine/model_admitted/SFT-CHEM-MOLECULAR-BOND-ANGLE-003-744ccd57c01a23f2.json;

empirical hash sha256:82eccf94040387b52a07e272058babcb8a2d6b9a272e51141a3a7f365f3b0063;

measurement receipt sha256:8cba26d29c398990114e3803676c6ccb709e1b24e59b77fc645b4a8ee360c25e.

114. Exact held-orientation dihedral and torsional-state relation

Claim identity: SFT-CHEM-DIHEDRAL-TORSIONAL-STATE-004

Question and theorem. A molecular dihedral is an ordered four-site periodic carrier with held orientation, not a signed continuum scalar. At the registered 24-sector resolution, structural EmptyOne, each positive sector successor and recurrent One exactly generate the coordinate cycle. Complete cyclic neighbour order forces conformer minima and barriers; each barrier magnitude is the ordered positive Take to its adjacent conformer. The coordinate and operation seal before all fifty NIST angle and energy rows open.

named-ordered-four-site-carrier__held-cycle-orientation__generated-sector-successor-coordinate__complete-neighbour-conformer-state__local-barrier-and-ordered-positive-Take__terminal-One-identifies-anchor-class__value-free-coordinate-and-operation-seal__complete-two-rotor-fifty-row-vector

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-CONFIGURATION-ORDER-PATH-011
- SFT-CHEM-MOLECULAR-BOND-ANGLE-003

Generated grammar. Generate the literal product of ordered carrier, held orientation, sector coordinate, conformer state, barrier Take, recurrence, value-free prediction and record-completeness forms; decide all 256 from admitted exact structure and the value-free fifty-row identity surface.

Boundary: The complete two-path NIST CCCBDB gauche-ethanol internal-rotation surface at its registered 24-sector observation resolution: OH torsion atoms 1-2-3-4 and CH3 torsion atoms 3-2-1-5, each with 24 unique sector states plus one recurrent endpoint, retaining molecule, state, rotor type, orientation, conformer and barrier roles, source method/condition and both energy-unit inscriptions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *named-ordered-four-site-carrier__held-cycle-orientation__generated-sector-successor-coordinate__complete-neighbour-conformer-state__local-barrier-and-ordered-positive-Take__terminal-One-identifies-anchor-class__value-free-coordinate-and-operation-seal__complete-two-rotor-fifty-row-vector*. Closure is *finite_complete*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	named-ordered-four-site-carrier	unordered-or-three-site-angle	A dihedral cannot be identified without four ordered sites.
orientation	held-cycle-orientation	signed-conventional-angle	A signed scalar imports negative proof magnitude.
coordinate	generated-sector-successor-coordinate	continuum-rotation-coordinate	A continuum coordinate imports ungenerated positions.
state	complete-neighbour-conformer-state	named-conformer-without-neighbours	A conformer name without complete neighbours cannot force local stability.
barrier	local-barrier-and-ordered-positive-Take	imported-saddle-or-signed-difference	An imported saddle equation or signed residue is outside the exact domain.
recurrence	terminal-One-identifies-anchor-class	duplicated-terminal-configuration	Treating the complete turn endpoint as new duplicates one state.
prediction	value-free-coordinate-and-operation-seal	angle-or-energy-target-readable	Readable source values could select the coordinate or state result.
record	complete-two-rotor-fifty-row-vector	selected-extrema-or-one-rotor	A selected subset hides ordinary, adverse or second-rotor rows.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 28d4dba6677d918ad1fab28653d7bf57c5b5dc78db96664bea051734d4add0cb.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 4c3f501c73da182e00d6bb954d2cc08acb0bd1c6623fc727c7072a999fe5a575.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: d2fc9dd88756d4926a2d192c69cb68060a83c89875275fea5222c4abd6cd492e.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 15551efeb74ffa5f29bbd3019850f377e29303a05c54ff1046a170258f76cf50.

Independent reconstruction. Implementation hash

sha256: 556ed2254c24f8fe47ce01ccaa0f31fa3f257b3d5cfb1a1c929fad026d20a177; independent certificate sha256: 3d7157954bf8ef55724561993b633e7888f549bab79596db6948a97cc7101af1; external-validation hash sha256: f2d9453bb45b6ad996fdef1c31f1403286cc0357bfe270203c71fe84a9ad2e6c. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-CCCBDB-SRD101-ETHANOL-EXPERIMENTAL-ROTATIONAL-BARRIER

Comparison records:

- ethanol-torsion-1-sector-01: coordinate EmptyOne(identity='empty-One'); source angle 0 degree; energy 5.41 kJ mol⁻¹ / 452 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-02: coordinate PositiveRatio(numerator=PositiveCount(value=1), denominator=PositiveCount(value=24)); source angle 15 degree; energy 4.70 kJ mol⁻¹ / 393 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-03: coordinate PositiveRatio(numerator=PositiveCount(value=2), denominator=PositiveCount(value=24)); source angle 30 degree; energy 3.00 kJ mol⁻¹ / 251 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-04: coordinate PositiveRatio(numerator=PositiveCount(value=3), denominator=PositiveCount(value=24)); source angle 45 degree; energy 1.28 kJ mol⁻¹ / 107 cm⁻¹; coordinate match True

- ethanol-torsion-1-sector-05: coordinate PositiveRatio(numerator=PositiveCount(value=4), denominator=PositiveCount(value=24)); source angle 60 degree; energy 0.52 kJ mol⁻¹ / 43 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-06: coordinate PositiveRatio(numerator=PositiveCount(value=5), denominator=PositiveCount(value=24)); source angle 75 degree; energy 1.13 kJ mol⁻¹ / 94 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-07: coordinate PositiveRatio(numerator=PositiveCount(value=6), denominator=PositiveCount(value=24)); source angle 90 degree; energy 2.71 kJ mol⁻¹ / 227 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-08: coordinate PositiveRatio(numerator=PositiveCount(value=7), denominator=PositiveCount(value=24)); source angle 105 degree; energy 4.29 kJ mol⁻¹ / 359 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-09: coordinate PositiveRatio(numerator=PositiveCount(value=8), denominator=PositiveCount(value=24)); source angle 120 degree; energy 4.90 kJ mol⁻¹ / 410 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-10: coordinate PositiveRatio(numerator=PositiveCount(value=9), denominator=PositiveCount(value=24)); source angle 135 degree; energy 4.14 kJ mol⁻¹ / 346 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-11: coordinate PositiveRatio(numerator=PositiveCount(value=10), denominator=PositiveCount(value=24)); source angle 150 degree; energy 2.41 kJ mol⁻¹ / 202 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-12: coordinate PositiveRatio(numerator=PositiveCount(value=11), denominator=PositiveCount(value=24)); source angle 165 degree; energy 0.70 kJ mol⁻¹ / 59 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-13: coordinate PositiveRatio(numerator=PositiveCount(value=12), denominator=PositiveCount(value=24)); source angle 180 degree; energy 0.00 kJ mol⁻¹ / 0 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-14: coordinate PositiveRatio(numerator=PositiveCount(value=13), denominator=PositiveCount(value=24)); source angle 195 degree; energy 0.70 kJ mol⁻¹ / 59 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-15: coordinate PositiveRatio(numerator=PositiveCount(value=14), denominator=PositiveCount(value=24)); source angle 210 degree; energy 2.41 kJ mol⁻¹ / 202 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-16: coordinate PositiveRatio(numerator=PositiveCount(value=15), denominator=PositiveCount(value=24)); source angle 225 degree; energy 4.14 kJ mol⁻¹ / 346 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-17: coordinate PositiveRatio(numerator=PositiveCount(value=16), denominator=PositiveCount(value=24)); source angle 240 degree; energy 4.90 kJ mol⁻¹ / 410 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-18: coordinate PositiveRatio(numerator=PositiveCount(value=17), denominator=PositiveCount(value=24)); source angle 255 degree; energy 4.29 kJ mol⁻¹ / 359 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-19: coordinate PositiveRatio(numerator=PositiveCount(value=18), denominator=PositiveCount(value=24)); source angle 270 degree; energy 2.71 kJ mol⁻¹ / 227 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-20: coordinate PositiveRatio(numerator=PositiveCount(value=19), denominator=PositiveCount(value=24)); source angle 285 degree; energy 1.13 kJ mol⁻¹ / 94 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-21: coordinate PositiveRatio(numerator=PositiveCount(value=20), denominator=PositiveCount(value=24)); source angle 300 degree; energy 0.52 kJ mol⁻¹ / 43 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-22: coordinate PositiveRatio(numerator=PositiveCount(value=21), denominator=PositiveCount(value=24)); source angle 315 degree; energy 1.28 kJ mol⁻¹ / 107 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-23: coordinate PositiveRatio(numerator=PositiveCount(value=22), denominator=PositiveCount(value=24)); source angle 330 degree; energy 3.00 kJ mol⁻¹ / 251 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-24: coordinate PositiveRatio(numerator=PositiveCount(value=23), denominator=PositiveCount(value=24)); source angle 345 degree; energy 4.70 kJ mol⁻¹ / 393 cm⁻¹; coordinate match True
- ethanol-torsion-1-sector-25: coordinate PositiveRatio(numerator=PositiveCount(value=24), denominator=PositiveCount(value=24)); source angle 360 degree; energy 5.41 kJ mol⁻¹ / 452 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-01: coordinate EmptyOne(identity='empty-One'); source angle 0 degree; energy 15.92 kJ mol⁻¹ / 1331 cm⁻¹; coordinate match True

- ethanol-torsion-2-sector-02: coordinate PositiveRatio(numerator=PositiveCount(value=1), denominator=PositiveCount(value=24)); source angle 15 degree; energy 13.59 kJ mol⁻¹ / 1136 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-03: coordinate PositiveRatio(numerator=PositiveCount(value=2), denominator=PositiveCount(value=24)); source angle 30 degree; energy 7.96 kJ mol⁻¹ / 665 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-04: coordinate PositiveRatio(numerator=PositiveCount(value=3), denominator=PositiveCount(value=24)); source angle 45 degree; energy 2.33 kJ mol⁻¹ / 195 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-05: coordinate PositiveRatio(numerator=PositiveCount(value=4), denominator=PositiveCount(value=24)); source angle 60 degree; energy 0.00 kJ mol⁻¹ / 0 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-06: coordinate PositiveRatio(numerator=PositiveCount(value=5), denominator=PositiveCount(value=24)); source angle 75 degree; energy 2.33 kJ mol⁻¹ / 195 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-07: coordinate PositiveRatio(numerator=PositiveCount(value=6), denominator=PositiveCount(value=24)); source angle 90 degree; energy 7.96 kJ mol⁻¹ / 665 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-08: coordinate PositiveRatio(numerator=PositiveCount(value=7), denominator=PositiveCount(value=24)); source angle 105 degree; energy 13.59 kJ mol⁻¹ / 1136 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-09: coordinate PositiveRatio(numerator=PositiveCount(value=8), denominator=PositiveCount(value=24)); source angle 120 degree; energy 15.92 kJ mol⁻¹ / 1331 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-10: coordinate PositiveRatio(numerator=PositiveCount(value=9), denominator=PositiveCount(value=24)); source angle 135 degree; energy 13.59 kJ mol⁻¹ / 1136 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-11: coordinate PositiveRatio(numerator=PositiveCount(value=10), denominator=PositiveCount(value=24)); source angle 150 degree; energy 7.96 kJ mol⁻¹ / 665 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-12: coordinate PositiveRatio(numerator=PositiveCount(value=11), denominator=PositiveCount(value=24)); source angle 165 degree; energy 2.33 kJ mol⁻¹ / 195 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-13: coordinate PositiveRatio(numerator=PositiveCount(value=12), denominator=PositiveCount(value=24)); source angle 180 degree; energy 0.00 kJ mol⁻¹ / 0 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-14: coordinate PositiveRatio(numerator=PositiveCount(value=13), denominator=PositiveCount(value=24)); source angle 195 degree; energy 2.33 kJ mol⁻¹ / 195 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-15: coordinate PositiveRatio(numerator=PositiveCount(value=14), denominator=PositiveCount(value=24)); source angle 210 degree; energy 7.96 kJ mol⁻¹ / 665 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-16: coordinate PositiveRatio(numerator=PositiveCount(value=15), denominator=PositiveCount(value=24)); source angle 225 degree; energy 13.59 kJ mol⁻¹ / 1136 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-17: coordinate PositiveRatio(numerator=PositiveCount(value=16), denominator=PositiveCount(value=24)); source angle 240 degree; energy 15.92 kJ mol⁻¹ / 1331 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-18: coordinate PositiveRatio(numerator=PositiveCount(value=17), denominator=PositiveCount(value=24)); source angle 255 degree; energy 13.59 kJ mol⁻¹ / 1136 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-19: coordinate PositiveRatio(numerator=PositiveCount(value=18), denominator=PositiveCount(value=24)); source angle 270 degree; energy 7.96 kJ mol⁻¹ / 665 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-20: coordinate PositiveRatio(numerator=PositiveCount(value=19), denominator=PositiveCount(value=24)); source angle 285 degree; energy 2.33 kJ mol⁻¹ / 195 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-21: coordinate PositiveRatio(numerator=PositiveCount(value=20), denominator=PositiveCount(value=24)); source angle 300 degree; energy 0.00 kJ mol⁻¹ / 0 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-22: coordinate PositiveRatio(numerator=PositiveCount(value=21), denominator=PositiveCount(value=24)); source angle 315 degree; energy 2.33 kJ mol⁻¹ / 195 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-23: coordinate PositiveRatio(numerator=PositiveCount(value=22), denominator=PositiveCount(value=24)); source angle 330 degree; energy 7.96 kJ mol⁻¹ / 665 cm⁻¹; coordinate match True

- ethanol-torsion-2-sector-24: coordinate PositiveRatio(numerator=PositiveCount(value=23), denominator=PositiveCount(value=24)); source angle 345 degree; energy 13.59 kJ mol⁻¹ / 1136 cm⁻¹; coordinate match True
- ethanol-torsion-2-sector-25: coordinate PositiveRatio(numerator=PositiveCount(value=24), denominator=PositiveCount(value=24)); source angle 360 degree; energy 15.92 kJ mol⁻¹ / 1331 cm⁻¹; coordinate match True
- torsion 1 barrier position 1 Take conformer position 21 = 489/100 kJ mol⁻¹
- torsion 1 barrier position 1 Take conformer position 5 = 489/100 kJ mol⁻¹
- torsion 1 barrier position 9 Take conformer position 5 = 219/50 kJ mol⁻¹
- torsion 1 barrier position 9 Take conformer position 13 = 49/10 kJ mol⁻¹
- torsion 1 barrier position 17 Take conformer position 13 = 49/10 kJ mol⁻¹
- torsion 1 barrier position 17 Take conformer position 21 = 219/50 kJ mol⁻¹
- torsion 2 barrier position 1 Take conformer position 21 = 398/25 kJ mol⁻¹
- torsion 2 barrier position 1 Take conformer position 5 = 398/25 kJ mol⁻¹
- torsion 2 barrier position 9 Take conformer position 5 = 398/25 kJ mol⁻¹
- torsion 2 barrier position 9 Take conformer position 13 = 398/25 kJ mol⁻¹
- torsion 2 barrier position 17 Take conformer position 13 = 398/25 kJ mol⁻¹
- torsion 2 barrier position 17 Take conformer position 21 = 398/25 kJ mol⁻¹
- adverse complete_fifty_row_vector: True
- adverse all_coordinates_match_postseal_source_angles: True
- adverse all_source_records_preserved: True
- adverse angle_energy_and_extrema_values_absent_before_seal: True
- adverse complete_identity_condition_surface: True
- adverse six_conformer_states_forced: True
- adverse six_barrier_states_forced: True
- adverse twelve_adjacent_positive_barrier_Takes: True
- adverse both_energy_units_force_same_state_order: True
- adverse both_recurrence_endpoints_identify_anchor: True
- adverse source_zero_glyphs_are_EmptyOne: True
- adverse reversed_barrier_Take_rejected: True
- adverse negative_signed_coordinate_rejected: True
- adverse tampered_source_identity_rejected: True

Falsification boundary: The claim fails if any of fifty post-seal source angles differs from its generated EmptyOne/sector-part/recurrent-One coordinate; if a conformer or barrier fails complete cyclic neighbour order; if any barrier Take is non-positive or loses an adjacent minimum; if either ordered four-atom carrier, rotor, orientation, source condition, ordinary row, least-state row or energy-unit inscription is erased; if any source value is readable before sealing; or if a signed, continuum, fitted or selected-row form is admitted.

Measurement receipt: sha256:0f35b6cf04c8a176b7144e6f842f9e824e63e1332af212addd8dda83e5e85380.
 Isolation certificate: sha256:eb47d9617682410889eb62d03446178f9eef6c14208d69576e744ae89478334b.
 Custody certificate: sha256:4ad309b73ca351cbbb92c89e0eb4c4f1a9e96baa0e5c4bffa1a7a842246bb7e2.

Registered source descriptions:

- NIST: ()

Registered target identities:

- None.

Meaning. The quantitative property family derives bond length, dissociation, angle, torsion, dipole, polarizability, ionization, affinity, vibration, rotation, binding, magnetic response, formation energy and their common-carrier vector without per-property fitting. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; source glyph 0 denotes structural absence only
- no negative, irrational, imaginary, floating, signed or continuum proof magnitude
- no imported torsional potential, Fourier series, differential equation, saddle equation or fitted coefficient
- no measured angle, energy, conformer outcome or barrier magnitude in law, forcing or prediction
- no erased molecule, state, ordered atom, rotor type, orientation, path position, method or condition
- no selected extrema-only, one-rotor or one-energy-unit subset
- no claim beyond the registered ethanol two-rotor observation boundary

Immutable identities. Source manifest

sha256:052106709df64d2ce9d019a1386217d3dde7324f6cbf29273cf5d5408505f120; derivation seal
 sha256:f439aa74e8a243f16686f48192c509af565cd7e937f03987d11f4dc40b098992; engine receipt
 sha256:0a854aa8467aaa502b4160813108bff2c1638acb00a4ae4d48124f227b38272b at receipts/engine/model_admitted/SFT-CHEM-DIHEDRAL-TORSIONAL-STATE-004-0a854aa8467aaa50.json; empirical hash
 sha256:9c71d234c14ea4565d8b164b9b5bbaac1453c0063445c1f82b4720bb66b6ff68; measurement receipt
 sha256:0f35b6cf04c8a176b7144e6f842f9e824e63e1332af212addd8dda83e5e85380.

115. Exact molecular dipole organization and squared-magnitude relation

Claim identity: SFT-CHEM-MOLECULAR-DIPOLE-MAGNITUDE-005

Question and theorem. A molecular dipole is the retained orientation of a complete molecular charge-distinction carrier. Its molecular symmetry closes forbidden components to structural EmptyOne and retains every allowed component as an exact positive magnitude on a held axis. The orientation-free squared magnitude is exactly the Junction of those component squares. The relation seals before all nine NIST values open; the H2/D2 absence rows, water-isotopologue component organization and reported total magnitudes then test the same law without a signed proof scalar or an irrational square root.

complete-named-molecular-charge-carrier__symmetry-forced-component-support__held-orientation-positive-components__finite-distinct-axis-Junction__exact-squared-magnitude-relation__value-free-symmetry-and-relation-seal__complete-five-species-nine-row-vector__one-structural-law-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001
- SFT-PHYS-ATOMIC-FIELD-SPLITTING-TERMINAL-005

- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-STATE-SYMMETRY-DEGENERACY-005
- SFT-CHEM-NUCLEAR-ELECTRONIC-COMPOSITION-012
- SFT-CHEM-CONFIGURATION-ORDER-PATH-011
- SFT-CHEM-MOLECULAR-BOND-ANGLE-003

Generated grammar. Generate the literal product of carrier, symmetry, orientation, composition, magnitude, prediction, record-completeness and extension forms; decide all 256 forms using only admitted charge, geometry, symmetry, exact arithmetic and measurement-custody laws.

Boundary: The complete registered H₂, D₂, H₂O, D₂O and HDO gas-phase vector: two homonuclear total-absence records and seven water-isotopologue component/total magnitude records, retaining species, molecular state, geometry, charge-distinction carrier, symmetry, axis, magnitude definition, method, condition, source and uncertainty; every numerical inscription is withheld until after relation sealing.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: complete-named-molecular-charge-carrier__symmetry-forced-component-support__held-orientation-positive-components__finite-distinct-axis-Junction__exact-squared-magnitude-relation__value-free-symmetry-and-relation-seal__complete-five-species-nine-row-vector__one-structural-law-no-extra-rule. Closure is finite_complete; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-named-molecular-charge-carrier	answer-only-dipole-scalar	A scalar answer erases species, state, geometry, charge carrier and symmetry.
symmetry	symmetry-forced-component-support	component-list-selected-after-values	A value-selected component list cannot establish molecular organization.
orientation	held-orientation-positive-components	signed-direction-as-proof-number	A conventional sign imports a negative proof magnitude.
composition	finite-distinct-axis-Junction	continuum-vector-premise	A continuum vector space and norm are not generated Fold forms.
magnitude	exact-squared-magnitude-relation	irrational-square-root-or-fitted-norm	A square root or fitted norm can leave the exact rational domain.
prediction	value-free-symmetry-and-relation-seal	dipole-value-readable-before-seal	A readable component or total could select the relation.
record	complete-five-species-nine-row-vector	favorable-species-or-component-subset	A selected subset can conceal an absent row, extra component or inconsistent total.
extension	one-structural-law-no-extra-rule	species-coefficient-charge-fit-or-correction	A species coefficient, charge fit or residual correction is a free parameter.

Adverse controls. All mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:96002ca88a85b1f4ef1f851c6afb78901331be77d1725f5967efeee1ccc9ef58.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:6c39c4fdebab031d4f1f275562874df74dcabd120c0398f32e7e2462b0208d7c.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:5248b2d4ab1a1d1c3933a9dfdd2ed5af6e9cedb211f9d67fae781dc396c07454.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:7166c2b777a7a1544b14d1f23d6af6f5e58f2e91436b5a097a93a0853bd404b3.

Independent reconstruction. Implementation hash

sha256:641217e4153c6a92891715bbc356a9db240b6dcc9bd10f73cc7c6b17efb6e66; independent certificate sha256:543b5cf6506a04c50600e59b055383b90dff643ffe8c6c5820168b26b389d2dc; external-validation hash sha256:0e8518da3fe8f53025977d9ed38a6a950320ce572b1af163350a47f27bc6cb7b. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-NBS-JCP-59-2254-1973
- NIST-CCCBDB-SRD101-EXPERIMENTAL-DIPOLES

Comparison records:

- H2: source 0.000 D maps to native EmptyOne; match True
- D2: source 0.000 D maps to native EmptyOne; match True
- H2O: 1 post-seal component intervals give exact squared-magnitude [859329/250000, 5377761/1562500]; reported squared total [859329/250000, 5377761/1562500]; overlap True
- D2O: 1 post-seal component intervals give exact squared-magnitude [343620369/100000000, 345179241/100000000]; reported squared total [343620369/100000000, 345179241/100000000]; overlap True
- HDO: 2 post-seal component intervals give exact squared-magnitude [6853489/20000000, 34340297/100000000]; reported squared total [342583081/100000000, 343472089/100000000]; overlap True
- adverse complete_five_species_nine_row_vector: True
- adverse all_measurements_absent_from_prediction: True
- adverse symmetry_forces_component_counts: True
- adverse homonuclear_source_glyphs_map_only_to_EmptyOne: True
- adverse every_squared_magnitude_interval_overlaps: True
- adverse every_displaced_total_rejected: True
- adverse missing_component_is_not_computable: True
- adverse duplicate_axis_rejected: True
- adverse conventional_signed_direction_absent_from_prediction: True
- adverse tampered_registry_identity_rejected: True

Falsification boundary: The claim fails if either registered homonuclear total is not source absence; if H2O or D2O does not retain exactly one reported component, HDO does not retain exactly two, or any post-seal exact outward component-square Junction fails to overlap its separately reported squared-total interval; if any row, identity, state, geometry, symmetry, axis, method, condition or uncertainty is erased; if a value is readable before sealing; or if a signed, irrational, fitted or continuum rule is introduced.

Measurement receipt: sha256:c6af87f488728153073021e08c3ff5d8379f3724b6c45215aded5a0aaedbcf71.

Isolation certificate: sha256:a290c5f115ce816a33bc84fb5803a8ea0f2ff50c8c3ee59439f33aac34e8aa26.

Custody certificate: sha256:59c786d68d53f725f2d71392f0ac2d03030885e7cb71711e4b021219730f9082.

Registered source descriptions:

- National Bureau of Standards / Journal of Chemical Physics: ()
- National Institute of Standards and Technology: ()

Registered target identities:

- None.

Meaning. The quantitative property family derives bond length, dissociation, angle, torsion, dipole, polarizability, ionization, affinity, vibration, rotation, binding, magnetic response, formation energy and their common-carrier vector without per-property fitting. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero, negative, irrational, imaginary, floating, signed or continuum proof value
- no conventional vector-space premise, square-root magnitude or signed Cartesian direction in proof
- no imported charge model, Stark Hamiltonian, wavefunction, fitted partial charge or molecular coefficient
- no measured component, total, uncertainty or source inscription in the law, grammar or prediction
- no erased species, state, geometry, charge carrier, symmetry, axis, method or condition
- no omitted homonuclear absence row or water-isotopologue component/total row
- no claim beyond the registered five-species, nine-row gas-phase boundary

Immutable identities. Source manifest

sha256:4ed40bae8a1bef03f925374d1dc2c9928c3771d3402bbdaa3a7214d01b97b3fe; derivation seal sha256:3f26018ad55b11710333262bef1d101f961a5768516e08834e09d37a2bb59a98; engine receipt sha256:023e38bf6e35d0ae45934d92177d47236bd57f486ea73704ec632bbc95578147 at receipts/engine/model_admitted/SFT-CHEM-MOLECULAR-DIPOLE-MAGNITUDE-005-023e38bf6e35d0ae.json; empirical hash sha256:1bcbbf831f3e4f47c691bf6086c738f9251ca595939c9ced66e3f5e94e989075; measurement receipt sha256:c6af87f488728153073021e08c3ff5d8379f3724b6c45215aded5a0aaedbcf71.

116. Exact molecular polarizability response and isotropic-composition law

Claim identity: SFT-CHEM-MOLECULAR-POLARIZABILITY-006

Question and theorem. A molecular polarizability component is the exact positive ratio between a retained induced-dipole distinction and the retained external electric distinction that causes it. Repeating equal field acts preserves this ratio at every generated depth. At the admitted three-axis boundary, the isotropic response is the exact one-third Junction of the three held component responses. The complete identity and relation vector seals before every one of the 252 NIST molecular alpha inscriptions opens.

complete-named-molecular-response-carrier__exact-positive-response-over-field-ratio__held-axis-positive-component__three-axis-exact-one-third-Junction__equal-act-ratio-invariant-successor__value-free-identity-and-relation-seal__complete-NIST-non-atomic-vector__one-response-law-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ALGEBRA-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-SOURCE-RESPONSE-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001

- SFT-PHYS-ATOMIC-FIELD-SPLITTING-TERMINAL-005
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOLECULAR-DIPOLE-MAGNITUDE-005

Generated grammar. Generate the literal product of carrier, response, orientation, composition, successor, prediction, record-completeness and extension forms; decide all 256 forms from admitted exact response, electric distinction, three-axis composition and measurement-custody laws.

Boundary: The depth-independent exact response-ratio and three-axis isotropic-composition laws, tested at the finite complete NIST CCCBDB non-atomic boundary; all 252 molecular rows in source order, retaining formula, name, molecular state, conformation, field-response definition, method, condition, units, reference and comment. All alpha inscriptions remain withheld until after relation sealing.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: complete-named-molecular-response-carrier__exact-positive-response-over-field-ratio__held-axis-positive-component__three-axis-exact-one-third-Junction__equal-act-ratio-invariant-successor__value-free-identity-and-relation-seal__complete-NIST-non-atomic-vector__one-response-law-no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-named-molecular-response-carrier	answer-only-alpha-scalar	A scalar answer erases species, state, conformation, field, method and condition.
response	exact-positive-response-over-field-ratio	imported-continuum-derivative	A continuum derivative imports an ungenerated field neighborhood.
orientation	held-axis-positive-component	signed-Cartesian-component	A signed Cartesian scalar imports a negative proof value.
composition	three-axis-exact-one-third-Junction	continuum-tensor-trace-premise	A continuum tensor is not a generated Fold form.
successor	equal-act-ratio-invariant-successor	depth-specific-response-coefficient	A new coefficient at each field depth is a free parameter family.
prediction	value-free-identity-and-relation-seal	alpha-readable-before-seal	A readable alpha vector could select the law or boundary.
record	complete-NIST-non-atomic-vector	selected-species-or-favorable-subset	A selected subset can conceal ordinary, charged, isotopic or adverse rows.
extension	one-response-law-no-extra-rule	species-fit-or-residual-correction	A species coefficient or residual correction is a fitted parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: a601c3fc936537ddc2a8e5bada019f173b666c2112cd154886dfdebedcc4603f.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: ff5445445145ef4f22d661ed3574b7ccb5ba85d0a2c0d22d8b1c10f9e76f0393.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: e3a38e0225aa3d3db805a7581f21d886f88918192b581311cae0aa9ad26c88bd.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:8219eb1edd889fdb8c997dfc86e12cc40c533a3d63cbcdf3f811d9d85d68d.

Independent reconstruction. Implementation hash

sha256:3f9c67bc39d326653d7864eaa1b63c32e52e7147f761683bfbfbd2bf5d432ed8; independent certificate sha256:df55b2409a41936003cbbb607a2423b303fabe5509449c39f9ad00bc942f8f81; external-validation hash sha256:7ec585b80afcdb660f667c5d8499660b580deddd50c1a7c5603458d226727b2a. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-CCCBDB-SRD101-EXPERIMENTAL-POLARIZABILITIES

Comparison records:

- NIST-CCCBDB-PROP-006-MOLECULAR-001: 0.783 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-002: 0.791 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-003: 0.787 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-004: 0.469 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-005: 1.710 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-006: 1.953 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-007: 1.698 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-008: 1.562 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-009: 0.800 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-010: 1.160 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-011: 39.999 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-012: 1.750 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-013: 2.515 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-014: 4.610 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-015: 2.841 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-016: 3.616 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-017: 6.431 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-018: 5.453 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-019: 4.340 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-020: 2.593 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-021: 1.501 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-022: 2.998 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-023: 2.507 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-024: 2.910 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-025: 3.079 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-026: 2.550 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-027: 3.631 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-028: 5.090 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-029: 3.882 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-030: 8.749 A³; exact positive and identity-bound True

- NIST-CCCBDB-PROP-006-MOLECULAR-031: 4.770 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-032: 11.269 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-033: 5.824 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-034: 3.487 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-035: 2.103 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-036: 5.015 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-037: 2.770 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-038: 2.420 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-039: 2.810 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-040: 4.237 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-041: 4.414 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-042: 13.589 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-043: 4.297 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-044: 3.927 A³; exact positive and identity-bound True
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- NIST-CCCBDB-PROP-006-MOLECULAR-046: 7.483 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-047: 6.790 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-048: 8.700 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-049: 9.573 A³; exact positive and identity-bound True
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- NIST-CCCBDB-PROP-006-MOLECULAR-053: 17.293 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-054: 11.351 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-055: 14.380 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-056: 2.448 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-057: 4.400 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-058: 3.319 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-059: 2.540 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-060: 2.761 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-061: 2.801 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-062: 2.824 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-063: 4.777 A³; exact positive and identity-bound True
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- NIST-CCCBDB-PROP-006-MOLECULAR-065: 4.416 A³; exact positive and identity-bound True
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- NIST-CCCBDB-PROP-006-MOLECULAR-068: 6.659 A³; exact positive and identity-bound True

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- NIST-CCCBDB-PROP-006-MOLECULAR-071: 4.650 A³; exact positive and identity-bound True
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- NIST-CCCBDB-PROP-006-MOLECULAR-126: 8.000 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-127: 10.260 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-128: 10.120 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-129: 17.886 A³; exact positive and identity-bound True
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- NIST-CCCBDB-PROP-006-MOLECULAR-133: 5.990 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-134: 6.240 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-135: 7.230 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-136: 7.090 A³; exact positive and identity-bound True
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- NIST-CCCBDB-PROP-006-MOLECULAR-171: 7.880 A³; exact positive and identity-bound True
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- NIST-CCCBDB-PROP-006-MOLECULAR-180: 7.020 A³; exact positive and identity-bound True
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- NIST-CCCBDB-PROP-006-MOLECULAR-182: 10.090 A³; exact positive and identity-bound True

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- NIST-CCCBDB-PROP-006-MOLECULAR-247: 15.930 A³; exact positive and identity-bound True
- NIST-CCCBDB-PROP-006-MOLECULAR-248: 29.720 A³; exact positive and identity-bound True
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- NIST-CCCBDB-PROP-006-MOLECULAR-252: 79.000 A³; exact positive and identity-bound True
- tampered_value_rejected: True
- missing_row_rejected: True
- displaced_identity_rejected: True
- complete_reference_cohorts_retained: True

Falsification boundary: The claim fails if the exact response ratio changes under any generated equal-field successor; if the isotropic carrier is not exactly the one-third Junction of three distinct held positive components; if any of the 252 post-seal NIST molecular alpha records is absent, duplicated, displaced, non-positive or not bound to its formula, state, conformation, units,

reference and source row; if target content is readable before sealing; or if a signed, irrational, continuum, imported, fitted or species-specific correction rule is introduced.

Measurement receipt: sha256:28f13fb914a04cb6154fca9ee7785591af5e6fe696532f19effc53147f967595.

Isolation certificate: sha256:5beaece4f57a1dfa4357642c726dc19c55f2665ada8fece99915a86d1d606a6.

Custody certificate: sha256:cffa82084f7f721428203caf14e7c94377fc001a48beae61ce69789285167e26.

Registered source descriptions:

- National Institute of Standards and Technology: ()

Registered target identities:

- None.

Meaning. The quantitative property family derives bond length, dissociation, angle, torsion, dipole, polarizability, ionization, affinity, vibration, rotation, binding, magnetic response, formation energy and their common-carrier vector without per-property fitting. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero, negative, irrational, imaginary, floating or continuum proof value
- no signed Cartesian component, continuum derivative, tensor calculus or perturbation series in proof
- no imported wavefunction, Hamiltonian, molecular model, basis set or fitted response coefficient
- no measured alpha, source inscription, reference value or table score in the law, grammar or prediction
- no erased formula, name, state, conformation, field definition, method, condition, unit or reference
- no selected species, source cohort, favorable subset or dropped comment row
- no numerical prediction for an ungenerated molecule and no claim beyond the registered static-response boundary

Immutable identities. Source manifest

sha256:88d4baa7441fa80b497a51de721ca60e877ff01c9e0cbeee290d7a65701a8dad; derivation seal

sha256:39a861bd5b15a7819b7eb8483806601eaecc208a8fc0497e7d9df9462282f855; engine receipt

sha256:926c85abc6d212f0264d91a7aef9609633f7a6c0a16bbcfb4cdb877020c6d20c at receipts/engine/model_admitted/SFT-CHEM-MOLECULAR-POLARIZABILITY-006-926c85abc6d212f0.json; empirical hash

sha256:f9d17c8ad9e0158acad2e15b10ed286f88d648b5f2146be1be961be2506fade4; measurement receipt

sha256:28f13fb914a04cb6154fca9ee7785591af5e6fe696532f19effc53147f967595.

117. Exact molecular ionization-energy and terminal-state ordering law

Claim identity: SFT-CHEM-MOLECULAR-IONIZATION-ENERGY-007

Question and theorem. Molecular ionization retains the complete neutral carrier, held electron-removal orientation and resulting positive-ion state. Its required energy is the ordered positive Take from the bound initial state height to the separated ion-plus-electron terminal height. Adiabatic ionization is the least Take over complete generated terminal support; a vertical held-geometry terminal is a member of that support and therefore cannot lie below the adiabatic least Take. The relation and all nine identities seal before the NIST experimental energy vector opens.

complete-neutral-to-ionic-carrier__held-removal-and-positive-terminal-separation__ordered-positive-final-from-initial-Take__least-adiabatic-and-held-geometry-vertical-paths__vertical-not-below-adiabatic__value-free-carrier-and-operation-seal__complete-nine-diatomic-NIST-vector__one-ionization-law-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-ELECTRONIC-STATE-IDENTITY-001
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-NUCLEAR-ELECTRONIC-COMPOSITION-012
- SFT-CHEM-MOLECULAR-POLARIZABILITY-006

Generated grammar. Generate the literal product of carrier, transformation, magnitude, path, order, prediction, record and extension forms; decide all 256 forms from admitted exact energy, state-transition, order, ion and measurement-custody laws.

Boundary: The depth-independent ordered positive ionization Take and adiabatic/vertical terminal-state order, tested at the finite complete first-nine-neutral-diatomic boundary inherited value-free from PROP-006: D2, HD, H2, N2, CO, NO, O2, HF and F2, retaining initial state and conformation, resulting ionic state, held removed carrier, path, method, condition, units, uncertainty class and official source identity.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-neutral-to-ionic-carrier__held-removal-and-positive-terminal-separation__ordered-positive-final-from-initial-Take__least-adiabatic-and-held-geometry-vertical-paths__vertical-not-below-adiabatic__value-free-carrier-and-operation-seal__complete-nine-diatomic-NIST-vector__one-ionization-law-no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-neutral-to-ionic-carrier	ionization-answer-with-erased-states	An answer-only energy erases the neutral carrier and resulting ionic state.
transformation	held-removal-and-positive-terminal-separation	signed-electron-subtraction	A signed electron subtraction imports a negative proof magnitude.
magnitude	ordered-positive-final-from-initial-Take	imported-orbital-energy-quality	An imported orbital theorem does not force the complete many-carrier state difference.
path	least-adiabatic-and-held-geometry-vertical-paths	adiabatic-vertical-state-conflation	Erasing geometry custody conflates two different terminal-state classes.
order	vertical-not-below-adiabatic	unconstrained-vertical-reordering	A held-geometry member cannot be declared below the least complete-support member.
prediction	value-free-carrier-and-operation-seal	ionization-value-readable-before-seal	Readable energy values could select the law or carrier boundary.
record	complete-nine-diatomic-NIST-vector	selected-isotopologue-or-favorable-row	A selected subset can conceal isotope, spin or polarity dependence.
extension	one-ionization-law-no-extra-rule	species-fit-or-residual-correction	A species coefficient or residual correction is a fitted parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 25533a4f7302622511ff09d6853c16c61d98e40b58af573892aa41f209023c19.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 7e25dcad984fef29c995bd64ab24a1a7b8014cfb471cb270963b3a98ea028828.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 2786bb13a2dbd67ed9dc9d0280bd2306cf73ad911f90fe5f8ac2af2c93ee2e6e.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 6a22ae157af4f1ea1b4090402d35a836df77b5ff41805e457511ab14589f3ec7.

Independent reconstruction. Implementation hash

sha256: 8483c0e625b0d7cd021fe5d4234b82655bea0672a8acfdc82159f7ca1c0cb0d5; independent certificate sha256: 0e70373510d4868197e64b7c7e7026faaea9c015a38bcab42a99c3c1bd25a72e; external-validation hash sha256: 364b07f8a504f8ef6891258eddcda4de654d2d07374ff2b29461ff2155f98f7c. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-WEBBOOK-SRD69-GAS-PHASE-ION-THERMOCHEMISTRY
- NIST-CCCBDB-SRD101-EXPERIMENTAL-IONIZATION-ENERGY

Comparison records:

- NIST-CCCBDB-PROP-007-D2-ADIABATIC-IE: 15.46658 eV; exact positive and state-bound **True**
- NIST-CCCBDB-PROP-007-HD-ADIABATIC-IE: 15.44465 eV; exact positive and state-bound **True**
- NIST-CCCBDB-PROP-007-H2-ADIABATIC-IE: 15.42593 eV; exact positive and state-bound **True**
- NIST-CCCBDB-PROP-007-N2-ADIABATIC-IE: 15.581 eV; exact positive and state-bound **True**
- NIST-CCCBDB-PROP-007-CO-ADIABATIC-IE: 14.014 eV; exact positive and state-bound **True**
- NIST-CCCBDB-PROP-007-NO-ADIABATIC-IE: 9.2642 eV; exact positive and state-bound **True**
- NIST-CCCBDB-PROP-007-O2-ADIABATIC-IE: 12.0697 eV; exact positive and state-bound **True**
- NIST-CCCBDB-PROP-007-HF-ADIABATIC-IE: 16.03 eV; exact positive and state-bound **True**
- NIST-CCCBDB-PROP-007-F2-ADIABATIC-IE: 15.697 eV; exact positive and state-bound **True**
- tampered_energy_rejected: **True**
- missing_row_rejected: **True**
- displaced_identity_rejected: **True**
- all_explicit_uncertainties_preserved: **True**

Falsification boundary: The claim fails if a registered separated ion-plus-electron state does not lie at an exact positive Take from its retained neutral state; if a held-geometry vertical terminal lies below the least complete-support adiabatic terminal; if any of the nine post-seal NIST records is absent, duplicated, displaced, non-positive or detached from its initial and resulting states, method, condition, units and uncertainty class; if a target value is readable before sealing; or if a signed, continuum, imported, fitted or species-specific correction rule is introduced.

Measurement receipt: sha256: 5f34f945bbfaed43e612ba8ef56a9b0f8c0b13f92eedfbc2d08558dcb1e105b.

Isolation certificate: sha256: fb223df91a013e8425393f2108d00c498dd793e6bfd2b70cb90f29534d506afe.

Custody certificate: sha256: 891b662bbdb1ce9ac8c80082dc264e3cfd99a316dabef05fa886e97a6c81f1a7.

Registered source descriptions:

- National Institute of Standards and Technology: ()

Registered target identities:

- None.

Meaning. The quantitative property family derives bond length, dissociation, angle, torsion, dipole, polarizability, ionization, affinity, vibration, rotation, binding, magnetic response, formation energy and their common-carrier vector without per-property fitting. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero, negative, irrational, imaginary, floating or continuum proof value
- no signed electron energy, negative orbital scalar or erased removal orientation
- no imported Koopmans theorem, orbital equality, wavefunction, Hamiltonian or fitted molecular coefficient
- no measured ionization energy, uncertainty or source inscription in the law, grammar or prediction
- no conflation of adiabatic and vertical terminal-state custody
- no erased neutral carrier, initial state, conformation, resulting ionic state, method or condition
- no selected isotope, favorable row, dropped uncertainty or species correction

Immutable identities. Source manifest

sha256:824e2289168ff94dc574b118cc56245ede6669f1d8039d76b3a91fde76dc2088; derivation seal
sha256:983e2cabb27a118f417d92d636d7edd2a03eaa5f6f3b65f3bef5dbc3748994ef; engine receipt
sha256:e507ad134235d5ff6051762ee89a5321ac63ccf0cd1aad7c97cbf734a15ea2a at receipts/engine/
model_admitted/SFT-CHEM-MOLECULAR-IONIZATION-ENERGY-007-e507ad134235d5ff.json; empirical
hash sha256:a374b967becc96d3267fda9d7eda93b3b9edfaf12d35396dacba2b3e110b8fbd; measurement
receipt sha256:5f34f945bbfaed43e612ba8ef56a9b0f8c0b13f92eedfbc2d08558dcb1e105b.

118. Exact molecular electron-affinity state-order and magnitude law

Claim identity: SFT-CHEM-MOLECULAR-ELECTRON-AFFINITY-008

Question and theorem. Molecular electron affinity retains the complete neutral carrier, held electron-gain action and resulting anion state. Conventional sign is replaced by a held state-order orientation: a bound anion lies below the neutral-plus-electron state, while an unbound anion lies above it. In either orientation the magnitude is the exact positive higher-state Take lower-state; coincident states carry structural EmptyOne. The law and all 96 identities seal before the complete NIST experimental vector opens.

complete-neutral-electron-anion-carrier__held-electron-gain-orientation__held-order-plus-positive-state-Take__bound-unbound-and-EmptyOne-boundary__value-and-orientation-free-carrier-seal__complete-NIST-molecular-experimental-vector__complete-source-uncertainty-custody__one-affinity-law-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001

- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTRON-COUNT-SPIN-002
- SFT-CHEM-ELECTRONIC-STATE-IDENTITY-001
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-NUCLEAR-ELECTRONIC-COMPOSITION-012
- SFT-CHEM-MOLECULAR-IONIZATION-ENERGY-007

Generated grammar. Generate the literal product of carrier, gain, magnitude, boundary, prediction, record, uncertainty and extension forms; decide all 256 from admitted exact energy, order, ion, electron-transfer, state-transition and measurement-custody laws.

Boundary: The depth-independent held-order plus positive-Take relation for every generated neutral/anion state pair, tested against every explicit experimental molecular electron-affinity record in the complete 192-row CCCBDB catalog after structurally excluding the 30 single-element atomic carriers: 162 molecular pages, 96 measured molecular rows, 93 bound and three unbound orientations, with all 89 explicit uncertainties.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-neutral-electron-on-anion-carrier__held-electron-gain-orientation__held-order-plus-positive-state-Take__bound-unbound-and-EmptyOne-boundary__value-and-orientation-free-carrier-seal__complete-NIST-molecular-experimental-vector__complete-source-uncertainty-custody__one-affinity-law-no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-neutral-electron-anion-carrier	electron-affinity-answer-with-erased-states	An answer-only scalar erases the neutral molecule and resulting anion.
gain	held-electron-gain-orientation	signed-electron-addition	A signed addition imports a negative proof magnitude.
magnitude	held-order-plus-positive-state-Take	conventional-signed-affinity-scalar	One signed number conflates state order with energy magnitude.
boundary	bound-unbound-and-EmptyOne-boundary	bound-anion-only-domain	Discarding unbound anions selects favorable conventional signs.
prediction	value-and-orientation-free-carrier-seal	affinity-value-or-orientation-readable-before-seal	A readable sign or magnitude could select the representation or carrier subset.
record	complete-NIST-molecular-experimental-vector	selected-positive-or-favorable-molecular-row	A selected subset can conceal unbound carriers and source diversity.
uncertainty	complete-source-uncertainty-custody	central-value-with-erased-uncertainty	Dropping uncertainty weakens the empirical record.
extension	one-affinity-law-no-extra-rule	species-fit-or-sign-correction	A species coefficient or sign correction is a fitted parameter.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:415942e31adccd232623a4989bc8ad1a9cea46589a6ff6df36912bd575cfbc3b`.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: aa691bdbebc0ad6120a4d4f5e1eb36b72b5697b26eab73da56bdd327cc4588c2.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 40c705e6b79ccd7fac5812e049d5c55f8a0bf1f86e78eacf18327927cb1208f3.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: 5f441baa7f22e43634a6b3abef0605200f253ad0f10294506f64217da98371f7.

Independent reconstruction. Implementation hash

sha256: 49c3967d51803d132a131e3bd73f7eb67f86954e149b8993809b027b0dbb1c47; independent certificate sha256: dc44b9c1baf69f8f335698c2bb3d8ca1321bb6c2175dd4725586e2995e1f3e8f; external-validation hash sha256: 7e85c75d67e9bf9accffc0800cc2ffd5374b98f778d76947151c85aeca958c5b. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-CCCBDB-COMPLETE-ELECTRON-AFFINITY-CATALOG
- NIST-CCCBDB-EXPERIMENTAL-MOLECULAR-ELECTRON-AFFINITY
- NIST-WEBBOOK-SRD69-GAS-PHASE-ION-THERMOCHEMISTRY

Comparison records:

- NIST-CCCBDB-PROP-008-7580678-EA: source plus-or-unmarked glyph, 0.342 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-14452596-EA: source plus-or-unmarked glyph, 0.437 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-13597972-EA: source plus-or-unmarked glyph, 0.7 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-13766262-EA: source plus-or-unmarked glyph, 0.3 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-14452610-EA: source plus-or-unmarked glyph, 1.3 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-3315375-EA: source plus-or-unmarked glyph, 1.238 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-12011540-EA: source plus-or-unmarked glyph, 2.4 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-12070154-EA: source plus-or-unmarked glyph, 3.273 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-13774920-EA: source plus-or-unmarked glyph, 0.37 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-10043115-EA: source plus-or-unmarked glyph, 3.16 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-2074875-EA: source plus-or-unmarked glyph, 3.862 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-3352576-EA: source plus-or-unmarked glyph, 1.827669 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-12505770-EA: source plus-or-unmarked glyph, 2.832 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True

- NIST-CCCBDB-PROP-008-10102439-EA: source plus-or-unmarked glyph, 0.026 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-7782447-EA: source plus-or-unmarked glyph, 0.451 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-3889756-EA: source plus-or-unmarked glyph, 3.3 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-13967061-EA: source plus-or-unmarked glyph, 0.49999 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-12061700-EA: source plus-or-unmarked glyph, 2.272 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-7782414-EA: source plus-or-unmarked glyph, 3.005 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-25681792-EA: source plus-or-unmarked glyph, 0.43 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-7681494-EA: source plus-or-unmarked glyph, 0.52 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-14332537-EA: source plus-or-unmarked glyph, 1.05 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-1309484-EA: source plus-or-unmarked glyph, 1.62 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-32752946-EA: source plus-or-unmarked glyph, 1.46 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-14457648-EA: source plus-or-unmarked glyph, 2.6 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-13774942-EA: source plus-or-unmarked glyph, 1.2771 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-12597352-EA: source plus-or-unmarked glyph, 2.202 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-11128248-EA: source plus-or-unmarked glyph, 0.81 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-13967141-EA: source plus-or-unmarked glyph, 1.028 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-12185090-EA: source plus-or-unmarked glyph, 0.589 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-14452665-EA: source plus-or-unmarked glyph, 1.092 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-13940211-EA: source plus-or-unmarked glyph, 2.314 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-12251900-EA: source plus-or-unmarked glyph, 2.6 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-2944050-EA: source plus-or-unmarked glyph, 0.205 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-12033566-EA: source plus-or-unmarked glyph, 1.194 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True

- NIST-CCCBDB-PROP-008-23550450-EA: source plus-or-unmarked glyph, 1.67 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-13827322-EA: source plus-or-unmarked glyph, 1.125 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-16068965-EA: source plus-or-unmarked glyph, 2.285 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-7447418-EA: source plus-or-unmarked glyph, 0.593 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-7647145-EA: source plus-or-unmarked glyph, 0.727 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-14989298-EA: source plus-or-unmarked glyph, 1.589 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-14989301-EA: source plus-or-unmarked glyph, 2.276 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-7790898-EA: source plus-or-unmarked glyph, 2.86 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-7782505-EA: source plus-or-unmarked glyph, 2.5 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-9000139-EA: source plus-or-unmarked glyph, 1.974 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-13940222-EA: source plus-or-unmarked glyph, 2.21 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-12640890-EA: source plus-or-unmarked glyph, 1.456 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-12185170-EA: source plus-or-unmarked glyph, 1.94 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-7550358-EA: source plus-or-unmarked glyph, 0.66 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-7647156-EA: source plus-or-unmarked glyph, 0.788 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-15656196-EA: source plus-or-unmarked glyph, 2.353 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-7726956-EA: source plus-or-unmarked glyph, 2.42 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-2465567-EA: source plus-or-unmarked glyph, 0.652 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-2122487-EA: source plus-or-unmarked glyph, 2.923 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-13770406-EA: source plus-or-unmarked glyph, 0.771 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-12596600-EA: source plus-or-unmarked glyph, 2.68 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-2597446-EA: source plus-or-unmarked glyph, 0.313 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True

- NIST-CCCBDB-PROP-008-14332286-EA: source plus-or-unmarked glyph, 0.338 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-3170830-EA: source plus-or-unmarked glyph, 1.078 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-22400266-EA: source plus-or-unmarked glyph, 3.609 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-13840885-EA: source plus-or-unmarked glyph, 4.32 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-10024972-EA: source plus-or-unmarked glyph, 0.76 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-124389-EA: source minus glyph, 0.59999 eV positive magnitude, anion-above-neutral-unbound-autodetachment; exact state-bound True
- NIST-CCCBDB-PROP-008-10102440-EA: source plus-or-unmarked glyph, 2.273 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-10028156-EA: source plus-or-unmarked glyph, 2.1028 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-2154598-EA: source plus-or-unmarked glyph, 0.179 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-13825906-EA: source plus-or-unmarked glyph, 1.123 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-13765430-EA: source plus-or-unmarked glyph, 1.271 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-12164975-EA: source plus-or-unmarked glyph, 3.42 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-15941772-EA: source plus-or-unmarked glyph, 3.537 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-75150-EA: source plus-or-unmarked glyph, 0.79998 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-7446095-EA: source plus-or-unmarked glyph, 1.107 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-12597034-EA: source plus-or-unmarked glyph, 2.093 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-1605727-EA: source plus-or-unmarked glyph, 1.603 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-10049044-EA: source plus-or-unmarked glyph, 2.14 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-24968556-EA: source plus-or-unmarked glyph, 1.097 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-13283313-EA: source plus-or-unmarked glyph, 0.038 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-2229074-EA: source plus-or-unmarked glyph, 0.08 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-2143693-EA: source plus-or-unmarked glyph, 0.49 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True

- NIST-CCCBDB-PROP-008-50000-EA: source minus glyph, 0.65 eV positive magnitude, anion-above-neutral-unbound-autodetachment; exact state-bound True
- NIST-CCCBDB-PROP-008-12033497-EA: source plus-or-unmarked glyph, 3.937 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-7637072-EA: source plus-or-unmarked glyph, 2.65 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-2264213-EA: source plus-or-unmarked glyph, 1.869 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-13765441-EA: source plus-or-unmarked glyph, 1.406 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-865361-EA: source plus-or-unmarked glyph, 0.465 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-7446119-EA: source plus-or-unmarked glyph, 1.9 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-3170807-EA: source plus-or-unmarked glyph, 2.173 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-2669898-EA: source plus-or-unmarked glyph, 0.668 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-2143682-EA: source plus-or-unmarked glyph, 1.572 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-75058-EA: source plus-or-unmarked glyph, 0.0072 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-3170692-EA: source plus-or-unmarked glyph, 0.423 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-2143580-EA: source plus-or-unmarked glyph, 1.161 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-2025561-EA: source minus glyph, 0.26 eV positive magnitude, anion-above-neutral-unbound-autodetachment; exact state-bound True
- NIST-CCCBDB-PROP-008-670542-EA: source plus-or-unmarked glyph, 3.17 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-67641-EA: source plus-or-unmarked glyph, 0.0015 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- NIST-CCCBDB-PROP-008-2396012-EA: source plus-or-unmarked glyph, 1.097 eV positive magnitude, anion-below-neutral-bound-attachment; exact state-bound True
- tampered_magnitude_rejected: True
- flipped_orientation_rejected: True
- complete_catalog_and_molecular_pages_preserved: True
- all_bound_and_unbound_rows_preserved: True
- all_explicit_uncertainties_preserved: True
- source_minus_glyphs_are_not_negative_numbers: True

Falsification boundary: The claim fails if any retained neutral/anion pair cannot be represented by one held state-order orientation and an exact positive higher-state Take lower-state magnitude or structural EmptyOne; if any of the 192 catalog carriers, 30 atomic exclusions, 162 molecular pages, 96 measured molecular rows, 93 bound rows, three unbound rows or 89 explicit uncertainties changes or is omitted; if a source minus glyph enters as a negative SFT number; if a target orientation or magnitude is readable before sealing; or if an imported, fitted, species-specific or sign-correction rule is introduced.

Measurement receipt: sha256:637516dbed378ff1bc62ea1794c866cdc860954b7346139d1bdf2a1f6d0f4079.
 Isolation certificate: sha256:37992c2f38e57b3425b90656f060e40159e079bf3b693db7426fa89c4b0e036c.
 Custody certificate: sha256:9278c52a02cf3f9d7c92607d8db1dc23b90debbe7bd4866bbd0ebaa6f5904bf5.

Registered source descriptions:

- National Institute of Standards and Technology: ()

Registered target identities:

- None.

Meaning. The quantitative property family derives bond length, dissociation, angle, torsion, dipole, polarizability, ionization, affinity, vibration, rotation, binding, magnetic response, formation energy and their common-carrier vector without per-property fitting. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; coincident state heights use structural EmptyOne
- no negative, irrational, imaginary, floating, signed or continuum proof value
- no conventional signed electron-affinity scalar as an SFT proof object
- no imported orbital theorem, wavefunction, Hamiltonian or fitted species coefficient
- no measured affinity magnitude, sign orientation or uncertainty in the law, grammar or prediction
- no favorable-sign selection, dropped unbound row, omitted uncertainty or atomic/molecular conflation

Immutable identities. Source manifest

sha256:fb4f3a1b216fc8310410bdcf3cdf3ed24407b68b8cacc5137ea8be5b53fe5624; derivation seal
 sha256:f908901cf1f6b9f06a09bef3cf4219396a915295f5fc5bd92995eef8d8b2112a; engine receipt
 sha256:6643867ae60a9d82388857fe0b1a21d1fded8ada313ee22c884dab97f9358a54 at receipts/engine/
 model_admitted/SFT-CHEM-MOLECULAR-ELECTRON-AFFINITY-008-6643867ae60a9d82.json; empirical
 hash sha256:0431400eea00db0a51a3a7078d9d94d1c2f78b024bcfa9ba9a4f66af56f9a293; measurement
 receipt sha256:637516dbed378ff1bc62ea1794c866cdc860954b7346139d1bdf2a1f6d0f4079.

119. Exact molecular vibrational recurrence-frequency law

Claim identity: SFT-CHEM-VIBRATIONAL-FREQUENCY-009

Question and theorem. A molecular vibrational frequency is derived from a complete retained molecule, mode and symmetry carrier as the exact positive count of finite recurrences divided by a positive observation-interval count. A conventional reciprocal-centimeter inscription is a held unit translation applied only after that ratio exists. All 2,009 displayed NIST mode identities seal before 1,984 positive measurements and 25 structural measurement absences open; theoretical frequencies, ratios and fitted scale factors are excluded.

*complete-molecule-mode-symmetry-carrier__finite-generated-recurrence-count__exact-recurrence-over-interval-ratio__
 _held-distinct-mode-and-symmetry-support__post-recurrence-held-unit-translation__value-free-complete-mode-seal__
 complete-displayed-NIST-surface-with-gap-custody__one-recurrence-law-no-scale-factor*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001

- SFT-PHYS-WAVE-PERIOD-FREQUENCY-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-MOLECULAR-SPECTRUM-HIERARCHY-004
- SFT-PHYS-MOLECULAR-SPECTROSCOPY-TERMINAL-005
- SFT-CHEM-ROVIBRONIC-COMPOSITION-001
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-RESOLVED-ROVIBRONIC-SPIN-COMPOSITION-013
- SFT-CHEM-MOLECULAR-ELECTRON-AFFINITY-008

Generated grammar. Generate the literal product of carrier, recurrence, magnitude, mode, translation, prediction, record and extension forms; decide all 256 from admitted exact arithmetic, recurrence, wave-frequency, spectroscopy, molecular-transition and measurement-custody laws.

Boundary: The depth-independent exact recurrence-count/positive-interval relation for every generated finite mode, tested against the complete 2,009-row table actually displayed by the frozen official NIST CCCBDB page: 145 displayed molecules, 1,984 positive experimental frequencies and 25 displayed absences. The page's advertised 164 molecules/2,452 vibrations and its undisplayed difference of 19/443 remain explicit.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-molecule-mode-symmetry-carrier__finite-generated-recurrence-count__exact-recurrence-over-interval-ratio__held-distinct-mode-and-symmetry-support__post-recurrence-held-unit-translation__value-free-complete-mode-seal__complete-displayed-NIST-surface-with-gap-custody__one-recurrence-law-no-scale-factor`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-molecule-mode-symmetry-carrier	frequency-answer-with-erased-mode	An answer-only number erases the molecule, mode, symmetry and transition class.
recurrence	finite-generated-recurrence-count	continuum-sinusoid-premise	A continuum waveform imports an ungenerated mathematical model.
magnitude	exact-recurrence-over-interval-ratio	imported-frequency-scalar	A named frequency without recurrence and interval does not derive its quantity.
mode	held-distinct-mode-and-symmetry-support	merged-or-relabelled-mode-support	Merging modes destroys their molecular and symmetry identities.
translation	post-recurrence-held-unit-translation	frequency-unit-selects-law	A conventional unit cannot select the recurrence law.
prediction	value-free-complete-mode-seal	frequency-target-readable-before-seal	Readable source frequencies could select the relation or row subset.
record	complete-displayed-NIST-surface-with-gap-custody	favorable-measured-row-subset	Dropping displayed absences or the advertised/displayed count discrepancy hides source limitations.
extension	one-recurrence-law-no-scale-factor	fitted-scale-or-molecular-correction	A scale factor or species residual is a fitted parameter.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:d53690ff53e454ed989a61797b9f5779c14d750df418db387ec3fe0fab543136`.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:d73bac966906ebfddb886397b97048a63b2636e1a23f40bc4a80208dff61f1e9.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:a50f85e3b72063e0979ae65dfa354e0629ae0bad55165dfd5709cdcef990a456.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:729c88340f1d3e0ab43f2d70f0bf7101a0984c0b5a74a66d10287ff325e187ce.

Independent reconstruction. Implementation hash

sha256:0268813a680ce29fedefe8c56215d6474488e3e95d0f862eb61661badcab8e41; independent certificate sha256:2d2b81cef10ccc3024bef80d4383729c942f478d3cdd1558edf0449ef0f25cb2; external-validation hash sha256:5dfc88ca1804f7eb6ee59ba35362c2eb024a71301ec5d64fda2e515c3ff20eef. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-CCCBDB-COMPLETE-DISPLAYED-FUNDAMENTAL-FREQUENCY-SURFACE

Comparison records:

- NIST-CCCBDB-PROP-009-MODE-0001: 4161 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0002: 2269 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0003: 1625 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0004: 2733 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0005: 1827 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0006: 3126 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0007: 2330 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0008: 775 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0009: 3570 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0010: 1862 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0011: 2143 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0012: 1138 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0013: 1054 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0014: 3961 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0015: 1286 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0016: 1123 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0017: 894 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0018: 2886 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0019: 773 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0020: 954 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0021: 3219 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0022: 1497 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0023: 3301 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0024: 3312 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0025: 2089 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0026: 712 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0027: 3653 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0028: 2029 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0029: 477 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0030: 3657 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0031: 1595 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0032: 3756 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0033: 2850 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0034: 1826 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0035: 756 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0036: 2684 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0037: 1565 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0038: 1501 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0039: 3436 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0040: 1392 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0041: 1098 per centimeter; exact and mode-bound True
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- NIST-CCCBDB-PROP-009-MODE-0045: 2224 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0046: 1285 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0047: 589 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0048: 3578 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0049: 1362 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0050: 893 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0051: 1225 per centimeter; exact and mode-bound True
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- NIST-CCCBDB-PROP-009-MODE-0063: 1148 per centimeter; exact and mode-bound True
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- NIST-CCCBDB-PROP-009-MODE-0066: 3374 per centimeter; exact and mode-bound True
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- NIST-CCCBDB-PROP-009-MODE-0078: 503 per centimeter; exact and mode-bound True
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- NIST-CCCBDB-PROP-009-MODE-0080: 2782 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0081: 1746 per centimeter; exact and mode-bound True
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- NIST-CCCBDB-PROP-009-MODE-0096: 577 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0097: 656 per centimeter; exact and mode-bound True
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- NIST-CCCBDB-PROP-009-MODE-0099: 1700 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0100: 1263 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0101: 790 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0102: 596 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0103: 543 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0104: 2060 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0105: 585 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0106: 1898 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0107: 410 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0108: 213 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0109: 2621 per centimeter; exact and mode-bound True
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- NIST-CCCBDB-PROP-009-MODE-0111: 542 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0112: 924 per centimeter; exact and mode-bound True
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- NIST-CCCBDB-PROP-009-MODE-0114: 1928 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0115: 965 per centimeter; exact and mode-bound True
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- NIST-CCCBDB-PROP-009-MODE-0120: 888 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0121: 691 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0122: 1449 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0123: 480 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0124: 1523 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0125: 1018 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0126: 603 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0127: 364 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0128: 991 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0129: 423 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0130: 1089 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0131: 701 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0132: 1260 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0133: 509 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0134: 1032 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0135: 647 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0136: 907 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0137: 492 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0138: 690 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0139: 297 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0140: 935 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0141: 263 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0142: 2917 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0143: 1534 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0144: 3019 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0145: 1306 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0146: 3263 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0147: 3024 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0148: 2914 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0149: 1638 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0150: 1452 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0151: 1344 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0152: 1058 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0153: 1127 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0154: 1061 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0155: 3650 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0156: 3294 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0157: 1605 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0158: 1353 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0159: 1115 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0160: 895 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0161: 3359 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0162: 1295 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0163: 386 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0164: 3070 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0165: 2152 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0166: 1388 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0167: 1118 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0168: 588 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0169: 528 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0170: 3166 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0171: 977 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0172: 433 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0173: 3570 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0174: 2943 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0175: 1770 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0176: 1387 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0177: 1229 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0178: 1105 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0179: 625 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0180: 1033 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0181: 638 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0182: 3550 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0183: 1708 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0184: 1331 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0185: 1325 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0186: 879 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0187: 647 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0188: 579 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0189: 762 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0190: 456 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0191: 2196 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0192: 786 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0193: 2258 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0194: 1573 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0195: 573 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0196: 550 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0197: 61 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0198: 2930 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0199: 1464 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0200: 1049 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0201: 3006 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0202: 1467 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0203: 1182 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0204: 2948 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0205: 1508 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0206: 1111 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0207: 529 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0208: 1262 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0209: 3014 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0210: 1178 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0211: 1435 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0212: 1090 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0213: 3036 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0214: 1117 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0215: 700 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0216: 1372 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0217: 1152 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0218: 507 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0219: 1759 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0220: 1301 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0221: 928 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0222: 804 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0223: 633 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0224: 455 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0225: 303 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0226: 709 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0227: 152 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0228: 909 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0229: 435 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0230: 1283 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0231: 631 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0232: 3026 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0233: 1623 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0234: 1342 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0235: 1023 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0236: 2989 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0237: 1444 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0238: 940 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0239: 3105 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0240: 826 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0241: 3086 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0242: 1217 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0243: 949 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0244: 3322 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0245: 687 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0246: 490 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0247: 627 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0248: 482 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0249: 630 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0250: 231 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0251: 3398 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0252: 3329 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0253: 1642 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0254: 1275 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0255: 1076 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0256: 780 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0257: 377 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0258: 3350 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0259: 3314 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0260: 1628 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0261: 1275 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0262: 966 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0263: 2954 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0264: 2267 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0265: 1385 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0266: 920 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0267: 3009 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0268: 1448 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0269: 1041 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0270: 362 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0271: 2290 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0272: 2119 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0273: 692 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0274: 2241 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0275: 1154 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0276: 504 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0277: 263 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0278: 472 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0279: 107 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0280: 3681 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0281: 3000 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0282: 2844 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0283: 1477 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0284: 1455 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0285: 1345 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0286: 1060 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0287: 1033 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0288: 2960 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0289: 1477 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0290: 1165 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0291: 200 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0292: 3564 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0293: 3439 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0294: 2854 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0295: 1754 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0296: 1577 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0297: 1390 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0298: 1258 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0299: 1046 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0300: 581 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0301: 1021 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0302: 603 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0303: 289 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0304: 2843 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0305: 1745 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0306: 1338 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0307: 1065 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0308: 551 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0309: 801 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0310: 127 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0311: 1048 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0312: 2835 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0313: 1732 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0314: 1312 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0315: 339 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0316: 3115 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0317: 3080 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0318: 3052 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0319: 1654 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0320: 1380 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0321: 1306 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0322: 1156 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0323: 929 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0324: 483 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0325: 940 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0326: 863 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0327: 711 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0328: 3070 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0329: 1728 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0330: 1360 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0331: 926 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0332: 550 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0333: 714 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0334: 803 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0335: 611 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0336: 3154 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0337: 1302 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0338: 955 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0339: 438 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0340: 2275 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0341: 1227 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0342: 802 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0343: 522 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0344: 1214 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0345: 618 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0346: 463 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0347: 196 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0348: 1872 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0349: 778 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0350: 394 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0351: 190 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0352: 1186 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0353: 558 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0354: 508 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0355: 1337 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0356: 218 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0357: 1340 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0358: 551 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0359: 406 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0360: 3334 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0361: 2918 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0362: 2124 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0363: 1382 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0364: 931 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0365: 3008 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0366: 1452 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0367: 1053 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0368: 633 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0369: 328 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0370: 3015 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0371: 1443 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0372: 1073 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0373: 865 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0374: 3007 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0375: 1957 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0376: 1398 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0377: 3086 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0378: 999 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0379: 841 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0380: 355 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0381: 3158 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0382: 2909 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0383: 1653 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0384: 1483 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0385: 1110 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0386: 905 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0387: 996 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0388: 820 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0389: 2995 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0390: 1088 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0391: 569 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0392: 3124 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0393: 1043 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0394: 1011 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0395: 769 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0396: 3361 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0397: 2961 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0398: 2820 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0399: 1623 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0400: 1473 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0401: 1430 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0402: 1130 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0403: 1044 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0404: 780 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0405: 3427 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0406: 2985 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0407: 1485 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0408: 1335 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0409: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0410: 268 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0411: 3125 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0412: 3078 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0413: 3042 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0414: 2239 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0415: 1615 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0416: 1416 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0417: 1282 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0418: 1096 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0419: 869 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0420: 570 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0421: 242 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0422: 972 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0423: 954 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0424: 683 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0425: 362 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0426: 2935 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0427: 2275 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0428: 1395 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0429: 890 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0430: 582 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0431: 167 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0432: 1220 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0433: 367 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0434: 2968 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0435: 933 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0436: 337 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0437: 2275 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0438: 1318 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0439: 982 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0440: 366 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0441: 3014 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0442: 2923 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0443: 2716 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0444: 1743 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0445: 1433 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0446: 1395 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0447: 1352 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0448: 1114 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0449: 867 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0450: 509 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0451: 2964 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0452: 1431 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0453: 1102 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0454: 764 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0455: 150 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0456: 3006 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0457: 1467 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0458: 1267 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0459: 1146 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0460: 857 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0461: 3063 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0462: 1050 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0463: 1020 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0464: 3065 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0465: 1146 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0466: 797 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0467: 3006 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0468: 1459 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0469: 1159 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0470: 824 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0471: 3037 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0472: 2950 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0473: 1613 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0474: 1454 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0475: 1408 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0476: 1239 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0477: 991 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0478: 838 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0479: 627 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0480: 346 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0481: 2991 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0482: 1438 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0483: 1143 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0484: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0485: 186 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0486: 3563 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0487: 1216 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0488: 1136 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0489: 831 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0490: 548 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0491: 422 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0492: 379 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0493: 224 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0494: 3567 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0495: 1452 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0496: 1157 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0497: 882 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0498: 558 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0499: 506 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0500: 288 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0501: 3043 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0502: 2955 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0503: 1870 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0504: 1440 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0505: 1378 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0506: 1188 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0507: 1000 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0508: 826 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0509: 598 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0510: 420 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0511: 3004 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0512: 1437 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0513: 1054 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0514: 567 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0515: 123 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0516: 2530 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0517: 2088 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0518: 1183 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0519: 790 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0520: 833 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0521: 2596 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0522: 915 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0523: 1930 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0524: 973 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0525: 1760 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0526: 860 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0527: 2609 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0528: 949 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0529: 369 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0530: 1020 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0531: 2520 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0532: 1603 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0533: 1172 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0534: 2954 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0535: 1388 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0536: 995 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0537: 289 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0538: 2896 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0539: 1379 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0540: 2969 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0541: 1468 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0542: 1190 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0543: 2985 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0544: 1469 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0545: 822 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0546: 3338 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0547: 3079 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0548: 3015 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0549: 1482 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0550: 1237 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0551: 1211 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0552: 1090 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0553: 998 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0554: 856 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0555: 773 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0556: 3079 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0557: 3015 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0558: 1463 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0559: 1268 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0560: 1131 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0561: 1095 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0562: 904 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0563: 817 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0564: 3583 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0565: 3051 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0566: 2944 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0567: 1788 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0568: 1430 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0569: 1382 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0570: 1264 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0571: 1182 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0572: 989 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0573: 847 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0574: 657 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0575: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0576: 2996 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0577: 1430 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0578: 1048 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0579: 642 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0580: 534 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0581: 93 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0582: 3045 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0583: 2969 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0584: 2943 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0585: 1754 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0586: 1454 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0587: 1445 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0588: 1371 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0589: 1207 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0590: 1166 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0591: 925 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0592: 767 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0593: 318 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0594: 3012 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0595: 1443 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0596: 1168 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0597: 1032 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0598: 332 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0599: 130 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0600: 3160 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0601: 3128 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0602: 3086 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0603: 1561 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0604: 1433 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0605: 1371 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0606: 1218 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0607: 1130 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0608: 1096 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0609: 1024 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0610: 920 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0611: 900 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0612: 857 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0613: 890 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0614: 866 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0615: 765 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0616: 631 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0617: 592 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0618: 3008 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0619: 2941 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0620: 1678 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0621: 1468 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0622: 1430 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0623: 1287 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0624: 1176 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0625: 1017 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0626: 854 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0627: 657 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0628: 578 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0629: 340 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0630: 3008 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0631: 1455 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0632: 1136 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0633: 758 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0634: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0635: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0636: 3015 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0637: 2975 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0638: 2964 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0639: 1457 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0640: 1413 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0641: 1357 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0642: 1171 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0643: 1140 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0644: 868 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0645: 569 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0646: 468 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0647: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0648: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0649: 1362 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0650: 1135 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0651: 942 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0652: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0653: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0654: 2974 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0655: 1412 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0656: 1278 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0657: 828 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0658: 606 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0659: 225 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0660: 3042 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0661: 1450 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0662: 1230 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0663: 964 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0664: 540 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0665: 365 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0666: 1420 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0667: 809 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0668: 349 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0669: 68 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0670: 1117 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0671: 714 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0672: 1237 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0673: 620 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0674: 380 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0675: 1251 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0676: 523 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0677: 216 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0678: 3038 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0679: 1479 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0680: 1188 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0681: 1126 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0682: 1070 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0683: 3103 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0684: 854 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0685: 3025 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0686: 1438 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0687: 1029 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0688: 866 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0689: 3082 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0690: 1188 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0691: 739 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0692: 3090 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0693: 3013 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0694: 2991 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0695: 2954 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0696: 2871 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0697: 1650 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0698: 1470 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0699: 1420 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0700: 1378 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0701: 1297 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0702: 1171 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0703: 963 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0704: 920 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0705: 428 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0706: 2954 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0707: 1443 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0708: 1045 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0709: 991 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0710: 912 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0711: 578 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0712: 174 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0713: 3523 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0714: 3155 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0715: 3137 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0716: 3126 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0717: 1531 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0718: 1447 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0719: 1395 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0720: 1358 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0721: 1254 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0722: 1159 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0723: 1121 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0724: 1054 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0725: 1009 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0726: 924 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0727: 908 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0728: 879 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0729: 833 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0730: 745 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0731: 674 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0732: 623 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0733: 516 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0734: 3517 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0735: 3143 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0736: 3123 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0737: 3110 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0738: 1448 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0739: 1404 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0740: 1328 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0741: 1265 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0742: 1188 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0743: 1098 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0744: 1062 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0745: 1013 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0746: 988 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0747: 923 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0748: 898 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0749: 831 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0750: 758 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0751: 674 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0752: 663 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0753: 621 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0754: 539 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0755: 3042 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0756: 1132 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0757: 992 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0758: 1278 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0759: 1068 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0760: 925 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0761: 737 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0762: 3056 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0763: 1555 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0764: 1410 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0765: 1176 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0766: 657 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0767: 830 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0768: 340 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0769: 3653 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0770: 2984 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0771: 2939 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0772: 2900 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0773: 1490 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0774: 1464 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0775: 1412 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0776: 1371 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0777: 1256 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0778: 1091 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0779: 1028 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0780: 888 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0781: 417 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0782: 2991 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0783: 2910 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0784: 1446 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0785: 1275 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0786: 1161 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0787: 812 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0788: 251 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0789: 200 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0790: 2996 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0791: 2817 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0792: 1464 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0793: 1452 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0794: 1244 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0795: 928 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0796: 418 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0797: 2952 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0798: 1464 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0799: 1150 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0800: 203 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0801: 2925 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0802: 1464 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0803: 1179 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0804: 242 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0805: 2996 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0806: 2817 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0807: 1464 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0808: 1452 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0809: 1227 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0810: 1102 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0811: 3161 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0812: 3140 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0813: 1491 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0814: 1384 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0815: 1140 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0816: 1066 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0817: 995 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0818: 871 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0819: 838 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0820: 728 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0821: 603 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0822: 863 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0823: 745 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0824: 613 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0825: 3154 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0826: 3129 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0827: 1556 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0828: 1267 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0829: 1180 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0830: 1040 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0831: 873 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0832: 3001 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0833: 2935 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0834: 1882 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0835: 1475 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0836: 1427 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0837: 1319 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0838: 1199 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0839: 1093 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0840: 1005 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0841: 924 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0842: 891 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0843: 746 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0844: 513 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0845: 3028 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0846: 3000 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0847: 1184 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0848: 1139 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0849: 1046 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0850: 790 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0851: 490 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0852: 113 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0853: 3087 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0854: 3003 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0855: 2992 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0856: 1630 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0857: 1438 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0858: 1280 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0859: 1196 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0860: 894 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0861: 512 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0862: 1013 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0863: 908 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0864: 522 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0865: 162 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0866: 976 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0867: 912 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0868: 770 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0869: 3101 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0870: 3055 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0871: 2984 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0872: 1596 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0873: 1381 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0874: 1294 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0875: 990 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0876: 301 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0877: 2916 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0878: 2240 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0879: 1380 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0880: 725 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0881: 2938 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0882: 1382 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0883: 1152 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0884: 2973 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0885: 1456 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0886: 1054 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0887: 213 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0888: 2966 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0889: 1448 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0890: 1029 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0891: 371 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0892: 3061 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0893: 3010 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0894: 2965 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0895: 2905 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0896: 1970 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0897: 1453 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0898: 1437 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0899: 1389 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0900: 1330 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0901: 1132 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0902: 1074 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0903: 876 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0904: 859 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0905: 555 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0906: 210 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0907: 3061 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0908: 2965 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0909: 1453 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0910: 1110 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0911: 1011 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0912: 842 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0913: 523 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0914: 322 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0915: 3060 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0916: 2935 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0917: 1570 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0918: 1450 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0919: 1276 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0920: 1186 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0921: 1113 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0922: 886 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0923: 2955 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0924: 1276 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0925: 1081 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0926: 850 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0927: 325 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0928: 2955 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0929: 1074 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0930: 846 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0931: 635 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0932: 3047 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0933: 2916 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0934: 1427 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0935: 1294 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0936: 1214 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0937: 888 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0938: 740 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0939: 3010 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0940: 3002 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0941: 1743 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0942: 1437 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0943: 1410 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0944: 1036 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0945: 1002 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0946: 723 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0947: 3051 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0948: 1144 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0949: 937 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0950: 616 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0951: 3071 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0952: 1072 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0953: 889 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0954: 748 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0955: 360 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0956: 3086 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0957: 2999 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0958: 1353 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0959: 1174 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0960: 1125 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0961: 792 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0962: 290 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0963: 3345 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0964: 2985 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0965: 2840 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0966: 2860 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0967: 1622 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0968: 1487 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0969: 1465 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0970: 1378 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0971: 1397 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0972: 1016 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0973: 1086 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0974: 892 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-0975: 773 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0976: 403 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0977: 3412 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0978: 2924 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0979: 2906 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0980: 1455 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0981: 1238 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0982: 1293 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0983: 1117 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0984: 816 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0985: 259 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0986: 218 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0987: 3531 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0988: 3149 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0989: 3128 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0990: 1472 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0991: 1401 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0992: 1148 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0993: 1075 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0994: 1017 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0995: 882 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0996: 864 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0997: 692 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0998: 614 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-0999: 827 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1000: 722 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1001: 621 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1002: 475 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1003: 3143 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1004: 3119 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1005: 1519 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1006: 1424 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1007: 1288 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1008: 1134 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1009: 1049 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1010: 866 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1011: 3120 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1012: 3052 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1013: 3044 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1014: 2264 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1015: 1468 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1016: 1344 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1017: 1195 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1018: 1125 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1019: 1049 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1020: 941 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1021: 808 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1022: 736 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1023: 527 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1024: 222 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1025: 3094 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1026: 3040 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1027: 1442 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1028: 1180 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1029: 1070 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1030: 1088 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1031: 880 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1032: 821 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1033: 543 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1034: 222 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1035: 2947 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1036: 2254 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1037: 1422 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1038: 1328 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1039: 1023 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1040: 809 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1041: 508 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1042: 187 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1043: 2985 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1044: 1197 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1045: 760 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1046: 357 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1047: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1048: 2982 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1049: 1226 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1050: 951 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1051: 387 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1052: 2965 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1053: 2255 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1054: 1425 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1055: 1270 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1056: 917 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1057: 526 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1058: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1059: 3063 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1060: 3043 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1061: 1572 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1062: 1414 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1063: 1283 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1064: 1160 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1065: 1063 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1066: 964 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1067: 619 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1068: 936 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1069: 863 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1070: 751 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1071: 421 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1072: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1073: 760 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1074: 370 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1075: 3075 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1076: 3043 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1077: 1565 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1078: 1444 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1079: 1239 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1080: 1052 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1081: 1009 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1082: 664 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1083: 3082 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1084: 3053 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1085: 3049 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1086: 1572 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1087: 1465 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1088: 1155 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1089: 1065 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1090: 969 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1091: 679 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1092: 960 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1093: 398 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1094: 1033 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1095: 980 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1096: 804 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1097: 719 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1098: 347 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1099: 3047 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1100: 1569 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1101: 1411 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1102: 1356 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1103: 1224 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1104: 1158 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1105: 1071 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1106: 621 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1107: 3055 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1108: 1580 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1109: 1233 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1110: 1016 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1111: 602 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1112: 960 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1113: 350 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1114: 927 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1115: 3012 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1116: 1483 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1117: 1130 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1118: 1018 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1119: 983 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1120: 756 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1121: 3069 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1122: 1411 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1123: 1149 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1124: 1063 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1125: 3040 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1126: 1525 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1127: 1346 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1128: 704 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1129: 785 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1130: 418 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1131: 2981 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1132: 2904 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1133: 2895 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1134: 2809 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1135: 1743 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1136: 1460 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1137: 1416 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1138: 1390 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1139: 1376 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1140: 1335 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1141: 1093 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1142: 993 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1143: 848 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1144: 668 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1145: 271 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1146: 2992 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1147: 2942 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1148: 1451 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1149: 1250 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1150: 1118 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1151: 892 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1152: 660 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1153: 220 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1154: 135 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1155: 2977 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1156: 2962 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1157: 2887 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1158: 1476 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1159: 1462 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1160: 1392 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1161: 1158 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1162: 869 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1163: 369 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1164: 2967 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1165: 1451 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1166: 1278 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1167: 940 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1168: 216 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1169: 2973 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1170: 2968 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1171: 1472 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1172: 1192 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1173: 748 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1174: 268 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1175: 2968 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1176: 2887 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1177: 1464 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1178: 1378 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1179: 1338 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1180: 1054 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1181: 922 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1182: 3091 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1183: 3075 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1184: 2886 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1185: 1500 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1186: 1378 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1187: 1365 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1188: 1106 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1189: 994 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1190: 915 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1191: 802 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1192: 1100 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1193: 941 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1194: 700 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1195: 516 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1196: 2900 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1197: 925 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1198: 891 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1199: 664 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1200: 350 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1201: 3105 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1202: 3043 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1203: 1580 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1204: 1292 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1205: 1239 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1206: 1090 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1207: 959 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1208: 805 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1209: 3094 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1210: 3073 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1211: 3030 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1212: 1584 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1213: 1483 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1214: 1218 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1215: 1072 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1216: 1032 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1217: 991 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1218: 601 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1219: 980 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1220: 880 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1221: 373 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1222: 1007 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1223: 937 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1224: 744 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1225: 700 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1226: 403 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1227: 3087 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1228: 3042 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1229: 1581 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1230: 1442 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1231: 1362 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1232: 1227 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1233: 1143 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1234: 1079 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1235: 652 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1236: 3058 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1237: 2995 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1238: 2963 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1239: 2938 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1240: 2805 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1241: 1720 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1242: 1649 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1243: 1455 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1244: 1391 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1245: 1375 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1246: 1304 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1247: 1253 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1248: 1147 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1249: 1074 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1250: 973 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1251: 539 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1252: 464 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1253: 230 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1254: 2980 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1255: 1455 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1256: 1042 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1257: 973 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1258: 928 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1259: 780 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1260: 295 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1261: 173 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1262: 122 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1263: 3035 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1264: 3031 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1265: 2966 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1266: 2964 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1267: 1771 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1268: 1460 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1269: 1440 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1270: 1430 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1271: 1375 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1272: 1248 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1273: 1159 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1274: 1060 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1275: 980 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1276: 844 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1277: 639 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1278: 429 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1279: 303 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1280: 3005 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1281: 2994 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1282: 1460 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1283: 1430 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1284: 1187 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1285: 1036 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1286: 607 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1287: 187 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1288: 136 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1289: 110 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1290: 2989 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1291: 2941 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1292: 2911 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1293: 1661 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1294: 1470 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1295: 1416 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1296: 1366 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1297: 1064 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1298: 801 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1299: 383 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1300: 2970 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1301: 1459 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1302: 1076 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1303: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1304: 193 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1305: 2945 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1306: 1444 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1307: 1079 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1308: 890 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1309: 429 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1310: 196 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1311: 3086 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1312: 2980 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1313: 2893 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1314: 1458 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1315: 1381 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1316: 1282 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1317: 1043 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1318: 974 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1319: 430 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1320: 3062 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1321: 992 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1322: 1326 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1323: 673 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1324: 3068 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1325: 1010 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1326: 995 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1327: 703 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1328: 1310 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1329: 1150 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1330: 849 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1331: 3063 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1332: 1486 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1333: 1038 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1334: 3047 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1335: 1596 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1336: 1178 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1337: 606 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1338: 975 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1339: 410 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1340: 2951 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1341: 2862 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1342: 1468 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1343: 1153 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1344: 1004 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1345: 199 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1346: 1260 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1347: 941 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1348: 1231 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1349: 1142 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1350: 928 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1351: 2973 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1352: 2934 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1353: 1440 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1354: 883 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1355: 625 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1356: 2959 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1357: 2862 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1358: 1441 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1359: 1252 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1360: 1217 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1361: 893 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1362: 746 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1363: 3088 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1364: 3051 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1365: 3008 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1366: 1632 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1367: 1486 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1368: 1449 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1369: 1342 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1370: 1082 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1371: 986 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1372: 894 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1373: 641 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1374: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1375: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1376: 666 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1377: 494 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1378: 926 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1379: 907 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1380: 769 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1381: 613 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1382: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1383: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1384: 3104 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1385: 3075 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1386: 1550 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1387: 1314 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1388: 1232 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1389: 1077 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1390: 952 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1391: 795 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1392: 350 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1393: 3020 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1394: 2855 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1395: 1477 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1396: 1168 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1397: 970 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1398: 749 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1399: 476 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1400: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1401: 1255 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1402: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1403: 3020 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1404: 2855 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1405: 1477 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1406: 1412 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1407: 1309 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1408: 1168 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1409: 1068 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1410: 935 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1411: 524 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1412: 307 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1413: 3095 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1414: 3095 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1415: 3000 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1416: 3000 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1417: 3000 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1418: 2984 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1419: 2904 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1420: 1638 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1421: 1607 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1422: 1464 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1423: 1420 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1424: 1389 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1425: 1363 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1426: 1313 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1427: 1303 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1428: 1072 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1429: 1012 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1430: 953 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1431: 770 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1432: 530 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1433: 396 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1434: 196 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1435: 2949 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1436: 1451 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1437: 1034 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1438: 991 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1439: 907 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1440: 907 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1441: 757 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1442: 615 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1443: 290 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1444: 153 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1445: 3066 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1446: 2958 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1447: 2916 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1448: 2902 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1449: 2852 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1450: 1616 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1451: 1467 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1452: 1449 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1453: 1302 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1454: 1284 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1455: 1213 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1456: 1048 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1457: 964 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1458: 962 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1459: 904 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1460: 692 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1461: 603 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1462: 254 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1463: 3039 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1464: 2967 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1465: 2927 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1466: 1467 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1467: 1432 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1468: 1356 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1469: 1302 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1470: 1297 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1471: 1207 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1472: 1113 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1473: 1082 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1474: 937 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1475: 800 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1476: 702 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1477: 387 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1478: 2987 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1479: 2941 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1480: 2916 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1481: 1996 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1482: 1470 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1483: 1438 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1484: 1369 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1485: 1286 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1486: 973 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1487: 726 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1488: 349 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1489: 2941 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1490: 1470 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1491: 985 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1492: 617 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1493: 3051 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1494: 2941 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1495: 1470 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1496: 1068 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1497: 1004 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1498: 443 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1499: 187 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1500: 2941 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1501: 2916 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1502: 1470 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1503: 1369 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1504: 1191 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1505: 968 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1506: 848 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1507: 554 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1508: 187 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1509: 3000 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1510: 1423 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1511: 1160 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1512: 1037 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1513: 588 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1514: 3070 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1515: 1174 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1516: 919 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1517: 3066 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1518: 1166 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1519: 888 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1520: 300 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1521: 3000 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1522: 1521 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1523: 1415 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1524: 990 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1525: 872 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1526: 3067 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1527: 2985 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1528: 1431 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1529: 1150 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1530: 1049 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1531: 870 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1532: 780 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1533: 308 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1534: 2953 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1535: 2775 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1536: 1459 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1537: 1444 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1538: 1186 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1539: 828 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1540: 367 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1541: 2977 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1542: 1453 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1543: 1046 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1544: 255 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1545: 2981 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1546: 2953 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1547: 2776 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1548: 1471 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1549: 1444 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1550: 1409 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1551: 1275 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1552: 1103 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1553: 1043 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1554: 424 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1555: 281 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1556: 2974 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1557: 2953 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1558: 2940 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1559: 2876 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1560: 2757 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1561: 1752 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1562: 1474 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1563: 1449 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1564: 1399 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1565: 1381 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1566: 1329 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1567: 1182 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1568: 1138 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1569: 922 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1570: 800 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1571: 632 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1572: 395 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1573: 272 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1574: 212 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1575: 2990 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1576: 2982 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1577: 2885 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1578: 1467 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1579: 1452 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1580: 1437 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1581: 1290 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1582: 1109 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1583: 965 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1584: 942 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1585: 933 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1586: 351 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1587: 227 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1588: 75 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1589: 2965 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1590: 2904 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1591: 2880 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1592: 1477 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1593: 1394 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1594: 1177 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1595: 797 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1596: 426 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1597: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1598: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1599: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1600: 198 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1601: 2962 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1602: 2951 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1603: 2887 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1604: 1468 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1605: 1459 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1606: 1371 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1607: 1330 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1608: 1166 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1609: 961 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1610: 918 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1611: 367 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1612: 280 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1613: 2965 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1614: 2872 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1615: 2853 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1616: 1460 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1617: 1442 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1618: 1382 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1619: 1361 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1620: 1151 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1621: 1059 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1622: 837 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1623: 425 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1624: 2968 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1625: 2930 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1626: 1461 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1627: 1257 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1628: 948 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1629: 731 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1630: 194 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1631: 102 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1632: 2965 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1633: 2912 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1634: 1460 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1635: 1300 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1636: 1180 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1637: 803 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1638: 225 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1639: 2968 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1640: 2870 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1641: 2853 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1642: 1461 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1643: 1461 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1644: 1379 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1645: 1290 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1646: 1009 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1647: 964 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1648: 271 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1649: 3085 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1650: 3054 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1651: 3054 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1652: 2989 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1653: 1627 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1654: 1579 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1655: 1400 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1656: 1288 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1657: 1220 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1658: 1192 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1659: 934 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1660: 444 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1661: 355 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1662: 1013 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1663: 941 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1664: 901 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1665: 683 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1666: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1667: 90 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1668: 988 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1669: 928 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1670: 872 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1671: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1672: 217 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1673: 3099 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1674: 3045 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1675: 3019 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1676: 2953 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1677: 1629 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1678: 1432 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1679: 1295 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1680: 1255 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1681: 1186 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1682: 963 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1683: 590 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1684: 170 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1685: 3356 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1686: 2975 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1687: 2927 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1688: 2889 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1689: 2824 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1690: 1484 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1691: 1447 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1692: 1292 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1693: 1273 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1694: 1228 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1695: 1181 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1696: 974 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1697: 920 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1698: 901 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1699: 871 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1700: 832 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1701: 789 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1702: 579 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1703: 299 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1704: 2967 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1705: 2916 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1706: 2879 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1707: 2818 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1708: 1459 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1709: 1412 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1710: 1341 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1711: 1268 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1712: 1211 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1713: 1203 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1714: 1171 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1715: 1099 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1716: 1023 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1717: 883 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1718: 848 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1719: 600 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1720: 65 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1721: 3345 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1722: 2975 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1723: 2950 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1724: 2918 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1725: 2905 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1726: 2887 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1727: 1614 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1728: 1467 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1729: 1443 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1730: 1390 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1731: 1291 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1732: 1237 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1733: 1137 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1734: 1120 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1735: 1050 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1736: 965 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1737: 788 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1738: 752 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1739: 606 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1740: 448 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1741: 174 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1742: 3411 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1743: 2970 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1744: 2957 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1745: 1452 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1746: 1213 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1747: 1185 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1748: 1168 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1749: 1036 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1750: structural EmptyOne per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1751: 925 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1752: 912 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1753: 883 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1754: 788 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1755: 354 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1756: 240 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1757: 2968 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1758: 2856 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1759: 1444 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1760: 1397 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1761: 1305 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1762: 1128 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1763: 1015 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1764: 837 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1765: 435 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1766: 424 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1767: 2970 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1768: 2863 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1769: 1449 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1770: 1369 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1771: 1256 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1772: 1136 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1773: 1086 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1774: 881 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1775: 288 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1776: 2968 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1777: 2856 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1778: 1459 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1779: 1335 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1780: 1217 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1781: 1110 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1782: 853 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1783: 490 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1784: 2970 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1785: 2863 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1786: 1457 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1787: 1378 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1788: 1291 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1789: 1052 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1790: 889 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1791: 610 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1792: 274 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1793: 3099 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1794: 3081 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1795: 3081 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1796: 3043 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1797: 3034 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1798: 2806 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1799: 1728 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1800: 1614 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1801: 1603 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1802: 1492 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1803: 1460 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1804: 1387 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1805: 1314 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1806: 1276 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1807: 1202 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1808: 1168 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1809: 1158 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1810: 1074 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1811: 1026 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1812: 996 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1813: 825 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1814: 649 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1815: 617 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1816: 437 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1817: 224 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1818: 1003 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1819: 996 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1820: 978 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1821: 918 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1822: 852 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1823: 740 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1824: 688 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1825: 450 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1826: 400 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1827: 217 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1828: 111 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1829: 2995 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1830: 1002 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1831: 2978 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1832: 839 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1833: 1083 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1834: 912 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1835: 1151 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1836: 617 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1837: 1130 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1838: 2978 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1839: 1230 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1840: 853 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1841: 2970 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1842: 1182 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1843: 821 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1844: 665 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1845: 1036 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1846: 829 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1847: 2973 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1848: 2892 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1849: 2866 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1850: 2866 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1851: 1480 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1852: 1463 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1853: 1450 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1854: 1379 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1855: 1346 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1856: 1144 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1857: 1036 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1858: 867 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1859: 401 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1860: 179 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1861: 2967 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1862: 2908 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1863: 1462 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1864: 1346 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1865: 1269 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1866: 993 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1867: 766 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1868: 210 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1869: 131 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1870: 2965 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1871: 2930 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1872: 2930 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1873: 1469 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1874: 1303 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1875: 1170 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1876: 858 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1877: 727 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1878: 215 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1879: 88 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1880: 2965 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1881: 2879 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1882: 2866 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1883: 1476 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1884: 1456 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1885: 1389 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1886: 1379 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1887: 1269 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1888: 1073 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1889: 1024 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1890: 910 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1891: 406 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1892: 2909 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1893: 1381 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1894: 733 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1895: 198 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1896: 2955 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1897: 1451 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1898: 999 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1899: 335 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1900: 2975 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1901: 1461 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1902: 932 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1903: 203 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1904: 2959 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1905: 2876 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1906: 1475 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1907: 1372 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1908: 1256 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1909: 925 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1910: 418 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1911: 3341 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1912: 2947 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1913: 2931 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1914: 2917 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1915: 2892 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1916: 2850 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1917: 2730 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1918: 1476 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1919: 1452 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1920: 1444 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1921: 1386 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1922: 1365 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1923: 1346 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1924: 1266 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1925: 1164 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1926: 1035 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1927: 1006 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1928: 906 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1929: 853 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1930: 822 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1931: 743 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1932: 546 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1933: 432 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1934: 404 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1935: 246 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1936: 2940 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1937: 2902 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1938: 2850 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1939: 2803 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1940: 1468 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1941: 1460 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1942: 1436 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1943: 1332 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1944: 1318 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1945: 1285 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1946: 1258 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1947: 1191 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1948: 1146 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1949: 1115 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1950: 1052 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1951: 964 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1952: 874 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1953: 810 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1954: 445 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1955: 254 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1956: 2966 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1957: 2877 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1958: 2863 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1959: 2852 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1960: 1460 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1961: 1460 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1962: 1440 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-1963: 1397 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1964: 1366 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1965: 1305 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1966: 1140 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1967: 1065 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1968: 1007 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1969: 901 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1970: 371 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1971: 305 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1972: 2967 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1973: 2936 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1974: 2920 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1975: 1460 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1976: 1303 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1977: 1222 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1978: 1010 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1979: 824 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1980: 726 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1981: 216 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1982: 94 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1983: 61 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1984: 2966 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1985: 2908 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1986: 2908 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1987: 1460 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1988: 1305 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1989: 1283 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1990: 1168 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1991: 894 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1992: 740 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1993: 208 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1994: 125 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1995: 2967 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1996: 2884 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1997: 2867 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1998: 2851 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-1999: 1476 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-2000: 1465 per centimeter; exact and mode-bound True

- NIST-CCCBDB-PROP-009-MODE-2001: 1452 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-2002: 1387 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-2003: 1375 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-2004: 1249 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-2005: 1061 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-2006: 1035 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-2007: 888 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-2008: 470 per centimeter; exact and mode-bound True
- NIST-CCCBDB-PROP-009-MODE-2009: 139 per centimeter; exact and mode-bound True
- tampered_frequency_rejected: True
- complete_displayed_support_preserved: True
- measured_and_absent_rows_preserved: True
- complete_displayed_molecule_support_preserved: True
- source_advertised_displayed_gap_preserved: True
- all_measured_values_exact_and_positive: True
- all_unmeasured_values_structural_EmptyOne: True

Falsification boundary: The claim fails if a generated molecular vibrational frequency cannot be reconstructed as a positive finite recurrence count over a positive interval while retaining molecule, mode and symmetry; if equal-interval repetition changes that ratio; if any of the 2,009 displayed rows, 145 molecules, 1,984 measured values or 25 displayed absences is omitted, duplicated or displaced; if the advertised 164/2,452 versus displayed 145/2,009 boundary is hidden; if any target opens before sealing; or if a continuum, theoretical, fitted, scaled or species-specific rule enters.

Measurement receipt: sha256 : a888b6cef5c5891708aa014feb497752fcfe1f7b92799ddaf124f26900a60bec.

Isolation certificate: sha256 : f11c3ad63db79ee228b3332fb19208a39b0e640712642e1c56958b709c5db0f7.

Custody certificate: sha256 : 0da95428936af82a0b0c6a30aab1986373df4033bbdf4209ee7500e921e238ae.

Registered source descriptions:

- National Institute of Standards and Technology: ()

Registered target identities:

- None.

Meaning. The quantitative property family derives bond length, dissociation, angle, torsion, dipole, polarizability, ionization, affinity, vibration, rotation, binding, magnetic response, formation energy and their common-carrier vector without per-property fitting. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; missing source measurements use structural EmptyOne
- no negative, irrational, imaginary, floating, signed or continuum proof value
- no continuum sinusoid, differential equation or harmonic-oscillator premise
- no theoretical frequency, experimental/theoretical ratio, fitted scale factor or species residual
- no measured frequency in the law, grammar, candidate forcing or prediction
- no dropped source absence or concealed advertised/displayed source-count discrepancy

Immutable identities. Source manifest

sha256 : 79b187abbe85549b3f1dbd4584e9e5e5b7faf274cdd71b8dfda125c800c008d0; derivation seal

sha256 : 274e190cce1ac2290b6e7aea7ce6c6d51a15121efec7070616657807a615c9d; engine receipt

sha256:b4073dd9088629d88d36ff88678f287dc7973aa06a39013bb2ad0a5445433dd7 at receipts/engine/model_admitted/SFT-CHEM-VIBRATIONAL-FREQUENCY-009-b4073dd9088629d8.json; empirical hash sha256:4afabea9b3e9d4c0945ce2e8d90eb051f488c6396d78ca6fe5268d39c52f58f6; measurement receipt sha256:a888b6cef5c5891708aa014feb497752fcfe1f7b92799ddaf124f26900a60bec.

120. Exact molecular held-axis rotational-constant law

Claim identity: SFT-CHEM-ROTATIONAL-CONSTANT-010

Question and theorem. A molecular rotational constant is derived from a complete retained molecular state, finite generated geometry and held principal axis as the exact positive count of completed axis recurrences divided by a positive observation-interval count. Positive rotational ordinals force $J(J+1)$ levels and adjacent $2J$ gaps; the unexcited form is structural EmptyOne. Every value-free identity from the complete official NIST list-to-choice-to-route seals before 1,681 positive A/B/C measurements and 1,334 blank axis cells open. No rigid rotor, inertia equation, continuum angle or fitted molecular coefficient enters.

complete-finite-state-geometry-axis-carrier__held-principal-axis-and-equivalence-class__exact-axis-recurrence-over-interval-ratio__positive-JJplusOne-level-and-2J-gap__post-recurrence-held-unit-translation__value-free-complete-axis-identity-seal__complete-NIST-list-choice-result-and-axis-custody__one-axis-recurrence-law-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-PHYS-WAVE-PERIOD-FREQUENCY-001
- SFT-PHYS-MOLECULAR-SPECTRUM-HIERARCHY-004
- SFT-PHYS-MOLECULAR-SPECTROSCOPY-TERMINAL-005
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-CONFIGURATION-ORDER-PATH-011
- SFT-CHEM-ROVIBRONIC-COMPOSITION-001
- SFT-CHEM-RESOLVED-ROVIBRONIC-SPIN-COMPOSITION-013
- SFT-CHEM-EQUILIBRIUM-BOND-LENGTH-001
- SFT-CHEM-MOLECULAR-BOND-ANGLE-003
- SFT-CHEM-DIHEDRAL-TORSIONAL-STATE-004
- SFT-CHEM-VIBRATIONAL-FREQUENCY-009

Generated grammar. Generate the literal product of carrier, axis, magnitude, ladder, translation, prediction, record and extension forms; decide all 256 candidates from admitted exact geometry, recurrence, spectroscopy, molecular composition and measurement-custody laws.

Boundary: The depth-independent held-axis recurrence ratio and positive rotational ladder, tested against every row returned by all 1,193 elemental-composition queries generated from all 2,186 entries of the frozen official NIST complete species list: 1,832

returned charge/state choices, 1,005 displayed molecular property rows and all 3,015 A/B/C cells. The 83 listed compositions with no returned selectable row remain explicit and are not misreported as measured.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-finite-state-geometry-axis-carrier__held-principal-axis-and-equivalence-class__exact-axis-recurrence-over-interval-ratio__positive-JJplusOne-level-and-2J-gap__post-recurrence-held-unit-translation__value-free-complete-axis-identity-seal__complete-NIST-list-choice-result-and-axis-custody__one-axis-recurrence-law-no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-finite-state-geometry-axis-carrier	answer-only-rotational-number	An answer-only number erases molecular state, geometry and axis identity.
axis	held-principal-axis-and-equivalence-class	merged-or-relabelled-axis	Merging principal axes destroys distinct and symmetry-equivalent recurrence records.
magnitude	exact-axis-recurrence-over-interval-ratio	imported-rotational-constant-or-inertia-equation	A named constant or moment-of-inertia equation does not derive the Fold quantity.
ladder	positive-JJplusOne-level-and-2J-gap	continuum-or-free-angular-spectrum	A continuum spectrum or selected angular coefficient is not generated.
translation	post-recurrence-held-unit-translation	reciprocal-centimeter-selects-law	A conventional unit cannot select the axis-recurrence relation.
prediction	value-free-complete-axis-identity-seal	rotational-target-readable-before-seal	A readable A, B or C value could select the relation or row set.
record	complete-NIST-list-choice-result-and-axis-custody	favorable-molecule-or-present-axis-subset	Dropping unavailable species, blank axes or asymmetric rows hides the source boundary.
extension	one-axis-recurrence-law-no-extra-rule	fitted-inertia-geometry-or-species-correction	A fitted inertia, geometry coefficient or species residual is a free parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:632627a6ed5ac0c86cb24760c29576c4cf1f34aa12990774deebb5805affec82.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:5b841759faf3f49267be06e270740e77e8589d37ba5845fa2a4de115a9dec573.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:9cc423d822fef57a20f09850f4bc8fd67f7ebe50a1e689ab12f5db62da371f03.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:a82b832626432a596cad3e368686f03ec6ce981fde655830bc3327f4b0564005.

Independent reconstruction. Implementation hash

sha256:1ea930ebdaa857ea2872cd82e36e884008a0cfebc83be2ed78d280997d8a3e6a; independent certificate sha256:a575a810af9faca57df3264f65c2d5d7041a3623d1e70e8a2917278da8434896; external-validation hash sha256:f8a9e1e99aced75a77345f0915c846f3e719bbcd5f8f0cce864f3c839a049e41. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-CCCBDB-COMPLETE-ROTATIONAL-CONSTANT-SURFACE

Comparison records:

- NIST-CCCBDB-PROP-010-ROW-0001-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0001-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0001-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0002-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0002-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0002-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0003-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0003-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0003-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0004-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0004-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0004-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0005-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0005-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0005-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0006-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0006-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0006-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0007-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0007-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0007-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0008-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0008-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0008-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0009-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0009-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0009-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0010-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0010-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0010-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0011-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0011-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0011-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0012-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True

- [illegible]

- NIST-CCCBDB-PROP-010-ROW-0025-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0025-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0025-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0026-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0026-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0026-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0027-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0027-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0027-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0028-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0028-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0028-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0029-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0029-AXIS-B: 0.60860 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0029-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0030-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0030-AXIS-B: 0.67264 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0030-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0031-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0031-AXIS-B: 0.80900 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0031-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0032-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0032-AXIS-B: 1.09934 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0032-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0033-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0033-AXIS-B: 1.10463 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0033-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0034-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0034-AXIS-B: 1.17660 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0034-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0035-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0035-AXIS-B: 1.17891 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0035-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0036-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0036-AXIS-B: 1.21200 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0036-AXIS-C: 1.21200 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0037-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0037-AXIS-B: 1.21283 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0037-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0038-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0038-AXIS-B: 1.23607 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0038-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0039-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0039-AXIS-B: 1.30838 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0039-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0040-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0040-AXIS-B: 1.32041 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0040-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0041-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0041-AXIS-B: 1.42700 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0041-AXIS-C: 1.42700 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0042-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0042-AXIS-B: 1.43768 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0042-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0043-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0043-AXIS-B: 1.45683 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0043-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0044-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0044-AXIS-B: 1.47822 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0044-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0045-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0045-AXIS-B: 1.51210 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0045-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0046-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0046-AXIS-B: 1.58470 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0046-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0047-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0047-AXIS-B: 1.63740 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0047-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0048-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0048-AXIS-B: 1.66600 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0048-AXIS-C: 1.66600 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0049-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0049-AXIS-B: 1.67195 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0049-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0050-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0050-AXIS-B: 1.69000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0050-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0051-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0051-AXIS-B: 1.69130 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0051-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0052-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0052-AXIS-B: 1.74666 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0052-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0053-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0053-AXIS-B: 1.78111 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0053-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0054-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0054-AXIS-B: 1.82005 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0054-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0055-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0055-AXIS-B: 1.89974 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0055-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0056-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0056-AXIS-B: 1.92253 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0056-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0057-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0057-AXIS-B: 1.93176 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0057-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0058-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0058-AXIS-B: 1.97705 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0058-AXIS-C: 1.97705 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0059-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0059-AXIS-B: 1.98795 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0059-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0060-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0060-AXIS-B: 1.99824 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0060-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0061-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0061-AXIS-B: 3.41230 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0061-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0062-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0062-AXIS-B: 3.54823 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0062-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0063-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0063-AXIS-B: 3.75154 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0063-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0064-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0064-AXIS-B: 4.27660 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0064-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0065-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0065-AXIS-B: 4.75366 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0065-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0066-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0066-AXIS-B: 4.76315 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0066-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0067-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0067-AXIS-B: 4.90030 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0067-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0068-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0068-AXIS-B: 4.90327 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0068-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0069-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0069-AXIS-B: 5.36206 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0069-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0070-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0070-AXIS-B: 5.42543 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0070-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0071-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0071-AXIS-B: 5.44879 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0071-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0072-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0072-AXIS-B: 5.68309 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0072-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0073-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0073-AXIS-B: 5.82552 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0073-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0074-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0074-AXIS-B: 6.06538 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0074-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0075-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0075-AXIS-B: 6.13700 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0075-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0076-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0076-AXIS-B: 6.22000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0076-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0077-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0077-AXIS-B: 6.33070 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0077-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0078-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0078-AXIS-B: 6.38700 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0078-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0079-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0079-AXIS-B: 6.39378 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0079-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0080-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0080-AXIS-B: 6.42637 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0080-AXIS-C: 6.42637 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0081-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0081-AXIS-B: 6.67940 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0081-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0082-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0082-AXIS-B: 6.72590 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0082-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0083-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0083-AXIS-B: 6.76300 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0083-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0084-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0084-AXIS-B: 6.80800 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0084-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0085-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0085-AXIS-B: 7.19980 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0085-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0086-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0086-AXIS-B: 7.40700 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0086-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0087-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0087-AXIS-B: 7.50390 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0087-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0088-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0088-AXIS-B: 7.51373 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0088-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0089-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0089-AXIS-B: 7.66030 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0089-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0090-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0090-AXIS-B: 7.78000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0090-AXIS-C: 7.78000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0091-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0091-AXIS-B: 7.94410 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0091-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0092-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0092-AXIS-B: 8.07200 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0092-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0093-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0093-AXIS-B: 8.38510 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0093-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0094-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0094-AXIS-B: 8.46488 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0094-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0095-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0095-AXIS-B: 8.53904 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0095-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0096-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0096-AXIS-B: 9.13400 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0096-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0097-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0097-AXIS-B: 9.56200 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0097-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0098-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0098-AXIS-B: 9.60025 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0098-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0099-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0099-AXIS-B: 9.95661 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0099-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0100-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0100-AXIS-B: 10.02090 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0100-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0101-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0101-AXIS-B: 10.31992 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0101-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0102-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0102-AXIS-B: 10.59342 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0102-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0103-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0103-AXIS-B: 11.01020 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0103-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0104-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0104-AXIS-B: 12.02575 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0104-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0105-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0105-AXIS-B: 12.37400 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0105-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0106-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0106-AXIS-B: 14.45988 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0106-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0107-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0107-AXIS-B: 15.35000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0107-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0108-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0108-AXIS-B: 16.66792 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0108-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0109-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0109-AXIS-B: 16.77000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0109-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0110-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0110-AXIS-B: 16.79430 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0110-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0111-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0111-AXIS-B: 17.11429 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0111-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0112-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0112-AXIS-B: 17.33805 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0112-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0113-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0113-AXIS-B: 18.91080 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0113-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0114-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0114-AXIS-B: 19.12087 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0114-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0115-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0115-AXIS-B: 20.95373 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0115-AXIS-C: 20.95373 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0116-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0116-AXIS-B: 30.44360 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0116-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0117-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0117-AXIS-B: 33.55866 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0117-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0118-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0118-AXIS-B: 45.65500 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0118-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0119-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0119-AXIS-B: 60.85300 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0119-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0120-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0120-AXIS-B: 17899.44000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0120-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0121-AXIS-A: 2.51967 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0121-AXIS-B: 0.68341 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0121-AXIS-C: 0.68341 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0122-AXIS-A: 2.65569 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0122-AXIS-B: 0.60646 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0122-AXIS-C: 0.55728 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0123-AXIS-A: 2.86509 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0123-AXIS-B: 2.86509 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0123-AXIS-C: 2.86509 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0124-AXIS-A: 2.96842 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0124-AXIS-B: 2.96842 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0124-AXIS-C: 2.81732 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0125-AXIS-A: 3.44000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0125-AXIS-B: 0.75460 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0125-AXIS-C: 0.72370 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0126-AXIS-A: 4.45236 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0126-AXIS-B: 4.45236 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0126-AXIS-C: 3.93000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0127-AXIS-A: 4.61046 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0127-AXIS-B: 0.91689 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0127-AXIS-C: 0.76315 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0128-AXIS-A: 4.82800 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0128-AXIS-B: 1.00120 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0128-AXIS-C: 0.82820 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0129-AXIS-A: 5.14280 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0129-AXIS-B: 5.14280 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0129-AXIS-C: 3.11420 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0130-AXIS-A: 5.21451 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0130-AXIS-B: 5.21451 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0130-AXIS-C: 2.58520 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0131-AXIS-A: 5.24120 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0131-AXIS-B: 5.24120 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0131-AXIS-C: 5.24120 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0132-AXIS-A: 5.48950 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0132-AXIS-B: 4.51580 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0132-AXIS-C: 2.44316 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0133-AXIS-A: 6.24800 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0133-AXIS-B: 6.09612 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0133-AXIS-C: 3.03582 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0134-AXIS-A: 6.54490 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0134-AXIS-B: 1.15555 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0134-AXIS-C: 0.97909 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0135-AXIS-A: 7.02000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0135-AXIS-B: 6.53220 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0135-AXIS-C: 3.33700 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0136-AXIS-A: 7.87411 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0136-AXIS-B: 7.87411 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0136-AXIS-C: 3.87880 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0137-AXIS-A: 7.90934 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0137-AXIS-B: 1.08303 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0137-AXIS-C: 0.94864 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0138-AXIS-A: 8.09898 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0138-AXIS-B: 7.02363 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0138-AXIS-C: 3.70257 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0139-AXIS-A: 8.14230 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0139-AXIS-B: 7.75743 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0139-AXIS-C: 3.89853 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0140-AXIS-A: 8.87400 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0140-AXIS-B: 1.34200 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0140-AXIS-C: 1.14800 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0141-AXIS-A: 9.12000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0141-AXIS-B: 8.08700 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0141-AXIS-C: 4.22500 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0142-AXIS-A: 9.40530 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0142-AXIS-B: 1.29530 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0142-AXIS-C: 1.13420 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0143-AXIS-A: 9.44430 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0143-AXIS-B: 9.44430 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0143-AXIS-C: 6.19600 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0144-AXIS-A: 9.57789 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0144-AXIS-B: 9.57789 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0144-AXIS-C: 4.74202 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0145-AXIS-A: 10.00021 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0145-AXIS-B: 1.30419 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0145-AXIS-C: 1.14986 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0146-AXIS-A: 10.34657 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0146-AXIS-B: 9.03572 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0146-AXIS-C: 4.72680 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0147-AXIS-A: 13.16134 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0147-AXIS-B: 1.01142 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0147-AXIS-C: 0.93327 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0148-AXIS-A: 13.34236 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0148-AXIS-B: 6.48776 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0148-AXIS-C: 4.28997 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0149-AXIS-A: 13.60000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0149-AXIS-B: 4.41000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0149-AXIS-C: 3.33000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0150-AXIS-A: 15.41996 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0150-AXIS-B: 7.27296 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0150-AXIS-C: 4.84530 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0151-AXIS-A: 18.47900 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0151-AXIS-B: 1.41150 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0151-AXIS-C: 1.30710 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0152-AXIS-A: 19.80540 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0152-AXIS-B: 11.24890 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0152-AXIS-C: 7.23860 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0153-AXIS-A: 20.35652 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0153-AXIS-B: 1.11803 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0153-AXIS-C: 1.05632 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0154-AXIS-A: 22.36500 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0154-AXIS-B: 1.49440 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0154-AXIS-C: 1.40080 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0155-AXIS-A: 23.41398 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0155-AXIS-B: 9.10311 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0155-AXIS-C: 6.40655 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0156-AXIS-A: 23.72000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0156-AXIS-B: 12.94000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0156-AXIS-C: 8.16000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0157-AXIS-A: 27.87700 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0157-AXIS-B: 14.51200 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0157-AXIS-C: 9.28500 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0158-AXIS-A: 29.03590 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0158-AXIS-B: 12.42298 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0158-AXIS-C: 8.46921 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0159-AXIS-A: 41.64000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0159-AXIS-B: 7.24800 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0159-AXIS-C: 6.00800 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0160-AXIS-A: 73.05775 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0160-AXIS-B: 8.41517 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0160-AXIS-C: 7.21927 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0161-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0161-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0161-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0162-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0162-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0162-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0163-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0163-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0163-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0164-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0164-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0164-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0165-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0165-AXIS-B: 0.04611 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0165-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0166-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0166-AXIS-B: 0.05674 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0166-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0167-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0167-AXIS-B: 0.05975 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0167-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0168-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0168-AXIS-B: 0.07343 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0168-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0169-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0169-AXIS-B: 0.09050 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0169-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0170-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0170-AXIS-B: 0.09287 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0170-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0171-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0171-AXIS-B: 0.09634 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0171-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0172-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0172-AXIS-B: 0.10179 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0172-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0173-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0173-AXIS-B: 0.10874 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0173-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0174-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0174-AXIS-B: 0.11295 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0174-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0175-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0175-AXIS-B: 0.12863 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0175-AXIS-C: 0.12863 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0176-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0176-AXIS-B: 0.14455 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0176-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0177-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0177-AXIS-B: 0.14990 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0177-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0178-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0178-AXIS-B: 0.15168 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0178-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0179-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0179-AXIS-B: 0.15223 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0179-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0180-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0180-AXIS-B: 0.15471 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0180-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0181-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0181-AXIS-B: 0.16148 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0181-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0182-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0182-AXIS-B: 0.17127 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0182-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0183-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0183-AXIS-B: 0.17668 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0183-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0184-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0184-AXIS-B: 0.17825 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0184-AXIS-C: 0.17825 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0185-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0185-AXIS-B: 0.18887 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0185-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0186-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0186-AXIS-B: 0.18979 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0186-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0187-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0187-AXIS-B: 0.19201 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0187-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0188-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0188-AXIS-B: 0.19250 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0188-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0189-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0189-AXIS-B: 0.20350 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0189-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0190-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0190-AXIS-B: 0.21806 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0190-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0191-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0191-AXIS-B: 0.23900 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0191-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0192-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0192-AXIS-B: 0.24393 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0192-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0193-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0193-AXIS-B: 0.24399 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0193-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0194-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0194-AXIS-B: 0.24502 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0194-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0195-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0195-AXIS-B: 0.25212 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0195-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0196-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0196-AXIS-B: 0.25610 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0196-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0197-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0197-AXIS-B: 0.25791 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0197-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0198-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0198-AXIS-B: 0.26440 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0198-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0199-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0199-AXIS-B: 0.26500 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0199-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0200-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0200-AXIS-B: 0.26797 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0200-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0201-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0201-AXIS-B: 0.26950 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0201-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0202-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0202-AXIS-B: 0.27381 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0202-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0203-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0203-AXIS-B: 0.27600 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0203-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0204-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0204-AXIS-B: 0.27990 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0204-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0205-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0205-AXIS-B: 0.27994 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0205-AXIS-C: 0.27994 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0206-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0206-AXIS-B: 0.29547 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0206-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0207-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0207-AXIS-B: 0.29673 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0207-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0208-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0208-AXIS-B: 0.30353 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0208-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0209-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0209-AXIS-B: 0.30362 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0209-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0210-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0210-AXIS-B: 0.33850 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0210-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0211-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0211-AXIS-B: 0.34484 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0211-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0212-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0212-AXIS-B: 0.35329 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0212-AXIS-C: 0.35329 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0213-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0213-AXIS-B: 0.35952 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0213-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0214-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0214-AXIS-B: 0.36480 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0214-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0215-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0215-AXIS-B: 0.36578 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0215-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0216-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0216-AXIS-B: 0.37580 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0216-AXIS-C: 0.37580 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0217-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0217-AXIS-B: 0.37940 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0217-AXIS-C: 0.37940 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0218-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0218-AXIS-B: 0.39500 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0218-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0219-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0219-AXIS-B: 0.40212 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0219-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0220-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0220-AXIS-B: 0.42463 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0220-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0221-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0221-AXIS-B: 0.42670 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0221-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0222-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0222-AXIS-B: 0.43690 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0222-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0223-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0223-AXIS-B: 0.44450 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0223-AXIS-C: 0.44450 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0224-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0224-AXIS-B: 0.44454 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0224-AXIS-C: 0.44454 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0225-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0225-AXIS-B: 0.48570 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0225-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0226-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0226-AXIS-B: 0.50158 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0226-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0227-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0227-AXIS-B: 0.51343 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0227-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0228-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0228-AXIS-B: 0.51648 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0228-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0229-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0229-AXIS-B: 0.51922 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0229-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0230-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0230-AXIS-B: 0.53101 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0230-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0231-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0231-AXIS-B: 0.53541 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0231-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0232-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0232-AXIS-B: 0.54551 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0232-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0233-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0233-AXIS-B: 0.54825 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0233-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0234-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0234-AXIS-B: 0.55217 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0234-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0235-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0235-AXIS-B: 0.55248 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0235-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0236-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0236-AXIS-B: 0.56650 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0236-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0237-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0237-AXIS-B: 0.57300 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0237-AXIS-C: 0.57300 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0238-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0238-AXIS-B: 0.57484 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0238-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0239-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0239-AXIS-B: 0.58121 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0239-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0240-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0240-AXIS-B: 0.58490 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0240-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0241-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0241-AXIS-B: 0.62254 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0241-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0242-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0242-AXIS-B: 0.62346 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0242-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0243-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0243-AXIS-B: 0.63404 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0243-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0244-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0244-AXIS-B: 0.63600 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0244-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0245-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0245-AXIS-B: 0.63654 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0245-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0246-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0246-AXIS-B: 0.63937 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0246-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0247-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0247-AXIS-B: 0.64139 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0247-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0248-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0248-AXIS-B: 0.64984 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0248-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0249-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0249-AXIS-B: 0.66625 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0249-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0250-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0250-AXIS-B: 0.67705 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0250-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0251-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0251-AXIS-B: 0.68428 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0251-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0252-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0252-AXIS-B: 0.69714 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0252-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0253-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0253-AXIS-B: 0.70260 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0253-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0254-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0254-AXIS-B: 0.70652 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0254-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0255-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0255-AXIS-B: 0.71173 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0255-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0256-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0256-AXIS-B: 0.71780 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0256-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0257-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0257-AXIS-B: 0.72082 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0257-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0258-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0258-AXIS-B: 0.72675 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0258-AXIS-C: 0.72675 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0259-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0259-AXIS-B: 0.72850 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0259-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0260-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0260-AXIS-B: 0.73110 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0260-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0261-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0261-AXIS-B: 0.73370 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0261-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0262-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0262-AXIS-B: 0.76960 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0262-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0263-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0263-AXIS-B: 0.77548 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0263-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0264-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0264-AXIS-B: 0.78648 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0264-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0265-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0265-AXIS-B: 0.79059 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0265-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0266-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0266-AXIS-B: 0.79478 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0266-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0267-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0267-AXIS-B: 0.79887 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0267-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0268-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0268-AXIS-B: 0.82005 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0268-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0269-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0269-AXIS-B: 0.88929 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0269-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0270-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0270-AXIS-B: 1.01500 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0270-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0271-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0271-AXIS-B: 1.05871 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0271-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0272-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0272-AXIS-B: 1.20568 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0272-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0273-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0273-AXIS-B: 1.34526 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0273-AXIS-C: 1.34526 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0274-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0274-AXIS-B: 1.41626 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0274-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0275-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0275-AXIS-B: 1.48893 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0275-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0276-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0276-AXIS-B: 1.51674 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0276-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0277-AXIS-A: 1.43000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0277-AXIS-B: 0.17200 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0277-AXIS-C: 0.17200 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0278-AXIS-A: 1.87387 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0278-AXIS-B: 0.36640 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0278-AXIS-C: 0.36640 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0279-AXIS-A: 2.60679 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0279-AXIS-B: 0.33395 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0279-AXIS-C: 0.33395 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0280-AXIS-A: 2.84070 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0280-AXIS-B: 0.47858 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0280-AXIS-C: 0.47858 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0281-AXIS-A: 2.84470 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0281-AXIS-B: 0.22261 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0281-AXIS-C: 0.22261 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0282-AXIS-A: 3.42541 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0282-AXIS-B: 0.43188 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0282-AXIS-C: 0.41316 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0283-AXIS-A: 4.25730 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0283-AXIS-B: 0.82338 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0283-AXIS-C: 0.79273 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0284-AXIS-A: 4.61997 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0284-AXIS-B: 0.54765 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0284-AXIS-C: 0.48864 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0285-AXIS-A: 4.89866 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0285-AXIS-B: 0.23251 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0285-AXIS-C: 0.23242 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0286-AXIS-A: 4.93675 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0286-AXIS-B: 0.47493 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0286-AXIS-C: 0.46620 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0287-AXIS-A: 4.94732 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0287-AXIS-B: 0.47839 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0287-AXIS-C: 0.47023 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0288-AXIS-A: 5.19800 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0288-AXIS-B: 0.44340 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0288-AXIS-C: 0.44340 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0289-AXIS-A: 5.20590 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0289-AXIS-B: 0.93168 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0289-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0290-AXIS-A: 5.24359 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0290-AXIS-B: 0.24288 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0290-AXIS-C: 0.24016 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0291-AXIS-A: 5.68000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0291-AXIS-B: 0.44958 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0291-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0292-AXIS-A: 6.37031 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0292-AXIS-B: 0.84124 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0292-AXIS-C: 0.83911 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0293-AXIS-A: 6.62400 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0293-AXIS-B: 1.01400 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0293-AXIS-C: 0.87600 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0294-AXIS-A: 6.74030 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0294-AXIS-B: 0.50975 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0294-AXIS-C: 0.49502 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0295-AXIS-A: 7.57917 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0295-AXIS-B: 0.56425 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0295-AXIS-C: 0.52365 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0296-AXIS-A: 7.58660 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0296-AXIS-B: 0.24628 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0296-AXIS-C: 0.23830 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0297-AXIS-A: 8.78179 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0297-AXIS-B: 0.87919 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0297-AXIS-C: 0.84490 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0298-AXIS-A: 8.85500 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0298-AXIS-B: 0.70240 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0298-AXIS-C: 0.64880 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0299-AXIS-A: 9.00561 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0299-AXIS-B: 0.47349 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0299-AXIS-C: 0.46261 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0300-AXIS-A: 9.72718 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0300-AXIS-B: 0.59040 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0300-AXIS-C: 0.55544 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0301-AXIS-A: 10.35600 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0301-AXIS-B: 0.86560 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0301-AXIS-C: 0.82700 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0302-AXIS-A: 10.64180 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0302-AXIS-B: 1.13887 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0302-AXIS-C: 1.02909 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0303-AXIS-A: 11.05232 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0303-AXIS-B: 0.47697 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0303-AXIS-C: 0.45619 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0304-AXIS-A: 15.55000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0304-AXIS-B: 1.22100 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0304-AXIS-C: 1.12600 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0305-AXIS-A: 15.75930 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0305-AXIS-B: 0.60477 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0305-AXIS-C: 0.58138 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0306-AXIS-A: 19.64720 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0306-AXIS-B: 0.90036 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0306-AXIS-C: 0.86092 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0307-AXIS-A: 20.46362 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0307-AXIS-B: 0.50424 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0307-AXIS-C: 0.49120 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0308-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0308-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0308-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0309-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0309-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0309-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0310-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0310-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0310-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0311-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0311-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0311-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0312-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0312-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0312-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0313-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0313-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0313-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0314-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0314-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0314-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0315-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0315-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0315-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0316-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0316-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0316-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0317-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0317-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0317-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0318-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0318-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0318-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0319-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0319-AXIS-B: 0.03737 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0319-AXIS-C: 0.03737 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0320-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0320-AXIS-B: 0.03968 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0320-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0321-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0321-AXIS-B: 0.05683 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0321-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0322-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0322-AXIS-B: 0.06087 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0322-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0323-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0323-AXIS-B: 0.08122 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0323-AXIS-C: 0.08122 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0324-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0324-AXIS-B: 0.08211 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0324-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0325-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0325-AXIS-B: 0.08279 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0325-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0326-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0326-AXIS-B: 0.08992 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0326-AXIS-C: 0.08992 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0327-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0327-AXIS-B: 0.09447 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0327-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0328-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0328-AXIS-B: 0.10910 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0328-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0329-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0329-AXIS-B: 0.11364 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0329-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0330-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0330-AXIS-B: 0.11416 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0330-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0331-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0331-AXIS-B: 0.11770 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0331-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0332-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0332-AXIS-B: 0.11781 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0332-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0333-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0333-AXIS-B: 0.15125 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0333-AXIS-C: 0.15125 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0334-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0334-AXIS-B: 0.15247 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0334-AXIS-C: 0.15247 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0335-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0335-AXIS-B: 0.15920 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0335-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0336-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0336-AXIS-B: 0.16067 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0336-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0337-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0337-AXIS-B: 0.16710 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0337-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0338-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0338-AXIS-B: 0.16763 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0338-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0339-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0339-AXIS-B: 0.17926 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0339-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0340-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0340-AXIS-B: 0.18587 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0340-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0341-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0341-AXIS-B: 0.18878 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0341-AXIS-C: 0.18494 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0342-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0342-AXIS-B: 0.18961 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0342-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0343-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0343-AXIS-B: 0.19917 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0343-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0344-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0344-AXIS-B: 0.19963 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0344-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0345-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0345-AXIS-B: 0.20286 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0345-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0346-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0346-AXIS-B: 0.27799 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0346-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0347-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0347-AXIS-B: 0.27971 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0347-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0348-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0348-AXIS-B: 0.28878 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0348-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0349-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0349-AXIS-B: 0.31180 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0349-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0350-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0350-AXIS-B: 0.32376 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0350-AXIS-C: 0.32376 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0351-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0351-AXIS-B: 0.32920 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0351-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0352-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0352-AXIS-B: 0.33880 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0352-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0353-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0353-AXIS-B: 0.35205 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0353-AXIS-C: 0.35205 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0354-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0354-AXIS-B: 0.35540 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0354-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0355-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0355-AXIS-B: 0.35585 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0355-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0356-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0356-AXIS-B: 0.36647 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0356-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0357-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0357-AXIS-B: 0.38952 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0357-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0358-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0358-AXIS-B: 0.39021 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0358-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0359-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0359-AXIS-B: 0.39880 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0359-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0360-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0360-AXIS-B: 0.41360 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0360-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0361-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0361-AXIS-B: 0.41670 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0361-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0362-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0362-AXIS-B: 0.41901 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0362-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0363-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0363-AXIS-B: 0.42961 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0363-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0364-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0364-AXIS-B: 0.43057 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0364-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0365-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0365-AXIS-B: 0.43145 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0365-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0366-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0366-AXIS-B: 0.46550 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0366-AXIS-C: 0.46550 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0367-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0367-AXIS-B: 0.47058 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0367-AXIS-C: 0.47058 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0368-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0368-AXIS-B: 0.48720 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0368-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0369-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0369-AXIS-B: 0.48940 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0369-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0370-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0370-AXIS-B: 0.51307 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0370-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0371-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0371-AXIS-B: 0.51820 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0371-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0372-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0372-AXIS-B: 0.55540 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0372-AXIS-C: 0.55540 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0373-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0373-AXIS-B: 0.57500 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0373-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0374-AXIS-A: 0.47029 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0374-AXIS-B: 0.09174 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0374-AXIS-C: 0.07911 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0375-AXIS-A: 0.47043 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0375-AXIS-B: 0.09182 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0375-AXIS-C: 0.07918 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0376-AXIS-A: 0.47149 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0376-AXIS-B: 0.08585 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0376-AXIS-C: 0.07447 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0377-AXIS-A: 0.48746 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0377-AXIS-B: 0.09743 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0377-AXIS-C: 0.08108 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0378-AXIS-A: 0.49304 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0378-AXIS-B: 0.09415 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0378-AXIS-C: 0.07894 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0379-AXIS-A: 0.53610 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0379-AXIS-B: 0.23410 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0379-AXIS-C: 0.18025 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0380-AXIS-A: 0.56257 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0380-AXIS-B: 0.14817 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0380-AXIS-C: 0.12276 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0381-AXIS-A: 0.59406 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0381-AXIS-B: 0.25421 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0381-AXIS-C: 0.19073 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0382-AXIS-A: 0.61108 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0382-AXIS-B: 0.09432 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0382-AXIS-C: 0.08159 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0383-AXIS-A: 0.67024 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0383-AXIS-B: 0.67024 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0383-AXIS-C: 0.41770 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0384-AXIS-A: 0.73299 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0384-AXIS-B: 0.36108 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0384-AXIS-C: 0.26773 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0385-AXIS-A: 0.73801 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0385-AXIS-B: 0.15749 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0385-AXIS-C: 0.12955 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0386-AXIS-A: 0.75840 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0386-AXIS-B: 0.70690 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0386-AXIS-C: 0.44641 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0387-AXIS-A: 0.79964 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0387-AXIS-B: 0.09835 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0387-AXIS-C: 0.08747 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0388-AXIS-A: 0.82360 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0388-AXIS-B: 0.26025 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0388-AXIS-C: 0.21272 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0389-AXIS-A: 0.85500 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0389-AXIS-B: 0.73787 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0389-AXIS-C: 0.47026 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0390-AXIS-A: 0.89897 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0390-AXIS-B: 0.30728 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0390-AXIS-C: 0.22835 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0391-AXIS-A: 0.93259 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0391-AXIS-B: 0.31042 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0391-AXIS-C: 0.23229 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0392-AXIS-A: 0.94789 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0392-AXIS-B: 0.18299 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0392-AXIS-C: 0.16284 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0393-AXIS-A: 0.95890 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0393-AXIS-B: 0.17661 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0393-AXIS-C: 0.16165 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0394-AXIS-A: 0.96656 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0394-AXIS-B: 0.83579 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0394-AXIS-C: 0.49300 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0395-AXIS-A: 0.97425 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0395-AXIS-B: 0.28173 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0395-AXIS-C: 0.24881 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0396-AXIS-A: 1.00273 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0396-AXIS-B: 0.72803 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0396-AXIS-C: 0.46018 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0397-AXIS-A: 1.02078 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0397-AXIS-B: 0.29432 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0397-AXIS-C: 0.22783 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0398-AXIS-A: 1.05934 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0398-AXIS-B: 0.29184 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0398-AXIS-C: 0.26014 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0399-AXIS-A: 1.06748 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0399-AXIS-B: 0.11075 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0399-AXIS-C: 0.10225 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0400-AXIS-A: 1.08153 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0400-AXIS-B: 0.29828 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0400-AXIS-C: 0.26103 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0401-AXIS-A: 1.11539 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0401-AXIS-B: 0.15148 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0401-AXIS-C: 0.13315 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0402-AXIS-A: 1.13465 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0402-AXIS-B: 0.30650 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0402-AXIS-C: 0.27019 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0403-AXIS-A: 1.14220 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0403-AXIS-B: 0.31135 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0403-AXIS-C: 0.27406 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0404-AXIS-A: 1.16386 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0404-AXIS-B: 0.31190 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0404-AXIS-C: 0.27136 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0405-AXIS-A: 1.20318 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0405-AXIS-B: 0.31237 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0405-AXIS-C: 0.27351 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0406-AXIS-A: 1.29384 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0406-AXIS-B: 0.33545 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0406-AXIS-C: 0.29644 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0407-AXIS-A: 1.36617 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0407-AXIS-B: 0.78947 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0407-AXIS-C: 0.55793 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0408-AXIS-A: 1.39467 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0408-AXIS-B: 0.19066 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0408-AXIS-C: 0.17328 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0409-AXIS-A: 1.39811 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0409-AXIS-B: 0.16875 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0409-AXIS-C: 0.15034 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0410-AXIS-A: 1.39893 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0410-AXIS-B: 0.12358 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0410-AXIS-C: 0.11355 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0411-AXIS-A: 1.40312 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0411-AXIS-B: 0.21004 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0411-AXIS-C: 0.18915 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0412-AXIS-A: 1.53700 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0412-AXIS-B: 0.31039 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0412-AXIS-C: 0.27132 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0413-AXIS-A: 1.59213 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0413-AXIS-B: 0.35378 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0413-AXIS-C: 0.30854 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0414-AXIS-A: 1.65847 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0414-AXIS-B: 0.29061 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0414-AXIS-C: 0.24660 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0415-AXIS-A: 1.66415 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0415-AXIS-B: 0.19466 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0415-AXIS-C: 0.17419 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0416-AXIS-A: 1.67558 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0416-AXIS-B: 0.12385 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0416-AXIS-C: 0.11532 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0417-AXIS-A: 1.70615 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0417-AXIS-B: 0.35651 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0417-AXIS-C: 0.31205 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0418-AXIS-A: 1.73710 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0418-AXIS-B: 0.33197 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0418-AXIS-C: 0.27799 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0419-AXIS-A: 1.74733 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0419-AXIS-B: 0.30333 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0419-AXIS-C: 0.25845 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0420-AXIS-A: 1.75200 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0420-AXIS-B: 0.35578 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0420-AXIS-C: 0.31279 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0421-AXIS-A: 1.75575 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0421-AXIS-B: 0.43905 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0421-AXIS-C: 0.34847 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0422-AXIS-A: 1.76846 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0422-AXIS-B: 0.36343 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0422-AXIS-C: 0.31046 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0423-AXIS-A: 1.80189 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0423-AXIS-B: 0.34632 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0423-AXIS-C: 0.29022 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0424-AXIS-A: 1.87324 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0424-AXIS-B: 0.20114 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0424-AXIS-C: 0.18164 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0425-AXIS-A: 1.88770 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0425-AXIS-B: 0.33901 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0425-AXIS-C: 0.30354 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0426-AXIS-A: 1.96078 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0426-AXIS-B: 0.36347 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0426-AXIS-C: 0.30579 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0427-AXIS-A: 2.02736 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0427-AXIS-B: 0.34417 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0427-AXIS-C: 0.29354 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0428-AXIS-A: 2.04421 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0428-AXIS-B: 0.38202 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0428-AXIS-C: 0.34205 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0429-AXIS-A: 2.05989 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0429-AXIS-B: 0.20353 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0429-AXIS-C: 0.18513 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0430-AXIS-A: 2.15431 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0430-AXIS-B: 0.35481 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0430-AXIS-C: 0.30415 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0431-AXIS-A: 2.25750 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0431-AXIS-B: 0.39558 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0431-AXIS-C: 0.33623 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0432-AXIS-A: 2.34900 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0432-AXIS-B: 0.21451 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0432-AXIS-C: 0.19649 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0433-AXIS-A: 2.35147 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0433-AXIS-B: 0.39601 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0433-AXIS-C: 0.33810 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0434-AXIS-A: 2.42556 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0434-AXIS-B: 0.37939 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0434-AXIS-C: 0.32802 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0435-AXIS-A: 2.48487 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0435-AXIS-B: 0.35014 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0435-AXIS-C: 0.30627 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0436-AXIS-A: 2.58548 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0436-AXIS-B: 0.40211 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0436-AXIS-C: 0.34745 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0437-AXIS-A: 2.59355 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0437-AXIS-B: 0.41580 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0437-AXIS-C: 0.35762 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0438-AXIS-A: 2.61900 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0438-AXIS-B: 0.33401 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0438-AXIS-C: 0.29537 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0439-AXIS-A: 2.80700 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0439-AXIS-B: 0.14416 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0439-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0440-AXIS-A: 2.90938 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0440-AXIS-B: 0.19176 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0440-AXIS-C: 0.17991 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0441-AXIS-A: 2.95050 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0441-AXIS-B: 0.41977 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0441-AXIS-C: 0.36753 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0442-AXIS-A: 2.99439 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0442-AXIS-B: 0.37035 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0442-AXIS-C: 0.32901 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0443-AXIS-A: 3.09855 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0443-AXIS-B: 0.41778 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0443-AXIS-C: 0.36749 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0444-AXIS-A: 3.12640 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0444-AXIS-B: 0.31156 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0444-AXIS-C: 0.29962 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0445-AXIS-A: 3.17525 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0445-AXIS-B: 0.39507 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0445-AXIS-C: 0.35052 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0446-AXIS-A: 3.48598 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0446-AXIS-B: 0.28681 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0446-AXIS-C: 0.26423 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0447-AXIS-A: 3.55348 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0447-AXIS-B: 0.44525 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0447-AXIS-C: 0.39479 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0448-AXIS-A: 4.81000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0448-AXIS-B: 0.29632 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0448-AXIS-C: 0.29632 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0449-AXIS-A: 5.17340 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0449-AXIS-B: 0.25021 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0449-AXIS-C: 0.25021 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0450-AXIS-A: 5.21000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0450-AXIS-B: 0.28506 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0450-AXIS-C: 0.28506 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0451-AXIS-A: 5.24631 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0451-AXIS-B: 0.32192 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0451-AXIS-C: 0.32192 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0452-AXIS-A: 5.28000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0452-AXIS-B: 0.30684 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0452-AXIS-C: 0.30684 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0453-AXIS-A: 6.37775 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0453-AXIS-B: 0.38247 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0453-AXIS-C: 0.36009 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0454-AXIS-A: 6.71950 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0454-AXIS-B: 0.32233 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0454-AXIS-C: 0.31589 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0455-AXIS-A: 8.00120 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0455-AXIS-B: 0.43364 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0455-AXIS-C: 0.41040 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0456-AXIS-A: 9.11200 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0456-AXIS-B: 0.37711 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0456-AXIS-C: 0.36176 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0457-AXIS-A: 9.37000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0457-AXIS-B: 0.34335 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0457-AXIS-C: 0.33076 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0458-AXIS-A: 9.60847 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0458-AXIS-B: 0.31767 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0458-AXIS-C: 0.30710 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0459-AXIS-A: 9.69038 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0459-AXIS-B: 0.41383 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0459-AXIS-C: 0.39615 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0460-AXIS-A: 10.41120 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0460-AXIS-B: 0.33789 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0460-AXIS-C: 0.32909 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0461-AXIS-A: 12.65023 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0461-AXIS-B: 0.34580 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0461-AXIS-C: 0.34578 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0462-AXIS-A: 20.38064 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0462-AXIS-B: 0.40142 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0462-AXIS-C: 0.39299 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0463-AXIS-A: 20.47007 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0463-AXIS-B: 0.35280 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0463-AXIS-C: 0.34634 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0464-AXIS-A: 20.93973 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0464-AXIS-B: 0.27907 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0464-AXIS-C: 0.27507 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0465-AXIS-A: 26.34993 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0465-AXIS-B: 0.35940 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0465-AXIS-C: 0.35392 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0466-AXIS-A: 30.63777 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0466-AXIS-B: 0.36930 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0466-AXIS-C: 0.36410 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0467-AXIS-A: 41.46000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0467-AXIS-B: 0.36348 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0467-AXIS-C: 0.35913 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0468-AXIS-A: 45.27301 per centimeter; exact and axis-bound True

- [illegible]

- [illegible]

- NIST-CCCBDB-PROP-010-ROW-0493-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0494-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0494-AXIS-B: 0.05203 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0494-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0495-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0495-AXIS-B: 0.06816 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0495-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0496-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0496-AXIS-B: 0.08730 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0496-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0497-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0497-AXIS-B: 0.10759 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0497-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0498-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0498-AXIS-B: 0.11340 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0498-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0499-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0499-AXIS-B: 0.11570 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0499-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0500-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0500-AXIS-B: 0.12626 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0500-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0501-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0501-AXIS-B: 0.12794 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0501-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0502-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0502-AXIS-B: 0.13343 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0502-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0503-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0503-AXIS-B: 0.13401 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0503-AXIS-C: 0.13401 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0504-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0504-AXIS-B: 0.13717 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0504-AXIS-C: 0.13717 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0505-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0505-AXIS-B: 0.14641 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0505-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0506-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0506-AXIS-B: 0.15174 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0506-AXIS-C: 0.15174 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0507-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0507-AXIS-B: 0.15710 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0507-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0508-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0508-AXIS-B: 0.19183 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0508-AXIS-C: 0.17704 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0509-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0509-AXIS-B: 0.25048 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0509-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0510-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0510-AXIS-B: 0.26084 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0510-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0511-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0511-AXIS-B: 0.35582 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0511-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0512-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0512-AXIS-B: 0.36362 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0512-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0513-AXIS-A: 0.08247 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0513-AXIS-B: 0.08247 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0513-AXIS-C: 0.05003 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0514-AXIS-A: 0.11015 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0514-AXIS-B: 0.11015 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0514-AXIS-C: 0.05932 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0515-AXIS-A: 0.11108 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0515-AXIS-B: 0.11108 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0515-AXIS-C: 0.11108 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0516-AXIS-A: 0.15529 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0516-AXIS-B: 0.07410 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0516-AXIS-C: 0.05011 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0517-AXIS-A: 0.16968 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0517-AXIS-B: 0.09415 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0517-AXIS-C: 0.06539 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0518-AXIS-A: 0.18457 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0518-AXIS-B: 0.04649 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0518-AXIS-C: 0.04112 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0519-AXIS-A: 0.21453 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0519-AXIS-B: 0.10728 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0519-AXIS-C: 0.07583 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0520-AXIS-A: 0.23471 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0520-AXIS-B: 0.23052 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0520-AXIS-C: 0.14072 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0521-AXIS-A: 0.24111 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0521-AXIS-B: 0.24111 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0521-AXIS-C: 0.13840 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0522-AXIS-A: 0.24907 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0522-AXIS-B: 0.11379 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0522-AXIS-C: 0.07802 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0523-AXIS-A: 0.25979 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0523-AXIS-B: 0.25171 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0523-AXIS-C: 0.14780 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0524-AXIS-A: 0.26414 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0524-AXIS-B: 0.11591 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0524-AXIS-C: 0.08046 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0525-AXIS-A: 0.26912 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0525-AXIS-B: 0.15247 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0525-AXIS-C: 0.10699 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0526-AXIS-A: 0.27232 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0526-AXIS-B: 0.09395 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0526-AXIS-C: 0.08574 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0527-AXIS-A: 0.27792 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0527-AXIS-B: 0.26609 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0527-AXIS-C: 0.15468 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0528-AXIS-A: 0.28316 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0528-AXIS-B: 0.26824 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0528-AXIS-C: 0.15895 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0529-AXIS-A: 0.28573 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0529-AXIS-B: 0.26661 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0529-AXIS-C: 0.15726 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0530-AXIS-A: 0.28736 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0530-AXIS-B: 0.27876 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0530-AXIS-C: 0.16521 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0531-AXIS-A: 0.28998 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0531-AXIS-B: 0.27028 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0531-AXIS-C: 0.15976 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0532-AXIS-A: 0.30075 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0532-AXIS-B: 0.15566 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0532-AXIS-C: 0.11106 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0533-AXIS-A: 0.30466 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0533-AXIS-B: 0.27959 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0533-AXIS-C: 0.15397 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0534-AXIS-A: 0.30996 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0534-AXIS-B: 0.07469 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0534-AXIS-C: 0.06456 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0535-AXIS-A: 0.31662 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0535-AXIS-B: 0.29896 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0535-AXIS-C: 0.17247 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0536-AXIS-A: 0.33135 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0536-AXIS-B: 0.07562 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0536-AXIS-C: 0.06547 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0537-AXIS-A: 0.33727 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0537-AXIS-B: 0.22260 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0537-AXIS-C: 0.14810 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0538-AXIS-A: 0.33898 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0538-AXIS-B: 0.16500 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0538-AXIS-C: 0.11318 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0539-AXIS-A: 0.33907 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0539-AXIS-B: 0.28404 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0539-AXIS-C: 0.16378 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0540-AXIS-A: 0.34139 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0540-AXIS-B: 0.16215 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0540-AXIS-C: 0.11700 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0541-AXIS-A: 0.34520 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0541-AXIS-B: 0.34520 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0541-AXIS-C: 0.18925 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0542-AXIS-A: 0.34854 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0542-AXIS-B: 0.34854 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0542-AXIS-C: 0.17399 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0543-AXIS-A: 0.34855 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0543-AXIS-B: 0.15697 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0543-AXIS-C: 0.10806 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0544-AXIS-A: 0.35297 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0544-AXIS-B: 0.17156 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0544-AXIS-C: 0.11554 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0545-AXIS-A: 0.35628 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0545-AXIS-B: 0.35628 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0545-AXIS-C: 0.19493 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0546-AXIS-A: 0.35630 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0546-AXIS-B: 0.17019 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0546-AXIS-C: 0.11502 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0547-AXIS-A: 0.36669 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0547-AXIS-B: 0.34731 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0547-AXIS-C: 0.17837 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0548-AXIS-A: 0.36737 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0548-AXIS-B: 0.20676 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0548-AXIS-C: 0.18656 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0549-AXIS-A: 0.36823 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0549-AXIS-B: 0.32308 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0549-AXIS-C: 0.17752 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0550-AXIS-A: 0.37470 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0550-AXIS-B: 0.34589 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0550-AXIS-C: 0.18068 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0551-AXIS-A: 0.37694 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0551-AXIS-B: 0.23059 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0551-AXIS-C: 0.15106 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0552-AXIS-A: 0.37773 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0552-AXIS-B: 0.31669 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0552-AXIS-C: 0.17766 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0553-AXIS-A: 0.38167 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0553-AXIS-B: 0.31273 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0553-AXIS-C: 0.16966 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0554-AXIS-A: 0.38203 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0554-AXIS-B: 0.37824 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0554-AXIS-C: 0.22055 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0555-AXIS-A: 0.38421 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0555-AXIS-B: 0.08490 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0555-AXIS-C: 0.06947 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0556-AXIS-A: 0.38857 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0556-AXIS-B: 0.16225 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0556-AXIS-C: 0.13165 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0557-AXIS-A: 0.39405 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0557-AXIS-B: 0.39204 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0557-AXIS-C: 0.19617 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0558-AXIS-A: 0.39442 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0558-AXIS-B: 0.11067 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0558-AXIS-C: 0.09513 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0559-AXIS-A: 0.40178 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0559-AXIS-B: 0.39140 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0559-AXIS-C: 0.22451 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0560-AXIS-A: 0.43006 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0560-AXIS-B: 0.40782 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0560-AXIS-C: 0.22737 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0561-AXIS-A: 0.43401 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0561-AXIS-B: 0.40361 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0561-AXIS-C: 0.20883 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0562-AXIS-A: 0.43728 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0562-AXIS-B: 0.08038 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0562-AXIS-C: 0.07137 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0563-AXIS-A: 0.44035 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0563-AXIS-B: 0.38181 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0563-AXIS-C: 0.20411 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0564-AXIS-A: 0.44070 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0564-AXIS-B: 0.15883 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0564-AXIS-C: 0.13542 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0565-AXIS-A: 0.44331 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0565-AXIS-B: 0.17264 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0565-AXIS-C: 0.12407 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0566-AXIS-A: 0.44504 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0566-AXIS-B: 0.35172 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0566-AXIS-C: 0.19599 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0567-AXIS-A: 0.45359 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0567-AXIS-B: 0.10968 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0567-AXIS-C: 0.09593 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0568-AXIS-A: 0.45855 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0568-AXIS-B: 0.45855 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0568-AXIS-C: 0.22927 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0569-AXIS-A: 0.45857 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0569-AXIS-B: 0.15383 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0569-AXIS-C: 0.11504 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0570-AXIS-A: 0.47802 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0570-AXIS-B: 0.17077 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0570-AXIS-C: 0.14424 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0571-AXIS-A: 0.48379 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0571-AXIS-B: 0.16964 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0571-AXIS-C: 0.14328 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0572-AXIS-A: 0.50668 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0572-AXIS-B: 0.12871 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0572-AXIS-C: 0.10459 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0573-AXIS-A: 0.51044 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0573-AXIS-B: 0.18596 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0573-AXIS-C: 0.14354 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0574-AXIS-A: 0.51713 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0574-AXIS-B: 0.21226 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0574-AXIS-C: 0.18635 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0575-AXIS-A: 0.51790 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0575-AXIS-B: 0.26181 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0575-AXIS-C: 0.17341 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0576-AXIS-A: 0.53569 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0576-AXIS-B: 0.17142 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0576-AXIS-C: 0.13637 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0577-AXIS-A: 0.54271 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0577-AXIS-B: 0.22426 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0577-AXIS-C: 0.19331 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0578-AXIS-A: 0.54724 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0578-AXIS-B: 0.12530 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0578-AXIS-C: 0.10183 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0579-AXIS-A: 0.55588 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0579-AXIS-B: 0.23054 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0579-AXIS-C: 0.19864 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0580-AXIS-A: 0.57495 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0580-AXIS-B: 0.20024 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0580-AXIS-C: 0.15277 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0581-AXIS-A: 0.57747 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0581-AXIS-B: 0.31067 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0581-AXIS-C: 0.27998 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0582-AXIS-A: 0.57781 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0582-AXIS-B: 0.16722 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0582-AXIS-C: 0.14619 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0583-AXIS-A: 0.57920 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0583-AXIS-B: 0.22212 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0583-AXIS-C: 0.16415 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0584-AXIS-A: 0.60121 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0584-AXIS-B: 0.22289 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0584-AXIS-C: 0.19852 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0585-AXIS-A: 0.60810 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0585-AXIS-B: 0.15249 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0585-AXIS-C: 0.12175 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0586-AXIS-A: 0.64762 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0586-AXIS-B: 0.22940 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0586-AXIS-C: 0.18163 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0587-AXIS-A: 0.65668 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0587-AXIS-B: 0.26508 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0587-AXIS-C: 0.18851 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0588-AXIS-A: 0.66665 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0588-AXIS-B: 0.23065 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0588-AXIS-C: 0.17694 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0589-AXIS-A: 0.67226 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0589-AXIS-B: 0.24906 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0589-AXIS-C: 0.19586 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0590-AXIS-A: 0.68547 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0590-AXIS-B: 0.21205 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0590-AXIS-C: 0.17267 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0591-AXIS-A: 0.70393 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0591-AXIS-B: 0.19782 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0591-AXIS-C: 0.15418 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0592-AXIS-A: 0.73499 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0592-AXIS-B: 0.02070 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0592-AXIS-C: 0.02021 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0593-AXIS-A: 0.76158 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0593-AXIS-B: 0.20818 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0593-AXIS-C: 0.16352 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0594-AXIS-A: 0.80238 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0594-AXIS-B: 0.02977 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0594-AXIS-C: 0.02885 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0595-AXIS-A: 0.85366 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0595-AXIS-B: 0.14421 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0595-AXIS-C: 0.14104 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0596-AXIS-A: 0.85904 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0596-AXIS-B: 0.07938 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0596-AXIS-C: 0.07577 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0597-AXIS-A: 0.86159 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0597-AXIS-B: 0.18728 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0597-AXIS-C: 0.15363 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0598-AXIS-A: 0.86831 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0598-AXIS-B: 0.04131 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0598-AXIS-C: 0.03973 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0599-AXIS-A: 0.87063 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0599-AXIS-B: 0.13712 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0599-AXIS-C: 0.13218 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0600-AXIS-A: 0.88066 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0600-AXIS-B: 0.12683 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0600-AXIS-C: 0.11840 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0601-AXIS-A: 0.89106 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0601-AXIS-B: 0.20650 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0601-AXIS-C: 0.16786 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0602-AXIS-A: 0.90018 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0602-AXIS-B: 0.12503 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0602-AXIS-C: 0.11708 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0603-AXIS-A: 0.90555 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0603-AXIS-B: 0.15165 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0603-AXIS-C: 0.13632 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0604-AXIS-A: 0.91459 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0604-AXIS-B: 0.05410 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0604-AXIS-C: 0.05156 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0605-AXIS-A: 0.92465 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0605-AXIS-B: 0.14222 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0605-AXIS-C: 0.13783 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0606-AXIS-A: 0.92825 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0606-AXIS-B: 0.15808 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0606-AXIS-C: 0.14217 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0607-AXIS-A: 0.93378 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0607-AXIS-B: 0.13876 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0607-AXIS-C: 0.12979 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0608-AXIS-A: 0.96154 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0608-AXIS-B: 0.28942 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0608-AXIS-C: 0.22188 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0609-AXIS-A: 0.97122 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0609-AXIS-B: 0.09939 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0609-AXIS-C: 0.09328 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0610-AXIS-A: 0.99102 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0610-AXIS-B: 0.08024 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0610-AXIS-C: 0.08024 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0611-AXIS-A: 1.01806 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0611-AXIS-B: 0.28259 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0611-AXIS-C: 0.22061 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0612-AXIS-A: 1.04530 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0612-AXIS-B: 0.18325 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0612-AXIS-C: 0.16553 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0613-AXIS-A: 1.05739 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0613-AXIS-B: 0.12939 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0613-AXIS-C: 0.12065 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0614-AXIS-A: 1.12096 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0614-AXIS-B: 0.16146 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0614-AXIS-C: 0.14602 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0615-AXIS-A: 1.12109 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0615-AXIS-B: 0.08985 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0615-AXIS-C: 0.08504 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0616-AXIS-A: 1.12945 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0616-AXIS-B: 0.14014 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0616-AXIS-C: 0.13103 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0617-AXIS-A: 1.28152 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0617-AXIS-B: 0.09591 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0617-AXIS-C: 0.09076 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0618-AXIS-A: 1.29640 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0618-AXIS-B: 0.04502 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0618-AXIS-C: 0.04366 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0619-AXIS-A: 1.29823 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0619-AXIS-B: 0.15035 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0619-AXIS-C: 0.13799 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0620-AXIS-A: 1.32898 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0620-AXIS-B: 0.12499 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0620-AXIS-C: 0.11679 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0621-AXIS-A: 1.37000 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0621-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0621-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0622-AXIS-A: 1.48814 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0622-AXIS-B: 0.14136 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0622-AXIS-C: 0.13255 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0623-AXIS-A: 1.57954 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0623-AXIS-B: 0.15542 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0623-AXIS-C: 0.14152 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0624-AXIS-A: 1.63701 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0624-AXIS-B: 0.16583 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0624-AXIS-C: 0.15057 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0625-AXIS-A: 1.67754 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0625-AXIS-B: 0.15827 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0625-AXIS-C: 0.14442 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0626-AXIS-A: 1.75571 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0626-AXIS-B: 0.16917 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0626-AXIS-C: 0.15430 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0627-AXIS-A: 1.75779 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0627-AXIS-B: 0.10870 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0627-AXIS-C: 0.10229 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0628-AXIS-A: 1.81004 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0628-AXIS-B: 0.13885 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0628-AXIS-C: 0.12884 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0629-AXIS-A: 1.84540 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0629-AXIS-B: 0.15990 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0629-AXIS-C: 0.14720 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0630-AXIS-A: 2.11390 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0630-AXIS-B: 0.18177 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0630-AXIS-C: 0.16710 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0631-AXIS-A: 2.26941 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0631-AXIS-B: 0.16099 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0631-AXIS-C: 0.15009 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0632-AXIS-A: 2.55797 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0632-AXIS-B: 0.11065 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0632-AXIS-C: 0.11065 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0633-AXIS-A: 2.82048 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0633-AXIS-B: 0.09331 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0633-AXIS-C: 0.09025 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0634-AXIS-A: 2.85199 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0634-AXIS-B: 0.12499 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0634-AXIS-C: 0.11962 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0635-AXIS-A: 4.90414 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0635-AXIS-B: 0.13197 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0635-AXIS-C: 0.12851 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0636-AXIS-A: 5.12267 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0636-AXIS-B: 0.14634 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0636-AXIS-C: 0.14204 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0637-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0637-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0637-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0638-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0638-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0638-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0639-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0639-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0639-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0640-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0640-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0640-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0641-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0641-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0641-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0642-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0642-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0642-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0643-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0643-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0643-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0644-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0644-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0644-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0645-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0645-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True

- [illegible]

- [illegible]

- [illegible]

- [illegible]

- [illegible]

- [illegible]

- [illegible]

- NIST-CCCBDB-PROP-010-ROW-0734-AXIS-B: 0.06998 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0734-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0735-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0735-AXIS-B: 0.07162 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0735-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0736-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0736-AXIS-B: 0.07321 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0736-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0737-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0737-AXIS-B: 0.07914 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0737-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0738-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0738-AXIS-B: 0.08950 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0738-AXIS-C: 0.08950 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0739-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0739-AXIS-B: 0.11930 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0739-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0740-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0740-AXIS-B: 0.13786 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0740-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0741-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0741-AXIS-B: 0.13947 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0741-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0742-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0742-AXIS-B: 0.15325 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0742-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0743-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0743-AXIS-B: 0.24660 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0743-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0744-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0744-AXIS-B: 0.28756 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0744-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0745-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0745-AXIS-B: 1.52325 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0745-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0746-AXIS-A: 0.01070 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0746-AXIS-B: 0.01070 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0746-AXIS-C: 0.01070 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0747-AXIS-A: 0.02124 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0747-AXIS-B: 0.02124 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0747-AXIS-C: 0.02124 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0748-AXIS-A: 0.05713 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0748-AXIS-B: 0.05713 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0748-AXIS-C: 0.05713 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0749-AXIS-A: 0.06724 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0749-AXIS-B: 0.06546 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0749-AXIS-C: 0.04831 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0750-AXIS-A: 0.08224 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0750-AXIS-B: 0.08224 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0750-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0751-AXIS-A: 0.10409 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0751-AXIS-B: 0.10409 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0751-AXIS-C: 0.10409 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0752-AXIS-A: 0.11628 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0752-AXIS-B: 0.07820 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0752-AXIS-C: 0.06439 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0753-AXIS-A: 0.12211 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0753-AXIS-B: 0.08199 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0753-AXIS-C: 0.07199 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0754-AXIS-A: 0.12859 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0754-AXIS-B: 0.05673 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0754-AXIS-C: 0.04954 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0755-AXIS-A: 0.13162 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0755-AXIS-B: 0.05115 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0755-AXIS-C: 0.03681 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0756-AXIS-A: 0.13739 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0756-AXIS-B: 0.08802 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0756-AXIS-C: 0.07451 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0757-AXIS-A: 0.13778 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0757-AXIS-B: 0.13778 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0757-AXIS-C: 0.13778 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0758-AXIS-A: 0.14938 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0758-AXIS-B: 0.10003 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0758-AXIS-C: 0.09907 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0759-AXIS-A: 0.15471 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0759-AXIS-B: 0.14328 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0759-AXIS-C: 0.13933 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0760-AXIS-A: 0.15808 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0760-AXIS-B: 0.15159 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0760-AXIS-C: 0.08547 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0761-AXIS-A: 0.16925 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0761-AXIS-B: 0.16886 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0761-AXIS-C: 0.09012 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0762-AXIS-A: 0.17126 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0762-AXIS-B: 0.16922 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0762-AXIS-C: 0.16868 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0763-AXIS-A: 0.17178 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0763-AXIS-B: 0.16145 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0763-AXIS-C: 0.16030 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0764-AXIS-A: 0.17214 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0764-AXIS-B: 0.16760 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0764-AXIS-C: 0.16281 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0765-AXIS-A: 0.17879 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0765-AXIS-B: 0.11325 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0765-AXIS-C: 0.09275 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0766-AXIS-A: 0.18142 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0766-AXIS-B: 0.12486 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0766-AXIS-C: 0.07393 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0767-AXIS-A: 0.18489 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0767-AXIS-B: 0.14133 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0767-AXIS-C: 0.10431 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0768-AXIS-A: 0.18571 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0768-AXIS-B: 0.17730 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0768-AXIS-C: 0.17461 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0769-AXIS-A: 0.18709 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0769-AXIS-B: 0.17541 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0769-AXIS-C: 0.17541 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0770-AXIS-A: 0.18960 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0770-AXIS-B: 0.18960 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0770-AXIS-C: 0.09480 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0771-AXIS-A: 0.19130 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0771-AXIS-B: 0.11126 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0771-AXIS-C: 0.11126 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0772-AXIS-A: 0.19242 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0772-AXIS-B: 0.19242 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0772-AXIS-C: 0.19242 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0773-AXIS-A: 0.19880 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0773-AXIS-B: 0.19880 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0773-AXIS-C: 0.09940 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0774-AXIS-A: 0.20082 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0774-AXIS-B: 0.14590 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0774-AXIS-C: 0.09343 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0775-AXIS-A: 0.20145 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0775-AXIS-B: 0.19363 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0775-AXIS-C: 0.09871 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0776-AXIS-A: 0.20465 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0776-AXIS-B: 0.17357 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0776-AXIS-C: 0.12711 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0777-AXIS-A: 0.20764 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0777-AXIS-B: 0.04216 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0777-AXIS-C: 0.03502 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0778-AXIS-A: 0.20830 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0778-AXIS-B: 0.19890 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0778-AXIS-C: 0.10180 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0779-AXIS-A: 0.20937 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0779-AXIS-B: 0.20238 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0779-AXIS-C: 0.10288 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0780-AXIS-A: 0.20943 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0780-AXIS-B: 0.04917 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0780-AXIS-C: 0.04084 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0781-AXIS-A: 0.21382 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0781-AXIS-B: 0.19730 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0781-AXIS-C: 0.10261 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0782-AXIS-A: 0.21460 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0782-AXIS-B: 0.21460 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0782-AXIS-C: 0.10730 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0783-AXIS-A: 0.21571 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0783-AXIS-B: 0.06800 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0783-AXIS-C: 0.05340 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0784-AXIS-A: 0.22087 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0784-AXIS-B: 0.16006 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0784-AXIS-C: 0.09624 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0785-AXIS-A: 0.22308 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0785-AXIS-B: 0.13632 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0785-AXIS-C: 0.10741 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0786-AXIS-A: 0.22798 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0786-AXIS-B: 0.22275 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0786-AXIS-C: 0.12969 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0787-AXIS-A: 0.22890 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0787-AXIS-B: 0.15703 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0787-AXIS-C: 0.09852 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0788-AXIS-A: 0.23076 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0788-AXIS-B: 0.07926 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0788-AXIS-C: 0.06328 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0789-AXIS-A: 0.23315 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0789-AXIS-B: 0.19036 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0789-AXIS-C: 0.10475 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0790-AXIS-A: 0.23356 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0790-AXIS-B: 0.23356 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0790-AXIS-C: 0.16163 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0791-AXIS-A: 0.23673 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0791-AXIS-B: 0.23269 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0791-AXIS-C: 0.13369 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0792-AXIS-A: 0.23994 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0792-AXIS-B: 0.23994 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0792-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0793-AXIS-A: 0.24329 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0793-AXIS-B: 0.24107 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0793-AXIS-C: 0.13174 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0794-AXIS-A: 0.24648 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0794-AXIS-B: 0.17599 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0794-AXIS-C: 0.12975 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0795-AXIS-A: 0.24968 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0795-AXIS-B: 0.13685 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0795-AXIS-C: 0.09926 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0796-AXIS-A: 0.25347 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0796-AXIS-B: 0.11286 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0796-AXIS-C: 0.07806 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0797-AXIS-A: 0.25919 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0797-AXIS-B: 0.14941 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0797-AXIS-C: 0.11638 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0798-AXIS-A: 0.26109 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0798-AXIS-B: 0.25132 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0798-AXIS-C: 0.14325 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0799-AXIS-A: 0.26825 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0799-AXIS-B: 0.18073 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0799-AXIS-C: 0.10793 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0800-AXIS-A: 0.27306 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0800-AXIS-B: 0.12685 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0800-AXIS-C: 0.08661 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0801-AXIS-A: 0.27566 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0801-AXIS-B: 0.12056 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0801-AXIS-C: 0.08804 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0802-AXIS-A: 0.28106 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0802-AXIS-B: 0.27437 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0802-AXIS-C: 0.14248 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0803-AXIS-A: 0.28443 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0803-AXIS-B: 0.13928 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0803-AXIS-C: 0.09514 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0804-AXIS-A: 0.29712 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0804-AXIS-B: 0.18577 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0804-AXIS-C: 0.11424 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0805-AXIS-A: 0.30357 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0805-AXIS-B: 0.20394 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0805-AXIS-C: 0.15950 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0806-AXIS-A: 0.30456 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0806-AXIS-B: 0.30025 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0806-AXIS-C: 0.15117 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0807-AXIS-A: 0.31221 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0807-AXIS-B: 0.13388 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0807-AXIS-C: 0.09522 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0808-AXIS-A: 0.31229 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0808-AXIS-B: 0.09630 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0808-AXIS-C: 0.09094 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0809-AXIS-A: 0.31391 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0809-AXIS-B: 0.04696 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0809-AXIS-C: 0.04235 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0810-AXIS-A: 0.31512 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0810-AXIS-B: 0.30843 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0810-AXIS-C: 0.15580 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0811-AXIS-A: 0.31837 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0811-AXIS-B: 0.11998 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0811-AXIS-C: 0.09161 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0812-AXIS-A: 0.32085 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0812-AXIS-B: 0.31397 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0812-AXIS-C: 0.15864 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0813-AXIS-A: 0.32530 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0813-AXIS-B: 0.31268 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0813-AXIS-C: 0.15917 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0814-AXIS-A: 0.32644 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0814-AXIS-B: 0.31939 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0814-AXIS-C: 0.16134 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0815-AXIS-A: 0.33526 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0815-AXIS-B: 0.32175 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0815-AXIS-C: 0.16409 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0816-AXIS-A: 0.33558 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0816-AXIS-B: 0.10900 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0816-AXIS-C: 0.09025 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0817-AXIS-A: 0.33756 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0817-AXIS-B: 0.14286 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0817-AXIS-C: 0.11411 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0818-AXIS-A: 0.33821 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0818-AXIS-B: 0.06970 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0818-AXIS-C: 0.05977 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0819-AXIS-A: 0.34174 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0819-AXIS-B: 0.32797 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0819-AXIS-C: 0.16729 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0820-AXIS-A: 0.34496 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0820-AXIS-B: 0.12930 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0820-AXIS-C: 0.09715 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0821-AXIS-A: 0.34591 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0821-AXIS-B: 0.15359 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0821-AXIS-C: 0.11552 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0822-AXIS-A: 0.35113 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0822-AXIS-B: 0.11299 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0822-AXIS-C: 0.09278 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0823-AXIS-A: 0.35801 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0823-AXIS-B: 0.14072 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0823-AXIS-C: 0.10109 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0824-AXIS-A: 0.35976 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0824-AXIS-B: 0.16033 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0824-AXIS-C: 0.11870 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0825-AXIS-A: 0.36064 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0825-AXIS-B: 0.21741 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0825-AXIS-C: 0.17400 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0826-AXIS-A: 0.36163 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0826-AXIS-B: 0.13601 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0826-AXIS-C: 0.09869 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0827-AXIS-A: 0.36651 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0827-AXIS-B: 0.08677 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0827-AXIS-C: 0.07910 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0828-AXIS-A: 0.39334 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0828-AXIS-B: 0.11302 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0828-AXIS-C: 0.08914 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0829-AXIS-A: 0.39344 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0829-AXIS-B: 0.15703 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0829-AXIS-C: 0.11469 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0830-AXIS-A: 0.39672 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0830-AXIS-B: 0.11419 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0830-AXIS-C: 0.09578 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0831-AXIS-A: 0.40009 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0831-AXIS-B: 0.15562 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0831-AXIS-C: 0.11332 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0832-AXIS-A: 0.40384 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0832-AXIS-B: 0.09263 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0832-AXIS-C: 0.07532 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0833-AXIS-A: 0.41543 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0833-AXIS-B: 0.14098 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0833-AXIS-C: 0.10517 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0834-AXIS-A: 0.41985 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0834-AXIS-B: 0.07778 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0834-AXIS-C: 0.07189 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0835-AXIS-A: 0.44688 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0835-AXIS-B: 0.04173 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0835-AXIS-C: 0.04096 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0836-AXIS-A: 0.45702 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0836-AXIS-B: 0.07788 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0836-AXIS-C: 0.07187 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0837-AXIS-A: 0.47759 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0837-AXIS-B: 0.05794 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0837-AXIS-C: 0.05381 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0838-AXIS-A: 0.50909 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0838-AXIS-B: 0.10212 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0838-AXIS-C: 0.09811 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0839-AXIS-A: 0.52243 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0839-AXIS-B: 0.08867 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0839-AXIS-C: 0.07692 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0840-AXIS-A: 0.52268 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0840-AXIS-B: 0.10745 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0840-AXIS-C: 0.09308 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0841-AXIS-A: 0.52446 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0841-AXIS-B: 0.11208 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0841-AXIS-C: 0.10650 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0842-AXIS-A: 0.59196 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0842-AXIS-B: 0.09689 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0842-AXIS-C: 0.08603 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0843-AXIS-A: 0.60397 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0843-AXIS-B: 0.07087 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0843-AXIS-C: 0.06979 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0844-AXIS-A: 0.65897 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0844-AXIS-B: 0.07090 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0844-AXIS-C: 0.06730 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0845-AXIS-A: 0.69657 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0845-AXIS-B: 0.09814 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0845-AXIS-C: 0.08728 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0846-AXIS-A: 0.72984 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0846-AXIS-B: 0.10827 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0846-AXIS-C: 0.09438 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0847-AXIS-A: 0.77994 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0847-AXIS-B: 0.07440 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0847-AXIS-C: 0.07059 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0848-AXIS-A: 0.78948 per centimeter; exact and axis-bound True

- [illegible]

- [illegible]

- [illegible]

- [illegible]

- [illegible]

- [illegible]

- [illegible]

- NIST-CCCBDB-PROP-010-ROW-0937-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0937-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0937-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0938-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0938-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0938-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0939-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0939-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0939-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0940-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0940-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0940-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0941-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0941-AXIS-B: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0941-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0942-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0942-AXIS-B: 0.05485 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0942-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0943-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0943-AXIS-B: 0.06086 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0943-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0944-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0944-AXIS-B: 0.07399 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0944-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0945-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0945-AXIS-B: 0.09041 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0945-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0946-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0946-AXIS-B: 0.09173 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0946-AXIS-C: 0.09173 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0947-AXIS-A: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0947-AXIS-B: 0.09557 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0947-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0948-AXIS-A: 0.02177 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0948-AXIS-B: 0.02177 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0948-AXIS-C: 0.01186 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0949-AXIS-A: 0.04424 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0949-AXIS-B: 0.03707 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0949-AXIS-C: 0.03707 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0950-AXIS-A: 0.05043 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0950-AXIS-B: 0.04072 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0950-AXIS-C: 0.02253 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0951-AXIS-A: 0.05359 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0951-AXIS-B: 0.01834 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0951-AXIS-C: 0.01366 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0952-AXIS-A: 0.06438 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0952-AXIS-B: 0.04774 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0952-AXIS-C: 0.02741 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0953-AXIS-A: 0.07259 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0953-AXIS-B: 0.01957 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0953-AXIS-C: 0.01546 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0954-AXIS-A: 0.08532 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0954-AXIS-B: 0.04186 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0954-AXIS-C: 0.03293 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0955-AXIS-A: 0.09107 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0955-AXIS-B: 0.09107 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0955-AXIS-C: 0.09107 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0956-AXIS-A: 0.09448 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0956-AXIS-B: 0.02878 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0956-AXIS-C: 0.02206 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0957-AXIS-A: 0.09480 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0957-AXIS-B: 0.04186 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0957-AXIS-C: 0.02904 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0958-AXIS-A: 0.09549 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0958-AXIS-B: 0.06185 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0958-AXIS-C: 0.06185 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0959-AXIS-A: 0.10338 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0959-AXIS-B: 0.10338 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0959-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0960-AXIS-A: 0.10405 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0960-AXIS-B: 0.04113 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0960-AXIS-C: 0.02948 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0961-AXIS-A: 0.11220 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0961-AXIS-B: 0.11220 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0961-AXIS-C: 0.11220 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0962-AXIS-A: 0.11842 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0962-AXIS-B: 0.11842 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0962-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0963-AXIS-A: 0.12302 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0963-AXIS-B: 0.04053 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0963-AXIS-C: 0.03069 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0964-AXIS-A: 0.12312 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0964-AXIS-B: 0.08071 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0964-AXIS-C: 0.06691 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0965-AXIS-A: 0.12323 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0965-AXIS-B: 0.10716 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0965-AXIS-C: 0.09259 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0966-AXIS-A: 0.12329 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0966-AXIS-B: 0.12249 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0966-AXIS-C: 0.06779 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0967-AXIS-A: 0.12564 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0967-AXIS-B: 0.10452 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0967-AXIS-C: 0.10452 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0968-AXIS-A: 0.12592 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0968-AXIS-B: 0.05273 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0968-AXIS-C: 0.03743 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0969-AXIS-A: 0.13236 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0969-AXIS-B: 0.04293 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0969-AXIS-C: 0.03244 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0970-AXIS-A: 0.13318 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0970-AXIS-B: 0.10026 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0970-AXIS-C: 0.06387 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0971-AXIS-A: 0.13994 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0971-AXIS-B: 0.08348 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0971-AXIS-C: 0.05852 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0972-AXIS-A: 0.14255 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0972-AXIS-B: 0.12043 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0972-AXIS-C: 0.10629 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0973-AXIS-A: 0.14336 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0973-AXIS-B: 0.10559 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0973-AXIS-C: 0.08474 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0974-AXIS-A: 0.14704 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0974-AXIS-B: 0.14495 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0974-AXIS-C: 0.10862 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0975-AXIS-A: 0.14992 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0975-AXIS-B: 0.14661 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0975-AXIS-C: 0.08458 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0976-AXIS-A: 0.15031 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0976-AXIS-B: 0.07567 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0976-AXIS-C: 0.05030 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0977-AXIS-A: 0.15068 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0977-AXIS-B: 0.14801 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0977-AXIS-C: 0.08482 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0978-AXIS-A: 0.16774 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0978-AXIS-B: 0.05235 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0978-AXIS-C: 0.04022 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0979-AXIS-A: 0.16899 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0979-AXIS-B: 0.10344 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0979-AXIS-C: 0.07552 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0980-AXIS-A: 0.17129 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0980-AXIS-B: 0.10502 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0980-AXIS-C: 0.06896 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0981-AXIS-A: 0.17223 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0981-AXIS-B: 0.05154 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0981-AXIS-C: 0.03973 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0982-AXIS-A: 0.17460 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0982-AXIS-B: 0.05218 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0982-AXIS-C: 0.04018 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0983-AXIS-A: 0.17525 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0983-AXIS-B: 0.09465 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0983-AXIS-C: 0.08348 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0984-AXIS-A: 0.18643 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0984-AXIS-B: 0.05263 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0984-AXIS-C: 0.04106 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0985-AXIS-A: 0.18771 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0985-AXIS-B: 0.08652 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0985-AXIS-C: 0.05928 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0986-AXIS-A: 0.18809 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0986-AXIS-B: 0.09859 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0986-AXIS-C: 0.09859 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0987-AXIS-A: 0.18848 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0987-AXIS-B: 0.08737 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-0987-AXIS-C: 0.05970 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0988-AXIS-A: 0.18864 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0988-AXIS-B: 0.05160 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0988-AXIS-C: 0.04051 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0989-AXIS-A: 0.18892 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0989-AXIS-B: 0.08575 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0989-AXIS-C: 0.05897 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0990-AXIS-A: 0.18905 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0990-AXIS-B: 0.03319 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0990-AXIS-C: 0.02823 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0991-AXIS-A: 0.18923 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0991-AXIS-B: 0.05260 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0991-AXIS-C: 0.04115 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0992-AXIS-A: 0.18949 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0992-AXIS-B: 0.05103 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0992-AXIS-C: 0.04019 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0993-AXIS-A: 0.19111 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0993-AXIS-B: 0.08397 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0993-AXIS-C: 0.05834 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0994-AXIS-A: 0.21792 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0994-AXIS-B: 0.12233 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0994-AXIS-C: 0.07834 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0995-AXIS-A: 0.22090 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0995-AXIS-B: 0.11180 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0995-AXIS-C: 0.08040 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0996-AXIS-A: 0.22963 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0996-AXIS-B: 0.08272 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0996-AXIS-C: 0.06081 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0997-AXIS-A: 0.24243 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0997-AXIS-B: 0.07789 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0997-AXIS-C: 0.06627 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0998-AXIS-A: 0.25169 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0998-AXIS-B: 0.07162 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0998-AXIS-C: 0.05992 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0999-AXIS-A: 0.27165 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0999-AXIS-B: 0.07001 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-0999-AXIS-C: 0.05567 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-1000-AXIS-A: 0.27325 per centimeter; exact and axis-bound True

- NIST-CCCBDB-PROP-010-ROW-1000-AXIS-B: 0.06824 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-1000-AXIS-C: 0.05461 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-1001-AXIS-A: 0.27423 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-1001-AXIS-B: 0.12955 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-1001-AXIS-C: 0.08948 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-1002-AXIS-A: 0.31175 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-1002-AXIS-B: 0.13971 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-1002-AXIS-C: 0.09645 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-1003-AXIS-A: 0.32588 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-1003-AXIS-B: 0.02898 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-1003-AXIS-C: 0.02730 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-1004-AXIS-A: 0.33273 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-1004-AXIS-B: 0.07365 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-1004-AXIS-C: 0.06174 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-1005-AXIS-A: 2727.42310 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-1005-AXIS-B: 2727.42310 per centimeter; exact and axis-bound True
- NIST-CCCBDB-PROP-010-ROW-1005-AXIS-C: structural EmptyOne per centimeter; exact and axis-bound True
- tampered_rotational_constant_rejected: True
- tampered_blank_axis_rejected: True
- complete_axis_support_preserved: True
- complete_molecular_row_support_preserved: True
- measured_and_absent_axis_cells_preserved: True
- complete_list_choice_query_boundary_preserved: True
- unreturned_composition_boundary_preserved: True
- all_measured_values_exact_and_positive: True
- all_blank_axis_values_structural_EmptyOne: True

Falsification boundary: The claim fails if a generated molecular rotational constant cannot be reconstructed as a positive finite held-axis recurrence count over a positive interval while preserving molecular state and geometry; if equal repetition changes the ratio; if the $J(J+1)$ level or $2J$ gap fails at any positive ordinal; if any of the 2,186 listed species, 1,193 composition queries, 1,832 returned choices, 83 unreturned composition boundaries, 1,005 displayed molecular rows, 1,681 measurements or 1,334 blank axis cells is concealed, duplicated or displaced; if any target opens before sealing; or if an imported, continuum, fitted or species-specific rule enters.

Measurement receipt: sha256:479dc58099daa7d32e37907f6a826a41375344b49a94334bb28036753223b930.

Isolation certificate: sha256:276196dad32d7a74e2c9049b4ca8226361f125a8b1bb2baa205187e82d4eac09.

Custody certificate: sha256:7993a39cfcf069d40396617b6ed5d6df83f6c84205c2c5264e987e1746b21732.

Registered source descriptions:

- National Institute of Standards and Technology: ()

Registered target identities:

- None.

Meaning. The quantitative property family derives bond length, dissociation, angle, torsion, dipole, polarizability, ionization, affinity, vibration, rotation, binding, magnetic response, formation energy and their common-carrier vector without per-property fitting. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; missing source measurements and the unexcited form use structural EmptyOne
- no negative, irrational, imaginary, floating, signed or continuum proof value
- no rigid-rotor equation, freely rotating continuum body, trigonometric angle or imported moment of inertia
- no measured rotational constant in the law, grammar, candidate forcing or prediction
- no fitted geometry, inertia, molecular scale, species correction or residual
- no dropped blank A/B/C cell, unreturned composition, returned state or retrieval batch

Immutable identities. Source manifest

sha256:d8a29bc9b6f09d2e23d409957435f557123fc0d51a510e0e14a040f34d7d3d38; derivation seal
 sha256:3610d9626aafdb8b84a15297e5f16a5ddd08fbaae1a3718384ce163118e153b1; engine receipt
 sha256:63886feb3b4de292c6af6642a40d8e23d9f9d8cec031ad62ee28fe2b443c0292 at
 receipts/engine/model_admitted/SFT-CHEM-ROTATIONAL-CONSTANT-010-63886feb3b4de292.json;
 empirical hash sha256:583ce16f9edffcf43c398f8345491f3b95cba4bce7cef23d306365f4d9616f1e;
 measurement receipt sha256:479dc58099daa7d32e37907f6a826a41375344b49a94334bb28036753223b930.

121. Exact intermolecular binding by retained state order

Claim identity: SFT-CHEM-INTERMOLECULAR-BINDING-011

Question and theorem. An intermolecular binding magnitude is derived only after every named constituent state, the exact separated-constituent composition, the bound composite state and a finite separation organization remain held. The higher separated state Takes the strictly lower bound composite state, yielding one exact positive relation. A record without strict bound order is structural EmptyOne. All 1,299 value-free identities seal before the complete 1,297-row CCCBDB calculated vector and two reported experimental water-cluster dissociation rows open; all 96 signed adverse source inscriptions remain explicit and never become negative SFT numbers.

named-constituents-and-bound-composite-carrier__finite-held-separation-organization__exact-positive-constituent-state-composition__ordered-positive-separated-Take-bound__strict-bound-order-or-structural-EmptyOne__value-free-complete-dimer-cluster-identity-seal__complete-computed-measured-adverse-and-scope-custody__one-state-order-law-with-depth-independent-constituent-extension

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-GEOMETRY-001

- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-MOL-SUPRAMOLECULAR-001
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-CONFIGURATION-ORDER-PATH-011
- SFT-CHEM-BOND-DISSOCIATION-ENERGY-002
- SFT-CHEM-MOLECULAR-DIPOLE-MAGNITUDE-005
- SFT-CHEM-MOLECULAR-POLARIZABILITY-006
- SFT-CHEM-VIBRATIONAL-FREQUENCY-009
- SFT-CHEM-ROTATIONAL-CONSTANT-010

Generated grammar. Generate the literal product of carrier, separation, composition, magnitude, absence, prediction, record and extension forms; decide all 256 candidates from admitted exact state order, molecular identity, intermolecular support, path composition and target-custody laws.

Boundary: The depth-independent exact state-order relation for every finite named constituent tuple, tested against all 11 official CCCBDB hydrogen-bonded dimers, all 1,297 linked method/basis value rows including every signed adverse inscription, two reported water-cluster dissociation values with uncertainty, and the preserved complete 62-page NIST ion-cluster thermochemistry scope record. Calculated, measured, unavailable and adverse evidence classes remain separate.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `named-constituents-and-bound-composite-carrier__finite-held-separation-organization__exact-positive-constituent-state-composition__ordered-positive-separated-Take-bound__strict-bound-order-or-structural-EmptyOne__value-free-complete-dimer-cluster-identity-seal__complete-computed-measured-adverse-and-scope-custody__one-state-order-law-with-depth-independent-constituent-extension`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	<code>named-constituents-and-bound-composite-carrier</code>	<code>answer-only-binding-number-with-erased-components</code>	An answer-only magnitude cannot distinguish an intermolecular composite from an internal molecular state.
separation	<code>finite-held-separation-organization</code>	<code>continuum-distance-or-imported-potential-coordinate</code>	A continuum coordinate or potential surface imports an ungenerated mathematical model.
composition	<code>exact-positive-constituent-state-composition</code>	<code>constituent-energies-erased-or-merged</code>	Erasing constituent states destroys the separated endpoint and prevents reconstruction.
magnitude	<code>ordered-positive-separated-Take-bound</code>	<code>signed-free-energy-or-negative-binding-number</code>	A signed answer leaves the positive Fold domain and obscures state order.
absence	<code>strict-bound-order-or-structural-EmptyOne</code>	<code>assume-every-generated-aggregate-is-bound</code>	A proximity or calculation label does not force a stable lower composite state.
prediction	<code>value-free-complete-dimer-cluster-identity-seal</code>	<code>binding-target-readable-before-seal</code>	A readable binding value could select the law, source rows or an interaction coefficient.
record	<code>complete-computed-measured-adverse-and-scope-custody</code>	<code>favorable-dimer-method-or-measurement-subset</code>	Selecting only positive or agreeing rows conceals adverse and unavailable source records.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
extension	one-state-order-law-with-depth-independent-constituent-extension	fitted-interaction-coefficient-or-species-correction	A fitted coefficient or species residual lets the target choose the result.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:3d0cd1fb895b18404438bb3fb1b7f4338a0379b2182de8c3fddeb7c8cb6fc5c1.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:176425b0e74acb7d87e416c3621c6ca06dae0c9e137c9fa54de5894016171848.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b7d933ca4f11e98dfe9de48da35a64af3aecb4e3a9eb70c82f5fe3f0b1c85d6f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:704fffe32a354b873618ef7b285259922176a972972bc4fadd9c6a39832ece5b.

Independent reconstruction. Implementation hash

sha256:021c7486deaa21b86baf522182ae02d68605952773bea0f90c224b40f7a575fb; independent certificate sha256:fe537538b80543b6d3bfc0b71659b446b9c17b5bed61cdede4b56cfbe88ec0a8; external-validation hash sha256:1f5b03b448020ad2686e4bd6c6b666914e5f9f6c364767a479a84257030e988a. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-CCCBDB-SRD-101-DIMER-BINDING
- NIST-FARADAY-C8FD00092A-WATER-CLUSTER-DISSOCIATION
- NIST-JPCRD-1.555757-ION-CLUSTER-SCOPE

Comparison records:

- CCCBDB-DIMER-25655838-METHOD-1-BASIS-20: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-1-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-1-BASIS-8: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-1-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-1-BASIS-3: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-1-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-1-BASIS-19: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-1-BASIS-10: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-1-BASIS-22: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-1-BASIS-32: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-1-BASIS-5: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-1-BASIS-25: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-1-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-1-BASIS-26: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-1-BASIS-35: exact-positive-calculated-binding; identity and exact post-seal result True

- [illegible]

- [illegible]

- [illegible]

- [illegible]

- CCCBDB-DIMER-25655838-METHOD-8-BASIS-13: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-8-BASIS-14: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-8-BASIS-15: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-8-BASIS-16: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-8-BASIS-38: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-61-BASIS-38: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-3-BASIS-12: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-3-BASIS-13: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-3-BASIS-14: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-3-BASIS-15: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-3-BASIS-38: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-81-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-81-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-81-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-81-BASIS-22: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-81-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-91-BASIS-2: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-91-BASIS-1: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-91-BASIS-4: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-91-BASIS-22: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-91-BASIS-18: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-88-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-88-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-88-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-88-BASIS-22: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-88-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-98-BASIS-2: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-98-BASIS-1: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-98-BASIS-4: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True

- CCCBDB-DIMER-25655838-METHOD-98-BASIS-22: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-98-BASIS-18: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-89-BASIS-2: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-89-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-89-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-89-BASIS-22: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-89-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-99-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-99-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-99-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-99-BASIS-22: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-99-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-86-BASIS-2: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-86-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-86-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-86-BASIS-22: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-86-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-96-BASIS-2: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-96-BASIS-1: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-96-BASIS-4: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-96-BASIS-22: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-96-BASIS-18: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-87-BASIS-2: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-87-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-87-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-87-BASIS-22: exact-positive-calculated-binding; identity and exact post-seal result True

- CCCBDB-DIMER-25655838-METHOD-87-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-97-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-97-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-97-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-97-BASIS-22: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-97-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-83-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-83-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-83-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-83-BASIS-22: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-83-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-93-BASIS-2: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-93-BASIS-1: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-93-BASIS-4: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-93-BASIS-22: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-93-BASIS-18: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-82-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-82-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-82-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-82-BASIS-22: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-82-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-92-BASIS-1: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-92-BASIS-4: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-25655838-METHOD-92-BASIS-22: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-1-BASIS-20: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-1-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-1-BASIS-8: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-1-BASIS-3: exact-positive-calculated-binding; identity and exact post-seal result True

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- CCCBDB-DIMER-22-METHOD-91-BASIS-1: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-91-BASIS-4: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-91-BASIS-22: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-91-BASIS-18: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-88-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-88-BASIS-1: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-88-BASIS-4: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-88-BASIS-22: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-88-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-98-BASIS-2: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-98-BASIS-1: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-98-BASIS-4: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-98-BASIS-22: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-98-BASIS-18: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-89-BASIS-2: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-89-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-89-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-89-BASIS-22: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-89-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-99-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-99-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-99-BASIS-4: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-99-BASIS-22: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-99-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-86-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-86-BASIS-1: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-86-BASIS-4: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True

- CCCBDB-DIMER-22-METHOD-86-BASIS-22: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-86-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-96-BASIS-2: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-96-BASIS-1: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-96-BASIS-4: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-96-BASIS-22: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-96-BASIS-18: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-87-BASIS-2: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-87-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-87-BASIS-4: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-87-BASIS-22: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-87-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-97-BASIS-2: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-97-BASIS-1: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-97-BASIS-4: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-97-BASIS-22: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-97-BASIS-18: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-83-BASIS-2: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-83-BASIS-1: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-83-BASIS-4: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-83-BASIS-22: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-83-BASIS-18: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-93-BASIS-2: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-93-BASIS-1: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True

- CCCBDB-DIMER-22-METHOD-93-BASIS-4: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-93-BASIS-22: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-93-BASIS-18: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-82-BASIS-2: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-82-BASIS-1: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-22-METHOD-82-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-1-BASIS-20: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-1-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-1-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-1-BASIS-4: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-1-BASIS-22: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-1-BASIS-32: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-1-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-1-BASIS-26: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-1-BASIS-35: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-8-BASIS-19: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-8-BASIS-32: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-8-BASIS-25: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-68-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-68-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-68-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-68-BASIS-22: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-68-BASIS-6: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-68-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-69-BASIS-2: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-69-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-69-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-69-BASIS-22: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-69-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-3-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-33-METHOD-3-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True

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- CCCBDB-DIMER-99-METHOD-93-BASIS-22: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-99-METHOD-82-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-99-METHOD-82-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-1-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-1-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-1-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-1-BASIS-10: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-1-BASIS-22: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-1-BASIS-32: signed-adverse-source-record-structural-EmptyOne; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-1-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-1-BASIS-35: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-8-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-8-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-8-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-8-BASIS-19: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-8-BASIS-22: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-8-BASIS-32: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-8-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-68-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-68-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-68-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-68-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-61-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-61-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-61-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-61-BASIS-22: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-61-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-69-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-69-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-69-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-69-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-3-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-3-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-3-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-3-BASIS-7: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-3-BASIS-22: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-102-METHOD-3-BASIS-32: exact-positive-calculated-binding; identity and exact post-seal result True

- [illegible]

- [illegible]

- [illegible]

- [illegible]

- [illegible]

- [illegible]

- [illegible]

- CCCBDB-DIMER-135-METHOD-83-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-135-METHOD-83-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-135-METHOD-83-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-135-METHOD-83-BASIS-18: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-135-METHOD-93-BASIS-2: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-135-METHOD-93-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-135-METHOD-93-BASIS-4: exact-positive-calculated-binding; identity and exact post-seal result True
- CCCBDB-DIMER-135-METHOD-82-BASIS-1: exact-positive-calculated-binding; identity and exact post-seal result True
- NIST-WATER-CLUSTER-H2O-DIMER-D0-2018: exact-positive-reported-experimental-cluster-dissociation; identity and exact post-seal result True
- NIST-WATER-CLUSTER-D2O-DIMER-D0-2018: exact-positive-reported-experimental-cluster-dissociation; identity and exact post-seal result True
- tampered_positive_value_rejected: True
- signed_adverse_cannot_become_negative_sft_number: True
- missing_row_rejected: True
- complete_dimer_catalogue_retained: True
- computed_and_experimental_classes_not_conflated: True
- all_96_signed_adverse_rows_retained: True

Falsification boundary: The claim fails if a retained bound composite below its exact named separated-constituent state does not yield their exact positive Take; if appending the same constituent state to both endpoints changes the binding; if a non-bound record requires a negative or numerical-zero SFT value; if any of the 11 dimers, 1,297 linked calculated values, 1,201 positive rows, 96 signed adverse rows, 269 displayed DNF inscriptions, two reported cluster dissociation values or the complete wider ion-cluster scope record is concealed or misclassified; if any target opens before sealing; or if a potential, continuum coordinate, fitted coefficient or species correction enters the executable law.

Measurement receipt: sha256:86eb3581f5872524a10efdf84f72cf7edf7070069b16e2e771bbcb0e5f0c69f5.
Isolation certificate: sha256:211ed203b2fce0b0591bca3d5692981e82b7148f8934cac9fe7603a42f8c83da.
Custody certificate: sha256:a55547fcde3293ce0983162afac904402bdddecc8656483cdf7823f90dee29cc.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The quantitative property family derives bond length, dissociation, angle, torsion, dipole, polarizability, ionization, affinity, vibration, rotation, binding, magnetic response, formation energy and their common-carrier vector without per-property fitting. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; an unbound or unavailable native result is structural EmptyOne
- no negative, irrational, imaginary, floating, signed or continuum SFT proof value
- no imported intermolecular potential, continuum distance field or force-family equation
- no measured or calculated binding target in the law, candidate grammar, forcing or prediction
- no fitted interaction coefficient, species correction, basis correction or favorable-row selection
- no conflation of calculated CCCBDB rows with reported experimental dissociation values

- no deletion of signed adverse, unavailable, unparsed-compendium or source-scope records

Immutable identities. Source manifest

sha256:6c978602d3c31960ff487130b285ddd8e10451870b45fa45e79a58ee9d4eda6f; derivation seal
 sha256:47864a988007eda16c806bc1d4a00392bd2805d42cd467883ec0f77272313ad6; engine receipt
 sha256:84a558db081597ed77db5cc0dc4e19dd467ed57378a066985d7d20cd8ee47f7e at receipts/engine/
 model_admitted/SFT-CHEM-INTERMOLECULAR-BINDING-011-84a558db081597ed.json; empirical hash
 sha256:9dec83058dc73c42c62b4d304d7bc2b8b7d8cd2d72a43e021f5b79147fb9926b; measurement receipt
 sha256:86eb3581f5872524a10efdf84f72cf7edf7070069b16e2e771bbcb0e5f0c69f5.

122. Exact molecular magnetic-response relation

Claim identity: SFT-CHEM-MOLECULAR-MAGNETIC-RESPONSE-012

Question and theorem. A complete molecular state retains spin, orbital and rotational support with two opposed orientations as held labels. Equal complementary support closes to structural EmptyOne; unmatched support retains a positive count and its generating orientation. Molecular moment is the exact positive response count per positive angular recurrence, and susceptibility is the exact positive induced response per positive applied-field act. All 174 NIST g-factor and magnetic-susceptibility cells seal without their values or orientations before the 136 printed magnitudes and 38 blanks open. No fitted g-factor, continuum derivative or species coefficient enters.

complete-molecular-state-and-angular-carrier__opposed-held-orientation-labels__pairwise-closure-to-EmptyOne-or-positive-excess__positive-response-per-angular-recurrence__positive-induced-response-per-field-act__value-free-complete-magnetic-identity-seal__complete-g-factor-susceptibility-and-unavailable-custody__one-ratio-law-with-depth-independent-repetition

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-FIELD-MAGNETIC-RELATIVITY-003
- SFT-PHYS-ELECTRON-DIRAC-G-FACTOR-002
- SFT-PHYS-ATOMIC-FIELD-SPLITTING-TERMINAL-005
- SFT-CHEM-ELECTRON-COUNT-SPIN-002
- SFT-CHEM-ORBITAL-SUPPORT-OCCUPANCY-003
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-STATE-SYMMETRY-DEGENERACY-005
- SFT-CHEM-MOLECULAR-EXCLUSION-EXCHANGE-006
- SFT-CHEM-JOINT-CORRELATION-DISSOCIATION-007
- SFT-CHEM-RESOLVED-ROVIBRONIC-SPIN-COMPOSITION-013
- SFT-CHEM-ROTATIONAL-CONSTANT-010

• SFT-CHEM-INTERMOLECULAR-BINDING-011

Generated grammar. Generate the literal product of carrier, orientation, closure, moment, susceptibility, prediction, record and extension forms; decide all 256 candidates from admitted exact molecular-state, spin, field and response laws.

Boundary: Every generated finite molecular state with positive angular and field counts, plus all 267 molecules represented by the 215 NIST SRD 114/115/117 holding groups, every accessible triatomic/hydrocarbon constants page, every officially linked unavailable diatomic page, and the complete 162-page NIST diatomic reference-data table.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: complete-molecular-state-and-angular-carrier__opposed-held-orientation-labels__pairwise-closure-to-EmptyOne-or-positive-excess__positive-response-per-angular-recurrence__positive-induced-response-per-field-act__value-free-complete-magnetic-identity-seal__complete-g-factor-susceptibility-and-unavailable-custody__one-ratio-law-with-depth-independent-repetition. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-molecular-state-and-angular-carrier	answer-only-magnetic-number	A bare number erases molecule, state and angular support.
orientation	opposed-held-orientation-labels	signed-direction-as-proof-number	A signed proof scalar imports negative numbers and loses the generating side.
closure	pairwise-closure-to-EmptyOne-or-positive-excess	assume-every-carrier-has-net-moment	Paired complementary support can close and must not be assigned a numerical zero.
moment	positive-response-per-angular-recurrence	fitted-or-species-g-coefficient	A fitted g-factor lets the measured molecule select the law.
susceptibility	positive-induced-response-per-field-act	continuum-derivative-field-law	A continuum derivative imports an ungenerated field model.
prediction	value-free-complete-magnetic-identity-seal	moment-or-susceptibility-readable-before-seal	Readable targets could select the relation or molecule subset.
record	complete-g-factor-susceptibility-and-unavailable-custody	favorable-or-accessible-molecule-subset	Selecting positive, convenient or currently accessible pages conceals blanks and source failures.
extension	one-ratio-law-with-depth-independent-repetition	species-correction-or-new-field-coefficient	A species residual creates a free parameter for every molecule.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 8a28bc87f19760a2a3bcdeca06c169daacee30c224ae0e3589d6bc9bcc96b29e.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 88351ed82e3b356141418fb4a634017b5c8363c76440ab70669c749056fb68ac.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 4d44bf45950d68536bf8b8524c301aeb44eb1136896de006f932a3141782e040.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: f198ddcdc8e48dd7e488e8d3419a4d2f84a7300140aa7e2fd06f914d8bc70723.

Independent reconstruction. Implementation hash

sha256: c8b12af9bed0f67987a44e08a1e706c96ffd39da79b98e11e8a8336da0232254; independent certificate sha256: f6b3fd7cc82ddcd41ba902568061de46341edb0cd82c7ed58da2f19bcbda77f6; external-validation hash sha256: 96f60ab4dafd5ff8ecde559be7d79d0aaa621b0f47c753d68eb11b4f57bfa346. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-MOLECULAR-MICROWAVE-SPECTRAL-DATABASES-SRD-114-115-117
- NIST-JPCRD-3-609-1974-DIATOMIC-MICROWAVE-SPECTRAL-TABLES

Comparison records:

- NIST-MOLSPEC-PROP-012-TRIATOMIC-P125-T003-R004-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result **True**
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P125-T003-R004-C003:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result **True**
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P125-T003-R004-C004:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result **True**
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P125-T003-R004-C005:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result **True**
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P130-T005-R004-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result **True**
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P130-T005-R004-C003:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result **True**
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P130-T005-R004-C004:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result **True**
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P136-T002-R005-C002: blank-source-cell-structural-EmptyOne; identity and exact post-seal result **True**
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P136-T002-R005-C003:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result **True**
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P138-T003-R003-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result **True**
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P138-T003-R004-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result **True**
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P138-T003-R005-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result **True**
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P138-T003-R006-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result **True**
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P138-T003-R007-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result **True**
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P140-T002-R003-C004:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result **True**
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P140-T002-R003-C005:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result **True**
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P141-T002-R016-C002: blank-source-cell-structural-EmptyOne; identity and exact post-seal result **True**

- NIST-MOLSPEC-PROP-012-TRIATOMIC-P141-T002-R016-C003: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P141-T002-R016-C004: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P141-T002-R016-C005: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P141-T002-R016-C006: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P141-T002-R016-C007: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P141-T002-R016-C008: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P144-T001-R003-C007: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P144-T001-R004-C007: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P144-T001-R005-C007: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P144-T001-R006-C007: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P144-T001-R007-C007: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P148-T002-R009-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P148-T002-R009-C003: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P148-T002-R009-C004: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P148-T002-R009-C005: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P148-T002-R009-C006: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P148-T002-R010-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P148-T002-R010-C003: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P148-T002-R010-C004: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P148-T002-R010-C005: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P148-T002-R010-C006: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P148-T002-R011-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True

- NIST-MOLSPEC-PROP-012-TRIATOMIC-P148-T002-R011-C003: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P148-T002-R011-C004: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P148-T002-R011-C005: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P148-T002-R011-C006: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P149-T002-R010-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P149-T002-R010-C003: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P149-T002-R011-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P149-T002-R011-C003: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P149-T002-R012-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P149-T002-R012-C003: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P151-T011-R004-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P151-T011-R004-C003: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P151-T011-R004-C004: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P151-T011-R005-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P151-T011-R005-C003: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P151-T011-R005-C004: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P151-T011-R006-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P151-T011-R006-C003: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P151-T011-R006-C004: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P151-T011-R007-C002: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P151-T011-R007-C003: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P151-T011-R007-C004: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True

- NIST-MOLSPEC-PROP-012-TRIATOMIC-P151-T011-R008-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P152-T005-R002-C003:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P152-T005-R002-C004:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P152-T005-R002-C005:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P152-T005-R002-C006:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P152-T005-R006-C003:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P152-T005-R006-C004:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P152-T005-R006-C005:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P152-T005-R006-C006:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P152-T005-R009-C003:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P152-T005-R009-C004:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P152-T005-R009-C005:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P152-T005-R009-C006:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P164-T004-R007-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P166-T002-R011-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P166-T002-R011-C003:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P166-T002-R011-C004:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P166-T002-R011-C005: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P166-T002-R011-C006: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P166-T002-R012-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P166-T002-R012-C003: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P166-T002-R012-C004: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True

- NIST-MOLSPEC-PROP-012-TRIATOMIC-P166-T002-R012-C005: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P166-T002-R012-C006: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P168-T004-R003-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P168-T004-R003-C003: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P168-T004-R004-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P168-T004-R005-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P168-T004-R006-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P168-T004-R007-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P173-T006-R011-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P173-T006-R012-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P173-T006-R013-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P173-T006-R014-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-TRIATOMIC-P173-T006-R015-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P183-T001-R017-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P186-T006-R014-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P186-T006-R014-C003: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P186-T006-R014-C004: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P186-T006-R014-C005: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P187-T003-R006-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P187-T003-R007-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P187-T003-R008-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P187-T003-R009-C002: exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True

- NIST-MOLSPEC-PROP-012-HYDROCARBON-P187-T003-R010-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P192-T009-R011-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P192-T009-R011-C003: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P192-T009-R011-C004: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P192-T009-R012-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P192-T009-R013-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P192-T009-R014-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P192-T009-R015-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P192-T011-R014-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P192-T011-R014-C003: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P192-T011-R014-C004: blank-source-cell-structural-EmptyOne; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P192-T011-R015-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P192-T011-R016-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P192-T011-R017-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P192-T011-R018-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P192-T017-R013-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P192-T017-R014-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P192-T017-R015-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P192-T017-R016-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P192-T017-R017-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P197-T001-R014-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P197-T001-R015-C002:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True

- [illegible]

- NIST-MOLSPEC-PROP-012-HYDROCARBON-P198-T011-R014-C003:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P198-T011-R014-C004:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-HYDROCARBON-P198-T011-R014-C005:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P051-L024-V01:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P051-L025-V01:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P052-L026-V01:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P053-L025-MAG:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P061-L045-V01:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P079-L036-V01:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P079-L036-V02:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P079-L053-V01:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P084-L056-V01:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P084-L058-V01:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P090-L050-V01:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P096-L052-V01:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P096-L052-V02:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P098-L039-V01:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P100-L050-V01:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P104-L034-V01:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P104-L034-V02:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P104-L034-V03:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P113-L035-V01:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True

- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P135-L052-V01:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P139-L026-V01:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P146-L046-V01:
exact-positive-molecular-magnetic-magnitude-with-held-source-orientation; identity and exact post-seal result True
- tampered_magnitude_rejected: True
- blank_cell_is_structural_EmptyOne: True
- signed_source_orientation_not_negative_proof_number: True
- complete_174_cell_vector_retained: True
- all_38_blank_cells_retained: True
- complete_diatomic_pdf_vector_retained: True
- quadrupole_chi_frequency_tensors_excluded: True

Falsification boundary: The claim fails if opposing directions require signed proof numbers; if equal support does not close structurally; if a retained molecular moment or susceptibility cannot be expressed as the declared exact positive count ratio; if equal repetition changes susceptibility; if any of the 174 registered cells, 136 printed values, 38 blanks, 215 holding groups, 121 unavailable linked pages or 162 PDF pages is concealed; if a quadrupole chi is relabelled as susceptibility; if targets open before sealing; or if a continuum, fitted or species-specific rule enters.

Measurement receipt: sha256:31a51628edd0b05258827df82b501e3419defac226b9cb76552eef8c8dbc52b9.
Isolation certificate: sha256:6a7f1d4daba2729ebc29be961bf5e1ec1ed0d21cc6390521c7f645b39a6921d7.
Custody certificate: sha256:1b9a1f2690226ad34f7d2bd011d54c4fe5f750bc9f8c387d2d323a2660bf6335.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The quantitative property family derives bond length, dissociation, angle, torsion, dipole, polarizability, ionization, affinity, vibration, rotation, binding, magnetic response, formation energy and their common-carrier vector without per-property fitting. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; balanced support and blank source cells use structural EmptyOne
- no negative, irrational, imaginary, floating or continuum proof value
- no signed direction scalar; source signs remain external held orientations
- no continuum field derivative, differential susceptibility equation or imported magnetic potential
- no measured moment or susceptibility in law, grammar, candidate forcing or prediction
- no fitted g-factor, species coefficient, residual or selected molecule subset
- no nuclear quadrupole chi tensor misclassified as magnetic susceptibility

Immutable identities. Source manifest

sha256:ad9062d2a3abc5b30476d2c680ebee00381d46a198d469f5469bde1962659ab; derivation seal
sha256:5346830af9d861d3a63064259fe092f1aa23dc6fe29092e588b64c915d3e4d8a; engine receipt
sha256:cf1bf113ca5c59ba033a73dc62255bb85b7589e64a1767784326c112129bf1b3 at receipts/engine/model_admitted/SFT-CHEM-MOLECULAR-MAGNETIC-RESPONSE-012-cf1bf113ca5c59ba.json; empirical
hash sha256:a2eeba0561cbe6bd02ff5fb33a9118752eff963bc80cd7657b29f389c852fea4; measurement
receipt sha256:31a51628edd0b05258827df82b501e3419defac226b9cb76552eef8c8dbc52b9.

123. Exact molecular formation-energy relation

Claim identity: SFT-CHEM-MOLECULAR-FORMATION-ENERGY-013

Question and theorem. A molecular formation-energy record is the exact relation between one completely identified product state and the exact composition of its named constituent reference states under a held phase and temperature reference. Unequal endpoints force one held above/below orientation and one exact positive separation; equality is structural EmptyOne. All 2,098 value-free identities seal before the complete NIST vector of 1,485 printed experimental values and 613 blank cells opens.

complete-product-reference-state-carrier__exact-named-constituent-reference-composition__held-product-reference-state-order__exact-positive-state-separation__equality-or-unmeasured-as-distinct-structural-EmptyOne__value-free-complete-formation-identity-seal__complete-values-blanks-orientations-and-reference-custody__one-relation-with-depth-in-dependent-shared-state-extension

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-STATE-RELATION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-CONFIGURATION-ORDER-PATH-011
- SFT-CHEM-BOND-DISSOCIATION-ENERGY-002
- SFT-CHEM-INTERMOLECULAR-BINDING-011

Generated grammar. Generate the literal product of carrier, reference, orientation, magnitude, absence, prediction, record and extension forms; decide all 256 candidates only from admitted exact composition, state order, conservation, path and target-custody laws.

Boundary: The depth-independent exact product/reference relation for every finite named constituent tuple, tested against all 2,098 cells from the complete official CCCBDB formation-energy property surface and the preserved thermodynamic reference-state page. Printed values, blanks and source orientations remain explicit.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-product-reference-state-carrier__exact-named-constituent-reference-composition__held-product-reference-state-order__exact-positive-state-separation__equality-or-unmeasured-as-distinct-structural-EmptyOne__value-free-complete-formation-identity-seal__complete-values-blanks-orientations-and-reference-custody__one-relation-with-depth-independent-shared-state-extension`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	<code>complete-product-reference-state-carrier</code>	<code>answer-only-formation-number</code>	A bare number erases product, phase, reference and condition.
reference	<code>exact-named-constituent-reference-composition</code>	<code>imported-or-fitted-atomic-reference</code>	An imported or fitted reference lets external values select the relation.
orientation	<code>held-product-reference-state-order</code>	<code>signed-formation-proof-number</code>	A signed proof number imports negatives and loses state order.
magnitude	<code>exact-positive-state-separation</code>	<code>species-coefficient-or-target-value</code>	A species coefficient or target value is a fitted parameter.
absence	<code>equality-or-unmeasured-as-distinct-structural-EmptyOne</code>	<code>numerical-zero-for-equality-or-blank</code>	Numerical zero is not an SFT value and a source blank is not a measured equality.
prediction	<code>value-free-complete-formation-identity-seal</code>	<code>formation-values-readable-before-seal</code>	Readable values could select the law, reference or species subset.
record	<code>complete-values-blanks-orientations-and-reference-custody</code>	<code>favorable-species-temperature-or-sign-subset</code>	Selecting only convenient values conceals blanks and opposite orientations.
extension	<code>one-relation-with-depth-independent-shared-state-extension</code>	<code>new-coefficient-per-added-constituent</code>	A new coefficient per constituent destroys zero-parameter closure.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:b109d0ce29eb231a4fcb0602cda3855138226e03d20686cd7e40da9a252c9cc6.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:a5ba3478effed23df2a18be405a80d3ebadedee12a2e55add0dc1f82f1c6429d.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:4ab7413dcf34ec4b1d73eeb0fe876b5bc80faaff4336e3a83e3dcd06e5934f45.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:1e45d5abd9e4473b83a3c57d47f3c8b3c34ae62317cfc472c9295f084c23bfa.

Independent reconstruction. Implementation hash

sha256:d6679d543ebdbd6d539098695cf08948f483c262f43eecea0f70ba571fcec269; independent certificate sha256:a02b3cf7aae3466827e2456271bd48fa0c03ad59269d782aa68919d588cf8530; external-validation hash sha256:bdedec16a14c5e347ec62e63ff0c466e98d1159282e1953ce8c3f7e5dfe74150. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- NIST-CCCBDB-SRD-101-COMPLETE-EXPERIMENTAL-FORMATION-ENERGY
- NIST-CCCBDB-THERMODYNAMIC-REFERENCE-STATES

Comparison records:

- NIST-CCCBDB-PROP-013-ROW-0001-REFERENCE-1: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0001-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0002-REFERENCE-1: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0002-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0003-REFERENCE-1: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0003-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0004-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0004-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0005-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0005-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0006-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0006-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0007-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0007-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0008-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0008-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0009-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0009-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0010-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0010-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0011-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0011-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0012-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0012-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0013-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0013-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0014-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0014-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0015-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0015-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0016-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0016-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0017-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0017-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0018-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0018-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0019-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0019-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0020-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0020-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0021-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0021-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0022-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0022-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0023-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0023-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0024-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0024-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0025-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0025-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0026-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0026-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0027-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0027-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0028-REFERENCE-1: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0028-REFERENCE-2: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0029-REFERENCE-1: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0029-REFERENCE-2: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0030-REFERENCE-1: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0030-REFERENCE-2: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0031-REFERENCE-1: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0031-REFERENCE-2: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0032-REFERENCE-1: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0032-REFERENCE-2: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0033-REFERENCE-1: printed-equality-structural-EmptyOne; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0033-REFERENCE-2: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0034-REFERENCE-1: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0034-REFERENCE-2: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0035-REFERENCE-1: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0035-REFERENCE-2: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0036-REFERENCE-1: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0036-REFERENCE-2: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0037-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0037-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0038-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0038-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0039-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0039-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0040-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0040-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0041-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0041-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0042-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0042-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0043-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0043-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0044-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0044-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0045-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0045-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0046-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0046-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0047-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0047-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0048-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0048-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0049-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0049-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0050-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0050-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0051-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0051-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0052-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0052-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0053-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0053-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0054-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0054-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0055-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0055-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0056-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0056-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0057-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0057-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0058-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0058-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0059-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0059-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0060-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0060-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0061-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0061-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0062-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0062-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0063-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0063-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0064-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0064-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0065-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0065-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0066-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0066-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0067-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0067-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0068-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0068-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0069-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0069-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0070-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0070-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0071-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0071-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0072-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0072-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0073-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0073-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0074-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0074-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0075-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0075-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0076-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0076-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0077-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0077-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0078-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0078-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0079-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0079-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0080-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0080-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0081-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0081-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0082-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0082-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0083-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0083-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0084-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0084-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0085-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0085-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0086-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0086-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0087-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0087-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0088-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0088-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0089-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0089-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0090-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0090-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0091-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0091-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0092-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0092-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0093-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0093-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0094-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0094-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0095-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0095-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0096-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0096-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0097-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0097-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0098-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0098-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0099-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0099-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0100-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0100-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0101-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0101-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0102-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0102-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0103-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0103-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0104-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0104-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0105-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0105-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0106-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0106-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0107-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0107-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0108-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0108-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0109-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0109-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0110-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0110-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0111-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0111-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0112-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0112-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0113-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0113-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0114-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0114-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0115-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0115-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0116-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0116-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0117-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0117-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0118-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0118-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0119-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0119-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0120-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0120-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0121-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0121-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0122-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0122-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0123-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0123-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0124-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0124-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0125-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0125-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0126-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0126-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0127-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0127-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0128-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0128-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0129-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0129-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0130-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0130-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0131-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0131-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0132-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0132-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0133-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0133-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0134-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0134-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0135-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0135-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0136-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0136-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0137-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0137-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0138-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0138-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0139-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0139-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0140-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0140-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0141-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0141-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0142-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0142-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0143-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0143-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0144-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0144-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0145-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0145-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0146-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0146-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0147-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0147-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0148-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0148-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0149-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0149-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0150-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0150-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0151-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0151-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0152-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0152-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0153-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0153-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0154-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0154-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0155-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0155-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0156-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0156-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0157-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0157-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0158-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0158-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0159-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0159-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0160-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0160-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0161-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0161-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0162-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0162-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0163-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0163-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0164-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0164-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0165-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0165-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0166-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0166-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0167-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0167-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0168-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0168-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0169-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0169-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0170-REFERENCE-1: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0170-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0171-REFERENCE-1: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0171-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0172-REFERENCE-1: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0172-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0173-REFERENCE-1: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0173-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0174-REFERENCE-1: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0174-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0175-REFERENCE-1: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0175-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0176-REFERENCE-1: unmeasured-structural-EmptyOne; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0176-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0177-REFERENCE-1: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0177-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0178-REFERENCE-1: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0178-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0179-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0179-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0180-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0180-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0181-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0181-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0182-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0182-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0183-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0183-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0184-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0184-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0185-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0185-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0186-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0186-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0187-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0187-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0188-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0188-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0189-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0189-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0190-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0190-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0191-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0191-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0192-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0192-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0193-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0193-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0194-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0194-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0195-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0195-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0196-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0196-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0197-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0197-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0198-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0198-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0199-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0199-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0200-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0200-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0201-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0201-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0202-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0202-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0203-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0203-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0204-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0204-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0205-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0205-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0206-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0206-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0207-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0207-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0208-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0208-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0209-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0209-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0210-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0210-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0211-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0211-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0212-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0212-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0213-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0213-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0214-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0214-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0215-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0215-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0216-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0216-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0217-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0217-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0218-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0218-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0219-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0219-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0220-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0220-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0221-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0221-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0222-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0222-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0223-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0223-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0224-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0224-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0225-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0225-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0226-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0226-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0227-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0227-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0228-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0228-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0229-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0229-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0230-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0230-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0231-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0231-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0232-REFERENCE-1: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0232-REFERENCE-2: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0233-REFERENCE-1: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0233-REFERENCE-2: printed-equality-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0234-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0234-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0235-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0235-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0236-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0236-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0237-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0237-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0238-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0238-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0239-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0239-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0240-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0240-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0241-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0241-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0242-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0242-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0243-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0243-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0244-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0244-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0245-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0245-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0246-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0246-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0247-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0247-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0248-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0248-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0249-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0249-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0250-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0250-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0251-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0251-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0252-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0252-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0253-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0253-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0254-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0254-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0255-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0255-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0256-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0256-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0257-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0257-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0258-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0258-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0259-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0259-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0260-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0260-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0261-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0261-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0262-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0262-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0263-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0263-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0264-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0264-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0265-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0265-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0266-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0266-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0267-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0267-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0268-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0268-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0269-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0269-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0270-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0270-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0271-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0271-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0272-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0272-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0273-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0273-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0274-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0274-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0275-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0275-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0276-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0276-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0277-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0277-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0278-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0278-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0279-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0279-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0280-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0280-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0281-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0281-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0282-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0282-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0283-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0283-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0284-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0284-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0285-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0285-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0286-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0286-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0287-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0287-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0288-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0288-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0289-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0289-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0290-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0290-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0291-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0291-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0292-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0292-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0293-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0293-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0294-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0294-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0295-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0295-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0296-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0296-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0297-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0297-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0298-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0298-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0299-REFERENCE-1: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0299-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0300-REFERENCE-1: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0300-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0301-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0301-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0302-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0302-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0303-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0303-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0304-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0304-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0305-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0305-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0306-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0306-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0307-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0307-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0308-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0308-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0309-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0309-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0310-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0310-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0311-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0311-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0312-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0312-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0313-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0313-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0314-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0314-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0315-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0315-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0316-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0316-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0317-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0317-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0318-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0318-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0319-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0319-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0320-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0320-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0321-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0321-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0322-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0322-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0323-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0323-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0324-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0324-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0325-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0325-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0326-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0326-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0327-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0327-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0328-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0328-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0329-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0329-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0330-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0330-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0331-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0331-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0332-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0332-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0333-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0333-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0334-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0334-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0335-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0335-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0336-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0336-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0337-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0337-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0338-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0338-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0339-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0339-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0340-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0340-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0341-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0341-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0342-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0342-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0343-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0343-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0344-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0344-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0345-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0345-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0346-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0346-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0347-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0347-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0348-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0348-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0349-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0349-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0350-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0350-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0351-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0351-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0352-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0352-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0353-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0353-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0354-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0354-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0355-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0355-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0356-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0356-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0357-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0357-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0358-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0358-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0359-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0359-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0360-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0360-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0361-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0361-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0362-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0362-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0363-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0363-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0364-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0364-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0365-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0365-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0366-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0366-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0367-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0367-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0368-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0368-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0369-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0369-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0370-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0370-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0371-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0371-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0372-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0372-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0373-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0373-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0374-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0374-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0375-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0375-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0376-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0376-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0377-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0377-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0378-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0378-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0379-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0379-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0380-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0380-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0381-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0381-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0382-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0382-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0383-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0383-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0384-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0384-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0385-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0385-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0386-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0386-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0387-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0387-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0388-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0388-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0389-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0389-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0390-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0390-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0391-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0391-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0392-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0392-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0393-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0393-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0394-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0394-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0395-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0395-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0396-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0396-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0397-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0397-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0398-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0398-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0399-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0399-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0400-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0400-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0401-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0401-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0402-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0402-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0403-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0403-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0404-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0404-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0405-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0405-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0406-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0406-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0407-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0407-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0408-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0408-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0409-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0409-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0410-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0410-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0411-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0411-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0412-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0412-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0413-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0413-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0414-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0414-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0415-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0415-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0416-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0416-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0417-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0417-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0418-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0418-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0419-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0419-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0420-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0420-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0421-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0421-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0422-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0422-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0423-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0423-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0424-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0424-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0425-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0425-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0426-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0426-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0427-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0427-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0428-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0428-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0429-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0429-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0430-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0430-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0431-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0431-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0432-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0432-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0433-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0433-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0434-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0434-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0435-REFERENCE-1: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0435-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0436-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0436-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0437-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0437-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0438-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0438-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0439-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0439-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0440-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0440-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0441-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0441-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0442-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0442-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0443-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0443-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0444-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0444-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0445-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0445-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0446-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0446-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0447-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0447-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0448-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0448-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0449-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0449-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0450-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0450-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0451-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0451-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0452-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0452-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0453-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0453-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0454-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0454-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0455-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0455-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0456-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0456-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0457-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0457-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0458-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0458-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0459-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0459-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0460-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0460-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0461-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0461-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0462-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0462-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0463-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0463-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0464-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0464-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0465-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0465-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0466-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0466-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0467-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0467-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0468-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0468-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0469-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0469-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0470-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0470-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0471-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0471-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0472-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0472-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0473-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0473-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0474-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0474-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0475-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0475-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0476-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0476-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0477-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0477-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0478-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0478-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0479-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0479-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0480-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0480-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0481-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0481-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0482-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0482-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0483-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0483-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0484-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0484-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0485-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0485-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0486-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0486-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0487-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0487-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0488-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0488-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0489-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0489-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0490-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0490-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0491-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0491-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0492-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0492-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0493-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0493-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0494-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0494-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0495-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0495-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0496-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0496-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0497-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0497-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0498-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0498-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0499-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0499-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0500-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0500-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0501-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0501-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0502-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0502-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0503-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0503-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0504-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0504-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0505-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0505-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0506-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0506-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0507-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0507-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0508-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0508-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0509-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0509-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0510-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0510-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0511-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0511-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0512-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0512-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0513-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0513-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0514-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0514-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0515-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0515-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0516-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0516-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0517-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0517-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0518-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0518-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0519-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0519-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0520-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0520-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0521-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0521-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0522-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0522-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0523-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0523-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0524-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0524-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0525-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0525-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0526-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0526-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0527-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0527-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0528-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0528-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0529-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0529-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0530-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0530-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0531-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0531-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0532-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0532-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0533-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0533-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0534-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0534-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0535-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0535-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0536-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0536-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0537-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0537-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0538-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0538-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0539-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0539-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0540-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0540-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0541-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0541-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0542-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0542-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0543-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0543-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0544-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0544-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0545-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0545-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0546-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0546-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0547-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0547-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0548-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0548-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0549-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0549-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0550-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0550-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0551-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0551-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0552-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0552-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0553-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0553-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0554-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0554-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0555-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0555-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0556-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0556-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0557-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0557-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0558-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0558-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0559-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0559-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0560-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0560-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0561-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0561-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0562-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0562-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0563-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0563-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0564-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0564-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0565-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0565-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0566-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0566-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0567-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0567-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0568-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0568-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0569-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0569-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0570-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0570-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0571-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0571-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0572-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0572-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0573-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0573-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0574-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0574-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0575-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0575-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0576-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0576-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0577-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0577-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0578-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0578-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0579-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0579-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0580-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0580-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0581-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0581-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0582-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0582-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0583-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0583-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0584-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0584-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0585-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0585-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0586-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0586-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0587-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0587-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0588-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0588-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0589-REFERENCE-1: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0589-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0590-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0590-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0591-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0591-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0592-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0592-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0593-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0593-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0594-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0594-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0595-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0595-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0596-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0596-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0597-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0597-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0598-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0598-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0599-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0599-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0600-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0600-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0601-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0601-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0602-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0602-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0603-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0603-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0604-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0604-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0605-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0605-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0606-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0606-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0607-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0607-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0608-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0608-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0609-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0609-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0610-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0610-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0611-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0611-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0612-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0612-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0613-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0613-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0614-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0614-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0615-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0615-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0616-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0616-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0617-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0617-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0618-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0618-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0619-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0619-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0620-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0620-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0621-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0621-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0622-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0622-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0623-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0623-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0624-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0624-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0625-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0625-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0626-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0626-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0627-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0627-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0628-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0628-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0629-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0629-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0630-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0630-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0631-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0631-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0632-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0632-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0633-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0633-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0634-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0634-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0635-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0635-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0636-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0636-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0637-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0637-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0638-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0638-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0639-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0639-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0640-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0640-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0641-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0641-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0642-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0642-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0643-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0643-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0644-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0644-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0645-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0645-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0646-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0646-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0647-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0647-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0648-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0648-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0649-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0649-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0650-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0650-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0651-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0651-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0652-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0652-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0653-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0653-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0654-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0654-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0655-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0655-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0656-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0656-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0657-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0657-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0658-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0658-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0659-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0659-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0660-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0660-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0661-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0661-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0662-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0662-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0663-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0663-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0664-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0664-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0665-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0665-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0666-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0666-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0667-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0667-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0668-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0668-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0669-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0669-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0670-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0670-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0671-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0671-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0672-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0672-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0673-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0673-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0674-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0674-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0675-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0675-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0676-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0676-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0677-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0677-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0678-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0678-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0679-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0679-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0680-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0680-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0681-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0681-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0682-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0682-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0683-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0683-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0684-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0684-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0685-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0685-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0686-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0686-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0687-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0687-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0688-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0688-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0689-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0689-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0690-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0690-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0691-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0691-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0692-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0692-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0693-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0693-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0694-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0694-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0695-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0695-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0696-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0696-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0697-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0697-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0698-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0698-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0699-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0699-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0700-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0700-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0701-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0701-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0702-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0702-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0703-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0703-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0704-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0704-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0705-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0705-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0706-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0706-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0707-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0707-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0708-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0708-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0709-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0709-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0710-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0710-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0711-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0711-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0712-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0712-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0713-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0713-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0714-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0714-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0715-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0715-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0716-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0716-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0717-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0717-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0718-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0718-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0719-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0719-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0720-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0720-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0721-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0721-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0722-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0722-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0723-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0723-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0724-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0724-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0725-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0725-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0726-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0726-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0727-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0727-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0728-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0728-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0729-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0729-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0730-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0730-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0731-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0731-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0732-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0732-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0733-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0733-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0734-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0734-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0735-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0735-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0736-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0736-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0737-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0737-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0738-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0738-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0739-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0739-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0740-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0740-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0741-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0741-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0742-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0742-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0743-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0743-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0744-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0744-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0745-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0745-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0746-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0746-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0747-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0747-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0748-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0748-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0749-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0749-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0750-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0750-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0751-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0751-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0752-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0752-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0753-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0753-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0754-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0754-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0755-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0755-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0756-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0756-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0757-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0757-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0758-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0758-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0759-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0759-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0760-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0760-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0761-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0761-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0762-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0762-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0763-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0763-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0764-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0764-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0765-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0765-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0766-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0766-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0767-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0767-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0768-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0768-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0769-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0769-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0770-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0770-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0771-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0771-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0772-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0772-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0773-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0773-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0774-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0774-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0775-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0775-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0776-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0776-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0777-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0777-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0778-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0778-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0779-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0779-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0780-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0780-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0781-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0781-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0782-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0782-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0783-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0783-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0784-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0784-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0785-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0785-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0786-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0786-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0787-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0787-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0788-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0788-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0789-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0789-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0790-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0790-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0791-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0791-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0792-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0792-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0793-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0793-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0794-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0794-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0795-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0795-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0796-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0796-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0797-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0797-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0798-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0798-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0799-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0799-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0800-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0800-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0801-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0801-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0802-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0802-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0803-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0803-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0804-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0804-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0805-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0805-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0806-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0806-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0807-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0807-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0808-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0808-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0809-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0809-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0810-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0810-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0811-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0811-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0812-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0812-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0813-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0813-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0814-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0814-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0815-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0815-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0816-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0816-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0817-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0817-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0818-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0818-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0819-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0819-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0820-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0820-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0821-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0821-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0822-REFERENCE-1: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0822-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0823-REFERENCE-1: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0823-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0824-REFERENCE-1: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0824-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0825-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0825-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0826-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0826-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0827-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0827-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0828-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0828-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0829-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0829-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0830-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0830-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0831-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0831-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0832-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0832-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0833-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0833-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0834-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0834-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0835-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0835-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0836-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0836-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0837-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0837-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0838-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0838-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0839-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0839-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0840-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0840-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0841-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0841-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0842-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0842-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0843-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0843-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0844-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0844-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0845-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0845-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0846-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0846-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0847-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0847-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0848-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0848-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0849-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0849-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0850-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0850-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0851-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0851-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0852-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0852-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0853-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0853-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0854-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0854-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0855-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0855-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0856-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0856-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0857-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0857-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0858-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0858-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0859-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0859-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0860-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0860-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0861-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0861-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0862-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0862-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0863-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0863-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0864-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0864-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0865-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0865-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0866-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0866-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0867-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0867-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0868-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0868-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0869-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0869-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0870-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0870-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0871-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0871-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0872-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0872-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0873-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0873-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0874-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0874-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0875-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0875-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0876-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0876-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0877-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0877-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0878-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0878-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0879-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0879-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0880-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0880-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0881-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0881-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0882-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0882-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0883-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0883-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0884-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0884-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0885-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0885-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0886-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0886-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0887-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0887-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0888-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0888-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0889-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0889-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0890-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0890-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0891-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0891-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0892-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0892-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0893-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0893-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0894-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0894-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0895-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0895-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0896-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0896-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0897-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0897-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0898-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0898-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0899-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0899-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0900-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0900-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0901-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0901-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0902-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0902-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0903-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0903-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0904-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0904-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0905-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0905-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0906-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0906-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0907-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0907-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0908-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0908-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0909-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0909-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0910-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0910-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0911-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0911-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0912-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0912-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0913-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0913-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0914-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0914-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0915-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0915-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0916-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0916-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0917-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0917-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0918-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0918-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0919-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0919-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0920-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0920-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0921-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0921-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0922-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0922-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0923-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0923-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0924-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0924-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0925-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0925-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0926-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0926-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0927-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0927-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0928-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0928-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0929-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0929-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0930-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0930-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0931-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0931-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0932-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0932-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0933-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0933-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0934-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0934-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0935-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0935-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0936-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0936-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0937-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0937-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0938-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0938-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0939-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0939-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0940-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0940-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0941-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0941-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0942-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0942-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0943-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0943-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0944-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0944-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0945-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0945-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0946-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0946-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0947-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0947-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0948-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0948-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0949-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0949-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0950-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0950-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0951-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0951-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0952-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0952-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0953-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0953-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0954-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0954-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0955-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0955-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0956-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0956-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0957-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0957-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0958-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0958-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0959-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0959-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0960-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0960-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0961-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0961-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0962-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0962-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0963-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0963-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0964-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0964-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0965-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0965-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0966-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0966-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0967-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0967-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0968-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0968-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0969-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0969-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0970-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0970-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0971-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0971-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0972-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0972-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0973-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0973-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0974-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0974-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0975-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0975-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0976-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0976-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0977-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0977-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0978-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0978-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0979-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0979-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0980-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0980-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0981-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0981-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0982-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0982-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0983-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0983-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0984-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0984-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0985-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0985-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0986-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0986-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0987-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0987-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0988-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0988-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0989-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0989-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0990-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-0990-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0991-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0991-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0992-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0992-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0993-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0993-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0994-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0994-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0995-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0995-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0996-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0996-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0997-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0997-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0998-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0998-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0999-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-0999-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1000-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1000-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1001-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-1001-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1002-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1002-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1003-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1003-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1004-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1004-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1005-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1005-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1006-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1006-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1007-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1007-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1008-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1008-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1009-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1009-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1010-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1010-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1011-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1011-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1012-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-1012-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1013-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1013-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1014-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1014-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1015-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1015-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1016-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1016-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1017-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1017-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1018-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1018-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1019-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1019-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1020-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1020-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1021-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1021-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1022-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1022-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1023-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-1023-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1024-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1024-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1025-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1025-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1026-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1026-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1027-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1027-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1028-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1028-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1029-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1029-REFERENCE-2: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1030-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1030-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1031-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1031-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1032-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1032-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1033-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1033-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1034-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-1034-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1035-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1035-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1036-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1036-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1037-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1037-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1038-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1038-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1039-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1039-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1040-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1040-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1041-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1041-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1042-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1042-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1043-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1043-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1044-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1044-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1045-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True

- NIST-CCCBDB-PROP-013-ROW-1045-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1046-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1046-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1047-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1047-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1048-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1048-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1049-REFERENCE-1: exact-positive-formation-separation; identity and exact post-seal result True
- NIST-CCCBDB-PROP-013-ROW-1049-REFERENCE-2: unmeasured-structural-EmptyOne; identity and exact post-seal result True
- tampered_magnitude_rejected: True
- blank_cell_is_structural_EmptyOne: True
- printed_external_equality_is_structural_EmptyOne: True
- signed_source_orientations_are_never_negative_proof_values: True
- complete_2098_cell_vector_retained: True
- all_613_blank_cells_retained: True
- all_22_printed_equalities_retained: True
- both_temperature_reference_axes_retained: True

Falsification boundary: The claim fails if exact product/reference state order does not force the stated held orientation and positive separation; if shared extension changes the relation; if equality requires numerical zero; if any of 2,098 cells, 1,485 values, 613 blanks, 756 below, 707 above, 22 equal records, 2,186 listed species, 83 unreturned compositions or the complete reference-state page is omitted; if any target opens before seal; or if a measured value, imported convention, fitted reference or species coefficient enters the law.

Measurement receipt: sha256:709fe2e224a6e6b8846d0c897e61b24be140d0dd75c83319a11f4e3832cb444b.

Isolation certificate: sha256:26d0eb21e23dd0df3adc29d468f1d650671a68fdb7d6cb0b32b00b5c4797c6d1.

Custody certificate: sha256:d078e7511bbe53af75e9ad4295d1029f96f2a19f3c4c158f5948367a395917d6.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The quantitative property family derives bond length, dissociation, angle, torsion, dipole, polarizability, ionization, affinity, vibration, rotation, binding, magnetic response, formation energy and their common-carrier vector without per-property fitting. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; equality and missing source measurement are distinct structural EmptyOne records

- no negative, irrational, imaginary, floating, signed or continuum SFT proof value
- no measured formation energy or source orientation in the law, grammar, forcing or prediction
- no imported thermodynamic convention value, fitted atomic reference or species coefficient
- no selected molecule, temperature axis, sign class, favorable value or completed-row subset
- no deletion of blanks, exact source equalities, unreturned compositions or reference-state custody

Immutable identities. Source manifest

sha256:e732580540799d09622f4a0c494e786f1d4f53c408ef835b6f560119703727fb; derivation seal
 sha256:d18dba63b67b4597e4bbbd28ae854f3602f1e6ba133e1e7e7a97dc19da4b136e; engine receipt
 sha256:f85de5707fd9bebc25ad8fc8dfdcbb8e927bef17fb3c2227ee37a11fbfb5cc at receipts/engine/
 model_admitted/SFT-CHEM-MOLECULAR-FORMATION-ENERGY-013-f85de5707fd9bebc.json; empirical
 hash sha256:89c993978eb40153c58015419f540eac4d4c0e668f9c0f51632a413a94a3252e; measurement
 receipt sha256:709fe2e224a6e6b8846d0c897e61b24be140d0dd75c83319a11f4e3832cb444b.

124. Complete zero-parameter cross-property molecular vector

Claim identity: SFT-CHEM-CROSS-PROPERTY-MOLECULAR-VECTOR-014

Question and theorem. One complete retained structural molecular carrier supplies every applicable property projection through the already admitted PROP-001 through PROP-013 relations. Each result remains attached to its named generating law and no property receives a fitted coefficient or target-derived carrier. All 9,025 source identities across all 13 property families seal before withheld target payloads or hashes open; 676 exact carriers support multiple property families and retain all 6,676 overlapping rows.

one-complete-structural-carrier__complete-applicable-property-support__named-admitted-relation-projections__zero-parameter-shared-carrier-projection__structural-EmptyOne-and-unjoined-custody-retained__complete-value-free-identity-seal__complete-13-family-source-row-custody__append-only-depth-independent-projection-extension

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-CHEM-EQUILIBRIUM-BOND-LENGTH-001
- SFT-CHEM-BOND-DISSOCIATION-ENERGY-002
- SFT-CHEM-MOLECULAR-BOND-ANGLE-003
- SFT-CHEM-DIHEDRAL-TORSIONAL-STATE-004
- SFT-CHEM-MOLECULAR-DIPOLE-MAGNITUDE-005
- SFT-CHEM-MOLECULAR-POLARIZABILITY-006
- SFT-CHEM-MOLECULAR-IONIZATION-ENERGY-007
- SFT-CHEM-MOLECULAR-ELECTRON-AFFINITY-008
- SFT-CHEM-VIBRATIONAL-FREQUENCY-009
- SFT-CHEM-ROTATIONAL-CONSTANT-010

- SFT-CHEM-INTERMOLECULAR-BINDING-011
- SFT-CHEM-MOLECULAR-MAGNETIC-RESPONSE-012
- SFT-CHEM-MOLECULAR-FORMATION-ENERGY-013

Generated grammar. Generate the literal product of carrier, support, projection, parameter, absence, prediction, record and extension forms; decide all 256 candidates from admitted property relations, exact compositional laws and the complete value-free cross-property identity surface.

Boundary: The depth-independent append-only projection law for every finite exact structural carrier, tested against all 9,025 registered PROP-001 through PROP-013 source rows, all 1,104 carriers, all 676 multi-property carriers, all 6,676 overlap rows and every explicit nonjoined or bound-composite custody row.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `one-complete-structural-carrier__complete-applicable-property-support__named-admitted-relation-projections__zero-parameter-shared-carrier-projection__structural-EmptyOne-and-unjoined-custody-retained__complete-value-free-identity-seal__complete-13-family-source-row-custody__append-only-depth-independent-projection-extension`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-complete-structural-carrier	separate-carrier-per-property	Separate carriers permit property-specific identity drift.
support	complete-applicable-property-support	selected-or-favorable-property-subset	A selected subset can hide absent or disagreeing properties.
projection	named-admitted-relation-projections	answer-only-property-values	Bare answers erase the relation that generated each result.
parameter	zero-parameter-shared-carrier-projection	per-property-fit-or-correction	A separate coefficient lets each target select its output.
absence	structural-EmptyOne-and-unjoined-custody-retained	delete-unmeasured-or-inapplicable-row	Deletion makes the vector appear more complete than the source.
prediction	complete-value-free-identity-seal	target-payload-or-hash-readable-before-seal	Target content or even its hash can select the carrier or subset.
record	complete-13-family-source-row-custody	single-species-showcase	A showcase species does not establish complete cross-property custody.
extension	append-only-depth-independent-projection-extension	recompute-or-refit-existing-properties	Refitting after a new property destroys prior invariance.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: dae2850b94b5c30d058d20b41d57c3a2edd53206f1f6f481d7543df62c7ef7da.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 0f2cd4b0548d46a0f71345fbde05a3f6b1f23e0b648cf4c6947872f5fef08350.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: a6a0630b90cac830af3a457c66e809c22b90845bef8f153008231429707cb570.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:6990e31fce589fa467850fdb670822bc8e69723498f8f0c6f3b23c01382cb6d3.

Independent reconstruction. Implementation hash

sha256:5ef8d40ff0804ad13b71fa5ca52256852f825f9f082f6dba604083a427a2476e; independent certificate sha256:00db245c1a6e63e213071a9ab5c6364a423d7f8042be629d14efc7171f6a741a; external-validation hash sha256:afb479a6cb3e32fcde8a4f0b5ff11b5d67902f6ac36f526e58dde5027f343a6e. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- SFT-V3-ADMITTED-PROP-001
- SFT-V3-ADMITTED-PROP-002
- SFT-V3-ADMITTED-PROP-003
- SFT-V3-ADMITTED-PROP-004
- SFT-V3-ADMITTED-PROP-005
- SFT-V3-ADMITTED-PROP-006
- SFT-V3-ADMITTED-PROP-007
- SFT-V3-ADMITTED-PROP-008
- SFT-V3-ADMITTED-PROP-009
- SFT-V3-ADMITTED-PROP-010
- SFT-V3-ADMITTED-PROP-011
- SFT-V3-ADMITTED-PROP-012
- SFT-V3-ADMITTED-PROP-013

Comparison records:

- SFT-CHEM-PROP-014-PROP-001-ROW-0001: PROP-001 on exact-formula:H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-001-ROW-0002: PROP-001 on exact-formula:D2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-002-ROW-0001: PROP-002 on exact-formula:H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-002-ROW-0002: PROP-002 on exact-formula:H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-002-ROW-0003: PROP-002 on exact-formula:H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-002-ROW-0004: PROP-002 on exact-formula:H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-002-ROW-0005: PROP-002 on exact-formula:D2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-002-ROW-0006: PROP-002 on exact-formula:D2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-002-ROW-0007: PROP-002 on exact-formula:D2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-002-ROW-0008: PROP-002 on exact-formula:D2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-003-ROW-0001: PROP-003 on exact-formula:BF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-003-ROW-0002: PROP-003 on exact-formula:XeF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-003-ROW-0003: PROP-003 on exact-formula:XeF4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-003-ROW-0004: PROP-003 on exact-formula:XeF4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-004-ROW-0001: PROP-004 on source-species-label:ethanol; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-004-ROW-0002: PROP-004 on source-species-label:ethanol; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-004-ROW-0003: PROP-004 on source-species-label:ethanol; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-004-ROW-0004: PROP-004 on source-species-label:ethanol; post-seal payload identity True

- [illegible]

- SFT-CHEM-PROP-014-PROP-004-ROW-0043: PROP-004 on source-species-label:ethanol; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-004-ROW-0044: PROP-004 on source-species-label:ethanol; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-004-ROW-0045: PROP-004 on source-species-label:ethanol; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-004-ROW-0046: PROP-004 on source-species-label:ethanol; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-004-ROW-0047: PROP-004 on source-species-label:ethanol; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-004-ROW-0048: PROP-004 on source-species-label:ethanol; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-004-ROW-0049: PROP-004 on source-species-label:ethanol; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-004-ROW-0050: PROP-004 on source-species-label:ethanol; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-005-ROW-0001: PROP-005 on exact-formula:H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-005-ROW-0002: PROP-005 on exact-formula:D2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-005-ROW-0003: PROP-005 on exact-formula:H2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-005-ROW-0004: PROP-005 on exact-formula:H2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-005-ROW-0005: PROP-005 on exact-formula:D2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-005-ROW-0006: PROP-005 on exact-formula:D2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-005-ROW-0007: PROP-005 on exact-formula:HDO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-005-ROW-0008: PROP-005 on exact-formula:HDO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-005-ROW-0009: PROP-005 on exact-formula:HDO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0001: PROP-006 on exact-formula:D2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0002: PROP-006 on exact-formula:HD; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0003: PROP-006 on exact-formula:H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0004: PROP-006 on exact-formula:H2+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0005: PROP-006 on exact-formula:N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0006: PROP-006 on exact-formula:CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0007: PROP-006 on exact-formula:NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0008: PROP-006 on exact-formula:O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0009: PROP-006 on exact-formula:HF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0010: PROP-006 on exact-formula:F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0011: PROP-006 on exact-formula:NaLi; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0012: PROP-006 on exact-formula:MgO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0013: PROP-006 on exact-formula:HCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0014: PROP-006 on exact-formula:Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0015: PROP-006 on exact-formula:CaO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0016: PROP-006 on exact-formula:HBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0017: PROP-006 on exact-formula:Br2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0018: PROP-006 on exact-formula:HI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0019: PROP-006 on exact-formula:BeH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0020: PROP-006 on exact-formula:HCN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0021: PROP-006 on exact-formula:H2O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-006-ROW-0022: PROP-006 on exact-formula:N2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0023: PROP-006 on exact-formula:CO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0024: PROP-006 on exact-formula:NO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0025: PROP-006 on exact-formula:O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0026: PROP-006 on exact-formula:FCN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0027: PROP-006 on exact-formula:H2S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0028: PROP-006 on exact-formula:OCS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0029: PROP-006 on exact-formula:SO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0030: PROP-006 on exact-formula:CS2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0031: PROP-006 on exact-formula:H2Se; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0032: PROP-006 on exact-formula:CSe2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0033: PROP-006 on exact-formula:BrCN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0034: PROP-006 on exact-formula:C2H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0035: PROP-006 on exact-formula:NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0036: PROP-006 on exact-formula:C2N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0037: PROP-006 on exact-formula:H2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0038: PROP-006 on exact-formula:BF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0039: PROP-006 on exact-formula:NF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0040: PROP-006 on exact-formula:PH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0041: PROP-006 on exact-formula:PF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0042: PROP-006 on exact-formula:P4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0043: PROP-006 on exact-formula:SO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0044: PROP-006 on exact-formula:F2SO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0045: PROP-006 on exact-formula:HCCCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0046: PROP-006 on exact-formula:ClF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0047: PROP-006 on exact-formula:CCl2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0048: PROP-006 on exact-formula:BCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0049: PROP-006 on exact-formula:Cl2CS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0050: PROP-006 on exact-formula:PCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0051: PROP-006 on exact-formula:AsH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0052: PROP-006 on exact-formula:AsCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0053: PROP-006 on exact-formula:As4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0054: PROP-006 on exact-formula:BBr3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0055: PROP-006 on exact-formula:PBr3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0056: PROP-006 on exact-formula:CH4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0057: PROP-006 on exact-formula:CH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0058: PROP-006 on exact-formula:HCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0059: PROP-006 on exact-formula:CH3F; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-006-ROW-0060: PROP-006 on exact-formula:CH₂F₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0061: PROP-006 on exact-formula:CHF₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0062: PROP-006 on exact-formula:CF₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0063: PROP-006 on exact-formula:SiH₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0064: PROP-006 on exact-formula:SiF₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0065: PROP-006 on exact-formula:CH₃Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0066: PROP-006 on exact-formula:SiH₃Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0067: PROP-006 on exact-formula:CH₂FCI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0068: PROP-006 on exact-formula:CH₂Cl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0069: PROP-006 on exact-formula:CHF₂Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0070: PROP-006 on exact-formula:CHFCl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0071: PROP-006 on exact-formula:CF₃Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0072: PROP-006 on exact-formula:CHCl₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0073: PROP-006 on exact-formula:CF₂Cl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0074: PROP-006 on exact-formula:CFCl₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0075: PROP-006 on exact-formula:CCl₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0076: PROP-006 on exact-formula:SiCl₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0077: PROP-006 on exact-formula:TiCl₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0078: PROP-006 on exact-formula:GeH₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0079: PROP-006 on exact-formula:GeF₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0080: PROP-006 on exact-formula:GeCl₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0081: PROP-006 on exact-formula:CH₃Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0082: PROP-006 on exact-formula:CF₃Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0083: PROP-006 on exact-formula:CH₂Br₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0084: PROP-006 on exact-formula:CHBr₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0085: PROP-006 on exact-formula:CH₃I; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0086: PROP-006 on exact-formula:C₂H₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0087: PROP-006 on exact-formula:C₄H₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0088: PROP-006 on exact-formula:N₂H₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0089: PROP-006 on exact-formula:CH₃CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0090: PROP-006 on exact-formula:CH₃OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0091: PROP-006 on exact-formula:CHONH₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0092: PROP-006 on exact-formula:N₂O₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0093: PROP-006 on exact-formula:PF₅; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0094: PROP-006 on exact-formula:CH₃SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0095: PROP-006 on exact-formula:C₂H₃Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0096: PROP-006 on exact-formula:CH₂CCl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0097: PROP-006 on exact-formula:CHClCHCl; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-006-ROW-0098: PROP-006 on exact-formula:CHClCHCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0099: PROP-006 on exact-formula:CHClCCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0100: PROP-006 on exact-formula:C2H3Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0101: PROP-006 on exact-formula:CH2CCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0102: PROP-006 on exact-formula:CH3CCH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0103: PROP-006 on exact-formula:CH3NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0104: PROP-006 on exact-formula:C3H2N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0105: PROP-006 on exact-formula:CH3CHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0106: PROP-006 on exact-formula:C2H4O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0107: PROP-006 on exact-formula:CH3NO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0108: PROP-006 on exact-formula:C2H4S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0109: PROP-006 on exact-formula:SF6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0110: PROP-006 on exact-formula:CCl3CHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0111: PROP-006 on exact-formula:SeF6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0112: PROP-006 on exact-formula:B2H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0113: PROP-006 on exact-formula:C2H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0114: PROP-006 on exact-formula:CH2CHCHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0115: PROP-006 on exact-formula:CH3OCHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0116: PROP-006 on exact-formula:CH3COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0117: PROP-006 on exact-formula:C2H5F; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0118: PROP-006 on exact-formula:CH3CF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0119: PROP-006 on exact-formula:CF3CH2F; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0120: PROP-006 on exact-formula:CF3CHF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0121: PROP-006 on exact-formula:C2F6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0122: PROP-006 on exact-formula:Si2H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0123: PROP-006 on exact-formula:CH3CH2Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0124: PROP-006 on exact-formula:CH2FCH2Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0125: PROP-006 on exact-formula:CH3CHCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0126: PROP-006 on exact-formula:CH2ClCH2Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0127: PROP-006 on exact-formula:CH2ClCHCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0128: PROP-006 on exact-formula:CH3CCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0129: PROP-006 on exact-formula:Al2Cl6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0130: PROP-006 on exact-formula:CHCl2CHF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0131: PROP-006 on exact-formula:C2H5Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0132: PROP-006 on exact-formula:C3H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0133: PROP-006 on exact-formula:CH2CHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0134: PROP-006 on exact-formula:C2H5CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0135: PROP-006 on exact-formula:C3H4N2; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-006-ROW-0136: PROP-006 on exact-formula:C3H4N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0137: PROP-006 on exact-formula:CH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0138: PROP-006 on exact-formula:CH3CH2OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0139: PROP-006 on exact-formula:CH3CONH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0140: PROP-006 on exact-formula:C4H4O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0141: PROP-006 on exact-formula:CH3SCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0142: PROP-006 on exact-formula:CH3CH2SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0143: PROP-006 on exact-formula:C4H4S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0144: PROP-006 on exact-formula:C3H5Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0145: PROP-006 on exact-formula:CH2ClCH2OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0146: PROP-006 on exact-formula:CH2CHCHCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0147: PROP-006 on exact-formula:CHCCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0148: PROP-006 on exact-formula:CH3CCCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0149: PROP-006 on exact-formula:CH3NHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0150: PROP-006 on exact-formula:CH3CH2NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0151: PROP-006 on exact-formula:C4H5N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0152: PROP-006 on exact-formula:C4H5N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0153: PROP-006 on exact-formula:CH3CH2CHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0154: PROP-006 on exact-formula:CH3COCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0155: PROP-006 on exact-formula:C2H6O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0156: PROP-006 on exact-formula:C4H4O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0157: PROP-006 on exact-formula:CH3SOCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0158: PROP-006 on exact-formula:C3H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0159: PROP-006 on exact-formula:C5H5N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0160: PROP-006 on exact-formula:CHOCHCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0161: PROP-006 on exact-formula:HCOOC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0162: PROP-006 on exact-formula:CH3CH2COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0163: PROP-006 on exact-formula:CH3COOCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0164: PROP-006 on exact-formula:C3F8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0165: PROP-006 on exact-formula:C2H6O2S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0166: PROP-006 on exact-formula:CH2ClCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0167: PROP-006 on exact-formula:CH3CHBrCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0168: PROP-006 on exact-formula:CH3CH2CH2Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0169: PROP-006 on exact-formula:CH2CHCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0170: PROP-006 on exact-formula:CH2C(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0171: PROP-006 on exact-formula:CH3CHCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0172: PROP-006 on exact-formula:C6H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0173: PROP-006 on exact-formula:CH3CH(CH3)CN; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-006-ROW-0174: PROP-006 on exact-formula:CH₃CH₂CH₂CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0175: PROP-006 on exact-formula:B₃N₃H₆; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0176: PROP-006 on exact-formula:C₃H₇OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0177: PROP-006 on exact-formula:CH₃CHOHCH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0178: PROP-006 on exact-formula:C₃H₇NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0179: PROP-006 on exact-formula:C₄H₆O₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0180: PROP-006 on exact-formula:C₃H₆O₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0181: PROP-006 on exact-formula:C₆H₄O₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0182: PROP-006 on exact-formula:C₄H₄N₂O₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0183: PROP-006 on exact-formula:C₆H₅F; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0184: PROP-006 on exact-formula:C₆F₆; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0185: PROP-006 on exact-formula:C₆H₅Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0186: PROP-006 on exact-formula:C₆H₄Cl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0187: PROP-006 on exact-formula:C₆H₅Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0188: PROP-006 on exact-formula:C₅H₈; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0189: PROP-006 on exact-formula:NH₂CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0190: PROP-006 on exact-formula:N(CH₃)₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0191: PROP-006 on exact-formula:CH₃CH(NH₂)CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0192: PROP-006 on exact-formula:C₆H₅CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0193: PROP-006 on exact-formula:CH₃COCH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0194: PROP-006 on exact-formula:CHOCH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0195: PROP-006 on exact-formula:C₃H₈O₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0196: PROP-006 on exact-formula:C₃H₇ONO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0197: PROP-006 on exact-formula:C₆H₅OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0198: PROP-006 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0199: PROP-006 on exact-formula:CH₃CH(CH₃)CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0200: PROP-006 on exact-formula:C₆H₅NH₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0201: PROP-006 on exact-formula:C₅H₈O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0202: PROP-006 on exact-formula:C₄H₈O₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0203: PROP-006 on exact-formula:C₄H₈O₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0204: PROP-006 on exact-formula:C₃H₈O₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0205: PROP-006 on exact-formula:C₆H₅CHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0206: PROP-006 on exact-formula:C₆H₅NO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0207: PROP-006 on exact-formula:CH₂ClCH₂CH₂CH₃; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-006-ROW-0208: PROP-006 on exact-formula:CH3CHClCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0209: PROP-006 on exact-formula:CH3CCl(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0210: PROP-006 on exact-formula:C5H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0211: PROP-006 on exact-formula:C5H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0212: PROP-006 on exact-formula:CH2CHCH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0213: PROP-006 on exact-formula:C6H5CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0214: PROP-006 on exact-formula:C5H9N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0215: PROP-006 on exact-formula:C4H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0216: PROP-006 on exact-formula:C4H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0217: PROP-006 on exact-formula:C4H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0218: PROP-006 on exact-formula:C4H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0219: PROP-006 on exact-formula:C2H5SC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0220: PROP-006 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0221: PROP-006 on exact-formula:C6H5CHCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0222: PROP-006 on exact-formula:C5H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0223: PROP-006 on exact-formula:C5H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0224: PROP-006 on exact-formula:C6H5OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0225: PROP-006 on exact-formula:C5H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0226: PROP-006 on exact-formula:C5H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0227: PROP-006 on exact-formula:C5H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0228: PROP-006 on exact-formula:C5H11N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0229: PROP-006 on exact-formula:C6H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0230: PROP-006 on exact-formula:C5H11Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0231: PROP-006 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0232: PROP-006 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0233: PROP-006 on exact-formula:CH3C6H4CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0234: PROP-006 on exact-formula:C10H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0235: PROP-006 on exact-formula:C10H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0236: PROP-006 on exact-formula:C5H12O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0237: PROP-006 on exact-formula:C5H12O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0238: PROP-006 on exact-formula:C6H14; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0239: PROP-006 on exact-formula:C3H7OC3H7; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0240: PROP-006 on exact-formula:C12H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0241: PROP-006 on exact-formula:C6H15N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0242: PROP-006 on exact-formula:C7H16; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0243: PROP-006 on exact-formula:C14H10; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-006-ROW-0244: PROP-006 on exact-formula:C14H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0245: PROP-006 on exact-formula:C8H18; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0246: PROP-006 on exact-formula:(CH3)3CC(CH3)3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0247: PROP-006 on exact-formula:C10H16; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0248: PROP-006 on exact-formula:C16H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0249: PROP-006 on exact-formula:C9H20; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0250: PROP-006 on exact-formula:C10H22; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0251: PROP-006 on exact-formula:C24H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-006-ROW-0252: PROP-006 on exact-formula:C60; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-007-ROW-0001: PROP-007 on exact-formula:D2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-007-ROW-0002: PROP-007 on exact-formula:HD; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-007-ROW-0003: PROP-007 on exact-formula:H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-007-ROW-0004: PROP-007 on exact-formula:N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-007-ROW-0005: PROP-007 on exact-formula:CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-007-ROW-0006: PROP-007 on exact-formula:NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-007-ROW-0007: PROP-007 on exact-formula:O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-007-ROW-0008: PROP-007 on exact-formula:HF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-007-ROW-0009: PROP-007 on exact-formula:F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0001: PROP-008 on exact-formula:LiH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0002: PROP-008 on exact-formula:Li2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0003: PROP-008 on exact-formula:BeH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0004: PROP-008 on exact-formula:BH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0005: PROP-008 on exact-formula:B2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0006: PROP-008 on exact-formula:CH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0007: PROP-008 on exact-formula:BC; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0008: PROP-008 on exact-formula:C2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0009: PROP-008 on exact-formula:NH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0010: PROP-008 on exact-formula:BN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0011: PROP-008 on exact-formula:CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0012: PROP-008 on exact-formula:OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0013: PROP-008 on exact-formula:BO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0014: PROP-008 on exact-formula:NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0015: PROP-008 on exact-formula:O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0016: PROP-008 on exact-formula:CF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0017: PROP-008 on exact-formula:NF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0018: PROP-008 on exact-formula:FO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0019: PROP-008 on exact-formula:F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0020: PROP-008 on exact-formula:Na2; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-008-ROW-0021: PROP-008 on exact-formula:NaF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0022: PROP-008 on exact-formula:MgH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0023: PROP-008 on exact-formula:MgO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0024: PROP-008 on exact-formula:Al₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0025: PROP-008 on exact-formula:AlO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0026: PROP-008 on exact-formula:SiH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0027: PROP-008 on exact-formula:Si₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0028: PROP-008 on exact-formula:SiF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0029: PROP-008 on exact-formula:PH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0030: PROP-008 on exact-formula:P₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0031: PROP-008 on exact-formula:PO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0032: PROP-008 on exact-formula:HS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0033: PROP-008 on exact-formula:AlS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0034: PROP-008 on exact-formula:CS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0035: PROP-008 on exact-formula:NS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0036: PROP-008 on exact-formula:S₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0037: PROP-008 on exact-formula:SO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0038: PROP-008 on exact-formula:SF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0039: PROP-008 on exact-formula:LiCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0040: PROP-008 on exact-formula:NaCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0041: PROP-008 on exact-formula:MgCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0042: PROP-008 on exact-formula:ClO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0043: PROP-008 on exact-formula:ClF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0044: PROP-008 on exact-formula:Cl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0045: PROP-008 on exact-formula:ZnF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0046: PROP-008 on exact-formula:HSe; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0047: PROP-008 on exact-formula:SeO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0048: PROP-008 on exact-formula:Se₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0049: PROP-008 on exact-formula:LiBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0050: PROP-008 on exact-formula:NaBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0051: PROP-008 on exact-formula:BrO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0052: PROP-008 on exact-formula:Br₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0053: PROP-008 on exact-formula:CH₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0054: PROP-008 on exact-formula:C₂H; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0055: PROP-008 on exact-formula:NH₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0056: PROP-008 on exact-formula:N₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0057: PROP-008 on exact-formula:HCO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0058: PROP-008 on exact-formula:HNO; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-008-ROW-0059: PROP-008 on exact-formula:HO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0060: PROP-008 on exact-formula:NCO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0061: PROP-008 on exact-formula:BO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0062: PROP-008 on exact-formula:N2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0063: PROP-008 on exact-formula:CO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0064: PROP-008 on exact-formula:NO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0065: PROP-008 on exact-formula:O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0066: PROP-008 on exact-formula:CF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0067: PROP-008 on exact-formula:SiH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0068: PROP-008 on exact-formula:PH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0069: PROP-008 on exact-formula:PO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0070: PROP-008 on exact-formula:SCN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0071: PROP-008 on exact-formula:CS2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0072: PROP-008 on exact-formula:SO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0073: PROP-008 on exact-formula:S3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0074: PROP-008 on exact-formula:CCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0075: PROP-008 on exact-formula:OCIO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0076: PROP-008 on exact-formula:GeH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0077: PROP-008 on exact-formula:BH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0078: PROP-008 on exact-formula:CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0079: PROP-008 on exact-formula:CCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0080: PROP-008 on exact-formula:H2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0081: PROP-008 on exact-formula:NO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0082: PROP-008 on exact-formula:BF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0083: PROP-008 on exact-formula:CF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0084: PROP-008 on exact-formula:SiH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0085: PROP-008 on exact-formula:H2CS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0086: PROP-008 on exact-formula:SO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0087: PROP-008 on exact-formula:CCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0088: PROP-008 on exact-formula:C2H3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0089: PROP-008 on exact-formula:CH3O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0090: PROP-008 on exact-formula:CH3CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0091: PROP-008 on exact-formula:CH3CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0092: PROP-008 on exact-formula:CH3OO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0093: PROP-008 on exact-formula:C2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0094: PROP-008 on exact-formula:C6N4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0095: PROP-008 on exact-formula:CH3COCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-008-ROW-0096: PROP-008 on exact-formula:C6H5; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-009-ROW-0001: PROP-009 on exact-formula:H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0002: PROP-009 on exact-formula:BH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0003: PROP-009 on exact-formula:AlH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0004: PROP-009 on exact-formula:CH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0005: PROP-009 on exact-formula:C2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0006: PROP-009 on exact-formula:NH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0007: PROP-009 on exact-formula:N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0008: PROP-009 on exact-formula:P2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0009: PROP-009 on exact-formula:OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0010: PROP-009 on exact-formula:BO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0011: PROP-009 on exact-formula:CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0012: PROP-009 on exact-formula:SO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0013: PROP-009 on exact-formula:SO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0014: PROP-009 on exact-formula:HF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0015: PROP-009 on exact-formula:CF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0016: PROP-009 on exact-formula:NF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0017: PROP-009 on exact-formula:F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0018: PROP-009 on exact-formula:HCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0019: PROP-009 on exact-formula:ClF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0020: PROP-009 on exact-formula:BH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0021: PROP-009 on exact-formula:NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0022: PROP-009 on exact-formula:NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0023: PROP-009 on exact-formula:NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0024: PROP-009 on exact-formula:HCN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0025: PROP-009 on exact-formula:HCN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0026: PROP-009 on exact-formula:HCN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0027: PROP-009 on exact-formula:HNC; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0028: PROP-009 on exact-formula:HNC; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0029: PROP-009 on exact-formula:HNC; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0030: PROP-009 on exact-formula:H2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0031: PROP-009 on exact-formula:H2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0032: PROP-009 on exact-formula:H2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0033: PROP-009 on exact-formula:HBO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0034: PROP-009 on exact-formula:HBO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0035: PROP-009 on exact-formula:HBO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0036: PROP-009 on exact-formula:HNO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0037: PROP-009 on exact-formula:HNO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-0038: PROP-009 on exact-formula:HNO; post-seal payload identity True

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- SFT-CHEM-PROP-014-PROP-009-ROW-1558: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1559: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1560: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1561: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1562: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1563: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1564: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1565: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1566: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1567: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1568: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1569: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1570: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1571: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1572: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1573: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1574: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1575: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1576: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1577: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1578: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1579: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-009-ROW-1580: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1581: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1582: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1583: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1584: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1585: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1586: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1587: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1588: PROP-009 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1589: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1590: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1591: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1592: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1593: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1594: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1595: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1596: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1597: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1598: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1599: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1600: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1601: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-009-ROW-1602: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1603: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1604: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1605: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1606: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1607: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1608: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1609: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1610: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1611: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1612: PROP-009 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1613: PROP-009 on exact-formula:CH3CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1614: PROP-009 on exact-formula:CH3CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1615: PROP-009 on exact-formula:CH3CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1616: PROP-009 on exact-formula:CH3CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1617: PROP-009 on exact-formula:CH3CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1618: PROP-009 on exact-formula:CH3CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1619: PROP-009 on exact-formula:CH3CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1620: PROP-009 on exact-formula:CH3CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1621: PROP-009 on exact-formula:CH3CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1622: PROP-009 on exact-formula:CH3CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1623: PROP-009 on exact-formula:CH3CH2CH2CH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-009-ROW-1624: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1625: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1626: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1627: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1628: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1629: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1630: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1631: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1632: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1633: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1634: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1635: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1636: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1637: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1638: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1639: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1640: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1641: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1642: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1643: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1644: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-009-ROW-1645: PROP-009 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True

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- SFT-CHEM-PROP-014-PROP-010-ROW-0357: PROP-010 on exact-formula:H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0358: PROP-010 on exact-formula:BH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0359: PROP-010 on exact-formula:BH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0360: PROP-010 on exact-formula:BH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0361: PROP-010 on exact-formula:C2H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0362: PROP-010 on exact-formula:C2H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0363: PROP-010 on exact-formula:C2H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0364: PROP-010 on exact-formula:B2H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0365: PROP-010 on exact-formula:B2H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0366: PROP-010 on exact-formula:B2H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0367: PROP-010 on exact-formula:SiH4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0368: PROP-010 on exact-formula:SiH4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0369: PROP-010 on exact-formula:SiH4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0370: PROP-010 on exact-formula:SbH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0371: PROP-010 on exact-formula:SbH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0372: PROP-010 on exact-formula:SbH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0373: PROP-010 on exact-formula:CH3NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0374: PROP-010 on exact-formula:CH3NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0375: PROP-010 on exact-formula:CH3NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0376: PROP-010 on exact-formula:PH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0377: PROP-010 on exact-formula:PH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0378: PROP-010 on exact-formula:PH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0379: PROP-010 on exact-formula:BH2NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0380: PROP-010 on exact-formula:BH2NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0381: PROP-010 on exact-formula:BH2NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0382: PROP-010 on exact-formula:C2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0383: PROP-010 on exact-formula:C2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0384: PROP-010 on exact-formula:C2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0385: PROP-010 on exact-formula:ND3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0386: PROP-010 on exact-formula:ND3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0387: PROP-010 on exact-formula:ND3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0388: PROP-010 on exact-formula:SiH3+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0389: PROP-010 on exact-formula:SiH3+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0390: PROP-010 on exact-formula:SiH3+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0391: PROP-010 on exact-formula:CH4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0392: PROP-010 on exact-formula:CH4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0393: PROP-010 on exact-formula:CH4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-0394: PROP-010 on exact-formula:D2S; post-seal payload identity True

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- SFT-CHEM-PROP-014-PROP-010-ROW-1231: PROP-010 on exact-formula:CH3OCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1232: PROP-010 on exact-formula:CH3OCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1233: PROP-010 on exact-formula:CH3OCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1234: PROP-010 on exact-formula:CH2CHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1235: PROP-010 on exact-formula:CH2CHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1236: PROP-010 on exact-formula:CH2CHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1237: PROP-010 on exact-formula:CH2F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1238: PROP-010 on exact-formula:CH2F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1239: PROP-010 on exact-formula:CH2F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1240: PROP-010 on exact-formula:FSN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1241: PROP-010 on exact-formula:FSN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1242: PROP-010 on exact-formula:FSN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1243: PROP-010 on exact-formula:CH2CHSH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1244: PROP-010 on exact-formula:CH2CHSH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1245: PROP-010 on exact-formula:CH2CHSH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1246: PROP-010 on exact-formula:CCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1247: PROP-010 on exact-formula:CCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1248: PROP-010 on exact-formula:CCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1249: PROP-010 on exact-formula:H2O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1250: PROP-010 on exact-formula:H2O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1251: PROP-010 on exact-formula:H2O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1252: PROP-010 on exact-formula:OCIO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1253: PROP-010 on exact-formula:OCIO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1254: PROP-010 on exact-formula:OCIO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1255: PROP-010 on exact-formula:NH2BHNH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1256: PROP-010 on exact-formula:NH2BHNH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1257: PROP-010 on exact-formula:NH2BHNH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1258: PROP-010 on exact-formula:CH2NCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1259: PROP-010 on exact-formula:CH2NCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1260: PROP-010 on exact-formula:CH2NCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1261: PROP-010 on exact-formula:SiC2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1262: PROP-010 on exact-formula:SiC2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1263: PROP-010 on exact-formula:SiC2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1264: PROP-010 on exact-formula:NHF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1265: PROP-010 on exact-formula:NHF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1266: PROP-010 on exact-formula:NHF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1267: PROP-010 on exact-formula:C3H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1268: PROP-010 on exact-formula:C3H5; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-010-ROW-1269: PROP-010 on exact-formula:C3H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1270: PROP-010 on exact-formula:C2H3Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1271: PROP-010 on exact-formula:C2H3Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1272: PROP-010 on exact-formula:C2H3Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1273: PROP-010 on exact-formula:CH3CHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1274: PROP-010 on exact-formula:CH3CHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1275: PROP-010 on exact-formula:CH3CHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1276: PROP-010 on exact-formula:F2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1277: PROP-010 on exact-formula:F2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1278: PROP-010 on exact-formula:F2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1279: PROP-010 on exact-formula:SO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1280: PROP-010 on exact-formula:SO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1281: PROP-010 on exact-formula:SO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1282: PROP-010 on exact-formula:CH3NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1283: PROP-010 on exact-formula:CH3NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1284: PROP-010 on exact-formula:CH3NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1285: PROP-010 on exact-formula:CH3NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1286: PROP-010 on exact-formula:CH3NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1287: PROP-010 on exact-formula:CH3NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1288: PROP-010 on exact-formula:CH2CHF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1289: PROP-010 on exact-formula:CH2CHF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1290: PROP-010 on exact-formula:CH2CHF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1291: PROP-010 on exact-formula:CH2NOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1292: PROP-010 on exact-formula:CH2NOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1293: PROP-010 on exact-formula:CH2NOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1294: PROP-010 on exact-formula:CFCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1295: PROP-010 on exact-formula:CFCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1296: PROP-010 on exact-formula:CFCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1297: PROP-010 on exact-formula:NF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1298: PROP-010 on exact-formula:NF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1299: PROP-010 on exact-formula:NF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1300: PROP-010 on exact-formula:CHONH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1301: PROP-010 on exact-formula:CHONH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1302: PROP-010 on exact-formula:CHONH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1303: PROP-010 on exact-formula:BHF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1304: PROP-010 on exact-formula:BHF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1305: PROP-010 on exact-formula:BHF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1306: PROP-010 on exact-formula:HCOOH; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-010-ROW-1307: PROP-010 on exact-formula:HCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1308: PROP-010 on exact-formula:HCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1309: PROP-010 on exact-formula:H2COO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1310: PROP-010 on exact-formula:H2COO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1311: PROP-010 on exact-formula:H2COO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1312: PROP-010 on exact-formula:FOO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1313: PROP-010 on exact-formula:FOO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1314: PROP-010 on exact-formula:FOO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1315: PROP-010 on exact-formula:SiH3Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1316: PROP-010 on exact-formula:SiH3Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1317: PROP-010 on exact-formula:SiH3Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1318: PROP-010 on exact-formula:ClNO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1319: PROP-010 on exact-formula:ClNO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1320: PROP-010 on exact-formula:ClNO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1321: PROP-010 on exact-formula:CF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1322: PROP-010 on exact-formula:CF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1323: PROP-010 on exact-formula:CF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1324: PROP-010 on exact-formula:HFCO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1325: PROP-010 on exact-formula:HFCO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1326: PROP-010 on exact-formula:HFCO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1327: PROP-010 on exact-formula:HNO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1328: PROP-010 on exact-formula:HNO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1329: PROP-010 on exact-formula:HNO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1330: PROP-010 on exact-formula:CH3SeH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1331: PROP-010 on exact-formula:CH3SeH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1332: PROP-010 on exact-formula:CH3SeH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1333: PROP-010 on exact-formula:FNO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1334: PROP-010 on exact-formula:FNO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1335: PROP-010 on exact-formula:FNO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1336: PROP-010 on exact-formula:PO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1337: PROP-010 on exact-formula:PO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1338: PROP-010 on exact-formula:PO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1339: PROP-010 on exact-formula:O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1340: PROP-010 on exact-formula:O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1341: PROP-010 on exact-formula:O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1342: PROP-010 on exact-formula:CH2CCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1343: PROP-010 on exact-formula:CH2CCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1344: PROP-010 on exact-formula:CH2CCH2; post-seal payload identity True

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- SFT-CHEM-PROP-014-PROP-010-ROW-1383: PROP-010 on exact-formula:HNCNH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1384: PROP-010 on exact-formula:HN3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1385: PROP-010 on exact-formula:HN3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1386: PROP-010 on exact-formula:HN3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1387: PROP-010 on exact-formula:HOBBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1388: PROP-010 on exact-formula:HOBBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1389: PROP-010 on exact-formula:HOBBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1390: PROP-010 on exact-formula:HOI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1391: PROP-010 on exact-formula:HOI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1392: PROP-010 on exact-formula:HOI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1393: PROP-010 on exact-formula:HOCO+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1394: PROP-010 on exact-formula:HOCO+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1395: PROP-010 on exact-formula:HOCO+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1396: PROP-010 on exact-formula:HNCO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1397: PROP-010 on exact-formula:HNCO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1398: PROP-010 on exact-formula:HNCO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1399: PROP-010 on exact-formula:HCCO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1400: PROP-010 on exact-formula:HCCO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1401: PROP-010 on exact-formula:HCCO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1402: PROP-010 on exact-formula:HNCS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1403: PROP-010 on exact-formula:HNCS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1404: PROP-010 on exact-formula:HNCS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1405: PROP-010 on exact-formula:CH3CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1406: PROP-010 on exact-formula:CH3CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1407: PROP-010 on exact-formula:CH3CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1408: PROP-010 on exact-formula:C3H7SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1409: PROP-010 on exact-formula:C3H7SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1410: PROP-010 on exact-formula:C3H7SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1411: PROP-010 on exact-formula:NH2CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1412: PROP-010 on exact-formula:NH2CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1413: PROP-010 on exact-formula:NH2CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1414: PROP-010 on exact-formula:C2H8N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1415: PROP-010 on exact-formula:C2H8N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1416: PROP-010 on exact-formula:C2H8N2; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-010-ROW-1417: PROP-010 on exact-formula:C2H6O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1418: PROP-010 on exact-formula:C2H6O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1419: PROP-010 on exact-formula:C2H6O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1420: PROP-010 on exact-formula:CHClCHCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1421: PROP-010 on exact-formula:CHClCHCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1422: PROP-010 on exact-formula:CHClCHCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1423: PROP-010 on exact-formula:CH3CHCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1424: PROP-010 on exact-formula:CH3CHCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1425: PROP-010 on exact-formula:CH3CHCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1426: PROP-010 on exact-formula:CH3SCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1427: PROP-010 on exact-formula:CH3SCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1428: PROP-010 on exact-formula:CH3SCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1429: PROP-010 on exact-formula:CH3ONO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1430: PROP-010 on exact-formula:CH3ONO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1431: PROP-010 on exact-formula:CH3ONO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1432: PROP-010 on exact-formula:C(CH3)3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1433: PROP-010 on exact-formula:C(CH3)3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1434: PROP-010 on exact-formula:C(CH3)3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1435: PROP-010 on exact-formula:CH3CHCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1436: PROP-010 on exact-formula:CH3CHCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1437: PROP-010 on exact-formula:CH3CHCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1438: PROP-010 on exact-formula:CCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1439: PROP-010 on exact-formula:CCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1440: PROP-010 on exact-formula:CCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1441: PROP-010 on exact-formula:CH2CH(CH3)2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1442: PROP-010 on exact-formula:CH2CH(CH3)2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1443: PROP-010 on exact-formula:CH2CH(CH3)2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1444: PROP-010 on exact-formula:AlCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1445: PROP-010 on exact-formula:AlCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1446: PROP-010 on exact-formula:AlCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1447: PROP-010 on exact-formula:C2Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1448: PROP-010 on exact-formula:C2Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1449: PROP-010 on exact-formula:C2Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1450: PROP-010 on exact-formula:BF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1451: PROP-010 on exact-formula:BF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1452: PROP-010 on exact-formula:BF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1453: PROP-010 on exact-formula:ZnF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1454: PROP-010 on exact-formula:ZnF2; post-seal payload identity True

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- SFT-CHEM-PROP-014-PROP-010-ROW-1531: PROP-010 on exact-formula:C4H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1532: PROP-010 on exact-formula:C4H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1533: PROP-010 on exact-formula:C4H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1534: PROP-010 on exact-formula:CF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1535: PROP-010 on exact-formula:CF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1536: PROP-010 on exact-formula:CF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1537: PROP-010 on exact-formula:SiHCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1538: PROP-010 on exact-formula:SiHCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1539: PROP-010 on exact-formula:SiHCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1540: PROP-010 on exact-formula:CHCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1541: PROP-010 on exact-formula:CHCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1542: PROP-010 on exact-formula:CHCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1543: PROP-010 on exact-formula:P4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1544: PROP-010 on exact-formula:P4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1545: PROP-010 on exact-formula:P4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1546: PROP-010 on exact-formula:S4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1547: PROP-010 on exact-formula:S4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1548: PROP-010 on exact-formula:S4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1549: PROP-010 on exact-formula:SOCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1550: PROP-010 on exact-formula:SOCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1551: PROP-010 on exact-formula:SOCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1552: PROP-010 on exact-formula:ClSSCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1553: PROP-010 on exact-formula:ClSSCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1554: PROP-010 on exact-formula:ClSSCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1555: PROP-010 on exact-formula:CH3CHCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1556: PROP-010 on exact-formula:CH3CHCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1557: PROP-010 on exact-formula:CH3CHCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1558: PROP-010 on exact-formula:CH3SOCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1559: PROP-010 on exact-formula:CH3SOCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1560: PROP-010 on exact-formula:CH3SOCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1561: PROP-010 on exact-formula:SiHF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1562: PROP-010 on exact-formula:SiHF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1563: PROP-010 on exact-formula:SiHF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1564: PROP-010 on exact-formula:CH2CCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1565: PROP-010 on exact-formula:CH2CCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1566: PROP-010 on exact-formula:CH2CCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1567: PROP-010 on exact-formula:CH3CH(CH3)CH3; post-seal payload identity True

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- SFT-CHEM-PROP-014-PROP-010-ROW-1641: PROP-010 on exact-formula:CH₂CF₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1642: PROP-010 on exact-formula:B₄H₁₀; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1643: PROP-010 on exact-formula:B₄H₁₀; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1644: PROP-010 on exact-formula:B₄H₁₀; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1645: PROP-010 on exact-formula:CH₃COF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1646: PROP-010 on exact-formula:CH₃COF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1647: PROP-010 on exact-formula:CH₃COF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1648: PROP-010 on exact-formula:NH₂CONH₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1649: PROP-010 on exact-formula:NH₂CONH₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1650: PROP-010 on exact-formula:NH₂CONH₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1651: PROP-010 on exact-formula:CH₃SeCH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1652: PROP-010 on exact-formula:CH₃SeCH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1653: PROP-010 on exact-formula:CH₃SeCH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1654: PROP-010 on exact-formula:CH₃COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1655: PROP-010 on exact-formula:CH₃COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1656: PROP-010 on exact-formula:CH₃COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1657: PROP-010 on exact-formula:CH₃CONH₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1658: PROP-010 on exact-formula:CH₃CONH₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1659: PROP-010 on exact-formula:CH₃CONH₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1660: PROP-010 on exact-formula:C₃H₇N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1661: PROP-010 on exact-formula:C₃H₇N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1662: PROP-010 on exact-formula:C₃H₇N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1663: PROP-010 on exact-formula:CHClCHCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1664: PROP-010 on exact-formula:CHClCHCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1665: PROP-010 on exact-formula:CHClCHCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1666: PROP-010 on exact-formula:C₃H₆S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1667: PROP-010 on exact-formula:C₃H₆S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1668: PROP-010 on exact-formula:C₃H₆S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1669: PROP-010 on exact-formula:CF₂O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1670: PROP-010 on exact-formula:CF₂O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1671: PROP-010 on exact-formula:CF₂O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1672: PROP-010 on exact-formula:CH₂ClCH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1673: PROP-010 on exact-formula:CH₂ClCH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1674: PROP-010 on exact-formula:CH₂ClCH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1675: PROP-010 on exact-formula:C₃H₆O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1676: PROP-010 on exact-formula:C₃H₆O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1677: PROP-010 on exact-formula:C₃H₆O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1678: PROP-010 on exact-formula:C₄H₆; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-010-ROW-1679: PROP-010 on exact-formula:C4H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1680: PROP-010 on exact-formula:C4H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1681: PROP-010 on exact-formula:HNO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1682: PROP-010 on exact-formula:HNO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1683: PROP-010 on exact-formula:HNO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1684: PROP-010 on exact-formula:ClOOCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1685: PROP-010 on exact-formula:ClOOCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1686: PROP-010 on exact-formula:ClOOCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1687: PROP-010 on exact-formula:FNO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1688: PROP-010 on exact-formula:FNO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1689: PROP-010 on exact-formula:FNO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1690: PROP-010 on exact-formula:CH3CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1691: PROP-010 on exact-formula:CH3CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1692: PROP-010 on exact-formula:CH3CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1693: PROP-010 on exact-formula:ClNO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1694: PROP-010 on exact-formula:ClNO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1695: PROP-010 on exact-formula:ClNO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1696: PROP-010 on exact-formula:CH3NO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1697: PROP-010 on exact-formula:CH3NO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1698: PROP-010 on exact-formula:CH3NO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1699: PROP-010 on exact-formula:CH2FCH2Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1700: PROP-010 on exact-formula:CH2FCH2Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1701: PROP-010 on exact-formula:CH2FCH2Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1702: PROP-010 on exact-formula:NO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1703: PROP-010 on exact-formula:NO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1704: PROP-010 on exact-formula:NO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1705: PROP-010 on exact-formula:ClF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1706: PROP-010 on exact-formula:ClF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1707: PROP-010 on exact-formula:ClF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1708: PROP-010 on exact-formula:C3H7OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1709: PROP-010 on exact-formula:C3H7OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1710: PROP-010 on exact-formula:C3H7OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1711: PROP-010 on exact-formula:CH2FCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1712: PROP-010 on exact-formula:CH2FCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1713: PROP-010 on exact-formula:CH2FCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1714: PROP-010 on exact-formula:CH2ClCHO; post-seal payload identity True

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- SFT-CHEM-PROP-014-PROP-010-ROW-1753: PROP-010 on exact-formula:ClONO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1754: PROP-010 on exact-formula:ClONO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1755: PROP-010 on exact-formula:ClONO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1756: PROP-010 on exact-formula:C4H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1757: PROP-010 on exact-formula:C4H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1758: PROP-010 on exact-formula:C4H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1759: PROP-010 on exact-formula:N2F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1760: PROP-010 on exact-formula:N2F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1761: PROP-010 on exact-formula:N2F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1762: PROP-010 on exact-formula:CH3OCHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1763: PROP-010 on exact-formula:CH3OCHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1764: PROP-010 on exact-formula:CH3OCHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1765: PROP-010 on exact-formula:C3H4O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1766: PROP-010 on exact-formula:C3H4O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1767: PROP-010 on exact-formula:C3H4O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1768: PROP-010 on exact-formula:C4H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1769: PROP-010 on exact-formula:C4H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1770: PROP-010 on exact-formula:C4H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1771: PROP-010 on exact-formula:C2H2F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1772: PROP-010 on exact-formula:C2H2F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1773: PROP-010 on exact-formula:C2H2F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1774: PROP-010 on exact-formula:CH2I2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1775: PROP-010 on exact-formula:CH2I2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1776: PROP-010 on exact-formula:CH2I2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1777: PROP-010 on exact-formula:CH2CHCHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1778: PROP-010 on exact-formula:CH2CHCHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1779: PROP-010 on exact-formula:CH2CHCHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1780: PROP-010 on exact-formula:CH2BrI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1781: PROP-010 on exact-formula:CH2BrI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1782: PROP-010 on exact-formula:CH2BrI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1783: PROP-010 on exact-formula:C3H6O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1784: PROP-010 on exact-formula:C3H6O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1785: PROP-010 on exact-formula:C3H6O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1786: PROP-010 on exact-formula:CH2ClCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1787: PROP-010 on exact-formula:CH2ClCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1788: PROP-010 on exact-formula:CH2ClCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1789: PROP-010 on exact-formula:ONNO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1790: PROP-010 on exact-formula:ONNO; post-seal payload identity True

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- SFT-CHEM-PROP-014-PROP-010-ROW-1829: PROP-010 on exact-formula:CH₂FCH₂Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1830: PROP-010 on exact-formula:CH₂FCH₂Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1831: PROP-010 on exact-formula:TiO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1832: PROP-010 on exact-formula:TiO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1833: PROP-010 on exact-formula:TiO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1834: PROP-010 on exact-formula:CH₃CH₂Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1835: PROP-010 on exact-formula:CH₃CH₂Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1836: PROP-010 on exact-formula:CH₃CH₂Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1837: PROP-010 on exact-formula:C₂H₄F₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1838: PROP-010 on exact-formula:C₂H₄F₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1839: PROP-010 on exact-formula:C₂H₄F₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1840: PROP-010 on exact-formula:HOCH₂CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1841: PROP-010 on exact-formula:HOCH₂CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1842: PROP-010 on exact-formula:HOCH₂CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1843: PROP-010 on exact-formula:CH₂ClCHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1844: PROP-010 on exact-formula:CH₂ClCHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1845: PROP-010 on exact-formula:CH₂ClCHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1846: PROP-010 on exact-formula:CH₂CCHCH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1847: PROP-010 on exact-formula:CH₂CCHCH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1848: PROP-010 on exact-formula:CH₂CCHCH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1849: PROP-010 on exact-formula:CH₂FI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1850: PROP-010 on exact-formula:CH₂FI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1851: PROP-010 on exact-formula:CH₂FI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1852: PROP-010 on exact-formula:CBr₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1853: PROP-010 on exact-formula:CBr₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1854: PROP-010 on exact-formula:CBr₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1855: PROP-010 on exact-formula:C₃H₄O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1856: PROP-010 on exact-formula:C₃H₄O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1857: PROP-010 on exact-formula:C₃H₄O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1858: PROP-010 on exact-formula:CH₂BrF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1859: PROP-010 on exact-formula:CH₂BrF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1860: PROP-010 on exact-formula:CH₂BrF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1861: PROP-010 on exact-formula:CH₂CHCHCH₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1862: PROP-010 on exact-formula:CH₂CHCHCH₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1863: PROP-010 on exact-formula:CH₂CHCHCH₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1864: PROP-010 on exact-formula:CH₃CHNOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1865: PROP-010 on exact-formula:CH₃CHNOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1866: PROP-010 on exact-formula:CH₃CHNOH; post-seal payload identity True

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- SFT-CHEM-PROP-014-PROP-010-ROW-1905: PROP-010 on exact-formula:H2CCCCCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1906: PROP-010 on exact-formula:C3H2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1907: PROP-010 on exact-formula:C3H2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1908: PROP-010 on exact-formula:C3H2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1909: PROP-010 on exact-formula:C3H5ClO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1910: PROP-010 on exact-formula:C3H5ClO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1911: PROP-010 on exact-formula:C3H5ClO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1912: PROP-010 on exact-formula:C6H14; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1913: PROP-010 on exact-formula:C6H14; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1914: PROP-010 on exact-formula:C6H14; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1915: PROP-010 on exact-formula:C5H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1916: PROP-010 on exact-formula:C5H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1917: PROP-010 on exact-formula:C5H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1918: PROP-010 on exact-formula:CH2ClCH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1919: PROP-010 on exact-formula:CH2ClCH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1920: PROP-010 on exact-formula:CH2ClCH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1921: PROP-010 on exact-formula:C(NH2)H2CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1922: PROP-010 on exact-formula:C(NH2)H2CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1923: PROP-010 on exact-formula:C(NH2)H2CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1924: PROP-010 on exact-formula:C3H8S2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1925: PROP-010 on exact-formula:C3H8S2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1926: PROP-010 on exact-formula:C3H8S2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1927: PROP-010 on exact-formula:C3H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1928: PROP-010 on exact-formula:C3H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1929: PROP-010 on exact-formula:C3H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1930: PROP-010 on exact-formula:C2H3OC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1931: PROP-010 on exact-formula:C2H3OC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1932: PROP-010 on exact-formula:C2H3OC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1933: PROP-010 on exact-formula:CH2CHOCHCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1934: PROP-010 on exact-formula:CH2CHOCHCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1935: PROP-010 on exact-formula:CH2CHOCHCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1936: PROP-010 on exact-formula:C6H14; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1937: PROP-010 on exact-formula:C6H14; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1938: PROP-010 on exact-formula:C6H14; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-010-ROW-1939: PROP-010 on exact-formula:C4H4N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1940: PROP-010 on exact-formula:C4H4N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1941: PROP-010 on exact-formula:C4H4N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1942: PROP-010 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1943: PROP-010 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1944: PROP-010 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1945: PROP-010 on exact-formula:CH2C(CH3)OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1946: PROP-010 on exact-formula:CH2C(CH3)OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1947: PROP-010 on exact-formula:CH2C(CH3)OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1948: PROP-010 on exact-formula:CHOCH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1949: PROP-010 on exact-formula:CHOCH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1950: PROP-010 on exact-formula:CHOCH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1951: PROP-010 on exact-formula:C5H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1952: PROP-010 on exact-formula:C5H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1953: PROP-010 on exact-formula:C5H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1954: PROP-010 on exact-formula:CH2N4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1955: PROP-010 on exact-formula:CH2N4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1956: PROP-010 on exact-formula:CH2N4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1957: PROP-010 on exact-formula:CH3CF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1958: PROP-010 on exact-formula:CH3CF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1959: PROP-010 on exact-formula:CH3CF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1960: PROP-010 on exact-formula:C5H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1961: PROP-010 on exact-formula:C5H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1962: PROP-010 on exact-formula:C5H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1963: PROP-010 on exact-formula:C3H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1964: PROP-010 on exact-formula:C3H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1965: PROP-010 on exact-formula:C3H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1966: PROP-010 on exact-formula:CH3CCl(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1967: PROP-010 on exact-formula:CH3CCl(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1968: PROP-010 on exact-formula:CH3CCl(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1969: PROP-010 on exact-formula:C5H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1970: PROP-010 on exact-formula:C5H10; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-010-ROW-1971: PROP-010 on exact-formula:C5H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1972: PROP-010 on exact-formula:CH2(SH)CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1973: PROP-010 on exact-formula:CH2(SH)CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1974: PROP-010 on exact-formula:CH2(SH)CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1975: PROP-010 on exact-formula:CH3CH(SH)CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1976: PROP-010 on exact-formula:CH3CH(SH)CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1977: PROP-010 on exact-formula:CH3CH(SH)CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1978: PROP-010 on exact-formula:C5H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1979: PROP-010 on exact-formula:C5H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1980: PROP-010 on exact-formula:C5H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1981: PROP-010 on exact-formula:CH2CHCHClCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1982: PROP-010 on exact-formula:CH2CHCHClCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1983: PROP-010 on exact-formula:CH2CHCHClCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1984: PROP-010 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1985: PROP-010 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1986: PROP-010 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1987: PROP-010 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1988: PROP-010 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1989: PROP-010 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1990: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1991: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1992: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1993: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1994: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1995: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1996: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1997: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1998: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-1999: PROP-010 on exact-formula:CH2ClCHCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2000: PROP-010 on exact-formula:CH2ClCHCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2001: PROP-010 on exact-formula:CH2ClCHCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2002: PROP-010 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2003: PROP-010 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2004: PROP-010 on exact-formula:C6H10; post-seal payload identity True

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- SFT-CHEM-PROP-014-PROP-010-ROW-2043: PROP-010 on exact-formula:C4H2N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2044: PROP-010 on exact-formula:C6H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2045: PROP-010 on exact-formula:C6H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2046: PROP-010 on exact-formula:C6H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2047: PROP-010 on exact-formula:C6H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2048: PROP-010 on exact-formula:C6H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2049: PROP-010 on exact-formula:C6H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2050: PROP-010 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2051: PROP-010 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2052: PROP-010 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2053: PROP-010 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2054: PROP-010 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2055: PROP-010 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2056: PROP-010 on exact-formula:CH2CHCH2CH2Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2057: PROP-010 on exact-formula:CH2CHCH2CH2Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2058: PROP-010 on exact-formula:CH2CHCH2CH2Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2059: PROP-010 on exact-formula:C6H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2060: PROP-010 on exact-formula:C6H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2061: PROP-010 on exact-formula:C6H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2062: PROP-010 on exact-formula:C4N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2063: PROP-010 on exact-formula:C4N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2064: PROP-010 on exact-formula:C4N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2065: PROP-010 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2066: PROP-010 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2067: PROP-010 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2068: PROP-010 on exact-formula:C4H5N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2069: PROP-010 on exact-formula:C4H5N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2070: PROP-010 on exact-formula:C4H5N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2071: PROP-010 on exact-formula:C4H6O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2072: PROP-010 on exact-formula:C4H6O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2073: PROP-010 on exact-formula:C4H6O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2074: PROP-010 on exact-formula:CH3C(SCH3)HCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2075: PROP-010 on exact-formula:CH3C(SCH3)HCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2076: PROP-010 on exact-formula:CH3C(SCH3)HCH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-010-ROW-2077: PROP-010 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2078: PROP-010 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2079: PROP-010 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2080: PROP-010 on exact-formula:CH2CClCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2081: PROP-010 on exact-formula:CH2CClCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2082: PROP-010 on exact-formula:CH2CClCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2083: PROP-010 on exact-formula:C6H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2084: PROP-010 on exact-formula:C6H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2085: PROP-010 on exact-formula:C6H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2086: PROP-010 on exact-formula:C4H9N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2087: PROP-010 on exact-formula:C4H9N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2088: PROP-010 on exact-formula:C4H9N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2089: PROP-010 on exact-formula:C6H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2090: PROP-010 on exact-formula:C6H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2091: PROP-010 on exact-formula:C6H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2092: PROP-010 on exact-formula:C6H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2093: PROP-010 on exact-formula:C6H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2094: PROP-010 on exact-formula:C6H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2095: PROP-010 on exact-formula:C3H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2096: PROP-010 on exact-formula:C3H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2097: PROP-010 on exact-formula:C3H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2098: PROP-010 on exact-formula:C2H4N4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2099: PROP-010 on exact-formula:C2H4N4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2100: PROP-010 on exact-formula:C2H4N4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2101: PROP-010 on exact-formula:CHOCHCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2102: PROP-010 on exact-formula:CHOCHCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2103: PROP-010 on exact-formula:CHOCHCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2104: PROP-010 on exact-formula:C5H11; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2105: PROP-010 on exact-formula:C5H11; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2106: PROP-010 on exact-formula:C5H11; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2107: PROP-010 on exact-formula:CH3N5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2108: PROP-010 on exact-formula:CH3N5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2109: PROP-010 on exact-formula:CH3N5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2110: PROP-010 on exact-formula:CH3CClCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2111: PROP-010 on exact-formula:CH3CClCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2112: PROP-010 on exact-formula:CH3CClCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2113: PROP-010 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2114: PROP-010 on exact-formula:C6H12; post-seal payload identity True

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- SFT-CHEM-PROP-014-PROP-010-ROW-2189: PROP-010 on exact-formula:C6H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2190: PROP-010 on exact-formula:C6H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2191: PROP-010 on exact-formula:PBr3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2192: PROP-010 on exact-formula:PBr3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2193: PROP-010 on exact-formula:PBr3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2194: PROP-010 on exact-formula:CHBr3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2195: PROP-010 on exact-formula:CHBr3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2196: PROP-010 on exact-formula:CHBr3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2197: PROP-010 on exact-formula:SiCl3CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2198: PROP-010 on exact-formula:SiCl3CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2199: PROP-010 on exact-formula:SiCl3CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2200: PROP-010 on exact-formula:CF3Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2201: PROP-010 on exact-formula:CF3Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2202: PROP-010 on exact-formula:CF3Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2203: PROP-010 on exact-formula:AsCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2204: PROP-010 on exact-formula:AsCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2205: PROP-010 on exact-formula:AsCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2206: PROP-010 on exact-formula:C3O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2207: PROP-010 on exact-formula:C3O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2208: PROP-010 on exact-formula:C3O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2209: PROP-010 on exact-formula:CH3CCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2210: PROP-010 on exact-formula:CH3CCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2211: PROP-010 on exact-formula:CH3CCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2212: PROP-010 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2213: PROP-010 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2214: PROP-010 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2215: PROP-010 on exact-formula:GeF4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2216: PROP-010 on exact-formula:GeF4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2217: PROP-010 on exact-formula:GeF4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2218: PROP-010 on exact-formula:C6H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2219: PROP-010 on exact-formula:C6H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2220: PROP-010 on exact-formula:C6H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2221: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2222: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2223: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2224: PROP-010 on exact-formula:F3PO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2225: PROP-010 on exact-formula:F3PO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2226: PROP-010 on exact-formula:F3PO; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-010-ROW-2227: PROP-010 on exact-formula:SeO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2228: PROP-010 on exact-formula:SeO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2229: PROP-010 on exact-formula:SeO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2230: PROP-010 on exact-formula:C5H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2231: PROP-010 on exact-formula:C5H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2232: PROP-010 on exact-formula:C5H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2233: PROP-010 on exact-formula:CF3I; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2234: PROP-010 on exact-formula:CF3I; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2235: PROP-010 on exact-formula:CF3I; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2236: PROP-010 on exact-formula:CI4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2237: PROP-010 on exact-formula:CI4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2238: PROP-010 on exact-formula:CI4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2239: PROP-010 on exact-formula:CBr4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2240: PROP-010 on exact-formula:CBr4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2241: PROP-010 on exact-formula:CBr4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2242: PROP-010 on exact-formula:CCl4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2243: PROP-010 on exact-formula:CCl4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2244: PROP-010 on exact-formula:CCl4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2245: PROP-010 on exact-formula:Cl3PO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2246: PROP-010 on exact-formula:Cl3PO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2247: PROP-010 on exact-formula:Cl3PO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2248: PROP-010 on exact-formula:CFCI3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2249: PROP-010 on exact-formula:CFCI3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2250: PROP-010 on exact-formula:CFCI3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2251: PROP-010 on exact-formula:Si(CH3)4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2252: PROP-010 on exact-formula:Si(CH3)4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2253: PROP-010 on exact-formula:Si(CH3)4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2254: PROP-010 on exact-formula:SO2Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2255: PROP-010 on exact-formula:SO2Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2256: PROP-010 on exact-formula:SO2Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2257: PROP-010 on exact-formula:CH3CCl2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2258: PROP-010 on exact-formula:CH3CCl2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2259: PROP-010 on exact-formula:CH3CCl2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2260: PROP-010 on exact-formula:CBrClF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2261: PROP-010 on exact-formula:CBrClF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2262: PROP-010 on exact-formula:CBrClF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2263: PROP-010 on exact-formula:CHClCCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2264: PROP-010 on exact-formula:CHClCCl2; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-010-ROW-2265: PROP-010 on exact-formula:CHClCCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2266: PROP-010 on exact-formula:CF2Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2267: PROP-010 on exact-formula:CF2Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2268: PROP-010 on exact-formula:CF2Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2269: PROP-010 on exact-formula:SiF4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2270: PROP-010 on exact-formula:SiF4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2271: PROP-010 on exact-formula:SiF4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2272: PROP-010 on exact-formula:C(CH3)3SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2273: PROP-010 on exact-formula:C(CH3)3SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2274: PROP-010 on exact-formula:C(CH3)3SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2275: PROP-010 on exact-formula:C2H6O2S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2276: PROP-010 on exact-formula:C2H6O2S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2277: PROP-010 on exact-formula:C2H6O2S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2278: PROP-010 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2279: PROP-010 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2280: PROP-010 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2281: PROP-010 on exact-formula:C6H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2282: PROP-010 on exact-formula:C6H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2283: PROP-010 on exact-formula:C6H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2284: PROP-010 on exact-formula:SO2F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2285: PROP-010 on exact-formula:SO2F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2286: PROP-010 on exact-formula:SO2F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2287: PROP-010 on exact-formula:CH3CF2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2288: PROP-010 on exact-formula:CH3CF2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2289: PROP-010 on exact-formula:CH3CF2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2290: PROP-010 on exact-formula:H2SO4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2291: PROP-010 on exact-formula:H2SO4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2292: PROP-010 on exact-formula:H2SO4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2293: PROP-010 on exact-formula:SeF4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2294: PROP-010 on exact-formula:SeF4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2295: PROP-010 on exact-formula:SeF4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2296: PROP-010 on exact-formula:CH2CClCHCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2297: PROP-010 on exact-formula:CH2CClCHCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2298: PROP-010 on exact-formula:CH2CClCHCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2299: PROP-010 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2300: PROP-010 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2301: PROP-010 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2302: PROP-010 on exact-formula:HClO4; post-seal payload identity True

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- SFT-CHEM-PROP-014-PROP-010-ROW-2455: PROP-010 on exact-formula:C2H3N3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2456: PROP-010 on exact-formula:C2H3N3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2457: PROP-010 on exact-formula:C2H3N3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2458: PROP-010 on exact-formula:H2NCH2COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2459: PROP-010 on exact-formula:H2NCH2COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2460: PROP-010 on exact-formula:H2NCH2COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2461: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2462: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2463: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2464: PROP-010 on exact-formula:C3H6N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2465: PROP-010 on exact-formula:C3H6N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2466: PROP-010 on exact-formula:C3H6N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2467: PROP-010 on exact-formula:CH2CHCONH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2468: PROP-010 on exact-formula:CH2CHCONH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2469: PROP-010 on exact-formula:CH2CHCONH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2470: PROP-010 on exact-formula:C4H6O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2471: PROP-010 on exact-formula:C4H6O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2472: PROP-010 on exact-formula:C4H6O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2473: PROP-010 on exact-formula:C5H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2474: PROP-010 on exact-formula:C5H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2475: PROP-010 on exact-formula:C5H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2476: PROP-010 on exact-formula:BrF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2477: PROP-010 on exact-formula:BrF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2478: PROP-010 on exact-formula:BrF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2479: PROP-010 on exact-formula:C5H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2480: PROP-010 on exact-formula:C5H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2481: PROP-010 on exact-formula:C5H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2482: PROP-010 on exact-formula:C5H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2483: PROP-010 on exact-formula:C5H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2484: PROP-010 on exact-formula:C5H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2485: PROP-010 on exact-formula:CH3NO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2486: PROP-010 on exact-formula:CH3NO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2487: PROP-010 on exact-formula:CH3NO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2488: PROP-010 on exact-formula:CH3OCH2CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2489: PROP-010 on exact-formula:CH3OCH2CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2490: PROP-010 on exact-formula:CH3OCH2CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2491: PROP-010 on exact-formula:HOONO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2492: PROP-010 on exact-formula:HOONO2; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-010-ROW-2493: PROP-010 on exact-formula:HOONO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2494: PROP-010 on exact-formula:ClONO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2495: PROP-010 on exact-formula:ClONO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2496: PROP-010 on exact-formula:ClONO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2497: PROP-010 on exact-formula:N2O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2498: PROP-010 on exact-formula:N2O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2499: PROP-010 on exact-formula:N2O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2500: PROP-010 on exact-formula:C4H9F; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2501: PROP-010 on exact-formula:C4H9F; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2502: PROP-010 on exact-formula:C4H9F; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2503: PROP-010 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2504: PROP-010 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2505: PROP-010 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2506: PROP-010 on exact-formula:CH2FCH2CH2F; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2507: PROP-010 on exact-formula:CH2FCH2CH2F; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2508: PROP-010 on exact-formula:CH2FCH2CH2F; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2509: PROP-010 on exact-formula:C2H5SC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2510: PROP-010 on exact-formula:C2H5SC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2511: PROP-010 on exact-formula:C2H5SC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2512: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2513: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2514: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2515: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2516: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2517: PROP-010 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2518: PROP-010 on exact-formula:CH2CHOCHCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2519: PROP-010 on exact-formula:CH2CHOCHCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2520: PROP-010 on exact-formula:CH2CHOCHCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2521: PROP-010 on exact-formula:C5H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2522: PROP-010 on exact-formula:C5H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2523: PROP-010 on exact-formula:C5H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2524: PROP-010 on exact-formula:HCOOC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2525: PROP-010 on exact-formula:HCOOC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2526: PROP-010 on exact-formula:HCOOC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2527: PROP-010 on exact-formula:CH2CHCH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2528: PROP-010 on exact-formula:CH2CHCH2CH2CH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-010-ROW-2529: PROP-010 on exact-formula:CH₂CHCH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2530: PROP-010 on exact-formula:C₄H₁₀O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2531: PROP-010 on exact-formula:C₄H₁₀O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2532: PROP-010 on exact-formula:C₄H₁₀O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2533: PROP-010 on exact-formula:C₃H₂N₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2534: PROP-010 on exact-formula:C₃H₂N₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2535: PROP-010 on exact-formula:C₃H₂N₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2536: PROP-010 on exact-formula:CHOOCHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2537: PROP-010 on exact-formula:CHOOCHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2538: PROP-010 on exact-formula:CHOOCHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2539: PROP-010 on exact-formula:C₅H₈; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2540: PROP-010 on exact-formula:C₅H₈; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2541: PROP-010 on exact-formula:C₅H₈; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2542: PROP-010 on exact-formula:CH₃CH₂CH₂CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2543: PROP-010 on exact-formula:CH₃CH₂CH₂CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2544: PROP-010 on exact-formula:CH₃CH₂CH₂CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2545: PROP-010 on exact-formula:C₅H₈; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2546: PROP-010 on exact-formula:C₅H₈; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2547: PROP-010 on exact-formula:C₅H₈; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2548: PROP-010 on exact-formula:C₅H₁₀O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2549: PROP-010 on exact-formula:C₅H₁₀O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2550: PROP-010 on exact-formula:C₅H₁₀O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2551: PROP-010 on exact-formula:C₄H₁₀O₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2552: PROP-010 on exact-formula:C₄H₁₀O₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2553: PROP-010 on exact-formula:C₄H₁₀O₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2554: PROP-010 on exact-formula:CH₃NHCH₂COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2555: PROP-010 on exact-formula:CH₃NHCH₂COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2556: PROP-010 on exact-formula:CH₃NHCH₂COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2557: PROP-010 on exact-formula:C₄H₈Cl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2558: PROP-010 on exact-formula:C₄H₈Cl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2559: PROP-010 on exact-formula:C₄H₈Cl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2560: PROP-010 on exact-formula:C₅H₉N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2561: PROP-010 on exact-formula:C₅H₉N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2562: PROP-010 on exact-formula:C₅H₉N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2563: PROP-010 on exact-formula:C₄H₁₀O₂; post-seal payload identity True

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- SFT-CHEM-PROP-014-PROP-010-ROW-2602: PROP-010 on exact-formula:C4H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2603: PROP-010 on exact-formula:C4H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2604: PROP-010 on exact-formula:C4H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2605: PROP-010 on exact-formula:C4H10O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2606: PROP-010 on exact-formula:C4H10O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2607: PROP-010 on exact-formula:C4H10O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2608: PROP-010 on exact-formula:C5H12S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2609: PROP-010 on exact-formula:C5H12S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2610: PROP-010 on exact-formula:C5H12S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2611: PROP-010 on exact-formula:C4H9NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2612: PROP-010 on exact-formula:C4H9NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2613: PROP-010 on exact-formula:C4H9NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2614: PROP-010 on exact-formula:C3H5ClO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2615: PROP-010 on exact-formula:C3H5ClO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2616: PROP-010 on exact-formula:C3H5ClO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2617: PROP-010 on exact-formula:C2H4N2O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2618: PROP-010 on exact-formula:C2H4N2O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2619: PROP-010 on exact-formula:C2H4N2O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2620: PROP-010 on exact-formula:C2H3NO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2621: PROP-010 on exact-formula:C2H3NO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2622: PROP-010 on exact-formula:C2H3NO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2623: PROP-010 on exact-formula:C5H6S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2624: PROP-010 on exact-formula:C5H6S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2625: PROP-010 on exact-formula:C5H6S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2626: PROP-010 on exact-formula:C5H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2627: PROP-010 on exact-formula:C5H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2628: PROP-010 on exact-formula:C5H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2629: PROP-010 on exact-formula:C4H9NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2630: PROP-010 on exact-formula:C4H9NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2631: PROP-010 on exact-formula:C4H9NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2632: PROP-010 on exact-formula:C3H8O2S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2633: PROP-010 on exact-formula:C3H8O2S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2634: PROP-010 on exact-formula:C3H8O2S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2635: PROP-010 on exact-formula:C5H12O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2636: PROP-010 on exact-formula:C5H12O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2637: PROP-010 on exact-formula:C5H12O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2638: PROP-010 on exact-formula:N(CH3)2CONH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2639: PROP-010 on exact-formula:N(CH3)2CONH2; post-seal payload identity True

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- SFT-CHEM-PROP-014-PROP-010-ROW-2830: PROP-010 on exact-formula:C3H6S3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2831: PROP-010 on exact-formula:C3H6S3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2832: PROP-010 on exact-formula:C3H6S3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2833: PROP-010 on exact-formula:C8H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2834: PROP-010 on exact-formula:C8H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2835: PROP-010 on exact-formula:C8H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2836: PROP-010 on exact-formula:C5H9N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2837: PROP-010 on exact-formula:C5H9N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2838: PROP-010 on exact-formula:C5H9N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2839: PROP-010 on exact-formula:CH(CN)3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2840: PROP-010 on exact-formula:CH(CN)3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2841: PROP-010 on exact-formula:CH(CN)3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2842: PROP-010 on exact-formula:S8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2843: PROP-010 on exact-formula:S8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2844: PROP-010 on exact-formula:S8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2845: PROP-010 on exact-formula:CF3CCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2846: PROP-010 on exact-formula:CF3CCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2847: PROP-010 on exact-formula:CF3CCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2848: PROP-010 on exact-formula:C12H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2849: PROP-010 on exact-formula:C12H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2850: PROP-010 on exact-formula:C12H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2851: PROP-010 on exact-formula:C14H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2852: PROP-010 on exact-formula:C14H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2853: PROP-010 on exact-formula:C14H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2854: PROP-010 on exact-formula:C6H4Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2855: PROP-010 on exact-formula:C6H4Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2856: PROP-010 on exact-formula:C6H4Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2857: PROP-010 on exact-formula:C13H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2858: PROP-010 on exact-formula:C13H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2859: PROP-010 on exact-formula:C13H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2860: PROP-010 on exact-formula:C3F6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2861: PROP-010 on exact-formula:C3F6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2862: PROP-010 on exact-formula:C3F6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2863: PROP-010 on exact-formula:SF6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2864: PROP-010 on exact-formula:SF6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2865: PROP-010 on exact-formula:SF6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2866: PROP-010 on exact-formula:C6H4Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2867: PROP-010 on exact-formula:C6H4Cl2; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-010-ROW-2868: PROP-010 on exact-formula:C6H4Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2869: PROP-010 on exact-formula:C10H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2870: PROP-010 on exact-formula:C10H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2871: PROP-010 on exact-formula:C10H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2872: PROP-010 on exact-formula:C2F6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2873: PROP-010 on exact-formula:C2F6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2874: PROP-010 on exact-formula:C2F6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2875: PROP-010 on exact-formula:BrF5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2876: PROP-010 on exact-formula:BrF5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2877: PROP-010 on exact-formula:BrF5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2878: PROP-010 on exact-formula:C10H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2879: PROP-010 on exact-formula:C10H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2880: PROP-010 on exact-formula:C10H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2881: PROP-010 on exact-formula:C8H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2882: PROP-010 on exact-formula:C8H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2883: PROP-010 on exact-formula:C8H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2884: PROP-010 on exact-formula:ClF5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2885: PROP-010 on exact-formula:ClF5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2886: PROP-010 on exact-formula:ClF5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2887: PROP-010 on exact-formula:CH3COC6H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2888: PROP-010 on exact-formula:CH3COC6H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2889: PROP-010 on exact-formula:CH3COC6H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2890: PROP-010 on exact-formula:CF3CHF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2891: PROP-010 on exact-formula:CF3CHF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2892: PROP-010 on exact-formula:CF3CHF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2893: PROP-010 on exact-formula:C7H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2894: PROP-010 on exact-formula:C7H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2895: PROP-010 on exact-formula:C7H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2896: PROP-010 on exact-formula:C7H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2897: PROP-010 on exact-formula:C7H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2898: PROP-010 on exact-formula:C7H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2899: PROP-010 on exact-formula:PF5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2900: PROP-010 on exact-formula:PF5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2901: PROP-010 on exact-formula:PF5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2902: PROP-010 on exact-formula:C9H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2903: PROP-010 on exact-formula:C9H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2904: PROP-010 on exact-formula:C9H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2905: PROP-010 on exact-formula:C6H5NO2; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-010-ROW-2906: PROP-010 on exact-formula:C6H5NO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2907: PROP-010 on exact-formula:C6H5NO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2908: PROP-010 on exact-formula:C5H10S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2909: PROP-010 on exact-formula:C5H10S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2910: PROP-010 on exact-formula:C5H10S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2911: PROP-010 on exact-formula:C6H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2912: PROP-010 on exact-formula:C6H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2913: PROP-010 on exact-formula:C6H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2914: PROP-010 on exact-formula:C7H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2915: PROP-010 on exact-formula:C7H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2916: PROP-010 on exact-formula:C7H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2917: PROP-010 on exact-formula:C3H8O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2918: PROP-010 on exact-formula:C3H8O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2919: PROP-010 on exact-formula:C3H8O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2920: PROP-010 on exact-formula:C7H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2921: PROP-010 on exact-formula:C7H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2922: PROP-010 on exact-formula:C7H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2923: PROP-010 on exact-formula:C5H11N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2924: PROP-010 on exact-formula:C5H11N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2925: PROP-010 on exact-formula:C5H11N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2926: PROP-010 on exact-formula:C2ClF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2927: PROP-010 on exact-formula:C2ClF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2928: PROP-010 on exact-formula:C2ClF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2929: PROP-010 on exact-formula:C5H11N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2930: PROP-010 on exact-formula:C5H11N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2931: PROP-010 on exact-formula:C5H11N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2932: PROP-010 on exact-formula:C6H5OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2933: PROP-010 on exact-formula:C6H5OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2934: PROP-010 on exact-formula:C6H5OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2935: PROP-010 on exact-formula:CH3CH(NH2)COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2936: PROP-010 on exact-formula:CH3CH(NH2)COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2937: PROP-010 on exact-formula:CH3CH(NH2)COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2938: PROP-010 on exact-formula:CHF2CHF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2939: PROP-010 on exact-formula:CHF2CHF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2940: PROP-010 on exact-formula:CHF2CHF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2941: PROP-010 on exact-formula:C6H5CHCH2; post-seal payload identity True

- [illegible]

- SFT-CHEM-PROP-014-PROP-010-ROW-2980: PROP-010 on exact-formula:N2O4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2981: PROP-010 on exact-formula:N2O4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2982: PROP-010 on exact-formula:N2O4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2983: PROP-010 on exact-formula:C5H8O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2984: PROP-010 on exact-formula:C5H8O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2985: PROP-010 on exact-formula:C5H8O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2986: PROP-010 on exact-formula:C4H2O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2987: PROP-010 on exact-formula:C4H2O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2988: PROP-010 on exact-formula:C4H2O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2989: PROP-010 on exact-formula:NH2CH2CH2COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2990: PROP-010 on exact-formula:NH2CH2CH2COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2991: PROP-010 on exact-formula:NH2CH2CH2COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2992: PROP-010 on exact-formula:CH3CH(CH3)ONO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2993: PROP-010 on exact-formula:CH3CH(CH3)ONO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2994: PROP-010 on exact-formula:CH3CH(CH3)ONO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2995: PROP-010 on exact-formula:C5H4O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2996: PROP-010 on exact-formula:C5H4O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2997: PROP-010 on exact-formula:C5H4O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2998: PROP-010 on exact-formula:C5H4O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-2999: PROP-010 on exact-formula:C5H4O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-3000: PROP-010 on exact-formula:C5H4O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-3001: PROP-010 on exact-formula:C4H4O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-3002: PROP-010 on exact-formula:C4H4O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-3003: PROP-010 on exact-formula:C4H4O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-3004: PROP-010 on exact-formula:C3H2O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-3005: PROP-010 on exact-formula:C3H2O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-3006: PROP-010 on exact-formula:C3H2O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-3007: PROP-010 on exact-formula:C6H12O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-3008: PROP-010 on exact-formula:C6H12O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-3009: PROP-010 on exact-formula:C6H12O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-3010: PROP-010 on exact-formula:C2H5NO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-3011: PROP-010 on exact-formula:C2H5NO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-3012: PROP-010 on exact-formula:C2H5NO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-3013: PROP-010 on exact-formula:IF5; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-010-ROW-3014: PROP-010 on exact-formula:IF5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-010-ROW-3015: PROP-010 on exact-formula:IF5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0001: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0002: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0003: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0004: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0005: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0006: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0007: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0008: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0009: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0010: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0011: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0012: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0013: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0014: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0015: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0016: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0017: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0018: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0019: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0020: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0021: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0022: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0023: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0024: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0025: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0026: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0027: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0028: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0029: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0030: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0031: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0032: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0033: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0034: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0035: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0036: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0037: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0038: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0039: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0040: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0041: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0042: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0043: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0044: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0045: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0046: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0047: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0048: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0049: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0050: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0051: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0052: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0053: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0054: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0055: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0056: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0057: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0058: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0059: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0060: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0061: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0062: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0063: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0064: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0065: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0066: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0067: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0068: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0069: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0070: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0071: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0072: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0073: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0074: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0075: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0076: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0077: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0078: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0079: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0080: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0081: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0082: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0083: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0084: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0085: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0086: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0087: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0088: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0089: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0090: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0091: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0092: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0093: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0094: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0095: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0096: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0097: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0098: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0099: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0100: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0101: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0102: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0103: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0104: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0105: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0106: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0107: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0108: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0109: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0110: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0111: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0112: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0113: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0114: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0115: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0116: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0117: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0118: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0119: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0120: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0121: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0122: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0123: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0124: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0125: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0126: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0127: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0128: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0129: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0130: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0131: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0132: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0133: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0134: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0135: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0136: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0137: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0138: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0139: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0140: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0141: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0142: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0143: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0144: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0145: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0146: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0147: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0148: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0149: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0150: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0151: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0152: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0153: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0154: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0155: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0156: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0157: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0158: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0159: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0160: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0161: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0162: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0163: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0164: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0165: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0166: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0167: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0168: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0169: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0170: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0171: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0172: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0173: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0174: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0175: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0176: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0177: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0178: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0179: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0180: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0181: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0182: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0183: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0184: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0185: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0186: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0187: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0188: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0189: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0190: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0191: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0192: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0193: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0194: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0195: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0196: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0197: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0198: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0199: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0200: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0201: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0202: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0203: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0204: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0205: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0206: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0207: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0208: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0209: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0210: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0211: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0212: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0213: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0214: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0215: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0216: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0217: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0218: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0219: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0220: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0221: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0222: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0223: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0224: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0225: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0226: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0227: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0228: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0229: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0230: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0231: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0232: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0233: PROP-011 on bound-composite-formula:H2OH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0234: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0235: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0236: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0237: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0238: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0239: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0240: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0241: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0242: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0243: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0244: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0245: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0246: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0247: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0248: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0249: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0250: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0251: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0252: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0253: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0254: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0255: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0256: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0257: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0258: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0259: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0260: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0261: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0262: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0263: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0264: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0265: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0266: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0267: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0268: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0269: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0270: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0271: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0272: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0273: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0274: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0275: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0276: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0277: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0278: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0279: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0280: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0281: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0282: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0283: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0284: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0285: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0286: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0287: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0288: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0289: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0290: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0291: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0292: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0293: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0294: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0295: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0296: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0297: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0298: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0299: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0300: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0301: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0302: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0303: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0304: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0305: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0306: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0307: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0308: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0309: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0310: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0311: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0312: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0313: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0314: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0315: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0316: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0317: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0318: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0319: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0320: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0321: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0322: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0323: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0324: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0325: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0326: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0327: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0328: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0329: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0330: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0331: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0332: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0333: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0334: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0335: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0336: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0337: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0338: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0339: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0340: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0341: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0342: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0343: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0344: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0345: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0346: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0347: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0348: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0349: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0350: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0351: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0352: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0353: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0354: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0355: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0356: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0357: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0358: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0359: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0360: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0361: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0362: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0363: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0364: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0365: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0366: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0367: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0368: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0369: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0370: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0371: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0372: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0373: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0374: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0375: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0376: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0377: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0378: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0379: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0380: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0381: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0382: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0383: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0384: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0385: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0386: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0387: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0388: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0389: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0390: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0391: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0392: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0393: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0394: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0395: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0396: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0397: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0398: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0399: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0400: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0401: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0402: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0403: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0404: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0405: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0406: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0407: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0408: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0409: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0410: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0411: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0412: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0413: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0414: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0415: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0416: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0417: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0418: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0419: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0420: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0421: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0422: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0423: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0424: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0425: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0426: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0427: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0428: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0429: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0430: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0431: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0432: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0433: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0434: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0435: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0436: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0437: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0438: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0439: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0440: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0441: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0442: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0443: PROP-011 on bound-composite-formula:NH3NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0444: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0445: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0446: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0447: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0448: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0449: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0450: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0451: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0452: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0453: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0454: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0455: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0456: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0457: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0458: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0459: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0460: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0461: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0462: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0463: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0464: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0465: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0466: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0467: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0468: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0469: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0470: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0471: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
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- SFT-CHEM-PROP-014-PROP-011-ROW-0481: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0482: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0483: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0484: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0485: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0486: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
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- SFT-CHEM-PROP-014-PROP-011-ROW-0488: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
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- SFT-CHEM-PROP-014-PROP-011-ROW-0491: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
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- SFT-CHEM-PROP-014-PROP-011-ROW-0499: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0500: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0501: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0502: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0503: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0504: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0505: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0506: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0507: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0508: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0509: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0510: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0511: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0512: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0513: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0514: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0515: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0516: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0517: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0518: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0519: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0520: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0521: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0522: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0523: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0524: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0525: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0526: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0527: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0528: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0529: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0530: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0531: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0532: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0533: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0534: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0535: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0536: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0537: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
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- SFT-CHEM-PROP-014-PROP-011-ROW-0539: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
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- SFT-CHEM-PROP-014-PROP-011-ROW-0541: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0542: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0543: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0544: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0545: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0546: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0547: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0548: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0549: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0550: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0551: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
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- SFT-CHEM-PROP-014-PROP-011-ROW-0557: PROP-011 on bound-composite-formula:H2ONH3; post-seal payload identity True
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- SFT-CHEM-PROP-014-PROP-011-ROW-0569: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
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- SFT-CHEM-PROP-014-PROP-011-ROW-0576: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
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- SFT-CHEM-PROP-014-PROP-011-ROW-0582: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0583: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
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- SFT-CHEM-PROP-014-PROP-011-ROW-0587: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
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- SFT-CHEM-PROP-014-PROP-011-ROW-0589: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0590: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0591: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0592: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0593: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0594: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0595: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0596: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
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- SFT-CHEM-PROP-014-PROP-011-ROW-0599: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0600: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0601: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
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- SFT-CHEM-PROP-014-PROP-011-ROW-0603: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0604: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0605: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
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- SFT-CHEM-PROP-014-PROP-011-ROW-0609: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
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- SFT-CHEM-PROP-014-PROP-011-ROW-0612: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0613: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0614: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0615: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0616: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0617: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0618: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0619: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
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- SFT-CHEM-PROP-014-PROP-011-ROW-0622: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0623: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0624: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0625: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0626: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0627: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0628: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0629: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0630: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0631: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0632: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0633: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0634: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0635: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0636: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0637: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0638: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0639: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0640: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0641: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0642: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0643: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0644: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0645: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0646: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0647: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0648: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0649: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0650: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0651: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0652: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0653: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0654: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0655: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0656: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0657: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0658: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0659: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0660: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0661: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0662: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0663: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0664: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0665: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0666: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0667: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0668: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0669: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0670: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0671: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0672: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0673: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0674: PROP-011 on bound-composite-formula:H2OH2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0675: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0676: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0677: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0678: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0679: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0680: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0681: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0682: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0683: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0684: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0685: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0686: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0687: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0688: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0689: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0690: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0691: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0692: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0693: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0694: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0695: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0696: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0697: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0698: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0699: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0700: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0701: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0702: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0703: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0704: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0705: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0706: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0707: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0708: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0709: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0710: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0711: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0712: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0713: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0714: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0715: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0716: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0717: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0718: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0719: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0720: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0721: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0722: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0723: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0724: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0725: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0726: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0727: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0728: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0729: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0730: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0731: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0732: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0733: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0734: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0735: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0736: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0737: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0738: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0739: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0740: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0741: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0742: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0743: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0744: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0745: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0746: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0747: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0748: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0749: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0750: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0751: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0752: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0753: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0754: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0755: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0756: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0757: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0758: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0759: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0760: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0761: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0762: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0763: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0764: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0765: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0766: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0767: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0768: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0769: PROP-011 on bound-composite-formula:H2OCH3OH; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0770: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0771: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0772: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0773: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0774: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0775: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0776: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0777: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0778: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0779: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0780: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0781: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0782: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0783: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0784: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0785: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0786: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0787: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0788: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0789: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0790: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0791: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0792: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0793: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0794: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0795: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0796: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0797: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0798: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0799: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0800: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0801: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0802: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0803: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0804: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0805: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0806: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0807: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0808: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0809: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0810: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0811: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0812: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0813: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0814: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0815: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0816: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0817: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0818: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0819: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0820: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0821: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0822: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0823: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0824: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0825: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0826: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0827: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0828: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0829: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0830: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0831: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0832: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0833: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0834: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0835: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0836: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0837: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0838: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0839: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0840: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0841: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0842: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0843: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0844: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0845: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0846: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0847: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0848: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0849: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0850: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0851: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0852: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0853: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0854: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0855: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0856: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0857: PROP-011 on bound-composite-formula:CH3OHH2O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0858: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0859: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0860: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0861: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0862: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0863: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0864: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0865: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0866: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0867: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0868: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0869: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0870: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0871: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0872: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0873: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0874: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0875: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0876: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0877: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0878: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0879: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0880: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0881: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0882: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0883: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0884: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0885: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0886: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0887: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0888: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0889: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0890: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0891: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0892: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0893: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0894: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0895: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0896: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0897: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0898: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0899: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0900: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0901: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0902: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0903: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0904: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0905: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0906: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0907: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0908: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0909: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0910: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0911: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0912: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0913: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0914: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0915: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0916: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0917: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0918: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0919: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
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- SFT-CHEM-PROP-014-PROP-011-ROW-0921: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0922: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0923: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0924: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0925: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0926: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0927: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0928: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0929: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0930: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0931: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0932: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0933: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0934: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0935: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0936: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0937: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0938: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0939: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0940: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0941: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0942: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0943: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0944: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0945: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0946: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0947: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0948: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0949: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0950: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0951: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0952: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0953: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0954: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0955: PROP-011 on bound-composite-formula:HCOOHH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0956: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0957: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0958: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0959: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0960: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0961: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0962: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0963: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0964: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0965: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0966: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0967: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0968: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0969: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0970: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0971: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0972: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0973: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0974: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0975: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0976: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0977: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0978: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0979: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0980: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0981: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0982: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0983: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0984: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0985: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0986: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0987: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0988: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0989: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-0990: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0991: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0992: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0993: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0994: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0995: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0996: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0997: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0998: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-0999: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1000: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1001: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1002: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1003: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1004: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1005: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1006: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1007: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1008: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1009: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1010: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1011: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-1012: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1013: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1014: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1015: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1016: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1017: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1018: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1019: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1020: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1021: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1022: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1023: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1024: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1025: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1026: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1027: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1028: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1029: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1030: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1031: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1032: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1033: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-1034: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1035: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1036: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1037: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1038: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1039: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1040: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1041: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1042: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1043: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1044: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1045: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1046: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1047: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1048: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1049: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1050: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1051: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1052: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1053: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1054: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1055: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-1056: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1057: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1058: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1059: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1060: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1061: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1062: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1063: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1064: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1065: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1066: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1067: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1068: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1069: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1070: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1071: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1072: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1073: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1074: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1075: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1076: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1077: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-1078: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1079: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1080: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1081: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1082: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1083: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1084: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1085: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1086: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1087: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1088: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1089: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1090: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1091: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1092: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1093: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1094: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1095: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1096: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1097: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1098: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1099: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-1100: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1101: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1102: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1103: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1104: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1105: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1106: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1107: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1108: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1109: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1110: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1111: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1112: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1113: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1114: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1115: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1116: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1117: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1118: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1119: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1120: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1121: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-1122: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1123: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1124: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1125: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1126: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1127: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1128: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1129: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1130: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1131: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1132: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1133: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1134: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1135: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1136: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1137: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1138: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1139: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1140: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1141: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1142: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1143: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-1144: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1145: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1146: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1147: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1148: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1149: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1150: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1151: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1152: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1153: PROP-011 on bound-composite-formula:H2OHCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1154: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1155: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1156: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1157: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1158: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1159: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1160: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1161: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1162: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1163: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1164: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1165: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-1166: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1167: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1168: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1169: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1170: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1171: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1172: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1173: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1174: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1175: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1176: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1177: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1178: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1179: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1180: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1181: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1182: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1183: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1184: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1185: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1186: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1187: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-1188: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1189: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1190: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1191: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1192: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1193: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1194: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1195: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1196: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1197: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1198: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1199: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1200: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1201: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1202: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1203: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1204: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1205: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1206: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1207: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1208: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1209: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-1210: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1211: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1212: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1213: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1214: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1215: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1216: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1217: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1218: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1219: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1220: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1221: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1222: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1223: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1224: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1225: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1226: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1227: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1228: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1229: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1230: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1231: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-1232: PROP-011 on bound-composite-formula:H2OCH3OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1233: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1234: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1235: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1236: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1237: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1238: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1239: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1240: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1241: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1242: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1243: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1244: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1245: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1246: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1247: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1248: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1249: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1250: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1251: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1252: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1253: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-1254: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1255: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1256: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1257: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1258: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1259: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1260: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1261: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1262: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1263: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1264: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1265: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1266: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1267: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1268: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1269: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1270: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1271: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1272: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1273: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1274: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1275: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-011-ROW-1276: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1277: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1278: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1279: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1280: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1281: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1282: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1283: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1284: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1285: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1286: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1287: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1288: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1289: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1290: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1291: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1292: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1293: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1294: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1295: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1296: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-011-ROW-1297: PROP-011 on bound-composite-formula:HCONH2CN2H4; post-seal payload identity True

- [illegible]

- [illegible]

- [illegible]

- [illegible]

- SFT-CHEM-PROP-014-PROP-012-ROW-0150: PROP-012 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0151: PROP-012 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0152: PROP-012 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0153: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P051-L024-V01; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0154: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P051-L025-V01; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0155: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P052-L026-V01; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0156: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P053-L025-MAG; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0157: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P061-L045-V01; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0158: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P079-L036-V01; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0159: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P079-L036-V02; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0160: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P079-L053-V01; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0161: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P084-L056-V01; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0162: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P084-L058-V01; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0163: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P090-L050-V01; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0164: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P096-L052-V01; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0165: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P096-L052-V02; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0166: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P098-L039-V01; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0167: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P100-L050-V01; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0168: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P104-L034-V01; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0169: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P104-L034-V02; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0170: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P104-L034-V03; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0171: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P113-L035-V01; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0172: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P135-L052-V01; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-012-ROW-0173: PROP-012 on
unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P139-L026-V01; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-012-ROW-0174: PROP-012 on unjoined-source-target:NIST-MOLSPEC-PROP-012-DIATOMIC-PDF-P146-L046-V01; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0001: PROP-013 on exact-formula:MgH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0002: PROP-013 on exact-formula:MgH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0003: PROP-013 on exact-formula:NeH+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0004: PROP-013 on exact-formula:NeH+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0005: PROP-013 on exact-formula:CH2NH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0006: PROP-013 on exact-formula:CH2NH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0007: PROP-013 on exact-formula:DF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0008: PROP-013 on exact-formula:DF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0009: PROP-013 on exact-formula:HF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0010: PROP-013 on exact-formula:HF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0011: PROP-013 on exact-formula:D2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0012: PROP-013 on exact-formula:D2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0013: PROP-013 on exact-formula:HDO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0014: PROP-013 on exact-formula:HDO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0015: PROP-013 on exact-formula:H2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0016: PROP-013 on exact-formula:H2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0017: PROP-013 on exact-formula:CO-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0018: PROP-013 on exact-formula:CO-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0019: PROP-013 on exact-formula:LiOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0020: PROP-013 on exact-formula:LiOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0021: PROP-013 on exact-formula:Br-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0022: PROP-013 on exact-formula:Br-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0023: PROP-013 on exact-formula:HBO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0024: PROP-013 on exact-formula:HBO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0025: PROP-013 on exact-formula:OH-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0026: PROP-013 on exact-formula:OH-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0027: PROP-013 on exact-formula:CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0028: PROP-013 on exact-formula:CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0029: PROP-013 on exact-formula:H2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0030: PROP-013 on exact-formula:H2CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0031: PROP-013 on exact-formula:BeOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0032: PROP-013 on exact-formula:BeOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0033: PROP-013 on exact-formula:DCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0034: PROP-013 on exact-formula:DCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0035: PROP-013 on exact-formula:HCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0036: PROP-013 on exact-formula:HCl; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0037: PROP-013 on exact-formula:C2H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0038: PROP-013 on exact-formula:C2H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0039: PROP-013 on exact-formula:BH4-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0040: PROP-013 on exact-formula:BH4-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0041: PROP-013 on exact-formula:CH4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0042: PROP-013 on exact-formula:CH4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0043: PROP-013 on exact-formula:ND3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0044: PROP-013 on exact-formula:ND3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0045: PROP-013 on exact-formula:NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0046: PROP-013 on exact-formula:NH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0047: PROP-013 on exact-formula:HBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0048: PROP-013 on exact-formula:HBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0049: PROP-013 on exact-formula:D2S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0050: PROP-013 on exact-formula:D2S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0051: PROP-013 on exact-formula:CH3NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0052: PROP-013 on exact-formula:CH3NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0053: PROP-013 on exact-formula:H2S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0054: PROP-013 on exact-formula:H2S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0055: PROP-013 on exact-formula:N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0056: PROP-013 on exact-formula:N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0057: PROP-013 on exact-formula:Xe; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0058: PROP-013 on exact-formula:Xe; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0059: PROP-013 on exact-formula:D2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0060: PROP-013 on exact-formula:D2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0061: PROP-013 on exact-formula:O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0062: PROP-013 on exact-formula:O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0063: PROP-013 on exact-formula:H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0064: PROP-013 on exact-formula:H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0065: PROP-013 on exact-formula:Kr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0066: PROP-013 on exact-formula:Kr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0067: PROP-013 on exact-formula:Ne; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0068: PROP-013 on exact-formula:Ne; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0069: PROP-013 on exact-formula:Ar; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0070: PROP-013 on exact-formula:Ar; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0071: PROP-013 on exact-formula:He; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0072: PROP-013 on exact-formula:He; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0073: PROP-013 on exact-formula:HD; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0074: PROP-013 on exact-formula:HD; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0075: PROP-013 on exact-formula:PH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0076: PROP-013 on exact-formula:PH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0077: PROP-013 on exact-formula:BO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0078: PROP-013 on exact-formula:BO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0079: PROP-013 on exact-formula:HO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0080: PROP-013 on exact-formula:HO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0081: PROP-013 on exact-formula:HI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0082: PROP-013 on exact-formula:HI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0083: PROP-013 on exact-formula:H2Se; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0084: PROP-013 on exact-formula:H2Se; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0085: PROP-013 on exact-formula:SiH4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0086: PROP-013 on exact-formula:SiH4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0087: PROP-013 on exact-formula:B2H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0088: PROP-013 on exact-formula:B2H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0089: PROP-013 on exact-formula:DO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0090: PROP-013 on exact-formula:DO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0091: PROP-013 on exact-formula:OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0092: PROP-013 on exact-formula:OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0093: PROP-013 on exact-formula:HCO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0094: PROP-013 on exact-formula:HCO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0095: PROP-013 on exact-formula:C2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0096: PROP-013 on exact-formula:C2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0097: PROP-013 on exact-formula:Na-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0098: PROP-013 on exact-formula:Na-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0099: PROP-013 on exact-formula:AsH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0100: PROP-013 on exact-formula:AsH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0101: PROP-013 on exact-formula:F; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0102: PROP-013 on exact-formula:F; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0103: PROP-013 on exact-formula:BH3-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0104: PROP-013 on exact-formula:BH3-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0105: PROP-013 on exact-formula:LiO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0106: PROP-013 on exact-formula:LiO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0107: PROP-013 on exact-formula:BH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0108: PROP-013 on exact-formula:BH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0109: PROP-013 on exact-formula:K; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0110: PROP-013 on exact-formula:K; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0111: PROP-013 on exact-formula:GeH4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0112: PROP-013 on exact-formula:GeH4; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0113: PROP-013 on exact-formula:NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0114: PROP-013 on exact-formula:NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0115: PROP-013 on exact-formula:N2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0116: PROP-013 on exact-formula:N2H4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0117: PROP-013 on exact-formula:LiH-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0118: PROP-013 on exact-formula:LiH-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0119: PROP-013 on exact-formula:I; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0120: PROP-013 on exact-formula:I; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0121: PROP-013 on exact-formula:HNO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0122: PROP-013 on exact-formula:HNO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0123: PROP-013 on exact-formula:Na; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0124: PROP-013 on exact-formula:Na; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0125: PROP-013 on exact-formula:Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0126: PROP-013 on exact-formula:Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0127: PROP-013 on exact-formula:PH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0128: PROP-013 on exact-formula:PH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0129: PROP-013 on exact-formula:C2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0130: PROP-013 on exact-formula:C2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0131: PROP-013 on exact-formula:Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0132: PROP-013 on exact-formula:Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0133: PROP-013 on exact-formula:KH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0134: PROP-013 on exact-formula:KH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0135: PROP-013 on exact-formula:NaH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0136: PROP-013 on exact-formula:NaH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0137: PROP-013 on exact-formula:BeH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0138: PROP-013 on exact-formula:BeH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0139: PROP-013 on exact-formula:MgH-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0140: PROP-013 on exact-formula:MgH-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0141: PROP-013 on exact-formula:AlH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0142: PROP-013 on exact-formula:AlH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0143: PROP-013 on exact-formula:Zn; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0144: PROP-013 on exact-formula:Zn; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0145: PROP-013 on exact-formula:HCN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0146: PROP-013 on exact-formula:HCN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0147: PROP-013 on exact-formula:Li-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0148: PROP-013 on exact-formula:Li-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0149: PROP-013 on exact-formula:Li+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0150: PROP-013 on exact-formula:Li+; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0151: PROP-013 on exact-formula:BeO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0152: PROP-013 on exact-formula:BeO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0153: PROP-013 on exact-formula:PH-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0154: PROP-013 on exact-formula:PH-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0155: PROP-013 on exact-formula:DS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0156: PROP-013 on exact-formula:DS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0157: PROP-013 on exact-formula:B+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0158: PROP-013 on exact-formula:B+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0159: PROP-013 on exact-formula:HS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0160: PROP-013 on exact-formula:HS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0161: PROP-013 on exact-formula:LiH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0162: PROP-013 on exact-formula:LiH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0163: PROP-013 on exact-formula:GeH2-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0164: PROP-013 on exact-formula:GeH2-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0165: PROP-013 on exact-formula:Be+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0166: PROP-013 on exact-formula:Be+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0167: PROP-013 on exact-formula:H2-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0168: PROP-013 on exact-formula:H2-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0169: PROP-013 on exact-formula:CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0170: PROP-013 on exact-formula:CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0171: PROP-013 on exact-formula:Mg; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0172: PROP-013 on exact-formula:Mg; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0173: PROP-013 on exact-formula:Na+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0174: PROP-013 on exact-formula:Na+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0175: PROP-013 on exact-formula:Al+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0176: PROP-013 on exact-formula:Al+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0177: PROP-013 on exact-formula:Mg+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0178: PROP-013 on exact-formula:Mg+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0179: PROP-013 on exact-formula:K+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0180: PROP-013 on exact-formula:K+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0181: PROP-013 on exact-formula:K-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0182: PROP-013 on exact-formula:K-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0183: PROP-013 on exact-formula:Ne+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0184: PROP-013 on exact-formula:Ne+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0185: PROP-013 on exact-formula:Li; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0186: PROP-013 on exact-formula:Li; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0187: PROP-013 on exact-formula:Si+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0188: PROP-013 on exact-formula:Si+; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0189: PROP-013 on exact-formula:Ar+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0190: PROP-013 on exact-formula:Ar+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0191: PROP-013 on exact-formula:AsH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0192: PROP-013 on exact-formula:AsH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0193: PROP-013 on exact-formula:Ca; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0194: PROP-013 on exact-formula:Ca; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0195: PROP-013 on exact-formula:Ti+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0196: PROP-013 on exact-formula:Ti+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0197: PROP-013 on exact-formula:ND2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0198: PROP-013 on exact-formula:ND2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0199: PROP-013 on exact-formula:NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0200: PROP-013 on exact-formula:NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0201: PROP-013 on exact-formula:HNC; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0202: PROP-013 on exact-formula:HNC; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0203: PROP-013 on exact-formula:CN-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0204: PROP-013 on exact-formula:CN-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0205: PROP-013 on exact-formula:SiH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0206: PROP-013 on exact-formula:SiH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0207: PROP-013 on exact-formula:N2H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0208: PROP-013 on exact-formula:N2H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0209: PROP-013 on exact-formula:CN+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0210: PROP-013 on exact-formula:CN+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0211: PROP-013 on exact-formula:Li2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0212: PROP-013 on exact-formula:Li2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0213: PROP-013 on exact-formula:H; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0214: PROP-013 on exact-formula:H; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0215: PROP-013 on exact-formula:Cu-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0216: PROP-013 on exact-formula:Cu-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0217: PROP-013 on exact-formula:D; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0218: PROP-013 on exact-formula:D; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0219: PROP-013 on exact-formula:C2H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0220: PROP-013 on exact-formula:C2H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0221: PROP-013 on exact-formula:MgH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0222: PROP-013 on exact-formula:MgH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0223: PROP-013 on exact-formula:AsH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0224: PROP-013 on exact-formula:AsH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0225: PROP-013 on exact-formula:Se; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0226: PROP-013 on exact-formula:Se; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0227: PROP-013 on exact-formula:SiH-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0228: PROP-013 on exact-formula:SiH-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0229: PROP-013 on exact-formula:N2H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0230: PROP-013 on exact-formula:N2H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0231: PROP-013 on exact-formula:O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0232: PROP-013 on exact-formula:O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0233: PROP-013 on exact-formula:AlH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0234: PROP-013 on exact-formula:AlH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0235: PROP-013 on exact-formula:PH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0236: PROP-013 on exact-formula:PH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0237: PROP-013 on exact-formula:BN-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0238: PROP-013 on exact-formula:BN-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0239: PROP-013 on exact-formula:Ga; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0240: PROP-013 on exact-formula:Ga; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0241: PROP-013 on exact-formula:SiH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0242: PROP-013 on exact-formula:SiH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0243: PROP-013 on exact-formula:AlH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0244: PROP-013 on exact-formula:AlH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0245: PROP-013 on exact-formula:S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0246: PROP-013 on exact-formula:S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0247: PROP-013 on exact-formula:Al-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0248: PROP-013 on exact-formula:Al-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0249: PROP-013 on exact-formula:C2H3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0250: PROP-013 on exact-formula:C2H3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0251: PROP-013 on exact-formula:Sn; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0252: PROP-013 on exact-formula:Sn; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0253: PROP-013 on exact-formula:As; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0254: PROP-013 on exact-formula:As; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0255: PROP-013 on exact-formula:Si-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0256: PROP-013 on exact-formula:Si-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0257: PROP-013 on exact-formula:P; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0258: PROP-013 on exact-formula:P; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0259: PROP-013 on exact-formula:BH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0260: PROP-013 on exact-formula:BH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0261: PROP-013 on exact-formula:BeH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0262: PROP-013 on exact-formula:BeH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0263: PROP-013 on exact-formula:Be; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0264: PROP-013 on exact-formula:Be; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0265: PROP-013 on exact-formula:Al; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0266: PROP-013 on exact-formula:Al; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0267: PROP-013 on exact-formula:LiN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0268: PROP-013 on exact-formula:LiN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0269: PROP-013 on exact-formula:Cu; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0270: PROP-013 on exact-formula:Cu; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0271: PROP-013 on exact-formula:NH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0272: PROP-013 on exact-formula:NH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0273: PROP-013 on exact-formula:SiH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0274: PROP-013 on exact-formula:SiH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0275: PROP-013 on exact-formula:Ge; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0276: PROP-013 on exact-formula:Ge; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0277: PROP-013 on exact-formula:H2CC-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0278: PROP-013 on exact-formula:H2CC-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0279: PROP-013 on exact-formula:CH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0280: PROP-013 on exact-formula:CH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0281: PROP-013 on exact-formula:BeN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0282: PROP-013 on exact-formula:BeN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0283: PROP-013 on exact-formula:CH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0284: PROP-013 on exact-formula:CH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0285: PROP-013 on exact-formula:CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0286: PROP-013 on exact-formula:CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0287: PROP-013 on exact-formula:BH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0288: PROP-013 on exact-formula:BH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0289: PROP-013 on exact-formula:Si; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0290: PROP-013 on exact-formula:Si; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0291: PROP-013 on exact-formula:Ti-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0292: PROP-013 on exact-formula:Ti-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0293: PROP-013 on exact-formula:N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0294: PROP-013 on exact-formula:N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0295: PROP-013 on exact-formula:Ti; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0296: PROP-013 on exact-formula:Ti; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0297: PROP-013 on exact-formula:CH-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0298: PROP-013 on exact-formula:CH-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0299: PROP-013 on exact-formula:C2-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0300: PROP-013 on exact-formula:C2-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0301: PROP-013 on exact-formula:V; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0302: PROP-013 on exact-formula:V; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0303: PROP-013 on exact-formula:B-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0304: PROP-013 on exact-formula:B-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0305: PROP-013 on exact-formula:BN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0306: PROP-013 on exact-formula:BN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0307: PROP-013 on exact-formula:B; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0308: PROP-013 on exact-formula:B; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0309: PROP-013 on exact-formula:C2H; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0310: PROP-013 on exact-formula:C2H; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0311: PROP-013 on exact-formula:C-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0312: PROP-013 on exact-formula:C-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0313: PROP-013 on exact-formula:CH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0314: PROP-013 on exact-formula:CH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0315: PROP-013 on exact-formula:Be2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0316: PROP-013 on exact-formula:Be2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0317: PROP-013 on exact-formula:C; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0318: PROP-013 on exact-formula:C; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0319: PROP-013 on exact-formula:BC; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0320: PROP-013 on exact-formula:BC; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0321: PROP-013 on exact-formula:C2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0322: PROP-013 on exact-formula:C2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0323: PROP-013 on exact-formula:B2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0324: PROP-013 on exact-formula:B2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0325: PROP-013 on exact-formula:OH+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0326: PROP-013 on exact-formula:OH+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0327: PROP-013 on exact-formula:HSe-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0328: PROP-013 on exact-formula:HSe-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0329: PROP-013 on exact-formula:HS-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0330: PROP-013 on exact-formula:HS-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0331: PROP-013 on exact-formula:H2+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0332: PROP-013 on exact-formula:H2+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0333: PROP-013 on exact-formula:H+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0334: PROP-013 on exact-formula:H+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0335: PROP-013 on exact-formula:CH+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0336: PROP-013 on exact-formula:CH+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0337: PROP-013 on exact-formula:C+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0338: PROP-013 on exact-formula:C+; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0339: PROP-013 on exact-formula:CH3PH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0340: PROP-013 on exact-formula:CH3PH2; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0341: PROP-013 on exact-formula:CH₂PH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0342: PROP-013 on exact-formula:CH₂PH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0343: PROP-013 on exact-formula:HBS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0344: PROP-013 on exact-formula:HBS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0345: PROP-013 on exact-formula:P₂H₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0346: PROP-013 on exact-formula:P₂H₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0347: PROP-013 on exact-formula:Ar₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0348: PROP-013 on exact-formula:Ar₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0349: PROP-013 on exact-formula:CH₃S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0350: PROP-013 on exact-formula:CH₃S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0351: PROP-013 on exact-formula:HCCL; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0352: PROP-013 on exact-formula:HCCL; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0353: PROP-013 on exact-formula:Al₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0354: PROP-013 on exact-formula:Al₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0355: PROP-013 on exact-formula:SiGe; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0356: PROP-013 on exact-formula:SiGe; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0357: PROP-013 on exact-formula:H₂F₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0358: PROP-013 on exact-formula:H₂F₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0359: PROP-013 on exact-formula:SiH₃F; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0360: PROP-013 on exact-formula:SiH₃F; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0361: PROP-013 on exact-formula:NaF-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0362: PROP-013 on exact-formula:NaF-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0363: PROP-013 on exact-formula:LiF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0364: PROP-013 on exact-formula:LiF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0365: PROP-013 on exact-formula:KF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0366: PROP-013 on exact-formula:KF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0367: PROP-013 on exact-formula:F₂-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0368: PROP-013 on exact-formula:F₂-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0369: PROP-013 on exact-formula:NaF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0370: PROP-013 on exact-formula:NaF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0371: PROP-013 on exact-formula:CaF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0372: PROP-013 on exact-formula:CaF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0373: PROP-013 on exact-formula:AlF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0374: PROP-013 on exact-formula:AlF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0375: PROP-013 on exact-formula:LiCl-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0376: PROP-013 on exact-formula:LiCl-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0377: PROP-013 on exact-formula:NaCl-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0378: PROP-013 on exact-formula:NaCl-; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0379: PROP-013 on exact-formula:KOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0380: PROP-013 on exact-formula:KOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0381: PROP-013 on exact-formula:MgF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0382: PROP-013 on exact-formula:MgF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0383: PROP-013 on exact-formula:Cl₂-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0384: PROP-013 on exact-formula:Cl₂-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0385: PROP-013 on exact-formula:KCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0386: PROP-013 on exact-formula:KCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0387: PROP-013 on exact-formula:MgCl-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0388: PROP-013 on exact-formula:MgCl-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0389: PROP-013 on exact-formula:CH₃OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0390: PROP-013 on exact-formula:CH₃OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0391: PROP-013 on exact-formula:ClF-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0392: PROP-013 on exact-formula:ClF-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0393: PROP-013 on exact-formula:LiCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0394: PROP-013 on exact-formula:LiCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0395: PROP-013 on exact-formula:NaOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0396: PROP-013 on exact-formula:NaOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0397: PROP-013 on exact-formula:NaCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0398: PROP-013 on exact-formula:NaCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0399: PROP-013 on exact-formula:CaOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0400: PROP-013 on exact-formula:CaOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0401: PROP-013 on exact-formula:BeF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0402: PROP-013 on exact-formula:BeF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0403: PROP-013 on exact-formula:SiH₃Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0404: PROP-013 on exact-formula:SiH₃Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0405: PROP-013 on exact-formula:H₂O₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0406: PROP-013 on exact-formula:H₂O₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0407: PROP-013 on exact-formula:PO-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0408: PROP-013 on exact-formula:PO-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0409: PROP-013 on exact-formula:MgO-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0410: PROP-013 on exact-formula:MgO-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0411: PROP-013 on exact-formula:BF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0412: PROP-013 on exact-formula:BF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0413: PROP-013 on exact-formula:CaCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0414: PROP-013 on exact-formula:CaCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0415: PROP-013 on exact-formula:SiO-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0416: PROP-013 on exact-formula:SiO-; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0417: PROP-013 on exact-formula:SiF₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0418: PROP-013 on exact-formula:SiF₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0419: PROP-013 on exact-formula:SiO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0420: PROP-013 on exact-formula:SiO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0421: PROP-013 on exact-formula:HOF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0422: PROP-013 on exact-formula:HOF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0423: PROP-013 on exact-formula:MgOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0424: PROP-013 on exact-formula:MgOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0425: PROP-013 on exact-formula:CH₃Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0426: PROP-013 on exact-formula:CH₃Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0427: PROP-013 on exact-formula:DOCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0428: PROP-013 on exact-formula:DOCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0429: PROP-013 on exact-formula:HOCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0430: PROP-013 on exact-formula:HOCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0431: PROP-013 on exact-formula:GeF₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0432: PROP-013 on exact-formula:GeF₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0433: PROP-013 on exact-formula:HPO₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0434: PROP-013 on exact-formula:HPO₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0435: PROP-013 on exact-formula:MgCl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0436: PROP-013 on exact-formula:MgCl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0437: PROP-013 on exact-formula:PF₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0438: PROP-013 on exact-formula:PF₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0439: PROP-013 on exact-formula:AlCl₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0440: PROP-013 on exact-formula:AlCl₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0441: PROP-013 on exact-formula:ClF₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0442: PROP-013 on exact-formula:ClF₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0443: PROP-013 on exact-formula:NH₂OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0444: PROP-013 on exact-formula:NH₂OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0445: PROP-013 on exact-formula:S₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0446: PROP-013 on exact-formula:S₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0447: PROP-013 on exact-formula:CH₂F₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0448: PROP-013 on exact-formula:CH₂F₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0449: PROP-013 on exact-formula:CH₃SiH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0450: PROP-013 on exact-formula:CH₃SiH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0451: PROP-013 on exact-formula:PO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0452: PROP-013 on exact-formula:PO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0453: PROP-013 on exact-formula:SiF₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0454: PROP-013 on exact-formula:SiF₄; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0455: PROP-013 on exact-formula:CH3SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0456: PROP-013 on exact-formula:CH3SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0457: PROP-013 on exact-formula:AlS-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0458: PROP-013 on exact-formula:AlS-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0459: PROP-013 on exact-formula:CH2OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0460: PROP-013 on exact-formula:CH2OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0461: PROP-013 on exact-formula:CuF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0462: PROP-013 on exact-formula:CuF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0463: PROP-013 on exact-formula:F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0464: PROP-013 on exact-formula:F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0465: PROP-013 on exact-formula:Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0466: PROP-013 on exact-formula:Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0467: PROP-013 on exact-formula:SO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0468: PROP-013 on exact-formula:SO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0469: PROP-013 on exact-formula:SF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0470: PROP-013 on exact-formula:SF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0471: PROP-013 on exact-formula:H2S2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0472: PROP-013 on exact-formula:H2S2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0473: PROP-013 on exact-formula:CH3O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0474: PROP-013 on exact-formula:CH3O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0475: PROP-013 on exact-formula:CaO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0476: PROP-013 on exact-formula:CaO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0477: PROP-013 on exact-formula:TiO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0478: PROP-013 on exact-formula:TiO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0479: PROP-013 on exact-formula:H2CS-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0480: PROP-013 on exact-formula:H2CS-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0481: PROP-013 on exact-formula:BeCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0482: PROP-013 on exact-formula:BeCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0483: PROP-013 on exact-formula:AlO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0484: PROP-013 on exact-formula:AlO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0485: PROP-013 on exact-formula:KO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0486: PROP-013 on exact-formula:KO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0487: PROP-013 on exact-formula:Si2H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0488: PROP-013 on exact-formula:Si2H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0489: PROP-013 on exact-formula:NaO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0490: PROP-013 on exact-formula:NaO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0491: PROP-013 on exact-formula:P2-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0492: PROP-013 on exact-formula:P2-; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0493: PROP-013 on exact-formula:CuCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0494: PROP-013 on exact-formula:CuCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0495: PROP-013 on exact-formula:ClO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0496: PROP-013 on exact-formula:ClO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0497: PROP-013 on exact-formula:Na₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0498: PROP-013 on exact-formula:Na₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0499: PROP-013 on exact-formula:HCF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0500: PROP-013 on exact-formula:HCF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0501: PROP-013 on exact-formula:SiS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0502: PROP-013 on exact-formula:SiS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0503: PROP-013 on exact-formula:FO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0504: PROP-013 on exact-formula:FO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0505: PROP-013 on exact-formula:H₂CS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0506: PROP-013 on exact-formula:H₂CS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0507: PROP-013 on exact-formula:MgS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0508: PROP-013 on exact-formula:MgS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0509: PROP-013 on exact-formula:CaS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0510: PROP-013 on exact-formula:CaS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0511: PROP-013 on exact-formula:K₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0512: PROP-013 on exact-formula:K₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0513: PROP-013 on exact-formula:VO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0514: PROP-013 on exact-formula:VO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0515: PROP-013 on exact-formula:S₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0516: PROP-013 on exact-formula:S₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0517: PROP-013 on exact-formula:PCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0518: PROP-013 on exact-formula:PCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0519: PROP-013 on exact-formula:Na₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0520: PROP-013 on exact-formula:Na₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0521: PROP-013 on exact-formula:SiCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0522: PROP-013 on exact-formula:SiCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0523: PROP-013 on exact-formula:P₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0524: PROP-013 on exact-formula:P₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0525: PROP-013 on exact-formula:PS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0526: PROP-013 on exact-formula:PS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0527: PROP-013 on exact-formula:MgO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0528: PROP-013 on exact-formula:MgO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0529: PROP-013 on exact-formula:SCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0530: PROP-013 on exact-formula:SCl; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0531: PROP-013 on exact-formula:PN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0532: PROP-013 on exact-formula:PN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0533: PROP-013 on exact-formula:BCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0534: PROP-013 on exact-formula:BCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0535: PROP-013 on exact-formula:NaLi; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0536: PROP-013 on exact-formula:NaLi; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0537: PROP-013 on exact-formula:HCP; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0538: PROP-013 on exact-formula:HCP; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0539: PROP-013 on exact-formula:As2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0540: PROP-013 on exact-formula:As2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0541: PROP-013 on exact-formula:AlS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0542: PROP-013 on exact-formula:AlS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0543: PROP-013 on exact-formula:Nf; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0544: PROP-013 on exact-formula:Nf; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0545: PROP-013 on exact-formula:BS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0546: PROP-013 on exact-formula:BS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0547: PROP-013 on exact-formula:Si2-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0548: PROP-013 on exact-formula:Si2-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0549: PROP-013 on exact-formula:CS-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0550: PROP-013 on exact-formula:CS-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0551: PROP-013 on exact-formula:CF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0552: PROP-013 on exact-formula:CF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0553: PROP-013 on exact-formula:BeS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0554: PROP-013 on exact-formula:BeS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0555: PROP-013 on exact-formula:NS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0556: PROP-013 on exact-formula:NS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0557: PROP-013 on exact-formula:CS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0558: PROP-013 on exact-formula:CS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0559: PROP-013 on exact-formula:Mg2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0560: PROP-013 on exact-formula:Mg2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0561: PROP-013 on exact-formula:MgN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0562: PROP-013 on exact-formula:MgN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0563: PROP-013 on exact-formula:HCS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0564: PROP-013 on exact-formula:HCS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0565: PROP-013 on exact-formula:CuO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0566: PROP-013 on exact-formula:CuO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0567: PROP-013 on exact-formula:Ca2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0568: PROP-013 on exact-formula:Ca2; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0569: PROP-013 on exact-formula:Al₂⁺; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0570: PROP-013 on exact-formula:Al₂⁺; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0571: PROP-013 on exact-formula:GaP; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0572: PROP-013 on exact-formula:GaP; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0573: PROP-013 on exact-formula:GaP; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0574: PROP-013 on exact-formula:GaP; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0575: PROP-013 on exact-formula:SiN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0576: PROP-013 on exact-formula:SiN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0577: PROP-013 on exact-formula:AlN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0578: PROP-013 on exact-formula:AlN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0579: PROP-013 on exact-formula:CCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0580: PROP-013 on exact-formula:CCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0581: PROP-013 on exact-formula:Cu₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0582: PROP-013 on exact-formula:Cu₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0583: PROP-013 on exact-formula:CP; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0584: PROP-013 on exact-formula:CP; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0585: PROP-013 on exact-formula:Si₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0586: PROP-013 on exact-formula:Si₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0587: PROP-013 on exact-formula:AlC; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0588: PROP-013 on exact-formula:AlC; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0589: PROP-013 on exact-formula:BSi; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0590: PROP-013 on exact-formula:BSi; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0591: PROP-013 on exact-formula:SiF⁺; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0592: PROP-013 on exact-formula:SiF⁺; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0593: PROP-013 on exact-formula:SiC; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0594: PROP-013 on exact-formula:SiC; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0595: PROP-013 on exact-formula:ClO⁻; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0596: PROP-013 on exact-formula:ClO⁻; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0597: PROP-013 on exact-formula:HOBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0598: PROP-013 on exact-formula:HOBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0599: PROP-013 on exact-formula:CCl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0600: PROP-013 on exact-formula:CCl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0601: PROP-013 on exact-formula:BeF₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0602: PROP-013 on exact-formula:BeF₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0603: PROP-013 on exact-formula:SiH₂F₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0604: PROP-013 on exact-formula:SiH₂F₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0605: PROP-013 on exact-formula:BHF₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0606: PROP-013 on exact-formula:BHF₂; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0607: PROP-013 on exact-formula:MgF₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0608: PROP-013 on exact-formula:MgF₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0609: PROP-013 on exact-formula:Be(OH)₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0610: PROP-013 on exact-formula:Be(OH)₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0611: PROP-013 on exact-formula:FBO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0612: PROP-013 on exact-formula:FBO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0613: PROP-013 on exact-formula:BF₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0614: PROP-013 on exact-formula:BF₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0615: PROP-013 on exact-formula:SiF₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0616: PROP-013 on exact-formula:SiF₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0617: PROP-013 on exact-formula:PF₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0618: PROP-013 on exact-formula:PF₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0619: PROP-013 on exact-formula:CH₂F₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0620: PROP-013 on exact-formula:CH₂F₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0621: PROP-013 on exact-formula:MgCl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0622: PROP-013 on exact-formula:MgCl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0623: PROP-013 on exact-formula:CO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0624: PROP-013 on exact-formula:CO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0625: PROP-013 on exact-formula:HCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0626: PROP-013 on exact-formula:HCOOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0627: PROP-013 on exact-formula:HFCO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0628: PROP-013 on exact-formula:HFCO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0629: PROP-013 on exact-formula:BeCl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0630: PROP-013 on exact-formula:BeCl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0631: PROP-013 on exact-formula:SiO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0632: PROP-013 on exact-formula:SiO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0633: PROP-013 on exact-formula:SiH₂Cl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0634: PROP-013 on exact-formula:SiH₂Cl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0635: PROP-013 on exact-formula:BO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0636: PROP-013 on exact-formula:BO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0637: PROP-013 on exact-formula:SO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0638: PROP-013 on exact-formula:SO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0639: PROP-013 on exact-formula:SF₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0640: PROP-013 on exact-formula:SF₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0641: PROP-013 on exact-formula:PO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0642: PROP-013 on exact-formula:PO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0643: PROP-013 on exact-formula:CH₂FCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0644: PROP-013 on exact-formula:CH₂FCl; post-seal payload identity True

- [illegible]

- SFT-CHEM-PROP-014-PROP-013-ROW-0683: PROP-013 on exact-formula:CH₂CHOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0684: PROP-013 on exact-formula:CH₂CHOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0685: PROP-013 on exact-formula:KI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0686: PROP-013 on exact-formula:KI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0687: PROP-013 on exact-formula:C₃H₈; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0688: PROP-013 on exact-formula:C₃H₈; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0689: PROP-013 on exact-formula:H₂NCO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0690: PROP-013 on exact-formula:H₂NCO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0691: PROP-013 on exact-formula:CH₂Cl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0692: PROP-013 on exact-formula:CH₂Cl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0693: PROP-013 on exact-formula:IF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0694: PROP-013 on exact-formula:IF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0695: PROP-013 on exact-formula:LiI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0696: PROP-013 on exact-formula:LiI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0697: PROP-013 on exact-formula:HNO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0698: PROP-013 on exact-formula:HNO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0699: PROP-013 on exact-formula:SiH₃Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0700: PROP-013 on exact-formula:SiH₃Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0701: PROP-013 on exact-formula:FNO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0702: PROP-013 on exact-formula:FNO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0703: PROP-013 on exact-formula:ClCO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0704: PROP-013 on exact-formula:ClCO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0705: PROP-013 on exact-formula:BrF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0706: PROP-013 on exact-formula:BrF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0707: PROP-013 on exact-formula:S₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0708: PROP-013 on exact-formula:S₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0709: PROP-013 on exact-formula:SSO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0710: PROP-013 on exact-formula:SSO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0711: PROP-013 on exact-formula:C₂H₄O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0712: PROP-013 on exact-formula:C₂H₄O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0713: PROP-013 on exact-formula:CaBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0714: PROP-013 on exact-formula:CaBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0715: PROP-013 on exact-formula:CH₂CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0716: PROP-013 on exact-formula:CH₂CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0717: PROP-013 on exact-formula:CH₃CH₂NH₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0718: PROP-013 on exact-formula:CH₃CH₂NH₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0719: PROP-013 on exact-formula:CH₃CH₂SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0720: PROP-013 on exact-formula:CH₃CH₂SH; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0721: PROP-013 on exact-formula:Se2-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0722: PROP-013 on exact-formula:Se2-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0723: PROP-013 on exact-formula:CH3SCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0724: PROP-013 on exact-formula:CH3SCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0725: PROP-013 on exact-formula:CH3Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0726: PROP-013 on exact-formula:CH3Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0727: PROP-013 on exact-formula:CH3NHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0728: PROP-013 on exact-formula:CH3NHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0729: PROP-013 on exact-formula:SCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0730: PROP-013 on exact-formula:SCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0731: PROP-013 on exact-formula:CH3CH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0732: PROP-013 on exact-formula:CH3CH2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0733: PROP-013 on exact-formula:CH3CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0734: PROP-013 on exact-formula:CH3CO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0735: PROP-013 on exact-formula:SiH3I; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0736: PROP-013 on exact-formula:SiH3I; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0737: PROP-013 on exact-formula:BrCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0738: PROP-013 on exact-formula:BrCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0739: PROP-013 on exact-formula:CH3I; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0740: PROP-013 on exact-formula:CH3I; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0741: PROP-013 on exact-formula:AlBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0742: PROP-013 on exact-formula:AlBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0743: PROP-013 on exact-formula:ICl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0744: PROP-013 on exact-formula:ICl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0745: PROP-013 on exact-formula:CH2CHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0746: PROP-013 on exact-formula:CH2CHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0747: PROP-013 on exact-formula:C2H3Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0748: PROP-013 on exact-formula:C2H3Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0749: PROP-013 on exact-formula:CH3CHNH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0750: PROP-013 on exact-formula:CH3CHNH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0751: PROP-013 on exact-formula:F2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0752: PROP-013 on exact-formula:F2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0753: PROP-013 on exact-formula:FOO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0754: PROP-013 on exact-formula:FOO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0755: PROP-013 on exact-formula:HSSSH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0756: PROP-013 on exact-formula:HSSSH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0757: PROP-013 on exact-formula:Br2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0758: PROP-013 on exact-formula:Br2; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0759: PROP-013 on exact-formula:NO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0760: PROP-013 on exact-formula:NO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0761: PROP-013 on exact-formula:FCN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0762: PROP-013 on exact-formula:FCN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0763: PROP-013 on exact-formula:NF₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0764: PROP-013 on exact-formula:NF₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0765: PROP-013 on exact-formula:IBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0766: PROP-013 on exact-formula:IBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0767: PROP-013 on exact-formula:HCCOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0768: PROP-013 on exact-formula:HCCOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0769: PROP-013 on exact-formula:CH₂NCH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0770: PROP-013 on exact-formula:CH₂NCH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0771: PROP-013 on exact-formula:CH₃CHS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0772: PROP-013 on exact-formula:CH₃CHS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0773: PROP-013 on exact-formula:ClNO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0774: PROP-013 on exact-formula:ClNO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0775: PROP-013 on exact-formula:C₃H₆; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0776: PROP-013 on exact-formula:C₃H₆; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0777: PROP-013 on exact-formula:SeO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0778: PROP-013 on exact-formula:SeO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0779: PROP-013 on exact-formula:I₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0780: PROP-013 on exact-formula:I₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0781: PROP-013 on exact-formula:CH₃CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0782: PROP-013 on exact-formula:CH₃CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0783: PROP-013 on exact-formula:AlI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0784: PROP-013 on exact-formula:AlI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0785: PROP-013 on exact-formula:ClS₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0786: PROP-013 on exact-formula:ClS₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0787: PROP-013 on exact-formula:Cl₂O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0788: PROP-013 on exact-formula:Cl₂O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0789: PROP-013 on exact-formula:N₂O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0790: PROP-013 on exact-formula:N₂O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0791: PROP-013 on exact-formula:C₂H₄S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0792: PROP-013 on exact-formula:C₂H₄S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0793: PROP-013 on exact-formula:CCl₂-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0794: PROP-013 on exact-formula:CCl₂-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0795: PROP-013 on exact-formula:CH₃CHCH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0796: PROP-013 on exact-formula:CH₃CHCH₃; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0797: PROP-013 on exact-formula:CS₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0798: PROP-013 on exact-formula:CS₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0799: PROP-013 on exact-formula:NaCN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0800: PROP-013 on exact-formula:NaCN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0801: PROP-013 on exact-formula:OCIO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0802: PROP-013 on exact-formula:OCIO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0803: PROP-013 on exact-formula:ClOO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0804: PROP-013 on exact-formula:ClOO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0805: PROP-013 on exact-formula:C₃H₇; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0806: PROP-013 on exact-formula:C₃H₇; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0807: PROP-013 on exact-formula:CS₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0808: PROP-013 on exact-formula:CS₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0809: PROP-013 on exact-formula:HCCF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0810: PROP-013 on exact-formula:HCCF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0811: PROP-013 on exact-formula:C₂H₅N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0812: PROP-013 on exact-formula:C₂H₅N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0813: PROP-013 on exact-formula:NH₂CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0814: PROP-013 on exact-formula:NH₂CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0815: PROP-013 on exact-formula:ClCN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0816: PROP-013 on exact-formula:ClCN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0817: PROP-013 on exact-formula:Se₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0818: PROP-013 on exact-formula:Se₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0819: PROP-013 on exact-formula:O₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0820: PROP-013 on exact-formula:O₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0821: PROP-013 on exact-formula:S₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0822: PROP-013 on exact-formula:S₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0823: PROP-013 on exact-formula:PBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0824: PROP-013 on exact-formula:PBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0825: PROP-013 on exact-formula:CH₃NC; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0826: PROP-013 on exact-formula:CH₃NC; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0827: PROP-013 on exact-formula:C₃H₅; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0828: PROP-013 on exact-formula:C₃H₅; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0829: PROP-013 on exact-formula:CH₃CCCH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0830: PROP-013 on exact-formula:CH₃CCCH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0831: PROP-013 on exact-formula:CH₂CCCH₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0832: PROP-013 on exact-formula:CH₂CCCH₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0833: PROP-013 on exact-formula:CH₂NN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0834: PROP-013 on exact-formula:CH₂NN; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0835: PROP-013 on exact-formula:HCCCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0836: PROP-013 on exact-formula:HCCCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0837: PROP-013 on exact-formula:BBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0838: PROP-013 on exact-formula:BBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0839: PROP-013 on exact-formula:SiBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0840: PROP-013 on exact-formula:SiBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0841: PROP-013 on exact-formula:CF₂⁺; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0842: PROP-013 on exact-formula:CF₂⁺; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0843: PROP-013 on exact-formula:C₃H₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0844: PROP-013 on exact-formula:C₃H₄; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0845: PROP-013 on exact-formula:HN₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0846: PROP-013 on exact-formula:HN₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0847: PROP-013 on exact-formula:NBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0848: PROP-013 on exact-formula:NBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0849: PROP-013 on exact-formula:C₃H₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0850: PROP-013 on exact-formula:C₃H₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0851: PROP-013 on exact-formula:N₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0852: PROP-013 on exact-formula:N₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0853: PROP-013 on exact-formula:CBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0854: PROP-013 on exact-formula:CBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0855: PROP-013 on exact-formula:CNN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0856: PROP-013 on exact-formula:CNN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0857: PROP-013 on exact-formula:SiC₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0858: PROP-013 on exact-formula:SiC₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0859: PROP-013 on exact-formula:C₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0860: PROP-013 on exact-formula:C₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0861: PROP-013 on exact-formula:CHCl₂⁺; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0862: PROP-013 on exact-formula:CHCl₂⁺; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0863: PROP-013 on exact-formula:C₃H₅⁺; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0864: PROP-013 on exact-formula:C₃H₅⁺; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0865: PROP-013 on exact-formula:C₃H₃⁺; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0866: PROP-013 on exact-formula:C₃H₃⁺; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0867: PROP-013 on exact-formula:BrO⁺; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0868: PROP-013 on exact-formula:BrO⁺; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0869: PROP-013 on exact-formula:H₂CCCCCH₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0870: PROP-013 on exact-formula:H₂CCCCCH₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0871: PROP-013 on exact-formula:AlF₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0872: PROP-013 on exact-formula:AlF₃; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0873: PROP-013 on exact-formula:SiHF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0874: PROP-013 on exact-formula:SiHF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0875: PROP-013 on exact-formula:BF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0876: PROP-013 on exact-formula:BF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0877: PROP-013 on exact-formula:SiF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0878: PROP-013 on exact-formula:SiF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0879: PROP-013 on exact-formula:AlF2Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0880: PROP-013 on exact-formula:AlF2Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0881: PROP-013 on exact-formula:H3BO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0882: PROP-013 on exact-formula:H3BO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0883: PROP-013 on exact-formula:PF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0884: PROP-013 on exact-formula:PF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0885: PROP-013 on exact-formula:Li2F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0886: PROP-013 on exact-formula:Li2F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0887: PROP-013 on exact-formula:BClF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0888: PROP-013 on exact-formula:BClF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0889: PROP-013 on exact-formula:Na2F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0890: PROP-013 on exact-formula:Na2F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0891: PROP-013 on exact-formula:AlFCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0892: PROP-013 on exact-formula:AlFCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0893: PROP-013 on exact-formula:CaF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0894: PROP-013 on exact-formula:CaF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0895: PROP-013 on exact-formula:CHF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0896: PROP-013 on exact-formula:CHF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0897: PROP-013 on exact-formula:CF2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0898: PROP-013 on exact-formula:CF2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0899: PROP-013 on exact-formula:AlCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0900: PROP-013 on exact-formula:AlCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0901: PROP-013 on exact-formula:Na2Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0902: PROP-013 on exact-formula:Na2Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0903: PROP-013 on exact-formula:F2SO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0904: PROP-013 on exact-formula:F2SO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0905: PROP-013 on exact-formula:CH3CHF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0906: PROP-013 on exact-formula:CH3CHF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0907: PROP-013 on exact-formula:SiHCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0908: PROP-013 on exact-formula:SiHCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0909: PROP-013 on exact-formula:CaCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0910: PROP-013 on exact-formula:CaCl2; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0911: PROP-013 on exact-formula:CHF₂Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0912: PROP-013 on exact-formula:CHF₂Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0913: PROP-013 on exact-formula:CF₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0914: PROP-013 on exact-formula:CF₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0915: PROP-013 on exact-formula:B₂O₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0916: PROP-013 on exact-formula:B₂O₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0917: PROP-013 on exact-formula:SF₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0918: PROP-013 on exact-formula:SF₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0919: PROP-013 on exact-formula:CH₃COF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0920: PROP-013 on exact-formula:CH₃COF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0921: PROP-013 on exact-formula:CH₃COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0922: PROP-013 on exact-formula:CH₃COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0923: PROP-013 on exact-formula:COFCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0924: PROP-013 on exact-formula:COFCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0925: PROP-013 on exact-formula:BCl₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0926: PROP-013 on exact-formula:BCl₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0927: PROP-013 on exact-formula:S₂F₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0928: PROP-013 on exact-formula:S₂F₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0929: PROP-013 on exact-formula:SO₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0930: PROP-013 on exact-formula:SO₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0931: PROP-013 on exact-formula:SiCl₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0932: PROP-013 on exact-formula:SiCl₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0933: PROP-013 on exact-formula:C₂H₆O₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0934: PROP-013 on exact-formula:C₂H₆O₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0935: PROP-013 on exact-formula:CaBr₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0936: PROP-013 on exact-formula:CaBr₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0937: PROP-013 on exact-formula:CH₃OCHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0938: PROP-013 on exact-formula:CH₃OCHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0939: PROP-013 on exact-formula:CH₂CF₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0940: PROP-013 on exact-formula:CH₂CF₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0941: PROP-013 on exact-formula:CH₃CHFCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0942: PROP-013 on exact-formula:CH₃CHFCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0943: PROP-013 on exact-formula:TiO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0944: PROP-013 on exact-formula:TiO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0945: PROP-013 on exact-formula:PCl₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0946: PROP-013 on exact-formula:PCl₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0947: PROP-013 on exact-formula:CHFCl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0948: PROP-013 on exact-formula:CHFCl₂; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0949: PROP-013 on exact-formula:CH3CHOHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0950: PROP-013 on exact-formula:CH3CHOHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0951: PROP-013 on exact-formula:CF2Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0952: PROP-013 on exact-formula:CF2Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0953: PROP-013 on exact-formula:CuF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0954: PROP-013 on exact-formula:CuF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0955: PROP-013 on exact-formula:C3H7OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0956: PROP-013 on exact-formula:C3H7OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0957: PROP-013 on exact-formula:CH3COCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0958: PROP-013 on exact-formula:CH3COCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0959: PROP-013 on exact-formula:Li2O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0960: PROP-013 on exact-formula:Li2O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0961: PROP-013 on exact-formula:CH3CONH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0962: PROP-013 on exact-formula:CH3CONH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0963: PROP-013 on exact-formula:NH2CONH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0964: PROP-013 on exact-formula:NH2CONH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0965: PROP-013 on exact-formula:BeBr2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0966: PROP-013 on exact-formula:BeBr2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0967: PROP-013 on exact-formula:CCl2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0968: PROP-013 on exact-formula:CCl2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0969: PROP-013 on exact-formula:CH3COCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0970: PROP-013 on exact-formula:CH3COCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0971: PROP-013 on exact-formula:CH3OC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0972: PROP-013 on exact-formula:CH3OC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0973: PROP-013 on exact-formula:C2H2O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0974: PROP-013 on exact-formula:C2H2O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0975: PROP-013 on exact-formula:HOCH2CH2NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0976: PROP-013 on exact-formula:HOCH2CH2NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0977: PROP-013 on exact-formula:CH3CH2CHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0978: PROP-013 on exact-formula:CH3CH2CHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0979: PROP-013 on exact-formula:CH3CHCHF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0980: PROP-013 on exact-formula:CH3CHCHF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0981: PROP-013 on exact-formula:GeCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0982: PROP-013 on exact-formula:GeCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0983: PROP-013 on exact-formula:ClF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0984: PROP-013 on exact-formula:ClF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0985: PROP-013 on exact-formula:CH3SOCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0986: PROP-013 on exact-formula:CH3SOCH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-0987: PROP-013 on exact-formula:CH₃CHClCH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0988: PROP-013 on exact-formula:CH₃CHClCH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0989: PROP-013 on exact-formula:CH₃CH(CH₃)CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0990: PROP-013 on exact-formula:CH₃CH(CH₃)CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0991: PROP-013 on exact-formula:HNO₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0992: PROP-013 on exact-formula:HNO₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0993: PROP-013 on exact-formula:CH₃CHCl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0994: PROP-013 on exact-formula:CH₃CHCl₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0995: PROP-013 on exact-formula:NF₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0996: PROP-013 on exact-formula:NF₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0997: PROP-013 on exact-formula:CH₂ClCH₂Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0998: PROP-013 on exact-formula:CH₂ClCH₂Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-0999: PROP-013 on exact-formula:CH₂ClCH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1000: PROP-013 on exact-formula:CH₂ClCH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1001: PROP-013 on exact-formula:CH₃OOCH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1002: PROP-013 on exact-formula:CH₃OOCH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1003: PROP-013 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1004: PROP-013 on exact-formula:CH₃CH₂CH₂CH₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1005: PROP-013 on exact-formula:C₃H₆O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1006: PROP-013 on exact-formula:C₃H₆O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1007: PROP-013 on exact-formula:CH₃CH₂Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1008: PROP-013 on exact-formula:CH₃CH₂Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1009: PROP-013 on exact-formula:SeO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1010: PROP-013 on exact-formula:SeO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1011: PROP-013 on exact-formula:FNO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1012: PROP-013 on exact-formula:FNO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1013: PROP-013 on exact-formula:XeF₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1014: PROP-013 on exact-formula:XeF₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1015: PROP-013 on exact-formula:GeO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1016: PROP-013 on exact-formula:GeO₂; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1017: PROP-013 on exact-formula:CHCl₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1018: PROP-013 on exact-formula:CHCl₃; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1019: PROP-013 on exact-formula:C₃H₆O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1020: PROP-013 on exact-formula:C₃H₆O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1021: PROP-013 on exact-formula:C₃H₄O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-1022: PROP-013 on exact-formula:C3H4O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1023: PROP-013 on exact-formula:CH2CHCHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1024: PROP-013 on exact-formula:CH2CHCHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1025: PROP-013 on exact-formula:CH3CH(NH2)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1026: PROP-013 on exact-formula:CH3CH(NH2)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1027: PROP-013 on exact-formula:C3H6O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1028: PROP-013 on exact-formula:C3H6O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1029: PROP-013 on exact-formula:CH3CHSHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1030: PROP-013 on exact-formula:CH3CHSHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1031: PROP-013 on exact-formula:CH3NO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1032: PROP-013 on exact-formula:CH3NO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1033: PROP-013 on exact-formula:NH2CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1034: PROP-013 on exact-formula:NH2CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1035: PROP-013 on exact-formula:C3H7SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1036: PROP-013 on exact-formula:C3H7SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1037: PROP-013 on exact-formula:CH3ONO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1038: PROP-013 on exact-formula:CH3ONO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1039: PROP-013 on exact-formula:C2H5Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1040: PROP-013 on exact-formula:C2H5Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1041: PROP-013 on exact-formula:CH3SCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1042: PROP-013 on exact-formula:CH3SCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1043: PROP-013 on exact-formula:C3H6O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1044: PROP-013 on exact-formula:C3H6O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1045: PROP-013 on exact-formula:CH3SSCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1046: PROP-013 on exact-formula:CH3SSCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1047: PROP-013 on exact-formula:N(CH3)3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1048: PROP-013 on exact-formula:N(CH3)3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1049: PROP-013 on exact-formula:CH3CHNOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1050: PROP-013 on exact-formula:CH3CHNOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1051: PROP-013 on exact-formula:CH2BrCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1052: PROP-013 on exact-formula:CH2BrCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1053: PROP-013 on exact-formula:CH2C(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1054: PROP-013 on exact-formula:CH2C(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1055: PROP-013 on exact-formula:C2H8N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1056: PROP-013 on exact-formula:C2H8N2; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-1057: PROP-013 on exact-formula:ClSSCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1058: PROP-013 on exact-formula:ClSSCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1059: PROP-013 on exact-formula:C3H5Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1060: PROP-013 on exact-formula:C3H5Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1061: PROP-013 on exact-formula:CH3CHCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1062: PROP-013 on exact-formula:CH3CHCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1063: PROP-013 on exact-formula:C3H5Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1064: PROP-013 on exact-formula:C3H5Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1065: PROP-013 on exact-formula:CH2SHCH2SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1066: PROP-013 on exact-formula:CH2SHCH2SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1067: PROP-013 on exact-formula:CH3CHCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1068: PROP-013 on exact-formula:CH3CHCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1069: PROP-013 on exact-formula:C2H5I; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1070: PROP-013 on exact-formula:C2H5I; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1071: PROP-013 on exact-formula:C3H5Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1072: PROP-013 on exact-formula:C3H5Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1073: PROP-013 on exact-formula:CHClCHCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1074: PROP-013 on exact-formula:CHClCHCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1075: PROP-013 on exact-formula:CHClCHCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1076: PROP-013 on exact-formula:CHClCHCl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1077: PROP-013 on exact-formula:CH2CHCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1078: PROP-013 on exact-formula:CH2CHCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1079: PROP-013 on exact-formula:CH2CCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1080: PROP-013 on exact-formula:CH2CCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1081: PROP-013 on exact-formula:CH2Br2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1082: PROP-013 on exact-formula:CH2Br2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1083: PROP-013 on exact-formula:CH2BrF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1084: PROP-013 on exact-formula:CH2BrF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1085: PROP-013 on exact-formula:ClNO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1086: PROP-013 on exact-formula:ClNO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1087: PROP-013 on exact-formula:CH3CSNH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1088: PROP-013 on exact-formula:CH3CSNH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1089: PROP-013 on exact-formula:CH2FCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1090: PROP-013 on exact-formula:CH2FCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1091: PROP-013 on exact-formula:C3H4O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1092: PROP-013 on exact-formula:C3H4O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1093: PROP-013 on exact-formula:C2F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1094: PROP-013 on exact-formula:C2F2; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-1095: PROP-013 on exact-formula: NH_2CSNH_2 ; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1096: PROP-013 on exact-formula: NH_2CSNH_2 ; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1097: PROP-013 on exact-formula: C_4H_8 ; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1098: PROP-013 on exact-formula: C_4H_8 ; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1099: PROP-013 on exact-formula: $\text{C}_3\text{H}_6\text{S}$; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1100: PROP-013 on exact-formula: $\text{C}_3\text{H}_6\text{S}$; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1101: PROP-013 on exact-formula: $\text{C}_2\text{H}_5\text{CN}$; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1102: PROP-013 on exact-formula: $\text{C}_2\text{H}_5\text{CN}$; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1103: PROP-013 on exact-formula: $\text{C}(\text{CH}_3)_3$; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1104: PROP-013 on exact-formula: $\text{C}(\text{CH}_3)_3$; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1105: PROP-013 on exact-formula: CH_2BrI ; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1106: PROP-013 on exact-formula: CH_2BrI ; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1107: PROP-013 on exact-formula: $\text{Zn}(\text{CH}_3)_2$; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1108: PROP-013 on exact-formula: $\text{Zn}(\text{CH}_3)_2$; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1109: PROP-013 on exact-formula: P_4 ; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1110: PROP-013 on exact-formula: P_4 ; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1111: PROP-013 on exact-formula: $\text{C}_3\text{H}_6\text{S}$; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1112: PROP-013 on exact-formula: $\text{C}_3\text{H}_6\text{S}$; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1113: PROP-013 on exact-formula: N_2F_2 ; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1114: PROP-013 on exact-formula: N_2F_2 ; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1115: PROP-013 on exact-formula: B_4H_{10} ; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1116: PROP-013 on exact-formula: B_4H_{10} ; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1117: PROP-013 on exact-formula: N_2F_2 ; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1118: PROP-013 on exact-formula: N_2F_2 ; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1119: PROP-013 on exact-formula: $\text{CH}_3\text{CHCH}_2\text{CH}_3$; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1120: PROP-013 on exact-formula: $\text{CH}_3\text{CHCH}_2\text{CH}_3$; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1121: PROP-013 on exact-formula: CCl_3 ; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1122: PROP-013 on exact-formula: CCl_3 ; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1123: PROP-013 on exact-formula: $\text{CH}_2\text{CH}(\text{CH}_3)_2$; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1124: PROP-013 on exact-formula: $\text{CH}_2\text{CH}(\text{CH}_3)_2$; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1125: PROP-013 on exact-formula: NO_3 ; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1126: PROP-013 on exact-formula: NO_3 ; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1127: PROP-013 on exact-formula: $\text{C}_3\text{H}_7\text{N}$; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1128: PROP-013 on exact-formula: $\text{C}_3\text{H}_7\text{N}$; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1129: PROP-013 on exact-formula: $\text{C}_2\text{H}_3\text{Br}$; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1130: PROP-013 on exact-formula: $\text{C}_2\text{H}_3\text{Br}$; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1131: PROP-013 on exact-formula: KCN ; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1132: PROP-013 on exact-formula: KCN ; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-1133: PROP-013 on exact-formula:BrNO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1134: PROP-013 on exact-formula:BrNO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1135: PROP-013 on exact-formula:CH2CHCHCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1136: PROP-013 on exact-formula:CH2CHCHCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1137: PROP-013 on exact-formula:INO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1138: PROP-013 on exact-formula:INO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1139: PROP-013 on exact-formula:CH2I2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1140: PROP-013 on exact-formula:CH2I2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1141: PROP-013 on exact-formula:C2H3I; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1142: PROP-013 on exact-formula:C2H3I; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1143: PROP-013 on exact-formula:CH3CCCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1144: PROP-013 on exact-formula:CH3CCCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1145: PROP-013 on exact-formula:C4H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1146: PROP-013 on exact-formula:C4H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1147: PROP-013 on exact-formula:CH2CCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1148: PROP-013 on exact-formula:CH2CCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1149: PROP-013 on exact-formula:CHCCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1150: PROP-013 on exact-formula:CHCCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1151: PROP-013 on exact-formula:C3H3N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1152: PROP-013 on exact-formula:C3H3N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1153: PROP-013 on exact-formula:BrCN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1154: PROP-013 on exact-formula:BrCN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1155: PROP-013 on exact-formula:C4H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1156: PROP-013 on exact-formula:C4H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1157: PROP-013 on exact-formula:C4H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1158: PROP-013 on exact-formula:C4H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1159: PROP-013 on exact-formula:ICN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1160: PROP-013 on exact-formula:ICN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1161: PROP-013 on exact-formula:C2Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1162: PROP-013 on exact-formula:C2Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1163: PROP-013 on exact-formula:C4H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1164: PROP-013 on exact-formula:C4H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1165: PROP-013 on exact-formula:HCCBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1166: PROP-013 on exact-formula:HCCBr; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1167: PROP-013 on exact-formula:C2H3CCH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1168: PROP-013 on exact-formula:C2H3CCH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1169: PROP-013 on exact-formula:C2N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1170: PROP-013 on exact-formula:C2N2; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-1171: PROP-013 on exact-formula:HCCCN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1172: PROP-013 on exact-formula:HCCCN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1173: PROP-013 on exact-formula:C4H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1174: PROP-013 on exact-formula:C4H2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1175: PROP-013 on exact-formula:C4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1176: PROP-013 on exact-formula:C4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1177: PROP-013 on exact-formula:C6H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1178: PROP-013 on exact-formula:C6H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1179: PROP-013 on exact-formula:SiF4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1180: PROP-013 on exact-formula:SiF4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1181: PROP-013 on exact-formula:F3PO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1182: PROP-013 on exact-formula:F3PO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1183: PROP-013 on exact-formula:GeF4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1184: PROP-013 on exact-formula:GeF4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1185: PROP-013 on exact-formula:CF4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1186: PROP-013 on exact-formula:CF4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1187: PROP-013 on exact-formula:CF3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1188: PROP-013 on exact-formula:CF3OH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1189: PROP-013 on exact-formula:B2O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1190: PROP-013 on exact-formula:B2O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1191: PROP-013 on exact-formula:TiCl4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1192: PROP-013 on exact-formula:TiCl4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1193: PROP-013 on exact-formula:SF4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1194: PROP-013 on exact-formula:SF4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1195: PROP-013 on exact-formula:SO2F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1196: PROP-013 on exact-formula:SO2F2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1197: PROP-013 on exact-formula:HSO3F; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1198: PROP-013 on exact-formula:HSO3F; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1199: PROP-013 on exact-formula:CH3CF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1200: PROP-013 on exact-formula:CH3CF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1201: PROP-013 on exact-formula:H2SO4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1202: PROP-013 on exact-formula:H2SO4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1203: PROP-013 on exact-formula:CF3Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1204: PROP-013 on exact-formula:CF3Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1205: PROP-013 on exact-formula:CHF2CH2F; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1206: PROP-013 on exact-formula:CHF2CH2F; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1207: PROP-013 on exact-formula:SiCl4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1208: PROP-013 on exact-formula:SiCl4; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-1209: PROP-013 on exact-formula:CF3Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1210: PROP-013 on exact-formula:CF3Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1211: PROP-013 on exact-formula:CF3I; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1212: PROP-013 on exact-formula:CF3I; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1213: PROP-013 on exact-formula:Cl3PO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1214: PROP-013 on exact-formula:Cl3PO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1215: PROP-013 on exact-formula:CH3CF2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1216: PROP-013 on exact-formula:CH3CF2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1217: PROP-013 on exact-formula:SiCl3CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1218: PROP-013 on exact-formula:SiCl3CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1219: PROP-013 on exact-formula:B3N3H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1220: PROP-013 on exact-formula:B3N3H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1221: PROP-013 on exact-formula:CF2Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1222: PROP-013 on exact-formula:CF2Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1223: PROP-013 on exact-formula:C2HF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1224: PROP-013 on exact-formula:C2HF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1225: PROP-013 on exact-formula:C3H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1226: PROP-013 on exact-formula:C3H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1227: PROP-013 on exact-formula:CHBrF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1228: PROP-013 on exact-formula:CHBrF2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1229: PROP-013 on exact-formula:SiBr4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1230: PROP-013 on exact-formula:SiBr4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1231: PROP-013 on exact-formula:AlBr3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1232: PROP-013 on exact-formula:AlBr3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1233: PROP-013 on exact-formula:CH3COOCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1234: PROP-013 on exact-formula:CH3COOCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1235: PROP-013 on exact-formula:HCOOC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1236: PROP-013 on exact-formula:HCOOC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1237: PROP-013 on exact-formula:C3H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1238: PROP-013 on exact-formula:C3H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1239: PROP-013 on exact-formula:H2NCH2COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1240: PROP-013 on exact-formula:H2NCH2COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1241: PROP-013 on exact-formula:C2H6O2S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1242: PROP-013 on exact-formula:C2H6O2S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1243: PROP-013 on exact-formula:SO2Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1244: PROP-013 on exact-formula:SO2Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1245: PROP-013 on exact-formula:C3H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1246: PROP-013 on exact-formula:C3H8O2; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-1247: PROP-013 on exact-formula:C3H4O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1248: PROP-013 on exact-formula:C3H4O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1249: PROP-013 on exact-formula:C4H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1250: PROP-013 on exact-formula:C4H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1251: PROP-013 on exact-formula:C3H6O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1252: PROP-013 on exact-formula:C3H6O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1253: PROP-013 on exact-formula:C3H4O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1254: PROP-013 on exact-formula:C3H4O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1255: PROP-013 on exact-formula:C4H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1256: PROP-013 on exact-formula:C4H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1257: PROP-013 on exact-formula:CFCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1258: PROP-013 on exact-formula:CFCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1259: PROP-013 on exact-formula:C4H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1260: PROP-013 on exact-formula:C4H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1261: PROP-013 on exact-formula:C4H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1262: PROP-013 on exact-formula:C4H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1263: PROP-013 on exact-formula:C3H7NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1264: PROP-013 on exact-formula:C3H7NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1265: PROP-013 on exact-formula:BrF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1266: PROP-013 on exact-formula:BrF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1267: PROP-013 on exact-formula:C4H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1268: PROP-013 on exact-formula:C4H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1269: PROP-013 on exact-formula:C4H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1270: PROP-013 on exact-formula:C4H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1271: PROP-013 on exact-formula:CH3COCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1272: PROP-013 on exact-formula:CH3COCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1273: PROP-013 on exact-formula:C4H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1274: PROP-013 on exact-formula:C4H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1275: PROP-013 on exact-formula:C2H6N2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1276: PROP-013 on exact-formula:C2H6N2O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1277: PROP-013 on exact-formula:Si(CH3)4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1278: PROP-013 on exact-formula:Si(CH3)4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1279: PROP-013 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1280: PROP-013 on exact-formula:CHOCH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1281: PROP-013 on exact-formula:XeF4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1282: PROP-013 on exact-formula:XeF4; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-1283: PROP-013 on exact-formula:CHOCH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1284: PROP-013 on exact-formula:CHOCH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1285: PROP-013 on exact-formula:BBr3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1286: PROP-013 on exact-formula:BBr3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1287: PROP-013 on exact-formula:C3H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1288: PROP-013 on exact-formula:C3H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1289: PROP-013 on exact-formula:C3H7NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1290: PROP-013 on exact-formula:C3H7NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1291: PROP-013 on exact-formula:F3NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1292: PROP-013 on exact-formula:F3NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1293: PROP-013 on exact-formula:C4H8O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1294: PROP-013 on exact-formula:C4H8O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1295: PROP-013 on exact-formula:C6H14; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1296: PROP-013 on exact-formula:C6H14; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1297: PROP-013 on exact-formula:CH3CCl(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1298: PROP-013 on exact-formula:CH3CCl(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1299: PROP-013 on exact-formula:C6H14; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1300: PROP-013 on exact-formula:C6H14; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1301: PROP-013 on exact-formula:CH3CCl2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1302: PROP-013 on exact-formula:CH3CCl2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1303: PROP-013 on exact-formula:C6H14; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1304: PROP-013 on exact-formula:C6H14; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1305: PROP-013 on exact-formula:C6H14; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1306: PROP-013 on exact-formula:C6H14; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1307: PROP-013 on exact-formula:C5H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1308: PROP-013 on exact-formula:C5H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1309: PROP-013 on exact-formula:C6H14; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1310: PROP-013 on exact-formula:C6H14; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1311: PROP-013 on exact-formula:CH3CHClCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1312: PROP-013 on exact-formula:CH3CHClCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1313: PROP-013 on exact-formula:CH2ClCHClCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1314: PROP-013 on exact-formula:CH2ClCHClCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1315: PROP-013 on exact-formula:CH2ClCH2CH2CH3; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-1316: PROP-013 on exact-formula:CH2ClCH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1317: PROP-013 on exact-formula:C5H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1318: PROP-013 on exact-formula:C5H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1319: PROP-013 on exact-formula:CH2C(CH3)OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1320: PROP-013 on exact-formula:CH2C(CH3)OCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1321: PROP-013 on exact-formula:CH2ClCHCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1322: PROP-013 on exact-formula:CH2ClCHCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1323: PROP-013 on exact-formula:C5H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1324: PROP-013 on exact-formula:C5H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1325: PROP-013 on exact-formula:PBr3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1326: PROP-013 on exact-formula:PBr3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1327: PROP-013 on exact-formula:C4H8O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1328: PROP-013 on exact-formula:C4H8O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1329: PROP-013 on exact-formula:CH3CCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1330: PROP-013 on exact-formula:CH3CCl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1331: PROP-013 on exact-formula:C2H3OC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1332: PROP-013 on exact-formula:C2H3OC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1333: PROP-013 on exact-formula:CH2CHCONH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1334: PROP-013 on exact-formula:CH2CHCONH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1335: PROP-013 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1336: PROP-013 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1337: PROP-013 on exact-formula:CH3NO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1338: PROP-013 on exact-formula:CH3NO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1339: PROP-013 on exact-formula:C(CH3)3NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1340: PROP-013 on exact-formula:C(CH3)3NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1341: PROP-013 on exact-formula:COBr2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1342: PROP-013 on exact-formula:COBr2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1343: PROP-013 on exact-formula:SiI4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1344: PROP-013 on exact-formula:SiI4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1345: PROP-013 on exact-formula:CHOCHCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1346: PROP-013 on exact-formula:CHOCHCHCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1347: PROP-013 on exact-formula:C(CH3)3SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1348: PROP-013 on exact-formula:C(CH3)3SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1349: PROP-013 on exact-formula:C3H5ClO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1350: PROP-013 on exact-formula:C3H5ClO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1351: PROP-013 on exact-formula:C6H12; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-1352: PROP-013 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1353: PROP-013 on exact-formula:CH3C(NH2)HCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1354: PROP-013 on exact-formula:CH3C(NH2)HCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1355: PROP-013 on exact-formula:C(NH2)H2C(CH3)HCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1356: PROP-013 on exact-formula:C(NH2)H2C(CH3)HCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1357: PROP-013 on exact-formula:C3O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1358: PROP-013 on exact-formula:C3O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1359: PROP-013 on exact-formula:CH2(SH)CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1360: PROP-013 on exact-formula:CH2(SH)CH(CH3)CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1361: PROP-013 on exact-formula:CH3CH(SH)CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1362: PROP-013 on exact-formula:CH3CH(SH)CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1363: PROP-013 on exact-formula:CCl4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1364: PROP-013 on exact-formula:CCl4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1365: PROP-013 on exact-formula:C(NH2)H2CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1366: PROP-013 on exact-formula:C(NH2)H2CH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1367: PROP-013 on exact-formula:C4H6O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1368: PROP-013 on exact-formula:C4H6O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1369: PROP-013 on exact-formula:CH3C(SCH3)HCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1370: PROP-013 on exact-formula:CH3C(SCH3)HCH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1371: PROP-013 on exact-formula:C4H9SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1372: PROP-013 on exact-formula:C4H9SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1373: PROP-013 on exact-formula:C2H5SC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1374: PROP-013 on exact-formula:C2H5SC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1375: PROP-013 on exact-formula:CH3CH2CH2Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1376: PROP-013 on exact-formula:CH3CH2CH2Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1377: PROP-013 on exact-formula:C5H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1378: PROP-013 on exact-formula:C5H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1379: PROP-013 on exact-formula:C4H6O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1380: PROP-013 on exact-formula:C4H6O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1381: PROP-013 on exact-formula:C6H12; post-seal payload identity True

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- SFT-CHEM-PROP-014-PROP-013-ROW-1420: PROP-013 on exact-formula:C5H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1421: PROP-013 on exact-formula:C3H8S2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1422: PROP-013 on exact-formula:C3H8S2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1423: PROP-013 on exact-formula:CH2CClCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1424: PROP-013 on exact-formula:CH2CClCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1425: PROP-013 on exact-formula:CH2CHCH2CH2Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1426: PROP-013 on exact-formula:CH2CHCH2CH2Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1427: PROP-013 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1428: PROP-013 on exact-formula:C6H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1429: PROP-013 on exact-formula:C5H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1430: PROP-013 on exact-formula:C5H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1431: PROP-013 on exact-formula:CH2CHCH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1432: PROP-013 on exact-formula:CH2CHCH2CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1433: PROP-013 on exact-formula:ClFO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1434: PROP-013 on exact-formula:ClFO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1435: PROP-013 on exact-formula:CHClCHCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1436: PROP-013 on exact-formula:CHClCHCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1437: PROP-013 on exact-formula:CHClCCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1438: PROP-013 on exact-formula:CHClCCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1439: PROP-013 on exact-formula:CHClCHCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1440: PROP-013 on exact-formula:CHClCHCH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1441: PROP-013 on exact-formula:C3H3NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1442: PROP-013 on exact-formula:C3H3NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1443: PROP-013 on exact-formula:CH2CHOCHCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1444: PROP-013 on exact-formula:CH2CHOCHCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1445: PROP-013 on exact-formula:C5H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1446: PROP-013 on exact-formula:C5H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1447: PROP-013 on exact-formula:CH3CBr3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1448: PROP-013 on exact-formula:CH3CBr3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1449: PROP-013 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1450: PROP-013 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1451: PROP-013 on exact-formula:C4H9N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1452: PROP-013 on exact-formula:C4H9N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1453: PROP-013 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1454: PROP-013 on exact-formula:C6H10; post-seal payload identity True

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- SFT-CHEM-PROP-014-PROP-013-ROW-1493: PROP-013 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1494: PROP-013 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1495: PROP-013 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1496: PROP-013 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1497: PROP-013 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1498: PROP-013 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1499: PROP-013 on exact-formula:C3H3NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1500: PROP-013 on exact-formula:C3H3NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1501: PROP-013 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1502: PROP-013 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1503: PROP-013 on exact-formula:C6H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1504: PROP-013 on exact-formula:C6H6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1505: PROP-013 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1506: PROP-013 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1507: PROP-013 on exact-formula:N2O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1508: PROP-013 on exact-formula:N2O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1509: PROP-013 on exact-formula:C4H6S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1510: PROP-013 on exact-formula:C4H6S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1511: PROP-013 on exact-formula:C3H6N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1512: PROP-013 on exact-formula:C3H6N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1513: PROP-013 on exact-formula:C4H6N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1514: PROP-013 on exact-formula:C4H6N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1515: PROP-013 on exact-formula:C4H6S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1516: PROP-013 on exact-formula:C4H6S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1517: PROP-013 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1518: PROP-013 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1519: PROP-013 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1520: PROP-013 on exact-formula:C5H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1521: PROP-013 on exact-formula:CH2CHSCHCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1522: PROP-013 on exact-formula:CH2CHSCHCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1523: PROP-013 on exact-formula:C6H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1524: PROP-013 on exact-formula:C6H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1525: PROP-013 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1526: PROP-013 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1527: PROP-013 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1528: PROP-013 on exact-formula:C6H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1529: PROP-013 on exact-formula:C4H5N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1530: PROP-013 on exact-formula:C4H5N; post-seal payload identity True

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- SFT-CHEM-PROP-014-PROP-013-ROW-1645: PROP-013 on exact-formula:C5H9N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1646: PROP-013 on exact-formula:C5H9N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1647: PROP-013 on exact-formula:SF5CF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1648: PROP-013 on exact-formula:SF5CF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1649: PROP-013 on exact-formula:PF6-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1650: PROP-013 on exact-formula:PF6-; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1651: PROP-013 on exact-formula:S2F10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1652: PROP-013 on exact-formula:S2F10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1653: PROP-013 on exact-formula:C3F8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1654: PROP-013 on exact-formula:C3F8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1655: PROP-013 on exact-formula:PF5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1656: PROP-013 on exact-formula:PF5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1657: PROP-013 on exact-formula:B2F4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1658: PROP-013 on exact-formula:B2F4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1659: PROP-013 on exact-formula:C2F6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1660: PROP-013 on exact-formula:C2F6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1661: PROP-013 on exact-formula:Al2Cl6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1662: PROP-013 on exact-formula:Al2Cl6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1663: PROP-013 on exact-formula:SF6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1664: PROP-013 on exact-formula:SF6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1665: PROP-013 on exact-formula:C3F6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1666: PROP-013 on exact-formula:C3F6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1667: PROP-013 on exact-formula:CF3CF2Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1668: PROP-013 on exact-formula:CF3CF2Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1669: PROP-013 on exact-formula:SF5Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1670: PROP-013 on exact-formula:SF5Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1671: PROP-013 on exact-formula:CF3COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1672: PROP-013 on exact-formula:CF3COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1673: PROP-013 on exact-formula:IF7; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1674: PROP-013 on exact-formula:IF7; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1675: PROP-013 on exact-formula:C6F6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1676: PROP-013 on exact-formula:C6F6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1677: PROP-013 on exact-formula:CF2ClCF2Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1678: PROP-013 on exact-formula:CF2ClCF2Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1679: PROP-013 on exact-formula:IF5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1680: PROP-013 on exact-formula:IF5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1681: PROP-013 on exact-formula:CF3OF; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1682: PROP-013 on exact-formula:CF3OF; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-1683: PROP-013 on exact-formula:CF2ClCFC12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1684: PROP-013 on exact-formula:CF2ClCFC12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1685: PROP-013 on exact-formula:Fe(CO)5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1686: PROP-013 on exact-formula:Fe(CO)5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1687: PROP-013 on exact-formula:C2F4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1688: PROP-013 on exact-formula:C2F4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1689: PROP-013 on exact-formula:C3H8O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1690: PROP-013 on exact-formula:C3H8O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1691: PROP-013 on exact-formula:C3H3N3O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1692: PROP-013 on exact-formula:C3H3N3O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1693: PROP-013 on exact-formula:C2H3NO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1694: PROP-013 on exact-formula:C2H3NO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1695: PROP-013 on exact-formula:C6H12O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1696: PROP-013 on exact-formula:C6H12O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1697: PROP-013 on exact-formula:C2H4CO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1698: PROP-013 on exact-formula:C2H4CO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1699: PROP-013 on exact-formula:C2ClF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1700: PROP-013 on exact-formula:C2ClF3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1701: PROP-013 on exact-formula:CF3CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1702: PROP-013 on exact-formula:CF3CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1703: PROP-013 on exact-formula:B2Cl4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1704: PROP-013 on exact-formula:B2Cl4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1705: PROP-013 on exact-formula:C2H6O3S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1706: PROP-013 on exact-formula:C2H6O3S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1707: PROP-013 on exact-formula:C4H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1708: PROP-013 on exact-formula:C4H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1709: PROP-013 on exact-formula:C3H6O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1710: PROP-013 on exact-formula:C3H6O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1711: PROP-013 on exact-formula:C3H5ClO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1712: PROP-013 on exact-formula:C3H5ClO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1713: PROP-013 on exact-formula:NH2COOC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1714: PROP-013 on exact-formula:NH2COOC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1715: PROP-013 on exact-formula:C4H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1716: PROP-013 on exact-formula:C4H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1717: PROP-013 on exact-formula:C4H10O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1718: PROP-013 on exact-formula:C4H10O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1719: PROP-013 on exact-formula:BrF5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1720: PROP-013 on exact-formula:BrF5; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-1721: PROP-013 on exact-formula:C4H10O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1722: PROP-013 on exact-formula:C4H10O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1723: PROP-013 on exact-formula:NH2CH2CH2COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1724: PROP-013 on exact-formula:NH2CH2CH2COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1725: PROP-013 on exact-formula:C3H2O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1726: PROP-013 on exact-formula:C3H2O3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1727: PROP-013 on exact-formula:CH3CH(NH2)COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1728: PROP-013 on exact-formula:CH3CH(NH2)COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1729: PROP-013 on exact-formula:C3H8O2S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1730: PROP-013 on exact-formula:C3H8O2S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1731: PROP-013 on exact-formula:C4H10O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1732: PROP-013 on exact-formula:C4H10O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1733: PROP-013 on exact-formula:C2H4N2O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1734: PROP-013 on exact-formula:C2H4N2O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1735: PROP-013 on exact-formula:C5H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1736: PROP-013 on exact-formula:C5H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1737: PROP-013 on exact-formula:CH3NHCH2COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1738: PROP-013 on exact-formula:CH3NHCH2COOH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1739: PROP-013 on exact-formula:C4H6O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1740: PROP-013 on exact-formula:C4H6O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1741: PROP-013 on exact-formula:C5H9NO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1742: PROP-013 on exact-formula:C5H9NO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1743: PROP-013 on exact-formula:PCl5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1744: PROP-013 on exact-formula:PCl5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1745: PROP-013 on exact-formula:C8H18O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1746: PROP-013 on exact-formula:C8H18O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1747: PROP-013 on exact-formula:C4H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1748: PROP-013 on exact-formula:C4H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1749: PROP-013 on exact-formula:C4H10O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1750: PROP-013 on exact-formula:C4H10O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1751: PROP-013 on exact-formula:C4H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1752: PROP-013 on exact-formula:C4H8O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1753: PROP-013 on exact-formula:ClCOClCO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1754: PROP-013 on exact-formula:ClCOClCO; post-seal payload identity True

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- SFT-CHEM-PROP-014-PROP-013-ROW-1793: PROP-013 on exact-formula:C5H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1794: PROP-013 on exact-formula:C5H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1795: PROP-013 on exact-formula:C5H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1796: PROP-013 on exact-formula:C5H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1797: PROP-013 on exact-formula:C5H12O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1798: PROP-013 on exact-formula:C5H12O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1799: PROP-013 on exact-formula:NH2CONHC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1800: PROP-013 on exact-formula:NH2CONHC2H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1801: PROP-013 on exact-formula:C5H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1802: PROP-013 on exact-formula:C5H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1803: PROP-013 on exact-formula:C6H12O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1804: PROP-013 on exact-formula:C6H12O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1805: PROP-013 on exact-formula:C10H22; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1806: PROP-013 on exact-formula:C10H22; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1807: PROP-013 on exact-formula:C5H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1808: PROP-013 on exact-formula:C5H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1809: PROP-013 on exact-formula:C6H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1810: PROP-013 on exact-formula:C6H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1811: PROP-013 on exact-formula:C5H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1812: PROP-013 on exact-formula:C5H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1813: PROP-013 on exact-formula:C9H20; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1814: PROP-013 on exact-formula:C9H20; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1815: PROP-013 on exact-formula:C4H8OS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1816: PROP-013 on exact-formula:C4H8OS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1817: PROP-013 on exact-formula:(CH3)3CC(CH3)3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1818: PROP-013 on exact-formula:(CH3)3CC(CH3)3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1819: PROP-013 on exact-formula:C5H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1820: PROP-013 on exact-formula:C5H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1821: PROP-013 on exact-formula:NH(CH3)CONH(CH3); post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1822: PROP-013 on exact-formula:NH(CH3)CONH(CH3); post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1823: PROP-013 on exact-formula:N(CH3)2CONH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1824: PROP-013 on exact-formula:N(CH3)2CONH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1825: PROP-013 on exact-formula:C8H18; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1826: PROP-013 on exact-formula:C8H18; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1827: PROP-013 on exact-formula:C4H10OS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1828: PROP-013 on exact-formula:C4H10OS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1829: PROP-013 on exact-formula:C4H8Cl2; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-1830: PROP-013 on exact-formula:C4H8Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1831: PROP-013 on exact-formula:C4H8Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1832: PROP-013 on exact-formula:C4H8Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1833: PROP-013 on exact-formula:C4H7NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1834: PROP-013 on exact-formula:C4H7NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1835: PROP-013 on exact-formula:C4H6OS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1836: PROP-013 on exact-formula:C4H6OS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1837: PROP-013 on exact-formula:C7H16; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1838: PROP-013 on exact-formula:C7H16; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1839: PROP-013 on exact-formula:C4H8Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1840: PROP-013 on exact-formula:C4H8Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1841: PROP-013 on exact-formula:C7H16; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1842: PROP-013 on exact-formula:C7H16; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1843: PROP-013 on exact-formula:C5H8O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1844: PROP-013 on exact-formula:C5H8O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1845: PROP-013 on exact-formula:C4H8Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1846: PROP-013 on exact-formula:C4H8Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1847: PROP-013 on exact-formula:C4H4O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1848: PROP-013 on exact-formula:C4H4O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1849: PROP-013 on exact-formula:C7H16; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1850: PROP-013 on exact-formula:C7H16; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1851: PROP-013 on exact-formula:C4H8Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1852: PROP-013 on exact-formula:C4H8Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1853: PROP-013 on exact-formula:C3H5Cl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1854: PROP-013 on exact-formula:C3H5Cl3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1855: PROP-013 on exact-formula:C5H11Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1856: PROP-013 on exact-formula:C5H11Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1857: PROP-013 on exact-formula:C4H7NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1858: PROP-013 on exact-formula:C4H7NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1859: PROP-013 on exact-formula:CHCl2CHCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1860: PROP-013 on exact-formula:CHCl2CHCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1861: PROP-013 on exact-formula:CCl3CHCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1862: PROP-013 on exact-formula:CCl3CHCl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1863: PROP-013 on exact-formula:C2H5NO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1864: PROP-013 on exact-formula:C2H5NO3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1865: PROP-013 on exact-formula:C5H4O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1866: PROP-013 on exact-formula:C5H4O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1867: PROP-013 on exact-formula:C5H10O; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-1868: PROP-013 on exact-formula:C5H10O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1869: PROP-013 on exact-formula:C2Cl6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1870: PROP-013 on exact-formula:C2Cl6; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1871: PROP-013 on exact-formula:C4H6OS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1872: PROP-013 on exact-formula:C4H6OS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1873: PROP-013 on exact-formula:C10H16; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1874: PROP-013 on exact-formula:C10H16; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1875: PROP-013 on exact-formula:C6H15N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1876: PROP-013 on exact-formula:C6H15N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1877: PROP-013 on exact-formula:CH3CH(CH3)ONO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1878: PROP-013 on exact-formula:CH3CH(CH3)ONO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1879: PROP-013 on exact-formula:C4H7NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1880: PROP-013 on exact-formula:C4H7NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1881: PROP-013 on exact-formula:C5H8O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1882: PROP-013 on exact-formula:C5H8O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1883: PROP-013 on exact-formula:C5H12S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1884: PROP-013 on exact-formula:C5H12S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1885: PROP-013 on exact-formula:C5H12S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1886: PROP-013 on exact-formula:C5H12S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1887: PROP-013 on exact-formula:C8H16; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1888: PROP-013 on exact-formula:C8H16; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1889: PROP-013 on exact-formula:C3H4O2S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1890: PROP-013 on exact-formula:C3H4O2S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1891: PROP-013 on exact-formula:C4H11NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1892: PROP-013 on exact-formula:C4H11NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1893: PROP-013 on exact-formula:C5H12S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1894: PROP-013 on exact-formula:C5H12S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1895: PROP-013 on exact-formula:C5H12S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1896: PROP-013 on exact-formula:C5H12S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1897: PROP-013 on exact-formula:C6H4O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1898: PROP-013 on exact-formula:C6H4O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1899: PROP-013 on exact-formula:C3H7ONO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1900: PROP-013 on exact-formula:C3H7ONO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1901: PROP-013 on exact-formula:C7H14; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1902: PROP-013 on exact-formula:C7H14; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1903: PROP-013 on exact-formula:C5H12S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1904: PROP-013 on exact-formula:C5H12S; post-seal payload identity True

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- SFT-CHEM-PROP-014-PROP-013-ROW-1943: PROP-013 on exact-formula:C4H10S2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1944: PROP-013 on exact-formula:C4H10S2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1945: PROP-013 on exact-formula:C6H6O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1946: PROP-013 on exact-formula:C6H6O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1947: PROP-013 on exact-formula:C5H10S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1948: PROP-013 on exact-formula:C5H10S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1949: PROP-013 on exact-formula:C5H11N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1950: PROP-013 on exact-formula:C5H11N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1951: PROP-013 on exact-formula:C5H5NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1952: PROP-013 on exact-formula:C5H5NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1953: PROP-013 on exact-formula:C6H5CHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1954: PROP-013 on exact-formula:C6H5CHO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1955: PROP-013 on exact-formula:C5H5NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1956: PROP-013 on exact-formula:C5H5NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1957: PROP-013 on exact-formula:C6H7N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1958: PROP-013 on exact-formula:C6H7N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1959: PROP-013 on exact-formula:C2Cl4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1960: PROP-013 on exact-formula:C2Cl4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1961: PROP-013 on exact-formula:C2H6N2O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1962: PROP-013 on exact-formula:C2H6N2O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1963: PROP-013 on exact-formula:C5H9N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1964: PROP-013 on exact-formula:C5H9N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1965: PROP-013 on exact-formula:C5H9N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1966: PROP-013 on exact-formula:C5H9N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1967: PROP-013 on exact-formula:C9H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1968: PROP-013 on exact-formula:C9H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1969: PROP-013 on exact-formula:C5H9N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1970: PROP-013 on exact-formula:C5H9N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1971: PROP-013 on exact-formula:N2O4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1972: PROP-013 on exact-formula:N2O4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1973: PROP-013 on exact-formula:N2O5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1974: PROP-013 on exact-formula:N2O5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1975: PROP-013 on exact-formula:CH3C6H4CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1976: PROP-013 on exact-formula:CH3C6H4CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1977: PROP-013 on exact-formula:CH3C6H4CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1978: PROP-013 on exact-formula:CH3C6H4CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1979: PROP-013 on exact-formula:C5H10S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1980: PROP-013 on exact-formula:C5H10S; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-1981: PROP-013 on exact-formula:CFCICFCI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1982: PROP-013 on exact-formula:CFCICFCI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1983: PROP-013 on exact-formula:CFCICCFI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1984: PROP-013 on exact-formula:CFCICCFI; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1985: PROP-013 on exact-formula:C6H4Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1986: PROP-013 on exact-formula:C6H4Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1987: PROP-013 on exact-formula:C4H6OS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1988: PROP-013 on exact-formula:C4H6OS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1989: PROP-013 on exact-formula:C6H4Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1990: PROP-013 on exact-formula:C6H4Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1991: PROP-013 on exact-formula:C6H5CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1992: PROP-013 on exact-formula:C6H5CH2CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1993: PROP-013 on exact-formula:C6H4Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1994: PROP-013 on exact-formula:C6H4Cl2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1995: PROP-013 on exact-formula:C4H5NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1996: PROP-013 on exact-formula:C4H5NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1997: PROP-013 on exact-formula:C4H5NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1998: PROP-013 on exact-formula:C4H5NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-1999: PROP-013 on exact-formula:C6H5COC6H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2000: PROP-013 on exact-formula:C6H5COC6H5; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2001: PROP-013 on exact-formula:C6H5CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2002: PROP-013 on exact-formula:C6H5CH3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2003: PROP-013 on exact-formula:C6H5O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2004: PROP-013 on exact-formula:C6H5O; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2005: PROP-013 on exact-formula:C6H5Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2006: PROP-013 on exact-formula:C6H5Cl; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2007: PROP-013 on exact-formula:C3H6S3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2008: PROP-013 on exact-formula:C3H6S3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2009: PROP-013 on exact-formula:C6H5NO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2010: PROP-013 on exact-formula:C6H5NO2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2011: PROP-013 on exact-formula:C2H6N2O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2012: PROP-013 on exact-formula:C2H6N2O2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2013: PROP-013 on exact-formula:C5H6S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2014: PROP-013 on exact-formula:C5H6S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2015: PROP-013 on exact-formula:C2H4N2S2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2016: PROP-013 on exact-formula:C2H4N2S2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2017: PROP-013 on exact-formula:C5H6S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2018: PROP-013 on exact-formula:C5H6S; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-2019: PROP-013 on exact-formula:C6H5NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2020: PROP-013 on exact-formula:C6H5NH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2021: PROP-013 on exact-formula:C6H12N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2022: PROP-013 on exact-formula:C6H12N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2023: PROP-013 on exact-formula:C3H4S3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2024: PROP-013 on exact-formula:C3H4S3; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2025: PROP-013 on exact-formula:S8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2026: PROP-013 on exact-formula:S8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2027: PROP-013 on exact-formula:C5H7N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2028: PROP-013 on exact-formula:C5H7N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2029: PROP-013 on exact-formula:C6H5Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2030: PROP-013 on exact-formula:C6H5Br; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2031: PROP-013 on exact-formula:C4H5NS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2032: PROP-013 on exact-formula:C4H5NS; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2033: PROP-013 on exact-formula:C6H5SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2034: PROP-013 on exact-formula:C6H5SH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2035: PROP-013 on exact-formula:C5H7N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2036: PROP-013 on exact-formula:C5H7N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2037: PROP-013 on exact-formula:C5H7N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2038: PROP-013 on exact-formula:C5H7N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2039: PROP-013 on exact-formula:C5H7N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2040: PROP-013 on exact-formula:C5H7N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2041: PROP-013 on exact-formula:C6H5CHCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2042: PROP-013 on exact-formula:C6H5CHCH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2043: PROP-013 on exact-formula:C5H7N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2044: PROP-013 on exact-formula:C5H7N; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2045: PROP-013 on exact-formula:C10H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2046: PROP-013 on exact-formula:C10H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2047: PROP-013 on exact-formula:C9H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2048: PROP-013 on exact-formula:C9H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2049: PROP-013 on exact-formula:C13H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2050: PROP-013 on exact-formula:C13H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2051: PROP-013 on exact-formula:C12H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2052: PROP-013 on exact-formula:C12H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2053: PROP-013 on exact-formula:C6H5NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2054: PROP-013 on exact-formula:C6H5NO; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2055: PROP-013 on exact-formula:C14H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2056: PROP-013 on exact-formula:C14H10; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-2057: PROP-013 on exact-formula:C6H5CH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2058: PROP-013 on exact-formula:C6H5CH2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2059: PROP-013 on exact-formula:C6H5CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2060: PROP-013 on exact-formula:C6H5CN; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2061: PROP-013 on exact-formula:C14H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2062: PROP-013 on exact-formula:C14H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2063: PROP-013 on exact-formula:C16H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2064: PROP-013 on exact-formula:C16H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2065: PROP-013 on exact-formula:C7H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2066: PROP-013 on exact-formula:C7H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2067: PROP-013 on exact-formula:CH6N4S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2068: PROP-013 on exact-formula:CH6N4S; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2069: PROP-013 on exact-formula:C12H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2070: PROP-013 on exact-formula:C12H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2071: PROP-013 on exact-formula:C19H16; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2072: PROP-013 on exact-formula:C19H16; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2073: PROP-013 on exact-formula:C24H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2074: PROP-013 on exact-formula:C24H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2075: PROP-013 on exact-formula:C8H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2076: PROP-013 on exact-formula:C8H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2077: PROP-013 on exact-formula:C6H5CCCH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2078: PROP-013 on exact-formula:C6H5CCCH; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2079: PROP-013 on exact-formula:C10H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2080: PROP-013 on exact-formula:C10H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2081: PROP-013 on exact-formula:C10H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2082: PROP-013 on exact-formula:C10H10; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2083: PROP-013 on exact-formula:C7H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2084: PROP-013 on exact-formula:C7H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2085: PROP-013 on exact-formula:C18H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2086: PROP-013 on exact-formula:C18H12; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2087: PROP-013 on exact-formula:C12H10N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2088: PROP-013 on exact-formula:C12H10N2; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2089: PROP-013 on exact-formula:C12H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2090: PROP-013 on exact-formula:C12H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2091: PROP-013 on exact-formula:C8H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2092: PROP-013 on exact-formula:C8H8; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2093: PROP-013 on exact-formula:C(CN)4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2094: PROP-013 on exact-formula:C(CN)4; post-seal payload identity True

- SFT-CHEM-PROP-014-PROP-013-ROW-2095: PROP-013 on exact-formula:C6N4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2096: PROP-013 on exact-formula:C6N4; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2097: PROP-013 on exact-formula:C60; post-seal payload identity True
- SFT-CHEM-PROP-014-PROP-013-ROW-2098: PROP-013 on exact-formula:C60; post-seal payload identity True
- tampered_target_payload_hash_rejected: True
- complete_13_property_families_retained: True
- complete_9025_source_rows_retained: True
- complete_1104_structural_carriers_retained: True
- all_676_multi_property_carriers_retained: True
- all_6676_overlap_rows_retained: True
- maximum_eight_family_H2_carrier_retained: True
- nonjoined_rows_not_guessed_into_formula_carriers: True

Falsification boundary: The claim fails if one property requires a separately fitted carrier or coefficient; if adding a lawful property changes any existing projection; if any of 13 families, 9,025 rows, 1,104 carriers, 676 overlap carriers or 6,676 overlap rows is omitted; if a nonjoinable row is guessed into a species group; if a target payload or hash opens before the complete identity seal; or if any favorable subset replaces the complete source custody surface.

Measurement receipt: sha256: fe95f21e782bb724d9413e9bdc5c39859df7f268924bdcc2fb7ba5e1566150f1.

Isolation certificate: sha256: 0111ee1e06ff26035a817fcd985226cf43530c72794f383b174e499c2b664518.

Custody certificate: sha256: f2b3b1b7039d4c01fa7287da4ebcadbbeec4d002a57ff0a63cae8d83bd96d62a.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The quantitative property family derives bond length, dissociation, angle, torsion, dipole, polarizability, ionization, affinity, vibration, rotation, binding, magnetic response, formation energy and their common-carrier vector without per-property fitting. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; absent or inapplicable native results remain structural EmptyOne
- no negative, irrational, imaginary, floating or continuum SFT proof value
- no target payload, target hash, presence flag or source orientation before the complete identity seal
- no per-property fit, correction, residual, coefficient or target-derived structural field
- no selected species, showcase molecule, favorable property family or agreeing-row subset
- no guessed synonym, formula, species join or conflation of a bound composite with a constituent molecule
- no deletion of single-property, nonjoinable, absent, adverse or bound-composite custody rows

Immutable identities. Source manifest

sha256: 0b5869727ab9a43ad55a272fe5247cdff51bd207889c89fb71af6486af56c7c7; derivation seal

sha256: 1acb5527e39915f77f49e357469c4843eecfe01b6581c6d8b50c76774909479f; engine receipt

sha256: c3953d770e1837fc70536ba386e631a93a812b8e1008fe71d1635af1386999b8 at receipts/engine/model_admitted/SFT-CHEM-CROSS-PROPERTY-MOLECULAR-VECTOR-014-c3953d770e1837fc.json;

empirical hash sha256: 02cc9554e34a899cd1a0ad5b6e973eae5e71482e5c89bbe4bee6eb2bedf307a4;

measurement receipt sha256: fe95f21e782bb724d9413e9bdc5c39859df7f268924bdcc2fb7ba5e1566150f1.

20. Statistical, thermodynamic, phase and transport chemistry - 19 admitted claims

Finite support, exact ratios and held transfer directions reconstruct thermodynamic and transport relations. This entire registered subcategory is included; no proper subset is presented as field completion.

125. Finite chemical microstate-support law

Claim identity: SFT-CHEM-FINITE-MICROSTATE-SUPPORT-001

Question and theorem. A chemical thermodynamic support is the complete generated finite tuple of held chemical microstates, partitioned so every state occurs exactly once in one named macro-observation fibre. Multiplicity is the exact positive fibre count and statistical weight is that count over the complete finite support count. All 387 external identities seal before 330 direct state records and 57 calorimetric rows open.

complete-generated-finite-chemical-support__complete-held-chemical-microstate-identity__disjoint-exhaustive-macro-observation-partition__exact-positive-fibre-count__exact-fibre-count-over-complete-support-count__complete-value-free-state-and-calorimetric-identity-seal__complete-387-row-external-structure-custody__depth-independent-one-state-finite-successor

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-PHYS-THERMO-MICRO-MACRO-001
- SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001
- SFT-PHYS-THERMO-STATE-RELATION-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-CROSS-PROPERTY-MOLECULAR-VECTOR-014

Generated grammar. Generate the literal product of support, identity, partition, multiplicity, weight, prediction, record and extension forms; decide all 256 candidates only from admitted exact state, observation, counting and thermodynamic carrier laws.

Boundary: Every nonempty finite generated chemical-state tuple and its complete disjoint observation partition, with one-state finite successor closure. Completed infinity and continuum ensembles are outside the grammar. External structure is tested on all 330 direct CaH⁺ state rows and all 57 NIST water calorimetric rows.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *complete-generated-finite-chemical-support__complete-held-chemical-microstate-identity__disjoint-exhaustive-macro-observation-partition__exact-positive-fibre-count__exact-fibre-count-over-complete-support-count__complete-value-free-state-and-calorimetric-identity-seal__complete-387-row-external-structure-custody__depth-independent-one-state-finite-successor*. Closure is

depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
support	complete-generated-finite-chemical-support	selected-or-completed-infinite-state-set	A selected or completed-infinite set is neither generated nor exhaustible.
identity	complete-held-chemical-microstate-identity	answer-only-state-count	A bare count erases composition, phase, condition and internal-state identity.
partition	disjoint-exhaustive-macro-observation-partition	overlapping-or-omitting-observation-classes	Overlap duplicates predecessors and omission deletes them.
multiplicity	exact-positive-fibre-count	floating-or-assumed-degeneracy	An assumed degeneracy imports an ungenerated statistical parameter.
weight	exact-fibre-count-over-complete-support-count	imported-distribution-or-partition-function	An imported distribution lets a prior select state weights.
prediction	complete-value-free-state-and-calorimetric-identity-seal	population-temperature-or-calorimetric-target-readable-before-seal	Target access can select the state support or partition.
record	complete-387-row-external-structure-custody	selected-population-or-calorimetric-showcase	A selected example cannot establish complete source custody.
extension	depth-independent-one-state-finite-successor	continuum-or-completed-infinity-closure	A completed continuum is not a finite generated successor.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256: 6572d31d70c74976313000e5bf8a08b713816419110371e85415da892fb3f767.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: fbdccf18d0cdab5ddb2ad9cea3230bc0270c618a33c39af0c9837ace021e14cd.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: cc68a1b05a06e3ac7880bd13665d65aa954944705a0c708f096979e3a06b158.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: c3034c6a32a8a2e7166ef3b370a8a37bb0b2dda0e0f52f0fc58dd2ce80e79934.

Independent reconstruction. Implementation hash

sha256: 8aa18c57ce9413e78e65bbc63653aac4111ef815b99ef34946dfbd41d7e3b668; independent certificate sha256: 678653cd310c1b50a38f09c12e3fe2d42c276b47f15c3244efce1f6b64dee79e; external-validation hash sha256: 5f7458def3484dd3a50c6bd6002b69c36a4f075631650df677692287048ff32c. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-MDS2-3389-CAH-PLUS-QUANTUM-JUMP-THERMOMETRY
- NIST-CHEMISTRY-WEBBOOK-SRD69-WATER-GAS-CALORIMETRIC-TABLE

Comparison records:

- SFT-CHEM-THERMO-001-POPULATION-0001: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0002: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0003: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0004: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0005: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0006: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0007: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0008: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0009: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0010: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0011: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0012: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0013: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0014: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0015: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0016: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0017: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0018: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0019: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0020: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0021: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0022: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True

- SFT-CHEM-THERMO-001-POPULATION-0023: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0024: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0025: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0026: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0027: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0028: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0029: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0030: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0031: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0032: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0033: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0034: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0035: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0036: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0037: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0038: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0039: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0040: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0041: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0042: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0043: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0044: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True

- SFT-CHEM-THERMO-001-POPULATION-0045: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0046: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0047: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0048: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0049: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0050: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0051: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0052: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0053: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0054: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0055: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0056: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0057: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0058: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0059: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0060: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0061: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0062: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0063: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0064: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0065: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0066: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True

- SFT-CHEM-THERMO-001-POPULATION-0067: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0068: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0069: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0070: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0071: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0072: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0073: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0074: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0075: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0076: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0077: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0078: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0079: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0080: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0081: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0082: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0083: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0084: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0085: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0086: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0087: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0088: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True

- SFT-CHEM-THERMO-001-POPULATION-0089: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0090: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0091: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0092: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0093: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0094: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0095: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0096: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0097: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
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- SFT-CHEM-THERMO-001-POPULATION-0099: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0100: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0101: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0102: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0103: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0104: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0105: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0106: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0107: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0108: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0109: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0110: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True

- SFT-CHEM-THERMO-001-POPULATION-0111: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0112: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0113: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0114: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0115: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0116: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
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- SFT-CHEM-THERMO-001-POPULATION-0119: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0120: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0121: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0122: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0123: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0124: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0125: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0126: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0127: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0128: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0129: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0130: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0131: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0132: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True

- SFT-CHEM-THERMO-001-POPULATION-0133: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0134: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0135: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0136: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
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- SFT-CHEM-THERMO-001-POPULATION-0140: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0141: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0142: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0143: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0144: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0145: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0146: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0147: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0148: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
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- SFT-CHEM-THERMO-001-POPULATION-0154: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True

- SFT-CHEM-THERMO-001-POPULATION-0155: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
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- SFT-CHEM-THERMO-001-POPULATION-0162: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
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- SFT-CHEM-THERMO-001-POPULATION-0168: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0169: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0170: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
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- SFT-CHEM-THERMO-001-POPULATION-0173: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
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- SFT-CHEM-THERMO-001-POPULATION-0175: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0176: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True

- SFT-CHEM-THERMO-001-POPULATION-0177: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0178: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0179: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
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- SFT-CHEM-THERMO-001-POPULATION-0181: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0182: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0183: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0184: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0185: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0186: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
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- SFT-CHEM-THERMO-001-POPULATION-0193: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0194: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0195: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0196: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0197: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0198: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True

- SFT-CHEM-THERMO-001-POPULATION-0199: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0200: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0201: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0202: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0203: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0204: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0205: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0206: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0207: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0208: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
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- SFT-CHEM-THERMO-001-POPULATION-0210: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
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- SFT-CHEM-THERMO-001-POPULATION-0212: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0213: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0214: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0215: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0216: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0217: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0218: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0219: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0220: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True

- SFT-CHEM-THERMO-001-POPULATION-0221: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0222: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0223: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0224: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0225: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0226: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0227: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0228: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0229: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0230: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0231: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0232: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0233: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0234: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0235: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0236: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0237: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0238: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0239: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0240: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0241: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0242: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True

- SFT-CHEM-THERMO-001-POPULATION-0243: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0244: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0245: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0246: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0247: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0248: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0249: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0250: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0251: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0252: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0253: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0254: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0255: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0256: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0257: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0258: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0259: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0260: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0261: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0262: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0263: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0264: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True

- SFT-CHEM-THERMO-001-POPULATION-0265: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0266: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0267: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0268: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0269: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0270: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0271: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0272: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0273: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0274: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0275: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0276: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0277: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0278: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0279: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0280: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0281: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0282: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0283: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0284: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0285: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0286: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True

- SFT-CHEM-THERMO-001-POPULATION-0287: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0288: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0289: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0290: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0291: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0292: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0293: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0294: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0295: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0296: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0297: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0298: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0299: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0300: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0301: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0302: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0303: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0304: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0305: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0306: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0307: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0308: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True

- SFT-CHEM-THERMO-001-POPULATION-0309: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0310: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0311: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0312: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0313: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0314: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0315: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0316: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0317: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0318: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0319: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0320: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0321: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0322: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0323: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0324: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0325: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0326: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0327: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0328: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0329: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-POPULATION-0330: direct-molecular-state-population-and-transition-record; identity and post-seal payload binding True

- SFT-CHEM-THERMO-001-CALORIMETRIC-0001: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0002: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0003: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0004: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0005: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0006: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0007: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0008: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0009: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0010: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0011: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0012: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0013: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0014: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0015: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0016: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0017: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0018: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0019: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0020: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0021: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0022: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True

- SFT-CHEM-THERMO-001-CALORIMETRIC-0023: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0024: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0025: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0026: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0027: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0028: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0029: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0030: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0031: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0032: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0033: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0034: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0035: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0036: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0037: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0038: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0039: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0040: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0041: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0042: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0043: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0044: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True

- SFT-CHEM-THERMO-001-CALORIMETRIC-0045: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0046: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0047: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0048: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0049: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0050: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0051: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0052: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0053: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0054: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0055: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0056: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- SFT-CHEM-THERMO-001-CALORIMETRIC-0057: evaluated-finite-calorimetric-state-row; identity and post-seal payload binding True
- tampered_target_payload_hash_rejected: True
- all_330_direct_state_rows_retained: True
- all_57_calorimetric_rows_retained: True
- all_387_external_rows_retained: True
- duplicated_1700_regime_boundary_retained: True
- state_and_calorimetric_classes_not_conflated: True
- prediction_contains_no_target_payload_hash: True
- continuum_and_completed_infinity_not_used: True

Falsification boundary: The claim fails if any generated state is omitted, duplicated or assigned to more than one fibre; if multiplicity or weight requires a distribution prior or fitted coefficient; if finite successor changes a previous assignment; if any target value opens before the 387-identity seal; if any of 330 state rows or 57 calorimetric rows is omitted or the duplicated 1700 K regime-boundary record is silently deduplicated; or if a continuum, completed infinity or external thermodynamic equation enters the proof.

Measurement receipt: sha256:1f0fd3924a9d7e1cf4a24f19846858486ca57470c0a0bde1f9138af1d18f818d.

Isolation certificate: sha256:8e30bb8726d79f3c8cbb1bbf3d2c2dadff9c7ad8959d8a296c8312db90757015.

Custody certificate: sha256:27053ca61ffa04c60a0cae032e54342ccb1e743dacb62dd2be1bca5ba7c0a430.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The thermodynamic extension derives finite microstate support, temperature, energy, heat/work, entropy, enthalpy, free direction, chemical potential, activity, fugacity, phases, colligative response, solvation and coupled transport while keeping external scalar inscriptions downstream. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; native absence is structural EmptyOne and external zero glyphs remain source inscriptions
- no negative, irrational, imaginary, floating, signed or continuum SFT proof value
- no completed infinity, continuum ensemble, partition-function prior or assumed probability distribution
- no temperature, population, calorimetric value, target payload or target hash in law, grammar or prediction
- no fitted coefficient, species parameter, imported statistical equation or target-selected state partition
- no selected species, favorable state rows, deduplication of the NIST regime-boundary row or omitted source row

Immutable identities. Source manifest

sha256:bb35d09e3801c1e7af9d86a2c983701de59e5473bb92581f9fa2b6dded0b4d9f; derivation seal sha256:be4be31bbcb5a77375ea49b37e53eee06dfb9ad3746b2bb129c340d44af06aa7; engine receipt sha256:ec8803582bbf3daeeaa2e7583a7d534153ea73225d80ebae365c30d3ddf95d572 at receipts/engine/model_admitted/SFT-CHEM-FINITE-MICROSTATE-SUPPORT-001-ec8803582bbf3dae.json; empirical hash sha256:2103ffc847f47452b25459c1af8bbec72a6fbb577afb8a251cdbf0e32356335c; measurement receipt sha256:1f0fd3924a9d7e1cf4a24f19846858486ca57470c0a0bde1f9138af1d18f818d.

126. Chemical temperature correspondence

Claim identity: SFT-CHEM-TEMPERATURE-CORRESPONDENCE-002

Question and theorem. Chemistry consumes the admitted Physics temperature carrier without rescaling it. Composition, phase, equilibrium reference and thermometric route remain held chemical identities, while equilibrated contexts share one exact carrier and composition-dependent consequences attach without a fitted conversion. Three value-free identities seal before the exact SI carrier and two independent thermometry intervals open.

chemistry-consumes-admitted-physics-temperature-carrier__held-chemical-composition-identity__held-phase-and-equilibrium-reference__identity-preserving-temperature-correspondence__one-common-carrier-across-equilibrated-routes__complete-value-free-thermometric-identity-seal__complete-three-row-two-route-value-vector__append-only-composition-consequence-with-common-carrier

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-PHYS-THERMO-TEMPERATURE-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-PHYS-THERMO-STATE-RELATION-001
- SFT-PHYS-VALIDATION-THERMAL-EQUILIBRIUM-044

- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-CROSS-PROPERTY-MOLECULAR-VECTOR-014
- SFT-CHEM-FINITE-MICROSTATE-SUPPORT-001

Generated grammar. Generate the literal product of authority, composition, condition, relation, equilibrium, prediction, record and extension forms; decide all 256 candidates only from admitted Physics temperature/equilibrium and exact Chemistry state-support laws.

Boundary: Every finite set of held chemical contexts sharing one admitted Physics temperature carrier at one held equilibrium reference, with append-only composition consequences. External testing retains the exact SI carrier, the complete acoustic-argon interval and the complete independent Johnson-noise interval.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: chemistry-consumes-admitted-physics-temperature-carrier__held-chemical-composition-identity__held-phase-and-equilibrium-reference__identity-preserving-temperature-correspondence__one-common-carrier-across-equilibrated-routes__complete-value-free-thermometric-identity-seal__complete-three-row-two-route-value-vector__append-only-composition-consequence-with-common-carrier. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
authority	chemistry-consumes-admitted-physics-temperature-carrier	chemistry-redefines-temperature	A Chemistry-specific temperature law duplicates and can contradict the admitted Physics carrier.
composition	held-chemical-composition-identity	composition-erased-temperature-scalar	Erasing composition loses the chemical state to which a consequence belongs.
condition	held-phase-and-equilibrium-reference	phase-and-equilibrium-reference-erased	Without phase and reference, distinct chemical states are conflated.
relation	identity-preserving-temperature-correspondence	fitted-chemical-temperature-conversion	A conversion coefficient lets composition or a target rescale temperature.
equilibrium	one-common-carrier-across-equilibrated-routes	route-or-composition-specific-temperature	Separate route temperatures violate common equilibrium correspondence.
prediction	complete-value-free-thermometric-identity-seal	thermometric-values-readable-before-seal	Measured centers or uncertainties could select the correspondence.
record	complete-three-row-two-route-value-vector	selected-thermometry-route	One favorable route cannot establish route-independent correspondence.
extension	append-only-composition-consequence-with-common-carrier	composition-dependent-rescaling	A new composition-specific scale is a free parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256: fb67c928165a3608eb3bd3072fb5af10813067075d6c3e1fd70d59c26fdc9d4b.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 08d3ba62c776a8ad7906a5924a83390d6dba4dba38c8525b71cbde2af57b2729.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 0e8bdbce6f749f4638c5f435c46f2b6c4b3c15645a1887c0aee7057a4bcb06a4.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : bb74aa92f5277635131d945ed866a633064999cf80bf5900ee6f87eefce3df95.

Independent reconstruction. Implementation hash

sha256 : 4c33a49fb5d141a2624b3feb9676d8af4556f8b7becf1d1bfab4748148051f63; independent certificate
sha256 : be9c0e3d29df4410a00941676955090c2347ce37f5ea1c2a6d49579dd03bfff36; external-validation
hash sha256 : 67a93a07db649b404656b7bd99723ae2c2ea1222b13287bd0f01a98d98616b3d. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were
preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-CODATA-2022-BOLTZMANN
- NIM-NIST-ACOUSTIC-BOLTZMANN-2017
- NIST-JOHNSON-NOISE-BOLTZMANN-2011

Comparison records:

- Exact SI common carrier: 13806490/10³⁰ joule per kelvin.
- Acoustic argon interval: [13806456,13806512]/10³⁰ joule per kelvin; exact carrier contained.
- Johnson-noise interval: [13806340,13806680]/10³⁰ joule per kelvin; exact carrier contained.
- Acoustic composition/condition: pure argon gas at the triple point of water retained.
- Acoustic kinetic-temperature and electronic Johnson temperature-response relations retained.
- tampered_acoustic_interval_rejected: **True**
- complete_three_target_rows_retained: **True**
- two_physically_distinct_routes_retained: **True**
- all_exact_interval_relations_pass: **True**
- prediction_contains_no_withheld_target_hash: **True**

Falsification boundary: The claim fails if Chemistry requires a different temperature carrier, composition-dependent scale or
route-specific correction; if adding a composition changes the common carrier; if any target value opens before all three identities
seal; if the exact SI carrier lies outside either complete measured interval; if argon, gas phase, TPW reference, kinetic-temperature
relation, Johnson response, uncertainty endpoint or source limitation is omitted; or if target values select the law.

Measurement receipt: sha256 : 66a07d4f83a2894410537fecdd50d589489dd805c58455ce4eba1dd952007af61.

Isolation certificate: sha256 : 6e5aafa523a662e61e27efc16c7894592cfb844d5e80d994527245561ed45786.

Custody certificate: sha256 : 3c74b8d6b1e90e26e41944cfcb1b761608a14ecf3331c16b1a28749d02d9fc43.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The thermodynamic extension derives finite microstate support, temperature, energy, heat/work, entropy, enthalpy, free
direction, chemical potential, activity, fugacity, phases, colligative response, solvation and coupled transport while keeping
external scalar inscriptions downstream. This particular result is closed only at its exact registered grammar and evidence
boundary.

Explicit exclusions.

- no numerical zero; external zero glyphs remain source inscriptions and native absence is structural EmptyOne
- no negative, irrational, imaginary, floating, signed or continuum SFT proof value

- no Chemistry-specific redefinition, rescaling, calibration coefficient or fitted temperature conversion
- no measured center, uncertainty, interval endpoint, relation flag or target hash in law, grammar or prediction
- no selected thermometry route, omitted uncertainty endpoint, composition erasure or target-selected equilibrium
- no imported continuum distribution, conventional thermodynamic equation or composition-specific free parameter

Immutable identities. Source manifest

sha256:37717e5466341dc7a3e37339d984834f5d3cdd11c72327a8143d86ad73799440; derivation seal sha256:73bf9031ce11215cfdd165d86caa1308cfb54d81c41e14bd66e50b036ae72bdd; engine receipt sha256:1f16cb99330ed8077b47621df7cdf3122bb975c5e47d6320fc67cdac852ee6a at receipts/engine/model_admitted/SFT-CHEM-TEMPERATURE-CORRESPONDENCE-002-1f16cb99330ed807.json; empirical hash sha256:5668f6846184af470b6b83671ff17e6d2abbce2aaaed3dae86bfe4370f44870b; measurement receipt sha256:66a07d4f83a2894410537fec5d50d589489dd805c58455ce4eba1dd952007af61.

127. Internal-energy composition law

Claim identity: SFT-CHEM-INTERNAL-ENERGY-COMPOSITION-003

Question and theorem. A complete chemical internal-energy state retains composition, molecular state, phase and environment. Its content is the exact composition of nonempty named positive parts. Change direction is a held rise/fall label and its magnitude is exact positive separation; equality is structural EmptyOne. All 13 NIST state identities seal before the complete 14-column internal-energy vector opens.

complete-held-chemical-internal-energy-state__nonempty-exact-positive-named-part-composition__held-transfer-orientation-plus-positive-magnitude__structural-EmptyOne-equality__complete-same-orientation-step-composition__complete-value-free-thermochemical-identity-seal__complete-13-row-14-column-state-vector__depth-independent-append-only-positive-part-successor

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-STATE-RELATION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-MOLECULAR-FORMATION-ENERGY-013
- SFT-CHEM-FINITE-MICROSTATE-SUPPORT-001
- SFT-CHEM-TEMPERATURE-CORRESPONDENCE-002

Generated grammar. Generate the literal product of carrier, parts, orientation, equality, path, prediction, record and extension forms; decide all 256 candidates only from admitted exact composition, order, energy-conservation and finite-state laws.

Boundary: Every finite nonempty named positive energy-part tuple and every finite same-orientation chemical-state path, with append-only successor closure. External testing uses every row and every column returned by the frozen NIST one-bar water query from 300 to 400 K, including both phase-boundary states.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *complete-held-chemical-internal-energy-state__nonempty-exact-positive-named-part-composition__held-transfer-or*

ientation-plus-positive-magnitude__structural-EmptyOne-equality__complete-same-orientation-step-composition__complete-value-free-thermochemical-identity-seal__complete-13-row-14-column-state-vector__depth-independent-append-only-positive-part-successor. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-held-chemical-internal-energy-state	answer-only-signed-energy-scalar	A signed scalar erases composition, state, phase and environment.
parts	nonempty-exact-positive-named-part-composition	fitted-total-or-unnamed-contribution	A fitted total or unnamed term can absorb any target discrepancy.
orientation	held-transfer-orientation-plus-positive-magnitude	negative-proof-magnitude	A negative proof magnitude conflates direction with content.
equality	structural-EmptyOne-equality	numerical-zero-energy-change	Numerical zero is not an SFT number or proof object.
path	complete-same-orientation-step-composition	endpoint-only-or-signed-cancellation	Endpoint-only cancellation deletes intermediate transfer provenance.
prediction	complete-value-free-thermochemical-identity-seal	internal-energy-target-readable-before-seal	A target value could select the parts or orientation.
record	complete-13-row-14-column-state-vector	selected-state-or-single-phase-vector	A selected state can hide phase and unfavorable rows.
extension	depth-independent-append-only-positive-part-successor	refit-prior-parts-after-successor	Refitting earlier parts destroys compositional invariance.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : e9f9f220534c3ab22f9ac7b0e43ba0366d5bb8daeca378b1c3463660116f568c.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : ac25cd920ca675a39afb321f926b0536aa9556c87516c2521c361a900b926f36.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 78a09cd905908aec930c816bf4003e9d3805f8b80b3ee4885f5e47b267c5959b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 375bcb57f0477b03ee3b23feaa7a6f9924c85f62f4dc5dad2f3ec753201f07c8.

Independent reconstruction. Implementation hash

sha256 : 60648f8c4830ff31c7de322ef0b9810c39d70085cc4ed96fe2f212e3f84c68cd; independent certificate sha256 : b3db337270b11c7c6c087fc082ee74d7769a2dc8fe6341b548495cd3f2130005; external-validation hash sha256 : 804e5c36be9a74cdf420b821315cf48bae694c37fe90e718515045467c81954. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-CHEMISTRY-WEBBOOK-SRD69-WATER-FLUID-PROPERTIES

Comparison records:

- row 1: U=2.0276785 kJ/mol; phase=liquid
- row 2: U=2.7806497 kJ/mol; phase=liquid
- row 3: U=3.5336292 kJ/mol; phase=liquid
- row 4: U=4.2870125 kJ/mol; phase=liquid
- row 5: U=5.0410950 kJ/mol; phase=liquid
- row 6: U=5.7961490 kJ/mol; phase=liquid
- row 7: U=6.5524638 kJ/mol; phase=liquid
- row 8: U=7.3103658 kJ/mol; phase=liquid
- row 9: U=7.5195656 kJ/mol; phase=liquid
- row 10: U=45.138243 kJ/mol; phase=vapor
- row 11: U=45.343651 kJ/mol; phase=vapor
- row 12: U=45.623168 kJ/mol; phase=vapor
- row 13: U=45.899550 kJ/mol; phase=vapor
- complete exact path: 4055357/2000000 to 917991/20000 kJ/mol; all 12 increments positive and additive
- both 372.75593 K liquid/vapour internal-energy states retained
- tapered_internal_energy_order_rejected: True
- complete_13_row_vector_retained: True
- complete_14_column_vector_retained: True
- both_phase_boundary_states_retained: True
- external_signed_inscriptions_retained_but_not_used_as_proof: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if internal energy requires a fitted or unnamed contribution; if change direction requires a negative proof magnitude; if exact successive same-direction steps do not compose to the direct separation; if appending a part changes an earlier part; if any target opens before all 13 identities seal; if any of 13 rows, 14 columns, 9 liquid rows, 4 vapour rows or either 372.75593 K phase-boundary state is omitted; if the complete direct internal-energy path is not exact and increasing; or if target values select the law.

Measurement receipt: sha256:10d47d00656ff6e7c6e4825df5e2ecea992c7af18723f14a8bfd6829c7ba0087.

Isolation certificate: sha256:3aa7d673eb7c6dfe04610d25d507f4436de059fa54cef774e934ccbdebaf6d3f.

Custody certificate: sha256:6c7a0c0d983cc4421563099b63812f13c8ab5a9881f7fcbb401462e2c27ce741.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The thermodynamic extension derives finite microstate support, temperature, energy, heat/work, entropy, enthalpy, free direction, chemical potential, activity, fugacity, phases, colligative response, solvation and coupled transport while keeping external scalar inscriptions downstream. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; equal energy states use structural EmptyOne
- no negative, irrational, imaginary, floating, signed or continuum SFT proof value
- no measured state value, phase, temperature, target payload or target hash in law, grammar or prediction

- no fitted constituent contribution, residual energy, imported thermodynamic equation or species coefficient
- no endpoint-only signed cancellation, selected temperature row, selected phase or deleted phase-boundary state
- external signed glyphs remain source inscriptions and never enter an SFT proof magnitude

Immutable identities. Source manifest

sha256:361da94f50690efc06eab1e727badee5b779e77d38b46ed1f6422c519a02be8e; derivation seal
 sha256:f033540222aa11dd17dbed6a32280756a144d1c9e01b5efbd08966de57d6bf00; engine receipt
 sha256:6ebff46c4f5eb74fd58280b2336b0f99c69245a9835381caee0511c1ce0c4fcc at receipts/engine/model_admitted/SFT-CHEM-INTERNAL-ENERGY-COMPOSITION-003-6ebff46c4f5eb74f.json; empirical
 hash sha256:f0a9d5201d5bdb1a31cd88cdadd1726e6ed5d80fad093542b9e99bf4a157c58f; measurement
 receipt sha256:10d47d00656ff6e7c6e4825df5e2ecea992c7af18723f14a8bfd6829c7ba0087.

128. Chemical heat and work transfer partition

Claim identity: SFT-CHEM-HEAT-WORK-TRANSFER-PARTITION-004

Question and theorem. For one held chemical boundary and path, carrier identity closed by the receiving macro-observation forces the heat class, while a retained organized source-to-response identity forces the work class. Direction is held separately from exact positive content; every transfer remains in a disjoint exhaustive path record, and an absent class is structural EmptyOne.

complete-held-chemical-transfer-path__observation-forced-closed-or-held-carrier-class__held-direction-plus-exact-positive-content__disjoint-exhaustive-heat-work-path-partition__per-class-positive-composition-plus-EmptyOne-absence__complete-value-free-transfer-identity-seal__complete-13-row-joint-calorimetric-expansion-vector__depth-independent-append-only-transfer-successor

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-THERMO-HEAT-WORK-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-STATE-RELATION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-MOLECULAR-FORMATION-ENERGY-013
- SFT-CHEM-FINITE-MICROSTATE-SUPPORT-001
- SFT-CHEM-TEMPERATURE-CORRESPONDENCE-002
- SFT-CHEM-INTERNAL-ENERGY-COMPOSITION-003

Generated grammar. Generate the literal product of carrier, class, orientation, partition, composition, prediction, record and extension forms; decide all 256 candidates only from admitted Fold observation, conservation, composition, Physics heat/work and chemical internal-energy laws.

Boundary: Every finite nonempty chemical transfer path across one held boundary, with each exact positive transfer classified only by retained or closed carrier identity and with append-only successor closure. External testing retains every column and row of the frozen NIST one-bar water surface.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-held-chemical-transfer-path__observation-forced-closed-or-held-carrier-class__held-direction-plus-exact-positive-content__disjoint-exhaustive-heat-work-path-partition__per-class-positive-composition-plus-EmptyOne-absence__complete-value-free-transfer-identity-seal__complete-13-row-joint-calorimetric-expansion-vector__depth-independent-append-only-transfer-successor`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	<code>complete-held-chemical-transfer-path</code>	<code>answer-only-signed-net-energy</code>	A signed answer erases the boundary, path and transfer provenance.
class	<code>observation-forced-closed-or-held-carrier-class</code>	<code>imported-or-arbitrary-heat-work-label</code>	An arbitrary label does not explain the heat/work distinction.
orientation	<code>held-direction-plus-exact-positive-content</code>	<code>negative-or-signed-transfer-content</code>	A signed proof value conflates direction with content.
partition	<code>disjoint-exhaustive-heat-work-path-partition</code>	<code>merged-or-overlapping-transfer-classes</code>	A merged or overlapping record cannot retain which distinction observation closed.
composition	<code>per-class-positive-composition-plus-EmptyOne-absence</code>	<code>signed-net-cancellation-or-numerical-zero</code>	Signed cancellation deletes transfer history and numerical zero is not an SFT number.
prediction	<code>complete-value-free-transfer-identity-seal</code>	<code>calorimetric-or-work-target-readable-before-seal</code>	Target values could select the partition.
record	<code>complete-13-row-joint-calorimetric-expansion-vector</code>	<code>selected-calorimetric-or-expansion-row</code>	A selected row can hide phase and unfavorable records.
extension	<code>depth-independent-append-only-transfer-successor</code>	<code>repartition-prior-path-after-successor</code>	Repartitioning earlier transfers destroys path invariance.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 4e83a2f0ca0302f4ad9fdf21d8f8a0aa5fddb76eecd291c9b0fe6a5074ad23.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: bb9a37bc7bee65cb2078e2e75c575e407c16c4552a160b83a095b90835629fa8.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: a3f6a20cad2355da9ecf3a82deb8ca150d31d9d0ef0d0ab49eb38fe708c46bd4.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: b965bcc294fa4000754d6417e81c5eb8ce5efaa93154fa6b55922b9bf091d36a.

Independent reconstruction. Implementation hash

sha256: f9e7542e4f2be91f39960e9e527abdf9a9fb973e325abf9c94a8ad71a2b01eb6; independent certificate sha256: 3c44ba86dbf39f169bcbca25a773e2732ac1b9383710a66e47100fb88323fa7a; external-validation hash sha256: 6a7ef47c237ba1d45b90eb27ac1e3f0f582ff34a8563668552e75b17f1ad6ade. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- NIST-CHEMISTRY-WEBBOOK-SRD69-WATER-FLUID-PROPERTIES

Comparison records:

- row 1: $C_p=75.315341 \text{ J/(mol K)}$; $pV=18077521/10000000000 \text{ kJ/mol}$; $H-U=18077/10000000 \text{ kJ/mol}$
- row 2: $C_p=75.290216 \text{ J/(mol K)}$; $pV=4533817/2500000000 \text{ kJ/mol}$; $H-U=3627/2000000 \text{ kJ/mol}$
- row 3: $C_p=75.313511 \text{ J/(mol K)}$; $pV=18207793/10000000000 \text{ kJ/mol}$; $H-U=569/312500 \text{ kJ/mol}$
- row 4: $C_p=75.369660 \text{ J/(mol K)}$; $pV=18293583/10000000000 \text{ kJ/mol}$; $H-U=18293/10000000 \text{ kJ/mol}$
- row 5: $C_p=75.453278 \text{ J/(mol K)}$; $pV=4597911/2500000000 \text{ kJ/mol}$; $H-U=2299/1250000 \text{ kJ/mol}$
- row 6: $C_p=75.564505 \text{ J/(mol K)}$; $pV=9250669/5000000000 \text{ kJ/mol}$; $H-U=18501/10000000 \text{ kJ/mol}$
- row 7: $C_p=75.706300 \text{ J/(mol K)}$; $pV=18622293/10000000000 \text{ kJ/mol}$; $H-U=9311/5000000 \text{ kJ/mol}$
- row 8: $C_p=75.882932 \text{ J/(mol K)}$; $pV=375087/200000000 \text{ kJ/mol}$; $H-U=9377/5000000 \text{ kJ/mol}$
- row 9: $C_p=75.938370 \text{ J/(mol K)}$; $pV=18792693/10000000000 \text{ kJ/mol}$; $H-U=2349/1250000 \text{ kJ/mol}$
- row 10: $C_p=37.443823 \text{ J/(mol K)}$; $pV=30516561/10000000 \text{ kJ/mol}$; $H-U=381457/125000 \text{ kJ/mol}$
- row 11: $C_p=36.943657 \text{ J/(mol K)}$; $pV=31155601/10000000 \text{ kJ/mol}$; $H-U=77889/25000 \text{ kJ/mol}$
- row 12: $C_p=36.484609 \text{ J/(mol K)}$; $pV=16015133/5000000 \text{ kJ/mol}$; $H-U=1601513/500000 \text{ kJ/mol}$
- row 13: $C_p=36.170298 \text{ J/(mol K)}$; $pV=6579653/2000000 \text{ kJ/mol}$; $H-U=3289827/1000000 \text{ kJ/mol}$
- all 13 direct calorimetric values exact positive
- all 13 pressure-volume and independent state-record work magnitudes exact positive and mutually consistent within displayed resolution
- both 372.75593 K liquid/vapour records retained
- `tampered_nonpositive_calorimetric_record_rejected`: True
- `complete_13_row_vector_retained`: True
- `complete_14_column_vector_retained`: True
- `both_phase_boundary_states_retained`: True
- `calorimetric_and_expansion_work_vectors_distinct_and_complete`: True
- `prediction_contains_no_withheld_target_hash`: True

Falsification boundary: The claim fails if one transfer requires an arbitrary rather than observation-forced heat/work class; if a transfer occurs in both or neither class; if direction requires a negative proof value; if an absent class requires numerical zero; if appending a transfer changes a prior record; if any target opens before all identities seal; if any returned row or column is omitted; if any direct calorimetric or expansion-work magnitude is not positive; if the independently recorded enthalpy/internal-energy separation and pressure-volume work do not agree within their exact displayed-resolution bounds; or if target values select the law.

Measurement receipt: sha256:5c4d5b41dc0a49a39782902cfb86b12ef8d3c32c5c68575f58addabcf1d8fad1.

Isolation certificate: sha256:81bf95c82940d94e342be09f53b8a4ca8f123034b509969760140f0c8721c13e.

Custody certificate: sha256:aef9620369263c5178beabd7c918f35eae5782576694e6ee81d53b659a2eea1a.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The thermodynamic extension derives finite microstate support, temperature, energy, heat/work, entropy, enthalpy, free direction, chemical potential, activity, fugacity, phases, colligative response, solvation and coupled transport while keeping external scalar inscriptions downstream. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; an absent transfer class is structural EmptyOne
- no negative, irrational, imaginary, floating, signed or continuum SFT proof value
- no imported heat/work equation, fitted coefficient, target-selected class or measured transfer value
- no target payload, temperature, phase, heat capacity, volume, enthalpy or internal energy before prediction seal
- no signed net cancellation, merged transfer classes, omitted path record or selected external row
- external signed glyphs remain source inscriptions and never enter an SFT proof magnitude

Immutable identities. Source manifest

sha256:c217ff2d8372d1fc8b8a8a98b42a92682bb89995eaa5523d5e65378c843bd5ac; derivation seal
 sha256:d771ebde216f4c1a7e29960dd1ce034c6eb5a77db871cf6f557d1bb78f7438; engine receipt
 sha256:3afeefa6b1dbde4d0a0000391a9865ad98d601460775ae79a5df5d9178494fb7 at receipts/engine/model_admitted/SFT-CHEM-HEAT-WORK-TRANSFER-PARTITION-004-3afeefa6b1dbde4d.json; empirical
 hash sha256:51bb64c053ab4db06b2ada473501729cc34f315e34bc66c00608e5f56fef97; measurement
 receipt sha256:5c4d5b41dc0a49a39782902cfb86b12ef8d3c32c5c68575f58addabcf1d8fad1.

129. Chemical entropy and multiplicity correspondence

Claim identity: SFT-CHEM-ENTROPY-MULTIPLICITY-CORRESPONDENCE-005

Question and theorem. Chemical entropy is the complete ledger of finite microstates merged by each held macro-observation: exact positive class multiplicity, exact support part, retained members and every unresolved pair. Singleton certainty is structural EmptyOne. Conventional scalar entropy remains a post-seal external inscription; no logarithm or irrational proof value enters the law.

complete-finite-chemical-microstate-support__disjoint-exhaustive-chemical-observation-classes__exact-positive-class-count-and-whole-part__complete-unresolved-distinction-ledger__structural-EmptyOne-singleton-certainty__complete-value-free-entropy-phase-identity-seal__complete-13-row-entropy-phase-transition-vector__depth-independent-one-microstate-ledger-successor

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-THERMO-ENTROPY-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-CHEM-FINITE-MICROSTATE-SUPPORT-001
- SFT-CHEM-TEMPERATURE-CORRESPONDENCE-002
- SFT-CHEM-INTERNAL-ENERGY-COMPOSITION-003
- SFT-CHEM-HEAT-WORK-TRANSFER-PARTITION-004

Generated grammar. Generate the literal product of support, partition, multiplicity, entropy, certainty, prediction, record and extension forms; decide all 256 candidates only from admitted exact information-support, observation, combinatorial and finite chemical microstate laws.

Boundary: Every finite nonempty generated chemical microstate support and total macro-observation partition, with exact positive class counts, complete within-class pair ledgers and one-microstate successor closure. External testing retains every entropy and phase row of the frozen NIST one-bar water surface.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-finite-chemical-microstate-support__disjoint-exhaustive-chemical-observation-classes__exact-positive-class-count-and-whole-part__complete-unresolved-distinction-ledger__structural-EmptyOne-singleton-certainty__complete-value-free-entropy-phase-identity-seal__complete-13-row-entropy-phase-transition-vector__depth-independent-one-microstate-ledger-successor`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
support	complete-finite-chemical-microstate-support	selected-or-completed-infinite-chemical-support	A selected or completed-infinite support cannot establish complete multiplicity.
partition	disjoint-exhaustive-chemical-observation-classes	overlapping-or-partial-macroclasses	Overlap or omission destroys exact observation multiplicity.
multiplicity	exact-positive-class-count-and-whole-part	fitted-probability-or-floating-weight	A fit imports a distribution and precision choice.
entropy	complete-unresolved-distinction-ledger	logarithmic-or-irrational-scalar-proof	A logarithmic scalar imports forbidden irrational proof values and erases provenance.
certainty	structural-EmptyOne-singleton-certainty	numerical-zero-singleton-entropy	Numerical zero is not an SFT proof value.
prediction	complete-value-free-entropy-phase-identity-seal	entropy-or-phase-target-readable-before-seal	Observed entropy could select multiplicity or ordering.
record	complete-13-row-entropy-phase-transition-vector	selected-entropy-or-single-phase-row	A selected row can hide a phase transition or adverse orientation.
extension	depth-independent-one-microstate-ledger-successor	resample-or-refit-after-new-microstate	Resampling changes prior evidence and cannot prove every finite support.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:2fd6df01f2f6645b918c4f0a42239b0aba93dcd0823d396b5fb301de9af0950d.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:00e0520a37f438e24cdfa26b3b4ea34524c63daa20954e348a468a4061bc6753.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:9c90cac199bcb345699c5ac52eab8a7e188ed66099aa4e20873cb42178eec7f5.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:08c5c80e1a4e9b120426e638ae944da76c9c0da6feee765064e024a4818a88f9.

Independent reconstruction. Implementation hash

sha256: a9f57b40167436d77fcbe78d6d6b39bf97d2db9f1356e675961c8470d46c43e5; independent certificate sha256: a82c23d9648a68fe44ce81cd969bb6f8c3afb9f42a12f9e086e71b3030d44035; external-validation hash sha256: 909555fd3b19fab42fdb277f8364a9ed86fa30823efcfd15b8aeda69f1e8622c. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-CHEMISTRY-WEBBOOK-SRD69-WATER-FLUID-PROPERTIES

Comparison records:

- row 1: S=7.0811251 J/(mol K); phase=liquid
- row 2: S=9.5501254 J/(mol K); phase=liquid
- row 3: S=11.940759 J/(mol K); phase=liquid
- row 4: S=14.259066 J/(mol K); phase=liquid
- row 5: S=16.510249 J/(mol K); phase=liquid
- row 6: S=18.698988 J/(mol K); phase=liquid
- row 7: S=20.829618 J/(mol K); phase=liquid
- row 8: S=22.906217 J/(mol K); phase=liquid
- row 9: S=23.469536 J/(mol K); phase=liquid
- row 10: S=132.57164 J/(mol K); phase=vapor
- row 11: S=133.28721 J/(mol K); phase=vapor
- row 12: S=134.24049 J/(mol K); phase=vapor
- row 13: S=135.16001 J/(mol K); phase=vapor
- complete exact entropy path: 70811251/1000000 to 13516001/100000 J/(mol K); all 12 increments positive and additive
- phase entropy jump: 13637763/125000 J/(mol K); independent enthalpy/temperature record 4066845420/37275593
- both 372.75593 K liquid/vapour entropy states retained
- tampered_nonpositive_entropy_record_rejected: **True**
- complete_13_row_vector_retained: **True**
- complete_14_column_vector_retained: **True**
- both_phase_boundary_states_retained: **True**
- complete_entropy_path_and_phase_jump_retained: **True**
- prediction_contains_no_withheld_target_hash: **True**

Falsification boundary: The claim fails if chemical entropy requires a logarithmic or irrational proof scalar rather than the complete unresolved-distinction ledger; if class multiplicity is fitted; if a microstate occurs in both or neither class; if singleton certainty requires numerical zero; if successor extension changes prior identities; if any target opens before all identities seal; if any of thirteen entropy or phase rows is omitted; if the complete external entropy path or phase-transition entropy is not exact positive; if the independently recorded phase enthalpy/temperature ratio disagrees with the entropy jump beyond exact displayed-resolution bounds; or if target values select the law.

Measurement receipt: sha256: 4db60da8563cfa7761a962cc7bf5332d5f6625e2727dfcce0105a7c7200c2e23.
Isolation certificate: sha256: c4efc7f66476dd0582b52d2d2692705062e84f4ad16ab0b5bb3a8f5b4f428e37.
Custody certificate: sha256: 495902901cdc6ab1ad35f0b7d46122a597cdc43001fe6da93e638ce2d76be078.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The thermodynamic extension derives finite microstate support, temperature, energy, heat/work, entropy, enthalpy, free direction, chemical potential, activity, fugacity, phases, colligative response, solvation and coupled transport while keeping external scalar inscriptions downstream. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; singleton certainty is structural EmptyOne
- no negative, irrational, imaginary, logarithmic, floating or continuum SFT proof value
- no imported statistical distribution, partition function, fitted probability, log base or entropy convention
- no target payload, scalar entropy, phase, temperature or transition value before prediction seal
- no selected support, overlapping class, omitted distinction pair, selected phase row or deleted boundary state
- external scalar entropy inscriptions remain correspondence records and never become SFT proof values

Immutable identities. Source manifest

sha256:d46fbbfe25da937c84b5d171b51b3ef93b24a418857d699b7d8037b8b4fbfd5e; derivation seal
 sha256:7549259d8e53077892d79f442434d9caf372660e616ed47ffdf84bf016061559; engine receipt
 sha256:be32db30904fd7974578c60b3e656371d428f718d2b9dec8097277d135f32ce1 at receipts/engine/model_admitted/SFT-CHEM-ENTROPY-MULTIPLICITY-CORRESPONDENCE-005-be32db30904fd797.json
 ; empirical hash sha256:bfc089916b3d21a70856ef8494996da93a4066d16bf52f04bca2fb10642b7a6f;
 measurement receipt sha256:4db60da8563cfa7761a962cc7bf5332d5f6625e2727dfcce0105a7c7200c2e23.

130. Enthalpy-equivalent chemical state relation

Claim identity: SFT-CHEM-ENTHALPY-EQUIVALENT-STATE-006

Question and theorem. An enthalpy-equivalent chemical state is the complete held composition, molecular state, phase and environment carrier whose exact positive content composes admitted internal energy with every retained organized environment-transfer part. If no such part is generated its place is structural EmptyOne. State orientation is held separately from exact positive separation.

complete-held-state-and-environment-carrier__retained-exact-positive-internal-content__observation-forced-organized-environment-transfer-parts__exact-positive-composition-plus-EmptyOne-absence__held-state-orientation-plus-positive-separation__complete-value-free-enthalpy-state-identity-seal__complete-13-row-enthalpy-component-vector__depth-independent-append-only-environment-part-successor

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-THERMO-HEAT-WORK-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-STATE-RELATION-001

- SFT-CHEM-MOLECULAR-FORMATION-ENERGY-013
- SFT-CHEM-INTERNAL-ENERGY-COMPOSITION-003
- SFT-CHEM-HEAT-WORK-TRANSFER-PARTITION-004
- SFT-CHEM-ENTROPY-MULTIPLICITY-CORRESPONDENCE-005

Generated grammar. Generate the literal product of state, internal, environment, composition, orientation, prediction, record and extension forms; decide all 256 candidates only from admitted exact composition, energy conservation, heat/work observation and chemical internal-energy laws.

Boundary: Every finite complete chemical state with exact positive internal content and every finite tuple of uniquely named retained organized environment-transfer parts, including structural EmptyOne absence and append-only successor closure.

External testing retains all rows and columns of the frozen NIST one-bar water surface.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-held-state-and-environment-carrier__retained-exact-positive-internal-content__observation-forced-organized-environment-transfer-parts__exact-positive-composition-plus-EmptyOne-absence__held-state-orientation-plus-positive-separation__complete-value-free-enthalpy-state-identity-seal__complete-13-row-enthalpy-component-vector__depth-independent-append-only-environment-part-successor`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
state	complete-held-state-and-environment-carrier	answer-only-signed-enthalpy-scalar	A signed scalar erases composition, phase, molecular state and environment.
internal	retained-exact-positive-internal-content	fitted-or-unnamed-internal-content	A fitted internal term can absorb any target discrepancy.
environment	observation-forced-organized-environment-transfer-parts	imported-pressure-volume-equation	An imported equation would select the environment correction rather than derive its role.
composition	exact-positive-composition-plus-EmptyOne-absence	signed-cancellation-or-numerical-zero-correction	Signs erase transfer provenance and numerical zero is not an SFT number.
orientation	held-state-orientation-plus-positive-separation	negative-enthalpy-proof-magnitude	A negative proof magnitude conflates relation direction with content.
prediction	complete-value-free-enthalpy-state-identity-seal	enthalpy-target-readable-before-seal	Target values could select an environment part or state order.
record	complete-13-row-enthalpy-component-vector	selected-enthalpy-or-single-phase-row	A selected row can hide phase and unfavorable records.
extension	depth-independent-append-only-environment-part-successor	refit-prior-state-after-environment-successor	Refitting prior content destroys compositional invariance.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:80a926b4621cb12ddc5ac7d7ca253e0d11f2b8c47ad3ac819a5924fc384ba52c`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:02e5082b7011e51d90a170ebfaf79de534275e71f43b0b6c882347c387d14767`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:dc3ee4a2d41ddcf2768cb2c32145867e7e91f90f2f3060dca1b2553f07a99bc0.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:19004c2bb651b15e01bfa9952eff5ec70445d968ceec5de2b02451229f237451.

Independent reconstruction. Implementation hash

sha256:0ddb94891f25c3102989f4cf13996785f8a568e5b6a40e2630b234a37fca4332; independent certificate sha256:bf6d0af822c6a9ed428ead68851822dbc315a8ee7fc51127c3fb6b7ba10feaa8; external-validation hash sha256:6220d02785bee0dc696c875fab8179a0d79748615ffe2c7a37bd734266cc0da4. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-CHEMISTRY-WEBBOOK-SRD69-WATER-FLUID-PROPERTIES

Comparison records:

- row 1: H=2.0294862 kJ/mol; U=2.0276785 kJ/mol; environment=18077/10000000 kJ/mol
- row 2: H=2.7824632 kJ/mol; U=2.7806497 kJ/mol; environment=3627/2000000 kJ/mol
- row 3: H=3.5354500 kJ/mol; U=3.5336292 kJ/mol; environment=569/312500 kJ/mol
- row 4: H=4.2888418 kJ/mol; U=4.2870125 kJ/mol; environment=18293/10000000 kJ/mol
- row 5: H=5.0429342 kJ/mol; U=5.0410950 kJ/mol; environment=2299/1250000 kJ/mol
- row 6: H=5.7979991 kJ/mol; U=5.7961490 kJ/mol; environment=18501/10000000 kJ/mol
- row 7: H=6.5543260 kJ/mol; U=6.5524638 kJ/mol; environment=9311/5000000 kJ/mol
- row 8: H=7.3122412 kJ/mol; U=7.3103658 kJ/mol; environment=9377/5000000 kJ/mol
- row 9: H=7.5214448 kJ/mol; U=7.5195656 kJ/mol; environment=2349/1250000 kJ/mol
- row 10: H=48.189899 kJ/mol; U=45.138243 kJ/mol; environment=381457/125000 kJ/mol
- row 11: H=48.459211 kJ/mol; U=45.343651 kJ/mol; environment=77889/25000 kJ/mol
- row 12: H=48.826194 kJ/mol; U=45.623168 kJ/mol; environment=1601513/500000 kJ/mol
- row 13: H=49.189377 kJ/mol; U=45.899550 kJ/mol; environment=3289827/1000000 kJ/mol
- complete exact enthalpy path: 10147431/5000000 to 49189377/1000000 kJ/mol; all 12 increments positive and additive
- all 13 environment components agree with pressure-volume records within exact displayed resolution
- both 372.75593 K liquid/vapour enthalpy states retained
- tampered_nonpositive_environment_component_rejected: True
- complete_13_row_vector_retained: True
- complete_14_column_vector_retained: True
- both_phase_boundary_states_retained: True
- complete_enthalpy_and_component_vectors_retained: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if an enthalpy-equivalent state requires an imported thermodynamic equation, signed scalar, fitted environment correction or lost state identity; if absence requires numerical zero; if direction requires negative content; if successor extension changes a prior part; if any target opens before all identities seal; if any of thirteen enthalpy/component rows is omitted; if the complete enthalpy path or environment contributions are not exact positive; if the independent pressure-volume and enthalpy/internal-energy component records disagree beyond exact displayed-resolution bounds; or if targets select the law.

Measurement receipt: sha256:82e1e19d3e2363208efc1c300c15929bb8bcc926794199d862c966a6974e6690.
 Isolation certificate: sha256:1eab2c82deb391bb06915384a469db720fb45720bf9e771425aa6446e6ef76c2.
 Custody certificate: sha256:c2fa13806128d50ef1e57d14d4290cf0fa3b5d93f5cce9eed6151a4da632352d.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The thermodynamic extension derives finite microstate support, temperature, energy, heat/work, entropy, enthalpy, free direction, chemical potential, activity, fugacity, phases, colligative response, solvation and coupled transport while keeping external scalar inscriptions downstream. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; absent environment contribution is structural EmptyOne
- no negative, irrational, imaginary, floating, signed or continuum SFT proof value
- no imported enthalpy or pressure-volume equation, fitted correction or target-selected environment part
- no target payload, enthalpy, internal energy, pressure, volume, phase or temperature before prediction seal
- no signed cancellation, unnamed contribution, selected state row, selected phase or deleted boundary state
- external conventional component relations remain post-seal correspondence records only

Immutable identities. Source manifest

sha256:c61b1a94ba0f2fcf859bba87b5d339abccf8673697678a3ad90bcde8285b8f31; derivation seal
 sha256:7ac5e9fcd904d896e31fc52cfbac88fd8f2f18dcc470f2c46a3baefc072b9a90; engine receipt
 sha256:5b82f6813b158e70e196ccabedafa13a2c287064aa34d91a9eaa64282c8c698f at receipts/engine/model_admitted/SFT-CHEM-ENTHALPY-EQUIVALENT-STATE-006-5b82f6813b158e70.json; empirical
 hash sha256:6e328f11086977b9a93ce3dfe4631c17996f495986f0897dbf1d5f8b9b3cf8cf; measurement
 receipt sha256:82e1e19d3e2363208efc1c300c15929bb8bcc926794199d862c966a6974e6690.

131. Free-energy-equivalent reaction direction law

Claim identity: SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007

Question and theorem. Reaction direction is the held orientation forced by strict exact product order between complete forward and reverse retained-energy and closed-distinction accounts. Equality is equilibrium with structural EmptyOne separations; incomparable accounts halt rather than receiving a fitted trade-off. All 64 paired JANAF row identities seal before Gibbs, log-K, temperature or direction values open.

complete-forward-and-reverse-path-accounts__exact-positive-retained-energy-requirements__exact-positive-closed-distinction-counts__strict-exact-product-support-order__held-forward-reverse-or-EmptyOne-equilibrium__complete-value-free-reaction-state-identity-seal__complete-64-row-two-direction-crossing-vector__depth-independent-common-account-successor

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001

- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-THERMO-STATE-RELATION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-INTERNAL-ENERGY-COMPOSITION-003
- SFT-CHEM-HEAT-WORK-TRANSFER-PARTITION-004
- SFT-CHEM-ENTROPY-MULTIPLICITY-CORRESPONDENCE-005
- SFT-CHEM-ENTHALPY-EQUIVALENT-STATE-006

Generated grammar. Generate the literal product of paths, energy, distinctions, order, orientation, prediction, record and extension forms; decide all 256 candidates only from admitted exact energy, entropy, observation, reaction-direction and equilibrium laws.

Boundary: Every finite pair of forward/reverse reaction path accounts on one held boundary and condition, ordered only by exact positive retained-energy and closed-distinction support, with common-successor closure. External testing retains every finite common NIST-JANAF NO₂/N₂O₄ gas row and every direction.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-forward-and-reverse-path-accounts__exact-positive-retained-energy-requirements__exact-positive-closed-distinction-counts__strict-exact-product-support-order__held-forward-reverse-or-EmptyOne-equilibrium__complete-value-free-reaction-state-identity-seal__complete-64-row-two-direction-crossing-vector__depth-independent-common-account-successor`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
paths	complete-forward-and-reverse-path-accounts	single-path-or-answer-only-direction	One path or an answer label cannot establish comparative direction.
energy	exact-positive-retained-energy-requirements	signed-or-fitted-free-energy-scalar	A signed or fitted scalar imports an equation and erases energy provenance.
distinctions	exact-positive-closed-distinction-counts	entropy-logarithm-or-omitted-distinction-cost	A logarithm or omitted cost imports a convention and loses closed distinctions.
order	strict-exact-product-support-order	weighted-sum-or-target-selected-comparison	A weighted scalar requires a free trade-off and may select the target.
orientation	held-forward-reverse-or-EmptyOne-equilibrium	negative-direction-value-or-forced-tie-break	A sign or tie-break invents content not present in the accounts.
prediction	complete-value-free-reaction-state-identity-seal	Gibbs-logK-or-direction-target-readable-before-seal	Target values could select the ordering rule.
record	complete-64-row-two-direction-crossing-vector	selected-temperature-or-single-direction-showcase	A selected row can hide a direction reversal or adverse record.
extension	depth-independent-common-account-successor	refit-order-after-shared-successor	Refitting destroys invariance under a common retained addition.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:606fc1c721222f9b47efb82f53bcb9aa9d6a1ebde9c081874db5ba3a3993386c.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:5adfd61c8d83a048a08e78a34fe2ed3ae7b04629a03233641f88c51078de5970.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:08802dfefb69a977d4c492e1acf4def01de341239bd9409f075e7511c32ae830b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:83e84fac56f05faea06d15de3226e865ffae6297e46b6a805107ad4eb135606c.

Independent reconstruction. Implementation hash

sha256:c0fd1c02d286a77db2825af20db087b2276dcfa067c7770c1d90aee61cf375b5; independent certificate sha256:afb17e383160d882c382d1a5756cc472416e78ca155854240c3ad5e3f70d4732; external-validation hash sha256:c2d5eaca5fa5a82dae93be7494c53ccab1beb058fc452f5068db053d1c0f8f96. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-JANAF-N-007-NO2-GAS
- NIST-JANAF-N-032-N2O4-GAS

Comparison records:

- row 1: T=100 K; |reaction Gibbs|=39189/1000 kJ/mol; direction=reverse; external log10 K=-20469/1000
- row 2: T=200 K; |reaction Gibbs|=2748/125 kJ/mol; direction=reverse; external log10 K=-2871/500
- row 3: T=250 K; |reaction Gibbs|=13197/1000 kJ/mol; direction=reverse; external log10 K=-2757/1000
- row 4: T=5963/20 K; |reaction Gibbs|=4729/1000 kJ/mol; direction=reverse; external log10 K=-207/250
- row 5: T=300 K; |reaction Gibbs|=1101/250 kJ/mol; direction=reverse; external log10 K=-383/500
- row 6: T=350 K; |reaction Gibbs|=4361/1000 kJ/mol; direction=forward; external log10 K=163/250
- row 7: T=400 K; |reaction Gibbs|=3271/250 kJ/mol; direction=forward; external log10 K=427/250
- row 8: T=450 K; |reaction Gibbs|=21757/1000 kJ/mol; direction=forward; external log10 K=1263/500
- row 9: T=500 K; |reaction Gibbs|=243/8 kJ/mol; direction=forward; external log10 K=1587/500
- row 10: T=600 K; |reaction Gibbs|=47451/1000 kJ/mol; direction=forward; external log10 K=4131/1000
- row 11: T=700 K; |reaction Gibbs|=64323/1000 kJ/mol; direction=forward; external log10 K=24/5
- row 12: T=800 K; |reaction Gibbs|=81003/1000 kJ/mol; direction=forward; external log10 K=5289/1000
- row 13: T=900 K; |reaction Gibbs|=97509/1000 kJ/mol; direction=forward; external log10 K=283/50
- row 14: T=1000 K; |reaction Gibbs|=28463/250 kJ/mol; direction=forward; external log10 K=5947/1000
- row 15: T=1100 K; |reaction Gibbs|=65023/500 kJ/mol; direction=forward; external log10 K=247/40
- row 16: T=1200 K; |reaction Gibbs|=18263/125 kJ/mol; direction=forward; external log10 K=6359/1000
- row 17: T=1300 K; |reaction Gibbs|=162031/1000 kJ/mol; direction=forward; external log10 K=6511/1000
- row 18: T=1400 K; |reaction Gibbs|=177841/1000 kJ/mol; direction=forward; external log10 K=1659/250
- row 19: T=1500 K; |reaction Gibbs|=193537/1000 kJ/mol; direction=forward; external log10 K=337/50
- row 20: T=1600 K; |reaction Gibbs|=209129/1000 kJ/mol; direction=forward; external log10 K=1707/250
- row 21: T=1700 K; |reaction Gibbs|=112311/500 kJ/mol; direction=forward; external log10 K=6903/1000

- row 22: T=1800 K; |reaction Gibbs|=12001/50 kJ/mol; direction=forward; external log10 K=3483/500
- row 23: T=1900 K; |reaction Gibbs|=31916/125 kJ/mol; direction=forward; external log10 K=7019/1000
- row 24: T=2000 K; |reaction Gibbs|=33819/125 kJ/mol; direction=forward; external log10 K=7067/1000
- row 25: T=2100 K; |reaction Gibbs|=57139/200 kJ/mol; direction=forward; external log10 K=7107/1000
- row 26: T=2200 K; |reaction Gibbs|=150381/500 kJ/mol; direction=forward; external log10 K=357/50
- row 27: T=2300 K; |reaction Gibbs|=63151/200 kJ/mol; direction=forward; external log10 K=1793/250
- row 28: T=2400 K; |reaction Gibbs|=330677/1000 kJ/mol; direction=forward; external log10 K=7197/1000
- row 29: T=2500 K; |reaction Gibbs|=345531/1000 kJ/mol; direction=forward; external log10 K=7219/1000
- row 30: T=2600 K; |reaction Gibbs|=360319/1000 kJ/mol; direction=forward; external log10 K=7239/1000
- row 31: T=2700 K; |reaction Gibbs|=75009/200 kJ/mol; direction=forward; external log10 K=7257/1000
- row 32: T=2800 K; |reaction Gibbs|=38971/100 kJ/mol; direction=forward; external log10 K=7269/1000
- row 33: T=2900 K; |reaction Gibbs|=404317/1000 kJ/mol; direction=forward; external log10 K=7283/1000
- row 34: T=3000 K; |reaction Gibbs|=418867/1000 kJ/mol; direction=forward; external log10 K=3647/500
- row 35: T=3100 K; |reaction Gibbs|=433361/1000 kJ/mol; direction=forward; external log10 K=3651/500
- row 36: T=3200 K; |reaction Gibbs|=111951/250 kJ/mol; direction=forward; external log10 K=7309/1000
- row 37: T=3300 K; |reaction Gibbs|=92439/200 kJ/mol; direction=forward; external log10 K=7317/1000
- row 38: T=3400 K; |reaction Gibbs|=59567/125 kJ/mol; direction=forward; external log10 K=3661/500
- row 39: T=3500 K; |reaction Gibbs|=49083/100 kJ/mol; direction=forward; external log10 K=293/40
- row 40: T=3600 K; |reaction Gibbs|=252537/500 kJ/mol; direction=forward; external log10 K=916/125
- row 41: T=3700 K; |reaction Gibbs|=259637/500 kJ/mol; direction=forward; external log10 K=7331/1000
- row 42: T=3800 K; |reaction Gibbs|=533431/1000 kJ/mol; direction=forward; external log10 K=7333/1000
- row 43: T=3900 K; |reaction Gibbs|=547543/1000 kJ/mol; direction=forward; external log10 K=3667/500
- row 44: T=4000 K; |reaction Gibbs|=56161/100 kJ/mol; direction=forward; external log10 K=7333/1000
- row 45: T=4100 K; |reaction Gibbs|=14391/25 kJ/mol; direction=forward; external log10 K=7333/1000
- row 46: T=4200 K; |reaction Gibbs|=589629/1000 kJ/mol; direction=forward; external log10 K=3667/500
- row 47: T=4300 K; |reaction Gibbs|=603577/1000 kJ/mol; direction=forward; external log10 K=7331/1000
- row 48: T=4400 K; |reaction Gibbs|=617487/1000 kJ/mol; direction=forward; external log10 K=7331/1000
- row 49: T=4500 K; |reaction Gibbs|=15784/25 kJ/mol; direction=forward; external log10 K=7329/1000
- row 50: T=4600 K; |reaction Gibbs|=129039/200 kJ/mol; direction=forward; external log10 K=3663/500
- row 51: T=4700 K; |reaction Gibbs|=131799/200 kJ/mol; direction=forward; external log10 K=7323/1000
- row 52: T=4800 K; |reaction Gibbs|=672761/1000 kJ/mol; direction=forward; external log10 K=7321/1000
- row 53: T=4900 K; |reaction Gibbs|=68649/100 kJ/mol; direction=forward; external log10 K=7317/1000
- row 54: T=5000 K; |reaction Gibbs|=700187/1000 kJ/mol; direction=forward; external log10 K=3657/500
- row 55: T=5100 K; |reaction Gibbs|=713851/1000 kJ/mol; direction=forward; external log10 K=7311/1000
- row 56: T=5200 K; |reaction Gibbs|=363741/500 kJ/mol; direction=forward; external log10 K=1827/250
- row 57: T=5300 K; |reaction Gibbs|=741081/1000 kJ/mol; direction=forward; external log10 K=913/125
- row 58: T=5400 K; |reaction Gibbs|=15093/20 kJ/mol; direction=forward; external log10 K=7301/1000
- row 59: T=5500 K; |reaction Gibbs|=384093/500 kJ/mol; direction=forward; external log10 K=1459/200

- row 60: T=5600 K; |reaction Gibbs|=781693/1000 kJ/mol; direction=forward; external log10 K=1823/250
- row 61: T=5700 K; |reaction Gibbs|=198793/250 kJ/mol; direction=forward; external log10 K=7287/1000
- row 62: T=5800 K; |reaction Gibbs|=808621/1000 kJ/mol; direction=forward; external log10 K=7283/1000
- row 63: T=5900 K; |reaction Gibbs|=822041/1000 kJ/mol; direction=forward; external log10 K=7277/1000
- row 64: T=6000 K; |reaction Gibbs|=104429/125 kJ/mol; direction=forward; external log10 K=7273/1000
- complete vector: 5 reverse rows followed by 59 forward rows
- exact equilibrium-crossing bracket: 300 K reverse; 350 K forward
- tampered_direction_rejected: True
- complete_64_row_vector_retained: True
- both_direction_classes_retained: True
- single_crossing_bracket_retained: True
- complete_two_species_eight_column_rows_retained: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if direction requires a fitted scalar trade-off, imported free-energy equation, sign-valued SFT proof or arbitrary tie-break; if exact equal accounts do not yield structural equilibrium; if common extension changes direction; if any target opens before all 64 identities seal; if any common JANAF row or species column is omitted; if Gibbs and log-K records disagree in direction; if all five reverse, all fifty-nine forward or the 300-to-350 K crossing bracket are not retained; or if targets select the law.

Measurement receipt: sha256:9aec02fc1628bc00abcbf34aa1be47e311cadcd76d1ce5200a7757652fa654b1.

Isolation certificate: sha256:59d077a1159627b2479cf9a6c3124e1aacd14982fbef6b3f9877823b2b11f383.

Custody certificate: sha256:69204eb9424349edb43174bb7bdb841d1fc3243e0a4a40c544d4ee5dd00678f0.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The thermodynamic extension derives finite microstate support, temperature, energy, heat/work, entropy, enthalpy, free direction, chemical potential, activity, fugacity, phases, colligative response, solvation and coupled transport while keeping external scalar inscriptions downstream. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; exact equality uses structural EmptyOne separations
- no negative, irrational, imaginary, logarithmic, floating, signed or continuum SFT proof value
- no imported Gibbs/free-energy equation, fitted axis weight, equilibrium parameter or target-derived orientation
- no temperature, formation Gibbs, log-K, direction, target payload or target hash before prediction seal
- no selected temperature, deleted direction row, interpolated crossing or omitted source column
- external signed Gibbs and log-K inscriptions remain post-seal correspondence records only

Immutable identities. Source manifest

sha256:b607e984bad0b7d5b77e31915cf67f6610211731aa39a9c4090797aa0b814886; derivation seal

sha256:8a15904e4938c2c3bac1c8ab1435d1e6f27f1e2e34c2ed91f96f20547d035793; engine receipt

sha256:f4cd9bc049a27671c9c722771d41956cc393a30419980ad2efa6b09ba7c62945 at receipts/engine/model_admitted/SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007-f4cd9bc049a27671.json;

empirical hash sha256:7fe1617328243e35638ba4132cc6d677c40148053b52d7c1108ccda978028f85;

measurement receipt sha256:9aec02fc1628bc00abcbf34aa1be47e311cadcd76d1ce5200a7757652fa654b1.

132. Chemical-potential-equivalent component law

Claim identity: SFT-CHEM-CHEMICAL-POTENTIAL-EQUIVALENT-COMPONENT-008

Question and theorem. The chemical-potential-equivalent of one held component is its complete exact phase-specific marginal addition account. At one fixed environment, strict product order directs paired exchange and exact equality is equilibrium with EmptyOne separations, without requiring equal bulk phase compositions.

complete-distinct-phase-component-addition-accounts__same-held-component-and-fixed-environment__paired-one-component-exchange-conserving-total-carrier__exact-positive-energy-and-closed-distinction-increments__strict-product-order-with-EmptyOne-equilibrium__complete-value-free-74-row-identity-seal__complete-four-system-74-row-VLE-vector-with-eight-endpoints__depth-independent-common-context-successor

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-PHYS-THERMO-STATE-RELATION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-MIXTURE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-SOLUTION-EQUILIBRIUM-001
- SFT-CHEM-INTERFACE-TRANSFER-001
- SFT-CHEM-INTERNAL-ENERGY-COMPOSITION-003
- SFT-CHEM-HEAT-WORK-TRANSFER-PARTITION-004
- SFT-CHEM-ENTROPY-MULTIPLICITY-CORRESPONDENCE-005
- SFT-CHEM-ENTHALPY-EQUIVALENT-STATE-006
- SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007

Generated grammar. Generate the literal product of accounts, identity, exchange, support, relation, prediction, record and extension forms; decide all 256 candidates only from admitted exact composition, transfer, energy, distinction and equilibrium laws.

Boundary: Every finite pair of distinct phase-specific addition accounts for the same held component at one held environment, with paired one-component exchange, exact positive marginal support and common-context successor closure. External testing preserves the complete NIST TRC ThermoML record: four binary systems, 74 matched interior VLE records and eight unmatched endpoints.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: complete-distinct-phase-component-addition-accounts__same-held-component-and-fixed-environment__paired-one-component-exchange-conserving-total-carrier__exact-positive-energy-and-closed-distinction-increments__strict-product-order-with-EmptyOne-equilibrium__complete-value-free-74-row-identity-seal__complete-four-system-74-row-VLE-vector-with-eight-endpoints__depth-indep

endent-common-context-successor. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
accounts	complete-distinct-phase-component-addition-accounts	bulk-phase-value-or-answer-only-potential	A bulk value or named answer does not identify the marginal component change.
identity	same-held-component-and-fixed-environment	changed-component-or-unheld-environment	Changing the component or environment destroys the exchange comparison.
exchange	paired-one-component-exchange-conserving-total-carrier	created-erased-or-unpaired-component	An unpaired move violates the admitted component carrier conservation.
support	exact-positive-energy-and-closed-distinction-increments	signed-logarithmic-or-fitted-chemical-potential	A signed scalar, logarithm or fit imports an unforced chemical-potential model.
relation	strict-product-order-with-EmptyOne-equilibrium	weighted-sum-tie-break-or-equal-composition-rule	A weight, tie-break or equal-composition premise adds content not forced by exchange.
prediction	complete-value-free-74-row-identity-seal	compound-condition-or-composition-readable-before-seal	Withheld mixture values could select the component relation.
record	complete-four-system-74-row-VLE-vector-or-with-eight-endpoints	selected-mixture-row-or-deleted-endpoint	Selecting a system or dropping a source boundary can hide adverse phase behavior.
extension	depth-independent-common-context-successor	refit-after-common-context-successor	Refitting after a shared context destroys structural invariance.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:370d0e92a7670c895b1faf7cf5e3c14d08f12d944516ffbfcc2925309552e45e.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:5285f20ecc230ca7ae8debe4ebf17afe66740a16b50643f456dcdc453bec1760.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:afee82ee3748a5a5b8e3265db30005d2094dea9db637bccc213df96288bccde5.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e109c1e88bc216f06e5185af42315cc1678ef8fa764af11e7cf36d087a20279e.

Independent reconstruction. Implementation hash

sha256:b20a330c97f879e6ab0d42486cd1354d4af2aa95fc92da3a334bd26a177be954; independent certificate sha256:8168697bc510d8a2618cace817eb6cc6dd1f42e4ee21f297c4a5732d63122c74; external-validation hash sha256:d889f4d501e02926eb7ed92187081640314ec895b0cb96faa7468908afb1048e. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- NIST-TRC-THERMOML-JCED-2019-9B00414

Comparison records:

- SFT-CHEM-THERMO-008-VLE-0001: system=1; T=350.93 K; P=101.3 kPa; x=0.9807; y=0.9622
- SFT-CHEM-THERMO-008-VLE-0002: system=1; T=350.51 K; P=101.3 kPa; x=0.9641; y=0.9319
- SFT-CHEM-THERMO-008-VLE-0003: system=1; T=350.03 K; P=101.3 kPa; x=0.9385; y=0.8936
- SFT-CHEM-THERMO-008-VLE-0004: system=1; T=349.86 K; P=101.3 kPa; x=0.9276; y=0.8783
- SFT-CHEM-THERMO-008-VLE-0005: system=1; T=349.5 K; P=101.3 kPa; x=0.8994; y=0.8468
- SFT-CHEM-THERMO-008-VLE-0006: system=1; T=349.2 K; P=101.3 kPa; x=0.8674; y=0.8161
- SFT-CHEM-THERMO-008-VLE-0007: system=1; T=348.96 K; P=101.3 kPa; x=0.8291; y=0.7861
- SFT-CHEM-THERMO-008-VLE-0008: system=1; T=348.8 K; P=101.3 kPa; x=0.7892; y=0.7598
- SFT-CHEM-THERMO-008-VLE-0009: system=1; T=348.69 K; P=101.3 kPa; x=0.6996; y=0.7098
- SFT-CHEM-THERMO-008-VLE-0010: system=1; T=348.76 K; P=101.3 kPa; x=0.6566; y=0.6867
- SFT-CHEM-THERMO-008-VLE-0011: system=1; T=348.89 K; P=101.3 kPa; x=0.6079; y=0.6656
- SFT-CHEM-THERMO-008-VLE-0012: system=1; T=349.18 K; P=101.3 kPa; x=0.5397; y=0.6329
- SFT-CHEM-THERMO-008-VLE-0013: system=1; T=349.44 K; P=101.3 kPa; x=0.4971; y=0.6112
- SFT-CHEM-THERMO-008-VLE-0014: system=1; T=349.72 K; P=101.3 kPa; x=0.4581; y=0.5913
- SFT-CHEM-THERMO-008-VLE-0015: system=1; T=350.77 K; P=101.3 kPa; x=0.3601; y=0.5324
- SFT-CHEM-THERMO-008-VLE-0016: system=1; T=352.88 K; P=101.3 kPa; x=0.2403; y=0.4356
- SFT-CHEM-THERMO-008-VLE-0017: system=1; T=357.09 K; P=101.3 kPa; x=0.1076; y=0.2627
- SFT-CHEM-THERMO-008-VLE-0018: system=1; T=359.18 K; P=101.3 kPa; x=0.0647; y=0.1774
- SFT-CHEM-THERMO-008-VLE-0019: system=1; T=361.26 K; P=101.3 kPa; x=0.0294; y=0.0896
- SFT-CHEM-THERMO-008-VLE-0020: system=1; T=362.31 K; P=101.3 kPa; x=0.0141; y=0.0455
- SFT-CHEM-THERMO-008-VLE-0021: system=2; T=364.68 K; P=101.3 kPa; x=0.8857; y=0.9432
- SFT-CHEM-THERMO-008-VLE-0022: system=2; T=366.36 K; P=101.3 kPa; x=0.7756; y=0.9002
- SFT-CHEM-THERMO-008-VLE-0023: system=2; T=369.38 K; P=101.3 kPa; x=0.5892; y=0.8513
- SFT-CHEM-THERMO-008-VLE-0024: system=2; T=372.39 K; P=101.3 kPa; x=0.4323; y=0.8143
- SFT-CHEM-THERMO-008-VLE-0025: system=2; T=375.42 K; P=101.3 kPa; x=0.3304; y=0.7785
- SFT-CHEM-THERMO-008-VLE-0026: system=2; T=378.39 K; P=101.3 kPa; x=0.2627; y=0.7421
- SFT-CHEM-THERMO-008-VLE-0027: system=2; T=381.38 K; P=101.3 kPa; x=0.2105; y=0.7052
- SFT-CHEM-THERMO-008-VLE-0028: system=2; T=384.4 K; P=101.3 kPa; x=0.1738; y=0.6649
- SFT-CHEM-THERMO-008-VLE-0029: system=2; T=387.39 K; P=101.3 kPa; x=0.1451; y=0.6222
- SFT-CHEM-THERMO-008-VLE-0030: system=2; T=390.41 K; P=101.3 kPa; x=0.1219; y=0.5759
- SFT-CHEM-THERMO-008-VLE-0031: system=2; T=393.4 K; P=101.3 kPa; x=0.1019; y=0.5267
- SFT-CHEM-THERMO-008-VLE-0032: system=2; T=396.41 K; P=101.3 kPa; x=0.0839; y=0.4734
- SFT-CHEM-THERMO-008-VLE-0033: system=2; T=399.42 K; P=101.3 kPa; x=0.0687; y=0.4201
- SFT-CHEM-THERMO-008-VLE-0034: system=2; T=402.4 K; P=101.3 kPa; x=0.0546; y=0.3619
- SFT-CHEM-THERMO-008-VLE-0035: system=2; T=405.41 K; P=101.3 kPa; x=0.041; y=0.2961
- SFT-CHEM-THERMO-008-VLE-0036: system=2; T=408.43 K; P=101.3 kPa; x=0.0294; y=0.2295
- SFT-CHEM-THERMO-008-VLE-0037: system=2; T=411.42 K; P=101.3 kPa; x=0.0187; y=0.1582
- SFT-CHEM-THERMO-008-VLE-0038: system=2; T=414.43 K; P=101.3 kPa; x=0.0091; y=0.0835

- SFT-CHEM-THERMO-008-VLE-0039: system=3; T=365.14 K; P=101.3 kPa; x=0.9032; y=0.9729
- SFT-CHEM-THERMO-008-VLE-0040: system=3; T=367.52 K; P=101.3 kPa; x=0.7971; y=0.9545
- SFT-CHEM-THERMO-008-VLE-0041: system=3; T=371.68 K; P=101.3 kPa; x=0.5801; y=0.9239
- SFT-CHEM-THERMO-008-VLE-0042: system=3; T=375.84 K; P=101.3 kPa; x=0.4194; y=0.9003
- SFT-CHEM-THERMO-008-VLE-0043: system=3; T=380.01 K; P=101.3 kPa; x=0.3305; y=0.8761
- SFT-CHEM-THERMO-008-VLE-0044: system=3; T=384.17 K; P=101.3 kPa; x=0.2662; y=0.8468
- SFT-CHEM-THERMO-008-VLE-0045: system=3; T=388.33 K; P=101.3 kPa; x=0.2206; y=0.8141
- SFT-CHEM-THERMO-008-VLE-0046: system=3; T=392.49 K; P=101.3 kPa; x=0.1824; y=0.7767
- SFT-CHEM-THERMO-008-VLE-0047: system=3; T=396.66 K; P=101.3 kPa; x=0.1541; y=0.7403
- SFT-CHEM-THERMO-008-VLE-0048: system=3; T=400.82 K; P=101.3 kPa; x=0.1269; y=0.6947
- SFT-CHEM-THERMO-008-VLE-0049: system=3; T=404.98 K; P=101.3 kPa; x=0.1036; y=0.6442
- SFT-CHEM-THERMO-008-VLE-0050: system=3; T=409.14 K; P=101.3 kPa; x=0.0839; y=0.5896
- SFT-CHEM-THERMO-008-VLE-0051: system=3; T=413.3 K; P=101.3 kPa; x=0.0669; y=0.5296
- SFT-CHEM-THERMO-008-VLE-0052: system=3; T=417.47 K; P=101.3 kPa; x=0.0518; y=0.4619
- SFT-CHEM-THERMO-008-VLE-0053: system=3; T=421.63 K; P=101.3 kPa; x=0.0386; y=0.3873
- SFT-CHEM-THERMO-008-VLE-0054: system=3; T=425.79 K; P=101.3 kPa; x=0.0267; y=0.3018
- SFT-CHEM-THERMO-008-VLE-0055: system=3; T=429.95 K; P=101.3 kPa; x=0.0166; y=0.2101
- SFT-CHEM-THERMO-008-VLE-0056: system=3; T=434.12 K; P=101.3 kPa; x=0.0076; y=0.1077
- SFT-CHEM-THERMO-008-VLE-0057: system=4; T=364.99 K; P=101.3 kPa; x=0.9241; y=0.9764
- SFT-CHEM-THERMO-008-VLE-0058: system=4; T=366.98 K; P=101.3 kPa; x=0.8302; y=0.9521
- SFT-CHEM-THERMO-008-VLE-0059: system=4; T=370.6 K; P=101.3 kPa; x=0.6731; y=0.9144
- SFT-CHEM-THERMO-008-VLE-0060: system=4; T=374.22 K; P=101.3 kPa; x=0.5339; y=0.8809
- SFT-CHEM-THERMO-008-VLE-0061: system=4; T=377.84 K; P=101.3 kPa; x=0.4217; y=0.8475
- SFT-CHEM-THERMO-008-VLE-0062: system=4; T=381.47 K; P=101.3 kPa; x=0.328; y=0.813
- SFT-CHEM-THERMO-008-VLE-0063: system=4; T=385.09 K; P=101.3 kPa; x=0.2642; y=0.7766
- SFT-CHEM-THERMO-008-VLE-0064: system=4; T=388.71 K; P=101.3 kPa; x=0.2163; y=0.7364
- SFT-CHEM-THERMO-008-VLE-0065: system=4; T=392.34 K; P=101.3 kPa; x=0.1768; y=0.6944
- SFT-CHEM-THERMO-008-VLE-0066: system=4; T=395.96 K; P=101.3 kPa; x=0.1442; y=0.6469
- SFT-CHEM-THERMO-008-VLE-0067: system=4; T=399.58 K; P=101.3 kPa; x=0.1187; y=0.598
- SFT-CHEM-THERMO-008-VLE-0068: system=4; T=403.2 K; P=101.3 kPa; x=0.0971; y=0.5431
- SFT-CHEM-THERMO-008-VLE-0069: system=4; T=406.83 K; P=101.3 kPa; x=0.0766; y=0.4806
- SFT-CHEM-THERMO-008-VLE-0070: system=4; T=410.46 K; P=101.3 kPa; x=0.06; y=0.4159
- SFT-CHEM-THERMO-008-VLE-0071: system=4; T=414.07 K; P=101.3 kPa; x=0.0447; y=0.3433
- SFT-CHEM-THERMO-008-VLE-0072: system=4; T=417.69 K; P=101.3 kPa; x=0.0315; y=0.2668
- SFT-CHEM-THERMO-008-VLE-0073: system=4; T=421.32 K; P=101.3 kPa; x=0.0201; y=0.1871
- SFT-CHEM-THERMO-008-VLE-0074: system=4; T=424.94 K; P=101.3 kPa; x=0.0092; y=0.0951
- complete vector: four binary systems and 74 matched multicomponent equilibrium rows
- source boundary: eight unmatched pure-component temperature endpoints retained

- phase relation vector: 66 gas-greater and 8 liquid-greater rows; one system-one crossing retained
- tampered_complement_rejected: True
- complete_74_row_vector_retained: True
- complete_four_system_surface_retained: True
- eight_unpaired_endpoints_retained: True
- both_external_composition_relations_retained: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if a component changes identity during exchange; if donor and receiver do not share one finite environment; if total component carrier is not conserved; if direction requires a signed scalar, logarithm, fitted activity coefficient or arbitrary tie-break; if equal complete accounts do not yield equilibrium or are incorrectly required to have equal bulk compositions; if a common context changes the relation; if any target opens before all 74 identities seal; if any of four systems, 74 matched interiors, eight unmatched endpoints, source conditions or uncertainties is omitted; or if measured values select the law.

Measurement receipt: sha256:e238388ad64631b2bfda5fc6fdf68bd5a49f92f1f878fa8b4e51725a8be53c84.

Isolation certificate: sha256:97b034ed2e9de0dcc2adca880f979bfe606f6b81b85813aadbbba4cb90b9e82c.

Custody certificate: sha256:81a4cd1e40e26a45084d9b28f3736e0fb1db488e5d570aa150d7c85dd6db2af2.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The thermodynamic extension derives finite microstate support, temperature, energy, heat/work, entropy, enthalpy, free direction, chemical potential, activity, fugacity, phases, colligative response, solvation and coupled transport while keeping external scalar inscriptions downstream. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; native absence and exact equality separations use structural EmptyOne
- no negative, irrational, imaginary, logarithmic, floating, signed or continuum SFT proof value
- no imported chemical-potential equation, activity coefficient, fugacity model, fitted weight or target-derived account
- no compound, temperature, pressure, liquid composition, gas composition, equilibrium result or target hash before prediction seal
- no selected mixture, selected row, deleted endpoint, fitted composition or model-correlated value used as measurement
- external decimal, zero and signed comparison glyphs remain post-seal source inscriptions only

Immutable identities. Source manifest

sha256:6619fc7ad4443ea90310358fa48bdb14424aa2e9412097f6284e1252352c1f08; derivation seal

sha256:517bfae6ab0c989e960bad25c9d709a9ab6a83538c6e19aec603de90c88a91c4; engine receipt

sha256:766e2849ce3f2f88465ac16a1309c55925e2738ec60b53f5aa3ac9a10d3605bc at receipts/engine/model_admitted/SFT-CHEM-CHEMICAL-POTENTIAL-EQUIVALENT-COMPONENT-008-766e2849ce3f2f88.

json; empirical hash sha256:48b9001c64c0f4764e588cb974e94328957892b4f00ceefd3796be245b9050e4;

measurement receipt sha256:e238388ad64631b2bfda5fc6fdf68bd5a49f92f1f878fa8b4e51725a8be53c84.

133. Activity and non-ideal composition law

Claim identity: SFT-CHEM-ACTIVITY-NONIDEAL-COMPOSITION-009

Question and theorem. Solution activity is the exact positive accessible component-exchange support count over held reference support at complete composition and condition. Non-ideality is the exact joint-versus-independent support relation, without a fitted activity coefficient; absent components are structural EmptyOne.

complete-component-exchange-support-account__complete-held-condition-and-component-coordinates__exact-accessible-support-over-reference-support__exact-joint-versus-independent-support-relation__structural-EmptyOne-absent-component__complete-value-free-204-row-identity-seal__complete-nine-dataset-204-row-vector-with-68-EmptyOne-boundaries__depth-independent-exact-support-replication

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-MIXTURE-001
- SFT-CHEM-STOICH-SOLUTION-001
- SFT-CHEM-SOLUTION-EQUILIBRIUM-001
- SFT-CHEM-FINITE-MICROSTATE-SUPPORT-001
- SFT-CHEM-TEMPERATURE-CORRESPONDENCE-002
- SFT-CHEM-INTERNAL-ENERGY-COMPOSITION-003
- SFT-CHEM-ENTROPY-MULTIPLICITY-CORRESPONDENCE-005
- SFT-CHEM-CHEMICAL-POTENTIAL-EQUIVALENT-COMPONENT-008

Generated grammar. Generate the literal product of carrier, composition, activity, interaction, absence, prediction, record and extension forms; decide all 256 candidates only from admitted exact support-count, component-exchange, composition, condition and structural-absence laws.

Boundary: Every finite solution component account with complete phase, environment and exact component coordinates, accessible/reference/independent support counts and exact-replication closure. External testing preserves the complete NIST TRC ThermoML record: six compounds, nine binary/ternary datasets, 204 direct isopiestic activity rows and all 68 absent-component boundaries.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *complete-component-exchange-support-account__complete-held-condition-and-component-coordinates__exact-accessible-support-over-reference-support__exact-joint-versus-independent-support-relation__structural-EmptyOne-absent-component__complete-value-free-204-row-identity-seal__complete-nine-dataset-204-row-vector-with-68-EmptyOne-boundaries__depth-independent-exact-support-replication*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-component-exchange-support-account	bulk-answer-or-unbound-activity-number	An unbound number erases the component exchange support it measures.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
composition	complete-held-condition-and-component-coordinates	selected-or-erased-composition-and-condition	Erasing composition or condition makes distinct solution states indistinguishable.
activity	exact-accessible-support-over-reference-support	logarithm-fugacity-or-fitted-coefficient	A logarithm, fugacity model or fitted coefficient imports an unforced scalar model.
interaction	exact-joint-versus-independent-support-relation	ideal-mixture-prior-or-target-derived-correction	Assumed ideality or a target-derived correction selects the relation externally.
absence	structural-EmptyOne-absent-component	numerical-zero-component-coordinate	Numerical zero would introduce a non-SFT number for an absent component.
prediction	complete-value-free-204-row-identity-seal	activity-composition-or-condition-readable-before-seal	Withheld values could select the support relation.
record	complete-nine-dataset-204-row-vector-with-68-EmptyOne-boundaries	selected-system-or-deleted-absence-boundary	Selecting a solution or dropping absent-component rows hides the complete evidence surface.
extension	depth-independent-exact-support-replication	refit-after-support-replication	Refitting after exact replication destroys the count relation.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:05f7e83d0f7d921921eaf57bdd6ddbb9c91bc0848fc8dbaeee759d57431eb9cd.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:510ab37f5269a853a135c73636fab9979082d8998a0856b984d7cc486c6035d7.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:c104ad2dfe8d5c17b40ee2c4e42127d20806f73fde7aa32230faf581cb4d0a76.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:73de1f3007eea2010fa6868c1073d5640c444ec9e7672d9b3d75ee603875c401.

Independent reconstruction. Implementation hash

sha256:45be4364607c8c9198d53020fbc94159dab951d08c518af3d52fe18e839ceabc; independent certificate sha256:a2f2511463015bdf3d29106a3a35c9b1564f18f759761d68955123a3fb6f2c4c; external-validation hash sha256:4ccb77ec5e92f5f0bc559db7db1cfff720bc4eff580f02c65962d52ecea0b43f. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-TRC-THERMOML-JCED-2019-9B00694

Comparison records:

- SFT-CHEM-THERMO-009-ACTIVITY-0001: dataset=1; T=298.15 K; water activity=0.9206; composition entries=2
- SFT-CHEM-THERMO-009-ACTIVITY-0002: dataset=1; T=298.15 K; water activity=0.9206; composition entries=2
- SFT-CHEM-THERMO-009-ACTIVITY-0003: dataset=1; T=298.15 K; water activity=0.9206; composition entries=2
- SFT-CHEM-THERMO-009-ACTIVITY-0004: dataset=1; T=298.15 K; water activity=0.9206; composition entries=2
- SFT-CHEM-THERMO-009-ACTIVITY-0005: dataset=1; T=298.15 K; water activity=0.942; composition entries=2
- SFT-CHEM-THERMO-009-ACTIVITY-0006: dataset=1; T=298.15 K; water activity=0.942; composition entries=2

- [illegible]

- [illegible]

- [illegible]

- [illegible]

- SFT-CHEM-THERMO-009-ACTIVITY-0159: dataset=5; T=298.15 K; water activity=0.9947; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0160: dataset=5; T=298.15 K; water activity=0.9933; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0161: dataset=5; T=298.15 K; water activity=0.9921; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0162: dataset=5; T=298.15 K; water activity=0.9902; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0163: dataset=5; T=298.15 K; water activity=0.9464; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0164: dataset=5; T=298.15 K; water activity=0.9538; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0165: dataset=5; T=298.15 K; water activity=0.9579; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0166: dataset=5; T=298.15 K; water activity=0.9658; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0167: dataset=5; T=298.15 K; water activity=0.9687; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0168: dataset=5; T=298.15 K; water activity=0.9738; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0169: dataset=5; T=298.15 K; water activity=0.9817; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0170: dataset=5; T=298.15 K; water activity=0.9839; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0171: dataset=6; T=298.15 K; water activity=0.9206; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0172: dataset=6; T=298.15 K; water activity=0.942; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0173: dataset=6; T=298.15 K; water activity=0.9504; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0174: dataset=6; T=298.15 K; water activity=0.9594; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0175: dataset=6; T=298.15 K; water activity=0.9653; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0176: dataset=6; T=298.15 K; water activity=0.9699; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0177: dataset=6; T=298.15 K; water activity=0.9731; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0178: dataset=6; T=298.15 K; water activity=0.9752; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0179: dataset=6; T=298.15 K; water activity=0.9798; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0180: dataset=7; T=298.15 K; water activity=0.978; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0181: dataset=7; T=298.15 K; water activity=0.977; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0182: dataset=7; T=298.15 K; water activity=0.9758; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0183: dataset=7; T=298.15 K; water activity=0.9732; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0184: dataset=7; T=298.15 K; water activity=0.9692; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0185: dataset=7; T=298.15 K; water activity=0.9661; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0186: dataset=7; T=298.15 K; water activity=0.9549; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0187: dataset=7; T=298.15 K; water activity=0.9418; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0188: dataset=8; T=298.15 K; water activity=0.9984; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0189: dataset=8; T=298.15 K; water activity=0.998; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0190: dataset=8; T=298.15 K; water activity=0.9974; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0191: dataset=8; T=298.15 K; water activity=0.9969; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0192: dataset=8; T=298.15 K; water activity=0.9964; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0193: dataset=8; T=298.15 K; water activity=0.9947; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0194: dataset=8; T=298.15 K; water activity=0.9933; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0195: dataset=8; T=298.15 K; water activity=0.9921; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0196: dataset=8; T=298.15 K; water activity=0.9902; composition entries=1

- SFT-CHEM-THERMO-009-ACTIVITY-0197: dataset=9; T=298.15 K; water activity=0.9464; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0198: dataset=9; T=298.15 K; water activity=0.9538; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0199: dataset=9; T=298.15 K; water activity=0.9579; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0200: dataset=9; T=298.15 K; water activity=0.9658; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0201: dataset=9; T=298.15 K; water activity=0.9687; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0202: dataset=9; T=298.15 K; water activity=0.9738; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0203: dataset=9; T=298.15 K; water activity=0.9817; composition entries=1
- SFT-CHEM-THERMO-009-ACTIVITY-0204: dataset=9; T=298.15 K; water activity=0.9839; composition entries=1
- complete vector: five binary and four ternary datasets, 204 direct isopiestic activity rows
- absence boundary: 68 external zero glyph rows translated only to structural EmptyOne
- activity range: 4603/5000 through 624/625
- tampered_activity_rejected: True
- complete_204_row_vector_retained: True
- complete_nine_dataset_surface_retained: True
- all_68_EmptyOne_boundaries_retained: True
- binary_and_ternary_surfaces_retained: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if activity is detached from component, phase, composition or condition; if accessible support is not an exact part of held reference support; if non-ideality requires a logarithm, fugacity model, fitted coefficient, assumed ideality or target-derived correction; if an absent component becomes numerical zero; if exact replication changes activity or relation; if any target opens before all 204 identities seal; if any of six compounds, nine datasets, 204 rows, 68 absence boundaries, conditions, uncertainties or methods is omitted; or if correlated model values enter as measurements.

Measurement receipt: sha256:c65ba24ddb530aaf3f3f757b6e1fc79a80406e2e816d785a5c04917edb9ca0e6.

Isolation certificate: sha256:003156173a172fc004eb43f3fbb17d46d617ad0bd5862f10a3d888e622bb215f.

Custody certificate: sha256:d77b2e55ba636ac0fc69cd5207f4bc16b66c55be0489e479b20958380bdbd563.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The thermodynamic extension derives finite microstate support, temperature, energy, heat/work, entropy, enthalpy, free direction, chemical potential, activity, fugacity, phases, colligative response, solvation and coupled transport while keeping external scalar inscriptions downstream. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; an absent component is structural EmptyOne and source zero glyphs remain external inscriptions
- no negative, irrational, imaginary, logarithmic, floating, signed or continuum SFT proof value
- no imported activity-coefficient equation, fugacity model, ideal-mixture prior, fitted interaction parameter or target-derived correction
- no compound, temperature, molality, activity, uncertainty, absence flag or target hash before prediction seal
- no selected solution, selected composition, deleted absence row or correlated/regressed/model-calculated value used as measurement

- every external decimal and zero glyph remains a post-seal source inscription

Immutable identities. Source manifest

sha256:84521965a64b0a21e9ec9b09b800b2f312668416358b3ea2dec80d693ca27759; derivation seal sha256:5c72191a874eacffd09411aa6c60b315dc02cf5e7795de5dc300a22fe6d23f37; engine receipt sha256:dde514594020e0de01d67c6dac4357c5f46c427a3512e28aefb37f0e7595b2ca at receipts/engine/model_admitted/SFT-CHEM-ACTIVITY-NONIDEAL-COMPOSITION-009-dde514594020e0de.json; empirical hash sha256:2b43fefed80f0cc2655a017236382ae03d4757e57538663d56dcebbbc2f65263e; measurement receipt sha256:c65ba24ddb530aaf3f3f757b6e1fc79a80406e2e816d785a5c04917edb9ca0e6.

134. Fugacity-equivalent real-gas mixture law

Claim identity: SFT-CHEM-FUGACITY-EQUIVALENT-GAS-MIXTURE-010

Question and theorem. The fugacity-equivalent relation is the exact positive accessible/reference component exchange-support fraction at complete held gas-mixture state; phase equilibrium is exact component exchange-support balance without an imported equation or fit.

complete-gas-component-exchange-account__complete-held-pressure-temperature-composition-state__held-component-resolved-exchange-support__exact-accessible-over-reference-support-relation__exact-component-exchange-support-balance__complete-value-free-94-state-identity-seal__complete-21-dataset-176-point-94-state-record__depth-independent-common-support-replication

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-MIXTURE-001
- SFT-CHEM-SOLUTION-EQUILIBRIUM-001
- SFT-CHEM-FINITE-MICROSTATE-SUPPORT-001
- SFT-CHEM-TEMPERATURE-CORRESPONDENCE-002
- SFT-CHEM-INTERNAL-ENERGY-COMPOSITION-003
- SFT-CHEM-HEAT-WORK-TRANSFER-PARTITION-004
- SFT-CHEM-ENTROPY-MULTIPLICITY-CORRESPONDENCE-005
- SFT-CHEM-ENTHALPY-EQUIVALENT-STATE-006
- SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007
- SFT-CHEM-CHEMICAL-POTENTIAL-EQUIVALENT-COMPONENT-008
- SFT-CHEM-ACTIVITY-NONIDEAL-COMPOSITION-009

Generated grammar. Generate the literal product of carrier, state, component, relation, equilibrium, prediction, record and extension forms; decide all 256 candidates only from admitted finite-support, component-exchange, phase-equilibrium, held-coordinate and exact-replication laws.

Boundary: Every finite gas-mixture component exchange account with held component, gas phase, pressure, temperature, composition, accessible/reference/independent support and partner-phase support. External testing preserves the complete NIST TRC ThermoML source: five compounds, 21 datasets, 176 raw points, seven direct pressure datasets, three paired vapor-composition datasets and all 94 equilibrium states.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-gas-component-exchange-account__complete-held-pressure-temperature-composition-state__held-component-resolved-exchange-support__exact-accessible-over-reference-support-relation__exact-component-exchange-support-balance__complete-value-free-94-state-identity-seal__complete-21-dataset-176-point-94-state-record__depth-independent-common-support-replication`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-gas-component-exchange-account	unbound-fugacity-number	An unbound scalar erases the component exchange support and gas state it names.
state	complete-held-pressure-temperature-composition-state	erased-pressure-temperature-or-composition	Erasing a held coordinate merges distinguishable real-gas states.
component	held-component-resolved-exchange-support	bulk-mixture-answer-without-component-identity	A bulk answer cannot preserve which component exchanges across the phase boundary.
relation	exact-accessible-over-reference-support-relation	imported-fugacity-eos-or-fitted-correction	An imported equation or fitted correction lets an external model select the answer.
equilibrium	exact-component-exchange-support-balance	assumed-ideal-gas-or-target-derived-balance	Assumed ideality or target-derived adjustment does not force phase exchange.
prediction	complete-value-free-94-state-identity-seal	real-gas-values-readable-before-seal	Readable values could select or tune the generated law.
record	complete-21-dataset-176-point-94-state-record	selected-mixture-or-deleted-pressure-only-state	Selection would hide the complete favorable, adverse and incomplete-measurement surface.
extension	depth-independent-common-support-replication	refit-after-exact-support-replication	Refitting after replication destroys the exact support relation.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: ac1871624cffa810330738c255b2c7f7b25512d1e59cc28c669f6565bb350eb3.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 866bd5cc1bbd9dd74d51b8822eacdbf9a47c90e379a87c04332539c0749e79c8.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 83a10a8cfd98436d1090cacb37321de60df488712e347320987b2db6ec6c5513.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: f776c0ac8c935e379bc803e2b52ed6d4ad2d8b984f228d594eaa036aaf003cef.

Independent reconstruction. Implementation hash

sha256: aff16368172ab56f6c3146b19c314ce64cf191c25bb2d53c1bbef69e27fa5c4a; independent certificate
sha256: 9cdac9b2c2474847459481325622dfcb31e302dc2c177091efb90c3df09d10e8; external-validation

hash sha256:6368dead58ae5ef87980d6f71dbc72d39bad155fd0928805be94d5300938b2b1. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-TRC-THERMOML-FPE-2019-485-145-152

Comparison records:

- SFT-CHEM-THERMO-010-REAL-GAS-0001: pressure dataset=12; P=925 kPa; gas composition=0.952; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0002: pressure dataset=12; P=2878 kPa; gas composition=0.979; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0003: pressure dataset=12; P=3704 kPa; gas composition=0.98; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0004: pressure dataset=12; P=4823 kPa; gas composition=0.982; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0005: pressure dataset=12; P=5762 kPa; gas composition=0.978; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0006: pressure dataset=12; P=6511 kPa; gas composition=0.981; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0007: pressure dataset=12; P=7150 kPa; gas composition=0.981; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0008: pressure dataset=12; P=7772 kPa; gas composition=0.982; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0009: pressure dataset=14; P=3000 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0010: pressure dataset=14; P=5050 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0011: pressure dataset=14; P=5570 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0012: pressure dataset=14; P=5950 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0013: pressure dataset=14; P=6630 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0014: pressure dataset=15; P=1535 kPa; gas composition=0.977; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0015: pressure dataset=15; P=2147 kPa; gas composition=0.987; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0016: pressure dataset=15; P=2792 kPa; gas composition=0.988; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0017: pressure dataset=15; P=3343 kPa; gas composition=0.98; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0018: pressure dataset=15; P=4062 kPa; gas composition=0.98; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0019: pressure dataset=15; P=4709 kPa; gas composition=0.976; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0020: pressure dataset=15; P=5544 kPa; gas composition=0.971; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0021: pressure dataset=15; P=7060 kPa; gas composition=0.97; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0022: pressure dataset=15; P=7804 kPa; gas composition=0.933; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0023: pressure dataset=15; P=2026 kPa; gas composition=0.979; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0024: pressure dataset=15; P=3666 kPa; gas composition=0.983; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0025: pressure dataset=15; P=5207 kPa; gas composition=0.98; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0026: pressure dataset=15; P=5994 kPa; gas composition=0.976; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0027: pressure dataset=15; P=6978 kPa; gas composition=0.968; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0028: pressure dataset=15; P=7774 kPa; gas composition=0.955; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0029: pressure dataset=15; P=8449 kPa; gas composition=0.915; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0030: pressure dataset=15; P=2535 kPa; gas composition=0.973; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0031: pressure dataset=15; P=3632 kPa; gas composition=0.976; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0032: pressure dataset=15; P=4635 kPa; gas composition=0.974; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0033: pressure dataset=15; P=5753 kPa; gas composition=0.97; coordinates=2

- SFT-CHEM-THERMO-010-REAL-GAS-0034: pressure dataset=15; P=6726 kPa; gas composition=0.965; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0035: pressure dataset=15; P=7840 kPa; gas composition=0.95; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0036: pressure dataset=15; P=8605 kPa; gas composition=0.918; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0037: pressure dataset=15; P=8776 kPa; gas composition=0.901; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0038: pressure dataset=17; P=3910 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0039: pressure dataset=17; P=5480 kPa; gas composition=unreported; coordinates=2
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- SFT-CHEM-THERMO-010-REAL-GAS-0041: pressure dataset=17; P=4350 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0042: pressure dataset=17; P=6050 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0043: pressure dataset=17; P=7780 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0044: pressure dataset=17; P=4750 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0045: pressure dataset=17; P=6550 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0046: pressure dataset=17; P=8850 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0047: pressure dataset=18; P=2680 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0048: pressure dataset=18; P=5580 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0049: pressure dataset=18; P=6380 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0050: pressure dataset=18; P=2970 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0051: pressure dataset=18; P=6190 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0052: pressure dataset=18; P=7020 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0053: pressure dataset=18; P=3260 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0054: pressure dataset=18; P=6670 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0055: pressure dataset=18; P=7590 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0056: pressure dataset=19; P=1617 kPa; gas composition=0.9977; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0057: pressure dataset=19; P=2456 kPa; gas composition=0.9976; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0058: pressure dataset=19; P=3453 kPa; gas composition=0.9977; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0059: pressure dataset=19; P=5508 kPa; gas composition=0.9959; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0060: pressure dataset=19; P=6279 kPa; gas composition=0.994; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0061: pressure dataset=19; P=7364 kPa; gas composition=0.9901; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0062: pressure dataset=19; P=8129 kPa; gas composition=0.9853; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0063: pressure dataset=19; P=9006 kPa; gas composition=0.9727; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0064: pressure dataset=19; P=2244 kPa; gas composition=0.9964; coordinates=2
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- SFT-CHEM-THERMO-010-REAL-GAS-0066: pressure dataset=19; P=3753 kPa; gas composition=0.9956; coordinates=2
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- SFT-CHEM-THERMO-010-REAL-GAS-0068: pressure dataset=19; P=6303 kPa; gas composition=0.9928; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0069: pressure dataset=19; P=7941 kPa; gas composition=0.9875; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0070: pressure dataset=19; P=8767 kPa; gas composition=0.981; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0071: pressure dataset=19; P=9345 kPa; gas composition=0.973; coordinates=2

- SFT-CHEM-THERMO-010-REAL-GAS-0072: pressure dataset=19; P=9999 kPa; gas composition=0.9561; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0073: pressure dataset=19; P=1874 kPa; gas composition=0.9941; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0074: pressure dataset=19; P=3079 kPa; gas composition=0.9941; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0075: pressure dataset=19; P=3990 kPa; gas composition=0.9935; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0076: pressure dataset=19; P=5044 kPa; gas composition=0.9926; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0077: pressure dataset=19; P=5936 kPa; gas composition=0.9914; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0078: pressure dataset=19; P=7032 kPa; gas composition=0.9885; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0079: pressure dataset=19; P=8056 kPa; gas composition=0.9843; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0080: pressure dataset=19; P=9151 kPa; gas composition=0.9775; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0081: pressure dataset=19; P=10088 kPa; gas composition=0.9636; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0082: pressure dataset=19; P=10780 kPa; gas composition=0.9352; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0083: pressure dataset=21; P=1890 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0084: pressure dataset=21; P=4300 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0085: pressure dataset=21; P=6390 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0086: pressure dataset=21; P=9280 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0087: pressure dataset=21; P=2110 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0088: pressure dataset=21; P=4820 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0089: pressure dataset=21; P=7030 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0090: pressure dataset=21; P=10100 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0091: pressure dataset=21; P=2310 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0092: pressure dataset=21; P=5260 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0093: pressure dataset=21; P=7660 kPa; gas composition=unreported; coordinates=2
- SFT-CHEM-THERMO-010-REAL-GAS-0094: pressure dataset=21; P=10750 kPa; gas composition=unreported; coordinates=2
- complete source: five compounds, 21 datasets and 176 raw points
- complete equilibrium vector: 94 states, 59 paired vapor compositions and 35 pressure-only states
- pressure range: 925 through 10780 kPa
- temperature range: 7078/25 through 31317/100 K
- reported vapor-composition range: 901/1000 through 9977/10000
- tampered_pressure_rejected: True
- complete_94_state_vector_retained: True
- complete_21_dataset_176_point_source_retained: True
- all_59_paired_states_retained: True
- all_35_pressure_only_states_retained: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if the exchange relation is detached from component, phase, pressure, temperature or composition; if accessible/reference support is not exact positive; if equilibrium is not exact component exchange balance; if any fugacity equation, equation of state, logarithm, ideal-gas assumption, fit or target-derived correction enters; if exact replication changes a relation; if a target opens before all 94 identities seal; if any of five compounds, 21 datasets, 176 points, 94 equilibrium states, 59 paired composition states, 35 pressure-only states, conditions, uncertainties or methods is omitted; or if a correlated/model-calculated value is treated as measurement.

Measurement receipt: sha256:a933e1dd241c989affe5433d8c4edd91fd31605d61d831001b6ebd16a38314c7.
 Isolation certificate: sha256:56e9848a4a537a989b5cdab2395286938324f75a7a3b075f1ebaa231f8fd774e.
 Custody certificate: sha256:7679b465a3c666daef014780957163159c9dd6ff3cd91b2f07b906a4c42b3ee3.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The thermodynamic extension derives finite microstate support, temperature, energy, heat/work, entropy, enthalpy, free direction, chemical potential, activity, fugacity, phases, colligative response, solvation and coupled transport while keeping external scalar inscriptions downstream. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; absent support is structural EmptyOne and source zero glyphs remain external inscriptions
- no negative, irrational, imaginary, logarithmic, floating, signed or continuum SFT proof value
- no imported fugacity equation, ideal-gas law, equation of state, compressibility or virial fit
- no fitted interaction parameter, selected mixture, target-derived correction or measured-value lookup
- no compound, temperature, pressure, composition, uncertainty or target hash before prediction seal
- no correlated, regressed or model-calculated value used as measurement

Immutable identities. Source manifest

sha256:51d5c134d4607183e7f1a0ca52bad988e40a12c0613731e297993476c6cb73fc; derivation seal
 sha256:437aba2cedae4ca028646969b1e6b01a69d3c5ccfe0eba069e01f55c197aaf73; engine receipt
 sha256:e767c09f6945e2ef91f63ff127007e663055b2d8215b06401e1d37f9db634ad2 at receipts/engine/model_admitted/SFT-CHEM-FUGACITY-EQUIVALENT-GAS-MIXTURE-010-e767c09f6945e2ef.json;
 empirical hash sha256:c647be6e7fa5ef4228248401a096bed58b7f28e79284367ba530153d0d43360b;
 measurement receipt sha256:a933e1dd241c989affe5433d8c4edd91fd31605d61d831001b6ebd16a38314c7.

135. Chemical phase-rule structural relation

Claim identity: SFT-CHEM-PHASE-RULE-STRUCTURAL-011

Question and theorem. Independent phase-rule degree support is the finite carrier word remaining after one exact cancellation per coexisting phase from all component carriers and two held environmental carriers; complete cancellation is EmptyOne.

complete-component-phase-coordinate-account__complete-held-independent-component-support__one-exact-coordination-cancellation-per-coexisting-phase__two-held-environment-coordinate-carriers__exact-carrier-cancellation-relation__structural-EmptyOne-invariant-state__complete-value-free-18-row-identity-seal__depth-independent-joint-component-phase-successor

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001

- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-MIXTURE-001
- SFT-CHEM-STOICH-SOLUTION-001
- SFT-CHEM-SOLUTION-EQUILIBRIUM-001
- SFT-CHEM-FINITE-MICROSTATE-SUPPORT-001
- SFT-CHEM-TEMPERATURE-CORRESPONDENCE-002
- SFT-CHEM-INTERNAL-ENERGY-COMPOSITION-003
- SFT-CHEM-HEAT-WORK-TRANSFER-PARTITION-004
- SFT-CHEM-ENTROPY-MULTIPLICITY-CORRESPONDENCE-005
- SFT-CHEM-ENTHALPY-EQUIVALENT-STATE-006
- SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007
- SFT-CHEM-CHEMICAL-POTENTIAL-EQUIVALENT-COMPONENT-008
- SFT-CHEM-ACTIVITY-NONIDEAL-COMPOSITION-009
- SFT-CHEM-FUGACITY-EQUIVALENT-GAS-MIXTURE-010

Generated grammar. Generate the literal product of carrier, components, phases, environment, relation, absence, observation and extension forms; decide all 256 candidates only from admitted finite-support, composition, exact cancellation, equilibrium, structural-absence and successor laws.

Boundary: Every finite positive component word, positive coexisting-phase word and exactly two held environmental coordinate carriers for which phase cancellation does not exceed complete carrier support. External testing preserves the complete 18-row component/phase identity product for one through four components, all 14 positive degree-support outcomes, four invariant EmptyOne outcomes, the full IUPAC term and the complete 32-page NIST source.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: complete-component-phase-coordinate-account__complete-held-independent-component-support__one-exact-coordinate-cancellation-per-coexisting-phase__two-held-environment-coordinate-carriers__exact-carrier-cancellation-relation__structural-EmptyOne-invariant-state__complete-value-free-18-row-identity-seal__depth-independent-joint-component-phase-successor. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-component-phase-coordinate-account	unbound-degree-number	A detached number erases which component, phase and coordinate supports produced it.
components	complete-held-independent-component-support	bulk-substance-with-erase d-independent-components	Erasing independent component identities changes the available coordinate support.
phases	one-exact-coordinate-cancellation-per-coexisting-phase	phase-label-without-coexistence-constraint	A phase name alone does not record the equilibrium identification it imposes.
environment	two-held-environment-coordinate-carriers	free-or-continuum-intensive-coordinate-space	A free continuum imports unenumerated degrees and an additional premise.
relation	exact-carrier-cancellation-relation	imported-subtractive-phase-rule-equation	Importing a subtractive formula introduces signed arithmetic instead of forcing the structure.
absence	structural-EmptyOne-invariant-state	numerical-zero-degree-count	Numerical zero is not an SFT value.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
observation	complete-value-free-18-row-identity-seal	degree-outcome-readable-before-seal	A readable degree outcome could select the cancellation relation.
extension	depth-independent-joint-component-phase-successor	recalculate-with-free-exception	A special-case exception would destroy structural closure.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:c97ceace9d4bfbefb0676881399ef81bd4406598da9d25694a1661bd1acaa.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:17b3d69b3d13c330858ab63eb72b9d42aeffa9d04bce6034cb704e3269a46af5.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:223b781cd755e95466fca62a859f8d580bd22ebf373837d0b9c87493c0bda0.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:23edf09057e08e0a2e30301f99cb1409dcb663ae52215bf3d95a9e146c5151b9.

Independent reconstruction. Implementation hash

sha256:eb53a9026c2fdb0c9f523c680ff34d0f69df96283d78b90982ca4f9bb66ae00b; independent certificate sha256:12904bed53eb1911588d4df8827cd19110241d62ebd3208ae3bb66d3e87a4f30; external-validation hash sha256:75fca9b19bde30778f1178ca6d72d3952ade8a3c027027bd1725b2d2b5b0dc09. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLDBOOK-P04533-PHASE-RULE
- NIST-PHASE-DIAGRAM-GENERAL-DISCUSSION-GLOSSARY

Comparison records:

- SFT-CHEM-THERMO-011-PHASE-RULE-001: components=1; phases=1; degree support=2
- SFT-CHEM-THERMO-011-PHASE-RULE-002: components=1; phases=2; degree support=1
- SFT-CHEM-THERMO-011-PHASE-RULE-003: components=1; phases=3; degree support=EmptyOne
- SFT-CHEM-THERMO-011-PHASE-RULE-004: components=2; phases=1; degree support=3
- SFT-CHEM-THERMO-011-PHASE-RULE-005: components=2; phases=2; degree support=2
- SFT-CHEM-THERMO-011-PHASE-RULE-006: components=2; phases=3; degree support=1
- SFT-CHEM-THERMO-011-PHASE-RULE-007: components=2; phases=4; degree support=EmptyOne
- SFT-CHEM-THERMO-011-PHASE-RULE-008: components=3; phases=1; degree support=4
- SFT-CHEM-THERMO-011-PHASE-RULE-009: components=3; phases=2; degree support=3
- SFT-CHEM-THERMO-011-PHASE-RULE-010: components=3; phases=3; degree support=2
- SFT-CHEM-THERMO-011-PHASE-RULE-011: components=3; phases=4; degree support=1
- SFT-CHEM-THERMO-011-PHASE-RULE-012: components=3; phases=5; degree support=EmptyOne
- SFT-CHEM-THERMO-011-PHASE-RULE-013: components=4; phases=1; degree support=5
- SFT-CHEM-THERMO-011-PHASE-RULE-014: components=4; phases=2; degree support=4

- SFT-CHEM-THERMO-011-PHASE-RULE-015: components=4; phases=3; degree support=3
- SFT-CHEM-THERMO-011-PHASE-RULE-016: components=4; phases=4; degree support=2
- SFT-CHEM-THERMO-011-PHASE-RULE-017: components=4; phases=5; degree support=1
- SFT-CHEM-THERMO-011-PHASE-RULE-018: components=4; phases=6; degree support=EmptyOne
- complete structure vector: 18 component/phase identities across one through four components
- degree support: 14 exact positive outcomes and four external zero glyphs translated only to EmptyOne
- complete sources: IUPAC Gold Book P04533 and the byte-preserved 32-page NIST phase-diagram glossary
- tampered_degree_support_rejected: True
- complete_18_row_vector_retained: True
- all_14_positive_targets_retained: True
- all_four_EmptyOne_boundaries_retained: True
- complete_sources_retained: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if any component, phase or held environmental carrier is erased; if a phase does not cancel exactly one available carrier; if complete cancellation becomes numerical zero; if an imported phase-rule equation, subtraction, signed count, continuum coordinate, selected diagram or target-derived correction enters; if the joint component/phase successor changes degree support; if any target opens before all 18 identities seal; if any of 18 rows, 14 positive outcomes, four EmptyOne boundaries, full IUPAC term or complete 32-page NIST source is omitted; or if a target degree outcome is tampered.

Measurement receipt: sha256:c3ae4acca66dce6f5a7c43a515813367ee583cb24495e8ad283bbaa58e14d966.

Isolation certificate: sha256:caa8c6c931fe5b77fd0df70610298af01a6e3376f9d1993df53a767d2ea31096.

Custody certificate: sha256:c85ffabdf82b1bed9e1497451e969f2963b74cfa86a01ad12324ca46179f3048.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The thermodynamic extension derives finite microstate support, temperature, energy, heat/work, entropy, enthalpy, free direction, chemical potential, activity, fugacity, phases, colligative response, solvation and coupled transport while keeping external scalar inscriptions downstream. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; complete cancellation is structural EmptyOne
- no subtraction, signed count, negative, irrational, imaginary, floating or continuum SFT proof value
- no imported Gibbs phase-rule or Gibbs-Duhem equation
- no free intensive coordinate, selected phase diagram or target-derived correction
- no degree outcome, source equation fragment or target hash before prediction seal

Immutable identities. Source manifest

sha256:0c62588596f6a18b4700df6fe80704189da1ae4206adacd759982d3bf39fb308; derivation seal

sha256:b8d2ca6d453bf99f364338c058dd89fbd5c588beda5b44560e8e94d529ae3c68; engine receipt

sha256:e98e83e7f17426c14efde71543d97e559531afa80a123e73257b1be1bb500ad5 at receipts/engine

/model_admitted/SFT-CHEM-PHASE-RULE-STRUCTURAL-011-e98e83e7f17426c1.json; empirical hash

sha256:c638498e9d9b046f451f6ffff13e29851198de19e277bac4bb01c3fbd0ff8b33a; measurement receipt

sha256:c3ae4acca66dce6f5a7c43a515813367ee583cb24495e8ad283bbaa58e14d966.

136. One-component finite phase-boundary relation

Claim identity: SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012

Question and theorem. A one-component two-phase boundary is the exact finite ordered word of exchange-balanced coexistence points with one independent held coordinate support and exact temperature-pressure co-order.

complete-one-component-two-phase-point__two-distinct-held-coexisting-phases__exact-component-exchange-support-balance__one-independent-held-coordinate-support__exact-positive-temperature-pressure-co-order__finite-ordered-co-existence-word__complete-value-free-15-point-identity-seal__depth-independent-append-and-common-replication

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-MIXTURE-001
- SFT-CHEM-SOLUTION-EQUILIBRIUM-001
- SFT-CHEM-FINITE-MICROSTATE-SUPPORT-001
- SFT-CHEM-TEMPERATURE-CORRESPONDENCE-002
- SFT-CHEM-HEAT-WORK-TRANSFER-PARTITION-004
- SFT-CHEM-ENTROPY-MULTIPLICITY-CORRESPONDENCE-005
- SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007
- SFT-CHEM-CHEMICAL-POTENTIAL-EQUIVALENT-COMPONENT-008
- SFT-CHEM-FUGACITY-EQUIVALENT-GAS-MIXTURE-010
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011

Generated grammar. Generate the literal product of carrier, phase, balance, degree, response, boundary, prediction and extension forms; decide all 256 candidates only from admitted component identity, exchange balance, phase-rule cancellation, exact order and finite-word successor laws.

Boundary: Every finite positive word of one-component, two-distinct-phase, exactly exchange-balanced coexistence points whose held temperature and pressure supports are strictly co-ordered in source succession. External testing preserves both compounds, all three direct datasets, all fifteen measured points, all twelve adjacent successions and the complete 21-dataset 176-point parent source.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *complete-one-component-two-phase-point__two-distinct-held-coexisting-phases__exact-component-exchange-support-balance__one-independent-held-coordinate-support__exact-positive-temperature-pressure-co-order__finite-ordered-coexistence-word__complete-value-free-15-point-identity-seal__depth-independent-append-and-common-replication*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-one-component-two-phase-point	unbound-temperature-pressure-pair	A detached pair erases the component, phases and exchange support that make it a boundary point.
phase	two-distinct-held-coexisting-phases	single-phase-state-or-erased-phase-pair	A single phase does not define a coexistence boundary.
balance	exact-component-exchange-support-balance	assumed-equilibrium-or-target-derived-equality	Assumption or target adjustment does not force coexistence.
degree	one-independent-held-coordinate-support	free-two-coordinate-continuum	A free plane contradicts the single remaining carrier forced by one component and two phases.
response	exact-positive-temperature-pressure-co-order	imported-differential-equation-or-fitted-slope	A differential equation or fitted slope imports a continuum model.
boundary	finite-ordered-coexistence-word	interpolated-continuum-curve	Interpolation invents unmeasured points and an unenumerated continuum.
prediction	complete-value-free-15-point-identity-seal	coexistence-values-readable-before-seal	Readable temperatures or pressures could select the relation.
extension	depth-independent-append-and-common-replication	refit-after-appending-or-replication	Refitting destroys exact succession and support provenance.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : a8bb616f623396c0755bfbe375224a6d1243cd1d70a2de0c828012f03b238b8.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 4d93404e8dc969847f2e255cba33207b379112bb2f515f82f9651540dcca3c4.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 7cf15a63db0c0fc886a8753a30c34af8251858f3ed1e4f81e2c73063f91113a8.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 46c03a504b44cf132f9018c39b95843037e55d2285b599fb39eb7234f5a2552e.

Independent reconstruction. Implementation hash

sha256 : 09e9c774170ed54e70f806dfaae7088a8d8be594c46767cf2df13946ead5c76d; independent certificate
sha256 : ac538517f78e57a04b5d4dc8635219375f127bbc20f3d3dc72774a2dcaab8584; external-validation
hash sha256 : 9cf05bf1eecf4dbfdfab5630ce99331255a3556b59b8d24cf8628d25fbe7172c. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-TRC-THERMOML-FPE-2019-485-145-152

Comparison records:

- SFT-CHEM-THERMO-012-COEXISTENCE-001: dataset=5; T=283.14 K; P=14.6 kPa
- SFT-CHEM-THERMO-012-COEXISTENCE-002: dataset=5; T=293.14 K; P=23.5 kPa
- SFT-CHEM-THERMO-012-COEXISTENCE-003: dataset=5; T=303.12 K; P=36.2 kPa
- SFT-CHEM-THERMO-012-COEXISTENCE-004: dataset=5; T=313.13 K; P=54.1 kPa
- SFT-CHEM-THERMO-012-COEXISTENCE-005: dataset=5; T=323.13 K; P=78.8 kPa

- SFT-CHEM-THERMO-012-COEXISTENCE-006: dataset=8; T=283.16 K; P=14.4 kPa
- SFT-CHEM-THERMO-012-COEXISTENCE-007: dataset=8; T=293.15 K; P=23.4 kPa
- SFT-CHEM-THERMO-012-COEXISTENCE-008: dataset=8; T=303.15 K; P=36.3 kPa
- SFT-CHEM-THERMO-012-COEXISTENCE-009: dataset=8; T=313.2 K; P=54.4 kPa
- SFT-CHEM-THERMO-012-COEXISTENCE-010: dataset=8; T=323.17 K; P=78.9 kPa
- SFT-CHEM-THERMO-012-COEXISTENCE-011: dataset=11; T=283.15 K; P=1.51 kPa
- SFT-CHEM-THERMO-012-COEXISTENCE-012: dataset=11; T=293.15 K; P=2.72 kPa
- SFT-CHEM-THERMO-012-COEXISTENCE-013: dataset=11; T=303.15 K; P=4.72 kPa
- SFT-CHEM-THERMO-012-COEXISTENCE-014: dataset=11; T=313.15 K; P=7.86 kPa
- SFT-CHEM-THERMO-012-COEXISTENCE-015: dataset=11; T=323.16 K; P=12.6 kPa
- complete vector: three direct one-component datasets and fifteen coexistence points
- ordered boundary: all twelve adjacent source successions increase in exact temperature and pressure
- temperature range: 14157/50 through 32317/100 K
- pressure range: 151/100 through 789/10 kPa
- tampered_pressure_rejected: True
- complete_15_point_vector_retained: True
- all_three_datasets_retained: True
- both_compounds_and_parallel_dataset_retained: True
- all_12_successions_coordered: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if a point loses component, phase, temperature, pressure or exchange-support identity; if the phase pair is not distinct or exchange balanced; if one-component two-phase support does not leave one independent carrier; if a later stable liquid-vapor point does not preserve exact temperature-pressure co-order; if any differential equation, equation of state, interpolation, fit, selected compound/interval or target-derived correction enters; if targets open before all fifteen identities seal; if either compound, any of three datasets, fifteen points, twelve adjacent successions, uncertainties, methods or the complete parent source is omitted; or if any measured coordinate is tampered.

Measurement receipt: sha256:ba0132d14d2259c28f391293653b71ff8cd492672f40d0dab07242d4977a30a0.

Isolation certificate: sha256:5a83e46eb13193e8ef39ee3c4056c936e5bf2477750bb896cded6110f253ce3b.

Custody certificate: sha256:dca882e3e325888f2988c2321d0f097038af4b5161b33798ad5ae7228cc9e53.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The thermodynamic extension derives finite microstate support, temperature, energy, heat/work, entropy, enthalpy, free direction, chemical potential, activity, fugacity, phases, colligative response, solvation and coupled transport while keeping external scalar inscriptions downstream. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; exchange-balance separation is structural EmptyOne
- no negative, irrational, imaginary, logarithmic, floating, signed or continuum SFT proof value
- no imported Clausius-Clapeyron, differential, equation-of-state or vapor-pressure equation

- no interpolation, regression, fitted slope, selected compound/segment or target-derived correction
- no compound, phase, temperature, pressure, uncertainty or target hash before prediction seal

Immutable identities. Source manifest

sha256:ca0fa5249fe9a873f6a01bce0a3ec6860a33ea33a2dc4c493850d872d4314215; derivation seal
 sha256:af0d387e485e7fc7951bfff10d80e1e9854ca846e64d050cffaabfbbd215e1f75; engine receipt
 sha256:6ec44113a3770dcd2f4e8e479ded39ebf04ba4ee6303a5a8009b127f5e21dc1f at receipts/engine/
 model_admitted/SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012-6ec44113a3770dcd.json; empirical
 hash sha256:4d9eb4f31ad722eab35590cd4ff23a54a1c54e1a601ef90fb81a5b8a88fbd72; measurement
 receipt sha256:ba0132d14d2259c28f391293653b71ff8cd492672f40d0dab07242d4977a30a0.

137. Finite exact multicomponent phase-diagram relation

Claim identity: SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013

Question and theorem. A binary or higher multicomponent two-phase diagram is the exact finite word of complete coexistence records. Each phase is a held component-coordinate word closing exactly to the One; both phases retain the same ordered component identities, every component preserves exact exchange-support balance, and the phase-rule relation leaves component-count degree support. External zero glyphs denote structural EmptyOne only. No lever rule, Gibbs triangle, convex hull, continuum, interpolation, regression or fit enters.

complete-multicomponent-two-phase-point__complete-exact-phase-words-closing-to-One__two-distinct-held-phase-composition-words__componentwise-exact-exchange-support-balance__exact-phase-rule-component-count-support__structural-EmptyOne-absent-coordinate__complete-value-free-116-record-identity-seal__depth-independent-append-and-exchange-replication

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-MIXTURE-001
- SFT-CHEM-STOICH-SOLUTION-001
- SFT-CHEM-SOLUTION-EQUILIBRIUM-001
- SFT-CHEM-FINITE-MICROSTATE-SUPPORT-001
- SFT-CHEM-CHEMICAL-POTENTIAL-EQUIVALENT-COMPONENT-008
- SFT-CHEM-ACTIVITY-NONIDEAL-COMPOSITION-009
- SFT-CHEM-FUGACITY-EQUIVALENT-GAS-MIXTURE-010
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011
- SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012

Generated grammar. Generate the literal product of carrier, composition, phase, balance, degree, absence, prediction and extension forms; decide all 256 candidates only from admitted exact composition, component exchange, phase-rule cancellation, EmptyOne and finite-word successor laws.

Boundary: Every finite positive word of binary-or-higher, two-distinct-phase coexistence points whose complete exact phase-coordinate words close to the One and whose componentwise exchange supports are equal. External testing preserves all five binary dataset pairs and 65 points, the complete 51-point ternary dataset, all six companion pure datasets and the complete 17-dataset, 187-point parent source.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-multicomponent-two-phase-point__complete-exact-phase-words-closing-to-One__two-distinct-held-phase-composition-words__componentwise-exact-exchange-support-balance__exact-phase-rule-component-count-support__structural-EmptyOne-absent-coordinate__complete-value-free-116-record-identity-seal__depth-independent-append-and-exchange-replication`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	<code>complete-multicomponent-two-phase-point</code>	<code>unbound-coordinate-cloud</code>	Detached coordinates erase component, phase and coexistence identity.
composition	<code>complete-exact-phase-words-closing-to-One</code>	<code>selected-coordinate-or-unclosed-sum</code>	A selected coordinate or unclosed collection does not specify a composition.
phase	<code>two-distinct-held-phase-composition-words</code>	<code>single-bulk-composition-or-erased-phase</code>	A bulk composition cannot identify two coexisting states.
balance	<code>componentwise-exact-exchange-support-balance</code>	<code>imported-lever-rule-or-target-derived-tie-line</code>	An imported construction or target correction does not force coexistence.
degree	<code>exact-phase-rule-component-count-support</code>	<code>free-continuum-diagram-plane</code>	A continuum plane imports unenumerated states and free coordinates.
absence	<code>structural-EmptyOne-absent-coordinate</code>	<code>numerical-zero-composition</code>	Numerical zero is not an SFT value.
prediction	<code>complete-value-free-116-record-identity-seal</code>	<code>coexistence-values-readable-before-seal</code>	Readable values could select the relation.
extension	<code>depth-independent-append-and-exchange-replication</code>	<code>redraw-refit-or-interpolate-after-extension</code>	Redrawing or fitting destroys exact finite provenance.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:809b36f7bae263ed6e92f2e46bf5009744523b4ca45bfe52bd818c74cd77531e.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:cff1c1d52df54bb84070068a33af03af60a7000aa4b1e24d1c84eb658e20502c.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:e9e316d3edd54e72091bfedc774b19181b1e3a5aeab77f4eec052b63f4fe2eda.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:925703c913af2f5b672ffd36ead47b9786f5f5da560bc541e9146aa2986775ae.

Independent reconstruction. Implementation hash

sha256:ccd25da4607e211976241f40cad02fb352257ac65b5d3d670ac60c1c56bbade; independent certificate
sha256:426a86ceec0c8226359512d3ffdf668f82f795dd2bfe8f1b7ec81c5d6c530b863; external-validation

hash sha256:1ae7c2da6b7689f2890ef1bb0b1c1f43a2b8d6df48c56057684468ef565d5d49. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- NIST-TRC-THERMOML-JCT-2012-47-260-266

Comparison records:

- SFT-CHEM-THERMO-013-BINARY-001: class=binary; datasets=7,8; source-point=1; target=sha256:1329ec5638c8159cf4f17bef30ee0bfe66dafab2c7a899e65c0b60cc548177c0
- SFT-CHEM-THERMO-013-BINARY-002: class=binary; datasets=9,10; source-point=1; target=sha256:5e7f75893c1a1d279a1fd54387adaaf097b5c06d0f7f6d15a803d684079dc5e9
- SFT-CHEM-THERMO-013-BINARY-003: class=binary; datasets=9,10; source-point=2; target=sha256:a42e7abbbbf091a28a9d8a38a9499ec5cce51b0a1f7c9c4866d22098846e7ea
- SFT-CHEM-THERMO-013-BINARY-004: class=binary; datasets=9,10; source-point=3; target=sha256:4f849265cf224b68702a793eb297deebf2e4fa9058be3d5728ae9b89b6208e8
- SFT-CHEM-THERMO-013-BINARY-005: class=binary; datasets=9,10; source-point=4; target=sha256:663f948debdf208593d1237092f9b6601441309c1d97c9a115f2ce712162b10d
- SFT-CHEM-THERMO-013-BINARY-006: class=binary; datasets=9,10; source-point=5; target=sha256:88649495039f61cc070045c41ba312fdb0f3e030c6709f716c874413161ffdd6
- SFT-CHEM-THERMO-013-BINARY-007: class=binary; datasets=9,10; source-point=6; target=sha256:9782a0f3bba2475ac0319fcb65af31bf26a8a93f2e802683bd18a7ce694c7aa5
- SFT-CHEM-THERMO-013-BINARY-008: class=binary; datasets=9,10; source-point=7; target=sha256:a5eb6e103b7f6d412ce92256c7f0cdbd2b77155c7c1331213ee62a664b4fd35e
- SFT-CHEM-THERMO-013-BINARY-009: class=binary; datasets=9,10; source-point=8; target=sha256:fa591fac2e423bfe3a637d072fe31824fa3244a3f4ac86c6fea3f460f84e7cce
- SFT-CHEM-THERMO-013-BINARY-010: class=binary; datasets=9,10; source-point=9; target=sha256:b4f43341bbe3267e9943ff2eb5f7ffa201e22e02d4b74b7de0488ba25394b06c
- SFT-CHEM-THERMO-013-BINARY-011: class=binary; datasets=9,10; source-point=10; target=sha256:cd42cfc5bb2952fcac5bfd3441cc4ad39154982b09983955407016afd598b438
- SFT-CHEM-THERMO-013-BINARY-012: class=binary; datasets=9,10; source-point=11; target=sha256:58f80b07e5ac35feadef94fabce48a9c8cba1a484c113abe8ac8284d0f98f919
- SFT-CHEM-THERMO-013-BINARY-013: class=binary; datasets=9,10; source-point=12; target=sha256:2a42cb54e31105432d40dad1b190e00049b7ecc66d6fbf10307fb6d649b818fb
- SFT-CHEM-THERMO-013-BINARY-014: class=binary; datasets=9,10; source-point=13; target=sha256:32e67df96dd45c2b692b79c9fd2d08f809bca7b2796a96095f37fd3307c2d52c
- SFT-CHEM-THERMO-013-BINARY-015: class=binary; datasets=9,10; source-point=14; target=sha256:6e2c6c3d4e4be08deb11fba01f93def82bae24613b69b8756ffa6013c95f6f68
- SFT-CHEM-THERMO-013-BINARY-016: class=binary; datasets=9,10; source-point=15; target=sha256:ab2a48730f0be7560afc57f15d334ea7bcc3526afaa327d065e52740782337d3
- SFT-CHEM-THERMO-013-BINARY-017: class=binary; datasets=9,10; source-point=16; target=sha256:eb1aa5914698cefc1666d824baa4aae32917237982d608ff48a7bfde743488e6
- SFT-CHEM-THERMO-013-BINARY-018: class=binary; datasets=9,10; source-point=17; target=sha256:0f14b63017316ef916f8043211b0803cf80cd28770ff552ae601949ef4a20fef
- SFT-CHEM-THERMO-013-BINARY-019: class=binary; datasets=9,10; source-point=18; target=sha256:22ccf7ffbc593991b14b57889e15de6e48fe697c1497c9b8d7933096ff4c6f8d

- SFT-CHEM-THERMO-013-BINARY-020: class=binary; datasets=9,10; source-point=19; target=sha256:9643542b05c1028183d39fa7f221e82893250919d8965a70c3d03bc4e415a3a1
- SFT-CHEM-THERMO-013-BINARY-021: class=binary; datasets=9,10; source-point=20; target=sha256:b4be24638637d935248c68b119a936caa71053f4236787a5f7e8a0355a6382fc
- SFT-CHEM-THERMO-013-BINARY-022: class=binary; datasets=9,10; source-point=21; target=sha256:71f4d16c8a68fde81fc37ae0a920cca6a5df2bcafb95fda51c2604c257124061
- SFT-CHEM-THERMO-013-BINARY-023: class=binary; datasets=11,12; source-point=1; target=sha256:ea76efcd6bf93f24905cec1fc6aae0a18410ff75591db5e6f2daca6c0547b806
- SFT-CHEM-THERMO-013-BINARY-024: class=binary; datasets=13,14; source-point=1; target=sha256:669bdea6327b89f4f2950f1f9d618ae2e7668d17fc5369a93d4e8447dbb5d41b
- SFT-CHEM-THERMO-013-BINARY-025: class=binary; datasets=13,14; source-point=2; target=sha256:7d7eedd69e68b9b5b4bfc25b96dd045f0283537c1a9b24c786ce3c748db322ea
- SFT-CHEM-THERMO-013-BINARY-026: class=binary; datasets=13,14; source-point=3; target=sha256:460057658a14508017cdb380b3f4c76a10563e299f74470566400301e9169f7c
- SFT-CHEM-THERMO-013-BINARY-027: class=binary; datasets=13,14; source-point=4; target=sha256:df19cfa319761c6821957d471c678a3ea83ad105e9f1f61060aa62dd9ee08dd9
- SFT-CHEM-THERMO-013-BINARY-028: class=binary; datasets=13,14; source-point=5; target=sha256:335bbab17d61f01c0ccc1c929dac937f68f27cbf687717ab3995666e955a104c
- SFT-CHEM-THERMO-013-BINARY-029: class=binary; datasets=13,14; source-point=6; target=sha256:745aeddc3bd9fb9e7c1052c47316df9aace68bd982c7cf5817b005cedf107908
- SFT-CHEM-THERMO-013-BINARY-030: class=binary; datasets=13,14; source-point=7; target=sha256:8c9bdca05fbbaffb1fc7d2fdc11e56bcf2f424ec1052899817a8ea909adc7aa4
- SFT-CHEM-THERMO-013-BINARY-031: class=binary; datasets=13,14; source-point=8; target=sha256:a44123a824d767a3189f87541766a0128b5bcc0d42eb1b3d0b1c5958924a9abc
- SFT-CHEM-THERMO-013-BINARY-032: class=binary; datasets=13,14; source-point=9; target=sha256:b574bc3a00ecac6186e7ad3d2bc104129d61e7d21df9c41944f97d4d027864a7
- SFT-CHEM-THERMO-013-BINARY-033: class=binary; datasets=13,14; source-point=10; target=sha256:77941149a2a1051a6c0fddf960f6899ab100e36f4255b84ad5d729ddd9501e5
- SFT-CHEM-THERMO-013-BINARY-034: class=binary; datasets=13,14; source-point=11; target=sha256:7942a920e96efe92ae01ea7486994f8c590a45998d57cd04593de47933e81fd1
- SFT-CHEM-THERMO-013-BINARY-035: class=binary; datasets=13,14; source-point=12; target=sha256:4175f76d3968a7c0b79b5606265c281c47e781c9e663cb5f1270eb308b1c0df7
- SFT-CHEM-THERMO-013-BINARY-036: class=binary; datasets=13,14; source-point=13; target=sha256:9cbe961f878520c3f88a10ab74a24733e410b6e978be702a9288892750e1a87e
- SFT-CHEM-THERMO-013-BINARY-037: class=binary; datasets=13,14; source-point=14; target=sha256:0a91d959c16750d69777bd30e9c0d0e956f222da5f0ffc095b0af279cf2302c6
- SFT-CHEM-THERMO-013-BINARY-038: class=binary; datasets=13,14; source-point=15; target=sha256:f111f674b46217b0a3e701ab7702d5f7545764897d0c70ef296c1e015414e3ba
- SFT-CHEM-THERMO-013-BINARY-039: class=binary; datasets=13,14; source-point=16; target=sha256:a19b5b673ac53a6a1907a5d27a0218198cbd147c41c9038c002ac78a70f2902d
- SFT-CHEM-THERMO-013-BINARY-040: class=binary; datasets=13,14; source-point=17; target=sha256:a42f67e528ffdc57df071c1b0dc53be29c9d15798c85a9dd1bdf1e03cd4110fd
- SFT-CHEM-THERMO-013-BINARY-041: class=binary; datasets=13,14; source-point=18; target=sha256:42ee5a74736df58f848edf07001c24b4fa43329d2ff3462bab24d409c9ea14b9e

- SFT-CHEM-THERMO-013-BINARY-042: class=binary; datasets=13,14; source-point=19; target=sha256:6af39851ac262fd2e38fda66b3028d2537e3c5a6412dcc546205e7653ea0961f
- SFT-CHEM-THERMO-013-BINARY-043: class=binary; datasets=13,14; source-point=20; target=sha256:e7672da34cd9e3bac70e50aeae84359e5d8883fd43c1cfbe555a7289d857d25f
- SFT-CHEM-THERMO-013-BINARY-044: class=binary; datasets=13,14; source-point=21; target=sha256:aedfefc0e742c01070f8ba23f7d4736a97d81cf23511e40b2feccf05386bf213
- SFT-CHEM-THERMO-013-BINARY-045: class=binary; datasets=15,16; source-point=1; target=sha256:863d6a8dc5a5643bcc942541ef3e701cda3aacb3cad805b0453c0dbcbbb00621
- SFT-CHEM-THERMO-013-BINARY-046: class=binary; datasets=15,16; source-point=2; target=sha256:558705d3c23f01cec0bea86116b729a0334829de6c9d2585c17fb6203052b23c
- SFT-CHEM-THERMO-013-BINARY-047: class=binary; datasets=15,16; source-point=3; target=sha256:da802ad6e59533aa16cc99aaf9b534156a3bb7d255868d86e485555057e8dbd0
- SFT-CHEM-THERMO-013-BINARY-048: class=binary; datasets=15,16; source-point=4; target=sha256:5b8087d5b40a289e97a3c5dc09fb87799187904ec6c3b3a82de4b7f459c7ec3e
- SFT-CHEM-THERMO-013-BINARY-049: class=binary; datasets=15,16; source-point=5; target=sha256:4ac158393236f615c9600f781689b95bc9738de854c086ad1623e2bff4a50d4f
- SFT-CHEM-THERMO-013-BINARY-050: class=binary; datasets=15,16; source-point=6; target=sha256:a74eb0a00f37474019ca44c910dd391c3934724f88f96f95ed124a860d49ad97
- SFT-CHEM-THERMO-013-BINARY-051: class=binary; datasets=15,16; source-point=7; target=sha256:0d33545b912064dea8b917dff4992c7094aa392beac66ff39dfa20db9279859a
- SFT-CHEM-THERMO-013-BINARY-052: class=binary; datasets=15,16; source-point=8; target=sha256:42a9577d91c18faa944618554c1ea61918cbff94ca26fa33486abbab9a80eeb3
- SFT-CHEM-THERMO-013-BINARY-053: class=binary; datasets=15,16; source-point=9; target=sha256:aee39e15814c4952d500dbd51f1997b45286771413aae175473f4e3fc8fa8de0
- SFT-CHEM-THERMO-013-BINARY-054: class=binary; datasets=15,16; source-point=10; target=sha256:74b26e1dfc26f95f50a3783dd89f777ff46f2054e8e7d34cdfc54beb4e7e1956
- SFT-CHEM-THERMO-013-BINARY-055: class=binary; datasets=15,16; source-point=11; target=sha256:904de526ed147fad4de7ae531217fbeb004a15371c95674a8aae1f2940515dcb
- SFT-CHEM-THERMO-013-BINARY-056: class=binary; datasets=15,16; source-point=12; target=sha256:8fbf6db900fcad7fa90ce83a1650745b211afbc0f52c971289111074dff2c41
- SFT-CHEM-THERMO-013-BINARY-057: class=binary; datasets=15,16; source-point=13; target=sha256:d855e27aff1d9efb9b2d48bbaa602893bbe3699ecef932f4e8be6d1556fa6e16
- SFT-CHEM-THERMO-013-BINARY-058: class=binary; datasets=15,16; source-point=14; target=sha256:d079049a4ffcf3bca80d0669391db1ab54f5d27ea88944081cce95fb79382a51
- SFT-CHEM-THERMO-013-BINARY-059: class=binary; datasets=15,16; source-point=15; target=sha256:27b6b8d20a8f32714dafc574b021083cfe8137c24dbca870354250a930f9b9a0
- SFT-CHEM-THERMO-013-BINARY-060: class=binary; datasets=15,16; source-point=16; target=sha256:25e7d690ebdb13f0d057794b8c6e07965010d8753e9b1b28430f792c64f68fa5
- SFT-CHEM-THERMO-013-BINARY-061: class=binary; datasets=15,16; source-point=17; target=sha256:8da2db6366b4c0c2beebe68910f88a8bdc33bae74e8077f40e2a1f8f261567f4
- SFT-CHEM-THERMO-013-BINARY-062: class=binary; datasets=15,16; source-point=18; target=sha256:9a5d74dbeda5bd393e1350a8741fe82c5adf693f0965c8f8e900fc515bbbafc8
- SFT-CHEM-THERMO-013-BINARY-063: class=binary; datasets=15,16; source-point=19; target=sha256:d26965bd6dfb8317abd320ea940f454b6e1bf50f3439e0db6e632b6996c87037

- SFT-CHEM-THERMO-013-BINARY-064: class=binary; datasets=15,16; source-point=20; target=sha256:28c158ca6fb3e9cf587afb3051b84a58e6d41f97e4e8499e85b9f4cc6449cd76
- SFT-CHEM-THERMO-013-BINARY-065: class=binary; datasets=15,16; source-point=21; target=sha256:35af82a3b636895f368071e6c8e6e16ab9860547c218d8e07db1f115493c60f2
- SFT-CHEM-THERMO-013-TERNARY-066: class=ternary; datasets=17; source-point=1; target=sha256:2757cb4a9018e10f55b1d5695512d3805396457a92c5887b429938fb3758ddbb
- SFT-CHEM-THERMO-013-TERNARY-067: class=ternary; datasets=17; source-point=2; target=sha256:762177eae55bd66117547e9878ef5cbaa3a5fedacc2fc8097b07d1a356697588
- SFT-CHEM-THERMO-013-TERNARY-068: class=ternary; datasets=17; source-point=3; target=sha256:eae964e282bf8c10b07692f5bff5be1814279e5305b87a093520262a0b72aafb
- SFT-CHEM-THERMO-013-TERNARY-069: class=ternary; datasets=17; source-point=4; target=sha256:8ef03040c1270fb8a313ac829508107f0b8f007d3fd52ef1b8441c108002836d
- SFT-CHEM-THERMO-013-TERNARY-070: class=ternary; datasets=17; source-point=5; target=sha256:ed3537f6dd5cf0fe9ab8e8c818109c7d3e9b9dac4f4e64f36618ce394a5e6459
- SFT-CHEM-THERMO-013-TERNARY-071: class=ternary; datasets=17; source-point=6; target=sha256:6bf18fd6cc4f92d16c6f1246f5476de79debe29ccfd0555953d6aafc5830d3a
- SFT-CHEM-THERMO-013-TERNARY-072: class=ternary; datasets=17; source-point=7; target=sha256:1a4717e2146c14464ea6979c4efd7da791e6b78aaf90acb121cee86a5dbec849
- SFT-CHEM-THERMO-013-TERNARY-073: class=ternary; datasets=17; source-point=8; target=sha256:88cb0e1d224e52ab99caa7bf1c76f60cae7b5d1f8f67639cff02864ab8db3c8e
- SFT-CHEM-THERMO-013-TERNARY-074: class=ternary; datasets=17; source-point=9; target=sha256:6cb49f04dba30bf6b8f5fe0c92739fb8ded5ae6e2d4e5d4d594b0ca61b9ba70b
- SFT-CHEM-THERMO-013-TERNARY-075: class=ternary; datasets=17; source-point=10; target=sha256:a02d2f0c72d70295fc7ec25829a929bdab4d37830b88e4aad6205a70d33ceb5a
- SFT-CHEM-THERMO-013-TERNARY-076: class=ternary; datasets=17; source-point=11; target=sha256:392447e422173fdd43087eec7c68c7f4c6ce70345a956ed5e960163e897c6582
- SFT-CHEM-THERMO-013-TERNARY-077: class=ternary; datasets=17; source-point=12; target=sha256:16b3fe7606e7f87e53498f74b631a1404035dde06dde4b83e0cf185130f8537d
- SFT-CHEM-THERMO-013-TERNARY-078: class=ternary; datasets=17; source-point=13; target=sha256:e5d3b0c5c7669225daf844755a71800f066f498eea386eec4f6425f9b5d8d58a
- SFT-CHEM-THERMO-013-TERNARY-079: class=ternary; datasets=17; source-point=14; target=sha256:72e88a53598688105de2f35b186c85359c72e4bd212e4ee52c1c61e9f4d04e9a
- SFT-CHEM-THERMO-013-TERNARY-080: class=ternary; datasets=17; source-point=15; target=sha256:24b7a63b5caa9e77a0770661d5da6b95eee2c5512e19cb33c1a385d8b9a54f7a
- SFT-CHEM-THERMO-013-TERNARY-081: class=ternary; datasets=17; source-point=16; target=sha256:320a53a9b1eddf641ab2d7ac8dca5e26cf704b55c5840b8200b0b55460bef217
- SFT-CHEM-THERMO-013-TERNARY-082: class=ternary; datasets=17; source-point=17; target=sha256:c0150cdf22d9ea13916d3f694d08d8d26e3c067c3cdac5d62c594f03a7c552ed
- SFT-CHEM-THERMO-013-TERNARY-083: class=ternary; datasets=17; source-point=18; target=sha256:b69f406984767a227f162b0e855bc393f7800752a072680f0c07a70cd42bcf75
- SFT-CHEM-THERMO-013-TERNARY-084: class=ternary; datasets=17; source-point=19; target=sha256:06702238033ed92bb4d605304825ac2a41374e2f2cb822d7fe4df2c56f02f41b
- SFT-CHEM-THERMO-013-TERNARY-085: class=ternary; datasets=17; source-point=20; target=sha256:6f39f9a7ad5bc39bcb62ada4e72652ab6e7439517a7da04ea7c2b38b45f989ec

- SFT-CHEM-THERMO-013-TERNARY-086: class=ternary; datasets=17; source-point=21; target=sha256:fb9fec649a3a698f67a5c284a135ff5f852345c7214f748a1359b93321a4d45b
- SFT-CHEM-THERMO-013-TERNARY-087: class=ternary; datasets=17; source-point=22; target=sha256:5cae64b0f063ffd25cef0e3dee79ab30e44f2a972a9bd4df6c8c2f1b3aeb09a3
- SFT-CHEM-THERMO-013-TERNARY-088: class=ternary; datasets=17; source-point=23; target=sha256:8698653fbb815d803d34a9e996d63b5ac60a4c089245937b55ca5770f70f5a08
- SFT-CHEM-THERMO-013-TERNARY-089: class=ternary; datasets=17; source-point=24; target=sha256:e140d68f87ff6632c37a2c216e01dcf8723a6d6b0879625b374d2ddcc3b8260
- SFT-CHEM-THERMO-013-TERNARY-090: class=ternary; datasets=17; source-point=25; target=sha256:b020afd16eac39abaf04582c0babf0cf4bf0153b4e67dccfc2a42ec5883e68e1
- SFT-CHEM-THERMO-013-TERNARY-091: class=ternary; datasets=17; source-point=26; target=sha256:a12bb95dfc9f8e3a3bde22f0e270d4093210862178bbcac62585d9657bfff2a6
- SFT-CHEM-THERMO-013-TERNARY-092: class=ternary; datasets=17; source-point=27; target=sha256:8f91017ee8fd966cf501e6127382fc7f50e71173430ac5d9e8640b017d91257e
- SFT-CHEM-THERMO-013-TERNARY-093: class=ternary; datasets=17; source-point=28; target=sha256:f17275d160829449a411f8a531f5a16a46bf1cf692052d2ef6dfb6303fc239a6
- SFT-CHEM-THERMO-013-TERNARY-094: class=ternary; datasets=17; source-point=29; target=sha256:77e7a72e9645d2a08802692fe9be63fec9140d3315c570205bdbd882db5533f6
- SFT-CHEM-THERMO-013-TERNARY-095: class=ternary; datasets=17; source-point=30; target=sha256:f0508fd44575e5043f85b5bcaf157048f152b48ce522d051cc27ee82b2f36dd1
- SFT-CHEM-THERMO-013-TERNARY-096: class=ternary; datasets=17; source-point=31; target=sha256:e29dff373152b34906e1fc58249eac1409603e7555197ca507c1fa5551604b61
- SFT-CHEM-THERMO-013-TERNARY-097: class=ternary; datasets=17; source-point=32; target=sha256:59b05297f7bbbf173022657e6cb7ed458912493730e14ddf7b33294ad87541d
- SFT-CHEM-THERMO-013-TERNARY-098: class=ternary; datasets=17; source-point=33; target=sha256:b249f514e8622569a1b17d497a05f6ae2444df3ca994043479ddeb2d7e4ab52e
- SFT-CHEM-THERMO-013-TERNARY-099: class=ternary; datasets=17; source-point=34; target=sha256:b840896722954778d21a8a46169902cb35be011c86c21453313da14a737f6332
- SFT-CHEM-THERMO-013-TERNARY-100: class=ternary; datasets=17; source-point=35; target=sha256:955092ebe237be39a260668f82c3c57d2c49caa63ae6af8087fe018a507e6360
- SFT-CHEM-THERMO-013-TERNARY-101: class=ternary; datasets=17; source-point=36; target=sha256:b88283627476d601f5f030d870297910d811e107dc6e10c80b5031252b25709d
- SFT-CHEM-THERMO-013-TERNARY-102: class=ternary; datasets=17; source-point=37; target=sha256:11f81e66af5b544e0a5d4c1762f4090f07f016aeb2022c68a3c12c9848b35e77
- SFT-CHEM-THERMO-013-TERNARY-103: class=ternary; datasets=17; source-point=38; target=sha256:b083331d7608d867d5a7ddb540a6451b6493ff24d4aba87af3dd7bb5e99a1b38
- SFT-CHEM-THERMO-013-TERNARY-104: class=ternary; datasets=17; source-point=39; target=sha256:261918af7139d54803bec802affba3cc9d10a9ab49a1eed075d8cdae4d8196f0
- SFT-CHEM-THERMO-013-TERNARY-105: class=ternary; datasets=17; source-point=40; target=sha256:ade687febfad4b64e232943fd74f79ad968f75b1519b473e483407fbc382b46c
- SFT-CHEM-THERMO-013-TERNARY-106: class=ternary; datasets=17; source-point=41; target=sha256:3e4daf5becdffa243fc44675d11b14b57f2c75ecc808e4ae0a536c73f4518319
- SFT-CHEM-THERMO-013-TERNARY-107: class=ternary; datasets=17; source-point=42; target=sha256:04ceb15e66554d818ca169416055d2d3103c1df74f0f4adc3f9faebddd4d204

- SFT-CHEM-THERMO-013-TERNARY-108: class=ternary; datasets=17; source-point=43; target=sha256:417c51eb0d058d3a008689ca33e0100c41edc7a482788fa420783d9a01a9f239
- SFT-CHEM-THERMO-013-TERNARY-109: class=ternary; datasets=17; source-point=44; target=sha256:100a041fd117100ef85ec098a82df80d9718441fc65b6dd37431954edc98051c
- SFT-CHEM-THERMO-013-TERNARY-110: class=ternary; datasets=17; source-point=45; target=sha256:65918294562afa26bd2f523d52c94f6ca3214b417d7760359440625027792128
- SFT-CHEM-THERMO-013-TERNARY-111: class=ternary; datasets=17; source-point=46; target=sha256:5796c6a76cfc8eed4c3ab86ad6b22c2710383198db2beb9fe4aacc92d58a175f
- SFT-CHEM-THERMO-013-TERNARY-112: class=ternary; datasets=17; source-point=47; target=sha256:36f2148e4c664cf0482b5440e58217f9453015aac47c0d816d193aee53f83845
- SFT-CHEM-THERMO-013-TERNARY-113: class=ternary; datasets=17; source-point=48; target=sha256:e9435d54ef185188378a632181f836d22d1f0c21b18556b90c709e1a990efa7
- SFT-CHEM-THERMO-013-TERNARY-114: class=ternary; datasets=17; source-point=49; target=sha256:a1d80a318f3e448bb15e6128d536bda24075d057484a0eaf47e7ec12671af964
- SFT-CHEM-THERMO-013-TERNARY-115: class=ternary; datasets=17; source-point=50; target=sha256:374475e6b84f296db93cf4cecc11917ab2773c55b80d11fe73fa3e2cf14dc29d
- SFT-CHEM-THERMO-013-TERNARY-116: class=ternary; datasets=17; source-point=51; target=sha256:74f98579f3fbb2e2ed0ed2ce4b108d5821c1655d35356b74b854439c7ba37ba7
- complete vector: 65 binary and 51 ternary coexistence records
- complete phase support: 566 exact component coordinates with 12 EmptyOne boundaries
- source coverage: five binary dataset pairs, one complete ternary dataset, six companion pure datasets
- temperature range: 6993/20 through 37237/100 K
- pressure range: 1013/10 through 1013/10 kPa
- tampered_ternary_composition_rejected: True
- complete_116_record_vector_retained: True
- all_binary_and_ternary_records_retained: True
- all_dataset_pairs_retained: True
- all_compositions_close_to_One: True
- all_absence_boundaries_are_EmptyOne: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if either phase loses a component or does not close exactly to the One; if phase identities are not distinct; if componentwise exchange support is unbalanced; if two-phase degree support is not the component count; if an external zero becomes a number rather than EmptyOne; if any imported geometric or continuum phase construction, equation of state, interpolation, regression, fit, selection or target-derived correction enters; if targets open before all 116 identities seal; if any of five binary pairs, 65 binary points, 51 ternary points, six companion datasets, uncertainties, methods, provenance or the complete 17-dataset 187-point source is omitted; or if any measured coordinate is tampered.

Measurement receipt: sha256:665b92f0695956b12597300209b4284141acedeb973346df55959c890f7a4cde.
Isolation certificate: sha256:117009c9c5440c5900ec50ac1dd6c6aa8500ec130760fd8f23bdebdbe9eef98c.
Custody certificate: sha256:e2e2ae2563e6b42a73c187e56a8fab3230e3d363cafcc88f2931a2a64fdb02b8.

Registered source descriptions:

- National Institute of Standards and Technology ThermoML Archive: ()

Registered target identities:

- None.

Meaning. The thermodynamic extension derives finite microstate support, temperature, energy, heat/work, entropy, enthalpy, free direction, chemical potential, activity, fugacity, phases, colligative response, solvation and coupled transport while keeping external scalar inscriptions downstream. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; source zero glyphs translate only to structural EmptyOne
- no negative, irrational, imaginary, logarithmic, floating, signed or continuum SFT proof value
- no imported lever rule, tie-line equation, Gibbs triangle, convex hull or equation of state
- no continuum phase diagram, interpolation, regression, fitted interaction parameter or target-derived correction
- no compound, phase, temperature, pressure, composition, uncertainty or target hash before prediction seal
- no selected system, dataset, phase, point, azeotropic region or favorable diagram segment
- every source decimal remains a post-seal external inscription and the complete source remains preserved

Immutable identities. Source manifest

sha256:923f3f18bd85efbadda1ef88ee8e819bbe6b8a4060cbac6f982cc0a1f5552a0e; derivation seal
 sha256:f527a58166e8a44092a56d524cb2a7f2c43150e771b801a2a6475411c2932c3c; engine receipt
 sha256:fbd07842719bd0ba06c1c54d03e5ad0376f3b1f790a937264a5fcf513b5f4393 at receipts/engine/model_admitted/SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013-fbd07842719bd0ba.json; empirical
 hash sha256:3238f75d83dae3a7cd99000f8f2794c3ac37ff3e97fb7232d7ec9a9355f91b49; measurement
 receipt sha256:665b92f0695956b12597300209b4284141acedeb973346df55959c890f7a4cde.

138. Exact colligative composition-response law

Claim identity: SFT-CHEM-COLLIGATIVE-COMPOSITION-RESPONSE-014

Question and theorem. A colligative response is the exact held relation produced when a distinct solute particle identity is retained while solvent exchange remains the transmitted carrier. Solute retention forces a held boiling, freezing or osmotic orientation; measured magnitude is an exact positive reference-response separation opened only after the law seals. Pure solvent uses structural EmptyOne. No conventional response equation, constant, dissociation factor, continuum, regression or fit enters.

complete-solvent-solute-particle-account__distinct-held-solvent-and-solute-identities__exact-solvent-transmission-and-solute-retention-boundary__held-boiling-freezing-osmotic-orientation-label__exact-positive-reference-response-separation__structural-EmptyOne-pure-solvent-boundary__complete-value-free-276-record-identity-seal__depth-independent-common-replication-and-record-append

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-MIXTURE-001
- SFT-CHEM-STOICH-SOLUTION-001
- SFT-CHEM-SOLUTION-EQUILIBRIUM-001

- SFT-CHEM-TEMPERATURE-CORRESPONDENCE-002
- SFT-CHEM-ENTROPY-MULTIPLICITY-CORRESPONDENCE-005
- SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007
- SFT-CHEM-CHEMICAL-POTENTIAL-EQUIVALENT-COMPONENT-008
- SFT-CHEM-ACTIVITY-NONIDEAL-COMPOSITION-009
- SFT-CHEM-FUGACITY-EQUIVALENT-GAS-MIXTURE-010
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011
- SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013

Generated grammar. Generate the literal product of carrier, identity, boundary, direction, magnitude, absence, prediction and extension forms; decide all 256 candidates only from admitted particle distinction, component exchange, phase balance, exact order, EmptyOne and finite successor laws.

Boundary: Every finite positive boiling, freezing or osmotic record with distinct held solvent/solute identities, an exact composition coordinate or EmptyOne reference, exact positive environment and response support, and a declared solvent-transmission/solute-retention boundary. External testing preserves all 16 boiling datasets and 144 rows, six freezing datasets and 37 rows, six osmotic datasets and 95 rows, and all three complete sources.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-solvent-solute-particle-account__distinct-held-solvent-and-solute-identities__exact-solvent-transmission-and-solute-retention-boundary__held-boiling-freezing-osmotic-orientation-label__exact-positive-reference-response-separation__structural-EmptyOne-pure-solvent-boundary__complete-value-free-276-record-identity-seal__depth-independent-common-replication-and-record-append`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-solvent-solute-particle-account	unbound-response-number	A detached number erases the solvent, solute, condition and response boundary.
identity	distinct-held-solvent-and-solute-identities	anonymous-particle-count-or-erased-solvent	Anonymous particles cannot distinguish the transmitted solvent from the held solute.
boundary	exact-solvent-transmission-and-solute-retention-boundary	imported-colligative-equation-or-dissociation-factor	An imported equation or factor selects a response instead of forcing it.
direction	held-boiling-freezing-osmotic-orientation-label	signed-temperature-or-pressure-displacement	A signed displacement imports negative arithmetic.
magnitude	exact-positive-reference-response-separation	linear-constant-fit-or-target-correction	A fitted constant or target correction adds an unforced parameter.
absence	structural-EmptyOne-pure-solvent-boundary	numerical-zero-solute-coordinate	Numerical zero is not an SFT value.
prediction	complete-value-free-276-record-identity-seal	response-values-readable-before-seal	Readable values could select the response law.
extension	depth-independent-common-replication-and-record-append	refit-after-particle-or-response-replication	Refitting destroys exact support provenance.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : f2c5b6cfae5f52b524291305bd9a8a0e89dcdb418486b4b61263d06f8e5136abe.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : b356604b939339bfe3ef8a4c4a5ce94003cfc48ee71399feece6c2dc749cab5c.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 2b9025759f0fc06c4b04d0bcfa37c22af6e9614ae7d0f78a6326ce010bafd101.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 9c205dbcdcb7664e04ade28b507ec90b165b5372542562b774b4fdd6ff84a854.

Independent reconstruction. Implementation hash

sha256 : 2ee3c73e79da1c4ed92c05a57beffffdf22bddff0e9905fbc107bb1afbb9b6ac8; independent certificate sha256 : 88bd2b1d99fa5283ade6a9cfac4d95d499c114c05804dc33f6db3b1ddf6d0db5; external-validation hash sha256 : 32ac6a27f3e91b1c9f0eec9e8951c42afbea67664ec184ab738433e70ec96f9d. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-TRC-THERMOML-FPE-2013-337-60-66
- NIST-TRC-THERMOML-FPE-2016-420-14-19
- NIST-TRC-THERMOML-JCT-2009-41-1439-1445

Comparison records:

- SFT-CHEM-THERMO-014-BOILING-0001: class=boiling; dataset=1; point=1;
target=sha256:8cc1743155505ec8765965184d79d33712016f1d75a0aa925d93c0710dd64c20
- SFT-CHEM-THERMO-014-BOILING-0002: class=boiling; dataset=1; point=2;
target=sha256:4579c5d6d936ac0cb4c974fe19722a652bb4804a9c3277d9ab7935bd19447c62
- SFT-CHEM-THERMO-014-BOILING-0003: class=boiling; dataset=1; point=3;
target=sha256:73fc03d2ae6be21e326a85dc7d32b758b977850144bddc3f72a66acc42f5c588
- SFT-CHEM-THERMO-014-BOILING-0004: class=boiling; dataset=1; point=4;
target=sha256:8fdd1a4609ddef48eefd8ab793009e907667468107c251038bc693d089feaf2
- SFT-CHEM-THERMO-014-BOILING-0005: class=boiling; dataset=1; point=5;
target=sha256:834bb80523e0c511b031e103b52c03e99a1f206db9dace432f1a4ebf1d2b65d0
- SFT-CHEM-THERMO-014-BOILING-0006: class=boiling; dataset=1; point=6;
target=sha256:735009feedb4f3c04306e75f10199bc06aedd82dfa33af72f39f62999c16c58f
- SFT-CHEM-THERMO-014-BOILING-0007: class=boiling; dataset=1; point=7;
target=sha256:1f9f5b109594ad3ce6a942940ea6294e9602b123dc24d7834634e0900c6a4984
- SFT-CHEM-THERMO-014-BOILING-0008: class=boiling; dataset=1; point=8;
target=sha256:5d37cbb305e6f6b0e1d272e12013397bf60166f1194ccfcee31a5ba9fdde057
- SFT-CHEM-THERMO-014-BOILING-0009: class=boiling; dataset=1; point=9;
target=sha256:6a7e160e2d25aee36cfb34879455c70949ea91b6f924e871a545391d2984f057
- SFT-CHEM-THERMO-014-BOILING-0010: class=boiling; dataset=2; point=1;
target=sha256:4837fb7fcbca9462ebb878bc096b5c220512b6d0773a2bb903ba4069a0f6cc81
- SFT-CHEM-THERMO-014-BOILING-0011: class=boiling; dataset=2; point=2;
target=sha256:00df4da8b8744754e565842d39d041c480cb73471ab0d8bccadd637b6cb58270

- SFT-CHEM-THERMO-014-BOILING-0012: class=boiling; dataset=2; point=3;
target=sha256:f2cda30b2896ed1468cb9390fd63e31ef5053f566444e1dc7e9c40798d4608c3
- SFT-CHEM-THERMO-014-BOILING-0013: class=boiling; dataset=2; point=4;
target=sha256:817083c37c7e17718e39c497c0cd42a0156106d99a6dae3e118b792de65bf1fe
- SFT-CHEM-THERMO-014-BOILING-0014: class=boiling; dataset=2; point=5;
target=sha256:d636506787223bd2bc42b423936b16a67104d7b1ac376bdd33945f1cac2732c
- SFT-CHEM-THERMO-014-BOILING-0015: class=boiling; dataset=2; point=6;
target=sha256:560d610ddc882db85bf7ab46326f172b923506e6101f916aa76f13f320f39786
- SFT-CHEM-THERMO-014-BOILING-0016: class=boiling; dataset=2; point=7;
target=sha256:626f9e6bce1668c7f04e8e29db73f4d4f5a02dc039ee791ca21161d14c37cf95
- SFT-CHEM-THERMO-014-BOILING-0017: class=boiling; dataset=2; point=8;
target=sha256:55e208265e26737a70b79bfcc1183f4cb33467c036b5e4c983f0934bcb116cb9
- SFT-CHEM-THERMO-014-BOILING-0018: class=boiling; dataset=2; point=9;
target=sha256:9914bdf2d3c06fea6fdeb20ed7764f315a1f8c3f5dba79b90cb16bbe78b6fba4
- SFT-CHEM-THERMO-014-BOILING-0019: class=boiling; dataset=3; point=1;
target=sha256:c5c3c3b615f5d820ba026880969d0fe9597164ac2c187ad41ae9d34ec8b7a01d
- SFT-CHEM-THERMO-014-BOILING-0020: class=boiling; dataset=3; point=2;
target=sha256:a7d9d7d021cbd267546cc0e8a43ed2de35291d6fd17b1ff56e99b2b9d0eccf25
- SFT-CHEM-THERMO-014-BOILING-0021: class=boiling; dataset=3; point=3;
target=sha256:415beb1ebbc34e470d5842e2c22311109650f198fd066355542d448a45122af
- SFT-CHEM-THERMO-014-BOILING-0022: class=boiling; dataset=3; point=4;
target=sha256:2f66b4db3580c661c201e92c58d4e0b054dae75cc54c6e73ef068e85caecb44
- SFT-CHEM-THERMO-014-BOILING-0023: class=boiling; dataset=3; point=5;
target=sha256:267bff39ef07a0c9c33483c31ba7520d2c799ada0680352d0b83779e063caf8c
- SFT-CHEM-THERMO-014-BOILING-0024: class=boiling; dataset=3; point=6;
target=sha256:833f7a61238b8b1dcbab0b35c6792da1dadfe707923fd13be1819d33fd95835df
- SFT-CHEM-THERMO-014-BOILING-0025: class=boiling; dataset=3; point=7;
target=sha256:1e4c96bce4daf311c37253627d38fc0d9fb11521d49b05c57dc73ed35b1915d4
- SFT-CHEM-THERMO-014-BOILING-0026: class=boiling; dataset=3; point=8;
target=sha256:821fb370afbf43177574824d8b304833828c545c548474e7947f51315d4f03bc
- SFT-CHEM-THERMO-014-BOILING-0027: class=boiling; dataset=3; point=9;
target=sha256:0b13c4504c09db09d8b67d0a79ebe413678b9bfff883a931bb96b94029273ee8e
- SFT-CHEM-THERMO-014-BOILING-0028: class=boiling; dataset=4; point=1;
target=sha256:56af2d6fe39d74077e710c524fef14a4cc874302d36166f5f26077e3c1eab5eb
- SFT-CHEM-THERMO-014-BOILING-0029: class=boiling; dataset=4; point=2;
target=sha256:efcdc26ffa3253624a575847c97a5ee382151d8a1342ff23dca5fddd5dc84f15
- SFT-CHEM-THERMO-014-BOILING-0030: class=boiling; dataset=4; point=3;
target=sha256:d9aa7019a4093423748625a4f3bc2ea27da0184ac65c3cf638691511279d7ff3
- SFT-CHEM-THERMO-014-BOILING-0031: class=boiling; dataset=4; point=4;
target=sha256:dc14befe209d3c2e6c1ad77571d3da719c0c5845fe7666d418ece4174a5cdc34
- SFT-CHEM-THERMO-014-BOILING-0032: class=boiling; dataset=4; point=5;
target=sha256:95f9ea6fd68dce8ccc963dbb46588e058bdd0b7dc935d7e3d923ba699b29b668
- SFT-CHEM-THERMO-014-BOILING-0033: class=boiling; dataset=4; point=6;
target=sha256:97d0203a427e4adabc1c94ea5147c114278ff1df72b17337a37616ceac10752f

- SFT-CHEM-THERMO-014-BOILING-0034: class=boiling; dataset=4; point=7;
target=sha256:86d2dc1c29ce236e34cf03eac9c1ac7c6c642e29319bb68d4541ab8f5394b8ea
- SFT-CHEM-THERMO-014-BOILING-0035: class=boiling; dataset=4; point=8;
target=sha256:6b6d7b0944a282cba69ae61370be72d8ceee3fce9bd3abab33dd6182f478d93e
- SFT-CHEM-THERMO-014-BOILING-0036: class=boiling; dataset=4; point=9;
target=sha256:2a2d3b17a5e315f1fc982bb5a49272943bc5a4350390811d1ec88029f9607e91
- SFT-CHEM-THERMO-014-BOILING-0037: class=boiling; dataset=5; point=1;
target=sha256:a72fc7f2f9af7c01d7cab53103a8f57a65e7d07e64b19726f82b561c018f95f7
- SFT-CHEM-THERMO-014-BOILING-0038: class=boiling; dataset=5; point=2;
target=sha256:c03cde08913c3572509139da4abea1b751b3c1ae8bf424e68104dab205fdefda
- SFT-CHEM-THERMO-014-BOILING-0039: class=boiling; dataset=5; point=3;
target=sha256:4d867db411d8057c89947d7fe3995433e11db0c464b22286e78e2f0cb715a50f
- SFT-CHEM-THERMO-014-BOILING-0040: class=boiling; dataset=5; point=4;
target=sha256:832868e96dc6ea6e2a76c062ec92eb4908093c6baf1bf0d483497a88e36756d0
- SFT-CHEM-THERMO-014-BOILING-0041: class=boiling; dataset=5; point=5;
target=sha256:a328934d53af65f4cbbf92d9207d28dc43dc6cb5c35a3059fd0e5a2d42b11e33
- SFT-CHEM-THERMO-014-BOILING-0042: class=boiling; dataset=5; point=6;
target=sha256:047d459be8ed0026af29fe7d2c405be096ae4721989a5d484e39b1e3f20864c3
- SFT-CHEM-THERMO-014-BOILING-0043: class=boiling; dataset=5; point=7;
target=sha256:9fc09cdd0eeee2c0cd4e67bec661c13d1a5475b301675410256ee77b770d1df5
- SFT-CHEM-THERMO-014-BOILING-0044: class=boiling; dataset=5; point=8;
target=sha256:26b6bb782ca9f8126d9340e61470710f9c68c417b162dcf5475aa77347e76c15
- SFT-CHEM-THERMO-014-BOILING-0045: class=boiling; dataset=5; point=9;
target=sha256:adc9c120a21f8b2872a9f0886a8e3d5258680005d0b9196b1ab66e77e998df17
- SFT-CHEM-THERMO-014-BOILING-0046: class=boiling; dataset=6; point=1;
target=sha256:e810436f848e8f8f6e25748f100ebe1ec95df751257b4b5491f163286881d955
- SFT-CHEM-THERMO-014-BOILING-0047: class=boiling; dataset=6; point=2;
target=sha256:a7277df13cb70fc3602dcf887dd25c7dea3f0087b41ca2ad9c0e258bb0aa304
- SFT-CHEM-THERMO-014-BOILING-0048: class=boiling; dataset=6; point=3;
target=sha256:78bd7ea0d9df8e51287d47241369b4f53d144a07dcbb51422296b166723432ee
- SFT-CHEM-THERMO-014-BOILING-0049: class=boiling; dataset=6; point=4;
target=sha256:5b3bd5fab943a5d2507e0b400508fde18634483cb1db8adfe62a502cd088bac1
- SFT-CHEM-THERMO-014-BOILING-0050: class=boiling; dataset=6; point=5;
target=sha256:e6c741873d8f5c6bcd3d035e8c7380fe93d93109c3519ccf3b3fe4f13b74e595
- SFT-CHEM-THERMO-014-BOILING-0051: class=boiling; dataset=6; point=6;
target=sha256:e42f6aaf65b6632f25a3ff568eaa113005976c7b71b2606458241c7c77b9e791
- SFT-CHEM-THERMO-014-BOILING-0052: class=boiling; dataset=6; point=7;
target=sha256:1e75863c6e53dd4843536ac6b39322094aed42226486b52a0114945308c7c043
- SFT-CHEM-THERMO-014-BOILING-0053: class=boiling; dataset=6; point=8;
target=sha256:cbffc6468798f6cae9e98b935e145c7eed8aa45041f1a0a38108964565bb490e
- SFT-CHEM-THERMO-014-BOILING-0054: class=boiling; dataset=6; point=9;
target=sha256:03774fb45ab4c2a614dfb5993fd296e7f52f38a994ecd0df4acc3fe2e87d2d2e
- SFT-CHEM-THERMO-014-BOILING-0055: class=boiling; dataset=7; point=1;
target=sha256:8ab9b3455c2b88ff6922487ce2dabd4d9062f9fcccc70ebf08502d6004a7042f

- SFT-CHEM-THERMO-014-BOILING-0056: class=boiling; dataset=7; point=2;
target=sha256:9ab6ae14b8cd2b724550b20008e35696fce10dbc72fbd171fcd7d32401051f53
- SFT-CHEM-THERMO-014-BOILING-0057: class=boiling; dataset=7; point=3;
target=sha256:30945b388e5ee0d31d5273be30c11da6884deb3f99a4b7d44d630be1d19cb96d
- SFT-CHEM-THERMO-014-BOILING-0058: class=boiling; dataset=7; point=4;
target=sha256:36f0329405bab3356c34521c08aebd75fef8036398e5a02645ee426e69c9936b
- SFT-CHEM-THERMO-014-BOILING-0059: class=boiling; dataset=7; point=5;
target=sha256:8341182c96e1e0102a4f9aacf1700406adef8afa3ff21971bb3846d9180seeda
- SFT-CHEM-THERMO-014-BOILING-0060: class=boiling; dataset=7; point=6;
target=sha256:06f3ebf2f20b844ba7ee4b8c5a4d8404f132e96c97a91be88001718397676191
- SFT-CHEM-THERMO-014-BOILING-0061: class=boiling; dataset=7; point=7;
target=sha256:b43bfe38d518fa346dcfcd280bf5bc09f966fe77e5f8d0790017a46fcb62e9ed
- SFT-CHEM-THERMO-014-BOILING-0062: class=boiling; dataset=7; point=8;
target=sha256:8eb287909fc71c5e019f6c61b9a845fe823770fd123582d6c84ce561d5e227ff
- SFT-CHEM-THERMO-014-BOILING-0063: class=boiling; dataset=7; point=9;
target=sha256:a3366a5d6c68f44e1ef562e95cf54e144adaa738317dbc01b458f213db1ed3be
- SFT-CHEM-THERMO-014-BOILING-0064: class=boiling; dataset=8; point=1;
target=sha256:abd3d97401c9607aa73ee2c74d197c101b608016716a45cd3eb94d8421bec13a
- SFT-CHEM-THERMO-014-BOILING-0065: class=boiling; dataset=8; point=2;
target=sha256:df13df8f840a5042a48878d3da72f04c3b470eaf581a9a805352dd02d8d01987
- SFT-CHEM-THERMO-014-BOILING-0066: class=boiling; dataset=8; point=3;
target=sha256:b0f325bfea820123bd9e46f3f0d70b1c6bd7a81f4aba6ed2f136131af3f928b8
- SFT-CHEM-THERMO-014-BOILING-0067: class=boiling; dataset=8; point=4;
target=sha256:7b5ed857a20a4fd665d6ef234b94efed27f39e5ca3f97d4eee75240e76981a5e
- SFT-CHEM-THERMO-014-BOILING-0068: class=boiling; dataset=8; point=5;
target=sha256:9f82638affbb7ae6c9d1b3f74fe25a65c22274ec1763ea7c828ac55f222276e
- SFT-CHEM-THERMO-014-BOILING-0069: class=boiling; dataset=8; point=6;
target=sha256:11ca36ca63dfd43da0de88c146115217bfbcdf0f416946cc0dcd56d98de0f862
- SFT-CHEM-THERMO-014-BOILING-0070: class=boiling; dataset=8; point=7;
target=sha256:bfa75b45a8d7e3279a6ba3be06814f6edbc16e9a75cb8196596c59039c6473f0
- SFT-CHEM-THERMO-014-BOILING-0071: class=boiling; dataset=8; point=8;
target=sha256:adb53ba0bd72406452859685bc82bec5fcfe6d6905d7e896d2260aae15c98fbf
- SFT-CHEM-THERMO-014-BOILING-0072: class=boiling; dataset=8; point=9;
target=sha256:28be6706986d351560b5fec1043af52c0ecf8b8c59bb7c18613c6691148a2953
- SFT-CHEM-THERMO-014-BOILING-0073: class=boiling; dataset=9; point=1;
target=sha256:5a42552fab31843189d64d64e7724c4ba7a663c22000231a5acd255b670cbf5f
- SFT-CHEM-THERMO-014-BOILING-0074: class=boiling; dataset=9; point=2;
target=sha256:4b4b45eea2c2e1b8da2d421dc50616ca9604787767f4efaa4da2108b11d73d1a
- SFT-CHEM-THERMO-014-BOILING-0075: class=boiling; dataset=9; point=3;
target=sha256:2d00f0961f86ff21109a2ef78376c4ad443bcba452dcb00b7af411f4374e47ef
- SFT-CHEM-THERMO-014-BOILING-0076: class=boiling; dataset=9; point=4;
target=sha256:e8887693d1fbd3c171f298a197ce5c0a7d33ba11ecd7c7740fa707568ee794d
- SFT-CHEM-THERMO-014-BOILING-0077: class=boiling; dataset=9; point=5;
target=sha256:fc873edd423adfb8ced9e105972ded6d88ad1a470ab7f4d68879f218a70084f

- SFT-CHEM-THERMO-014-BOILING-0078: class=boiling; dataset=9; point=6;
target=sha256:90e727fb521bc30bfbeac3b8eff039141876162e5f1a5a58b643e53d98406678
- SFT-CHEM-THERMO-014-BOILING-0079: class=boiling; dataset=9; point=7;
target=sha256:0d85a270c8f56755db29dba11d696081fb3af052f00073b254c577208c242762
- SFT-CHEM-THERMO-014-BOILING-0080: class=boiling; dataset=9; point=8;
target=sha256:d96add9efe5fcb69fe2b2138b3f52aeb637fdb3acc260793dd18a4b7f099af09
- SFT-CHEM-THERMO-014-BOILING-0081: class=boiling; dataset=9; point=9;
target=sha256:a529c99f79379f3d85e215359b937ac8147ccd28c63da6461bab7689d422f5de
- SFT-CHEM-THERMO-014-BOILING-0082: class=boiling; dataset=10; point=1;
target=sha256:2ca24bbbf4924e6a9a47fed6e7b88d1fe5d2cf291bdac4a3dfd6c04bab8946f
- SFT-CHEM-THERMO-014-BOILING-0083: class=boiling; dataset=10; point=2;
target=sha256:3b45c9e567d2960c2441612259289bfed17fe7ad2b78d1475dcd283519c8a352
- SFT-CHEM-THERMO-014-BOILING-0084: class=boiling; dataset=10; point=3;
target=sha256:5d5f1ff49a58fb6e00d853d10065f964c3c378c1e01e0a5a96f658e1b04b7b77
- SFT-CHEM-THERMO-014-BOILING-0085: class=boiling; dataset=10; point=4;
target=sha256:d4d694169a74aead64cf41181581384744fb801bf75d4e5f1d4d5a5f23b68d8
- SFT-CHEM-THERMO-014-BOILING-0086: class=boiling; dataset=10; point=5;
target=sha256:78d5b9d93c7002888e7f569f7e6f28e5458102f919ca8c6947c6b43b403f030c
- SFT-CHEM-THERMO-014-BOILING-0087: class=boiling; dataset=10; point=6;
target=sha256:8d59e009d5c5566a61a47a9e5930b9ffef26fd10000eb01f308fa1d940b6ce10
- SFT-CHEM-THERMO-014-BOILING-0088: class=boiling; dataset=10; point=7;
target=sha256:7d895739b84c7137cd6005d2269653a4e687c7c083e8c5ebfc1c9a61cb0d0dbc
- SFT-CHEM-THERMO-014-BOILING-0089: class=boiling; dataset=10; point=8;
target=sha256:7b51b3ad40fa93e41ffb028a82343ded1c65e2e2d9555ff7c96f60d4e5a6cd4e
- SFT-CHEM-THERMO-014-BOILING-0090: class=boiling; dataset=10; point=9;
target=sha256:2ef41244cea459bc27ebe466ea422dbc230301982eaa7884cb2e92d5c31a78b5
- SFT-CHEM-THERMO-014-BOILING-0091: class=boiling; dataset=11; point=1;
target=sha256:8f72c80b7c67b362b8f5c5cd49e4d0ca9e1165125bd2d828be742ee0b220601b
- SFT-CHEM-THERMO-014-BOILING-0092: class=boiling; dataset=11; point=2;
target=sha256:4dd9b3e19bb3f20215b8cf29caeced0ba372519b776ccaf5093726e02ef279df
- SFT-CHEM-THERMO-014-BOILING-0093: class=boiling; dataset=11; point=3;
target=sha256:7c92070d1753446d921500a0639c46d691b0a03f459e01189b8b640b3d7565fc
- SFT-CHEM-THERMO-014-BOILING-0094: class=boiling; dataset=11; point=4;
target=sha256:74873b9578a3aade04ca35bd78efbed700e2c8ca285bfab1a86380398a89b730
- SFT-CHEM-THERMO-014-BOILING-0095: class=boiling; dataset=11; point=5;
target=sha256:4b1951791919d2de8ac487030b17876065d05c7e1bf096e4cd2d75b50466bb4a
- SFT-CHEM-THERMO-014-BOILING-0096: class=boiling; dataset=11; point=6;
target=sha256:d0b1243f92cdaa640121f356d78d7549340d209f6b6d84f6ad57b3d478969445
- SFT-CHEM-THERMO-014-BOILING-0097: class=boiling; dataset=11; point=7;
target=sha256:2b2a79a35213cc1a90ff041c9ab6a61d84b99fb94e7e3f0c7288cb530d784be4
- SFT-CHEM-THERMO-014-BOILING-0098: class=boiling; dataset=11; point=8;
target=sha256:f7362b829d6407d40558256c7f86df697a7045524a360d21d50197483b419faf
- SFT-CHEM-THERMO-014-BOILING-0099: class=boiling; dataset=11; point=9;
target=sha256:a92d580a60f6fe50a0048faf5e933955ce00d28adc2f97a5ccdf35419b9b333d

- SFT-CHEM-THERMO-014-BOILING-0100: class=boiling; dataset=12; point=1;
target=sha256:b9e12b2718d4e660183fb482463d0be874c05ef7f52c856822de0ff14dee7401
- SFT-CHEM-THERMO-014-BOILING-0101: class=boiling; dataset=12; point=2;
target=sha256:225f1570dfb6b88e7b99ac83ce415beb43dba5d71e4f6cde5972516d09e7cf52
- SFT-CHEM-THERMO-014-BOILING-0102: class=boiling; dataset=12; point=3;
target=sha256:c5b903512edae0d6d79aa33c961162dc851ac89aa0488cd4837ceaaeebdc7fb4
- SFT-CHEM-THERMO-014-BOILING-0103: class=boiling; dataset=12; point=4;
target=sha256:56e0a8203447ddc1af935899783dabc7a34faf86e9cb03e004d950579b3afde3
- SFT-CHEM-THERMO-014-BOILING-0104: class=boiling; dataset=12; point=5;
target=sha256:c9eb49af6b97977dbddaf731215a66ba2c1bca7a727f5acf3d7d06328e4d8939
- SFT-CHEM-THERMO-014-BOILING-0105: class=boiling; dataset=12; point=6;
target=sha256:ac75c44284806bbc0a16e5b69483016b709f65825d0fcd8f957b3e238bf15708
- SFT-CHEM-THERMO-014-BOILING-0106: class=boiling; dataset=12; point=7;
target=sha256:78d1ffebababed5e48d33e4f6302178ccd576740da39e6b60fd7146c80d276ce
- SFT-CHEM-THERMO-014-BOILING-0107: class=boiling; dataset=12; point=8;
target=sha256:d46d36b85049a755d39c75fe2cf0827763b968de82bdcf05bb7e91a7de05fcc0
- SFT-CHEM-THERMO-014-BOILING-0108: class=boiling; dataset=12; point=9;
target=sha256:825736eef2ab67ad77c480fbe0d4f852ca6d19a1742c1ca1b2865392c76018b5
- SFT-CHEM-THERMO-014-BOILING-0109: class=boiling; dataset=13; point=1;
target=sha256:28659ac67f95c5c223883cb745d1dd396c71475837f8edd4030040c496a6d33c
- SFT-CHEM-THERMO-014-BOILING-0110: class=boiling; dataset=13; point=2;
target=sha256:a8dcb5d43a956cf66fa26e80c78aee80d11e254de9f1562de2aa5f90e303226d
- SFT-CHEM-THERMO-014-BOILING-0111: class=boiling; dataset=13; point=3;
target=sha256:44007baee22d2296cb3119e9565f3f2adc5cd44cdc8c9bb6accef46820676215
- SFT-CHEM-THERMO-014-BOILING-0112: class=boiling; dataset=13; point=4;
target=sha256:987e7c2f8b5dffb52dc71cc1b12677ebcfe374d467e8c18b0ceba9ab5a5c9ad0f
- SFT-CHEM-THERMO-014-BOILING-0113: class=boiling; dataset=13; point=5;
target=sha256:84e91921ab6bd59ac10b0cd1ffd776ef8ee62d1017b75eb41f1540d9c895b6f3
- SFT-CHEM-THERMO-014-BOILING-0114: class=boiling; dataset=13; point=6;
target=sha256:4c76b0f651d4c11bef17dda496fff0af5df03f637abf6e209296817b1c68874c
- SFT-CHEM-THERMO-014-BOILING-0115: class=boiling; dataset=13; point=7;
target=sha256:ded25298351b88c65ddefbaf8069e33b80687ad41a1eb1e5544f9a90e35c48
- SFT-CHEM-THERMO-014-BOILING-0116: class=boiling; dataset=13; point=8;
target=sha256:e3a33ba0425cfff02a4bafc7cba87174f314597d0f54ba8f7595742210e569ad
- SFT-CHEM-THERMO-014-BOILING-0117: class=boiling; dataset=13; point=9;
target=sha256:6bed0e91cda5fbc69d5a5f0ed32698f818fa6a19b8ff6d047f66edb278a47f34
- SFT-CHEM-THERMO-014-BOILING-0118: class=boiling; dataset=14; point=1;
target=sha256:e3db9d50d461767beebf080022eb930ea1165b52aec145e7f74376e997600df4
- SFT-CHEM-THERMO-014-BOILING-0119: class=boiling; dataset=14; point=2;
target=sha256:bc24c99c1ac1f970e3ae1ff706c2a74a68a3137ffc2944c48dd5f595cdf8b4f
- SFT-CHEM-THERMO-014-BOILING-0120: class=boiling; dataset=14; point=3;
target=sha256:7bbd2fcd9df404bfb612563c14b3fd38a73e26a21c090c3c99b25f02df9270dc
- SFT-CHEM-THERMO-014-BOILING-0121: class=boiling; dataset=14; point=4;
target=sha256:7d47ffef8d3ba4fbc0b3bffdabdf3374123646b556e8bc93cd767d8347efc1d7

- SFT-CHEM-THERMO-014-BOILING-0122: class=boiling; dataset=14; point=5;
target=sha256:6094e471b501383e415f92377ec0a0f7425e1b7c1dd7c018bb33f9beb980f3dc
- SFT-CHEM-THERMO-014-BOILING-0123: class=boiling; dataset=14; point=6;
target=sha256:99843ff9df54a0cf8508b855f061f0744ed3b43f4f900717dc25c93f04285345
- SFT-CHEM-THERMO-014-BOILING-0124: class=boiling; dataset=14; point=7;
target=sha256:120d423ad825e0edc3138bdccb0c6fb4e3d20e95aad5668a1b73c4da6028672
- SFT-CHEM-THERMO-014-BOILING-0125: class=boiling; dataset=14; point=8;
target=sha256:89109c935093532bb0e66beaac04e717a8bf3911bccf0a5eaef4c6a66468af15
- SFT-CHEM-THERMO-014-BOILING-0126: class=boiling; dataset=14; point=9;
target=sha256:828c0ab1ddc1cec0c6ca9675771c8e40ccb94333d896819fb62571770a343f4
- SFT-CHEM-THERMO-014-BOILING-0127: class=boiling; dataset=15; point=1;
target=sha256:d92301eb40439ea47771b3a9d284bf7414170cec979d32fe40ed3e9b30bca48f
- SFT-CHEM-THERMO-014-BOILING-0128: class=boiling; dataset=15; point=2;
target=sha256:20f639d25ff5e28f73fa2c776e21ccf237efde332c876836bf5310f56d8bfc1f
- SFT-CHEM-THERMO-014-BOILING-0129: class=boiling; dataset=15; point=3;
target=sha256:729c3e1991550274e01b4090d9b09da7374b58249cb88454376fef3d66d561a1
- SFT-CHEM-THERMO-014-BOILING-0130: class=boiling; dataset=15; point=4;
target=sha256:08bcbb14864aed62f519c4bbb466911da9703df5d211e6daf932dc97248d3184
- SFT-CHEM-THERMO-014-BOILING-0131: class=boiling; dataset=15; point=5;
target=sha256:4c7ecba8d88e269394222b00a37f410bcbde764630827f4848971280055154a5
- SFT-CHEM-THERMO-014-BOILING-0132: class=boiling; dataset=15; point=6;
target=sha256:2cdfb2966f1e890f81a14d594a8046fe8b42d993ca2c627ad98e4e5b4eb5cfbe
- SFT-CHEM-THERMO-014-BOILING-0133: class=boiling; dataset=15; point=7;
target=sha256:2584bd8056d469c521b3a7228e7b9245c1ab547551ee66f7651bea57de2ee1e2
- SFT-CHEM-THERMO-014-BOILING-0134: class=boiling; dataset=15; point=8;
target=sha256:917cb05169b61f81710d9c2cc1db3a6e14fb1cd8b2dc50d0ccced0383976fdb
- SFT-CHEM-THERMO-014-BOILING-0135: class=boiling; dataset=15; point=9;
target=sha256:c7c8acd28f93446fc77ee5923dcac4ae9d136927962c05dca8a27ace23ae7e
- SFT-CHEM-THERMO-014-BOILING-0136: class=boiling; dataset=16; point=1;
target=sha256:ef75f8db61078cc8a30ae95214ad5567c54f4dd52c974713fe2e6cbaabfb8821
- SFT-CHEM-THERMO-014-BOILING-0137: class=boiling; dataset=16; point=2;
target=sha256:c33d45fbc2e0c7b50e7d78c5689e5d14de2103d0aa799ce1bb10db94c5904667
- SFT-CHEM-THERMO-014-BOILING-0138: class=boiling; dataset=16; point=3;
target=sha256:40fb760c36786bc68d670c05c85b37564d29c1eb240c035e5a3167d8e77968d5
- SFT-CHEM-THERMO-014-BOILING-0139: class=boiling; dataset=16; point=4;
target=sha256:dd01ea7232154e63899d856bbba0a509cfee7052aaf25982c98555ecdc3c8c89
- SFT-CHEM-THERMO-014-BOILING-0140: class=boiling; dataset=16; point=5;
target=sha256:0186d2a848019981670dc824d794e3823be9b66eec243022593741f2acf3c0df
- SFT-CHEM-THERMO-014-BOILING-0141: class=boiling; dataset=16; point=6;
target=sha256:cc26ea389798300970dc3963c1d14831e72b0c724751a138e96dccacc9741b8d
- SFT-CHEM-THERMO-014-BOILING-0142: class=boiling; dataset=16; point=7;
target=sha256:a0cef668b5907b6a1b4404ddc0190f78609de53649aafd5376a3d6deeb5b6808
- SFT-CHEM-THERMO-014-BOILING-0143: class=boiling; dataset=16; point=8;
target=sha256:d7ab8cd98acc3ba16f91c02b0daafd864e29a856933b8fe8d8da8b559eed7f10

- SFT-CHEM-THERMO-014-BOILING-0144: class=boiling; dataset=16; point=9;
target=sha256:31dba2f9fa47ab4f4fef69c7e3ace9fd8932e6675077b471ebdb04fa2efd9a67
- SFT-CHEM-THERMO-014-FREEZING-0145: class=freezing; dataset=1; point=1;
target=sha256:4530548d566ad324d26d53a073ddc37a1ba610f7051b77cbbf4d3644e3c2b849
- SFT-CHEM-THERMO-014-FREEZING-0146: class=freezing; dataset=1; point=2;
target=sha256:25e53ea60ba9f1dea7e1d453a02952a4ab1a40345b7713edf151f743dc0fda0d
- SFT-CHEM-THERMO-014-FREEZING-0147: class=freezing; dataset=1; point=3;
target=sha256:2ecd7d3dc17ff88b19c2c716cf1ceb0b8e6116e686588dfc6b78581cf646f20a
- SFT-CHEM-THERMO-014-FREEZING-0148: class=freezing; dataset=1; point=4;
target=sha256:46f298592a4f2eb415420507c9255675c95d2013a233eace09537604a8daf310
- SFT-CHEM-THERMO-014-FREEZING-0149: class=freezing; dataset=1; point=5;
target=sha256:54a267e4be569126fb240240926932db8d979da14d48737d36867278033a55f8
- SFT-CHEM-THERMO-014-FREEZING-0150: class=freezing; dataset=1; point=6;
target=sha256:44cd08b65e9817bea052f93f36545c01040753ea3dd3a0b6858ec8c25a239b63
- SFT-CHEM-THERMO-014-FREEZING-0151: class=freezing; dataset=2; point=1;
target=sha256:3f2fed03a453962f1d58a650b4ee59298b231ff54b379f092b148bf23fdbe2a
- SFT-CHEM-THERMO-014-FREEZING-0152: class=freezing; dataset=2; point=2;
target=sha256:d333093ebd329adac384863f64f4a62446baeb766b86cf878caa9908726d5dd5
- SFT-CHEM-THERMO-014-FREEZING-0153: class=freezing; dataset=2; point=3;
target=sha256:a403e67270b95c2de6788f7c6b84fe75ed1201b41fe34ffd17f5beef88035256
- SFT-CHEM-THERMO-014-FREEZING-0154: class=freezing; dataset=3; point=1;
target=sha256:5bbe0553f8a7263b1ff60da7be23096d0e47f22c4d6caea458e4d3900f5732d5
- SFT-CHEM-THERMO-014-FREEZING-0155: class=freezing; dataset=3; point=2;
target=sha256:1649b29a4441cd0363e6a3294200fcb19e0c663ee6eb3c2099984add2c52fea7
- SFT-CHEM-THERMO-014-FREEZING-0156: class=freezing; dataset=3; point=3;
target=sha256:f6cd077614dd1005ec990258628c80a7c60853ff693e0dab547de9f780295609
- SFT-CHEM-THERMO-014-FREEZING-0157: class=freezing; dataset=3; point=4;
target=sha256:11ab3809911371079fcd8056e08aad80017c6ab2037cfba760673a0cf399e641
- SFT-CHEM-THERMO-014-FREEZING-0158: class=freezing; dataset=4; point=1;
target=sha256:e72523ee3456450a8a240b4d016b7d781365048edc775c9dd2ea56eba8ae7da6
- SFT-CHEM-THERMO-014-FREEZING-0159: class=freezing; dataset=4; point=2;
target=sha256:3c9d931b313c358823229bf65d63b0f3b2aa7e94a8e8b5281849a4561981d7a
- SFT-CHEM-THERMO-014-FREEZING-0160: class=freezing; dataset=4; point=3;
target=sha256:4f9893045b07ea288561457e29e407440eca27eb3ecffcc014dca1bbfb20450d
- SFT-CHEM-THERMO-014-FREEZING-0161: class=freezing; dataset=4; point=4;
target=sha256:34b459d4ba5d8783a23c9bb34fe8aa99d86a8c2f4725b6a00885f9445c9f0ae5
- SFT-CHEM-THERMO-014-FREEZING-0162: class=freezing; dataset=4; point=5;
target=sha256:0fa2f28a661748190b65443871dbcbcd5fb3c59f469700223310eb51281ee0ae
- SFT-CHEM-THERMO-014-FREEZING-0163: class=freezing; dataset=4; point=6;
target=sha256:426ecbc0a85ec6f9bf4a48e67d796594b31809ff9e13aa494cb7589045703824
- SFT-CHEM-THERMO-014-FREEZING-0164: class=freezing; dataset=4; point=7;
target=sha256:240826d1274eb629b1f27f1b500695acf9504b07345e005e2212a248b229b5e3
- SFT-CHEM-THERMO-014-FREEZING-0165: class=freezing; dataset=4; point=8;
target=sha256:b959ef02415918eeb39e60a69430bdb98ee1ce9b2f92b7f830d45378be4ac0c8

- SFT-CHEM-THERMO-014-FREEZING-0166: class=freezing; dataset=4; point=9;
target=sha256:c9e4432d03e34fd90a7b76dc9c4d6022b4d6285eeefbc36c3dc951fc6d1f7bf2
- SFT-CHEM-THERMO-014-FREEZING-0167: class=freezing; dataset=5; point=1;
target=sha256:f72c296f37ba3578a47a6e42d6c730e3ace27bb4022d68a3d83d47d6b13c6a1c
- SFT-CHEM-THERMO-014-FREEZING-0168: class=freezing; dataset=5; point=2;
target=sha256:46dca09114141b3ecee033f027d00c9b3196077ba78fbb4645adf81225514282
- SFT-CHEM-THERMO-014-FREEZING-0169: class=freezing; dataset=5; point=3;
target=sha256:28a4297f46756bdb82aa61de8ae61a3a3d5b7cdc78f3fc00fda64d1b3fa24a93
- SFT-CHEM-THERMO-014-FREEZING-0170: class=freezing; dataset=5; point=4;
target=sha256:2c5c661b0d3dad8353fc97bdadaf3396de805b484c1a1f1e847ae3573eea3449
- SFT-CHEM-THERMO-014-FREEZING-0171: class=freezing; dataset=5; point=5;
target=sha256:8aa047bc1370766e27d2781dd20113bb0744cc6f1d7ff8dcab09603de0c41a3a
- SFT-CHEM-THERMO-014-FREEZING-0172: class=freezing; dataset=5; point=6;
target=sha256:1c57a07cc8f8466d131b12d3b005a261cac4e55fbac687d68a0e27be257884e5
- SFT-CHEM-THERMO-014-FREEZING-0173: class=freezing; dataset=5; point=7;
target=sha256:29f52282eb98972bd6ab4a8414678efc881d42b8d21aa300a067feecbdb89dd
- SFT-CHEM-THERMO-014-FREEZING-0174: class=freezing; dataset=5; point=8;
target=sha256:92db399107bdbb1d8447576a7edc7a5d268ee09234e9085c644847517d5d2c28
- SFT-CHEM-THERMO-014-FREEZING-0175: class=freezing; dataset=5; point=9;
target=sha256:1e65f48e0eb85808c81064a891f7c6638be0e6164af6e5684e2f06cb902059c4
- SFT-CHEM-THERMO-014-FREEZING-0176: class=freezing; dataset=6; point=1;
target=sha256:a2fd4eb7a276db127113d7678ff9c332e8fa5b0d9593544fdc7f9e0d4b606a59
- SFT-CHEM-THERMO-014-FREEZING-0177: class=freezing; dataset=6; point=2;
target=sha256:a7639c9c1d4baecaf1c2079d97bb0b2c755d43769ef3c61fa4f374e903c36430
- SFT-CHEM-THERMO-014-FREEZING-0178: class=freezing; dataset=6; point=3;
target=sha256:07ac6bd12147ecf21a6386388ea1ebcae29f186e6fe9cc0aca9b43dbe8ac789a
- SFT-CHEM-THERMO-014-FREEZING-0179: class=freezing; dataset=6; point=4;
target=sha256:8bae98d1adf9e9e7df1681bcddeab947046ad7bf0140812a8cf364cdd3f84322
- SFT-CHEM-THERMO-014-FREEZING-0180: class=freezing; dataset=6; point=5;
target=sha256:63bd9f994bc812b36b2953029728dfb420b00eebebd7d4cefc4d4939a9fe924c
- SFT-CHEM-THERMO-014-FREEZING-0181: class=freezing; dataset=6; point=6;
target=sha256:0e5069a370c4fb214157076d79c08007f789c084c13d51531e6baad3a65e8b1
- SFT-CHEM-THERMO-014-OSMOTIC-0182: class=osmotic; dataset=1; point=1;
target=sha256:2af53b20ea3ef363df724807c2ccdc43e7370d94b302eaf6e25e41569e2a9021
- SFT-CHEM-THERMO-014-OSMOTIC-0183: class=osmotic; dataset=1; point=2;
target=sha256:a65558b5236de0507567db607ff4e1548f081e0ca435132238b08ccc013a3a48
- SFT-CHEM-THERMO-014-OSMOTIC-0184: class=osmotic; dataset=1; point=3;
target=sha256:7d9c0457b47cc15d600a6a97a37e2a949bbf5858bcac02d8d403c73a6104ef0a
- SFT-CHEM-THERMO-014-OSMOTIC-0185: class=osmotic; dataset=1; point=4;
target=sha256:6f12c3b31cdd50a10e8c9b377aed7d612673fc200dd994af4e4749fc26780e34
- SFT-CHEM-THERMO-014-OSMOTIC-0186: class=osmotic; dataset=1; point=5;
target=sha256:e31e0684530c5be6bee76392c935d0c256adcf727ef25e2b7fe3bd956c347b1b
- SFT-CHEM-THERMO-014-OSMOTIC-0187: class=osmotic; dataset=1; point=6;
target=sha256:04ada80aad0a05e42dcb00248eab3e9eef32c495b5eb5d26e8dad6bd4c56a17b

- SFT-CHEM-THERMO-014-OSMOTIC-0188: class=osmotic; dataset=1; point=7;
target=sha256:7779979421660a75b2369ad3314c297db91bf17b8340e642a5b1551816051a55
- SFT-CHEM-THERMO-014-OSMOTIC-0189: class=osmotic; dataset=1; point=8;
target=sha256:256cc1a50d37c2966a136d8f03fd529be405e1260314af976e7b1e782158b39b
- SFT-CHEM-THERMO-014-OSMOTIC-0190: class=osmotic; dataset=1; point=9;
target=sha256:d4ab2e02bbff37577522d1cd208416651f5dda6443ed65e2395e0796881f4619
- SFT-CHEM-THERMO-014-OSMOTIC-0191: class=osmotic; dataset=1; point=10;
target=sha256:fdd773a83580b2eaadd80e0c3dfe98134021d75540c9f0c89de67bf32634fd8e
- SFT-CHEM-THERMO-014-OSMOTIC-0192: class=osmotic; dataset=1; point=11;
target=sha256:e5857d478b765c08eddaede7b4db77223efcc676b7dfa786f8ab2008b8708780
- SFT-CHEM-THERMO-014-OSMOTIC-0193: class=osmotic; dataset=1; point=12;
target=sha256:2747fb2d57d74d45ba838e5512211a4d9f16d90cc616eb115eb6d01761c3c8ce
- SFT-CHEM-THERMO-014-OSMOTIC-0194: class=osmotic; dataset=1; point=13;
target=sha256:403985187afc2eade514f7adf5eae2b4ae327d4d5d09ffc2cbce7609ca129e66
- SFT-CHEM-THERMO-014-OSMOTIC-0195: class=osmotic; dataset=1; point=14;
target=sha256:8d97eb0a7495c71e9d792315722dfc0b59012ba91164e4bf9e43bf07b5f631b1
- SFT-CHEM-THERMO-014-OSMOTIC-0196: class=osmotic; dataset=1; point=15;
target=sha256:7a8dc7c15c1baf093fe1e94132b1e16fc81807255ab073e3363d8674d1e04f07
- SFT-CHEM-THERMO-014-OSMOTIC-0197: class=osmotic; dataset=1; point=16;
target=sha256:14ed41ea08701f97f90d16fa9ceb9dbbe17eb8004202f2952d367f5fa5b8ab90
- SFT-CHEM-THERMO-014-OSMOTIC-0198: class=osmotic; dataset=1; point=17;
target=sha256:4a17c66efd180fda2a51b7f2608e5654d4e1b5f4369397d9d796f580766541e4
- SFT-CHEM-THERMO-014-OSMOTIC-0199: class=osmotic; dataset=2; point=1;
target=sha256:3deb87a628f9d0faf821efc7d1aa38d01b6c6f1bb84b287c5343398c137a561d
- SFT-CHEM-THERMO-014-OSMOTIC-0200: class=osmotic; dataset=2; point=2;
target=sha256:7a5e57530c795efa37f9b5b02522a915f874200fb8f6fc0f3b8547d04101d75b
- SFT-CHEM-THERMO-014-OSMOTIC-0201: class=osmotic; dataset=2; point=3;
target=sha256:e01b933a1b2138e5f00f65b4c3d84348a8bde845ca1650426de4a3c57c366b6b
- SFT-CHEM-THERMO-014-OSMOTIC-0202: class=osmotic; dataset=2; point=4;
target=sha256:f242240848a7da647f38a73a7876fe52112a6f38e9ddc66bfb13a850ce1e2dca
- SFT-CHEM-THERMO-014-OSMOTIC-0203: class=osmotic; dataset=2; point=5;
target=sha256:a37b499a34f6e15677f7f4240faec930af9a892dd96fafb56aede8ee57164e4b
- SFT-CHEM-THERMO-014-OSMOTIC-0204: class=osmotic; dataset=2; point=6;
target=sha256:9558b589775498a9f5650c81723b1899cb7cc74ed66942acca19cfc8b61baf64
- SFT-CHEM-THERMO-014-OSMOTIC-0205: class=osmotic; dataset=2; point=7;
target=sha256:e53355dde4c57b90ec622ee9ab9f7080697a7ad1366a339525182ea28a840422
- SFT-CHEM-THERMO-014-OSMOTIC-0206: class=osmotic; dataset=2; point=8;
target=sha256:8e33ae3f6a11af0455a5fd5a91bb3f017bf33c9c46517860ce281a895a0d90fb
- SFT-CHEM-THERMO-014-OSMOTIC-0207: class=osmotic; dataset=2; point=9;
target=sha256:c36bd3f820d8373f4696c029bcfa074fe98fc57a3935064fd5c2b86befbd0d71
- SFT-CHEM-THERMO-014-OSMOTIC-0208: class=osmotic; dataset=2; point=10;
target=sha256:2f52baf60c1557d226e404736a7e07ce64aafeb4ccd59f9b95a527595c9e2ce8
- SFT-CHEM-THERMO-014-OSMOTIC-0209: class=osmotic; dataset=2; point=11;
target=sha256:4e7b6a0571486f04f5168566ef17d1bb08b3e1ffa333c44304026d01c742b73f

- SFT-CHEM-THERMO-014-OSMOTIC-0210: class=osmotic; dataset=2; point=12;
target=sha256:4591fd76bc85b93c3dcf8fe961815a54d08975ad000534c1da15d21d1567ba88
- SFT-CHEM-THERMO-014-OSMOTIC-0211: class=osmotic; dataset=2; point=13;
target=sha256:9b86552dab07ac120ad150cecf6861a3fa99163fdc27ce22de8a0c9d8999fca8
- SFT-CHEM-THERMO-014-OSMOTIC-0212: class=osmotic; dataset=2; point=14;
target=sha256:3892e81db343f20fe9e282572e2089bda76855bd0b8c090ef4276be33d9b6acb
- SFT-CHEM-THERMO-014-OSMOTIC-0213: class=osmotic; dataset=3; point=1;
target=sha256:adb63cc7b5787c99271416ccb01fa0d6ef175f99d42481207ec8161dc1680166
- SFT-CHEM-THERMO-014-OSMOTIC-0214: class=osmotic; dataset=3; point=2;
target=sha256:a0727d9a60161b8d68534d003cad3b777ac7438bfbe5c21a613e1eb9601963fa
- SFT-CHEM-THERMO-014-OSMOTIC-0215: class=osmotic; dataset=3; point=3;
target=sha256:5bc1b094b4f62afdb12162b535303bfc07ae51daad4d7ff45e5a3d6fd453cade
- SFT-CHEM-THERMO-014-OSMOTIC-0216: class=osmotic; dataset=3; point=4;
target=sha256:fa6be34f43304df3c9eab5c615db0cbd1b3fd1d46c75ec7685a57902d823b9f4
- SFT-CHEM-THERMO-014-OSMOTIC-0217: class=osmotic; dataset=3; point=5;
target=sha256:7d8059ce74d21233ef1c2a3870219dade0ad89b5f5b27310e7c0b301c9946107
- SFT-CHEM-THERMO-014-OSMOTIC-0218: class=osmotic; dataset=3; point=6;
target=sha256:f10189051b1e73a35a3c99184045856bd7c2926773d91e37e5c9d0dedd4d93c0
- SFT-CHEM-THERMO-014-OSMOTIC-0219: class=osmotic; dataset=3; point=7;
target=sha256:7918237caa47402286c0930a92bd67f2f884a1d7df0386d10032ca364210125a
- SFT-CHEM-THERMO-014-OSMOTIC-0220: class=osmotic; dataset=3; point=8;
target=sha256:3f8580e926a262aa7d5f7bedce1637cf9656db989bc02aa63987b7a68714c9df
- SFT-CHEM-THERMO-014-OSMOTIC-0221: class=osmotic; dataset=3; point=9;
target=sha256:86a8c192ad9183566ef8fa9fcbca38571d624a778708e45248a6322a895e3e17
- SFT-CHEM-THERMO-014-OSMOTIC-0222: class=osmotic; dataset=3; point=10;
target=sha256:85a0e893af5d8d8ac5dd0830b957196f57cd7f0189c4ab8981a945a177c3e9f9
- SFT-CHEM-THERMO-014-OSMOTIC-0223: class=osmotic; dataset=3; point=11;
target=sha256:378c94a09d8561511b0b6336752deef830276bb9dea7cc5ebdae659d838dffab
- SFT-CHEM-THERMO-014-OSMOTIC-0224: class=osmotic; dataset=3; point=12;
target=sha256:3b8b18c7d3c30c8a9ff3b1c637de95f35230e400a971f256d341a7744b4a9c47
- SFT-CHEM-THERMO-014-OSMOTIC-0225: class=osmotic; dataset=3; point=13;
target=sha256:1797f791b156de118abad46c93606fb3a955953e1933c64327967ba5299c020b
- SFT-CHEM-THERMO-014-OSMOTIC-0226: class=osmotic; dataset=3; point=14;
target=sha256:1db2d1005089d9a4030df4a6ec07c026651f7b29fb3633e837d3a2814438b9c9
- SFT-CHEM-THERMO-014-OSMOTIC-0227: class=osmotic; dataset=3; point=15;
target=sha256:31d784dbf08ebfc3a33487f29aa91b99024a718608a50f3584b13d94f7ec3b6a
- SFT-CHEM-THERMO-014-OSMOTIC-0228: class=osmotic; dataset=3; point=16;
target=sha256:f30bf107c0ae5ec16f12bbc02fc7c0f863b9508d54a7055c7a9f11961d8e485f
- SFT-CHEM-THERMO-014-OSMOTIC-0229: class=osmotic; dataset=4; point=1;
target=sha256:0b1e041262393c5b25a2a1ee4a4414ee3a304fa0126ab0b81421365a5c430a73
- SFT-CHEM-THERMO-014-OSMOTIC-0230: class=osmotic; dataset=4; point=2;
target=sha256:30b225ffb5ca29acc8f329e50cbb94f242f940cdad18b6abced67c859fe30d74
- SFT-CHEM-THERMO-014-OSMOTIC-0231: class=osmotic; dataset=4; point=3;
target=sha256:e793220b6b9618d3ce13c6d09f5503f0e1d4dc7f69312a0adc9c11214d6a7f0c

- SFT-CHEM-THERMO-014-OSMOTIC-0232: class=osmotic; dataset=4; point=4;
target=sha256:371cb08cc7b81c5041e8a4ff5a0f47c565a7e1f0f834dabffb7c25b66aecf427
- SFT-CHEM-THERMO-014-OSMOTIC-0233: class=osmotic; dataset=4; point=5;
target=sha256:3d4b64cc0dbc2b8d27993f57c5aba59c465c998f8f336680106d0f0bc1d54a05
- SFT-CHEM-THERMO-014-OSMOTIC-0234: class=osmotic; dataset=4; point=6;
target=sha256:bf39187f084c16099486158b3dde534f4a6f8ef1bcf4e9dc4ac186a9c720e3eb
- SFT-CHEM-THERMO-014-OSMOTIC-0235: class=osmotic; dataset=4; point=7;
target=sha256:5a714a6a497387b18cc98126e1a42c04f8c59892fe91b433f869b8cca3b8472a
- SFT-CHEM-THERMO-014-OSMOTIC-0236: class=osmotic; dataset=4; point=8;
target=sha256:067acd9ec30cecec35626c52e8ea89999ed58d14dd8e4f447d6b44481ed5bfda
- SFT-CHEM-THERMO-014-OSMOTIC-0237: class=osmotic; dataset=4; point=9;
target=sha256:1c8addf2a8b4d1e3620a58da5951349f0937ee9534aec8ba36d85836c85fe760
- SFT-CHEM-THERMO-014-OSMOTIC-0238: class=osmotic; dataset=4; point=10;
target=sha256:c8039198772c33d97690b7064a830d8dc69361c89c3b739e05d09e10cd22b5ce
- SFT-CHEM-THERMO-014-OSMOTIC-0239: class=osmotic; dataset=4; point=11;
target=sha256:bbdcda6f608fbc4ce6a8569a27c3dc01a219af47c777ccb8722ada7ad53cbbbe
- SFT-CHEM-THERMO-014-OSMOTIC-0240: class=osmotic; dataset=4; point=12;
target=sha256:3e7290d6f098bd5bb4d0df12f7b3db6575e6b75fdac3bb226269f3eaac6f321d
- SFT-CHEM-THERMO-014-OSMOTIC-0241: class=osmotic; dataset=4; point=13;
target=sha256:72f43b4db2dab0ca73bf43a80996e3691769205ea6e1ff5ef5d4363115356cbb
- SFT-CHEM-THERMO-014-OSMOTIC-0242: class=osmotic; dataset=4; point=14;
target=sha256:022c2a9ac28b89525c0de2830e9ced48ebf86ee794bd3e1e8cafcdacac115c68
- SFT-CHEM-THERMO-014-OSMOTIC-0243: class=osmotic; dataset=4; point=15;
target=sha256:2d008e7978c6b98c7611a43b8c798068297d1593d612fab33abbbe45a2c02aba
- SFT-CHEM-THERMO-014-OSMOTIC-0244: class=osmotic; dataset=4; point=16;
target=sha256:4b3c9baeb324517cc82961622ee9cca82c6736257fa646e4cf18dfd1a87f2002
- SFT-CHEM-THERMO-014-OSMOTIC-0245: class=osmotic; dataset=4; point=17;
target=sha256:b8541542520a13bf6599581b369878e1c23e781bd4fe63636bd190cd0b02bb78
- SFT-CHEM-THERMO-014-OSMOTIC-0246: class=osmotic; dataset=5; point=1;
target=sha256:23b413353eb4334290bfb65a071d3e1a6344467a1528dcf3ec80f75996f269f2
- SFT-CHEM-THERMO-014-OSMOTIC-0247: class=osmotic; dataset=5; point=2;
target=sha256:a1b54af1c62993840790db8769aac1681634f615a57a616d71b62b8b2b4da71b
- SFT-CHEM-THERMO-014-OSMOTIC-0248: class=osmotic; dataset=5; point=3;
target=sha256:1d30ad0af258a57dcdb4f4a11f5fce8417f4860b40ad6fdecf94e7770c84adcd
- SFT-CHEM-THERMO-014-OSMOTIC-0249: class=osmotic; dataset=5; point=4;
target=sha256:5cb145abc3fa6b58098bcfca165a181b0672f8a8f0d929733a68fc83379205a8
- SFT-CHEM-THERMO-014-OSMOTIC-0250: class=osmotic; dataset=5; point=5;
target=sha256:24b50dfa847a62d8e9e8afaa6e76554f6d091f55ab12e415437f57f52b6c1da4
- SFT-CHEM-THERMO-014-OSMOTIC-0251: class=osmotic; dataset=5; point=6;
target=sha256:94601fe259aa1d530725fe17e4e789acdfe2427a8da4de2608f7827475d19f74
- SFT-CHEM-THERMO-014-OSMOTIC-0252: class=osmotic; dataset=5; point=7;
target=sha256:5dcd38d0ea7876c3408d582754840e0c652bcc325deb8ca374e3364c173df0ff
- SFT-CHEM-THERMO-014-OSMOTIC-0253: class=osmotic; dataset=5; point=8;
target=sha256:8407df2b75812e730b41e304b896edc03494bb11e45f8a608c76f2b792c3a537

- SFT-CHEM-THERMO-014-OSMOTIC-0254: class=osmotic; dataset=5; point=9;
target=sha256:c8aec4daa0b96593c5181564608aed4662fb352047761116d72e9ffba2d19c95
- SFT-CHEM-THERMO-014-OSMOTIC-0255: class=osmotic; dataset=5; point=10;
target=sha256:e83e1fa3b75b1d3b86d642a0784b53aadd67d64633c38b02121c495c29ccfdd0
- SFT-CHEM-THERMO-014-OSMOTIC-0256: class=osmotic; dataset=5; point=11;
target=sha256:c2b1446107c0bdadb84dfa84aa677c5a6fcbb01e621213ee4c20d128e6fe8e31
- SFT-CHEM-THERMO-014-OSMOTIC-0257: class=osmotic; dataset=5; point=12;
target=sha256:7591b24323c93be5b3456d19411c60f174e189167a645b0cbb0dbe032779ee85
- SFT-CHEM-THERMO-014-OSMOTIC-0258: class=osmotic; dataset=5; point=13;
target=sha256:8752a3e2743124b9052cd1518d9f28ad77ddbba8e514db323338dcfe490d4463
- SFT-CHEM-THERMO-014-OSMOTIC-0259: class=osmotic; dataset=5; point=14;
target=sha256:7bdc04c264882736847c56917cce16e6fc16964018ddef7c1ccaac724cbcf
- SFT-CHEM-THERMO-014-OSMOTIC-0260: class=osmotic; dataset=5; point=15;
target=sha256:29e08a5a5b938bda1c89f51b2b6dc3a85a864fcea0a1afc9418d4c71e44d30cf
- SFT-CHEM-THERMO-014-OSMOTIC-0261: class=osmotic; dataset=5; point=16;
target=sha256:8a60dba075f61f1368d556e921fb8a6dda0dd6b585577e61c64d7ed2c3847919
- SFT-CHEM-THERMO-014-OSMOTIC-0262: class=osmotic; dataset=6; point=1;
target=sha256:8b5212d8befc0d56b98012630c6a327b32ac5b20d1f928d9e62a5764457e67d4
- SFT-CHEM-THERMO-014-OSMOTIC-0263: class=osmotic; dataset=6; point=2;
target=sha256:021a3e17f9b23bbdcc3f5b8b9d9a04fdf9a8003117c616e2aa91276ce652b06d
- SFT-CHEM-THERMO-014-OSMOTIC-0264: class=osmotic; dataset=6; point=3;
target=sha256:003121d567fa13129338b85be203b86107525392f55ee4dd61a8116b4ae4c04e
- SFT-CHEM-THERMO-014-OSMOTIC-0265: class=osmotic; dataset=6; point=4;
target=sha256:ca15551612ed0f6b672350482519ada8f84469dbc8a54f14c6d4dc24d763cd1f
- SFT-CHEM-THERMO-014-OSMOTIC-0266: class=osmotic; dataset=6; point=5;
target=sha256:53b451a0349e2bc6bf5602973035f375127e8d701bef9314a994e3c47b79423f
- SFT-CHEM-THERMO-014-OSMOTIC-0267: class=osmotic; dataset=6; point=6;
target=sha256:7728133367362ca1040cef2283d0beaeb8a9f0d03ca95e8b773ef347bdd1a27c
- SFT-CHEM-THERMO-014-OSMOTIC-0268: class=osmotic; dataset=6; point=7;
target=sha256:bf1932ff9a8f68a938c5052803594d2575f35092d31e85c0625e7cad1e68743c
- SFT-CHEM-THERMO-014-OSMOTIC-0269: class=osmotic; dataset=6; point=8;
target=sha256:6d89385216f70c75030518e4e86fdf5a5c98ae45ad011c86ee3adfbfbf2769794
- SFT-CHEM-THERMO-014-OSMOTIC-0270: class=osmotic; dataset=6; point=9;
target=sha256:f4918c5be4a857b5ea99c30d0e57c3fc8bb8915e91ce655a4d0282d8f7525798
- SFT-CHEM-THERMO-014-OSMOTIC-0271: class=osmotic; dataset=6; point=10;
target=sha256:db4764666d1ff960814113bd3ab3689e2b03c0303085c16d79e8ed1a019428f6
- SFT-CHEM-THERMO-014-OSMOTIC-0272: class=osmotic; dataset=6; point=11;
target=sha256:d5eb6020029e787ce5177f396a26fc2b3ff00baec9e204bdb6100472ea199d11
- SFT-CHEM-THERMO-014-OSMOTIC-0273: class=osmotic; dataset=6; point=12;
target=sha256:c965b2ddb7b7bc0f55b7875529eb920615944e277bb17167a7634d0ba7fb7fb7
- SFT-CHEM-THERMO-014-OSMOTIC-0274: class=osmotic; dataset=6; point=13;
target=sha256:b35b0f3812f4a05c4747266ff9beaab3f58c2b85e33741d3acb21b36c2b90ac8
- SFT-CHEM-THERMO-014-OSMOTIC-0275: class=osmotic; dataset=6; point=14;
target=sha256:7c8fb72d22507a8614d5456e05a2c5c8e228a50e9d1ec49804a9687ce1b05f54

- SFT-CHEM-THERMO-014-OSMOTIC-0276: class=osmotic; dataset=6; point=15; target=sha256:153dff493146a4137e9537e608622b14742592e65e93b2a4611d0dd06befb6c7
- complete vector: 144 boiling, 37 freezing and 95 osmotic response records
- exact ranges: {'boiling': {'minimum_composition': '17/40000', 'maximum_composition': '353/12500', 'minimum_response': '2881/10', 'maximum_response': '1233/4'}, 'freezing': {'minimum_composition': '0', 'maximum_composition': '1', 'minimum_response': '25501/100', 'maximum_response': '9139/25'}, 'osmotic': {'minimum_composition': '151/5000', 'maximum_composition': '20577/5000', 'minimum_response': '359/1000', 'maximum_response': '957/1000'}}
- source coverage: all 28 datasets and 276 points across three complete NIST ThermoML sources
- tampered_response_rejected: True
- complete_276_record_vector_retained: True
- all_three_response_classes_retained: True
- all_28_datasets_retained: True
- absence_translated_only_to_EmptyOne: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if solvent or solute identity is erased or duplicated; if the boundary does not retain solute while transmitting solvent exchange; if boiling, freezing or osmotic orientation requires signed magnitude; if pure solvent is represented by numerical zero; if any conventional colligative equation, constant, factor, continuum, interpolation, regression, fit, selection or target correction enters; if targets open before all 276 identities seal; if any of 144 boiling, 37 freezing or 95 osmotic records, any uncertainty, method, compound, dataset or complete source is omitted; or if any measured coordinate or response is tampered.

Measurement receipt: sha256:cf6ad68ff1b1a1d0e123cc709a065cd14acfe78d1fae446f5a743467dbd20c64.

Isolation certificate: sha256:17305396cde704b4467c952e766eada120d137404e815f6aa40d8802be29c9bb.

Custody certificate: sha256:23ea54acadb0541a2cdc9e85a516c5163505a89146a537a3341ad503a045e59d.

Registered source descriptions:

- National Institute of Standards and Technology ThermoML Archive: ()

Registered target identities:

- None.

Meaning. The thermodynamic extension derives finite microstate support, temperature, energy, heat/work, entropy, enthalpy, free direction, chemical potential, activity, fugacity, phases, colligative response, solvation and coupled transport while keeping external scalar inscriptions downstream. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; absent solute composition is structural EmptyOne
- no negative, irrational, imaginary, logarithmic, floating, signed or continuum SFT proof value
- no imported van't Hoff factor, dissociation parameter, Raoult law, boiling/freezing constant or osmotic equation
- no molality/molarity response equation, interpolation, regression, fit, selected system/row or target correction
- no compound, solvent, solute, phase, temperature, pressure, composition, response, uncertainty or target hash before prediction seal
- every source decimal remains a post-seal external inscription and every source row remains preserved

Immutable identities. Source manifest

sha256:51286051ce2085b8185e2c60c6b700e34e7663bc1f6e199a6b572a2df45728d8; derivation seal

sha256:76d7c134378eb95ec6ae17bd16bc25ff30d1e3e0af2ee1eb29c8854a2e91a9b9; engine receipt

sha256:f8c933f2777154b409d33968b254da03d8786cab4bb3dfc87e7686d5aef17177 at receipts/engine/model_admitted/SFT-CHEM-COLLIGATIVE-COMPOSITION-RESPONSE-014-f8c933f2777154b4.json;

empirical hash sha256:5b9b80327b176c82d6b226852799c47a3eb2fc8f61eb3766fadbe2c333ee5a5c;
 measurement receipt sha256:cf6ad68ff1b1a1d0e123cc709a065cd14acfe78d1fae446f5a743467dbd20c64.

139. Exact solvation and dissolution free-order law

Claim identity: SFT-CHEM-SOLVATION-DISSOLUTION-FREE-ORDER-015

Question and theorem. Solvation and dissolution are exact state-transfer relations whose carrier retains the solute, every solvent, the distinct source and destination states, and the condition. Free order is represented by a held state orientation plus exact positive magnitude, never a signed SFT value; solubility is exact positive condition-bound composition capacity. No force field, continuum solvent, partition or activity fit, solubility-product equation, logarithm, correlation, regression or fitted parameter enters.

complete-solute-solvent-state-condition-account__distinct-held-solute-and-solvent-identities__held-distinct-source-and-destination-states__exact-retained-condition-or-EmptyOne-reference__held-free-order-orientation-plus-positive-magnitude__structural-EmptyOne-only-for-absence__complete-value-free-799-record-identity-seal__depth-independent-support-replication-and-record-append

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-MIXTURE-001
- SFT-CHEM-STOICH-SOLUTION-001
- SFT-CHEM-SOLUTION-EQUILIBRIUM-001
- SFT-CHEM-INTERMOLECULAR-BINDING-011
- SFT-CHEM-ENTROPY-MULTIPLICITY-CORRESPONDENCE-005
- SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007
- SFT-CHEM-CHEMICAL-POTENTIAL-EQUIVALENT-COMPONENT-008
- SFT-CHEM-ACTIVITY-NONIDEAL-COMPOSITION-009
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013
- SFT-CHEM-COLLIGATIVE-COMPOSITION-RESPONSE-014

Generated grammar. Generate the literal product of carrier, identity, state, condition, order, absence, prediction and extension forms; decide all 256 candidates only from admitted Fold identity, exact state transition, exact order, solution recurrence, EmptyOne and finite-successor laws.

Boundary: Every finite solvation or dissolution record with one held solute, one or more distinct held solvents, distinct source and destination states, an exact condition or structural reference absence, and a held order/capacity. External testing preserves all 642 FreeSolv experimental rows and all 157 direct NIST ThermoML solubility rows.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *complete-solute-solvent-state-condition-account__distinct-held-solute-and-solvent-identities__held-distinct-source-and-destination-states__exact-retained-condition-or-EmptyOne-reference__held-free-*

order-orientation-plus-positive-magnitude__structural-EmptyOne-only-for-absence__complete-value-free-799-record-identity-seal__depth-independent-support-replication-and-record-append. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-solute-solvent-state-condition-account	detached-solvation-or-solubility-number	A detached number erases the transfer carrier.
identity	distinct-held-solute-and-solvent-identities	anonymous-or-collapsed-components	Collapsed components cannot identify what is transferred into which solvent.
state	held-distinct-source-and-destination-states	erased-source-or-destination-state	Erasing either endpoint makes transfer direction undefined.
condition	exact-retained-condition-or-EmptyOne-reference	unbound-or-fitted-condition	An unbound or fitted condition does not identify a measurement state.
order	held-free-order-orientation-plus-positive-magnitude	signed-free-energy-proof-value	Signed arithmetic imports a negative SFT value.
absence	structural-EmptyOne-only-for-absence	numerical-zero-capacity-or-condition	Numerical zero is not an SFT value.
prediction	complete-value-free-799-record-identity-seal	compound-condition-or-value-readable-before-seal	Readable targets could select the law.
extension	depth-independent-support-replication-and-record-append	refit-after-replication-or-record-append	Refitting destroys exact provenance.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256: da0291c82e364fb4f4650bea9bde25cb3bdd7eee5eb7b10b5161e4771e152ec4.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 332d926901dd672e1ea164e117a965e7b52ec3a2a806d00f219bb3b4dff099e1.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: ab508b13f5285443c6d955fe95bd7a3ae899aaec94641d78e021c4265ab1e2a9.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: 875124c6f77386d94b99032237a5110b18e26adc010c2f46b28ca1d8dd757215.

Independent reconstruction. Implementation hash

sha256: e4ea6e71451bfc92863a46f67f5d8d67c66fd783f573c1725b04543a9b2b6402; independent certificate sha256: 54ffb1b8e77f63f459aaa27a242aa9314030a3df9f68bdda70d7b85213927319; external-validation hash sha256: be7ae81e52e1003a627c3f8f21432ec2de7eece0540eac2b13e43cdc5b07f86d. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- FREESOLV-0.52-COMMIT-6C7D19B
- NIST-TRC-THERMOML-JCED-2016-61-1470-1476

Comparison records:

- SFT-CHEM-THERMO-015-SOLVATION-0001: class=solvation;
target=sha256:38a52da26be00e66b1cbfe1da753b7758caa85909b0aac3bae8cbc700b085fbf
- SFT-CHEM-THERMO-015-SOLVATION-0002: class=solvation;
target=sha256:87e3f3bc9c008884e29dde1b4002e338e71c6c6f6c4a1d28683cefd560e805fd
- SFT-CHEM-THERMO-015-SOLVATION-0003: class=solvation;
target=sha256:e41229c89cf1147d201c534228fb969e94410f8f0797e6b2a0a02ffd2682d92c
- SFT-CHEM-THERMO-015-SOLVATION-0004: class=solvation;
target=sha256:07433de6ae23f824d441f3eecd8cdf0dbab2224fed7f20f76d00e83028462083
- SFT-CHEM-THERMO-015-SOLVATION-0005: class=solvation;
target=sha256:db183f2271cf720f649aa1bcbbe42b12307fd0229c916aca49afc6fb6b193f92
- SFT-CHEM-THERMO-015-SOLVATION-0006: class=solvation;
target=sha256:e7924142d69d2d82cec5c06ba066695e971942a6b2ebbb933363d32eb70f9c9b
- SFT-CHEM-THERMO-015-SOLVATION-0007: class=solvation;
target=sha256:6ef9b6a8ea02818c5629324112701529ea68a753a3d7e74a16f721c82be436e9
- SFT-CHEM-THERMO-015-SOLVATION-0008: class=solvation;
target=sha256:37e818c9be03f34d2f087fc7cae926a6d8baab45efc0e5d2efb9e4ca6587ace5
- SFT-CHEM-THERMO-015-SOLVATION-0009: class=solvation;
target=sha256:78130011e92fae0ebfec67de1f3713ea13bcef06d82565c92c94616a7a032417
- SFT-CHEM-THERMO-015-SOLVATION-0010: class=solvation;
target=sha256:8b8028cc8927b0f0ccc7e9d0837607a96340437785d22b210c1c8eedd4eefd57
- SFT-CHEM-THERMO-015-SOLVATION-0011: class=solvation;
target=sha256:ae0d095b003f4b6bc94a0e7daf941dd973474e2322cc61e710e308d70ff06214
- SFT-CHEM-THERMO-015-SOLVATION-0012: class=solvation;
target=sha256:0edc58af58eb1df1673f2fa36d1689315d630c9d67cf41ec3a9c81d95a5f145a
- SFT-CHEM-THERMO-015-SOLVATION-0013: class=solvation;
target=sha256:989fbfae2a999ddc1ce5d55e51e52f18b3c01f19f929bd195042f70d61993922
- SFT-CHEM-THERMO-015-SOLVATION-0014: class=solvation;
target=sha256:8e8c16f2df5be14aef2b57eca8369fef0069bc1126b509842f5f6baa1cdd3263
- SFT-CHEM-THERMO-015-SOLVATION-0015: class=solvation;
target=sha256:05c02b5040da61bc8647fd809251793ec5bd980d5484d7d1c04d006c64c1a7f4
- SFT-CHEM-THERMO-015-SOLVATION-0016: class=solvation;
target=sha256:02527faf8dd770b6984dcf5d5c04d42b476252fce7edd36d48314ce7dec6f4f9
- SFT-CHEM-THERMO-015-SOLVATION-0017: class=solvation;
target=sha256:b3386e8c92dbd3bc1e4285fa860527b9e7c9d70dd0d5c5f948842d4dbb19891d
- SFT-CHEM-THERMO-015-SOLVATION-0018: class=solvation;
target=sha256:ecbe4ddf69b3b710b81dc3db87dc06c2f0d4d4d5cef4115bcabcb93c227fadd3
- SFT-CHEM-THERMO-015-SOLVATION-0019: class=solvation;
target=sha256:c4874c65b5db1f62b191b5b09f15d6464524741496de1cac22fd545505fcec3f
- SFT-CHEM-THERMO-015-SOLVATION-0020: class=solvation;
target=sha256:dafc69e445b95376a8f766fdc41e7ec4c30f9a33ca4936f70c63b6823fa41fd8
- SFT-CHEM-THERMO-015-SOLVATION-0021: class=solvation;
target=sha256:5336cc334bfaaa6c113ba79e01676297292eeb9128eca049d393541f5e068de3
- SFT-CHEM-THERMO-015-SOLVATION-0022: class=solvation;
target=sha256:b43563122783b2d18d040f17d50bc4ad2d20cd56fe93458ad43937eb797d74e3

- SFT-CHEM-THERMO-015-SOLVATION-0023: class=solvation;
target=sha256:4ebfe93def9c1615c91499743dbccd24a269f682fcb08bf91d166029006d47c5
- SFT-CHEM-THERMO-015-SOLVATION-0024: class=solvation;
target=sha256:fef8050866ff5b5815f4b396c300468f278e2e3fd8c73201b2d85dddb43ca33a
- SFT-CHEM-THERMO-015-SOLVATION-0025: class=solvation;
target=sha256:16c141fcb027b53211aa85b0507db7287e98baf667f75f15bd2dcb55ae36ed33
- SFT-CHEM-THERMO-015-SOLVATION-0026: class=solvation;
target=sha256:32c19b7d46f01dfdbd0426698cc19dff80c3e8dde78e405a02da9b0035eb00d
- SFT-CHEM-THERMO-015-SOLVATION-0027: class=solvation;
target=sha256:ca3d7260c55ad5f1f44f7a372a1ca3d1e15f1c7c0a282a7853d5b41e86da88bf
- SFT-CHEM-THERMO-015-SOLVATION-0028: class=solvation;
target=sha256:e9042e340a81deb81b19242bd2e62545e78c7b5a8dd23286ff9a1a77e849f0c1
- SFT-CHEM-THERMO-015-SOLVATION-0029: class=solvation;
target=sha256:b85fd18c8e68770c06931d8c008d7fea012a8569a3b88aec08c5046d2cc0d35
- SFT-CHEM-THERMO-015-SOLVATION-0030: class=solvation;
target=sha256:c5b92859383c000ac1a0e9deb8979dfe345d894bdfc990823f65004501e79280
- SFT-CHEM-THERMO-015-SOLVATION-0031: class=solvation;
target=sha256:175f5ebdbaddcb8174764af34d9726e2939bbf912a810a1ae627c7b2be65fc31
- SFT-CHEM-THERMO-015-SOLVATION-0032: class=solvation;
target=sha256:b9187548f76879ae6371706cc5e8ee98cbdf2869928b96b19666e44ea7856d80
- SFT-CHEM-THERMO-015-SOLVATION-0033: class=solvation;
target=sha256:1b5252989d19c700a9b41f501b149df72358cc4effd70de6eada2eb335986dae
- SFT-CHEM-THERMO-015-SOLVATION-0034: class=solvation;
target=sha256:4564959233cfd430e76b831f669cd0eb4d4bc9ed7b0c74199547b772af8e3167
- SFT-CHEM-THERMO-015-SOLVATION-0035: class=solvation;
target=sha256:46dc17bf915cdc31da565d75189fd117c22af8b27cf0670eead615a2e4c6cef9
- SFT-CHEM-THERMO-015-SOLVATION-0036: class=solvation;
target=sha256:6a5733705bd2d83fc4aef2cf92ec1d138ded2b739318aef2216c6bc0c5de8a86
- SFT-CHEM-THERMO-015-SOLVATION-0037: class=solvation;
target=sha256:8230df4259403e1b4c5bb779b90ea1964a16dace4d8d80faf50b7bc0ab2bccd1
- SFT-CHEM-THERMO-015-SOLVATION-0038: class=solvation;
target=sha256:81382c366eb147d449cf9df8889e89f4958e1e10b1f648853145b91549b080fd
- SFT-CHEM-THERMO-015-SOLVATION-0039: class=solvation;
target=sha256:e454ed38d8d6e1c45e5855db8f0f43b8161513f5247078583e4b273448569e5c
- SFT-CHEM-THERMO-015-SOLVATION-0040: class=solvation;
target=sha256:6a4e076e4d4af8fa4986447b76d5ae9beb7de1efe695d291205edbcfbef11c0b
- SFT-CHEM-THERMO-015-SOLVATION-0041: class=solvation;
target=sha256:a130afbe18bc90efa4beb8129b6c4b611e1efa575e97dc87089fba8b0bb8875
- SFT-CHEM-THERMO-015-SOLVATION-0042: class=solvation;
target=sha256:8ae1246d65c91bd958e9c4ef9e80ba95572ba1cbea8e993a23170678f2f102e2
- SFT-CHEM-THERMO-015-SOLVATION-0043: class=solvation;
target=sha256:c7e88b0ac9a3653865a5445fcf3e923fc92e789f7a1136a1b075e1671b3fadad
- SFT-CHEM-THERMO-015-SOLVATION-0044: class=solvation;
target=sha256:29c13e803c22f6975d1ef965a4f1213a801328ae8a04a826bc40284095a2df6b

- SFT-CHEM-THERMO-015-SOLVATION-0045: class=solvation;
target=sha256:a443504639b1d8fc400c700b6a97c1bf32ac3714f5206ac27b59299e513e75f2
- SFT-CHEM-THERMO-015-SOLVATION-0046: class=solvation;
target=sha256:355ef24e613472186f1b5478a230535ecb2f410d1a038b6509bf29180a57b178
- SFT-CHEM-THERMO-015-SOLVATION-0047: class=solvation;
target=sha256:09e48450e75cac1a07bede2c3c4794007aa4302f2b56a9c4ce187efc9e6de0df
- SFT-CHEM-THERMO-015-SOLVATION-0048: class=solvation;
target=sha256:8e8077dd0d18b26930869562f24693ed09fb3d560e704f75d6de91f9fc4c6cfd
- SFT-CHEM-THERMO-015-SOLVATION-0049: class=solvation;
target=sha256:ce9b4c4abaf248992086fc87dcf1ab4574760d7cd26bdf5d9aadbebc19db7a9c
- SFT-CHEM-THERMO-015-SOLVATION-0050: class=solvation;
target=sha256:36de35eb0d2bdb82f038bad8d6ce75c0926fcfd286d4a37b5553bb24bd11b670
- SFT-CHEM-THERMO-015-SOLVATION-0051: class=solvation;
target=sha256:c6370152b1d0f70edf76657744f5016712c97f8ba1591475c4a6d5382bee3101
- SFT-CHEM-THERMO-015-SOLVATION-0052: class=solvation;
target=sha256:835cb7561244118279b9188f922c4971b85660f7b7a64d4cbcd75f4bc90c64f4
- SFT-CHEM-THERMO-015-SOLVATION-0053: class=solvation;
target=sha256:319a1321842435490e89ef84dcf15e242341a205497ca86b2c0745acc0e61af1
- SFT-CHEM-THERMO-015-SOLVATION-0054: class=solvation;
target=sha256:b114701bb2f90159a68a97471efe997c72e21716f0ca9d505359d892c4a46dff
- SFT-CHEM-THERMO-015-SOLVATION-0055: class=solvation;
target=sha256:45219d6e32dfb9d66d3e1176281bbd355012435a04941bfa8c34de6a591003aa
- SFT-CHEM-THERMO-015-SOLVATION-0056: class=solvation;
target=sha256:5d7283b61c231e4bb950af6c8a41266c60402b6a69abc47e7e3e4a368463c426
- SFT-CHEM-THERMO-015-SOLVATION-0057: class=solvation;
target=sha256:6c6df68e1e52e7219c2b6ae39a6ccdc3564b31602b683bb868360726b39a4617
- SFT-CHEM-THERMO-015-SOLVATION-0058: class=solvation;
target=sha256:849dd83c72cea96a227b0683532c2169a9037c5a19325b1b4a10c8d34f22d79f
- SFT-CHEM-THERMO-015-SOLVATION-0059: class=solvation;
target=sha256:45eb6bd6213943b6a6cbcd7d8de9c55cbafa47622042b86b49448cbfda607314e
- SFT-CHEM-THERMO-015-SOLVATION-0060: class=solvation;
target=sha256:ee4dd8ed7e21ee12a1e44bc12dc0583e62a4a5d9c76d99b0e69e073156f6adaa
- SFT-CHEM-THERMO-015-SOLVATION-0061: class=solvation;
target=sha256:ab43b3605b6dec5a6bf1a8bbda7c4e25e59d165e5f9cbeaca2f04368e27e0dd0
- SFT-CHEM-THERMO-015-SOLVATION-0062: class=solvation;
target=sha256:193c7ee45885e7ecd96fbb1752036983e6e6ffeda0d0d45cabdb439483e4d842
- SFT-CHEM-THERMO-015-SOLVATION-0063: class=solvation;
target=sha256:59de80fe094743fcdeb1f786e975eafa691f87a49438de7754d5fde51736a762
- SFT-CHEM-THERMO-015-SOLVATION-0064: class=solvation;
target=sha256:fb85323e3e6d80cf63fcf9e7d8fb413b85288c7f37ea6da371128dcc583e72db
- SFT-CHEM-THERMO-015-SOLVATION-0065: class=solvation;
target=sha256:297d7a333ca522fa06576200a5a89d1d6d5b2747bc2af2188e74fbc4baeb7079
- SFT-CHEM-THERMO-015-SOLVATION-0066: class=solvation;
target=sha256:bd242def4341519aa9f0fa5983addf1631194715a1b44fe4cd5bab1a3af95a10

- SFT-CHEM-THERMO-015-SOLVATION-0067: class=solvation;
target=sha256:3e81246415d67469a8192f393fab441e49421129b6e4a5937f9eb8ce275a9d3a
- SFT-CHEM-THERMO-015-SOLVATION-0068: class=solvation;
target=sha256:fd211108aa96159175b0f2ea45617ec4182f4ca9696bc3e75365804c3865b013
- SFT-CHEM-THERMO-015-SOLVATION-0069: class=solvation;
target=sha256:817b1b891aed01902511fbdf5b22440a7a1f59ad7a79d0056498a9b5625890d5
- SFT-CHEM-THERMO-015-SOLVATION-0070: class=solvation;
target=sha256:0193b86e0efc05d0b9445f3da1800e2b1021eb32581dd06d8e272b888a0556f4
- SFT-CHEM-THERMO-015-SOLVATION-0071: class=solvation;
target=sha256:e8bd2ad51d5f6958e0d3ad1276c17304ba51b6f3d8449efa8d7d8fbe2bce7240
- SFT-CHEM-THERMO-015-SOLVATION-0072: class=solvation;
target=sha256:cec968999c1fbc3e9231eed1c63a9c9cbd94e742cb6d47169b4416d97069058f
- SFT-CHEM-THERMO-015-SOLVATION-0073: class=solvation;
target=sha256:e17fc2a69f08a2da4864e2746f6b2105f612dcbcd99ca2a74488e96c5d02edf1
- SFT-CHEM-THERMO-015-SOLVATION-0074: class=solvation;
target=sha256:dabe417ab22a19933f7cd89345e6fa2d773320812e7ea0fd840ce851b8b5946f
- SFT-CHEM-THERMO-015-SOLVATION-0075: class=solvation;
target=sha256:908b39daca5bac1df015834bd02235bef0dd7a9c6b9c4dcf9ddbdc760be30286
- SFT-CHEM-THERMO-015-SOLVATION-0076: class=solvation;
target=sha256:4af9d7735581d6813e9e3029af510e10061e469897e8b8687c41a6198b890103
- SFT-CHEM-THERMO-015-SOLVATION-0077: class=solvation;
target=sha256:a2c8b2a12dd4bfd626a7f776f4068acac0a3fa51c49231916579616a7774379a
- SFT-CHEM-THERMO-015-SOLVATION-0078: class=solvation;
target=sha256:0f4706d0cf6888a128814ab940436f59709387f4ef4ff61ce11eda3e75fe021e
- SFT-CHEM-THERMO-015-SOLVATION-0079: class=solvation;
target=sha256:0e6a2e40a1c4ab4f2b1606cd4fabca0b38142f8ac66542aed161c48cf83cea5b
- SFT-CHEM-THERMO-015-SOLVATION-0080: class=solvation;
target=sha256:f08a67089581fafa94839428707b716baebdeb0ab1a2eaae737f1c4229c562de
- SFT-CHEM-THERMO-015-SOLVATION-0081: class=solvation;
target=sha256:039f5c9c1a49013e8a3944ef809eea5b9dbfb046bc3c26133e6f5b299eb511c6
- SFT-CHEM-THERMO-015-SOLVATION-0082: class=solvation;
target=sha256:e5bef837a33f7e0609977d82ceb084e666217aa3f5b39454ec8d5e9d5dac141a
- SFT-CHEM-THERMO-015-SOLVATION-0083: class=solvation;
target=sha256:063122f317836b1a7192cdc9349a724ea385d6b5fb905dc92f1285dab8a21615
- SFT-CHEM-THERMO-015-SOLVATION-0084: class=solvation;
target=sha256:1549282e697f0b3b4c5fc480f710bcfbe0f3770f0dd1c18d19b60f4200d7567f
- SFT-CHEM-THERMO-015-SOLVATION-0085: class=solvation;
target=sha256:b464bb072bcf0279c13aecb3da4172f2cffbe0e6020bf62c7ddec61489fbafc7
- SFT-CHEM-THERMO-015-SOLVATION-0086: class=solvation;
target=sha256:2e38bf9cf1f15373623c25236cdf8a990ac0d57c6930b248be7bb16b51212381
- SFT-CHEM-THERMO-015-SOLVATION-0087: class=solvation;
target=sha256:a5143922ab28ef52c528adf78402181a6c428c859ec23ddc42314a2e080361c6
- SFT-CHEM-THERMO-015-SOLVATION-0088: class=solvation;
target=sha256:d9b6e9994d2c358b87ffa50ac98dbe20bdcf6bf64a23737307d08c2409cc8abc

- SFT-CHEM-THERMO-015-SOLVATION-0089: class=solvation;
target=sha256:debde99a95a75032044e0a60b7af2257ae96db60db60b8bdb9227dfa36462880
- SFT-CHEM-THERMO-015-SOLVATION-0090: class=solvation;
target=sha256:b7b8ddc376a64479fcbcc987cae18827ef32aa259a075ab878575388d22f0d48
- SFT-CHEM-THERMO-015-SOLVATION-0091: class=solvation;
target=sha256:df7d5c69863453f256648b569879b9239dde78a5af16d099bc76669110f5a436
- SFT-CHEM-THERMO-015-SOLVATION-0092: class=solvation;
target=sha256:04bb1d4473abd1ffd1075cd1a8358570efbc529d4efa10c33cd80333d4e26f4c
- SFT-CHEM-THERMO-015-SOLVATION-0093: class=solvation;
target=sha256:06e057ae97cb8e6fbd4eb4a937ce8cd5e9ed4256d3b534ba4f194304a7ccc66
- SFT-CHEM-THERMO-015-SOLVATION-0094: class=solvation;
target=sha256:4570e4092ad497d5c84987a3f519b1bc8e15dca4dab1cf26bc316b78bec619cf
- SFT-CHEM-THERMO-015-SOLVATION-0095: class=solvation;
target=sha256:748a109fb104d834e6a1befed23955b1f0e5ac5cb86f1a8e5f58a011a35c3758
- SFT-CHEM-THERMO-015-SOLVATION-0096: class=solvation;
target=sha256:a9c99974da1c5b3af450173b4e0800eb08e2b34b119795af1027ab9330c91975
- SFT-CHEM-THERMO-015-SOLVATION-0097: class=solvation;
target=sha256:e0ca96fdb1f39f3317a900dc22e8ed9b8a866450527cb73dabd1317a67cb9950
- SFT-CHEM-THERMO-015-SOLVATION-0098: class=solvation;
target=sha256:49a6967a5e8544ed3d46477927122393ff6fae1606bd396145daaa08e4c0cbf3
- SFT-CHEM-THERMO-015-SOLVATION-0099: class=solvation;
target=sha256:3f039d6f50963afa4afcbcaa02a1b9ef137de88801408fbaef816678fb94bc47
- SFT-CHEM-THERMO-015-SOLVATION-0100: class=solvation;
target=sha256:c2a4fd84eb73769d7ce3e8a3da59fe76c914446b2918d5785d82810644032532
- SFT-CHEM-THERMO-015-SOLVATION-0101: class=solvation;
target=sha256:a6447256076dbfc3acc1f1d8c1609ec013a9bec421f20298a5223eaa8e9fb1bd
- SFT-CHEM-THERMO-015-SOLVATION-0102: class=solvation;
target=sha256:593dc4e821f66fe7858eb0499c5bf97e319fd033d7c88c547521b92c79c4e6b0
- SFT-CHEM-THERMO-015-SOLVATION-0103: class=solvation;
target=sha256:53582e8c9d1d76b7aa08138663c5b743a468402635a1a0865d58f8ce70ec27ca
- SFT-CHEM-THERMO-015-SOLVATION-0104: class=solvation;
target=sha256:33df939b6ce00d5d0ac553e97581b0a5b4a414e3373cb2f9a882559b7410cbea
- SFT-CHEM-THERMO-015-SOLVATION-0105: class=solvation;
target=sha256:f10f3dd21daa7d2bd06fad45bdf72f80cac8e065b0547a92794339b3dca13414
- SFT-CHEM-THERMO-015-SOLVATION-0106: class=solvation;
target=sha256:8d90775d850366664b3919f0c3f5e3a6153c8e429b6c79d0d50719c3cc1ef41d
- SFT-CHEM-THERMO-015-SOLVATION-0107: class=solvation;
target=sha256:178d428884abb13d0d2b98d5872d68f5916bb9562f7e330a51b47d4cfa6e0151
- SFT-CHEM-THERMO-015-SOLVATION-0108: class=solvation;
target=sha256:180ab609994a684fcd085201898b943411d4838b4991895776fac0b84d2c2ff
- SFT-CHEM-THERMO-015-SOLVATION-0109: class=solvation;
target=sha256:33a3faa9f719cd27dda2bad2fdcf74fac14623ef71325e0b42c1f8929ef57512
- SFT-CHEM-THERMO-015-SOLVATION-0110: class=solvation;
target=sha256:3d6438c6d3eba2772f0542816751352344b4b56ddff0a82b7c54e5f5690aafbc

- SFT-CHEM-THERMO-015-SOLVATION-0111: class=solvation;
target=sha256:4309783aef125b171e738f8d36dfd467a8110eb65c08d926363c364b99f42816
- SFT-CHEM-THERMO-015-SOLVATION-0112: class=solvation;
target=sha256:bbb63dea0e97046a8baab018d25e5e048dae62305b67b1171c9ed82b5a1555c3
- SFT-CHEM-THERMO-015-SOLVATION-0113: class=solvation;
target=sha256:e10d75e1c0ff9ce65b52543e2153a803b237d572c04dfa1c6efdbe37d413baea
- SFT-CHEM-THERMO-015-SOLVATION-0114: class=solvation;
target=sha256:5bafaba8e1b4598f3241829c883a37d18aad9bc9a1492776679474cff2d71c79
- SFT-CHEM-THERMO-015-SOLVATION-0115: class=solvation;
target=sha256:4a9faed816334e18ec57775203987d01c539faf6d28a63d270aacd6753637184
- SFT-CHEM-THERMO-015-SOLVATION-0116: class=solvation;
target=sha256:91201ad7ecf1459d7edc4d9eab37606e2a83d6d5ba27a59008841238154cb1c5
- SFT-CHEM-THERMO-015-SOLVATION-0117: class=solvation;
target=sha256:544009d0423617674c6821a03ce1c63e01ff85b2fe86dbb8c4261a90ea1a0930
- SFT-CHEM-THERMO-015-SOLVATION-0118: class=solvation;
target=sha256:f9b58a9002bf72f980ecf59cdc8c68bf81a8cf8a231e861cbad37245c8ddaa55
- SFT-CHEM-THERMO-015-SOLVATION-0119: class=solvation;
target=sha256:c1fd5f6da932d0a51878e69a292977429332aada1ec5236f6838afc7cc9b223a
- SFT-CHEM-THERMO-015-SOLVATION-0120: class=solvation;
target=sha256:6b687ac4b22b984000aa39069cfc0a386c7ac64874e8991664e9d9c2c85400a8
- SFT-CHEM-THERMO-015-SOLVATION-0121: class=solvation;
target=sha256:4b212ae50d15def1769adaf562837ad1f46569d41d4f0805d9030d766c6a7965
- SFT-CHEM-THERMO-015-SOLVATION-0122: class=solvation;
target=sha256:43c00b7e00a14b503829a3543a7fa8da8d954bffe37bc527a500f6c9d31675ae
- SFT-CHEM-THERMO-015-SOLVATION-0123: class=solvation;
target=sha256:09074e5f8a647161f914b5508b1a48c0d7847a9369b3f1072cbea707ec216d78
- SFT-CHEM-THERMO-015-SOLVATION-0124: class=solvation;
target=sha256:84c5adb51b90ee84d1eaf2350451c449a794a55201e07f874c4c3b6f65933328
- SFT-CHEM-THERMO-015-SOLVATION-0125: class=solvation;
target=sha256:9739ae98588137a447731be43b6dce2be50a43fbbe77b4fada0bfda9a1d64f4c
- SFT-CHEM-THERMO-015-SOLVATION-0126: class=solvation;
target=sha256:818488c8e045060690406cd14750548126328479f49157e225b845755bff9ef4
- SFT-CHEM-THERMO-015-SOLVATION-0127: class=solvation;
target=sha256:7983236041e52872f60067a2a5315ebbc56eeffdb846a8546a91aa6f70f952
- SFT-CHEM-THERMO-015-SOLVATION-0128: class=solvation;
target=sha256:d640587af94404006bbb8d38a44f7ea776721121931e2328f0a7d22874f9f9f
- SFT-CHEM-THERMO-015-SOLVATION-0129: class=solvation;
target=sha256:21c1596b0680b5ba792ed7c2cfa9999c73bac1a86dc6f4074d45fd47b24e6b17
- SFT-CHEM-THERMO-015-SOLVATION-0130: class=solvation;
target=sha256:f279594ca5fa86219ca56734cdcc7c555bf58ceadd01fea55a4fcdae187e3cd9
- SFT-CHEM-THERMO-015-SOLVATION-0131: class=solvation;
target=sha256:b67e5b1b84eb25289e70a530404c2129ec1ed3b7d14730b8af141bfc3672a791
- SFT-CHEM-THERMO-015-SOLVATION-0132: class=solvation;
target=sha256:9e76d398387edf735433af6c22184292a6c62755f9c99f473b2ef9654e8f2ba

- SFT-CHEM-THERMO-015-SOLVATION-0133: class=solvation;
target=sha256:9a6ba79100fa0ca85e964235ddbade87164086bcc49e6de4f22a933e678bcc1c
- SFT-CHEM-THERMO-015-SOLVATION-0134: class=solvation;
target=sha256:688e1ca9c74e934edc3896508e67164e2606d81cef1ce44077a4c44b0c189be0
- SFT-CHEM-THERMO-015-SOLVATION-0135: class=solvation;
target=sha256:c166b8fababe7effcacc225f9557abbc763d2336d0cadb427faabfad4a32e727
- SFT-CHEM-THERMO-015-SOLVATION-0136: class=solvation;
target=sha256:24e1cf2d6163b3364deb10a6710f306a6aa0b3713be875f56fdd95968387fafc
- SFT-CHEM-THERMO-015-SOLVATION-0137: class=solvation;
target=sha256:655421fc4810c8b77fa5e64fb1a4fa67d0c4cd396652294f64a8112914a6f808
- SFT-CHEM-THERMO-015-SOLVATION-0138: class=solvation;
target=sha256:c48591f23efdb49a1f24a1b18c75125c086bbf5526c30adafe8dc05ef29e8805
- SFT-CHEM-THERMO-015-SOLVATION-0139: class=solvation;
target=sha256:317d9c20b7e3f08ba96281338ac352e439cf693f5bb21e6f7f14ff771c9e34d6
- SFT-CHEM-THERMO-015-SOLVATION-0140: class=solvation;
target=sha256:50ac9168ea2b7dda796dc648d2401a59657d4d9f639c98e41d34407fd00dfd74
- SFT-CHEM-THERMO-015-SOLVATION-0141: class=solvation;
target=sha256:6a69ce88ffd924865c36031b2a6b40a770851527bbcf2d4201951006ae184b90
- SFT-CHEM-THERMO-015-SOLVATION-0142: class=solvation;
target=sha256:282fed2634639315bbfdd12a414026cab3fb43a2bb235ac5e487a365fb3d19d0
- SFT-CHEM-THERMO-015-SOLVATION-0143: class=solvation;
target=sha256:3db12f52901cf32a77647adf934c005bfdbc6bc8c29a28bc5f5b0a4336eb3a5
- SFT-CHEM-THERMO-015-SOLVATION-0144: class=solvation;
target=sha256:849ab2d180f28e42b3f22605122ffd328cec7617d8a0c14264d5770a786b78f3
- SFT-CHEM-THERMO-015-SOLVATION-0145: class=solvation;
target=sha256:13cbb416cdc27763db52ef1309bb91b3c3427342bf7cf995f93f98f0e460b46d
- SFT-CHEM-THERMO-015-SOLVATION-0146: class=solvation;
target=sha256:b5e971ef2ffe326f2fffc2cb3f65c66af8ff8d478ecce014b887b7e0537da42b
- SFT-CHEM-THERMO-015-SOLVATION-0147: class=solvation;
target=sha256:da92dd32629aa57ac02846411685e58168cafc924bcef7c6ece89cea97e3bd8
- SFT-CHEM-THERMO-015-SOLVATION-0148: class=solvation;
target=sha256:e19eea141861207bcb899f0e8e476b81396f20637a1bdd9484e2580906588bca
- SFT-CHEM-THERMO-015-SOLVATION-0149: class=solvation;
target=sha256:0b2944521885bcd0946aaee599da57cbf7dbec939782f77736a2966a48ac11a0
- SFT-CHEM-THERMO-015-SOLVATION-0150: class=solvation;
target=sha256:a905f7965566b4f50d6e7817476065b33007a421ea7431bedfcc07930429b1c6
- SFT-CHEM-THERMO-015-SOLVATION-0151: class=solvation;
target=sha256:94bf2bc17de9c79735c1a30079ab387ebcbd892cf90dd49ed32f2e62acde8710
- SFT-CHEM-THERMO-015-SOLVATION-0152: class=solvation;
target=sha256:407890fc3a1a64de69c6196dd65607a32717768d27764292dcf911864e1ec497
- SFT-CHEM-THERMO-015-SOLVATION-0153: class=solvation;
target=sha256:545094dc52f1b7ff151928ee5bace5e8e9ffe18d44bd84b99a28cb8f94bde3b2
- SFT-CHEM-THERMO-015-SOLVATION-0154: class=solvation;
target=sha256:a0d2395ee191b7a93d9378bea38e283f72837a8a60c527569d1c46912f9635ec

- SFT-CHEM-THERMO-015-SOLVATION-0155: class=solvation;
target=sha256:f4a43324c1a49da598c64dfa7cc71c90ed1939e348ea6d4b736f1815884c927e
- SFT-CHEM-THERMO-015-SOLVATION-0156: class=solvation;
target=sha256:882e0b8cb3459c982618632bca4c3e0b0c687a77b768691f6588e763b559d958
- SFT-CHEM-THERMO-015-SOLVATION-0157: class=solvation;
target=sha256:b3bbfd101e7008ec97d99076e7d58bcf0a9e5904d36513e80b622888344326bb
- SFT-CHEM-THERMO-015-SOLVATION-0158: class=solvation;
target=sha256:481d4c7e9f7e121eb50e97e85bfa2e8ec88d8af5e2ee3dc42e9331e77f7dce77
- SFT-CHEM-THERMO-015-SOLVATION-0159: class=solvation;
target=sha256:feadf1ca833d96a017059bab1d4df96e6f4b59062d47b8c4e5c5c00d53270bd3
- SFT-CHEM-THERMO-015-SOLVATION-0160: class=solvation;
target=sha256:b2c1cc17b76d3e8f0708e4e5251efaa506d9326f8ae77a0f9c110bb03224fc51
- SFT-CHEM-THERMO-015-SOLVATION-0161: class=solvation;
target=sha256:0be0cc8099513aa0278ff7137bc1a017bb5ae59ae72188acbd0a71dcfc809379
- SFT-CHEM-THERMO-015-SOLVATION-0162: class=solvation;
target=sha256:e9736d5d79513d40c113b08167eb5ee6db3ee1fa5f81e3d11c421a3481f213bf
- SFT-CHEM-THERMO-015-SOLVATION-0163: class=solvation;
target=sha256:6b15417d81dacdbd40b112629f68cbc289c840c289f1bd2a52d260d68f6c7ae5
- SFT-CHEM-THERMO-015-SOLVATION-0164: class=solvation;
target=sha256:290ddeb354895ab257e3cf0fa63c4a9c67a58365e80739cb7c3746767c26532c
- SFT-CHEM-THERMO-015-SOLVATION-0165: class=solvation;
target=sha256:8fd9269cf7f29a4ab22322a9c22480ec19f76ef42d3188f1ed1386b45a294a8
- SFT-CHEM-THERMO-015-SOLVATION-0166: class=solvation;
target=sha256:13502a8907b266d7564a9c91cad1da348ca3fe71d616b7de4d82f2aa4f6a307c
- SFT-CHEM-THERMO-015-SOLVATION-0167: class=solvation;
target=sha256:484357d63c0b033e9ed17b1008197c8d1b226186ee35b419bc854d869a97769c
- SFT-CHEM-THERMO-015-SOLVATION-0168: class=solvation;
target=sha256:5b703da4b42cb7f208adc04ee608f1e92ea4c65fdab9eb700c0642831a2c3ee1
- SFT-CHEM-THERMO-015-SOLVATION-0169: class=solvation;
target=sha256:110693ac51057f211e06d4cba9bc0766e7ca0bdd4e6ab84b0f73d69dfc29aed1
- SFT-CHEM-THERMO-015-SOLVATION-0170: class=solvation;
target=sha256:91ced05526cb8f3efae40fc13fde521d5a464237355dc7524ae7669cf4e2a840
- SFT-CHEM-THERMO-015-SOLVATION-0171: class=solvation;
target=sha256:345f4b23c446149a0c9cc6623b809c02ee6b46d980dcca3da126b34c3825ea56
- SFT-CHEM-THERMO-015-SOLVATION-0172: class=solvation;
target=sha256:edca50bfb48ddf27df2fa362edb66d6ddc9cadcd51d58f71dcbec4ddd7fdce96
- SFT-CHEM-THERMO-015-SOLVATION-0173: class=solvation;
target=sha256:b4d2356502cbd5cbcff5b6332991740772940d863c3ecd93d303eb2b191f775f
- SFT-CHEM-THERMO-015-SOLVATION-0174: class=solvation;
target=sha256:d9b495b71284b1556b8ff09e7833ab396083aea74852d00aabce7fd40304147c
- SFT-CHEM-THERMO-015-SOLVATION-0175: class=solvation;
target=sha256:612935e86d55502edb6c9ccc90981f3eb76ef996d50cd7f561c2d1519fb0f889
- SFT-CHEM-THERMO-015-SOLVATION-0176: class=solvation;
target=sha256:3321c0f3031092d08b267068d35a9bd47972aa8722c9e017c1453e93c16186f4

- SFT-CHEM-THERMO-015-SOLVATION-0177: class=solvation;
target=sha256:1e593d65ce0c3e26980cfe48a3c19c3357b3a1d085ebb29547e5c554cbd68dc6
- SFT-CHEM-THERMO-015-SOLVATION-0178: class=solvation;
target=sha256:7240ec6baae71a06f914e639294008a9d72cf8200584f80b37cddc8120e33060
- SFT-CHEM-THERMO-015-SOLVATION-0179: class=solvation;
target=sha256:9aab580e0528897fdeac7a73463e7a94dbfa2e8643000b2ba535af6e75475f37
- SFT-CHEM-THERMO-015-SOLVATION-0180: class=solvation;
target=sha256:3ca6e6b1a700ed30e15f17f87c548c2d6a864fc3db2906ae1a270b5f8804a1e4
- SFT-CHEM-THERMO-015-SOLVATION-0181: class=solvation;
target=sha256:8588ec0576b6884aeb6377ac6fb2962cb50a57b762aff2516c28e6e2b1472f8e
- SFT-CHEM-THERMO-015-SOLVATION-0182: class=solvation;
target=sha256:1c89986d9982a0442fb032c2ca2236725c22f0f3e0b13d99360ed66d28d96565
- SFT-CHEM-THERMO-015-SOLVATION-0183: class=solvation;
target=sha256:c13d08d0bf632d4ad1e41b71744341cee15f67100519805a42e3992fcf1d46dc
- SFT-CHEM-THERMO-015-SOLVATION-0184: class=solvation;
target=sha256:f759c47f23d413d4212d12cfe9ffe97542d7ca9372686bf762d2cfacc1ab1047
- SFT-CHEM-THERMO-015-SOLVATION-0185: class=solvation;
target=sha256:d785cf140475e25a59bf1edabdec1c277852c889dcd08c77841a166555caea74
- SFT-CHEM-THERMO-015-SOLVATION-0186: class=solvation;
target=sha256:5a9b966c60618d8c15a7c3dab10b92a6ab24a338c36cccc009024f15c09f3164
- SFT-CHEM-THERMO-015-SOLVATION-0187: class=solvation;
target=sha256:2d4cc29f860d3f1f5a084a6802eb5036e5739e48e55c5b5f1d662bdd79deaa83
- SFT-CHEM-THERMO-015-SOLVATION-0188: class=solvation;
target=sha256:c594b29d97bd467f920d8824b74557ed3fe244a48ebbc1dea4ccd440786350f2
- SFT-CHEM-THERMO-015-SOLVATION-0189: class=solvation;
target=sha256:32a5fd59f8230d80c912c6c32ddad3635a172f1920132cedbf92a56ac50a5f3c
- SFT-CHEM-THERMO-015-SOLVATION-0190: class=solvation;
target=sha256:158f54ecd0a42879c0918dac47708a829d7113bb5b10624ec365a6abc026bc12
- SFT-CHEM-THERMO-015-SOLVATION-0191: class=solvation;
target=sha256:a08108d2afe2f4c94780d17c87fec2b70bf2e76acc0ff748b29981f6e149ddd9
- SFT-CHEM-THERMO-015-SOLVATION-0192: class=solvation;
target=sha256:fa2bce443e15432379fc45372d78285e32a79326fa67cce2a73e44b95f2432d9
- SFT-CHEM-THERMO-015-SOLVATION-0193: class=solvation;
target=sha256:37a7111c0e3d0130acedbe39d4730c94b506a5703871486371ce5ce5cb3ceb13
- SFT-CHEM-THERMO-015-SOLVATION-0194: class=solvation;
target=sha256:118bd45f7860d200fa84a00f668ed346621b2c18c3d6f6fd5108963e82cab52f
- SFT-CHEM-THERMO-015-SOLVATION-0195: class=solvation;
target=sha256:553ce0dc4897abf4b39deb20dbe79b16a1e708e81eca53e448c46ae0a3b6c316
- SFT-CHEM-THERMO-015-SOLVATION-0196: class=solvation;
target=sha256:1da3a7332cd5d425f752cc0186a0c54ae6653a13320cbf6ccb17f8faa34d0f91
- SFT-CHEM-THERMO-015-SOLVATION-0197: class=solvation;
target=sha256:c989b9a82862e10fcdc36017abb9dc7fb6dd78b1b72fb54e2e9bc3f32a2d4536
- SFT-CHEM-THERMO-015-SOLVATION-0198: class=solvation;
target=sha256:242a27e8bf436a3627310a33e382077551f6d6cdc768e7c84319f670df6f7491

- SFT-CHEM-THERMO-015-SOLVATION-0199: class=solvation;
target=sha256:d5aa2708b17d0067642b8b55c49ba36a7c4ba465d1dec5db077312ff055ead48
- SFT-CHEM-THERMO-015-SOLVATION-0200: class=solvation;
target=sha256:2a7b47fdecf43f6d1406c60d6e8340e3fceb4d905d7e91dd4478b4ea06410b21
- SFT-CHEM-THERMO-015-SOLVATION-0201: class=solvation;
target=sha256:28b6b0b2fb27a022193edfcb119b8f0780dfce108a20c2d77b63eb3ac16db463
- SFT-CHEM-THERMO-015-SOLVATION-0202: class=solvation;
target=sha256:499d8ba27caace0d2449d6afac689438406259729bf310a098a19bd75599b4d8
- SFT-CHEM-THERMO-015-SOLVATION-0203: class=solvation;
target=sha256:592e48cf9df3c7ac2fc614e9558a9835f1ca147a059e43976b1c4357463e04a4
- SFT-CHEM-THERMO-015-SOLVATION-0204: class=solvation;
target=sha256:ba07fbd9e468024d94fd6bb645ceadfa527bb2c3680c04c83f561ddfa387026f
- SFT-CHEM-THERMO-015-SOLVATION-0205: class=solvation;
target=sha256:1053e24d626da013933aa35d92a74633564bb9f5a4ac39649386713eb7e041fb
- SFT-CHEM-THERMO-015-SOLVATION-0206: class=solvation;
target=sha256:dc132aebfa84b59f101db537b825eb667ab1497ac0150e171f79dbc0d5e52b65
- SFT-CHEM-THERMO-015-SOLVATION-0207: class=solvation;
target=sha256:22e7a0faec39c3ec404ece880d0296d566b46177bd1eab2ec060d775488178a9
- SFT-CHEM-THERMO-015-SOLVATION-0208: class=solvation;
target=sha256:708e928749695f08172386c58da8ebc0e6fc7695bd97ca6e141c279599884a5a
- SFT-CHEM-THERMO-015-SOLVATION-0209: class=solvation;
target=sha256:7d3e5b49ba1affe1cf84f41e0483d1aed87323a3fa5ad4ad31137d7556407aa0
- SFT-CHEM-THERMO-015-SOLVATION-0210: class=solvation;
target=sha256:7ad6a4271143c666c02a0677dc0d48d78e58a49072b13dbcb7697645d12eae52
- SFT-CHEM-THERMO-015-SOLVATION-0211: class=solvation;
target=sha256:bb9d09671668ec499b4ba0808e35bf65ec8ffeb566220aa914529fac79e194c2
- SFT-CHEM-THERMO-015-SOLVATION-0212: class=solvation;
target=sha256:e0ca49c87e489fd7fd9d8dc781d923991696f5ab8a51f34f98c18a85671a8e23
- SFT-CHEM-THERMO-015-SOLVATION-0213: class=solvation;
target=sha256:a07856788702bed96db2e29ad9249ec84fc76a6b70fe5358215d875e7b1cfa41
- SFT-CHEM-THERMO-015-SOLVATION-0214: class=solvation;
target=sha256:d5d32b11029ca748a0e5d874806331d3f82fb163b695d83d3ba2e03f49df97ec
- SFT-CHEM-THERMO-015-SOLVATION-0215: class=solvation;
target=sha256:7bd7e73b72da41af9b0ad71ee70011cc3bb300cf6da72878261c29b575105cce
- SFT-CHEM-THERMO-015-SOLVATION-0216: class=solvation;
target=sha256:6207cb7645a5d5a7cea4fcf8664c9d245685e8fbe055337a6ddc27f90df2e8c
- SFT-CHEM-THERMO-015-SOLVATION-0217: class=solvation;
target=sha256:784f6ff133e62fc37257031bb4ba27ba22a8c58c24ef71f8fdf798afea17f3c5
- SFT-CHEM-THERMO-015-SOLVATION-0218: class=solvation;
target=sha256:0745a4acb3ee0fe12a6ab03c969fe03c6bdced5322ff8768a508119dc09c0183
- SFT-CHEM-THERMO-015-SOLVATION-0219: class=solvation;
target=sha256:38a0c51c7b857d9d595509bbc873877076fd154d6254d765ac0efbe303fa4f20
- SFT-CHEM-THERMO-015-SOLVATION-0220: class=solvation;
target=sha256:b8098e45203981f05638a6d4f32fc59febddc6f66dbfa8e89754923b4de2ee0d

- SFT-CHEM-THERMO-015-SOLVATION-0221: class=solvation;
target=sha256:4d242c716536b0f687594d104dd32cb2be4efc5317282bb01fa6b86960c67b6c
- SFT-CHEM-THERMO-015-SOLVATION-0222: class=solvation;
target=sha256:1e730d92bf5ba1c805d70673d945dab66456ce309e39885b44271cfd008f61ed
- SFT-CHEM-THERMO-015-SOLVATION-0223: class=solvation;
target=sha256:2b0e4eb47de5b638579c52650ae4e8a53c9ccda06b2c203eb70277fd10aa67
- SFT-CHEM-THERMO-015-SOLVATION-0224: class=solvation;
target=sha256:a23431a5d78a080806f33d1427987780963cdd9d2092c02bf680e080ce90f29f
- SFT-CHEM-THERMO-015-SOLVATION-0225: class=solvation;
target=sha256:039086ede33e07b7b17f688ad3ad1af729f83a5fe239eb391e9a505801a07536
- SFT-CHEM-THERMO-015-SOLVATION-0226: class=solvation;
target=sha256:2dd379160b1426c3b07686cf6085eaf53d41b9f5ff743720ef9e37ff39c758bc
- SFT-CHEM-THERMO-015-SOLVATION-0227: class=solvation;
target=sha256:e06a4582c5fba9f6b63402731ef0d94c9ecb052fe2bf0694391c197c4d927ab1
- SFT-CHEM-THERMO-015-SOLVATION-0228: class=solvation;
target=sha256:b2563e92b9afa2fdd0ffdfbe1a20aaf5a5015b3ccff8a3026a5a7b1e1b3766aa
- SFT-CHEM-THERMO-015-SOLVATION-0229: class=solvation;
target=sha256:56f27136b0e087295c048740416e5a318a8466073caf20d6ca1be5a97354c9ca
- SFT-CHEM-THERMO-015-SOLVATION-0230: class=solvation;
target=sha256:6a8749408ffccde886983a40ed2420764db714bb41b1af936fe29d68ac323793
- SFT-CHEM-THERMO-015-SOLVATION-0231: class=solvation;
target=sha256:f2ec30e7048f7d706089690939584c497bd3303cd908d935b259eca34652306d
- SFT-CHEM-THERMO-015-SOLVATION-0232: class=solvation;
target=sha256:0d562f60b1ca057097b0f44137a788d7ffa8e3661fad4010f9567b0d1b3c1d41
- SFT-CHEM-THERMO-015-SOLVATION-0233: class=solvation;
target=sha256:1112dccef8a6e182d8b44da7f2929d473ba1e442888dd8cfed07e72940d6995f
- SFT-CHEM-THERMO-015-SOLVATION-0234: class=solvation;
target=sha256:b387ddc629c411b4713a07db08a2d143786a10d9b47df86490ad74dc96dfa5ff
- SFT-CHEM-THERMO-015-SOLVATION-0235: class=solvation;
target=sha256:73b24502549f21203ccd9fbfe731e3a6d333de209e969e8f10e45772862f047c
- SFT-CHEM-THERMO-015-SOLVATION-0236: class=solvation;
target=sha256:b869b1410b400b527154bef80925abd3d7fa31bb9cc1faed965e6091c7b15ed3
- SFT-CHEM-THERMO-015-SOLVATION-0237: class=solvation;
target=sha256:6a68d076a60e3d2cab911f98886bfb648c7027f82e3bc3c6fdd28b293144bd55
- SFT-CHEM-THERMO-015-SOLVATION-0238: class=solvation;
target=sha256:baced9f1b79be5deec92c012942e083090af068e97cb178fedc60eacea1f474
- SFT-CHEM-THERMO-015-SOLVATION-0239: class=solvation;
target=sha256:7d017130fc613a073ddb1b4d9f216915af816cc0698778fe3e7558c12aa69c1f
- SFT-CHEM-THERMO-015-SOLVATION-0240: class=solvation;
target=sha256:4f96ed91e5edabd3229384dfdf965bbba6f19b0eef318b5e6531502dc9f3f1d9
- SFT-CHEM-THERMO-015-SOLVATION-0241: class=solvation;
target=sha256:f78ae5bf33a0f5b67c553a54ebff4fc9f5ddb171917b8efd4dc67e2a4a58b2daa
- SFT-CHEM-THERMO-015-SOLVATION-0242: class=solvation;
target=sha256:7eb70935ca7214e0a42837efb1b7afa9ff0bf4e977622c7dba00ce4c5eaf9590

- SFT-CHEM-THERMO-015-SOLVATION-0243: class=solvation;
target=sha256:9945bfb5d52487d9b2a4c7c4c99542992b86e507d28ff4449e37b8e3e45b3e6c
- SFT-CHEM-THERMO-015-SOLVATION-0244: class=solvation;
target=sha256:ccb55b3a584ef98e4ca098ca5250c626f69365dbecdc643fc90bd857cbb5cb1
- SFT-CHEM-THERMO-015-SOLVATION-0245: class=solvation;
target=sha256:6cbe702fb880bfdab7692ac217f1b5d27624a04f41772c56c1e227149499cb61
- SFT-CHEM-THERMO-015-SOLVATION-0246: class=solvation;
target=sha256:3dcec529432e060f82e01ba579d61f9c7fc92b06451b1ba20055fff63e498293
- SFT-CHEM-THERMO-015-SOLVATION-0247: class=solvation;
target=sha256:80309b0b6669b244df66fcb79cc76bc35d7853946fb9062865bc953f02bd31e4
- SFT-CHEM-THERMO-015-SOLVATION-0248: class=solvation;
target=sha256:7baef0ca18ecf6a2de40787e28e91ca36d5a80abc42636070cba11d96c4d70a7
- SFT-CHEM-THERMO-015-SOLVATION-0249: class=solvation;
target=sha256:96ea4553d46b77bfb3121ef7c114444d807bece6a89092e28a620ec32760ceef
- SFT-CHEM-THERMO-015-SOLVATION-0250: class=solvation;
target=sha256:ec5afb5bdea0157f9be2e8f5980f9069b22b48fba9a9b073e3d535d1e64a6b3e
- SFT-CHEM-THERMO-015-SOLVATION-0251: class=solvation;
target=sha256:fdb65b553bbbe8dc4f5ae2d467eae883590e66871ee9192b1b025effc0ec1f78
- SFT-CHEM-THERMO-015-SOLVATION-0252: class=solvation;
target=sha256:84d765233901058261703217c74d16df47e1e85de7fbd9b21d7d9dff30aebb70
- SFT-CHEM-THERMO-015-SOLVATION-0253: class=solvation;
target=sha256:ed6dff0938210bcfea435e6166b99049e08191b33ef2f1367eb76c76982a41c5
- SFT-CHEM-THERMO-015-SOLVATION-0254: class=solvation;
target=sha256:3ed752d2db56fc6e8dfa90d91ce7e553e22aed88ebf9f1ac76793932023eb31f
- SFT-CHEM-THERMO-015-SOLVATION-0255: class=solvation;
target=sha256:ee5f25a8f13cca980b84dd93bb397db65224a73121b241099c0a7edab91dd8e
- SFT-CHEM-THERMO-015-SOLVATION-0256: class=solvation;
target=sha256:8516264c493284d93ac622f56358a30ebcd32e05b8d14b5c35d9fe3a4070f74d
- SFT-CHEM-THERMO-015-SOLVATION-0257: class=solvation;
target=sha256:900808371d364ff9ca8e8a7b2e20fcf7b4902d04c7825e65c3042dfbc819123a
- SFT-CHEM-THERMO-015-SOLVATION-0258: class=solvation;
target=sha256:20d62f6554bb64a1a1f85b31620187ba32c56f588ffb5f282a405af9103e4088
- SFT-CHEM-THERMO-015-SOLVATION-0259: class=solvation;
target=sha256:6cda09019972d6af07076531f2e48edd53f434a041a7740e022e9c40b46b5fae
- SFT-CHEM-THERMO-015-SOLVATION-0260: class=solvation;
target=sha256:1e8e062c0f50941fd61dc13b799c8a6316808468ac75946d87582bb81dff9887
- SFT-CHEM-THERMO-015-SOLVATION-0261: class=solvation;
target=sha256:8888aced5090e7fetc6da68209e65e03c5c6d3f06b6bba5f8d0f69d478916cf
- SFT-CHEM-THERMO-015-SOLVATION-0262: class=solvation;
target=sha256:57dcd22c14926adacbb2c550b968a351399ca551ab0001e043cb245f30d93f3
- SFT-CHEM-THERMO-015-SOLVATION-0263: class=solvation;
target=sha256:796c69347fd638ffab5cdccfb80851933a3cd1509e54ffc695efd1e0e3cbda11
- SFT-CHEM-THERMO-015-SOLVATION-0264: class=solvation;
target=sha256:24c7c6e8cf9cb04f176c1aac365f196aa157ee1507772ed5412f3b21528cfdbc

- SFT-CHEM-THERMO-015-SOLVATION-0265: class=solvation;
target=sha256:1a526bea285ba4fc954bc87667b4a8eb4a9f603bb60d3bcde63d4ede5a8fa4d8
- SFT-CHEM-THERMO-015-SOLVATION-0266: class=solvation;
target=sha256:2db777321cefe4eb163a81da69d706859bde69e444ddb4296b3474ebe7cd48ee
- SFT-CHEM-THERMO-015-SOLVATION-0267: class=solvation;
target=sha256:247822c2bede26868bc61c043f724a921454d3a66b2aa65f70f0a1ccefb02619
- SFT-CHEM-THERMO-015-SOLVATION-0268: class=solvation;
target=sha256:e4dfb24f48dd9b795dbd4d901bf81aca18456c6a34a5c5718ff4d5db5fc7b8e2
- SFT-CHEM-THERMO-015-SOLVATION-0269: class=solvation;
target=sha256:2b6a568e3909a30c7bac75915ecddb2a4f0d5f1366ad20d93448bb0198eebb12
- SFT-CHEM-THERMO-015-SOLVATION-0270: class=solvation;
target=sha256:217d2927c26352ccd58e0bdc1713cc883317d451149f3bca6a69eeef048f4bdb
- SFT-CHEM-THERMO-015-SOLVATION-0271: class=solvation;
target=sha256:c309c14d0b57118b76042a325a0607ebc1a2e0450ae02a9794e5e5ef705770ff
- SFT-CHEM-THERMO-015-SOLVATION-0272: class=solvation;
target=sha256:67049c892d202bba5c243a846e2f2b7ae3eef256f6d556a47539f5291987b04e
- SFT-CHEM-THERMO-015-SOLVATION-0273: class=solvation;
target=sha256:3197576110045eb139d63c9cc8e7dd7f7d5f235ebe49f94c7c0f987e1bae1c1e
- SFT-CHEM-THERMO-015-SOLVATION-0274: class=solvation;
target=sha256:3e11b06462a7e040507fd5809805f6fc447a1800a98854639a7fe8c4318486a0
- SFT-CHEM-THERMO-015-SOLVATION-0275: class=solvation;
target=sha256:71892fc5dc86270e97dfa727713efe8cef8202c961f54dbc369cfbec69d9b506
- SFT-CHEM-THERMO-015-SOLVATION-0276: class=solvation;
target=sha256:0ba79526d60e1eb005f73e87ac05aca3cbbd9755f818c8c8d008f96f8bf5621a
- SFT-CHEM-THERMO-015-SOLVATION-0277: class=solvation;
target=sha256:7b10321906f289e541145e681ac9ff50cc91b48465930108d90280a63d63c57d
- SFT-CHEM-THERMO-015-SOLVATION-0278: class=solvation;
target=sha256:cabc025d491f79c6e098309578d756994e495214fb4d9dba939164a99229102c
- SFT-CHEM-THERMO-015-SOLVATION-0279: class=solvation;
target=sha256:5eb2264b33d06400e7e4d1666858be413c6387a4b9590945851b985d797b6a6f
- SFT-CHEM-THERMO-015-SOLVATION-0280: class=solvation;
target=sha256:1fbe771f2d686004f6f552bf14a3c5835dfad07518ddba31192140ec7c6e888e
- SFT-CHEM-THERMO-015-SOLVATION-0281: class=solvation;
target=sha256:78a91693008202b3d5a27545649ce1c1fc596d38cdbc81defee92b2966d0bd9
- SFT-CHEM-THERMO-015-SOLVATION-0282: class=solvation;
target=sha256:2e7bd1e074042a740680e3655ebc4976ce9f1636ed6cf89538d77ac02ce15232
- SFT-CHEM-THERMO-015-SOLVATION-0283: class=solvation;
target=sha256:2d90552fe2bbbcc15c8c08220020e1c7d2316063674b759a2bc4f0a4b7997144
- SFT-CHEM-THERMO-015-SOLVATION-0284: class=solvation;
target=sha256:85abdde60aa8ed7dab3e5891cec3b203fd243e99ae140043d7422afdbc8604d
- SFT-CHEM-THERMO-015-SOLVATION-0285: class=solvation;
target=sha256:87bb30ad7bade194ab9af326eb55610fd83d95e39bedb552d552456806369b91
- SFT-CHEM-THERMO-015-SOLVATION-0286: class=solvation;
target=sha256:7440b4c2f459b08f80ff79f6393b7ebd9d632eccf9178849e0dbae8e32e352a3

- SFT-CHEM-THERMO-015-SOLVATION-0287: class=solvation;
target=sha256:80f8a7aa081691718e30b8f96a0f6cbb0858c4290c7c1023eb250f1b09a2e521
- SFT-CHEM-THERMO-015-SOLVATION-0288: class=solvation;
target=sha256:7bc99fd58e0a5524776db328f52204d14cfb9c4c2ceeb81976e1506aed4adbb8
- SFT-CHEM-THERMO-015-SOLVATION-0289: class=solvation;
target=sha256:47d74383e70b7cd4bdb85d71e869080219a8b0dddc927b1c8b81fca94ca59351
- SFT-CHEM-THERMO-015-SOLVATION-0290: class=solvation;
target=sha256:4a9856ae2d0e5cfb259a3d79aa6143b0cd86e0d8582f6f96ebca3cc6e5b015f2
- SFT-CHEM-THERMO-015-SOLVATION-0291: class=solvation;
target=sha256:61b3f6e126c490aa38f72d99da1eee6c3b63e34ac98c3233f7f79076199a7ee5
- SFT-CHEM-THERMO-015-SOLVATION-0292: class=solvation;
target=sha256:d9e23e145754c0de2a3f582110aff9e2b5dd8932797b6c41893aee3d0e870fbc
- SFT-CHEM-THERMO-015-SOLVATION-0293: class=solvation;
target=sha256:7bf0fe9fd184f66b5b9165b02b84a0037f39613a492c5d1736f799edb7539b71
- SFT-CHEM-THERMO-015-SOLVATION-0294: class=solvation;
target=sha256:612d29ae5c34f9f7c37c06c4d29d30144251ab10c33285b6f3c8893bbcc52bd8
- SFT-CHEM-THERMO-015-SOLVATION-0295: class=solvation;
target=sha256:5011bb52f1d14a435d56c7b7eba1d0638fd8aae27ddf5062ea1da9d32ed207b5
- SFT-CHEM-THERMO-015-SOLVATION-0296: class=solvation;
target=sha256:1ad07b2db7629b5f061d26f09110f8df5aeb78470c3b3081b88c7429df958180
- SFT-CHEM-THERMO-015-SOLVATION-0297: class=solvation;
target=sha256:a12e7d47604463e060fc3f8ffd28f198922e4fe25bd5f5e4bd68673fbe13a8f7
- SFT-CHEM-THERMO-015-SOLVATION-0298: class=solvation;
target=sha256:892ee942f0fdb3daf3ef1f326762895908b285996bb3adba0c8b4dbdcaf71642
- SFT-CHEM-THERMO-015-SOLVATION-0299: class=solvation;
target=sha256:3bb62401a5d3ff791beae8e2601d4557bf4bdc5119ae57b985bc771e887ef403
- SFT-CHEM-THERMO-015-SOLVATION-0300: class=solvation;
target=sha256:9fe438bdbc42b1eb9ff3bfb187e4b2786e1ab6c190b0bf44922e71e07701c8a2
- SFT-CHEM-THERMO-015-SOLVATION-0301: class=solvation;
target=sha256:63dc99a33885e3722ef8ccf5cac2275372054f248fdec8e02ffd9f5875fa22e5
- SFT-CHEM-THERMO-015-SOLVATION-0302: class=solvation;
target=sha256:d4837abf67772bea420c6dcae1c211d44f161ff8d3e96e025f51b29fc2c52e4e
- SFT-CHEM-THERMO-015-SOLVATION-0303: class=solvation;
target=sha256:6da6ee864ded5e2b197fc0ad54e8d569202585819f2159ac99df413ca774f6e2
- SFT-CHEM-THERMO-015-SOLVATION-0304: class=solvation;
target=sha256:a8f893412801081b68e16bea7155126e854dbf4607a24eb37ac082c58b3acfb
- SFT-CHEM-THERMO-015-SOLVATION-0305: class=solvation;
target=sha256:66778171af91daf18538554d21c53646e4e66786c01481d81ebc60fe11c04896
- SFT-CHEM-THERMO-015-SOLVATION-0306: class=solvation;
target=sha256:b520c429e20601ee7e730d71567afb96ad71528be1e5b549327abc821d8fd309
- SFT-CHEM-THERMO-015-SOLVATION-0307: class=solvation;
target=sha256:0626a49e70b0d2b1f7c3d646c7330c9a553b0454060183858bf9dce5f96de52c
- SFT-CHEM-THERMO-015-SOLVATION-0308: class=solvation;
target=sha256:c3efd48cb0fd8efdc3de38dec78db8119bf3b512598d526f71d0f6598283af1

- SFT-CHEM-THERMO-015-SOLVATION-0309: class=solvation;
target=sha256:7f7998cff444a443b9f2c7172f3fdea5637cd39f2babaa01deb03933dcd113
- SFT-CHEM-THERMO-015-SOLVATION-0310: class=solvation;
target=sha256:74d19e45f1d369410e5a172b7f8b0b3501ea103fc5267b946b6c37dd4f858890
- SFT-CHEM-THERMO-015-SOLVATION-0311: class=solvation;
target=sha256:22c837da39d329d70aba2af701c35fc24776ce8b82ac07175f021ce9c4fb8a94
- SFT-CHEM-THERMO-015-SOLVATION-0312: class=solvation;
target=sha256:69706303c1c9732a0610a26f4919f3441cb4e080f1036dae9e2093f491806302
- SFT-CHEM-THERMO-015-SOLVATION-0313: class=solvation;
target=sha256:c300eca57f2556bf88b37e21bb65c0b5ea817a18cb7f90834f32e5d4e1c958d6
- SFT-CHEM-THERMO-015-SOLVATION-0314: class=solvation;
target=sha256:d1a5049ab86dcb11952fc063b243abbbada4f92af92ab7ec7dc4d178488ecf28
- SFT-CHEM-THERMO-015-SOLVATION-0315: class=solvation;
target=sha256:05259d3f8878fbeat6080e600a0f760eb7e2da0f623ee3369b9c843ab4b69231
- SFT-CHEM-THERMO-015-SOLVATION-0316: class=solvation;
target=sha256:5d8797a77e6e96c23f8348905052d03ba69caf117cd85ca23a268fd90709d4a5
- SFT-CHEM-THERMO-015-SOLVATION-0317: class=solvation;
target=sha256:1ce50e2440979c87bed84e4e8af99eb9ae5e02b10e9cb762cbb043aa4f9b77e0
- SFT-CHEM-THERMO-015-SOLVATION-0318: class=solvation;
target=sha256:48dd903f29d5ce04a46f3361fc411aa70728df1bdd9acf2476b29011d3cb848a
- SFT-CHEM-THERMO-015-SOLVATION-0319: class=solvation;
target=sha256:8ffdedac3a23fbcf4f6768746355288bbbedf96692202e08e6ab8634d861652fd
- SFT-CHEM-THERMO-015-SOLVATION-0320: class=solvation;
target=sha256:eb782ead589f8d844085c44ad5b42822fb0c2ec7eafbbeca1f3b78cfaf9464c9
- SFT-CHEM-THERMO-015-SOLVATION-0321: class=solvation;
target=sha256:9aa08174340d8ea3d68cf32d3bc5efa48ecbd655bb1283869269bcea44f4c317
- SFT-CHEM-THERMO-015-SOLVATION-0322: class=solvation;
target=sha256:63c01cf80dadcfbfef49bdd992181d2fed3d82f4c6807e1cddc369f89e9b4929
- SFT-CHEM-THERMO-015-SOLVATION-0323: class=solvation;
target=sha256:70e78122f9c879a09bf308b2d088e23397fbd74abf242e9ada3061f2fae0b90
- SFT-CHEM-THERMO-015-SOLVATION-0324: class=solvation;
target=sha256:679625d093bfbfc3c339eb40245a6d0bdfd5fe1daa41fca2ea643f2fc0743e3e6
- SFT-CHEM-THERMO-015-SOLVATION-0325: class=solvation;
target=sha256:6d2f26c0fcf278dc7e3a0a716cbad9a6c2d9a46ee23ad2c9508d399814a5ca46
- SFT-CHEM-THERMO-015-SOLVATION-0326: class=solvation;
target=sha256:53ac8353c3521cd6a9bd83d6b573ede3ad722d67c88450efff9f5e833435ea23
- SFT-CHEM-THERMO-015-SOLVATION-0327: class=solvation;
target=sha256:04e27e6327e797f84244e58cc7a756557ab315a575a6707565d525b42fa4d70a
- SFT-CHEM-THERMO-015-SOLVATION-0328: class=solvation;
target=sha256:9a40f33dbfaa1ac81c24f93cc70aaf64201b70c50fdf3e33ca3cd2de968597d7
- SFT-CHEM-THERMO-015-SOLVATION-0329: class=solvation;
target=sha256:ffdf2a5ecf5449c557bebbc6a8314dd6f8c2923f0cfd92e3f39dfac95dc0772
- SFT-CHEM-THERMO-015-SOLVATION-0330: class=solvation;
target=sha256:0f6b729a731d9d728c56b91eae627da49ad8cdef2ca66f782f687a1d04ade02c

- SFT-CHEM-THERMO-015-SOLVATION-0331: class=solvation;
target=sha256:dfef0aede57bbebfff10d81c5a44e062305198e4548f5444d8823d14ef30f8ba4
- SFT-CHEM-THERMO-015-SOLVATION-0332: class=solvation;
target=sha256:8fc0128f2c4707e2f93e62979a3e68459fe5759723b792fc717a6bc034c07ce8
- SFT-CHEM-THERMO-015-SOLVATION-0333: class=solvation;
target=sha256:e1e256fc82c5b5a2cf052374abf4a6936ffc55415ceba527fe4c4af4cef7e4b9
- SFT-CHEM-THERMO-015-SOLVATION-0334: class=solvation;
target=sha256:5fd7c0ae27f1f5368218b0f0cf7839186e1b6305ec1080d7a1bc45ec8de029bc
- SFT-CHEM-THERMO-015-SOLVATION-0335: class=solvation;
target=sha256:1203fd41bd09f914b5e00e51d7e319d39bb8f8bbb9e074c6bbcb07e795af1d1
- SFT-CHEM-THERMO-015-SOLVATION-0336: class=solvation;
target=sha256:000c32e82682727be977886160f2dcaab7fd0fccc1a298767df0f3c4a0c794f
- SFT-CHEM-THERMO-015-SOLVATION-0337: class=solvation;
target=sha256:01f3477b16d78b6a6091dbd5e957b5ad8864584d30f8735ecf83a0d17ddda90
- SFT-CHEM-THERMO-015-SOLVATION-0338: class=solvation;
target=sha256:dc692da9c4bb8288e445752b6b9070cc3d7dd8c5784a1ab14d4502c32adc10e8
- SFT-CHEM-THERMO-015-SOLVATION-0339: class=solvation;
target=sha256:718f95b5ebf51243bd2473f8875cb0eef61e98ddaeb65342982a491b60f3440e
- SFT-CHEM-THERMO-015-SOLVATION-0340: class=solvation;
target=sha256:b64f28641ff665e6151be638ba2ffddb39bf7faf43b14b3e2861e50b9d4b4375
- SFT-CHEM-THERMO-015-SOLVATION-0341: class=solvation;
target=sha256:97f1a33b48bce05e58a6c6a8edc8e19552a7e4a7348a977d0a044e109ad3cfa6
- SFT-CHEM-THERMO-015-SOLVATION-0342: class=solvation;
target=sha256:c3aa48de52b4df2f2c6766b30f925c2cc46be19aba8819164d437cc4a8860166
- SFT-CHEM-THERMO-015-SOLVATION-0343: class=solvation;
target=sha256:524d8322c67a52dbc00ebe25cd51ca62bb7ffd40b37bef2799911647d11acc8f
- SFT-CHEM-THERMO-015-SOLVATION-0344: class=solvation;
target=sha256:1349402fb6bd4c20c5dd1262b8067e1c3252e7c09f12d4b8aaefcf3c9753bcb3
- SFT-CHEM-THERMO-015-SOLVATION-0345: class=solvation;
target=sha256:d9166a96bddec0c4822b3a33a103047390c29b98b7d4e968b7f949e72c552202
- SFT-CHEM-THERMO-015-SOLVATION-0346: class=solvation;
target=sha256:c4ec000dc328ab2f70402f5ac783573f22b19681d3c94f576b6c6779d6a86ec3
- SFT-CHEM-THERMO-015-SOLVATION-0347: class=solvation;
target=sha256:98a61543af6de6332b18f5a50380a76910794ee33534e570a717718d7760500b
- SFT-CHEM-THERMO-015-SOLVATION-0348: class=solvation;
target=sha256:4d6caab97368203660a366afc534b4403b39311e76e3f686d55429a105694c45
- SFT-CHEM-THERMO-015-SOLVATION-0349: class=solvation;
target=sha256:c5e5ba085855f32960ce14c19eb79217ff799ccd97225ed4499e0dc3ccb2fac4
- SFT-CHEM-THERMO-015-SOLVATION-0350: class=solvation;
target=sha256:a5169e2724428ebcb7add2508ad48dd1f9d338e2f9a05bd7685f22365f1fbdde
- SFT-CHEM-THERMO-015-SOLVATION-0351: class=solvation;
target=sha256:3fc8bf879b02de25c8f5d3b7675e455e495b3acd48d2925459bbeb7e1b625b84
- SFT-CHEM-THERMO-015-SOLVATION-0352: class=solvation;
target=sha256:39572a86a4ac3e5760a0c9be695ee543ab21186c6b7837713d489575c962d814

- SFT-CHEM-THERMO-015-SOLVATION-0353: class=solvation;
target=sha256:a3af94944b1aaaf18dc1dc75a2b3b7686aaaab6fff060a263c126dd7c6472063
- SFT-CHEM-THERMO-015-SOLVATION-0354: class=solvation;
target=sha256:50f6294e40f43d5c9e52f08cb34c297b44cbf0e3e581226ff38f8964841a7905
- SFT-CHEM-THERMO-015-SOLVATION-0355: class=solvation;
target=sha256:a9818a9da14e5b565ca6de76725d2faaf601487626043c2101ffaeb4c13e5847
- SFT-CHEM-THERMO-015-SOLVATION-0356: class=solvation;
target=sha256:c140ba003b6c3d3acc9eac1c50ac807c5a4c872ace7d0c3c2bf3d66823da5981
- SFT-CHEM-THERMO-015-SOLVATION-0357: class=solvation;
target=sha256:edb2b42f0b0fc8e22cd4c6955bb9822aed0292b3b0bfb3b42a18d23c16f15289
- SFT-CHEM-THERMO-015-SOLVATION-0358: class=solvation;
target=sha256:4e79fe5167d3e796123c4dae3dad11a3a6d10f7d3c70091f5ac2a78b08d861a0
- SFT-CHEM-THERMO-015-SOLVATION-0359: class=solvation;
target=sha256:908f911b70f1d16e95cf27b6192aa1a781ac832c8ec74b4bcd5bf001dc9c8ce
- SFT-CHEM-THERMO-015-SOLVATION-0360: class=solvation;
target=sha256:56479c51b751170909a20d4e2a58a55233557b5d371b977e27fb081a00bd42c4
- SFT-CHEM-THERMO-015-SOLVATION-0361: class=solvation;
target=sha256:8e7742137123125a637baec8b813601c789f797381f1c3f3ac58fb5fa0586633
- SFT-CHEM-THERMO-015-SOLVATION-0362: class=solvation;
target=sha256:e611cca5ffebbd5e30010f394bc20dfeea45b38ef3ee97a728a08cb0ba75d053
- SFT-CHEM-THERMO-015-SOLVATION-0363: class=solvation;
target=sha256:0d7be14fcef300012a7999bfec2b6038247765d2ed4628010dc1093b778b35bd
- SFT-CHEM-THERMO-015-SOLVATION-0364: class=solvation;
target=sha256:ab2c3e1257ab848d2ca11c4f6ee12c1669bca8c7f945922143c9df87142a1a4e
- SFT-CHEM-THERMO-015-SOLVATION-0365: class=solvation;
target=sha256:6a6669f2aff9886497cc4164cb6602e1a5d22e6aaf0e27c5e0c66f930b514179
- SFT-CHEM-THERMO-015-SOLVATION-0366: class=solvation;
target=sha256:9b48f7c16403705ec921cada0cd3f1ed82724ac0dd0c0cb619cbd7ccd3b600f9
- SFT-CHEM-THERMO-015-SOLVATION-0367: class=solvation;
target=sha256:92089be72a1de524cb0603ec768ab804040299a233a7e6483cd92b72173d7e8d
- SFT-CHEM-THERMO-015-SOLVATION-0368: class=solvation;
target=sha256:bdb9c1187201e78c3a16454cdd69e22ffb20ccd570e85c8ec330a8424492242a
- SFT-CHEM-THERMO-015-SOLVATION-0369: class=solvation;
target=sha256:159828be39a13bf01a571d58b9d02d29804d32fe8d77f4335aecf4cf097c0471
- SFT-CHEM-THERMO-015-SOLVATION-0370: class=solvation;
target=sha256:a20b88a823b0a98ed938cb911da7044550b2b75e944da30be7f32ad993a089ed
- SFT-CHEM-THERMO-015-SOLVATION-0371: class=solvation;
target=sha256:f52216e56be810dae8504c7c8d51a4ee09357189e2f4d8b41b551d449019568f
- SFT-CHEM-THERMO-015-SOLVATION-0372: class=solvation;
target=sha256:b098af580edd27ecc635fe1f08ab79179beaf6e1b39902038b00ad7da7b2d92f
- SFT-CHEM-THERMO-015-SOLVATION-0373: class=solvation;
target=sha256:59586ba21e6ad3aead6945b78b9e10e6469e2d9e10a62ed51be6325d4e9df783
- SFT-CHEM-THERMO-015-SOLVATION-0374: class=solvation;
target=sha256:22f1b5651560014ea6ae691b4bd6138f2a63fe80a3a508c84acf54f7cdef5d94

- SFT-CHEM-THERMO-015-SOLVATION-0375: class=solvation;
target=sha256:23a599bebd64481c544b62ff635846b02d231da4dedf46d17e390f6204b5bb32
- SFT-CHEM-THERMO-015-SOLVATION-0376: class=solvation;
target=sha256:2938468b30d15cbca8ac5f101e67957bade3a7f829b200f3cfd20a7d1f88230
- SFT-CHEM-THERMO-015-SOLVATION-0377: class=solvation;
target=sha256:00289c33ad61a913e53c7cda3f05ca11b498a6bca314f723c93811d0b8fc98c9
- SFT-CHEM-THERMO-015-SOLVATION-0378: class=solvation;
target=sha256:9be65601ca195bbb8767eb04d2624519424d45f17c9f042686e6a795b83354a1
- SFT-CHEM-THERMO-015-SOLVATION-0379: class=solvation;
target=sha256:202b22428f31e38e9c64afeda7344090bcb9faeaaec9e8f0863221a392ff975
- SFT-CHEM-THERMO-015-SOLVATION-0380: class=solvation;
target=sha256:64a7448aaada48c8a8a5fc7a8b8f9eabd8bbfdadaea3e7b72dc74a800826b299
- SFT-CHEM-THERMO-015-SOLVATION-0381: class=solvation;
target=sha256:6fff5759807a4f80c4f6f916302a386e7fa17bb07483c24a38ce8bad09547d10
- SFT-CHEM-THERMO-015-SOLVATION-0382: class=solvation;
target=sha256:2bc9ebb40d94d7bc937ee0bc59a4f0cc0f6fe20c62a90371afaf66e23b833dd3
- SFT-CHEM-THERMO-015-SOLVATION-0383: class=solvation;
target=sha256:e0d18ea4353ff7209cece6ef45e98ad9b84383f9cefa0f628dc7f7f362ee7d7d
- SFT-CHEM-THERMO-015-SOLVATION-0384: class=solvation;
target=sha256:8e8c46b3417426f67ae07d9ea31991c1ce48a97e0d5e8ed84e5889bbd26d4116
- SFT-CHEM-THERMO-015-SOLVATION-0385: class=solvation;
target=sha256:adc14929787a4a29cd98cad7359ba9ed5c30373a4fcc056695f994b2025bad25
- SFT-CHEM-THERMO-015-SOLVATION-0386: class=solvation;
target=sha256:367742f5f84c8bbdaa4451c4baff3321acb6289c1e86bec9128fab3ba1b35dc8
- SFT-CHEM-THERMO-015-SOLVATION-0387: class=solvation;
target=sha256:742f951fac2e1353fc6c81706d3029813973f6f03377e80f7cdebb18d79ab6a5
- SFT-CHEM-THERMO-015-SOLVATION-0388: class=solvation;
target=sha256:9685da7aa4d68c1af539e1d53ee69133122abaa2b7475f820a2e68a4d44c2218
- SFT-CHEM-THERMO-015-SOLVATION-0389: class=solvation;
target=sha256:805f99accd179366d19018ae0bbd5fcf8c277743aa6a0764a9a05a2dfa940566
- SFT-CHEM-THERMO-015-SOLVATION-0390: class=solvation;
target=sha256:46238df8a9ca4004c9214f82118f8f2429e20f956b0acd54550dfcf8176c29c6
- SFT-CHEM-THERMO-015-SOLVATION-0391: class=solvation;
target=sha256:f3aa26b57b1747884c9431ccda74108fc36385edd741cb30f4a2b11101e5d9c4
- SFT-CHEM-THERMO-015-SOLVATION-0392: class=solvation;
target=sha256:ea0cde43b3c46adedf5f00562b6663b741fd2a2874ecab5aa9c94004362193e9
- SFT-CHEM-THERMO-015-SOLVATION-0393: class=solvation;
target=sha256:d734e66fbd22c0e177c941c7c0b72eba7727bf02f9ceba3223466249849205d8
- SFT-CHEM-THERMO-015-SOLVATION-0394: class=solvation;
target=sha256:ed152ebaf73feabccd48c82d509a73c1665b9fb15732f4eba02e84528cafa071
- SFT-CHEM-THERMO-015-SOLVATION-0395: class=solvation;
target=sha256:c7579ac8b546ef4f9e087c9e0456a268b666b1e1752dfaf89b59933f76204de1
- SFT-CHEM-THERMO-015-SOLVATION-0396: class=solvation;
target=sha256:37cec92863ea0906fb59bdb4e378bb0a81519aca11438aa65bcba81fc68864b9

- SFT-CHEM-THERMO-015-SOLVATION-0397: class=solvation;
target=sha256:73d0a0e506df3449b5b1751ba2a93b3ff07f8d64d5f85afb4b5ac88332e8ee
- SFT-CHEM-THERMO-015-SOLVATION-0398: class=solvation;
target=sha256:73ed6687ba17a0651ff4400fdcba304ee2c465824d6ae064a06acb170c322e9e
- SFT-CHEM-THERMO-015-SOLVATION-0399: class=solvation;
target=sha256:faf1942775a4925faee7df09b1a165576e141b4990aa12f64d26b0c70125a186
- SFT-CHEM-THERMO-015-SOLVATION-0400: class=solvation;
target=sha256:adfb1a5e1e77810040c17ad734cd5cdd917082b72e10a6fb6502ea160377e0e2
- SFT-CHEM-THERMO-015-SOLVATION-0401: class=solvation;
target=sha256:27e6b05307439aa9b7d02f396f6281f446cc0e9ea661cec915e53ee8e26ce19e
- SFT-CHEM-THERMO-015-SOLVATION-0402: class=solvation;
target=sha256:414d65f8e1046ddfd8264ccd257e593a0942584ef4c52aadcc6a14d1d447a6fa
- SFT-CHEM-THERMO-015-SOLVATION-0403: class=solvation;
target=sha256:1b500e400f55b1c8c66e0b610448d698d75b30dc0f93d1fd20affecffb135e26
- SFT-CHEM-THERMO-015-SOLVATION-0404: class=solvation;
target=sha256:3f4af2fc6e1a7119d73569d27b95d0d08551a68cf2398df43de7be2ef8803f03
- SFT-CHEM-THERMO-015-SOLVATION-0405: class=solvation;
target=sha256:9fc8ff6e78ea7125c4a205381428c16f0b1182b4f9848a606c8ba21e0fca0c7b
- SFT-CHEM-THERMO-015-SOLVATION-0406: class=solvation;
target=sha256:d7aa8c3d9be2492f86a35fd540d3aef48d9ffaa4ddd2c3dbc8e7fb55b6071b42
- SFT-CHEM-THERMO-015-SOLVATION-0407: class=solvation;
target=sha256:714baadef166a89b48c26753c91d7c9e2be20aa13b7459e4bcfb0656452fb54f
- SFT-CHEM-THERMO-015-SOLVATION-0408: class=solvation;
target=sha256:b81f499bc3d30cebeb5943dcfb2e346f564bd33cc565553bf601872576323979
- SFT-CHEM-THERMO-015-SOLVATION-0409: class=solvation;
target=sha256:a92797cf0989226f1e28c1e03c3ddf1eaf8897da2baa872f64dee17517f2869f
- SFT-CHEM-THERMO-015-SOLVATION-0410: class=solvation;
target=sha256:fb1fdbfd3a9029f840941f89913c7aed2652e71357e1c7b3fd1ae0cc30db6d8e
- SFT-CHEM-THERMO-015-SOLVATION-0411: class=solvation;
target=sha256:d523ed0e91ea4c393dd8c960b5141e7dfcf86a35391fcf74462c32b7ed928204
- SFT-CHEM-THERMO-015-SOLVATION-0412: class=solvation;
target=sha256:96303cac3ce1ae057bfa5f47276108a034656ea79039019bea1b3531ea0aaf8d
- SFT-CHEM-THERMO-015-SOLVATION-0413: class=solvation;
target=sha256:1ae981d4ac048668a9fc6a3a880f867eeeb56e4cea90e4b85e93733f8e532fc7
- SFT-CHEM-THERMO-015-SOLVATION-0414: class=solvation;
target=sha256:cad6886e3d479ad8f158c7191b76035db59cb7937785bb540a34c334a4e6fad4
- SFT-CHEM-THERMO-015-SOLVATION-0415: class=solvation;
target=sha256:263a725c832c0ca5e93c916f8a028f56d1b114e17660ae331bea4ee826db0caa
- SFT-CHEM-THERMO-015-SOLVATION-0416: class=solvation;
target=sha256:a5bb475307dc42ccc37bfea1a740506328a8f61f691e670d425cb236f8564b15
- SFT-CHEM-THERMO-015-SOLVATION-0417: class=solvation;
target=sha256:17ab89034489d2879c46bca5afa12d433cba01d3b3c64a39920bf1f18b43b93f
- SFT-CHEM-THERMO-015-SOLVATION-0418: class=solvation;
target=sha256:069d165b60d1487f6dd09b9598509edc6360162bd62b5d820b095e5ceaa84367

- SFT-CHEM-THERMO-015-SOLVATION-0419: class=solvation;
target=sha256:aac036e6ab464087b333e0de8b73ffd76b224f4c500ca8ef61c372c25ff91677
- SFT-CHEM-THERMO-015-SOLVATION-0420: class=solvation;
target=sha256:7a0010a0c4812b313ac1f30980c3dd3fb5abdf6781379aec9c20945126a311f8
- SFT-CHEM-THERMO-015-SOLVATION-0421: class=solvation;
target=sha256:93631ebdb8b4f584d81afc030cc6d8de066efb5b7988ea528d902b5e4d86aafa
- SFT-CHEM-THERMO-015-SOLVATION-0422: class=solvation;
target=sha256:3ac287b603c7f206093aa4cd69f1e6820f1086c4b8854567fa236bfb19b68d4f
- SFT-CHEM-THERMO-015-SOLVATION-0423: class=solvation;
target=sha256:ad2156fd209e2c09dc72747e4a69e54d77393cd4417863f33ee431c587769d68
- SFT-CHEM-THERMO-015-SOLVATION-0424: class=solvation;
target=sha256:94acfab5a92cadb960f426dde1bde941d4bcfab62c0f7b1e03a59928683c9d77
- SFT-CHEM-THERMO-015-SOLVATION-0425: class=solvation;
target=sha256:dac883e84b9875fba8c3d10266abada56da8eb3cbc40efc6acf93763852af84c
- SFT-CHEM-THERMO-015-SOLVATION-0426: class=solvation;
target=sha256:c2d405bd4a4e1f11d43ccf1b551b30612f5b5c4c1fe20dc19e1933d4686e45ed
- SFT-CHEM-THERMO-015-SOLVATION-0427: class=solvation;
target=sha256:d306498f12e94ea43e38a4dff3e2b664e0fa9bde2b6bbf2773b7bce92291c15
- SFT-CHEM-THERMO-015-SOLVATION-0428: class=solvation;
target=sha256:be1528e44916c7c1feaf39a762e839714193510d7d61246502dec5448f1a6f5
- SFT-CHEM-THERMO-015-SOLVATION-0429: class=solvation;
target=sha256:9b11303a94a54a7f4a127c57552c740b2ed78c7b0cfe131def3616c58a2b54b7
- SFT-CHEM-THERMO-015-SOLVATION-0430: class=solvation;
target=sha256:20be2a75041c709d191b09822931931926d9d6d1482f7e639b19580d24fad243
- SFT-CHEM-THERMO-015-SOLVATION-0431: class=solvation;
target=sha256:8ba0936e5526775ceec1c0ac21237a4d380f64eb78e1fc67d750d0816e9806d7
- SFT-CHEM-THERMO-015-SOLVATION-0432: class=solvation;
target=sha256:fc90555e2e66bf3f68e76d926d8f2af01f6a3b780530b19b551c7105a3cdea6
- SFT-CHEM-THERMO-015-SOLVATION-0433: class=solvation;
target=sha256:6e492dc2d0e65c465f7cd71b78857f60880e6d5fd4655ad39be42c459cd51e91
- SFT-CHEM-THERMO-015-SOLVATION-0434: class=solvation;
target=sha256:c2ac2ed0af3fa39c171cc302d090a370a2d7ce3dca7828dc3e71fbb8c7644dfe
- SFT-CHEM-THERMO-015-SOLVATION-0435: class=solvation;
target=sha256:63ea3f82b3a1564a5fe7aad86db9c9f7c03e6944b7b2a2c43844fb6b3ae6d2d3
- SFT-CHEM-THERMO-015-SOLVATION-0436: class=solvation;
target=sha256:d704bf25f14c1667e2d346b8fe2032c3d3d5f63a95a34e3cef5f50cee79c0aa8
- SFT-CHEM-THERMO-015-SOLVATION-0437: class=solvation;
target=sha256:c70df610adf4a4672e339c62a23ccf288d5c44033f689d059a95baeae901ab31
- SFT-CHEM-THERMO-015-SOLVATION-0438: class=solvation;
target=sha256:bc9e9a7aec35ab8097d7d4c6f0fc2b4d10c86bcf82af53f9701816dbf828e10a
- SFT-CHEM-THERMO-015-SOLVATION-0439: class=solvation;
target=sha256:abff78321176e7776d819a084b7c53918bb4429696ec0d84985416a73ffb7d77
- SFT-CHEM-THERMO-015-SOLVATION-0440: class=solvation;
target=sha256:e36460d8a60d3a8d85ea8ba8245ec9d9ae6dae67e4a81fc01f062377586905d9

- SFT-CHEM-THERMO-015-SOLVATION-0441: class=solvation;
target=sha256:b271aaa856c2c7e3e8cfc7ddd1a1ed263e8c50724d176057767dc35720b7b1e2
- SFT-CHEM-THERMO-015-SOLVATION-0442: class=solvation;
target=sha256:83aa64b9d383f2929bb1f80b9495a0349b59539f174358deaa3f22deb7370aa2
- SFT-CHEM-THERMO-015-SOLVATION-0443: class=solvation;
target=sha256:a6214e8d1812d8281c538b644ea60c5c7ad2f5140c5acc80379527330fbf38ac
- SFT-CHEM-THERMO-015-SOLVATION-0444: class=solvation;
target=sha256:01d1331d42f4103ca81b64b7e2681f5209b282d3396eca2b56dc24ced8c953ea
- SFT-CHEM-THERMO-015-SOLVATION-0445: class=solvation;
target=sha256:12ad1493bbe01978b9b28a710af8b9500d6a0a02730ba8e81067b6db29937f3a
- SFT-CHEM-THERMO-015-SOLVATION-0446: class=solvation;
target=sha256:b652dedaea587fe0f9738ce079f33ba94e4f9882cbd02253b99c3f99b934b2d0
- SFT-CHEM-THERMO-015-SOLVATION-0447: class=solvation;
target=sha256:e1d60fcb9c602d662d9eb6ed9c526c5aa1cc7c8b9600ed1d470aecb8a278e2cf
- SFT-CHEM-THERMO-015-SOLVATION-0448: class=solvation;
target=sha256:1c5f43eaac5927eca687524596e57879d3ef798efe8894ffec299188321f7efb
- SFT-CHEM-THERMO-015-SOLVATION-0449: class=solvation;
target=sha256:2a84adb43d834e1df317c14b6ed208fed1a882e04bfd2663a0c700f7ccec6e7
- SFT-CHEM-THERMO-015-SOLVATION-0450: class=solvation;
target=sha256:c54aa44cd5dd63d1afbc0f9eca238365f85fda9aa4378c6768474a16e1f75add
- SFT-CHEM-THERMO-015-SOLVATION-0451: class=solvation;
target=sha256:dfa3ba990921672b0cf1e00af3401b19504c16247d0f4f9eea2c4e9b403fd757
- SFT-CHEM-THERMO-015-SOLVATION-0452: class=solvation;
target=sha256:1a67ac1b40709306950f5f55ef859a383dc821a56592a7ac9b93bc07bc3fc892
- SFT-CHEM-THERMO-015-SOLVATION-0453: class=solvation;
target=sha256:80744f40cf89b3f593e2a11180eed2e283dd887f45ff76f8d10078e8af93446a
- SFT-CHEM-THERMO-015-SOLVATION-0454: class=solvation;
target=sha256:c02b946e45c6913f20deebf2970a9add06cb3fe0f0efc244c0ba78ff26e90c8d
- SFT-CHEM-THERMO-015-SOLVATION-0455: class=solvation;
target=sha256:bfd11378b1f8d8b2404118d9975852471452787fd2c51dd0017fdcf86aa20dc6
- SFT-CHEM-THERMO-015-SOLVATION-0456: class=solvation;
target=sha256:b740cec416899d8e1041a7e233b96955263d69712df8ac66b4e85008f7a6a13b
- SFT-CHEM-THERMO-015-SOLVATION-0457: class=solvation;
target=sha256:5185587693c02ac7efcbc0c80be5e9baf2a78a1320fd07f9edde98e9bf0cc692
- SFT-CHEM-THERMO-015-SOLVATION-0458: class=solvation;
target=sha256:952a7e343fefbd20cd4e811128248cd18d327208343a60533c3685b9d207c952
- SFT-CHEM-THERMO-015-SOLVATION-0459: class=solvation;
target=sha256:f1aa615c74087d90832e12fb2bac6399d9caac37d890c7c94d980c2c0ed3fd3b
- SFT-CHEM-THERMO-015-SOLVATION-0460: class=solvation;
target=sha256:0ea390fce2aa75f6ce13032c95e73384cfaa1fbb42204e932f7844a5b1384f82
- SFT-CHEM-THERMO-015-SOLVATION-0461: class=solvation;
target=sha256:1eda162f6dd73e1b82b577d4945b996db251d8c83825ea48ef0979d39938e37d
- SFT-CHEM-THERMO-015-SOLVATION-0462: class=solvation;
target=sha256:5756b15d7149928dfe6fdb32655929d6cbf7bcb45ce1a0d709f4f14d5d06243

- SFT-CHEM-THERMO-015-SOLVATION-0463: class=solvation;
target=sha256:71747e7ee1fceb9b701237a3513071bedcf8085aa7866772345699d39d347f23
- SFT-CHEM-THERMO-015-SOLVATION-0464: class=solvation;
target=sha256:a6f54401e816ecfeb017c72cf6881c2fdceaea5323cf61c3db0a7636942d2711
- SFT-CHEM-THERMO-015-SOLVATION-0465: class=solvation;
target=sha256:1312c4ca9db3ac5a50af4813eed0d4f43e5e35a887cb0798ed87380ad53fa187
- SFT-CHEM-THERMO-015-SOLVATION-0466: class=solvation;
target=sha256:26c2512cc45be9f3e66b72e680e712efd8c5e1f0771f307eda818536bd10380b
- SFT-CHEM-THERMO-015-SOLVATION-0467: class=solvation;
target=sha256:d64906e342a6e61e3709fa3ad64810c804474b0a2df7ba2be2898a64975c92ec
- SFT-CHEM-THERMO-015-SOLVATION-0468: class=solvation;
target=sha256:dfc2dea0f92e0650682da663141f63f810577628cbede713a3b1896ba01196fb
- SFT-CHEM-THERMO-015-SOLVATION-0469: class=solvation;
target=sha256:ea28d86d02f78957784ea2856bd9a0b83408660265dadc51b74cc1a320b952ab
- SFT-CHEM-THERMO-015-SOLVATION-0470: class=solvation;
target=sha256:2edfbd305dd064784ad83e61cf933c876afd36dd057fd2e6986f3184a07c71ea
- SFT-CHEM-THERMO-015-SOLVATION-0471: class=solvation;
target=sha256:776efbf122c05a3235972743b95192564f2a576bf0146ca839870b9ee8d55d5f
- SFT-CHEM-THERMO-015-SOLVATION-0472: class=solvation;
target=sha256:2c227d66962ca0c9d48ee584377a7913a7a940bdad3a5a712102a8af354368ba
- SFT-CHEM-THERMO-015-SOLVATION-0473: class=solvation;
target=sha256:ae68a491b0d69fbfe833523ac30d6fbcf32e5d6169ef301c2d720d2d8cb9338
- SFT-CHEM-THERMO-015-SOLVATION-0474: class=solvation;
target=sha256:f819f346db2bec69cba54ffc4a8d5a4c407e3ea221abad5be2d3a42d4df758fa
- SFT-CHEM-THERMO-015-SOLVATION-0475: class=solvation;
target=sha256:4ee99d7696e16da3278af71f847edbf6ec8dc2d4984db55d8230d637196a23cf
- SFT-CHEM-THERMO-015-SOLVATION-0476: class=solvation;
target=sha256:3eaf074cd05768299bd9ed8ec99445f19fe53652674890ba325cf97d46355911
- SFT-CHEM-THERMO-015-SOLVATION-0477: class=solvation;
target=sha256:af6c7bcdbac4beec0407858beab99a7679032cd1867ff9cfe4012f64f34e0766
- SFT-CHEM-THERMO-015-SOLVATION-0478: class=solvation;
target=sha256:f9ba38a15e352041b4e9840c31ea6376ca499ca7e2bbcd92c4b704701c1b558e
- SFT-CHEM-THERMO-015-SOLVATION-0479: class=solvation;
target=sha256:07969bcee11df6fb54a5140eac10d17ca83966d509c11c4aa524fe6088c87f23
- SFT-CHEM-THERMO-015-SOLVATION-0480: class=solvation;
target=sha256:06c4a7d9919c71e7891013ff5680ed5d5d74c1c19053f3d9a3e79aa920475b6c
- SFT-CHEM-THERMO-015-SOLVATION-0481: class=solvation;
target=sha256:cd9f2a1e9d3f29cbaa5dc3605c026937cb0e21d450ebd3d2ef54526fcc4142ab
- SFT-CHEM-THERMO-015-SOLVATION-0482: class=solvation;
target=sha256:97fb37aaaf0c6068874491ed5cec3bb4c5d5772daae76aff615a44c5c654eabf
- SFT-CHEM-THERMO-015-SOLVATION-0483: class=solvation;
target=sha256:cd8c560f32b00c2f25f2971006e5e623254ec5fedc1df3b793df28d49a135872
- SFT-CHEM-THERMO-015-SOLVATION-0484: class=solvation;
target=sha256:f9a4814d148f253a495cf4d618064a53f86b9fdefc8d2b4682208e6b56a6e12d

- SFT-CHEM-THERMO-015-SOLVATION-0485: class=solvation;
target=sha256:084d5120a2694d1ec723ff1b9fb0984ac042e142076872daa8b80c76695670df
- SFT-CHEM-THERMO-015-SOLVATION-0486: class=solvation;
target=sha256:68ff38940199a94c24c4ad5efb67bf78d97e61eaddbac542664220fbe3357a78
- SFT-CHEM-THERMO-015-SOLVATION-0487: class=solvation;
target=sha256:4a40adf5469c224b0ab481e0442d1c4bfd1b9bde123184978c1fcea053a984
- SFT-CHEM-THERMO-015-SOLVATION-0488: class=solvation;
target=sha256:2f70e4550169424d63ad33e5b5ecfdcf8866ee28e25b21041fd061a28fdc023e
- SFT-CHEM-THERMO-015-SOLVATION-0489: class=solvation;
target=sha256:4f4b127cc1c00fce10672f8dc1392acec0d491f98df4abd3ce7ce00d32ff8e34
- SFT-CHEM-THERMO-015-SOLVATION-0490: class=solvation;
target=sha256:1a72f451f9dc835cdfc6d31ae153855c3ea1e0d1f4d9bb025837953b88b8f41c
- SFT-CHEM-THERMO-015-SOLVATION-0491: class=solvation;
target=sha256:68857ccd53092112fa66cc6dd4f96b0017df3953fde0e809ec9b7ad9a46c3a06
- SFT-CHEM-THERMO-015-SOLVATION-0492: class=solvation;
target=sha256:23a660e0101c5a98ac8b0be0fa5bdf7608bbe985fdf40c3f9672a70a51925be7
- SFT-CHEM-THERMO-015-SOLVATION-0493: class=solvation;
target=sha256:3e5e3deac044d90929a115113077c2bb3ff8d18a65c1cf0f5dd3d490ea0338c9
- SFT-CHEM-THERMO-015-SOLVATION-0494: class=solvation;
target=sha256:f9e486aeaeb0ba836887e6e195e8d6890a7529b3441036ccb08e822d84b85922
- SFT-CHEM-THERMO-015-SOLVATION-0495: class=solvation;
target=sha256:90ec6de5d2cfcec99f7915973090c80f5b0a13a4628ca0720593b700f7d076a3
- SFT-CHEM-THERMO-015-SOLVATION-0496: class=solvation;
target=sha256:fb769b0c9c37555fcd7a09b211fa2b1dfff9cdc5f5f5fae8bc4f640270840318
- SFT-CHEM-THERMO-015-SOLVATION-0497: class=solvation;
target=sha256:1ef6a2e730799c840c876951b69a44e4b9bc49ad026b903500b8d4c297aea9fa
- SFT-CHEM-THERMO-015-SOLVATION-0498: class=solvation;
target=sha256:cde5bfea98ab0fb68ed8fe7df9b4468c801dd7facf0cdb125398a73b7649eb19
- SFT-CHEM-THERMO-015-SOLVATION-0499: class=solvation;
target=sha256:e3ce5aa26c1f2082a55147f7dd577683ee3f1513a945e63140c083decbe40df
- SFT-CHEM-THERMO-015-SOLVATION-0500: class=solvation;
target=sha256:6e55ed4786ca332fcf43a1b3ea32136c2d92edd1e7c8d18a824b7efcac127904
- SFT-CHEM-THERMO-015-SOLVATION-0501: class=solvation;
target=sha256:9fa08496df95f97148f1e35278c2bd4fda51986edd7af7bfb090f2a64441a51f
- SFT-CHEM-THERMO-015-SOLVATION-0502: class=solvation;
target=sha256:6cd9c69632bae8bc6b269a8286e0dbc251ab849119a8a38177f323d2052bfa38
- SFT-CHEM-THERMO-015-SOLVATION-0503: class=solvation;
target=sha256:b80d1996e7c8bb151a1c965d22f0a8f3d4805148128bae7d3329f5dae3d5e9b6
- SFT-CHEM-THERMO-015-SOLVATION-0504: class=solvation;
target=sha256:fad5420e6d672c3b14350389080c1a8d4073dd9331fb43e471d923c5fb3314dc
- SFT-CHEM-THERMO-015-SOLVATION-0505: class=solvation;
target=sha256:2e3baf3b22dc2ef69fdacf91d3f9514445808a9091415ea6e6eba9cf427d5b57
- SFT-CHEM-THERMO-015-SOLVATION-0506: class=solvation;
target=sha256:77973cb0fa854367592dde7e479b373537e1002c336d066f79539b8d5bea3c24

- SFT-CHEM-THERMO-015-SOLVATION-0507: class=solvation;
target=sha256:92c36dc95e9c4d31d1e9c9bcf60380768a34156dcb46f948c823596a4b8bc054
- SFT-CHEM-THERMO-015-SOLVATION-0508: class=solvation;
target=sha256:f9e36ad1f32a3cf5e56e045a42f6f5cfd957d869655eb256b09f2ae9c1812de7
- SFT-CHEM-THERMO-015-SOLVATION-0509: class=solvation;
target=sha256:ed0dbb0658ccd5d721ba91b1009f41d6b62cf45155d06914a9e59e5a4669ced6
- SFT-CHEM-THERMO-015-SOLVATION-0510: class=solvation;
target=sha256:00b2424be7bf34aacb1852d9fc3bc80ba63c817226d855767a4753a1b89d4125
- SFT-CHEM-THERMO-015-SOLVATION-0511: class=solvation;
target=sha256:27f7000a444850545a09e5181d4d2301604c55dcc2eb781cfc56dbe81941059d
- SFT-CHEM-THERMO-015-SOLVATION-0512: class=solvation;
target=sha256:6dcea8ea6bfd442bd3476b8cc59ee725e31fb603ceac9e17447eb9da2ace0663
- SFT-CHEM-THERMO-015-SOLVATION-0513: class=solvation;
target=sha256:ecac01ad1788709dc07a5362b945cec65a62cfcb8264766aef37d6c74f2d2b84
- SFT-CHEM-THERMO-015-SOLVATION-0514: class=solvation;
target=sha256:283178beedf643f533ba078c4292aebc9bd69a6302cc5d1f86c80228bcab5df8
- SFT-CHEM-THERMO-015-SOLVATION-0515: class=solvation;
target=sha256:002bd0b6642f816d8cb2e7d2abe460427ee22fac739c86887b1019780fa79bb7
- SFT-CHEM-THERMO-015-SOLVATION-0516: class=solvation;
target=sha256:dbd8cf0c4772c6584446585aeb58cd2e26ef46ba4c1b5cd89c5970558a49f11f
- SFT-CHEM-THERMO-015-SOLVATION-0517: class=solvation;
target=sha256:e7814a02204898ec22cd750a982d71787341af88852761eeb0f0939c7238e69c
- SFT-CHEM-THERMO-015-SOLVATION-0518: class=solvation;
target=sha256:2483db875cb98b5f425a5cd8e7da10fb4eb4e54b8bfff7c0cb7f9a235270656eb
- SFT-CHEM-THERMO-015-SOLVATION-0519: class=solvation;
target=sha256:71d103302c6d164912dc727c12ff8df5fdf7a4ae712bd8e67588db505a04f691
- SFT-CHEM-THERMO-015-SOLVATION-0520: class=solvation;
target=sha256:7d2cd20b520474c9be587177620d673bc3dc92da97276dae16bbfa42660ed6a9
- SFT-CHEM-THERMO-015-SOLVATION-0521: class=solvation;
target=sha256:aa925ff8c3486dc6678f32be244224626e876762c13406be59dba4aa0a7f6e2a
- SFT-CHEM-THERMO-015-SOLVATION-0522: class=solvation;
target=sha256:dfa450b4bbce4476cd4c4ea3d77594be721764127180d056ff19ac62c5a38829
- SFT-CHEM-THERMO-015-SOLVATION-0523: class=solvation;
target=sha256:5820baa4d760b19a81bb6975d7e403a135ff3669b7c3eba786a3aef4c501bab4
- SFT-CHEM-THERMO-015-SOLVATION-0524: class=solvation;
target=sha256:90ad38073c1affd87acbbd712f8e144d11de237ac0c8a20e13caabe9bb469f8
- SFT-CHEM-THERMO-015-SOLVATION-0525: class=solvation;
target=sha256:1800e9b925d044a49ce3f77b9cd9c3523857544bf4df8c61221260b926f4cc70
- SFT-CHEM-THERMO-015-SOLVATION-0526: class=solvation;
target=sha256:dd6cb6ddae4c0ef4b6431f265107cd5501d67bd026f8c7a9ddf8d42a507ed847
- SFT-CHEM-THERMO-015-SOLVATION-0527: class=solvation;
target=sha256:e1cb77c801d3b58406f65ba159f1fd90a8b1c5f33ccd3ceb28f27e4b3165db1e
- SFT-CHEM-THERMO-015-SOLVATION-0528: class=solvation;
target=sha256:8dd978dd9359244807c683c8010e28dc64cba56fb8e4c61b16accd554d613459

- SFT-CHEM-THERMO-015-SOLVATION-0529: class=solvation;
target=sha256:375c328b9f0dce2ca43bfee8f9270721bfd0891f25c72e2dcfc20ffb5b83b1ae
- SFT-CHEM-THERMO-015-SOLVATION-0530: class=solvation;
target=sha256:b40c3a392b045732f071d30c26c0f2183963a66bc1357f31830836cb740fa44c
- SFT-CHEM-THERMO-015-SOLVATION-0531: class=solvation;
target=sha256:3819838a7ee0dc68859a413a65bf506e44aafc34b7a499afa01756caa0a8867a
- SFT-CHEM-THERMO-015-SOLVATION-0532: class=solvation;
target=sha256:52726e9b28f5993477a68a206f842a7d0ef97bda2e5affd20c70120e58ac14a5
- SFT-CHEM-THERMO-015-SOLVATION-0533: class=solvation;
target=sha256:3394bb808a2379b406771ae9cee17f1f612c5d260bdb3f3fe534539fb06ff2c5
- SFT-CHEM-THERMO-015-SOLVATION-0534: class=solvation;
target=sha256:55974fc65c5306cdab5cde020d70148d5be10f5b294f859812ba05a421357ab6
- SFT-CHEM-THERMO-015-SOLVATION-0535: class=solvation;
target=sha256:000e03405ca189762f8d512a8b65fe1fa6dcf453f1ac66da2cec4c9bdfd4d724
- SFT-CHEM-THERMO-015-SOLVATION-0536: class=solvation;
target=sha256:ec9d6c3cd72d8b626d4dbf8aa074a5ac1056df4de3b9147e8cdc942623631ad2
- SFT-CHEM-THERMO-015-SOLVATION-0537: class=solvation;
target=sha256:c645544150d57a60fb621b82bf6451e1d13e4675ba3e33b2fdc3b026c4001d23
- SFT-CHEM-THERMO-015-SOLVATION-0538: class=solvation;
target=sha256:c3011ca82ea3e9ed1557a727ed7ae4139e14d5ffc6360dc09e74f0e1a2c1766a
- SFT-CHEM-THERMO-015-SOLVATION-0539: class=solvation;
target=sha256:bd4f916334590f4a2480fce194024a4e3a7bedffb77f34c1ecfb0e916740c3a
- SFT-CHEM-THERMO-015-SOLVATION-0540: class=solvation;
target=sha256:8a8ef414ee92f8201b7cdc97cf0bd99a76a3772a567b9a37bbca30830673a327
- SFT-CHEM-THERMO-015-SOLVATION-0541: class=solvation;
target=sha256:a8c9084a7b40e7dbef271e22c2019333564eb67bdfe4184f8290a176779baac8
- SFT-CHEM-THERMO-015-SOLVATION-0542: class=solvation;
target=sha256:1fd028802cbbb1a09e6acbe8b6d92d113b42a1faa36706b35d7c42761b652873
- SFT-CHEM-THERMO-015-SOLVATION-0543: class=solvation;
target=sha256:4f88aca1ec4daa9042987c2d28f1925b8944a5c16bcbb7a56c3f955ee67b1b1
- SFT-CHEM-THERMO-015-SOLVATION-0544: class=solvation;
target=sha256:f306191583c07be8d7911385ef1b487f905a71768905195e169c8775c5d5d226
- SFT-CHEM-THERMO-015-SOLVATION-0545: class=solvation;
target=sha256:1949e9959ecde602021c7f69f5e1574f3fe1864699c6365e6760913fbe0703c8
- SFT-CHEM-THERMO-015-SOLVATION-0546: class=solvation;
target=sha256:2a57319b692105ce08560c22febf163e61bf2f232f42afe746b3da49ac504687
- SFT-CHEM-THERMO-015-SOLVATION-0547: class=solvation;
target=sha256:09d01a32fd6a4a7673ee15c537e980e172fb3941d5346e3e1c10d670d4aa4d83
- SFT-CHEM-THERMO-015-SOLVATION-0548: class=solvation;
target=sha256:3f79351a4995bb1b2106aeeb72902b4dccc03b190f43b98b2b94a51c878129b
- SFT-CHEM-THERMO-015-SOLVATION-0549: class=solvation;
target=sha256:ef636480a9a3b36da8506ef77d94f3c3853f80d903535dd354ec22a58218f7eb
- SFT-CHEM-THERMO-015-SOLVATION-0550: class=solvation;
target=sha256:bd8503ed4fb1662966ba5641cef9e5853a81a08f7fbfd745c728cbc52c3e168

- SFT-CHEM-THERMO-015-SOLVATION-0551: class=solvation;
target=sha256:a30b896cd7a89550a140c662f564d855cc2e60d21f48a5813da256420da03e5c
- SFT-CHEM-THERMO-015-SOLVATION-0552: class=solvation;
target=sha256:2e570ec3c3aed1428eaf2764ad59aea4767466eba9d26f257dc02eb247d60c93
- SFT-CHEM-THERMO-015-SOLVATION-0553: class=solvation;
target=sha256:bf27d467b847fc8aa8d52107fb460098f0aefdee0294915d6fd2291a8d72ccd3
- SFT-CHEM-THERMO-015-SOLVATION-0554: class=solvation;
target=sha256:e71a2b868655f34d7edfc72b9cf6649a97ab9fe019738fb6a0b1794d7732b6bf
- SFT-CHEM-THERMO-015-SOLVATION-0555: class=solvation;
target=sha256:e6f04d5bf765066a9d0f1b41df56a2d568a442aaf890cf0579b5184447c53543
- SFT-CHEM-THERMO-015-SOLVATION-0556: class=solvation;
target=sha256:995ddd2e0b61e370fdbd6eb4462ab1f47ac953e70e89878966274275a20b736d
- SFT-CHEM-THERMO-015-SOLVATION-0557: class=solvation;
target=sha256:9cf5aecdc3ac0e0ab8b41dfc1fbc916252f19abea236da219692d00a4e19e9ec2
- SFT-CHEM-THERMO-015-SOLVATION-0558: class=solvation;
target=sha256:8078d65299392621ab1add9c65ef3089d2cd25d58f61e7717e95974768a68d9b
- SFT-CHEM-THERMO-015-SOLVATION-0559: class=solvation;
target=sha256:7fb67b0eca141b7d4a4aac34570a72eb16c71003cf4749b9a4a4c15c27b9aac8
- SFT-CHEM-THERMO-015-SOLVATION-0560: class=solvation;
target=sha256:a98dc0058529d8069e83d9635889f9d05f457eaa54cf2c4869a430954a051607
- SFT-CHEM-THERMO-015-SOLVATION-0561: class=solvation;
target=sha256:a9b0cd373acbfcae821269057e2c81de12734baca91bbcd8a41ad11129292777a
- SFT-CHEM-THERMO-015-SOLVATION-0562: class=solvation;
target=sha256:866cc0f0b59404534fbedab9aceff07e1bd49616bec95c4bbdee5ef0ad1a9739
- SFT-CHEM-THERMO-015-SOLVATION-0563: class=solvation;
target=sha256:d87b94b01c6d0fd931c014f3c35a94e062338819dcb42285ab6dd4e78b829637
- SFT-CHEM-THERMO-015-SOLVATION-0564: class=solvation;
target=sha256:15bae5c766428004d7fbf8451b8e22240917f5c90b9bcb8c5b1047ba5e4f477f
- SFT-CHEM-THERMO-015-SOLVATION-0565: class=solvation;
target=sha256:45eb2065f5267e0ad718cf1e40863a891a5675a0864f393443fe91bcbe69d3f3
- SFT-CHEM-THERMO-015-SOLVATION-0566: class=solvation;
target=sha256:2361aef4e2c26400391a56ad242ec0b0b9426dcf146e5c7b944e5779979cfa55
- SFT-CHEM-THERMO-015-SOLVATION-0567: class=solvation;
target=sha256:0a564e78b12f545c98c25bc9d01107d77c2d79188c46e14f987eb53768374ca6
- SFT-CHEM-THERMO-015-SOLVATION-0568: class=solvation;
target=sha256:072275d5612c7f9f00fac504c0814468d5a912cf477bf85076ec3a7a844de1f6
- SFT-CHEM-THERMO-015-SOLVATION-0569: class=solvation;
target=sha256:f48968c9aab7bfa5bdba04ef7c675e794d7bb0c2cef85e9b1e0a7af24a28e406
- SFT-CHEM-THERMO-015-SOLVATION-0570: class=solvation;
target=sha256:48180dd9b4f9a46c0f86a4319f9a5016deaf906095fe587f4a9720137c91871a
- SFT-CHEM-THERMO-015-SOLVATION-0571: class=solvation;
target=sha256:f61784a5f9d64f7ac3089c03db241304f770fb39d694d1da3b150df7ffd28dc4
- SFT-CHEM-THERMO-015-SOLVATION-0572: class=solvation;
target=sha256:69e100a34a746bdd6b81cb54d5972c026193b19bb99bfb26fdaf08df05204b

- SFT-CHEM-THERMO-015-SOLVATION-0573: class=solvation;
target=sha256:8e424e9ddd1a266bb514af2fd75e6a82b02849d1ac3d5eca7689a2f4771b2049
- SFT-CHEM-THERMO-015-SOLVATION-0574: class=solvation;
target=sha256:40aae116a792d599490318544ca14a10830d68aa5318186b9b6e8a4d4cca5baf
- SFT-CHEM-THERMO-015-SOLVATION-0575: class=solvation;
target=sha256:a88964678e2b52e910ea4e8e3ad5d03b7d7e42cb5b15e14920464b84933f04c7
- SFT-CHEM-THERMO-015-SOLVATION-0576: class=solvation;
target=sha256:eabf0b37721839fa5a60828c72553583e35b97c76e6fa05d846ae4cb2ea1b5de
- SFT-CHEM-THERMO-015-SOLVATION-0577: class=solvation;
target=sha256:c47f3ec28f83202478f24c788a65ea2ee78a24a509a52d9189bac9f652fcef8d
- SFT-CHEM-THERMO-015-SOLVATION-0578: class=solvation;
target=sha256:53d4d81a32252b7ad1ea30446e9a315a76f49e10efbe281d32ca58f0a9478b46
- SFT-CHEM-THERMO-015-SOLVATION-0579: class=solvation;
target=sha256:8375aaef3a96cfc9e0ea786b4e02111b113e4603fab74fb6410f6d1df187af75
- SFT-CHEM-THERMO-015-SOLVATION-0580: class=solvation;
target=sha256:e5125369490ff74dbdee7047c5a5ab51b2558978ec917efab12a9a5d8c97789b
- SFT-CHEM-THERMO-015-SOLVATION-0581: class=solvation;
target=sha256:5b0c6ebdef32c75b6127fb7e40a1b29a12b1ed4d2f621bb415a526c3179dcae9
- SFT-CHEM-THERMO-015-SOLVATION-0582: class=solvation;
target=sha256:81f740e38b04c756f06c03cd8c5839bc18d3a67badf23cdd91c0b08b9b1e686e
- SFT-CHEM-THERMO-015-SOLVATION-0583: class=solvation;
target=sha256:81bfb5c5c71d95efc696f253a45568acd1c030a96dfa6e3b7419b7060b5eddc
- SFT-CHEM-THERMO-015-SOLVATION-0584: class=solvation;
target=sha256:5e9b29714fc03b88973debf079753a9c74c75e0074a933999b9a3bcae95daea6
- SFT-CHEM-THERMO-015-SOLVATION-0585: class=solvation;
target=sha256:5af0a8a0b8de5f5cf686b48c6181466863444de78e70f23653e029e82f33a304
- SFT-CHEM-THERMO-015-SOLVATION-0586: class=solvation;
target=sha256:98fa566d66cac614cbeb46f47f81d049c36356e03f9ba14a265438d9e20ba4c3
- SFT-CHEM-THERMO-015-SOLVATION-0587: class=solvation;
target=sha256:c4401b2e5bb02bf5637223ab7e749c03cb9f3a744631ecb4560cd2b02719665e
- SFT-CHEM-THERMO-015-SOLVATION-0588: class=solvation;
target=sha256:6a4e018297aa79aa0a8f283f3a6ed2be99acc534187dd8b59727855308f41d6c
- SFT-CHEM-THERMO-015-SOLVATION-0589: class=solvation;
target=sha256:3c3812a7bd144d4426a454d2187624caa156e64f4c51193ced3593a640f0e04f
- SFT-CHEM-THERMO-015-SOLVATION-0590: class=solvation;
target=sha256:ab907bb8986c8612ae1fc83b69ea9d9ec3170cd26cdf18fbed59c63e56264419
- SFT-CHEM-THERMO-015-SOLVATION-0591: class=solvation;
target=sha256:be28c72967bebf9c97cb2951b2ad21d0596ee88c97f62843d8a67ec5c3ba7edd9
- SFT-CHEM-THERMO-015-SOLVATION-0592: class=solvation;
target=sha256:e879c1409e4cd151fcea85058a18c06dbd865aa7e40d8f0bbb26bfe5d254c07d
- SFT-CHEM-THERMO-015-SOLVATION-0593: class=solvation;
target=sha256:48292c01eb1f80aaf007819fe60fcd18c674dd203d508beae3b1620dc406e87
- SFT-CHEM-THERMO-015-SOLVATION-0594: class=solvation;
target=sha256:797bc0a6e2001af26106da9d5bec8a7270e3ac9000959ca84e2d183f6cf82933

- SFT-CHEM-THERMO-015-SOLVATION-0595: class=solvation;
target=sha256:8ccd410c814d2b0a0fdaa6004c76c6ad7e6d7346055a35f11f86ad0749fb0f62
- SFT-CHEM-THERMO-015-SOLVATION-0596: class=solvation;
target=sha256:af58cb701c48af50f3b148fd05b27ce3023aa17faa62d4f381c7e072e20f3df3
- SFT-CHEM-THERMO-015-SOLVATION-0597: class=solvation;
target=sha256:59828190547ceb883f53d223da8b67fa38c9cf55e4540653c7ec393c62eb8e40
- SFT-CHEM-THERMO-015-SOLVATION-0598: class=solvation;
target=sha256:0b73ba183dd8bf2339ae3c53c5ef4d3102b7be99cdb6f74accd724096ca5797e
- SFT-CHEM-THERMO-015-SOLVATION-0599: class=solvation;
target=sha256:b7a718b84fd1ba176310a293d48ab63574284a018020f23231bc6e818ecad862
- SFT-CHEM-THERMO-015-SOLVATION-0600: class=solvation;
target=sha256:8e811ae36a17317378b861802f094d9fb2956c08e22a99cd7d6cc640af68882c
- SFT-CHEM-THERMO-015-SOLVATION-0601: class=solvation;
target=sha256:567ceb32735244b1b2f51cc3480945721a9c84a5606daa8b7cafd0b77867f8bf
- SFT-CHEM-THERMO-015-SOLVATION-0602: class=solvation;
target=sha256:82f7e4d22a5146fd77105d8810cd8eb91b1e546ab3cc90f5301dfb3e836e9411
- SFT-CHEM-THERMO-015-SOLVATION-0603: class=solvation;
target=sha256:b689624901fba46e7373e35d38b19f643071adc368d0043a05ebb1dd1b7330cd
- SFT-CHEM-THERMO-015-SOLVATION-0604: class=solvation;
target=sha256:83e95a7ca446c27a70616334dd17edc38525167ffd4943bee3b447fd1ffec2b7
- SFT-CHEM-THERMO-015-SOLVATION-0605: class=solvation;
target=sha256:427a959151d72e2878476ff376dda3d3f90f4f97c8002db412337f8368bfc5a
- SFT-CHEM-THERMO-015-SOLVATION-0606: class=solvation;
target=sha256:cc5bf206c32e4ce1b0693a4902e3542aa7357d60878c09f754ea4b602e3f4318
- SFT-CHEM-THERMO-015-SOLVATION-0607: class=solvation;
target=sha256:ac5000676aae53f61565d2ccc3118539cd4fe9f928bd7e4100b4b25411372041
- SFT-CHEM-THERMO-015-SOLVATION-0608: class=solvation;
target=sha256:429cb90f8b2d7b3b25edd0748b1461ac2f1ed358fd06005e8520cf5f9d60c621
- SFT-CHEM-THERMO-015-SOLVATION-0609: class=solvation;
target=sha256:1c5355e2369310acc60a68ca344a66ac035ff1fcc60a37301f95267428ac514d
- SFT-CHEM-THERMO-015-SOLVATION-0610: class=solvation;
target=sha256:53ba1eb2e48f231bb50bbc24b291fd2593b54af441eec456d09ab2b0c5edba1d
- SFT-CHEM-THERMO-015-SOLVATION-0611: class=solvation;
target=sha256:8758be040fe8389d9ec8aed2108bf5bda3810e5c03411408e68ac989659fc44e
- SFT-CHEM-THERMO-015-SOLVATION-0612: class=solvation;
target=sha256:70ae8e411b0ae87bd9600ab3a3c20465dd084056b93ebaf95d0a209f70e27122
- SFT-CHEM-THERMO-015-SOLVATION-0613: class=solvation;
target=sha256:71b6dc8c66ac8b0981b189ccc5379a12ae78a4b32b16069471c771b04742176e
- SFT-CHEM-THERMO-015-SOLVATION-0614: class=solvation;
target=sha256:ca972e52c034b96ba318ce5e6ca9974b42263f170d0c22ac8f221519d2d1a0a4
- SFT-CHEM-THERMO-015-SOLVATION-0615: class=solvation;
target=sha256:419868fbc11a5d36af271026db0d9205f7e5f6885778cfb8ca98247e07e33d5b
- SFT-CHEM-THERMO-015-SOLVATION-0616: class=solvation;
target=sha256:a44e3e3ba952b96dbb226a849f84506742f9b05d5e294b37e068d92dbe64249b

- SFT-CHEM-THERMO-015-SOLVATION-0617: class=solvation;
target=sha256:c416b1c4da4ed8c2d78b55ec34a6f923d13c51b965e4eeb30623af9fe12e0bf4
- SFT-CHEM-THERMO-015-SOLVATION-0618: class=solvation;
target=sha256:2d425fea97d30dacd0555e02a34b188d086de36a132db75ef341862fe1b526fb
- SFT-CHEM-THERMO-015-SOLVATION-0619: class=solvation;
target=sha256:fa63629e90c355f9fef7a9f2e99869f80f143a1b7429d6b625cf5128116797b5
- SFT-CHEM-THERMO-015-SOLVATION-0620: class=solvation;
target=sha256:ab43852863838102cf75d7977badf2b1a89a63471b12ad1fa48d2f2ccae90154
- SFT-CHEM-THERMO-015-SOLVATION-0621: class=solvation;
target=sha256:9a4f8582887a7144f169b974240c24e4a4358d3f4a01fc1a1faa15438792da85
- SFT-CHEM-THERMO-015-SOLVATION-0622: class=solvation;
target=sha256:db09b9f25b1feb566e36d5563adeb5f608c4c167c4f0c9c35304b91e9d39d690
- SFT-CHEM-THERMO-015-SOLVATION-0623: class=solvation;
target=sha256:9713358d13a057c93b576ae104441b476314e66b14f514bca318cd1f4bb99cc5
- SFT-CHEM-THERMO-015-SOLVATION-0624: class=solvation;
target=sha256:41beb5f392c410e25c66dded705d498b1b56a810f9bc44787758aaa81cf1ffc1
- SFT-CHEM-THERMO-015-SOLVATION-0625: class=solvation;
target=sha256:ca653e324536a57d9a5682170eb0026821c487cf85a5e69473fb4ff47f77347c
- SFT-CHEM-THERMO-015-SOLVATION-0626: class=solvation;
target=sha256:ee04828b4246678bdb2f3463f4491d72a57ca55a2b6f45d9c469e7f4511639d6
- SFT-CHEM-THERMO-015-SOLVATION-0627: class=solvation;
target=sha256:ea5ab821729e33197cf13092376d67c6fde1200350c94cf6625e0eced22fce6a
- SFT-CHEM-THERMO-015-SOLVATION-0628: class=solvation;
target=sha256:395a8cac8180e85ff933e715168c94fe8ade343aa033ab28cc8a803b1f1c3cc4
- SFT-CHEM-THERMO-015-SOLVATION-0629: class=solvation;
target=sha256:7a821555ce293374768c39905dd0452b29efd898b568ff5fa88dea6b6fad3d06
- SFT-CHEM-THERMO-015-SOLVATION-0630: class=solvation;
target=sha256:21d254c8a0938c4e18f8e8354122e2b304913f6e1e573ba457969da4f456ce6
- SFT-CHEM-THERMO-015-SOLVATION-0631: class=solvation;
target=sha256:65e995648eab9050b773c3099246ac952f5908dbcf54a6f17c775d9669a5d6cf
- SFT-CHEM-THERMO-015-SOLVATION-0632: class=solvation;
target=sha256:b01dbfed63dbe49db2d7060469527306a8efb528a250e52208e30394d2ca98f7
- SFT-CHEM-THERMO-015-SOLVATION-0633: class=solvation;
target=sha256:2ff80fe5c1b26325c6653ce9ce9dbffdbd9dbef97f8110241ce9480f5d64d58
- SFT-CHEM-THERMO-015-SOLVATION-0634: class=solvation;
target=sha256:1820f95ae7cd9b5d47f4033f41724771d731516e57ea399d0b2d3b3722713c06
- SFT-CHEM-THERMO-015-SOLVATION-0635: class=solvation;
target=sha256:c8c64d399244531074f5d3ec1c8222c90d12a378e6cb2db44b267dfb48d9a2ed
- SFT-CHEM-THERMO-015-SOLVATION-0636: class=solvation;
target=sha256:cca89439b75236f23ab034a1d69973f4c89c9b3736e9dc856427134418aba83d
- SFT-CHEM-THERMO-015-SOLVATION-0637: class=solvation;
target=sha256:77e566a1805c32848e5c1c81f4d8a17b8e3bd0eb40fd086965ff982c048308f4
- SFT-CHEM-THERMO-015-SOLVATION-0638: class=solvation;
target=sha256:fdc90cb602c2997e63f86dfe8c213ccda4c61b21da230ff8573d0425cc3a60a7

- SFT-CHEM-THERMO-015-SOLVATION-0639: class=solvation;
target=sha256:0f7efbb75cdc6396a0beae4a1ac66bb29d4f1ec3b7fee760b9f3807af1326ef7
- SFT-CHEM-THERMO-015-SOLVATION-0640: class=solvation;
target=sha256:daa13cc1d1c0b1b70e9d83963e7d6ebba4a24b8b922f6143da3661ebe16b355c
- SFT-CHEM-THERMO-015-SOLVATION-0641: class=solvation;
target=sha256:03bcbdb6bca8c6222402777eadf5dd1b7c1820c6bfb9540c766d172c4c739ffd
- SFT-CHEM-THERMO-015-SOLVATION-0642: class=solvation;
target=sha256:031adb452ff1e234fd854459e460d92aac646b85524f3f5b1b6710db6cfd2207
- SFT-CHEM-THERMO-015-DISSOLUTION-0001: class=dissolution;
target=sha256:2137df4538d3425954696d7ff78d69cee8fee7613bf7804085c10b775b5c6c91
- SFT-CHEM-THERMO-015-DISSOLUTION-0002: class=dissolution;
target=sha256:539708b011a2bbbf210652e7c6c6e58d9ac866404e17819701d56f00ad20892
- SFT-CHEM-THERMO-015-DISSOLUTION-0003: class=dissolution;
target=sha256:a37cef3fd3994c62a4cd1c5f1fca4f1ac1bfaec431a966893e3e903baba9af1d
- SFT-CHEM-THERMO-015-DISSOLUTION-0004: class=dissolution;
target=sha256:c6912dec38464fb95fea10cd73439bf963873aa6dc65629209564541df54cd93
- SFT-CHEM-THERMO-015-DISSOLUTION-0005: class=dissolution;
target=sha256:33c6cb12766e218791d8546348033ff04c6f8997d5ae13ebc89f30a3ca00e8dc
- SFT-CHEM-THERMO-015-DISSOLUTION-0006: class=dissolution;
target=sha256:b035a55b02e194abd0f498d41314c45a30e3c2ffff5e25fc3b71666eebda6b8d
- SFT-CHEM-THERMO-015-DISSOLUTION-0007: class=dissolution;
target=sha256:6c69ddf727a57509e6ff20e20d5ac2a3e1b4e9051c7ffb36d9de8decfb0fc44d
- SFT-CHEM-THERMO-015-DISSOLUTION-0008: class=dissolution;
target=sha256:dedf3ad590a859384136da3ad4323d4ccd0438868dc7db2dfed97e1b4d28345f
- SFT-CHEM-THERMO-015-DISSOLUTION-0009: class=dissolution;
target=sha256:b43e1bfb44290c02b76c6cdf327108cdc53e90182802b81e3a749747d80348d
- SFT-CHEM-THERMO-015-DISSOLUTION-0010: class=dissolution;
target=sha256:1dc3ab45da03ec649681742409fc2d41f2c0e14318d55835c38a90c21dd6ecee
- SFT-CHEM-THERMO-015-DISSOLUTION-0011: class=dissolution;
target=sha256:a30245ee5a482c8ab90608fcb6581446186324f81c592af99c9f37eb2b4f3464
- SFT-CHEM-THERMO-015-DISSOLUTION-0012: class=dissolution;
target=sha256:001507a976cd51110214d30291518673450d76c759456c78092aae5ec1542695
- SFT-CHEM-THERMO-015-DISSOLUTION-0013: class=dissolution;
target=sha256:e4283334a88a632a7cff24ce59f3b47bd4e87919190997523593e599974100ac
- SFT-CHEM-THERMO-015-DISSOLUTION-0014: class=dissolution;
target=sha256:30c09dc5ec8f13cce616d5d9918fb7e12754f256f616bcb94924db70fa6c5524
- SFT-CHEM-THERMO-015-DISSOLUTION-0015: class=dissolution;
target=sha256:1466c48bc1d9f9c831e1840e302e1d8ec0dc03c259253a9d0d7cf202ca125e76
- SFT-CHEM-THERMO-015-DISSOLUTION-0016: class=dissolution;
target=sha256:972562ebe74195569e15685d96497e34b3691eb6d5225855d6ef7494324746aa
- SFT-CHEM-THERMO-015-DISSOLUTION-0017: class=dissolution;
target=sha256:29d661ae65047d50995f5ff17eaf2733ebe8655ff2260f3b6720a54e029439c2
- SFT-CHEM-THERMO-015-DISSOLUTION-0018: class=dissolution;
target=sha256:2bd485d7437edf5ffb7177883db5ad0d9b0eaffad6cf8aa1d1608a749838369b

- SFT-CHEM-THERMO-015-DISSOLUTION-0019: class=dissolution;
target=sha256:5b1a4672e65f538c298613878d331ab0f559079c11c4666f6ae5f312eb147b00
- SFT-CHEM-THERMO-015-DISSOLUTION-0020: class=dissolution;
target=sha256:e6416e05d8d882357be557e43c5922f7c951386e0ab657ec6ac31981754c9155
- SFT-CHEM-THERMO-015-DISSOLUTION-0021: class=dissolution;
target=sha256:572682783181095461bf6735e8ade1594f9a30771a118224a7dac74b966de081
- SFT-CHEM-THERMO-015-DISSOLUTION-0022: class=dissolution;
target=sha256:f04496676245eb2ecc93677bcf72111457afb03343219f34019cca5ac30a9bfd
- SFT-CHEM-THERMO-015-DISSOLUTION-0023: class=dissolution;
target=sha256:2f686d9860509781456418d090dd7f66ae66dc3a7ea713f852c823ded3c5d147
- SFT-CHEM-THERMO-015-DISSOLUTION-0024: class=dissolution;
target=sha256:92b5c9e552ed69f32ef50dc28a9b0aabc974ac22bdc9f675a4cb5ca2284f4c93
- SFT-CHEM-THERMO-015-DISSOLUTION-0025: class=dissolution;
target=sha256:ce89cfcb4a5c01516bdd3577ce21ae334b6bf1de7da038a312d7df0934e31861
- SFT-CHEM-THERMO-015-DISSOLUTION-0026: class=dissolution;
target=sha256:e4aecdd1f47024fb665df6e1ca0e0842e9daac1a1abef9246df06941e1756ce75
- SFT-CHEM-THERMO-015-DISSOLUTION-0027: class=dissolution;
target=sha256:6bf0443baa2f3757c102e5d14464b73b1b7f5316c19be0f6516420dfaf55047
- SFT-CHEM-THERMO-015-DISSOLUTION-0028: class=dissolution;
target=sha256:0f260cc32d81145af455489bcf83d14bbcd3cf5ffb08245669482748d12496a2
- SFT-CHEM-THERMO-015-DISSOLUTION-0029: class=dissolution;
target=sha256:50d3684e6f30809aaba93c3b54bb31e6ba89ff7cbd53b17731b7ab44b51f8810
- SFT-CHEM-THERMO-015-DISSOLUTION-0030: class=dissolution;
target=sha256:9a884a5e4d57b3811ad3e3c84af5c2a309d4ea29cd096a5348dc6c724e8da9e0
- SFT-CHEM-THERMO-015-DISSOLUTION-0031: class=dissolution;
target=sha256:aa3743aab31e144f112a8b444b15fd6868a188860251064cdc4a539d00dd9411
- SFT-CHEM-THERMO-015-DISSOLUTION-0032: class=dissolution;
target=sha256:359045cc37af4ef212855e6ac8f05bf82e038c2df1aa74b0c4599eaa36f5a6d
- SFT-CHEM-THERMO-015-DISSOLUTION-0033: class=dissolution;
target=sha256:669c40822cbaa00787781963e4e8bea2e3b8c7fbd739f90a1be77476bd0953e9
- SFT-CHEM-THERMO-015-DISSOLUTION-0034: class=dissolution;
target=sha256:608b12faa4f87393cc845a15e866dbf42f6a0cac9830f42496a3c6c3bb8e4c0f
- SFT-CHEM-THERMO-015-DISSOLUTION-0035: class=dissolution;
target=sha256:5eefe17c05944da3ac28187127ff4f616668069502de5990804ce4b395855eb0
- SFT-CHEM-THERMO-015-DISSOLUTION-0036: class=dissolution;
target=sha256:7ffe3c26ad63c9444f60dcf1dc3585f433605c12aca2f17a097e1119a903607a
- SFT-CHEM-THERMO-015-DISSOLUTION-0037: class=dissolution;
target=sha256:2afe7f1313718ae4814003d936f998b7048a19f534f60c693280f2fdc10f1ecb
- SFT-CHEM-THERMO-015-DISSOLUTION-0038: class=dissolution;
target=sha256:6657a33b94688537632db55f5a6da11fa2ed5d7d1cb33764b5a8dc69ae8c78f2
- SFT-CHEM-THERMO-015-DISSOLUTION-0039: class=dissolution;
target=sha256:7107375557e123b89d34c691bfb209e93b50203815ef101a0e4c75b2335087f2
- SFT-CHEM-THERMO-015-DISSOLUTION-0040: class=dissolution;
target=sha256:7a13f3eae452ebf91784b2fc544dcf7b04cac990e50f936aa64e7b1f1eecf569

- SFT-CHEM-THERMO-015-DISSOLUTION-0041: class=dissolution;
target=sha256:a14c07baaa294c7ef20e536ad75c2ee796d75e176863405a131cac9de317dd2c
- SFT-CHEM-THERMO-015-DISSOLUTION-0042: class=dissolution;
target=sha256:b16b54a34dae574624bd169e5687af92be63d23e1a7dd9477e2d0ad0293e0f66
- SFT-CHEM-THERMO-015-DISSOLUTION-0043: class=dissolution;
target=sha256:befd243acf74282aa33f6a94e1fb28f61b2833db8d31c070c74eb814d2018920
- SFT-CHEM-THERMO-015-DISSOLUTION-0044: class=dissolution;
target=sha256:44ff6a65ffadfb63c4c42ed5e161b3b33a937cfd33838c8fa2c69609d960f2fa
- SFT-CHEM-THERMO-015-DISSOLUTION-0045: class=dissolution;
target=sha256:84199550b8e26965697f6947273aea513297983dcc8caa1ac13bb86f73bf0c96
- SFT-CHEM-THERMO-015-DISSOLUTION-0046: class=dissolution;
target=sha256:4d2316dc60aca3699d1d57e3a8963e749401ee9792d8999dcf31c174ba4de7a6
- SFT-CHEM-THERMO-015-DISSOLUTION-0047: class=dissolution;
target=sha256:5d687028d64eeabcf6586275c77af5e26156f020b1946b2ca3000241669fb71
- SFT-CHEM-THERMO-015-DISSOLUTION-0048: class=dissolution;
target=sha256:3941604203a1d71e189ab32e3b98d0fc0f5164b0fc62f8485371cd8288150bc9
- SFT-CHEM-THERMO-015-DISSOLUTION-0049: class=dissolution;
target=sha256:67e537804c1a763e0f2277b9982753aa0e9a0e06027102e891407141c5f9ba7
- SFT-CHEM-THERMO-015-DISSOLUTION-0050: class=dissolution;
target=sha256:d0ea4d9977af40d7bdef544d2741477014e41c8d8227c2d4f31d91b54e9eb1d4
- SFT-CHEM-THERMO-015-DISSOLUTION-0051: class=dissolution;
target=sha256:dc6f3f9dba8c1d4aa39ec4a7805bfb2bc24855d9feda70078b62abe73bfc6e55
- SFT-CHEM-THERMO-015-DISSOLUTION-0052: class=dissolution;
target=sha256:54ef97de60eca5a79b48165fd5235c13859314c09cf75a576468641f2c5b1c97
- SFT-CHEM-THERMO-015-DISSOLUTION-0053: class=dissolution;
target=sha256:15efb9b63c95cbe6b2a4bfa4804fe0721a28605cd30ebbe634cd7434b541aedef
- SFT-CHEM-THERMO-015-DISSOLUTION-0054: class=dissolution;
target=sha256:4b1e3723e2159e6be065828f9ab45c01095a463da6639f6307d7185579b07edd
- SFT-CHEM-THERMO-015-DISSOLUTION-0055: class=dissolution;
target=sha256:304a3cfc9d204b6758c06ef691dbbfc0727bf16f7a32e365ed8c4882553b66cb
- SFT-CHEM-THERMO-015-DISSOLUTION-0056: class=dissolution;
target=sha256:d8b78f0ed641fcfc37a7fe0c48a0e291cc2cb8d1df2a03ec122bd3c7382a64fd
- SFT-CHEM-THERMO-015-DISSOLUTION-0057: class=dissolution;
target=sha256:90aeb7957d5956a79e8f884227f34093653327341efc29f35815e049677294b7
- SFT-CHEM-THERMO-015-DISSOLUTION-0058: class=dissolution;
target=sha256:08a8c6feed1c25aa5403311b95321f2495821c70741456f4be9e48d4a39dc9f6
- SFT-CHEM-THERMO-015-DISSOLUTION-0059: class=dissolution;
target=sha256:1ac72fa5aebc946f8b70531f03d51769308ab807a1881d42b77529e9be4e4548
- SFT-CHEM-THERMO-015-DISSOLUTION-0060: class=dissolution;
target=sha256:f4477a40f5f92d4dfb90affe5b1331c825646ba1eb3d8f9d2e9a660a0458e825
- SFT-CHEM-THERMO-015-DISSOLUTION-0061: class=dissolution;
target=sha256:bbd10bce4fbc648f7ac4234aecef42c8671ba3bb23f793587b354cbdbebe467d
- SFT-CHEM-THERMO-015-DISSOLUTION-0062: class=dissolution;
target=sha256:698ac9ca70fe3ddfe3be7c2c66a7d8d21b88a11f3bd7bc203d8f5c8971d04c28

- SFT-CHEM-THERMO-015-DISSOLUTION-0063: class=dissolution;
target=sha256:76a24cf26c0774ad8b875a50948c8c71ce6d91c6773716f0282fba4aecf87baa
- SFT-CHEM-THERMO-015-DISSOLUTION-0064: class=dissolution;
target=sha256:1f1dba88d983570c95ace38b4cf159879a5e00418e2aaae04c0d31b45721bd60
- SFT-CHEM-THERMO-015-DISSOLUTION-0065: class=dissolution;
target=sha256:8ae714b14e589b2cfce50134568f36aba58032a4ff0efb793add9973d2d1ea1d
- SFT-CHEM-THERMO-015-DISSOLUTION-0066: class=dissolution;
target=sha256:794cc8d38b6bd676838b4e983f8c8740b02045146262754a2e630ca7d68f2b26
- SFT-CHEM-THERMO-015-DISSOLUTION-0067: class=dissolution;
target=sha256:e61e61f09562ff06ac2caae81a7ad87850938b3c1a57ee8f8f41117ebe1be8f1
- SFT-CHEM-THERMO-015-DISSOLUTION-0068: class=dissolution;
target=sha256:f2afd707d404776aefe28616ada0c85845ddecd20791637b8dab5ae3c564bdc
- SFT-CHEM-THERMO-015-DISSOLUTION-0069: class=dissolution;
target=sha256:62e46fb92878341a35ac85de5921b301ed1b529e58011d2b6627a8571cd26520
- SFT-CHEM-THERMO-015-DISSOLUTION-0070: class=dissolution;
target=sha256:483d65ce76a2059f8f8a1e27f9ab32e027837cf57a73cbdd6acdc6b9675136ad
- SFT-CHEM-THERMO-015-DISSOLUTION-0071: class=dissolution;
target=sha256:deb7f25c6ac59128fc7da579970dc4df0f369e16552879b0b2a02ce20819544e
- SFT-CHEM-THERMO-015-DISSOLUTION-0072: class=dissolution;
target=sha256:65734a45069a6cf869653b7be032a6cbdddc43ac2a70c710ae63d7b954c58513
- SFT-CHEM-THERMO-015-DISSOLUTION-0073: class=dissolution;
target=sha256:47b16b7314ab018c91e01ed285bd23e0657e58369c030435be8deee76713fd6c
- SFT-CHEM-THERMO-015-DISSOLUTION-0074: class=dissolution;
target=sha256:5a09bd972acc3bd7ff297c667bb055eca8968de8f7301cb02e18f6e8525266aa
- SFT-CHEM-THERMO-015-DISSOLUTION-0075: class=dissolution;
target=sha256:6989e16327b19e5149ae3aab10dc26a6d04960a161fa0eaa0df30c2a55cd2b68
- SFT-CHEM-THERMO-015-DISSOLUTION-0076: class=dissolution;
target=sha256:1da66c3cfc86997a4281643b23b5eaad422cec36bd0e06da4f1b32a4549365e5
- SFT-CHEM-THERMO-015-DISSOLUTION-0077: class=dissolution;
target=sha256:272c25d1271a7ffe2cfbca23aa4f46bcf7da9a0da840730b5a110c8e480cff2
- SFT-CHEM-THERMO-015-DISSOLUTION-0078: class=dissolution;
target=sha256:8b2df1a0763a185ba9a0a6d72f6562d7a3c19ac02b212047bad4a883d93cc592
- SFT-CHEM-THERMO-015-DISSOLUTION-0079: class=dissolution;
target=sha256:e880223a3d14057225ae897f20c67639f4a1477c8008daf0aefbfb579b26802b
- SFT-CHEM-THERMO-015-DISSOLUTION-0080: class=dissolution;
target=sha256:571553452faae755e134d40b2c1d0e4775cc9e6a4f87db464a8ad0f458711d01
- SFT-CHEM-THERMO-015-DISSOLUTION-0081: class=dissolution;
target=sha256:6056d8f1609b18648c6d95752fd5ee351104edb06968f76c0bd424a72ca6782e
- SFT-CHEM-THERMO-015-DISSOLUTION-0082: class=dissolution;
target=sha256:59aff76e10a40b4e5ebff5d998934b0cb0f1c1dbfdb2d5c5360b84f38051f8fc
- SFT-CHEM-THERMO-015-DISSOLUTION-0083: class=dissolution;
target=sha256:5942d9efd03b9c0570cdc0b08901ee9c4367d8f52e51c031b851b17f31dc7d7b
- SFT-CHEM-THERMO-015-DISSOLUTION-0084: class=dissolution;
target=sha256:bcc565a5d128f95f6eb03a88e313749c471b35f4674b86f884edef9448a3fdcb

- SFT-CHEM-THERMO-015-DISSOLUTION-0085: class=dissolution;
target=sha256:fdb82737c9865f00d3e718f49bef95feceb9097bb11bdc4ad7115d14f4cbf52e
- SFT-CHEM-THERMO-015-DISSOLUTION-0086: class=dissolution;
target=sha256:f36edd4a8a0bff5f5e26c81b6e7323729d989e20c1e878a30599cfb772623bbd
- SFT-CHEM-THERMO-015-DISSOLUTION-0087: class=dissolution;
target=sha256:155a2dd7904e02f75885f79e6f172a10adcac7119a02861945d89bc1717563f0
- SFT-CHEM-THERMO-015-DISSOLUTION-0088: class=dissolution;
target=sha256:dd26ea648a05fea25723b7b1cc742c22b57df531f37b9d5bb51b974a5710edaa
- SFT-CHEM-THERMO-015-DISSOLUTION-0089: class=dissolution;
target=sha256:b59b52d194613d37cd428e32f179c0035e45740606b9f5d20a010a55c1076d34
- SFT-CHEM-THERMO-015-DISSOLUTION-0090: class=dissolution;
target=sha256:2a2b025ae64c9919351944c2a5bcea6bc0a248f9c596b6a37424e32405db29fa
- SFT-CHEM-THERMO-015-DISSOLUTION-0091: class=dissolution;
target=sha256:997990c28c3645ae63067b3a6ea94f90cd65051ea7939720e81d75bd25b3215c
- SFT-CHEM-THERMO-015-DISSOLUTION-0092: class=dissolution;
target=sha256:3efddb9e334159924abf462c65356e949af4bf0bcbe6464fc49cc4d839063bb0
- SFT-CHEM-THERMO-015-DISSOLUTION-0093: class=dissolution;
target=sha256:d4372952ea772540298a3396e2fc3bd6f96380d8612eda40de409b41c0fbe5ad
- SFT-CHEM-THERMO-015-DISSOLUTION-0094: class=dissolution;
target=sha256:eb76a47b7d2418ced9896338669078df2879e6c2593a7442a79cac19b6e4eacd
- SFT-CHEM-THERMO-015-DISSOLUTION-0095: class=dissolution;
target=sha256:9121e203cb3f2a150133c26949bd700cb3912435f47c585d1cd0a20b912a8fec
- SFT-CHEM-THERMO-015-DISSOLUTION-0096: class=dissolution;
target=sha256:310fe5de6a96cf53bcb0cce3250c20e882551081636d5e4abbbf860d54808856
- SFT-CHEM-THERMO-015-DISSOLUTION-0097: class=dissolution;
target=sha256:8b448d00aa0476ccc778c6353a20be8ae14c662d017c2cdd0f60fee1bec654a2
- SFT-CHEM-THERMO-015-DISSOLUTION-0098: class=dissolution;
target=sha256:bbda2c934913e7388997e43ec22e81a6c7b4875a690120ef8e02313a437815be
- SFT-CHEM-THERMO-015-DISSOLUTION-0099: class=dissolution;
target=sha256:31479b91558a7befdc104073ceab60ad2a9c2be2b1bf2aa0ef11c9a83dea1703
- SFT-CHEM-THERMO-015-DISSOLUTION-0100: class=dissolution;
target=sha256:b4ea4c25cff2a73b6fa30c0df692ebb4c1c5711541eedc7b368bc7ac7d2f199d
- SFT-CHEM-THERMO-015-DISSOLUTION-0101: class=dissolution;
target=sha256:e8d70f809fd11e7a34d7ff3adcd9031dc2db3fe4181bc789fce10a63ac071796
- SFT-CHEM-THERMO-015-DISSOLUTION-0102: class=dissolution;
target=sha256:78efbf681f9a9588be395277f2495d832940055e8402995ba07faaa31db40cff
- SFT-CHEM-THERMO-015-DISSOLUTION-0103: class=dissolution;
target=sha256:0a35c8175ac4b8af1fb8f50ad2432fc5fa18aab06f654d807460233427adc9a6
- SFT-CHEM-THERMO-015-DISSOLUTION-0104: class=dissolution;
target=sha256:1a3a03200af6dd81b744cf78d1756e90189838620330f5dd3816e9ed865e8bc2
- SFT-CHEM-THERMO-015-DISSOLUTION-0105: class=dissolution;
target=sha256:12f8be0c532594cc113a3b030d9d30459d5a3a2b91ad3ffb3982775c14df3011
- SFT-CHEM-THERMO-015-DISSOLUTION-0106: class=dissolution;
target=sha256:484940c91bad992d507fa43b959e87d9b17147b653f9fd9e0c50819b38e9c324

- SFT-CHEM-THERMO-015-DISSOLUTION-0107: class=dissolution;
target=sha256:48e48d43baf49cc7058dc067adbdc9ce3c2ba2d8da224eed6e602e335096e891
- SFT-CHEM-THERMO-015-DISSOLUTION-0108: class=dissolution;
target=sha256:e6d04b97b11f3b366610610d634cc981726ce19196331c3426b19c52c9e2888a
- SFT-CHEM-THERMO-015-DISSOLUTION-0109: class=dissolution;
target=sha256:13960a0c43cca3df4ed228372d5742d2832356577f14fa4b98ed4b3cd51cd8d8
- SFT-CHEM-THERMO-015-DISSOLUTION-0110: class=dissolution;
target=sha256:c9b7d409d5a2052ed51849ecaa3d414d947ca2fadccb3b687671fefa2ca04919
- SFT-CHEM-THERMO-015-DISSOLUTION-0111: class=dissolution;
target=sha256:5e86f7d61ec5d3837f5416765eb4ad248cb2e64f6d7dea5b5dbdfb5eabb437b1
- SFT-CHEM-THERMO-015-DISSOLUTION-0112: class=dissolution;
target=sha256:752aeb48005d1d09a09f32d0d7ae1fff20d78584724f36bed218bd86e4fdecbe
- SFT-CHEM-THERMO-015-DISSOLUTION-0113: class=dissolution;
target=sha256:2229e6627f0d61ae1ff825dc0ac3736ce7c2410550259a3b8b2c9b4f5c3e0048
- SFT-CHEM-THERMO-015-DISSOLUTION-0114: class=dissolution;
target=sha256:5b54b38e71ecbfce4f7d6f4343a0344c24a00ca96b925be1fda0a81c58d90954
- SFT-CHEM-THERMO-015-DISSOLUTION-0115: class=dissolution;
target=sha256:d4ff9b19d87bb778a3b026de84842f484bb1dfae883a2ab1d7418a2e67c1a93b
- SFT-CHEM-THERMO-015-DISSOLUTION-0116: class=dissolution;
target=sha256:0ab32c7314ab8544966977164e62cbc2ecb0ed11d085a39e6dcff7db30a459d4
- SFT-CHEM-THERMO-015-DISSOLUTION-0117: class=dissolution;
target=sha256:e48cdf19be123e392e3b04b8ecfd23dcc934c06d8d0554655034eae1c362d1c
- SFT-CHEM-THERMO-015-DISSOLUTION-0118: class=dissolution;
target=sha256:42ad52c7ad8e643c1c6e510eee99ac3cefd88eaa2d832355486db41ae1449287
- SFT-CHEM-THERMO-015-DISSOLUTION-0119: class=dissolution;
target=sha256:c0d9b6c64994d061f5f00cfe78cdab7e2b9b858884d5ed72d8441bef6b88f536
- SFT-CHEM-THERMO-015-DISSOLUTION-0120: class=dissolution;
target=sha256:0b5762ea50acd0c08ef09d0e20638dc1465d56293dc4bb854eb9eef36bdd0bbd
- SFT-CHEM-THERMO-015-DISSOLUTION-0121: class=dissolution;
target=sha256:635e036da1aaa35448d02d767f823ae493d2126905a045cba7664581638f132
- SFT-CHEM-THERMO-015-DISSOLUTION-0122: class=dissolution;
target=sha256:de06163ed4159f19bfbfa50aa1a13e58b5236a23814a96c357442596e23d8f20
- SFT-CHEM-THERMO-015-DISSOLUTION-0123: class=dissolution;
target=sha256:45e5dabfd1eb538ba743dcdba03fa48a65da0ca44d6ad79b9fbf255e012d71d5
- SFT-CHEM-THERMO-015-DISSOLUTION-0124: class=dissolution;
target=sha256:6af289c85f6d944ab6f8debd0fd13b6b6cd30cec9ca85e619db8efdaf5bf6180
- SFT-CHEM-THERMO-015-DISSOLUTION-0125: class=dissolution;
target=sha256:b528d10518a6ac8d10c1a759d36d5ee0f4912bda390e4f30a0583440ca1b5c82
- SFT-CHEM-THERMO-015-DISSOLUTION-0126: class=dissolution;
target=sha256:ae708d1fc83439d5d0bf73a2860c9255e163395aebf95c5027cc85c2e31151f9
- SFT-CHEM-THERMO-015-DISSOLUTION-0127: class=dissolution;
target=sha256:8f5615f6dee6ae5a4eea257d8ade254bc6d9a506126a7506dffe63cf62589dce
- SFT-CHEM-THERMO-015-DISSOLUTION-0128: class=dissolution;
target=sha256:cd0adca5935a08d57c7cbeed3f57209ce4b17970fcd8c80825210e470927897e

- SFT-CHEM-THERMO-015-DISSOLUTION-0129: class=dissolution;
target=sha256:351e9b41896a3197f4ee58ae4be134e5e7450500a9e3ec28c4bdb5b1dc34e61e
- SFT-CHEM-THERMO-015-DISSOLUTION-0130: class=dissolution;
target=sha256:c6aab1c258d94e256a1cbd997681be07288c6e897ce1114eb1d860dd99d1f4ef
- SFT-CHEM-THERMO-015-DISSOLUTION-0131: class=dissolution;
target=sha256:399ce6debb8b4fb41635c0ff2f3b31237544299845e062bc1fcdce3b6eb67279
- SFT-CHEM-THERMO-015-DISSOLUTION-0132: class=dissolution;
target=sha256:c9a0718c5b25828bbf6c11badf0d902782801510401f911a5a10fb0f2d766e8f
- SFT-CHEM-THERMO-015-DISSOLUTION-0133: class=dissolution;
target=sha256:dd3c01d572179b7a6da8fc65279e52ff834f1758114a7a416713e3d802d6e2be
- SFT-CHEM-THERMO-015-DISSOLUTION-0134: class=dissolution;
target=sha256:965a9ae9b1f81619152b0794860394bf4f118db61ed4986fa40ddca32c1a0353
- SFT-CHEM-THERMO-015-DISSOLUTION-0135: class=dissolution;
target=sha256:b67c5408d25f2411db74c3d0c5f659e997791ed791b4ceb3cbac4abb2d0b79c4
- SFT-CHEM-THERMO-015-DISSOLUTION-0136: class=dissolution;
target=sha256:b186475d74583a9048c11da6e21a459eae94db4f3d43243db9787e3f20d2514a
- SFT-CHEM-THERMO-015-DISSOLUTION-0137: class=dissolution;
target=sha256:21ae0ba2001612b06f877b287c5a7e08f983ce578406855dc078d99c068ac34a
- SFT-CHEM-THERMO-015-DISSOLUTION-0138: class=dissolution;
target=sha256:7d8408501d6c2fade5dae08c852fc23264c82107d2e9f9b9ec6591eb313e63ab
- SFT-CHEM-THERMO-015-DISSOLUTION-0139: class=dissolution;
target=sha256:f4aed6ae246ea70e938508582572a741ce0a57d14f3e977cb82468a2234c3eb4
- SFT-CHEM-THERMO-015-DISSOLUTION-0140: class=dissolution;
target=sha256:1c6109360562d7f0bb4aa4e76df22b418f5ee1b7e4ae01e4cacad980b1818b00
- SFT-CHEM-THERMO-015-DISSOLUTION-0141: class=dissolution;
target=sha256:1ac9167dd15ffb364753b6af3f9709ffe65388536cb7b3773c71b6119369474f
- SFT-CHEM-THERMO-015-DISSOLUTION-0142: class=dissolution;
target=sha256:f4a78e800555ec17bb020e921e930ca5f77a6082219690db54fe7d38a4b42aa8
- SFT-CHEM-THERMO-015-DISSOLUTION-0143: class=dissolution;
target=sha256:12d2bbb05f61c5d6cc052d8eba4c1d758a0edda5077d6b9d9e8c1abe075bc6d6
- SFT-CHEM-THERMO-015-DISSOLUTION-0144: class=dissolution;
target=sha256:6af13eefe5a7ddf2343b43a6606075c3ce238b690940973ec92532c07b7d0743
- SFT-CHEM-THERMO-015-DISSOLUTION-0145: class=dissolution;
target=sha256:8a0f6f62c84e6cb436a43ebc7c6e98265ceeaaf9b06784392110a555989b683a
- SFT-CHEM-THERMO-015-DISSOLUTION-0146: class=dissolution;
target=sha256:3129874b47041f74e7885813bbdb20b9b1bacc6f006f779c7a80a3a136649cb7
- SFT-CHEM-THERMO-015-DISSOLUTION-0147: class=dissolution;
target=sha256:e083a5cca334ab842b08612d522674a81831ca7351761b631da403ddf7397c5d
- SFT-CHEM-THERMO-015-DISSOLUTION-0148: class=dissolution;
target=sha256:9bd111c3220128cc7c20b2a6b39d582fc08987339ea9c4a2d801723b63c95c59
- SFT-CHEM-THERMO-015-DISSOLUTION-0149: class=dissolution;
target=sha256:011f36cf949dd5d085e12bf495161642217f41767ac108d2ed6ca318eed44df9
- SFT-CHEM-THERMO-015-DISSOLUTION-0150: class=dissolution;
target=sha256:2d986d8805fa4d144d4aa3ea30b084ee586c804f69f9b5b0ba3bf04066748794

- SFT-CHEM-THERMO-015-DISSOLUTION-0151: class=dissolution;
target=sha256:5ae925a3523714e2127ff184c002d4af418144d27ff88105d937ca6ed2713247
- SFT-CHEM-THERMO-015-DISSOLUTION-0152: class=dissolution;
target=sha256:33d5f96e54df19d5f05a7436fe75978ba043961caa01af6257ebb95753d03c6b
- SFT-CHEM-THERMO-015-DISSOLUTION-0153: class=dissolution;
target=sha256:e4bbc84bc7621eef0e6d2fed62d4ef1a5e895d4458ea5a9c4a87341883cf5ab3
- SFT-CHEM-THERMO-015-DISSOLUTION-0154: class=dissolution;
target=sha256:35265aa8ed8f87fe1314af41ac86c6959f601b0a3805d499ad205a66b6d66e0b
- SFT-CHEM-THERMO-015-DISSOLUTION-0155: class=dissolution;
target=sha256:bbe3d53db983a95cd14be68ca93458b8c966f6f66568637240f312e87a4ffa19
- SFT-CHEM-THERMO-015-DISSOLUTION-0156: class=dissolution;
target=sha256:f9b1e6f87f33258391f3571e95ded1c6bac7a8210728a81b940998be6af0249d
- SFT-CHEM-THERMO-015-DISSOLUTION-0157: class=dissolution;
target=sha256:bf3f31daa86f2af8998dc98f59d1c3c2e6d269d3507426dc6434c873619c8514
- complete vector: 642 experimental solvation and 157 direct dissolution records
- exact ranges: {'solvation_magnitude_kcal_per_mol': {'minimum': '1/100', 'maximum': '2547/100'},
'solvation_uncertainty_kcal_per_mol': {'minimum': '3/100', 'maximum': '193/100'}, 'solubility_mole_fraction': {'minimum':
'3/20000', 'maximum': '3807/100000'}, 'solubility_uncertainty': {'minimum': '1/12500', 'maximum': '119/6250'},
'dissolution_temperature_K': {'minimum': '29821/100', 'maximum': '35989/100'}}
- source coverage: immutable complete FreeSolv v0.52 revision and complete seven-dataset NIST ThermoML source
- tampered_negative_solubility_rejected: True
- complete_799_record_vector_retained: True
- all_642_solvation_and_157_dissolution_rows_retained: True
- favorable_opposed_and_EmptyOne_solvation_orientations_retained: True
- all_seven_dissolution_datasets_retained: True
- all_mixed_solvent_records_retained: True
- all_absent_condition_coordinates_translated_to_EmptyOne: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if any solute, solvent, source state, destination state or condition is erased; if a signed or numerical-zero SFT proof value enters; if any force field, continuum-solvent model, partition/activity fit, solubility-product equation, logarithm, correlation, regression, selection or target correction enters; if target content opens before all 799 identities seal; if any of 642 FreeSolv or 157 direct NIST rows or their complete source provenance is omitted; if calculated/correlated companions are counted as measurements; or if any measured value, orientation, condition or uncertainty is tampered.

Measurement receipt: sha256:94dd705a94b95b08ded5788f7adc96100ba0f23aad1d7b75edf77a84a9cb374f.
Isolation certificate: sha256: bcb2fa38222326f4c8fc400ec7488859b3082633051ae0df1bc2e5c57d01f0b7.
Custody certificate: sha256: a49e41ed07c551ed8990084701984d75bab0d8d63e0f27e8d9446db05b2c5939.

Registered source descriptions:

- MobleyLab FreeSolv curated experimental hydration free-energy database: ()
- National Institute of Standards and Technology ThermoML Archive: ()

Registered target identities:

- None.

Meaning. The thermodynamic extension derives finite microstate support, temperature, energy, heat/work, entropy, enthalpy, free direction, chemical potential, activity, fugacity, phases, colligative response, solvation and coupled transport while keeping external scalar inscriptions downstream. This particular result is closed only at its exact registered grammar and evidence

boundary.

Explicit exclusions.

- no numerical zero; absent condition or coincident separation is structural EmptyOne
- no negative, irrational, imaginary, logarithmic, floating, signed or continuum SFT proof value
- no imported force field, continuum-solvent model, partition coefficient, activity-coefficient fit or solubility-product equation
- no correlation, interpolation, regression, fitted parameter, selected compound/solvent/state/condition/row or target correction
- no compound, solute, solvent, state, condition, value, uncertainty, reference or target hash before prediction seal
- every source row and companion field remains preserved; calculated and correlated companions are never measurements

Immutable identities. Source manifest

sha256:595ba60cb8f08813b26f6d6136988e76f1ad246098e74e6db794fc3809ff0556; derivation seal
 sha256:4052ce5a767c914dbe4d9e1b72dc94d2c99ed03eade3df656746640c55cec38e; engine receipt
 sha256:b14e9d25bec95b67d11287094e2589aaad277f46a2d01302e5b1055014bfaf3 at receipts/engine/
 model_admitted/SFT-CHEM-SOLVATION-DISSOLUTION-FREE-ORDER-015-b14e9d25bec95b67.json;
 empirical hash sha256:aa9f2a4d7f27afe76c3db6a6c7ee65dba4d4c5a489be948b8f42495e2144ad27;
 measurement receipt sha256:94dd705a94b95b08ded5788f7adc96100ba0f23aad1d7b75edf77a84a9cb374f.

140. Exact counted molecular-diffusion law

Claim identity: SFT-CHEM-MOLECULAR-DIFFUSION-RELATION-016

Question and theorem. Molecular diffusion is counted adjacent-cell redistribution of a retained molecular identity within a complete constituent, phase, condition, time and spatial carrier. Binary, self and tracer observations share this relation. Measured diffusion magnitude is exact positive post-seal support; no stochastic premise, continuum field, Fick law, Brownian/random-walk model, Stokes-Einstein relation, activation fit or transport coefficient is imported.

complete-molecular-transition-condition-account__distinct-held-migrating-and-medium-identities__counted-adjacent-generated-cell-transition__complete-global-constituent-conservation__exact-positive-tick-cell-and-path-support__exact-positive-postseal-diffusion-support__complete-value-free-164-record-identity-seal__depth-independent-common-replication-and-record-append

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-CPLX-TIME-SPACE-001
- SFT-COMP-CPLX-RANDOMNESS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001

- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-MAT-MICRO-DIFFUSION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013
- SFT-CHEM-SOLVATION-DISSOLUTION-FREE-ORDER-015

Generated grammar. Generate the literal product of carrier, identity, adjacency, conservation, resource, magnitude, prediction and extension forms; decide all 256 candidates only from admitted identity, adjacency, state-transition, resource, conservation, structural absence and finite-successor laws.

Boundary: Every finite binary, self or tracer diffusion record with complete constituent identities, held phase, exact conditions, counted adjacent cells, transitions and ticks. External testing preserves every direct diffusion point from three complete NIST sources and every non-diffusion companion dataset as excluded provenance.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-molecular-transition-condition-account__distinct-held-migrating-and-medium-identities__counted-adjacent-generated-cell-transition__complete-global-constituent-conservation__exact-positive-tick-cell-and-path-support__exact-positive-postseal-diffusion-support__complete-value-free-164-record-identity-seal__depth-independent-common-replication-and-record-append`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-molecular-transition-condition-account	detached-diffusion-number-or-continuum-field	A detached number or field erases molecular transitions.
identity	distinct-held-migrating-and-medium-identities	anonymous-particle-or-erased-medium	Anonymous transport cannot preserve chemical ownership.
adjacency	counted-adjacent-generated-cell-transition	unbounded-continuous-displacement	A continuous displacement imports an ungenerated continuum.
conservation	complete-global-constituent-conservation	created-lost-or-merged-constituent	Creation, loss or merging violates the complete material account.
resource	exact-positive-tick-cell-and-path-support	unrecorded-time-space-or-random-premise	Unrecorded resources or random premises cannot reconstruct transport.
magnitude	exact-positive-postseal-diffusion-support	imported-Fick-Brownian-Stokes-Einstein-or-fit	An imported equation or fit selects the magnitude.
prediction	complete-value-free-164-record-identity-seal	species-condition-method-or-value-readable-before-seal	Readable targets could select the law.
extension	depth-independent-common-replication-and-record-append	refit-after-transition-replication-or-record-append	Refitting destroys exact transition provenance.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:6090ff8c6e10e83ca7eacd4248abceb8cf86616a419ac15d60c93c1a2257dc3a`.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:12533f782acfe04b5ca5e9bc3dcd0ac25d42d8c3959460eff1652fe3180de1ec.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:977c97f2cdac10cac4dd364914c4dac5d60921c82d00a4a44a7fedb6668c0014.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:cf3593abbd85e70255e7d1ac4467bc9f9e6588bb00d25789ad8a93e4deb19413.

Independent reconstruction. Implementation hash

sha256:633536b524cffe4e5564ecc0aec337bd514df3705225fd7973bfbdb18250f6b05; independent certificate sha256:03d85ddfa85bb7e54f97286fec213d6997a625ceb73594d51c851a5cfb605a76; external-validation hash sha256:7ac9ad864fe87f09cdf3f692c720283b8423e5c00af94f7004ae86682ce3603a. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-TRC-THERMOML-JCED-2011-56-4840-4848
- NIST-TRC-THERMOML-FPE-2017-437-34-42
- NIST-TRC-THERMOML-FPE-2008-271-43-52

Comparison records:

- SFT-CHEM-THERMO-016-BINARY-0001: class=binary; dataset=1; point=1; target=sha256:738c405360048881bcbce85c7991cb1949932e577d045657356c1d0c8bfca7df
- SFT-CHEM-THERMO-016-BINARY-0002: class=binary; dataset=1; point=2; target=sha256:6796b5e28c0a5c1d08780ef300492b14d00105fcb96b1da315b3538af59965ad
- SFT-CHEM-THERMO-016-BINARY-0003: class=binary; dataset=1; point=3; target=sha256:8eb7a979cbfb6c8cf4d134f8cbb89a3758c0261db92287af88ea0660c1c92ebd
- SFT-CHEM-THERMO-016-BINARY-0004: class=binary; dataset=1; point=4; target=sha256:52e94db9ba7edd1d3544eee1ce1fbc633ab77c1b811cadb8400229e504af49fe
- SFT-CHEM-THERMO-016-BINARY-0005: class=binary; dataset=1; point=5; target=sha256:27a636267d3892e7cdca6cab188d292266a1955c24b571515521ada818c2f512
- SFT-CHEM-THERMO-016-BINARY-0006: class=binary; dataset=1; point=6; target=sha256:e8adc9ebe40b98876d5dd2182f013f5c47da9fcf7957d27c38d72f8f3b0945dc
- SFT-CHEM-THERMO-016-BINARY-0007: class=binary; dataset=1; point=7; target=sha256:4c0dc01df9c91faf57920ecabfc917ca61655266e61295cd30ca3028c55630aa
- SFT-CHEM-THERMO-016-BINARY-0008: class=binary; dataset=1; point=8; target=sha256:31108f34700ae3af1a1e3b1d4a5e80a3b591a909cf59dd1aaa129dd14f8aa216
- SFT-CHEM-THERMO-016-BINARY-0009: class=binary; dataset=1; point=9; target=sha256:ebd5b270910583454c1caa2329ec91868074e35088d6c6d39748c33869e54b02
- SFT-CHEM-THERMO-016-BINARY-0010: class=binary; dataset=1; point=10; target=sha256:af68805fd3b36cbdfdad62f9ceab5f438bab403501cb1c24507d0f9c5f6a3473
- SFT-CHEM-THERMO-016-BINARY-0011: class=binary; dataset=1; point=11; target=sha256:7e981e2a02d027f551791ff5cae6f74510659ff3313c7c6b4276f4c17b66995a
- SFT-CHEM-THERMO-016-BINARY-0012: class=binary; dataset=1; point=12; target=sha256:6ec8f77984aa52b6c583ca5d17c702a4c38fbb04c0444ca34674f8fec2f99155
- SFT-CHEM-THERMO-016-BINARY-0013: class=binary; dataset=1; point=13; target=sha256:d0d917071ce4d6bed387634bdeb5d60c9fa3367345762b4a6542702ac8a9800e

- SFT-CHEM-THERMO-016-BINARY-0014: class=binary; dataset=1; point=14;
target=sha256:39b1f501ac13354b61e28919f515389a7ef1b1cff3cc3ff85cbfa0d408d9e3ca
- SFT-CHEM-THERMO-016-BINARY-0015: class=binary; dataset=1; point=15;
target=sha256:6573d81fb89bfac8bed1b1718add333c57801fe0538cdd2319769289444b5d0f
- SFT-CHEM-THERMO-016-BINARY-0016: class=binary; dataset=1; point=16;
target=sha256:8182483437031bd24d27dc5d36fa1e1764ae2c71b7b142f79a86f1288f95ab9d
- SFT-CHEM-THERMO-016-BINARY-0017: class=binary; dataset=1; point=17;
target=sha256:6127ac9ed97f66dbf7ece5572d4074caa2e7f2e8c8de74eaa52901a306ec2e47
- SFT-CHEM-THERMO-016-BINARY-0018: class=binary; dataset=1; point=18;
target=sha256:04d42c65e5c1d744e9e49836de36f1d186e4b7ef1dcd94f0813575fc1a9dca2e
- SFT-CHEM-THERMO-016-BINARY-0019: class=binary; dataset=1; point=19;
target=sha256:68be9f8523f3e0fc3850a1dcd5357769d9f2876b44ff58870ab31cd26ec2dfef
- SFT-CHEM-THERMO-016-BINARY-0020: class=binary; dataset=1; point=20;
target=sha256:510667c85363d0cd304a467fcf382d05fb03a111a0685a1c9771d8a3a88b23af
- SFT-CHEM-THERMO-016-BINARY-0021: class=binary; dataset=1; point=21;
target=sha256:69d0645f4f3ea691e99a8cddd2b34c0ba6896f62f3fe4e71b8f3522a96504489
- SFT-CHEM-THERMO-016-BINARY-0022: class=binary; dataset=1; point=22;
target=sha256:17129fecf54e14e259adee2d764cf36548828bff37f053e6055feeed86f6baea
- SFT-CHEM-THERMO-016-BINARY-0023: class=binary; dataset=1; point=23;
target=sha256:541b9eeef6f13627c93e59ad40e69381100082b2bcf1e57bf37c80b536a557270
- SFT-CHEM-THERMO-016-BINARY-0024: class=binary; dataset=1; point=24;
target=sha256:277a014f83e0165bf1d45fd44229d847d3032a9d90d230f8a481b9d743e91032
- SFT-CHEM-THERMO-016-BINARY-0025: class=binary; dataset=1; point=25;
target=sha256:990872ecd7ad7b72dc9cc478ea523c3dc26c8dda0d563eb10698dc7a58cb8138
- SFT-CHEM-THERMO-016-BINARY-0026: class=binary; dataset=1; point=26;
target=sha256:a1058073e462ed001483effee6763df5984616ce05c85d10f5edcab9ac45fb5e
- SFT-CHEM-THERMO-016-BINARY-0027: class=binary; dataset=1; point=27;
target=sha256:0905078d217ee3747c6564d0db7df788bd3e0bcd98ac6a3e146d19c57f2159fc
- SFT-CHEM-THERMO-016-BINARY-0028: class=binary; dataset=1; point=28;
target=sha256:884f16205e35f248009a973f1f8a87a058f4b2a02c25d346e06f64bf0a6e5e0b
- SFT-CHEM-THERMO-016-BINARY-0029: class=binary; dataset=1; point=29;
target=sha256:647595472a694e72e193bae550cde215d9abb1e1990ae15fa2549b7275e9d653
- SFT-CHEM-THERMO-016-BINARY-0030: class=binary; dataset=1; point=30;
target=sha256:da9c392e4db4e02d29c219559503d9f962305347aca2ba7a36e66045e778aae2
- SFT-CHEM-THERMO-016-BINARY-0031: class=binary; dataset=1; point=31;
target=sha256:e9e68beea546a7b3be2d3184d185e15154795708694d4b8d6b1a83ccab72e5d1
- SFT-CHEM-THERMO-016-BINARY-0032: class=binary; dataset=1; point=32;
target=sha256:900d3c4cb6744cdd735425778f50ec50d822a4f7fa6bd96dd36064820e838e36
- SFT-CHEM-THERMO-016-BINARY-0033: class=binary; dataset=1; point=33;
target=sha256:c4a8d0688de4ff94985c585d22afe035e40bceec0050a0afef12006d143f4f46e
- SFT-CHEM-THERMO-016-BINARY-0034: class=binary; dataset=1; point=34;
target=sha256:7d68741069fc0357e5c6bf5cb962f4be612038a986c9386af65ebbc3f36c9366
- SFT-CHEM-THERMO-016-BINARY-0035: class=binary; dataset=1; point=35;
target=sha256:8735792cb9b49ad7cef52abd66b15507bd476476db96b9e87b0092e8d4f8d8d3

- SFT-CHEM-THERMO-016-BINARY-0036: class=binary; dataset=1; point=36;
target=sha256:7fbb656edce0945c8ecd8393c3a1348505589b0cdbba8b2d60afa4bb83cbcafe
- SFT-CHEM-THERMO-016-BINARY-0037: class=binary; dataset=1; point=37;
target=sha256:5b443e1ba8508593badceacd50129d48575e65c0576272e0f738f579c36ba073
- SFT-CHEM-THERMO-016-BINARY-0038: class=binary; dataset=1; point=38;
target=sha256:d8bdb718a700efda773c70e530c0384f9b22fff6617f84eca4fa2b3e018e687f
- SFT-CHEM-THERMO-016-BINARY-0039: class=binary; dataset=1; point=39;
target=sha256:bfd385694af00fb091d5b2d51ede1464220f170ac20b6a6e196d88f27e0e2f6c
- SFT-CHEM-THERMO-016-BINARY-0040: class=binary; dataset=1; point=40;
target=sha256:dd81d822a9e5211c92dc52a20c95db94c712e0a74ee39d6231e5e388f2b52c75
- SFT-CHEM-THERMO-016-BINARY-0041: class=binary; dataset=1; point=41;
target=sha256:9d155c841d693e001dacaf10f9f2833d5bd0635076c0a5ffb85dd9990eeb8a77
- SFT-CHEM-THERMO-016-BINARY-0042: class=binary; dataset=1; point=42;
target=sha256:760236c3db2f38ea2da253151357354d1327ac46e28008f24070b66266043c6d
- SFT-CHEM-THERMO-016-BINARY-0043: class=binary; dataset=1; point=43;
target=sha256:96fba303cc76c0b12ad2c0d7eb69ea0bbb1b79d68ebc782e5a15e489eed2163
- SFT-CHEM-THERMO-016-BINARY-0044: class=binary; dataset=1; point=44;
target=sha256:aea88697ba420b05ed4f3ab8321417e440cd35e59140831c44f9fc95c8b85822
- SFT-CHEM-THERMO-016-BINARY-0045: class=binary; dataset=1; point=45;
target=sha256:12dc6efc7d0825e6085fa44e575be99b2c7230dfee7d735b04428fa2fc38030e
- SFT-CHEM-THERMO-016-BINARY-0046: class=binary; dataset=1; point=46;
target=sha256:6c64d0c25dabdc5e57ac79a6c3e1a709b95c9f0bc44941744ff905949ec25c9
- SFT-CHEM-THERMO-016-BINARY-0047: class=binary; dataset=1; point=47;
target=sha256:c6f18a65c7ef4fd0808b45fd5cad1cfc21923cafcf36285958b5d9fc28d38658
- SFT-CHEM-THERMO-016-BINARY-0048: class=binary; dataset=1; point=48;
target=sha256:860277bcae9fc06189d9a27a8eb3ca16feffc8840cb98bd5b0934bb1b027a868
- SFT-CHEM-THERMO-016-BINARY-0049: class=binary; dataset=1; point=49;
target=sha256:630d020c59995ba0ade3ca599a7e3cb25781f91b47cf9d65c13557340392dfd6
- SFT-CHEM-THERMO-016-BINARY-0050: class=binary; dataset=1; point=50;
target=sha256:6f2e473d0d670b1340cb3c80ee5a6d29590b398983d237023d2638a1096805a6
- SFT-CHEM-THERMO-016-BINARY-0051: class=binary; dataset=1; point=51;
target=sha256:b18575119734b408024004ceefc1122fac8f38efc3846208abaf80c9d75240b2
- SFT-CHEM-THERMO-016-BINARY-0052: class=binary; dataset=1; point=52;
target=sha256:bda429d68436d93c4cf1b363acf45e4b7c08db760e4a0302979177f40745409a
- SFT-CHEM-THERMO-016-BINARY-0053: class=binary; dataset=1; point=53;
target=sha256:220a963d193318a6f4b2aac427280f0b252bb335d120662e6fcfbfcf99d21f75
- SFT-CHEM-THERMO-016-BINARY-0054: class=binary; dataset=1; point=54;
target=sha256:3b6f2ba6295d6bf0e2611caf23711cc066a59097753b9cb17f376ffd23ded722
- SFT-CHEM-THERMO-016-BINARY-0055: class=binary; dataset=1; point=55;
target=sha256:ac75e595fe392d548bf42bce0d660f5adb62409f4086b96890516de5ddb0a74e
- SFT-CHEM-THERMO-016-BINARY-0056: class=binary; dataset=1; point=56;
target=sha256:4ec7fd6977d80d81913f7c7e8d210e52259ff2f6d6942c5fec84e431fba63b43
- SFT-CHEM-THERMO-016-BINARY-0057: class=binary; dataset=1; point=57;
target=sha256:375bd066f422ef78f9b9c680f37535cce43e070ff93e3cf0d3781f5b6a8389c6

- SFT-CHEM-THERMO-016-BINARY-0058: class=binary; dataset=1; point=58;
target=sha256:bc06f16024ac94c0301e22973500be29f11e8f217bdbc2976e54a0074bbfa5b5
- SFT-CHEM-THERMO-016-BINARY-0059: class=binary; dataset=1; point=59;
target=sha256:9ab4834fb7e03bc760d1512d0c7c30fb73bc45756e0809e495c8992261d0048a
- SFT-CHEM-THERMO-016-BINARY-0060: class=binary; dataset=1; point=60;
target=sha256:ed36d1ed443f94d82b375cba6f02aac025d3f3d5fb707e815b43bf7855e892a5
- SFT-CHEM-THERMO-016-BINARY-0061: class=binary; dataset=1; point=61;
target=sha256:c318cd412aca41125c98a3a5db21d7fc7573f385a9b809a412da66e5f4d32d73
- SFT-CHEM-THERMO-016-BINARY-0062: class=binary; dataset=1; point=62;
target=sha256:a60c15c843ca9efb02b3e2f70e3653721b704624439e8763ddb6c83fce4c8758
- SFT-CHEM-THERMO-016-BINARY-0063: class=binary; dataset=1; point=63;
target=sha256:0b9fb4ab6062abc408076d1d68a9755759296cb0c05279e1fee2473e7ffacede
- SFT-CHEM-THERMO-016-BINARY-0064: class=binary; dataset=1; point=64;
target=sha256:10138c38470fbefa8ae06cc620616a3072b02cec6d0b1ba37542c65059224ce9
- SFT-CHEM-THERMO-016-BINARY-0065: class=binary; dataset=1; point=65;
target=sha256:a28ac1994a5593d83d973b2f96444adb883cacf8d62d2a608e0b115fc87c4b3a
- SFT-CHEM-THERMO-016-BINARY-0066: class=binary; dataset=1; point=66;
target=sha256:14a65af1f8535e5383f4b7d17931b88dfc9fb2f67a2e97fc90456a6d4575b680
- SFT-CHEM-THERMO-016-BINARY-0067: class=binary; dataset=1; point=67;
target=sha256:4c96c268b4ab6a46a93d1f1305fd96040e510db0ace34d201322b68064aa776e
- SFT-CHEM-THERMO-016-BINARY-0068: class=binary; dataset=1; point=68;
target=sha256:8e0c9758c8b4f4208dad89edaeb62f42b027004c79e51782f054a618d4f7df3c
- SFT-CHEM-THERMO-016-BINARY-0069: class=binary; dataset=1; point=69;
target=sha256:0b1a97408f6164b068f7fbcec5e5ef2453458141030e6438ebe2741df060f977
- SFT-CHEM-THERMO-016-BINARY-0070: class=binary; dataset=1; point=70;
target=sha256:dbcb86e767168dd874b6b1c31783af4eb49cee478b33f840abf1bc9579603900
- SFT-CHEM-THERMO-016-BINARY-0071: class=binary; dataset=1; point=71;
target=sha256:c612cc8ad64116a37649d26c4b6f1ef508ba67efe9c2f9cd5b42d8dd364e719f
- SFT-CHEM-THERMO-016-BINARY-0072: class=binary; dataset=1; point=72;
target=sha256:61c43fd2ef63f0b93afa32912a06d7deecfe274c3d313952fa80b91f377338c9
- SFT-CHEM-THERMO-016-BINARY-0073: class=binary; dataset=1; point=73;
target=sha256:ab50fea2b64d7650e5127dac1ccb1c71898f44880c4c2620a747fdffd7056b64
- SFT-CHEM-THERMO-016-BINARY-0074: class=binary; dataset=1; point=74;
target=sha256:f04eaf3d948eaa49079b6705fdab0de375390a6c0fd32debc1957e3e02cff5a0
- SFT-CHEM-THERMO-016-BINARY-0075: class=binary; dataset=1; point=75;
target=sha256:6b7aa8b82764008c0c914d58242f102c064e1784fe74e659ffe9d325d550aec4
- SFT-CHEM-THERMO-016-BINARY-0076: class=binary; dataset=1; point=76;
target=sha256:39f106625bb3ecf711a37a4400b0725165cd4e268d8dd5848f9907026a042a67
- SFT-CHEM-THERMO-016-BINARY-0077: class=binary; dataset=1; point=77;
target=sha256:93ae387683e0e89f1dcf7d99e3e660856b324bc26d4d7f11afde1b06376a8ee5
- SFT-CHEM-THERMO-016-BINARY-0078: class=binary; dataset=1; point=78;
target=sha256:e0a675ac3f07320a435804d4b288d2cd7526c795f14fc61fc42462679e30765e
- SFT-CHEM-THERMO-016-BINARY-0079: class=binary; dataset=1; point=79;
target=sha256:e7ccac103593ad5b236b28d41b4e4d1193452f130671f96d182903360571d3a4

- SFT-CHEM-THERMO-016-BINARY-0080: class=binary; dataset=1; point=80;
target=sha256:47c31324b1ad88a55eeaffa9d8e52b94542e3cf09715a3a617d48ee54b52ff24
- SFT-CHEM-THERMO-016-BINARY-0081: class=binary; dataset=1; point=81;
target=sha256:5b7e7498eac25a9c0354851fb4aa4de92b0399b2a97c5e6fc122a2ccf46e6d29
- SFT-CHEM-THERMO-016-BINARY-0082: class=binary; dataset=1; point=82;
target=sha256:a8e9f576c25b26f588a200a6475f9ce88b93afb0e5376b748d417631150dd230
- SFT-CHEM-THERMO-016-BINARY-0083: class=binary; dataset=1; point=83;
target=sha256:5c488447f7e1401ce36b34bb5b92fad734b09bb3e28f9c537cdb8786bb812d9
- SFT-CHEM-THERMO-016-BINARY-0084: class=binary; dataset=1; point=84;
target=sha256:b87e0956e570b527076a81c02d3075fe33659ac900ff59e7b201a5603ed44254
- SFT-CHEM-THERMO-016-BINARY-0085: class=binary; dataset=1; point=85;
target=sha256:6fc3b891095e9603bff861d4b01f9c45f2b6b85429445a9a0b17197e07a9684b
- SFT-CHEM-THERMO-016-BINARY-0086: class=binary; dataset=1; point=86;
target=sha256:f717ab9cdc646f8268daf489618e1477d7460059eb6b1c45d033f66ac1d2bfd2
- SFT-CHEM-THERMO-016-BINARY-0087: class=binary; dataset=1; point=87;
target=sha256:9c906178b80012a8c40d1a68ddfb1162373b9861eb08a53f47e12a4429a4547f
- SFT-CHEM-THERMO-016-BINARY-0088: class=binary; dataset=1; point=88;
target=sha256:330cedd3fe7e9154d2c0f46878cd0401ec2e95a7ae9288f3cbb726e22fb4d711
- SFT-CHEM-THERMO-016-BINARY-0089: class=binary; dataset=1; point=89;
target=sha256:bf39ee4e66c354114cc5299f631c0373f373ceae0d90c0a657c8339169d8de24
- SFT-CHEM-THERMO-016-BINARY-0090: class=binary; dataset=1; point=90;
target=sha256:572f93b3c7cfe70ced1e6b7233e8d0a1e5412af8e89b9e99ace3ac741b9bf2ef
- SFT-CHEM-THERMO-016-BINARY-0091: class=binary; dataset=1; point=91;
target=sha256:d32c0ac358502aa50a41e59591788dca2d60c43b0caaf947d65c3a63779ef60b
- SFT-CHEM-THERMO-016-BINARY-0092: class=binary; dataset=1; point=92;
target=sha256:402281f7e8f5aaee5fa6e1afa7e7f99d859e1159eda5bfd3d353c0871a3b81fb
- SFT-CHEM-THERMO-016-BINARY-0093: class=binary; dataset=1; point=93;
target=sha256:4b2818e14a6b522eb9aa490bd81eeb775976836b03fafd6c27d8e942f1b3878b
- SFT-CHEM-THERMO-016-BINARY-0094: class=binary; dataset=1; point=94;
target=sha256:07341a2abe1b421d1463fb2148333fc54d204df2234056181d7af0bfd0c0e3ab
- SFT-CHEM-THERMO-016-BINARY-0095: class=binary; dataset=1; point=95;
target=sha256:a302f7016c5b1862e2638c90f05b0ddbb709c59b7f65af205f8f474f1462e178
- SFT-CHEM-THERMO-016-BINARY-0096: class=binary; dataset=1; point=96;
target=sha256:e4fb7018139012a036b2415b9231ae152a7c47273c9fd639920ac0f3fe094f25
- SFT-CHEM-THERMO-016-BINARY-0097: class=binary; dataset=1; point=97;
target=sha256:47f211f5743a2bd531fed2e3f6290b4fbb084613ed783d0582d19ae40fa6ca20
- SFT-CHEM-THERMO-016-BINARY-0098: class=binary; dataset=1; point=98;
target=sha256:a6b67582c7b9e6beac5e64cc9a2bf03fbbe2cee0323753dfdd2dd8aefbfa75d5
- SFT-CHEM-THERMO-016-BINARY-0099: class=binary; dataset=1; point=99;
target=sha256:9b589b6782a1235c283adddfb3e7665f86be72f11f0e84a2280f527efec16fa
- SFT-CHEM-THERMO-016-BINARY-0100: class=binary; dataset=1; point=100;
target=sha256:7c6be00572d5fc20a85fd138feecb66bd31865f30793776458027cdcb633c659
- SFT-CHEM-THERMO-016-BINARY-0101: class=binary; dataset=1; point=101;
target=sha256:ff7468e2e29be43de98b0ac10a4d621704e569ab4288b890019ec8ac3c605147

- SFT-CHEM-THERMO-016-BINARY-0102: class=binary; dataset=1; point=102;
target=sha256:9c1be2076a29851643599be6c132ce921ebd07966e36022c05b2104df99a38d5
- SFT-CHEM-THERMO-016-BINARY-0103: class=binary; dataset=1; point=103;
target=sha256:940c62ecce8c2b096e3833a6d3d62c54e4bc87f454b85317a3201b82cbd8575b
- SFT-CHEM-THERMO-016-BINARY-0104: class=binary; dataset=1; point=104;
target=sha256:a7d9b53bfdfc676b72c3cb8e095901ab014da03d19152639ff7158aed4eca503
- SFT-CHEM-THERMO-016-BINARY-0105: class=binary; dataset=1; point=105;
target=sha256:d70c0d54999521a8227275031fec2b29f5066b5149a53cf0913cbef29fcaacaf
- SFT-CHEM-THERMO-016-BINARY-0106: class=binary; dataset=1; point=106;
target=sha256:27c3c5caf9c2bf10e7c76389104cf84c6915ecffb24c6c0a572029218eeef126
- SFT-CHEM-THERMO-016-BINARY-0107: class=binary; dataset=1; point=107;
target=sha256:5c3aeb85c1e4c3f9823cb9ac607fb28899a53fbcc8d42e3841675fe95c5c1019
- SFT-CHEM-THERMO-016-BINARY-0108: class=binary; dataset=1; point=108;
target=sha256:4edd8fafec48ed7297cc47fdce52c652456cd9721bb03014cd7a79d3c88aff65
- SFT-CHEM-THERMO-016-BINARY-0109: class=binary; dataset=1; point=109;
target=sha256:249411e6ab79eec494210af42cf272fd4de212a709a3d00072b48e2db563bb9d
- SFT-CHEM-THERMO-016-BINARY-0110: class=binary; dataset=1; point=110;
target=sha256:793739099b02abbff7234b37bfd51ae732ae0c7e20dc9d5be6be8edd99c3a612
- SFT-CHEM-THERMO-016-BINARY-0111: class=binary; dataset=1; point=111;
target=sha256:570409dfddf7e021d4db4338db3f558bc76a16a26ccc7a3e2b39145d056673b9
- SFT-CHEM-THERMO-016-BINARY-0112: class=binary; dataset=1; point=112;
target=sha256:87f05070a50634dfd867e96db41cdd94ed10d8c6621c52f378c1a26d20ac77f4
- SFT-CHEM-THERMO-016-BINARY-0113: class=binary; dataset=1; point=113;
target=sha256:d264b623885fcc8aff96aa63d29495c1c0b3ed35742ce44221a7cd0ccb4f5f96
- SFT-CHEM-THERMO-016-SELF-0114: class=self; dataset=1; point=1;
target=sha256:2fb96ad5923c2b3aca9ecb6c759d2ad0165e573472778743645b32c464306cb6
- SFT-CHEM-THERMO-016-SELF-0115: class=self; dataset=3; point=1;
target=sha256:a35af64dc657e1c42c4c65b7266fa90ffe9ca58a07e7ec42e032a17d4d1fc5ba
- SFT-CHEM-THERMO-016-SELF-0116: class=self; dataset=3; point=2;
target=sha256:32ef070e6807e14f2308cb76585ce105f613cdc54a4e1296cac6cc4b4d76d89b
- SFT-CHEM-THERMO-016-SELF-0117: class=self; dataset=3; point=3;
target=sha256:12bc1fee6993f502bd934b6b8977c74f52bf18fe1bcb0e0a1a3de4c20165dba
- SFT-CHEM-THERMO-016-TRACER-0118: class=tracer; dataset=5; point=1;
target=sha256:0b88d4ba113413eda078300d7549f3c82e94f64127e6c751121565490e3246f3
- SFT-CHEM-THERMO-016-TRACER-0119: class=tracer; dataset=5; point=2;
target=sha256:18b489389ce6cf8140f0f943d21f9976d257d962acba971afc07bab44f50346b
- SFT-CHEM-THERMO-016-TRACER-0120: class=tracer; dataset=5; point=3;
target=sha256:276496171bcf65f92b5e0b1239c3fd48f58f407da39a4e4b3e47e8c0d550c96c
- SFT-CHEM-THERMO-016-TRACER-0121: class=tracer; dataset=5; point=4;
target=sha256:ca84d389b69d8b4fd03cf1f60e57e48867c22151d3425fa9b2b31b6545d840c5
- SFT-CHEM-THERMO-016-TRACER-0122: class=tracer; dataset=5; point=5;
target=sha256:f41cb4aafcd97dd140ba8e0366328b21754255160138d50f1fa62769c2f608d
- SFT-CHEM-THERMO-016-TRACER-0123: class=tracer; dataset=5; point=6;
target=sha256:5a44c40c28bf35c9639d60c70bd80d8679a1a7c800a3f9edf0730097883d7d4c

- SFT-CHEM-THERMO-016-TRACER-0124: class=tracer; dataset=6; point=1;
target=sha256:e702f8a7e6a7b82ba7c824e22502cad424887ff28af7a25c80df2e44e6b1be57
- SFT-CHEM-THERMO-016-TRACER-0125: class=tracer; dataset=6; point=2;
target=sha256:b25d76cc8ba69b7818fe6fcaff1e5eb7a9982a7f6d09c8bda59bda25127ec7ff
- SFT-CHEM-THERMO-016-TRACER-0126: class=tracer; dataset=6; point=3;
target=sha256:8201df1ede9cce7d4735e9cc668c604ec16a1beb4b3ee4d5756b9a37b5a286c4
- SFT-CHEM-THERMO-016-TRACER-0127: class=tracer; dataset=6; point=4;
target=sha256:dea4713ec911004a093c4d9e20bee00ef9f69e9c7c86f730cb53832990326e00
- SFT-CHEM-THERMO-016-TRACER-0128: class=tracer; dataset=6; point=5;
target=sha256:df829d8d4f8a9a3a36a9bd57e23f444597b5e42079993bf633525fa546bbd802
- SFT-CHEM-THERMO-016-TRACER-0129: class=tracer; dataset=6; point=6;
target=sha256:abd6b42c58b485c05a0246ca3d123d4e9343f33a16edbde73bfaf405786553dc
- SFT-CHEM-THERMO-016-TRACER-0130: class=tracer; dataset=6; point=7;
target=sha256:4e6723ac07b67f42c1d7e2d7a6e9d83436e38bcff24753f635f9f4652445f90d
- SFT-CHEM-THERMO-016-TRACER-0131: class=tracer; dataset=6; point=8;
target=sha256:2e6da342caf9a9b0bb7888e48e3cfa4f921b513f2e921cda67c716b04368aa32
- SFT-CHEM-THERMO-016-TRACER-0132: class=tracer; dataset=6; point=9;
target=sha256:5028c2bf11aa96db6644b2b6663c0b799b1937c9f2fd756e515c1927a5b56a91
- SFT-CHEM-THERMO-016-TRACER-0133: class=tracer; dataset=6; point=10;
target=sha256:bd2a9ed1afdf5da54e29aead72252be96efca5f0688edfe6e14e7ea076b68c8
- SFT-CHEM-THERMO-016-TRACER-0134: class=tracer; dataset=6; point=11;
target=sha256:8bf0f5404dd06c4b9c225d23ba8cd4e2cef18d0dc3bb3231ce1e833dd822a79b
- SFT-CHEM-THERMO-016-TRACER-0135: class=tracer; dataset=13; point=1;
target=sha256:ec86e646e50a514f3bb6e3bc529446b3a342238018c9d64483b6b1fbb8f6370c
- SFT-CHEM-THERMO-016-TRACER-0136: class=tracer; dataset=13; point=2;
target=sha256:9deef0e724f933779f931a39ff25fa65f4aa295a4f87ebe12244fcb34453d275
- SFT-CHEM-THERMO-016-TRACER-0137: class=tracer; dataset=13; point=3;
target=sha256:dc4683ea1f4b395194ed88d72006221c08a8df515a713e8f9108a0cce39687d0
- SFT-CHEM-THERMO-016-TRACER-0138: class=tracer; dataset=13; point=4;
target=sha256:6bb28fe008c5b7768227373529ab1ef2c1028b9b31727cddb06596f34d6bcd33
- SFT-CHEM-THERMO-016-TRACER-0139: class=tracer; dataset=13; point=5;
target=sha256:e0e2dbf74fb1756fec468b43196788125d6e7c7356ad333348a09530f7d0a9f7
- SFT-CHEM-THERMO-016-BINARY-0140: class=binary; dataset=6; point=1;
target=sha256:b897c1c0226a80af4e64cd7bf4f0aa5d820b8bc302bd83076d71fa525ffaa523
- SFT-CHEM-THERMO-016-BINARY-0141: class=binary; dataset=6; point=2;
target=sha256:8b26edfcca317d0774757b0d8b63153ec17ad3e7e4f22f2b322de475d3d46d41
- SFT-CHEM-THERMO-016-BINARY-0142: class=binary; dataset=6; point=3;
target=sha256:0a25a168edbcea6ee22d24b25f972811905c486356b2104b17ae8940aab657f8
- SFT-CHEM-THERMO-016-BINARY-0143: class=binary; dataset=6; point=4;
target=sha256:2439c2e5146ac0a0a35dd32c83ab3ff60178b8504f0f4f631f9ea34b3aba483b
- SFT-CHEM-THERMO-016-BINARY-0144: class=binary; dataset=6; point=5;
target=sha256:8b62bc63906f7c0b20221fe6b90c544b402372396ed9692acae13354802ef4df
- SFT-CHEM-THERMO-016-BINARY-0145: class=binary; dataset=8; point=1;
target=sha256:b90aacd29951852d72ef74adc49d443eb83799ed99b49d8d307da96e470306ce

- SFT-CHEM-THERMO-016-BINARY-0146: class=binary; dataset=8; point=2;
target=sha256:f980cf61feac6d6c34725d74ef795368801634e0a86efbcefa90c63c23875843
- SFT-CHEM-THERMO-016-BINARY-0147: class=binary; dataset=8; point=3;
target=sha256:b5e4906ac317011950bdfd9f6a984c3f5764100e625bfae45e8e53799fe54e14
- SFT-CHEM-THERMO-016-BINARY-0148: class=binary; dataset=8; point=4;
target=sha256:9a1c4fd25cabe41e60382ce84a12e62a9fa03e2a7b59a8314c3068fac225fcbd
- SFT-CHEM-THERMO-016-BINARY-0149: class=binary; dataset=8; point=5;
target=sha256:d58f167df3ba0a6f1c41716d818dae54fb6ae080315ece45e6adce4908683399
- SFT-CHEM-THERMO-016-BINARY-0150: class=binary; dataset=10; point=1;
target=sha256:09b2f3e465f31baa8e39bc0de0e5f12d394fefa3de2b60443f44118dec7b3626
- SFT-CHEM-THERMO-016-BINARY-0151: class=binary; dataset=10; point=2;
target=sha256:48a0e9c59b75aff23ddf07c765ca0077b5fdfe07efa9c22a88804659d96e4fd0
- SFT-CHEM-THERMO-016-BINARY-0152: class=binary; dataset=10; point=3;
target=sha256:8eae46cae0450012a09c585e8c3f996b37e8172515a1256c4553aec63fb5b4c
- SFT-CHEM-THERMO-016-BINARY-0153: class=binary; dataset=10; point=4;
target=sha256:060f0d9e6d40c8e61f99439a8206eab53f4056901f4114fcd73e0090791d04ce
- SFT-CHEM-THERMO-016-BINARY-0154: class=binary; dataset=10; point=5;
target=sha256:687184e9f623872d8b34d31211820f99b210b1c82cefdb8270f69914ef41e62c
- SFT-CHEM-THERMO-016-BINARY-0155: class=binary; dataset=12; point=1;
target=sha256:4e649824257ab0b64c42bdf6b021b79b3c5552b3f9c71f63abb186c52fa26d2e
- SFT-CHEM-THERMO-016-BINARY-0156: class=binary; dataset=12; point=2;
target=sha256:4a5db5796925dce03a5617ba1a5b4399f5a7f29ca477bfeeadb1e7070fd71ff8
- SFT-CHEM-THERMO-016-BINARY-0157: class=binary; dataset=12; point=3;
target=sha256:4f71f2f5434dc3a21f472534e8cae7e9f86ff5fc6ae672b93b991758ff455f82
- SFT-CHEM-THERMO-016-BINARY-0158: class=binary; dataset=12; point=4;
target=sha256:cc05af7d9b7e5faed6f37a026be49de246ede5031222e1c34f35683feeffeef4
- SFT-CHEM-THERMO-016-BINARY-0159: class=binary; dataset=12; point=5;
target=sha256:eb27ddd521494abd7544ad0cc8f5b230280c3180f28bd6dd63edb96a234af230
- SFT-CHEM-THERMO-016-BINARY-0160: class=binary; dataset=14; point=1;
target=sha256:9cc22b493e21936f2b59320e2bc004524caa0adb3fee4fc3ef0bfb401bdf73b0
- SFT-CHEM-THERMO-016-BINARY-0161: class=binary; dataset=14; point=2;
target=sha256:7bf4f7dc5b0f351bc876582d4e67588bb61dd866e00b5d8e1b59f11c0b25cbb9
- SFT-CHEM-THERMO-016-BINARY-0162: class=binary; dataset=14; point=3;
target=sha256:458146bbd4061d6b8757bab701a172f81f6fe41ccfae504bb21bae95fa786eec
- SFT-CHEM-THERMO-016-BINARY-0163: class=binary; dataset=14; point=4;
target=sha256:3a89495fa394855229be5c4c8929a55a0dc4956cd76c9d41dd395249536ef62b
- SFT-CHEM-THERMO-016-BINARY-0164: class=binary; dataset=14; point=5;
target=sha256:2e4db97deabf78ac4e2cd36846d637bcbbe760a0e14a4d768c0fc6927c03fada
- complete vector: 138 binary, 4 self and 22 tracer diffusion records
- exact diffusion ranges: {'binary': {'minimum': '199/200000000000', 'maximum': '10329/1000000000000'}, 'self': {'minimum': '179/1000000000000', 'maximum': '5949/1000000000000'}, 'tracer': {'minimum': '179/10000', 'maximum': '5949/1000'}}
- source coverage: all 30 datasets and 373 points preserved from three complete NIST ThermoML sources
- tampered_negative_diffusion_rejected: True
- complete_164_record_vector_retained: True

- all_binary_self_tracer_rows_retained: True
- all_diffusion_datasets_retained: True
- all_methods_and_phases_retained: True
- all_absent_conditions_translated_to_EmptyOne: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if any migrating or medium identity, phase, condition, cell, transition or tick is erased; if adjacency or constituent conservation fails; if a random premise, signed/numerical-zero proof value, continuum, Fick/Brownian/random-walk/Stokes-Einstein model, activation/transport fit, interpolation, regression, selection or target correction enters; if target content opens before all 164 identities seal; if any direct binary, self or tracer row, uncertainty, method, dataset, complete source or companion provenance is omitted; or if any target is tampered.

Measurement receipt: sha256:63696cad12aef6d4a5dd536ae31babbc3032e8c6ba5867af3f191090fc7512b3.
Isolation certificate: sha256:2100503fd6e32b8b1dfade71fb46c3b5a9c00c540c600441c428b09f4f1e618a.
Custody certificate: sha256:d8b11bda796dc99f008f29b17bdb6aca15ff9d84f9913b0156dc58c725dd1678.

Registered source descriptions:

- National Institute of Standards and Technology ThermoML Archive: ()

Registered target identities:

- None.

Meaning. The thermodynamic extension derives finite microstate support, temperature, energy, heat/work, entropy, enthalpy, free direction, chemical potential, activity, fugacity, phases, colligative response, solvation and coupled transport while keeping external scalar inscriptions downstream. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; absent external condition coordinates are structural EmptyOne
- no negative, irrational, imaginary, logarithmic, floating, signed or continuum SFT proof value
- no imported Fick equation, Brownian continuum, random-walk probability, Stokes-Einstein relation or fitted activation/transport coefficient
- no interpolation, regression, fit, selected species/medium/phase/condition/method/dataset/row or target correction
- no species, medium, phase, composition, temperature, pressure, method, value, uncertainty or target hash before prediction seal
- every complete source and non-diffusion companion dataset remains preserved; companions never become diffusion measurements

Immutable identities. Source manifest

sha256:7b4a832e8ee821bce6398f5c7266c2ad66df2a5fdf81eae3f0082142757688db; derivation seal
sha256:918cbcce056c1375fabcd49ac7015bb1488d4e3f8e5fb43085a8dda06319c641; engine receipt
sha256:f7dfd76d3091935073b5eac5cd205ceb53134e52ff85e583d02d3e83d22a9569 at receipts/engine/model_admitted/SFT-CHEM-MOLECULAR-DIFFUSION-RELATION-016-f7dfd76d30919350.json; empirical hash sha256:3beae6e701abafc0ca41ebbf8453b31a8ea415f8b6b14a16bbe8939be29231; measurement receipt sha256:63696cad12aef6d4a5dd536ae31babbc3032e8c6ba5867af3f191090fc7512b3.

141. Exact composition-bound viscous transport law

Claim identity: SFT-CHEM-VISCOUS-TRANSPORT-RELATION-017

Question and theorem. Chemical viscosity is counted adjacent-layer momentum-packet exchange whose carrier retains every component, phase, condition and time/space resource. Chemistry owns the composition-bound relation while the momentum carrier is inherited explicitly from Physics. Measured viscosity is exact positive post-seal support; no continuum velocity gradient, Newtonian constitutive equation, Arrhenius/WLF/VFT form, logarithm or fitted coefficient is imported.

complete-composition-phase-condition-momentum-account__distinct-held-complete-component-identities__counted-adjacent-layer-momentum-exchange__held-opposed-layer-transfer-orientation__exact-positive-packet-exchange-tick-and-condition-support__exact-positive-postseal-viscosity-support__complete-value-free-425-record-identity-seal__depth-independent-common-replication-and-record-append

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-CPLX-TIME-SPACE-001
- SFT-PHYS-MECH-MOMENTUM-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-FLUID-VISCOSITY-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013
- SFT-CHEM-MOLECULAR-DIFFUSION-RELATION-016

Generated grammar. Generate the literal product of carrier, identity, transfer, orientation, resource, magnitude, prediction and extension forms; decide all 256 candidates only from admitted momentum, adjacency, transition, conservation, exact-resource, composition, EmptyOne and finite-successor laws.

Boundary: Every finite pure, binary or ternary viscosity record with complete component identities, phase, exact conditions, counted adjacent layers, momentum packets, exchanges and ticks. External testing preserves all 425 direct viscosity points and all companion datasets from three complete NIST sources.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *complete-composition-phase-condition-momentum-account__distinct-held-complete-component-identities__counted-adjacent-layer-momentum-exchange__held-opposed-layer-transfer-orientation__exact-positive-packet-exchange-tick-and-condition-support__exact-positive-postseal-viscosity-support__complete-value-free-425-record-identity-seal__depth-independent-common-replication-and-record-append*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-composition-phase-condition-momentum-account	detached-viscosity-number-or-continuum-field	A detached number or field erases momentum exchange.
identity	distinct-held-complete-component-identities	anonymous-or-collapsed-composition	Collapsed composition cannot own a chemical viscosity relation.
transfer	counted-adjacent-layer-momentum-exchange	continuous-velocity-gradient-premise	A continuum gradient imports an ungenerated carrier.
orientation	held-opposed-layer-transfer-orientation	signed-shear-or-stress-proof-magnitude	Signed shear imports prohibited arithmetic.
resource	exact-positive-packet-exchange-tick-and-condition-support	unrecorded-packet-exchange-time-or-condition	Unrecorded resources cannot reconstruct viscosity.
magnitude	exact-positive-post-seal-viscosity-support	imported-Newtonian-Arrhenius-WLF-VFT-or-fit	An imported constitutive or fitted law selects the magnitude.
prediction	complete-value-free-425-record-identity-seal	substance-composition-condition-method-or-value-readable-before-seal	Readable targets could select the law.
extension	depth-independent-common-replication-and-record-append	refit-after-exchange-replication-or-record-append	Refitting destroys exact exchange provenance.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:6cdc85dc279a9981e6d04c818d7607bc708fe116e05e7b79f10a3dcb445367b0.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:e1ca95407c2e13b5431ae3e67fa0ebbddb431d7fc2fbf2972eb4065a0ff510b0.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:96b43d968fa4c1ae90ad40e4b3632c90951d3b63a7b05e19b7ef12f8506f4d0b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:5cec5ed561661cd4bb33f1873162036d20dfa8ade499bba20f8341bc55f1a2e3.

Independent reconstruction. Implementation hash

sha256:32fc2c9391821cbe2257d182742869045e1bf89f58330bdeedf036f31ca72efaf; independent certificate sha256:68b22b29c07ade1145095453df20edc9148bc53462c0c2ecb5d061cd0db99345; external-validation hash sha256:255b896a9d3d36d578be15714c3e8489149603a5b8f4a69a6743f7324ed5bbff. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-TRC-THERMOML-FPE-2018-474-6-13
- NIST-TRC-THERMOML-JCED-2005-50-1038-1042
- NIST-TRC-THERMOML-FPE-2017-453-13-23

Comparison records:

- SFT-CHEM-THERMO-017-VISCOSITY-0001: dataset=2; point=1;
target=sha256:51a596ffaff23d0b5321b2bf3c405103bb29cc5a3a0617f4d6135f6663e95aff

- SFT-CHEM-THERMO-017-VISCOSITY-0002: dataset=2; point=2;
target=sha256:2c65ce3f38a2fbb4e712e99dbd399ce0658df6cd6e76393d1b9328a8529d744e
- SFT-CHEM-THERMO-017-VISCOSITY-0003: dataset=2; point=3;
target=sha256:b3a837c25a69a6ef9f76cab40dae48b4f936c5c3b538dfaa7316db24eab5359f
- SFT-CHEM-THERMO-017-VISCOSITY-0004: dataset=2; point=4;
target=sha256:befd6b51def7089c937ed235ef2f95203ff3c66ecf5906b7699286060a911583
- SFT-CHEM-THERMO-017-VISCOSITY-0005: dataset=2; point=5;
target=sha256:d074385de4615304defe411ce553b0bcbdad18bc5828fd64bd3377e5fe30ce6f
- SFT-CHEM-THERMO-017-VISCOSITY-0006: dataset=2; point=6;
target=sha256:659ce8a4e57638377dd3ad09e065d033b9954bf047bc1a295ac7e3f56a477da9
- SFT-CHEM-THERMO-017-VISCOSITY-0007: dataset=2; point=7;
target=sha256:b9fc19497ee3944cb3ac3b938f6fb52714a93e04d0349e9ab8acaf8eaa7d9904
- SFT-CHEM-THERMO-017-VISCOSITY-0008: dataset=2; point=8;
target=sha256:a8debf3431e4fb1f05a1a678285c0cff36a8c52960a06266d79289bedd28fb53
- SFT-CHEM-THERMO-017-VISCOSITY-0009: dataset=2; point=9;
target=sha256:9643e320ecfceb89bec4c3cce23c77bc56e5ae976d6338ae11474c95b7d6cf5
- SFT-CHEM-THERMO-017-VISCOSITY-0010: dataset=2; point=10;
target=sha256:a2672dd38b5939c9808c7add23a3b78a9ce747cebb882b5d9f17254058437982
- SFT-CHEM-THERMO-017-VISCOSITY-0011: dataset=2; point=11;
target=sha256:837943a109cbd821e4a580f191fe5dc99c70b784c8c238abd7cc6b6a5e191d39
- SFT-CHEM-THERMO-017-VISCOSITY-0012: dataset=2; point=12;
target=sha256:7481c28bb5ced9a3ec9d3a68ea98bff9106dbd21ea96904ee8424056fe0a0204
- SFT-CHEM-THERMO-017-VISCOSITY-0013: dataset=2; point=13;
target=sha256:51ba8830943c7f76f251d5220c8555bfc42a5f65646a59ad2d1e0e934e96c76
- SFT-CHEM-THERMO-017-VISCOSITY-0014: dataset=2; point=14;
target=sha256:a2dfc10d6d87e5deb9f5d847899c38824063116091fec5d3e2866c15179b2874
- SFT-CHEM-THERMO-017-VISCOSITY-0015: dataset=2; point=15;
target=sha256:a317f065287330ebd3daca481d92c8612975a13e79a529ab8266159494a4ba1
- SFT-CHEM-THERMO-017-VISCOSITY-0016: dataset=2; point=16;
target=sha256:f793fee9616599d87b5f1bf6aaaf3f6eb7777d16c575bb0a4571dbde43475f78
- SFT-CHEM-THERMO-017-VISCOSITY-0017: dataset=2; point=17;
target=sha256:0dde1f2e14fe499cb442ddac0fd9ac934541bf481dcb5c00167b51923089e947
- SFT-CHEM-THERMO-017-VISCOSITY-0018: dataset=2; point=18;
target=sha256:fa9f073d3f533e69100d24b57d4398e83dc9976ae2ff0b464e8285476435aac7
- SFT-CHEM-THERMO-017-VISCOSITY-0019: dataset=2; point=19;
target=sha256:2640e9d178fac3f5cf24ee49c1e19080e164754e171221ff703b2566494c79fd
- SFT-CHEM-THERMO-017-VISCOSITY-0020: dataset=2; point=20;
target=sha256:abbdcd67210d004c8512df9310a9ddcee5d62f41746904e7a88c604af5033226
- SFT-CHEM-THERMO-017-VISCOSITY-0021: dataset=2; point=21;
target=sha256:180a92d3734a39632a7cda42e2ede23b946bc6df46132afe8d9ad3ba67443793
- SFT-CHEM-THERMO-017-VISCOSITY-0022: dataset=2; point=22;
target=sha256:9ea24279efd35d57313c0ec6d0c19b1d36d18edda7f1469c724ea0bfdf38a853
- SFT-CHEM-THERMO-017-VISCOSITY-0023: dataset=2; point=23;
target=sha256:bbd9ef84def5797b7c5f84cae4049d300b3fb1fb368b51072903672ff6f86c07

- SFT-CHEM-THERMO-017-VISCOSITY-0024: dataset=2; point=24;
target=sha256:e2cf92bb612353387c0e09a90b979f9982c2950d3ba1f6606ef1a8f50a9c5573
- SFT-CHEM-THERMO-017-VISCOSITY-0025: dataset=2; point=25;
target=sha256:dc38f4d3503a5b64159f1d0ae683de4df61383e59ceea1e1519070a0718fa92d
- SFT-CHEM-THERMO-017-VISCOSITY-0026: dataset=2; point=26;
target=sha256:34bc0b774a0e8a860c5b6d63a9773481d4d056aac9c1cd294e336108f0455bfe
- SFT-CHEM-THERMO-017-VISCOSITY-0027: dataset=2; point=27;
target=sha256:996b6935bfd36b62d4afd1cd4279a58b75a6177758d8eaab1c8d1f3a180bbf7c
- SFT-CHEM-THERMO-017-VISCOSITY-0028: dataset=2; point=28;
target=sha256:a322e2d82caeed3fc029e0ce109c897119d89212f28afb51f1532f4d439ac537
- SFT-CHEM-THERMO-017-VISCOSITY-0029: dataset=2; point=29;
target=sha256:5c022153ebf25fe36a3e47d2968c1aa051cc6cbb7c724891e4c5a7768aa3890b
- SFT-CHEM-THERMO-017-VISCOSITY-0030: dataset=2; point=30;
target=sha256:e1c627ff4af877b88aa47b3373176d4686f6951282cf804abeb9b6edb9871d8b
- SFT-CHEM-THERMO-017-VISCOSITY-0031: dataset=2; point=31;
target=sha256:1a041e71dad6230c3de6741285f0af2843a562d5dbe8f6f26692fd03c7143eee
- SFT-CHEM-THERMO-017-VISCOSITY-0032: dataset=2; point=32;
target=sha256:9f0f738c3472e159e49508b532e4aa790f5b816d5772d1e1512e37317cac08b1
- SFT-CHEM-THERMO-017-VISCOSITY-0033: dataset=2; point=33;
target=sha256:700ef92032f65f9ed177bc8bd3f0afc0a368b22bdfd68848793d861118f4d6ba
- SFT-CHEM-THERMO-017-VISCOSITY-0034: dataset=2; point=34;
target=sha256:f3d213d9d2521944ecb4e7bfb722e5876275f6b4d115b2f0195f12bd6b13e77d
- SFT-CHEM-THERMO-017-VISCOSITY-0035: dataset=2; point=35;
target=sha256:8939220059409169c4232280a9bcb858eda52c8b87e55be58e9226d9039a17a6
- SFT-CHEM-THERMO-017-VISCOSITY-0036: dataset=2; point=36;
target=sha256:4a16dc3e15de1842128eaf3a3cfb7bfc9b9b526ca81893e32bd9501686a9eb33
- SFT-CHEM-THERMO-017-VISCOSITY-0037: dataset=2; point=37;
target=sha256:65f8527e3d87dd0878b3a0ccdc06f686441040406beb25b49d70711beb04d062
- SFT-CHEM-THERMO-017-VISCOSITY-0038: dataset=2; point=38;
target=sha256:27600182456376136098926845aef8fe58550e9fdaaa4391eed8ffc3439dfc9a
- SFT-CHEM-THERMO-017-VISCOSITY-0039: dataset=2; point=39;
target=sha256:689b76c5d692898fa0fc6cf55769269e69e1da6bc1dba58af71139ec5c8054dd
- SFT-CHEM-THERMO-017-VISCOSITY-0040: dataset=2; point=40;
target=sha256:2f03cdf16a88e133a6f3bdb4affd4ef2571c1b86c501b6a4af520b44f988842b
- SFT-CHEM-THERMO-017-VISCOSITY-0041: dataset=2; point=41;
target=sha256:74083332a7f8823446d8b25d62f24066c74a9cde26b85cde88444e911b9c0914
- SFT-CHEM-THERMO-017-VISCOSITY-0042: dataset=2; point=42;
target=sha256:f3216f591aa21406250900e8b9e6e14910a3d7a16ec055d8533d0a5a84ad91cd
- SFT-CHEM-THERMO-017-VISCOSITY-0043: dataset=2; point=43;
target=sha256:452ef4875f796131f5edff625a081d8ba81b423d56f911ca1d1719594db390aa
- SFT-CHEM-THERMO-017-VISCOSITY-0044: dataset=2; point=44;
target=sha256:091726ba37a4f9681ca53649353e6703bd96c912cb7b55679a6604d3a8022239
- SFT-CHEM-THERMO-017-VISCOSITY-0045: dataset=2; point=45;
target=sha256:078bf97404d6ea19bf4dc552a113c8e3c91d19b880f0a16e6a5037b1bae882ed

- SFT-CHEM-THERMO-017-VISCOSITY-0046: dataset=2; point=46;
target=sha256:f9afda29b470fc43a98ccfe96db9b06320d6c4268f5fd6241aa28a5cfc4e2fe7
- SFT-CHEM-THERMO-017-VISCOSITY-0047: dataset=2; point=47;
target=sha256:3070b404776a2e7f5b0595aa6565e981cbbe3038ca9bd20a8f3fc110bb36548c
- SFT-CHEM-THERMO-017-VISCOSITY-0048: dataset=2; point=48;
target=sha256:a33c2c62af6eef4cfa8a8b263b919076aec3f32074045ddfc81be440f45338ee
- SFT-CHEM-THERMO-017-VISCOSITY-0049: dataset=2; point=49;
target=sha256:d296f4abc337301081b1b0f80930856c292ec21ec646a8a87ea2879f0b5ece9b
- SFT-CHEM-THERMO-017-VISCOSITY-0050: dataset=2; point=50;
target=sha256:45ee45eca5bc2557f1b665d46121edb473d4bd75776c38c27bc020a6520cea96
- SFT-CHEM-THERMO-017-VISCOSITY-0051: dataset=1; point=1;
target=sha256:d6b90206e0ba002578b83911d10ae36ad288cf11bbbc29999b711ae4f0087f5c
- SFT-CHEM-THERMO-017-VISCOSITY-0052: dataset=1; point=2;
target=sha256:9360e599336fd3fef477efc70f2259f886f4efe53a216d6cc9d3b2e502328c59
- SFT-CHEM-THERMO-017-VISCOSITY-0053: dataset=1; point=3;
target=sha256:75661bd6e0ac04dd899cc026a2a20bedd2a550bcca5722b36068049eb357572b
- SFT-CHEM-THERMO-017-VISCOSITY-0054: dataset=1; point=4;
target=sha256:2a12797a38c89cae0e28997ca5b0a6358c23baa22750a5bad36ff25911f25a59
- SFT-CHEM-THERMO-017-VISCOSITY-0055: dataset=1; point=5;
target=sha256:89017f53e8883d727bd7f2d4e923affefb01dbc0d6938cab547163092cc993d0
- SFT-CHEM-THERMO-017-VISCOSITY-0056: dataset=3; point=1;
target=sha256:bc1910b24600d78ada2bc49b0522af3367b47bfee58128a68519e0ebffbf5a27
- SFT-CHEM-THERMO-017-VISCOSITY-0057: dataset=3; point=2;
target=sha256:764db2357d17518467020631146fe12b56176f1d5f18098a31dba51249a6617d
- SFT-CHEM-THERMO-017-VISCOSITY-0058: dataset=3; point=3;
target=sha256:b953c526964017ec2d92b4994b08c7ed5e015c058fcca79e329e06764c53b1f6
- SFT-CHEM-THERMO-017-VISCOSITY-0059: dataset=3; point=4;
target=sha256:c0c4547b5948c4ed0618d9aefbc40e72b11818f35e3c9cb1cccd9bdc9b155030
- SFT-CHEM-THERMO-017-VISCOSITY-0060: dataset=3; point=5;
target=sha256:67f64500d78b003811b90ba247c9fff6dcf29fa9f8ce23943587b962f3f86cff
- SFT-CHEM-THERMO-017-VISCOSITY-0061: dataset=3; point=6;
target=sha256:5759e3c3a6abeda86b4e685f933cfd1b4878426ba816229cc42b68eddcfb705
- SFT-CHEM-THERMO-017-VISCOSITY-0062: dataset=5; point=1;
target=sha256:98f265ab766c495b9c0596c82ed286a158dec1d5ea0aa0dd846c1a207c34ab52
- SFT-CHEM-THERMO-017-VISCOSITY-0063: dataset=5; point=2;
target=sha256:c04f52d419a2bb3b6ef3ecfd3df8e889a06dbf576ac154cbd4a52d4d903169db
- SFT-CHEM-THERMO-017-VISCOSITY-0064: dataset=5; point=3;
target=sha256:38bcff6b042133d591d9cc13665a0ddebb48ae10f24079e91a43817d7587bc4c
- SFT-CHEM-THERMO-017-VISCOSITY-0065: dataset=5; point=4;
target=sha256:d9aa4cb5e5c9da1ca0a1965a134e007862b218daa834c780d6f1cfaee40aa716
- SFT-CHEM-THERMO-017-VISCOSITY-0066: dataset=5; point=5;
target=sha256:3cc319adf52eff6529bd21e3aaa8a6c02b09ed75f992375835fc6978fd1a356b
- SFT-CHEM-THERMO-017-VISCOSITY-0067: dataset=5; point=6;
target=sha256:ac9d7a65b2cbb69d484ad7ace3718fd57b232323a0ab7354e7105fdfa73cdbdb

- SFT-CHEM-THERMO-017-VISCOSITY-0068: dataset=5; point=7;
target=sha256:c47756d8229200f83e0069fbf6e186123316f80b36bf4c03ff3acc76a7ea37c0
- SFT-CHEM-THERMO-017-VISCOSITY-0069: dataset=5; point=8;
target=sha256:afef405c463186e60d2568108979ca9f4f202b22c01880da76d7e3ea78d542ce
- SFT-CHEM-THERMO-017-VISCOSITY-0070: dataset=5; point=9;
target=sha256:095001197c64d3e43d72f41ac0b3ae101ab638d0943a87a446f75c1993536144
- SFT-CHEM-THERMO-017-VISCOSITY-0071: dataset=5; point=10;
target=sha256:4dcbaa0e368120ea981c19b922973d18537c324ef7782ae05f578e8ff0d2e282
- SFT-CHEM-THERMO-017-VISCOSITY-0072: dataset=5; point=11;
target=sha256:d5bd76ac6e0c35ef78411d1c2907f8d08383475eb6531984b68201f3294bd4ad
- SFT-CHEM-THERMO-017-VISCOSITY-0073: dataset=5; point=12;
target=sha256:603b0ab0a89c62a716d3848906dbfc32cbad2b06ecf80bd9e3ef8fdfa64c625f
- SFT-CHEM-THERMO-017-VISCOSITY-0074: dataset=5; point=13;
target=sha256:72f55ac368c6a5b45c3848d4ed604b2c99516ceb856b86c7f6a48376f54815fd
- SFT-CHEM-THERMO-017-VISCOSITY-0075: dataset=5; point=14;
target=sha256:b3fabff8c57c1df5959cdb575df18a2383ff35151bb50ee2ad1d34379d123da2
- SFT-CHEM-THERMO-017-VISCOSITY-0076: dataset=5; point=15;
target=sha256:31498cec4d360eed592e9ed41f538873ad6cc5f38c4d4448b076448968f265d
- SFT-CHEM-THERMO-017-VISCOSITY-0077: dataset=5; point=16;
target=sha256:efae41e7010f24c47000b33a3f6b8f947ade5e543d869c4b5cd639ef971170da
- SFT-CHEM-THERMO-017-VISCOSITY-0078: dataset=5; point=17;
target=sha256:a89f19a333ed83453a286838275ad02ba5d8a173fc3cda4464a73c5bc701249f
- SFT-CHEM-THERMO-017-VISCOSITY-0079: dataset=5; point=18;
target=sha256:f6dd20234ea98c93909e1ce176fc923f122f1cffa3e25e9b44f3752ca4173d15
- SFT-CHEM-THERMO-017-VISCOSITY-0080: dataset=5; point=19;
target=sha256:6326d8faa07fc9c04e1349fbc0388440316091386d671faac3a9374ec600b2
- SFT-CHEM-THERMO-017-VISCOSITY-0081: dataset=5; point=20;
target=sha256:371b858636b6867288c554a7bf9de94c49d4e4b3aaa450f46353ba5e58f23943
- SFT-CHEM-THERMO-017-VISCOSITY-0082: dataset=5; point=21;
target=sha256:f3f4fe3d8290fe17f25fcd5067e04e2bbf8a1f65f4c031c8e5bccff88ecf4e83
- SFT-CHEM-THERMO-017-VISCOSITY-0083: dataset=5; point=22;
target=sha256:22637f6297ae3f4a4597fe0827f95be7797e86ab2f125ee94ec486c9bc4a3833
- SFT-CHEM-THERMO-017-VISCOSITY-0084: dataset=5; point=23;
target=sha256:9dfda94e143aace343e7ce9718851244fb3650572cf711fd0366e4e217cc4d40
- SFT-CHEM-THERMO-017-VISCOSITY-0085: dataset=5; point=24;
target=sha256:81b36d2781c5bbc58f8eace7b85423b63108a2e31272c51fcdc2b28c1a2ec4d0
- SFT-CHEM-THERMO-017-VISCOSITY-0086: dataset=5; point=25;
target=sha256:4cb3718803350eef937365bc4a8011cdd0c24679f97d424d2b8f239be1d380fa
- SFT-CHEM-THERMO-017-VISCOSITY-0087: dataset=5; point=26;
target=sha256:085b0f65f60677b768fcb64ccfe6b2d06cfffcdab2862eedc6e4eae64f700a57c
- SFT-CHEM-THERMO-017-VISCOSITY-0088: dataset=5; point=27;
target=sha256:58a8ab9a2a232e194d74b61b84ce930e29f1ea37027440d2a584d972c323cfca
- SFT-CHEM-THERMO-017-VISCOSITY-0089: dataset=5; point=28;
target=sha256:2f2801f4b8d64be352e7c91d9d550eec391706bc08ba800c51339097f84f2ebd

- SFT-CHEM-THERMO-017-VISCOSITY-0090: dataset=5; point=29;
target=sha256:543c22788e87483b3052fef51d78297957997ba06bd68e68b66a7748842a3680
- SFT-CHEM-THERMO-017-VISCOSITY-0091: dataset=5; point=30;
target=sha256:7a94a1aaadd8ca3a6a66104dc1ed23e04469006b80e607f41d5007def8cf3ae8
- SFT-CHEM-THERMO-017-VISCOSITY-0092: dataset=5; point=31;
target=sha256:864247d9571c159721b05298a0cbbc6501b01fca67bda66fed3795825ac4e3ae
- SFT-CHEM-THERMO-017-VISCOSITY-0093: dataset=5; point=32;
target=sha256:4019ef74105ede006fa70c4d85577b464b4a59a75fa1edf0c0996ab2ee27007a
- SFT-CHEM-THERMO-017-VISCOSITY-0094: dataset=5; point=33;
target=sha256:2fc8d6d6972c05594ad841d5ec278d7a9adc436dab493cc37085a2e2fd319ec7
- SFT-CHEM-THERMO-017-VISCOSITY-0095: dataset=5; point=34;
target=sha256:c360294e16ffe0b8e78914cfae3c6fdc627f6348eb87047d81d8721584ce2ad3
- SFT-CHEM-THERMO-017-VISCOSITY-0096: dataset=5; point=35;
target=sha256:1ac6aa2e8ad2fddc5b055d92f21a3c7658c37895ab0f0fae66e6ecb56819298a
- SFT-CHEM-THERMO-017-VISCOSITY-0097: dataset=5; point=36;
target=sha256:ff3bae9b7b7aff1dcf9df2bd725bb7838f79c7b1b572b1e2c88d0daaff0606fa
- SFT-CHEM-THERMO-017-VISCOSITY-0098: dataset=5; point=37;
target=sha256:642b85abc34175f30a5d6e0e60265044914b31e67ce77cc698e57c183138907b
- SFT-CHEM-THERMO-017-VISCOSITY-0099: dataset=5; point=38;
target=sha256:b9e83c5b9314d518d3694dd5ea7469bbb9ae6c85b628bc67ea38a31a3ba19bb2
- SFT-CHEM-THERMO-017-VISCOSITY-0100: dataset=5; point=39;
target=sha256:edf1eb37ee2618bd9b70f30881547c675054b096b072e14249f7f6be5d3ef34f
- SFT-CHEM-THERMO-017-VISCOSITY-0101: dataset=5; point=40;
target=sha256:a3f787239d2d235820d23f0364295904f441fba1da8e78bacdf7ec8a4aae47ee
- SFT-CHEM-THERMO-017-VISCOSITY-0102: dataset=5; point=41;
target=sha256:de8ff29bc3ee9e2992719672e8f81b0be352b01bd951af3940e8556fb42cdd0a
- SFT-CHEM-THERMO-017-VISCOSITY-0103: dataset=5; point=42;
target=sha256:15fc90ea0611c50ba8599c3f9639039ac0573de7d1407cc88dc240c468601820
- SFT-CHEM-THERMO-017-VISCOSITY-0104: dataset=5; point=43;
target=sha256:08d3b1221bd043786a1c36337e0c68a399c30d07d9c838810dcb40690d92380a
- SFT-CHEM-THERMO-017-VISCOSITY-0105: dataset=5; point=44;
target=sha256:ac26527104deb9ae726e04bdeb8a869aea516074777d07dc7e09b8bdb9c2acc9
- SFT-CHEM-THERMO-017-VISCOSITY-0106: dataset=5; point=45;
target=sha256:ab7129e9f2585d797a53b36e2219800a30d37a1922a33f50fc5017bdb14152f9
- SFT-CHEM-THERMO-017-VISCOSITY-0107: dataset=5; point=46;
target=sha256:a1799aa9a138557b9f1b7d3e2c21184e1645b00749752c3fbc250c53fefa5854
- SFT-CHEM-THERMO-017-VISCOSITY-0108: dataset=5; point=47;
target=sha256:6936dcdd5ea2756ace5ac72e191f70d24007b8c27cfb3a24241d452cbad8ac87
- SFT-CHEM-THERMO-017-VISCOSITY-0109: dataset=5; point=48;
target=sha256:e32a72ff7bd69b29c2cc6d78e1b7ff93ae9a77e41538da3250c4edfae0437768
- SFT-CHEM-THERMO-017-VISCOSITY-0110: dataset=5; point=49;
target=sha256:ab30a1c6cb454edb44a80b6f06de73e963c85417f5afb76240de2f8a596d08bc
- SFT-CHEM-THERMO-017-VISCOSITY-0111: dataset=5; point=50;
target=sha256:aa1bbbbbdecbcf0f2262878ee89d23a012c11a66ae0718e21c1392a9df2b78ec7

- SFT-CHEM-THERMO-017-VISCOSITY-0112: dataset=5; point=51;
target=sha256:d2ab9bc9676a57410931fa76b37d11775283d47ecf025bfc0370c81ddf28871e
- SFT-CHEM-THERMO-017-VISCOSITY-0113: dataset=5; point=52;
target=sha256:cd6b4b4faa3813eca7e19132974df087731394a3583cca853b600e617a45e167
- SFT-CHEM-THERMO-017-VISCOSITY-0114: dataset=5; point=53;
target=sha256:91f7870ee3a2ca2d3a51765977b46d2e8d7fc0a70ec2d7d229d743a8e0c4effc
- SFT-CHEM-THERMO-017-VISCOSITY-0115: dataset=5; point=54;
target=sha256:4f30101fdaf8e45710198e84836ce136df55dec67bee658144b6b548b603c666
- SFT-CHEM-THERMO-017-VISCOSITY-0116: dataset=5; point=55;
target=sha256:d594b292fb9bee7b60075194970a0f0472161776e2885c1bd82ed11957a956c
- SFT-CHEM-THERMO-017-VISCOSITY-0117: dataset=5; point=56;
target=sha256:8c4ba8b0e6425cfc7e400b7ae205214d0094ebc83760ab49b22b713b863a4397
- SFT-CHEM-THERMO-017-VISCOSITY-0118: dataset=5; point=57;
target=sha256:02172581bf33c27cb7769db39a65a31278098a8638323ace4ac88599b52f935f
- SFT-CHEM-THERMO-017-VISCOSITY-0119: dataset=5; point=58;
target=sha256:50ca80c71c44369470b467ba442ceca22245abb36052c65d02b9c98c14759057
- SFT-CHEM-THERMO-017-VISCOSITY-0120: dataset=5; point=59;
target=sha256:3a5955170e7888d781043de5813bb5bfb2a4fbb7641085f702d0c0b72800275f
- SFT-CHEM-THERMO-017-VISCOSITY-0121: dataset=5; point=60;
target=sha256:b65dc8c0352951b1809b537fc467fe640ae4d20ebdc51a11d24ac08c5545339d
- SFT-CHEM-THERMO-017-VISCOSITY-0122: dataset=5; point=61;
target=sha256:ed052efc244fabe3c28efe9e8ba809c41dd22ebffca6078f8e3edd3d515f165c
- SFT-CHEM-THERMO-017-VISCOSITY-0123: dataset=5; point=62;
target=sha256:6de67fa4a112ff08e83ff95d4f20792dbec1b18695d57d69887d9f0f848279d3
- SFT-CHEM-THERMO-017-VISCOSITY-0124: dataset=5; point=63;
target=sha256:84a598bfa524c69d13758fb1d1e8559844420a6299a8e0004cfc4fbb4545f685
- SFT-CHEM-THERMO-017-VISCOSITY-0125: dataset=5; point=64;
target=sha256:b51dcda47f8119c3e848b701394cf2d9ecb8c9198410469bf49b9fcc30994444
- SFT-CHEM-THERMO-017-VISCOSITY-0126: dataset=5; point=65;
target=sha256:8806a4b2f1ac951b7afcaea17d3f77eeb8b4625619573d8fdb3263c2901392e
- SFT-CHEM-THERMO-017-VISCOSITY-0127: dataset=5; point=66;
target=sha256:544929309d82ec34ab21753dca1e8a3435975f7daaff0819a0190f993cb0c4f3
- SFT-CHEM-THERMO-017-VISCOSITY-0128: dataset=5; point=67;
target=sha256:b8dcc00a56d3486e226411fad006da845c816c2cdaa166710629d4e81a9ffbd
- SFT-CHEM-THERMO-017-VISCOSITY-0129: dataset=5; point=68;
target=sha256:cdc6d43c272e39685904dc54c9f1ecd8acd689839f388134a668dabfffe90aa5
- SFT-CHEM-THERMO-017-VISCOSITY-0130: dataset=5; point=69;
target=sha256:416694cde58a91d7f0c06620eb9c4b2a7dc9288391821644bb3626b38c767887
- SFT-CHEM-THERMO-017-VISCOSITY-0131: dataset=5; point=70;
target=sha256:c462d4bd49b261a77f69c750ff16f04074f0593beb20e1338dd2e36ce85edffcc
- SFT-CHEM-THERMO-017-VISCOSITY-0132: dataset=7; point=1;
target=sha256:d45cbd53a4249d1d25c7298d8fd17c24587bc5f7bc2388a0658a119a29bba928
- SFT-CHEM-THERMO-017-VISCOSITY-0133: dataset=7; point=2;
target=sha256:72fa993a1b6bf71a57a9b419353b6d383f524b9938e95cf70d3f05e278e1e602

- SFT-CHEM-THERMO-017-VISCOSITY-0134: dataset=7; point=3;
target=sha256:edfa538d730102ad23a19ff90615641b91bd265ceafad65a4b6893cf09c21301
- SFT-CHEM-THERMO-017-VISCOSITY-0135: dataset=7; point=4;
target=sha256:b4eef33607a84fe65bfb7d8ce67548be8509cadcd47bbfd652f36e6aed22dd3d5
- SFT-CHEM-THERMO-017-VISCOSITY-0136: dataset=7; point=5;
target=sha256:3fb9dab4876eb280615bd3281a7dbfd742d5266151e6048b7a23a4d688a831dc
- SFT-CHEM-THERMO-017-VISCOSITY-0137: dataset=7; point=6;
target=sha256:4891471712321ff3cfea58b390c945adabb949da95200662560f140d80835f5f
- SFT-CHEM-THERMO-017-VISCOSITY-0138: dataset=7; point=7;
target=sha256:3db9ba044af1156b74cfc700c43367f9f535c9d14e7c6d6068f0f2c1e0d95c4
- SFT-CHEM-THERMO-017-VISCOSITY-0139: dataset=7; point=8;
target=sha256:e4ba7826d9b2ab53c740db2aa4b4c7ad2262f6999c25e36ab50f177563f4f494
- SFT-CHEM-THERMO-017-VISCOSITY-0140: dataset=7; point=9;
target=sha256:0dad4f0b8d585feb784dd41576cdfef87b6c48fe99c8c2257f8c5a981f0c09be
- SFT-CHEM-THERMO-017-VISCOSITY-0141: dataset=7; point=10;
target=sha256:5948d1e2a1a59b583f246b3afb24ffd8fa06286d4fa184b9bcca6bba2fa9a876
- SFT-CHEM-THERMO-017-VISCOSITY-0142: dataset=7; point=11;
target=sha256:bf1a139cd3b932e908a881188b7aa5d2ef8580d1a2671902b84e876e3b81ffed
- SFT-CHEM-THERMO-017-VISCOSITY-0143: dataset=7; point=12;
target=sha256:2915aac5f295f1e779a4ab7fa58b3c33ca5810bf0b4bbf8ffbd8f8dc2ec4b414
- SFT-CHEM-THERMO-017-VISCOSITY-0144: dataset=7; point=13;
target=sha256:543f4486e119d727e197450f146301a68b5738957d47a86d99037905aac46fd4
- SFT-CHEM-THERMO-017-VISCOSITY-0145: dataset=7; point=14;
target=sha256:391d4cec98416d142b21fc5e7d38fde42dc504580c802d11239e795ab233f0e1
- SFT-CHEM-THERMO-017-VISCOSITY-0146: dataset=7; point=15;
target=sha256:4a1440a4de0147f277eb548302230dd6184b43627370e57bfe78bbd3c5f980ed
- SFT-CHEM-THERMO-017-VISCOSITY-0147: dataset=7; point=16;
target=sha256:2446376d42f8a51e4651c1fd2bef0ebae5a9ca9625607fda41d3800b6404378b
- SFT-CHEM-THERMO-017-VISCOSITY-0148: dataset=7; point=17;
target=sha256:f9ca353d0fe729a90283d99d5da3ffe4b9408eb247ff175b12a2723110bef28c
- SFT-CHEM-THERMO-017-VISCOSITY-0149: dataset=7; point=18;
target=sha256:ff82bd0b2f10e934d2df446bba8ea557ee90c3d681dd94a5661160e151ff4bdb
- SFT-CHEM-THERMO-017-VISCOSITY-0150: dataset=7; point=19;
target=sha256:69a8a07aa0214120328c5ecc928c82c3faf468e78da2bf571f7135d6ef771ca4
- SFT-CHEM-THERMO-017-VISCOSITY-0151: dataset=7; point=20;
target=sha256:8169f1624641784eeb3ea01ff19e3a4b73dc3c4fac0035adb2d1399c7e1d89d5
- SFT-CHEM-THERMO-017-VISCOSITY-0152: dataset=7; point=21;
target=sha256:8f7bf60eb2c2abe99f454b7f8470859f005c346e608e87da384a30bc648999f7
- SFT-CHEM-THERMO-017-VISCOSITY-0153: dataset=7; point=22;
target=sha256:a030a8d2086cc2aa6c7799ff5c9e804103c844951f6c7c8ef24b13591d266cba
- SFT-CHEM-THERMO-017-VISCOSITY-0154: dataset=7; point=23;
target=sha256:77b2614c0477ba48caf57dbc84ab094a18121cc2d4a2d651aeaff551cf5b74e8
- SFT-CHEM-THERMO-017-VISCOSITY-0155: dataset=7; point=24;
target=sha256:aa974de9e8d0ec3b1c1c023f4fb37c2ee7e498e6ed121245f25c4e674b148cc1

- SFT-CHEM-THERMO-017-VISCOSITY-0156: dataset=7; point=25;
target=sha256:115f3f39a824f546b64de64be1c2eafca206e3f1f946f64de433f3f1a575aef9
- SFT-CHEM-THERMO-017-VISCOSITY-0157: dataset=7; point=26;
target=sha256:f3af8892474b4abbb696741260d13550e27963672c943adeaf8b2192a8e14a53
- SFT-CHEM-THERMO-017-VISCOSITY-0158: dataset=7; point=27;
target=sha256:fb72fa027132bf8dc5b9e3dea34f32367479bec5588ba441ab88502af8af2490
- SFT-CHEM-THERMO-017-VISCOSITY-0159: dataset=7; point=28;
target=sha256:781c9b47a902ee843204adcd38488ac6247612ca6d2a644e771f2b8cab176df1
- SFT-CHEM-THERMO-017-VISCOSITY-0160: dataset=7; point=29;
target=sha256:b1da238aabace97a9eb03ff533d80c0b6eb09bbaa64a225cbf9d1feefd853237
- SFT-CHEM-THERMO-017-VISCOSITY-0161: dataset=7; point=30;
target=sha256:3fd0923ce55666cfc9545195c01366f308b2d0b83c646faa3f8b05aaf7ee267e
- SFT-CHEM-THERMO-017-VISCOSITY-0162: dataset=7; point=31;
target=sha256:0a9450827be0f2359832f2aeb1622fe46aaf7cfbd341e090b1f75102fdbfe5d4
- SFT-CHEM-THERMO-017-VISCOSITY-0163: dataset=7; point=32;
target=sha256:405ad2a94143f6f7f425e8a73794a839934f0b6e52b708bf461bc40034fc223b
- SFT-CHEM-THERMO-017-VISCOSITY-0164: dataset=7; point=33;
target=sha256:59b18400faf25fb6973ae603ea70674652695953b3afba372574d6da75934f8
- SFT-CHEM-THERMO-017-VISCOSITY-0165: dataset=7; point=34;
target=sha256:be8f41bfcb3dad6b11555d9248967d91b024602d04bd5182df56922f6250c69b
- SFT-CHEM-THERMO-017-VISCOSITY-0166: dataset=7; point=35;
target=sha256:020ba2b20d9cbe405fb0f643f5780e1413f8f758bf36ef20527e3685c5f00fbd
- SFT-CHEM-THERMO-017-VISCOSITY-0167: dataset=7; point=36;
target=sha256:97e6074e8066667de50f0ca0eeadbfeb454d084d5cee3cc1705582d2cf45d50c
- SFT-CHEM-THERMO-017-VISCOSITY-0168: dataset=7; point=37;
target=sha256:de2d52e7e9a19fa4d05efefcde5fc79782275f707c197b83c054094a7d49e17e
- SFT-CHEM-THERMO-017-VISCOSITY-0169: dataset=7; point=38;
target=sha256:b31ba8f8adea58709500890db3c3d5b258145d28370684eb4cdf7bde4177ea5
- SFT-CHEM-THERMO-017-VISCOSITY-0170: dataset=7; point=39;
target=sha256:51004c1329c1e77b9a101e8d606b44234e32dbcb6af8b279444f9f735fe197d
- SFT-CHEM-THERMO-017-VISCOSITY-0171: dataset=7; point=40;
target=sha256:cf09de493e5f5f28b37a610fe219295f44af2af9f6565c61e47007e04d6fb10b
- SFT-CHEM-THERMO-017-VISCOSITY-0172: dataset=7; point=41;
target=sha256:3ade657aee440df264438b7f8535b8d9cbbd9052a4720fac83ef8479d5804b29
- SFT-CHEM-THERMO-017-VISCOSITY-0173: dataset=7; point=42;
target=sha256:8f7dbe36e6a8e2a816fc38a0df2084cf7fc1cf4ef7728b8169056b8d119af8b3
- SFT-CHEM-THERMO-017-VISCOSITY-0174: dataset=7; point=43;
target=sha256:929fca6b35f896041c5a9b7ca6a981a385b42b9e1891298cc9fdf8cf0fcc6f1f
- SFT-CHEM-THERMO-017-VISCOSITY-0175: dataset=7; point=44;
target=sha256:915010f7a54778d0949e0c4b39453d5e62fe2bc00abe1d3c70ee6e95be1b3e75
- SFT-CHEM-THERMO-017-VISCOSITY-0176: dataset=7; point=45;
target=sha256:6dd13dab311a39cb0977e2ace6585338a26c29ec23de1a684a86d1335103ce0e
- SFT-CHEM-THERMO-017-VISCOSITY-0177: dataset=7; point=46;
target=sha256:bbeac4a464a9eab89b11a0fe0320f9c9f3b1642631c15b7e274f3f6f85624520

- SFT-CHEM-THERMO-017-VISCOSITY-0178: dataset=7; point=47;
target=sha256:a48d67a4d737acc4f4d6b53ec4a6a4528c9b276a114915c400134fb436d16728
- SFT-CHEM-THERMO-017-VISCOSITY-0179: dataset=7; point=48;
target=sha256:c9c3144163de3d15c322cd3730f8e265e512319bd03409350de735f180495772
- SFT-CHEM-THERMO-017-VISCOSITY-0180: dataset=7; point=49;
target=sha256:61678166639aa4543ce6df61091430932263dcaeb923886c449c13a38b9d5c44
- SFT-CHEM-THERMO-017-VISCOSITY-0181: dataset=7; point=50;
target=sha256:31e3826894bd47f280763625a4086292776984de400afa0ae26f7e956b942aa0
- SFT-CHEM-THERMO-017-VISCOSITY-0182: dataset=7; point=51;
target=sha256:207438492b59bb7ccf242624f1842f481571a9e915d514eaf51c2cf020cee109
- SFT-CHEM-THERMO-017-VISCOSITY-0183: dataset=7; point=52;
target=sha256:281c36bfe78377575b8a970b336f5cf0bd8cb4553a416f62f0f061aba58e9b6b
- SFT-CHEM-THERMO-017-VISCOSITY-0184: dataset=7; point=53;
target=sha256:b3457b769d65477357ced11349ca7fe2048c1d0616339af22ef5783bd79329db
- SFT-CHEM-THERMO-017-VISCOSITY-0185: dataset=7; point=54;
target=sha256:647cf486b663b27368d7e8cd7228217702b7e119098cf6e219f62a4c3c5d6069
- SFT-CHEM-THERMO-017-VISCOSITY-0186: dataset=7; point=55;
target=sha256:4daa11c35529fb150cd4ff0c3df61829fff7344e8524f57bb11fca10628ed1dd
- SFT-CHEM-THERMO-017-VISCOSITY-0187: dataset=7; point=56;
target=sha256:a7e5e94d62b4a9062ffd8affaaef17c53ab71dddabd4b930d1ac2d68a20211db
- SFT-CHEM-THERMO-017-VISCOSITY-0188: dataset=7; point=57;
target=sha256:3f558e33b4918b792bf94170a8412cf5dcef507ecddf2fe2442b7585a58a4242
- SFT-CHEM-THERMO-017-VISCOSITY-0189: dataset=7; point=58;
target=sha256:9f3d806090cd1a625212f7a929d1dea836b61762b51cbf76e1e453e1fa589515
- SFT-CHEM-THERMO-017-VISCOSITY-0190: dataset=7; point=59;
target=sha256:37b03dc4188b4198e4b2d84b6883d18ca3b129ec5e347bd27f4006c3fafd5465
- SFT-CHEM-THERMO-017-VISCOSITY-0191: dataset=7; point=60;
target=sha256:b2fdceaf9a5406ecb7008fa229b61458408cf5ab738c40e6a24d7146ae78ad32
- SFT-CHEM-THERMO-017-VISCOSITY-0192: dataset=7; point=61;
target=sha256:d2548d860749f76119917f890aa3d4f85357695ca8d0281ea1b0a65c30a94279
- SFT-CHEM-THERMO-017-VISCOSITY-0193: dataset=7; point=62;
target=sha256:840f522be359aaca8c27ae477c3d16f2d7ad6829213b359a0794820e9dce89e1
- SFT-CHEM-THERMO-017-VISCOSITY-0194: dataset=7; point=63;
target=sha256:2236e09844f7f7adf5a9dec64279382a3d55df5963b5210ae939bea58301aad1
- SFT-CHEM-THERMO-017-VISCOSITY-0195: dataset=7; point=64;
target=sha256:de900f985707a00fde788ec0f36dd8bd65985e6a7f4fe9c20bb1ce40155e2a27
- SFT-CHEM-THERMO-017-VISCOSITY-0196: dataset=7; point=65;
target=sha256:6030a0af673c33a9699f21cfbc8a5871d9aa5c8d6b84a751a1abef1577c9af40
- SFT-CHEM-THERMO-017-VISCOSITY-0197: dataset=7; point=66;
target=sha256:97b712294a14227a2b2ddf75d261a6af09fa9c9c52482c744e05948aa3533e88
- SFT-CHEM-THERMO-017-VISCOSITY-0198: dataset=7; point=67;
target=sha256:2b9d3981f8654e71adbe538fd7ff60de845f6b412b137d153f631c66854c2a64
- SFT-CHEM-THERMO-017-VISCOSITY-0199: dataset=7; point=68;
target=sha256:05f822bbd0ff8b84aa4c191247bbe06275536208b5c744e04d463e8d4dff5323

- SFT-CHEM-THERMO-017-VISCOSITY-0200: dataset=7; point=69;
target=sha256:5641d70ae4a88bef4c191de6650d3fd4a29da8a7564603cab2d5f61e7fa41e5f
- SFT-CHEM-THERMO-017-VISCOSITY-0201: dataset=7; point=70;
target=sha256:4e58d24fe0c56db0dbb71b80eb9beaee66f3277c904aefa502d90448361b42d4
- SFT-CHEM-THERMO-017-VISCOSITY-0202: dataset=7; point=71;
target=sha256:f34edfc9a67bf29c13ecc3ae16be8d6420ce6da60f0d940bc7b6d3e4fa421eaa
- SFT-CHEM-THERMO-017-VISCOSITY-0203: dataset=7; point=72;
target=sha256:3f488efe177a0721b9b2457934ac6c9a3fdeeb92e9f45506b21f9e3ebf81d9a4
- SFT-CHEM-THERMO-017-VISCOSITY-0204: dataset=7; point=73;
target=sha256:c680a52dfcfc3b40a160268cf0a0f775133769ec2b64d10a98676a3e3521fdec
- SFT-CHEM-THERMO-017-VISCOSITY-0205: dataset=7; point=74;
target=sha256:ff870736f9318b938f218dfcbe44ef4beadc9da1a42a68d511e85ad85274d5a
- SFT-CHEM-THERMO-017-VISCOSITY-0206: dataset=7; point=75;
target=sha256:f3304742b32bcf1334bf257c69a4cc6cad82d7947b936523ddf276ac58b63751
- SFT-CHEM-THERMO-017-VISCOSITY-0207: dataset=7; point=76;
target=sha256:96eb15a06b8e9eae56f2db3e46e7a264e765fd4abf9e77f62b35f9958765b9a
- SFT-CHEM-THERMO-017-VISCOSITY-0208: dataset=7; point=77;
target=sha256:7a2b6571cf987227eb310e982b8b4cebe95d0ce1fb18dd18ce8ebe0c711702c0
- SFT-CHEM-THERMO-017-VISCOSITY-0209: dataset=7; point=78;
target=sha256:78777b635f62cbb496baae1451231b0963c8d8cc0304cf8a2d6e473e08cf0220
- SFT-CHEM-THERMO-017-VISCOSITY-0210: dataset=2; point=1;
target=sha256:c069ec8e2f1060ee59be767e612e05cc0717c06907ba03277fc2191dc437a324
- SFT-CHEM-THERMO-017-VISCOSITY-0211: dataset=2; point=2;
target=sha256:60d79725fde3aeda5b826c20d0e0a5df7fa7587e0105709ec01837908a87412e
- SFT-CHEM-THERMO-017-VISCOSITY-0212: dataset=2; point=3;
target=sha256:33df884c75e221199215dfaf451e72ed51f3905a15d8fcf17e86f3502e354f3d
- SFT-CHEM-THERMO-017-VISCOSITY-0213: dataset=2; point=4;
target=sha256:ae1dceafc763be943d6cd29065041751fae83bb6729af70501c1aa3ae9f4ec35
- SFT-CHEM-THERMO-017-VISCOSITY-0214: dataset=2; point=5;
target=sha256:57e2b9504e4b1b30ab83534a87afc4f813fc614ca18c8fa1cc2d9130a14822e8
- SFT-CHEM-THERMO-017-VISCOSITY-0215: dataset=2; point=6;
target=sha256:35d7f727f7eda009118df87fa369b0f7028d0f169c9b1913a9b5a74d8fcd9a1d
- SFT-CHEM-THERMO-017-VISCOSITY-0216: dataset=2; point=7;
target=sha256:47b97d9c90f8844b6f44d5935592bebc3261ffdd57d6f0ff630ed2827fe4faeb
- SFT-CHEM-THERMO-017-VISCOSITY-0217: dataset=2; point=8;
target=sha256:84b3f55af2c00c78de7ed787843f8c7bf5905756a885062050fed1e1ee112ae4
- SFT-CHEM-THERMO-017-VISCOSITY-0218: dataset=2; point=9;
target=sha256:945c6acb3758341d3f294a2f6122cedd35de588cda3480eeebcfd53d8503d79e
- SFT-CHEM-THERMO-017-VISCOSITY-0219: dataset=2; point=10;
target=sha256:2ba2cf0abc178789eac6d18e6dc9fdc023fca00eb6c09113b70595cd12376cea
- SFT-CHEM-THERMO-017-VISCOSITY-0220: dataset=2; point=11;
target=sha256:986d2d8c30f819e345eb3a7339d8b9205ab772f00ecf27913da4bf077029e072
- SFT-CHEM-THERMO-017-VISCOSITY-0221: dataset=2; point=12;
target=sha256:64e61809ad627073629a9f3706e567361e87da5005209c9ecef4cd1d0c19c0

- SFT-CHEM-THERMO-017-VISCOSITY-0222: dataset=2; point=13;
target=sha256:55221872a3dbb857c40a2cdbf6c6e0daa1c1e0058eeee621351baedc3843c1e0
- SFT-CHEM-THERMO-017-VISCOSITY-0223: dataset=2; point=14;
target=sha256:486131ee935e6d95f7f8f3bf99847c20a413c8ffb73cb1898909455e1d3976a
- SFT-CHEM-THERMO-017-VISCOSITY-0224: dataset=2; point=15;
target=sha256:fb9c2392d371722f6f9c0f31e406c2d077a92d86c104156ac088600555487946
- SFT-CHEM-THERMO-017-VISCOSITY-0225: dataset=2; point=16;
target=sha256:eacd95eff9a1a3e54c0cd693a0d00438d3928103d9dd0a3a4176c456c6caac0c
- SFT-CHEM-THERMO-017-VISCOSITY-0226: dataset=2; point=17;
target=sha256:ad4ab19995f51bdb8797ef681bcc40dd08de79bf36e17cbfba2009f9f6ab0c22
- SFT-CHEM-THERMO-017-VISCOSITY-0227: dataset=2; point=18;
target=sha256:92ea7c4ec8c95ec7a9f722063b40724334a95e74ac97d981c7e17cd6834ebd5a
- SFT-CHEM-THERMO-017-VISCOSITY-0228: dataset=2; point=19;
target=sha256:79d841df78b8aac6c0b4971c22b78e396b9167e440b2151042226cb512a40a7e
- SFT-CHEM-THERMO-017-VISCOSITY-0229: dataset=2; point=20;
target=sha256:6595a36c0bb7eac1acb344ceb81073d846caa6db037784b743349b4e62344d6a
- SFT-CHEM-THERMO-017-VISCOSITY-0230: dataset=2; point=21;
target=sha256:81292c49f2ca227e22992787389cea73a051b00a808c7ce9af1d42b0d4f42220
- SFT-CHEM-THERMO-017-VISCOSITY-0231: dataset=2; point=22;
target=sha256:11426e63e9d0ee19eda42be72f8b43f8e2fbee5c43b78547a4e0f1cae1f1bb62
- SFT-CHEM-THERMO-017-VISCOSITY-0232: dataset=2; point=23;
target=sha256:09dc4dda8d3ca7601a499cbff1f4c9ebb9a6451210c4fcba4c5db75758bce859
- SFT-CHEM-THERMO-017-VISCOSITY-0233: dataset=2; point=24;
target=sha256:2e43391101615c18a6dad10a9b4172cb7ca8024645de8d0bd9c4301f01125d2f
- SFT-CHEM-THERMO-017-VISCOSITY-0234: dataset=2; point=25;
target=sha256:cb5fade30825f62617a9425b6d6340cc28ee1d7290df4761d15d7b10fe9d08ce
- SFT-CHEM-THERMO-017-VISCOSITY-0235: dataset=2; point=26;
target=sha256:b14a02e522f54a00b66adf691aeea316827a90806458cdf7ac669d0c34e10d40
- SFT-CHEM-THERMO-017-VISCOSITY-0236: dataset=2; point=27;
target=sha256:12cd7ad5204a1ad2c1186d49d2d56ed83020c3ffd8c42ea3f737a47937cae092
- SFT-CHEM-THERMO-017-VISCOSITY-0237: dataset=2; point=28;
target=sha256:b8005c4c1102a0f386eb0060a2385cf2f7200a773dc34c947125bb5d4d869e19
- SFT-CHEM-THERMO-017-VISCOSITY-0238: dataset=2; point=29;
target=sha256:7a756b10b7c5394ddf664553c038b6ae6599bd31e3a61042a8461180c929be85
- SFT-CHEM-THERMO-017-VISCOSITY-0239: dataset=2; point=30;
target=sha256:a238cc500dbecba936268efcdabdd788816aa21d1f9806cd964dae96f97f8e61
- SFT-CHEM-THERMO-017-VISCOSITY-0240: dataset=2; point=31;
target=sha256:7e51987a3889535ad9e9807b736a10698b8f5f64b9896a71f4a5ce26eb9cbf93
- SFT-CHEM-THERMO-017-VISCOSITY-0241: dataset=2; point=32;
target=sha256:6fcb623219c2801823c72fbd133c629d020aaafd9a8f712cc11d7fb1219d7eb0
- SFT-CHEM-THERMO-017-VISCOSITY-0242: dataset=2; point=33;
target=sha256:bf554149ecb887d76a5d904104300c679e2b44d220b8eee02731e7a475021d51
- SFT-CHEM-THERMO-017-VISCOSITY-0243: dataset=2; point=34;
target=sha256:64b5266fcce1525840af0020480eefdd04af8118857273ada631a7c5665cffad

- SFT-CHEM-THERMO-017-VISCOSITY-0244: dataset=2; point=35;
target=sha256:04ef2122e1cff18f41341c24f174449d49092aaf3939b4d1cd9f25a4f190a0e9
- SFT-CHEM-THERMO-017-VISCOSITY-0245: dataset=2; point=36;
target=sha256:efb677292d41b3318116494cd33a9602ca0106d599300d104f1f74867ba0285c
- SFT-CHEM-THERMO-017-VISCOSITY-0246: dataset=2; point=37;
target=sha256:4f2ea951f4016b5e670cf0266a2464f18935d06f932097b5caec4e6b648371f4
- SFT-CHEM-THERMO-017-VISCOSITY-0247: dataset=2; point=38;
target=sha256:07c464138d4e300199361bc7c7564e942e4011d7873edd38f6e8db2c3bd168e5
- SFT-CHEM-THERMO-017-VISCOSITY-0248: dataset=2; point=39;
target=sha256:c41aa04df6d0e7a518967153d28ab8eafeef95297d27610382696fe6f5e9d87
- SFT-CHEM-THERMO-017-VISCOSITY-0249: dataset=2; point=40;
target=sha256:080f554877b8db362764de99147bea239153e6f2de68d9e35d74be74fe12dc6a
- SFT-CHEM-THERMO-017-VISCOSITY-0250: dataset=2; point=41;
target=sha256:68de6f4ca78e63dd2192478ae37948341a403141e926a7acf5588d6bb7e6ae67
- SFT-CHEM-THERMO-017-VISCOSITY-0251: dataset=2; point=42;
target=sha256:b289f758706c5791a63771d6e848cce53e4d33476fab92720489d861b2b79b95
- SFT-CHEM-THERMO-017-VISCOSITY-0252: dataset=2; point=43;
target=sha256:6a8b15341b49eaa8975864d6c4cdef5900a86b5f46f4e17ad2f8332780cfc622
- SFT-CHEM-THERMO-017-VISCOSITY-0253: dataset=2; point=44;
target=sha256:dbd4f1ee21d03ac1bec0d6640f6850b773aef6fdf88807c3fd18fcad52f601bb
- SFT-CHEM-THERMO-017-VISCOSITY-0254: dataset=2; point=45;
target=sha256:c7a1fd522d1201282553a0efa68dc8dcd375a6e5715b64bfd0105482f73c5467
- SFT-CHEM-THERMO-017-VISCOSITY-0255: dataset=2; point=46;
target=sha256:a737e7abe48f207fa3b46cd40f43df216e7fbe066e45dc7047ced01e4007ebc0
- SFT-CHEM-THERMO-017-VISCOSITY-0256: dataset=2; point=47;
target=sha256:14d1ebf6e50b61e1596df7533810674279b344a6958d32a43156bab9e40c74a5
- SFT-CHEM-THERMO-017-VISCOSITY-0257: dataset=2; point=48;
target=sha256:ebf3f4b9f9ce7b4d34f92345e0383dd0336499f675720930c0a3429a7444d36b
- SFT-CHEM-THERMO-017-VISCOSITY-0258: dataset=2; point=49;
target=sha256:a99ef3f71c7a6c6263b75dd0e6adc593425bec0aab354b371f3140c722d23213
- SFT-CHEM-THERMO-017-VISCOSITY-0259: dataset=2; point=50;
target=sha256:f788b167ee609494af18a274016170c1c7e1df862281455346ecd897e5eed8eb
- SFT-CHEM-THERMO-017-VISCOSITY-0260: dataset=2; point=51;
target=sha256:0342e281dc5e53aba070f199fdf1a52661d8869fd2ef79b6113bc5dc5c876aa9
- SFT-CHEM-THERMO-017-VISCOSITY-0261: dataset=2; point=52;
target=sha256:939e01a1c2ec16282cb7e49714e4f3e75d681942f64d5ddf90c9aba714682732
- SFT-CHEM-THERMO-017-VISCOSITY-0262: dataset=2; point=53;
target=sha256:e2304a503b5be31a31dd1b5f72973844d3b5d1f19128fd9f72711b7e328b234c
- SFT-CHEM-THERMO-017-VISCOSITY-0263: dataset=2; point=54;
target=sha256:17d3cccdff74a9f48c5e564de2fff43db1d4a62c96b6dcb641525193c4638647a
- SFT-CHEM-THERMO-017-VISCOSITY-0264: dataset=2; point=55;
target=sha256:fd25bc819f4e5a997b4c4e14e6f1831c18d3b3762780d2d241d97a0568df2a2
- SFT-CHEM-THERMO-017-VISCOSITY-0265: dataset=2; point=56;
target=sha256:8d16b44e5fb283638402131907cd0bce8d9123ffb8459ad8daeba017fc42a1ef

- SFT-CHEM-THERMO-017-VISCOSITY-0266: dataset=2; point=57;
target=sha256:a149adb304c7278d12555b618ece2b4b4175b2bded3b2077ceec310a46edab78
- SFT-CHEM-THERMO-017-VISCOSITY-0267: dataset=2; point=58;
target=sha256:e1284a47b6012e42be9dfd4ca8a3182f7e530e9a68d5b8dffde6f7a6f5e5690f
- SFT-CHEM-THERMO-017-VISCOSITY-0268: dataset=2; point=59;
target=sha256:24427a8900fc41a3db63a53ff4805488217687774e4501f3aca9a2b2a4b0c294
- SFT-CHEM-THERMO-017-VISCOSITY-0269: dataset=2; point=60;
target=sha256:b51193d6ab892fd5c644cc2018b45ecebc52c7cb0e87f0ffa92b2c51b2acd593
- SFT-CHEM-THERMO-017-VISCOSITY-0270: dataset=2; point=61;
target=sha256:f0783283b5b74bffa79b82977b89b9aa4c81551cb0fabee125a92dfbed6b53f35
- SFT-CHEM-THERMO-017-VISCOSITY-0271: dataset=2; point=62;
target=sha256:a29a27d1e1e354fe4498deae46603694ea381d29f5e45953146b48700fe84dd8
- SFT-CHEM-THERMO-017-VISCOSITY-0272: dataset=2; point=63;
target=sha256:143b4441ec5f228af5a06d86220e3ec7f55f8e1bc42dd4a228d52d4d1a6259b9
- SFT-CHEM-THERMO-017-VISCOSITY-0273: dataset=2; point=64;
target=sha256:696eb95ccb8ea7f775c58f91b3142e6582aef29db209b1bec216513a0ea36e8a
- SFT-CHEM-THERMO-017-VISCOSITY-0274: dataset=2; point=65;
target=sha256:9e77b21f2611eaeda353f1e35709e1f5505d6ca3aeb3cf8d2e14a2e8c9f6832d
- SFT-CHEM-THERMO-017-VISCOSITY-0275: dataset=2; point=66;
target=sha256:73a84fad1b0d9fc569111ba94fe5cc0fca82ba2fe29c1f87e3b3be7df82b0d7
- SFT-CHEM-THERMO-017-VISCOSITY-0276: dataset=2; point=67;
target=sha256:5e7215464d0b4858b101b0997fe4d9692ecaded806a3ebef844fc8a72705ca35
- SFT-CHEM-THERMO-017-VISCOSITY-0277: dataset=2; point=68;
target=sha256:406cfdbfa982d1441ea33e6a1fd2c8ce528a7212678f9625bec1043229645ccc
- SFT-CHEM-THERMO-017-VISCOSITY-0278: dataset=2; point=69;
target=sha256:87de953264b1810910e426339243174f066c79c66fa65c4d219f33d5303fc899
- SFT-CHEM-THERMO-017-VISCOSITY-0279: dataset=2; point=70;
target=sha256:4b6ccad363f64fbc51777363851725bf109f450f07eca0e1a7908b2146afc276
- SFT-CHEM-THERMO-017-VISCOSITY-0280: dataset=2; point=71;
target=sha256:f3607d4a7626db892b9bd372bfac3723033594a48d5371e2bd7c057034ce7bfa
- SFT-CHEM-THERMO-017-VISCOSITY-0281: dataset=2; point=72;
target=sha256:93ff9afde729d3c3a54b74d71319b74f0ee3a3fb7f63040b39b4fa09e43ddba6
- SFT-CHEM-THERMO-017-VISCOSITY-0282: dataset=4; point=1;
target=sha256:aca451bc57b94393ba4b94e22af14ab7508d4b53f2f6b633979d8ef40e1fba02
- SFT-CHEM-THERMO-017-VISCOSITY-0283: dataset=4; point=2;
target=sha256:e1ad4617d02ebee030678bad24bc7e9b4ce09052d6ca13f4a7251e98f8da577
- SFT-CHEM-THERMO-017-VISCOSITY-0284: dataset=4; point=3;
target=sha256:6f687b07d8bb62504fedd2e25eaad8c59b6b7241150ca5644f7cf9d3903ec606
- SFT-CHEM-THERMO-017-VISCOSITY-0285: dataset=4; point=4;
target=sha256:3f6bf52f3ade3b92ae687aa77a441b00b9a40005d6793b93d81fae82a5470a9c
- SFT-CHEM-THERMO-017-VISCOSITY-0286: dataset=4; point=5;
target=sha256:7f1edf822ce0bd17ec211b6f0d13682c359a109fc11ec5170d0a033604778abc
- SFT-CHEM-THERMO-017-VISCOSITY-0287: dataset=4; point=6;
target=sha256:2cedf53c9477d7b098bfa97e2153ceaf68ebfb7d5e08639c7d5974f07f16c074

- SFT-CHEM-THERMO-017-VISCOSITY-0288: dataset=4; point=7;
target=sha256:34ed47367dd320fa7bdfa1c08d96fd3eb1c1c93f9623f9ab42ad16b4d4fe799d
- SFT-CHEM-THERMO-017-VISCOSITY-0289: dataset=4; point=8;
target=sha256:eda0c99dfd21038b2839d420cca7ad80e4903b599e6bcb8fad74a304db1afd30
- SFT-CHEM-THERMO-017-VISCOSITY-0290: dataset=4; point=9;
target=sha256:ffed83d389a6066221bfa2dfd772134a979b5eb69b7215df1069a567697eb176
- SFT-CHEM-THERMO-017-VISCOSITY-0291: dataset=4; point=10;
target=sha256:3378931eeb565d16f417575f500b980209148026d88c79c68b6a1bcef5230132
- SFT-CHEM-THERMO-017-VISCOSITY-0292: dataset=4; point=11;
target=sha256:2444d208f7b2531f1bd49cf2adbd6654ff20d567cd01494dd377fc9316a9de66
- SFT-CHEM-THERMO-017-VISCOSITY-0293: dataset=4; point=12;
target=sha256:ec0a1dcfa0caaf2778273e403e3294787eab688b713581db0307309698d5b725
- SFT-CHEM-THERMO-017-VISCOSITY-0294: dataset=4; point=13;
target=sha256:7005944e6eec90689075ae87f9d28665bd1c5f0ccbe76ffcedd74d6cd18866
- SFT-CHEM-THERMO-017-VISCOSITY-0295: dataset=4; point=14;
target=sha256:22ba8d19fa19398d0ff726c6162a651dca1cb11c411e9b6aa7cc28cb81c6b30d
- SFT-CHEM-THERMO-017-VISCOSITY-0296: dataset=4; point=15;
target=sha256:d23a99744bc0132a9f6989fb0e431e4a612b989c345413cadac877a3261a904f
- SFT-CHEM-THERMO-017-VISCOSITY-0297: dataset=4; point=16;
target=sha256:33209167d0be70685de8aae630b6c3d3e7a7ac8bd6f267a3974fe884bd86b002
- SFT-CHEM-THERMO-017-VISCOSITY-0298: dataset=4; point=17;
target=sha256:419a1ad7007ba64b29e34dbed41aaf27758fc4c6add44b5e1c1789d1ddd973e7
- SFT-CHEM-THERMO-017-VISCOSITY-0299: dataset=4; point=18;
target=sha256:4e757fd34fde22930eaea13e35b8a77a3fc398e62917d0a139888108141263e1
- SFT-CHEM-THERMO-017-VISCOSITY-0300: dataset=4; point=19;
target=sha256:77fc99d00ab28953bebb1a56a2929df3c83db5a04342f8bbc81d12a004814f1f
- SFT-CHEM-THERMO-017-VISCOSITY-0301: dataset=4; point=20;
target=sha256:553ba9c11f5c8fc4dc489ac5cbfbb110d325da4cda12b88ef93c378afce4f6bf
- SFT-CHEM-THERMO-017-VISCOSITY-0302: dataset=4; point=21;
target=sha256:5a7083229336f106812cb24ac49041c995dec8be910f4ede062bf71c6eadf0d6
- SFT-CHEM-THERMO-017-VISCOSITY-0303: dataset=4; point=22;
target=sha256:5298ac42003204d867f904dcb62870b89c796b1ca0b1941be96dcf7fe45ab217
- SFT-CHEM-THERMO-017-VISCOSITY-0304: dataset=4; point=23;
target=sha256:b900e1ba1e6aaf4eb2131af26002eb9be00cc9d0505d13f61946deb544b0a8c9
- SFT-CHEM-THERMO-017-VISCOSITY-0305: dataset=4; point=24;
target=sha256:95c6aed33bf402e7799484b1be6377a1467f109863c62c96eb29497d404215bf
- SFT-CHEM-THERMO-017-VISCOSITY-0306: dataset=4; point=25;
target=sha256:cfd13a41302be76d0da2b94712758227d9a60a02bb943a0b999a4ea0b92b5393
- SFT-CHEM-THERMO-017-VISCOSITY-0307: dataset=4; point=26;
target=sha256:fcdb73c1f329527e263ec02301980caa6eb692b8715a33764f7a89142864dea6
- SFT-CHEM-THERMO-017-VISCOSITY-0308: dataset=4; point=27;
target=sha256:8d2782ed23fcb85b481482e192ad3a1ed6e499c43568338cd1dd548f9737b7b
- SFT-CHEM-THERMO-017-VISCOSITY-0309: dataset=4; point=28;
target=sha256:2564b7946f59be32d2328851b83fa2cd6715de8da659b1fb5c3d3d492d32d856

- SFT-CHEM-THERMO-017-VISCOSITY-0310: dataset=4; point=29;
target=sha256:57bd2aaad2618fce287a1f8583ab7382428f9d86e0ab427eb36bda1053029059
- SFT-CHEM-THERMO-017-VISCOSITY-0311: dataset=4; point=30;
target=sha256:875266e6e9c50768ddcc7f6a8bcd21ce7122b83e114785340eafbef04d2e82ce
- SFT-CHEM-THERMO-017-VISCOSITY-0312: dataset=4; point=31;
target=sha256:b3792913ac8793b1dd5df78f5fa0fe3d931b02b9d2cdc5736e5caa025c85b23d
- SFT-CHEM-THERMO-017-VISCOSITY-0313: dataset=4; point=32;
target=sha256:beaee8d43f2acc97d1d278cdc4520ef217d3d302bf8308c6098faaff9b47f49e
- SFT-CHEM-THERMO-017-VISCOSITY-0314: dataset=4; point=33;
target=sha256:fb280be367d322d1d85b30cc3c2ec031334189fcfb696b48cf14c820a38841
- SFT-CHEM-THERMO-017-VISCOSITY-0315: dataset=4; point=34;
target=sha256:83dfc6c53aba29a1b80ece08443c41d362d136044026f6549d795ab1843720b1
- SFT-CHEM-THERMO-017-VISCOSITY-0316: dataset=4; point=35;
target=sha256:308ce4fd469528bd1307577f3bd19fddc29f38c8e96b17ff4da0565955f4b87f
- SFT-CHEM-THERMO-017-VISCOSITY-0317: dataset=4; point=36;
target=sha256:1c827ac6961a7966bfff023066976240cbb657dba179b925201af64815f70fc6
- SFT-CHEM-THERMO-017-VISCOSITY-0318: dataset=4; point=37;
target=sha256:bef292cecc78948fe9f7e2c8e1a2aceeaec12bb2a94af77eb10b3f64804cad42
- SFT-CHEM-THERMO-017-VISCOSITY-0319: dataset=4; point=38;
target=sha256:3a63b84f3dc0949a3c07e172c1f222f1356570946b323ed9d2e33e1dcf4a88e3
- SFT-CHEM-THERMO-017-VISCOSITY-0320: dataset=4; point=39;
target=sha256:9032858f63fae4216b480fe571a1407b05d1e2994bc1c4eaff1bc65bd70fc948
- SFT-CHEM-THERMO-017-VISCOSITY-0321: dataset=4; point=40;
target=sha256:685e8c1205829f41e9ec6abe27584b6183303c8108e44170b0b024e1342dc5e4
- SFT-CHEM-THERMO-017-VISCOSITY-0322: dataset=4; point=41;
target=sha256:97c156e50f20eb2ea98f0f9606e1a14c53e1e59503880a92ebf02af892a16c7c
- SFT-CHEM-THERMO-017-VISCOSITY-0323: dataset=4; point=42;
target=sha256:4ed5b80608fd1e0522c6d6ef691d6600cd365e66b0faa58743672529b7bbbe60
- SFT-CHEM-THERMO-017-VISCOSITY-0324: dataset=4; point=43;
target=sha256:3967ed9c373df8788c4da62cc121433af7d168676e9d965b1339f546ce507879
- SFT-CHEM-THERMO-017-VISCOSITY-0325: dataset=4; point=44;
target=sha256:b4d17fc1f0a73b15d89c30fc1d5973b5a2ab9b7e7cfed96ff3c827c8aa037e99
- SFT-CHEM-THERMO-017-VISCOSITY-0326: dataset=4; point=45;
target=sha256:795fe42c83ccb7e2324343577ee695d3cf431be3d8c70d0bf17c2c543552d91e
- SFT-CHEM-THERMO-017-VISCOSITY-0327: dataset=4; point=46;
target=sha256:c665b301ec1d776e9d226e10a412a197ee24588958958a14b7d3c8d0ae267b9c
- SFT-CHEM-THERMO-017-VISCOSITY-0328: dataset=4; point=47;
target=sha256:b48cbf29feb266f232f53e1422ab08ca304d9013da0ae8a0fc1af187e121e419
- SFT-CHEM-THERMO-017-VISCOSITY-0329: dataset=4; point=48;
target=sha256:f6ce6101d0072bf4c48d5d2105927f88c13a7dcea90cb893713049ab296aed3a
- SFT-CHEM-THERMO-017-VISCOSITY-0330: dataset=4; point=49;
target=sha256:57ea6a327f7765336cb422f9062fc33860dedf5bd21a76713ed6181447a7afe5
- SFT-CHEM-THERMO-017-VISCOSITY-0331: dataset=4; point=50;
target=sha256:0778ed6595048ea05f9b3ab7a0d3ff264edffa1df379b29019bbb5b5bea41d3

- SFT-CHEM-THERMO-017-VISCOSITY-0332: dataset=4; point=51;
target=sha256:39cfa8fb9b5e53328a7aeede8212fcb3105a272854990f049e23008ed7abf891
- SFT-CHEM-THERMO-017-VISCOSITY-0333: dataset=4; point=52;
target=sha256:4b17c9222addad214a9b3d66d27eb436fb5e3093aec3290a9b13c17ce259492b
- SFT-CHEM-THERMO-017-VISCOSITY-0334: dataset=4; point=53;
target=sha256:18f2fd27d74889b292f554eabfe4dca9d6c444507daa26aad70d1b2e4fe27c83
- SFT-CHEM-THERMO-017-VISCOSITY-0335: dataset=4; point=54;
target=sha256:30339c5b52dbe2e7340470f30143772ccbbedb7b40bae687cddf983f77951906
- SFT-CHEM-THERMO-017-VISCOSITY-0336: dataset=4; point=55;
target=sha256:82ef2d4ee95f8c47e42f6ec3643eb98e7a53df047e72fc8931379ae86b6b6cbe
- SFT-CHEM-THERMO-017-VISCOSITY-0337: dataset=4; point=56;
target=sha256:2cbfa2757b54c87f1099ad32bf1b458bb73446988cc37b7746baaacbf12558db
- SFT-CHEM-THERMO-017-VISCOSITY-0338: dataset=4; point=57;
target=sha256:d4b0a2b532539b7c3430fc055ac8e31ad7fa8147f88758a80a070080f2a1ec7d
- SFT-CHEM-THERMO-017-VISCOSITY-0339: dataset=4; point=58;
target=sha256:9e759cb8a2b06b6f329464839b62b6d0f596b33525a9956b20e8f3c816501d2c
- SFT-CHEM-THERMO-017-VISCOSITY-0340: dataset=4; point=59;
target=sha256:880f6e187d5140f4a96cc8dac3975c99b45c14bdf74d2f144724da4bc99c12f9
- SFT-CHEM-THERMO-017-VISCOSITY-0341: dataset=4; point=60;
target=sha256:081f7134bfa1d1812e5e55a7f16466952183be604c68898f727e52e097f3112e
- SFT-CHEM-THERMO-017-VISCOSITY-0342: dataset=4; point=61;
target=sha256:8a2b2a4cc28828fdec0fd405f8be3836a17cbce6ec25c623b0ca90351312c936
- SFT-CHEM-THERMO-017-VISCOSITY-0343: dataset=4; point=62;
target=sha256:739f0ff787a0570b7bdb7363647d3741a906d38edad907c18f2d8f6b8d0c7aaf
- SFT-CHEM-THERMO-017-VISCOSITY-0344: dataset=4; point=63;
target=sha256:0b87dca314fe330405dd9da2c87502f110dd06cf2a60661c9c6a70bbabc71d5f
- SFT-CHEM-THERMO-017-VISCOSITY-0345: dataset=4; point=64;
target=sha256:cbf6277f1bb4c571b2061ceb4dd87de45ad9181e8862642d775fbb6a7fdccc00
- SFT-CHEM-THERMO-017-VISCOSITY-0346: dataset=4; point=65;
target=sha256:ee9a5c30f88bf171d16276b96b18a77c94cbc2733f9a8c561df1a5d5762653ec
- SFT-CHEM-THERMO-017-VISCOSITY-0347: dataset=4; point=66;
target=sha256:7731dedc216d26edeb059daff96debe0c606714257fb5b3db70cb6c3f4f1d8a4
- SFT-CHEM-THERMO-017-VISCOSITY-0348: dataset=4; point=67;
target=sha256:9c8286eda43d0c0de20452d9a4db15caae819a16538716f56d3fb8acf8e708ef
- SFT-CHEM-THERMO-017-VISCOSITY-0349: dataset=4; point=68;
target=sha256:7185a1e364fceb31074add1bb0ce52b4cfce9dfb561fbfa73c450ff4fb31585a
- SFT-CHEM-THERMO-017-VISCOSITY-0350: dataset=4; point=69;
target=sha256:bc81f7dc4c1cf9881f090c6ca24dc896e248a88ef116d83ba459f8dbc853f953
- SFT-CHEM-THERMO-017-VISCOSITY-0351: dataset=4; point=70;
target=sha256:ffc3ccf82695bf3eb87e9eae10f9021ef462af23a9465e18047d6652cf2c3fdf
- SFT-CHEM-THERMO-017-VISCOSITY-0352: dataset=4; point=71;
target=sha256:ae1013d1fed5c2ab8a5ee1a2d9f095748ed657d6f8661b6e09a8fc6e7e2c84be
- SFT-CHEM-THERMO-017-VISCOSITY-0353: dataset=4; point=72;
target=sha256:30def384ce04daff0df5af5b3fbf9c007e6a21a5077793d9ec30dc8ebe9c5aee

- SFT-CHEM-THERMO-017-VISCOSITY-0354: dataset=6; point=1;
target=sha256:9cbbf7b64c7cda9b60d1f0cdabd564e977741a443df1c83e35182067186c8a9c
- SFT-CHEM-THERMO-017-VISCOSITY-0355: dataset=6; point=2;
target=sha256:a2c9bea3c81b9ff024dc09cfd0850d88ff25421427ff1c286a0a53c3bb18d014
- SFT-CHEM-THERMO-017-VISCOSITY-0356: dataset=6; point=3;
target=sha256:89b76918996a3744cc7fc4886f2964481b3f403b3e71560efa5a0b77d1dfc418
- SFT-CHEM-THERMO-017-VISCOSITY-0357: dataset=6; point=4;
target=sha256:497d958abc0e4c31316bcd6eeab8e45bb2d0139b11edc18585d548734dc33b7b
- SFT-CHEM-THERMO-017-VISCOSITY-0358: dataset=6; point=5;
target=sha256:7c96332cfaaf2ba783f191e31204864f27aaace55df3d9d97f9a7232d513ae2
- SFT-CHEM-THERMO-017-VISCOSITY-0359: dataset=6; point=6;
target=sha256:422ecf8b91db2d7bca5cb3af407009d683a5ac92ae6909b58245204d032ce1df
- SFT-CHEM-THERMO-017-VISCOSITY-0360: dataset=6; point=7;
target=sha256:397464e1ffc1aebad0350bf138b84a13508427360002271871fbc5eb2e54ea94
- SFT-CHEM-THERMO-017-VISCOSITY-0361: dataset=6; point=8;
target=sha256:34a831c859e812a7391fd1348c666ddb58e6df56e6aba1b7dd1170226c1b28a6
- SFT-CHEM-THERMO-017-VISCOSITY-0362: dataset=6; point=9;
target=sha256:6c85fe4f4ce07278921bf2fd3b1efab68c67a219536318ecb00439dc0f12a13c
- SFT-CHEM-THERMO-017-VISCOSITY-0363: dataset=6; point=10;
target=sha256:f5d4bae164be340a19da5b0892d57b1a74a8172be9c24048dcbac587e54323bb
- SFT-CHEM-THERMO-017-VISCOSITY-0364: dataset=6; point=11;
target=sha256:39af3fd6e0f8494fa5d6b008ff44882c1ae1bde8b5a76bc7577d084cefb5874
- SFT-CHEM-THERMO-017-VISCOSITY-0365: dataset=6; point=12;
target=sha256:33f0101e7ebca7a3ff26c1a90a4eb9892baafc71b3d467ebc5af732104b9a61
- SFT-CHEM-THERMO-017-VISCOSITY-0366: dataset=6; point=13;
target=sha256:1dec013b34a71a9b0ab6947cf46edb34f826825626c8e6d94a046bcd30d8506d
- SFT-CHEM-THERMO-017-VISCOSITY-0367: dataset=6; point=14;
target=sha256:d19dd98fbb54682f310311423497a17177f8e0021488b3edf78e77e7833291c6
- SFT-CHEM-THERMO-017-VISCOSITY-0368: dataset=6; point=15;
target=sha256:e1878d1026f45530710f70e0097939c5b8cc25c8ea7d9a728a8a8c1b0a187464
- SFT-CHEM-THERMO-017-VISCOSITY-0369: dataset=6; point=16;
target=sha256:9b732f7ffacd6b48f54fd628e4b7f1511028ba778287f9132a87a3d22649fcd1
- SFT-CHEM-THERMO-017-VISCOSITY-0370: dataset=6; point=17;
target=sha256:f9a4004e1b99555d54e7864b4c2b753dd181d0a9850da3937d8b7d9406b8a8b9
- SFT-CHEM-THERMO-017-VISCOSITY-0371: dataset=6; point=18;
target=sha256:3e4603434b1d66b1c6ecb022ecc4c63c3a3838a64ed47bc69b428629648ad8e5
- SFT-CHEM-THERMO-017-VISCOSITY-0372: dataset=6; point=19;
target=sha256:d60fa2ae05d2678e426ac0e58076821aa8724de9bdb4be48b7a6513b5479ad10
- SFT-CHEM-THERMO-017-VISCOSITY-0373: dataset=6; point=20;
target=sha256:a47aa8288d14abe2a44fe6f1973e7fc5198ea2a413785667366db81100516b07
- SFT-CHEM-THERMO-017-VISCOSITY-0374: dataset=6; point=21;
target=sha256:e75519ae24ad6ae236e2e6448261d11eac9babf1f6c085488f75f944291f843d
- SFT-CHEM-THERMO-017-VISCOSITY-0375: dataset=6; point=22;
target=sha256:096784dc89680cac35ad64e1efebd1de8c48a1696e853f950294220e40fae38f

- SFT-CHEM-THERMO-017-VISCOSITY-0376: dataset=6; point=23;
target=sha256:43b4ca26c97ff3e2c6b80f8101ceb6363d806c04899f08b87cfc7d0ca760fbb6
- SFT-CHEM-THERMO-017-VISCOSITY-0377: dataset=6; point=24;
target=sha256:2934b4b30a38d797eb9b006b7bb5c16e7ebb865114069f4f752363fc52a7377e
- SFT-CHEM-THERMO-017-VISCOSITY-0378: dataset=6; point=25;
target=sha256:fa82933251249047170a0ca28e105704877236898ca0bfb3bb62a3d49a09b621
- SFT-CHEM-THERMO-017-VISCOSITY-0379: dataset=6; point=26;
target=sha256:1c419f718c1a920895aebb4efc0f8a488d2f4da7cd388f1b6bf97047a53b759c
- SFT-CHEM-THERMO-017-VISCOSITY-0380: dataset=6; point=27;
target=sha256:0f69eecd147d667d373cd2f7351ca59a72e45e2912a6d1d7491d4d097598c2db
- SFT-CHEM-THERMO-017-VISCOSITY-0381: dataset=6; point=28;
target=sha256:1be20b2f8629bb4b3d0f712928da60d56347b4b72b6e60169c1ff7caa3f9827c
- SFT-CHEM-THERMO-017-VISCOSITY-0382: dataset=6; point=29;
target=sha256:75d28a8934089ea96c5f7aeed79fd4d092ed3d32012bd1830be92801348c4b68
- SFT-CHEM-THERMO-017-VISCOSITY-0383: dataset=6; point=30;
target=sha256:d56f7c56ba73109464b72623f9d8bf15b3db1d4663139eb108ab61d0bc5324ce
- SFT-CHEM-THERMO-017-VISCOSITY-0384: dataset=6; point=31;
target=sha256:19e0cfd85afd77b5f1848f4fc6ac27dabae9de0bb1f30c91a29a7153d5d496b
- SFT-CHEM-THERMO-017-VISCOSITY-0385: dataset=6; point=32;
target=sha256:0cdac91c4ea265d465ed6d5bdf5e860d70b8dbe9382eda542b81b7db6fd5d0e
- SFT-CHEM-THERMO-017-VISCOSITY-0386: dataset=6; point=33;
target=sha256:93fce8d7b1b18ca287be5c6adbe3068c58e00c775594ce0510ad177bf9cc4044
- SFT-CHEM-THERMO-017-VISCOSITY-0387: dataset=6; point=34;
target=sha256:cf4364d37b94cb10454f519b75b8e4fbefbb81c297251014f5ad554c80b00ce6
- SFT-CHEM-THERMO-017-VISCOSITY-0388: dataset=6; point=35;
target=sha256:80d136d0fc0b4e84fbb5cc5b72c6ddeb0df236b87fe37d2ab5ad7183e5f61491
- SFT-CHEM-THERMO-017-VISCOSITY-0389: dataset=6; point=36;
target=sha256:42bfd044424879d09faef1071912e19b67cb0858a357edd98dafc97bfe0d8be2
- SFT-CHEM-THERMO-017-VISCOSITY-0390: dataset=6; point=37;
target=sha256:c7fdf3571c2cdcdfe0ce5a8a832d295fa1a87f4dc9615909a2a09c4f582542e
- SFT-CHEM-THERMO-017-VISCOSITY-0391: dataset=6; point=38;
target=sha256:2c74cef00e81d943e57afe4df7333ef3b54af1fa2563ea13108906a78335f1e5
- SFT-CHEM-THERMO-017-VISCOSITY-0392: dataset=6; point=39;
target=sha256:a9fb1e2a2c0d04aa840c2e3e6233157d0be2c53118b65ccb1d9e04e6f472b98a
- SFT-CHEM-THERMO-017-VISCOSITY-0393: dataset=6; point=40;
target=sha256:a5c1051a9acce5fca35ae0baf40f4ebe6996de132bf1b60f79e871c68ec9a60
- SFT-CHEM-THERMO-017-VISCOSITY-0394: dataset=6; point=41;
target=sha256:927dac8d95a714769d63520c2dbe64ecc90c567ede2e7fe684db49880b3b5ea2
- SFT-CHEM-THERMO-017-VISCOSITY-0395: dataset=6; point=42;
target=sha256:97705d704b958e56e53d706008263040f46d132fc9500e81491164f643ad35b5
- SFT-CHEM-THERMO-017-VISCOSITY-0396: dataset=6; point=43;
target=sha256:7d2c265bb5c1b8ca68106ac0b97ba7f74b6da186c76e06b6612e586b64aa7a9b
- SFT-CHEM-THERMO-017-VISCOSITY-0397: dataset=6; point=44;
target=sha256:e8305c13101fded2e9857460d28bce7616b83f7814d68c58f10e6a35f3ad694e

- SFT-CHEM-THERMO-017-VISCOSITY-0398: dataset=6; point=45;
target=sha256:b92e84fe0ad9e62819e20e1fb89640307f585d578488e963cc2b1d58031e5526
- SFT-CHEM-THERMO-017-VISCOSITY-0399: dataset=6; point=46;
target=sha256:129d65623d2047c7fe69e86754a8bf3c119270bcf2d4dc8b58f21ebfc56c5ee7
- SFT-CHEM-THERMO-017-VISCOSITY-0400: dataset=6; point=47;
target=sha256:135491b56955289d2a40e43a799c0042bf744387b580d10f01390bce18e01ff1
- SFT-CHEM-THERMO-017-VISCOSITY-0401: dataset=6; point=48;
target=sha256:99edb33d74b3f97c594fd4099e5ecc9a4bd57cd5028dafd0b8acdf2604ca3c29
- SFT-CHEM-THERMO-017-VISCOSITY-0402: dataset=6; point=49;
target=sha256:35fe227109e5c7d616ba275d5e36667b65952d5e764bb9e3dba0e09ef17edc
- SFT-CHEM-THERMO-017-VISCOSITY-0403: dataset=6; point=50;
target=sha256:f5cbd3a91bc4802ee8ab0c4cedce93b79a0bb63792f8ae9f526ecfb27f1beb72
- SFT-CHEM-THERMO-017-VISCOSITY-0404: dataset=6; point=51;
target=sha256:2d0ffffbb3ea20c735c07af0eed68f19b417401afe7dadd79fbdea6de1f1c60
- SFT-CHEM-THERMO-017-VISCOSITY-0405: dataset=6; point=52;
target=sha256:8e182ce055312150516e21fa192c50849e5d0bc2c2db805ab21ec3547c7b8b55
- SFT-CHEM-THERMO-017-VISCOSITY-0406: dataset=6; point=53;
target=sha256:9c304d0c6e34defb33ed6fcab966e8657f72e690bd0b4de068016866287fa8b3
- SFT-CHEM-THERMO-017-VISCOSITY-0407: dataset=6; point=54;
target=sha256:caf89e351b5478e57a56b4185ee40a81117e103d639263528a360273acc01eb5
- SFT-CHEM-THERMO-017-VISCOSITY-0408: dataset=6; point=55;
target=sha256:a803598a63e41aa15aff157b93843068cf9317d4dd98d6d900bf57005644950a
- SFT-CHEM-THERMO-017-VISCOSITY-0409: dataset=6; point=56;
target=sha256:bd163cb472c29afa7822b6e6b33300fd4e4687b7d70ac1a4babc9b5546a3034e
- SFT-CHEM-THERMO-017-VISCOSITY-0410: dataset=6; point=57;
target=sha256:1505cdc0297a401e3018471f7ffe3be80bfbff2a7a9dc31f0b61fee1a9804d20
- SFT-CHEM-THERMO-017-VISCOSITY-0411: dataset=6; point=58;
target=sha256:4ab2cb59a6a45f3a7a847d8087cc3477c91598682ad469217d0382979f2017cd
- SFT-CHEM-THERMO-017-VISCOSITY-0412: dataset=6; point=59;
target=sha256:6059fa2e989e2e5a51afa4150f54cb3694f51ec4f3d6632793d63ec42fa1ac27
- SFT-CHEM-THERMO-017-VISCOSITY-0413: dataset=6; point=60;
target=sha256:b9fcbcb2c42094893d533bfaebcd9e64e819d20fc2ded73324d96dbd324f495
- SFT-CHEM-THERMO-017-VISCOSITY-0414: dataset=6; point=61;
target=sha256:015187f549842cb9e4b0632c52669ce740a02f22eab5dfe49ec5a419efc45833
- SFT-CHEM-THERMO-017-VISCOSITY-0415: dataset=6; point=62;
target=sha256:50f7d41d4879d5675764773a10ba191af410aa654ecb8ab6f9fdd15aa000c451
- SFT-CHEM-THERMO-017-VISCOSITY-0416: dataset=6; point=63;
target=sha256:77fa9294761e162ff380e82690b30c7057deda45a8bcf8839b768092e8e1693d
- SFT-CHEM-THERMO-017-VISCOSITY-0417: dataset=6; point=64;
target=sha256:a521b248c3140055e7594f03eee0bb033aae2ffd91480ba28ea2c6052b39fd54
- SFT-CHEM-THERMO-017-VISCOSITY-0418: dataset=6; point=65;
target=sha256:6d05206cd2a1c10cea8c4ae6dbcac7e7317697848043b5333e0cd42eda69773
- SFT-CHEM-THERMO-017-VISCOSITY-0419: dataset=6; point=66;
target=sha256:facfff7186336df775c6935d7f22760fd26b76468acea79c9e6fc2e6ed7c9a1c

- SFT-CHEM-THERMO-017-VISCOSITY-0420: dataset=6; point=67;
target=sha256:7f8698b6c10174dcc8d9a88e67e3dccc49a9422f29bc3fa83851e022c31e6e3b
- SFT-CHEM-THERMO-017-VISCOSITY-0421: dataset=6; point=68;
target=sha256:6b46dab5dba55ef2038465702f47b2868ff977c68e9999a6e3fa98f8236792f5
- SFT-CHEM-THERMO-017-VISCOSITY-0422: dataset=6; point=69;
target=sha256:be70e3a15acf6698ca6d7c01ef49cf80d86552f0f6e4a63cf503e110c15065f2
- SFT-CHEM-THERMO-017-VISCOSITY-0423: dataset=6; point=70;
target=sha256:e25bc3c48aaa9f461baf1ae2035e3f2e5fab60c138c347ad011ef86b9bd931c1
- SFT-CHEM-THERMO-017-VISCOSITY-0424: dataset=6; point=71;
target=sha256:15a40b1f984a73b75b2f3daf4c9c4c1f994b337fda721473b0e4c916c8264e93
- SFT-CHEM-THERMO-017-VISCOSITY-0425: dataset=6; point=72;
target=sha256:cf407501b9724354c94a98b7a7f0a74cab941c71e90d1db75570cc9aae0aa4fe
- complete vector: 11 pure, 364 binary and 50 ternary viscosity records
- exact viscosity range: {'minimum': '1/40000', 'maximum': '21/2000'}
- source coverage: all 17 datasets and 900 points preserved from three complete NIST ThermoML sources
- tampered_negative_viscosity_rejected: True
- complete_425_record_vector_retained: True
- all_pure_binary_ternary_rows_retained: True
- all_datasets_and_methods_retained: True
- all_absent_conditions_translated_to_EmptyOne: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if any component, phase, condition, layer, momentum packet, exchange or tick is erased; if adjacency or momentum conservation fails; if signed/numerical-zero proof value, continuum velocity gradient, Newtonian, Arrhenius/WLF/VFT form, fit, logarithm, interpolation, regression, selection or target correction enters; if target content opens before all 425 identities seal; if any direct viscosity row, uncertainty, method, dataset, complete source or companion provenance is omitted; or if any target is tampered.

Measurement receipt: sha256 : a22d48c997f6b09b64709211599e2fbc94d5ab7316b15fbc1bb633e7d8504e84.
Isolation certificate: sha256 : e498a09e9da4407874f6bf450e451d7544423a3d6a59a7835329749e9ca60a73.
Custody certificate: sha256 : f35a63341d52abeb42289882462c4715320d0ba15f3debcbdbb2f2fe76560247e.

Registered source descriptions:

- National Institute of Standards and Technology ThermoML Archive: ()

Registered target identities:

- None.

Meaning. The thermodynamic extension derives finite microstate support, temperature, energy, heat/work, entropy, enthalpy, free direction, chemical potential, activity, fugacity, phases, colligative response, solvation and coupled transport while keeping external scalar inscriptions downstream. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; absent external condition coordinates are structural EmptyOne
- no negative, irrational, imaginary, logarithmic, floating, signed or continuum SFT proof value
- no imported Newtonian constitutive equation, continuum velocity gradient, Arrhenius, WLF, VFT or fitted viscosity law
- no interpolation, regression, selected substance/mixture/phase/condition/method/dataset/row or target correction
- no substance, composition, phase, temperature, pressure, method, value, uncertainty or target hash before prediction seal

- every complete source and non-viscosity companion dataset remains preserved; companions never become viscosity measurements

Immutable identities. Source manifest

sha256:4791e9e3597298d4ce23a4790ab210dec4d2ff902e2208fce6f6bc2ff29227be; derivation seal sha256:4f6cc8c5422dc684acbf2540ab1dcd5aad88efdca7165350879df61bbb776ef; engine receipt sha256:c8111e85027bf3cde2f2e532e313e118d1f88278dde6cc7b2094a36741a1099c at receipts/engine/model_admitted/SFT-CHEM-VISCOUS-TRANSPORT-RELATION-017-c8111e85027bf3cd.json; empirical hash sha256:35832d1d371250eb81f19633a21a6bbbed3f251cd3adef37c0cd1fc0f56619b82; measurement receipt sha256:a22d48c997f6b09b64709211599e2fbc94d5ab7316b15fbc1bb633e7d8504e84.

142. Exact composition-bound thermal-conductivity law

Claim identity: SFT-CHEM-THERMAL-CONDUCTIVITY-RELATION-018

Question and theorem. Chemical thermal conductivity is counted adjacent-cell energy-packet transfer per exact tick, boundary and thermal-order separation while retaining every component, phase and condition. Chemistry owns composition-, species-, phase- and condition-dependence; the energy carrier is inherited explicitly from Physics. Measured conductivity is exact positive post-seal support, with direction held as a label rather than a signed magnitude.

complete-composition-phase-condition-energy-transfer-account__distinct-held-complete-component-and-phase-identities__counted-adjacent-cell-energy-packet-transfer__held-higher-to-lower-thermal-order-orientation__exact-positive-packet-transfer-tick-boundary-order-and-condition-support__exact-positive-postseal-thermal-conductivity-support__complete-value-free-655-record-identity-seal__depth-independent-common-replication-and-record-append

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-CPLX-TIME-SPACE-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-THERMO-TEMPERATURE-001
- SFT-PHYS-THERMO-HEAT-WORK-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013
- SFT-CHEM-MOLECULAR-DIFFUSION-RELATION-016
- SFT-CHEM-VISCOUS-TRANSPORT-RELATION-017

Generated grammar. Generate the literal product of carrier, identity, transfer, orientation, resource, magnitude, prediction and extension forms; decide all 256 candidates only from admitted energy, thermal-order, adjacency, transition, conservation, exact-resource, composition, phase, EmptyOne and finite-successor laws.

Boundary: Every finite pure, binary or ternary thermal-conductivity record with complete component identities, phase, exact conditions, counted adjacent cells, thermal orders, energy packets, transfers, ticks and boundary support. External testing preserves all 655 direct conductivity points and all companion properties from three complete NIST sources.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: complete-composition-phase-condition-energy-transfer-account__distinct-held-complete-component-and-phase-identities__counted-adjacent-cell-energy-packet-transfer__held-higher-to-lower-thermal-order-orientation__exact-positive-packet-transfer-tick-boundary-order-and-condition-support__exact-positive-postseal-thermal-conductivity-support__complete-value-free-655-record-identity-seal__depth-independent-common-replication-and-record-append. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-composition-phase-condition-energy-transfer-account	detached-conductivity-number-or-continuum-field	A detached number or field erases the chemical energy-transfer account.
identity	distinct-held-complete-component-and-phase-identities	anonymous-or-collapsed-composition-phase	Collapsed composition or phase cannot own a chemical conductivity relation.
transfer	counted-adjacent-cell-energy-packet-transfer	Fourier-gradient-or-continuum-carrier-premise	A continuum temperature gradient imports an ungenerated carrier.
orientation	held-higher-to-lower-thermal-order-orientation	signed-heat-flux-proof-magnitude	Signed heat flux imports prohibited arithmetic.
resource	exact-positive-packet-transfer-tick-boundary-order-and-condition-support	unrecorded-energy-transfer-time-boundary-order-or-condition	Unrecorded resources cannot reconstruct conductivity.
magnitude	exact-positive-postseal-thermal-conductivity-support	imported-Fourier-kinetic-mixing-temperature-fit-or-logarithm	An imported constitutive or fitted law selects the magnitude.
prediction	complete-value-free-655-record-identity-seal	substance-composition-phase-condition-method-or-value-readable-before-seal	Readable targets could select the law.
extension	depth-independent-common-replication-and-record-append	refit-after-transfer-replication-or-record-append	Refitting destroys exact transfer provenance.

Adverse controls. All mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256 : 2e9e0c6296090072c99e998f1a77802f431158ef0511d3ceaa162a46909c47aa.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : a88b6a8963333b0c6d33acf0922e42b2453b50268806322dea8862dedf87573f.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256 : d349dd63e7295a45ecfc7256af405571833485a087e730fc6e0d6dc99c663381.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:9794945ebed113e8a3dedc788a055bda7ac0d7568804ce4f77efb8fac5f22200.

Independent reconstruction. Implementation hash

sha256:d13bebeaa843c8c89c673ba47216854cd60819a2a55a5de01928cb0bba47eb16; independent certificate sha256:3e4a14cd7b0c98ed5ace4b6ea9b7019abb1a88ee24e20c68d733553201325a1c; external-validation hash sha256:cc559efb4d3466513805a2ce6b2b22ad9f7470a994d0b49ec578b24661f277b8. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-TRC-THERMOML-JCED-2013-58-663-670
- NIST-TRC-THERMOML-JCT-2019-133-135-142
- NIST-TRC-THERMOML-FPE-2018-477-78-86

Comparison records:

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0001: dataset=1; property=1; point=1; target=sha256:55244898022330db10565c64597c16e70be734cf6ca980fc88feee8f395ebeb
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0002: dataset=1; property=1; point=2; target=sha256:b29796e2c481a5222f265c26e2824949a37e4431acb95ee0444cfc06bf41da09
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0003: dataset=1; property=1; point=3; target=sha256:1f1331b14d253b8d23ba02453b024c9bf935f2d0b8032b48644ac0d3f637114a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0004: dataset=1; property=1; point=4; target=sha256:26f2d0b6733ed3f9239e9052ca96bae1ca57e4de13c6b46f2e6d6d6d26399fca
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0005: dataset=1; property=1; point=5; target=sha256:05a65a3a6fff6a7495d0d390ce5b4a109d044d095a589f5773af7421f7e5120e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0006: dataset=1; property=1; point=6; target=sha256:b8e05fc752df64b2fa3c38bd5ab9c8beef6f59e3b475cb8246c98d9176787623
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0007: dataset=1; property=1; point=7; target=sha256:fa27ac26f38be2985e62eccda5886521dd2a6345f4849ccf043215b32b16178c
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0008: dataset=2; property=1; point=1; target=sha256:dbd2688ccc40bbe685ddc7788581bb772ee6dd4a17e530244a153982ea5a6101
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0009: dataset=2; property=1; point=2; target=sha256:e818c2a9193efb68d4e52f288f6bbd2f423440bce40c2ee37a374c70981220dd
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0010: dataset=2; property=1; point=3; target=sha256:b1d830753c027e84c2638b8df59f2b02a582ac77598e2902052a2abdea51df60
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0011: dataset=2; property=1; point=4; target=sha256:636f3ca8b23df4656421145f85891bb8e8ea1fece5e3ca1ea7255c2d423f6959
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0012: dataset=2; property=1; point=5; target=sha256:2bd326a64cc66d40632fc5e47ae57bf8a0e936aa31225d1c06b1dc4747ae4168
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0013: dataset=2; property=1; point=6; target=sha256:c6c50bd8fa6d2c374623084297c86dca7c295da80e44e331e7e7e30828fcaa29
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0014: dataset=2; property=1; point=7; target=sha256:5a58f5394b663f603514d310921be9ef7ac7bc305305716d76c34f1adbd6de5e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0015: dataset=3; property=1; point=1; target=sha256:4d1d8c9ab716224ba93868991b81a387744a313a75b7852b5739f8e20ee68657
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0016: dataset=3; property=1; point=2; target=sha256:a3eb626a70b770bedd8aad9501701e9335ac5880fbb4b5acca6f450d91337fcd

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0017: dataset=3; property=1; point=3; target=sha256:d542255912e5031b13c243216b9d2f8f3c09bd940f682b6e810ac323294dae7a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0018: dataset=3; property=1; point=4; target=sha256:02983d65d72f23814a3a670a510eda8945b290074d0cff249f668afc3655fbfb
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0019: dataset=3; property=1; point=5; target=sha256:79253d0f6d33dfea70b9fb943b8db1f330c30a67745e015cefe5c7403017f372
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0020: dataset=3; property=1; point=6; target=sha256:a32cd2fetc121a33d26f699007191db74627550079a3d75e6cd6afc09425bf39
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0021: dataset=3; property=1; point=7; target=sha256:d25cccc488dee72beecf5189cf7c629bb19aef726d4fa9d3974f68edac016d3
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0022: dataset=3; property=1; point=8; target=sha256:41f2d48bd68fc5556ca5fa21cbd78f03fe9ce23a5f2ff4efafee5d6f2ffeb4d3
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0023: dataset=3; property=1; point=9; target=sha256:209a90e67ec8977c649505ccfd81d34246a0f2e9631e34c1e4bf7ab9b4eb4ff0
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0024: dataset=3; property=1; point=10; target=sha256:b922f47500d7bb7211f48e2febd99765c047105b60da46ee809be1d05db51bf8
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0025: dataset=4; property=1; point=1; target=sha256:2a97bfb2c9eebd1f284af60ecb70a9c1797e17a9fbe37b88d9a9848abf631bb9
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0026: dataset=4; property=1; point=2; target=sha256:ed7cf6b5cb4eb59790426d1b2eb1ede414f54c0bb318d4507e792fa407fd798b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0027: dataset=4; property=1; point=3; target=sha256:2019ad333975952d1f0e173cf8ac6d2dde0abfdb840e77fbf3c9bc56d7d46274
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0028: dataset=4; property=1; point=4; target=sha256:4a4d3760ff7c9d0a27d0a32088285aba32ac2e4fec7196b46a8debaa8027ff2d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0029: dataset=4; property=1; point=5; target=sha256:f36c30398353302f20b459bfb16d9d3b0df9c44ef55c0e508e5fd6d9066ff8f3
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0030: dataset=4; property=1; point=6; target=sha256:d8c566b135234c72283b5890b2708a8354e1ec2ad04d67827dddb453486ec439
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0031: dataset=4; property=1; point=7; target=sha256:ade5fb3317502901a702106e28239bc7edfe7c96a2f999239c7ab247497436f5
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0032: dataset=4; property=1; point=8; target=sha256:6bbb7272d0fa8f7db89f75b42199859cc753f9e89c0f2fd517bfd8517c5343ff
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0033: dataset=4; property=1; point=9; target=sha256:f8917ce3ac8a9c0281cc80da581859d31df7e1866e0a28bda586eee022554359
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0034: dataset=4; property=1; point=10; target=sha256:bd752d81e37072c971a18bb6a6bfc37c719781ad37380042865de3470c859f3f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0035: dataset=4; property=1; point=11; target=sha256:d59f5f017e7d5b8588da85b60982fcaae1efca49d948332512c494ea730849d6
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0036: dataset=4; property=1; point=12; target=sha256:b8424b960633553a90fa473e6700522d4a55c60b99df14443b2564aa74002038
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0037: dataset=4; property=1; point=13; target=sha256:cdd91e442c678641fb79a929b38855d99213e9df28a7ea1af264e1e05e274d7a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0038: dataset=4; property=1; point=14; target=sha256:f2479aafb7c73e5e11fd71eb055876efef5fd6a5d1752998bd6be85e00f61830

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0039: dataset=4; property=1; point=15; target=sha256:3014799e634955cd3a6dfa9e12c6badcc37658b14d81c797d790e6aeaba446e9
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0040: dataset=4; property=1; point=16; target=sha256:98ed6d804b4cda5f35d80211f1b3c06c97606c177a81420838dfb6e146bd08e1
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0041: dataset=4; property=1; point=17; target=sha256:1a71d6c908c33c9099901fb5064a5adc4980a18a63c927ad2b67f0c47c727b53
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0042: dataset=4; property=1; point=18; target=sha256:aada194b7b359aa35688a2452e65ea4df9c64fee2719187831618fb7f8c223f6
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0043: dataset=4; property=1; point=19; target=sha256:7bb84ab8efe5c804d1550ef7a30b826dbeb45b6ff2d26430fa654c99d3619065
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0044: dataset=4; property=1; point=20; target=sha256:7d106f01961a2d6e8b74bd07cbdf3fdb692da822737f70aca37b976fced52916
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0045: dataset=4; property=1; point=21; target=sha256:09534b0bc9f4c03a5ce3689c9a9aa5bf8da312916feb24043904a810bf09b52c
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0046: dataset=4; property=1; point=22; target=sha256:ce271f53cc0ddffe562e9fc06809ef821037bcffe45b041425cdc54e3330a8c2
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0047: dataset=4; property=1; point=23; target=sha256:0c664daf139b3b8966a44cd83fd0e0672ad30c109d098e57b31e928d30cccb88
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0048: dataset=4; property=1; point=24; target=sha256:ef1fad3cd63515b82d1c37bf1875a5b7ed301fd2a6789bb224e80ab1a69c2971
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0049: dataset=4; property=1; point=25; target=sha256:ffeed6019621cba953b63899aa85bb1c6eb84b96f80c522cb8348755788ff26
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0050: dataset=4; property=1; point=26; target=sha256:745290e6194f5b7e4abfeb2e39b60616d710cddf1ba25603550c20f7445564be
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0051: dataset=4; property=1; point=27; target=sha256:49feac1725919dd581aee77c42ded267c1a578bfe27853e1c702706d4c566c81
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0052: dataset=4; property=1; point=28; target=sha256:f75d8d8b86ede0b43b18f6cfebee68c28fa8769237a6ae85b0e0f29fc7d8747a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0053: dataset=4; property=1; point=29; target=sha256:fbcb446c4a23f8ea4b45843b0290ebc229404790939f4f07c9ede9178d1eede92
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0054: dataset=4; property=1; point=30; target=sha256:1b994f7db9e480ea269f4db36a701cc136bd76f161d0c5821567ef45ad7c3062
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0055: dataset=4; property=1; point=31; target=sha256:0a14836f9f00c34e28b85180d01cc775919fbd3f3118f9e1ca147a8078f3dc4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0056: dataset=4; property=1; point=32; target=sha256:d24aabfd386654bf7185532b19dd113812d029ea62f3a9fee513b2cedd83fe96
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0057: dataset=4; property=1; point=33; target=sha256:03441ba880204c85c6a3d28a7e5c4c2ebaadf1af640d422c1c55858e48be7d48
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0058: dataset=4; property=1; point=34; target=sha256:d3c571b5393ed9d2cb17666f32aa7eece8eae20167ebba94645cc402a777903e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0059: dataset=4; property=1; point=35; target=sha256:c47f00ce44f10d12eb2783376e380e50f05f38b3018dd1f40757d7800a74433b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0060: dataset=4; property=1; point=36; target=sha256:d82b44398bfe6c75b32fc62a79aefb6b691659cc3f4ea8ada79b134c1f73fa59

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0061: dataset=4; property=1; point=37; target=sha256:acc2ac8476e7ff67952c3f696e4c0f4ff8311053a76c3e207b424e47aacaec1c
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0062: dataset=4; property=1; point=38; target=sha256:4538911dab13ea954b51e6f4834e4322a539b7b807137811c84f25840926e2cc
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0063: dataset=4; property=1; point=39; target=sha256:c60c61c825d9438371ae8f8713965ccb9d16bf96ce2f740fb57097368d58eca5
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0064: dataset=4; property=1; point=40; target=sha256:ddb474e02dba5b0e86570ce02639e52285907249992db3ca9f3bf04232d2e060
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0065: dataset=4; property=1; point=41; target=sha256:7c02ad00e1c5a74e18ace33aaa62724e3a44ed08c623ea8600ad9b658a02d99
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0066: dataset=4; property=1; point=42; target=sha256:39a8197f49c7c897ce17dec64ad3ef36b3b6b996312e14efcfbb7aa2c517e52
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0067: dataset=4; property=1; point=43; target=sha256:cdd66574be2f4344fb0c67bd4c3672bb0715fcc2d77225a567567d62612c9972
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0068: dataset=4; property=1; point=44; target=sha256:dd9202448e59fd726035048b48fee2d25f72ed2224939490a8de9ba5150932
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0069: dataset=4; property=1; point=45; target=sha256:27cb8184b764e0288b99251612f22db2e8422efc48285d54f4937fa98b4745c2
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0070: dataset=4; property=1; point=46; target=sha256:f1cb3e5ef424646d2fc702562924ff23e6bc7bca46566b2b82efbda16a38bf5
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0071: dataset=4; property=1; point=47; target=sha256:4d9486ebe89f87a5838b6889b00b77942d70b793df5cda045dfa00563fc18bfd
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0072: dataset=4; property=1; point=48; target=sha256:aab1419df3f65d6968fb1ee228d921d1654e21fd6392109c2277145ac4f8e8
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0073: dataset=4; property=1; point=49; target=sha256:ff941d2e7b3277a7e647aa5a85060b9c5d5f9406519ae2a8ac480caf93e66d1c
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0074: dataset=4; property=1; point=50; target=sha256:3dd5b3d1fe227a2717f9d8c477ff9b5d10d661ee191429b3723505bc0cef5b15
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0075: dataset=4; property=1; point=51; target=sha256:ddd079370cfe894013d56ff89d497066c53e8f9fef07a4b84470983bede8749d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0076: dataset=4; property=1; point=52; target=sha256:700eb71a788e2404d61c6f8f0c93fbc13bf8534c32d4b096a96dbfc481b8d267
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0077: dataset=4; property=1; point=53; target=sha256:5547e583f69d27d384f27910cd6049024f7a83b1fa601e9d6ad612d82ccbbf53
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0078: dataset=4; property=1; point=54; target=sha256:539a96eb1e018281d49ec609df3f097579299b04b376d671be17bc5b8cf853d5
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0079: dataset=4; property=1; point=55; target=sha256:c18939a29e24ad179c05880d2a3698b0018feb2fc32b18ca243e71133deded6e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0080: dataset=4; property=1; point=56; target=sha256:0aabddb26ef469f9d61e509a9801c5f0d9772ae4456e2cbf6c7e335f754dac23
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0081: dataset=4; property=1; point=57; target=sha256:1b7b83188251b3529add25a87db698e4c38e7f1f5e305b096f41e4466aff82d8
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0082: dataset=4; property=1; point=58; target=sha256:ad7128bd75375e7d4dae769ac4262649ac45200ae8198f20ed335a5707a2eb16

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0083: dataset=4; property=1; point=59; target=sha256:455ced7ea1caa3b29201c1f5a44b874521772f3dc0a1235bd257f1d55f2b20c8
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0084: dataset=4; property=1; point=60; target=sha256:ad59109e77ae9149ddc4ccd3be3314a17384c659451d37c1ac2befb8d8add2ac
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0085: dataset=4; property=1; point=61; target=sha256:215c14ced897df5927db08e3e8ff392122ce2c5392be3d67fea7c972a0869e5e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0086: dataset=4; property=1; point=62; target=sha256:a862cb1f493f620fe560cd6d5c87be95cab15f54da8c669319c5df095d1ef455
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0087: dataset=4; property=1; point=63; target=sha256:33e912d149761e41d0fcb30b57e65781fe45df38313a2edd9a908ff4c23651aa
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0088: dataset=5; property=1; point=1; target=sha256:ad09d20db974c88a29a7a9a6e26bd5d3d90c1b7713847a55bc21e28fad74c1c4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0089: dataset=5; property=1; point=2; target=sha256:f3101103f1d0233a50c45ccde4a8455096faea788e6ba8c903db323b31f6206b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0090: dataset=5; property=1; point=3; target=sha256:f259266a844c048a829984c1f08ae07872f04c25c7d9e629b34f8dd862bd4dad
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0091: dataset=5; property=1; point=4; target=sha256:bd19c53e9d047a2c96c3aa15585fbdfb6bfd11299fbe7ed4ef6d2a782d8ea3dd
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0092: dataset=5; property=1; point=5; target=sha256:e05cfad7a166d5565921c4efce116e0c355ec8be83086255c6e366acb898aee1
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0093: dataset=5; property=1; point=6; target=sha256:54dd607fea4fa30431680678c815de4ab3626dea0f95d23534f547df694210f6
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0094: dataset=5; property=1; point=7; target=sha256:2365a4a862fa3987af7e729b5ea44f344d0833f2b09fbaf3c7e7ab2307efc6f9
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0095: dataset=5; property=1; point=8; target=sha256:bd219689e12f4d7962c64fec0127479998cfcce70e229edfbb304ac50a65c87f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0096: dataset=5; property=1; point=9; target=sha256:764928d1438964b851b188a2f14d94a403a07308cc166a7fd705eabf4a33f7e4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0097: dataset=5; property=1; point=10; target=sha256:d3aa11b8307772fe971ee03df439660f986c32b8799c13ae9291bac2d561bfff1
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0098: dataset=5; property=1; point=11; target=sha256:8f73ce754166509625758eebee278b6f1d781240f3ead268025aa82bd9a37ccf
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0099: dataset=5; property=1; point=12; target=sha256:e37e36c2b417296f78e80d02afef22cc3135320f99e2675d7a517d47563ba7a6
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0100: dataset=5; property=1; point=13; target=sha256:8818bb03b4112a32f3c326a9e672b0fc4a4de5f15819bcfa356ba1cc1559a2b4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0101: dataset=5; property=1; point=14; target=sha256:c60c1f5481074bfc67999affc32ce537c43e3584a34492363ef13a470b049dc1
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0102: dataset=5; property=1; point=15; target=sha256:feffa09ecaeb1ff77009852b1b3d96bb23dee94404ca93632d530788a76307a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0103: dataset=5; property=1; point=16; target=sha256:592741f6394561cc985a4608ff590b1fea8ed3207eaf3f7d32e26bc6ca7f50cb
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0104: dataset=5; property=1; point=17; target=sha256:4f933d6cffe9d1e3c3f5afeeaad234473f07a9a3a3b9764e26722447520ec73

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0105: dataset=5; property=1; point=18; target=sha256:1578ef607abb426189d09651a25a7fb4384f48d66743173b90d167ae046a99b3
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0106: dataset=5; property=1; point=19; target=sha256:add14dc61ce1c8b0d891bc68621ecf5e9ededfc66bedefa5fce83803607a0a7b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0107: dataset=5; property=1; point=20; target=sha256:2ea5414692a437b7948201166255e357a69976575a3dac7552c9fff65070644f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0108: dataset=5; property=1; point=21; target=sha256:1dfac1b425a788bac4d4992ffe1b122ddb7364d8246b871bcd2544ea958b73bc
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0109: dataset=5; property=1; point=22; target=sha256:e26fb1a6edef429c9afcf2e3da7b75bfe9c71a57255b1f44077a84576044f892
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0110: dataset=5; property=1; point=23; target=sha256:0a5a167fb5b9f715f8ea4aafc24dbc7dcae1777c6a0ac1facce063e5e6e97e1b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0111: dataset=5; property=1; point=24; target=sha256:a3306f0bb5ee6d728b817bc19a0cd1764a63189e4b4347d5c1f00a341875f4a8
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0112: dataset=5; property=1; point=25; target=sha256:f9fbe376da7811c88ac74ce0c8d419673573f1ebd52f4111060cbc3c987cc04b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0113: dataset=5; property=1; point=26; target=sha256:bdabf6f9066853f47d9d6e65b5bba8dcea1eb426f74f1583d3405216d004f78
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0114: dataset=5; property=1; point=27; target=sha256:adba281fdf18f014724a08c8d1ff875e98ef6c93d69b251d4a651bf100a23853
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0115: dataset=5; property=1; point=28; target=sha256:138a17b9fce0ec722f4a4a4b31f75507ed1b1ad48edeea079d40605702c8d427
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0116: dataset=5; property=1; point=29; target=sha256:1bc582ce40be2cb84d73b8f023237a555e591922088ad4b26cb8475e7299e8e8
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0117: dataset=5; property=1; point=30; target=sha256:a7a44c24271f0fddefd704ca6244e12a0ed46ce2d8155ac68c64db03e6ec94ea
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0118: dataset=5; property=1; point=31; target=sha256:85f178e18ecc9267de970461f21382589503b07a7aab5775bee320a7bec4865e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0119: dataset=5; property=1; point=32; target=sha256:152594436dafc5acb849357e19c9c5f425baac1507a1261620b4630f263de52
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0120: dataset=5; property=1; point=33; target=sha256:139b95f9b33ffddc7356849e3511b9a02ddac966ab08daf0110ddd008332d66
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0121: dataset=5; property=1; point=34; target=sha256:e9e601dafd85b563c75e0ac87b3718ecb2d86b959a47108d240d809a6b140d09
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0122: dataset=5; property=1; point=35; target=sha256:f9c17f156dff64e019bfd238907e217b0b46788afd194401d5e4e2f67e7bfccf
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0123: dataset=5; property=1; point=36; target=sha256:aea7152d06687e156aed1d9011c14b397a7c9d691a03c233f9b015ab5651e701
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0124: dataset=5; property=1; point=37; target=sha256:844152ae3dfa6939cb1d3028584f97af302bddedea7f77f690fd81fee548fcf
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0125: dataset=5; property=1; point=38; target=sha256:ee8f9527e0ba488e3b6b56c1a0b156d78b6b402e0896fecf116a17eddc7cf1ef
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0126: dataset=5; property=1; point=39; target=sha256:f9022c501da70986c7761ab32140981b615beec23229ccfb5ca5a1a9581ddfb3

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0127: dataset=5; property=1; point=40; target=sha256:e04ed17b90f48cc607e4d3bd386fa6eb2aafdb66b4feb17428c118ab65a98280
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0128: dataset=5; property=1; point=41; target=sha256:5f58a880db6f67d30259371ff763d08f0a41f1c532048066e0d53f1064bbb094
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0129: dataset=5; property=1; point=42; target=sha256:5d6bf6cbb1610816a18fb6e54cb6471b826dcdbcecb9478245653d10b256e5d3
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0130: dataset=5; property=1; point=43; target=sha256:c66295cfb4690d650de07448fa9b67c8ae112caab553857b199d9d4eb14ba06e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0131: dataset=5; property=1; point=44; target=sha256:38908f9ca58c882e0ab81020d5eae0b060a6324e4fe2041c048f6ffdf820261b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0132: dataset=5; property=1; point=45; target=sha256:99323383ef936c55cad7a282c7253b85cdcd67e383f5de259b4888cc8eb38ab1
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0133: dataset=5; property=1; point=46; target=sha256:642ecfd74b1168c7fef0625d4720c8f1438ef7dcf0b0056129f4fe29086f6a94
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0134: dataset=5; property=1; point=47; target=sha256:c395a9531072ac7e7449f000d119f836a1ad48203ae302758297f0e6c0bca822
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0135: dataset=5; property=1; point=48; target=sha256:34c7ac9ce557af4859372fb8d4cc80a29cb0794e45ae9c284bdb61913d9edf83
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0136: dataset=5; property=1; point=49; target=sha256:a241a1c00e9275278478a476f92dfd41afe2fe8a91dc352aab4f29a611b2e9c
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0137: dataset=5; property=1; point=50; target=sha256:41ce98cb24ab0b7f09806987cd9cb1baee0c44e7561182637d748fb077fdb86b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0138: dataset=5; property=1; point=51; target=sha256:157f8e262a15e52de9e3de40e5dbd228f9c6746f20fdf66d91fb4574b0137482
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0139: dataset=5; property=1; point=52; target=sha256:a84365c37eb3b8e4bd753ceba1360f8abdd09f61dba8727d1ed876c5b5c18cd
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0140: dataset=5; property=1; point=53; target=sha256:4b5ee75f4069e2d24b8bca0e80a71d83b047b8dfbf8a54703b8930fecfb97de4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0141: dataset=5; property=1; point=54; target=sha256:03d4d20850d28c2f5aeeabc9c907b55084300813dc9e7c72add1b12afb9bfaea
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0142: dataset=5; property=1; point=55; target=sha256:46c8b6baa760ae3bde117411e59afad7e083e912e5fe6aa576845d0fab465c4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0143: dataset=5; property=1; point=56; target=sha256:16e2398015435edf0b5aca703b140e2e54a0da22d3581dce4b87050006215820
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0144: dataset=5; property=1; point=57; target=sha256:878dceace0c82ecf24a19f49a4d7c3a1427aec7bafc3d63c37dec2725c3e3555
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0145: dataset=5; property=1; point=58; target=sha256:520106d1207fa326d8d388d89fab1b49cfb7f9609c93b40e8b659843dc66a345
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0146: dataset=5; property=1; point=59; target=sha256:f72c5d81c5b919487cf721ed33b5be32c58d3e18e3afedfe047a735d3909e5c8
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0147: dataset=5; property=1; point=60; target=sha256:6c1a43adcf227ac6423cf2f6ce8f6ddc5fee91048014b3ccaff5a8c55216fcb1
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0148: dataset=5; property=1; point=61; target=sha256:5a68fbf59583d22d9d02a44efba14ceb0f9eeb9d7796c1754890dde105d41c05

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0149: dataset=5; property=1; point=62; target=sha256:737ae38e8c193b88b2739b55786be3ddabc5434c8d0894ab81be2d72145eaa36
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0150: dataset=5; property=1; point=63; target=sha256:3094fc1d1765e60280e99f0facc06b7e13ef42b53f6abc4193991ae9c9fde02a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0151: dataset=6; property=1; point=1; target=sha256:621b08349db1c4e5e86785c95446177c779f80e11e54b6c8177b6b0ec63284b7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0152: dataset=6; property=1; point=2; target=sha256:f37c1525c03788b4906675dbf25eac11d5ffd717432493553e7cbb05622beb19
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0153: dataset=6; property=1; point=3; target=sha256:c288881b7f899e9994d2e7c1e61aa19c025ecf68135a4ee8d632cb61608ec60a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0154: dataset=6; property=1; point=4; target=sha256:72b87d611e4284d2d0fe003d757e1e96cb96270fbff820bd53f413c1bf5a4b14
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0155: dataset=6; property=1; point=5; target=sha256:215166ec974b288af0083838faf6cccd573a14659b13cf4e059b17180a6cf59a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0156: dataset=6; property=1; point=6; target=sha256:f37df0d0bce8986fcc2cf7be947a916911096f413ef94188a3b11cf278938fdb
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0157: dataset=6; property=1; point=7; target=sha256:08e8352cb65254a216527cb85428013f535ebc2cdef311da5741e51181430110
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0158: dataset=6; property=1; point=8; target=sha256:847ba8dc9e0424966f7e440141c96fd769680c01af95385a96891a3e5a9e7300
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0159: dataset=6; property=1; point=9; target=sha256:7be70d64262bce5503a426b1ce6c8921d21b7c224e093dfd0f5f01ca675c9436
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0160: dataset=6; property=1; point=10; target=sha256:1ae2a59c2152be31dcb54e5cdba84d98b789d29077992816fa4788afafa4b8a4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0161: dataset=6; property=1; point=11; target=sha256:ee5b0665401c336a572e19d0901abf808000d68d8cc6c6bbb3d0094ca21048
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0162: dataset=6; property=1; point=12; target=sha256:7ad816bab7f341e0ae70660bc4b2f2c7d801a0e23a738e50e947eb5dd212d075
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0163: dataset=6; property=1; point=13; target=sha256:2ec4bee48404fd674d6cc47dd6cc507de83431ad54ab3ac22e50b8deb088a3d8
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0164: dataset=6; property=1; point=14; target=sha256:6fcc1e0d83c172fb8469ff86d4471b47956c4022482a4b39e6a9d037b851dc6e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0165: dataset=6; property=1; point=15; target=sha256:1f97b226fb06fc5d782f6bc1733c808077a9c0c71d636780d4ad716fc757cddc
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0166: dataset=6; property=1; point=16; target=sha256:d1992571de1f849126b1a1b7d0d1ed5ba024cec4bb3623e715dab9fa266b2d60
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0167: dataset=6; property=1; point=17; target=sha256:36984ba4e15a9a0a67a6c22f39ad02be1e83cd226a11a0d7fcfb47e27cc9a5d7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0168: dataset=6; property=1; point=18; target=sha256:af444c082d4cfc5b0ac791dc9cfa6ee4d5e31f0362ff19829fe6bfcf3f4a2fe9
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0169: dataset=6; property=1; point=19; target=sha256:65925abd904e8bbbed5ea4df4052b75e71f51f85d549848242a0309d885717280
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0170: dataset=6; property=1; point=20; target=sha256:66b6523d06660f0a1377c08a44ae2c7483e5f0d1817f5807e75a973bd6c42935

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0171: dataset=6; property=1; point=21; target=sha256:a1dcf93babeb67cf704fef4feb3575e8916bddf4f9c5cca8f264b2692d5d4b3
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0172: dataset=6; property=1; point=22; target=sha256:c295da74d61970bd6433471e6d38b682ea9798ba5cff661833ac80a935a34f3f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0173: dataset=6; property=1; point=23; target=sha256:e12f6bbb8b0d44c4d346745134af5767ff8698ad3c8094f076a6195169dd667
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0174: dataset=6; property=1; point=24; target=sha256:bd2e1948c2e8dd374926d38a47f194373bc8d8928988f7bfaa787b191e9628cf
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0175: dataset=6; property=1; point=25; target=sha256:5b7a8ec784b137bae66e3ecd414a59a4a6543c05a3fd89b614d5b0f4faba6350
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0176: dataset=6; property=1; point=26; target=sha256:1839e09280b52c0ea551116ae6169c354423dd7a54cbbb86a0e1e5e24604f09b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0177: dataset=6; property=1; point=27; target=sha256:2065063dfb7d24a55cb6d1cb1d61d87ac0dbaa5d1f171155b9a8dacc11813102
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0178: dataset=6; property=1; point=28; target=sha256:b80e070a3381df670ef2e077c01711631cfa9f5141b7e90dd22cd9ff40644e59
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0179: dataset=6; property=1; point=29; target=sha256:73e69660a2ed6f7e61e1f4da1e2eb5bc4d0f31a0de8c12153f0c7b301431b99f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0180: dataset=6; property=1; point=30; target=sha256:dd28253043860b818f689b8afd694e50f9a6772e7fbfd535ef3401375b3ff90
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0181: dataset=6; property=1; point=31; target=sha256:29a0ee4a1fcef7ce6233eab682c557f6c7c8ed3d439ba9ae379b26248ad0eb52
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0182: dataset=6; property=1; point=32; target=sha256:ec3aff62a4ed513d9c66eb0b220f5ea5f041ab7901ffb0dd4ed5a94cfa2df098
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0183: dataset=6; property=1; point=33; target=sha256:c5692c8cefb9ac652b8719bc664db542b1d625e490f6537774e49cb625a4b9da
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0184: dataset=6; property=1; point=34; target=sha256:8bf877540c6ca2390f076363a56f6dd82495c05a814b4d5f533390135943a102
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0185: dataset=6; property=1; point=35; target=sha256:f5f44f6c1c1c450c9b4ebec688296ee80469cfedb6646cd54e5b44cff0af14f8
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0186: dataset=6; property=1; point=36; target=sha256:60dcc55d320ec6df69b36676a17a86de6edebefa51ff1a8d177143c28ecbb54a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0187: dataset=6; property=1; point=37; target=sha256:b2a570891d2358ca69184007083decd8520b7615d0eab756b7f8d4886f82c902
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0188: dataset=6; property=1; point=38; target=sha256:4b1128b0277cb7bb79e077077ed8b363a66c0480915c7149e02556418cd2a48f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0189: dataset=6; property=1; point=39; target=sha256:3d330981c461fe851aec991d8bfcc68f5e34ba67563e8d27666ae781ccd8bc62
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0190: dataset=6; property=1; point=40; target=sha256:cbb171e76d75facaa2e082dbd53ef0da92ea0128ee6619cb0bc29486a4415aee
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0191: dataset=6; property=1; point=41; target=sha256:e1a9b68261ef207ba57deb9c8cf8d843914c4409d950205043d9cf4b4f1aa9c3
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0192: dataset=6; property=1; point=42; target=sha256:1bf345cd1141e7dc0159ee1d2ba9cd50fddbbbf806e1f1e850558619853e79b

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0193: dataset=6; property=1; point=43; target=sha256:e9751233166b5f0b7ffdd709c9aa1d8ccd80004c7a1f106dba137e6130cd123e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0194: dataset=6; property=1; point=44; target=sha256:ebf78632e85b01b9ed01e3562f6cfa24c5be3d07c37a2ba345eaa6d53f89533c
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0195: dataset=6; property=1; point=45; target=sha256:7e9d9937e93e8e837976cc98609b79d674456cb75400e78f908ede41f2699b8d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0196: dataset=6; property=1; point=46; target=sha256:0061ecbd8c5f019794835ee69cb61b408474a3a7943bc5299c8f0eb6d17f780e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0197: dataset=6; property=1; point=47; target=sha256:4c5f28224743bb12cfd114b499833d447148eaaecee907c837de6fc8ac2681b6
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0198: dataset=6; property=1; point=48; target=sha256:e312134efc81649116b745e75e19b63e51438b8c8bafbdd36a27dfb542f05e25
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0199: dataset=6; property=1; point=49; target=sha256:598942604eeef05fe5a1df98943e2abfc249bef590170634474fd712932f6f4b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0200: dataset=6; property=1; point=50; target=sha256:8450311de44a678c86c9ed9e3b6be08883569e0b764b1ffc89e87af9a595f60e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0201: dataset=6; property=1; point=51; target=sha256:bb4ff4afe56b051afb45e23eff7ef865ed098e72b70b2d5bbadbe08fedd08f45
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0202: dataset=6; property=1; point=52; target=sha256:f8bf8d2aa4f14dacce0474bcecb1bbb294ef6b71ebe4d95b626a30c8ff274e3e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0203: dataset=6; property=1; point=53; target=sha256:b9b8da3e444b0dfd32bd4f536fe16ccbd4987526fab88dac770fda27ab58d6c7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0204: dataset=6; property=1; point=54; target=sha256:319b154feb74da3a426087ab6d7c4ff856da53148e8a834e72239750209d38e2
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0205: dataset=6; property=1; point=55; target=sha256:11dbd07cb2940e26420ac2e9d0d44bd13c915cf84d7a9bfa1f770b8a1a7a7484
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0206: dataset=6; property=1; point=56; target=sha256:fd922edb8bbbd6fe65e65c9ed1c7818353e437ea1c34d5bf4171ac4e520f7aed
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0207: dataset=6; property=1; point=57; target=sha256:35d545c2cd4cb628c51462b8aefeb6b9e6d8554cfc19c34d3ab1030eb7a451c4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0208: dataset=6; property=1; point=58; target=sha256:8acd8a03c767d5d85c012f73bc86d30c15998a69d291c3d69f16de84d10b83e4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0209: dataset=6; property=1; point=59; target=sha256:fd079e25cd13964f00afddf68ea747f8fbd4723aeceddcf0e04ffb6d34360b4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0210: dataset=6; property=1; point=60; target=sha256:2fa5aca837324f723674a866015789f3cac9af1262f1298459fc3423cbbe216f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0211: dataset=6; property=1; point=61; target=sha256:c8c128a03eb0b30ab0e204f3463ad55c3fdf61d773dec7c9cedf812c71cfb42
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0212: dataset=6; property=1; point=62; target=sha256:400435aa4acf6f518e9b14c0fc1e3162fa25a57b6dffaee2b3e836033041f995
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0213: dataset=6; property=1; point=63; target=sha256:a5a97a7bfd9020da6d68e7738cab0e0db425ba70c4db1bc4e2fbef820a6860da
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0214: dataset=7; property=1; point=1; target=sha256:a47ccfd029e6b1b8e0aa52688c9339ea22e9d2ede6fb380c9c76efb2f58b17fa

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0215: dataset=7; property=1; point=2; target=sha256:917d47cf4d695b608a2c74ac8365088db3aa3b9ce389a4731d864fdf6579082a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0216: dataset=7; property=1; point=3; target=sha256:16fa6d27a2bdee82f79686baa0f5919a681afe07c4962eec04e7ad154ae945b4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0217: dataset=7; property=1; point=4; target=sha256:df255d155bcb14a698abe442f8ac2c01160623a4ab58f6335f5de40d4ca8c86d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0218: dataset=7; property=1; point=5; target=sha256:64925acd5d554b67601bc849d1111bd018247bcb29b2650abc3e2fca36d29658
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0219: dataset=7; property=1; point=6; target=sha256:e363d5de28e0a6f948062cbd7b2cd0100536118c8ec578f32022fbbb9f6781b6
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0220: dataset=7; property=1; point=7; target=sha256:1216e9dc48630c591f4c58afbb8341c1423188ee30455d725d0b7bbfe008c171
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0221: dataset=7; property=1; point=8; target=sha256:cf1920895e0897b590c835c2cb83d2b6a7c571e2d9331dc1799e83a90f2259fc
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0222: dataset=7; property=1; point=9; target=sha256:878f4d9d09dcf98346799268d9db2925791794d0720b09b397fb249768b44d0a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0223: dataset=7; property=1; point=10; target=sha256:12089fc6d76778465c163bda08631a3db203217c28b1f268707eb593fff6be52
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0224: dataset=7; property=1; point=11; target=sha256:27c4e153741517c9f999fdab83999297f778a975a8e81fb2263af79cb625487f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0225: dataset=7; property=1; point=12; target=sha256:e964dc3d9e13da76e905be5c8e33501a502e35d879a2a2eb77c55c4d6f014ed7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0226: dataset=7; property=1; point=13; target=sha256:e3caf74bedd0e9f14bacacb5438e146b4c50a19adef44c67c33ca788116f38b7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0227: dataset=7; property=1; point=14; target=sha256:1e3b09ea6e6bf24965b5d0be42f8bf104151133b3f4d750384357f51533c84ee
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0228: dataset=7; property=1; point=15; target=sha256:96dedba0ca957554edc6e4f6ed49a290cad5650dd620eb980997b2d1c2f9e9e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0229: dataset=7; property=1; point=16; target=sha256:7a07332c57557e9c266dac7a1b0a660b7b859185646f9aa5a576f1590fad4800
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0230: dataset=7; property=1; point=17; target=sha256:19dab53a01d29b5a7e6289e5cdb220cea9bd57edd0e8a4997e4ce4202cbe8e89
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0231: dataset=7; property=1; point=18; target=sha256:4a568d6382d72ff244444e6e826eb0f8f01dc31812462950849e206a2324f16e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0232: dataset=7; property=1; point=19; target=sha256:a98cd65a143fef8ae65a5e39285672fdb2c4ce24e1b4bc4e90194f8b62de3ab2
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0233: dataset=7; property=1; point=20; target=sha256:e2df0e8548b3a35a4db13c196a95b16bb96bdbb02787a76c062df2bbadd10936
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0234: dataset=7; property=1; point=21; target=sha256:83f5d450ae3455411549d83fa9d1a03c97fc73b953bf33ca1373b59651c347f4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0235: dataset=7; property=1; point=22; target=sha256:c09e0ca7013fb12fe97f6f00c0912b542e10188e810c9bcf050322efdcc05e22
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0236: dataset=7; property=1; point=23; target=sha256:e13fc56d408fcd606ba8c5286589d427f2b72e999705d073b368643db77ac37

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0237: dataset=7; property=1; point=24; target=sha256:1b27f6a9037fc41601996b5666aa0d18e344561c0d5e0f068bb8dcf80df4a991
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0238: dataset=7; property=1; point=25; target=sha256:9799c1cf47f790f83422cf82bab9e6b5d3e36b98410701a71ba9145631ea5c8c
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0239: dataset=7; property=1; point=26; target=sha256:cc08ae2512ebd01ec57986f521d0132943113adbaba1fd6d683efe00b53fa865
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0240: dataset=7; property=1; point=27; target=sha256:cf793961a956e35ea139c9226706a89041dfea52ff83f5e8769bd7641ea5fd2c
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0241: dataset=7; property=1; point=28; target=sha256:8370fd688fe128310a893b4a9a9067ef005bc468b1c847cdf99bf9786fc7c9cf
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0242: dataset=7; property=1; point=29; target=sha256:498e4622fbcaee2468cfbf8c3dd4f2c919c4bcb49dcabf0b8c80794d15d75c53
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0243: dataset=7; property=1; point=30; target=sha256:06c2d01611ec381565d08e3390218ad48fe4c31b484720e6c66d57eaf14d4ba1
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0244: dataset=7; property=1; point=31; target=sha256:8cc40593e405c19e087c6be8268ab1e3d19a4f724316ddfe94e8140f00c75cca
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0245: dataset=7; property=1; point=32; target=sha256:04bac778b982e5cda0c658104616fc13222585ccc91602a5a9de3a825e38b414
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0246: dataset=7; property=1; point=33; target=sha256:0567b716ebe52b47664dd575a6aad3a7781e48cffcca299c7b6cac5fb0120390
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0247: dataset=7; property=1; point=34; target=sha256:86b43e6da074799f4651e05ba2fc136b3b6bb944b0fd044170b4dbdca72fb097
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0248: dataset=7; property=1; point=35; target=sha256:38df3cb6376adb666c383e2cfa368c350c41459545f636aec7b7eebc32d1b257
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0249: dataset=7; property=1; point=36; target=sha256:c412cdfaff98adf7af8db1b64b5fa692407d9649c8c9512d9f773048411417f4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0250: dataset=7; property=1; point=37; target=sha256:e5787f8c944e40a75c292a03090390090b98027ed291b25bee6b8d019430fcb7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0251: dataset=7; property=1; point=38; target=sha256:83c4a602402c9ae0d98d4aedec72780e4110ac80b68626ef0b9e6a403c7fa7b4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0252: dataset=7; property=1; point=39; target=sha256:74418eea7af926aebc8de98b193243ced0114eed5aefca6aadccca720aec8ba
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0253: dataset=7; property=1; point=40; target=sha256:468c00da0b44e3f9cb4d1d615cafc5a1ec7231e18d292d33ba41724a6d9b9631
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0254: dataset=7; property=1; point=41; target=sha256:12848c11264ce924ed194a8f70cb318eed1cedf6f8f20fa063b236b0827f6d4e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0255: dataset=7; property=1; point=42; target=sha256:735e13ae67da704d7e8dbbd3ca3728599fa2ae130702be85b14d3046ee9bdf05
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0256: dataset=7; property=1; point=43; target=sha256:7c41f0e5c88739f597edd2e61ff5006d5a37c9fd74c1275083b8666220c7dcec
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0257: dataset=7; property=1; point=44; target=sha256:11af80877ec45a0f89fe7c303871dbb4d183284779b1df35c88d68dc4337f31d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0258: dataset=7; property=1; point=45; target=sha256:a1fd51904001d403207e7e1224d99e39a71c5cb39e8f549e81b6087289f2e5e1

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0259: dataset=7; property=1; point=46; target=sha256:4990a2abc0b0b208936fce34122e7e6f814524f06f582c9dbc8b1f7ef1f6439e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0260: dataset=7; property=1; point=47; target=sha256:6a3f15830d259a12a50112e940663de3c185c9b889803753f94ebefcbbdeb2da
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0261: dataset=7; property=1; point=48; target=sha256:2c2d114a1aeac3d395d4bf8b614f800bab50d68b79ac92fab6a5864a57e6fc71
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0262: dataset=7; property=1; point=49; target=sha256:2b4b86077df55ffdf99171bb67a5f9aa8bc72f046a5ef8fc02b5fea2bf9c6eef
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0263: dataset=7; property=1; point=50; target=sha256:15e138568c8842c6fb343b877080d04a7c2e0ae033454e0948aef6c784d1ab0b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0264: dataset=7; property=1; point=51; target=sha256:fb98ab28489935d87655c86a3738ad4364876184b07b26fc65a6bea9a2c0d2f6
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0265: dataset=7; property=1; point=52; target=sha256:32035b4ffed593b3a4f001388ff4b1009d4468cfd83f24c00d74c64eb3e78c83
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0266: dataset=7; property=1; point=53; target=sha256:221e8981777c6dc6e9c9aa8e08ba1ee7893cc6c9a2826bfdd79344a1bf334f17
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0267: dataset=7; property=1; point=54; target=sha256:7abc663bc87d697172e0749f03cf7daded440b0aa5d8e113c3f25846b7f5b65a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0268: dataset=7; property=1; point=55; target=sha256:b6db80a4f27069cf5e438592ff8a752784cbab98cdea1e2b7568499766f3f8db
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0269: dataset=7; property=1; point=56; target=sha256:935a03994f60e76319d9665bb0c315a7a5b6f285dd2cacbe5f83c850895da62e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0270: dataset=7; property=1; point=57; target=sha256:269b9f30e29f67f58fafa6d843bd754edb29c0e6477bd92fb31c2b5b70a7e93c
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0271: dataset=7; property=1; point=58; target=sha256:f19accf4c68742b7f57ab09d816eae56e6e79cddc0f03b9a7b54228278c14e0e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0272: dataset=7; property=1; point=59; target=sha256:30f0e1d975ce427e4f60ca35705a212ac1b1b136fbc88c094deda1ce527c5962
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0273: dataset=7; property=1; point=60; target=sha256:013592a2ecdbcfeac8592186a2ebf98f1073bedb386d565f1ef7114c90d65176
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0274: dataset=7; property=1; point=61; target=sha256:eb71ca76877f17265bdc9232cd976d0174f9dcd7acc8983f920830a4e6756c94
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0275: dataset=7; property=1; point=62; target=sha256:d34b1fb5c8811f89b60e10930f055e717d7c76a3fe71103a4fe79b4d2922931b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0276: dataset=7; property=1; point=63; target=sha256:554a6707ba94e072a7fa74fe8f202e34a5827bd6c9afe1a3c47f20c3ee35a789
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0277: dataset=7; property=1; point=64; target=sha256:406e286537c96a0137aa1dc7731872cf408f1729acf2f17b9a3536ddac24d5f6
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0278: dataset=7; property=1; point=65; target=sha256:e6f4d5b12ee6451417086c691cd5cbe02102c89558722359721a26c730e9cb22
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0279: dataset=7; property=1; point=66; target=sha256:dcf549460aad568b0883da626cb99ede0d90a9934d0cc4a58403a8fcaf005e52
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0280: dataset=7; property=1; point=67; target=sha256:1c27949d2e28c0256c42f58943ba566e2382fbc066d11286b423a35b1ddcf6e2

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0281: dataset=7; property=1; point=68; target=sha256:ef8b5f37c5177c485c3ffb3e4d8b24d7c6face827c8e223d6ab7ebc0906c350b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0282: dataset=7; property=1; point=69; target=sha256:04861b151346d1e5f1ae2e5815f656362221d81a8435f40378c6adfb4547dc37
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0283: dataset=7; property=1; point=70; target=sha256:190a252c223dc21663b0f0b2dc6a60fbeb5efbef558bcfc1aaa679ea1f9904
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0284: dataset=7; property=1; point=71; target=sha256:cdf0e446b3b9ea599840053ba2c01d56e0fa54f98c04ed98ec941948dff6a1a8
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0285: dataset=7; property=1; point=72; target=sha256:6d430a0ecc08979cc55d1d05f7038d5adf9012cb6cdf6a7772cd9af53955e78b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0286: dataset=7; property=1; point=73; target=sha256:ff7a07b547ee0dadced4dd8633dbb65fcb91ca6e5dd60124beddd0ae729a80ff
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0287: dataset=7; property=1; point=74; target=sha256:5641cd4e32f48f6c84d526c0f76b7ebe1c70991e0563986075cb7723f279e63e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0288: dataset=7; property=1; point=75; target=sha256:96574ad17d4ae31a68ab1687dbf61d2bf8ab483a490e2d3b78234c98aeab3c1d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0289: dataset=7; property=1; point=76; target=sha256:1d1df2bccf1b1368eefd12cd847c824b53aacc00a6540f68cb555a8e3b805d40
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0290: dataset=7; property=1; point=77; target=sha256:5a2645dfd7e98937cfca3802ed3365f840fd6ccd54d58b9000b661de1fbb5b84
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0291: dataset=7; property=1; point=78; target=sha256:e9ed164e146c32b2f7d53b0dda13149432460ff3eece5f201fb90d34b3892ebd
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0292: dataset=7; property=1; point=79; target=sha256:cf60236c9f48bc81d1170aef21a211c74285396171e9481cbec10591152c0a3e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0293: dataset=7; property=1; point=80; target=sha256:1c5cbab6588a12b46a3135f7ff0db70fa53d680c9427a85b938e2b99332e0a8b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0294: dataset=7; property=1; point=81; target=sha256:d29ea79a013aff815865cc4ce931733132c44a63e79e70fe4062c5e311a8a532
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0295: dataset=7; property=1; point=82; target=sha256:1211a5dfa1849851ca4cd88cda995e25371046ca5a461d19750cad79d6c4c9c0
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0296: dataset=7; property=1; point=83; target=sha256:1b1e02d2c316fcd947fca6a050c5f72bd1f224a8e68b5abe380f2d63b1d97f02
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0297: dataset=7; property=1; point=84; target=sha256:7ae926a471f7ca6374f4e54beb4b14e0bdd960f572f147ad4ac84bff52cec05f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0298: dataset=7; property=1; point=85; target=sha256:a799f6740ad3c3aa814f5aefbd7dbc5a1a8447a3e4aa81d6688dae9e323555a8
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0299: dataset=7; property=1; point=86; target=sha256:b50482730784cf841fc575bccfb5244bb50c74512cdd98cc1858ae72798957c7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0300: dataset=7; property=1; point=87; target=sha256:750abc055262d3f46a9cf9860a6c77df566363fa08ea7126f2dd791038f9ad4f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0301: dataset=7; property=1; point=88; target=sha256:779d91f29ff9c1e7eac70d74ae5210aec6256bb51fb6c18adbaf2938a9e9e266
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0302: dataset=7; property=1; point=89; target=sha256:999b2956d93cb5344ab8c07f070341a1e17a89013c731f6929dfdc9b3c1fd40b

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0303: dataset=7; property=1; point=90; target=sha256:d1b1abfaf7ee7b9b902c980dd1f90f97214f9e4f7c20308cec9554ee0806b65e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0304: dataset=7; property=1; point=91; target=sha256:00144397a72abb8866ec533e99872949b492f85a7ba84fcc6e53a55520ddc4b4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0305: dataset=7; property=1; point=92; target=sha256:1b9041d375be0fc86b30a80a61ca6a2582c33e94dfb586805f538bb336a0d370
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0306: dataset=7; property=1; point=93; target=sha256:9bb32bb0b62bbda334dd5d326a4c5f5e88356b99c923e6a5c688020423cd68d4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0307: dataset=7; property=1; point=94; target=sha256:ddd5a1747dcf327b7d3b6595b8520a3454ddc03f872efdb935632c59b83618e3
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0308: dataset=7; property=1; point=95; target=sha256:d499014425041508c974fc9fe62e8c4ed61542f276c7e12e43cd5e5e48ea3619
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0309: dataset=7; property=1; point=96; target=sha256:31b4803fabac909f4a8b4850b41fd03aa4b347e425ac3d5822b34b898dfe02d0
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0310: dataset=7; property=1; point=97; target=sha256:8a8583a9f3f77e919dcab799733dd7e3833f429c9d7295be13dc9e85a1441233
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0311: dataset=7; property=1; point=98; target=sha256:fb9c7469d6afe139da7c1f39b21a8f1caf6fcb421328bf45540c5a711b4ff2bb
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0312: dataset=7; property=1; point=99; target=sha256:227f4df28da842704dd353907fa8d8aa55eae7ea386eb123481871caffc2a0ae
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0313: dataset=7; property=1; point=100; target=sha256:b43c1e0fd01255d86a580cd7fa76feec2c5ae4efa61e9b1701479e54a4fe1e63
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0314: dataset=7; property=1; point=101; target=sha256:4f4123bcc94eb61d5d5b99066a20f34c3eab38352396ea0aebbdba8d732bd009
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0315: dataset=7; property=1; point=102; target=sha256:5bd839abd6c28e584181b375706818eea8711643214b412a243a0609748cc932
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0316: dataset=7; property=1; point=103; target=sha256:9963ff1efff78bcc2f00d0c26f598d7673fba7fe73fd0fe5b908f2ee1f18a1c7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0317: dataset=7; property=1; point=104; target=sha256:c29a1c6eab22fa59f91968234acdf9ee2606b5b2a028b93d646d25c145395ae4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0318: dataset=7; property=1; point=105; target=sha256:6b501886d7dc3a8c15faf6d62d37fd6702b2c088f9911661d5bfe7f51870e053
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0319: dataset=7; property=1; point=106; target=sha256:be2e503a12fd4158a50cbfbcf9f0e071bb419326bc74a7df8f94abfcebfc5a57
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0320: dataset=7; property=1; point=107; target=sha256:848a99f76eb765ff9eb4ceb38c61c6b04c53562d9a7d96ef168ebcc23fbab00a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0321: dataset=7; property=1; point=108; target=sha256:571d0e63ef42edf72af7d5ceb7855ef40fa281b762e0d3b1b25e7be6098ae42b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0322: dataset=7; property=1; point=109; target=sha256:774abd3514f1708c2fd5f7a0eeac63eb2b2b71f15c9b68e397bda369d2ff17
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0323: dataset=7; property=1; point=110; target=sha256:323fb5372669ff86043ffc41ba6932025be4e16bec73c5f8f118bb7d79f0c94e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0324: dataset=7; property=1; point=111; target=sha256:9504b79da950bca1ae8cea16f80081e765c0f8848224d4f2f752860acedbd213

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0325: dataset=7; property=1; point=112; target=sha256:e89c2ac52dfbb6ddb5a503a1953544df86bd6d2a94a9b781573a9e7211d188b7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0326: dataset=7; property=1; point=113; target=sha256:9fddb27b0741be95f56a27cee908c0e6f6d9b3608dd8cbb4c144899bf7505816
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0327: dataset=7; property=1; point=114; target=sha256:57d635fd5b6bb9170cd7c0236a0e76b351237700d8218ce95a63ae3a977096f3
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0328: dataset=7; property=1; point=115; target=sha256:4f1a55f51ce99ef76ea5b5f38506282fbb7c154e43fdac8808ab0bea86aa6416
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0329: dataset=7; property=1; point=116; target=sha256:5507885703bd4864f03e5d79c235a3733d54e8b5e464de2933712c249687f047
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0330: dataset=7; property=1; point=117; target=sha256:7f7d34a49bb69a88337f5c6b3b9da9aeb3710ee1a7db4c5601e7e5de79869f49
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0331: dataset=7; property=1; point=118; target=sha256:fb6e8bca44e8845d13070e3ac79d521ad97fc37393510a04eacc71d34704084f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0332: dataset=7; property=1; point=119; target=sha256:68483efc393d7313a420cf42ffccb0479d724061789b62d2fce86ff8097aa529
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0333: dataset=7; property=1; point=120; target=sha256:99bca94563a471073824e3019d3458a36f351884b2924f019e27997453cdbefb
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0334: dataset=7; property=1; point=121; target=sha256:cb1a637338e0d8474b576d527b8a32367c973ecf2f7f871ed97b49eb33704748
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0335: dataset=7; property=1; point=122; target=sha256:68056065ac3bf3aaf756e6e503c96cd072a876a3ebb88f36773c16aeff3c2d5a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0336: dataset=7; property=1; point=123; target=sha256:6d4eb719e9637492099bf0c6912ef30d250021cf3b90ca4557e48c1f209f2ee0
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0337: dataset=7; property=1; point=124; target=sha256:813e5be9eb51fbb08a75812fe714be6fd3ff13cf2cc8a60f33c768fca3ddc63d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0338: dataset=7; property=1; point=125; target=sha256:39bb929bdd30e16c149ac979a8c3c91b6ca31c6fd0d9228eba9b8380d31f1e94
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0339: dataset=7; property=1; point=126; target=sha256:af349653ce611ab33df39533333ab8f2ee4d7f5ab2296181d69b397a97fa99be
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0340: dataset=7; property=1; point=127; target=sha256:cc1743673cc2e98dd52b7176e62c11ebcce6f93f50427593305536a1b54b82d9
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0341: dataset=7; property=1; point=128; target=sha256:bd8890138dd9177d8ae80d7de4f26fa3f580098a06938e51493d95b8edcdb912
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0342: dataset=7; property=1; point=129; target=sha256:dfd7f962876e2c521d2665ff30f31eb1223b8f12fdd9664016c39159a385db6b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0343: dataset=7; property=1; point=130; target=sha256:74f67fab9bee0390d4dc9f09ee06b98a50a2836b890abcbf54241232e82a4463
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0344: dataset=7; property=1; point=131; target=sha256:ef10df93ac5a29bb8dddac5951ee46cf00ad4fb72e551c09b573abd0cb102e11
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0345: dataset=7; property=1; point=132; target=sha256:ad6d7927ce434ee685de8609c69871e096a83887852280722f1a42fa46473cd6
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0346: dataset=7; property=1; point=133; target=sha256:a50ec5912aaec328a7bf4ea71f1e06f8508358fb1dfaf07d62eae266cb464636

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0347: dataset=7; property=1; point=134; target=sha256:a91ea1a9d30c70d4f9ff0fccb2c7f4d3536f1eb759349966f794fda7abbf4f98
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0348: dataset=7; property=1; point=135; target=sha256:2ad7b65a31f4e43168481c249ea77da278dbfef36c92e66cd1ddf117c446217c
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0349: dataset=7; property=1; point=136; target=sha256:5eb061fcb6939118f335285725c416830cbd2e07a68f7e635be5d695b949dc57
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0350: dataset=7; property=1; point=137; target=sha256:b05c460935f9d638d30c37a5014f0585c4d9569130c71537f451d33788bb1402
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0351: dataset=7; property=1; point=138; target=sha256:ffc80baff24e1d1ff4788cc1f504265daf441e271860a0b7590404f7c804c083
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0352: dataset=7; property=1; point=139; target=sha256:388b2a4e8a79558aa617d34407224bcb0202f6fb8756ed8608aa58620ce797a5
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0353: dataset=7; property=1; point=140; target=sha256:b14cbdadcb03c5f11577b0396a7ebc19ba7b225103b1677bb387449c57bf5715
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0354: dataset=7; property=1; point=141; target=sha256:a564897381453c4c2858d0526a4930fb917c39786e38e0fb6322de094b47bb02
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0355: dataset=7; property=1; point=142; target=sha256:4b2fe77e11cae5cca8f83d856d9910d66e54fe454f962d3c2676014d2d6baf81
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0356: dataset=7; property=1; point=143; target=sha256:53ae5cdf4d0102669071a6039c6b06013d86dec3dd7d2dec4e7584fb1a6af615
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0357: dataset=7; property=1; point=144; target=sha256:ccb0348f72f77874351967cd947e61fca8e8ac34f6e4d1da11f724f403b61169
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0358: dataset=7; property=1; point=145; target=sha256:5cd0c647b9433c5a9f43d0501bfd0199f6e97c3379c5e4ce24077b0f9c7751ec
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0359: dataset=7; property=1; point=146; target=sha256:ccb84cefed91218efe970ef65231afbbdb22b31200da2aed2170a5524c745866
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0360: dataset=7; property=1; point=147; target=sha256:d1a0dcf8366d8c70764e9f4dc2c77775a15970648e0e3e0e1874a32ede245bf4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0361: dataset=7; property=1; point=148; target=sha256:388c88cec66408d13414c0f1ffbe1db11713ee977fd0c21988320edca66c6f3f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0362: dataset=7; property=1; point=149; target=sha256:df7dd895c81518f526acef6d1bde5634757429b21e2c9f1c8a13e2f95be59116
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0363: dataset=7; property=1; point=150; target=sha256:fc76d4a79e5a7f08a5cfd5ad56991681ea670bde07e7603fa6ec6374118f36d1
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0364: dataset=7; property=1; point=151; target=sha256:348042f554adc52e43f45511bcf6c143d5f89ae15a3c4178f3495bb32efa56e8
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0365: dataset=7; property=1; point=152; target=sha256:3560fd166abe4adc22330c5e51ee532dd9bf2e0ea83ed6ad4393a0433ef86f8c
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0366: dataset=7; property=1; point=153; target=sha256:3eaaaf8615228f06c89f93425478e901ed100bc3a4ed6a3f0e5f568f1c82dd2a2
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0367: dataset=7; property=1; point=154; target=sha256:431c193cd2b0e124afbcbfdb3e1f97df3a51fb24c79fd3823ed4cc6c25affe22
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0368: dataset=7; property=1; point=155; target=sha256:8533b97e8203fc2a28bfe2fec7e4c96f206aee19cca64cff4c94214d3aac65b

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0369: dataset=7; property=1; point=156; target=sha256:f1ff72e04b5040363c365014da5d999a461d22c428b1985b56e70748cc0505e9
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0370: dataset=7; property=1; point=157; target=sha256:f5cbe72849ae57c08dbd770b5d880782efe865c35c78e0e3e9bd9924230e9e56
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0371: dataset=7; property=1; point=158; target=sha256:e742ca44da5a28e9add32634eb4b234ff437576202950dc68a6c27ab4454ab39
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0372: dataset=7; property=1; point=159; target=sha256:bc194f9116c4835a50fcdc9b0a1ca155c85cda06b29f7954c6eff7c07f9692d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0373: dataset=7; property=1; point=160; target=sha256:f70ed46065dafa47281c55d9d86298d164589bf77e871392616adfd735b37c23
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0374: dataset=7; property=1; point=161; target=sha256:2dfaddce5632e1acfe0f56fc14057350c30f372b40e0cd705a1c61f3594569
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0375: dataset=7; property=1; point=162; target=sha256:ba5d8234646a749fc4a041b6618ddb490f864e5bac2201728b01538a11664d40
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0376: dataset=7; property=1; point=163; target=sha256:5a0bd7aa2576cafb2f1045f13c4adac30ee978f26a3643ac0b2647c5f0c3b0e2
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0377: dataset=7; property=1; point=164; target=sha256:8a5d06a862762ac2d938efb9ca6c86917621ef82d623f5e33aa15acbb8f11e89
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0378: dataset=7; property=1; point=165; target=sha256:b0bbcd6c1e0890731a10b52f89cf97b73634a50bb66ea79de74d54b51016d35b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0379: dataset=7; property=1; point=166; target=sha256:99de29b0661eeae3168689d95bc4f3248dcae9e9a6e5d06a3f631c1ae02f78f0
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0380: dataset=7; property=1; point=167; target=sha256:35cfcd624b4461c2b5788ae00089f9612cf9321501e92cbc99f560d01a10ca2a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0381: dataset=7; property=1; point=168; target=sha256:491b4a19b753e03a613bfb50edb8781badabb281d7207638d5e5e732b66ffd71
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0382: dataset=7; property=1; point=169; target=sha256:e9330ca44fca70eca6a202331ef56a22762a1fbbd920338803f8da5925e1ac13
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0383: dataset=7; property=1; point=170; target=sha256:0c3a55265f58d5c53a67265695d9c01ebfdfe1d3f32d26d9205594b68c6d2179
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0384: dataset=7; property=1; point=171; target=sha256:e092c47aab0f1884aaceb6a6add13dea35658e5ab4d853afb5eb64feaa6eab75
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0385: dataset=7; property=1; point=172; target=sha256:0bc7b4a75758f28a2c1c6971e0cd6a171a78fa4331845cd1ac787b2b133fba7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0386: dataset=7; property=1; point=173; target=sha256:87684287a2faa7dc1885c0c0fe89817c307df3074d173664766d6d055f606b4f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0387: dataset=7; property=1; point=174; target=sha256:19c70ea005858d48c15430ecfeba322b627378522dc66074d6f18a2e98fca84c
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0388: dataset=7; property=1; point=175; target=sha256:65e2544ca721f5e730d10b236facc998d1ed0f53a9aa6023fe002b6b245b4762
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0389: dataset=7; property=1; point=176; target=sha256:723feef14afd0b557fba833ee95481707c12d9e5a67bcc090657d31b83f5a760
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0390: dataset=7; property=1; point=177; target=sha256:4521a90729c8fbd16c68fbb41157ab6ab0286fddca38c675d1968a79d15becfa

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0391: dataset=7; property=1; point=178; target=sha256:96706473b65334a356d810e1ecc1dafa2583e3a23938b5c88661b560040cec40
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0392: dataset=7; property=1; point=179; target=sha256:d675f34a803d127cbcf551b92f5795fa8e0a0de3146c8f2919fc5c5d50b138f8
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0393: dataset=7; property=1; point=180; target=sha256:9b027cd9c381e3bce481172b87ff05219826c8a5cb31aea2cfb3e178dc222ca6
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0394: dataset=7; property=1; point=181; target=sha256:4b83271809154ae9067fa7f1dbab5ddea3eb0cefade7b0f691f2a4f33d04ebb4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0395: dataset=7; property=1; point=182; target=sha256:30953295e1bccb2323e644bfaaf11de9665ec3f78e3993917bc9b9401a70f3f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0396: dataset=7; property=1; point=183; target=sha256:057cbaf882397e0820dc711cd206f1fc5dfe7c5d347c7e0f86dea07f78483a80
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0397: dataset=7; property=1; point=184; target=sha256:4f49bfa97d818df804395f32826c2f2d867dde5ae4a9ed45ca835642c68d750f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0398: dataset=7; property=1; point=185; target=sha256:062ec15d352110e8146feb6c89114de8a2f88541b44a56134604c003537da0b3
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0399: dataset=7; property=1; point=186; target=sha256:44eb5e1abd7b9e053cb2b9ff0a73e3ec54025d90fe8cac0fce3a66260a2a3e09
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0400: dataset=7; property=1; point=187; target=sha256:65c69e0b9694e5e3296b4b588958a5f7fa632cb0d0eb7c9007089413a837bcb8
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0401: dataset=7; property=1; point=188; target=sha256:c6d9a5bfae36931a8daa64b7e47ddb11ef3a88df331e364f749bc45439577b5c
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0402: dataset=7; property=1; point=189; target=sha256:45203182f93e520645693793b95d0d555a6e9e53645c37e10014bae519b9da81
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0403: dataset=7; property=1; point=190; target=sha256:91c111d91023d726786c03eab39e7aff7d5f681594058a56ffc35bf489afd013
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0404: dataset=7; property=1; point=191; target=sha256:b9085795e93093f4d3b5f9c48e54498aa960468890107ddcf30f07cb97d9a7d4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0405: dataset=7; property=1; point=192; target=sha256:5f9fa52fd17a50bddfd1df36bb389f74fe0f89a3a0d54ce90bb4013b5d9d50dc
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0406: dataset=7; property=1; point=193; target=sha256:8cc517f7e8020103152a4580a0067ff7c61d11c7cd5e74bcdec6636a0914ad37
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0407: dataset=7; property=1; point=194; target=sha256:7484ca2b128e757cb37db78968b23e3a43b7fffb8829888ac0eb6d269d56d0e4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0408: dataset=7; property=1; point=195; target=sha256:d760daf429ff9eea911fcc38e4fdd54b802ed81a901ee4e0b4e4145e89802868
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0409: dataset=7; property=1; point=196; target=sha256:876e1f5f5f3969a804b54a2d79c24a22f05586f6533862da4eba381277080878
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0410: dataset=7; property=1; point=197; target=sha256:facd2481bdab51b293a15d394d211dd940619734558ec33ce613b8de369e2a44
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0411: dataset=7; property=1; point=198; target=sha256:88ae12d498526e16fd9d92972977cef9728f59420ad7f434f232cbaf1a95fb4b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0412: dataset=7; property=1; point=199; target=sha256:fe7eeee53a6b82460e2cd2da31892556e4e0139495ee4ca1cbe3828a4bdce7de

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0413: dataset=7; property=1; point=200; target=sha256:7417892bfb80f0206de1df81bcd7e8b44b79b3d5bcec9e0a6367524707b99161
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0414: dataset=7; property=1; point=201; target=sha256:cf80f2eea014a9e762a115f0a6ff8b64f455c54ad5fc4c6182ab1aea1572afe7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0415: dataset=7; property=1; point=202; target=sha256:4b50e6bcc3e7545c7a27e1e973c7200e5fb599b2806c8d15329a22a58f1d47a4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0416: dataset=7; property=1; point=203; target=sha256:74bcec664b20d035b70917447214c732254fa0c3932467cde7ac73801cf7919c
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0417: dataset=7; property=1; point=204; target=sha256:6dfc1bd4ce168b159e15e24540e064a3e0d6f4d14b66c7f43f6666a4e03d73a8
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0418: dataset=7; property=1; point=205; target=sha256:33439a343da517f19dbcf29eca5fba741fdb25368b26d336c5582eebe060ff83
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0419: dataset=7; property=1; point=206; target=sha256:e55bcc8598bf420aafae5d00091884d35bc17238daf9df533749b14122327edf
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0420: dataset=7; property=1; point=207; target=sha256:cc38dcdb93ee5b81eb4c553cda92e906fc04ae2892619c4fca41f0c50ce148c4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0421: dataset=7; property=1; point=208; target=sha256:6f01234c651245784744417915e6df1a4fefb476f5fe3052a3ac52c1cfa6f19d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0422: dataset=7; property=1; point=209; target=sha256:2f36fb2be667a4cbaafd114324a88dcafd29e4a662a1583804787a13e74943f9
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0423: dataset=7; property=1; point=210; target=sha256:6ad1b4dd9b4653cd0258cb19c816c7341ce79c5ef76852d9475e67ef28d3395a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0424: dataset=7; property=1; point=211; target=sha256:98743a1c7572eda3c2712fd1d989b224bdbc578fabd0655f919aed4405cbe9
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0425: dataset=7; property=1; point=212; target=sha256:af7efafd2e6cc53420ea5f963d76e5ae3789c68f28ab2bc4895265258fa91b2e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0426: dataset=7; property=1; point=213; target=sha256:1defa2ebdf1ab93c9c26c64448f020964c0ae8eb0a4e47dff3cb21dfd16a7367
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0427: dataset=7; property=1; point=214; target=sha256:e24dd32f972920f47058d7487e78fbd5fe4c723c43d1d0e81657a4ae291b4f3c
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0428: dataset=7; property=1; point=215; target=sha256:f5b81521147e5ac29088c255077e3dac8baabe20d5a1a5e942cf9c276a879fa4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0429: dataset=7; property=1; point=216; target=sha256:f4ac5c1f1d9935a47d6df63f1296dfea79ab5d3d163ae31e217300f704f2af35
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0430: dataset=7; property=1; point=217; target=sha256:ca68ba7eee98e82f95717dbfed5bd139eff37168d35f2879be8968278826cdd9
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0431: dataset=7; property=1; point=218; target=sha256:b376b785f15abe59b9012354da8e71b4849db19922c9d144bbf5507dc2b40829
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0432: dataset=7; property=1; point=219; target=sha256:8c7a08ea534048fce8f2f49567eb3d0b89180f71405d62dcfff7b22c633ba35e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0433: dataset=7; property=1; point=220; target=sha256:29444c88557e213e4bb6b787faa59ee855062132e6319d0bcd1683dcd8a154e5
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0434: dataset=7; property=1; point=221; target=sha256:0383289ae4e64e8af9c7ce7f63d98036a2fb7836ae6b4aa9fe23123c90cc3f62

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0435: dataset=7; property=1; point=222; target=sha256:3092feabf194577f1bc65e119af8cf540d85857d6d0d21601fe2a9b85f7de8eb
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0436: dataset=7; property=1; point=223; target=sha256:c4b4d1221a0ebab18272b90e65fe07779b6b307b07fadedb4ff9d096bda9bf59
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0437: dataset=7; property=1; point=224; target=sha256:ef8dcdaf00eaea532c01c57ff0cc10b027e5b9814d18522dab920b13351bea30
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0438: dataset=7; property=1; point=225; target=sha256:61288d2a311e7ba76a9476e62d8f243f4ac3493a92f20db247e6f92e8ee38b26
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0439: dataset=7; property=1; point=226; target=sha256:92fdb9d2cad4b07b2d19f8badc2e9344352027db40ddcefd06085430d97f27
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0440: dataset=7; property=1; point=227; target=sha256:1106d5412cb1332a40b399fce16b9c01241c2b568f712f28e1c79024f3099cce
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0441: dataset=7; property=1; point=228; target=sha256:266551952dd2e6e25cee1b917054da938116bfb21959c29c91dc778e88b131ae
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0442: dataset=7; property=1; point=229; target=sha256:93417d9294a5fa310875abf8cf05f79f3dc6b46524718eae5d5ac3a7e8ea4ca7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0443: dataset=7; property=1; point=230; target=sha256:7ef45433eb9b766b8bbdfb6fec7516f29d6eb54bede57f3de973dbbebb9b8d99
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0444: dataset=7; property=1; point=231; target=sha256:217abd9c4d709f31f981dd2bd1a997fabaa6d8c701b197d8abdcf22b2af3a52d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0445: dataset=7; property=1; point=232; target=sha256:1cda35870b81eeb421018781fc1b46f980ede13479c06892026317cdadf74d53
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0446: dataset=7; property=1; point=233; target=sha256:421ab3100ae34db3b3031f752eadbc4056f9efe092ff8b450bd1b461424138e2
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0447: dataset=7; property=1; point=234; target=sha256:836cffddf3842c1b5a100234b87ce543c05cd3c24ddfe21b6b857d0d01391dbb
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0448: dataset=7; property=1; point=235; target=sha256:9086fe1711c3f877a21ccc5f3ea14da41f896dbe9e90f1b52b598289d89094ef
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0449: dataset=7; property=1; point=236; target=sha256:f36c5623bd485197e6a8bebe3e27e00972ed50a3f4095c8274bbe05ed7737b6
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0450: dataset=7; property=1; point=237; target=sha256:bf9355457aac8c6b5c6855dec4c320b72aab4d1299aa3dac2bf0de23cd875124
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0451: dataset=7; property=1; point=238; target=sha256:aadf8665d96162b18f8cc9cd0fbf2f8890082e6a98ccec55779251facc37ee7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0452: dataset=7; property=1; point=239; target=sha256:c3ddf8055725be30b5aa052aafae0e0959e2d48e6d5206140b8127997b9ea3b4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0453: dataset=7; property=1; point=240; target=sha256:a9a683df8bfabe99ceb074e1af1343b736a7ffe1a5e1951d4494172c7d3828de
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0454: dataset=7; property=1; point=241; target=sha256:e5ea82f77b1e84a4e531bf1f1dba9ea7a4ae6bd9ed9ccdda4ca2f2075144f24a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0455: dataset=7; property=1; point=242; target=sha256:9e1e5ab7ecef081ca6a3843990c12ebf937a14c7a4bdc582d862873720f7c99d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0456: dataset=7; property=1; point=243; target=sha256:7f4ec13909d2250a2797365434b3a622751d22b186d17394711aab3f280a53ec

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0457: dataset=7; property=1; point=244; target=sha256:5d63288907a19f0c64efcaaa3394bddd4b76f45771b3ebf3fe383f76439880a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0458: dataset=7; property=1; point=245; target=sha256:70d78bc8aa53b947c1452603cf36f574108f4087876cc4a11aea5d931b84e667
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0459: dataset=7; property=1; point=246; target=sha256:75060164989552d53db629263f9c49365a966adb9c1af338d31c3a85633faab4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0460: dataset=7; property=1; point=247; target=sha256:18afed7cf13db7711b49a7861be43a6a0bec5c7ff9f70ac5d0ce3d44dc8e696b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0461: dataset=7; property=1; point=248; target=sha256:d7c24a525829b1295775c8aece874dc57db0cab61789546f3bb1d293e3d70b0c
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0462: dataset=7; property=1; point=249; target=sha256:5162960ac0c6748852dc1ba104deba41507b8df0adc55642fffb2e440cb0b3d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0463: dataset=7; property=1; point=250; target=sha256:889fdb3a453cd6fcec78ba9184e2968d04080625d5a2b990ca2614a03b42f9ad
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0464: dataset=7; property=1; point=251; target=sha256:23b719e83b611b427a37da235232ea6f051d530e2fed6ddd918016d0616052d5
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0465: dataset=7; property=1; point=252; target=sha256:7997d742985dd1d8e8c32f3808d00e2b5aa5829381a6decc2bc6a4c58e11f121
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0466: dataset=7; property=1; point=253; target=sha256:3df9d2a06134d25e21ce872d08122b077690766e442104933cfe389a1d5e21e7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0467: dataset=7; property=1; point=254; target=sha256:1a6d13f86d7b1e9ba641172aaee2051a5e27d17337feceb239dac667f3fba5f2
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0468: dataset=7; property=1; point=255; target=sha256:9959bed487c4972412bc3a9676ef737a1e0611d81364b5fe158d613bc9c1b06a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0469: dataset=7; property=1; point=256; target=sha256:38ef75889fbfe75177ab2a1cfa702f709294facf1fb797dfa8df73a79ddb5503
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0470: dataset=7; property=1; point=257; target=sha256:220b216bca64ad586c50852c6f135bf0a9143e5076fd17ee3f6ed274f886604e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0471: dataset=7; property=1; point=258; target=sha256:5ed728e6b666f5e31e3f7db24b8743872ea2a841a17974347dc28adc391cbff1
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0472: dataset=7; property=1; point=259; target=sha256:1fbd0a3e329950178987ea16de865a2010e7b2fddd09b7e0456e1226fc868236
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0473: dataset=1; property=1; point=1; target=sha256:2d0def7407a7fa10e97ee062fb7cfb1f69abc1244e6f91ed54eb53d2b9a77f5a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0474: dataset=1; property=1; point=2; target=sha256:859eff3c8291129f3966954c2baa2774dcabc01de8f99a77dc1188229e08e2de
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0475: dataset=1; property=1; point=3; target=sha256:2f814f9736dd46a6c11636d20e3591e97b85dcf5fe604f8a639d6521838bba27
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0476: dataset=1; property=1; point=4; target=sha256:cd7e1016d4b05186da2f75c796b7c6e66c38f14f3b183b1da6b04e04a53c8926
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0477: dataset=1; property=1; point=5; target=sha256:82a4a63df6145b2ecbab4e9f43a68120d600702662b85379d073cf091af5ddf3
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0478: dataset=1; property=1; point=6; target=sha256:9710e18d0b54a37ec092189e379d2060086b45cbd64bc6148f77980e3f5d121e

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0479: dataset=1; property=1; point=7; target=sha256:733b24dd90099e0e80c1ef08a99684eb1e26146f19f3ac4afb6fbc0d9bd8080f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0480: dataset=1; property=1; point=8; target=sha256:64a6588b097845b8857348dfd8250175f2b4af2594fadbfbbaa53c4790891eeef
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0481: dataset=1; property=1; point=9; target=sha256:4c0465fd598602970d393b692e4dcccff05378bbf48a47906d80b4976c513868f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0482: dataset=2; property=1; point=1; target=sha256:bfb579c5836249f7dac970f4760a74209ae66b8f37c0ead4b4e9d32d999cd829
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0483: dataset=2; property=1; point=2; target=sha256:a9e951e9ab82a27656757e604d1a1494239c804522f23b51ad35c11fd2feef39
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0484: dataset=2; property=1; point=3; target=sha256:e9b2a3e87f8c913eea354c2dde91691689b02b22f6dae1592337a5c6b34d41bd
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0485: dataset=2; property=1; point=4; target=sha256:4699fd291ad79108701d7e0ece1497cc50b09fae5d4d96bfe2cb9d2826197b2d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0486: dataset=2; property=1; point=5; target=sha256:5c6db8ebe3b845ce2e3b944bb1d3a772f55bf4fddacd3b01fdd0eeb093e7b939
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0487: dataset=2; property=1; point=6; target=sha256:7a4c33fec8610ee745ca6c66d980f31e98d0bcd600cd7380f4b7cef8a7cdee2e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0488: dataset=2; property=1; point=7; target=sha256:c08ccdb69d56bfd92f87306ad8e189a234426fe1af6e4c82c704390dfae84a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0489: dataset=2; property=1; point=8; target=sha256:ff1f62a644f2d789e43107296521788cc7a0fb77ea2568e7736adda04b858661
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0490: dataset=2; property=1; point=9; target=sha256:758fd3eb513a8649104fcbc5a06892a67e22f20e1f25b26320afd6dbe92060e7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0491: dataset=3; property=1; point=1; target=sha256:0e0d10a6b687d1f570d108b6f78fd42191c1da708fc0e0baad83b4c982c318d6
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0492: dataset=3; property=1; point=2; target=sha256:32c5c3a21246af5dcb291ec4acc6a456b21ec264791c4d94d8f4596592ba3d0b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0493: dataset=3; property=1; point=3; target=sha256:c9361c29fed5b0336764f0099ef5628d4b4dc2830051e5e182b35360d90399d5
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0494: dataset=3; property=1; point=4; target=sha256:de6e6eb2b47708671f910c823ca49f9d638ba9a41dd10300cdf7144d6d58764
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0495: dataset=3; property=1; point=5; target=sha256:6b979905863ef7fe3cf112295cc707d0c2beb51806a0509334e6a8c7ca4899a2
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0496: dataset=3; property=1; point=6; target=sha256:b280e26e0ad4db9489925159ade273b7a0bfa0346dd903ca47461808a73bb5c6
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0497: dataset=3; property=1; point=7; target=sha256:6cbaaf527031be920a640b9488f31cdaf44b2c14e21e0859d6e4efd24003b64c
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0498: dataset=3; property=1; point=8; target=sha256:0dce115fa84631a1a41d4c955f5102b6f7138a818acbb9d5dee77985d3026b11
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0499: dataset=3; property=1; point=9; target=sha256:b8d2e7be1783fae84b2e5c71bb5af0fc457117e9792aebd6f29ef6905c23e8ba
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0500: dataset=3; property=1; point=10; target=sha256:a9680ab7066a2bf03f5129d6d1c0dc7006ce856224b1e95e5fbbf5af932523af

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0501: dataset=4; property=1; point=1; target=sha256:713f43d7968608953ad7f7b3d59ce2feffe852b56a1e5bfe4e44e9e700838b69
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0502: dataset=4; property=1; point=2; target=sha256:69ac38c5a8486c0e2b563f8b2f058916acfa2496907740d5eb2139769921ed66
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0503: dataset=4; property=1; point=3; target=sha256:3c3f131a505db8af0ef25ecc4ac6a41f4225fafac7ceb878388e2161c7b2112b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0504: dataset=4; property=1; point=4; target=sha256:8963053745f95ff10ece2adad593bc000792d25c62d932c8d79cf431bc02a762
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0505: dataset=4; property=1; point=5; target=sha256:561a637a99d24eaf4ab0ff5c1c0ee7f0a0682dd07674b68c0f7fb85cf51c3018
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0506: dataset=4; property=1; point=6; target=sha256:d55c98ce06d7d0532c99fda2107259fcc8543b79f85a7e5ff14714d0057790a6
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0507: dataset=5; property=1; point=1; target=sha256:497767405a0c6585ca734cd9547d51209fcc963f3b58a64a770056c8fe5a6681
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0508: dataset=5; property=1; point=2; target=sha256:e41c964587d5f3335efe892e1179fd6042bdda3eed938db61991813c5cc09b59
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0509: dataset=5; property=1; point=3; target=sha256:76f92d1e9f1051e1e12e98636b75ee91bbb04485156c6a39f5a4af89bae60c17
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0510: dataset=5; property=1; point=4; target=sha256:7c7e31ffead015fe9625e3df01e0906f74240da26b4ac7705c4591fc23319326
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0511: dataset=5; property=1; point=5; target=sha256:18238602c4caba995c53a85fe41547dc346647198db90ff9de59fb7d9c917566
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0512: dataset=5; property=1; point=6; target=sha256:b760d5c240e1087b1a04a8e3eee2614801615642a6caea62c7b5df30faedcd23
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0513: dataset=6; property=1; point=1; target=sha256:42b2744623bc925965fdd18cd50038440800d43d6eec00e0b80143431d903e49
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0514: dataset=6; property=1; point=2; target=sha256:93cae0c1457fe9ee89a6fe3478e667684a2b875c8d46c8e2c60f98eb26791776
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0515: dataset=6; property=1; point=3; target=sha256:3706e9d515a29a53d8e745f6911d779ce2dcb5d36d256236d7340a75499bcd56
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0516: dataset=6; property=1; point=4; target=sha256:e700d4b0143edbe1a1b325d240c01c434790c6c6c1af67c195309f6a038a1813
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0517: dataset=6; property=1; point=5; target=sha256:893cd47becc11a0dabee1cabdb213dd38ce6e95231e9e3fbb955e4373ab319b9
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0518: dataset=6; property=1; point=6; target=sha256:079105f216948aedffac63a6ecef23febcbcc891a091bdfb55480f22a2e26d9
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0519: dataset=7; property=1; point=1; target=sha256:fc3b2095a3af30e7268e01da914bd15ac06719b8763baed65f6f7463f552bf9d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0520: dataset=7; property=1; point=2; target=sha256:7798345577046324ff0158c0d83308438f39491057493daf8a774fef9d73651a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0521: dataset=7; property=1; point=3; target=sha256:7857d7e0b173cee6421637bc05bb3cdfc3ba4d4517e5bd7585cb445bfff2fa53
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0522: dataset=7; property=1; point=4; target=sha256:c2f767b6c85feda4a628e7a7809d5a69d8fbd30eceebbb04490d92e23395560

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0523: dataset=7; property=1; point=5; target=sha256:7e3fd382c59a6174addc224feacbd5406ab507238d7a80f0526f72499a472b2
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0524: dataset=7; property=1; point=6; target=sha256:8dae4be1563533e2e751247f739fc866760fb1a20f2518cd60f967b88603adcb
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0525: dataset=8; property=1; point=1; target=sha256:a92e1f4fffd842b981c01d58935c8632a52b8677259648313766e222a8a6cd0e4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0526: dataset=8; property=1; point=2; target=sha256:cb07cc8a63fb987a4183a63fc9ce0d33dfecf69276db670410c9f2386dcca6bd
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0527: dataset=8; property=1; point=3; target=sha256:0c77157df3cfb0da1d1b1162ba63bab34e42865fe2a132e8161261a4b18fd7b8
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0528: dataset=8; property=1; point=4; target=sha256:4eca0c40f5068875364c40f241415e20fd93a4b6cb86353316f9d6fea663b106
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0529: dataset=8; property=1; point=5; target=sha256:29be0bf47936e824254bd5a5d9411a9699da0fc60a8e703dd431b1f8c5e443fb
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0530: dataset=8; property=1; point=6; target=sha256:1c497c51130857fc6fed243877ebdad5dde6a9d745143099b1c727e045895cd9
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0531: dataset=9; property=1; point=1; target=sha256:12dbf20d5cb81ff1515cc76225ebe57114bc19546be9f630ed8eddac645f437c
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0532: dataset=9; property=1; point=2; target=sha256:79538ed4aa0dc3f7bfb7ce17593e21d1e2d26f89ca5b3f5fd88f28646ac30790
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0533: dataset=9; property=1; point=3; target=sha256:e9d6da868778b6ecca60ff0ec204ab440d5c94367631416de8460e7ca1ad187f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0534: dataset=9; property=1; point=4; target=sha256:f6bbc5b73d4287458675a59cecd86ad5cc47e0e68f385c8c84523ac633faec1a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0535: dataset=9; property=1; point=5; target=sha256:cf978ce9c7c5c41ddf4dafb89d222039ed74f15c45fe0e61b72dc311f790f2ff
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0536: dataset=9; property=1; point=6; target=sha256:21bae234dde0be7a1821fc4a611b3b29d86726bd319e8298e357ac32beefb82
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0537: dataset=10; property=1; point=1; target=sha256:4a719e0270fdf2e452b6a56b59228ba65bb52d662dde6b97d3d6fb0d917cae99
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0538: dataset=10; property=1; point=2; target=sha256:4540ef31274074a643de733e2109da65ec397aef1d1be277dbe904d80f512d3e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0539: dataset=10; property=1; point=3; target=sha256:a6c4c7766a35adad5249d25c5717104846c672ed4aab23051bab675a1636836f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0540: dataset=10; property=1; point=4; target=sha256:0689086c683d724896a7e3245f9424231cb880a711d3372a1cf71f96a24c1ea5
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0541: dataset=10; property=1; point=5; target=sha256:61d3302b10f7428b64136c09c4da9d400774a1ea0ab90f92a2b4e1f910937f16
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0542: dataset=10; property=1; point=6; target=sha256:2f1dd8940da5efb7bb731ca5e4f72d9fdaa126d4ac6297a3f60a685352fd44fe
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0543: dataset=11; property=1; point=1; target=sha256:19c6d91539628d949870c72305e15f4a2cb47aaff9d8658fc0bb62b032b4c4c6
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0544: dataset=11; property=1; point=2; target=sha256:e0089ae7f43c498fc689c641741f53cdb47e45133f33f898370e66a0bff8bbae

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0545: dataset=11; property=1; point=3; target=sha256:36c56141ef5a5b4984b4089188ccf01fbc90b777a9cafaba1773fb54ce761656
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0546: dataset=11; property=1; point=4; target=sha256:f8667d8aa49d4b632dc23d16bbf72081d425cb6feaeda764ce33d84d6318d884
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0547: dataset=11; property=1; point=5; target=sha256:7f950ef7d6825fa89dc147dd908337c4e93ca01bd0fc56ab83a7be800b16706d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0548: dataset=11; property=1; point=6; target=sha256:c568a3324247c1c66a68ea42dd3537b06b420b8505b59403f14a1e00343d2ddb
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0549: dataset=12; property=1; point=1; target=sha256:fb96aee19ff6537346f1d2678cc03d8b34bb92c404f25ebf1cc1fea27726a3b0
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0550: dataset=12; property=1; point=2; target=sha256:07e70b64fe462569ead296fcdf967be864886f3ad01848d0fc971a28d239540
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0551: dataset=12; property=1; point=3; target=sha256:40339b9a5357fedffa97968e7cc7b4592cd9bc7939eb10468a5a76e26450dd01
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0552: dataset=12; property=1; point=4; target=sha256:492443346d2b6bfb81b792bbe3ffbe780ed7fe7ce18e7a66b77d9c3177eef14
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0553: dataset=12; property=1; point=5; target=sha256:942400664a7a04631e4221d7fdf2f330f65600d3c1060dcf88467c0ea08290cc
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0554: dataset=12; property=1; point=6; target=sha256:6dfd4c8677e5fd89a68e4a0fc5fac59c14d140c016e97d334a813981546f0b0c
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0555: dataset=13; property=1; point=1; target=sha256:86f850eac1b5b1d8cf6991a78fdd95cc25e6868c957b2cd056f8bd39286a35c7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0556: dataset=13; property=1; point=2; target=sha256:2d837cb77adda02e4358d2fcdb08908462ec62b8265c86040719655e75c77ace
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0557: dataset=13; property=1; point=3; target=sha256:4bc0b13b98095a261500654ff71a0aba3886671465e5fba5afc81019e3a0fbef
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0558: dataset=13; property=1; point=4; target=sha256:8f14d60347240ef89f1918fff0ee427cb040fa58c8cd47905952005e4b97b2f1
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0559: dataset=13; property=1; point=5; target=sha256:9beba7b73f01a66220e64005a136c09f8761e0c6c903135e8f183a144e5327f7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0560: dataset=13; property=1; point=6; target=sha256:70627a9f018895b35f70464464cc9085d1077e4b05c538a1553d9be32d551067
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0561: dataset=14; property=1; point=1; target=sha256:99c59a0dea05f1bd1a454c5aad9a1bf3e83cfa728e120caf0ead91927d3d91
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0562: dataset=14; property=1; point=2; target=sha256:28d4117955b8de256188eb716b34ce2998ce05fa755c78c8fcc940aacb2993f0
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0563: dataset=14; property=1; point=3; target=sha256:605fc0713743f13e065aabbaf5654d98c27d77af25419c43477da2eed404155d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0564: dataset=14; property=1; point=4; target=sha256:5684599880d5fa3786cfa1be0e27fd9a356794f5217cd6efbaf43ab89a2b0943
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0565: dataset=14; property=1; point=5; target=sha256:a74b314a35f76c0a2746d18004003cbf898025f19e3692c15069b1c48371419
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0566: dataset=14; property=1; point=6; target=sha256:1a1bd4eb52a96b4d5c688084cc5ce7fea836528db9fb1350137860c03700ff5c

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0567: dataset=15; property=1; point=1; target=sha256:7aca769a8b7041fa6bcf6d77452fb68f9fdab4c0b9c808cf82b1534578f258f2
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0568: dataset=15; property=1; point=2; target=sha256:84312bd68b8ac3815b3ec7c387cf769081104188e6d681772e5e3957eaf36b39
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0569: dataset=15; property=1; point=3; target=sha256:6833a45bfa6905081eb9da61d1afc98b8079ee07732e1460bd4030e59f3ff52a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0570: dataset=15; property=1; point=4; target=sha256:ecf3a05a16d3963073a65689ff0409d0d56d064bda1a2c133e307c034120fe59
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0571: dataset=15; property=1; point=5; target=sha256:4e62479bbb3edfbfbf867147dd4c58bbc4f9b5bdc9045bc399c06fb1e86b6eba
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0572: dataset=15; property=1; point=6; target=sha256:e25f58fa37ad988cf6faa0d8e1d75a6a1fe9699da979565c2fc81bb31b0a3b59
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0573: dataset=16; property=1; point=1; target=sha256:eff9fb29d2fdf8817732fa6ba764a1fd73bf28b5d2329840a139e00a8911a2c1
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0574: dataset=16; property=1; point=2; target=sha256:ce5e2f3357551d7455958cecc2e9c4f8c508c23a6c82478a8e9647611a511970
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0575: dataset=16; property=1; point=3; target=sha256:bec1880a10a72834eff3dbce76fd2bb200f2da61690655b125bc355f3283a33b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0576: dataset=16; property=1; point=4; target=sha256:5e3adc5f38765602c56353bc6b23be32655c2260ee69b8f477c2b1695b6628b3
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0577: dataset=16; property=1; point=5; target=sha256:ac1d13b241e56d52f0ab5af51aacc170d38720d55a069f249216fe63967fcea8
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0578: dataset=16; property=1; point=6; target=sha256:df838ea75071be347ca4a905b71199d3543955c16e982c4203fba7511ed87f2d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0579: dataset=17; property=1; point=1; target=sha256:b02fb5e16bf85d97f2a7cd8a76444882dc71c34b69495d428c7ba918015dd36e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0580: dataset=17; property=1; point=2; target=sha256:35d86c4e694f9a881cec5a89655c2925cb4990e2cb40b102810e4c283b0f5920
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0581: dataset=17; property=1; point=3; target=sha256:59b5b0a546b58b6507ad068ba119ef43b32ce3a809c6d1b1331f8b5c7a1a182e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0582: dataset=17; property=1; point=4; target=sha256:49802a7f1bde4dfdd6addc515d47674d330b71e29153a915ede157bc667adf95
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0583: dataset=17; property=1; point=5; target=sha256:73a1c4fec1e32727d63779b8701359555d265c09fb546db7b24802a89db43ead
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0584: dataset=17; property=1; point=6; target=sha256:34ab295849de47afa698e8992ddbdeec9d4a7c76d7f0c15e21756b87cf20cf5a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0585: dataset=1; property=1; point=1; target=sha256:16355164a1f3805daa8c0c252fbd3b836768d674b8d863e50449a89b3aed51c2
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0586: dataset=1; property=1; point=2; target=sha256:973b32e65e8e40733074e0fa04143fc7a6b43de9e645d523d03a6f13f90b9ebb
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0587: dataset=1; property=1; point=3; target=sha256:b79f2845b33798d638a02a68ec745016af7b0ed25caefbcd3af99dacac9f9255
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0588: dataset=1; property=1; point=4; target=sha256:76502185c654347ccb2a4f6c5dd0408e7544a6df7c1a72e528f37af3c9005290

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0589: dataset=1; property=1; point=5; target=sha256:cf797a2ae18a9faa5fb2a34ba76c65c81c899772c531e2a07b57cdd0e26b4065
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0590: dataset=1; property=1; point=6; target=sha256:c4114b43aa6b3baf0442a43cab42c9c793399c58fd20dc624f7671a6ad23feca
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0591: dataset=1; property=1; point=7; target=sha256:1938b2f3ca8f57a4d94de04c250bcbde9f8003946e797af5caf9635e5f8ad31d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0592: dataset=1; property=1; point=8; target=sha256:a6b20334b70f70c9373f5a659887dd361edc3b0b807ba836ca715b80d1be9c08
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0593: dataset=6; property=1; point=1; target=sha256:e42541648af834e90651eb9434479e39bb868f266bc3c50215e42bdb49a7a6ea
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0594: dataset=6; property=1; point=2; target=sha256:ddc84d9fc9c0b85cf9ffc3c48d6b273f36bcbe9f2994ef1b24714edc8faf132f7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0595: dataset=6; property=1; point=3; target=sha256:a7100e23a1835c85915761f35e84bafddb2b69809c83fa41224ec7e4ec21cbab
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0596: dataset=6; property=1; point=4; target=sha256:1f936f15123c33341b6d703699daf35c93dfe104d775b8d90579626da7c87adc
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0597: dataset=6; property=1; point=5; target=sha256:dc5fe3b1eec81c516f45c6ed1cab966616e6401817959b6a81f414a26ac0b3a7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0598: dataset=6; property=1; point=6; target=sha256:03cf31811f42643a90f0e35640d5f10c6864214b1ff8f446dc5bb578eb1168cd
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0599: dataset=6; property=1; point=7; target=sha256:fe16e5dc97ce086adb54aad67be931298e88eb77ed5bfe4f2e3e7348d27e6b29
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0600: dataset=6; property=1; point=8; target=sha256:ef6f2cc0cb1b93e9395099af567e6085bc092ab119d3720f5b241b7328535c3e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0601: dataset=7; property=1; point=1; target=sha256:83903b76cc944eff2fb2ad7f098b5a408a40910da594ea0e390fb436718a049f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0602: dataset=7; property=1; point=2; target=sha256:ff8885c5b743f2f62cffff36c53b46fc50c911e3aaa9d55c6821d75bd58ea98e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0603: dataset=7; property=1; point=3; target=sha256:f5058c90091ce17e4e11f9b5d50bf6a1e507823e1c4b3bf7212b67fcc3a180d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0604: dataset=12; property=1; point=1; target=sha256:07a81c0cefe485425d8bc33067107611428a7529b2b74ac842e766144c3fd915
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0605: dataset=12; property=1; point=2; target=sha256:47e7e6bca03a0396c57f6a6f60ec3f5544036cef49a09b1b40e4b8694b268dc1
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0606: dataset=12; property=1; point=3; target=sha256:3cb0c06f74c9495d5d8987198649c7867c2b42a033b43b019bc9fe6a891d2a3a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0607: dataset=12; property=1; point=4; target=sha256:929874ea506e4e4182dfd8d3334c6aa327b21e630720f4c43a64fa31dbb26df9
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0608: dataset=12; property=1; point=5; target=sha256:fe556860bdb183cfb4795be6b5567c3473f062436c2960584cf178d04c67f86d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0609: dataset=12; property=1; point=6; target=sha256:7bf8f1be815d2db5eca7e3ae858a1fe6c8b3239b704900256728445f6977d36f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0610: dataset=12; property=1; point=7; target=sha256:8026948510b8d0c6e1994ba6b5320ee4db513b502df59b018516b4aeb10e3f4b

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0611: dataset=13; property=1; point=1; target=sha256:ee025a8c1a27e1556895a8c1f91a35ebf8ded5146e18a9dd19d815d406fef12b
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0612: dataset=13; property=1; point=2; target=sha256:31ffa73add88f6cff3c24b8532c897115d33cc342c6388d8b6a0255beeab7bd
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0613: dataset=13; property=1; point=3; target=sha256:01503f35b1ecc8feaf24c5fab88ff02453b4ef42e526eccc7cd6dd3191637a74
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0614: dataset=13; property=1; point=4; target=sha256:a32526a6351922d48feca77f17dff0aaa4c163ce6266312a471be5a746250694
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0615: dataset=18; property=1; point=1; target=sha256:43df4e9df603bc36c3117da83d87cc05bd7b6569ec0ebf5bcf0358f2ac50ab9f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0616: dataset=18; property=1; point=2; target=sha256:c004a1563880d57562f26d7406f895ece1b5b4aa61718e2c746844a23cf3f01f
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0617: dataset=18; property=1; point=3; target=sha256:b8274eaeef9ebd665e2db2adafb6f9a971fd2e5e051bc83bede0f82dabc82c2a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0618: dataset=18; property=1; point=4; target=sha256:d5936a59a04518ca3ae9688fbffa403fa877dcc35b19aadfec13581f58f40be
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0619: dataset=18; property=1; point=5; target=sha256:1088d4b63c6b771f6984eeea8e90e1d1f2ca770458714499d8e91435f0a5a65e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0620: dataset=19; property=1; point=1; target=sha256:f4227367bf9a1757a64ece7047143a3f30ab9191d37ca9a0b4f7e9d6d55bdc06
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0621: dataset=19; property=1; point=2; target=sha256:c174b4ee24c7d0f8f11657543320b03fd4f5d46d0dc989a07c8032ff4ef9c5df
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0622: dataset=19; property=1; point=3; target=sha256:9070431a61f95047683a1a7a8fcee382e1161d51a5a0997fa87c810ae861abff
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0623: dataset=19; property=1; point=4; target=sha256:0b5ebadc613ca213e8e4f0beca5da55c048bb6fb3b117f6a82f19a1c59359003
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0624: dataset=19; property=1; point=5; target=sha256:07fea602c5c2ae8915c33c6a1a536426b1eaf55274b68e7cbdf5c681565dcb
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0625: dataset=24; property=1; point=1; target=sha256:bee33c57574131f8e34bfd6ce67dd9a295b66ee21834e71136fe65235e56ea37
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0626: dataset=24; property=1; point=2; target=sha256:2cb417d3c1e96508df9f714e0074fbd563f5ac1a82944002fb9461d1f61bfe2a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0627: dataset=24; property=1; point=3; target=sha256:c1ee918de9f147bc49a0409f956ff047857ed5e0501d0976ee642c8cabf9e606
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0628: dataset=24; property=1; point=4; target=sha256:c3a6f07dfb39062fbaaa0106eb1bf3e22543e7a88868c42c62bf4cd48a45faf8
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0629: dataset=25; property=1; point=1; target=sha256:8eb091685a66225e6dc60b38eca51841985329d30bfebbbcc5576275b7e26444
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0630: dataset=25; property=1; point=2; target=sha256:7d7518ea245a431497d1d7fd1f6f0871339b7cbfe871b8913d2e3314932f4af7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0631: dataset=25; property=1; point=3; target=sha256:a032e79312c5217ab18307a4b56b606033e4f871cf5bd8d2af1275a4762a469e
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0632: dataset=25; property=1; point=4; target=sha256:8b0294eeb52a2362c092d769ce1e105e49db8519b905290a69aed9bb59ab469e

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0633: dataset=25; property=1; point=5; target=sha256:27b555815f4e1b8bd67f140d7ab8ac59c992245bd9ff5b82cf9522061eb5d7b0
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0634: dataset=25; property=1; point=6; target=sha256:903451c271b739037df4bd57ad2fde6e4c1ea01eb00d823bf871bceec827e2642
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0635: dataset=30; property=1; point=1; target=sha256:7e0684dc67fbf6b31cd59945e5f95001de577d12948f9cabdded39e78d99570b6
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0636: dataset=30; property=1; point=2; target=sha256:0b70440efb99999e68793e8cb9eee293f3e589f9a36dce892293186aeec4031a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0637: dataset=30; property=1; point=3; target=sha256:221e28d57f5d938bb43266df097710b1c7d5cac1526cfcc304ddaa77fbd5ccae
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0638: dataset=31; property=1; point=1; target=sha256:30c9e5ed76af30a9ac5f396427cee4333800741d9c32387fa94e91070d15020a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0639: dataset=31; property=1; point=2; target=sha256:927f2aebcf399380b4e88828d3be1f3f740bd814b7f753638b786f63b8aec8d5
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0640: dataset=31; property=1; point=3; target=sha256:fc6bcd0fab8dcc1059a130fc36deb433167b8da3fb0643a4dbb6c400f8facd2
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0641: dataset=31; property=1; point=4; target=sha256:145d230b3401a20361e01a59405db6aef3cb523766cb3e6827d639c362c4ec78
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0642: dataset=31; property=1; point=5; target=sha256:54eed520d8a53772c4474c5079b0f990efe4fd45450861fda99927472c2a378
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0643: dataset=31; property=1; point=6; target=sha256:38c98f4e2fc8f6883f694cf0814ecb506e3aaa3bdca258fe3447dcd3a6e2e6d4
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0644: dataset=31; property=1; point=7; target=sha256:a92006960d7167133e5414839f0aaaaf6ad9e94aca212137487b2f72c2689444
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0645: dataset=36; property=1; point=1; target=sha256:cb6559bf6b33cc13399037150917568507942eafd26edde20571cb6ded8c4682
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0646: dataset=36; property=1; point=2; target=sha256:d20b5b9dd1cb71f53633a64b038b913311a4ed04cf2f5297aa2959153692a145
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0647: dataset=36; property=1; point=3; target=sha256:abd333dad2576e5351cc3dcb288fb42783e294dadf9a6a099f6db568c48c5364
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0648: dataset=37; property=1; point=1; target=sha256:d7e030cf05564da97b6b1a06ea039285264e441e792c972da8b070c68ceb4348
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0649: dataset=37; property=1; point=2; target=sha256:a669c7658c2db8d374e7386d7e1b587874328f00c037be360fa12bf3fde27c94
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0650: dataset=37; property=1; point=3; target=sha256:1fe99d28333fa29c6c8867ae89bfc599854584837fe2bc581844908e9e8fbd6d
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0651: dataset=37; property=1; point=4; target=sha256:863d9a16dbdcde3f2374535543ca6fd991c83fb799e91127b719cbc01d24e69a
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0652: dataset=37; property=1; point=5; target=sha256:a4fa8f390e051712a7be04577bc40bca6a8e1f99406b257ad96ff51101dbdbd7
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0653: dataset=37; property=1; point=6; target=sha256:7868bbe7e1aaf56b2dcc228d9460cf38e722fc032a607c5cbad1f1dc5992a270
- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0654: dataset=37; property=1; point=7; target=sha256:aa84ecc620805336150b978851557b1e38301fc82d620910f49c8a57398a843b

- SFT-CHEM-THERMO-018-THERMAL-CONDUCTIVITY-0655: dataset=37; property=1; point=8; target=sha256:a2f97b4ad7309199cab0a14e0f1af9ccd5bbc0c7c6fdbed91dab43112c7711e0
- complete vector: 123 pure, 273 binary and 259 ternary thermal-conductivity records
- phase vector: 51 gas, 571 liquid and 33 crystalline records
- exact conductivity range: {'minimum': '1759/100000', 'maximum': '167/250'}
- source coverage: all 61 datasets and 679 points preserved from three complete NIST ThermoML sources
- tampered_negative_thermal_conductivity_rejected: True
- complete_655_record_vector_retained: True
- all_pure_binary_ternary_rows_retained: True
- all_gas_liquid_crystal_rows_retained: True
- all_datasets_and_methods_retained: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if any component, phase, condition, cell, thermal order, energy packet, transfer, tick or boundary is erased; if adjacency or energy conservation fails; if numerical-zero or signed proof value, continuum gradient, Fourier/kinetic/mixing/temperature fit, logarithm, interpolation, regression, selection or target correction enters; if target content opens before all 655 identities seal; if any direct conductivity row, uncertainty, method, dataset, complete source or companion provenance is omitted; or if any target is tampered.

Measurement receipt: sha256:87fe6cc9b8d2cd0e4c2e048ea89cde43ec8d92c2dc4837d9d0c091a67f8f3a31.
Isolation certificate: sha256:074270eeac558cae84510d10dc61a9a122f4167cd0de9c399015237b68ed16cd.
Custody certificate: sha256:0a9b2d13439a5f0ed2b119ffba55826301ba28cd3172c17990ff39d43f1bb07f.

Registered source descriptions:

- National Institute of Standards and Technology ThermoML Archive: ()

Registered target identities:

- None.

Meaning. The thermodynamic extension derives finite microstate support, temperature, energy, heat/work, entropy, enthalpy, free direction, chemical potential, activity, fugacity, phases, colligative response, solvation and coupled transport while keeping external scalar inscriptions downstream. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; absent external condition coordinates are structural EmptyOne
- no negative, irrational, imaginary, logarithmic, floating, signed or continuum SFT proof value
- no imported Fourier constitutive equation, continuum temperature gradient, kinetic-theory formula, fitted mixing or temperature law
- no interpolation, regression, selected substance/mixture/phase/condition/method/dataset/property/row or target correction
- no substance, composition, phase, temperature, pressure, method, value, uncertainty or target hash before prediction seal
- every complete source and non-conductivity companion remains preserved; companions never become conductivity measurements

Immutable identities. Source manifest

sha256:5a930b3abb017913d3513fab4dbfddb1948a41635595834c2dde394cf1455575; derivation seal
sha256:f104154ac56a9fba1725acc07fe0eac0b8da1f91f4f18c216597f4b167c731f6; engine receipt
sha256:08666e0eaf14fe6fcf6a532601c27e29c5aa331301a04b6783d05d0685b2c66f at receipts/engine/model_admitted/SFT-CHEM-THERMAL-CONDUCTIVITY-RELATION-018-08666e0eaf14fe6f.json;
empirical hash sha256:345801aa0f1dc6a16c8b2b73344bb1bb07a03dda20fc4523c9be87e4552e9c99;

measurement receipt sha256 : 87fe6cc9b8d2cd0e4c2e048ea89cde43ec8d92c2dc4837d9d0c091a67f8f3a31.

143. Exact coupled mass, heat and charge transport law

Claim identity: SFT-CHEM-COUPLED-MASS-HEAT-CHARGE-TRANSPORT-019

Question and theorem. Coupled chemical transport is one counted adjacent-cell transition ledger carrying distinct mass, heat and charge packets with exact positive resources and held per-carrier orientations. Every pairwise observation is a projection of the complete triad, not an imported Onsager matrix or continuum flux equation. Chemistry owns composition, phase, condition and cross-effect provenance; physical energy and charge carriers remain explicit dependencies.

*complete-composition-phase-mass-heat-charge-account__distinct-held-triad-and-pairwise-projections__counted-share
d-adjacent-cell-transition-ledger__held-per-carrier-transfer-orientations__exact-positive-packet-event-tick-boundary-a
nd-condition-support__exact-positive-postseal-pairwise-response-support__complete-value-free-232-record-pair-identi
ty-seal__depth-independent-common-replication-and-record-append*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-CPLX-TIME-SPACE-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-THERMO-TEMPERATURE-001
- SFT-PHYS-THERMO-HEAT-WORK-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-ELECTRON-COUNT-SPIN-002
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013
- SFT-CHEM-MOLECULAR-DIFFUSION-RELATION-016
- SFT-CHEM-VISCOUS-TRANSPORT-RELATION-017
- SFT-CHEM-THERMAL-CONDUCTIVITY-RELATION-018

Generated grammar. Generate the literal product of carrier, identity, transition, orientation, resource, magnitude, prediction and extension forms; decide all 256 candidates only from admitted mass, energy, charge, adjacency, transition, conservation,

exact-resource, composition, phase, EmptyOne and finite-successor laws.

Boundary: Every finite coupled chemical event with complete component and phase identities, the ordered mass/heat/charge triad, counted adjacent cells, per-carrier packets, shared transitions, ticks, boundary and conditions. External testing preserves 232 direct pairwise response rows and all companion properties from three complete NIST sources.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: complete-composition-phase-mass-heat-charge-account__distinct-held-triad-and-pairwise-projections__counted-shared-adjacent-cell-transition-ledger__held-per-carrier-transfer-orientations__exact-positive-packet-event-tick-boundary-and-condition-support__exact-positive-postseal-pairwise-response-support__complete-value-free-232-record-pair-identity-seal__depth-independent-common-replication-and-record-append. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-composition-phase-mass-heat-charge-account	detached-cross-coefficient-or-answer-vector	Detached coefficients erase the transported carriers.
identity	distinct-held-triad-and-pairwise-projections	anonymous-collapsed-carriers-or-pair	A collapsed carrier or pair cannot reconstruct the coupling.
transition	counted-shared-adjacent-cell-transition-ledger	imported-Onsager-matrix-gradient-or-flux-equation	A phenomenological matrix or continuum gradient imports the desired coupling.
orientation	held-per-carrier-transfer-orientations	signed-flux-proof-magnitudes	Signed mass, heat or charge magnitudes violate exact proof arithmetic.
resource	exact-positive-packet-event-tick-boundary-and-condition-support	unrecorded-packet-event-time-boundary-or-condition	Unrecorded resources cannot reconstruct the coupled response.
magnitude	exact-positive-postseal-pairwise-response-support	phenomenological-cross-coefficient-fit-or-target-value	A fitted cross coefficient selects the result.
prediction	complete-value-free-232-record-pair-identity-seal	substance-pair-property-condition-method-or-value-readable-before-seal	Readable pairwise targets could select the coupling law.
extension	depth-independent-common-replication-and-record-append	refit-after-event-replication-or-record-append	Refitting destroys shared-event provenance.

Adverse controls. All mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256 : 3399ec9910f453c0b7db9b005ef3555368247e69dd8b02215892fa42b99ef7d7.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 136edd4d5663fa87791e5ff5eaf6ae96ce63195ffe5d427641e2ae6b0b126a0b.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256 : 50ce5513251e071e3968b2d174a03921cb91dea97250ea8c4a05e0dfacc4a9ca.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256 : 2e3b725f3164e51632cbd64ee62ac88f3dcc6ff3be6c2544bfff13a4d6869572.

Independent reconstruction. Implementation hash

sha256:d8d20511c728466d61e7dac463b04ade10f989f12262f6a4aab497a5b541bfd9; independent certificate sha256:87d9652e69abd39c49462fd4bb34163cf24abf335805f9da46c2200c486c9bd9; external-validation hash sha256:fe8e1439f2421109e8afd5bac066f73a6f2f3428079cc81ff5caf757a260c887. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- NIST-TRC-THERMOML-JCED-2015-60-3621-3630
- NIST-TRC-THERMOML-JCED-2006-51-680-685
- NIST-TRC-THERMOML-JCED-2014-59-757-763

Comparison records:

- SFT-CHEM-THERMO-019-COUPLED-0001: pair=mass-heat; dataset=2; property=1; point=1; target=sha256:126509111036b5ffcb99240f6da0adfad7e78c4bc2d9ad2b9597d6efd953592
- SFT-CHEM-THERMO-019-COUPLED-0002: pair=mass-heat; dataset=2; property=1; point=2; target=sha256:281c639940d09b6602cce1cda5404a35dedfec3d1d28cad0d4b77c11732e3b7
- SFT-CHEM-THERMO-019-COUPLED-0003: pair=mass-heat; dataset=2; property=1; point=3; target=sha256:f4d3e5fa28fa3ea6e8486ceb259fdd77e7bd0d16a543077faff9f4b20dac508
- SFT-CHEM-THERMO-019-COUPLED-0004: pair=mass-heat; dataset=3; property=1; point=1; target=sha256:6efd316637d771423eb3dc3b86949d1d4db4bc4b99fb2f4a030dd22d1db284f8
- SFT-CHEM-THERMO-019-COUPLED-0005: pair=mass-heat; dataset=4; property=1; point=1; target=sha256:171e85bdd1cac4d1f775030c234a8542af4e9aee6bf83e2e7397064b706a8e09
- SFT-CHEM-THERMO-019-COUPLED-0006: pair=mass-heat; dataset=5; property=1; point=1; target=sha256:094dbd337dbae5a4ac323d475f9f41e5190b84229e7137405449825dce6eeb12
- SFT-CHEM-THERMO-019-COUPLED-0007: pair=mass-heat; dataset=7; property=1; point=1; target=sha256:3c512c2ce5cee2194b590880c4ca691b088a4141a2cd2e8d56f3c26f8074d74e
- SFT-CHEM-THERMO-019-COUPLED-0008: pair=mass-heat; dataset=7; property=1; point=2; target=sha256:db8380669dbdd5881b83ecdffd4d02adce8b12f9af7f1f9ea2b16b49d8a75911
- SFT-CHEM-THERMO-019-COUPLED-0009: pair=mass-heat; dataset=7; property=1; point=3; target=sha256:a853851e18b6d9b8cc875f7297ec606f7b0cebc99b172bd3aaf619d0a9fe0f4e
- SFT-CHEM-THERMO-019-COUPLED-0010: pair=mass-heat; dataset=7; property=1; point=4; target=sha256:2b7eefd4f756f7e3f7c31ea77087e2e3c169f5d24804b604fdc319a15a5a09b1
- SFT-CHEM-THERMO-019-COUPLED-0011: pair=mass-heat; dataset=7; property=1; point=5; target=sha256:0bfb9c840880114f6169cf97f8548c2db3234e9d386693d2b5211a3960b73bb2
- SFT-CHEM-THERMO-019-COUPLED-0012: pair=mass-heat; dataset=7; property=1; point=6; target=sha256:a419b569f3ed6049155697288287949936726e2a29d243a960d78a7914402119
- SFT-CHEM-THERMO-019-COUPLED-0013: pair=mass-heat; dataset=7; property=1; point=7; target=sha256:f5e5ad1db511fc12f44139ebbf750cd1950ede9f380701e5adb71635c57a53f
- SFT-CHEM-THERMO-019-COUPLED-0014: pair=mass-heat; dataset=7; property=1; point=8; target=sha256:272d8cb901436842257ee68e86af2c853fd890cf250154628121b24355ac18b4
- SFT-CHEM-THERMO-019-COUPLED-0015: pair=mass-heat; dataset=7; property=1; point=9; target=sha256:bb2e6b7f72d3a3abfb5e2df2ac8a34089e1080813b01bb78d3d7fa65aa07e442
- SFT-CHEM-THERMO-019-COUPLED-0016: pair=mass-heat; dataset=7; property=1; point=10; target=sha256:156d39ccd3e854208f771833bb6372ef4a74e6c6d811f38329b0ef5bcea703b0
- SFT-CHEM-THERMO-019-COUPLED-0017: pair=mass-heat; dataset=7; property=1; point=11; target=sha256:0367def1b1464ad50ddd467cf8fe49a677be6c88fd34ffd25a550e2d51e47971

- SFT-CHEM-THERMO-019-COUPLED-0018: pair=mass-heat; dataset=7; property=1; point=12; target=sha256:6b461f6314ae814515aac20a49cb3ca3d98a058dd8eb6d0d523ae00f786a420b
- SFT-CHEM-THERMO-019-COUPLED-0019: pair=mass-heat; dataset=7; property=1; point=13; target=sha256:de4aee25e2d9007a78899317abccd09c201645517dca3ae88e03b3eba4374aa9
- SFT-CHEM-THERMO-019-COUPLED-0020: pair=mass-heat; dataset=7; property=1; point=14; target=sha256:9d8c22927f64eb5d0d5ff95a580621d9232753c620caf28a3b96b543a2698f9e
- SFT-CHEM-THERMO-019-COUPLED-0021: pair=mass-heat; dataset=7; property=1; point=15; target=sha256:7a8c310c634b605e1f9a90da7d9f39b3462dbf9769f8732030bd77e7ff79ba98
- SFT-CHEM-THERMO-019-COUPLED-0022: pair=mass-heat; dataset=7; property=1; point=16; target=sha256:8ccf38c60359c58aa59e03f588ae7943247021fde6e4053d0d8cb3be2b2d88ab
- SFT-CHEM-THERMO-019-COUPLED-0023: pair=mass-charge; dataset=1; property=1; point=1; target=sha256:05e505b250e3eb8cbeae4df4fa2ae524aa219bcf6acf7d38e99802c40c6e2e96
- SFT-CHEM-THERMO-019-COUPLED-0024: pair=mass-charge; dataset=1; property=1; point=2; target=sha256:865b5cd1e314050518f0bead53f69c713fce0e4fccee30d905c336705c6519f4
- SFT-CHEM-THERMO-019-COUPLED-0025: pair=mass-charge; dataset=1; property=1; point=3; target=sha256:5947514964b801a10844534ab895be8c5ae05f49bdb926bac9ff3a608520ea6c
- SFT-CHEM-THERMO-019-COUPLED-0026: pair=mass-charge; dataset=1; property=1; point=4; target=sha256:ad641eee9cc4f32b3c03b01170146e605fb739ce687a4f37c2da64acf6cfdc3f
- SFT-CHEM-THERMO-019-COUPLED-0027: pair=mass-charge; dataset=1; property=1; point=5; target=sha256:3d36587be69452516d6c934043858af8dd806af4590823616f207ae759714abf
- SFT-CHEM-THERMO-019-COUPLED-0028: pair=mass-charge; dataset=1; property=1; point=6; target=sha256:c0a0f323498f05512c534efcefd367adacf0445c99200fc84cbd122d6a14c876
- SFT-CHEM-THERMO-019-COUPLED-0029: pair=mass-charge; dataset=1; property=1; point=7; target=sha256:898ee88cfd5013c735ed976cfd74ddae1416e8e8663c79e4b87077ff169b2f5
- SFT-CHEM-THERMO-019-COUPLED-0030: pair=mass-charge; dataset=1; property=1; point=8; target=sha256:83e63487b210d32f35aa522daafbdb6d8f5c7512ae9f0f00ea14522a9b42541f
- SFT-CHEM-THERMO-019-COUPLED-0031: pair=mass-charge; dataset=1; property=1; point=9; target=sha256:242b2bf1720f6b43a8ac57d715c0f38b72333d158b261835ae0db18fd23aa864
- SFT-CHEM-THERMO-019-COUPLED-0032: pair=mass-charge; dataset=1; property=1; point=10; target=sha256:5ca9a737c7df7e00d7bd3a7c9e327e2ce717caabeb9a43c84c2edc44b2777496
- SFT-CHEM-THERMO-019-COUPLED-0033: pair=mass-charge; dataset=1; property=1; point=11; target=sha256:555a8e24d122874735c1adab918aa820bfa3a10ef258ee736783e2571e18ab7f
- SFT-CHEM-THERMO-019-COUPLED-0034: pair=mass-charge; dataset=1; property=1; point=12; target=sha256:e3147e16d7f5e14bbd96ee515ef8fad2d5ec7f9b270fd9869267590227a002f9
- SFT-CHEM-THERMO-019-COUPLED-0035: pair=mass-charge; dataset=1; property=1; point=13; target=sha256:6466c9c610af90d707837ef332e287df67475d0ba86b68a9cbdf33b150c98056
- SFT-CHEM-THERMO-019-COUPLED-0036: pair=mass-charge; dataset=1; property=1; point=14; target=sha256:c98f579f9ea0b1523cf5aa5f9e87202c5ed02e8020df7d90069d24826c394904
- SFT-CHEM-THERMO-019-COUPLED-0037: pair=mass-charge; dataset=1; property=1; point=15; target=sha256:10d3c687cae73b8cff47605a5e14ece4d919b6ee782f9ef8cb3ad87c194a753e
- SFT-CHEM-THERMO-019-COUPLED-0038: pair=mass-charge; dataset=1; property=1; point=16; target=sha256:3c65e1f4c24200d6ffaadb284c4823db7c8c1d6584ebdab8613ae2869b165ba
- SFT-CHEM-THERMO-019-COUPLED-0039: pair=mass-charge; dataset=1; property=1; point=17; target=sha256:0d2d3bf38a522d7d8b0589880cd6651e48c96b392bd0e6b504598b0799bfff63

- SFT-CHEM-THERMO-019-COUPLED-0040: pair=mass-charge; dataset=1; property=1; point=18; target=sha256:3d0b84deb546fb56e90dc41ddd5263682d00dc031b6bf0e6e6945269964f3e46
- SFT-CHEM-THERMO-019-COUPLED-0041: pair=mass-charge; dataset=1; property=1; point=19; target=sha256:4f836d0856b9f0af0762aabd84b6b9387048dd4c9a4960ce424b62925e66d3df
- SFT-CHEM-THERMO-019-COUPLED-0042: pair=mass-charge; dataset=3; property=1; point=1; target=sha256:faa4524b0d555960b16ba44cafbfab74de0c0fd56e25558e723ad82aaa0d7d2
- SFT-CHEM-THERMO-019-COUPLED-0043: pair=mass-charge; dataset=3; property=1; point=2; target=sha256:a0710b0cbb815c9acb087725690dc7115e4537e06588b589e9b57ae963b881ef
- SFT-CHEM-THERMO-019-COUPLED-0044: pair=mass-charge; dataset=3; property=1; point=3; target=sha256:aed46b96066657ca5aaa5ef217b7c46042ee886fa18bbe44f73777e3fbac7537
- SFT-CHEM-THERMO-019-COUPLED-0045: pair=mass-charge; dataset=3; property=1; point=4; target=sha256:5fc0a73297209ea8a7580051ad9ffe9ba861ebb4605569a817175e002c3d4f2a
- SFT-CHEM-THERMO-019-COUPLED-0046: pair=mass-charge; dataset=3; property=1; point=5; target=sha256:4a2c80a82b0904d9b62646c9180d3e33bc315e8dd04d872c4899c20cb3b02821
- SFT-CHEM-THERMO-019-COUPLED-0047: pair=mass-charge; dataset=3; property=1; point=6; target=sha256:839ae1140d51ad4353108f2e5b85c9968ca6ce793ac5c6a0102b94b9a9256592
- SFT-CHEM-THERMO-019-COUPLED-0048: pair=mass-charge; dataset=3; property=1; point=7; target=sha256:529dfbd8e16c1dbb560f254894ed225769664973f285b74ea28b59b7a5ae506e
- SFT-CHEM-THERMO-019-COUPLED-0049: pair=mass-charge; dataset=3; property=1; point=8; target=sha256:052bf988c82f684d1972ae4a891d21150658a61b161cedc43ca1650f714e1570
- SFT-CHEM-THERMO-019-COUPLED-0050: pair=mass-charge; dataset=3; property=1; point=9; target=sha256:1d4b1f798896a040037670dfe8aab176d81a58631473d74d09bd95c2101d5a2f
- SFT-CHEM-THERMO-019-COUPLED-0051: pair=mass-charge; dataset=3; property=1; point=10; target=sha256:014edf32f433aa48d4e9594d5d6cd22022278c9556896ae2abd0c712ff31febe
- SFT-CHEM-THERMO-019-COUPLED-0052: pair=mass-charge; dataset=3; property=1; point=11; target=sha256:0d0f9d310c54199b829e57ae582c4384113e58cbe5e665c89074c4a97542cb39
- SFT-CHEM-THERMO-019-COUPLED-0053: pair=mass-charge; dataset=3; property=1; point=12; target=sha256:de8ce46cb782bf0ec8f036d572bf1376a29f2519e0e23b5ba7a6607cfce5e38
- SFT-CHEM-THERMO-019-COUPLED-0054: pair=mass-charge; dataset=3; property=1; point=13; target=sha256:3d50d9d7cfaa94ca39f8ee41e862dc676fc46a24db6355c59b7ddd14b8435dc8
- SFT-CHEM-THERMO-019-COUPLED-0055: pair=mass-charge; dataset=3; property=1; point=14; target=sha256:975d6a3af352c2141139616df1bb9a023a6c4d7be927c04fb7010bff74ba0245
- SFT-CHEM-THERMO-019-COUPLED-0056: pair=mass-charge; dataset=3; property=1; point=15; target=sha256:b5e66553317e967a2bc42be356a330165891870786421db420038b2f64b06d89
- SFT-CHEM-THERMO-019-COUPLED-0057: pair=mass-charge; dataset=3; property=1; point=16; target=sha256:1561c834b5fa99d704697cb99fd7013bacdd69542f0520a7f8bcbcb6631f5334
- SFT-CHEM-THERMO-019-COUPLED-0058: pair=mass-charge; dataset=3; property=1; point=17; target=sha256:a96a0dc4faaa967687b1d9360f87f6791bcf1978a6805aebc15e568686ae4151
- SFT-CHEM-THERMO-019-COUPLED-0059: pair=mass-charge; dataset=3; property=1; point=18; target=sha256:5e9093862f2e81fdf5bc276525e4501a3f510ca525be850463a294482d16cf4d
- SFT-CHEM-THERMO-019-COUPLED-0060: pair=mass-charge; dataset=3; property=1; point=19; target=sha256:9886fddd887eac7b416e35ef0f099185e8409de1ea34c13b4600f2756dc05232
- SFT-CHEM-THERMO-019-COUPLED-0061: pair=mass-charge; dataset=5; property=1; point=1; target=sha256:cfc08f09408210d4b7706bbaab364f75a3f70b6fbcd441a7c0dc85bca5a8259

- SFT-CHEM-THERMO-019-COUPLED-0062: pair=mass-charge; dataset=5; property=1; point=2; target=sha256:fd0d58ce1ddbfe35ea24f8ef9fec0919cee473b89b4b7ddb198e65e72a22da4d
- SFT-CHEM-THERMO-019-COUPLED-0063: pair=mass-charge; dataset=5; property=1; point=3; target=sha256:32e79c537c3452d397540907a685c3a345953e3c14220b5f89043dc677d72553
- SFT-CHEM-THERMO-019-COUPLED-0064: pair=mass-charge; dataset=5; property=1; point=4; target=sha256:d4cfcc3dd0d2b880aa1b09bc58faab394744f690c9aa86e6ea47ca674ce9869e
- SFT-CHEM-THERMO-019-COUPLED-0065: pair=mass-charge; dataset=5; property=1; point=5; target=sha256:a5cb16a41c7c7eed937dfad21cb0d8784cb1b0873cd231cfc08a5da7ad4a416c
- SFT-CHEM-THERMO-019-COUPLED-0066: pair=mass-charge; dataset=5; property=1; point=6; target=sha256:ff33d28ee8199e7e4c0ca43341bf5c0dddf50c1f1e20dc1d611c93a90f08eff
- SFT-CHEM-THERMO-019-COUPLED-0067: pair=mass-charge; dataset=5; property=1; point=7; target=sha256:ccad2bffa6452300304a3d215b3af5704863480c83af3ed0d745690fb47aced6
- SFT-CHEM-THERMO-019-COUPLED-0068: pair=mass-charge; dataset=5; property=1; point=8; target=sha256:be0de60312ab990e31d92a72cc88409d25e397605d45872766be41190ad9c931
- SFT-CHEM-THERMO-019-COUPLED-0069: pair=mass-charge; dataset=5; property=1; point=9; target=sha256:1902c422ae39382f127eb0348b6539167c94806bc58cf43aa2217d31276978aa
- SFT-CHEM-THERMO-019-COUPLED-0070: pair=mass-charge; dataset=5; property=1; point=10; target=sha256:b429b929c8eb943f2858c26dc3cfc0010d41574d2dba327f8d053e289c73dbf0
- SFT-CHEM-THERMO-019-COUPLED-0071: pair=mass-charge; dataset=5; property=1; point=11; target=sha256:aeb1f108cea1a8ac6eb1237f8f1e6505593f1bd03922a3faf0f723000c976cec
- SFT-CHEM-THERMO-019-COUPLED-0072: pair=mass-charge; dataset=5; property=1; point=12; target=sha256:f981a895e73268d3a88c063c8dfef71ab8fd14c4c5f3d3c4e92236ae9144bbf8
- SFT-CHEM-THERMO-019-COUPLED-0073: pair=mass-charge; dataset=5; property=1; point=13; target=sha256:fe1e3347f1959c30a64fbe0a67ee34cd119f82184cedf7dfce9e5fb96bb8566
- SFT-CHEM-THERMO-019-COUPLED-0074: pair=mass-charge; dataset=5; property=1; point=14; target=sha256:b8db4f8434dddad3a72ee53ec20629d3b75fc17c84ba80a637cab86205977960
- SFT-CHEM-THERMO-019-COUPLED-0075: pair=mass-charge; dataset=5; property=1; point=15; target=sha256:83759b54d6cdfada73041d3fdda8307add20128c34a30704bdb7c481ac852338
- SFT-CHEM-THERMO-019-COUPLED-0076: pair=mass-charge; dataset=5; property=1; point=16; target=sha256:f10904161c7a092d51bf58180537cef42899926d561bcd1d5d0cddd6081f9b5
- SFT-CHEM-THERMO-019-COUPLED-0077: pair=mass-charge; dataset=5; property=1; point=17; target=sha256:d2563210db04df3d9ac6d74fa9ee504028cad3d923b8a84edd441db31d21aa9b
- SFT-CHEM-THERMO-019-COUPLED-0078: pair=mass-charge; dataset=5; property=1; point=18; target=sha256:99f24edc4b7a5fed8dc42b8da13b581c40d3a883b1ef88f3e0fee1a488c62f06
- SFT-CHEM-THERMO-019-COUPLED-0079: pair=mass-charge; dataset=5; property=1; point=19; target=sha256:9999f796e14f9aa279cb920a115379bc55a70442d1e0ee1a33205a3f1669072d
- SFT-CHEM-THERMO-019-COUPLED-0080: pair=mass-charge; dataset=5; property=1; point=20; target=sha256:d4715f4c50acf652e0bfb7b74a144bf2ca5a57badb3708a4dfbe4fe4274302c0
- SFT-CHEM-THERMO-019-COUPLED-0081: pair=mass-charge; dataset=5; property=1; point=21; target=sha256:3cd210207df32f7293e642255610d67d3ac4e236bac1b6a1456711ffdc8df50
- SFT-CHEM-THERMO-019-COUPLED-0082: pair=mass-charge; dataset=5; property=1; point=22; target=sha256:74674341bdcd4a8f25ee5a1f806f565084ac2ae7966cb1dadec010b0bff29dca
- SFT-CHEM-THERMO-019-COUPLED-0083: pair=mass-charge; dataset=5; property=1; point=23; target=sha256:eb0b4f2af344dcf4ad0dd3cc75df60f74799f970f6fdd44f05dd025f3bff8822

- SFT-CHEM-THERMO-019-COUPLED-0084: pair=mass-charge; dataset=5; property=1; point=24; target=sha256:1995771e0133364414b956ed266e29a9ee20bb609cf792c59da833ef683edded
- SFT-CHEM-THERMO-019-COUPLED-0085: pair=mass-charge; dataset=5; property=1; point=25; target=sha256:478a42c1012dabfe7ff4e817ca7eb6d3cba1c269720df3cc4012b899e9f8a366
- SFT-CHEM-THERMO-019-COUPLED-0086: pair=mass-charge; dataset=5; property=1; point=26; target=sha256:5408bf38d032ca929c37799d09ca83e82547f2afbd879cd53f0eeecbf96cd94d
- SFT-CHEM-THERMO-019-COUPLED-0087: pair=mass-charge; dataset=5; property=1; point=27; target=sha256:4129e357b83a2910d3d6d1eafb0a6ba21bec7e4aa2287eb2c11e549613080de
- SFT-CHEM-THERMO-019-COUPLED-0088: pair=mass-charge; dataset=5; property=1; point=28; target=sha256:85896b152924d379dc6b7851a99c10087289d67d49b2aec22eb28ab4b1fca450
- SFT-CHEM-THERMO-019-COUPLED-0089: pair=mass-charge; dataset=5; property=1; point=29; target=sha256:cfa04ea9e7674e66a8a18cc02e1650b8831bce45d0e9bcbce005f0a4f5af1b06
- SFT-CHEM-THERMO-019-COUPLED-0090: pair=mass-charge; dataset=5; property=1; point=30; target=sha256:d59cf7c47b4ecde63a6225d109d6c143ca926ffc2a237b8e38c19cae9b8cc593
- SFT-CHEM-THERMO-019-COUPLED-0091: pair=mass-charge; dataset=5; property=1; point=31; target=sha256:242c9476c8ee3f68d850777c7bc5f0b64a4aa438602116aa5827d01bd3e7e377
- SFT-CHEM-THERMO-019-COUPLED-0092: pair=mass-charge; dataset=5; property=1; point=32; target=sha256:ff72fa6d35791d7f2274039bbbf5dfe39166407bec1162fd57b5be4eec6ef19b
- SFT-CHEM-THERMO-019-COUPLED-0093: pair=mass-charge; dataset=5; property=1; point=33; target=sha256:b395952906ec8782939bfcf98423d009faf3ac44d74fb2180ef6cc9c6764c0d8
- SFT-CHEM-THERMO-019-COUPLED-0094: pair=mass-charge; dataset=5; property=1; point=34; target=sha256:b838d1c154efe09df12a2bb8e49cebb403f05afa90b372e215a92b8359d0e7fd
- SFT-CHEM-THERMO-019-COUPLED-0095: pair=mass-charge; dataset=5; property=1; point=35; target=sha256:604e3da17c0f31369cf9dacb6bef813b8dbf2d64a957a1ce6c0c546942ed3e06
- SFT-CHEM-THERMO-019-COUPLED-0096: pair=mass-charge; dataset=5; property=1; point=36; target=sha256:dfab0ce375e952f4eb1c4ad92831bfb84011460cae23ca78661275d869e64fa9
- SFT-CHEM-THERMO-019-COUPLED-0097: pair=mass-charge; dataset=5; property=1; point=37; target=sha256:e75c0e1ed827ffe54c2447ce9125eaca0b0eae2cbeb0b4379c736f0c63810ae
- SFT-CHEM-THERMO-019-COUPLED-0098: pair=mass-charge; dataset=5; property=1; point=38; target=sha256:2fab8778687e6502d013a6f9ac9fa10a96c4b354c4b2c09177ad1ee463b092ff
- SFT-CHEM-THERMO-019-COUPLED-0099: pair=mass-charge; dataset=5; property=1; point=39; target=sha256:74d2ba723ccf2ffca1b2ec2bc1398763b7129201065d415e27cf1a82b75b4a7c
- SFT-CHEM-THERMO-019-COUPLED-0100: pair=mass-charge; dataset=7; property=1; point=1; target=sha256:bfab742c59423b7ae4fe9ec6ff1cb83c2ab58801376e258a832a2c4ccee62879
- SFT-CHEM-THERMO-019-COUPLED-0101: pair=mass-charge; dataset=7; property=1; point=2; target=sha256:315c91f065adaaf73fcb1a56e96ae552387690a4e8af06756b7fe372c46448e4
- SFT-CHEM-THERMO-019-COUPLED-0102: pair=mass-charge; dataset=7; property=1; point=3; target=sha256:57811c5fd6900d258793ae947f0e3a194e0132799e2642af3d02ee1f9702e677
- SFT-CHEM-THERMO-019-COUPLED-0103: pair=mass-charge; dataset=7; property=1; point=4; target=sha256:9f2570e1caa33d7dbc5408c6508f757d184251c70be479eeaa7c599829b58d4e
- SFT-CHEM-THERMO-019-COUPLED-0104: pair=mass-charge; dataset=7; property=1; point=5; target=sha256:322ad56eb1e695a3acff039c69e7be7700c188f9b37d7e65ccf51b75be64628b
- SFT-CHEM-THERMO-019-COUPLED-0105: pair=mass-charge; dataset=7; property=1; point=6; target=sha256:c5ca4c128467822e16191ff7cef209c1ec8b6cdab38208777c35ede46eea4849

- SFT-CHEM-THERMO-019-COUPLED-0106: pair=mass-charge; dataset=7; property=1; point=7; target=sha256:8acaa479c2c3a233d7391566af74d4ae76ad80ffa65b02c13f1199dea35a5aff
- SFT-CHEM-THERMO-019-COUPLED-0107: pair=mass-charge; dataset=7; property=1; point=8; target=sha256:8f85cfacd05dd42fdf1a5ccf98fb0ab1033b0325e4d7553cf7422326840ecd4c
- SFT-CHEM-THERMO-019-COUPLED-0108: pair=mass-charge; dataset=7; property=1; point=9; target=sha256:1833f4c0f2b12dc8f258b88b23dd7608dad42dde65157ecdc7659cc63f2c161d
- SFT-CHEM-THERMO-019-COUPLED-0109: pair=mass-charge; dataset=7; property=1; point=10; target=sha256:851debd2ad0bf78566d70753404e8084cbba9c370f0f4f2bfb1a4cd5da976f39
- SFT-CHEM-THERMO-019-COUPLED-0110: pair=mass-charge; dataset=7; property=1; point=11; target=sha256:d3226dff7028bd63469c4f3f97dcb3a702e75a3463ced0711673da780eaff855
- SFT-CHEM-THERMO-019-COUPLED-0111: pair=mass-charge; dataset=7; property=1; point=12; target=sha256:b6cc30f408132bdc101a88d41258e7cb2f829e1d7ead3f5a115ca1c1e0f9c5d2
- SFT-CHEM-THERMO-019-COUPLED-0112: pair=mass-charge; dataset=7; property=1; point=13; target=sha256:8fc450f32a4ca35e569173bd73090335f06f810135811e67acbbcc0292cafaef
- SFT-CHEM-THERMO-019-COUPLED-0113: pair=mass-charge; dataset=7; property=1; point=14; target=sha256:8907d5f7d6bd6144c8b55b5662c9f6442ae226aaa4d78dc1526da53768e0bdd5
- SFT-CHEM-THERMO-019-COUPLED-0114: pair=mass-charge; dataset=7; property=1; point=15; target=sha256:a086bf6063e1d1bb51859d12982de2df94750e99ba47f781b9ebdcd944137a59
- SFT-CHEM-THERMO-019-COUPLED-0115: pair=mass-charge; dataset=7; property=1; point=16; target=sha256:9100aa207b30edba06bc18376fe6363c01dd4a93e4ef48e63d1fc83a4d0b21f6
- SFT-CHEM-THERMO-019-COUPLED-0116: pair=mass-charge; dataset=7; property=1; point=17; target=sha256:b0a36efcdf327bf9eee092fc85c358f5f7cffe35d23cf4172c5faeb92951e48
- SFT-CHEM-THERMO-019-COUPLED-0117: pair=mass-charge; dataset=7; property=1; point=18; target=sha256:c0ec1dd5915ae9555443001e5f0124b6102ef624ca4526c4895d666d52168794
- SFT-CHEM-THERMO-019-COUPLED-0118: pair=mass-charge; dataset=7; property=1; point=19; target=sha256:5f2e0158508aa8f2cde64eca68b44917a82f860e7a604127e43324a424512e4f
- SFT-CHEM-THERMO-019-COUPLED-0119: pair=mass-charge; dataset=8; property=1; point=1; target=sha256:7071af79a410e10014541f5dce7f247b223876d3cbfc38fb4333fd328688eefe
- SFT-CHEM-THERMO-019-COUPLED-0120: pair=mass-charge; dataset=8; property=1; point=2; target=sha256:c7510aba6c95133ed22f43d449131b8ee63b8fa804dd42662ac0412bbce48d71
- SFT-CHEM-THERMO-019-COUPLED-0121: pair=mass-charge; dataset=8; property=1; point=3; target=sha256:504cd79902c26c5536987c9c4d5a8b11f320087dfd9e2cf6b9d62ff697ac2ae0
- SFT-CHEM-THERMO-019-COUPLED-0122: pair=mass-charge; dataset=8; property=1; point=4; target=sha256:9c3ccd514966ed2daf9c9d7559ecc2114bd1e9895b3bd3c8456035a4071a1c09
- SFT-CHEM-THERMO-019-COUPLED-0123: pair=mass-charge; dataset=8; property=1; point=5; target=sha256:271680361c56acc504753e45d08309666d063ebaa4d66ab4633e20b1a04a577
- SFT-CHEM-THERMO-019-COUPLED-0124: pair=mass-charge; dataset=8; property=1; point=6; target=sha256:bcc88ab48edbb24620c59ff899474df78735893357ead52c7776225b11edda5
- SFT-CHEM-THERMO-019-COUPLED-0125: pair=mass-charge; dataset=8; property=1; point=7; target=sha256:a8bc1be367e5bb5137a70ebe3b34b203fef313c70596466c51cd98d36d29e56
- SFT-CHEM-THERMO-019-COUPLED-0126: pair=mass-charge; dataset=8; property=1; point=8; target=sha256:3047b0b5d863753aeb3084440f6bb0cbf42360119857ef5c79028f6c360d6f57
- SFT-CHEM-THERMO-019-COUPLED-0127: pair=mass-charge; dataset=8; property=1; point=9; target=sha256:3d097e96dbbc9b18c33bdd927c4a2145ec34c9be8cfe942012f50498514bd9c8

- SFT-CHEM-THERMO-019-COUPLED-0128: pair=mass-charge; dataset=8; property=1; point=10; target=sha256:b494fa1547dc649989eca35eb287cce5f2fcb4d3b460ce0fe1d9b6938534583b
- SFT-CHEM-THERMO-019-COUPLED-0129: pair=mass-charge; dataset=8; property=1; point=11; target=sha256:609646fadf40c41b8fc57239242fec9b51dff4ec6adf168fbee520c3aa134263
- SFT-CHEM-THERMO-019-COUPLED-0130: pair=mass-charge; dataset=8; property=1; point=12; target=sha256:38fe400216a647b9f53a80df864eaeaf4352b9652a6357a5081663b66ab83c666
- SFT-CHEM-THERMO-019-COUPLED-0131: pair=mass-charge; dataset=8; property=1; point=13; target=sha256:0284ff244cdc8a54a4b6115d3a45e264204c2ecdbb38260cfca9528859e72e8a
- SFT-CHEM-THERMO-019-COUPLED-0132: pair=mass-charge; dataset=8; property=1; point=14; target=sha256:15e62d17e7b4ea762bff594e9cc39ca8cc5792f56f54e06e00667e0156c7072b
- SFT-CHEM-THERMO-019-COUPLED-0133: pair=mass-charge; dataset=8; property=1; point=15; target=sha256:f6b05af1eb7c58a00c5c448054b8a2ea80e3690ded369d6cb52f40da707eec6b
- SFT-CHEM-THERMO-019-COUPLED-0134: pair=mass-charge; dataset=8; property=1; point=16; target=sha256:74d388245cc1e7c99983ec2101303eb2e0a49dbb5090d69ace80754953f8ac31
- SFT-CHEM-THERMO-019-COUPLED-0135: pair=mass-charge; dataset=8; property=1; point=17; target=sha256:e7746f8c8fd8e3fd0be31c610548ab3c518b825ba9e80a7b6752fcad501b3f13
- SFT-CHEM-THERMO-019-COUPLED-0136: pair=mass-charge; dataset=8; property=1; point=18; target=sha256:e810ee13501369cf03fa8050bf211776e62a35b03a13728ceca447b2c8dae3f9
- SFT-CHEM-THERMO-019-COUPLED-0137: pair=mass-charge; dataset=8; property=1; point=19; target=sha256:c895b6d921524b1076e120d02b0a7546c0a3eb1f780895a19449869be4aa5c58
- SFT-CHEM-THERMO-019-COUPLED-0138: pair=mass-charge; dataset=9; property=1; point=1; target=sha256:bae7bfd075c91bc268d56c319483e43233a25f71e2f874a52b4e8b29211f732a
- SFT-CHEM-THERMO-019-COUPLED-0139: pair=mass-charge; dataset=9; property=1; point=2; target=sha256:3bc34b8f5e4d4adb20bb4f869ecb4f2563d6ac945e3ea7d7b55d1386a0709973
- SFT-CHEM-THERMO-019-COUPLED-0140: pair=mass-charge; dataset=9; property=1; point=3; target=sha256:0d2b91240e20efb15bd6836ff4e43775305cc793c2422fc354f6a5ffaa2a3912
- SFT-CHEM-THERMO-019-COUPLED-0141: pair=mass-charge; dataset=9; property=1; point=4; target=sha256:12a5d49a6ab9990c3a4eb1dd00b0e6eca8f114d9b8eed3accb3d66d630dde5b5
- SFT-CHEM-THERMO-019-COUPLED-0142: pair=mass-charge; dataset=9; property=1; point=5; target=sha256:0cd1f31bb73b0e93eb5a688b47b60fc404156247bcdec0a93309e8d40d8976ee
- SFT-CHEM-THERMO-019-COUPLED-0143: pair=mass-charge; dataset=9; property=1; point=6; target=sha256:82e1d6be5ca90b77373107400ee85db73a347f23025d0c477ea070ad3fe79685
- SFT-CHEM-THERMO-019-COUPLED-0144: pair=mass-charge; dataset=9; property=1; point=7; target=sha256:63962c920cbcacfc084931f690bbe0fbc5a1402b7580fc8d0e219e171ac9c95c
- SFT-CHEM-THERMO-019-COUPLED-0145: pair=mass-charge; dataset=9; property=1; point=8; target=sha256:3caf865308ed2be29c330512d261335eeaba045b87b3f462f5c99fbb8a4adf6c
- SFT-CHEM-THERMO-019-COUPLED-0146: pair=mass-charge; dataset=9; property=1; point=9; target=sha256:07157ae95838d93c29c81701a8d282a19112745d3fb5a54b5a4ef9fb2c99ab1a
- SFT-CHEM-THERMO-019-COUPLED-0147: pair=mass-charge; dataset=9; property=1; point=10; target=sha256:8e29ebbab9947a96b689da8e8d4f7e695d8729253255b59b558d926274d484ff
- SFT-CHEM-THERMO-019-COUPLED-0148: pair=mass-charge; dataset=9; property=1; point=11; target=sha256:c2c452f5cb5ce1cfba977d6aa69fdb8ce3e6e215d2638fc404d2d69e626484e0
- SFT-CHEM-THERMO-019-COUPLED-0149: pair=mass-charge; dataset=9; property=1; point=12; target=sha256:175521a9d71ef64b297f830726715165469de1439608ff4887e701d17fdbaf01

- SFT-CHEM-THERMO-019-COUPLED-0150: pair=mass-charge; dataset=10; property=1; point=1; target=sha256:63c3d4223a04b130a666d9c0b1a693bfa85c8009c5828e65e75abaf88e1f6793
- SFT-CHEM-THERMO-019-COUPLED-0151: pair=mass-charge; dataset=10; property=1; point=2; target=sha256:2a90402774d2201643e85a953e8d070eb3a4cd2c5700bbd483fcb6c13681f5d
- SFT-CHEM-THERMO-019-COUPLED-0152: pair=mass-charge; dataset=10; property=1; point=3; target=sha256:62dfadcb0725a1e8e6c1c2b25936b9d2edbce59cac3263fc739b6925b6f8db7c
- SFT-CHEM-THERMO-019-COUPLED-0153: pair=mass-charge; dataset=10; property=1; point=4; target=sha256:39d2e45143b8a2d775076f2b7e90faf46ec159affcb55ba0a4aa480a9edfca27
- SFT-CHEM-THERMO-019-COUPLED-0154: pair=mass-charge; dataset=10; property=1; point=5; target=sha256:2a1709b7353d6c9f858573fa48bdee755b3e5cd2d15e2bdd16485254bff44247
- SFT-CHEM-THERMO-019-COUPLED-0155: pair=mass-charge; dataset=10; property=1; point=6; target=sha256:dbd3270b545672f6a974c7a9e42a39892dc4b850aa34e04b4dcfcf348856b10
- SFT-CHEM-THERMO-019-COUPLED-0156: pair=mass-charge; dataset=10; property=1; point=7; target=sha256:fc435d64fa5e0df03d6670c3c490e15434e1df7cf04cec40c19b704f7d7f9d2c
- SFT-CHEM-THERMO-019-COUPLED-0157: pair=mass-charge; dataset=10; property=1; point=8; target=sha256:fe698c092026b93bee1c78b909d701306229170b85e0a045508839b89a3506cf
- SFT-CHEM-THERMO-019-COUPLED-0158: pair=mass-charge; dataset=10; property=1; point=9; target=sha256:70315c6bbf021a453ce350ec8007ffec3739926caf23f6780055413d28e50158
- SFT-CHEM-THERMO-019-COUPLED-0159: pair=mass-charge; dataset=10; property=1; point=10; target=sha256:8dc14ae9ab8ce6cf6c3883c2525cc00fa46731ec9b838bf41c81612a37a38936
- SFT-CHEM-THERMO-019-COUPLED-0160: pair=mass-charge; dataset=10; property=1; point=11; target=sha256:b84c7d01a2555fbde2ecb37e4d11314bb598de8e28600a9eda72663e1e173a6e
- SFT-CHEM-THERMO-019-COUPLED-0161: pair=mass-charge; dataset=10; property=1; point=12; target=sha256:cf66a7ebdea542129bcffd7f69196b4e2f574d23bbd0fa78785bb67de5ce1f74
- SFT-CHEM-THERMO-019-COUPLED-0162: pair=mass-charge; dataset=11; property=1; point=1; target=sha256:24e7cc29e6c5a756204500d8a898b461bd5c9888a7b321addae302723f219b08
- SFT-CHEM-THERMO-019-COUPLED-0163: pair=mass-charge; dataset=11; property=1; point=2; target=sha256:dc640d65524653652bc6ce06f974a63d8100dc541b3b0a26b5734334657b3e3d
- SFT-CHEM-THERMO-019-COUPLED-0164: pair=mass-charge; dataset=11; property=1; point=3; target=sha256:0f4f0b6cf1d48eb242c6d46de12e765e75e33be1e3253345c7e8c5bc53702e2d
- SFT-CHEM-THERMO-019-COUPLED-0165: pair=mass-charge; dataset=11; property=1; point=4; target=sha256:791991b377fc06df952bd182eba3082640199291fbb6035f43f034f607513b5e
- SFT-CHEM-THERMO-019-COUPLED-0166: pair=mass-charge; dataset=11; property=1; point=5; target=sha256:1fc604a8bb1b4df1a8defef7477543b9b6fdb1366590c415b7ce19b3ccdcaef
- SFT-CHEM-THERMO-019-COUPLED-0167: pair=mass-charge; dataset=11; property=1; point=6; target=sha256:9ab2fdbdaa50b8626cc2e4b2066784c73f8b279d7ff453bef0ccb6f76916a526
- SFT-CHEM-THERMO-019-COUPLED-0168: pair=mass-charge; dataset=11; property=1; point=7; target=sha256:2d31f713f6c9f4cf489d83bbc9c82899253d50528699af18208a5489900bb6ab
- SFT-CHEM-THERMO-019-COUPLED-0169: pair=heat-charge; dataset=1; property=1; point=1; target=sha256:3628b00862ba236abcf183ff5221075d5a89e3a6f5ae538270c5962cd42c9c6e
- SFT-CHEM-THERMO-019-COUPLED-0170: pair=heat-charge; dataset=1; property=1; point=2; target=sha256:8af52d8585fd94947da3939abfd8488d968d5abaf1bd97e8e85fc158d591ede7
- SFT-CHEM-THERMO-019-COUPLED-0171: pair=heat-charge; dataset=1; property=1; point=3; target=sha256:1386734c86eb3e894d25e4ce0ea928d4f71038a6e614fbbe55e7cad039d4e91e

- SFT-CHEM-THERMO-019-COUPLED-0172: pair=heat-charge; dataset=1; property=1; point=4; target=sha256:b1c5e078bb0ea99a0bab76565419838cb3ea96495c5e2e90c78360f46a2c895e
- SFT-CHEM-THERMO-019-COUPLED-0173: pair=heat-charge; dataset=1; property=1; point=5; target=sha256:02022d13102041a82477caa076b78cf812c82531fa0427b3b4dc8ec708c03fec
- SFT-CHEM-THERMO-019-COUPLED-0174: pair=heat-charge; dataset=1; property=1; point=6; target=sha256:4633ef98543eb527cdc975fa3b30f617bfdebe1f985dae8cd758fc6778c46a65
- SFT-CHEM-THERMO-019-COUPLED-0175: pair=heat-charge; dataset=1; property=1; point=7; target=sha256:88d85dfb4cd3d1bb56b3b88e55a1cdc8540b1a6f86fa29e86a97e96d4dee6311
- SFT-CHEM-THERMO-019-COUPLED-0176: pair=heat-charge; dataset=1; property=1; point=8; target=sha256:de05b04fd498885af9ac149f8134d123413c8fcf9901a9f05aa8fd4971be6c81
- SFT-CHEM-THERMO-019-COUPLED-0177: pair=heat-charge; dataset=1; property=1; point=9; target=sha256:31a45f94752574a57fe228651d975a352107a0d4d301c9769b1aa2331eb43766
- SFT-CHEM-THERMO-019-COUPLED-0178: pair=heat-charge; dataset=1; property=1; point=10; target=sha256:5061034907715bda25993f905693222fbda75f2efc3316c9e530f6a7157da0e4
- SFT-CHEM-THERMO-019-COUPLED-0179: pair=heat-charge; dataset=1; property=1; point=11; target=sha256:94e095764b8f266bb510a47888b71536415a118bd30a224c17d2864f77633738
- SFT-CHEM-THERMO-019-COUPLED-0180: pair=heat-charge; dataset=1; property=1; point=12; target=sha256:59398f7f4f08245affff5fc8a1d2457e89f3f61afd5014a67bbddda270739709
- SFT-CHEM-THERMO-019-COUPLED-0181: pair=heat-charge; dataset=1; property=1; point=13; target=sha256:c7025a2593a547536a7bf2db4a4d80748f14cd57aa892dd9d5a27276364ad1c4
- SFT-CHEM-THERMO-019-COUPLED-0182: pair=heat-charge; dataset=1; property=1; point=14; target=sha256:496fb9661ceeb7a4324f7f92199167048ce57e72c431a7abd099b0e4032bf58a
- SFT-CHEM-THERMO-019-COUPLED-0183: pair=heat-charge; dataset=1; property=1; point=15; target=sha256:a5664d15d83dd66cc80ad2b3686fa238a6860f638950d4ce77a05ebabe24e0b6
- SFT-CHEM-THERMO-019-COUPLED-0184: pair=heat-charge; dataset=1; property=1; point=16; target=sha256:861db492ce70aaace036e53e93a9318f777bddf17301c0253bbdc22b2895c18a
- SFT-CHEM-THERMO-019-COUPLED-0185: pair=heat-charge; dataset=1; property=1; point=17; target=sha256:d3c0e6a4370ac0c12cb91562f2fe0055b208003f4148863ecc4b25012ac65ae3
- SFT-CHEM-THERMO-019-COUPLED-0186: pair=heat-charge; dataset=1; property=1; point=18; target=sha256:fe38ec562d23cfec2728f516e7764a9554a8948dc585fb8d5458a9782274a5fd
- SFT-CHEM-THERMO-019-COUPLED-0187: pair=heat-charge; dataset=1; property=1; point=19; target=sha256:5515ce378d97c7268fa185c9e5970eb5508425f6290eeb37d775600d35aa26b5
- SFT-CHEM-THERMO-019-COUPLED-0188: pair=heat-charge; dataset=1; property=1; point=20; target=sha256:0d51a67fb5e74993bc6c009c98e500ab5046bfedc30424443746e21063ef69b2
- SFT-CHEM-THERMO-019-COUPLED-0189: pair=heat-charge; dataset=1; property=1; point=21; target=sha256:21d5d816375e69f1a0c37e765fbb649f9b37e8e95cc22a2747324fa86e2f0a7b
- SFT-CHEM-THERMO-019-COUPLED-0190: pair=heat-charge; dataset=1; property=1; point=22; target=sha256:cc7089e4ef166a0294f593220e43759d629267310d475b35fe6d696ce640559f
- SFT-CHEM-THERMO-019-COUPLED-0191: pair=heat-charge; dataset=1; property=1; point=23; target=sha256:3b5d489d2b33aa2b45ed30b796c78cd5aaaf5d2935ffcb0dc18e601464643c2
- SFT-CHEM-THERMO-019-COUPLED-0192: pair=heat-charge; dataset=1; property=1; point=24; target=sha256:f77cd642da33b9bd69b30f800cfc5cc2502b47af0cce5a9d25237d1a825030f7
- SFT-CHEM-THERMO-019-COUPLED-0193: pair=heat-charge; dataset=1; property=1; point=25; target=sha256:889295a59de1b8bdc06c8b3780f279f2fb58660011eba5f76632f8e780de8c2f

- SFT-CHEM-THERMO-019-COUPLED-0194: pair=heat-charge; dataset=1; property=1; point=26; target=sha256:e605e409856032b7aa3238d04c2f377acefaa7888f24efd2c0dd5c4bb1d3c2af
- SFT-CHEM-THERMO-019-COUPLED-0195: pair=heat-charge; dataset=1; property=1; point=27; target=sha256:bb30939103c4c9d14a7165647f0867cb392aa1276b642c5dec92792675a3f6c2
- SFT-CHEM-THERMO-019-COUPLED-0196: pair=heat-charge; dataset=1; property=1; point=28; target=sha256:837645bf156c438e8ad4c8d113368001e1acacbab7b83cde4d0fb41ed1452a8
- SFT-CHEM-THERMO-019-COUPLED-0197: pair=heat-charge; dataset=1; property=1; point=29; target=sha256:1b6aa737e0d3972eaf5b7ce690e52e23b41189f5161a872323e58fc9cd2ac40d
- SFT-CHEM-THERMO-019-COUPLED-0198: pair=heat-charge; dataset=1; property=1; point=30; target=sha256:7c0c452474026b7ae54d374446cc38c07b67e92268714bd2c867550a62f4db4f
- SFT-CHEM-THERMO-019-COUPLED-0199: pair=heat-charge; dataset=1; property=1; point=31; target=sha256:80222b8b5ead6339e0dbb6ab840124ec38cc6907f012820995be73a3bda90d8a
- SFT-CHEM-THERMO-019-COUPLED-0200: pair=heat-charge; dataset=1; property=1; point=32; target=sha256:e3fbf4ceff707571f72ba450db47f4bc72c7ee70d9ea4def9f7a74829c9f205e
- SFT-CHEM-THERMO-019-COUPLED-0201: pair=heat-charge; dataset=2; property=1; point=1; target=sha256:1bc48551e5ee6f773339f031e53a2ade8361bd872d9c53d0fb514b9d0ab3db4b
- SFT-CHEM-THERMO-019-COUPLED-0202: pair=heat-charge; dataset=2; property=1; point=2; target=sha256:7974c3e88faba14541675d2a637833587e7df3f4e0f061dc68ced8911d3e0cf3
- SFT-CHEM-THERMO-019-COUPLED-0203: pair=heat-charge; dataset=2; property=1; point=3; target=sha256:fbaa1c54e89820dc3108412fd54d0500b7cf79f004ae4323564a36237b443b9c
- SFT-CHEM-THERMO-019-COUPLED-0204: pair=heat-charge; dataset=2; property=1; point=4; target=sha256:9dadae68cff208ac924b9d32ae59d41299514dea7e310689d214862218c92027
- SFT-CHEM-THERMO-019-COUPLED-0205: pair=heat-charge; dataset=2; property=1; point=5; target=sha256:f7992ac81a0bbfd7c993810f6dc1a6817a3abefe8bbe6a947131e10dadfa8579
- SFT-CHEM-THERMO-019-COUPLED-0206: pair=heat-charge; dataset=2; property=1; point=6; target=sha256:a9a23bc69ff98e3322f214df0086ec290b2c9a33751e461d2a8bba1ab66b0e78
- SFT-CHEM-THERMO-019-COUPLED-0207: pair=heat-charge; dataset=2; property=1; point=7; target=sha256:7d275c834cc3b9b05e1dc66e3c6ae71d020b8b6903927afa080b526f3aa202a5
- SFT-CHEM-THERMO-019-COUPLED-0208: pair=heat-charge; dataset=2; property=1; point=8; target=sha256:39cee462a6650e240dd09e8bf9b5c640b0a3dc91dfd2820623b7f69314e68eda
- SFT-CHEM-THERMO-019-COUPLED-0209: pair=heat-charge; dataset=2; property=1; point=9; target=sha256:f2076eeef35f075eeb64e7f6ba2d77c177f27b243b3faa6e961318928f9c5d87
- SFT-CHEM-THERMO-019-COUPLED-0210: pair=heat-charge; dataset=2; property=1; point=10; target=sha256:77627edc02973b3e2750088781fcd9ab21eec68827a100e42407364eee65583c
- SFT-CHEM-THERMO-019-COUPLED-0211: pair=heat-charge; dataset=2; property=1; point=11; target=sha256:36f40c001d78a928298c6c8f835a539b3e30c979e5908d92057ec969b5e30a65
- SFT-CHEM-THERMO-019-COUPLED-0212: pair=heat-charge; dataset=2; property=1; point=12; target=sha256:378bfa3a4e0dad920c6d1e9482bf5807cd1d9dc18073c149442bdaa4c08d7483
- SFT-CHEM-THERMO-019-COUPLED-0213: pair=heat-charge; dataset=2; property=1; point=13; target=sha256:29cf74a55de15b50b4e51a4be6adbbdded4bcc156bf795a602945834eaff38c0
- SFT-CHEM-THERMO-019-COUPLED-0214: pair=heat-charge; dataset=2; property=1; point=14; target=sha256:2db0d1a78d89ec5aa3827983473c405d95480ccf02254fd84195136d4769394d
- SFT-CHEM-THERMO-019-COUPLED-0215: pair=heat-charge; dataset=2; property=1; point=15; target=sha256:53be5afffc62632950db18d4b811b08e7b117081c0f16ad976aa9d3c53011e8c

- SFT-CHEM-THERMO-019-COUPLED-0216: pair=heat-charge; dataset=2; property=1; point=16; target=sha256:84a61e961092e1b857adb3feac5c353b920045d00a398242a5a5b117de6868d7
- SFT-CHEM-THERMO-019-COUPLED-0217: pair=heat-charge; dataset=2; property=1; point=17; target=sha256:2d545b4e7a205cb9a069755609fe0c23f75a24cadd27b028e3ae56cabc322ab9
- SFT-CHEM-THERMO-019-COUPLED-0218: pair=heat-charge; dataset=2; property=1; point=18; target=sha256:8903f77bc084f84b13143621cb10dc273abdef68fbcadd5865c96f153dc0e60
- SFT-CHEM-THERMO-019-COUPLED-0219: pair=heat-charge; dataset=2; property=1; point=19; target=sha256:e093c64445925e3d6586cdf7a52453fe13fcc9cbab7f986068fb99d34594a0b4
- SFT-CHEM-THERMO-019-COUPLED-0220: pair=heat-charge; dataset=2; property=1; point=20; target=sha256:889e0cb60b220514fe37d329623f6d600b601714f04440403244665c1d82abce
- SFT-CHEM-THERMO-019-COUPLED-0221: pair=heat-charge; dataset=2; property=1; point=21; target=sha256:2785f6a5e8dbd15d3b1a44cc1732018bed23c6e14951b9a1245292c373214515
- SFT-CHEM-THERMO-019-COUPLED-0222: pair=heat-charge; dataset=2; property=1; point=22; target=sha256:5e7fc14a0d4a6b0e01128c5f34113b9b6d8ba9e9664ddf6b65f3f947969381ee
- SFT-CHEM-THERMO-019-COUPLED-0223: pair=heat-charge; dataset=2; property=1; point=23; target=sha256:52afe4fc45fbf8024db7c485043f873dfe06774842d223fe90724be44c1b82f7
- SFT-CHEM-THERMO-019-COUPLED-0224: pair=heat-charge; dataset=2; property=1; point=24; target=sha256:835e138279b06a7d8097ba9c8b7a84eeb669e46b61be7c70c2dbd130c995a29d
- SFT-CHEM-THERMO-019-COUPLED-0225: pair=heat-charge; dataset=2; property=1; point=25; target=sha256:a4a616da71773c621a47bed77dd3ad0baaeac9fbc92499e1545e4bde813d1477
- SFT-CHEM-THERMO-019-COUPLED-0226: pair=heat-charge; dataset=2; property=1; point=26; target=sha256:5cb0e8b9fd34e5ffe8f152078853d5f948561be3948fd29e21ea54e927ece4c8
- SFT-CHEM-THERMO-019-COUPLED-0227: pair=heat-charge; dataset=2; property=1; point=27; target=sha256:f7d7865a4d8958741b1cc3f8bb03d8a5faadb6abc26263a0c0cdb72b89dcd983
- SFT-CHEM-THERMO-019-COUPLED-0228: pair=heat-charge; dataset=2; property=1; point=28; target=sha256:ca76806f66a901d9e05110063901de8fec7b8c14edbdda790834040446acbf
- SFT-CHEM-THERMO-019-COUPLED-0229: pair=heat-charge; dataset=2; property=1; point=29; target=sha256:6cdece026f8bf44e38d90ea60481b1929ad19c7f299f99043b31a3e1d666ddb2
- SFT-CHEM-THERMO-019-COUPLED-0230: pair=heat-charge; dataset=2; property=1; point=30; target=sha256:22a8e1bb1fcd6078883cf461ee7e43941e6decfa762de33d9f87d64bcb1a2586
- SFT-CHEM-THERMO-019-COUPLED-0231: pair=heat-charge; dataset=2; property=1; point=31; target=sha256:d8807897f648eea8550c1bdd172c7ae60e17bb33fcb196fe67f3da64b861ca91
- SFT-CHEM-THERMO-019-COUPLED-0232: pair=heat-charge; dataset=2; property=1; point=32; target=sha256:8f0fca3b5844627187b0e5027ade7c40ff5ca019abf1e0b89af83c51ca8c0033
- complete pair vector: 22 mass-heat, 146 mass-charge and 64 heat-charge records
- complete property vector: 22 thermally forced binary diffusion, 31 charge-probed tracer diffusion, 147 electrical conductivity and 32 thermal conductivity records
- source coverage: all 23 datasets and 375 points preserved from three complete NIST ThermoML sources
- tampered_negative_coupled_response_rejected: True
- complete_232_record_vector_retained: True
- all_pairwise_surfaces_roles_and_properties_retained: True
- all_datasets_methods_and_EmptyOne_conditions_retained: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if any component, phase, mass/heat/charge carrier, packet, shared event, cell, tick, boundary, condition or pairwise projection is erased; if adjacency or carrier conservation fails; if numerical-zero or signed proof value, Onsager matrix, continuum gradient/flux equation, phenomenological coefficient, fit, logarithm, interpolation, regression, selection or target correction enters; if target content opens before all 232 identities seal; if any selected response row, uncertainty, method, dataset, complete source or companion provenance is omitted; or if any target is tampered.

Measurement receipt: sha256:942fd12c5db6339ec32adb75b5f4f7500b5b8467090fe8da4980d747567f1f3c.

Isolation certificate: sha256:600c014812e88b665a467dd1a04874d7c80def8662ffb83910a957c743bdda86.

Custody certificate: sha256:86ab17f97eb50ab4e6f02f3fe431ffc42dc7445cfc9e6b72af8ab9e2d3e5cb13.

Registered source descriptions:

- National Institute of Standards and Technology ThermoML Archive: ()

Registered target identities:

- None.

Meaning. The thermodynamic extension derives finite microstate support, temperature, energy, heat/work, entropy, enthalpy, free direction, chemical potential, activity, fugacity, phases, colligative response, solvation and coupled transport while keeping external scalar inscriptions downstream. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; absent external condition coordinates are structural EmptyOne
- no negative, irrational, imaginary, logarithmic, floating, signed or continuum SFT proof value
- no imported Onsager matrix, continuum gradient or flux equation, phenomenological cross coefficient or fitted transport law
- no interpolation, regression, selected substance/pair/property/phase/condition/method/dataset/row or target correction
- no substance, carrier pair, property, composition, phase, condition, method, value, uncertainty or target hash before prediction seal
- every complete source and companion property remains preserved; companions never become coupled measurements

Immutable identities. Source manifest

sha256:07c0a8489389c15fd631607da0e383821c97a136434d881c3398aeeba4514a43; derivation seal

sha256:7147e33503c1eea3a4ebdf3a9dd32b771a0447c3c13111288505cb8213d7e9b5; engine receipt

sha256:25a5d2fe2a127f733017316c3bd650bc4a18d0fdf118e8de6fbcfa007cf8e6b2 at receipts/engine/model_admitted/SFT-CHEM-COUPLED-MASS-HEAT-CHARGE-TRANSPORT-019-25a5d2fe2a127f73.json;

empirical hash sha256:171b3d01e1b3ef979808e163bad3d0dbd6c22087bb0fff8ae63e78591450f12d;

measurement receipt sha256:942fd12c5db6339ec32adb75b5f4f7500b5b8467090fe8da4980d747567f1f3c.

21. Quantitative kinetics and reaction dynamics - 13 admitted claims

Rates, barriers, branches, isotope effects and product states retain complete condition-specific custody. This entire registered subcategory is included; no proper subset is presented as field completion.

144. Exact elementary-transition rate relation

Claim identity: SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001

Question and theorem. An elementary chemical transition rate is the exact positive count of completed instances of one registered transition between distinct retained molecular states per positive reference tick and observation support, with conditions held and direction represented as a label.

complete-registered-elementary-transition-carrier__distinct-held-initial-and-terminal-molecular-states__positive-completed-transition-count__positive-reference-tick-count__complete-condition-and-positive-observation-support__exact-transition-count-per-tick-and-observation-support__complete-value-free-46-record-identity-seal__depth-independent-co

mmon-event-tick-replication-and-record-append

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-CPLX-TIME-SPACE-001
- SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-FINITE-MICROSTATE-SUPPORT-001
- SFT-CHEM-ENTROPY-MULTIPLICITY-CORRESPONDENCE-005
- SFT-CHEM-COUPLED-MASS-HEAT-CHARGE-TRANSPORT-019

Generated grammar. Generate the literal product of carrier, state, event, recurrence, boundary, magnitude, prediction and extension forms; decide all 256 candidates only from admitted state-transition, conservation, exact-resource, molecular-state, mechanism, statistical-support, EmptyOne and finite-successor laws.

Boundary: Every finite registered elementary chemical transition between distinct molecular states with complete conditions, positive completed-event count, positive tick count and positive observation support. External testing preserves all 46 rate rows from four complete declared NIST SRD 17 experimental records.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-registered-elementary-transition-carrier__distinct-held-initial-and-terminal-molecular-states__positive-completed-transition-count__positive-reference-tick-count__complete-condition-and-positive-observation-support__exact-transition-count-per-tick-and-observation-support__complete-value-free-46-record-identity-seal__depth-independent-common-event-tick-replication-and-record-append`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-registered-elementary-transition-carrier	detached-rate-number-or-continuum-field	A detached rate erases the transition that is counted.
state	distinct-held-initial-and-terminal-molecular-states	endpoint-erased-or-identity-change	Without distinct retained endpoints no completed transition is identifiable.
event	positive-completed-transition-count	continuous-change-primitive-or-uncounted-event	An uncounted continuum change is not an exact rate resource.
recurrence	positive-reference-tick-count	clock-free-or-continuum-time-derivative	A rate cannot be formed without exact recurrence support.
boundary	complete-condition-and-positive-observation-support	condition-or-observation-support-erased	Erased conditions or support make rate rows incomparable.
magnitude	exact-transition-count-per-tick-and-observation-support	imported-mass-action-order-arrhenius-fit-or-logarithm	A conventional or fitted rate law could select the result.
prediction	complete-value-free-46-record-identity-seal	reaction-condition-method-or-value-readable-before-seal	Readable target content could select the law.
extension	depth-independent-common-event-tick-replication-and-record-append	refit-after-event-resource-replication-or-record-append	Refitting destroys exact provenance.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 48291353ca37a7c1340174c29fba07c35f62c087c1f5a31b32a34118e6bac96d.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: a46626f333263521a5965b4a5bc43e8c7ff30731e0f8a4db0039ddf22568f1a9.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 519926578c279da068c3655fb46e01e48241efacf91360903f3e5747e69baf59.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 49b610a82e9c08ee82eb4bcdcf16654d4c719db30c6bbe74ae0f9db79f03642e.

Independent reconstruction. Implementation hash

sha256: ec1567aebcb3233a790d084a4ef7ac80b72c3a77ce57c60cb1adfe63c6995f89; independent certificate sha256: 933dad9c7c78861685cce07ad1b356145d6861a54eb75f5ae7f516736694daaf; external-validation hash sha256: 3b514a7f189383784e270cf8998203d72cff80d23a747dfa1a560dce54778670. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-SRD17-1972TIN-KOO3-15
- NIST-SRD17-1967WIL2017-1
- NIST-SRD17-1967AVR-KRA501-503-1
- NIST-SRD17-1962ASH-BUR685-691-1

Comparison records:

- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0001: source=NIST-SRD17-1972TIN-KOO3-15; row=1; target=sha256:929f6b4a516a3291436cb22468eb522ffe63f386f82e628465c2d3e32ad51d07
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0002: source=NIST-SRD17-1972TIN-KOO3-15; row=2; target=sha256:9b14a1a459ba7ca11871d752949c156c21bbd8bc84835a1870107f2cef5cb0c5
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0003: source=NIST-SRD17-1972TIN-KOO3-15; row=3; target=sha256:fda345c3ee5e3380c4f72e9d042ecb5b037784137cda615d11c3b4373d6ccaac
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0004: source=NIST-SRD17-1972TIN-KOO3-15; row=4; target=sha256:5ee87a3fbacca9899093a6b5552e4b0d80223386547f973f9a7addab729dff1c
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0005: source=NIST-SRD17-1967WIL2017-1; row=1; target=sha256:ce9722fc9514b10ee660426d7960e5f40df1c9f03154b79ab67e16dc88182c40
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0006: source=NIST-SRD17-1967WIL2017-1; row=2; target=sha256:bc4635fa3f494225437127c70f524acb60ab8bba072dff94f1c93428c9252b1
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0007: source=NIST-SRD17-1967WIL2017-1; row=3; target=sha256:4667dad06afd410d0e16648745c5883595b3c8ea0ed921ccd4ba278444304fec
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0008: source=NIST-SRD17-1967WIL2017-1; row=4; target=sha256:acd703daa42e474a22b6c26e39199ceec98da86b396a309d293ac5f5daafec06
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0009: source=NIST-SRD17-1967WIL2017-1; row=5; target=sha256:8e603fb590c65bdacb5637fbecdf4d11d729686eeef9a5ed04b531daf31f281
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0010: source=NIST-SRD17-1967WIL2017-1; row=6; target=sha256:4ba699bdccb22f2ddb09ada218466be758a6b9572ebc351045dce0bb3047c9d9
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0011: source=NIST-SRD17-1967WIL2017-1; row=7; target=sha256:ee166ef904d1dc669f89c710f950bf8fa4a6146effeb8a64f136336ae80fdc4c
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0012: source=NIST-SRD17-1967WIL2017-1; row=8; target=sha256:2fab09a80a747446c832865d0ba45db6bc84f4e5615b9b625ed7a2dcf7d4c517
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0013: source=NIST-SRD17-1967WIL2017-1; row=9; target=sha256:a95f213f0c2fbcce3ccffab9ffc99b839e8869cff656cab22c392ab4a01e41db
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0014: source=NIST-SRD17-1967WIL2017-1; row=10; target=sha256:7c8fc82074c1f279089f7581b7c63faed6557698af854907c9b8716e6feaa0e0
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0015: source=NIST-SRD17-1967WIL2017-1; row=11; target=sha256:ea199a090544b4a22d73994b555a151e4fa3aba89ffad7fe4cea687615d84706
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0016: source=NIST-SRD17-1967WIL2017-1; row=12; target=sha256:46269f2ce86b3de5bcb197643775e46b660c76ec546478657c8bbd541ce15a82
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0017: source=NIST-SRD17-1967WIL2017-1; row=13; target=sha256:89855a4435dd4ff0a4a1e0ff31f8796f8caf1c41b9857255f565d0895e4a5dd7
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0018: source=NIST-SRD17-1967AVR-KRA501-503-1; row=1; target=sha256:44e2bbb0a0cb51364a75a4c734978a0256c25c5f56ffd68a7ed714f0afcdef42
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0019: source=NIST-SRD17-1967AVR-KRA501-503-1; row=2; target=sha256:23396a8728a58e8438cd73fde13e6208918d2ce03698b35d08fb9105b2c2a01c
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0020: source=NIST-SRD17-1967AVR-KRA501-503-1; row=3; target=sha256:5942e298192e77683845c0271bb1ac74adeefb4ecb2c840f991e565185e73474
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0021: source=NIST-SRD17-1967AVR-KRA501-503-1; row=4; target=sha256:26266404472bdfb381007b5e15f9997ddda5da77eeb9ca410a23a5eaf105050
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0022: source=NIST-SRD17-1967AVR-KRA501-503-1; row=5; target=sha256:e252e90983f42d8f06bc785efb97365934499f502bc5cfb1082e5e4eb1da8171

- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0023: source=NIST-SRD17-1967AVR-KRA501-503-1; row=6; target=sha256:704792bb2b01662ba24451071c4ebbdcb39f574363ffc1b0a67b111728255f4
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0024: source=NIST-SRD17-1967AVR-KRA501-503-1; row=7; target=sha256:ae6966208af7113ad93888a10ec49129bc4f3aef3c4bb0feba20e43360779681
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0025: source=NIST-SRD17-1967AVR-KRA501-503-1; row=8; target=sha256:d41838bc9a606eb586045bca0c1d227d4e7cb701158285547d93e0bce1a39d39
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0026: source=NIST-SRD17-1967AVR-KRA501-503-1; row=9; target=sha256:17d16d44ccffebe2679382b17ef58abc5ad07e699391cfdad1a5aa9ef7953635
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0027: source=NIST-SRD17-1967AVR-KRA501-503-1; row=10; target=sha256:ee61ed799668d5f00eb2e7b7d8c4646008c17b177b94afa0d9b3f69bafb8a710
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0028: source=NIST-SRD17-1967AVR-KRA501-503-1; row=11; target=sha256:1a853d890a2a24c7a3e85a485f8e0b70f3c6c1de5c00e84af06269d20802639b
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0029: source=NIST-SRD17-1962ASH-BUR685-691-1; row=1; target=sha256:3afd73939f290543825d4a15a6ff1605480d8b307c35e97152c02f9d89d366af
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0030: source=NIST-SRD17-1962ASH-BUR685-691-1; row=2; target=sha256:428109e091897fa40e2eb8ffc21bd7fd6151a1be35197b9ee42943d941add8f0
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0031: source=NIST-SRD17-1962ASH-BUR685-691-1; row=3; target=sha256:d2469c34502faba06394453ac598b4dfdcd7e9bbeb1a2b00dd68384541363821
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0032: source=NIST-SRD17-1962ASH-BUR685-691-1; row=4; target=sha256:b66aee84ac2515f90f765e7e37c5e30409e2646f184c56941bdb030c0ee4a5d1
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0033: source=NIST-SRD17-1962ASH-BUR685-691-1; row=5; target=sha256:744f2e7b76b6ba3fc8aa3b04335790aca691dd26a377a83e0f420307d5858e33
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0034: source=NIST-SRD17-1962ASH-BUR685-691-1; row=6; target=sha256:31a5b07db16340c13f58926bee9f8d869476c4b7871a4b4113710e29bd8aec1c
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0035: source=NIST-SRD17-1962ASH-BUR685-691-1; row=7; target=sha256:f0444e397671b2a71d2b3887213ff2e6132fa18084d8677d07b6905255c84b70
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0036: source=NIST-SRD17-1962ASH-BUR685-691-1; row=8; target=sha256:bbf4f5d4a563673bccbe10a87f65b3aa0aa4fd0fc05b02eaa41ca92e1205000
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0037: source=NIST-SRD17-1962ASH-BUR685-691-1; row=9; target=sha256:3520520573e0a22b20e3ba6a2cf7bd3a00e1c69aaf43516f0b3c9f21829e9376
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0038: source=NIST-SRD17-1962ASH-BUR685-691-1; row=10; target=sha256:2ea4e5afe44abce47fa003a62d5cba39d03ddc44709ccc4eacd40814ca54824a
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0039: source=NIST-SRD17-1962ASH-BUR685-691-1; row=11; target=sha256:96556ac26c66b7f48b33ebf31910f459c48fe6c0e4532a4fa75fa6ef4dba83e3
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0040: source=NIST-SRD17-1962ASH-BUR685-691-1; row=12; target=sha256:a8af1b28a46634a5c753cdf01c5f6f0275e93baf15caaa36201692c0abd59a9a
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0041: source=NIST-SRD17-1962ASH-BUR685-691-1; row=13; target=sha256:2fd79a79f0c0462a1c6767e4efd052fda7b9eea3fff50d68bda69e42c9727a4a
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0042: source=NIST-SRD17-1962ASH-BUR685-691-1; row=14; target=sha256:362451b734747ec9d8064225dc27ad371214e5f831ead816442c50220a1b33be
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0043: source=NIST-SRD17-1962ASH-BUR685-691-1; row=15; target=sha256:dafe7000d14d39f17fddea09d471ff170605826aabfa902e13c0cdb1b60f7a08
- SFT-CHEM-KIN-001-ELEMENTARY-RATE-0044: source=NIST-SRD17-1962ASH-BUR685-691-1; row=16; target=sha256:494ac9fbed5ce93131c7e3f3b4d28f7538e322a4de1ea6338dbdb933d608ddac

receipt sha256:890740f09cef121794d93890cdca087a49f4bf17233757eae2d01b046246323e.

145. Exact condition-bound concentration-dependence relation

Claim identity: SFT-CHEM-CONCENTRATION-DEPENDENCE-RELATION-002

Question and theorem. Chemical concentration dependence is the complete source-ordered relation between one held reactant's exact positive concentration support and exact positive elementary-transition rate response, retaining every condition, method, uncertainty and unfavorable row; no exponent is fitted.

one-held-registered-reactant-identity__exact-positive-concentration-support-per-row__exact-positive-elementary-rate-response-per-row__complete-held-condition-method-and-uncertainty-record__complete-source-ordered-row-census__exact-condition-bound-concentration-rate-table__complete-value-free-9-row-identity-seal__depth-independent-complete-row-append-with-prior-trace-preserved

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-CPLX-TIME-SPACE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-KIN-ORDER-001
- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001

Generated grammar. Generate the literal product of reactant, intervention, response, condition, completeness, relation, prediction and extension forms; decide all 256 candidates only from admitted exact arithmetic, intervention, observation, transition-rate, condition, record-retention, EmptyOne and finite-successor laws.

Boundary: Every finite source-bound concentration/rate census for one registered reactant with exact positive concentration and rate supports and complete condition, method, uncertainty and row provenance. External testing preserves all nine rows, columns, uncertainties and notes in the complete primary-source Table 2.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *one-held-registered-reactant-identity__exact-positive-concentration-support-per-row__exact-positive-elementary-rate-response-per-row__complete-held-condition-method-and-uncertainty-record__complete-source-ordered-row-census__exact-condition-bound-concentration-rate-table__complete-value-free-9-row-identity-seal__depth-independent-complete-row-append-with-prior-trace-preserved*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
reactant	one-held-registered-reactant-identity	anonymous-or-changing-reactant	Rows for different reactants cannot define one dependence relation.
intervention	exact-positive-concentration-support-per-row	concentration-erased-or-continuum-variable	Erasing the intervention coordinate destroys the tested dependence.
response	exact-positive-elementary-rate-response-per-row	answer-only-rate-or-unregistered-change	A detached response cannot be traced to a transition.
condition	complete-held-condition-method-and-uncertainty-record	temperature-method-or-uncertainty-collapsed	Condition collapse can manufacture a dependency.
completeness	complete-source-ordered-row-census	selected-favorable-rows-or-averaged-answer	Selection or averaging erases adverse and distinct responses.
relation	exact-condition-bound-concentration-rate-table	imported-mass-action-power-order-fit-or-logarithm	An imported exponent or fit could choose the relation.
prediction	complete-value-free-9-row-identity-seal	species-condition-density-rate-or-value-readable-before-seal	Readable targets could select the law.
extension	depth-independent-complete-row-append-with-prior-trace-preserved	refit-after-complete-row-append	Refitting would alter prior evidence.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:b1d72b68a2bd1a80a2bb68aa54ace5d7954dc13b2286e0154a1bf02a2ab93f6d.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:ad8483b431872ab8037584d169405e18251dc018ce047adaf4e71af286c5c22b.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:6bb297594f50de4003c4fa2d010adb50d7e691aeda1dc9e4b77902a576c653ee.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:6032bd3464fc34dee7df17948b2aa3fa78ac6287a2ac0594cabe570385df1070.

Independent reconstruction. Implementation hash

sha256:317531a97e5760c0d280f33e3994e879c37e8b02f7b409f6064fe0f17ad08cd4; independent certificate sha256:067654f59953740f14337da0184c2cfe090f526ef2adc7336c655cb01eeed24; external-validation hash sha256:2f6d80b31308c06e713cb0b560de7d075f9394dff48d45f42794d94cf1788dd7. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- RSC-PCCP-2014-C3CP54664K-TABLE-2

Comparison records:

- SFT-CHEM-KIN-002-CONCENTRATION-RATE-0001: source-row=1;
target=sha256:eac12abf205401578aa223e1dce5d07b1d8e3f439430333cd7fcb091e6e7611a

- SFT-CHEM-KIN-002-CONCENTRATION-RATE-0002: source-row=2;
target=sha256:7efa618b0909482023f29f9a208efb273333c407d2d695c072a5198ceb73ac52
- SFT-CHEM-KIN-002-CONCENTRATION-RATE-0003: source-row=3;
target=sha256:9d6bb686927648daf0eccc2b6ea4d140478cacb824c8e3f6c47fa8a6e28da9
- SFT-CHEM-KIN-002-CONCENTRATION-RATE-0004: source-row=4;
target=sha256:6a19df2f6cb7a2f0968c18fdc7608660c67a96c4de76be386e84558f14666345
- SFT-CHEM-KIN-002-CONCENTRATION-RATE-0005: source-row=5;
target=sha256:ce1331fef57c9006f0632d3d0a6825499ceb727152552ed11e89b1a516850dfa
- SFT-CHEM-KIN-002-CONCENTRATION-RATE-0006: source-row=6;
target=sha256:d79e6516c70edee3c03a394e40dbeb25c5364290bb27add9f57362acff6ef130
- SFT-CHEM-KIN-002-CONCENTRATION-RATE-0007: source-row=7;
target=sha256:3b940c89eec7313e5b88af2af0137c4e02df0b0d3e127d9661514b3161b050a7
- SFT-CHEM-KIN-002-CONCENTRATION-RATE-0008: source-row=8;
target=sha256:74183858ba44e11cf0a4258ec472cd1ecb10d06767eb8149a497f56a50ce1265
- SFT-CHEM-KIN-002-CONCENTRATION-RATE-0009: source-row=9;
target=sha256:69ce9351174338dbb7cc1ba2f55ebf6d6aacf9c60378e91c98d4313a946a8171
- exact density range: {'minimum': '17/5', 'maximum': '17'}
- exact rate range: {'minimum': '3/1000000000000', 'maximum': '19/1000000000000'}
- complete primary Table 2: nine rows and 27 uncertainty coordinates; no fitted exponent in the Fold law
- tampered_negative_density_rejected: True
- complete_nine_row_vector_retained: True
- source_order_and_uncertainties_retained: True
- complete_pdf_and_table_retained: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if reactant, concentration, rate, condition, method, uncertainty, row or adverse response is erased; if a conventional/fitted law or prohibited value enters; if targets open before all nine identities seal; if any Table 2 row, column, uncertainty or note is omitted; or if tampering is accepted.

Measurement receipt: sha256:3a69de6a18e4209618a9536837c2fe25eda2f0a0e68f1c6439626d8fc3ecc9e4.

Isolation certificate: sha256:0fcd0c3b681801c6beb5eed5cd223ebefa4962b3b8970a91009361d5dd354f75.

Custody certificate: sha256:506f5306fd7a696342fe59ce064a4324604e99293c87f673c153f9d7d24dd833.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The kinetics extension derives exact rate, concentration and temperature dependence, activation, transition boundaries, competing and composed mechanisms, reversible equilibrium, turnover, diffusion limits, isotope effects and product-state dynamics. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; absence is structural EmptyOne
- no negative, irrational, imaginary, logarithmic, floating, signed or continuum SFT proof value
- no imported mass-action or power law, conventional reaction order, fitted exponent/coefficient or continuum derivative
- no interpolation, regression, averaging, favorable-row selection or target-derived correction

- no target content before the complete identity and prediction seal

Immutable identities. Source manifest

sha256:7fd4e66a905d41a4a80ebf3bfc305e67aec10a50fa3f121bee6795fc64891a9; derivation seal sha256:3b6174d582ffc07362b1baa999ee6a298648ed808d16f930e1dce1cb29b4ed64; engine receipt sha256:4892f45dbce65c1aee2563912c23758797804f838aec48ee3fd5adfdb8bbf3fd at receipts/engine/model_admitted/SFT-CHEM-CONCENTRATION-DEPENDENCE-RELATION-002-4892f45dbce65c1a.json; empirical hash sha256:d25ec73022f198001c29c7310757e97365ec3ac3a202fb060bb5902b54b440fa; measurement receipt sha256:3a69de6a18e4209618a9536837c2fe25eda2f0a0e68f1c6439626d8fc3ecc9e4.

146. Exact condition-bound temperature-dependence relation

Claim identity: SFT-CHEM-TEMPERATURE-DEPENDENCE-RELATION-003

Question and theorem. Chemical temperature dependence is the complete source-ordered relation between each held registered reaction's exact positive temperature support and exact positive elementary-transition rate response, retaining complete conditions, structural absences and unfavorable rows; no imported or fitted functional form enters.

held-registered-reaction-identity-per-row__exact-positive-temperature-support-per-row__exact-positive-elementary-rate-response-per-row__complete-held-density-bath-method-and-uncertainty-record__complete-two-reaction-source-ordered-row-census__exact-condition-bound-temperature-rate-table__complete-value-free-19-target-identity-seal__depth-independent-complete-row-append-with-prior-trace-preserved

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-CPLX-TIME-SPACE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-KIN-ORDER-001
- SFT-CHEM-FINITE-MICROSTATE-SUPPORT-001
- SFT-CHEM-TEMPERATURE-CORRESPONDENCE-002
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007
- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001
- SFT-CHEM-CONCENTRATION-DEPENDENCE-RELATION-002

Generated grammar. Generate the literal product of reaction, temperature, response, condition, completeness, relation, prediction and extension forms; decide all 256 candidates only from admitted exact arithmetic, temperature correspondence, finite-microstate, state-energy, free-order, transition-rate, record-retention, EmptyOne and finite-successor laws.

Boundary: Every finite source-bound temperature/rate census for registered reactions with exact positive temperature and rate supports and complete density, bath-gas, method, uncertainty, structural-absence and source-row provenance. External testing preserves all fourteen Table 1 condition rows and all nineteen measured reaction-rate targets.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-registered-reaction-identity-per-row__exact-positive-temperature-support-per-row__exact-positive-elementary-rate-response-per-row__complete-held-density-bath-method-and-uncertainty-record__complete-two-reaction-source-ordered-row-census__exact-condition-bound-temperature-rate-table__complete-value-free-19-target-identity-seal__depth-independent-complete-row-append-with-prior-trace-preserved. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
reaction	held-registered-reaction-identity-per-row	anonymous-collapsed-or-changing-reaction	Collapsing reaction identity can manufacture one temperature relation from distinct processes.
temperature	exact-positive-temperature-support-per-row	temperature-erased-signed-or-continuum-variable	Erasing the intervention coordinate destroys the tested dependence.
response	exact-positive-elementary-rate-response-per-row	answer-only-rate-or-unregistered-change	A detached response cannot be traced to a chemical transition.
condition	complete-held-density-bath-method-and-uncertainty-record	density-bath-method-or-uncertainty-collapsed	Condition collapse can manufacture an apparent temperature effect.
completeness	complete-two-reaction-source-ordered-row-census	selected-favorable-rows-reaction-or-averaged-answer	Selection or averaging erases repeated, absent and adverse observations.
relation	exact-condition-bound-temperature-rate-table	imported-arrhenius-exponential-logarithm-prefactor-or-activation-fit	An imported form or fitted value could choose the answer.
prediction	complete-value-free-19-target-identity-seal	temperature-density-rate-or-target-value-readable-before-seal	Readable targets could select the law.
extension	depth-independent-complete-row-append-with-prior-trace-preserved	refit-after-complete-row-append	Refitting would alter prior evidence.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:c981dcbfd037341f375666006fddad274af594480fa1e3a5af8340aab05f7406.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:093065893ef1f59dd7f2864c0679cd56177fe3b3bb5d8e98e98dce55bcc3c93b.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:d70c2138f920f441cfa5785d9a6353b9a51406b197af1569739866e693d65c47.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:d3f53d6c07bd67a22f90d528067c80f3ecdc77645c86e1f25ea073fda83e117b.

Independent reconstruction. Implementation hash

sha256:48fcc4a18b5915d667683138433f9867a7fe7e781b01d7b937e053e6412b880f; independent certificate sha256:695b124dc4aa8c633fa84ed746df47166f2b04d075b14c7fddd0ab84e1cfb6c6; external-validation hash sha256:ef5167d82355d90be705dbd3aaad9b39adcd33f33d7fd3b3ab8113e5ce58b644. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- ACS-JPCA-2015-JP505790M-COMplete-TEMPERATURE-RATE-SERIES

Comparison records:

- SFT-CHEM-KIN-003-TEMPERATURE-RATE-0001: condition-row=1; reaction=ethanol; target=sha256:5a9e25e36ee469cd4ff351a20373dd4ee73c005bf6274facbdf281d850103756
- SFT-CHEM-KIN-003-TEMPERATURE-RATE-0002: condition-row=2; reaction=ethanol; target=sha256:9a536857a270bde1a39eb19ecf348824a0cc73e68e0d50bd88a3afd17ce26db5
- SFT-CHEM-KIN-003-TEMPERATURE-RATE-0003: condition-row=3; reaction=ethanol; target=sha256:d6e80e1ee06b4672556f230f1c12c4e6c3868f9e802bc42eccdcad31bc0b10d
- SFT-CHEM-KIN-003-TEMPERATURE-RATE-0004: condition-row=3; reaction=propan-2-ol; target=sha256:e200de323fa0bd1d8a9e30e1b5f6e453951773c9622002c3b4a089a741a94c26
- SFT-CHEM-KIN-003-TEMPERATURE-RATE-0005: condition-row=4; reaction=ethanol; target=sha256:f3cd6c2d0993dbf656673df29c1652fabd8fe95198d2a60bc22b4cf9ca289af6
- SFT-CHEM-KIN-003-TEMPERATURE-RATE-0006: condition-row=4; reaction=propan-2-ol; target=sha256:0354bebf5cf0df0ebdda53ad56b9fabd96c87f86bc0f9876e13b467150585cde
- SFT-CHEM-KIN-003-TEMPERATURE-RATE-0007: condition-row=5; reaction=ethanol; target=sha256:d9b958acccebc72927028b2b0fe127d06e589d7562839d47c38243cb536dd03d
- SFT-CHEM-KIN-003-TEMPERATURE-RATE-0008: condition-row=6; reaction=propan-2-ol; target=sha256:e326e4650d3e19caaaa155df3b2de25851db0275444d33b23183b994735d92bc
- SFT-CHEM-KIN-003-TEMPERATURE-RATE-0009: condition-row=7; reaction=ethanol; target=sha256:84efafc147f14bb157b84ab2d82b8821d2ba7dd2e3d01844e6eeb6746ae622b4
- SFT-CHEM-KIN-003-TEMPERATURE-RATE-0010: condition-row=7; reaction=propan-2-ol; target=sha256:f8a3daa5a5e7e9c8e88e4320b95244e0c4336f8d8192fe5d9ac8b5cc15e10fdc
- SFT-CHEM-KIN-003-TEMPERATURE-RATE-0011: condition-row=8; reaction=ethanol; target=sha256:6c577cc87c3fdaf8177f1ab23450a417487f6d9212d5b6095296932e9eb76ac0
- SFT-CHEM-KIN-003-TEMPERATURE-RATE-0012: condition-row=9; reaction=ethanol; target=sha256:732ef7e2abc4a7a5a6fe5bb14f077574e7e8e0d29873af7aa08c67935c2177f0
- SFT-CHEM-KIN-003-TEMPERATURE-RATE-0013: condition-row=10; reaction=ethanol; target=sha256:4d8a6b511d84f76997cd5a8ea021a7bda575fb37c820b5656748902fc8242d05
- SFT-CHEM-KIN-003-TEMPERATURE-RATE-0014: condition-row=10; reaction=propan-2-ol; target=sha256:4fa5b9b3cedf2315e134cfe9d7a6761d80ef29d04f749b111a86d1e19065b961
- SFT-CHEM-KIN-003-TEMPERATURE-RATE-0015: condition-row=11; reaction=ethanol; target=sha256:1f062e2dfd311fa7c3195c272e0f33082c0f9a4e49ff917f22ad16d8452e0463
- SFT-CHEM-KIN-003-TEMPERATURE-RATE-0016: condition-row=12; reaction=ethanol; target=sha256:56056ab71870081bc5625c9f88327b5c8fac9f565fd6b6a1fa6bda4987135450
- SFT-CHEM-KIN-003-TEMPERATURE-RATE-0017: condition-row=12; reaction=propan-2-ol; target=sha256:5387248d072f53ac7d9a57643b4101b31a60ed8593f32b5e582aaf821fcebfe7

- SFT-CHEM-KIN-003-TEMPERATURE-RATE-0018: condition-row=13; reaction=ethanol; target=sha256:626ac62fa9fab813ea5d053332a0a4042f49854707d2cc1372f423bb008c879
- SFT-CHEM-KIN-003-TEMPERATURE-RATE-0019: condition-row=14; reaction=ethanol; target=sha256:b553af34bddeb5addef198b908c69650c338330f96bef4bde66a26fbfcc9af4d
- exact temperature range K: {'minimum': '54', 'maximum': '148'}
- exact rate range molecule⁻¹ cm³ s⁻¹: {'minimum': '17/1000000000000', 'maximum': '7/1000000000000'}
- complete primary measured Table 1: fourteen condition rows, nineteen measured targets, fifty-seven uncertainty coordinates and nine structural absences; no imported or fitted temperature law
- tampered_negative_temperature_rejected: True
- complete_nineteen_target_vector_retained: True
- source_order_reactions_and_uncertainties_retained: True
- complete_pdf_measured_table_and_structural_absences_retained: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if reaction, temperature, rate, density, bath gas, method, uncertainty, structural absence, source row or adverse response is erased; if an imported/fitted form or prohibited value enters; if targets open before all nineteen identities seal; if any measured Table 1 row or column is omitted; or if tampering is accepted.

Measurement receipt: sha256:1fa2904bea1116b77f81bacee7363bd37f989a8c8c76631cb7465481abb75369.

Isolation certificate: sha256:01f21d2ae37c7213e4950ba29a309be7b55dd7918c8a8cca83634db92e6246b5.

Custody certificate: sha256:4ec5417959a64d262bd47dfb91a3f372972998326de082545bfe6e7303bf329f.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The kinetics extension derives exact rate, concentration and temperature dependence, activation, transition boundaries, competing and composed mechanisms, reversible equilibrium, turnover, diffusion limits, isotope effects and product-state dynamics. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; absence is structural EmptyOne
- no negative, irrational, imaginary, logarithmic, floating, signed or continuum SFT proof value
- no imported Arrhenius, exponential or logarithmic law, fitted prefactor/activation value or continuum temperature derivative
- no interpolation, regression, averaging, selected reaction, condition, row or target-derived correction
- no target content before the complete identity and prediction seal

Immutable identities. Source manifest

sha256:e6fcfafcfc04477ebc763e86b543b8bb8668da567cb001dd544931093f915a98; derivation seal

sha256:9a366de5b8c65b757c3f758f93e8453d5d6fbd5b9d8ffa2b837250cddb67a61; engine receipt

sha256:6bc633600c0f5fdf8d6c3eed4c09ec23bff9eff88220c5aeea8aa9e5103fdbec at receipts/engine/model_admitted/SFT-CHEM-TEMPERATURE-DEPENDENCE-RELATION-003-6bc633600c0f5fdf.json;

empirical hash sha256:b16bd0bc0f76e48c1e6586eadf8e8c838a25896a3e38144eda3451260fef37c1;

measurement receipt sha256:1fa2904bea1116b77f81bacee7363bd37f989a8c8c76631cb7465481abb75369.

147. Exact generated-path activation-barrier value relation

Claim identity: SFT-CHEM-ACTIVATION-BARRIER-VALUE-RELATION-004

Question and theorem. For a complete generated path with retained species, torsion and state identities, exact relative support and structural least-state EmptyOne force the activation barrier as the greatest positive path support under held orientation.

complete-generated-discrete-path-state-word__exact-source-bound-state-support__highest-positive-generated-path-boundary__least-state-EmptyOne-relative-support__least-crossing-positive-support__held-species-torsion-and-state-identities__complete-source-ordered-profile-reference-and-adverse-record__complete-value-free-44-target-identity-seal-and-depth-independent-append

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-CPLX-TIME-SPACE-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-CONFIGURATION-ORDER-PATH-011
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007
- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001
- SFT-CHEM-CONCENTRATION-DEPENDENCE-RELATION-002
- SFT-CHEM-TEMPERATURE-DEPENDENCE-RELATION-003

Generated grammar. Generate the literal product of path, source, boundary, reference, minimality, identity, record and prediction forms; decide all 256 candidates only from admitted exact arithmetic, discrete path, order, state-energy, free-order, reaction-mechanism, structural activation-boundary, EmptyOne, record-retention and finite-successor laws.

Boundary: Every finite generated discrete path with held species/path/state identities, exact positive relative supports or structural EmptyOne, complete source provenance and one greatest positive boundary. External testing preserves the complete NIST SRD 101 experimental collection: 41 species, 44 torsion targets, 782 path states and one unresolved row.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *complete-generated-discrete-path-state-word__exact-source-bound-state-support__highest-positive-generated-path-boundary__least-state-EmptyOne-relative-support__least-crossing-positive-support__held-species-torsion-and-state-identities__complete-source-ordered-profile-reference-and-adverse-record__complete-value-free-44-target-identity-seal-and-depth-independent-append*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
path	complete-generated-discrete-path-state-word	endpoint-only-or-saddle-continuum-premise	Endpoints or a named saddle do not enumerate a crossing path.
source	exact-source-bound-state-support	free-fitted-or-imported-barrier-number	A detached number could select the result.
boundary	highest-positive-generated-path-boundary	arbitrary-named-transition-state	A named state need not be the path boundary.
reference	least-state-EmptyOne-relative-support	absolute-energy-origin-or-signed-difference	An arbitrary origin or negative scalar imports an inadmissible choice.
minimality	least-crossing-positive-support	nonminimal-added-support	Extra support does not define the crossing.
identity	held-species-torsion-and-state-identities	species-path-or-state-collapsed	Collapsing identity can manufacture a barrier across different paths.
record	complete-source-ordered-profile-reference-and-adverse-record	barrier-answer-only-or-selected-profile	A selected value cannot reconstruct the empirical path.
prediction	complete-value-free-44-target-identity-seal-and-depth-independent-append	barrier-value-readable-before-seal-or-refit-on-append	Target access or refitting can choose the answer.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:ad07ed9d36212b22986e58a9c0d0b28743341f8fc5917cfff1d916919e80eb14a.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:de9fe7f8b8035e92c04ed35023d3922a5503636dbb7de5a5e0ba2e1bea813a41.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b23129569841797ef9cdc42ce099d06de6cd655d3b4a57e809cfadf81f568d17.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:484a066bbe743358b4a49cb5de2cfba27303edb8da1c5213e1c18329482113fc.

Independent reconstruction. Implementation hash

sha256:442540f66e0f5d2f86075953ae1a8e25b3e4c9f173bb13cd1da810b0fc610af5; independent certificate sha256:6d93e945e5f4101bfc7b4b05aa4c772b971d1100b2be90e236ae12e20df95e86; external-validation hash sha256:5326b9d9af789f807dd773ed2fabe0431704e31ee08f2a256e07dc68c15ec403. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-SRD-101-CCCBDB-EXPERIMENTAL-INTERNAL-ROTATION-BARRIERS-COMPLETE

Comparison records:

- SFT-CHEM-KIN-004-BARRIER-0001: species=Methane selenol; torsion=1;
target=sha256:b6d547af7fb3a6781171efb126455c1bf32af407d946cee83b027ad6d21d22a8
- SFT-CHEM-KIN-004-BARRIER-0002: species=Glycine; torsion=1;
target=sha256:0b63dfa1c9c1f82b793debcc43dca5fdf8c86e2853c84fe6d91f692c04c4d2

- SFT-CHEM-KIN-004-BARRIER-0003: species=Acetamide; torsion=1;
target=sha256:81609b45b7cc85e75e646d0d4a33bceec37d74a0887aead174dc2b381c36e0b4
- SFT-CHEM-KIN-004-BARRIER-0004: species=Ethanol; torsion=1;
target=sha256:8ee01b852447f2139f965ceda2dd6fbd3debaf7784371222fbfb8f20066e32de
- SFT-CHEM-KIN-004-BARRIER-0005: species=Ethanol; torsion=2;
target=sha256:bc50ff0d503c68679a74bcc0341337f5a1abd24fa09484b3b87409b1ca8f8103
- SFT-CHEM-KIN-004-BARRIER-0006: species=Formic acid; torsion=1;
target=sha256:154e0614dda75a55ca430295541b822e09e68ffd711b0b866d6ab62b1f1d7625
- SFT-CHEM-KIN-004-BARRIER-0007: species=Acetic acid; torsion=2;
target=sha256:758e8447c87bc257591e72d9808e37bd0c760d6a5228eba1e2bab9f0e90dcd27
- SFT-CHEM-KIN-004-BARRIER-0008: species=Methyl alcohol; torsion=1;
target=sha256:202c815d0aec14029606f26c689afe5b3d88c34daf7669a9124fc418f4c21268
- SFT-CHEM-KIN-004-BARRIER-0009: species=Isopropyl alcohol; torsion=1;
target=sha256:e6611cd708ce63aca7a6a3b773a7df3d83b5d329d4d5c494655c20a5243687d0
- SFT-CHEM-KIN-004-BARRIER-0010: species=Ethane, 1,1,1-trichloro-; torsion=1;
target=sha256:e17b24a15a9156dcf3cbdf6ecfa141b1e15b8bba6a58c0eb3e5be9694b408d4
- SFT-CHEM-KIN-004-BARRIER-0011: species=Ethane; torsion=1;
target=sha256:65629616aae34af2f92598cce5df811ad94c296ba4aed1260a6dd0872e4ff8b4
- SFT-CHEM-KIN-004-BARRIER-0012: species=methyl amine; torsion=1;
target=sha256:0a24c514529084d57964402e1e38b8d17bc5011d049f45c0974cbd4d5f62e362
- SFT-CHEM-KIN-004-BARRIER-0013: species=Ethyl chloride; torsion=1;
target=sha256:d675feed3cf3bcd661a615e482d522e1f5254163c271e3a784b5b3a34ede6d42
- SFT-CHEM-KIN-004-BARRIER-0014: species=Acetaldehyde; torsion=1;
target=sha256:176cc6f5c204aea99c4dee78cc8058d0f5270c79e9bcb0221f2edbeb45218222
- SFT-CHEM-KIN-004-BARRIER-0015: species=Ethane, 1,1-difluoro-; torsion=1;
target=sha256:dfdd4140251eb43f615f5f23cbfb41614beaa81046f2f9ca1a769d5b1a73e6
- SFT-CHEM-KIN-004-BARRIER-0016: species=Propylene oxide; torsion=1;
target=sha256:ca88dfe7dd43f66a786f30c2285d938720a392611cc4cb3633ad714e94863ce5
- SFT-CHEM-KIN-004-BARRIER-0017: species=hexafluoroethane; torsion=1;
target=sha256:ae28bcc675fd010039f58c349a680dc4a5d697062c44b1cc420dbb7179fe2095
- SFT-CHEM-KIN-004-BARRIER-0018: species=2-Butanone; torsion=1;
target=sha256:357f278a05bf579f1c4a4a1ad388dfc524b5973fb58b8bb358e1a55af1a85658
- SFT-CHEM-KIN-004-BARRIER-0019: species=2-Butanone; torsion=2;
target=sha256:9e0004389ad62a181411907da6de6f8ed25b6f592d3c3915bb20778ddf2bf8a0
- SFT-CHEM-KIN-004-BARRIER-0020: species=2-Butanone; torsion=3;
target=sha256:04868a350ec57a379fe318bf52aa49a4c87c6bb2f3e14c3ea7b0163e28dee37d
- SFT-CHEM-KIN-004-BARRIER-0021: species=Styrene; torsion=1;
target=sha256:4d596ab31571bf867364d35f2c620a1c0859e2abe55ba3ac4d47a806138e5f27
- SFT-CHEM-KIN-004-BARRIER-0022: species=benzaldehyde; torsion=1;
target=sha256:fab68f1b62c60c7e15f53777e75af516eade5b8e57ac00bc506f932643f708bb
- SFT-CHEM-KIN-004-BARRIER-0023: species=Butane; torsion=1;
target=sha256:f32cfc7a01e5e0c1b7db77f9214b42d8aba797876368ca0d65cb791dbef8f28c
- SFT-CHEM-KIN-004-BARRIER-0024: species=1-Butyne; torsion=1;
target=sha256:ed167c6d4e1a1c5d233f7e3dba20fb01977f3495c40852ce969a115fe8fc6491

- SFT-CHEM-KIN-004-BARRIER-0025: species=Ethane, 1,2-dichloro-; torsion=1;
target=sha256:736907408c78399cf8ef9ebf6fec5972bc4a3e5f231e69af0948e466d9b8bc52
- SFT-CHEM-KIN-004-BARRIER-0026: species=chloroacetaldehyde; torsion=1;
target=sha256:3e340da947930e1043503a48eec8f8bb9d674e67edac8feae5eddf937f5a54d
- SFT-CHEM-KIN-004-BARRIER-0027: species=Ethanediol; torsion=1;
target=sha256:fe7bb0e1c7d95711b2121176b593f5e8fb3ced271f94820adc5dda46f1a52b4e
- SFT-CHEM-KIN-004-BARRIER-0028: species=phenol; torsion=1;
target=sha256:16a3e7e6f7da361966312794fcf7164371533d1b1b62d0a1c8a933850d1ef98f
- SFT-CHEM-KIN-004-BARRIER-0029: species=Propene; torsion=1;
target=sha256:c76d0c70f430c40cf95488e47969dac062bcdad027b7678276492cbbc50c94a8
- SFT-CHEM-KIN-004-BARRIER-0030: species=Dimethyl ether; torsion=1;
target=sha256:0f449cf5994584a97c60bb466780b5f48cd96e2f4f683582b7a37e5453d6d105
- SFT-CHEM-KIN-004-BARRIER-0031: species=Propanal; torsion=1;
target=sha256:1fe7041be3db4e84a88d1e738e22a53a7eca2c4ea05224df6a72d2df7c5fbafc
- SFT-CHEM-KIN-004-BARRIER-0032: species=Hydrazine; torsion=1;
target=sha256:8e59e5da34a0921e66a8dfec4230efc002bf8427be893a1c38744ebdb8f8b1d0
- SFT-CHEM-KIN-004-BARRIER-0033: species=fluoroethane; torsion=1;
target=sha256:72a721c825e7142e218f4415d219a70cc4f1ca6bd73d1971d8dae575340d10a5
- SFT-CHEM-KIN-004-BARRIER-0034: species=Ethane, 1,1,1-trifluoro-; torsion=1;
target=sha256:a622909ae9a8902add9b62cb28fb3edf55d2a09f088036ce4fa8c6f521bb84ce
- SFT-CHEM-KIN-004-BARRIER-0035: species=2-Butene, 1-chloro-; torsion=1;
target=sha256:78a2fded26102f11fd355c4d24294c80341234c669707aed80487d1181ed185b
- SFT-CHEM-KIN-004-BARRIER-0036: species=Nitric acid, ethyl ester; torsion=1;
target=sha256:039bfe3ee7a3b2bf4d039763fed5481bce61fb2182965085cf02a2d3ec2b3097
- SFT-CHEM-KIN-004-BARRIER-0037: species=methyl silane; torsion=1;
target=sha256:f4643ba0f54fd7f65f054d392b1fe9baca0f6be9cb293299e6f02b49f96ef375
- SFT-CHEM-KIN-004-BARRIER-0038: species=Methoxyacetonitrile; torsion=1;
target=sha256:25d5683a46c3cf37d84359c837d731ce3e259a4221f86b48e83a9869c0379895
- SFT-CHEM-KIN-004-BARRIER-0039: species=Methylketene; torsion=1;
target=sha256:82b5e65d875fd0f175cd648e3b3e7718d51f793d46311fb83fe6de08055b7ad1
- SFT-CHEM-KIN-004-BARRIER-0040: species=Hydrogen peroxide; torsion=1;
target=sha256:764842e3a04f836b461c65e4e84f0094c60a198b827e1a3bd5157c8ca8715eed
- SFT-CHEM-KIN-004-BARRIER-0041: species=Disulfur dichloride; torsion=1;
target=sha256:644998149a8f96778dbedafc7c2b8e34dc8cf40336a5569fe79cfd3665cae655
- SFT-CHEM-KIN-004-BARRIER-0042: species=Diboron tetrachloride; torsion=1;
target=sha256:f1cd8d5bd3893a0dd087beff7991dd7b2b855b6e9099cc4fa8b7622455ce40c0
- SFT-CHEM-KIN-004-BARRIER-0043: species=borane ammonia; torsion=1;
target=sha256:a9c2e9b2ca810ad12849f1820090de8d84efd50a60e40544b32e2571c4c9aa6a
- SFT-CHEM-KIN-004-BARRIER-0044: species=Diboron tetrafluoride; torsion=1;
target=sha256:778e76dd700e97f8ce08ed8bb52fd672d07d2c8562ea35a2a5134158f3f0e2b8
- exact barrier range kJ mol⁻¹: {'minimum': '29/100', 'maximum': '50577/100'}
- exact barrier range cm⁻¹: {'minimum': '24', 'maximum': '42279'}
- complete NIST SRD 101 experimental surface: forty-one species, forty-four torsion targets, seven hundred eighty-two path states and one unresolved row; no imported or fitted barrier law
- tampered_negative_path_support_rejected: True

- `complete_forty_four_target_vector_retained`: True
- `source_order_species_and_paths_retained`: True
- `structural_absences_unresolved_row_index_and_detail_pages_retained`: True
- `prediction_contains_no_withheld_target_hash`: True

Falsification boundary: The claim fails if a species, path, state, source, barrier, structural least state, absence, unresolved row or adverse profile is erased; if an imported/fitted continuum form or prohibited value enters; if targets open before all forty-four identities seal; if any NIST index species/detail page is omitted; or if tampering is accepted.

Measurement receipt: sha256:30af0c949970e611377de7f62339778ce0be2871c153413793275f0da9de524d.

Isolation certificate: sha256:262ca84e7a16512b63e1139982675e5c4bc5555ef222bc305714a7842dde002.

Custody certificate: sha256:d51ed94f65c8350291ea93c13a7874d12aa15059a2d31fdef0a9476c496a6ccc.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The kinetics extension derives exact rate, concentration and temperature dependence, activation, transition boundaries, competing and composed mechanisms, reversible equilibrium, turnover, diffusion limits, isotope effects and product-state dynamics. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; external zero-energy glyphs and absent coordinates become structural EmptyOne
- no negative, irrational, imaginary, logarithmic, floating, signed or continuum SFT proof value
- no imported transition-state or saddle-point continuum, Arrhenius law, fitted activation value or absolute energy origin
- no interpolation, regression, averaging, selected species/path/state/method/row or target-derived correction
- no barrier, profile, unit, note or target hash before the complete identity and prediction seal

Immutable identities. Source manifest

sha256:8752474b96533e9dd9d8c2437715a79adc010f998eac07f6510268595557770b; derivation seal

sha256:71e875aca0785c0755b6594c415b0c2f5460ab8ee5c77b270cc9f516983eaaae; engine receipt

sha256:50c0543ffc31753eb293d22583c527329ffd998725e840721130fea1b6b1b707 at receipts/engine/

model_admitted/SFT-CHEM-ACTIVATION-BARRIER-VALUE-RELATION-004-50c0543ffc31753e.json;

empirical hash sha256:72e181d9d8131ea17dcf4e7022154f8697b7e414f6dfd831c8b1fb2d03bf2084;

measurement receipt sha256:30af0c949970e611377de7f62339778ce0be2871c153413793275f0da9de524d.

148. Finite transition-state-equivalent Fold path-boundary law

Claim identity: SFT-CHEM-TRANSITION-STATE-EQUIVALENT-BOUNDARY-005

Question and theorem. A complete finite reaction path with held identities and one unique greatest exact positive support forces a transition-state-equivalent boundary carrier as the exact entry-boundary-exit partition; external signed barriers remain positive magnitudes plus held orientations only.

finite-generated-path-boundary-carrier__held-reaction-path-and-isotopologue-identities__unique-greatest-exact-positive-path-support__complete-entry-boundary-exit-partition__positive-magnitude-plus-held-orientation__complete-H2-D-2-signature-uncertainty-and-adverse-record__experimental-targets-separated-from-calculated-and-fitted-records__value-free-complete-isotopologue-identity-seal-and-depth-independent-append

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-CPLX-TIME-SPACE-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-CONFIGURATION-ORDER-PATH-011
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001
- SFT-CHEM-CONCENTRATION-DEPENDENCE-RELATION-002
- SFT-CHEM-TEMPERATURE-DEPENDENCE-RELATION-003
- SFT-CHEM-ACTIVATION-BARRIER-VALUE-RELATION-004

Generated grammar. Generate the literal product of carrier, identity, location, partition, orientation, observation, provenance and prediction forms; decide all 256 candidates only from admitted exact arithmetic, finite path, state-transition, order, retained-information, reaction-mechanism, temperature-dependence and activation-boundary laws.

Boundary: Every finite generated reaction path with held reaction/path/isotopologue identities, structural EmptyOne or exact positive supports, and one unique greatest support. External testing preserves the complete predeclared primary article and supplement, both measured H2/D2 signatures, all thirteen supplementary files, and all disclosed calculated/fitted records as separately classified provenance.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `finite-generated-path-boundary-carrier__held-reaction-path-and-isotopologue-identities__unique-greatest-exact-positive-path-support__complete-entry-boundary-exit-partition__positive-magnitude-plus-held-orientation__complete-H2-D2-signature-uncertainty-and-adverse-record__experimental-targets-separated-from-calculated-and-fitted-records__value-free-complete-isotopologue-identity-seal-and-depth-independent-append`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	finite-generated-path-boundary-carrier	saddle-point-or-continuum-transition-state-premise	A named continuum saddle imports geometry not generated by the Fold path.
identity	held-reaction-path-and-isotopologue-identities	isotopologue-or-reaction-identity-collapsed	Collapsed identities cannot preserve isotope-dependent observations.
location	unique-greatest-exact-positive-path-support	arbitrary-named-state-or-fitted-coordinate	A selected state or fitted coordinate could choose the boundary.
partition	complete-entry-boundary-exit-partition	endpoint-only-or-boundary-answer-only	Endpoints alone erase the states adjacent to crossing.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
orientation	positive-magnitude-plus-held-orientation	signed-or-negative-proof-scalar	A signed scalar imports a prohibited negative number.
observation	complete-H2-D2-signature-uncertainty-and-adverse-record	single-favorable-isotope-or-barrier-only	Selecting one isotope erases the measured contrast.
provenance	experimental-targets-separated-from-calculated-and-fitted-records	experimental-calculated-fitted-records-mixed	Mixing provenance can promote a fitted calculation to measurement.
prediction	value-free-complete-isotopologue-identity-seal-and-depth-independent-append	conventional-KIE-equation-target-access-or-refit	A conventional equation, target access or refit can select the outcome.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:025c100fb93976786bf9430e525305b36f44834c1f1b2ef1db150082083576f0.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:eb116ec9d0f733b592cda376862df7f5ae974d7b26e77709639ec892bc48f6e3.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b31a212d041dcbe98ceccfb9dc711978d21e3168391c730241927055d50ecc82.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:8838ff535f54184fd3e435fec754d54961f003c556ff9f6d9d1f660144d693f3.

Independent reconstruction. Implementation hash

sha256:8085af96fd1b83a16653ce11379ae1fcd9179cfd81fec6867ef72fe5bfff7792; independent certificate sha256:4bab37e5ed64f0ef887809476d66812972cd5b5ca9a5a4f4ea302564a92fff2e; external-validation hash sha256:db56c3139ad3d7f1c4b71689bb8c0e269645433195b198b8a3c5eb23094af349. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- KYRIAKOU-ET-AL-ACS-NANO-NN500703K-COMPLETE

Comparison records:

- KIN-005-H2-EXPERIMENTAL-BOUNDARY-SIGNATURE: isotope=H2;
target=sha256:63c2ee4cfd2f271ce2c2b0c957a8dd8f17e9efb0ba7568e4f14b30a9e06601f3
- KIN-005-D2-EXPERIMENTAL-BOUNDARY-SIGNATURE: isotope=D2;
target=sha256:7a9a1feffb75ecb0a38b6a0af779e7a38ce1b34bf80848178f5440eeaeae349a
- measured H2 apparent-barrier magnitude/orientation: 23/1000 eV; reverse-held-temperature-order
- measured D2 apparent-barrier magnitude/orientation: 9/200 eV; held-temperature-order
- complete primary article and thirteen supplementary files preserved; fitted and calculated records disclosed but excluded from experimental targets
- tampered_signed_orientation_mismatch_rejected: True
- complete_two_isotopologue_target_vector_retained: True
- opposite_measured_H2_D2_signatures_retained: True

- `calculated_and_fitted_disclosures_retained_but_not_measurements`: True
- `prediction_contains_no_withheld_target_hash`: True

Falsification boundary: The claim fails if the finite boundary is not unique, any generated state or held identity is erased, a signed source inscription becomes a negative SFT number, H2 or D2 is selected away, experimental and calculated provenance are mixed, any article/supplement record is omitted, targets open before both identities seal, or tampering is accepted.

Measurement receipt: `sha256:fc82312a51819b7431ae5cf24561b0d5bba058a5359379750e52fb7d608ee6eb`.

Isolation certificate: `sha256:1807f31e1747e2f546913dde410a1e56e704093dbdfc4ab74e770c39eb868603`.

Custody certificate: `sha256:512565587a05b3c928da9257fedc32f3419f5a2008410b7d553c1fc8e14318a6`.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The kinetics extension derives exact rate, concentration and temperature dependence, activation, transition boundaries, competing and composed mechanisms, reversible equilibrium, turnover, diffusion limits, isotope effects and product-state dynamics. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; structural absence is EmptyOne and external zero glyphs remain provenance only
- no negative, irrational, imaginary, logarithmic, floating, signed or continuum SFT proof value
- no imported transition-state geometry, saddle-point continuum, conventional KIE equation or Arrhenius law
- no fitted barrier, prefactor, model adjustment, selected isotope/surface/condition/row or target correction
- no isotope direction, barrier, uncertainty, temperature, exposure, coverage, curve or target hash before prediction seal

Immutable identities. Source manifest

`sha256:5cc54b971e77aec26c83d858a0c52b705c0774b4be592b93353d150ec1fe78a6`; derivation seal

`sha256:8248d6b608d5953e58c19e7c2b286f95a5f12846f98da29288e28ac47ea31383`; engine receipt

`sha256:9eee2a251c9da6dafdab46390daca0bd590d4b567de17896f18e2b83a8116a13` at `receipts/engine/model_admitted/SFT-CHEM-TRANSITION-STATE-EQUIVALENT-BOUNDARY-005-9eee2a251c9da6da.json`; empirical hash `sha256:ad3becefb0e38e77e194a77513f6cdfef9728d15a28ee4ab352ccebe2860f043c`; measurement receipt `sha256:fc82312a51819b7431ae5cf24561b0d5bba058a5359379750e52fb7d608ee6eb`.

149. Exact complete-support competing-channel branching relation

Claim identity: SFT-CHEM-COMPETING-CHANNEL-BRANCHING-006

Question and theorem. For one complete source-ordered set of registered product channels, exact retained support forces the complete support as its exact sum and every branch share as channel support over that complete support, with every weak and absent channel retained.

complete-registered-product-channel-support__exact-sum-of-complete-retained-support__exact-channel-support-over-complete-support__held-reaction-condition-product-and-source-row-identities__structural-EmptyOne-channel-retained-without-invented-value__complete-source-ordered-support-share-uncertainty-and-adverse-record__experimental-vector-separated-from-calculated-and-analysis-disclosures__value-free-complete-eight-channel-identity-seal-and-depth-in-dependent-successor

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-CONFIGURATION-ORDER-PATH-011
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001
- SFT-CHEM-CONCENTRATION-DEPENDENCE-RELATION-002
- SFT-CHEM-TEMPERATURE-DEPENDENCE-RELATION-003
- SFT-CHEM-ACTIVATION-BARRIER-VALUE-RELATION-004
- SFT-CHEM-TRANSITION-STATE-EQUIVALENT-BOUNDARY-005

Generated grammar. Generate the literal product of support, whole, relation, identity, absence, record, provenance and prediction forms; decide all 256 candidates only from admitted exact arithmetic, combinatorics, conditional information, reaction-mechanism, complete path, rate, activation and transition-boundary laws.

Boundary: Every finite complete source-ordered product-channel word with held reaction, condition, product and row identities, exact positive support or structural EmptyOne, at least two competing channels and one positive complete support. External testing preserves all eight experimental products, every uncertainty, the separate calculated column, the complete article, nineteen supplementary files and both supplement PDFs.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: complete-registered-product-channel-support__exact-sum-of-complete-retained-support__exact-channel-support-over-complete-support__held-reaction-condition-product-and-source-row-identities__structural-EmptyOne-channel-retained-without-invented-value__complete-source-ordered-support-share-uncertainty-and-adverse-record__experimental-vector-separated-from-calculated-and-analysis-disclosures__value-free-complete-eight-channel-identity-seal-and-depth-independent-successor. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
support	complete-registered-product-channel-support	selected-dominant-or-favorable-channel-subset	Selecting abundant products erases competing support.
whole	exact-sum-of-complete-retained-support	imported-probability-normalization-axiom	An imported normalization premise could choose the relation.
relation	exact-channel-support-over-complete-support	fitted-or-free-branching-ratio	A fitted fraction is not forced by the channel record.
identity	held-reaction-condition-product-and-source-row-identities	reaction-condition-or-product-identity-collapsed	Collapsed identity can combine unrelated products or conditions.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
absence	structural-EmptyOne-channel-retained-without-invented-value	unfavorable-null-or-unresolved-channel-omitted	Omission can inflate remaining shares.
record	complete-source-ordered-support-share-uncertainty-and-adverse-record	ratio-answer-only-or-normalized-table	An answer-only table cannot reconstruct source support.
provenance	experimental-vector-separated-from-calculated-and-analysis-disclosures	experimental-and-calculated-columns-mixed	Mixing columns can substitute theory for measurement.
prediction	value-free-complete-eight-channel-identity-seal-and-depth-independent-successor	branch-value-readable-before-seal-or-corrected-after-release	Target access or correction can select the result.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:085e9055fb705e2c4032420ef3ab49c640a2234adc05b6a2d4a765d4e5889d42.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:9d635a6d8c161ac988aed8b65c38a8ff810cd2ce00688ce09fbc8498ac2fcbba.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:5214cc0103dacbde1e632817a3d3c30910979ad3b85f5a2926a2638551b73389.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:42ee0f7bbfa81b4ef923c82e42e5724f6c34b73fee4142b4480404426086fb5a.

Independent reconstruction. Implementation hash

sha256:34474641f7116374241973357efa1c3fe5a20d2b1519dfe6530ef793034f333b; independent certificate sha256:cff74adb48045223b8a067fbfffb3c3113e7bd3a3e4887621086fef4887c203e5; external-validation hash sha256:2b35a5c9912f3ee4a2903c858a125afa5b3f79b60f080f2094bf85f1d604176b. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- COMMUNICATIONS-CHEMISTRY-S42004-024-01239-7-COMPLETE

Comparison records:

- KIN-006-PRODUCT-CHANNEL-01: product=1245HT (II);
target=sha256:8508a4512cfdbc6898bfc77cb6bf4d52e01acb38e7c775ab32e616298504393
- KIN-006-PRODUCT-CHANNEL-02: product=34DMCB (III);
target=sha256:04ec0f94fb3e1b2347d23882c3a86015679e114991ff3e53b5b8168b5fc562ca
- KIN-006-PRODUCT-CHANNEL-03: product=Fulvene (IV);
target=sha256:e59486612b0f53b11dcdaacceb7f70e4d8024f36c05e21d06c7690fe67afe7ef
- KIN-006-PRODUCT-CHANNEL-04: product=12HD5Y (V);
target=sha256:01185c800c984c1d7a4db291d403f5978ee1b6bace97e1eb00dd975b15fa7cf7
- KIN-006-PRODUCT-CHANNEL-05: product=2E13DB(VI);
target=sha256:2b433c671d5d88e17c01e6c437179dbf161ecc76d9eeffa1a94af197773d4c72

- KIN-006-PRODUCT-CHANNEL-06: product=Benzene (VII);
target=sha256:074082df972aea14df2762ddf63c2a83795c668aa2d95551250a858b470d6da3
- KIN-006-PRODUCT-CHANNEL-07: product=cis-13HD5Y (VIII);
target=sha256:6f0cb767e2b3b3d640bde79b7f6a32d0f67698daf0cb7363b844a5fcd60beca9
- KIN-006-PRODUCT-CHANNEL-08: product=trans-13HD5Y (XII);
target=sha256:cda6545276f0e7acdaba614f4158826453ff1f01b046daf0017e9e89d1896425
- exact experimental branching range: {'minimum': '3/50', 'maximum': '23/100'}
- exact experimental uncertainty range: {'minimum': '1/25', 'maximum': '2/25'}
- complete eight-product experimental vector reconstructs One; complete article, nineteen supplementary files, two PDFs and separate calculated/analysis provenance retained
- tampered_omitted_product_channel_rejected: True
- complete_eight_channel_target_vector_retained: True
- exact_whole_reconstructed_without_renormalization: True
- weak_and_unfavorable_channels_retained: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if a registered channel, weak product, uncertainty, source row or adverse disclosure is erased; if shares do not reconstruct One exactly; if an imported normalization, fit, renormalization or selected channel enters; if experimental and calculated columns mix; if targets open before all eight identities seal; or if tampering passes.

Measurement receipt: sha256:83e1a9c37c206a8142fdff555c7c423b5bc988d3d3c62fcd3e0a04354de8ad.

Isolation certificate: sha256:0b9827e7243d168ba387f45e42f0d831d8cb35e313c12803fec0c64598ccca91.

Custody certificate: sha256:5d65c46f6e176ae38f74af1a305d06161eb34136e45c9681e5e38a09d8642fcd.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The kinetics extension derives exact rate, concentration and temperature dependence, activation, transition boundaries, competing and composed mechanisms, reversible equilibrium, turnover, diffusion limits, isotope effects and product-state dynamics. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; absent reported support is structural EmptyOne and external zero glyphs remain provenance only
- no negative, irrational, imaginary, logarithmic, floating, signed or continuum SFT proof value
- no imported probability normalization, stochastic premise, branching equation or fitted branching ratio
- no renormalization, averaging, selected favorable channel, omitted weak/null/unresolved product or target correction
- no branch value, condition, uncertainty, spectrum, analysis result or target hash before all eight identities seal

Immutable identities. Source manifest

sha256:54a2dbd65abe6e171e28db4925a80a01b3b3e44fa7236842105fd31ef83f9d12; derivation seal

sha256:06ccfa040c12c2da0251409598ad712b376c74c0c1aa5c34fcea860e6ade7c45; engine receipt

sha256:21b108e96fa5c917f772307b1036cc4900f7d25ff1c144b9193a9899fc3d97eb at receipts/engine

/model_admitted/SFT-CHEM-COMPETING-CHANNEL-BRANCHING-006-21b108e96fa5c917.json; empirical

hash sha256:e66a1fe6d19f916c3fe76014ff4c3905b192059b9e9099d44aaecd94d363444d; measurement

receipt sha256:83e1a9c37c206a8142fdff555c7c423b5bc988d3d3c62fcd3e0a04354de8ad.

150. Exact complete-state sequential mechanism composition law

Claim identity: SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007

Question and theorem. A complete finite mechanism is the exact ordered word of every retained state occurrence and boundary-matched elementary transition occurrence, with every intermediate, condition and favorable, adverse or unresolved status retained exactly once and no imported evolution equation.

complete-source-ordered-state-and-transition-support__exact-entry-exit-boundary-matching-for-every-edge__every-intermediate-occurrence-retained-explicitly__finite-exact-ordered-edge-composition__held-state-and-transition-condition-boundaries__favorable-adverse-unresolved-and-parallel-status-retained__complete-article-supplement-PDB-raw-custody-and-control-record__value-free-seventeen-record-identity-seal-and-depth-independent-successor

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-INTERMEDIATE-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-CAT-PATHWAY-001
- SFT-CHEM-CONFIGURATION-ORDER-PATH-011
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001
- SFT-CHEM-CONCENTRATION-DEPENDENCE-RELATION-002
- SFT-CHEM-TEMPERATURE-DEPENDENCE-RELATION-003
- SFT-CHEM-ACTIVATION-BARRIER-VALUE-RELATION-004
- SFT-CHEM-TRANSITION-STATE-EQUIVALENT-BOUNDARY-005
- SFT-CHEM-COMPETING-CHANNEL-BRANCHING-006

Generated grammar. Generate the literal product of support, adjacency, intermediate, composition, condition, status, provenance and prediction forms; decide all 256 candidates only from admitted exact state-transition, path, mechanism, rate, activation-boundary and complete-channel laws.

Boundary: Every finite complete ordered mechanism word with at least two registered state occurrences, exactly one registered elementary edge between each adjacent occurrence, exact positive ordinals, held conditions and held measured, adverse or unresolved status. External testing binds seventeen source categories: five deposited structures, two late unresolved states, seven complete power-titration columns and three favorable, adverse or unresolved controls, together with the complete article, thirteen supplementary files, three PDFs, five PDB records and CXIDB 221 custody metadata.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: complete-source-ordered-state-and-transition-support__exact-entry-exit-boundary-matching-for-every-edge__every-intermediate-occurrence-retained-explicitly__finite-exact-ordered-edge-composition__he

ld-state-and-transition-condition-boundaries__favorable-adverse-unresolved-and-parallel-status-retained__complete-article-supplement-PDB-raw-custody-and-control-record__value-free-seventeen-record-identity-seal-and-depth-independent-successor. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
support	complete-source-or-ordered-state-and-transition-support	endpoint-only-or-selected-snapshot-support	Endpoints erase the mechanism path.
adjacency	exact-entry-exit-boundary-matching-for-every-edge	aggregate-start-to-finish-jump	An aggregate jump does not prove its elementary boundaries.
intermediate	every-intermediate-occurrence-retained-explicitly	implicit-eliminated-or-fitted-intermediate	An omitted intermediate cannot be reconstructed.
composition	finite-exact-ordered-edge-composition	imported-differential-exponential-or-lifetime-law	An imported evolution equation can select the answer.
condition	held-state-and-transition-condition-boundaries	condition-time-or-dose-collapsed	Collapsed conditions can join incompatible observations.
status	favorable-adverse-unresolved-and-parallel-status-retained	adverse-unresolved-or-parallel-record-omitted	Removing an unfavorable record overstates the path.
provenance	complete-article-supplement-PDB-raw-custody-and-control-record	structure-answer-without-source-custody	An answer-only path cannot be independently audited.
prediction	value-free-seventeen-record-identity-seal-and-depth-independent-successor	time-coordinate-occupancy-or-target-readable-before-seal	Target access can select the sequence.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:30c6da0438176805ea8be6e71373522d15a70f6760ae262ac234c869ba443295.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:10fd204ed24617bebf253bca36e4110c0ed722606b733a0ef22c27ee659bc87e.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:1f619513b4b7a56c73cb9514db4cd7f2100aa1f75a71d4b4840f963b28fea65d.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:aef10654e939363c6af1dc3447bb580d591434f118ae52ab2e087df9e99fe857.

Independent reconstruction. Implementation hash

sha256:9fc0d1986901d4960c82996dec9c3c03c72fa6fef8aab4a82e089988410ac044; independent certificate
sha256:1d287425ecd0af9e0cc162921a08734ef4e5cb35779e10210ad87bae86b42f0a; external-validation
hash sha256:bbb108ed195e299c26aac8f99a6c30caefa3d18c2531bfbb6c1d7d8bc1be6081. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NATURE-COMMUNICATIONS-S41467-024-49814-9-COMplete
- RCSB-PDB-8WZF-8WZV
- CXIDB-221

Comparison records:

- KIN-007-SEQUENTIAL-RECORD-01: class=deposited-structure-snapshot; PDB=8WZF; target=sha256:e5b9216ade2f9703fa7d8188aeb7a1849efc66c30755b54911ee13d59adff6d3
- KIN-007-SEQUENTIAL-RECORD-02: class=deposited-structure-snapshot; PDB=8WZG; target=sha256:c7ad093b4ddfd002673e44e7804cdab2b9d01717a16a8ba1a91fe77e06e254d
- KIN-007-SEQUENTIAL-RECORD-03: class=deposited-structure-snapshot; PDB=8WZR; target=sha256:293483938972a4a45a8bf53792505e5a2adf2b1b64a6a47e94816245f27ed48a
- KIN-007-SEQUENTIAL-RECORD-04: class=deposited-structure-snapshot; PDB=8WZT; target=sha256:16875e6cf1881bfa6c52f18d61021e22925439a0216aeabf16f30d064e4f56f1
- KIN-007-SEQUENTIAL-RECORD-05: class=deposited-structure-snapshot; PDB=8WZV; target=sha256:17403c4b28458c6965648ff5644978a5b899bba9c068ebb75029962515935df1
- KIN-007-SEQUENTIAL-RECORD-06: class=late-structure-observation; PDB=structural-EmptyOne-no-deposited-model; target=sha256:a387ee6052333dccc0102c4282dcef8b2ba929161a0b76b1761d8079defc6ede
- KIN-007-SEQUENTIAL-RECORD-07: class=late-structure-observation; PDB=structural-EmptyOne-no-deposited-model; target=sha256:6f3457beaa534bcd0c1444b95ab535faac527d33444bf8021536bcd3fdb37770
- KIN-007-SEQUENTIAL-RECORD-08: class=complete-power-titration-column; PDB=structural-EmptyOne-no-dedicated-deposit; target=sha256:46fcfa29190360f6364c8c37d3ace47ddfe0d6e91e2093ffb87741ffe303ed6a
- KIN-007-SEQUENTIAL-RECORD-09: class=complete-power-titration-column; PDB=structural-EmptyOne-no-dedicated-deposit; target=sha256:60269b0efc795dea76276f2d0ce4b1644f01051dff7273fc4af1c55de60a4bdf
- KIN-007-SEQUENTIAL-RECORD-10: class=complete-power-titration-column; PDB=structural-EmptyOne-no-dedicated-deposit; target=sha256:ad87a149c290416b479a27c96fb3c38169b08ac48f22f7c68366ff417b5e16e5
- KIN-007-SEQUENTIAL-RECORD-11: class=complete-power-titration-column; PDB=structural-EmptyOne-no-dedicated-deposit; target=sha256:9844ea33abbbfd9aff53fff9d4bd64c915a68f4f5b0bed11d07fab81e8f59925
- KIN-007-SEQUENTIAL-RECORD-12: class=complete-power-titration-column; PDB=structural-EmptyOne-no-dedicated-deposit; target=sha256:9b4b46da63cbdfb2b980c6f879a72ee747090ea37128ad3f55d035366894ac71
- KIN-007-SEQUENTIAL-RECORD-13: class=complete-power-titration-column; PDB=structural-EmptyOne-no-dedicated-deposit; target=sha256:374ab0db56d1e44d6bdf67b036bb213179b65fcb741fd90d635409276b8f2b8c
- KIN-007-SEQUENTIAL-RECORD-14: class=complete-power-titration-column; PDB=structural-EmptyOne-no-dedicated-deposit; target=sha256:fb4e57ec1768df7c939172494ad8517128e0daf472354aab4f73ebc6217e6922
- KIN-007-SEQUENTIAL-RECORD-15: class=negative-delay-control; PDB=structural-EmptyOne-no-dedicated-deposit; target=sha256:8da2d87dbb1e16f8f840f2571d83ce26b74ad98b3e5e006f7fbf253aa17736fd
- KIN-007-SEQUENTIAL-RECORD-16: class=interleaved-dark-adverse-control; PDB=structural-EmptyOne-no-dedicated-deposit; target=sha256:bdc8ff8196b8e7b058eba706fd7aba9611fc9d9a94c86fe0acf79bf20754bb19

- KIN-007-SEQUENTIAL-RECORD-17: class=unresolved-multiple-intermediate-disclosure; PDB=structural-EmptyOne-no-deposited-model; target=sha256:d37831653e25b986b3c8b3c346c7fa9ff3074be60779d899ca612cd00b9fb7fe
- exact deposited component vector: [{'C': 3, 'MN': 1, 'O': 3}, {'C': 2, 'MN': 1, 'O': 2}, {'C': 2, 'MN': 1, 'O': 2}, {'C': 2, 'MN': 1, 'O': 2}, {'C': 1, 'MN': 1, 'O': 1}]
- exact deposited occupancy vector: ('1', '9/10', '9/10', '17/20', '13/20')
- exact positive difference-density sigma range: {'minimum': '8/5', 'maximum': '69/5'}
- exact positive difference-density range: {'minimum': '33/500', 'maximum': '221/250'}
- complete time-resolved vector: five PDB deposits, two late unresolved records, seven power columns, favorable negative-delay control, adverse 1 ms interleaved-dark contamination record and unresolved late-stage boundary
- tampered_omitted_source_record_rejected: True
- tampered_broken_transition_boundary_rejected: True
- complete_seventeen_record_target_vector_retained: True
- external_zero_glyphs_remain_structural_EmptyOne: True
- adverse_and_unresolved_records_remain_visible: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if any state, elementary edge, intermediate, condition, PDB record, power column, late record, favorable control, adverse contamination record or unresolved disclosure is omitted; if adjacent edge boundaries fail; if an imported evolution law, fit, interpolation, steady state or target correction enters; if experimental and calculated records mix; if targets open before all seventeen identities seal; or if tampering passes.

Measurement receipt: sha256:28087e980c5b7f05c9ebbf47a4e79749681ebd83afe291fe2696f8f96b2f4c4b.

Isolation certificate: sha256:308de0c7dc7766fa7ad53a7dd91789a09d8ead82a08b3da101ad1eec9a57e843.

Custody certificate: sha256:a6b7daaedb60c3780321d41cc9473c65d8ca2575a13527c099f6c283ec90c5cf.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The kinetics extension derives exact rate, concentration and temperature dependence, activation, transition boundaries, competing and composed mechanisms, reversible equilibrium, turnover, diffusion limits, isotope effects and product-state dynamics. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; absent observations or deposits are structural EmptyOne and external zero glyphs remain source inscriptions only
- no negative, irrational, imaginary, logarithmic, floating, signed or continuum SFT proof value
- no imported differential equation, exponential decay, fitted lifetime, rate law or steady-state assumption
- no interpolation, averaging, inferred missing intermediate, selected time, selected power, omitted parallel path or target correction
- no time, power, coordinate, occupancy, density, resolution, assignment, adverse result or target hash before all seventeen identities seal
- experimental deposits, calculated QM/MM trajectories, favorable controls, adverse controls and unresolved records remain distinct

Immutable identities. Source manifest

sha256:9fd69a7423f274dee1c3cbbb2fa9148e5e3d18379191595bd42afe3a71868854; derivation seal

sha256:5ec80776a4ac2430c30464e7038800f7bf15ad56445738b409bb3276fca7ec09; engine receipt

sha256:be8cc14ef02150d594aa01d2ec92b07714665d9fbd8fd2be40c483cafc5444bd at receipts/engine/model_admitted/SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007-be8cc14ef02150d5.json;
 empirical hash sha256:d8648ff01dda54c5aef7076e87ac54e2dbc63a37ecabc8fab3fc3c1c0a69a216;
 measurement receipt sha256:28087e980c5b7f05c9ebbf47a4e79749681ebd83afe291fe2696f8f96b2f4c4b.

151. Exact complete-path parallel-mechanism composition law

Claim identity: SFT-CHEM-PARALLEL-MECHANISM-COMPOSITION-008

Question and theorem. For one registered reaction with one retained initial state, a parallel mechanism is forced as the complete source-ordered family of every distinct sequential path word. Each path preserves all states, elementary edges, intermediates, shared boundaries, terminal occurrences, conditions and statuses; no stochastic premise, fitted path weight, imported parallel-rate equation or favorable-path selection enters the composition.

complete-registered-parallel-path-support__exact-common-initial-state-boundary__complete-state-edge-word-for-every-path__finite-source-ordered-parallel-path-composition__shared-boundaries-retained-in-every-path-occurrence__all-path-statuses-and-unresolved-records-retained__complete-article-supplement-and-twenty-eight-sheet-cell-ledger__value-free-twenty-eight-sheet-identity-seal-and-depth-independent-path-successor

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-INTERMEDIATE-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-CAT-PATHWAY-001
- SFT-CHEM-CONFIGURATION-ORDER-PATH-011
- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001
- SFT-CHEM-CONCENTRATION-DEPENDENCE-RELATION-002
- SFT-CHEM-TEMPERATURE-DEPENDENCE-RELATION-003
- SFT-CHEM-ACTIVATION-BARRIER-VALUE-RELATION-004
- SFT-CHEM-TRANSITION-STATE-EQUIVALENT-BOUNDARY-005
- SFT-CHEM-COMPETING-CHANNEL-BRANCHING-006
- SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007

Generated grammar. Generate the literal product of support, origin, path, composition, sharing, status, provenance and prediction forms; decide all 256 candidates only from admitted Fold state-transition, graph, complete-channel and sequential-composition laws.

Boundary: Every finite complete source-ordered family of at least two distinct sequential mechanism words sharing one registered reaction and one exact initial state, with every path identity, source occurrence, state, elementary edge, intermediate, condition, status and terminal occurrence retained. External testing preserves the complete article, supplement, peer-review record, source workbook, all twenty-eight worksheets and all 18,158 registered rectangular cell positions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-registered-parallel-path-support__exact-common-initial-state-boundary__complete-state-edge-word-for-every-path__finite-source-ordered-parallel-path-composition__shared-boundaries-retained-in-every-path-occurrence__all-path-statuses-and-unresolved-records-retained__complete-article-supplement-and-twenty-eight-sheet-cell-ledger__value-free-twenty-eight-sheet-identity-seal-and-depth-independent-path-successor`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
support	complete-registered-parallel-path-support	selected-dominant-or-favorable-path-subset	Selecting one visible path erases competing mechanism support.
origin	exact-common-initial-state-boundary	unrelated-or-collapsed-initial-boundary	Paths without the same retained source do not form one parallel mechanism.
path	complete-state-edge-word-for-every-path	endpoint-only-path-summary	Endpoints erase the states, edges and intermediates that distinguish paths.
composition	finite-source-ordered-parallel-path-composition	imported-stochastic-weight-or-parallel-rate-equation	An imported probability or rate equation can select a favored path.
sharing	shared-boundaries-retained-in-every-path-occurrence	shared-prefix-terminal-or-occurrence-collapsed	Collapsing a shared occurrence destroys path-local reconstruction.
status	all-path-statuses-and-unresolved-records-retained	weak-adverse-unresolved-or-unassigned-path-omitted	Omission makes the remaining path family falsely complete.
provenance	complete-article-supplement-and-twenty-eight-sheet-cell-ledger	selected-plot-mean-or-answer-only-product-table	A selected plot or aggregate cannot reconstruct the complete evidence surface.
prediction	value-free-twenty-eight-sheet-identity-seal-and-depth-independent-path-successor	product-time-value-readable-before-seal-or-corrected-after-release	Target access or correction can select a mechanism.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 4284e164839068d7c0513b0103c71e86b6205533d5cb5a1a9a1d689969d1d527.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 3602442bbc92a3b010c04659eff9caf97c1f1e4bcde2113fa36a414500d8272.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: e7a4fcc41f03f28770e4713264c13c4466eb590d3a02ed983ed2b952d6e60cdb.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 5cbd04f180d0ae52de4f50472a7120d44454d4257ab035f8cdfd28a0071fc28b.

Independent reconstruction. Implementation hash

sha256:89c54fdb18551c93c3ea3555fddf946ce408ef46c9f122a1b430f7631eb88476; independent certificate sha256:df4dcfee5ac96395fc93cfe3dd92602b346d8977e58c6e31294a6a0442751543; external-validation hash sha256:ca1a7061bdaddc7fd4913a4304bbbcf3e83e55da7352b415f07d27f22976b1d0. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NATURE-COMMUNICATIONS-S41467-026-70199-4-SOURCE-DATA

Comparison records:

- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-001: sheet=Fig.2a; rectangular-cells=288; target=sha256:dd642b3d98f2ac6209a526e5ce6a3f328028e6b764abc81ba6463146176c2b2b
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-002: sheet=Fig.2b; rectangular-cells=800; target=sha256:090c49c133369b09c9fb708839940fd30baa3c327370970a9b5e3c689d70c843
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-003: sheet=Fig.2c; rectangular-cells=297; target=sha256:6f72dee379cb2f2d6f198f1ada14af150a45074125643f89ab0c3ca3f72a724a
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-004: sheet=Fig.2d; rectangular-cells=297; target=sha256:59e3e2385bd9c9be9d2ff58ad4ba8fcc788ce8fa693f8bb00c10534a8f70f5f7
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-005: sheet=Fig.2e; rectangular-cells=270; target=sha256:31bc353ec1f14c667dfe32db2012223d09d706cc4417adf43bef308bfd3b900
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-006: sheet=Fig.2f; rectangular-cells=270; target=sha256:a48f7f5f93dbbe7269914e0f9161a73d2d2280432bf383c60c2fffa3f162c43e
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-007: sheet=Fig.2g; rectangular-cells=297; target=sha256:b0915236569f438bef8525e26bb944e5173e3cde73984ea1e35bf40f500212aa
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-008: sheet=Fig.4b; rectangular-cells=270; target=sha256:1ccd265a295a76a292d3d8beb06a9b6821bd5f53bdd0deaab0877d810cd3e72a
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-009: sheet=Fig.5c; rectangular-cells=481; target=sha256:828a44a4a868eefe15776eefec83c34d0eebf18fa10a60727fe24cdce87db942
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-010: sheet=Supplementary Fig.4; rectangular-cells=90; target=sha256:8f109762cd2d3bc5b69c9d10fc44b08eedf6d0caeb19a6d0eee2709cfe9745c0
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-011: sheet=Supplementary Fig. 6a; rectangular-cells=105; target=sha256:3796858bfc6e5b19eb20c6e70b6a3fe2ad1814758713b9083120d6725db558c3
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-012: sheet=Supplementary Fig.6b; rectangular-cells=120; target=sha256:1fdccf59b1f50ec39506015ef347d2232d9f9cde47c121a0fd3d6988f28fe465
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-013: sheet=Supplementary Fig.6c; rectangular-cells=135; target=sha256:a9aa63cd2b0bc781ada23802aabbdfec23c8715e63af318929901897e1b6b1f5
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-014: sheet=Supplementary Fig.6d; rectangular-cells=108; target=sha256:2dba9b0ff282862e861cde30406b4a8325b43e144f5204408d7812cff5a619ac
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-015: sheet=Supplementary Fig.8a; rectangular-cells=144; target=sha256:7312975e650ea64d15265dc78479bac7e5f3a4755f740c5ca6ef874b072a51ba
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-016: sheet=Supplementary Fig.8b; rectangular-cells=117; target=sha256:f1f454fa988cbf7ae7caf38257500b9f49da2840cd4941a7eee219def1ab1cf0
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-017: sheet=Supplementary Fig.10; rectangular-cells=108; target=sha256:4395ad39e9615e058038e4992bf13b49e222e90d701827b464865720c299a16e
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-018: sheet=Supplementary Fig.12a; rectangular-cells=156; target=sha256:8ca85de97c132eb112e56e564a72310c91b62c2a6368c2a875062882b03fc0f2

- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-019: sheet=Supplementary Fig.12b; rectangular-cells=132; target=sha256:661e6e4f6b719e3b8accd296bb4c15f3344567509f150eee9f903a654d6401e2
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-020: sheet=Supplementary Fig.13; rectangular-cells=150; target=sha256:d95176b81087e7546d75789d24d8499f08904d103f3cc7e55afbfea702d8f906
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-021: sheet=Supplementary Fig.18a; rectangular-cells=110; target=sha256:2ecb3d083022919ee9898b474b45f3ec6341e68e65ddf5f1e182279ffcb573d6
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-022: sheet=Supplementary Fig.18b; rectangular-cells=110; target=sha256:41f5ce57121e660f1fb755fc79a0fa00672689d17a82df95d0b8c4ce7465b4eb
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-023: sheet=Supplementary Fig.20a; rectangular-cells=3240; target=sha256:5c4d134eac746ef1fee6579cb439ac12a244a4de5ec231bf2880c8808a3c5c83
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-024: sheet=Supplementary Fig.20b; rectangular-cells=3248; target=sha256:596ebff5ffbfad3d791a5e85cff0ec728459b42f5052184122cb56ee3d612d55
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-025: sheet=Supplementary Fig.20c; rectangular-cells=3248; target=sha256:06b339f02436a779e6fc2631c888ddb7920b8b5f99158ef341a4d3dc42c156fb
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-026: sheet=Supplementary Fig.20d; rectangular-cells=3240; target=sha256:9e2696e7a5825c1ac112db197ba17a01b24621655fb1dd4ced0ee8c5d24f52a8
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-027: sheet=Supplementary Fig.32; rectangular-cells=195; target=sha256:7c34e90aa740f0381ae08e8271f3b85b49ff5efb719275b84b94139724ae498e
- SFT-CHEM-KIN008-COMPLETE-SOURCE-SHEET-028: sheet=Supplementary Fig.35b; rectangular-cells=132; target=sha256:19fabd8b3560ecee47d31f9e8c05a052a16f92d41ecfa75ce07e61ebd5a9fb34
- complete parallel path word: 1-EP->1->2; 1-EP->7->8->9->2; weak trace 1-EP->2-EP->2
- complete primary parallel product-time surface: 385 raw observations across Figure 4b, Supplementary Figure 32 and Figure 5c
- complete workbook: 28 worksheets and 18,158 registered cell positions; 2,109 source zero glyphs retained only as structural observed absence
- source formulas, every replicate and unresolved Supplementary Figure 31 two-structure peak retained without selecting or fitting a path
- tampered_omitted_worksheet_rejected: True
- tampered_broken_common_initial_boundary_rejected: True
- complete_twenty_eight_sheet_target_vector_retained: True
- all_18158_registered_cell_positions_retained: True
- all_2109_external_zero_glyphs_remain_structural_EmptyOne: True
- unresolved_two-structure_peak_remains_visible: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if any registered path, state, edge, shared boundary, terminal occurrence, product, time, replicate, formula, worksheet cell, weak/adverse/unresolved/unassigned record or source file is omitted; if paths do not meet one common source; if an imported stochastic/rate law, fit, average, interpolation, selection or target correction enters; if values open before all twenty-eight identities seal; or if tampering passes.

Measurement receipt: sha256:5b07d88c2e3337fb7dae24459702de89d6e7bf8c2bbd17db26a5e76f0834eef0.
Isolation certificate: sha256:520a5efdb4d2476a9b50511a321f86a3412aee3836b6baecee9104f9071efdca.
Custody certificate: sha256:ce7f96b6423b719685049f1be8fc77c1ed654b8964b04b2718adb37db90f914b.

Registered source descriptions:

- Nature Communications primary reaction-network study: ()

Registered target identities:

- None.

Meaning. The kinetics extension derives exact rate, concentration and temperature dependence, activation, transition boundaries, competing and composed mechanisms, reversible equilibrium, turnover, diffusion limits, isotope effects and product-state dynamics. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; source glyph 0 is external observed absence and native absence is structural EmptyOne
- no negative, irrational, imaginary, logarithmic, floating, signed or continuum SFT proof value
- no imported parallel-reaction equation, stochastic premise, fitted or free path weight or steady-state assumption
- no selected favorable path, product, time, condition, replicate, mean, method, worksheet or row
- no averaging away replicates, interpolation, inferred path, renormalization, omitted weak/adverse/unresolved/unassigned record or target correction
- no product-time value, cell label, formula result or target hash before all twenty-eight worksheet identities seal
- source formulas remain disclosed provenance and never become Fold proof parameters

Immutable identities. Source manifest

sha256:af0cd97f035399f64a3ed8057f81a7f9c968c82d7eb41a8b1d85bc8ffbbbc18cd; derivation seal
 sha256:1e649eb6354eb017957d6ca13e43eef3c71eb943d3a6d61d2d1536481baf6d39; engine receipt
 sha256:bc2b4cf9b8725fda1618d867ddf5f00f72a1f0c1c175e212fe6ccb5b84a91f8 at receipts/engine/model_admitted/SFT-CHEM-PARALLEL-MECHANISM-COMPOSITION-008-bc2b4cf9b8725fda.json;
 empirical hash sha256:857518894f05b283cfe89a7dcca649b078abb236420125a8e011515e68307ff;
 measurement receipt sha256:5b07d88c2e3337fb7dae24459702de89d6e7bf8c2bbd17db26a5e76f0834eef0.

152. Exact reversible kinetic-equilibrium correspondence law

Claim identity: SFT-CHEM-REVERSIBLE-KINETIC-EQUILIBRIUM-CORRESPONDENCE-009

Question and theorem. For two distinct retained states, exact forward and reverse transition occurrences close one and the same two-state graph. The ordered directed edges are the kinetic record and the graph's retained recurrence support is the equilibrium support. Direction is a held orientation rather than a sign; no imported reversible-rate equation, equilibrium constant, stochastic weight, fitted balance or steady-state premise enters the relation.

both-held-directional-transition-occurrences__same-exact-two-state-closed-transition-graph__exact-ordered-directed-transition-word__closed-graph-recurrence-support__every-initial-terminal-composition-retained-separately__all-favorable-adverse-and-disagreement-records-retained__complete-article-supplement-movie-and-archive-ledger__value-free-164-record-identity-seal-and-depth-independent-pair-successor

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STATE-ENERGY-ORDER-004

- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001
- SFT-CHEM-CONCENTRATION-DEPENDENCE-RELATION-002
- SFT-CHEM-TEMPERATURE-DEPENDENCE-RELATION-003
- SFT-CHEM-ACTIVATION-BARRIER-VALUE-RELATION-004
- SFT-CHEM-TRANSITION-STATE-EQUIVALENT-BOUNDARY-005
- SFT-CHEM-COMPETING-CHANNEL-BRANCHING-006
- SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007
- SFT-CHEM-PARALLEL-MECHANISM-COMPOSITION-008

Generated grammar. Generate the literal product of support, graph, kinetics, equilibrium, composition, status, provenance and prediction forms; decide all 256 candidates only from admitted exact state-transition, energy-order, complete-channel, sequential and parallel mechanism laws.

Boundary: Every finite source-ordered family of complete two-state graphs containing two distinct registered states, one directed edge in each held orientation, one exact condition boundary and retained favorable, adverse or unresolved status. External testing binds all 164 pre-registered records: ten article pages, 144 supplementary pages, one additional-description page, one 73-frame movie and all eight source-data archive members.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `both-held-directional-transition-occurrences__same-exact-two-state-closed-transition-graph__exact-ordered-directed-transition-word__closed-graph-recurrence-support__every-initial-terminal-composition-on-retained-separately__all-favorable-adverse-and-disagreement-records-retained__complete-article-supplement-movie-and-archive-ledger__value-free-164-record-identity-seal-and-depth-independent-pair-successor`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
support	both-held-directional-transition-occurrences	selected-forward-or-reverse-direction	One direction cannot establish a reversible pair.
graph	same-exact-two-state-closed-transition-graph	separate-unrelated-forward-and-reverse-graphs	Unrelated paths cannot supply one kinetic-equilibrium correspondence.
kinetics	exact-ordered-directed-transition-word	endpoint-rate-fit-or-aggregate-only	A fit or endpoint erases ordered directed transition occurrences.
equilibrium	closed-graph-recurrence-support	imported-equilibrium-constant-steady-state-or-balance-law	An imported equilibrium premise selects the correspondence.
composition	every-initial-terminal-composition-retained-separately	terminal-compositions-collapsed-or-averaged	Averaging hides direction-specific observations.
status	all-favorable-adverse-and-disagreement-records-retained	direction-label-disagreement-or-adverse-row-omitted	Omission falsely regularizes the source.
provenance	complete-article-supplement-movie-and-archive-ledger	selected-kinetic-pages-or-answer-table	A selected excerpt cannot prove complete source custody.
prediction	value-free-164-record-identity-seal-and-depth-independent-pair-successor	state-time-composition-energy-or-target-readable-before-seal	Target access can select the graph.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:031648f865d496b31aa8663f8195801278ec0aa46f6cd8c110e0583711ee995e.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:f976f96c4c75e3cc0516ce7c42682d4564ac0f88354c33b3a174ac0469888306.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:36c518bcb50f7e57b7eb5186752deb869539c727c5238cdb6eb51b2415bc24d9.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:3cb62abc0f290d2a4464fe44e0ae90dfc42dd5086631cd904027281754d17655.

Independent reconstruction. Implementation hash

sha256:2a382a0a65490149ba9d7890876e2287c1856b0756e6fcf7f3879ff1f79f08f4; independent certificate sha256:30f30cbd38d18e1c4cd495345cf9005a12aa2a554c31c6ec95b408556075ba03; external-validation hash sha256:c673aac7fb41b1d700ffff840de11fd2f2f04efcdd33032d746bd19b84fe1686b. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NATURE-COMMUNICATIONS-S41467-023-40190-4-COMPLETE

Comparison records:

- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-001: document=article.pdf; record=page-001; target=sha256:e9d438246b3e44da503c988f9a5a5269c06d686b35a745e05dfbb1fb529b0619
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-002: document=article.pdf; record=page-002; target=sha256:02921c17fd700bce77f9270d87e6d8bdd290ffe5077ca46d354c2214cf88174b
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-003: document=article.pdf; record=page-003; target=sha256:2a94660b58f990ae358b19463072b519563eb5851c84c02c998b243b4f1b9eed
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-004: document=article.pdf; record=page-004; target=sha256:7d140c799edca95481a4ef3bf53bb936ed3ae1411d91c0cf2b3320ce2c17e5d6
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-005: document=article.pdf; record=page-005; target=sha256:3c32d8dd6dbd07d8e394b8a2f993b588742ad794cc15ca850068e3d701c7c521
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-006: document=article.pdf; record=page-006; target=sha256:5285a510b51d8accaf52c44b1c4b4d55a38fbb7425671f5da3d35a76761bbfa7
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-007: document=article.pdf; record=page-007; target=sha256:6134bf47bb133ed83529456d1538252747158d0f98d5d4a130384071e2a942fc
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-008: document=article.pdf; record=page-008; target=sha256:ba2e5ca6ef36c4ed553a57b017fc18043fa273691f3d533a05bad930a0b39bbd
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-009: document=article.pdf; record=page-009; target=sha256:f5ccc3d6d9037046dfd9a67aa20ccddcc8a1ca0410ee9a8f3a0798cb3f05e79e
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-010: document=article.pdf; record=page-010; target=sha256:38f631f02316ede043a8eed173dbc7c2eab9a25a76c59d594079f84ad61a7f8a
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-011: document=supplementary-information.pdf; record=page-001; target=sha256:fddcd25ea652adb9f6b250a4f982b447350e22a7795e3a7d5bf31470f3848f80
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-012: document=supplementary-information.pdf; record=page-002; target=sha256:f6ac5380fa71f72f53efd55a9fcf7561c1cd02c1d50da8eeb0365919214d87a0

- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-013: document=supplementary-information.pdf; record=page-003; target=sha256:e402af7e867e0bef1b5caadb23f2b2b79c835ece6b32001af43018022ac6edb7
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- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-027: document=supplementary-information.pdf; record=page-017; target=sha256:7f7220e130d105260a2f6eacebb58502922a946b320f5fa784aedcfc55667f30d
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-028: document=supplementary-information.pdf; record=page-018; target=sha256:a00cfdadd7e3144dc79322566568d349cad4271a426cadf33114b594dcdd6dab
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-029: document=supplementary-information.pdf; record=page-019; target=sha256:27502d678cdd89223e5f69f580eb8bb27756c99da07b6aaa515f918253318362
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- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-032: document=supplementary-information.pdf; record=page-022; target=sha256:246a0cce4496bc35303f1c72a73164f001e04a55c98536f19035e02dc56f01ad
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- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-041: document=supplementary-information.pdf; record=page-031; target=sha256:2b328dc06ea5a170b9fdeb7836c0ac86570ea7e7764d11cab6a83bf87ab81cc
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-042: document=supplementary-information.pdf; record=page-032; target=sha256:420f76e2c386354b83d313d5be6813874204af33d3115a6ac6202619e834ad61
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- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-048: document=supplementary-information.pdf; record=page-038; target=sha256:6c987358845e82a98590d01a2625bc9eaf5b5a077dc6dceb05510da1774ee3b1
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-049: document=supplementary-information.pdf; record=page-039; target=sha256:9a7bba3f48a588248fad43e2da594b76d119cae3c2673dcb20d2981867932d1b
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- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-051: document=supplementary-information.pdf; record=page-041; target=sha256:374582a5a43cd57c8a281257127fcb3631731eaf3114ca2ef0ec9732c0b73850
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- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-053: document=supplementary-information.pdf; record=page-043; target=sha256:f9157a214f5aaa64f24ecad2a17b26ebd2059f021c781d51e9d452d8ff38b8c
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-054: document=supplementary-information.pdf; record=page-044; target=sha256:580ce27b9459bbe8d5cb9e4a08e80eda0d79a1b159eb4d2d17b3017b27e08070
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- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-059: document=supplementary-information.pdf; record=page-049; target=sha256:58c514bf0dec9640304c63f9cebfd438b48ad9a18022664b8d5f3dfbcd43f0e3
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-060: document=supplementary-information.pdf; record=page-050; target=sha256:06243e4ea472a81fe97ba77c662d118661abbbd3770d91c143313758d2cba5f4
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- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-157: document=source-data.zip; record=xv666_revision.cif; target=sha256:08a741a39e8cb0c80ea8ff59b39ede0f2830c1f22f51dd3d34264488d2d53f8b
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-158: document=source-data.zip; record=__MACOSX/._xv666_revision.cif; target=sha256:0fb76643ce71699919053940bb843d7b461a1ba20da0cd6b89433eb911f46d1d
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-159: document=source-data.zip; record=xv772_revision.cif; target=sha256:0fa39e72d78aa02052f62dbf337c540a8865556ee9f0103caf7ff4bc0a8d2a74
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-160: document=source-data.zip; record=__MACOSX/._xv772_revision.cif; target=sha256:594e91ed05eb1893dada6b5d08eb50f3a584d232a2528a0be54d92d8cc358d7a
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-161: document=source-data.zip; record=xv804_revision.cif; target=sha256:7ce684473bbb1bf964d85b39d68722e679b2d6189ebf454855eb7be1843b6cd8
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-162: document=source-data.zip; record=__MACOSX/._xv804_revision.cif; target=sha256:64ce2bebb5a158538c2226668a7af0dbbd68c10a9f9d5dfc1eff0baae2da0212
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-163: document=source-data.zip; record=yv042_revision.cif; target=sha256:31b3b33405bcec903c5f4459a95d7f7cbe5e897c66e2e894dfd1070bec5f4b13
- SFT-CHEM-KIN009-COMPLETE-SOURCE-RECORD-164: document=source-data.zip; record=__MACOSX/._yv042_revision.cif; target=sha256:9f442cdee417879f2fe2c338748355c7123bf796e287b3ef9d9f252d5915e058
- same-pair forward record: 83/17 becomes 32/68 after 88 hours at the held condition

- same-pair reverse record: 98/2 becomes 71/29 after 82 hours at the held condition
- 68/32 and 71/29 terminal records remain separate and are not averaged
- all source fits, direction-label inconsistencies, 155 PDF pages, 73 movie frames and eight archive members remain post-seal evidence
- tampered_omitted_source_record_rejected: True
- tampered_broken_reverse_boundary_rejected: True
- complete_164_record_vector_retained: True
- all_155_pdf_pages_retained: True
- all_73_movie_frames_retained: True
- all_8_archive_members_retained: True
- adverse_composition_and_direction_disagreements_remain_visible: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if the two directions do not close the same exact state pair; if either directed occurrence, initial or terminal composition, condition, adverse disagreement, page, movie frame, archive member or source file is omitted; if an imported rate/equilibrium law, constant, stochastic weight, fit, average, interpolation, renormalization or target correction enters; if values open before all 164 identities seal; or if tampering passes.

Measurement receipt: sha256:79e8c5e46d3f7c6eccab4a4668ff68c5697cd033d66e3b4fcd26ebf1fb24a581.

Isolation certificate: sha256:606836d8bdd945a378b7aa1acb1ef08711bd9eb284a242b3dcdbd7c4011787c38.

Custody certificate: sha256:2f1b48d5870ed7afab7790e128beed380d4aac33fc4dc19d5e4cf57ef249abc9.

Registered source descriptions:

- Nature Communications primary reversible molecular-isomerization study: ()

Registered target identities:

- None.

Meaning. The kinetics extension derives exact rate, concentration and temperature dependence, activation, transition boundaries, competing and composed mechanisms, reversible equilibrium, turnover, diffusion limits, isotope effects and product-state dynamics. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; source glyph 0 is an external reference inscription and native absence is structural EmptyOne
- no negative, irrational, imaginary, logarithmic, floating, signed or continuum SFT proof value
- no imported reversible-rate equation, equilibrium law or constant, stochastic premise, direction weight or steady-state assumption
- no selected direction, time, state, condition, method, page, frame, archive member or row
- no refit, averaging, interpolation, renormalization, target correction or omission of disagreement/adverse/unresolved evidence
- no state, time, composition, rate, quantum yield, energy, uncertainty, fit, calculation, value or target hash before all 164 identities seal
- source fits, slopes, energies, equations and calculations remain post-seal provenance and never become Fold proof parameters

Immutable identities. Source manifest

sha256:a09738d181d023f513579d7089a2991bfd769907f8f3f6f7b2c71bebf6cebe40; derivation seal

sha256:a9b3265792bffa6c4d7637042ef56d8f74b1ee5667ea3c2735a06512d050e166; engine receipt

sha256:6b09df5f93ff4357e250a41622fc4330df40fab5daaab1dcda40cdaaa23f4f7e at receipts/engine/model_admitted/SFT-CHEM-REVERSIBLE-KINETIC-EQUILIBRIUM-CORRESPONDENCE-009-6b09df5f93ff4357.json; empirical hash

sha256:07444d79c2ebe8a2acc6aa3226679e992713035485ee14b607831f995dec4cff; measurement receipt

sha256:79e8c5e46d3f7c6eccab4a4668ff68c5697cd033d66e3b4fcd26ebf1fb24a581.

153. Exact catalytic turnover and cycle-frequency law

Claim identity: SFT-CHEM-CATALYTIC-TURNOVER-CYCLE-FREQUENCY-010

Question and theorem. A catalytic turnover is forced only when one complete ordered state-transition word carries the same held catalyst identity from its entry state through every intermediate and returns it to that exact entry state. The cycle frequency is the exact positive relation between counted complete return words and counted parts of one held observation interval. No turnover-frequency equation, continuum rate, Michaelis-Menten or steady-state premise, stochastic cycle weight or fitted catalyst efficiency is imported.

same-held-catalyst-identity-at-entry-through-every-state-and-return__complete-ordered-state-and-transition-word-closing-on-entry__one-complete-catalyst-return-word-is-one-turnover__completed-return-words-per-exact-positive-observation-partition__five-structural-states-with-four-separately-observed-statuses-distinguished__all-favorable-adverse-control-and-unresolved-records-retained__complete-497-record-article-supplement-movie-archive-ledger__value-free-497-record-identity-seal-and-depth-independent-cycle-successor

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001
- SFT-CHEM-CONCENTRATION-DEPENDENCE-RELATION-002
- SFT-CHEM-TEMPERATURE-DEPENDENCE-RELATION-003
- SFT-CHEM-ACTIVATION-BARRIER-VALUE-RELATION-004
- SFT-CHEM-TRANSITION-STATE-EQUIVALENT-BOUNDARY-005
- SFT-CHEM-COMPETING-CHANNEL-BRANCHING-006
- SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007
- SFT-CHEM-PARALLEL-MECHANISM-COMPOSITION-008
- SFT-CHEM-REVERSIBLE-KINETIC-EQUILIBRIUM-CORRESPONDENCE-009

Generated grammar. Generate the literal product of catalyst, cycle, turnover, frequency, state-status, evidence-status, provenance and prediction forms; decide all 256 candidates only from admitted exact state-transition, complete-channel, sequential, parallel and reversible-graph laws.

Boundary: Every finite source-ordered family of complete catalytic cycles with one held catalyst identity, two or more distinct registered states, one exact transition between each adjacent pair and a final transition returning to entry, one exact held condition and retained observed, structural, favorable, adverse, control or unresolved status. External testing binds all 497 pre-registered source records: article landing and unavailable-PDF response, 106 supplementary pages, one 1,604-frame movie, Zenodo metadata and all 387 members of seven archives.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *same-held-catalyst-identity-at-entry-through-every-state-and-return__complete-ordered-state-and-transition-wor*

d-closing-on-entry__one-complete-catalyst-return-word-is-one-turnover__completed-return-words-per-exact-positive-observation-partition__five-structural-states-with-four-separately-observed-statuses-distinguished__all-favorable-adverse-control-and-unresolved-records-retained__complete-497-record-article-supplement-movie-archive-ledger__value-free-497-record-identity-seal-and-depth-independent-cycle-successor. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
catalyst	same-held-catalyst-identity-at-entry-through-every-state-and-return	selected-participant-or-renamed-product	A selected participant does not establish catalyst persistence.
cycle	complete-ordered-state-and-transition-word-closing-on-entry	endpoint-pair-or-incomplete-intermediate-path	An open path is not a catalytic cycle.
turnover	one-complete-catalyst-return-word-is-one-turnover	imported-turnover-rate-equation-or-product-total	An imported rate or product total cannot define a completed cycle.
frequency	completed-return-words-per-exact-positive-observation-partition	fitted-continuum-rate-steady-state-or-stochastic-weight	A fit or stochastic premise imports the frequency law.
state	five-structural-states-with-four-separately-observed-statuses-distinguished	four-observed-conductance-levels-substituted-for-five-structural-states	Collapsing the structural intermediate breaks the complete transition word.
status	all-favorable-adverse-control-and-unresolved-records-retained	deactivation-insufficient-fit-control-or-unresolved-record-omitted	Omission regularizes the source surface.
provenance	complete-497-record-article-supplement-movie-archive-ledger	selected-turnover-table-pot-or-fitted-answer	A selected excerpt cannot establish complete custody.
prediction	value-free-497-record-identity-seal-and-depth-independent-cycle-successor	cycle-value-rate-frequency-condition-or-target-readable-before-seal	Target access can select the law.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 2ce1cf81fd9fb478914b81f89a8754ef3ec7bda0fc9c7015b2db3db9d31eacd7.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: e6eb21d97d89d1019ac7a127874faf4760b4d67d5a386403ce0f525d5092ea3b.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 1ef907300c126efd48f82b02ce545e3b7531117225e1791ccb270e5ed7557ffa.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 7b64a0247ea7d3022f9b791033c6fe2633b94538665cab0765a52be9721ec1bc.

Independent reconstruction. Implementation hash

sha256:837d6aa5fe7c4f188bae741b6453726e388427ab257f0ef9c386161d3fe235e4; independent certificate sha256:7e3c5716707798c6e2110a04dc72132d3b21b2454e2aab9a9c80b9ee2e0c0bb7; external-validation hash sha256:f59c549e01bfb6347204fe4dc36518e763c9cf7a6c936e1fa4bfd6fa01adfcfc. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NATURE-NANOTECHNOLOGY-S41565-021-00959-4-COMPLETE

Comparison records:

- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-001: document=article.html; record=complete-article-landing-html; target=sha256:4b891f64ac9e702e1b3ffd072da44de99625a06093cfb2ec0a9787ccfc9f297a
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-002: document=article.pdf; record=complete-returned-response; target=sha256:ab5950299104b5038ab1901b44ff7fc04eeffe1a5869a990c0c45ce323b5d0aa
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-003: document=supplementary-information.pdf; record=page-001; target=sha256:5b714f6c7688363fb725511d20042a1376900612cb0abc62db01d742f6607a59
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-004: document=supplementary-information.pdf; record=page-002; target=sha256:189f0038fe9d8ffe4ac5e803ad6d7675915f3931b0b0069fe33b8b1e30961ff6a
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-005: document=supplementary-information.pdf; record=page-003; target=sha256:3e4cf672e8ffaaa9adf71cee21f17600beb7c2b450e2bf6762e4ceb8d1201629
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-006: document=supplementary-information.pdf; record=page-004; target=sha256:049fe440d1b7e94131eedd962e347971f6cabee1bd06e041404923c21dc966e2
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-007: document=supplementary-information.pdf; record=page-005; target=sha256:8f30d8a3b2aa8224ed426980be76c991e7a2a1d1243b247724904c738f8d1643
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-008: document=supplementary-information.pdf; record=page-006; target=sha256:c336bc3771da3893368720378ec20a59bce8e4372ff585f1b597ae7a4c008085
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-009: document=supplementary-information.pdf; record=page-007; target=sha256:ea13aa956fa364d7c008003b8ea41fd6a5322bee3eafb7faa0f27e5a3690034
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-010: document=supplementary-information.pdf; record=page-008; target=sha256:c926ccc8c058cd5d00098b71b085dfa5109518292b41614452f510d1c4a72c84
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-011: document=supplementary-information.pdf; record=page-009; target=sha256:a6b9d53a4462a5a64c8876816c66aa9d9c11d5293d97474de03126ce56772883
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-012: document=supplementary-information.pdf; record=page-010; target=sha256:f79b3c92a9b591ca3279fdeef3388a10098a56dc943fb57e425dca9536c7a5bf
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-013: document=supplementary-information.pdf; record=page-011; target=sha256:49c1679532eda964b83930f00830c812cea9739ec78ce2e5603e1186d7cc3741
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-014: document=supplementary-information.pdf; record=page-012; target=sha256:dfb1890753f89f5cea89d2cfa44e49123570c685dd2136f0c455bea41999f612
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-015: document=supplementary-information.pdf; record=page-013; target=sha256:aac46cf43721ef1b61a37f05610775608078a0f430aa51d1f554bf580a37bc6d
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-016: document=supplementary-information.pdf; record=page-014; target=sha256:d5fd708f2108c020fd5b333eb1931eb307c1b47b54c74166c67bf2bd0ccbb2c6
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-017: document=supplementary-information.pdf; record=page-015; target=sha256:158d554cbb92db5de7b96d4823164ae862ca5946f3649f93e33f97098bd27f27
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-018: document=supplementary-information.pdf; record=page-016; target=sha256:3bdccac5bbe1f26a40818b0480a0b8fff8f424154359eb5ce7548cce58a5b87e

- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-019: document=supplementary-information.pdf; record=page-017; target=sha256:32147b17a5c587db14fbadd67ad521362822dfa96d4332ae6d9a7fc50141177d
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-020: document=supplementary-information.pdf; record=page-018; target=sha256:8b13e72c16273251b9fdd5488015065c2b3d0a9bd748b7c42eef3396f6036646
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-021: document=supplementary-information.pdf; record=page-019; target=sha256:c62fb7c8ece24b2fca195501cca7cbdc697d182c679558dff62644e54f0a9673
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-022: document=supplementary-information.pdf; record=page-020; target=sha256:1d687ba5219b839a91efd514c79d9ea559dcd7ce039cb72fb25d609047200a4
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-023: document=supplementary-information.pdf; record=page-021; target=sha256:fb1805b57b63976de96730874784488679592b92f994be728474627da17b78e
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-024: document=supplementary-information.pdf; record=page-022; target=sha256:57f9fbc203563ba3339673e169f9d44de980255cb25cc588f04de9714dc80eeb
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-025: document=supplementary-information.pdf; record=page-023; target=sha256:4946cdb2232f8af9fcc113d4a97a4980becf3c4c9f08be79e9b673c1c264b616
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-026: document=supplementary-information.pdf; record=page-024; target=sha256:12c4f0c2037411d4757090f599858f95f954506d9c94111cfc93e8291320c097
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-027: document=supplementary-information.pdf; record=page-025; target=sha256:e338f3509d8cca1ed06412ca3ca63b384efd35b51263d5d75ca539567469127d
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-028: document=supplementary-information.pdf; record=page-026; target=sha256:19a0b095687962fe67907650e220e43da0d19688696c9bac9f5934591bcc62b8
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-029: document=supplementary-information.pdf; record=page-027; target=sha256:86882503ce19129a1b2c4e6fc443db982e68ed6c67b70b526831fcdc239532ca
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-030: document=supplementary-information.pdf; record=page-028; target=sha256:2ad66ce5319522a10dea8fafad626db31df7d509016e5b101b51c2e800a54a7e
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-031: document=supplementary-information.pdf; record=page-029; target=sha256:a257346a0109a16ccdfdc376596c024e14e6f076db9741a6e59438be975f1364
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-032: document=supplementary-information.pdf; record=page-030; target=sha256:124597a5d156bb45e490c01314e70accdf29a37a247710f9a10a2f4df6ff7b27
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-033: document=supplementary-information.pdf; record=page-031; target=sha256:2ee128ea66acf87ce33203d0b82ea9a1b46ca4025b19727b72dd380732390d66
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-034: document=supplementary-information.pdf; record=page-032; target=sha256:a3a83f6700c67e89702ab0338822d05278f8c5ac2f1d18e341effde1629c3676
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-035: document=supplementary-information.pdf; record=page-033; target=sha256:7c5d97f3f9d8954fbc0ec616f2492a5a9c097018cb833f715d5639d6a8260f0e
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-036: document=supplementary-information.pdf; record=page-034; target=sha256:1e6caf1b4585164ce6524e07596381cbe1161d326f7e7999d65584550a69ca3d
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-037: document=supplementary-information.pdf; record=page-035; target=sha256:adcd44610a23451786ee3cd9f65666d76cd12d3403606a812e61d01fa91a0774
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-038: document=supplementary-information.pdf; record=page-036; target=sha256:544268746bee47546e763bab09aae6e89623eb1a6dcae2809705ca33b9dd4e5a
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-039: document=supplementary-information.pdf; record=page-037; target=sha256:665315b880e3d09d051412039c8b576089ed3f44e94c8f1698df3493d8eb0061
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-040: document=supplementary-information.pdf; record=page-038; target=sha256:e60afad43067a8f26b3f29a041ca02784b317d4acf11422e65a2a9f13e52114f

- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-041: document=supplementary-information.pdf; record=page-039; target=sha256:9597864a9dc45964d469a821aaaf21f5fa4728620ec71461da175aec7e146684
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-042: document=supplementary-information.pdf; record=page-040; target=sha256:6c5f48278a3d00d78334432df7600709821e02768a52d590a4aafdb02c36356a
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-043: document=supplementary-information.pdf; record=page-041; target=sha256:3c94faf64700cdf7a24200ec0986da7f124b1c0c371ad93f7e6c494cdb8ea5f0
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-044: document=supplementary-information.pdf; record=page-042; target=sha256:97451fbdc52a6a67ef49396bd3bfaffa07f184322a0313bb4839b6c94dd02c6c
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-045: document=supplementary-information.pdf; record=page-043; target=sha256:a123e09565cf6fe91f525020719408c77da133362f2a8a51da4e9463202490b9
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-046: document=supplementary-information.pdf; record=page-044; target=sha256:a355a0d17e15059cd5612a999f75c8027e36d54570eb9f68afaef687253cf379
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-047: document=supplementary-information.pdf; record=page-045; target=sha256:d50bc883ca3c1db9e084517ac0a43651e9e2d12d31631f0f63efca6a7e571519
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-048: document=supplementary-information.pdf; record=page-046; target=sha256:b60e8a35bcf2c515221085e61351fd54ab957003d157eabc5b38258d872571df
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-049: document=supplementary-information.pdf; record=page-047; target=sha256:96dfbcbab930e1da00656a907166e6ebcbbcb08b5b31036db60cba5596d9bef7
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-050: document=supplementary-information.pdf; record=page-048; target=sha256:2e50f9445edee5a51dedf220714d5c050dfaee09a8a87318eba80fe4ac075c9
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-051: document=supplementary-information.pdf; record=page-049; target=sha256:eee8c7e824d1dd1538a116b4caa597b4a9dc443c538ded9693924d1edce42e9f
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-052: document=supplementary-information.pdf; record=page-050; target=sha256:60857b1f04b94f51e53ee825f5419f53215ddf3bb43e1d8680afa209edebc469
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-053: document=supplementary-information.pdf; record=page-051; target=sha256:185347dea84f7953824e4943481a0ed6ef23259cbb089cb23e8acb0230a252f9
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-054: document=supplementary-information.pdf; record=page-052; target=sha256:be154627a1ca9f86deda805ead28e4adb787d8658e3625e3c4bbe1deab31d919
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-055: document=supplementary-information.pdf; record=page-053; target=sha256:357c5c4170a60a49d7d36942c8b7021e5507ba4c7fd10ca39fcac3175c36bc60
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-056: document=supplementary-information.pdf; record=page-054; target=sha256:bbed73fe5bed6d10fa3b0e13a93de5c0374281912310cf637d20c6513c09a365
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-057: document=supplementary-information.pdf; record=page-055; target=sha256:6f9c61b9e71ec45f15ea6ea6fd458c5b928171713405adf730b003af86d7f7b6
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-058: document=supplementary-information.pdf; record=page-056; target=sha256:083beea3096921eeb3e716a8db78db9e31ef68e906e23dcf8cade45267d6c61c
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-059: document=supplementary-information.pdf; record=page-057; target=sha256:499fb38ceef9fb50297332fb72055c25b28d66800f09aaa96e6454dc412bb9c5
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-060: document=supplementary-information.pdf; record=page-058; target=sha256:29ba225166180d07b402d1a671ff45253408f83bba8a72ef0ebb15a65d11a90e
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-061: document=supplementary-information.pdf; record=page-059; target=sha256:ff307e17e66275fbb490c99cc7ae4c6c317c52ff36a0f11c825c9b884e1e17f0
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-062: document=supplementary-information.pdf; record=page-060; target=sha256:27dc35b5df185bba05c73b2e744b602859aff64d023bbf5be246b2defc660523

- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-063: document=supplementary-information.pdf; record=page-061; target=sha256:c0c37ba2b46c4d94a4540e24816827051d0530d15dbb13fdeb4143a541b1bc1f
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-064: document=supplementary-information.pdf; record=page-062; target=sha256:d0a0bf28326ce932a2ddd0312fe673aaae5972b4c7262b7cc445096d7fa030e
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-065: document=supplementary-information.pdf; record=page-063; target=sha256:4e67c330847908f4e8a0ae2b3026501028d8241215ac3a8987355a5905fb92e4
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-066: document=supplementary-information.pdf; record=page-064; target=sha256:90348108fb63276a275ed7198ee90ba669e0c113c84f1ce5093cae1056b2754e
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-067: document=supplementary-information.pdf; record=page-065; target=sha256:1ab1fbdeb0434a14c6952503efb62c5c35e38d865227ffed53457f9467909f31
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-068: document=supplementary-information.pdf; record=page-066; target=sha256:560684f04954dad9929fb3aa2a809928c96fcf5bf8ada45b30858dd80ce797b6
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-069: document=supplementary-information.pdf; record=page-067; target=sha256:eca2c72ab84a0a0549c10da7cabe66b61d636f842f8e39d52311da561eccc504
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-070: document=supplementary-information.pdf; record=page-068; target=sha256:b3fb106f7b713aab5275d52cc2eda61cdac03fceb6338c2d733e500db995103
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-071: document=supplementary-information.pdf; record=page-069; target=sha256:7cf9711d29d10f1d04600b43956e75f2462b516c15cf4720f099202a2632c168
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-072: document=supplementary-information.pdf; record=page-070; target=sha256:60504035c8604f9c4fad63b2c3e5eeb274258fac4df20debc0438c54e1b94765
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-080: document=supplementary-information.pdf; record=page-078; target=sha256:af9b4c896b8832531a45489c8244b5fdf83a85b296c029dde1b371e7c969efe5
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-081: document=supplementary-information.pdf; record=page-079; target=sha256:adcd7d1a23a302bd6498ea98a0ecc797f5e32ce443d02b8be31fa20282d02e8b
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-083: document=supplementary-information.pdf; record=page-081; target=sha256:8a612000f62da1aa69e1c35e759b90c064026aebf415c076ba466c34d5d7814c
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-084: document=supplementary-information.pdf; record=page-082; target=sha256:015484657511961a71c51d9667607ab6ded572760a4ec64e8a1499eb88d8a3ac

- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-085: document=supplementary-information.pdf; record=page-083; target=sha256:9bed408e8d8d1e521e7d0eb3ea2c7fbfcd5c6c4532e584ad0080c73997981487
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-086: document=supplementary-information.pdf; record=page-084; target=sha256:064a21aef18adb37ef4d53da8178ccc9cfa19562686652bbe0cec2d56c63d8ed
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-087: document=supplementary-information.pdf; record=page-085; target=sha256:7dfb476fef4ce7dffdd6361306f2082784c2ca55932f9f6fa558199a74d0df51
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-088: document=supplementary-information.pdf; record=page-086; target=sha256:255f9962c83ae121082d3aee73f095d1f575d06a421b6066800cbfc7aa1a7b01
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-089: document=supplementary-information.pdf; record=page-087; target=sha256:3fc6cc868fcc33602b3df5c5a74af3380c4bf0d3e070c62d6769c5b4def982ad
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-090: document=supplementary-information.pdf; record=page-088; target=sha256:e9616a238e347c46f6f42c5dfb3050240aa551e0b7c9830d3fbb28ac56035879
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-091: document=supplementary-information.pdf; record=page-089; target=sha256:065d277d376ecb5df11948c04fdd6ffdf4e3bcce71ae1aa1b4979eb0e7d84594
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-092: document=supplementary-information.pdf; record=page-090; target=sha256:59f53e3b6cba84c02199d8e4b910a9b4319e9d7419a633ec246e4271da9321b6
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-093: document=supplementary-information.pdf; record=page-091; target=sha256:0cfc34a88e004d0f004d02736d4196cdaa5c84d77f22cbaf79dd7bd7912b0302
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-101: document=supplementary-information.pdf; record=page-099; target=sha256:d9adf25a373de96b2c9068f98c5ac8fbe9d0fa680dcb16598ea25f81a432b9f0
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-103: document=supplementary-information.pdf; record=page-101; target=sha256:7029a5c198f05a041b8302ba61c89d5d1cc37d2bdc9442da0e2974a91566766c
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-106: document=supplementary-information.pdf; record=page-104; target=sha256:8c33dc4a760df8d595211c6a750e7933abe314e180923a8c2cc406b7dac31bbb

- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-107: document=supplementary-information.pdf; record=page-105; target=sha256:f791a213c2c096e79c12248c365f748fd82fcc5aa520aee9a5b35a3f10067cdc
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-113: document=supplementary-data.zip; record=SI_source_data0708/Compounds_1_2_3/Compound_1; target=sha256:6142d431523908676c786c1e11db250db688a7a2e999f9cbc382c09ff6aa13dd
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-114: document=supplementary-data.zip; record=SI_source_data0708/Compounds_1_2_3/Compound_1/1; target=sha256:8b794a57155c1af26aa7e79a86a18fc5285d3ef926e7f07026586ee888d7481c
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-115: document=supplementary-data.zip; record=SI_source_data0708/Compounds_1_2_3/Compound_1/1/acq; target=sha256:a7321ab5d2b6c168c1729807437e68a1f456ae7d0c65eb4d5c2965eb98bc1b6f
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-116: document=supplementary-data.zip; record=SI_source_data0708/Compounds_1_2_3/Compound_1/1/acqus; target=sha256:93e16df7f0d57c4648285090529dd3cb3ac53971ae3dfe68438d22c7f2c774a4
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-117: document=supplementary-data.zip; record=SI_source_data0708/Compounds_1_2_3/Compound_1/1/audita.txt; target=sha256:8e4a7dc431dcb3996e625bfd5deda22d2109aab1bacfa50d69f04564b86fe2ab
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-118: document=supplementary-data.zip; record=SI_source_data0708/Compounds_1_2_3/Compound_1/1/fid; target=sha256:afefaf417920ec834337ddcb533e1eed04b602b6868a63d8222de76e97bd179d
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-121: document=supplementary-data.zip; record=SI_source_data0708/Compounds_1_2_3/Compound_1/1/pdata/1; target=sha256:ab9f2a71a1ba7b66d6d5e48593ef9e2a14bc251f924887f54c3a0ec1e3bc46ad
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-122: document=supplementary-data.zip; record=SI_source_data0708/Compounds_1_2_3/Compound_1/1/pdata/1/1i; target=sha256:2e4e2314d739eaaea8de5dce46bed2fb5ba062a628fe0a200a05b92786baddf3
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-123: document=supplementary-data.zip; record=SI_source_data0708/Compounds_1_2_3/Compound_1/1/pdata/1/1r; target=sha256:9621d7e78aba4ba1438b7b75db701d390ee3b7af365b8eaf2c2a7732438cd5a8

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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-126: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-127: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-136: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-137: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-138: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-139: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-140: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-142: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/acqu;
target=sha256:618741ab283fea2c28c14d7edab100193af872dbea16bef08a2c5c2e787f104a
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-143: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/acqus;
target=sha256:3d8ca7cfc6c40d13ac517000e4e7df852bb0d20f0b653391cb69b52ebd771fae
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-144: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/audita.txt;
target=sha256:ad6b70b22c677d8f74b2b85966c4e09d7cde4402aa7c7925777fe9fc207ab715
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-145: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/cpdprg2;
target=sha256:23637b95408ca7e33b384c807f253c7cc69f7d4fc72ad0340157732bce5a1816
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-146: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/format.temp;
target=sha256:a6ae15dd0c484db61948b66b49523d3ba89a8fed858a7421d973e492bfaa8ebb
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-147: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/fq1list;
target=sha256:92bfd6d41f1b00108f7aa51355d9ec189de412fc70c03985edd931568b607336
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-148: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/pdata/;
target=sha256:e0cfeff7dd0673bf11a9968a7972b54af25fff33a8220b40e07c2642e2899dc2
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-149: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/pdata/1/;
target=sha256:ca456188a0a83f952d4e954cf2c2b0281a85c362f6193d5d6aad3fbd1fb7c93c
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-150: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/pdata/1/1i;
target=sha256:43c7d55e428a901dc48e047f4d07c0c7e38be8f4faad7bdf537bb70e9c70b5a
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-151: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/pdata/1/1r;
target=sha256:42736e97cd316182fc7b7fdee26b217a78963e36d11544a940459811ef3e6322
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-152: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/pdata/1/auditp.txt;
target=sha256:237a8ab69b48d39f4ac966c73f00b16638843e88eb4edd990a11ed846ed5b111
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-153: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/pdata/1/intrng;
target=sha256:bfa1c05278a8325ba5c7f8cff57d7ae02e050a70c100f6a61d08cb2f73a9105c
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-154: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/pdata/1/outd;
target=sha256:3fc3cc30825a9489cb5b51afa3e9fd5cd0a51f26466cc12b5d1a345db87416e3
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-155: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/pdata/1/parm.txt;
target=sha256:e66a8dad7965bcc27acc81b9ec57e034faf5d8df5935d63fedf850264d16ea6b

- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-156: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/pdata/1/peaklist.xml;
target=sha256:33b019388748dc7e6276d0698d19880ee72bb4cc949f6bdb570e4b401f0eec0d
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-157: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/pdata/1/peakrng;
target=sha256:1cbdf97422c896df5836ac6cd377e65a0809d91f4fdb9db18f01ad54481f2ea8
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-158: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/pdata/1/peaks;
target=sha256:aabfb80fc6a3e313368ce899e06fe678ba9e3caa69958a814edc88586b2dac32
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-159: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/pdata/1/proc;
target=sha256:2c2724034568300549d460da4d841336b74c391a565634d217dd5d7be24b475d
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-160: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/pdata/1/procs;
target=sha256:fe7f58f2858572888e802f14730da0529fe29328faea86e17b15d5d35e20d5c8
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-161: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/pdata/1/title;
target=sha256:07fc99fce4cf3318e5b5aaaae9921c4b5ed4a9dd582ca0aa7cc6cf33452ebdb0
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-162: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/prosol_History;
target=sha256:2d794ce35dc66ddf642ec581ca96f31afe3c6f381c1a6e0fcc5596d5c372dc83
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-163: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/pulseprogram;
target=sha256:27f787e148de71e63b0bfe7de091a67055d097d1dc2506ae24876e649dee4537
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-164: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/scon2;
target=sha256:10c4025c9e10b6894181a67b57dbe0523ea32747f09eb25e0abb3cf8c62adfbe
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-165: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/shimvalues;
target=sha256:9b18f7624ef460898dad10ee03d9632a5bc8e1e1d087af98eb9bcfc5256dfd8d
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-166: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/specpar;
target=sha256:432d56626e4efbff8188b0bfa8c0ff01cbc4365d37e7065e085797767b922d58
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-167: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/uxnmr.info;
target=sha256:2efa12f04cd0938f38de94e01d63485312bb6d3d47cfb6f4ed049dbc0d5b450d
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-168: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_1/2/uxnmr.par;
target=sha256:348980bec6b49c780800a6d24b191b19b82920e637e009da29ad46bef5e98119
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-169: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/;
target=sha256:7e92f269f2b60296fe77f1c38460bc9f3cfb6733452fa741e5f09eab25279e74
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-170: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/;
target=sha256:e0a48d8a6f61383651c6c271fb0087b7167670f428a2ddaf55da60cc8f51cafa
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-171: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/acqu;
target=sha256:178f2f74fe946b7328630b05900c83bc37367e735414ffa381f14bfecfd535a1

- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-172: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/acqus;
target=sha256:ce00f8355c585d8726181bb23afd99806ca975498ef48ff8b064ec54702865d6
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-173: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/audita.txt;
target=sha256:461cc21c112d7258b25325d4c4ebfa13845ac0f45aeb686c27a305f4baa10dd1
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-174: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/fid;
target=sha256:c2c7554e5769e38cd8eb5700948b43f66666fadc9b732f82a4b50b9d8a8e62c6
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-175: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/format.temp;
target=sha256:5db92b06606bd751197c4b405fc3e1a0dfeec2ea7453cceb5487c97ad0bf299b
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-176: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/pdata/;
target=sha256:d7336999ae8b4e43899e1452f026035f2fe791e80e068bab24ca9b100cb5a1ca
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-177: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/pdata/1/;
target=sha256:e20e09b01ab3e2cfc4c46008504fdb8be175111b7006daace13281ffad90776
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-178: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/pdata/1/i;
target=sha256:2fb12e134b31fad6c51b6f53f3d7d47a11ab9ed0f7b6573899e161b38e79c259
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-179: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/pdata/1/1r;
target=sha256:8df9bc339fcc533dc5bf0834c03f76cf430901cb9df48a04cf6a995160ccd7b5
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-180: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/pdata/1/auditp.txt;
target=sha256:4f6888b7a0ba31fe6c6af5f32c5d3217fe86dad38d25c906605d9a0cdf2121
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-181: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/pdata/1/intrng;
target=sha256:7d54c75e99e609f4b954a2ffe6cb7bf7a289ac9d869415f5d5aedc25fdec1b72
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-182: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/pdata/1/outd;
target=sha256:1c41af27b67a0baf37a2c3b005520f54b0df731d95c6a4a56b963d416abd076
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-183: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/pdata/1/parm.txt;
target=sha256:f7ed20abfcd8e25232b1dfc68894609f47c4f60339b951e693bfb56ea0adf663
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-184: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/pdata/1/peaklist.xml;
target=sha256:f879e64f224da34d9515d6773c743889de3206e9f21368a41e882ff296ddb7c0
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-185: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/pdata/1/peakrng;
target=sha256:205cc5298450ef97e7ab6c39e6f2d10794b6125c926dff2abcb25406e373d63b
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-186: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/pdata/1/peaks;
target=sha256:9b465bb8ca4cd1bcbcb0c89d78ea242c2a532dbd4286875084be2a12205a0018
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-187: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/pdata/1/proc;
target=sha256:fb37e4b15eb0400ac882c73f773e6b93765f5dfe8bbd3eb6f6173b9722af02bd

- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-188: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/pdata/1/procs;
target=sha256:6a7b4c9d75dae0844340294b30b9c6ba1830c6aa9797931f5ed85d7ceb4ec338
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-189: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/pdata/1/title;
target=sha256:188c7bce0ddaf35316b2115ebf5a92b3f1101b486d8bdb18edb794cb680f6a81
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-190: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/prosol_History;
target=sha256:269cebbb6aaa3a24c61c56842b07067a7b40857ab835e39b09a662bad4c26788
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-191: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/pulseprogram;
target=sha256:29e0db46af0c2b757264edf837cb6a0ea054278ef47fe7866ec975c071756aa0
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-192: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/scon2;
target=sha256:944d1c05df1cdf9214cb605deaf5743519e67bfe7b2cbcd8afbd7887110ed965
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-193: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/shimvalues;
target=sha256:c3bcacc004ece6ed71bb6b2c0e114736de0ee5978a2e40b6742ec96d1accbb
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-194: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/specpar;
target=sha256:3af7178c227a29e97bfa5500d4f28da7f087be817d4c5ba11ab4f1ea1ddce587
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-195: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/uxnmr.info;
target=sha256:078d60683627eb442943567238b985831b78b84460854452fae258161bf0ebb3
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-196: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/1/uxnmr.par;
target=sha256:2a84cc769fada6acd722a91fc65fbb03a72c1df00ff4b3a3333db855d2ae1c9
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-197: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/;
target=sha256:d7aa373c460e0e8c0fe4514fb53d22fd4b08b9ec6a2df7044de419c116f33240
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-198: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/acq;
target=sha256:f23fdceca5c96ba2eb0f5554b5f76198996c1e6fcfaf24a81a7b58c52d562392
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-199: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/acqus;
target=sha256:a1c190f2122f4d8c73284a8863b2b9c01a542cb636640805256650e40b1bed3e
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-200: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/audita.txt;
target=sha256:91566dae01feb2cf0d0fc011318fba430bab5e02ce4d910b3a567e98cca384fc
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-201: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/cpdprg2;
target=sha256:a88122732be09954766ac0502e17ce48ffd7a224dc3759f06a8ffdf134ac54cc
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-202: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/fid;
target=sha256:0ce9011a176392e4e94f58dff7d5a010c3f2f1f808fa813712652b5e513e4da2
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-203: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/format.temp;
target=sha256:4e61608742d46604d622cdefff2f2ae30358d095090d8d42ae732fe9c59dddfd

- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-204: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/pdata/
target=sha256:c343ec651cb9d3a1e004cf1b92e4b88b29eff8ade9721dbb38c55733dbf43037
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-205: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/pdata/1/
target=sha256:1c6f9d1ffa0ea0c242d79e3facf30c7f1a751b9e10d12f8773308c4791ea29ba
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-206: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/pdata/1/1i;
target=sha256:8696960f439006264071a67229157bff7bf75392c3f5dd26a934da1fa9dd7921
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-207: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/pdata/1/1r;
target=sha256:e762e51437079ec9b1402997210cf9280172af5d1793d3c98c60579a85054057
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-208: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/pdata/1/auditp.txt;
target=sha256:703a396c3f504117fed6c2e949dff54646c3b5409888824514e10c06dd95565
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-209: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/pdata/1/intrng;
target=sha256:c3bd2bc0a4c7ad37505d3fe965c812d307ed38d9226e44c1b27a74cc739337f1
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-210: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/pdata/1/outd;
target=sha256:18436179651e9e436840957d29be2aec1500b547baee2e0375569616340c6806
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-211: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/pdata/1/parm.txt;
target=sha256:ce1005b665b544ac4137e60b254cf060e133fa1a997205ab3c387da1bf1c8692
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-212: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/pdata/1/peaklist.xml;
target=sha256:cdd6ff9c274a8b1a7a2ebc5a13e6fc726027c8244e401f3cc519d3b6a5dfe6f2
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-213: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/pdata/1/peakrng;
target=sha256:783a29c464bda5bbd1ffa37acacd8a12a0ccefd1160cac29a6c5030cbcd9112
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-214: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/pdata/1/peaks;
target=sha256:7d6ff7a2cce364ed7bc5871c16849ae3462f3578f21adc7469ba1c87696f5028
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-215: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/pdata/1/proc;
target=sha256:d5ff586574ea7503d6dad5eb8c5adcc83f4ae4620179a396068b596f68849e27
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-216: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/pdata/1/procs;
target=sha256:f628bf397a6a4482d3cf46821c7e1c329df1745728f2db6fc6994575422ee0
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-217: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/pdata/1/thumb.png;
target=sha256:ea6b17ce353975577be5eed150dbd05d2cabebfe903b8acc7a640c724320e93
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-218: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/pdata/1/title;
target=sha256:f40afcc8b13c36bb81a72354e99cdf22b68f541276b752f677ecb2d77ca0181a
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-219: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/prosol_History;
target=sha256:1911941882aac8d1ad9f2f90524d2334d36237dc5b20ed030386ee59acc6d185

- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-220: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/pulseprogram;
target=sha256:e291ced20bf40b2328669f85334601b10baab13060f8faecc88612457652f61b
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-221: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/scon2;
target=sha256:9ac831f03dda6e3fa4f92a83eeef5f8d6e8af3976a190df6752810d74ddf5590
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-222: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/shimvalues;
target=sha256:8da98f83e02d9c27754ca963125f3d394d8188539dd0f86dc57ab5967923fcd4
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-223: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/specpar;
target=sha256:7f3a76686b68dfe6ed4122e56bea7aceb644e466882da91338bd55e8a6e42c1f
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-224: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/uxnmr.info;
target=sha256:dd25785de5480576877b87f6660218d0bb20dfe2aa1b30fe872a710d7ad54310
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-225: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_2/2/uxnmr.par;
target=sha256:79f864dcc958c1788977d9f05721f946943f68576c07dc3ee8b7d235555fd90e
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-226: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/;
target=sha256:673732b97a84555ba9ad64b065cce8d44c7f0246e2ba173264322165503f7d96
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-227: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/;
target=sha256:080cc2dcfe662b047cc9e6c6163bc1774de4fcff1a462228bd1a74d7e7c664a8
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-228: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/acqu;
target=sha256:b92218f6869c3c6004e42272989a4c399e759bdec200080e7da210341ea340e9
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-229: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/acqus;
target=sha256:e4c990f304e61928b2945d5ae4186c49e0a34e6cc943983bb09d87d76eb8fdb
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-230: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/audita.txt;
target=sha256:524e93cc104722d480d2e50b4dd2326432766e754a871c800f1651a18a4c8d48
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-231: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/fid;
target=sha256:73666ce7735ac1628424e4c7cd1956685d62c82e8ca91e6310018225bdfdf6a45
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-232: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/format.temp;
target=sha256:e103131ac87819d18ce9f99ac897c49c4bc8c21977ca4929aa931637c5d72ee3
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-233: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/fq1list;
target=sha256:291c701a200b157ceb7df8fedccfa42a382e2f53cd36cfa68e90dc73edb7a2e3
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-234: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/pdata/;
target=sha256:ef2dc7d945463e4438e682d3ee6ea5bcc1f8b4b175cff63ef700ee3b9faa235e
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-235: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/pdata/1/;
target=sha256:afea09e2d45c8424ed80408cb4c5966035becfe1c76ace162b96b1e000b30776

- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-236: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/pdata/1/i;
target=sha256:6d6ba8d2ca5018a383c2655caf3dc2ed4ea5b5e4fe8294930462ed27ae0a74b7
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-237: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/pdata/1/1r;
target=sha256:2fc2e25ae528530d3def3f4a1a994b90c38346e34724dd09995ca95cfe23e762
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-238: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/pdata/1/auditp.txt;
target=sha256:29c5d4bc67b3ad11dd43d28950352427ef1648f7fa5077b4246c4725b9e4159f
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-239: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/pdata/1/intrng;
target=sha256:10c4795de2bfb93e61dab48d3f5d612fceeaa107eaa19791b0c58e55b89bb0ef3
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-240: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/pdata/1/outd;
target=sha256:834c43785aa66b814c3f1830de29a30d8c865b3fb543b2d95afec90c8ef89fca
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-241: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/pdata/1/parm.txt;
target=sha256:4718f5b88d1e0cf384684eb69d290e5c9553ee6a7def8c2f972038af84301515
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-242: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/pdata/1/peaklist.xml;
target=sha256:3c3a886b1c064db10ca267b7f0b2161538b05a57ced5525cd754e29d38c1329b
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-243: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/pdata/1/peakrng;
target=sha256:cda0a53f661065bfa15502120fc0980e3b92357051aedbee5891fc7eb748a68b
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-244: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/pdata/1/peaks;
target=sha256:2f94499f7213b7172941014775fcf31d3a109cf9d993c666844baedc67a38313
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-245: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/pdata/1/proc;
target=sha256:4aeade2019526120208bc086f406773ed0e1cd17690a28a831cacf42c5158c2d
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-246: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/pdata/1/procs;
target=sha256:1e0d58b18d88d2be2a30e7b267e7e5c99eb3fcf2646ba268717ffaf9fd6ab653
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-247: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/pdata/1/thumb.png;
target=sha256:4a10eb37f356b75966b0790a8d61cc7830f345cbdf2062f1b12fc4e607cb73b9
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-248: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/pdata/1/title;
target=sha256:be8d1b85388e88c0ea1623604ca7500c395f2baf6f1813f23dd54140475cfa1c
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-249: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/prosol_History;
target=sha256:ec5c979b2e56359f80dbefd8adf39cbd466e11405995957d75d0d9022935059e
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-250: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/pulseprogram;
target=sha256:319c2fde38fac9bae31d07064ae3d604554e1ceab01fc1e19a4abb4c06fcedfb
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-251: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/scon2;
target=sha256:a0e8d8da411da159973a5b310884e692e50b6f875caf90553d833fccaf4f7e0c

- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-252: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/shimvalues;
target=sha256:11fb127f560b20653c3e16e0bdec1611de4e9f0179dc0c8f4cc023fb8e415c08
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-253: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/specpar;
target=sha256:a5dc57811797610870a1323c1f89c7e48a0a7b15f5c6c3fc5566e8c50bb555a3
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-254: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/uxnmr.info;
target=sha256:8fff12a6d7b4ca2b40f92773d8a32935dbf0e70fd0858a570a258b32c3d3de48
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-255: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/1/uxnmr.par;
target=sha256:48a60081f6f12028f6a2185c6e096fb1dd98510b1b4d7ab4577b9b9e76ed128b
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-256: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/;
target=sha256:79cb58c26cae0011630c8255d339b74ba8bb1a45c3e85fea896f1db4d37ed34b
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-257: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/acqu;
target=sha256:dfe5f0286138bdb22e25517a1c4d9e2e20a5a9734b54f5ad0915ee83abea1b08
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-258: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/acqus;
target=sha256:bcbdc7bf1e46a396183e25ae9e49f1282ab9530dd977f64acd06adb5f13bf6d8
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-259: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/audita.txt;
target=sha256:2aad2a75a36065baac69f6cabbef36e4f721dc3a965adfd6fa9a9c1cb292856
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-260: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/cpdprg2;
target=sha256:7ff46c41c8bf800eae4c08a6e22754cd6416e071e2782becfa5bbf025a1ddb1e
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-261: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/fid;
target=sha256:352eedae984d16c771351cf926f6874292cdab542b1df5a213e9cad7dcedad6f
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-262: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/format.temp;
target=sha256:2b77c9679d08cdda04e4be4da0b7990675c7ef6775147efd851d9334ab9570c4
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-263: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/pdata/;
target=sha256:6a24e198c1553fb244aef5140ec5bb0e09df083ef671af594a222ba3bf0946a
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-264: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/pdata/1/;
target=sha256:d3f511680a7409f97711cf30f234cf61476e088ec10fbe079548fb114dbafe0b
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-265: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/pdata/1/i/;
target=sha256:4c86f8164ff619e8e90a390ba9b16655fc6cc059bed5315a6b3d49342a8cd7ac
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-266: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/pdata/1/1r;
target=sha256:116694082d57e918d02dd46cb51adda8b7cb4eaab40566a1a3d8fc6a7c6cfe02
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-267: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/pdata/1/auditp.txt;
target=sha256:b73cbbede149ce4cc36663073095a49a66a25f76dd683fc7a8b6f6583f40ba13

- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-268: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/pdata/1/intrng;
target=sha256:dbec13edaeb91337e869d930227cccd93694510d077170de5d0a759bc9d6df72
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-269: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/pdata/1/outd;
target=sha256:f393733d4fce4802a3cd3044e38a9680408016b76675a7d9e44422dfee87b00c
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-270: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/pdata/1/parm.txt;
target=sha256:febb6d79d284ee507fd83dc375b68af97d6ba5d98e2738825b14e661c993d3f
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-271: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/pdata/1/peaklist.xml;
target=sha256:6f53c312667b36082622fafc6b7a03d5e9382fc67bf03b749b17181a22abff44
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-272: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/pdata/1/peakrng;
target=sha256:7324ac4c756971108de76a67d12219aabce2d1910d1e4fb1f00e26426372a981
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-273: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/pdata/1/peaks;
target=sha256:8c8840b5deaf954d46d6a53b2808a95a90d7a52a0d6203b3a778b1d73b22a7b0
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-274: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/pdata/1/proc;
target=sha256:36860a5c15cf4748f017f061113565d0daeee416238166dd73fb613554a32177
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-275: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/pdata/1/procs;
target=sha256:83f07bbe933011f18d96155a78ce347bc344388ddc8284b082c78e047241f758
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-276: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/pdata/1/thumb.png;
target=sha256:07b02f003c1d03a6efccc56e6999073649dfe54922def29ef54408e572d8fcd
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-277: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/pdata/1/title;
target=sha256:65a68c01946f6a01f75d33f3e310e574fa091f2563b5f5c98b1c48c1590e7b8b
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-278: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/prosol_History;
target=sha256:00ac95da50cc403c1b39229a9a7e188c170e6d8c4160704fdee9f6857151b902
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-279: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/pulseprogram;
target=sha256:6d45f1f520854832d608afa38d21278b56b64f638e664e994d7e3caa611b5e9c
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-280: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/scon2;
target=sha256:7e82deac144da8a467956a32f582e303fafd2920bf752c372569c4f421f0a71f
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-281: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/shimvalues;
target=sha256:46be6b91cf14b9765b77a4f4d39585a104d4a2d96129b55b67570189df695b72
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-282: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/specpar;
target=sha256:169bebd9ab57bde805740c1886d59392c1abbe3ef06025e53b7699e15e6871f8
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-283: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/2/uxnmr.info;
target=sha256:b06e01cb2577a109e6a84978ab5b8b905a9e945ac15cb72b532cf74efa6137ba

- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-284: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/uxnmr.par;
target=sha256:e87bccde9063f9d7a4ec276fa57f92e0e49f98b6253f7906610ab0ae56831ac7
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-285: document=supplementary-data.zip;
record=SI_source_data0708/Compounds_1_2_3/Compound_3/HRMS_spectrum.png;
target=sha256:fab7d49f06cb8981431e5f22a3b8577b081a0f94c279ff66ed599547c9a11553
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-286: document=supplementary-data.zip;
record=SI_source_data0708/Fig.S1/;
target=sha256:5200da48a16ac73ac8748dd2d0d0cce1ef34c399ed79fda97202c4d956c33e92
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-287: document=supplementary-data.zip;
record=SI_source_data0708/Fig.S1/Fig.S1.png;
target=sha256:dc0e19b9b2d56859bd8600fee1ba37e132464a86aad1175a74d89fca1df77bbc
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-288: document=supplementary-data.zip;
record=SI_source_data0708/Fig.S10/;
target=sha256:15fc3e26509193f80cc795753562dabf234330df6076e63135ada782749e9ae6
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-290: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-291: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-292: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-293: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-294: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-295: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-296: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-297: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-298: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-299: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-300: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-301: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-302: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-303: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-304: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-305: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-306: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-307: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-308: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-309: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-310: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-311: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-312: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-314: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-315: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-317: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-320: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-323: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-335: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-365: document=supplementary-data.zip;
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- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-450: document=supplementary-data.zip;
record=SI_source_data0708/TableS5;
target=sha256:5689cfdb803b2d2f8c4bf97c404cc0bcd5d2835c17f41e8f67c01a805f7bafc5
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-451: document=supplementary-data.zip;
record=SI_source_data0708/TableS5/TableS5.png;
target=sha256:19b6b9293a9f3e41e7b7a61f56217e31ae4b793d66eb5c3714f20ece1dbbfd50
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-452: document=source-data-figure-1.zip; record=Fig. 1/NMR data of compounds 1, 2 and 3.zip; target=sha256:d6fd2d815608a835a55ed7552c1f05567083d6ed8c20d86fcb217a4c93d754b
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-453: document=source-data-figure-2.zip; record=Fig. 2/fig.2b.txt; target=sha256:e7550f4a258bf3e9a0fa50fad79e96cdd85e9ef8533d0b792f5a95d61e5f4cb4
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-454: document=source-data-figure-2.zip; record=Fig. 2/fig.2c.rar; target=sha256:82bf29984ef35d35245a402cfe026745342a4d2df96eb9a1d20be0b123e11f3f
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-455: document=source-data-figure-2.zip; record=Fig. 2/fig.2d.txt; target=sha256:4a2b1923696644f5ce0b4766742ab00626ce7292ca2e101cbfe385a08d5de305
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-456: document=source-data-figure-3.zip; record=Fig. 3/fig.3a.txt; target=sha256:142be0b03e85bcc1c1b42f1ec30f393c0a7cae430a4f3e22834fc1f0c868b41e
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-457: document=source-data-figure-3.zip; record=Fig. 3/fig.3b.txt; target=sha256:cbd4dd6fd76d47098b91243690c8b159cd2d40783d84cb448649dadaddfffb520
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-458: document=source-data-figure-3.zip; record=Fig. 3/fig.3c.txt; target=sha256:d377bbd0fb7768a3c4c81970b562ff55e93801b555316251aba1b50836f56c78
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-459: document=source-data-figure-3.zip; record=Fig. 3/fig.3d.txt; target=sha256:a4af5681455e2506f4a049da2bd6609457db899b1990e8d1da6be7e541cd23a4
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-460: document=source-data-figure-3.zip; record=Fig. 3/fig.3e.txt; target=sha256:69fbd18b59010f28835ded7e61762a81e9043954ef82fc7f1f1cd0d2546714a75
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-461: document=source-data-figure-4.zip; record=Fig. 4/fig.4a.txt; target=sha256:88880c2c12f1d795d67ebd60b9dc5641b4dcbbd8b08426e632d97e6a64bab1d0
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-462: document=source-data-figure-4.zip; record=Fig. 4/fig.4b.txt; target=sha256:bee1162903286e1f3c6fe26a097df172aa69ffd0ddfe72ade71fe64ca649b854
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-463: document=source-data-figure-4.zip; record=Fig. 4/fig.4c.txt; target=sha256:1cdac6b2f9c3519cb1722b918c39e55741f14bf91ef1f3abda156d9ab56686b5
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-464: document=source-data-figure-4.zip; record=Fig. 4/fig.4d.txt; target=sha256:4c54f65cf557771ae5c0e9dd622ed026e87b2097a310e0c176e9c3e10d9f0253
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-465: document=source-data-figure-4.zip; record=Fig. 4/fig.4e.txt; target=sha256:6471f6af59bb533f23596fef7a9258e6ee2c9fd42b20a02fdbc10de73948f90
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-466: document=source-data-figure-4.zip; record=Fig. 4/fig.4f.txt; target=sha256:ad6a0c772050363e855aa8819e588c9a9d0afa7dc479d907ec5ecd4f01159164

- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-467: document=source-data-figure-6.zip; record=Fig. 6/fig.6a.txt; target=sha256:9045f4334cb338b1edd8e670a50d1fe4edf4e63ca4df7f29a55bac76140997be
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-468: document=source-data-figure-6.zip; record=Fig. 6/fig.6b.txt; target=sha256:733be4e52c81588b34a786c1056f29e5b798ec9e5ca4c34b87a2a29c5c93b848
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-469: document=source-data-figure-6.zip; record=Fig. 6/fig.6c.txt; target=sha256:288e5292e94c63fef443f6ef549b316dfe4a76c040169b14cf15c90b48332c87
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-470: document=source-data-figure-6.zip; record=Fig. 6/fig.6f.txt; target=sha256:4b50fc00493aca484b8bf819222e4c5f6e2140237d755315fc5b8f20c4a97e6f
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-471: document=source-data-figure-6.zip; record=Fig. 6/fig.6g.xlsx; target=sha256:e22e792b152869e4a39b48152fb59c0e14698b88b879b56b98a1088c8eac487f
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-472: document=zenodo-complete-source-data.zip; record=Separate Source Data files/; target=sha256:955a929eb247f716b58931e368c9e2ed18e3cfca133b36fc5ac81e57a1fee459
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-473: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 1/; target=sha256:38917d6ca216bd539fb7b21d93774a7cbf141b4272abd47e461c73f5d22b9e74
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-474: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 1/NMR data of compounds 1, 2 and 3.zip; target=sha256:6db3d22025f2399f409f31c9eb59b5c29ca013677206bfee49a48cdf9c9ff706
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-475: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 2/; target=sha256:2cf9677d2e1f8e8ddeafe0226d322b3ed9b7caeb1b07ea0427e1f5181b8de472
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-476: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 2/Fig.2c.rar; target=sha256:7a48ec7b7d6c2ddc4ff9d089ad2bb2247e3ad8da5cb8fa5b0b9183faf0660023
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-477: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 2/fig.2b.txt; target=sha256:42ae9a34db21e4b39ed7279685139263c6ab4f62cfca4854148339b17e4a0674
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-478: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 2/fig.2d.txt; target=sha256:0b0d01031c491435e3fe0eeb17bee345bd0bb764bd4b18b7248b1fbf8dd4eac3
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-479: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 3/; target=sha256:cd4bd8f47e4320469e2a49d9bc55020eb5b227990110c17497522e63cee93fb8
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-480: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 3/fig.3a.txt; target=sha256:d7922eee358f649aaf47c4321222321ceef0bb03cbee74d64f7bf90148ef314e
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-481: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 3/fig.3b.txt; target=sha256:b73ad7af9ddd38d9f1b31083cc461617fe79ad6dfac5a36704d5a67835fc3aa1
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-482: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 3/fig.3c.txt; target=sha256:13902c7557dcb08e9601864048ee157b5eed6932d82dd6e238327e6ebc4b46e
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-483: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 3/fig.3d.txt; target=sha256:70d3d16ec10e992dbb10d9f0a2935acfb057460f758fff7ffd27d1daee07518f
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-484: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 3/fig.3e.txt; target=sha256:b9d77028ca73852ad7f785b45db7b9d51117cc692cd9bef4d9a658f36ab9d177
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-485: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 4/; target=sha256:59717efb0c456205058acb98e5726aab0d69e06eb6c2d2852fc5c41cfde40862
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-486: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 4/fig.4a.txt; target=sha256:2ff65a5fb420e91bf8d34c0eddc9cda652b097b4e0729cef0f5bae1a48f9802
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-487: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 4/fig.4b.txt; target=sha256:30019e41851c6a8521371130f7b0d3d5a4ec29713f9f0efd2c6ce0b5a9a1bae4
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-488: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 4/fig.4c.txt; target=sha256:f8a09e700920f5b25c0c93e1e788b28b1a54b76fac69c2aeddcbce72ad5f8111

- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-489: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 4/fig.4d.txt; target=sha256:66093a3142cc400962dbf8ef037b8add90e3500ec42e52e202316942c312fde8
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-490: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 4/fig.4e.txt; target=sha256:eea0961ea4372e745f7cbf953f02d835f3822f071bd1f727d0e5691eadb36965
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-491: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 4/fig.4f.txt; target=sha256:9294ef3605ebdbdc8c2f94c3c7e8e08a1d5c00be877b18798940dc20094749d
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-492: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 6/; target=sha256:f03339cf97b97d8b75d18401e16db72a05d33db4cc346d63cfeca9f01ce39ee3
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-493: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 6/Fig.6g.xlsx; target=sha256:905ecedb77bcd1c5192ecd4e6b49d9678f45ed027b9c8640fbb9bdd4629fb56c
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-494: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 6/fig.6a.txt; target=sha256:bcd7ab4624c0d4afca3b302b846b8156138daf65ef9b9298a68966fdc1a01d9c
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-495: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 6/fig.6b.txt; target=sha256:88cbd6b2ca251717c4002aca3a55370c84e72a190677586ae3d329a1f411abe6
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-496: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 6/fig.6c.txt; target=sha256:3af6fc4794150846dcb72cf40af78dbc0f8f909a28d211dd095cd0cc4fec0a90
- SFT-CHEM-KIN010-COMPLETE-SOURCE-RECORD-497: document=zenodo-complete-source-data.zip; record=Separate Source Data files/Fig. 6/fig.6f.txt; target=sha256:f66083920ea0a93c3d778759bb832bed67d3c1abefd9acc6514ee49be7ab8b35
- five-state structural cycle retained; four separately observed conductance states retained without substitution
- exact transition word: State 1 to 2 to 3 to 4 to 5 to State 1 catalyst return
- Table S1 TOF values retained exactly: 0.5, 4.6, 29.6, 39.0, 203.9, 615.6 and 2098.7 per second
- Table S2 and Table S3 remain independent and unaveraged; 385,617 raw trace rows remain byte-bound
- all source fits, external signed/zero inscriptions, 106 pages, 1,604 movie frames and 387 archive members remain post-seal evidence
- tampered_omitted_source_record_rejected: True
- tampered_broken_catalyst_return_rejected: True
- complete_497_record_vector_retained: True
- all_106_supplementary_pages_retained: True
- all_1604_movie_frames_retained: True
- all_387_archive_members_retained: True
- adverse_unavailable_pdf_and_insufficient_fit_records_visible: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if catalyst identity differs anywhere in the cycle; if the last transition does not return to the exact entry state; if any state, transition, condition, status, TOF row, independent rate-table value, raw trace, page, movie, archive member or unavailable-PDF adverse record is omitted; if four observed conductance levels are substituted for the five-state structural cycle; if a rate equation, TOF formula, continuum, fit, steady-state or stochastic premise, selection, average, interpolation or target correction enters; if target content opens before all 497 identities and the consequence seal; or if either omission or broken-return tampering passes.

Measurement receipt: sha256:8fc4e1370c2cdfe81b55ccd88319447fb91309cf4c96e5c69ff55d6bf962c316.

Isolation certificate: sha256:a8e291839769b931f33aa75ab4e6e96a3a3dd6ab4dfe460a2932359c6516ab26.

Custody certificate: sha256:bd316dec21292e9e4c7d5cbb4045fad9ea9539201e2bf102d0d09e33072959a9.

Registered source descriptions:

- Nature Nanotechnology single-molecule catalytic-cycle study: ()

Registered target identities:

- None.

Meaning. The kinetics extension derives exact rate, concentration and temperature dependence, activation, transition boundaries, competing and composed mechanisms, reversible equilibrium, turnover, diffusion limits, isotope effects and product-state dynamics. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero
- negative proof quantity
- irrational quantity
- imaginary quantity
- continuum time
- imported catalytic rate equation
- turnover-frequency formula
- Michaelis-Menten premise
- steady-state premise
- stochastic cycle weight
- fitted catalyst efficiency
- selected cycle or intermediate
- selected condition
- selected row, page, frame or archive member
- omitted deactivation, adverse or unresolved record
- average
- interpolation
- renormalization
- target-derived correction

Immutable identities. Source manifest

sha256:16a98caf797bc39041aed57f6fc8a965e0250f4f1545ef0e61735034d019e199; derivation seal
 sha256:eb28322961716d9ae55c74653a6f95b5a51ee88c63e50b2fb41bd575ece7016d; engine receipt
 sha256:0b28dd797f47297cfd8d3af3c3ab496f32cadda8a0a9600ce9110d9ef7e694d9 at receipts/engine/
 model_admitted/SFT-CHEM-CATALYTIC-TURNOVER-CYCLE-FREQUENCY-010-0b28dd797f47297c.json;
 empirical hash sha256:98489ea29450813ca1b9fd04ba5389cfc94c5ef5f0fdb425f83f7ab68a3ec13d;
 measurement receipt sha256:8fc4e1370c2cdf8e1b55ccd88319447fb91309cf4c96e5c69ff55d6bf962c316.

154. Exact diffusion-limited reaction boundary

Claim identity: SFT-CHEM-DIFFUSION-LIMITED-REACTION-BOUNDARY-011

Question and theorem. A diffusion-limited reaction occurrence is admissible only after one complete finite transport word carries the same held reactant identity to an encounter state that is exactly the reaction entry state. The exact completion relation counts completed reaction occurrences per positive held observation partition. No diffusion equation, continuum concentration field, Fick or Smoluchowski law, fitted coefficient or stochastic collision weight is imported.

complete-finite-transport-state-and-transition-word__transport-exit-is-exact-reaction-encounter-entry__reaction-admissible-only-after-complete-transport-word__completed-reaction-occurrences-per-exact-positive-observation-parts__all-separated-solvation-transport-encounter-and-product-stages-retained__all-favorable-adverse-control-and-unresolved-statuses-retained__complete-251-record-article-supplements-movies-and-dual-archive-ledger__value-free-251-record-i

identity-seal-and-depth-independent-path-successor

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-MOLECULAR-DIFFUSION-RELATION-016
- SFT-CHEM-COUPLED-MASS-HEAT-CHARGE-TRANSPORT-019
- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001
- SFT-CHEM-CONCENTRATION-DEPENDENCE-RELATION-002
- SFT-CHEM-TEMPERATURE-DEPENDENCE-RELATION-003
- SFT-CHEM-ACTIVATION-BARRIER-VALUE-RELATION-004
- SFT-CHEM-TRANSITION-STATE-EQUIVALENT-BOUNDARY-005
- SFT-CHEM-COMPETING-CHANNEL-BRANCHING-006
- SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007
- SFT-CHEM-PARALLEL-MECHANISM-COMPOSITION-008
- SFT-CHEM-REVERSIBLE-KINETIC-EQUILIBRIUM-CORRESPONDENCE-009
- SFT-CHEM-CATALYTIC-TURNOVER-CYCLE-FREQUENCY-010

Generated grammar. Generate the literal product of support, composition, limitation, rate, path, status, provenance and prediction forms; decide all 256 candidates only from admitted finite state-transition, transport and reaction composition laws.

Boundary: Every finite source-ordered transport-reaction family retaining the same reactant through two or more registered transport states, every adjacent transition, one exact transport exit/reaction-entry state, one held condition, one product state and every favorable, adverse, control or unresolved status. External testing binds all 251 pre-registered records: article landing, 43 PDF pages, two videos containing 1,350 frames, Figshare metadata and all 204 member identities from two byte-identical independently hosted archives.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-finite-transport-state-and-transition-word__transport-exit-is-exact-reaction-encounter-entry__reaction-admissible-only-after-complete-transport-word__completed-reaction-occurrences-per-exact-positive-observation-parts__all-separated-solvation-transport-encounter-and-product-stages-retained__all-favorable-adverse-control-and-unresolved-statuses-retained__complete-251-record-article-supplements-movies-and-dual-archive-ledger__value-free-251-record-identity-seal-and-depth-independent-path-successor`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
support	complete-finite-transport-state-and-transition-word	selected-initial-encounter-or-product-endpoint	Endpoints erase the path that limits reaction entry.
composition	transport-exit-is-exact-reaction-encounter-entry	transport-and-reaction-created-as-unrelated-processes	Unrelated processes do not establish a limiting boundary.
limitation	reaction-admissible-only-after-complete-transport-word	imported-Fick-Smoluchowski-continuum-or-collision-law	An imported diffusion premise selects the boundary.
rate	completed-reaction-occurrences-per-exact-positive-observation-parts	fitted-diffusion-coefficient-rate-constant-or-continuum-time	A fit or continuum quantity cannot define the exact completion relation.
path	all-separated-solvation-transport-encounter-and-product-stages-retained	solvation-transport-encounter-or-product-stage-omitted-or-collapsed	Omission destroys the complete transport-reaction word.
status	all-favorable-adverse-control-and-unresolved-statuses-retained	simulation-measurement-discrepancy-large-droplet-deviation-or-nonencounter-question-omitted	Omission falsely regularizes the evidence.
provenance	complete-251-record-article-supplements-movies-and-dual-archive-ledger	selected-fit-rate-table-or-trajectory	A selected excerpt cannot establish complete source custody.
prediction	value-free-251-record-identity-seal-and-depth-independent-path-successor	distance-time-velocity-yield-rate-fit-or-target-reachable-before-seal	Target access can select the law.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:230461f3f16c1e8d4dcc002e483ed4dcdb5e14facab643316a578720087e4396.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:1982efbde5e56155231d0cd706ba46a6387041fe50ff4468610806f0b91e867.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ba8cd124b9fe32e11d20d7ba7f09c2d1c981b9ef04ba141a2bb4a389466b9e95.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:a2561e2af70c72fc194a8e4935df682e05cbc394dd31c9c74d9a10b04c72a0e7.

Independent reconstruction. Implementation hash

sha256:e303ec15316d3be82616949fdb93059624065fe38a016f0abfd5a1b7333c1d0c; independent certificate sha256:8fd7279bc975b807492281bd69637db4e988d4230c51ef579d5aa862b5acd5fd; external-validation hash sha256:03a5a7f9b5ca613cb9c1e31149355123f0cfc3069136dac043d4b2b2a474a00c. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NATURE-COMMUNICATIONS-S41467-025-68008-5-COMPLETE

Comparison records:

- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-001: document=article.html; record=complete-article-landing.html; target=sha256:547b5c8c0576a95215f23b4e3f99e98bc65cf939279de70c08d1260a61dce79c

- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-002: document=article.pdf; record=page-001; target=sha256:735340b233d50d7ce6207feb3037de4b36bac9ee1b5fdc4149000497bc00d298
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-003: document=article.pdf; record=page-002; target=sha256:569f0f831069800aa84ee8ba29c31b7e0450b7bb8e634cee6a10d0769b5041ca
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-004: document=article.pdf; record=page-003; target=sha256:e4bdfd02ac3777007bc882ad320c2c24a81a017cb6c582ab9dd8f440cdcb0963
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-094: document=nature-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_X_MP2.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-095: document=nature-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_X_no_elec.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-096: document=nature-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_Y.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-098: document=nature-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_Y_MP2.dat; target=sha256:3aac83ce290762ca354fd825dba44667a03efac9212d26edc77f1cf13addddef0
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-099: document=nature-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_Y_no_elec.dat; target=sha256:843db96ef4743aae3615577e2f948f617ffde74ce507ca7710e2f196492f9887
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-100: document=nature-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_Z.dat; target=sha256:cbb8c1b434f2b22b1a60ffa16d31d3fd85b496934aca61755e9d94d9dc170ae0
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-101: document=nature-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_Z_DFT.dat; target=sha256:2c18902afa43e468b223abc86a1469c7e2aa891f27fd31c4d9d993b5cfd45b4f
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-102: document=nature-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_Z_MP2.dat; target=sha256:44dddf20528ed5e446e075e8fa4adb2c95a6bd7fde71f60c8c926a21b464aca5
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-103: document=nature-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_Z_no_elec.dat; target=sha256:7a910ba725ac2bb0de5e8b2562aef9aef962f16438941ae95a24af1245b6ee04
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-104: document=nature-source-data.zip; record=Source Data/SI/Fig_S2/README_FIG_S2.txt; target=sha256:17ac6429d39f2306dbe685c0f1411fa569b00163947a368ae0ff89f22f8ef18b
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-105: document=nature-source-data.zip; record=Source Data/SI/Fig_S3; target=sha256:735dcaf8dd569a1d52d0f82f70ed005e092603d083707b98d607a25729b39795
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-106: document=nature-source-data.zip; record=Source Data/SI/Fig_S3/charge_Li_DFT.dat; target=sha256:8440bd672d09bddbe24f6bdfd4d5f28a754f7d2d3cbe093c59f1c3ea46710523
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-107: document=nature-source-data.zip; record=Source Data/SI/Fig_S3/charge_Li_MP2.dat; target=sha256:86211e5210ff1c9bcbe6a3148fe7fb6cde8ee30c788e70d8a10b69173b72edf8
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-108: document=nature-source-data.zip; record=Source Data/SI/Fig_S3/README_FIG_S3.txt; target=sha256:b2e8f7e89d1b35525bf9d7e17478e1baf99debd200fef878fac451a2bb296bb2
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-109: document=nature-source-data.zip; record=Source Data/SI/Fig_S4; target=sha256:7720a6e6dfa17cf7b82d7b61e73816e5fb4bca7294e6e5b30f44f0549a43a4aa
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-110: document=nature-source-data.zip; record=Source Data/SI/Fig_S4/potential_Li+He.dat; target=sha256:741445c967d0e9ec2bdb31bac38bf2aa9afabbbf7964af77face3f362fbcf2104
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-111: document=nature-source-data.zip; record=Source Data/SI/Fig_S4/potential_Li+He_CCSD.dat; target=sha256:4d22ba2ebcd489f6825215dd7dd2758dace7b08fa9f9a51c951f5720f5748774
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-112: document=nature-source-data.zip; record=Source Data/SI/Fig_S4/README_FIG_S4.txt; target=sha256:af591a499a477f06b46a6c8f9fd4acc3ce1cb7584e05b28b4d7f0590ef920f85
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-113: document=nature-source-data.zip; record=Source Data/SI/Fig_S5; target=sha256:a0b8896998284446a7cc8acef01f833ee2c195f68a98ec603244cfac40effe02
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-114: document=nature-source-data.zip; record=Source Data/SI/Fig_S5/properties_LiBz2_theta10_noHe.dat; target=sha256:0d1ce9f189398ba941d8eaa1c45ffb141f80330f1fce667ef9a4cc9a86edafca
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-115: document=nature-source-data.zip; record=Source Data/SI/Fig_S5/README_FIG_S5.txt; target=sha256:5b85eb5249e8b335a7dc672f23725f5e253b9e4064f9877f1dea1d6a7c14cb15

- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-116: document=nature-source-data.zip; record=Source Data/SI/Fig_S6/; target=sha256:9fa7db0d911fab5465ba4255b094744b0e650e721636ec3ae2a5560d8b6d3063
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-117: document=nature-source-data.zip; record=Source Data/SI/Fig_S6/Bz-Bz_distance_He1800_theta0.dat; target=sha256:1d674fb86877501a3e8a30853d406c946492d1b7f832afe0513d1307f71120d6
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-118: document=nature-source-data.zip; record=Source Data/SI/Fig_S6/Bz-Bz_distance_He1800_theta30.dat; target=sha256:830a8a05bd0e6964c4e997af877c2258f60c4a09508304e1c910ba0febce9c15
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-119: document=nature-source-data.zip; record=Source Data/SI/Fig_S6/Bz-Bz_distance_He1800_theta60.dat; target=sha256:bd40cd811e8686f92aa9fd38e8a1bd9707dd4c9ff1159d76c45f50f06d22e0a9
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-120: document=nature-source-data.zip; record=Source Data/SI/Fig_S6/Bz-Bz_distance_He1800_theta90.dat; target=sha256:a701c035f5dbb92c6653c5492ed9fa2416fba3a0b55229eeab74dc500143c6fb
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-121: document=nature-source-data.zip; record=Source Data/SI/Fig_S6/LiBz2_distance_He1800_theta0.dat; target=sha256:eeb6f0dd5ea17687ba673bee4c113ed9b90264083a0fe9153fa02e1679228979
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-122: document=nature-source-data.zip; record=Source Data/SI/Fig_S6/LiBz2_distance_He1800_theta30.dat; target=sha256:871da9f091d7a2a6e825990cc8ca47bc015cfee0a07d09afdbf54a5621a19be2
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-123: document=nature-source-data.zip; record=Source Data/SI/Fig_S6/LiBz2_distance_He1800_theta60.dat; target=sha256:e3737095fa35b33a3234046f8c21cbece28492b55b8523ad8d59e551e95b34a1
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-124: document=nature-source-data.zip; record=Source Data/SI/Fig_S6/LiBz2_distance_He1800_theta90.dat; target=sha256:39bacf5828266d7f2eae516e75b66961ad79bffd8aa9768faaea9bc4b19ebd9
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-125: document=nature-source-data.zip; record=Source Data/SI/Fig_S6/Li_coordination_He1800_theta0.dat; target=sha256:36ab0d8beeee4de741feb66cb1a9a72e43b99ed5fa626978dbcf081d60965149
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-126: document=nature-source-data.zip; record=Source Data/SI/Fig_S6/Li_coordination_He1800_theta30.dat; target=sha256:a46136564a3331ce2411165ee7e5d129db251c5919f094318be2d111b6302942
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-127: document=nature-source-data.zip; record=Source Data/SI/Fig_S6/Li_coordination_He1800_theta60.dat; target=sha256:3596a29d7c076b4ad71ee05fd570dd04053dc197bd3dec60e89c65b15798e9df
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-128: document=nature-source-data.zip; record=Source Data/SI/Fig_S6/Li_coordination_He1800_theta90.dat; target=sha256:d7fd8017b777dd836a698754e775519feffa782db91a9eafcc91d1a8c5d34a37
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-129: document=nature-source-data.zip; record=Source Data/SI/Fig_S6/README_FIG_S6.txt; target=sha256:baefd0d6390e9299608fa8f897080f051195e1e5f8fea303aeff2ec96e90be79
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-130: document=nature-source-data.zip; record=Source Data/SI/Fig_S7/; target=sha256:44539ad312598534ee97dd97f47e448b79034125ed7a9d18a56931163b25e0fb
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-131: document=nature-source-data.zip; record=Source Data/SI/Fig_S7/charge_distant_Bz_CCD_NBO.dat; target=sha256:24c1fb47fbbf1f826a9df65f510dc9dc59712bc0bc4f9780470ecb1c4013fe
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-132: document=nature-source-data.zip; record=Source Data/SI/Fig_S7/charge_distant_Bz_CCD_RESP.dat;

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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-133: document=nature-source-data.zip; record=Source Data/SI/Fig_S7/charge_distant_Bz_DFT_NBO.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-134: document=nature-source-data.zip; record=Source Data/SI/Fig_S7/charge_distant_Bz_DFT_RESP.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-135: document=nature-source-data.zip; record=Source Data/SI/Fig_S7/charge_distant_Bz_MP2_NBO.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-136: document=nature-source-data.zip; record=Source Data/SI/Fig_S7/charge_distant_Bz_MP2_RESP.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-137: document=nature-source-data.zip; record=Source Data/SI/Fig_S7/charge_Li_CCD_NBO.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-138: document=nature-source-data.zip; record=Source Data/SI/Fig_S7/charge_Li_CCD_RESP.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-139: document=nature-source-data.zip; record=Source Data/SI/Fig_S7/charge_Li_DFT_NBO.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-140: document=nature-source-data.zip; record=Source Data/SI/Fig_S7/charge_Li_DFT_RESP.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-141: document=nature-source-data.zip; record=Source Data/SI/Fig_S7/charge_Li_MP2_NBO.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-142: document=nature-source-data.zip; record=Source Data/SI/Fig_S7/charge_Li_MP2_RESP.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-143: document=nature-source-data.zip; record=Source Data/SI/Fig_S7/charge_nearby_Bz_CCD_NBO.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-144: document=nature-source-data.zip; record=Source Data/SI/Fig_S7/charge_nearby_Bz_CCD_RESP.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-145: document=nature-source-data.zip; record=Source Data/SI/Fig_S7/charge_nearby_Bz_DFT_NBO.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-146: document=nature-source-data.zip; record=Source Data/SI/Fig_S7/charge_nearby_Bz_DFT_RESP.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-147: document=nature-source-data.zip; record=Source Data/SI/Fig_S7/charge_nearby_Bz_MP2_NBO.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-148: document=nature-source-data.zip; record=Source Data/SI/Fig_S7/charge_nearby_Bz_MP2_RESP.dat;

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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-149: document=nature-source-data.zip; record=Source Data/SI/Fig_S7/README_FIG_S7.txt;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-150: document=figshare-source-data.zip; record=Source Data/
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-151: document=figshare-source-data.zip; record=Source Data/Manuscript/;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-152: document=figshare-source-data.zip; record=Source Data/Manuscript/Fig1/;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-153: document=figshare-source-data.zip; record=Source Data/Manuscript/Fig1/Li-He3600_average_40ps.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-154: document=figshare-source-data.zip; record=Source Data/Manuscript/Fig1/LiBenzene2-He3600_average.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-155: document=figshare-source-data.zip; record=Source Data/Manuscript/Fig1/properties_LiBenzene2-He3600_theta10_final.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-156: document=figshare-source-data.zip; record=Source Data/Manuscript/Fig1/README_FIG1.txt;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-157: document=figshare-source-data.zip; record=Source Data/Manuscript/Fig2/;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-158: document=figshare-source-data.zip; record=Source Data/Manuscript/Fig2/0_2ps_LiBzCovariance.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-160: document=figshare-source-data.zip; record=Source Data/Manuscript/Fig2/13ps_LiBzCovariance.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-161: document=figshare-source-data.zip; record=Source Data/Manuscript/Fig2/303ps_LiBzCovariance.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-162: document=figshare-source-data.zip; record=Source Data/Manuscript/Fig2/3ps_LiBzCovariance.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-163: document=figshare-source-data.zip; record=Source Data/Manuscript/Fig2/63ps_LiBzCovariance.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-164: document=figshare-source-data.zip; record=Source Data/Manuscript/Fig2/Li+_Ion_signal.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-165: document=figshare-source-data.zip; record=Source Data/Manuscript/Fig2/README_FIG2.txt;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-166: document=figshare-source-data.zip; record=Source Data/Manuscript/Fig3/; target=sha256:47d93b596796955dbaace260571fb5265f4d815428e49e19dcbccb13ffdd6a8a
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-167: document=figshare-source-data.zip; record=Source Data/Manuscript/Fig3/Co(Li+Bz_Bz).dat.dat; target=sha256:f2839ad7b0abf6077c9148a9f88738bcd9e3e17a8a2b6852c8ec137c7d22926a
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-168: document=figshare-source-data.zip; record=Source Data/Manuscript/Fig3/Li+BzCoincidenceFiltered.dat; target=sha256:bcad50ccbca87db70c0138f4818c02efe5a031637c7b380d6c49289b2883086b
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-169: document=figshare-source-data.zip; record=Source Data/Manuscript/Fig3/README_FIG3.txt; target=sha256:39af7b83ddb195422284ede1beac87368939d2b60cedcd549ff16e66b9733aa
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-171: document=figshare-source-data.zip; record=Source Data/SI/; target=sha256:09c4f69981c3510e820b2a1a68f5aab8d7130e1d51966703e9d496ef1034a1e0
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-172: document=figshare-source-data.zip; record=Source Data/SI/Fig_S1/; target=sha256:9375132ca706d52d7d6a44b3d046aeda03ddc4ab3ebd85e01bd38547accb28c2
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-173: document=figshare-source-data.zip; record=Source Data/SI/Fig_S1/potential_Li-Bz_X.dat; target=sha256:c376b9a638e9a59d16c068b07d3ce807a9b94150188593ed10e7c62d971698b7
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-174: document=figshare-source-data.zip; record=Source Data/SI/Fig_S1/potential_Li-Bz_X_CCSD.dat; target=sha256:e74fb076469ce18b4169f77796f9781cf76b01f06bb83e2c191c65f8ebfdfab9
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-175: document=figshare-source-data.zip; record=Source Data/SI/Fig_S1/potential_Li-Bz_Z.dat; target=sha256:5cd179bc06dd4a5a39541d8d22243cfe15a99d698d0bb8787849ae5071061023
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-176: document=figshare-source-data.zip; record=Source Data/SI/Fig_S1/potential_Li-Bz_Z_CCSD.dat; target=sha256:91d9429f23647cc5cd195a269427e4a92a38d4cda5b10cb24e85015a0baed0c3
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-177: document=figshare-source-data.zip; record=Source Data/SI/Fig_S1/README_FIG_S1.txt; target=sha256:52fe05ab427d959355e3c14db90d4ca3eb69e80924ccb63ad68ae140f0a90bd9
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-178: document=figshare-source-data.zip; record=Source Data/SI/Fig_S10/; target=sha256:e9ca9ddac3af96807192542c8040fbd88266032ed972ab4b0d60d96c455a0ea
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-179: document=figshare-source-data.zip; record=Source Data/SI/Fig_S10/Co(Li+Bz_Bz).dat; target=sha256:76ae231577e36ef668c37b3d7558249bbf794d0685f195cf50f1ee7e79749e10
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-180: document=figshare-source-data.zip; record=Source Data/SI/Fig_S10/Li+BzImageFiltered.dat; target=sha256:6f445894ef0fa3f15a2793a2c39c873b23af94abcf98f3032ec4a26bb7b84353
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-181: document=figshare-source-data.zip; record=Source Data/SI/Fig_S10/Li+BzImageRaw.dat; target=sha256:1247692e137b914b2459fa365ecc7c70cedcc8a00742931c5dd5c2a299456057
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-182: document=figshare-source-data.zip; record=Source Data/SI/Fig_S10/Li+BzVelocityDistributions.txt; target=sha256:f0df774314b6f563dcb2eefdb60c0e6a297c536f327707e078a52a2e3c069c40
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-183: document=figshare-source-data.zip; record=Source Data/SI/Fig_S10/Li+Bz_Bz+_angular_covariance.dat; target=sha256:cfa90cf37874d5ee2538b72c789db627152c38b943b95b808909189a4e9f34a9

- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-184: document=figshare-source-data.zip; record=Source Data/SI/Fig_S10/Li+Bz_Bz+_velocity_covariance.dat;
target=sha256:27cb577b1c8c9dce99b466438bc3d5033462cb453108503f6b86e7e78b4266c4
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-185: document=figshare-source-data.zip; record=Source Data/SI/Fig_S10/MassSpectrum.dat; target=sha256:68d9bf2f1b08c31deecf97ccfa8ac4f23212f82afce1ea63a2ee4b07bbd81601
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-186: document=figshare-source-data.zip; record=Source Data/SI/Fig_S10/README_FIG_S10.txt;
target=sha256:186bddf653415763094a6dea0049f24709b719b4cbb8d9db74e2ca4b0629e3a0
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-187: document=figshare-source-data.zip; record=Source Data/SI/Fig_S11/; target=sha256:156c8c92488bc6654e5a3900c27758f603274abc453d0ef2ce3822ba6d2814b2
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-188: document=figshare-source-data.zip; record=Source Data/SI/Fig_S11/LeakValveYield.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-189: document=figshare-source-data.zip; record=Source Data/SI/Fig_S11/README_FIG_S11.txt;
target=sha256:408fa3737613166fe9be11e9a3ee456da12bd20408328aa5721d204bbf7b3d9d
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-190: document=figshare-source-data.zip; record=Source Data/SI/Fig_S12/; target=sha256:95b2deaf5c7b0cea879bf734de65f16040a1c92f31f78bfe34469a06a4f74e88
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-191: document=figshare-source-data.zip; record=Source Data/SI/Fig_S12/README_FIG_S12.txt;
target=sha256:b6bb73b44c6f11ee69c04af2d24a5bd3d3c7886556c9d49c73c7b67a5d95dfd7
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-192: document=figshare-source-data.zip; record=Source Data/SI/Fig_S12/R_values_doped_undoped.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-193: document=figshare-source-data.zip; record=Source Data/SI/Fig_S2/; target=sha256:e61415d63d796facc1a918b870e0e9e471457e4da4023f239a0515419a55d608
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-194: document=figshare-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_X.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-195: document=figshare-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_X_DFT.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-196: document=figshare-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_X_MP2.dat;
target=sha256:20e1328f573054f7e4b06a796234af6c19090c575bd310208cd39dd0f345bb6b
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-197: document=figshare-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_X_no_elec.dat;
target=sha256:c156caa8ad6f87e13d0c87548da8e77bb66eb261dd7d160f56ccb67c6a3c2ad8
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-198: document=figshare-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_Y.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-199: document=figshare-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_Y_DFT.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-200: document=figshare-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_Y_MP2.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-201: document=figshare-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_Y_no_elec.dat; target=sha256:022d4edc02921f940db3d9a22fe1eaceac0256cb1a79d381bce3e807892c382f
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-202: document=figshare-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_Z.dat; target=sha256:ba1b614e0d61e39d630e9501fce1f51bf8d6f01ce4952dcabc216c87e399257b
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-203: document=figshare-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_Z_DFT.dat; target=sha256:4a6860872dec7fe6a7db5319ee824ea6b4da860282baf0039b8b8135ac48fcd9
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-204: document=figshare-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_Z_MP2.dat; target=sha256:be225d8c40d5f972629deec5ea40fce943ee07edf1de48e3ad9e66798c8a2d09
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-205: document=figshare-source-data.zip; record=Source Data/SI/Fig_S2/potential_Li+Bz_Z_no_elec.dat; target=sha256:c1b0cd39a76482c3d8c0aed6e2bff454277277d753664b247520fa997239d4ab
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-206: document=figshare-source-data.zip; record=Source Data/SI/Fig_S2/README_FIG_S2.txt; target=sha256:9d74c646979ce9c4006ce79e902d79615a861a909f80fc9584024aeebf5a2423
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-207: document=figshare-source-data.zip; record=Source Data/SI/Fig_S3/; target=sha256:2b7e5715be18c7ecaee53425d4ca39956689fa3e5ee2e5b871def29860040344
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-208: document=figshare-source-data.zip; record=Source Data/SI/Fig_S3/charge_Li_DFT.dat; target=sha256:c6aa4ca8e628d27be34d5a9e30593ca2105f2e1d24047526b6afc50da16656ca
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-209: document=figshare-source-data.zip; record=Source Data/SI/Fig_S3/charge_Li_MP2.dat; target=sha256:c24c19f9193919bdc22c11d29a5c6d6cd980c4e203844a78c0eb44ea6a77bcaf
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-210: document=figshare-source-data.zip; record=Source Data/SI/Fig_S3/README_FIG_S3.txt; target=sha256:ad9e454dbc203450f4df88014e3465f72b3805fc0a95dd4f7fa5c54e79ee7c83
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-211: document=figshare-source-data.zip; record=Source Data/SI/Fig_S4/; target=sha256:a5d63d720471c8398096206eaa311299f8a2e24ec62b867ee7a225f43f250efe
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-212: document=figshare-source-data.zip; record=Source Data/SI/Fig_S4/potential_Li+He.dat; target=sha256:20b299d937bcd22b7b91e7e9123126abccf68d4f13ae8e182dce1d724b85d517
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-213: document=figshare-source-data.zip; record=Source Data/SI/Fig_S4/potential_Li+He_CCSD.dat; target=sha256:4450dbb9bfd9dd6547efa8d84cf782a425c84dfac716a441a29f19cf78c329f
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-214: document=figshare-source-data.zip; record=Source Data/SI/Fig_S4/README_FIG_S4.txt; target=sha256:dbabafe60e726ac9eacb47a99efe4e2aaa05a2f75d639d5725d5cf24eea477a9
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-215: document=figshare-source-data.zip; record=Source Data/SI/Fig_S5/; target=sha256:2a17cbe6dc4827ba55dcd4f254284474298bf6b3246a65d5376b3555c44d05d8
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-216: document=figshare-source-data.zip; record=Source Data/SI/Fig_S5/properties_LiBz2_theta10_noHe.dat; target=sha256:c50263b705284e1f384ed00653dc33e523a67bfe5d694390dd406839f203d94a
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-217: document=figshare-source-data.zip; record=Source Data/SI/Fig_S5/README_FIG_S5.txt; target=sha256:f20dab594185401e8cf751f7f29276a056ed90b78ee058aabde1312a0655799c
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-218: document=figshare-source-data.zip; record=Source Data/SI/Fig_S6/; target=sha256:87bd29e8b7c49a725cc5281b8c4b676a47d42ae88614f950deed724495c1f204

- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-219: document=figshare-source-data.zip; record=Source Data/SI/Fig_S6/Bz-Bz_distance_He1800_theta0.dat; target=sha256:323dfc6a6acc88f539781e2cb24339f754222de61dcaeb0c30961b42911b8c68
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-220: document=figshare-source-data.zip; record=Source Data/SI/Fig_S6/Bz-Bz_distance_He1800_theta30.dat; target=sha256:bbcefdbca3189f6eb91435fb250bae179605b3bfb90b462714084b57d08e9277
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-221: document=figshare-source-data.zip; record=Source Data/SI/Fig_S6/Bz-Bz_distance_He1800_theta60.dat; target=sha256:b43ba1978921f7e51142e87b1f6b3fe706ac57f1f2fbb46c175309c31f13a464
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-222: document=figshare-source-data.zip; record=Source Data/SI/Fig_S6/Bz-Bz_distance_He1800_theta90.dat; target=sha256:1e35c11feaf292b1e268eb57a3b389079dcb06154cd0914d2f72cc73cbc85509
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-223: document=figshare-source-data.zip; record=Source Data/SI/Fig_S6/LiBz2_distance_He1800_theta0.dat; target=sha256:3df4346b1e967e7e0761fc3ba35cfe00fa256cdca8ee4edafb7f843b116bd0bb
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-224: document=figshare-source-data.zip; record=Source Data/SI/Fig_S6/LiBz2_distance_He1800_theta30.dat; target=sha256:ff77cbd8c484215470ede50b2850047b994a3d09270368cfeda99dc6b18ecddd
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-225: document=figshare-source-data.zip; record=Source Data/SI/Fig_S6/LiBz2_distance_He1800_theta60.dat; target=sha256:4cd134fd294a948d438aed9ee09cd63ca5e1761660ad82ead5fa3cc4d48c91a
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-226: document=figshare-source-data.zip; record=Source Data/SI/Fig_S6/LiBz2_distance_He1800_theta90.dat; target=sha256:627cf8fd08378c977eb014e23a6c1affc4348351b6bc0e344edf3bb62d165559
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-227: document=figshare-source-data.zip; record=Source Data/SI/Fig_S6/Li_coordination_He1800_theta0.dat; target=sha256:2d6648f196c712c13109b6430853f41c36866046587e104e97ae7d598f2da789
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-228: document=figshare-source-data.zip; record=Source Data/SI/Fig_S6/Li_coordination_He1800_theta30.dat; target=sha256:3b288c80c561a3b7ac1c849442e96d9eea89350039ac7f40fb5160574c8bcc3d
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-229: document=figshare-source-data.zip; record=Source Data/SI/Fig_S6/Li_coordination_He1800_theta60.dat; target=sha256:e91d04c69eb426bc9f7eb2f4d4e66bdc2787bb82667c84d7008127d7fa42973e
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-230: document=figshare-source-data.zip; record=Source Data/SI/Fig_S6/Li_coordination_He1800_theta90.dat; target=sha256:8a52e75c2042ea76efdb4f68ca90a329c1349bc16149d0dfd586d16247211244
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-231: document=figshare-source-data.zip; record=Source Data/SI/Fig_S6/README_FIG_S6.txt; target=sha256:61ff57ebc7b1c1e117578a92315d20437f2f1af720081cf43c79c20ad87b2924
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-232: document=figshare-source-data.zip; record=Source Data/SI/Fig_S7/; target=sha256:a9d9009838cd03bd18b73b208df6c160a681daf4e0513ddb24e344a70a0fc9a0
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-233: document=figshare-source-data.zip; record=Source Data/SI/Fig_S7/charge_distant_Bz_CCD_NBO.dat; target=sha256:2c738393f60a5e6f263e132c74059b1ae3b113436b5d2eff40420c56700c6e72
- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-234: document=figshare-source-data.zip; record=Source Data/SI/Fig_S7/charge_distant_Bz_CCD_RESP.dat; target=sha256:f9ca559841ce30356ae49f8d3aa5caafce8e14f3a339243880f75f97d36efe9b

- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-235: document=figshare-source-data.zip; record=Source Data/SI/Fig_S7/charge_distant_Bz_DFT_NBO.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-236: document=figshare-source-data.zip; record=Source Data/SI/Fig_S7/charge_distant_Bz_DFT_RESP.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-237: document=figshare-source-data.zip; record=Source Data/SI/Fig_S7/charge_distant_Bz_MP2_NBO.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-238: document=figshare-source-data.zip; record=Source Data/SI/Fig_S7/charge_distant_Bz_MP2_RESP.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-239: document=figshare-source-data.zip; record=Source Data/SI/Fig_S7/charge_Li_CCD_NBO.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-240: document=figshare-source-data.zip; record=Source Data/SI/Fig_S7/charge_Li_CCD_RESP.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-241: document=figshare-source-data.zip; record=Source Data/SI/Fig_S7/charge_Li_DFT_NBO.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-242: document=figshare-source-data.zip; record=Source Data/SI/Fig_S7/charge_Li_DFT_RESP.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-243: document=figshare-source-data.zip; record=Source Data/SI/Fig_S7/charge_Li_MP2_NBO.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-244: document=figshare-source-data.zip; record=Source Data/SI/Fig_S7/charge_Li_MP2_RESP.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-245: document=figshare-source-data.zip; record=Source Data/SI/Fig_S7/charge_nearby_Bz_CCD_NBO.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-246: document=figshare-source-data.zip; record=Source Data/SI/Fig_S7/charge_nearby_Bz_CCD_RESP.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-247: document=figshare-source-data.zip; record=Source Data/SI/Fig_S7/charge_nearby_Bz_DFT_NBO.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-248: document=figshare-source-data.zip; record=Source Data/SI/Fig_S7/charge_nearby_Bz_DFT_RESP.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-249: document=figshare-source-data.zip; record=Source Data/SI/Fig_S7/charge_nearby_Bz_MP2_NBO.dat;
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- SFT-CHEM-KIN011-COMPLETE-SOURCE-RECORD-250: document=figshare-source-data.zip; record=Source Data/SI/Fig_S7/charge_nearby_Bz_MP2_RESP.dat;
target=sha256:bde95286f0be63e00d09e2ac540905bf460c39c00d98e2618a096db346a15053

- SFT-CHEM-KIN011-COMplete-SOURCE-RECORD-251: document=figshare-source-data.zip; record=Source Data/SI/Fig_S7/README_FIG_S7.txt; target=sha256:7e3029f97c9db334de820edbf9c29ff12d5325d76a69291d467c74f121a845da
- complete finite separated-reactant, solvation, transport, encounter and product path retained
- transport exit is exact reaction entry; reaction waits on complete finite transport word
- external velocity inscriptions retained: experiment 43 ± 5 m/s; simulation 14 m/s; discrepancy unreconciled
- external source rate estimate $5 \times 10^{12} \text{ M}^{-1} \text{ s}^{-1}$ retained only as post-seal provenance
- 43 PDF pages, 1,350 movie frames, 204 archive members and 11,512 key raw rows remain byte-bound
- tampered_omitted_source_record_rejected: True
- tampered_transport_exit_reaction_entry_mismatch_rejected: True
- complete_251_record_vector_retained: True
- all_43_pdf_pages_retained: True
- all_1350_movie_frames_retained: True
- all_204_archive_members_retained: True
- all_discrepancy_and_adverse_records_visible: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if the transported identity changes; if any transport state or adjacent transition is omitted; if transport exit differs from reaction encounter entry; if reaction is admitted before transport closure; if any page, video, archive member, raw row, radius/time row, velocity discrepancy, large-droplet deviation, resolution limit or reviewer adverse question is omitted; if a diffusion equation, continuum, Fick or Smoluchowski law, fitted coefficient, stochastic collision weight, selection, averaging, interpolation or target correction enters; if target content opens before all 251 identities and the consequence seal; or if omission or broken-boundary tampering passes.

Measurement receipt: sha256:385cd999a4d50661258e771a91217f0a879d4c9aa2a9796d09c39e4415504354.
Isolation certificate: sha256:2aab2aea74138301db1b3e8df50b24a16c63a52cca0f0bf5a52ffb5ce97315e5.
Custody certificate: sha256:e8098e46278ada2adfc6fd67a80d57bb8bd8855d28298d23ab09b36ad423dc07.

Registered source descriptions:

- Nature Communications time-resolved diffusion-limited reaction study: ()

Registered target identities:

- None.

Meaning. The kinetics extension derives exact rate, concentration and temperature dependence, activation, transition boundaries, competing and composed mechanisms, reversible equilibrium, turnover, diffusion limits, isotope effects and product-state dynamics. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero
- negative proof quantity
- irrational quantity
- imaginary quantity
- continuum field
- continuum concentration
- diffusion equation
- Fick law

- Smoluchowski law
- stochastic collision weight
- fitted diffusion coefficient
- fitted rate constant
- selected time, path, species or condition
- selected method, page, frame, member or row
- omitted adverse or unresolved record
- average
- interpolation
- renormalization
- target-derived correction

Immutable identities. Source manifest

sha256:361fb1893935343d3ce1e9f3499bb66e0dc8b891f3bc71c2bb35e901cec80e20; derivation seal sha256:b45b9f5133e8f624a98de623715e4e92f38a9121def6f4e30d7a2530512480eb; engine receipt sha256:9a0aab13e6029bd95143848a3cdcd088e2b4a24251abd0cd8e3b06c5787ed929 at receipts/engine/model_admitted/SFT-CHEM-DIFFUSION-LIMITED-REACTION-BOUNDARY-011-9a0aab13e6029bd9.json; empirical hash sha256:1b8d623b7aff08445036417fd794eea61778ef4e4f4fa0c8b5774c13bfb43d20; measurement receipt sha256:385cd999a4d50661258e771a91217f0a879d4c9aa2a9796d09c39e4415504354.

155. Exact kinetic isotope-effect relation

Claim identity: SFT-CHEM-KINETIC-ISOTOPE-EFFECT-RELATION-012

Question and theorem. For two distinct held isotopologue identities traversing the same complete reaction path under the same held condition, each event rate is independently forced as completed events per exact positive observation parts. Their ordered quotient is the exact kinetic isotope-effect relation; greater, lesser or equal direction remains a held label. No kinetic-isotope equation, numerical mass premise, mass-frequency law, continuum, fitted exponent or statistical weight is imported.

two-distinct-held-isotopologue-identities__same-complete-reaction-path-and-condition__completed-events-per-exact-positive-observation-parts__ordered-quotient-of-two-exact-positive-event-rates__positive-ratio-plus-held-greater-less-or-equal-orientation__complete-90-rate-ratio-3-decay-and-adverse-vector__experimental-vectors-separated-from-source-models-and-review-questions__value-free-71-record-identity-seal-and-depth-independent-pair-successor

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-ELEM-ISOTOPE-001
- SFT-CHEM-NUCLEAR-ELECTRONIC-COMPOSITION-012
- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001
- SFT-CHEM-CONCENTRATION-DEPENDENCE-RELATION-002
- SFT-CHEM-TEMPERATURE-DEPENDENCE-RELATION-003

- SFT-CHEM-ACTIVATION-BARRIER-VALUE-RELATION-004
- SFT-CHEM-TRANSITION-STATE-EQUIVALENT-BOUNDARY-005
- SFT-CHEM-COMPETING-CHANNEL-BRANCHING-006
- SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007
- SFT-CHEM-PARALLEL-MECHANISM-COMPOSITION-008
- SFT-CHEM-REVERSIBLE-KINETIC-EQUILIBRIUM-CORRESPONDENCE-009
- SFT-CHEM-CATALYTIC-TURNOVER-CYCLE-FREQUENCY-010
- SFT-CHEM-DIFFUSION-LIMITED-REACTION-BOUNDARY-011

Generated grammar. Generate the literal product of identity, path, rate, relation, orientation, observation, provenance and prediction forms; decide all 256 candidates only from admitted exact isotope identity, state-transition, mechanism and rate laws.

Boundary: Every finite source-ordered family of complete ordered isotopologue pairs retaining two distinct held isotope identities on one identical reaction path and condition, two independently counted exact positive event rates, one exact positive quotient and every favorable, adverse, control or unresolved status. External testing binds all 71 pre-registered records: article landing, 47 PDF pages and all 23 complete source-data worksheets containing 923,260 nonempty cells, ninety explicit rate-ratio rows and three direct decay KIE inscriptions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: two-distinct-held-isotopologue-identities__same-complete-reaction-path-and-condition__completed-events-per-exact-positive-observation-parts__ordered-quotient-of-two-exact-positive-event-rates__positive-ratio-plus-held-greater-less-or-equal-orientation__complete-90-rate-ratio-3-decay-and-adverse-vector__experimental-vectors-separated-from-source-models-and-review-questions__value-free-71-record-identity-seal-and-depth-independent-pair-successor. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
identity	two-distinct-held-isotopologue-identities	isotopologue-identities-collapsed-or-replaced-by-numerical-mass	Collapsed or numerical identities cannot preserve which reaction carrier changed.
path	same-complete-reaction-path-and-condition	different-truncated-or-answer-only-paths-compared	Different or truncated paths do not isolate an isotope relation.
rate	completed-events-per-exact-positive-observation-parts	imported-rate-law-continuum-time-or-fitted-rate	An imported or fitted rate can select the result.
relation	ordered-quotient-of-two-exact-positive-event-rates	imported-KIE-mass-frequency-exponent-or-statistical-law	An imported isotope-effect model is not forced by the retained pair.
orientation	positive-ratio-plus-held-greater-less-or-equal-orientation	signed-difference-negative-or-zero-number	A signed difference imports forbidden numerical structure.
observation	complete-90-rate-ratio-3-decay-and-adverse-vector	selected-average-or-single-favorable-isotope-effect	Selection or averaging erases replicates, normal, inverse and equal cases.
provenance	experimental-vectors-separated-from-source-models-and-review-questions	experimental-fitted-calculated-and-interpretive-records-mixed	Mixed provenance lets an interpretation masquerade as measurement.
prediction	value-free-71-record-identity-seal-and-depth-independent-pair-successor	rate-ratio-value-workbook-or-target-readable-before-seal	Target access can select the law.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:e2258430ee76f2af16bdb91ad52f88bddbe12b36afe211365dec9a771bcc630b.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:4eff5532afc3fecdd37dc62269a3ca1e050a04f255aa23a1f1b475d2c4894a50.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:33c3033508e143d8440f06e3ce528c706ee2a573bc8975b40ea913555f24bba1.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:216b87f5d926f90ad996e353f7d0102858e987c5ed402c7a1a26776dd9187360.

Independent reconstruction. Implementation hash

sha256:0799de228267d231cc71c36fb1bd13857a6f866e8a4c3a3034ee949e7b8bd7f2; independent certificate sha256:942fb1d54f4c04de6ef76731b61df8aa25fb5950b233ca96d35ff08b2a94b657; external-validation hash sha256:7d483b175daff39afa61d9bc6573d34581f60052ace0985963dbc622be71e246. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NATURE-COMMUNICATIONS-S41467-024-44753-X-COMPLETE

Comparison records:

- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-001: document=article.html; record=complete-article-landing-html; target=sha256:58ee4e71a257cd0d506a62601fbb08aede324b0b8ac0ac30313b99f3d8a46cb0
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-002: document=article.pdf; record=article.pdf-page-1; target=sha256:fecaae8e360c80e5994ceb3a51f1ce01f11d164e2217e947f7524f5096cf8db7
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-003: document=article.pdf; record=article.pdf-page-2; target=sha256:10055b4dc39ded2fd71ce66f55ac733cb0395523fc127768d8eda0f6c6f50532
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-004: document=article.pdf; record=article.pdf-page-3; target=sha256:3c7e1562e18352da0bbe4839789c69dab28a4aaec0838b34f8d2c6578e8611c
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-005: document=article.pdf; record=article.pdf-page-4; target=sha256:261a717de7257a6d835907faeaa4fe731de17464ed65eea7d3cb7c7fbf983995
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-006: document=article.pdf; record=article.pdf-page-5; target=sha256:8b1dfdda0423fe516f5c9341f61e5b0e13a014944f70830d3fced2e213f3cd60
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-007: document=article.pdf; record=article.pdf-page-6; target=sha256:bdc3ca51914870803d48a149186c1eae8cba989ef92e2a496b8fe85dcf371ba
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-008: document=article.pdf; record=article.pdf-page-7; target=sha256:3092f3c87303061f3a32a7252b3c4f2c42e72d1e30c164747e4cddb271f36627
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-009: document=article.pdf; record=article.pdf-page-8; target=sha256:422d2054ee14cf12543501bc4f3d63cfc804edf7e37fe9aa6e966ef29f0d4cf
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-010: document=supplementary-information.pdf; record=supplementary-information.pdf-page-1; target=sha256:49fb297040f69b8776e103a458351d243b5d75b4a3fdc3dce2331a27e23b5393
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-011: document=supplementary-information.pdf; record=supplementary-information.pdf-page-2; target=sha256:37506bfb14bb98796f8d1056548b0624dc3aded87a501b120fb744cfc41c90f

- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-012: document=supplementary-information.pdf; record=supplementary-information.pdf-page-3; target=sha256:e9d291c918b7d385cf76ec34f7f8479cfef9aebfdca6defbe58e18f10ca52514
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-013: document=supplementary-information.pdf; record=supplementary-information.pdf-page-4; target=sha256:e54ecc0f42ae0ed9217586860db6edb628bc92f8365cac9f44addf90e7eb0cd5
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-014: document=supplementary-information.pdf; record=supplementary-information.pdf-page-5; target=sha256:2b20ac3aa14f84bc87412bff87100a59a4e9cd9df7da0bbf9e3f9c3d6cbfced
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-015: document=supplementary-information.pdf; record=supplementary-information.pdf-page-6; target=sha256:c6eb1009b876b7805126b87f2676ffa5a93bf8e9199c8efe57ff882692da6a4b
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-016: document=supplementary-information.pdf; record=supplementary-information.pdf-page-7; target=sha256:1615d7cef43a032ededcf33fd26969b6ba91efcebb4728872c403dd8e038ba20
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-017: document=supplementary-information.pdf; record=supplementary-information.pdf-page-8; target=sha256:dfe124f60bfc13ced5aa83a631673835eb6ec4945f2246e3a689b87e256c01e2
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-018: document=supplementary-information.pdf; record=supplementary-information.pdf-page-9; target=sha256:f813ff47d1529c3a13ca102421c4a41ecc3afa6e10483afb12a06d955e1c4149
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-019: document=supplementary-information.pdf; record=supplementary-information.pdf-page-10; target=sha256:a78d7a9e7a33fb2cbd0c2816a3da1a855af458ea46d0eb9c605c0459b41e6b50
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- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-025: document=reporting-summary.pdf; record=reporting-summary.pdf-page-5; target=sha256:80e9ac600dca03d2828e2a432e8cabeeedafa963d09f1146f233fe0d68dd9977
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- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-028: document=reporting-summary.pdf;
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- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-044: document=reporting-summary.pdf; record=reporting-summary.pdf-page-24; target=sha256:230778ec6d5b7a760a2786a4591b5ecb3ec24417a1a70e07a7e1ea719da3d48d
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- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-046: document=reporting-summary.pdf; record=reporting-summary.pdf-page-26; target=sha256:6d31c7cc365add420d5e2871ab5bc3d59b5028bc18b8c3df7b488b5410871ecc
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-047: document=reporting-summary.pdf; record=reporting-summary.pdf-page-27; target=sha256:f41f6fc9a01f0bc8a2d978a4a7f7c0f96762f30d3b7c2999bdc5918d3fc89934
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-048: document=reporting-summary.pdf; record=reporting-summary.pdf-page-28; target=sha256:e9c4bed163fcd226dd0049e471f73468a648cc25452f760654658be793b53055
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-049: document=source-data.xlsx; record=Figure 3a; target=sha256:7a561528a3fc99e4ae8692d10648e3e834d30d8b1a6a66bfef507aeb060eba3f
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-050: document=source-data.xlsx; record=Figure 3b; target=sha256:e58ca86a0768e633a68f90ad69731e99999be20f263000cf1c38aebd2544fd2c
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-051: document=source-data.xlsx; record=Figure 3c; target=sha256:f11a4ad0d7de6395e62b4611ede78baf40b526d206178d3e94520f18c06f396
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-052: document=source-data.xlsx; record=Figure 4a; target=sha256:26b3ee3c1f8a2e5a72591dee0a009d4af539f69a363615e2c8fabab74b4665c0
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-053: document=source-data.xlsx; record=Figure 4b; target=sha256:eaba0890378322fce36cad05c7b0dc02fc58cd5c3fc881bb17bab54bb0e799c8
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- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-055: document=source-data.xlsx; record=Figure 4d; target=sha256:327bb092f782f242aceb0d04c2645bc97db4ed640553f55c37aeff0e9cbdd897
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-056: document=source-data.xlsx; record=Figure 4e; target=sha256:7bb0830fce126e5979f583bf46836fa573c6a4f3ba48873d7ceb58d28319fd0b
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-057: document=source-data.xlsx; record=Figure 4f; target=sha256:aa29b7f3a7e149f94169ba7cf2a96e32c23a5d5146a0ff606b62f2f65557cd3e
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-058: document=source-data.xlsx; record=Figure 4g; target=sha256:c348297f4a3cf6fa8ca6f977e0b802857fa26d0c619511d4a62054a8636fb07b
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-059: document=source-data.xlsx; record=Figure 4h; target=sha256:8c1dcbcf3df772bd161cb545fb444aa75385113bd8cde756f9b501d1fb71f65b
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-060: document=source-data.xlsx; record=Supplementary Figure 1; target=sha256:bcaac4c404f1a181cca82f67d2e540a0d140b1e0a8f86e166adf956290c496dc
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-061: document=source-data.xlsx; record=Supplementary Figure 2; target=sha256:e49a5dd28035ce1f9b72d16f96137cf333ce828057077e4680534ee7f8d90256
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-062: document=source-data.xlsx; record=Supplementary Figure 3; target=sha256:5061ec985e8ccd74cc2213433bb92f05b5ae16d41663611dcf8352a3eb382672
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-063: document=source-data.xlsx; record=Supplementary Figure 4; target=sha256:797488c478de8ba273938a5c21a53e2b099644e3a2302d6792d57cbaef6590e4

- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-064: document=source-data.xlsx; record=Supplementary Figure 5; target=sha256:bc34028c83f09f559469ecb10c3ae4b068f43e6285a3a73a2ef4c2bc28349422
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-065: document=source-data.xlsx; record=Supplementary Figure 6; target=sha256:10d836dc6a8e8e98bcc0b1fc997396a28277802a43c20229533486ff27902d32
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-066: document=source-data.xlsx; record=Supplementary Figure 8; target=sha256:4f099f5aad5c9b5573f7097f8a06aa3ff29fcd3e4e07ebe0acd2598ece1b91e3
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-067: document=source-data.xlsx; record=Supplementary Figure 9; target=sha256:d85d135f42c0f57b6fe37841d13cbd0abcbeec3cbbb9c68dee91b49f2dfc028
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-068: document=source-data.xlsx; record=Supplementary Figure 10; target=sha256:bb5c037554be6e7f56a5c1132b8b7286efa12bacaabe25e32c6f352a01485d5b
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-069: document=source-data.xlsx; record=Supplementary Figure 11; target=sha256:967490a5b431fda05acf67a30aede6f2435596a1cf079a72307ac0131fd84125
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-070: document=source-data.xlsx; record=Supplementary Figure 12; target=sha256:8b3ea806ceb27455a7442ff76e848a039c3259ab9b59c68ef8a1b21236654919
- SFT-CHEM-KIN012-COMPLETE-SOURCE-RECORD-071: document=source-data.xlsx; record=Supplementary Figure 13; target=sha256:a446778232d5786c67aa29340e686b9cac7152500bd42f52bede893a8f99a7c1
- two distinct held isotope identities retained on one complete reaction path and condition
- two exact positive event rates force one ordered quotient with held greater, lesser or equal orientation
- complete external vector: 90 explicit rate ratios and direct decay KIE inscriptions 2.11, 0.827 and 0.55
- complete source surface: 47 PDF pages, 23 worksheets, 923,260 nonempty cells and 39,002 populated rows
- normal, inverse and near-unity cases, all three experiments, replicates, controls, limitations and reviewer challenges retained
- tampered_omitted_source_record_rejected: True
- tampered_mismatched_isotopologue_path_rejected: True
- complete_71_record_vector_retained: True
- all_47_pdf_pages_retained: True
- all_23_source_data_worksheets_retained: True
- all_923260_nonempty_cells_retained: True
- complete_ratio_decay_replicate_direction_and_adverse_surface_visible: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if either isotope identity, any path role or condition is erased or differs between a pair; if either rate is not independently counted from exact positive events and observation parts; if an ordered quotient or its held orientation is altered; if any of the 71 source records, 47 pages, 23 worksheets, 923,260 nonempty cells, ninety explicit ratios, three decay KIEs, replicate, normal, inverse, near-equal, reviewer challenge or source limitation is omitted; if a KIE equation, numerical mass, mass-frequency law, continuum, fit, exponent, statistical weight, selection, averaging, digitization or target correction enters; if target content opens before all 71 identities and the consequence seal; or if omission or mismatched-path tampering passes.

Measurement receipt: sha256:926ae605cde3648b24eca249cb052e0108bab375501349ee7a1f040dba144f98.
Isolation certificate: sha256:67c96ebb242cc0e988556a3e2a20e9e21355acb72094e1dbfbff323dc0da6b4a.
Custody certificate: sha256:9701379a64749612a84e8b1eeeff226973e2e0e2ed0c792f3b4b902cb9be9128.

Registered source descriptions:

- Nature Communications complete kinetic isotope-effect article and source-data surface: ()

Registered target identities:

- None.

Meaning. The kinetics extension derives exact rate, concentration and temperature dependence, activation, transition boundaries, competing and composed mechanisms, reversible equilibrium, turnover, diffusion limits, isotope effects and product-state dynamics. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero
- negative proof quantity
- irrational quantity
- imaginary quantity
- numerical isotope mass
- kinetic isotope-effect equation
- mass-frequency law
- transition-state premise
- continuum time
- fitted exponent
- statistical weight
- selected isotopologue, reaction or condition
- selected page, worksheet or row
- omitted replicate, adverse or unresolved record
- average
- digitization
- interpolation
- renormalization
- target-derived correction

Immutable identities. Source manifest

sha256:01933d5d0f4df684dec9c640ba49f8905ae536da7282feaefa915f52f5fcfff5; derivation seal sha256:8156e952435b384fd561cbf4d250180f52c21b174552ebc0eeb02729bbc53711; engine receipt sha256:e78e85c742634420e497ccc0b312982f65edd547fa58e040c49c08fe8f80c1be at receipts/engine/model_admitted/SFT-CHEM-KINETIC-ISOTOPE-EFFECT-RELATION-012-e78e85c742634420.json; empirical hash sha256:ea05d6a7967b86e1f1563aef617dce9b9bbfa1ab4cd64c67af5cde687e1f8421; measurement receipt sha256:926ae605cde3648b24eca249cb052e0108bab375501349ee7a1f040dba144f98.

156. Exact reaction-dynamics scattering and product-state law

Claim identity: SFT-CHEM-REACTION-DYNAMICS-SCATTERING-PRODUCT-STATE-013

Question and theorem. One finite held incoming reaction channel and preparation compose with the complete finite support of distinct outgoing joint product-state words. Completed events in each retained state per complete positive event support force exact positive state shares, while each incoming/outgoing orientation remains a held relation. No scattering equation, cross-section law, angular continuum, probability amplitude, fitted potential or distribution is imported.

finite-held-incoming-channel-and-preparation__complete-finite-distinct-outgoing-product-state-support__joint-ordered-coproduct-state-word__completed-state-events-per-complete-positive-event-support__held-incoming-outgoing-orientation-relation__complete-51-record-36-page-14-worksheet-978591-cell-vector__experiment-theory-fit-estimate-tentative-control-and-review-status-separated__value-free-51-record-identity-seal-and-depth-independent-scattering-successor

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-ROVIBRONIC-COMPOSITION-001
- SFT-CHEM-RESOLVED-ROVIBRONIC-SPIN-COMPOSITION-013
- SFT-CHEM-MOLECULAR-QUANTUM-MEASUREMENT-REDUCTION-014
- SFT-CHEM-COMPETING-CHANNEL-BRANCHING-006
- SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007
- SFT-CHEM-PARALLEL-MECHANISM-COMPOSITION-008
- SFT-CHEM-DIFFUSION-LIMITED-REACTION-BOUNDARY-011
- SFT-CHEM-KINETIC-ISOTOPE-EFFECT-RELATION-012

Generated grammar. Generate the literal product of incoming, outgoing, composition, quantity, orientation, observation, provenance and prediction forms; decide all 256 candidates only from admitted finite channel, state, transition, mechanism and exact-count laws.

Boundary: Every finite held incoming reaction channel and preparation; every complete finite, source-ordered support of distinct joint coproduct-state words; exact positive completed-event shares; held incoming/outgoing orientations; and every favorable, adverse, control, fitted, normalized, estimated, tentative or unresolved status. External testing binds all 51 pre-registered records: article landing, 36 PDF pages and all 14 source-data worksheets containing 978,591 nonempty cells and 6,408 key state-resolved product, branching and scattering cells.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `finite-held-incoming-channel-and-preparation__complete-finite-distinct-outgoing-product-state-support__joint-ordered-coproduct-state-word__completed-state-events-per-complete-positive-event-support__held-incoming-outgoing-orientation-relation__complete-51-record-36-page-14-worksheet-978591-cell-vector__experiment-theory-fit-estimate-tentative-control-and-review-status-separated__value-free-51-record-identity-seal-and-depth-independent-scattering-successor`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
incoming	finite-held-incoming-channel-and-preparation	incoming-carriers-erased-or-replaced-by-continuum-beam	Erased or continuous input cannot retain which finite reaction channel was prepared.
outgoing	complete-finite-distinct-outgoing-product-state-support	selected-product-or-continuous-outgoing-distribution	A selected or continuous outcome erases finite alternative product states.
composition	joint-ordered-coproduct-state-word	coproduct-states-separated-or-marginalized	Marginal product states erase which coproduct pair occurred together.
quantity	completed-state-events-per-complete-positive-event-support	imported-cross-section-probability-amplitude-or-fitted-density	An imported or fitted distribution can select the answer.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
orientation	held-incoming-outgoing-orientation-relation	signed-angle-continuum-or-numerical-zero-direction	A continuum angle or signed scalar imports forbidden numerical structure.
observation	complete-51-record-36-page-14-worksheet-978591-cell-vector	selected-normalized-or-favorable-product-vector	Selection or normalization can erase weak, absent, adverse or discrepant channels.
provenance	experiment-theory-fit-estimate-tentative-control-and-review-status-separated	experimental-theoretical-fitted-estimated-and-tentative-records-mixed	Mixed provenance lets a model or correction masquerade as observation.
prediction	value-free-51-record-identity-seal-and-depth-independent-scattering-successor	product-values-workbook-or-target-readable-before-seal	Target access can select the channel or orientation law.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : c3368122ce69604fa482097f98a6f24cfdc39a48881caf520316891513ab63b0.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 005b5a6989d16edcce3dd0ab053d1cd39097f02ab6f2453fc91a1686dd62fb6c.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 5ed5def66516cd6ad6f9f1a3fdd09bd19a0cbb828f703d6c4c039d9d425412dc.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 2b981718d9e7635b0a69e481ae2b19b0c93cb95e5cd80c7cbfa8607a99beacb0.

Independent reconstruction. Implementation hash

sha256 : 79927404d80ebd867b3ec6c2b4d332013fd19c13dbfd2242ce6e177d825bc5af; independent certificate sha256 : 304263ad8ca7dce9f21aaa298852ae0f05beef610bf3eea49f2ee8c2d5d26753; external-validation hash sha256 : 4018f0f6cca90eb82d37bcf71aa602f08c730067dc67be5f051a54743b8fa0df. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NATURE-COMMUNICATIONS-S41467-025-66587-X-COMplete

Comparison records:

- SFT-CHEM-KIN013-COMplete-SOURCE-RECORD-001: document=article.html; record=complete-article-landing.html; target=sha256:a29fa09ac3f11e8d4ce3eb0d7f3aa7f2b3efb9889d2ef0494054685a92e031de
- SFT-CHEM-KIN013-COMplete-SOURCE-RECORD-002: document=article.pdf; record=article.pdf-page-1; target=sha256:e827d6b8900ac17ce5db46ac355919512fc58563bf3e9ea42877ab6e824f2042
- SFT-CHEM-KIN013-COMplete-SOURCE-RECORD-003: document=article.pdf; record=article.pdf-page-2; target=sha256:ac5c317a05d2439cfc4844af364923449df081c4208d92f3800253ed2516bf61
- SFT-CHEM-KIN013-COMplete-SOURCE-RECORD-004: document=article.pdf; record=article.pdf-page-3; target=sha256:cedbc80274240aea1caf1ef4f96185cf1e8530badaded8e0f6d2e926930cae4b
- SFT-CHEM-KIN013-COMplete-SOURCE-RECORD-005: document=article.pdf; record=article.pdf-page-4; target=sha256:18dd63f90b935a99ddb6166436782514c289d3078a9eb663526884a240641e4b

- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-006: document=article.pdf; record=article.pdf-page-5; target=sha256:e5e35cb5ab2aeb71852de20787fc43fa49180ebe73a27004bf38820d55884259
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-007: document=article.pdf; record=article.pdf-page-6; target=sha256:ba908bdd0256e1b87b7b5200931f28bd8ea00099882cb88a87124b1ad70eb147
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-008: document=article.pdf; record=article.pdf-page-7; target=sha256:6d9f757923f877623378cd6870eca0114e17ccc570b3fbcad34dbed7ecfeb022
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-009: document=article.pdf; record=article.pdf-page-8; target=sha256:20e0158be5cd85ad47c83288892385004c7f45e8bfac702558ecf9fd635ad30c
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-010: document=article.pdf; record=article.pdf-page-9; target=sha256:8b168671c4c734effc0a66462c84d41530f2b7edd1bd9318f56c5533e5238c61
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-011: document=supplementary-information.pdf; record=supplementary-information.pdf-page-1; target=sha256:352d1d82da946f82f13665116f58f5e68df32d6e2d2148e019e9e7f9be8d3560
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-012: document=supplementary-information.pdf; record=supplementary-information.pdf-page-2; target=sha256:369720e7482e8d8953d22912b4f53a3396c3e2a110add2fe02c31f7dc4118e08
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-013: document=supplementary-information.pdf; record=supplementary-information.pdf-page-3; target=sha256:061c243ffd2966d46c2bcd0ed90c1497a97c15f771e93063c252a938a75c3d37
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-014: document=supplementary-information.pdf; record=supplementary-information.pdf-page-4; target=sha256:921177ad280861bb87f6b06155eaf07e29e10147198a9aebc4efcd37cd5e157a
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-015: document=supplementary-information.pdf; record=supplementary-information.pdf-page-5; target=sha256:e833ec23a0c04e23fe7a4a761567e3ca46fa7af8d93eff256bd4880394b88fd4
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-016: document=supplementary-information.pdf; record=supplementary-information.pdf-page-6; target=sha256:ba0fad8a4103991cf9da5c73d487257cf1a8b5ff386961f75fcc5ac44d16c862
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-017: document=supplementary-information.pdf; record=supplementary-information.pdf-page-7; target=sha256:fde6e781679dff82102b1648679ca62517b36686f477036f9bd8722baaa4c4c9
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-018: document=supplementary-information.pdf; record=supplementary-information.pdf-page-8; target=sha256:085d43a7d33d780a462f816541f1d4439636027931b03c4f1f3ea7aead645182
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-019: document=supplementary-information.pdf; record=supplementary-information.pdf-page-9; target=sha256:1523226119d343ebf93b2de659f3f8007bf2a979cb12e1bcf3b55a808e1606fc
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-020: document=supplementary-information.pdf; record=supplementary-information.pdf-page-10; target=sha256:de05021eeaf66a61eff4d6b026e47b9a9d194ae6b32f26eada1f941272efa698
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-021: document=supplementary-information.pdf; record=supplementary-information.pdf-page-11; target=sha256:ca2ab0de9c95321e9ed9d5f1611ce7d90b0fffb34b40587663d11a2b280a2b3c
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-022: document=transparent-peer-review.pdf; record=transparent-peer-review.pdf-page-1; target=sha256:4e0ed0d7ae313088a60150ad4688772386ebc886d22eabb35221732f7e133d4d
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-023: document=transparent-peer-review.pdf; record=transparent-peer-review.pdf-page-2;

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- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-024: document=transparent-peer-review.pdf; record=transparent-peer-review.pdf-page-3; target=sha256:728e148c4d784bc811f37a57a6e87eec4ea755ca2444efbf77487f49b3c1f6cd
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-025: document=transparent-peer-review.pdf; record=transparent-peer-review.pdf-page-4; target=sha256:deb0eb0527e6a76ae3eb60ffa07f969cded8eccb574476546902b8f976ae53c1
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-026: document=transparent-peer-review.pdf; record=transparent-peer-review.pdf-page-5; target=sha256:4ff2befb080911d435af89f025a7db0c5c7d83fb3a8b54ec1e5d01f4840e6cec
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-027: document=transparent-peer-review.pdf; record=transparent-peer-review.pdf-page-6; target=sha256:becbacc3eae5c69a0ca984bd24a54fcd6b73f3a57ce7614017a9835290d6549b
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-028: document=transparent-peer-review.pdf; record=transparent-peer-review.pdf-page-7; target=sha256:49bb0712d577541c202842608770ac4d370f94b6636d93ec7693f85427a4e5db
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-029: document=transparent-peer-review.pdf; record=transparent-peer-review.pdf-page-8; target=sha256:4eac19b97cb74e27d46225e4a39780353339a70eaa9567cb35cb9a50a4b34774
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-030: document=transparent-peer-review.pdf; record=transparent-peer-review.pdf-page-9; target=sha256:e66522f37faf9004c2df7e79e17ef696af5e6d8162b68f36705cbe57f1c548c8
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-031: document=transparent-peer-review.pdf; record=transparent-peer-review.pdf-page-10; target=sha256:8b79b187871d99dbc9abd22e1aff1462c452dfab52210dff99dbc2e609868aae
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-032: document=transparent-peer-review.pdf; record=transparent-peer-review.pdf-page-11; target=sha256:670331609eae5c7168f49ed775cc63851e3d8baf69b2981714edf4e1308dc8d0
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-033: document=transparent-peer-review.pdf; record=transparent-peer-review.pdf-page-12; target=sha256:babdb6ecf44e442dd4d6364101d302e47d09b77801976520e50f2a9925fce6185
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-034: document=transparent-peer-review.pdf; record=transparent-peer-review.pdf-page-13; target=sha256:3f76495e5985d11acea32f4cb3dfa52fde3e9b9ecda60984442748e7d7d1780c
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-035: document=transparent-peer-review.pdf; record=transparent-peer-review.pdf-page-14; target=sha256:fce13d971220e106d0c750e13f011e83ec8de8579ca9949f6ce3b3c79892e573
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-036: document=transparent-peer-review.pdf; record=transparent-peer-review.pdf-page-15; target=sha256:33df7a8e663fc02e75c106b80477670986a3317b618529e90949567b7f61a556
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-037: document=transparent-peer-review.pdf; record=transparent-peer-review.pdf-page-16; target=sha256:851cca9a74068f2c36ccbd35cf79e771ca30c2f96a7bbd524b2836e7b4c8b158
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-038: document=source-data.xlsx; record=fig 1; target=sha256:85ead855a055eecb16af5db34798a35e92011b02d538d864fe69c28a593ac4d0
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-039: document=source-data.xlsx; record=fig2; target=sha256:7d9676ad04d2bf8817b2bf5159100ebab9ae79dfe4ff66e74b6e2a7ad789023c

- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-040: document=source-data.xlsx; record=fig3; target=sha256:be6262181e27918cdb7e191198cf50272f526647b363867229c7b49d4b145dbc
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-041: document=source-data.xlsx; record=fig4; target=sha256:866fa7b8bc9edc93b6d81c585e6017fcaeaaf81fe69af0531f51939f8057cc08
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-042: document=source-data.xlsx; record=Supplementary Fig. 2A; target=sha256:da72e08428300b3b15989c6c5a80dcc4373285bde0573fc45f525a48e6cd485c
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-043: document=source-data.xlsx; record=Supplementary Fig. 2B; target=sha256:eafb4b0ab23a862cb1246f46b4b159a7015a9e240fa333b2c87c0f32a1f2fa62
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-044: document=source-data.xlsx; record=Supplementary Fig. 2C; target=sha256:74ed28535ccfd12b499f9958043b880a24466d9d199d09fcf8f0f7968f0f38ed
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-045: document=source-data.xlsx; record=Supplementary Fig. 3; target=sha256:c8b3fe05ee28a2d166656f60abec021e571440223442371936363356cb387b3a
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-046: document=source-data.xlsx; record=Supplementary Fig. 4; target=sha256:60c0781da0b3596341e9d043c319b738a7a937cb2a36e51484da451218fee72a
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-047: document=source-data.xlsx; record=Supplementary Fig. 5; target=sha256:0621ea1aff906bd15add374ef1f2a576c788052f9a22116b24b26ad47c7db8ac
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-048: document=source-data.xlsx; record=Supplementary Fig. 6; target=sha256:c69a349512baea922f419c29dc971eda7fad9d6916e5d28a4de7f9224d60a9eb
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-049: document=source-data.xlsx; record=Supplementary Fig. 7; target=sha256:49ce1ccabf1f134a036201af7ec900fa675b88d184ff765068d980ab6cefcea8
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-050: document=source-data.xlsx; record=Supplementary Fig. 8; target=sha256:4f6aa5baa7953b2547c5e2c4fee1f85e3f91d0f65ffabb9ca0547c121fe1bb23
- SFT-CHEM-KIN013-COMPLETE-SOURCE-RECORD-051: document=source-data.xlsx; record=Supplementary Fig. 9; target=sha256:6c940f91ddb4c29e4f4b311ddd8bc97d0c99d39886dcf218f0a7d4866f8a35a8
- finite held F+CH₄ incoming preparation and complete joint CH₃(vi)+HF(v) outgoing product-state support retained
- exact positive state shares and held same/transverse/opposed incoming-outgoing orientation relations reconstructed
- source external vector retains 40% ground-state CH₃ and 57% experimental versus 58% theoretical umbrella-excited flux inscriptions
- source orientation progression retains (0_0,3) forward, (2_2,2) sideways and (0_0,2) backward inscriptions
- 36 PDF pages, 14 worksheets, 978,591 nonempty cells and 6,408 key state-resolved cells remain byte-bound
- fits, normalizations, tentative HF(v=1), sharper theoretical forward peak, theory limitations, detection limitations and full reviewer challenges remain visible
- tampered_omitted_source_record_rejected: True
- tampered_incoming_outgoing_reaction_mismatch_rejected: True
- complete_51_record_vector_retained: True
- all_36_pdf_pages_retained: True
- all_14_source_data_worksheets_retained: True
- all_978591_cells_and_6408_key_state_cells_retained: True
- complete_fit_normalization_tentative_limitation_discrepancy_and_review_surface_visible: True
- prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if any incoming carrier, preparation, outgoing product carrier, product-state coordinate or joint pair identity is erased; if outgoing support is selected, duplicated or incomplete; if a state share is not forced from exact positive completed events and complete positive support; if orientation becomes a signed angle or continuum; if any of the 51

source records, 36 pages, 14 worksheets, 978,591 nonempty cells, 6,408 key state-resolved cells, weak or tentative channel, fitted or normalized record, experimental/theoretical discrepancy, limitation, reviewer challenge or correction record is omitted; if a scattering equation, cross-section law, amplitude, potential, continuum, fit, selection, averaging, interpolation or target correction enters; if target content opens before all 51 identities and the consequence seal; or if omission or mismatched-reaction tampering passes.

Measurement receipt: sha256:aefb1f824ee1e73e79824dff52258d6e7bb71c0b431794f5ad028be8264f84e1.

Isolation certificate: sha256:d72236ea29470aac8ef95210feb2f157fca4dbaa488049e17056f863dbb71c08.

Custody certificate: sha256:887c67664c38a1206126904c7d92621a69aa5888f86a0a53cd4d3f558a70117f.

Registered source descriptions:

- Nature Communications complete pair-correlated product-state and scattering study: ()

Registered target identities:

- None.

Meaning. The kinetics extension derives exact rate, concentration and temperature dependence, activation, transition boundaries, competing and composed mechanisms, reversible equilibrium, turnover, diffusion limits, isotope effects and product-state dynamics. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero
- negative proof quantity
- irrational quantity
- imaginary quantity
- scattering equation
- cross-section law
- angular continuum
- probability amplitude
- fitted potential or distribution
- imported energy or momentum equation
- selected product state, channel or orientation
- selected page, worksheet, cell or row
- omitted weak, adverse, reviewer or unresolved record
- average
- digitization
- interpolation
- renormalization
- target-derived correction

Immutable identities. Source manifest

sha256:c17710e4bc85dee6f36839f24d359b51693dbf81d0fc3e1bb988fffd1c0491d3; derivation seal

sha256:22d4fb87347a192cda404a89b583b39616c3b1c41e7849f60ac3d80aac45c897; engine receipt

sha256:8faabcf7374b20bedc92888804c83aec6e4fea13c9c6a3a56504466db2a4b2a7 at receipts/engine/model_admitted/SFT-CHEM-REACTION-DYNAMICS-SCATTERING-PRODUCT-STATE-013-8faabcf7374b20be.json; empirical hash

sha256:9866023815ae2ecd13eacb18d11ef95258a88eba12ba651feb5d2661127063e0; measurement receipt

sha256:aefb1f824ee1e73e79824dff52258d6e7bb71c0b431794f5ad028be8264f84e1.

22. Inorganic, coordination, solid-state and organometallic chemistry - 17 admitted claims

Coordination, spin, colour, magnetism, metal-carbon support, clusters and defects remain exact typed organizations. This entire registered subcategory is included; no proper subset is presented as field completion.

157. Coordination entity and retained central-ligand identity law

Claim identity: SFT-CHEM-COORDINATION-ENTITY-RETAINED-IDENTITY-001

Question and theorem. A coordination entity is the complete retained carrier of one central occurrence, every distinct ligand occurrence and every positive central-ligand incidence. Central and surrounding roles are forced by common incidence; repeated ligand labels never erase occurrences, and the next complete ligand preserves every prior identity.

complete-coordination-entity-carrier__incidence-forced-central-and-surrounding-roles__central-and-every-ligand-occurrence-retained__positive-central-ligand-incidence-trace__complete-gap-free-ligand-support__structural-EmptyOne-only__complete-20-record-identity-sealed-structure-vector__next-ligand-preserves-entire-prior-entity

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-ELECTRONIC-STATE-IDENTITY-001
- SFT-CHEM-MULTICENTRE-DELOCALIZED-SUPPORT-008

Generated grammar. Generate the literal product of carrier, role, identity, attachment, composition, absence, observation and extension forms. Decide all 256 forms only by admitted Fold identity, finite incidence, exact composition, source-boundary and successor laws.

Boundary: Every positive finite single-central coordination support with one or more distinct attached ligand occurrences, tested after seal against all twenty registered IUPAC and NIST definition, structure, value, reference and limitation records.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *complete-coordination-entity-carrier__incidence-forced-central-and-surrounding-roles__central-and-every-ligand-occurrence-retained__positive-central-ligand-incidence-trace__complete-gap-free-ligand-support__structural-EmptyOne-only__complete-20-record-identity-sealed-structure-vector*

`__next-ligand-preserves-entire-prior-entity`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-coordination-entity-carrier	formula-or-name-only	A name does not retain the entity's generated constituents.
role	incidence-forced-central-and-surrounding-roles	conventional-central-and-ligand-names-imported	Imported role names cannot force which occurrence is common to every incidence.
identity	central-and-every-ligand-occurrence-retained	element-or-ligand-occurrences-collapsed	Collapsing repeated labels erases distinct occurrences.
attachment	positive-central-ligand-incidence-trace	proximity-or-continuum-distance-only	Proximity alone does not retain a generated attachment.
composition	complete-gap-free-ligand-support	selected-or-average-ligand-support	Selection or averaging erases members of the coordination entity.
absence	structural-EmptyOne-only	numerical-zero-or-negative-vacancy	Numerical zero and signed vacancy are not Fold-native forms.
observation	complete-20-record-identity-sealed-structure-vector	source-structure-visible-before-seal	Observed structures could select the law.
extension	next-ligand-preserves-entire-prior-entity	successor-replaces-central-or-prior-ligand	Replacement destroys retained identity.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 3254bec33aa01d96b62e74217f4d37ae318000c91abbcb4ad4ef37c90694368f.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: f66cb6ca88fb85372b37bf07edebb7f4da470cf35b55436d89ffd8bdf6aac60f.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: e1e2b0d99f3d534c6bd98ece387c012a856e7f1fd79bdcc4ac386298fc90dba3.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: a8393883a2f33f9528f53f16d5fc1843af1f820b2c3ba30856be27c6abe75e2a.

Independent reconstruction. Implementation hash

sha256: 8dc7f3501d87b96658691eb9d8d501d6640ef3e834ccb46378233179d272e6ef; independent certificate sha256: cc7b6d5dbd45c62de9235ea48faabb75d9dfaf1014d1dc61878604971f75bb42; external-validation hash sha256: 8d62f4298ece0e57faf6552f89ff52dbb2ce3489a668e88f7f210b6a126b47ec. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-C01330-C00930-L03518
- NIST-CCCBDB-SRD101-FECO5-FERROCENE

Comparison records:

- SFT-CHEM-INORG001-STRUCTURE-001: coordination-entity-current-term=An assembly consisting of a central atom (usually metallic) to which is attached a surrounding array of other groups of atoms (ligands).; target=sha256:3a7da8adc69d1c87fd0614cec9875af2ed93b4dbef5dd25011f817e805cc06a9
- SFT-CHEM-INORG001-STRUCTURE-002: central-atom-current-term=The atom in a coordination entity which binds other atoms or groups of atoms (ligands) to itself, thereby occupying a central position in the coordination entity.; target=sha256:4851ffe8dd614ffe3c2f5fbf04e2d6c7d3f1a1170d3b4e6ecdaed4ad74bf8169
- SFT-CHEM-INORG001-STRUCTURE-003: ligands-current-term=In an inorganic coordination entity, the atoms or groups joined to the central atom.; target=sha256:56b06c26e7d6743816d781eca1d5e803481a4a850878deb5a3f8c79a6b7fc100
- SFT-CHEM-INORG001-STRUCTURE-004: FeCO5-complete-page=sha256:62d6da495b385e07a80ea58d986b40ed11973dd639863055e38e23493222159d; target=sha256:e5165e952f88b4d2a96f874ffc87f402eb2de733cd5d5f95d6f023c615fa6f1f
- SFT-CHEM-INORG001-STRUCTURE-005: FeCO5-entity-identity=Fe(CO)5; Iron pentacarbonyl; target=sha256:02cb4c39fc95cbeaa3811b8e8bc9e68eed0836409a148a4ab66974fadf4d605b
- SFT-CHEM-INORG001-STRUCTURE-006: FeCO5-point-group=D3h; target=sha256:c4fe369559689ea1fedba6af7ca05b7852b94601ce88048911782e0189ef527b
- SFT-CHEM-INORG001-STRUCTURE-007: FeCO5-C-Fe-link-count=5; target=sha256:704923826a608671c3f0e7389f3b1a974f1fb1933afd09ee0e0e0b6e518299a4
- SFT-CHEM-INORG001-STRUCTURE-008: FeCO5-C-O-link-count=5; target=sha256:b97521e3ffad2973f5d373e10da344f1b00c563d68fe67d8ea659c93ce77a5d5
- SFT-CHEM-INORG001-STRUCTURE-009: FeCO5-internal-coordinate-record=rCO=1.145 angstrom; source comment average of equatorial and axial; target=sha256:ebc85bce2e69080b3beb6ec77b5c7c23a35372f085e13a12076f65526fd4499d
- SFT-CHEM-INORG001-STRUCTURE-010: FeCO5-coordinate-table=Fe1 and ten retained C/O occurrences; five Fe-C distances 1.8240 angstrom; target=sha256:648250ea3a5015d5481d3eb60a645268ccb629f37cddb39d3f6e3965b3dc687c
- SFT-CHEM-INORG001-STRUCTURE-011: FeCO5-reference-record=1976Hellwege(II/7); Structure Data of Free Polyatomic Molecules; target=sha256:26a47c937ae0f90063391f64c24d4f078ce9d77b4fb548e74e1583bcb8dd0361
- SFT-CHEM-INORG001-STRUCTURE-012: FeCO5-rotational-data-status=No experimental rotational constants available; target=sha256:33358eb55da85fddc21c779359387ec9b375b50a2bfe42b5d3784c59e88ef6d9
- SFT-CHEM-INORG001-STRUCTURE-013: ferrocene-complete-page=sha256:b6f9f0c8bdd15df1d166a3b2b9cf7c8517d72529976ff062caaca36d44ad97f0; target=sha256:1b793e05355b93fdfee04531cdd24942c046983577f8fed2e588ecddffbe896
- SFT-CHEM-INORG001-STRUCTURE-014: ferrocene-entity-identity=Fe(C5H5)2; ferrocene; target=sha256:977428f3cde22ff19c844d74bd691aea7d4fc023f470b039b6beb9cbe374e7f
- SFT-CHEM-INORG001-STRUCTURE-015: ferrocene-point-group=D5d; target=sha256:c98537d7ac440f0f6672c60118b3b86d8d27b653b0be2ec224b1d3427a7a49b0
- SFT-CHEM-INORG001-STRUCTURE-016: ferrocene-C-Fe-link-count=10; target=sha256:f0af41727320e3b3781574407f721e9997c662f4afd3d8a81c3d9e4231edf736
- SFT-CHEM-INORG001-STRUCTURE-017: ferrocene-C-C-link-count=10; target=sha256:d210cb041fa0fca9abf20c401da9a48536269b7e342272f740cb3753195e2507
- SFT-CHEM-INORG001-STRUCTURE-018: ferrocene-H-C-link-count=10; target=sha256:4607075154506ce6c1b4c808ffbb223a28770d076f9f12b18f3c7d8f77a85386
- SFT-CHEM-INORG001-STRUCTURE-019: ferrocene-coordinate-data-status=No coordinate data available; target=sha256:8a8863f7c1cda3d15ece9c78fc772bb8f2fbd8c17004b93942521cd46595cee
- SFT-CHEM-INORG001-STRUCTURE-020: ferrocene-rotational-data-status=No experimental rotational constants available; target=sha256:d8db1b99e3a2f8626d9716dca71b89dd3dd6fed144077f88663500a38b2d2e26
- control tampered_omitted_source_record_rejected: True
- control tampered_central_occurrence_mismatch_rejected: True

- control numerical_zero_rejected: True
- control complete_20_record_vector_retained: True
- control reported_absence_and_disclaimer_surface_visible: True
- control prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if a coordination entity does not retain its central occurrence, any ligand occurrence or any central-ligand incidence; if repeated ligand labels collapse occurrences; if adding one ligand replaces prior identity; if any of twenty registered authority records changes or is omitted; if NIST Fe(CO)₅ or ferrocene lacks the retained Fe-ligand structure; if reported absences or disclaimers are suppressed; or if numerical zero, signed, continuum, fitted or target-derived proof structure enters the law.

Measurement receipt: sha256:03b32d277c08d7f342ba2de0f99ce5c7b1fba32bf8c5144574cdfd1bbbcd068.

Isolation certificate: sha256:a536252215beca4eb9715080490d30d5178581f4c3b79acacc4b78a73cfcb11a.

Custody certificate: sha256:b27eeeff4c6ed065361384f537099ecb58fa97f22d5d120130d1ca525147b49.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01330/json> (sha256:4019d4af2537787bbcb95ca7d94446b8d03a73751075c1401e1d74babd77145b)
- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C00930/json> (sha256:0b54c0a0c7f9e4f4dafbc8ef72b10f983fe84029905b93d0e3abf8d4db89567d)
- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/L03518/json> (sha256:e92c4fe3e4fadefef66f21481b61fdd9bab374bfda4a4ce59149866693f03fac)
- National Institute of Standards and Technology: <https://cccbdb.nist.gov/expgeom2x.asp?casno=13463406&charge=0> (sha256:62d6da495b385e07a80ea58d986b40ed11973dd639863055e38e23493222159d)
- National Institute of Standards and Technology: <https://cccbdb.nist.gov/expgeom2x.asp?casno=102545&charge=0> (sha256:b6f9f0c8bdd15df1d166a3b2b9cf7c8517d72529976ff062caaca36d44ad97f0)

Registered target identities:

- SFT-CHEM-INORG001-STRUCTURE-001 from IUPAC-GOLD-BOOK-C01330
- SFT-CHEM-INORG001-STRUCTURE-002 from IUPAC-GOLD-BOOK-C00930
- SFT-CHEM-INORG001-STRUCTURE-003 from IUPAC-GOLD-BOOK-L03518
- SFT-CHEM-INORG001-STRUCTURE-004 from NIST-CCCBDB-SRD101-FECO5
- SFT-CHEM-INORG001-STRUCTURE-005 from NIST-CCCBDB-SRD101-FECO5
- SFT-CHEM-INORG001-STRUCTURE-006 from NIST-CCCBDB-SRD101-FECO5
- SFT-CHEM-INORG001-STRUCTURE-007 from NIST-CCCBDB-SRD101-FECO5
- SFT-CHEM-INORG001-STRUCTURE-008 from NIST-CCCBDB-SRD101-FECO5
- SFT-CHEM-INORG001-STRUCTURE-009 from NIST-CCCBDB-SRD101-FECO5
- SFT-CHEM-INORG001-STRUCTURE-010 from NIST-CCCBDB-SRD101-FECO5
- SFT-CHEM-INORG001-STRUCTURE-011 from NIST-CCCBDB-SRD101-FECO5
- SFT-CHEM-INORG001-STRUCTURE-012 from NIST-CCCBDB-SRD101-FECO5
- SFT-CHEM-INORG001-STRUCTURE-013 from NIST-CCCBDB-SRD101-FERROCENE
- SFT-CHEM-INORG001-STRUCTURE-014 from NIST-CCCBDB-SRD101-FERROCENE
- SFT-CHEM-INORG001-STRUCTURE-015 from NIST-CCCBDB-SRD101-FERROCENE
- SFT-CHEM-INORG001-STRUCTURE-016 from NIST-CCCBDB-SRD101-FERROCENE
- SFT-CHEM-INORG001-STRUCTURE-017 from NIST-CCCBDB-SRD101-FERROCENE

- SFT-CHEM-INORG001-STRUCTURE-018 from NIST-CCCBDB-SRD101-FERROCENE
- SFT-CHEM-INORG001-STRUCTURE-019 from NIST-CCCBDB-SRD101-FERROCENE
- SFT-CHEM-INORG001-STRUCTURE-020 from NIST-CCCBDB-SRD101-FERROCENE

Meaning. The inorganic extension derives coordination identity and number, denticity, geometry, isomerism, splitting, spin, colour, magnetism, organometallic support, inverse redox transforms, cluster bonding, local solid organization, defects and coupled inorganic reaction networks. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero
- negative, irrational, imaginary, floating, signed or continuum proof value
- imported coordination-number, valence-bond, ligand-field, molecular-orbital or geometry model
- observed IUPAC definition or NIST structure record before prediction seal
- fitted bond length, selected complex, omitted record or target-derived correction

Immutable identities. Source manifest

sha256:4005ad04a1068adf42e713e5c30a08c2744c8feb41d79fa4c82873192ec8a424; derivation seal
 sha256:f4e20ff7170f7fb214d7af331b5d36802966686add2564649be2afc8a41a81d4; engine receipt
 sha256:bf8c0a082163ff1e531df0754a15edff05eb59c8c881f602af3bc4b581c030b6 at receipts/engine/model_admitted/SFT-CHEM-COORDINATION-ENTITY-RETAINED-IDENTITY-001-bf8c0a082163ff1e.js
 on; empirical hash sha256:f9185228d2ae4254e6d60f0be3e8881f663984c49983a4dcd01404b96c6281aa;
 measurement receipt sha256:03b32d277c08d7f342ba2de0f99ce5c7b1fba32bf8c5144574cdffd1bbbcd068.

158. Exact coordination number from generated central-ligand incidence

Claim identity: SFT-CHEM-COORDINATION-NUMBER-INCIDENCE-COUNT-002

Question and theorem. For an admitted complete coordination entity, coordination number is exactly the positive cardinality of distinct direct ligand-incidence traces on the retained central occurrence. Equal ligand-group labels do not collapse occurrences; structural absence contributes no incidence; the next direct attachment preserves all prior traces and adds exactly one.

admitted-complete-coordination-entity__every-distinct-direct-central-incidence__positive-count-of-generated-incidence-traces__occurrence-distinct-incidences-counted-once__structural-EmptyOne-outside-positive-support__complete-23-record-value-free-identity-seal__direct-inorganic-incidence-sense-held__next-incidence-preserves-prior-and-adds-one

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-STOICH-COMPOSITION-001

• SFT-CHEM-COORDINATION-ENTITY-RETAINED-IDENTITY-001

Generated grammar. Generate the literal product of carrier, membership, quantity, identity, absence, observation, boundary and extension forms. Decide all 256 forms solely from admitted complete coordination identity, exact finite incidence and source-boundary laws.

Boundary: Every admitted positive finite single-central coordination entity and its complete distinct direct-incidence support, tested after seal against all twenty-three registered IUPAC and NIST definition, structure, count and limitation records.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: admitted-complete-coordination-entity__every-distinct-direct-central-incidence__positive-count-of-generated-incidence-traces__occurrence-distinct-incidences-counted-once__structural-EmptyOne-outside-positive-support__complete-23-record-value-free-identity-seal__direct-inorganic-incidence-sense-held__next-incidence-preserves-prior-and-adds-one. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	admitted-complete-coordination-entity	formula-or-central-name-only	A name does not expose complete incident support.
membership	every-distinct-direct-central-incidence	nearby-or-selected-ligands	Proximity or selection does not retain the entity's direct incidence boundary.
quantity	positive-count-of-generated-incidence-traces	imported-coordination-number-table	A conventional table selects the answer.
identity	occurrence-distinct-incidences-counted-once	repeated-ligand-labels-merged	Merging equal group labels undercounts distinct occurrences.
absence	structural-EmptyOne-outside-positive-support	numerical-zero-or-negative-vacancy	Numerical zero and signed vacancy are forbidden proof forms.
observation	complete-23-record-value-free-identity-seal	observed-counts-readable-before-seal	Observed 3, 4 or 5 could select a counting rule.
boundary	direct-inorganic-incidence-sense-held	general-crystal-pi-and-sigma-senses-mixed	Mixing distinct source senses makes the count ambiguous.
extension	next-incidence-preserves-prior-and-adds-one	next-ligand-recounts-or-replaces-support	Replacement does not preserve the prior count proof.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 06240c8ae23e0d02a4dbf631e88ae9be3ba86d6c1c70da935995a6a1fa406e74.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: b858b1d443e38816bc4d9e47eb36c5e480a86198ca26dc7b875e94d32dd359b8.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: e954b1f1e39dd1334dd16f0d9ad3ad29f95fab3fcbbbeb41847549e663825f49.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: be953d5c33caebf0ec4bc2bf110273f93907a3f6e89f73a2d5a28f279cef7d94.

Independent reconstruction. Implementation hash

sha256:5683ddca7f0db8012cdebf267e3d0f4718374599beaaba4a3aff6981d68f7b65; independent certificate sha256:42228dbd9e2b81592e605444b8150a5cdab6695b31ad80c998eab85d65a16671; external-validation hash sha256:ad26c6a9ea74dd6c60c622565ea9e0e703bc373d818d997e0f5d2b8e880b664e. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-C01331
- NIST-CCCBDB-SRD101-SCF3-TICL4-FECO5

Comparison records:

- SFT-CHEM-INORG002-COUNT-001:
complete-current-term-record=sha256:5c995bf40117e65eb042dc5d585692c56443ef3dc93805466779053da415fb5a;
target=sha256:de6cff135985df03966a6bd0b1d90d48b50e15071f49a6de09f6871626549267
- SFT-CHEM-INORG002-COUNT-002: general-direct-link-definition=The coordination number of a specified atom in a chemical species is the number of other atoms directly linked to that specified atom. For example, the coordination number of carbon in methane is four, and it is five in protonated methane, CH_5^+ . (The term is used in a different sense in the crystallographic description of ionic crystals.);
target=sha256:96e448e97df309e6484fad8ba2a952664af4f8c98f844f95657413f6964e34e2
- SFT-CHEM-INORG002-COUNT-003: inorganic-sigma-link-definition=In an inorganic coordination entity, the number of σ bonds between ligands and the central atom. π bonds are not considered in determining the coordination number.;
target=sha256:084ed54301da5690d47287844031867d37b062da578e5716a102105b64c66e6d
- SFT-CHEM-INORG002-COUNT-004: crystallographic-sense-boundary=term is used in a different sense in the crystallographic description of ionic crystals; target=sha256:82adb87cf4027b63293a83644c674f5caf39d3bb33e953f8938e0c9c18c243b3
- SFT-CHEM-INORG002-COUNT-005: pi-link-exclusion-boundary= π -bonds are not considered in determining the coordination number; target=sha256:eea983d5c69d7d87ea8d4a42612c22247e57cb7a69594c2d2df3fedd74d39978
- SFT-CHEM-INORG002-COUNT-006:
ScF3-complete-page=sha256:63e6ea2c88b3aef4137c15a7986358be95c411aff96a9789ec421641e20963dc;
target=sha256:c59b1131599e8796e1a489588a0981d5f8f4acac22ff7006075ea5e49724cf58
- SFT-CHEM-INORG002-COUNT-007: ScF3-entity-identity=ScF3; Scandium trifluoride;
target=sha256:cdbecf03f6600c848e327ef34fe57774a34005e29fee79086f7cd55609dd065
- SFT-CHEM-INORG002-COUNT-008: ScF3-point-group=D3h;
target=sha256:9d5790428817e53d772def92d48c64e43305cb69570fb82714c43b7c9b0af08f
- SFT-CHEM-INORG002-COUNT-009: ScF3-direct-link-count=3;
target=sha256:98219d648b3c3621ace7cd6266d5e9e4dece1d0575b3c386f659b866eb31026e
- SFT-CHEM-INORG002-COUNT-010: ScF3-coordinate-data-status=No coordinate data available;
target=sha256:e0462795b914801c2c516223edbcf22bce63108db936ae12ee56ec93565d54ba
- SFT-CHEM-INORG002-COUNT-011: ScF3-rotational-data-status=No experimental rotational constants available;
target=sha256:663b359de45b54b0450630de072dde511ffd94fc9b6c45c564affea7f4a7c26f
- SFT-CHEM-INORG002-COUNT-012:
TiCl4-complete-page=sha256:3cdd6c8814bb2c9efc172afc3d83cc81e45c78a94e17d6fcbffd593b5de26358;
target=sha256:0a12725386a4a356d10f7b0e8219e4c7952b155250f71a31d470b2e2c0c3e502
- SFT-CHEM-INORG002-COUNT-013: TiCl4-entity-identity=TiCl4; Titanium tetrachloride;
target=sha256:4dd98b6572c4aa783e3857239aabdd8faa256276b6ce75685632231dcfc1f5bd
- SFT-CHEM-INORG002-COUNT-014: TiCl4-point-group=Td;
target=sha256:2f7eb512f6e315e9b38d1da25116004a0c12a6216aff9ceaaa05851e914ba7b2

- SFT-CHEM-INORG002-COUNT-015: TiCl_4 -direct-link-count=4;
target=sha256:5099c2657ef01bf0db4c9a20eb6271e71b4e791998d66aa1247da875d5ffab9b
- SFT-CHEM-INORG002-COUNT-016: TiCl_4 -internal-coordinate-record=rTiCl experimental internal-coordinate record retained;
target=sha256:9973871025153442ba7249795c86d5d6077a066b0a76e7d26fabb87e670be970
- SFT-CHEM-INORG002-COUNT-017: TiCl_4 -rotational-data-status=No experimental rotational constants available;
target=sha256:f4286df5f02118f8258eb9bd7e0d4a3ab29792e5aef89d39e53f1aabc1e4162f
- SFT-CHEM-INORG002-COUNT-018:
FeCO5-complete-page=sha256:35d46ce135ea052860c42e24dce03a4211336c0038a8a64813b24136b5b1e4b4;
target=sha256:d2963459bfd513967f7621bbdc8616879ec8b622d4637091da1be36109de0ee
- SFT-CHEM-INORG002-COUNT-019: FeCO5-entity-identity=Fe(CO)5; Iron pentacarbonyl;
target=sha256:73ffa90daa703d430d6f3734e4530aebc464c701f1a52da3b52b3f2ec2603317
- SFT-CHEM-INORG002-COUNT-020: FeCO5-point-group=D3h;
target=sha256:79a8b11cd39e31073e8dbb828b7643f7d96187bcc5f431e1ce923f11b311275f
- SFT-CHEM-INORG002-COUNT-021: FeCO5-direct-link-count=5;
target=sha256:60bd64b803b5cf973d6f237e54204a4f9493a31f5e0dd77140f07f5ce3347b2e
- SFT-CHEM-INORG002-COUNT-022: FeCO5-internal-coordinate-record=rCO=1.145 angstrom; C-Fe direct-link support retained; target=sha256:bbc40e706200e4f41897898442b1c5b9914b5f8969c43062a22e5be7470f41fe
- SFT-CHEM-INORG002-COUNT-023: FeCO5-rotational-data-status=No experimental rotational constants available;
target=sha256:1d8dbfcb1b942029569ece87cfe6f20c9908afe2f2eff22d26d585b383589fd0
- control tampered_omitted_source_record_rejected: True
- control tampered_central_occurrence_mismatch_rejected: True
- control numerical_zero_rejected: True
- control complete_23_record_vector_retained: True
- control boundary_absence_and_disclaimer_surface_visible: True
- control prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if coordination number differs from the positive cardinality of complete distinct direct incidences on the retained central occurrence; if equal ligand-group labels collapse occurrences; if absence is counted; if the successor does not preserve prior traces and add one; if any of twenty-three records changes or is omitted; if the IUPAC general, inorganic sigma, pi-exclusion or crystallographic senses are mixed; if NIST ScF_3 , TiCl_4 and Fe(CO)5 do not retain direct-link counts 3, 4 and 5; or if any imported, fitted, continuum, numerical-zero or target-derived structure enters the law.

Measurement receipt: sha256:4475eb52fac887d3123c960e3a66a8d05b0f697640f53e48f58336a528a767e5.
Isolation certificate: sha256:d4ae8a4aa1384bb8f4f60e184c7b509a3444d81f523a566256b7001e9c0e98f8.
Custody certificate: sha256:eb6b52a31c3a7bb6e7a446167cdb62e4055ce1a3c54a7d884c57132310c470db.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01331/json>
(sha256:5c995bf40117e65eb042dc5d585692c56443ef3dc93805466779053da415fb5a)
- National Institute of Standards and Technology: <https://cccbdb.nist.gov/expgeom2x.asp?casno=13709472&charge=0>
(sha256:63e6ea2c88b3aef4137c15a7986358be95c411aff96a9789ec421641e20963dc)
- National Institute of Standards and Technology: <https://cccbdb.nist.gov/expgeom2x.asp?casno=7550450&charge=0>
(sha256:3cdd6c8814bb2c9efc172afc3d83cc81e45c78a94e17d6fcbffd593b5de26358)
- National Institute of Standards and Technology: <https://cccbdb.nist.gov/expgeom2x.asp?casno=13463406&charge=0>
(sha256:35d46ce135ea052860c42e24dce03a4211336c0038a8a64813b24136b5b1e4b4)

Registered target identities:

- SFT-CHEM-INORG002-COUNT-001 from IUPAC-GOLD-BOOK-C01331

- SFT-CHEM-INORG002-COUNT-002 from IUPAC-GOLD-BOOK-C01331
- SFT-CHEM-INORG002-COUNT-003 from IUPAC-GOLD-BOOK-C01331
- SFT-CHEM-INORG002-COUNT-004 from IUPAC-GOLD-BOOK-C01331
- SFT-CHEM-INORG002-COUNT-005 from IUPAC-GOLD-BOOK-C01331
- SFT-CHEM-INORG002-COUNT-006 from NIST-CCCBDB-SRD101-SCF3
- SFT-CHEM-INORG002-COUNT-007 from NIST-CCCBDB-SRD101-SCF3
- SFT-CHEM-INORG002-COUNT-008 from NIST-CCCBDB-SRD101-SCF3
- SFT-CHEM-INORG002-COUNT-009 from NIST-CCCBDB-SRD101-SCF3
- SFT-CHEM-INORG002-COUNT-010 from NIST-CCCBDB-SRD101-SCF3
- SFT-CHEM-INORG002-COUNT-011 from NIST-CCCBDB-SRD101-SCF3
- SFT-CHEM-INORG002-COUNT-012 from NIST-CCCBDB-SRD101-TICL4
- SFT-CHEM-INORG002-COUNT-013 from NIST-CCCBDB-SRD101-TICL4
- SFT-CHEM-INORG002-COUNT-014 from NIST-CCCBDB-SRD101-TICL4
- SFT-CHEM-INORG002-COUNT-015 from NIST-CCCBDB-SRD101-TICL4
- SFT-CHEM-INORG002-COUNT-016 from NIST-CCCBDB-SRD101-TICL4
- SFT-CHEM-INORG002-COUNT-017 from NIST-CCCBDB-SRD101-TICL4
- SFT-CHEM-INORG002-COUNT-018 from NIST-CCCBDB-SRD101-FECO5
- SFT-CHEM-INORG002-COUNT-019 from NIST-CCCBDB-SRD101-FECO5
- SFT-CHEM-INORG002-COUNT-020 from NIST-CCCBDB-SRD101-FECO5
- SFT-CHEM-INORG002-COUNT-021 from NIST-CCCBDB-SRD101-FECO5
- SFT-CHEM-INORG002-COUNT-022 from NIST-CCCBDB-SRD101-FECO5
- SFT-CHEM-INORG002-COUNT-023 from NIST-CCCBDB-SRD101-FECO5

Meaning. The inorganic extension derives coordination identity and number, denticity, geometry, isomerism, splitting, spin, colour, magnetism, organometallic support, inverse redox transforms, cluster bonding, local solid organization, defects and coupled inorganic reaction networks. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero
- negative, irrational, imaginary, floating, signed or continuum proof value
- imported coordination-number table, valence rule, geometry catalogue, ligand-field or bonding model
- observed 3, 4 or 5 count before prediction seal
- selected complex, omitted boundary or limitation row, fitted distance or target-derived correction

Immutable identities. Source manifest

sha256:0f526a103c27901373b6fa074a0202b0c6819c86f92271f136c5da2591be7eb3; derivation seal
 sha256:3fd5e0e5da8c00ea83d664dd2aaa104139cdeb201feaccd911d63675b5bb06d; engine receipt
 sha256:7b69ed0faf732d306aa10eac8ff74cb7e0cdad9debc2c45b3ab75da4352f8 at receipts/engine/model_admitted/SFT-CHEM-COORDINATION-NUMBER-INCIDENCE-COUNT-002-7b69ed0faf732d30.json
 ; empirical hash sha256:232a9caaecff04a6ed2dd32c8fa3ac1c0c366ad000e58e4205e52f091578b402;
 measurement receipt sha256:4475eb52fac887d3123c960e3a66a8d05b0f697640f53e48f58336a528a767e5.

159. Exact ligand denticity and chelation from connected donor-site topology

Claim identity: SFT-CHEM-LIGAND-DENTICITY-CHELATION-TOPOLOGY-003

Question and theorem. For one retained ligand carrier attached to one retained central occurrence, denticity is the positive cardinality of its complete distinct donor-site incidences. One site is open; the next separate site is the first successor that closes a carrier-centre path and forces chelation; every later site preserves closure and increments denticity once.

one-retained-ligand-carrier-occurrence__every-distinct-donor-site-on-that-carrier__same-retained-central-occurrence__positive-count-of-generated-donor-incidences__first-closed-carrier-centre-path-forces-chelation__separate-site-and-attachment-topologies-held__complete-24-record-value-free-identity-seal__next-site-preserves-prior-and-adds-one

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-COORDINATION-ENTITY-RETAINED-IDENTITY-001
- SFT-CHEM-COORDINATION-NUMBER-INCIDENCE-COUNT-002

Generated grammar. Generate the literal product of carrier, membership, centre, quantity, chelation, boundary, observation and extension forms. Decide all 256 forms solely from admitted coordination identity, exact direct-incidence count, graph connectivity, information retention and source-boundary laws.

Boundary: Every positive finite complete donor-site topology for one retained ligand carrier and one retained central occurrence, tested after seal against all twenty-four registered surfaces from six complete current IUPAC term records.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *one-retained-ligand-carrier-occurrence__every-distinct-donor-site-on-that-carrier__same-retained-central-occurrence__positive-count-of-generated-donor-incidences__first-closed-carrier-centre-path-forces-chelation__separate-site-and-attachment-topologies-held__complete-24-record-value-free-identity-seal__next-site-preserves-prior-and-adds-one*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-retained-ligand-carrier-occurrence	chemical-label-only	A repeated chemical label does not distinguish a given ligand occurrence.
membership	every-distinct-donor-site-on-that-carrier	selected-or-nearby-donor-sites	Selection and proximity do not preserve complete binding support.
centre	same-retained-central-occurrence	donor-sites-across-centres-merged	Merging centres destroys the one-centre relation.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
quantity	positive-count-of-generated-donor-incidences	imported-denticity-name-or-table	A conventional name or table selects the count.
chelation	first-closed-carrier-centre-path-forces-chelation	two-or-more-label-imported-as-threshold	A named threshold does not explain the topology.
boundary	separate-site-and-attachment-topologies-held	eta-kappa-and-separate-sites-collapsed	Collapsing attachment modes confuses atoms with separate binding sites.
observation	complete-24-record-value-free-identity-seal	source-topologies-readable-before-seal	Observed classifications could select the law.
extension	next-site-preserve-s-prior-and-adds-one	next-site-recounts-or-replaces-prior-support	Replacement destroys the retained carrier topology.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:d2e626a588cd93ee88531f913bb162e4b4e64f3f30bd5ba51f4a74cd667b728a.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:540ae6ba1bb32c4f5e9a97e3ed0a8d63bcd22f1e8bed904307358c8fc051b4f0.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:5375d603fc8724cc403e86b72b7b09f3781cde795357ac3ae381627e4e5f67ab.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:bf68ddd68284b11091a5a2c2d7a4a21af1b5cb64235b0bfb63751c97cf63121b.

Independent reconstruction. Implementation hash

sha256:ffba3598813889f85c6338b1936c5788a87bf1a97f99a08cdad008ebd1bb9ff4; independent certificate sha256:4acb68a799b1aadfb9db6309143322369dfac97aeb59b223f6cbfc70c3069fa4; external-validation hash sha256:353f12609a8788226ddd82c06c75dda79812c86b2e54cc67205259f17b7fce49. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-B00649
- IUPAC-GOLD-BOOK-C01012
- IUPAC-GOLD-BOOK-D01594
- IUPAC-GOLD-BOOK-H01881
- IUPAC-GOLD-BOOK-K03366
- IUPAC-GOLD-BOOK-L03518

Comparison records:

- SFT-CHEM-INORG003-TOPOLOGY-001:
binding-sites::complete-current-term-record=sha256:4f1d275678db299b664a1a00c5ca5cd19fbf012f3dc7077973476ac31b2676d9;
target=sha256:8ef5994d03619da1c8873b26c71fedf1b07950dcef8ced3696647df3e3a904f6
- SFT-CHEM-INORG003-TOPOLOGY-002: binding-sites::complete-definition-surface=[{"id":1,"links":[{"term":"coordination",
"url":"https://goldbook.iupac.org/terms/view/C01329"},{"term":"hydrogen
bonding", "url":"https://goldbook.iupac.org/terms/view/H02899"},{"term":"ion

pair", "url": "https://goldbook.iupac.org/terms/view/I03231"}, {"term": "molecular entity", "url": "https://goldbook.iupac.org/terms/view/M03986"}, {"term": "substrate", "url": "https://goldbook.iupac.org/terms/view/S06083"}], "sources": ["PAC, 1994, 66, 1077. 'Glossary of terms used in physical organic chemistry (IUPAC Recommendations 1994)' on page 1089 (https://doi.org/10.1351/pac199466051077)"], "text": "A specific region (or atom) in a molecular entity that is capable of entering into a stabilizing interaction with another molecular entity. An example of such an interaction is that of an active site in an enzyme with its substrate. Typical forms of interaction are by hydrogen bonding, coordination and ion pair formation. Two binding sites in different molecular entities are said to be complementary if their interaction is stabilizing."}; target=sha256:ec82e47628f36a7dc70fbcdcf2a51d8980e34101d010ba85d121efb3fa48b82b

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Falsification boundary: The claim fails if denticity differs from the positive cardinality of complete distinct donor incidences for one retained ligand carrier and centre; if equal chemical labels merge separate carrier occurrences; if one site is called chelation; if the first multiple-site connected support does not close a carrier-centre path; if any of twenty-four records changes or is omitted; if kappa, eta or separate-site boundaries collapse; or if any imported, fitted, continuum, numerical-zero or target-derived structure enters the law.

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Meaning. The inorganic extension derives coordination identity and number, denticity, geometry, isomerism, splitting, spin, colour, magnetism, organometallic support, inverse redox transforms, cluster bonding, local solid organization, defects and coupled inorganic reaction networks. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero
- negative, irrational, imaginary, floating, signed or continuum proof value
- imported denticity table, chelate taxonomy, coordination-number rule, geometry catalogue or bonding model
- observed IUPAC definition, example or exclusion before prediction seal
- selected term, omitted boundary, fitted topology or target-derived correction

Immutable identities. Source manifest

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160. Exact coordination geometry from complete held orientation and adjacency

Claim identity: SFT-CHEM-COORDINATION-GEOMETRY-HELD-ORIENTATION-004

Question and theorem. For one retained central occurrence, coordination geometry is the complete exact signature of every directly attached ligand occurrence, its three-generated-axis held orientation word and every generated boundary adjacency. Count is retained but cannot alone select geometry; space rank three and boundary rank two are forced.

one-retained-centre-and-every-direct-ligand-occurrence__complete-direct-incidence-support__three-generated-axis-held-orientation-word__complete-generated-boundary-adjacency-trace__exact-complete-position-adjacency-signature__forced-generator-three-and-boundary-rank-two__sealed-complete-authority-surfaces-including-adverse-identity-rows__next-position-preserves-prior-and-adds-its-complete-relations

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOLECULAR-BOND-ANGLE-003
- SFT-CHEM-COORDINATION-ENTITY-RETAINED-IDENTITY-001
- SFT-CHEM-COORDINATION-NUMBER-INCIDENCE-COUNT-002
- SFT-CHEM-LIGAND-DENTICITY-CHELATION-TOPOLOGY-003

Generated grammar. Generate the literal product of carrier, incidence, orientation, adjacency, identity, rank, observation and extension forms; decide all 256 forms solely from admitted retained-occurrence, exact incidence, generator-three, boundary-rank-two, graph, information-retention and measured-value-boundary laws.

Boundary: Every positive finite complete coordination entity with one retained centre, every directly attached ligand occurrence, one exact three-axis Fold orientation word per ligand, every generated boundary adjacency, all fifty-three value-free registered authority surfaces, all four preserved adverse identity rows and all reported or absent NIST coordinate surfaces.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *one-retained-centre-and-every-direct-ligand-occurrence__complete-direct-incidence-support__three-generated-axis-held-orientation-word__complete-generated-boundary-adjacency-trace__exact-complete-position-adjacency-signature__forced-generator-three-and-boundary-rank-two__sealed-complete-authority-surfaces-including-adverse-identity-rows__next-position-preserves-prior-and-adds-its-complete-relations*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-retained-centre-and-every-direct-ligand-occurrence	chemical-formula-only	A formula does not retain the central or ligand occurrences.
incidence	complete-direct-in-cidence-support	coordination-number-alone-selects-shape	Equal counts can carry different orientations and adjacency; count alone cannot select geometry.
orientation	three-generated-axis-held-orientation-word	imported-shape-name-or-continuum-coordinate	A conventional name or continuum coordinate imports the result.
adjacency	complete-generated-boundary-adjacency-trace	selected-or-inferred-neighbour-pairs	Selecting neighbours omits part of the boundary relation.
identity	exact-complete-position-adjacency-signature	orientation-or-occurrence-collapse	Collapsing an occurrence or orientation destroys a distinguishable geometry.
rank	forced-generator-three-and-boundary-rank-two	free-dimensional-or-polyhedral-rank	A free rank imports geometric structure.
observation	sealed-complete-authority-surfaces-including-adverse-identity-rows	selected-favourable-geometry-rows	Selected rows cannot test the full source boundary.
extension	next-position-preserved-prior-and-adds-its-complete-relations	new-position-replaces-or-recounts-prior-geometry	Replacement erases the prior geometry.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 836b5409fd4f0dac8102b4150cf7e91af925158e0a301d6e87b9ad869a4e820e.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 3520c5bc29c8d47286e0dcdb62c240382b9bb0913863d99be61b44deb015600b.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: a3292f18e0415e96aabd4b2afb904288694e903424640f1fc8c4c65cddce6f56.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 2f6f49c1a6587712dc6a2b44590f8b561ab555a4de447177c44c2d5053f9f774.

Independent reconstruction. Implementation hash

sha256: 581054c84a17dfec89ee61cf7e0c651d9f3f62af0c9c7bfa7f342552f8cefa60; independent certificate sha256: 681b846c26da2c54459dc119bedc0cc740d45b7d44f78506d3e42e5d6e9f1fcb; external-validation hash sha256: 1e8f6199a34880b0ab7b78728863de081f8361d04d7d1e31f4ba2e7ca4efe031. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-C01332
- NIST-CCCBDB-COMPLETE-EXPERIMENTAL-GEOMETRY

Comparison records:

- complete external positive-incidence vector: (2, 3, 4, 5, 6)
- complete external point-group inscription vector: ('C 2v', 'D 3h', 'T d', 'D 3h', 'O h')
- reported internal-coordinate absence rows retained: 3

- four original formaldehyde target-identity mismatches preserved as adverse rows
- control tampered_omitted_source_record_rejected: True
- control numerical_zero_rejected: True
- control equal_count_different_orientation_not_merged: True
- control all_four_original_adverse_identity_rows_retained: True
- control complete_53_record_vector_retained: True
- control prediction_contains_no_withheld_target_hash: True

Falsification boundary: The claim fails if any direct ligand occurrence, orientation word or generated adjacency is omitted or collapsed; if coordination count alone selects geometry; if rank differs from forced three-space/two-boundary support; if a successor rewrites prior geometry; if any of fifty-three source surfaces, four adverse target-identity mismatches, five positive incidence counts, point-group inscriptions, reported coordinate rows, absent coordinate/reference rows or tampered control is omitted or mismatched; or if an imported table, continuum proof scalar, numerical zero, fitted shape or target-derived correction enters the forcing law.

Measurement receipt: sha256:e3582c7c5b9c3e6c2106fd89af89d8193acc0b9968ed42475105a1046f3f22e4.

Isolation certificate: sha256:16dcc93e54c0fcf6030372021f19963fe02f170271a075c1ffc87641df6bdc23.

Custody certificate: sha256:b1de8fc4dcaa933c411b0e32baad736e0072aeec0df533aa590f6fe389c.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The inorganic extension derives coordination identity and number, denticity, geometry, isomerism, splitting, spin, colour, magnetism, organometallic support, inverse redox transforms, cluster bonding, local solid organization, defects and coupled inorganic reaction networks. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero; external glyph 0 is an inscription and native absence is EmptyOne
- negative, irrational, imaginary, floating, signed or continuum proof value
- imported geometry table, shape name, point group, angle, distance, field model or coordination-count lookup
- selected favourable rows or deleted adverse/absent rows
- source outcome, fitted orientation or target-derived condition

Immutable identities. Source manifest

sha256:7427d389fa431d4ce7155c3d78365cef555af5fc8771afb3008718ed7878755; derivation seal sha256:2e2b045ae41e97402b7ba4bee09d7474c530c34b01996c94a8f9be689e72aa71; engine receipt sha256:bc7f8d5d5235f114984798515944893fc3be3c17a229192cb6e12728ce9d459a at receipts/engine/model_admitted/SFT-CHEM-COORDINATION-GEOMETRY-HELD-ORIENTATION-004-bc7f8d5d5235f114.json; empirical hash sha256:353c302ab28d3b570a9aac41879ed8370cf30bf4c63d256b8a57650dc0f999e3; measurement receipt sha256:e3582c7c5b9c3e6c2106fd89af89d8193acc0b9968ed42475105a1046f3f22e4.

161. Exact coordination-form equivalence and native isomer distinction classes

Claim identity: SFT-CHEM-COORDINATION-ISOMERISM-EQUIVALENCE-005

Question and theorem. Exact coordination-form equivalence is the exhaustive occurrence-bijection preservation of complete composition, attachment, adjacency and three-axis two-fibre words. The first failed invariant forces attachment, mirror-complement or remaining orientation-adjacency distinction without importing an isomer catalogue.

two-complete-retained-coordination-forms__same-complete-composition-multiset__enumerate-every-occurrence-bijection__complete-attachment-and-adjacency-preservation__complete-three-axis-two-fibre-relation__first-failed-invariant-

forces-native-class__complete-17-surface-sealed-vector-with-linkage-addendum__successor-preserves-prior-signatures-and-adds-relations

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-CHEM-STEREO-DIASTEREOMER-001
- SFT-CHEM-COORDINATION-GEOMETRY-HELD-ORIENTATION-004

Generated grammar. Generate the literal product of carrier, composition, bijection, graph, orientation, native-class, observation and successor forms; decide all 256 candidates solely from admitted finite-form, exact-combinatorial, graph, information-retention, stereochemical and INORG-004 geometry dependencies.

Boundary: Every pair of positive finite complete coordination forms with the same complete composition multiset, every occurrence permutation, every retained attachment and adjacency, every three-axis word using only fibre-one, fibre-two or structural EmptyOne, the generated global fibre complement, all seventeen sealed IUPAC surfaces, both preserved Gold Book identity redirects and the explicitly absent literal linkage term.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *two-complete-retained-coordination-forms__same-complete-composition-multiset__enumerate-every-occurrence-bijection__complete-attachment-and-adjacency-preservation__complete-three-axis-two-fibre-relation__first-failed-invariant-forces-native-class__complete-17-surface-sealed-vector-with-linkage-addendum__successor-preserves-prior-signatures-and-adds-relations*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	two-complete-retained-coordination-forms	names-or-formulae-only	Names do not retain complete forms.
composition	same-complete-composition-multiset	different-compositions-called-isomers	Isomer comparison requires the same complete composition.
bijection	enumerate-every-occurrence-bijection	selected-correspondence-between-occurrences	A selected mapping can manufacture equivalence.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
graph	complete-attachment-and-adjacency-preservation	partial-or-named-connectivity	Partial connectivity omits attachment distinctions.
orientation	complete-three-axis-two-fibre-relation	shape-name-point-group-or-continuum-transform	Imported transformations select the result.
class	first-failed-invariant-forces-native-class	imported-isomer-catalogue	A catalogue does not force the first failed invariant.
observation	complete-17-surface-sealed-vector-with-linkage-addendum	selected-favourable-term	Selected definitions cannot test all classes.
extension	successor-preserve-prior-signatures-and-adds-relations	new-occurrence-reclassifies-prior-subform	Rewriting loses the prior class trace.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:66971362d01c4bf8e1c451dc6d2a2097b2d6b1c552c66b3f1aa8f75b70d898fb.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:c723dde7680fd8706fa94bffd1d3608d70a049a519fe4641b12071d4acc68ea.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:7ff9603de75798987adc47ad7c72867df0c68e3418a5e0a169532db09961f8a5.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:5a8af17393a9535939d4757b26d4fd9df34953bcd5e259578bf0148ec1ea76.

Independent reconstruction. Implementation hash

sha256:7f904adb8713cf9132881bab0a0eb2c672357647230c3b4850cca452f37585e; independent certificate sha256:0f0b80428959622115ff3f966b31946f242027d731450843533862e346d32f7a; external-validation hash sha256:3e65213230b98a933be374fcf46c148662e7f4e551eae3a498000b4defb9e77. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-I03294
- IUPAC-GOLD-BOOK-C01093
- IUPAC-GOLD-BOOK-E02069
- IUPAC-RED-BOOK-2005

Comparison records:

- IUPAC general isomer relation retained
- IUPAC relative-position same-side/opposite-side distinction retained
- IUPAC mirror-image non-superposable distinction retained
- IUPAC Red Book point-of-ligation and two isomeric donor-attachment modes retained
- two registered-to-presented Gold Book identity redirects retained
- explicit literal linkage-isomer term absence in the complete extracted Red Book retained

- control tampered_omitted_source_record_rejected: True
- control numerical_zero_rejected: True
- control third_fibre_label_rejected: True
- control different_composition_rejected: True
- control both_presented_identity_redirects_preserved: True
- control explicit_linkage_literal_absence_preserved: True
- control complete_17_record_vector_retained: True
- control prediction_contains_no_withheld_target_hash_or_inscription: True

Falsification boundary: The claim fails if any occurrence, attachment, adjacency, orientation cell or occurrence bijection is omitted; if a third fibre or numerical zero enters a proof object; if different composition is admitted as an isomer pair; if a selected mapping, imported isomer name or continuum transform chooses equivalence; if linkage, geometric or mirror classes are asserted without their forced failed invariant; if a successor erases a prior relation; if any of seventeen source surfaces, either redirected presented identity or the explicit literal-term absence is omitted or altered; or if any source outcome enters the candidate generator or eliminator.

Measurement receipt: sha256:cdb22fe1dc979cfc77bd768e94b51a9cccf2d5e195a90f0cde716c123acdd618.

Isolation certificate: sha256:8f001f2a59e6fab0cb118c7134684458264262fc37586711f8f7721999226628.

Custody certificate: sha256:5d1ca6ee3c1558ea5f76bd271d9c0d90da899a005215898727c985a4e5014382.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The inorganic extension derives coordination identity and number, denticity, geometry, isomerism, splitting, spin, colour, magnetism, organometallic support, inverse redox transforms, cluster bonding, local solid organization, defects and coupled inorganic reaction networks. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero; external glyph 0 is an inscription and native absence is EmptyOne
- negative, irrational, imaginary, floating, signed or continuum proof value or transformation
- a third fibre label; three axes are positions containing only the two forced fibres or EmptyOne
- imported linkage, geometric, optical, cis-trans, enantiomer, diastereomer, point-group or shape catalogue
- selected occurrence map, favourable row, hidden redirect or absent term, fitted class or target-derived condition

Immutable identities. Source manifest

sha256:9360b77c463251da1ccf2649dfcc129aece3fd7e52527d8d11c0910e7f5e281d; derivation seal

sha256:7b94dbb324b286a214c8d91f285507b3838fa522d83208130f66d4cca9a0666e; engine receipt

sha256:f10fd2daccd977ff082e2bd425c3a548ce350809e15baf1bd5f529f94a719898 at receipts/engine/model_admitted/SFT-CHEM-COORDINATION-ISOMERISM-EQUIVALENCE-005-f10fd2daccd977ff.json;

empirical hash sha256:e2e2d0617c19717952cd046c9c9b88b44215d2fef1c3be0f0746239ded6464af;

measurement receipt sha256:cdb22fe1dc979cfc77bd768e94b51a9cccf2d5e195a90f0cde716c123acdd618.

162. Exact ligand-incidence state partition and balanced splitting

Claim identity: SFT-CHEM-LIGAND-STATE-SPLITTING-006

Question and theorem. Generator three and boundary rank two force five second-rank state supports. Complete ligand incidence partitions those supports into exact split levels: six direct-axis occurrences force multiplicities three and two, structural separation two-thirds and balance distances two-fifths and three-fifths; four complete-axis occurrences force the reversed two and three partition, unit separation and reversed complementary balance.

generated-rank-three-boundary-two-support__complete-ligand-support-incidence__complete-equivalence-partition-by-incidence__positive-cardinality-of-each-partition-block__exact-positive-normalized-incidence-separation__complementary-positive-two-block-distances__complete-sealed-definition-and-spectrum-surfaces__same-partition-law-and-removed-ligand-remerging

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-CHEM-ELECTRONIC-STATE-IDENTITY-001
- SFT-CHEM-ORBITAL-SUPPORT-OCCUPANCY-003
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-STATE-SYMMETRY-DEGENERACY-005
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-SELECTION-RULE-STRUCTURE-010
- SFT-CHEM-COORDINATION-GEOMETRY-HELD-ORIENTATION-004
- SFT-CHEM-COORDINATION-ISOMERISM-EQUIVALENCE-005

Generated grammar. Generate the complete literal product of carrier, interaction, partition, multiplicity, separation, balance, observation and extension choices; decide all 256 candidates solely from admitted exact arithmetic, generator-three, space-rank, boundary-rank, state-equivalence, geometry and prior INORG receipts.

Boundary: Every finite three-axis ligand word formed only from the two forced Fold fibres and structural EmptyOne; the complete two-contrast plus three-boundary-pair state support; every exact interaction-signature partition; all thirty-two registered definition, development, ancillary, adverse absence and law-sealed blind spectrum surfaces.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *generated-rank-three-boundary-two-support__complete-ligand-support-incidence__complete-equivalence-partition-by-incidence__positive-cardinality-of-each-partition-block__exact-positive-normalized-incidence-separation__complementary-positive-two-block-distances__complete-sealed-definition-and-spectrum-surfaces__same-partition-law-and-removed-ligand-remerging*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	generated-rank-three-boundary-two-support	imported-orbital-or-field-table	A named table imports the desired splitting.
interaction	complete-ligand-support-incidence	free-field-strength-or-fitted-parameter	A free strength can fit any spectrum.
partition	complete-equivalence-partition-by-incidence	asserted-level-catalogue	An asserted catalogue does not derive degeneracy removal.
multiplicity	positive-cardinality-of-each-partition-block	free-degeneracy-label	A free degeneracy loses member conservation.
separation	exact-positive-normalized-incidence-separation	signed-floating-or-dimensional-gap	Signed or fitted gaps violate the exact Fold domain.
balance	complementary-positive-two-block-distances	opposite-signed-shift-premise	Signed displacements import negative values.
observation	complete-sealed-definition-and-spectrum-surfaces	selected-peak-or-favourable-spectrum	Selected peaks can manufacture correspondence.
extension	same-partition-law-and-removed-ligand-remerging	species-specific-exception-or-reclassification	An exception permits outcomes to select the law.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:e478e458e59bf2d84a283159d4f028110587ff565d5f33f1fa2e39fef5958e9b.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:1d1533f13270b69d6905fe412a120b236af2c31ce594a9003ad79e470dceb8dd.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b0fc797514dc0dbedb68131cd4762a7d7e48a708faff0de53b156742e9c8e983.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:18dc29834a0328c06e614c9d815b5e3203664aadf55afd0418e2127add028c2b.

Independent reconstruction. Implementation hash

sha256:c04362add35b01f9aad45d2a6839155fccffabebccefcece0a3d7af89be233e21; independent certificate sha256:a555afecb1eaf3ed863933c519c3d226f071a8cf0e7e04a7ffc6e0ee7687cb99; external-validation hash sha256:69a3a7774f16b1437b3efd8648ce259c854446011730884f3e81043d11e474ba. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-LT06764
- IUPAC-L03517
- NIST-WEBBOOK-C13007926-UVVIS-COMplete
- NIST-WEBBOOK-C12082085-UVVIS-COMplete
- NIST-WEBBOOK-C12108111-UVVIS-COMplete
- NIST-WEBBOOK-C13939065-UVVIS-COMplete
- NIST-WEBBOOK-C14040110-UVVIS-COMplete

- NIST-WEBBOOK-C12146360-UVVIS-COMplete

Comparison records:

- forced five-support partition: two held contrasts plus three boundary pairs
- six direct-axis multiplicities 3 and 2; structural separation 2/3; balance distances 2/5 and 3/5
- four complete-axis multiplicities 2 and 3; structural separation 1; balance distances 3/5 and 2/5
- law-sealed blind NIST spectrum interior maxima (('48361/200', '77817/200'),) with exact separation (('3682/25'),)
- two law-sealed absent NIST spectrum rows and twelve development ancillary captures preserved
- IUPAC degeneracy-removal and ligand-attachment/reduced-symmetry surfaces preserved; conventional variable parameters remain downstream only
- control omitted_source_row_rejected: True
- control numerical_zero_rejected: True
- control tampered_six_direct_axis_multiplicity_rejected: True
- control all_32_target_hashes_bound_postseal: True
- control prediction_contains_no_target_payload_hash_or_peak_value: True
- control two_adverse_blind_absences_preserved: True
- control twelve_development_ancillary_rows_preserved: True

Falsification boundary: The claim fails if generator three and boundary rank two do not generate exactly five supports; if any ligand word, support, incidence or partition block is omitted; if equal signatures split or unequal signatures merge; if member multiplicity is not conserved; if the six-direct-axis and four-complete-axis partitions, exact separations or complementary balance identities fail; if numerical zero, a third fibre, signed displacement, field parameter, orbital table, selected peak, smoothing or fitted tolerance enters forcing; if any of thirty-two source surfaces is removed; if either adverse spectrum absence or ancillary capture is hidden; or if the law-sealed blind spectrum lacks the preregistered positive distinguishability condition.

Measurement receipt: sha256:8153f034d35e423c941465ac1d443f0eb61ccb0e902da801131a79a982f40b31.

Isolation certificate: sha256:80aa8f6eb10c6b080a8625153449296dcf609676779b416e8bf4411cb965deda.

Custody certificate: sha256:03ab544f16f84d8b538e79758aef4bab2ed82c77cd3f84157ea7c62c03804730.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The inorganic extension derives coordination identity and number, denticity, geometry, isomerism, splitting, spin, colour, magnetism, organometallic support, inverse redox transforms, cluster bonding, local solid organization, defects and coupled inorganic reaction networks. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero; external glyph 0 is only an inscription and native absence is EmptyOne
- negative, irrational, imaginary, floating, signed or continuum proof values
- imported orbital, crystal-field, ligand-field, point-group, geometry or spectrochemical tables
- free field strength, fitted splitting parameter, dimensional wavelength anchor, selected peak, smoothing threshold or target-derived tolerance
- deleted ancillary, adverse, absent, unresolved or boundary evidence

Immutable identities. Source manifest

sha256:f8409ec7cfad4abb9d9840577eebe5d2ec302ec4948b852f396dd38bdb7d50c3; derivation seal

sha256:67da654d5d4378614cef358cf80e9a7fda9cd554dc462a4b5636a0f2aefc2614; engine receipt

sha256:848c37693954460e03928ce0db1329df5fcb381ed8847ad7a56087d36bb383c5 at receipts/engine/model_admitted/SFT-CHEM-LIGAND-STATE-SPLITTING-006-848c37693954460e.json; empirical hash sha256:c33fcb00c39d8ecff554fa68308f767a594f79870e65050b67e590ef67104c2; measurement receipt sha256:8153f034d35e423c941465ac1d443f0eb61ccb0e902da801131a79a982f40b31.

163. Exact complex spin-state extrema, order and crossover

Claim identity: SFT-CHEM-COMPLEX-SPIN-STATE-ORDER-007

Question and theorem. The forced three-plus-two split support has ten complete six-electron occupancy signatures and unique low- and high-spin extrema. Exact pair-closure plus split-crossing path counts force high-before-low, coincidence and low-before-high under diluted, first-retained and second-retained interaction recurrence.

one-complex-electron-support__forced-three-plus-two-support__complete-pair-single-occupancy-census__least-crossing-then-least-unmatched-extremum__greatest-unmatched-then-least-crossing-extremum__counted-pair-closure-plus-split-crossing-paths__complete-weak-boundary-strong-trichotomy__monotone-dilution-with-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-INVERSE-SQUARE-001
- SFT-CHEM-ELECTRONIC-STATE-IDENTITY-001
- SFT-CHEM-ELECTRON-COUNT-SPIN-002
- SFT-CHEM-ORBITAL-SUPPORT-OCCUPANCY-003
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-STATE-SYMMETRY-DEGENERACY-005
- SFT-CHEM-COORDINATION-GEOMETRY-HELD-ORIENTATION-004
- SFT-CHEM-LIGAND-STATE-SPLITTING-006

Generated grammar. Generate the literal product of carrier, split, enumeration, low extremum, high extremum, cost, order and extension choices; decide all 256 candidates only from admitted support, occupancy, state-order, INORG-006 split and geometric-dilution dependencies.

Boundary: Every symmetry-distinct pair/single occupancy of six electron occurrences across the forced three-plus-two support; the two complete extrema; every structural EmptyOne or positive interaction recurrence; all three frozen INORG-007 external source identities and their favorable, conventional and transport-mismatch surfaces.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *one-complex-electron-support__forced-three-plus-two-support__complete-pair-single-occupancy-census__least-crossing-then-least-unmatched-extremum__greatest-unmatched-then-least-crossing-extremum__counted-pair-closure-plus-split-crossing-paths__complete-weak-boundary-strong-trichotomy__monotone-dilution-with-no-extra-rule*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-complex-electron-support	cross-complex-state-mixture	States from different carriers do not share one exact order.
split	forced-three-plus-two-support	imported-orbital-or-field-table	A conventional table would select the state grammar.
enumeration	complete-pair-single-occupancy-censuses	selected-configurations	Selected configurations cannot establish an extremum.
low	least-crossing-the-n-least-unmatched-extremum	named-low-spin-assumption	A name does not force an occupancy.
high	greatest-unmatched-then-least-crossing-extremum	named-high-spin-assumption	A name does not force an occupancy.
cost	counted-pair-closure-plus-split-crossing-paths	fitted-pairing-or-field-energy	A fitted energy imports a free parameter and target choice.
order	complete-weak-boundary-strong-trichotomy	asserted-ground-state-label	An asserted label does not compare both generated extrema.
extension	monotone-dilution-with-no-extra-rule	species-temperature-or-distance-fit	A species fit can manufacture a crossover.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256 : e2d07179d6de685df74a9720a311a4f8318a718eeb76d2a79c5d7a4bdf21420a.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 7d14707c6a60f8fa0a4bb5a178b01a78976b3cdeadb033283263d611a022efff.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256 : 4e3e24b835cd6e673bea64e4eeb7ea051c4fcb1da426304c1838dad0610ba47c.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256 : 22593e0323b5d4976d727a40cbd2e9a430a8ac6915bb809d773e9679e6536e4e.

Independent reconstruction. Implementation hash

sha256 : c2b60b959e0ebc594d2b4bae4d4c86461ec25b4cab690ec7ac429ad30b4265b8; independent certificate sha256 : f9d40e8f72c146294583e6174c2da00e44653fdc83824cbc6ffd56e3816e6f44; external-validation hash sha256 : d3a60593dd5ec6b3634e898fba17a3f2475d9b702b7b2980bb8173dd25afb6e0. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-HT06789
- IUPAC-LT07067
- IUPAC-ST07104

Comparison records:

- complete six-electron split-support census: 10 signatures
- forced low signature lower pairs 3, spin width 1, crossing EmptyOne

- forced high signature lower pair 1, lower singles 2, upper singles 2, spin width 5, crossings 2
- exact recurrence order vector high-before-low, crossover coincidence, low-before-high
- IUPAC crossover vector 1016/5 pm at 115 K low-spin 1A1; 2199/10 pm at 227 K high-spin 5T2
- registered HT06789 transport returned LT06788 combined low/high definition and is preserved
- control omitted_source_row_rejected: True
- control numerical_zero_rejected: True
- control transport_mismatch_preserved: True
- control all_three_target_hashes_bound_postseal: True
- control prediction_contains_no_target_payload_or_dimensional_value: True
- control tampered_order_vector_rejected: True

Falsification boundary: The claim fails if the complete six-electron quotient census is not ten; if either extremum is nonunique; if the exact low and high signatures, spin widths or crossing counts differ; if the weak/boundary/strong order vector differs; if numerical zero, signed spin, a field or pairing parameter, a dimensional fit or measured target enters forcing; if any of the three registered external rows or the HT06789 transport mismatch is removed; or if the complete post-seal crossover row does not retain low spin at the shorter reported distance and high spin at the longer reported distance.

Measurement receipt: sha256:ea99957df818216580f5999f4483559a6988faca44f961c5f35e588498092203.
Isolation certificate: sha256:a8e96bb21202a5103bfc5df6af47e78d5f2dd9eb122815bda889f777d43d727c.
Custody certificate: sha256:6e889a09681ea49744f21026ce0a2acdd58481f547da9e43d9260adda5aa1a82.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The inorganic extension derives coordination identity and number, denticity, geometry, isomerism, splitting, spin, colour, magnetism, organometallic support, inverse redox transforms, cluster bonding, local solid organization, defects and coupled inorganic reaction networks. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero; native absence is structural EmptyOne
- negative, irrational, imaginary, floating, signed-spin or continuum proof values
- imported crystal-field, ligand-field, orbital, Hamiltonian, pairing-energy or spectrochemical models
- free or fitted field strength, crossover threshold, temperature, distance, species rule or tolerance
- deleted favorable, conventional, absent, adverse or transport-mismatch evidence

Immutable identities. Source manifest

sha256:864275fb2ff01a987afb3955df23406cfa0a464987bdf8d62b30ebc6aafe0a1; derivation seal
sha256:b2cc403ba3b72ababc2a5f15a219a28993685d1bb46e6a55526c372f6a996069; engine receipt
sha256:86e888fefab0bd9d8e195edeaf51cf5ec7e9822c849f49afbb0acbb6f4068c3a at receipts/engine/model_admitted/SFT-CHEM-COMPLEX-SPIN-STATE-ORDER-007-86e888fefab0bd9d.json; empirical hash
sha256:955a28dc21785995e5d354fb2f9abf7b24555ebcfa9cc63c2f72ff96bc6ac842; measurement receipt
sha256:ea99957df818216580f5999f4483559a6988faca44f961c5f35e588498092203.

164. Exact inorganic electronic-transition and selective-colour law

Claim identity: SFT-CHEM-INORGANIC-COLOUR-ELECTRONIC-TRANSITION-008

Question and theorem. Two retained carrier kinds force four directed complex-transition classes. Each transition retains endpoint identities and an exact positive order gap; selective absorption and the complementary retained observation class force inorganic

colour without a conventional colour wheel.

one-retained-coordination-entity__complete-ligand-metal-carrier-pair__two-retained-state-identities__held-source-target-direction__positive-state-order-separation__proper-absorbed-distinction-class__retained-complement-observation-class__complete-spectra-with-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-CHEM-ELECTRONIC-STATE-IDENTITY-001
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-SELECTION-RULE-STRUCTURE-010
- SFT-CHEM-COORDINATION-ENTITY-RETAINED-IDENTITY-001
- SFT-CHEM-COORDINATION-GEOMETRY-HELD-ORIENTATION-004
- SFT-CHEM-LIGAND-STATE-SPLITTING-006
- SFT-CHEM-COMPLEX-SPIN-STATE-ORDER-007

Generated grammar. Generate the literal product of carrier, endpoints, state, direction, gap, absorption, colour and extension choices; decide all 256 candidates only from admitted distinction, order, transition, selection-rule and INORG-004/006/007 dependencies.

Boundary: Every ordered ligand/metal carrier pair in one retained coordination entity; every positive exact state successor; every positive proper absorbed subset of a finite complete incident observation support; four frozen definitions and four immutable complete NIST spectra with original custody classes preserved.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *one-retained-coordination-entity__complete-ligand-metal-carrier-pair__two-retained-state-identities__held-source-target-direction__positive-state-order-separation__proper-absorbed-distinction-class__retained-complement-observation-class__complete-spectra-with-no-extra-rule*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-retained-coordination-entity	anonymous-complex-transition	An anonymous transition erases which retained complex carries it.
endpoints	complete-ligand-metal-carrier-pair	imported-orbital-symbol-pair	Conventional orbital names would select the endpoint grammar.
state	two-retained-state-identities	energy-only-endpoints	Energy-only endpoints lose electronic-state identity.
direction	held-source-target-direction	signed-charge-displacement	A signed scalar imports negative quantity and erases endpoint direction.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
gap	positive-state-order-separation	floating-or-dimensional-energy-gap	A dimensional or floating gap imports an unforced scale.
absorption	proper-absorbed-distinction-class	selected-colour-name	A named colour cannot establish what distinction was absorbed.
colour	retained-complement-observation-classes	conventional-colour-wheel	A colour wheel imports a continuum convention into forcing.
extension	complete-spectra-with-no-extra-rule	species-peak-or-threshold-exception	A selected peak or smoothing threshold can manufacture agreement.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256: 6b47458870348ee8710f24a9e456c06ac1aa61565b585a34ad96634d597bb431.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 6e17855367e1b8418aa3ffc834fb5f49cf902e8ea3b3eaa96fea201de8b462c6.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 21bc634dde5de24bd5869754ef74fdc5166c83efaf450d08ee38f4d10e3f2e39.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: 303b56d8f61dac28d30f00350787031e42fa1aa61b360604c4ee35f33330a40f.

Independent reconstruction. Implementation hash

sha256: 8c5c851821f98c29035cfd800a0ec7403230d5624c75ef1f555b9673bde94ed1; independent certificate sha256: 4014cc4b9dd96f842714f8b320f0f4347a80c27c8c810a211694404e459d6241; external-validation hash sha256: ce9ec6cda0446ac5380436faf80eaa2ac8c0fb504709453243a25454349417d6. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-08245
- IUPAC-L03519
- IUPAC-L03520
- IUPAC-M03868
- NIST-WEBBOOK-C13007926-UVVIS-COMplete
- NIST-WEBBOOK-C12082085-UVVIS-COMplete
- NIST-WEBBOOK-C12108111-UVVIS-COMplete
- NIST-WEBBOOK-C12146360-UVVIS-COMplete

Comparison records:

- four forced directed carrier transition classes
- exact state-order gap and proper absorbed/retained colour partition
- four IUPAC transition definitions with absent metal-to-metal definition surface preserved
- four complete NIST spectra: point counts 224, 79, 80, 73; interior maxima counts 2, 2, 1, 2
- exact interior maxima positions (('22823/100', '5419/20'), ('51021/200', '6239/20'), ('63393/200'), ('48361/200', '77817/200'))
- three development-observed and one originally law-sealed blind spectrum reused without recapture

- control omitted_source_row_rejected: True
- control numerical_zero_rejected: True
- control all_eight_target_hashes_bound_postseal: True
- control shared_spectra_not_recaptured: True
- control metal_to_metal_definition_absence_preserved: True
- control prediction_contains_no_target_payload_peak_or_wavelength: True

Falsification boundary: The claim fails if the two carrier kinds do not generate exactly four directed classes; if endpoint identity, direction or positive order gap is lost; if absorbed and retained distinctions do not form a complete nonempty partition; if numerical zero, a signed/continuum quantity, orbital model, colour wheel, selected peak, smoothing, fitted threshold or dimensional target enters forcing; if any of the eight rows or original custody classes is removed; or if any complete external spectrum lacks positive exact selective-maximum support under the registered no-smoothing/no-threshold rule.

Measurement receipt: sha256:26c6e297ec77de3ffcb73a650ce4d3cb1e3af9b55f6f5134a024da7f491c791a.
Isolation certificate: sha256:1694c2eb2362efe3372641dc9b8f1c01cd825fbd3b3a8b1ca935e22057360b99.
Custody certificate: sha256:c3f7ff22fe0fb737cdc4b0cf8b1b56b2b832e4faa93c502004751743d789697e.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The inorganic extension derives coordination identity and number, denticity, geometry, isomerism, splitting, spin, colour, magnetism, organometallic support, inverse redox transforms, cluster bonding, local solid organization, defects and coupled inorganic reaction networks. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero
- negative irrational imaginary signed or continuum proof values
- orbital or ligand-field model
- colour wheel or wavelength-colour table
- selected peak smoothing threshold or dimensional fit
- recaptured or relabeled shared evidence

Immutable identities. Source manifest

sha256:5bd4a9f72b54e497084cad8b6441de2382b8e364b2685faad9cd4828ba999a30; derivation seal
sha256:fa615d52642febbfb31877c24d9fad276391626f9b27ab71364cfb0ae88aa0fc; engine receipt
sha256:aa45a1a796871c130839367088a65e6bfa18323f0da03bd63c834e5a79717997 at receipts/engine/model_admitted/SFT-CHEM-INORGANIC-COLOUR-ELECTRONIC-TRANSITION-008-aa45a1a796871c13.j
son; empirical hash sha256:4958cbb181e57c24602861e3e01077bfe334d99b6bc18868721cb210de993d00;
measurement receipt sha256:26c6e297ec77de3ffcb73a650ce4d3cb1e3af9b55f6f5134a024da7f491c791a.

165. Exact inorganic unpaired-support magnetic-state law

Claim identity: SFT-CHEM-INORGANIC-MAGNETIC-STATE-009

Question and theorem. Complete paired support closes to EmptyOne and the held diamagnetic/repelled class; positive complete unpaired support forces its exact cardinality, successor spin width and held paramagnetic/drawn class.

one-retained-complex-carrier__complete-unpaired-occurrence-support__pairwise-closure-to-EmptyOne__exact-unpaired-support-count__unpaired-successor-spin-width__held-drawn-or-repelled-field-relation__EmptyOne-diamagnetic-positive-paramagnetic__unpaired-successor-with-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-CHEM-ELECTRON-COUNT-SPIN-002
- SFT-CHEM-ORBITAL-SUPPORT-OCCUPANCY-003
- SFT-CHEM-MOLECULAR-EXCLUSION-EXCHANGE-006
- SFT-CHEM-MOLECULAR-MAGNETIC-RESPONSE-012
- SFT-CHEM-COORDINATION-ENTITY-RETAINED-IDENTITY-001
- SFT-CHEM-LIGAND-STATE-SPLITTING-006
- SFT-CHEM-COMPLEX-SPIN-STATE-ORDER-007

Generated grammar. Generate the literal product of carrier, support, balance, moment, width, direction, classification and extension choices; decide all 256 candidates solely from admitted exact spin, exclusion, occupancy, magnetic-response and INORG-006/007 dependencies.

Boundary: Every finite complete unpaired electron occurrence support on one retained coordination entity; structural EmptyOne or positive cardinality; both held field relations; all three frozen IUPAC definitions and all 174 shared sealed NIST magnetic-response cells.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `one-retained-complex-carrier__complete-unpaired-occurrence-support__pairwise-closure-to-EmptyOne__exact-unpaired-support-count__unpaired-successor-spin-width__held-drawn-or-repelled-field-relation__EmptyOne-diamagnetic-positive-paramagnetic__unpaired-successor-with-no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-retained-complex-carrier	free-magnetic-number	A free magnetic number is detached from the inorganic state.
support	complete-unpaired-occurrence-support	selected-or-averaged-spin	A selected or averaged spin loses electron occurrences.
balance	pairwise-closure-to-EmptyOne	numerical-zero-paired-state	Numerical zero is not a native state or proof magnitude.
moment	exact-unpaired-support-count	square-root-spin-only-formula	A square-root formula imports irrational continuum magnitude.
width	unpaired-successor-spin-width	asserted-multiplicity	An asserted multiplicity can contradict complete support.
direction	held-drawn-or-repelled-field-relation	signed-susceptibility-proof	A signed scalar imports negative proof quantity.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
classification	EmptyOne-diamagnetic-positive-paramagnetic	species-magnetic-lookup	A species lookup lets observed class select the rule.
extension	unpaired-successor-with-no-extra-rule	fitted-g-factor-or-complex-exception	A fitted factor or species exception destroys zero-parameter closure.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:b226eee1438c3e2d29e5804d416f16fba95222a086d98367609f011867567781.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:78c1f7148fd8972db9581f40f6b8f36412609e8d70d05441ff18652295172089.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:271d65b4781a0989f6b5737d5d4512361e3d0856b41be1870e0db5e07426ad0c.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:fbd5ee6331f4b24c286212d49fbccef4a189cd37fa96908db88d6dabd8afc4c3.

Independent reconstruction. Implementation hash

sha256:a91bfa43154c4847ffcdf90542e959039c3baac83f9ddba3ec7f719e964c521c; independent certificate sha256:c975d7767580253f6571b2591f85d55a1f41b9c691cf4686aeefa5cec0c4d859; external-validation hash sha256:97e57e978cd721f68b8b07755c298742f1cde340b078394e8f38faa5690d8f92. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-M03689
- IUPAC-P04404
- IUPAC-D01668
- NIST-MOLECULAR-MICROWAVE-SPECTRAL-DATABASES-SRD-114-115-117
- NIST-JPCRD-3-609-1974-DIATOMIC-MICROWAVE-SPECTRAL-TABLES

Comparison records:

- balanced EmptyOne support: spin width 1, repelled-from-field, diamagnetic
- four unpaired occurrences: exact moment support 4, spin width 5, drawn-into-field, paramagnetic
- unpaired successor: moment support 5 and spin width 6
- three complete IUPAC magnetic definitions
- complete NIST vector: 174 cells, 136 exact magnitudes, 38 structural absences
- orientation classes {'source-aligned': 6, 'source-opposed': 62, 'source-orientation-unspecified': 68, 'structural-absence': 38}; exact vector sha256:a550f5ae61aaa0ede0f8d580bc8735726194db361ae466e928a9908451a6f06b
- control omitted_source_row_rejected: **True**
- control numerical_zero_rejected: **True**
- control all_177_target_hashes_bound_postseal: **True**
- control all_136_magnitudes_preserved: **True**
- control all_38_absences_preserved: **True**

- control shared_sources_not_recaptured: True
- control prediction_contains_no_target_value_or_orientation: True

Falsification boundary: The claim fails if unpaired support is incomplete or duplicated; if balanced support does not close to structural EmptyOne; if positive moment support or spin width differs from the complete unpaired count and its successor; if the held field/class pair differs; if forbidden arithmetic, a square-root formula, fitted g-factor, signed proof scalar or species lookup enters forcing; if any of 177 rows, 136 exact magnitudes, 38 structural absences or source orientations is removed; or if shared evidence is recaptured or relabeled.

Measurement receipt: sha256: abbdac72dd4fe28046f0c51fa3fe2a25652403a75c131db24bc21c71cc8a59cf.
Isolation certificate: sha256: 34562683f384839c08163f401a35d0de10c22e25b8ed45dc26d92ae407a91125.
Custody certificate: sha256: 202cb48c558a928027ddc3074c9ebd8419f0a41e25f0c70c774c44e478b6c61b.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The inorganic extension derives coordination identity and number, denticity, geometry, isomerism, splitting, spin, colour, magnetism, organometallic support, inverse redox transforms, cluster bonding, local solid organization, defects and coupled inorganic reaction networks. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero
- negative irrational imaginary signed or continuum proof values
- spin-only square-root formula or signed susceptibility proof scalar
- fitted g-factor or species lookup
- recaptured, selected or relabeled shared evidence

Immutable identities. Source manifest

sha256: 0bb0f8679858958bd69378a0d19d54f54eda9df58fe1eebe2e7a586b7a85cc14; derivation seal
sha256: b5ffd946d45a8a4c19abd2704f447aec709f24f956c858f1632ee263f0ab889b; engine receipt
sha256: cabca6204189701f435c36647c70dd5ecc6dfef4e03a746ac984b2fe3ed6a7a6 at receipts/engine/model_admitted/SFT-CHEM-INORGANIC-MAGNETIC-STATE-009-cabca6204189701f.json; empirical hash
sha256: 3fa22ca288965f7f774fd172bdccae251d408490c33334cf575afd7b340e61c3; measurement receipt
sha256: abbdac72dd4fe28046f0c51fa3fe2a25652403a75c131db24bc21c71cc8a59cf.

166. Exact direct metal-carbon organometallic bond law

Claim identity: SFT-CHEM-ORGANOMETALLIC-METAL-CARBON-BOND-010

Question and theorem. Positive complete direct admitted-centre/carbon incidence support with held bond electrons uniquely forces organometallic bond topology and its exact cardinality; absent direct support remains structural EmptyOne.

one-retained-chemical-entity__retained-admitted-centre-occurrence__retained-carbon-occurrence__direct-centre-carbon-incidence__complete-held-bond-electron-support__complete-positive-direct-incidence-support__positive-support-or-organometallic-EmptyOne-otherwise__direct-incidence-successor-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-ELECTRON-COUNT-SPIN-002
- SFT-CHEM-ORBITAL-SUPPORT-OCCUPANCY-003
- SFT-CHEM-COORDINATION-ENTITY-RETAINED-IDENTITY-001
- SFT-CHEM-COORDINATION-GEOMETRY-HELD-ORIENTATION-004

Generated grammar. Generate the literal product of carrier, centre, carbon, topology, electron, multiplicity, classification and extension choices; decide all 256 candidates solely from admitted chemical-entity, element, bond, composition, electron-support and coordination dependencies.

Boundary: Every finite one-entity support of distinct direct admitted-centre/carbon incidences with positive held electron support; structural EmptyOne absence; all twelve frozen IUPAC criterion, scope, example, context and exclusion surfaces.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `one-retained-chemical-entity__retained-admitted-centre-occurrence__retained-carbon-occurrence__direct-centre-carbon-incidence__complete-held-bond-electron-support__complete-positive-direct-incidence-support__positive-support-organometallic-EmptyOne-otherwise__direct-incidence-success-or-no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-retained-chemical-entity	free-compound-name	A name does not retain the bonded entity.
centre	retained-admitted-centre-occurrence	imported-conventional-metal-list	A conventional list imports the classification boundary.
carbon	retained-carbon-occurrence	selected-formula-fragment	A selected formula fragment need not be a bonded carbon occurrence.
topology	direct-centre-carbon-incidence	proximity-or-name-association	Proximity or naming does not force a bond.
electrons	complete-held-bond-electron-support	assumed-valence-number	An assumed valence number loses the bond support.
multiplicity	complete-positive-direct-incidence-support	single-selected-bond	Selecting one bond loses additional incidences.
classification	positive-support-organometallic-EmptyOne-otherwise	species-or-name-lookup	A lookup lets the observed label choose the law.
extension	direct-incidence-successor-no-extra-rule	fitted-species-exception	A species exception destroys zero-parameter closure.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:1381c39401d708de40fb8820ae5735f176d2a7358ae65c7ef722b53e734e5c1f.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:5e00c3df6f454bbc5a0763f911fd5632d9dc15b2f547d2c3e765ab279a576d3f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:d461b8f5715c4b11ea78e4a6ef3519682316b51fef944992007f07198999f11b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:a67dd6a446837438015839ae90cf04ecde526a2d922b8f13da075cd31b10c67a.

Independent reconstruction. Implementation hash

sha256:ab61bd01cdebaa7c83d4abd7827e059b2462bb7c7ebc3b8fde994ff57b44475c; independent certificate sha256:b97a85a4e7d03d244ea37ec85d7cc5684cbb3ec5da304b1d5805745e49d2ef39; external-validation hash sha256:7495312c8d90798c1578f143cdf80019a2d3aefa3ec76f86af729c8d858a5b6b. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-O04328

Comparison records:

- structural absence remains EmptyOne; one direct incidence forces count one and organometallic class; successor count two
- complete IUPAC vector: twelve distinct criterion, scope, example, context and exclusion surfaces
- six named example surfaces and the no-direct-structural-evidence exclusion retained
- complete exact target vector sha256:068d1e8f09b989098ad4e3baf5e506db49f46b6040acb397df7f7b506b282697
- control omitted_source_row_rejected: True
- control numerical_zero_rejected: True
- control all_twelve_target_hashes_bound_postseal: True
- control all_six_named_examples_preserved: True
- control direct-evidence-exclusion-preserved: True
- control sources_not_recaptured: True
- control prediction_contains_no_definition_example_or_target_payload: True

Falsification boundary: The claim fails if the chemical entity, centre, carbon, direct adjacency or held electron support is lost or duplicated; if structural absence becomes numerical zero or a positive class; if the exact count differs from complete direct incidence support; if a name, conventional list, species lookup, observed example or fitted exception selects the law; if any of twelve IUPAC surfaces is removed or changed; or if prediction accesses outcomes before sealing.

Measurement receipt: sha256:e2f26196014eb87a633b3bd7a480451017bba3594e7786b4e2a3e43592747d11.

Isolation certificate: sha256:976dfadf9d034750878ae54aded184f08f781b7029eee9e50efa1845132b9bb.

Custody certificate: sha256:aaba6d7931b5d99c3bcf26638261102fb854d33e814b35f9ec28d7430709c24e.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The inorganic extension derives coordination identity and number, denticity, geometry, isomerism, splitting, spin, colour, magnetism, organometallic support, inverse redox transforms, cluster bonding, local solid organization, defects and coupled inorganic reaction networks. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero
- negative irrational imaginary signed or continuum proof values
- conventional metal list
- species or compound-name lookup
- observed example or fitted exception

Immutable identities. Source manifest

sha256:ad060ab94ad3ea40bb25f8e2330b71c1262faec3ce28501c011129a025d72d97; derivation seal
 sha256:4e40f0ca47b3187da2446eacac6904c271de76c3b1675f45cfb9442290a4d3c2; engine receipt
 sha256:52d096e861416551b25a99cb16f11af5879e0d00e360c200c5f3b3fd19ab9a41 at receipts/engine/model_admitted/SFT-CHEM-ORGANOMETALLIC-METAL-CARBON-BOND-010-52d096e861416551.json;
 empirical hash sha256:890326b07c69c00798173e8199f3d3664275f975e14d1110d26290c5fcb54240;
 measurement receipt sha256:e2f26196014eb87a633b3bd7a480451017bba3594e7786b4e2a3e43592747d11.

167. Exact organometallic electron-support capacity and accounting law

Claim identity: SFT-CHEM-ORGANOMETALLIC-ELECTRON-ACCOUNTING-011

Question and theorem. Exact admitted support widths 2, 6 and 10 force capacity 18; complete complementary nonbonded and bond electron support yields an exact held capacity relation and pair successor until capacity halt.

one-retained-organometallic-entity__complete-nonbonded-and-bond-pair-support__forced-two-plus-six-plus-ten-capacity__complete-complementary-fibre-pairing__exact-nonbonded-plus-bond-partition__capacity-complete-or-incomplete-held-relation__postseal-stable-complex-correspondence__pair-successor-until-capacity-then-halt

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-CHEM-ELECTRON-COUNT-SPIN-002
- SFT-CHEM-ORBITAL-SUPPORT-OCCUPANCY-003
- SFT-CHEM-MOLECULAR-EXCLUSION-EXCHANGE-006
- SFT-CHEM-MOLECULAR-MAGNETIC-RESPONSE-012
- SFT-CHEM-ORGANOMETALLIC-METAL-CARBON-BOND-010

Generated grammar. Generate the literal product of carrier, support, capacity, balance, partition, relation, comparison and extension choices; decide all 256 candidates only from admitted spin, exclusion, orbital-support, magnetic-response and INORG-010 dependencies.

Boundary: Every finite complementary-pair support partitioned into nonbonded and bond occurrences on one retained organometallic entity, up to the exact generated s+p+d capacity; all four frozen IUPAC scope, component and total surfaces.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `one-retained-organometallic-entity__complete-nonbonded-and-bond-pair-support__forced-two-plus-six-plus-ten-capacity__complete-complementary-fibre-pairing__exact-nonbonded-plus-bond-partition__capacity-complete-or-incomplete-held-relation__postseal-stable-complex-correspondence__pair-successor-until-capacity-then-halt`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-retained-organometallic-entity	free-electron-number	A free number is detached from the complex.
support	complete-nonbonded-and-bond-pair-support	selected-valence-electrons	Selected electrons lose complete occurrence support.
capacity	forced-two-plus-six-plus-ten-capacity	imported-eighteen-rule	Importing eighteen assumes the target.
balance	complete-complementary-fibre-pairing	signed-spin-sum	A signed sum imports negative proof values.
partition	exact-nonbonded-plus-bond-partition	oxidation-state-bookkeeping	Conventional oxidation bookkeeping can change labels without support.
relation	capacity-complete-or-incomplete-held-relation	species-stability-lookup	A species lookup lets observed stability choose the rule.
comparison	postseal-stable-complex-correspondence	measured-total-selects-capacity	An observed count cannot select the capacity.
extension	pair-successor-until-capacity-then-halt	complex-specific-exception	An exception destroys zero-parameter closure.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:146b87bdd4b5aab8d04517bdaab4138b84e84526f6d4f677376ca0c5b2360cc0.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:e518381cfa2d79272094ad0906b3b281396edbdd160b7af30a47a7924fd1fb7d.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:3f9d0285d263ba69b4b24f11b24bad880b506ddf95660f02c0effc87990ca3bf.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:5f3dabfe4f7582721494c27ca6815d74fb795788f1c8725ebf0af62cf18ed9ae.

Independent reconstruction. Implementation hash

sha256:d466f8cc395117f46ec8d56e7874c59ed7a41af8df44f78ff0f6085a7642c131; independent certificate sha256:e57b24ecec0331b22bfed18ae53c1e48c11b23356b28d8672b8bb0a80b3300dc; external-validation hash sha256:6f3f1b4fffc9e0a8d96e0a8754a409044256ea7732e2cd820a6a2bcec7aa2f0b. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-E01913

Comparison records:

- forced support widths 2, 6 and 10; exact capacity 18
- complete paired account 18; exact partial-to-successor count 2 to 4; capacity overflow halts
- complete IUPAC four-row stable-complex, component-account, total and analogy vector
- complete exact target vector sha256:66dd87043b883a6c276d622768861e0b3a7e5ce6cd1f335cb3ad3cccdc45a0de
- control omitted_source_row_rejected: True
- control numerical_zero_rejected: True
- control all_four_target_hashes_bound_postseal: True
- control capacity_overflow_rejected: True
- control sources_not_recaptured: True
- control prediction_contains_no_definition_observed_eighteen_or_target_payload: True

Falsification boundary: The claim fails if capacity is imported rather than regenerated as 2+6+10; if any electron occurrence is missing, duplicated, unpaired or assigned to both account parts; if the exact total or capacity relation differs; if support exceeds capacity without halting; if observed eighteen, oxidation-state bookkeeping or a species lookup selects the law; if any of four IUPAC surfaces is removed; or if outcomes open before the prediction seal.

Measurement receipt: sha256:64b9733f276add0558095e29378978932dfee86f7fc0b6bba51980727057aa56.

Isolation certificate: sha256:7255429087ad1de92fde5f0d79a872b0eed4522400643a0f13dbc59850406daf.

Custody certificate: sha256:26a5fd7da8f3a81df2d1a5125ec53546ff0aaafd108aef29411a349697a25e18.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The inorganic extension derives coordination identity and number, denticity, geometry, isomerism, splitting, spin, colour, magnetism, organometallic support, inverse redox transforms, cluster bonding, local solid organization, defects and coupled inorganic reaction networks. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero
- negative irrational imaginary signed or continuum proof values
- imported eighteen-electron rule
- oxidation-state bookkeeping
- species stability lookup
- observed target total or fit

Immutable identities. Source manifest

sha256:67f72af14813c58d569417c19b5d92c2311c62ffba8ebb6c2bfdafda2a4d6903; derivation seal

sha256:30dc00e1f8d08522b68a30b0b352076e90c8c7427de1a04727321201892d8a42; engine receipt

sha256:39e83309e44945b8cf34698ec31d140041412cdd19451d214598fcf3c7c5011b at receipts/engine/model_admitted/SFT-CHEM-ORGANOMETALLIC-ELECTRON-ACCOUNTING-011-39e83309e44945b8.json;

empirical hash sha256:d607de61fd2b88c77dad5ecb6aa1323adbb1e85ca4980b0cf3a97a86edffd802;

measurement receipt sha256:64b9733f276add0558095e29378978932dfee86f7fc0b6bba51980727057aa56.

168. Exact oxidative-addition/reductive-elimination inverse law

Claim identity: SFT-CHEM-OXIDATIVE-ADDITION-REDUCTIVE-ELIMINATION-012

Question and theorem. Oxidative addition conserves two carriers while replacing their bond with two metal incidences and exact held transfer two or one-plus-one; reductive elimination is the exact inverse.

one-retained-organometallic-process__complete-two-carrier-source-bond__two-generated-metal-carrier-incidences__exact-carrier-and-metal-conservation__held-two-electron-transfer-orientation__complete-one-metal-two-or-two-metal-one-one-partition__exact-reductive-inverse-correspondence__trace-composition-with-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-ORGANOMETALLIC-METAL-CARBON-BOND-010
- SFT-CHEM-ORGANOMETALLIC-ELECTRON-ACCOUNTING-011

Generated grammar. Generate the literal product of process carrier, source, product, conservation, transfer, partition, reverse and extension alternatives; decide all 256 forms solely from admitted transition, conservation, covalent-bond and INORG-010/011 dependencies.

Boundary: Every finite one- or two-metal exact trace containing one two-carrier source bond, two product incidences, the complete transfer partition and its exact inverse; all five frozen IUPAC oxidative, split, radical-scope and reductive surfaces.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *one-retained-organometallic-process__complete-two-carrier-source-bond__two-generated-metal-carrier-incidences__exact-carrier-and-metal-conservation__held-two-electron-transfer-orientation__complete-one-metal-two-or-two-metal-one-one-partition__exact-reductive-inverse-correspondence__trace-composition-with-no-extra-rule*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-retained-organometallic-process	anonymous-reaction-name	A name loses the process carrier.
source	complete-two-carrier-source-bond	selected-reactant-fragments	Selected fragments lose the complete covalent source bond.
product	two-generated-metal-carrier-incidences	asserted-oxidation-state-change	A signed oxidation-state change is not a topology.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
conservation	exact-carrier-and-metal-conservation	formal-charge-bookkeeping	Formal charge bookkeeping can lose occurrences.
transfer	held-two-electron-transfer-orientation	signed-electron-loss	Signed loss imports negative proof values.
partition	complete-one-metal-two-or-two-metal-one-one-partition	single-conventional-mechanism	One conventional mechanism omits the split-metal form.
reverse	exact-reductive-inverse-correspondence	separate-reductive-rule	A separate rule need not invert the process.
extension	trace-composition-with-no-extra-rule	species-specific-exception	An exception destroys zero-parameter closure.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : b6d81a12f5f3ca520ab7754a21782f1430c905b5bbff9ab2dd8b29b9b47a6632.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 87e2a33631f3aeb63805f9caa890df1aa7304cae07933ddd562843748420342f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 2a20514d16a5f907e7b0d117251c1002007c30799a32a31d99967d8d1d9e399c.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : e33cf8852dd6f8d4a5225d26581778a1dd775c316669ab9a8902bc5fd305f594.

Independent reconstruction. Implementation hash

sha256 : 5d073f4f903c48778e7f8e7485ad6c3ca39d237cefe99b3d522dc8890f1f23af; independent certificate
sha256 : de2d02b4f2a285e5347657265d5582a8312de47830fce5f51a7b68b117dc6972; external-validation
hash sha256 : 2175e7953b32c870ef6db2192b58fc2cb2264dfb26490f67975cfe63a751442d. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-O04367
- IUPAC-R05223

Comparison records:

- single-metal transfer distribution (2); two-metal distribution (1,1)
- two carriers and two product incidences retained; exact reductive inverse restores source bond
- complete five-row two-source IUPAC vector including radical scope
- complete exact target vector sha256:c97b7e93b37ef5fc8aac82a1a7ce483a88d74edf7c17b9200588b4597d4f5825
- control omitted_source_row_rejected: True
- control numerical_zero_rejected: True
- control all_five_target_hashes_bound_postseal: True
- control third_metal_partition_rejected: True
- control sources_not_recaptured: True
- control prediction_contains_no_definition_formula_or_target: True

Falsification boundary: The claim fails if either carrier or any metal occurrence is lost or duplicated; if the source bond is not replaced by exactly two incidences; if the exact transfer partition differs from two or one-plus-one; if signed oxidation arithmetic enters proof; if reductive elimination is not the exact inverse; if any of five IUPAC surfaces is removed; or if outcomes open before prediction sealing.

Measurement receipt: sha256:1d10d9a1c5d1fa83c995079fe193b4cb8651d964d5dbcc37ada0275568b05848.

Isolation certificate: sha256:62a5a85295d8ea79c547ed56002651b34574c5774fa99d757d9510a7b7a6abbb.

Custody certificate: sha256:e1da42feed7253c014ca8e85c767918ec2e618b61011cb51ef22844fe3164b5c.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The inorganic extension derives coordination identity and number, denticity, geometry, isomerism, splitting, spin, colour, magnetism, organometallic support, inverse redox transforms, cluster bonding, local solid organization, defects and coupled inorganic reaction networks. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero
- negative irrational imaginary signed or continuum proof values
- signed oxidation-state arithmetic
- formal-charge substitution
- observed mechanism or fit

Immutable identities. Source manifest

sha256:128b5e1d45ad04a26285f3380e287a80b528336bb85033ff973adc2421e6cdda; derivation seal

sha256:d25930f609eec1fa3f39d160b4a2f2268a62bfae8cb21aa4364c4c40f57dba9e; engine receipt

sha256:9dad1784e9aa6524686d39e0c757c65b9dba2355973f668470cf4639e9d2316a at receipts/engine/model_admitted/SFT-CHEM-OXIDATIVE-ADDITION-REDUCTIVE-ELIMINATION-012-9dad1784e9aa6524.json; empirical hash

sha256:5d8c754e643ebf2392169d34147d279342348f2635694c1cc5cc37633fc41dd0; measurement receipt

sha256:1d10d9a1c5d1fa83c995079fe193b4cb8651d964d5dbcc37ada0275568b05848.

169. Exact insertion, migratory-insertion and elimination adjacency-trace law

Claim identity: SFT-CHEM-INSERTION-ELIMINATION-PATHWAY-013

Question and theorem. Insertion, exact extrusion, migratory insertion and elimination are complete carrier-preserving before/after adjacency traces with exact removed/added support.

complete-retained-carrier-support__complete-before-after-adjacency-difference__xz-plus-y-to-x-y-z-adjacency-law__migration-plus-insertion-composition__extrusion-exact-insertion-inverse__complete-two-eliminand-centre-trace__held-unsaturation-ring-or-carbene-boundary__adjacency-trace-composition-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-ORGANOMETALLIC-METAL-CARBON-BOND-010
- SFT-CHEM-OXIDATIVE-ADDITION-REDUCTIVE-ELIMINATION-012

Generated grammar. Generate the literal product of carrier, trace, insertion, migration, reverse, elimination, product and extension choices; decide all 256 candidates only from admitted graph, transition, conservation, mechanism and INORG-010/012 dependencies.

Boundary: Every finite complete adjacency transformation for insertion, exact extrusion, migratory insertion and distinct- or single-centre elimination; all ten frozen IUPAC definition, composition, scope, absent-example and product-boundary surfaces.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-retained-carrier-support__complete-before-after-adjacency-difference__xz-plus-y-to-x-y-z-adjacency-law__migration-plus-insertion-composition__extrusion-exact-insertion-inverse__complete-two-eliminand-centre-trace__held-unsaturation-ring-or-carbene-boundary__adjacency-trace-composition-no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-retained-carrier-support	anonymous-products	Anonymous products lose occurrence identity.
trace	complete-before-after-adjacency-difference	endpoint-only-reaction	Endpoints alone do not certify the transformation.
insertion	xz-plus-y-to-x-y-z-adjacency-law	imported-product-formula	A product formula can hide adjacency.
migration	migration-plus-insertion-composition	named-migratory-step	A name does not force composition.
reverse	extrusion-exact-insertion-inverse	insertion-elimination-conflation	Conflating reverse classes loses the exact boundary.
elimination	complete-two-eliminand-centre-trace	selected-leaving-group	Selection loses one eliminand or centre.
product	held-unsaturation-ring-or-carbene-boundary	imported-unsaturation-name	A conventional name is not a generated adjacency.
extension	adjacency-trace-composition-no-extra-rule	mechanism-specific-exception	An exception destroys closure.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8f6271ac7a4981344298f6c4f45e6a81f596ce88254956b9340f4afd7f77d5b2`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3bcfac2ffcfa23cdf4d7f5ff51ceb469759c7d9093129cc082b66b7d951d736e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:8d6b9b4c2c2b73e6fd7db423cc213cacc6afe5e797d637948ba971157b5aea53.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e970003a88fa5912ac776eb9900a0a78d2a3e0188942ea82dc64c7a98d37e3.

Independent reconstruction. Implementation hash

sha256:ca328a9dedb8150b9d722da3d09400d4d1057e3ef3551453418ab288d0c4ad38; independent certificate sha256:dc46fc8ccb4ab099035caccfe0b3795a47981100f93daa79e09912d726bbad7a; external-validation hash sha256:90dc18fab808b9a7918e53bd0958f69936f4940b586d62e201ad03322bd56050. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-M03924
- IUPAC-I03058
- IUPAC-E02038

Comparison records:

- insertion adjacency counts before/after 1/2 and removed/added 1/2; extrusion exact inverse
- migratory composition migration plus insertion; distinct- and single-centre elimination retained
- complete ten-row three-source IUPAC vector including host-lattice scope and absent rendered example
- complete exact target vector sha256:8e5ccd6525a21ac3a2294d9c62e6d4a480f8cab994dc5472d73cd28f667f9f9e
- control omitted_source_row_rejected: True
- control numerical_zero_rejected: True
- control all_ten_target_hashes_bound_postseal: True
- control absent_rendered_example_preserved: True
- control sources_not_recaptured: True
- control prediction_contains_no_definition_product_or_target: True

Falsification boundary: The claim fails if any carrier, centre, eliminand or before/after adjacency is lost, duplicated or invented; if removed/added support differs from the exact graph difference; if insertion, extrusion, migratory composition or elimination boundaries are conflated; if an observed product or named mechanism selects the law; if any of ten IUPAC surfaces, the host-lattice scope row or absent rendered example is removed; or if outcomes open before sealing.

Measurement receipt: sha256:9aef7008f13027a4896eb67d50a824db9a7e634de536438d547a4a1837b252f6.

Isolation certificate: sha256:3b169b952d29580b598bced0da65d83a8ac94033d96dc0d8b3b7f86e36b952f7.

Custody certificate: sha256:a6610ce7b9387d65c2aec93fb2fb699eff95a66880e4b22944f033e5ea6c102a.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The inorganic extension derives coordination identity and number, denticity, geometry, isomerism, splitting, spin, colour, magnetism, organometallic support, inverse redox transforms, cluster bonding, local solid organization, defects and coupled inorganic reaction networks. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero
- negative irrational imaginary signed or continuum proof values

- product-formula lookup
- named-mechanism selection
- omission of adverse scope or absence rows

Immutable identities. Source manifest

sha256:16debb733794c0600a8e5b6d848a019b27d661b9d9c5b1f7f03b9c5cc4604f51; derivation seal
 sha256:799d1b919a67366f3ec28b5b800359a9fc033459d2522e9f6a0d0a5c4209b828; engine receipt
 sha256:d78ae550b9001c0abdf4ae53a676d047c060d850bbfe81e021b74c2ce2d800e2 at receipts/engine/model_admitted/SFT-CHEM-INSERTION-ELIMINATION-PATHWAY-013-d78ae550b9001c0a.json;
 empirical hash sha256:78ed8cd3d925e26a92ed53d981e162883d9daf4d8b32117b8f3f6a728857e600;
 measurement receipt sha256:9aef7008f13027a4896eb67d50a824db9a7e634de536438d547a4a1837b252f6.

170. Exact finite connected metal-cluster and metal-bonding law

Claim identity: SFT-CHEM-METAL-CLUSTER-BONDING-014

Question and theorem. A cluster is complete finite connected multicentre support with distinct direct, bridging and held grouping relation classes and positive counts or EmptyOne.

one-retained-cluster-entity__complete-finite-multicentre-support__complete-connected-cluster-relation-graph__explicit-direct-metal-bond-subgraph__complete-bridging-ligand-path-support__held-grouping-relation-without-bond-necessity__positive-count-or-EmptyOne-per-relation-class__connected-centre-successor-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-COORDINATION-ENTITY-RETAINED-IDENTITY-001
- SFT-CHEM-COORDINATION-GEOMETRY-HELD-ORIENTATION-004
- SFT-CHEM-ORGANOMETALLIC-METAL-CARBON-BOND-010

Generated grammar. Generate the literal product of carrier, centres, connectivity, direct, bridge, grouping, count and extension alternatives; decide all 256 forms from admitted graph, bond, composition and coordination dependencies.

Boundary: Every finite connected graph of at least two metal centres with direct, bridging or held grouping relations and explicit bridge occurrences; all ten frozen cluster and iron-sulfur-cluster topology, example and charge-inscription surfaces.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *one-retained-cluster-entity__complete-finite-multicentre-support__complete-connected-cluster-relation-graph__explicit-direct-metal-bond-subgraph__complete-bridging-ligand-path-support__held-grouping-relation-without-bond-necessity__positive-count-or-EmptyOne-per-relation-class__connected-centre-successor-no-extra-rule*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-retained-cluster-entity	free-cluster-name	A name loses centre support.
centres	complete-finite-multicentre-support	selected-metal-pair	A selected pair loses remaining centres.
connectivity	complete-connected-cluster-relation-graph	proximity-only-list	A list does not prove connected support.
direct	explicit-direct-metal-bond-subgraph	assumed-metal-metal-bond	Assumption confuses grouping with bonding.
bridge	complete-bridging-ligand-path-support	implicit-ligand-bridge	An implicit bridge loses ligand occurrences.
grouping	held-grouping-relation-without-bond-necessity	bond-required-cluster-definition	Requiring a bond rejects lawful held grouping.
counts	positive-count-or-EmptyOne-per-relation-class	numerical-zero-missing-mode	Numerical zero is not structural absence.
extension	connected-centre-successor-no-extra-rule	catalogue-specific-exception	A catalogue destroys closure.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:cd28075ad4bbb757ac21bab231623686649ad77440b72929361ada97cc64d341.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:736d3021fd7640c4f5d9a2683ce42a7c183f3fe95a3f92a9d87b3bd9c13772ba.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:41cb99f723caba4345d662c36178059437d552b4cbe97052415795fa7d54b3da.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:273e15f298298e569200c9e1580473ddb8fa7ffc521918aac588a2dc65b1c3af.

Independent reconstruction. Implementation hash

sha256:f4802c5c4cf1cb11d8bb057388890bbd3b5504c52aad4b93164bd65ca10e35eb; independent certificate sha256:1bb69ae78d7d03cb6d0cdef9f2c579b76e8b72966b4adab52955fe6850af3706; external-validation hash sha256:482c085a340b9d15f0100998809e08f3b6828c004016e1c926ecbb787cfc9f0e. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-CT06769
- IUPAC-IT06779

Comparison records:

- two-centre direct, two-centre two-bridge and held grouping cluster witnesses; successor centre count three
- complete ten-row two-source cluster and iron-sulfur vector; external charge inscriptions retained downstream
- complete exact target vector sha256:a96d298d931b9f18de9fad29050c343e88a40991823aa72e6fb3147cc9ddb99e
- control omitted_source_row_rejected: **True**
- control numerical_zero_rejected: **True**

- control all_ten_target_hashes_bound_postseal: True
- control disconnected_cluster_rejected: True
- control signed_charge_retained_downstream_only: True
- control sources_not_recaptured: True
- control prediction_contains_no_definition_example_charge_or_target: True

Falsification boundary: The claim fails if centre support is finite-incomplete, duplicated or disconnected; if direct, bridging and held grouping relations are conflated; if bridge occurrences are implicit; if absent classes use numerical zero; if an observed example or charge inscription selects the law; if any of ten IUPAC surfaces is removed; or if outcomes open before sealing.

Measurement receipt: sha256:bccc01412a818af69db687a19d4c57918651102b3fc9ea3ce4586b60225cd1f1.

Isolation certificate: sha256:fb118521ece96f22876cd00060a4d0988b074200c2193dbcd7720295a79adf57.

Custody certificate: sha256:fdc997ab0b730b941bffc03df9bb98e3d5cd18dbc52d2a48e1221a3b09c46bba.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The inorganic extension derives coordination identity and number, denticity, geometry, isomerism, splitting, spin, colour, magnetism, organometallic support, inverse redox transforms, cluster bonding, local solid organization, defects and coupled inorganic reaction networks. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero
- negative irrational imaginary signed continuum fitted or free values
- bond-required cluster assumption
- cluster catalogue or charge inscription

Immutable identities. Source manifest

sha256:1d001ed1ea9a03f9d60af3e5993f6cab33be48fdd49a9fef966043f6e73ecd6; derivation seal
 sha256:bd35c140a8515b5e42c107584eb5a327e5d7910b2abb8b356e7cdc2e37b058b8; engine receipt
 sha256:700577fb0682ff8f314409c1d5c5be7fbcacc966d351c9b4c3d7a3e0d4c42ffa at receipts/engine/model_admitted/SFT-CHEM-METAL-CLUSTER-BONDING-014-700577fb0682ff8f.json; empirical hash
 sha256:1b46c354cb2009801573bf097fb242f5aa70ab5bf5bfbed8e9136acbdba46e28; measurement receipt
 sha256:bccc01412a818af69db687a19d4c57918651102b3fc9ea3ce4586b60225cd1f1.

171. Exact solid-state formula, local coordination and ownership law

Claim identity: SFT-CHEM-SOLID-STATE-LOCAL-COORDINATION-015

Question and theorem. One finite complete local solid motif forces a primitive positive formula, complete local adjacency, generated repeat rank one through three, exact constituent support or EmptyOne, and a Chemistry-to-Materials ownership handoff.

one-finite-local-chemical-motif__complete-species-occurrence-multiset__exact-primitive-positive-count-ratio__complete-local-bond-adjacency-support__generated-repeat-axis-rank-one-two-or-three__positive-second-constituent-support-or-EmptyOne__chemistry-local-materials-bulk-handoff__local-occurrence-successor-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-COORDINATION-ENTITY-RETAINED-IDENTITY-001
- SFT-CHEM-COORDINATION-GEOMETRY-HELD-ORIENTATION-004
- SFT-CHEM-METAL-CLUSTER-BONDING-014

Generated grammar. Generate the literal product of carrier, composition, formula, bonds, network, constituent, ownership and extension alternatives; decide all 256 forms from the admitted exact-arithmetic, graph, generator-three, stoichiometry, bond, coordination and cluster dependencies.

Boundary: Every positive finite complete local occurrence motif with exact local adjacency, primitive positive formula, generated repeat rank one, two or three, and explicit Chemistry-to-Materials handoff; all ten frozen coordination-network and returned mixed-crystal surfaces.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `one-finite-local-chemical-motif__complete-species-occurrence-multiset__exact-primitive-positive-count-ratio__complete-local-bond-adjacency-support__generated-repeat-axis-rank-one-two-or-three__positive-second-constituent-support-or-EmptyOne__chemistry-local-materials-bulk-handoff__local-occurrence-successor-no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-finite-local-chemical-motif	bulk-material-response	Bulk response is outside Chemistry ownership.
composition	complete-species-occurrence-multiset	nominal-formula-label	A label can omit occurrences.
formula	exact-primitive-positive-count-ratio	unreduced-or-fitted-stoichiometry	Unreduced or fitted coefficients are not unique.
bonds	complete-local-bond-adjacency-support	averaged-coordination-number	An average loses local adjacency.
network	generated-repeat-axis-rank-one-two-or-three	continuum-infinite-lattice	A continuum lattice imports infinity.
constituent	positive-second-constituent-support-or-EmptyOne	real-valued-solid-solution-fraction	A real fraction imports continuum fitting.
ownership	chemistry-local-materials-bulk-handoff	chemistry-claims-bulk-response	Bulk response belongs to Materials.
extension	local-occurrence-successor-no-extra-rule	crystal-family-exception	A catalogue exception destroys closure.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:5f80bdf4263711c69268ce5c7533cb6100018336e5bc743f50273a069a1419e0.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:8bb5f157d743587937ff2c6fee467c417645d7a213bdd9a39108ffe2fea187ea.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ba24eb619dbb63c0b75788e6e0f439b41d4530e8eb5062bdc38df7eab82624a1.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e6022b5615070514098747177630bb454d2c1489f4897e042aa3a1172556be2a.

Independent reconstruction. Implementation hash

sha256:4db397fc91f8b95c14ac53ee1a151891be33af2ea7fafd3e88bd4d53b8eb7e2e; independent certificate sha256:51a0ab541d80acd07fec164d5d065e732cdf9894905687366659665589b349d6; external-validation hash sha256:68ad3d77a457217efdbd1417706dc9e3b2b619055ac0f0f4b294081dc8e8c5b8. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-12823
- IUPAC-S05735

Comparison records:

- primitive local formula one-to-one; host/guest formula two-to-one; successor formula three-to-two
- generated repeat rank three, one exact second constituent and structural EmptyOne absence
- complete ten-row two-source coordination-network and returned mixed-crystal vector
- complete exact target vector sha256:aa9839553c992235bef03939c82cab2a8ab344cab678b3bf8a3130e2e55faa14
- control omitted_source_row_rejected: **True**
- control numerical_zero_rejected: **True**
- control all_ten_target_hashes_bound_postseal: **True**
- control rank_four_rejected: **True**
- control returned_mixed_crystal_identity_mismatch_preserved: **True**
- control sources_not_recaptured: **True**
- control prediction_contains_no_definition_or_target_payload: **True**

Falsification boundary: The claim fails if any local occurrence or bond is omitted, if the formula is not the primitive positive count ratio, if repeat rank exceeds generator three, if absence uses numerical zero, if Chemistry claims bulk response, if the returned mixed-crystal identity mismatch is hidden, if any of ten source surfaces is removed, or if outcomes open before prediction sealing.

Measurement receipt: sha256:288bd0274c8c1802e6cba4337fc42e30a9bba97d811bdf8136394fb611cd4491.

Isolation certificate: sha256:b546e7fe44990aeeab45612fc7bf23c5982dd44af8126dc95a8ff27e9340c31a.

Custody certificate: sha256:d1deaeb1fab49420513f3dd9dac7471aa1fa0409ddd19559773bf12d4d8771ad.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The inorganic extension derives coordination identity and number, denticity, geometry, isomerism, splitting, spin, colour, magnetism, organometallic support, inverse redox transforms, cluster bonding, local solid organization, defects and coupled inorganic reaction networks. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero
- negative irrational imaginary signed continuum fitted free or imported values
- nominal formula substituted for complete occurrences
- infinite continuum lattice
- bulk material response
- observed coordination-network or mixed-crystal outcomes

Immutable identities. Source manifest

sha256:38630554bbccf8741d58ccb295a7393bd62c1bde9afb31f6ffd6fa41d0257720; derivation seal sha256:0fd53e52477bc5fde9bf456229babf51f7cf248faae7cd1db55524139c2ed83f; engine receipt sha256:c4c74ec1c8d623ccb4245420514693f5f61e324edc4642b83f108685aaec5cf3 at receipts/engine/model_admitted/SFT-CHEM-SOLID-STATE-LOCAL-COORDINATION-015-c4c74ec1c8d623cc.json; empirical hash sha256:154036027d4519246185a4276073c81011401bf82a5c453bfa9a9682c818fc05; measurement receipt sha256:288bd0274c8c1802e6cba4337fc42e30a9bba97d811bdf8136394fb611cd4491.

172. Exact defect chemistry and non-stoichiometry law

Claim identity: SFT-CHEM-DEFECT-NONSTOICHIOMETRY-016

Question and theorem. A defect is an exact reference/observed local-motif interruption; vacancies are EmptyOne occupancies and non-stoichiometry is separate positive missing and added support.

finite-complete-reference-and-observed-motif__complete-reference-site-and-species-support__reference-site-with-structural-EmptyOne__separate-positive-missing-and-added-supports__complete-vacancy-substitution-interstitial-classes__exact-reference-observed-primitive-formulas__reference-membership-forces-origin-class__site-occurrence-successor-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-SOLID-STATE-LOCAL-COORDINATION-015

Generated grammar. Generate the literal product of carrier, reference, vacancy, difference, class, composition, origin and extension alternatives; decide all 256 forms from admitted exact arithmetic, discrete support, stoichiometry and local-solid coordination.

Boundary: Every finite complete reference-site and observed local motif with explicit vacancy, substitution and interstitial supports, exact positive reconciliation and primitive formulas; all fifteen frozen solid-solution, lattice-defect, noncrystal-scope, extrinsic and intrinsic-defect surfaces including two definition-note surfaces.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `finite-complete-reference-and-observed-motif__complete-reference-site-and-species-support__reference-site-with-structural-EmptyOne__separate-positive-missing-and-added-supports__complete-vacancy-substitution-interstitial-classes__exact-reference-observed-primitive-formulas__reference-membership-forces-origin-class__site-occurrence-successor-no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	finite-complete-reference-and-observed-motif	continuum-defect-density	A continuum density loses the exact local defect occurrences.
reference	complete-reference-site-and-species-support	nominal-formula-only	A nominal formula does not identify the occupied sites.
vacancy	reference-site-with-structural-EmptyOne	numerical-zero-occupancy	Numerical zero is not a Fold form.
difference	separate-positive-missing-and-added-supports	signed-stoichiometric-subtraction	Signed subtraction imports negative values.
class	complete-vacancy-substitution-interstitial-classes	selected-defect-name	A selected name omits alternative interruptions.
composition	exact-reference-observed-primitive-formulas	fitted-nonstoichiometric-variable	A fitted variable is neither exact nor unique.
origin	reference-membership-forces-origin-class	assumed-intrinsic-or-extrinsic-label	An assumed label does not follow from composition.
extension	site-occurrence-successor-no-extra-rule	catalogue-specific-defect-exception	A catalogue exception destroys closure.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:8204d74d409522f72bc1236cf75cc481113aa8427940146a91a6cd57f8a91f3c.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:50d69b795270f36966a19d26a08327d440fed2a62626c621c84ec3812b23e421.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b359a66e7784432771172cf0ce102b7e5a99f2e2300c7f3a7c818d6aa395f8f0.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:dc70b6328870fea37e1dfc1f1f28e1d629fd4a7337cf515e125911b05d41aed7.

Independent reconstruction. Implementation hash

sha256:5da013a3936df7e25f1a93e2c3ebd2d33d485edc7a9491fdb3bc37fc622aa8fa; independent certificate sha256:87ede2363f8309a705cc4420fed8e7432f95bc11b6f756ced28983dbed3df7ac; external-validation hash sha256:eec0f9a0fb499aff91e06842e06e413641bc2f45f11581fc3fd4a6f8f4bf0d74. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-S05735
- IUPAC-13807
- IUPAC-09464
- IUPAC-14011
- IUPAC-14026

Comparison records:

- vacancy formula one-to-two with one positive missing A and structural EmptyOne occupancy
- native interstitial formula three-to-two; foreign substitution formula one-to-one-to-two
- complete fifteen-row five-source defect vector with two definition-note surfaces
- V1 two-surface adverse record preserved and V2 correction separately sealed
- complete exact target vector sha256:4662e91b77bb6e4b448c32569191e9f358560f01af41ee33df865552ff79153f
- control omitted_source_row_rejected: True
- control numerical_zero_rejected: True
- control all_fifteen_target_hashes_bound_postseal: True
- control v1_two_missing_note_surfaces_preserved: True
- control v2_two_note_surfaces_complete: True
- control two_scope_or_identity_distinctions_preserved: True
- control sources_not_recaptured: True
- control prediction_contains_no_definition_note_or_target_payload: True

Falsification boundary: The claim fails if a site or occurrence is omitted, if vacancy is numerical zero, if missing and added support is signed or fitted, if primitive formulas or defect/origin classes do not reconstruct, if either V1 missing note surface is hidden, if the V1 predecessor is overwritten, if either scope mismatch is concealed, if any of fifteen rows is removed, or if outcomes open before sealing.

Measurement receipt: sha256:263c667c873091304b7bdba6d8f4c929bc64e3a53fc8c44fce863fc581b3ad3c.

Isolation certificate: sha256:20a029f19f2043cbe65b00947b0ee5bd9af1a1a995385133b9837e7f04c70bd8.

Custody certificate: sha256:65e5ce354f826d0309a7e0bcf7ece8e61e19a9dfcf1fb0eddb75d65b7a377d04.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The inorganic extension derives coordination identity and number, denticity, geometry, isomerism, splitting, spin, colour, magnetism, organometallic support, inverse redox transforms, cluster bonding, local solid organization, defects and coupled inorganic reaction networks. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero vacancy
- negative irrational imaginary signed continuum fitted free or imported values
- signed stoichiometric subtraction
- fitted non-stoichiometric variable
- selected defect catalogue
- external dimensional zero inscription

- observed definition, note, example or identity redirect

Immutable identities. Source manifest

sha256:ec86c7392506d59b4c7b221f682bdc95e8b3811a1c99e5b8dfaead0e46e1ee7a; derivation seal
 sha256:609c18f495a3bc04fe6714411a0900ad07b41aa10b6f55bd5865e5793bc0d7a6; engine receipt
 sha256:a3ffe6b3e015e29ebb712dba07c85b3d95b0ee8f54ed0a8ce86d808a718da2e1 at receipts/engine/model_admitted/SFT-CHEM-DEFECT-NONSTOICHIOMETRY-016-a3ffe6b3e015e29e.json; empirical hash
 sha256:1ec54fab0a82b7b69c43c6076dd22bf49d0f6cfa2868c8e65c4fe161ecbf6ddb; measurement receipt
 sha256:263c667c873091304b7bdba6d8f4c929bc64e3a53fc8c44fce863fc581b3ad3c.

173. Exact inorganic acid/base, redox and reaction-network law

Claim identity: SFT-CHEM-INORGANIC-ACID-BASE-REDOX-NETWORK-017

Question and theorem. A complete inorganic reaction network retains every species, Lewis pair/adduct joining, conserved redox transfer and ordered transition path.

complete-finite-species-transition-network__complete-held-species-identity-support__provider-acceptor-two-occurrence-pair-transfer__retained-provider-acceptor-adduct-composition__positive-complete-held-electron-transfer__one-donor-acceptor-conserved-transfer__complete-ordered-transition-path__transition-successor-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-OPERATIONAL-PROCESS-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-OXIDATIVE-ADDITION-REDUCTIVE-ELIMINATION-012

Generated grammar. Generate the literal product of carrier, species, acid-base, adduct, redox, coupling, path and extension alternatives; decide all 256 forms from admitted graph, information conservation, operational process, Lewis, redox, stoichiometry and reaction dependencies.

Boundary: Every finite complete inorganic species-transition network with Lewis pair/adduct and/or redox transfer steps and a complete ordered path; all fifteen frozen Lewis acid, Lewis base, Lewis adduct, oxidation and reduction surfaces.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *complete-finite-species-transition-network__complete-held-species-identity-support__provider-acceptor-two-occurrence-pair-transfer__retained-provider-acceptor-adduct-composition__positive-complete-held-electron-transfer__one-donor-acceptor-conserved-transfer__complete-ordered-transition-path__transition-successor-no-extra-rule*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-finite-species-transition-network	selected-inorganic-equation	One selected equation does not close a reaction network.
species	complete-held-species-identity-support	untracked-reagent-labels	Untracked labels cannot establish conservation.
acid-base	provider-acceptor-two-occurrence-pair-transfer	assumed-acid-and-base-names	Names alone do not establish pair transfer.
adduct	retained-provider-acceptor-adduct-composition	unstructured-acid-base-mixture	A mixture loses the joining record.
redox	positive-complete-held-electron-transfer	signed-oxidation-number-arithmetic	A signed inscription is not a native transfer record.
coupling	one-donor-acceptor-conserved-transfer	independent-oxidation-and-reduction-labels	Independent labels can violate carrier conservation.
path	complete-ordered-transition-path	unordered-reaction-list	An unordered list loses operational composition.
extension	transition-success-or-no-extra-rule	named-reaction-exception	A named exception destroys closure.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256: 42f2fb7327e9b9718efbccd0e5b6bf2c43eec0d8381f7285e4d604a2c50beff8.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 3314f98cb68462b24dc837e35b5cb4eda91046781c14948eb732deec1ac3f037.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 13cdfb4e9ed2a765a33a31901fec864cfce7f4ab41918717fb461aa1c7cca934.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: d4a117393d81d2286a282a2cc3f7e4923730b3cdc51b7d48b8ff67215ea98ac9.

Independent reconstruction. Implementation hash

sha256: b992d5c22ba42e5384bf2d8d7757fff59066458a41b5e0575ffec2d389e9be21; independent certificate sha256: a7f227ab4462eb16956af873be99733802c98e8bcaf38dc5bb7bae9b45a03bec; external-validation hash sha256: 30ebf3d5a131efb37d9fc3c28fa47157ac03bdac606b6a83927b4d55d5c4292c. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-L03508
- IUPAC-L03511
- IUPAC-L03510
- IUPAC-O04362
- IUPAC-R05222

Comparison records:

- Lewis network: three species, one step, two retained pair occurrences and one retained adduct

- redox network: two species, one two-occurrence conserved transfer and exact reverse
- joint successor: five species and two ordered steps
- complete fifteen-row five-source Lewis/adduct/oxidation/reduction vector
- one introduced but unrendered acid example retained as explicit absence
- complete exact target vector sha256:b0343b9905c2624253ce4fd64237dd9d8f7bfbc7acf37e2c80e80f2bb60c539
- control omitted_source_row_rejected: True
- control numerical_zero_rejected: True
- control all_fifteen_target_hashes_bound_postseal: True
- control missing_network_species_rejected: True
- control rendered_acid_example_absence_preserved: True
- control signed_oxidation_number_downstream_only: True
- control sources_not_recaptured: True
- control prediction_contains_no_definition_example_criterion_or_target_payload: True

Falsification boundary: The claim fails if a species, pair occurrence, adduct or redox carrier is omitted; if oxidation and reduction do not share one transfer support; if reversal does not swap the same endpoints and carriers; if path order is incomplete; if signed oxidation numbers enter forcing; if the absent rendered acid example is manufactured or removed; if any of fifteen rows is lost; or if outcomes open before prediction sealing.

Measurement receipt: sha256:c0dad75ce6d7c17f2a298b31d9d54af0f0ac6e9cd0fc4ff3dafd9606559c9277.

Isolation certificate: sha256:6ce33b81c2c3f49d1c38393f7ac69ad3946c72857b297e1a17945a3fc63a7173.

Custody certificate: sha256:f811effc7f1420f424c1e3d4c21686d0829827455d1813d6eea30744ea4e274b.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The inorganic extension derives coordination identity and number, denticity, geometry, isomerism, splitting, spin, colour, magnetism, organometallic support, inverse redox transforms, cluster bonding, local solid organization, defects and coupled inorganic reaction networks. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- numerical zero
- negative irrational imaginary signed continuum fitted free or imported values
- signed oxidation-number arithmetic
- independent oxidation and reduction carrier records
- unstructured acid/base mixture
- observed source definition, named criterion or example structure

Immutable identities. Source manifest

sha256:42c9578674be63d7e02cab5a924fcd9ebdcae3b62c377bfaf866b3abb12fdaa2; derivation seal

sha256:e2d20f935ac430df9bb6080d855baff697b74166b45fd96918f1c76067aa0a2b; engine receipt

sha256:e70073623d4c8b9da5476274af3b012bb74c0e85282d335c5b289bf4f45c17eb at receipts/engine/model_admitted/SFT-CHEM-INORGANIC-ACID-BASE-REDOX-NETWORK-017-e70073623d4c8b9d.json;

empirical hash sha256:efffab803b0c0b4c20cab560d00fb226287ad918ef9d6d66724ab2f8785f45e;

measurement receipt sha256:c0dad75ce6d7c17f2a298b31d9d54af0f0ac6e9cd0fc4ff3dafd9606559c9277.

23. Organic structure, stereochemistry and reaction mechanisms - 16 admitted claims

Conjugation, aromatic recurrence, conformers and every registered reaction family are completely enumerated. This entire registered subcategory is included; no proper subset is presented as field completion.

174. Exact connected alternating conjugated-support law

Claim identity: SFT-CHEM-CONJUGATED-SUPPORT-001

Question and theorem. A conjugated Fold support is one finite connected molecular subcarrier in which every adjacent incidence retains one of the two forced Fold fibres, every successive incidence carries the opposed fibre through a shared atom occurrence, and every atom and incidence is retained. A fresh occurrence has one unique opposed-fibre successor.

one-retained-molecular-subcarrier__complete-connected-adjacency-path__two-forced-held-support-fibres__exact-opposed-adjacent-fibre-recurrence__complete-atom-and-incidence-support__shared-centre-support-propagation__structure-sealed-before-external-observation__opposed-fibre-successor-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001

Generated grammar. Generate the literal product of carrier, connectivity, fibres, alternation, coverage, propagation, observation and extension alternatives; decide all 256 forms from admitted form enforcement, exact graph, distinction, conservation, molecular, bond-order and organic-carrier dependencies.

Boundary: Every finite complete molecular path of at least three distinct atom occurrences and its complete adjacent incidence support; all ten frozen terminology, experimental structure, separated-support control, vibrational and UV-visible surfaces.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *one-retained-molecular-subcarrier__complete-connected-adjacency-path__two-forced-held-support-fibres__exact-opposed-adjacent-fibre-recurrence__complete-atom-and-incidence-support__shared-centre-support-propagation__structure-sealed-before-external-observation__opposed-fibre-successor-no-extra-rule*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-retained-molecular-subcarrier	selected-unbound-bond-marks	Selected marks do not define one molecular subcarrier.
connectivity	complete-connected-adjacency-path	disconnected-support-fragments	Disconnected fragments cannot propagate one support distinction.
fibres	two-forced-held-support-fibres	bond-support-label-erased	Erasure merges the two required support distinctions.
alternation	exact-opposed-adjacent-fibre-recurrence	arbitrary-adjacent-fibre-repetition	Repeated adjacent fibres close the alternating path.
coverage	complete-atom-and-incidence-support	favourable-subpath-only	A selected subpath can hide a failing incidence.
propagation	shared-centre-support-propagation	independent-bond-pairs	Independent pairs do not share a propagation centre.
observation	structure-sealed-before-external-observation	spectrum-or-name-selects-structure	An external name or spectrum cannot select the Fold law.
extension	opposed-fibre-successor-no-extra-rule	named-conjugation-exception	A named exception introduces an unforced rule.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:ea3d597fa8c957906e43bfe880a880a005ce3620609a64ca1c902d8a39f96ca4.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:74e78231b7c9efae1e60628976a62a7a2b903369f65b7fe18e031547d6af3410.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:845971aabc3fa5f9ea9b1fb58b0a105bc7d3be3cd9e7c3b25fe6f7b720e1f3fc.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:f6070dde7827458587172cae6fed94256029c85053a997a0a27ee12f120f27c7.

Independent reconstruction. Implementation hash

sha256:06fc9c2662e145b4f4feb3b72e81917a0cf237d2b295f49809e535896d618ab8; independent certificate sha256:392980e76d317e217839d5b993ffa4a731e1910b49c45f5e39b32ee6617a4dab; external-validation hash sha256:43d8bdf4d13a8e7756e7fe50cf1de92ac924d56d38ec260f583e7d32f5e77c1e. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-C01267
- IUPAC-08786
- IUPAC-D01583
- NIST-CCCBDB-106990-NEUTRAL-EXPERIMENTAL
- NIST-CCCBDB-591935-NEUTRAL-GEOMETRY
- NIST-WEBBOOK-C106990-SPECTRA
- NIST-WEBBOOK-C106990-UVVIS-JCAMP-INDEX-0

Comparison records:

- Fold witness: three atom occurrences, two opposed fibres; successor four atoms and three fibres

- IUPAC surfaces: two independent alternating-single/multiple records plus intervening-bond and nonlocal-support records
- NIST butadiene rCC inscriptions 1.476 and 1.337 angstrom; separated-control inscriptions 1.511 and 1.339 angstrom
- complete NIST vibrational table retained with 26 header/data rows
- complete NIST UV-visible JCAMP vector retained: 502 of 502 points, 200.6193 to 333.0485 nm, external maximum log-epsilon 4.47885
- one V1 uppercase-table parser overrun preserved; two tables corrected in claim-specific V2 without recapture
- complete exact target vector sha256:5003f9d54a3860ef359a8ec17f8c42f7ecfb80c7cd9dfbbebaef0c8e90ce94b1
- control omitted_source_row_rejected: True
- control numerical_zero_rejected: True
- control all_ten_target_hashes_bound_postseal: True
- control repeated_adjacent_fibre_rejected: True
- control incomplete_incidence_rejected: True
- control duplicate_occurrence_rejected: True
- control separated_control_not_classified_from_bond_counts_alone: True
- control external_signed_control_inscription_downstream_and_preserved: True
- control v1_parser_overrun_preserved_and_corrected_separately: True
- control sources_not_recaptured: True
- control prediction_contains_no_definition_coordinate_spectrum_or_target_payload: True

Falsification boundary: The claim fails if its path is disconnected; if any atom, incidence or support fibre is omitted; if adjacent fibres repeat; if an external name, definition, coordinate or spectrum selects the survivor; if the separated-double-bond control is misclassified from bond counts alone; if any of ten rows or any of 502 UV-visible points is lost; if the external signed control inscription is erased or made native; if the V1 parser overrun is hidden; or if outcomes open before the value-free prediction seal.

Measurement receipt: sha256:9724d1d1bdec3603477a974f7de8fa6bfee4c28c2d9f816b984f709a493d2a79.

Isolation certificate: sha256:a808b3b1eedd3674ca4f4ce06e88952fbaab01b8aa8c70bd8d6c9dde00b85d52.

Custody certificate: sha256:ea733a7109e3a1072b88db4327bc34ef58658a26b03db5473d3305e793b2f226.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The organic extension derives conjugation, resonance, aromatic recurrence and its measured stability excess, antiaromatic/nonaromatic distinction, conformer classes and populations, substitution, addition, elimination and composition-retaining rearrangement with complete adverse evidence. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; structural absence is EmptyOne
- no negative irrational imaginary signed continuum fitted free or imported parameter
- no conventional single/double-bond notation used to select the two Fold fibres
- no resonance notation or electron-count rule imported into forcing
- no selected favourable subpath or omitted incidence
- no definition, molecular name, coordinate, vibration or spectrum opened to select the law
- the preserved V1 parser overrun cannot be hidden or treated as evidence

Immutable identities. Source manifest

sha256:746be20ff4d5d25b250d7b1249597fb8f7398fc9e9f8b12cce6dc4de3d8a2c21; derivation seal sha256:8b8ea1ac27e8179b9c5f6c343235279d6a2240d59f287d9c995c06bb1a77cbe2; engine receipt sha256:a125efec090d44d39f62cb3e59e7cc2a0cc9bab6174c74d4f4b654b3d63fd602 at receipts/engine/model_admitted/SFT-CHEM-CONJUGATED-SUPPORT-001-a125efec090d44d3.json; empirical hash sha256:a74eb5c2c98c7b303d5082461d83e1050792cdb9527b7a761057c4c2cad98c2c; measurement receipt sha256:9724d1d1bdec3603477a974f7de8fa6bfee4c28c2d9f816b984f709a493d2a79.

175. Exact one-carrier multiple-representation equivalence law

Claim identity: SFT-CHEM-RESONANCE-EQUIVALENT-REPRESENTATION-002

Question and theorem. A Fold representation-equivalence class retains exactly one molecular carrier, every atom occurrence and every adjacency while retaining at least two distinct encoding identities. In the minimal generated pair, every support incidence in the second encoding is the exact opposed Fold-fibre complement of the first; this relation creates neither a second species nor an equilibrium process.

one-retained-molecular-carrier__complete-equal-atom-occurrence-support__complete-equal-adjacency-support__multiple-distinct-encoding-identities__complete-opposed-fibre-complement__one-carrier-many-representations__representation-relation-only__shared-complement-successor-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-ELECTRONIC-STATE-IDENTITY-001
- SFT-CHEM-CONJUGATED-SUPPORT-001

Generated grammar. Generate the literal product of carrier, atom-support, adjacency, encoding-number, fibre-relation, identity, process and extension alternatives; decide all 256 forms from admitted form enforcement, exact graph, distinction, conservation, molecular identity, bond-order, electronic-state and conjugated-support dependencies.

Boundary: Every finite complete connected alternating molecular representation pair related by global Fold-fibre complement. Larger finite representation classes are generated by composition of exact local pair relations while retaining the same carrier, atoms and adjacency. The empirical boundary is the four complete preregistered IUPAC records.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *one-retained-molecular-carrier__complete-equal-atom-occurrence-support__complete-equal-adjacency-support__multiple-distinct-encoding-identities__complete-opposed-fibre-complement__one-carrier-many-representations__representation-relation-only__shared-complement-successor-no-extra-rule*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-retained-molecular-carrier	different-molecular-carriers	Different carriers are different molecular identities.
atoms	complete-equal-atom-occurrence-support	atom-occurrence-change	Adding, deleting or replacing an occurrence changes the carrier description.
adjacency	complete-equal-adjacency-support	connectivity-change	Changed adjacency is a distinct constitution, not an encoding alternative.
encodings	multiple-distinct-encoding-identities	one-encoding-only	One inscription supplies no representation equivalence class.
fibres	complete-opposed-fibre-complement	partial-or-arbitrary-label-change	An arbitrary partial change does not preserve the generated Fold relation.
identity	one-carrier-many-representations	representations-counted-as-species	Formal encodings do not create new chemical carriers.
process	representation-relation-only	equilibrium-or-transition-imported	A representation relation does not imply a physical interconversion process.
extension	shared-complement-successor-no-extra-rule	named-resonance-exception	A named exception adds an unforced classification rule.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:152162b54a4c8d977d699a0727f97e05ddfe6383ae89e9385f8f1ac8ea80a636.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:e7a2b5b60dd12d089a291d8d0af1e75cbc0c6116bd921fa18a4144081275c0a5.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:f34ad0563fb74648d861b946cd0252f5c67be84342326313699eff366115dfdb.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:d70fc47c99e2a1d6c52e708668603ef0449c4b6a628e65ba004527696bc97e51.

Independent reconstruction. Implementation hash

sha256:672828073e0b7df2d72f2a558f361ffc783b9bab9dd3db199f0c0abedde8259; independent certificate sha256:0b2813f8a627ccd0bf81541b882c0b6f16554c4a6f2b00c6c1fc09d834d47f53; external-validation hash sha256:3b0d8ce14a52ee321bbb0f8be59bc397fda2ba249b4980678c1031a99610452a. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-R05326
- IUPAC-08205
- IUPAC-C01309
- IUPAC-D01583

Comparison records:

- Fold witness: one carrier, three atom occurrences, two incidences and two exact complementary encodings
- successor: four atom occurrences and three incidences with complete prior prefixes and complement retained
- IUPAC resonance surface: one molecular entity represented through contributing structures
- IUPAC resonance-form surface: at least two formal structures where one structure is insufficient

- IUPAC contributing-structure surface: formal encodings are not separate species
- IUPAC resonance-form control: representation arrow is not an equilibrium arrow
- IUPAC delocalization surface: support is not localized between two atoms
- external wavefunction, coefficient and signed-charge inscriptions preserved downstream only
- one V1 incomplete record-search result preserved; corrected across all four records in claim-specific V2 without recapture
- complete exact target vector sha256:dd8eb366ccd5e0095d5831392f20b7a0a89464e1b7b3b8dda73a5979fb3ce90b
- control omitted_source_row_rejected: True
- control numerical_zero_rejected: True
- control all_four_target_hashes_bound_postseal: True
- control different_carrier_rejected: True
- control changed_adjacency_rejected: True
- control partial_noncomplement_rejected: True
- control identical_encoding_rejected: True
- control external_wavefunction_coefficient_and_charge_downstream_and_preserved: True
- control v1_search_scope_error_preserved_and_corrected_separately: True
- control sources_not_recaptured: True
- control prediction_contains_no_definition_note_example_wavefunction_coefficient_charge_or_payload: True

Falsification boundary: The claim fails if equivalent encodings change the carrier, atom occurrences or adjacency; if the minimal pair is not a complete opposed-fibre complement; if formal encodings are counted as multiple species or an equilibrium; if any of four complete records is omitted; if external wavefunction, coefficient or signed-charge language enters native forcing; if the V1 search-scope error is hidden; or if outcomes open before the value-free prediction seal.

Measurement receipt: sha256:d757d819b42fd630ef6e26cd9c88ede628de9b7dd0c9cbd9594ef5897a9ea219.
Isolation certificate: sha256:5e2b842f795de670fd2cd3705905da6aaee6c5aeeb0c6916c0a8a4180980e65d.
Custody certificate: sha256:ef6c80388da8496e9b36c81806d2cd8ff9452cbdb8ff2006ce1e6270e4e779ed.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The organic extension derives conjugation, resonance, aromatic recurrence and its measured stability excess, antiaromatic/nonaromatic distinction, conformer classes and populations, substitution, addition, elimination and composition-retaining rearrangement with complete adverse evidence. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; structural absence is EmptyOne
- no negative irrational imaginary signed continuum fitted free or imported parameter
- no Lewis bond mark charge coefficient wavefunction or named resonance rule used to select the survivor
- no physical interconversion equilibrium time or stochastic mixing imported into representation equivalence
- no target definition note example or outcome opened to select the law
- the preserved V1 incomplete search-scope result cannot be hidden or treated as external absence

Immutable identities. Source manifest

sha256:343681d313e6d4b0f96b085441de153fef5d85b68f9e2a7f31e398eab79e970f; derivation seal

sha256:8cfbbe91a0942f12e4f1e489aaf07f7b9170c536afca92b2ec6ad90c024292be; engine receipt sha256:f2c0e4ec6a2c1223b27e05831466da02ceb92ff44b121759fa09ff673c96222d at receipts/engine/model_admitted/SFT-CHEM-RESONANCE-EQUIVALENT-REPRESENTATION-002-f2c0e4ec6a2c1223.json; empirical hash sha256:160d4a199dfb2dd6f67ff1ca174da9504f6cdf5728ce464446a6f907d7adeb1d; measurement receipt sha256:d757d819b42fd630ef6e26cd9c88ede628de9b7dd0c9cbd9594ef5897a9ea219.

176. Exact aromatic recurrence, support sequence and positive stability law

Claim identity: SFT-CHEM-AROMATIC-RECURRENCE-STABILITY-003

Question and theorem. An aromatic Fold carrier is one complete generated molecular cycle retaining both Fold boundary fibres and at least one complete layer of their four ordered pair cells. The primitive support count is six; every lawful successor adds the same four-cell layer, forcing ten, fourteen and every later positive successor. Opening the complete recurrence requires one retained positive transfer, so the closed recurrence precedes its opened localized reference without an imported electron-count or measured-energy rule.

one-complete-molecular-carrier__complete-generated-cycle__complete-two-fibre-boundary__complete-four-ordered-pair-cell-layer__explicit-complete-first-return-trace__positive-recurrence-opening-gap__value-free-seal-and-complete-blind-vector__complete-four-cell-successor-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-CHEM-MULTICENTRE-DELOCALIZED-SUPPORT-008
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-MOLECULAR-FORMATION-ENERGY-013
- SFT-CHEM-CONJUGATED-SUPPORT-001
- SFT-CHEM-RESONANCE-EQUIVALENT-REPRESENTATION-002

Generated grammar. Generate the literal product of carrier, cycle, Fold boundary, pair-cell layer, return trace, stability order, observation custody and extension alternatives; decide all 256 forms solely from admitted Fold assembly, exact graph, finite dynamics, generator three, multicentre support, energy order, formation-energy accounting, conjugation and representation-equivalence dependencies.

Boundary: Every positive finite complete molecular cycle; both Fold boundary fibres; every positive number of complete four-ordered-pair-cell layers; explicit first-return traces; and the complete nine-target authority surface containing six development-observed correspondence records and three separately outcome-unopened blind NIST CCCBDB experimental-data pages.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: one-complete-molecular-carrier__complete-generated-cycle__complete-two-fibre-boundary__complete-four-ordered-pair-cell-layer__explicit-complete-first-return-trace__positive-recurrence-opening-gap__value-free-seal-and-complete-blind-vector__complete-four-cell-successor-no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-complete-molecular-carrier	aromatic-name-or-answer-only	A name does not retain the molecular support that is claimed to recur.
cycle	complete-generated-cycle	open-path-or-selected-ring	An open path or selected example does not establish cyclic first return.
boundary	complete-two-fibre-boundary	one-fibre-or-duplicated-boundary	An incomplete or duplicated boundary erases one forced Fold distinction.
layer	complete-four-ordered-pair-cell-layer	selected-transition-cell-subset	A selected subset can be tuned to a familiar count and is not closed under pair-cell perturbation.
return	explicit-complete-first-return-trace	period-label-without-trace	A named cycle length cannot prove return or preserve its path.
stability	positive-recurrence-opening-gap	measured-energy-or-name-selected-order	A measured energy or conventional aromatic label cannot select the structural order.
observation	value-free-seal-and-complete-blind-vector	selected-or-preopened-value-row	A selected or preopened value vector can feed the desired order back into the law.
extension	complete-four-cell-successor-no-extra-rule	imported-electron-count-or-species-rule	An imported count formula or species exception is an added selector.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:935f9aa7ca5faacd45d5e339e379aeb1bc6564e988e570f8ec8034bab647703e.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:c45c54d01082611072335278b23e29f9c731e980a73b00b614ccc626d7f62003.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:211dcf37e2fc914bdbb435f631635b8b7ee7e5257befb1dadba7c3e02df35ccb.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:bf4a86f8a63e2725408da8b6a9afbceebe1a70aff5f97355febb97527cbbffcf.

Independent reconstruction. Implementation hash

sha256:81177b27e06e5986f12963e391e2514fafa3172ca1c0ceb6aa3dcb45c6373823; independent certificate sha256:a8e280705ec87491bbcd9ab35f0cec2568091b021f633433cb10500130f8a4f0; external-validation hash sha256:9544092f0a992b95095056dd0532ca855fc2866a4b3cebdff395dc8595ea0a0cc. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-A00441
- IUPAC-A00442
- IUPAC-R05333
- NIST-WEBBOOK-C71432-THERMOCHEMISTRY
- NIST-WEBBOOK-C110838-THERMOCHEMISTRY
- NIST-WEBBOOK-C110827-THERMOCHEMISTRY
- NIST-CCCBDB-SRD101-C71432-COMPLETE-EXPERIMENTAL-DATA

- NIST-CCCBDB-SRD101-C110838-COMPLETE-EXPERIMENTAL-DATA
- NIST-CCCBDB-SRD101-C110827-COMPLETE-EXPERIMENTAL-DATA

Comparison records:

- Fold recurrence support sequence forced without imported electron-count rule: 6, 10, 14
- complete authority surface: nine sources; six transparently development-observed and three independently outcome-unopened before seal
- blind CCCBDB Hfg(298.15 K) external strings: benzene 82.93, cyclohexene -4.32, cyclohexane -123.14 kJ mol⁻¹
- blind exact transfer magnitudes: isolated 118.82, cyclic threefold 206.07, localized threefold reference 356.46 kJ mol⁻¹
- blind recurrence-stability excess: 150.39 kJ mol⁻¹; conservative uncertainty 6.60; positive lower envelope 143.79
- complete blind CCCBDB surface retained: 59 scientific tables and 353 rows
- complete development WebBook surface retained: 9 thermochemistry tables and 121 rows
- complete exact target vector sha256:506ad2097630d820a2302d69a68c70b757844c24159a486bb25731e882589a12
- control omitted_source_row_rejected: True
- control numerical_zero_rejected: True
- control all_nine_target_hashes_bound_postseal: True
- control incomplete_pair_cell_layer_rejected: True
- control duplicated_boundary_fibre_rejected: True
- control open_two_centre_cycle_rejected: True
- control development_observed_rows_not_mislabelled_blind: True
- control three_independent_outcomes_opened_only_after_seal: True
- control blind_recurrence_stability_excess_clears_conservative_envelope: True
- control external_signed_and_absent_inscriptions_downstream_and_preserved: True
- control blind_sources_not_recaptured: True
- control prediction_contains_no_huckel_electron_count_enthalpy_value_sign_uncertainty_or_payload: True

Falsification boundary: The claim fails if the carrier or cycle is incomplete; if either Fold boundary fibre or any ordered pair cell is missing; if the primitive/successor support counts differ from six, ten and fourteen; if recurrence lacks an explicit first return; if opening it requires no positive transfer; if a measured value or imported electron-count rule selects the law; if any of nine records or any complete scientific table row is omitted; if the blind three-source vector was opened before sealing; if its exact localized threefold transfer does not exceed the cyclic transfer by a positive amount clearing the conservative uncertainty envelope; or if signed and absent external inscriptions are erased or treated as native numbers.

Measurement receipt: sha256:3a55eefbd981f0b771bcf3a7fd6910fa27f0cb88f9d58dbfd55984e804b5001b.
Isolation certificate: sha256:91344eeb9389f8aed8634e03f3c143c758d5cf3ec52a920b8ddc0a388c4e9dba.
Custody certificate: sha256:aba70a073c9555c47907c13250d134e3035e702d62dc0576b8751584e78acfd5.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The organic extension derives conjugation, resonance, aromatic recurrence and its measured stability excess, antiaromatic/nonaromatic distinction, conformer classes and populations, substitution, addition, elimination and composition-retaining rearrangement with complete adverse evidence. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; structural absence is EmptyOne
- no negative irrational imaginary continuum fitted free or imported native parameter
- no Hückel $4n+2$ rule electron count molecular name or measured energy used to select the survivor
- no selected ordered-pair-cell subset or omitted cycle incidence
- the six inspected family records are development-observed correspondence and cannot be called blind
- the three independent CCCBDB outcomes remain inaccessible until after the value-free law and target seals
- external signed enthalpy strings remain downstream held inscriptions and never become native signed arithmetic

Immutable identities. Source manifest

sha256:5adce4d4decdf695a59074bfbcb825603f87d8742ce44a737c1e3345276042ab2; derivation seal
 sha256:22326aba7c2f2f4505f2a7718f5253cf35b7f4c9b12b959ef802c2c04463d765; engine receipt
 sha256:5339c34ad1f819bbefa0ebed15404358964ec90488544bdc060c5feff9ea9b28 at receipts/engine/
 model_admitted/SFT-CHEM-AROMATIC-RECURRENCE-STABILITY-003-5339c34ad1f819bb.json;
 empirical hash sha256:b989b7c7cd47248a8c09860be10992193e6815dd1bf9be9af1c2aeddeb0d61a2;
 measurement receipt sha256:3a55eefbd981f0b771bcf3a7fd6910fa27f0cb88f9d58dbfd55984e804b5001b.

177. Exact aromatic, antiaromatic and nonaromatic distinction law

Claim identity: SFT-CHEM-ANTIAROMATIC-NONAROMATIC-DISTINCTION-004

Question and theorem. For one retained molecular cycle, the generated Fold recurrence has exactly three structural classes: complete planar closure, a planar complete support with frustrated return, or a structurally broken plane/conjugation with EmptyOne recurrence. Their primitive supports are six, four and EmptyOne respectively; the complete classes append all four ordered pair cells at each successor, and two positive distinction transfers force the closed, broken, frustrated stability order without an imported electron-count or measured-energy rule.

one-retained-same-cycle-carrier__complete-three-class-census__held-planar-or-broken-geometry__held-complete-or-broken-conjugation__closed-frustrated-or-EmptyOne-return__positive-two-step-stability-order__value-free-complete-comparative-seal__complete-four-cell-successors-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-FOLD-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-CHEM-MULTICENTRE-DELOCALIZED-SUPPORT-008
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-CONJUGATED-SUPPORT-001
- SFT-CHEM-AROMATIC-RECURRENCE-STABILITY-003

Generated grammar. Generate the literal product of carrier, class census, geometry, support, recurrence boundary, stability order, observation custody and extension alternatives; decide all 256 forms solely from the admitted Fold assembly, exact arithmetic, graph, order, dynamics, multicentre support, state-energy, conjugation and aromatic-recurrence dependencies.

Boundary: Every positive finite complete molecular cycle on one retained carrier; both held geometry states; both held conjugation states; complete two-fibre closure, complete four-cell frustrated return and structural EmptyOne after an explicit break; all three same-cycle classes; every positive complete four-cell successor; and the complete five-target authority surface.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: one-retained-same-cycle-carrier__complete-three-class-census__held-planar-or-broken-geometry__held-complete-or-broken-conjugation__closed-frustrated-or-EmptyOne-return__positive-two-step-stability-order__value-free-complete-comparative-seal__complete-four-cell-successors-no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-retained-same-cycle-carrier	different-or-name-only-carriers	Different carriers cannot establish alternatives of the same cycle.
census	complete-three-class-census	selected-favourable-category	Selecting a familiar label omits lawful same-cycle alternatives.
geometry	held-planar-or-broken-geometry	planarity-assumed-or-erased	Erasing geometry merges a frustrated planar recurrence with a recurrence avoided by an out-of-plane break.
support	held-complete-or-broken-conjugation	conjugation-name-only	A name does not distinguish complete support from a broken recurrence path.
boundary	closed-frustrated-or-EmptyOne-return	electron-count-label-imported	An imported count label does not derive return, frustration or structural absence.
order	positive-two-step-stability-order	measured-energy-selected-order	Measured values cannot choose which structural alternative is stable.
observation	value-free-complete-comparative-seal	selected-or-preopened-comparison	A selected outcome can conceal an adverse same-cycle alternative.
extension	complete-four-cell-successors-no-extra-rule	named-ring-or-extra-count-rule	A species exception or imported count recurrence adds a free selector.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:d176c1387ce23303fbbc8c6dc07f3b393a745b7c7a55169a80b29a3c1708e948.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:609c69cb835989c3d24ef6d5a2325751b7385ea118dad35a8aafd1601bc464b6.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:94f61dfe71b7ed0a238246258bd85860d3fff22a00cbce57fca5cd2d2c3de1d1.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:ee8e18877e321c36d2291e4a72de93eb8bea3ba740c9de24a3bece1fbb5e2554.

Independent reconstruction. Implementation hash

sha256:8385e986613b79d8ca1201092e25822e271727daeec668b9c3e3ed20de0efaea; independent certificate sha256:c29b2a028e65eefca73dbcd797387080e288e75d999674d418224749164edc25; external-validation hash sha256:744b63a4e21be0cfa846f2dee9582f4175eadb2671719ac9d64efffa657cff72. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-A00441
- IUPAC-AT06987

- NIST-WEBBOOK-C71432-THERMOCHEMISTRY
- NIST-CCCBDB-1120532-NEUTRAL-EXPERIMENTAL
- NIST-CCCBDB-629209-NEUTRAL-EXPERIMENTAL

Comparison records:

- Fold same-cycle base supports forced without imported electron-count rule: closed 6, broken structural EmptyOne, frustrated 4
- Fold same-cycle successors: closed 10, broken structural EmptyOne, frustrated 8
- complete authority surface: five sources; three development-observed and two independently outcome-unopened before seal
- blind structure vector: cyclobutadiene D2H true minimum and D4h false-square control; cyclooctatetraene D2D with 1.337/1.470 angstrom bond alternation and opposed z-coordinate signs
- energy vector: cyclobutadiene ionization 8.160 +/- 0.030 eV; its absent formation-enthalpy row preserved without fabrication
- energy vector: benzene 82.93 +/- 0.50 and cyclooctatetraene 297.60 +/- 1.40 kJ mol⁻¹ formation enthalpy
- exact repeated-CH-unit formation-enthalpy gap 14027/600 kJ mol⁻¹; uncertainty 31/120; positive lower gap 578/25
- complete blind CCCBDB surface retained: 36 scientific tables and 172 rows; development WebBook surface: 3 tables and 54 rows
- complete exact target vector sha256:6668de6da30756b1bd5ec4fb2e7ebee0c8ca7f206cb03d2e17a7a45282385828
- control omitted_source_row_rejected: True
- control numerical_zero_rejected_by_fold_program: True
- control all_five_target_hashes_bound_postseal: True
- control inconsistent_antiaromatic_break_rejected: True
- control missing_nonaromatic_break_rejected: True
- control development_observed_rows_not_mislabelled_blind: True
- control two_independent_outcomes_opened_only_after_seal: True
- control complete_blind_structure_vector_present: True
- control positive_repeated_unit_energy_gap_clears_uncertainty: True
- control absent_cyclobutadiene_hfg_preserved_not_fabricated: True
- control external_signed_zero_and_absent_inscriptions_downstream: True
- control prediction_contains_no_class_name_geometry_energy_value_sign_uncertainty_or_payload: True

Falsification boundary: The claim fails if the same molecular carrier or cycle is changed; if any of the three classes is omitted; if complete closure, complete frustration or structural EmptyOne is relabelled; if the primitive supports differ from six, four and EmptyOne or the successors from ten, eight and EmptyOne; if the two positive order steps disappear; if an electron-count, named species or measured value selects the law; if any of five source surfaces or any scientific row is omitted; if either blind outcome was opened before sealing; if the returned structural and energy vector does not preserve planar distorted antiaromatic, nonplanar bond-alternating nonaromatic and the positive aromatic/nonaromatic repeated-unit energy separation; or if the absent cyclobutadiene formation-enthalpy row or any signed/zero external inscription is erased or fabricated.

Measurement receipt: sha256:10b67a6107bb8e767c33f13478c868fb1032e7954c5927fa1caa3139b281a6c7.
Isolation certificate: sha256:53c26ac779d8ff4270424aeead2c2a35d80615068c0f7f9ab8ef9036aeef6c15.
Custody certificate: sha256:05d85ad88760978a9609e820a08e409f4914d0fac30e942f47b5ecd56d4fa186.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The organic extension derives conjugation, resonance, aromatic recurrence and its measured stability excess, antiaromatic/nonaromatic distinction, conformer classes and populations, substitution, addition, elimination and composition-retaining rearrangement with complete adverse evidence. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; structural absence is EmptyOne
- no negative irrational imaginary continuum fitted free or imported native parameter
- no Hückel $4n+2$ or $4n$ premise electron-count lookup molecular-name selector or measured energy in survivor selection
- no omission of the aromatic antiaromatic or nonaromatic same-cycle alternative
- no relabelling a broken plane or conjugation as complete frustrated recurrence
- three inspected records remain development-observed correspondence and are not called blind
- two NIST CCCBDB outcomes remain inaccessible until after the law and value-free prediction seal
- external signed decimal zero and absent inscriptions remain downstream records and never native proof arithmetic

Immutable identities. Source manifest

sha256:1794864551fd4234178fcce230fc1340e1f70d70ec411e63bce9ade75b444cc8; derivation seal
 sha256:62bd9b8fb6a429399457107249cc2f82a5ee601b4aaf31d6cdaaee58a92decdf; engine receipt
 sha256:3ba25097797fbac99c2575eb5554d30f6a475c4f74116b452716ba04c802953c at receipts/engine/model_admitted/SFT-CHEM-ANTIAROMATIC-NONAROMATIC-DISTINCTION-004-3ba25097797fbac9.json; empirical hash sha256:4f7130196b3eda24b12c369c159c0c5705179e62c314f09ad9b99e1ff40096db;
 measurement receipt sha256:10b67a6107bb8e767c33f13478c868fb1032e7954c5927fa1caa3139b281a6c7.

178. Exact finite conformer generation and equivalence law

Claim identity: SFT-CHEM-CONFORMER-GENERATION-EQUIVALENCE-005

Question and theorem. For any positive finite connected molecular graph with a complete ordered rotor census and finite held torsion alphabets, every conformer assignment is generated exactly once by the Cartesian state product. Exact equivalence is the orbit relation of every exhaustively generated atom-type and bond-preserving graph automorphism; the disjoint orbit quotient is the complete conformer census, without coordinate tolerance, measured-energy selection or species rule.

complete-finite-molecular-graph__complete-ordered-rotor-census__complete-cartesian-state-generation__exhaustive-graph-automorphism-action__exact-automorphism-orbit-equivalence__complete-disjoint-orbit-quotient__value-free-operational-census-seal__finite-product-successor-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-FOLD-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001

- SFT-CHEM-CONFIGURATION-ORDER-PATH-011
- SFT-CHEM-DIHEDRAL-TORSIONAL-STATE-004

Generated grammar. Generate the literal product of carrier, rotor, state generation, symmetry, equivalence, quotient, observation and extension alternatives; decide all 256 forms solely from admitted exact finite graph, combinatoric, order, process, molecular, isomeric and torsional-state dependencies.

Boundary: Every positive finite connected atom-labelled graph; every complete ordered four-site rotor; each positive finite held state alphabet with exact reversal involution; all Cartesian assignments; every atom-type and bond-preserving position bijection; all induced rotor actions; the complete disjoint orbit quotient; and the four frozen development-observed authority records.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-finite-molecular-graph__complete-ordered-rotor-census__complete-cartesian-state-generation__exhaustive-graph-automorphism-action__exact-automorphism-orbit-equivalence__complete-disjoint-orbit-quotient__value-free-operational-census-seal__finite-product-successor-no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-finite-molecular-graph	name-or-coordinate-list	A name or selected coordinates omit molecular identity and adjacency.
rotor	complete-ordered-rotor-census	selected-single-angle	One selected angle omits other rotors and held states.
generation	complete-cartesian-state-generation	energy-picked-conformer	Measured energy cannot choose which assignments are generated.
symmetry	exhaustive-graph-automorphism-action	named-symmetry-assumption	A named symmetry can omit or add an equivalence action.
equivalence	exact-automorphism-orbit-equivalence	coordinate-tolerance-clustering	A tolerance or continuum metric introduces a fitted boundary.
quotient	complete-disjoint-orbit-quotient	selected-representative-list	A selected list can duplicate or omit lawful conformers.
observation	value-free-operational-census-seal	source-readable-generator	External conformer names or energies could select the quotient.
extension	finite-product-successor-no-extra-rule	species-exception-or-extra-rule	A species exception cannot establish a general finite algorithm.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:0611fbeb251d4f59ac0a173fc58f185aa81e89a43f6b30934c85b60e5d1c27ecc`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3e179ed0ed078768bc5962ef15eb9d7443f645bddb2961a63d541c47e6eeae6b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e62ca0b5c72d0d97e65f461a694390f91af76bba9a0bfd03f1782a02b951ef70`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:79d8c3d0dd9e24447d07871d00fbb2414c471de8c4e98b4893075e6d921b5c94`.

Independent reconstruction. Implementation hash

`sha256:3d1075bd610a2dfef31f171ba23d69b3b0e5329462aa9354aade3e46322ec7cf`; independent certificate `sha256:99fcd5d2c894d404b986a962b036bc5bb9169efa2207efc009380eaaeaaefae`; external-validation hash `sha256:ab61840d76da04924a650f333aa327e5341b356d11651d1646990acdfed039a`. The independent

program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-C01262
- IUPAC-C01258
- IUPAC-C01259
- NIST-CCCBDB-106978-NEUTRAL-EXPERIMENTAL

Comparison records:

- four-site Fold census: three raw held assignments; two complete graph automorphisms; two disjoint classes of sizes one and two
- all four external authority records disclosed as development-observed and not claimed blind
- complete external butane class vector: Anti and Gauche; adverse Gauche false-minimum row retained
- complete NIST CCCBDB surface retained: 19 scientific tables and 105 rows
- external internal-rotation barrier inscription 16.6 kJ mol^{-1} preserved downstream and not used by the generator
- complete exact target vector sha256:7a5c28d75f4b64c160ff172e97e96c164ce5fd251af0576e324547fa33d3f322
- control omitted_source_rejected: `True`
- control numerical_zero_rejected: `True`
- control all_four_target_hashes_bound_postseal: `True`
- control development_observed_not_mislabelled_blind: `True`
- control complete_assignment_partition: `True`
- control complete_graph_action_census: `True`
- control incomplete_reversal_rejected: `True`
- control adverse_gauche_false_row_preserved: `True`
- control all_external_inscriptions_preserved: `True`
- control prediction_contains_no_returned_conformer_energy_or_payload: `True`

Falsification boundary: The claim fails if an atom, bond, rotor, held state, Cartesian assignment or preserving graph bijection is omitted; if equivalence uses a coordinate tolerance or measured energy; if any assignment occurs in no class or more than one class; if the four-site witness does not generate three assignments, two graph actions and two classes of sizes one and two; if any of four authority records or any of the 19 tables and 105 CCCBDB rows is omitted; if Anti/Gauche is selected rather than compared after derivation; or if the adverse Gauche false-minimum row, signed values, conventional zeros or absent cells are erased.

Measurement receipt: sha256:b38a34852427957ef59c255fcbedb4717c071ef2bf5c6046729e19e1198fe640.

Isolation certificate: sha256:3a460c31d6f7753ec613a0c1dc4e917cb3f5666fce9b7d4a12830e6f35dc0e89.

Custody certificate: sha256:051ef7aba3d63bf5dda1c8d51a4286d8cd7306bccdeb333067a4e59c8d40982f.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The organic extension derives conjugation, resonance, aromatic recurrence and its measured stability excess, antiaromatic/nonaromatic distinction, conformer classes and populations, substitution, addition, elimination and composition-retaining rearrangement with complete adverse evidence. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero; structural absence is EmptyOne
- no negative irrational imaginary continuum tolerance fitted free or imported native parameter
- no measured energy molecular name coordinate library or selected conformer used to generate or quotient assignments
- no omitted graph automorphism rotor state or adverse external row
- all four external records are disclosed development-observed correspondence and are not called blind
- external signed decimal zero and absent inscriptions remain downstream records and never native arithmetic

Immutable identities. Source manifest

sha256:746660d0bb993dc4d5631870db630ce9399fef236256ea2b246c2a8e405ba28a; derivation seal sha256:50efb123f2af0321157392bd58d9e05caa64d2ff4c4511475dd4b7fd243faebb; engine receipt sha256:9731b0ba4ed8c8186e8ba78f051da4a84543f8006fd7e34d9f8351c7f03a03f4 at receipts/engine/model_admitted/SFT-CHEM-CONFORMER-GENERATION-EQUIVALENCE-005-9731b0ba4ed8c818.json; empirical hash sha256:7918ad77f8a3b91dcbb79fe0544c345a8fead17177e5130074128455848c0604; measurement receipt sha256:b38a34852427957ef59c255fcbddb4717c071ef2bf5c6046729e19e1198fe640.

179. Exact conditioned conformer population and ordering law

Claim identity: SFT-CHEM-CONFORMER-POPULATION-ORDERING-006

Question and theorem. For every complete finite conformer quotient, retained observation condition and positive finite trace, each conformer population is exactly its deterministic class-occurrence count divided by the complete trace boundary. An unobserved class is structural EmptyOne. Relative energy order uses structural least state plus positive exact Take, while every condition and observation-timescale remains retained.

complete-conformer-equivalence-census__held-observation-condition__positive-finite-observation-boundary__exact-occurrence-count-ratio__unobserved-class-EmptyOne__positive-Take-energy-and-count-orders__value-free-conditioned-vector-seal__one-observation-successor-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-DIHEDRAL-TORSIONAL-STATE-004
- SFT-CHEM-CONFORMER-GENERATION-EQUIVALENCE-005

Generated grammar. Generate the literal product of carrier, condition, finite boundary, population, absence, order, observation and extension alternatives; decide all 256 forms solely from admitted Fold and ORG-005 dependencies.

Boundary: Every positive finite conformer equivalence census; every positive finite deterministic observation trace; one held condition and timescale; exact occurrence fractions; structural EmptyOne for no occurrence; positive relative-energy magnitudes and ordered Take; all fourteen frozen external target identities and every preserved response.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-conformer-equivalence-census__held-observation-condition__positive-finite-observation-boundary__exact-recurrence-count-ratio__unobserved-class-EmptyOne__positive-Take-energy-and-count-orders__value-free-conditioned-vector-seal__one-observation-successor-no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-conformer-equivalence-census	selected-conformer-name	A selected name omits the complete derived equivalence quotient.
condition	held-observation-condition	condition-erased-population	Population without physical condition conflates distinct deterministic traces.
boundary	positive-finite-observation-boundary	infinite-or-unspecified-average	An unspecified limit imports continuum or unbounded averaging.
population	exact-recurrence-count-ratio	random-or-fitted-probability	A random law or fitted distribution is not forced by the observed process.
absence	unobserved-class-EmptyOne	numerical-zero-population	Numerical zero is not a native Fold form.
order	positive-Take-energy-and-count-orders	signed-energy-difference-or-imported-distribution	A signed difference or imported exponential law adds forbidden arithmetic.
observation	value-free-conditioned-vector-seal	target-readable-condition-or-value	External temperature, duration, energy or population could select the law.
extension	one-observation-successor-no-extra-rule	recomputed-history-or-extra-rule	Discarding history or adding a smoothing rule changes the declared population.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : dbf00b2e14e0985a395b48ad5fc42c403396c1f613947d332d25d9fc211a9833.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 4d0481e2ae084c31d19fb511582b3190512a6eea6815af97bd20ffe64904ed92.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : f0d62f745b0104e1ce0a08a41b230efd8c7f1d9a16db9ab4350bb164935d9a5d.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : f27d69679274e9b257e03a0bcda6f6d88227b1629ab288d84681d0d3134a3177d.

Independent reconstruction. Implementation hash

sha256 : 993a7b032c6a9d724724224c0f2ab33c6bd5afcc55aedc1eb0dbbf62af7e7af8; independent certificate sha256 : 97bf02c9ce497fc631a26a49bf02e3da9c7e0a4776089d1efa68639356e2a935; external-validation hash sha256 : 5d00dcb14c560374d90728f4f26d4ba8d703dd58b51b1371bce874ac6dbce86e. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-C01262
- IUPAC-C01259
- IUPAC-CT01038

- NIST-CCCBDB-106978-NEUTRAL-EXPERIMENTAL
- NIST-CCCBDB-106978-INTERNAL-ROTATION
- NIST-CCCBDB-109660-INTERNAL-ROTATION-1
- OSTI-22415852-PRIMARY-RECORD
- OSTI-22415852-PUBLIC-FULLTEXT-ENDPOINT
- PUBMED-25591384-COMPLETE-FULLTEXT-PROVIDER-SURFACE
- ACS-JP404315T-COMPLETE-PRIMARY-AND-SI-SURFACE
- ACS-FIGSHARE-EXACT-RESOURCE-DOI-JP404315T
- ACS-FIGSHARE-4012420-JP404315T-SI-001
- CORE-43583184-DOWNLOAD-PDF-LEGACY
- CORE-43583184-DOWNLOAD-CURRENT

Comparison records:

- exact Fold witness populations 3/4 and 1/4; one-observation successor 3/5 and 2/5
- blind 298.5 K ordered n-pentane vector: tt 0.33 ± 0.03 , tg 0.51 ± 0.01 , pm 0.02 ± 0.01 , pp 0.14 ± 0.01
- blind displayed ordered vector is exactly $33/100 + 51/100 + 1/50 + 7/50 = 1$
- blind intramolecular energy order: tt structural reference, tg 480, pp 658, pm 3263 cal mol⁻¹; positive gaps 480, 178, 2605
- blind Etg at 300 K: 441 ± 114 cal mol⁻¹; external signed temperature variation -1.9 ± 0.3 cal K⁻¹ mol⁻¹ retained downstream
- complete ACS supporting surface: 8 tables and 224 measured condition/coupling rows; complete article: 22 spectra
- all failed provider routes, recorder failure and isotropic 201/200 rounded-display adverse row preserved
- complete exact target vector sha256:af48a1e80265f4eb1eb710a5a0c9a33801f94c6b2b73d8e9e5eb964056e32c78
- control omitted_source_rejected: True
- control numerical_zero_rejected: True
- control all_fourteen_target_hashes_bound_postseal: True
- control exact_witness_populations: True
- control exact_successor_populations: True
- control blind_ordered_population_sum_exact: True
- control blind_population_order_matches_reconstruction: True
- control blind_positive_energy_gaps_complete: True
- control condition_and_timescale_retained: True
- control all_adverse_routes_preserved: True
- control all_224_acs_measurement_rows_preserved: True
- control prediction_contains_no_opened_value_or_payload: True

Falsification boundary: The claim fails if any ORG-005 conformer class, observation, condition, timescale or source response is omitted; if a probability is fitted, randomized, imported or detached from its exact positive finite boundary; if native numerical zero or signed energy subtraction is used; if the 256-form census has other than one survivor; if the blind 298.5 K population vector, 22-spectrum condition surface, printed energy vector, all 224 ACS measurement rows, failed provider routes or the isotropic 201/200 rounded-display adverse row is erased or altered.

Measurement receipt: sha256:3db3549bf6bd34d53d6973f63a5463283badb8b3bb47a085c390917865414ea7.
 Isolation certificate: sha256:f0c9bdf42921f15d2d9741919bc0030b82bef7d468ff1e5556583e456b12f00c.
 Custody certificate: sha256:e0293fd8a351e782c7942da0f764079817474e6956bb7df33b050350589eae8f.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The organic extension derives conjugation, resonance, aromatic recurrence and its measured stability excess, antiaromatic/nonaromatic distinction, conformer classes and populations, substitution, addition, elimination and composition-retaining rearrangement with complete adverse evidence. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no native numerical zero; structural absence and the least-state reference are EmptyOne
- no negative irrational imaginary continuum fitted free random or imported native parameter
- no Boltzmann exponential measured temperature species label energy or population used to generate the law
- all external signed zero negative decimal uncertainty and rounded inscriptions remain downstream evidence
- no condition observation-timescale source response failed capture adverse row or rounding discrepancy omitted
- no external result opened before its applicable value-free source and prediction seal

Immutable identities. Source manifest

sha256:ea3ec53a1b100b61c5a74ad28c4bbe1c421a64a9d8d8a316d6c68375385f0ae9; derivation seal
 sha256:cefd89d5d60b34e3099385846815aa397bf4201144c5738a09e850af53930c07; engine receipt
 sha256:6668695f63755775127464869702a2e9defabbd3bd60182cca72c614f12985d1 at receipts/engine/model_admitted/SFT-CHEM-CONFORMER-POPULATION-ORDERING-006-6668695f63755775.json;
 empirical hash sha256:cc9db7ffaa577fb15b4ae6c31c85d53a247268993b4305752a3007656a192a30;
 measurement receipt sha256:3db3549bf6bd34d53d6973f63a5463283badb8b3bb47a085c390917865414ea7.

180. Exact entering/leaving carrier substitution family law

Claim identity: SFT-CHEM-NUCLEOPHILIC-SUBSTITUTION-FAMILY-007

Question and theorem. For every complete finite registered exchange carrier, one incoming held pair forms the unique centre-entering bond while the two occurrences of the displaced centre-leaving bond remain held on the leaving carrier. Every atom and electron occurrence is conserved. The complete generated path family consists of a one-transition exchange and cleavage followed by formation; formation before cleavage violates the unique exchange slot.

*complete-source-ordered-reaction-carrier__complete-source-bonds-and-free-entering-pair__every-held-pair-occurrenc
 e-conserved__one-bond-cleaved-and-one-bond-formed__complete-one-transition-and-cleavage-first-path-family__ever
 y-state-edge-status-and-source-record-retained__value-free-structure-and-mechanism-target-seal__fresh-retained-subs
 trate-successor-no-extra-rule*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001
- SFT-CHEM-CONJUGATED-SUPPORT-001
- SFT-CHEM-RESONANCE-EQUIVALENT-REPRESENTATION-002
- SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007
- SFT-CHEM-PARALLEL-MECHANISM-COMPOSITION-008

Generated grammar. Generate the literal product of carrier, source, support, change, path, record, observation and extension alternatives; decide all 256 forms only from admitted exact graph, conservation, bond and transition laws.

Boundary: Every positive finite complete exchange carrier with distinct centre, retained, entering and leaving occurrences; all source bonds and free pairs; every atom and held electron occurrence; the single exchange slot; both generated path orders; every source record and the complete nine-target external surface.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-source-ordered-reaction-carrier__complete-source-bonds-and-free-entering-pair__every-held-pair-occurrence-conserved__one-bond-cleaved-and-one-bond-formed__complete-one-transition-and-cleavage-first-path-family__every-state-edge-status-and-source-record-retained__value-free-structure-and-mechanism-target-seal__fresh-retained-substrate-successor-no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-source-ordered-reaction-carrier	selected-product-label-or-disconnected-fragments	A named answer cannot reconstruct one complete reaction carrier.
source	complete-source-bonds-and-free-entering-pair	reactant-or-bond-support-omitted	Omission prevents exact reconstruction of what changes.
support	every-held-pair-occurrence-conserved	electrons-created-destroyed-or-collapsed-to-charge-number	A scalar charge or incomplete pair loses the transferred occurrences.
change	one-bond-cleaved-and-one-bond-formed	endpoint-only-product-or-extra-bond-change	An endpoint or extra edit does not prove the minimal substitution map.
path	complete-one-transition-and-cleavage-first-path-family	named-mechanism-rate-law-or-unordered-snapshots	A conventional name or imported kinetic law can select a path.
record	every-state-edge-status-and-source-record-retained	intermediate-adverse-or-mechanism-row-omitted	Selected records falsely collapse the path family.
observation	value-free-structure-and-mechanism-target-seal	substrate-product-mechanism-readable-before-seal	External structures or terminology could select the survivor.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
extension	fresh-retained-substrate-successor-no-extra-rule	molecule-specific-exception-or-recomputed-prefix	A special substrate rule is not depth-independent.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256: fcf85ccb37f220acfed618f6f5c32ec97f78706ba979582092879fd5ae996949.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: aaccbe0cf8bec56b1dc1a7e869373b2a6c63d5a3e969343a37d85f6f4ca7e044.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 134e6adaac3587112751abcc0bf1dc299acef1f3de726acc25d6e55c0fb1375d.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: 68853c8cd7fbe6861b7de6eab4d243ac1db44bb0352bcb716f933963fb38aa0b.

Independent reconstruction. Implementation hash

sha256: ee3917008c755d08a81ee18899dad31c21f2f09b4a7a3478c210d1c0ae776d3b; independent certificate sha256: dee1b9268693ee13f42f60890322fa5c10e36ce1f8abaf42ce90269b4d5265d0; external-validation hash sha256: 4c1747959ec1966fc14b2526057e185558f1897ba71d0ccadfd6b005146a2f92. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-N04249
- IUPAC-08191
- IUPAC-E02130
- IUPAC-S06078
- IUPAC-H02809
- PUBCHEM-PUG-BROMOMETHANE-STRUCTURE
- PUBCHEM-PUG-HYDROXIDE-STRUCTURE
- PUBCHEM-PUG-METHANOL-STRUCTURE
- PUBCHEM-PUG-BROMIDE-STRUCTURE

Comparison records:

- complete nine-record external structure and mechanism vector retained
- seven exact IUPAC and PubChem payloads opened only after the ORG-007 seal; two development-observed IUPAC rows disclosed
- source formula inventory {'Br': 1, 'C': 1, 'H': 4, 'O': 1}; terminal formula inventory {'Br': 1, 'C': 1, 'H': 4, 'O': 1}; exact conservation **True**
- connectivity CBr plus [OH-] becomes CO plus [Br-]
- post-seal IUPAC records preserve entering-group bond formation, elementary-or-stepwise replacement, and both-electron heterolysis
- V1 development-snapshot hash transcription error preserved; V2 correction changes no source, target, prediction, law or outcome and performs no recapture
- deliberately omitted-row and tampered-label controls rejected

Falsification boundary: The claim fails if any carrier, source bond, free pair, atom or electron occurrence is omitted; if more or fewer than one centre bond is replaced; if formation-before-cleavage is accepted; if either generated path family is omitted; if the census has other than 256 forms and one survivor; if any of nine external records is erased; if the post-seal IUPAC entering-group, substitution or heterolysis records fail their declared relations; if the PubChem source and terminal formula inventories differ; or if C-Br to C-O connectivity replacement is absent.

Measurement receipt: sha256:8d78610af3fe9c19b535e725245880d6490a4a0f9f1013ebd86f74a17c86cc1e.

Isolation certificate: sha256:acf3e362bc6acccc247083c5af9d598f0a88b600c690ed32a6b9eae0535ae503.

Custody certificate: sha256:3ac0e70009001d3a242d802f02cbc47a42a262637572850bc2c08fe427ed5781.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The organic extension derives conjugation, resonance, aromatic recurrence and its measured stability excess, antiaromatic/nonaromatic distinction, conformer classes and populations, substitution, addition, elimination and composition-retaining rearrangement with complete adverse evidence. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no native numerical zero; structural absence is EmptyOne
- no negative irrational imaginary continuum fitted free random or imported native parameter
- no conventional nucleophilic SN1 SN2 rate energy solvent or substrate rule used to generate the survivor
- no external substrate product mechanism structure formula charge or source response opened before its applicable seal
- the two development-observed IUPAC records are disclosed and never relabelled blind
- all nine source rows and the V1 identity-hash transcription correction remain preserved

Immutable identities. Source manifest

sha256:7655a510e36ef573279d047a0d856190163a8a4f8738be2073be02aa0e5612b7; derivation seal

sha256:d6c329d38d6f3799bd1f17529a93ea6717b6a6cd7985fd376bda8445066f80e7; engine receipt

sha256:7f79c743cc120a8dfb19266933a69ef142f1cfc52806ae6677920b4f6f73b55b at receipts/engine/model_admitted/SFT-CHEM-NUCLEOPHILIC-SUBSTITUTION-FAMILY-007-7f79c743cc120a8d.json;

empirical hash sha256:6a3637ed9ef35351b1618849696a90d742f956f12e967cfb69a2647a2f76be98;

measurement receipt sha256:8d78610af3fe9c19b535e725245880d6490a4a0f9f1013ebd86f74a17c86cc1e.

181. Exact acceptor-driven aromatic substitution family law

Claim identity: SFT-CHEM-ELECTROPHILIC-SUBSTITUTION-FAMILY-008

Question and theorem. The structural-EmptyOne entering acceptor receives the complete aromatic donor pair; the entering bond forms, the leaving bond closes, and the complete leaving pair restores the opened aromatic incidence through the complete one- or two-transition family.

complete-aromatic-source-ordered-carrier__structural-EmptyOne-entering-acceptor-slot__complete-aromatic-donor-pair-transfer__entering-bond-formed-leaving-bond-cleaved__complete-concerted-and-addition-then-restoration-path-family__leaving-pair-restores-aromatic-incidence__value-free-aromatic-structure-and-mechanism-target-seal__fresh-aromatic-successor-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001
- SFT-CHEM-CONJUGATED-SUPPORT-001
- SFT-CHEM-RESONANCE-EQUIVALENT-REPRESENTATION-002
- SFT-CHEM-AROMATIC-RECURRENCE-STABILITY-003
- SFT-CHEM-NUCLEOPHILIC-SUBSTITUTION-FAMILY-007

Generated grammar. Generate the literal product of carrier, acceptor, donor, change, path, recurrence, observation and extension alternatives; decide all 256 forms solely from admitted Fold, aromatic and ORG-007 support laws.

Boundary: Every positive finite aromatic carrier with complete donor, recurrence, entering-acceptor and leaving supports; all one- and two-transition paths; every source record; and the complete four-target, 452-page external surface.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-aromatic-source-ordered-carrier__structural-EmptyOne-entering-acceptor-slot__complete-aromatic-donor-pair-transfer__entering-bond-formed-leaving-bond-cleaved__complete-concerted-and-addition-then-restoration-path-family__leaving-pair-restores-aromatic-incidence__value-free-aromatic-structure-and-mechanism-target-seal__fresh-aromatic-successor-no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-aromatic-source-ordered-carrier	selected-product-label-or-nonaromatic-fragments	A named answer or disconnected fragments cannot reconstruct one aromatic reaction carrier.
acceptor	structural-EmptyOne-entering-acceptor-slot	incoming-pair-imported-or-acceptor-slot-filled	Giving the acceptor a pre-existing pair changes which carrier supplies the new bond.
donor	complete-aromatic-donor-pair-transfer	donor-pair-erased-created-or-collapsed-to-charge	A scalar charge or incomplete pair loses the transferred occurrences.
change	entering-bond-formed-leaving-bond-cleaved	endpoint-only-product-or-extra-support-change	An endpoint or extra edit does not prove the local exchange.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
path	complete-concerted-and-addition-then-restoration-path-family	named-rate-law-cleavage-first-or-unordered-snapshots	A conventional name, kinetic law or wrong order can select a mechanism.
recurrence	leaving-pair-restores-aromatic-incidence	aromatic-support-destroyed-or-fitted-restoration	An unclosed or fitted recurrence does not return the carrier to its exact class.
observation	value-free-aromatic-structure-and-mechanism-target-seal	substrate-product-mechanism-readable-before-seal	External structures or terminology could select the survivor.
extension	fresh-aromatic-successor-no-extra-rule	substrate-specific-exception-or-recomputed-prefix	A special aromatic substrate rule is not depth-independent.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:99f596213d121bb8742f42ada55e184ffaf133177e25320786f17090b6f40ede.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:a1856309d25e527286f50640a1c4df186611acafa974c242d69116d804b51d62.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:73fea1dd0a577468d6192db9705cbd9e4af4b14e7c949267ac45273765d0513c.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:f21fbfa77ea89505835834b2f6db7293fdb1699771609146c61b3cd108a5e30a.

Independent reconstruction. Implementation hash

sha256:c9932ae5ec274544adec908d64acdc2fb9663f69b42a287a7614897b3d165e42; independent certificate sha256:4ea584fee7657d69ffd975f39cb7ea67dad13e433494d43bfff92ba74b4f842f; external-validation hash sha256:817c78f19cfdc2109086268b2ddae6bfb29473faa07e0069dbeb3ab1fdb91211. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-E02020
- IUPAC-08167
- NATURE-CHEMISTRY-S41557-022-01101-0-ARTICLE
- NATURE-CHEMISTRY-41557-2022-1101-MOESM1

Comparison records:

- complete four-source electrophilic substitution vector retained
- one 13,165,129-byte, 452-page supplementary PDF opened only after seal
- all 25 Table S1 rows retained: 14 displayed zero-yield, 9 displayed full-yield, and intermediate 12 and 93 rows
- EAS-like C-electrophile/nucleophilic-phenol coupling, dearomatized support and rearomatization all present
- radical, phenoxonium, starting-material, inconsistent-condition, not-promoted and inseparable-mixture controls retained
- complete target vector sha256:ed54355fd3feb17562d89a8411647e97b9acd33bd1ad872e2ec1f7099a98a2d2

Falsification boundary: The claim fails if any carrier, pair, atom, bond change, recurrence state or path is omitted; if cleavage-first or a nonunique 256-form census passes; if any of four sources or 452 supplementary pages is erased; if the IUPAC donor/acceptor

relations fail; if the complete PDF lacks EAS-like donor/acceptor coupling, dearomatized support and rearomatization; or if any of 25 optimization rows, including 14 displayed zero-yield rows and mechanism controls, is omitted.

Measurement receipt: sha256:9f525b88cb394c4c60e2428acbce429c30ac857ffa264dd6071ed16496e49dca.

Isolation certificate: sha256:f099a4397bbe02d819e76cad9c7c042e472b29d5172cf7e7ab0c6f4c3ba5a7d6.

Custody certificate: sha256:8505e301d20457fbb768e25305d5125b0fc7e93fd5c3f8321d30e2a4c05cf52b.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The organic extension derives conjugation, resonance, aromatic recurrence and its measured stability excess, antiaromatic/nonaromatic distinction, conformer classes and populations, substitution, addition, elimination and composition-retaining rearrangement with complete adverse evidence. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no native numerical zero; structural acceptor absence is EmptyOne
- no negative irrational imaginary continuum fitted free random or imported native parameter
- no conventional mechanism or external target used to generate the survivor
- all 452 PDF pages and every adverse row retained

Immutable identities. Source manifest

sha256:51b6d989f01c0cab59c9ee58f5e11dc0f4d6b15694cc72e229a06fdc0ba1acf0; derivation seal

sha256:9325b68be749a62ac59946d913919d55c748ac1cad0bea503af8c8a52b9182e7; engine receipt

sha256:e2c208084ac61cb0dc0ba6eeafd12ad74b3cd556f27a40952c83213e4087d548 at receipts/engine/model_admitted/SFT-CHEM-ELECTROPHILIC-SUBSTITUTION-FAMILY-008-e2c208084ac61cb0.json;

empirical hash sha256:314e43ebe29f2529b2a3c73d912969ed88f597f10199a8939dddac74d5260d41;

measurement receipt sha256:9f525b88cb394c4c60e2428acbce429c30ac857ffa264dd6071ed16496e49dca.

182. Exact complete addition-reaction family law

Claim identity: SFT-CHEM-ADDITION-REACTION-FAMILY-009

Question and theorem. Every complete addition transform retains every source carrier, atom and held support in one product, forms exactly two new cross-component adjacencies, relocates a positive finite multiplicity family with its base incidences retained, and generates every componentwise site class.

complete-multicarrier-source-and-single-product__every-reactant-atom-occurrence-retained__every-held-support-occurrence-retained__exact-two-new-cross-component-adjacencies__positive-finite-reduced-layer-family-with-base-retained__complete-componentwise-same-adjacent-and-nonadjacent-site-family__value-free-addition-product-vector-seal__fresh-unchanged-carrier-successor-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001
- SFT-CHEM-NUCLEOPHILIC-SUBSTITUTION-FAMILY-007
- SFT-CHEM-ELECTROPHILIC-SUBSTITUTION-FAMILY-008

Generated grammar. Generate the literal product of carrier, atom, support, adjacency, multiplicity, site, observation and extension alternatives; decide all 256 forms solely from admitted Fold conservation and transition laws.

Boundary: Every positive finite multicarrier source becoming one product; every exact atom and held-support occurrence; every two-adjacency cross-carrier orientation; every positive finite reduced multiplicity family; every componentwise same, adjacent and non-adjacent site class; all V1--V9 records; all 48 independent ORD payloads; and all 28 source-selected product vectors.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-multicarrier-source-and-single-product__every-reactant-atom-occurrence-retained__every-held-support-occurrence-retained__exact-two-new-cross-component-adjacencies__positive-finite-reduced-layer-family-with-base-retained__complete-componentwise-same-adjacent-and-nonadjacent-site-family__value-free-addition-product-vector-seal__fresh-unchanged-carrier-successor-no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-multicarrier-source-and-single-product	selected-product-label-or-reactant-fragment-omitted	A named product or omitted reactant cannot reconstruct the complete source.
atoms	every-reactant-atom-occurrence-retained	atom-created-erased-or-collapsed-to-formula	A formula or incomplete occurrence list loses exact identity.
supports	every-held-support-occurrence-retained	support-created-erased-or-collapsed-to-order-number	A scalar order or partial support cannot reconstruct the transform.
adjacency	exact-two-new-cross-component-adjacencies	changed-support-count-confused-with-new-adjacency-count	Relocated multiplicity support is not an additional new atom adjacency.
multiplicity	positive-finite-reduced-layer-family-with-base-retained	exactly-one-layer-or-one-component-only	A fixed layer or component restriction omits complete additions.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
sites	complete-component wise-same-adjacent-and-nonadjacent-site-family	adjacent-only-or-species-selected-site	One preferred site distance omits lawful addition forms.
observation	value-free-addition-product-vector-seal	product-equation-or-database-row-readable-before-seal	An external product could select the survivor.
extension	fresh-unchanged-carrier-successor-no-extra-rule	reaction-specific-exception-or-recomputed-prefix	A species exception is not depth-independent.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:b4d30f68770bf25a8533b46853006fac234e03ffa393c50ce99208d6114a5527.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:fada8ee64fdef84223b4741479fa3c2ff07a513187bd07c5bdc3ecd74d7be29.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:96f91d21ebe376c29b29b34568f6afd6418bef7a382e25a98ee596df431d5ff4.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:b8d40804af5109f053a3c08bb6eb3d167edce2f0b8be09e1c06a78c4340f7fc9.

Independent reconstruction. Implementation hash

sha256:e19b0d5ea2b8ae95042e0e93b5b89171cec1deaceae18380dfefeb89390c42788e; independent certificate sha256:d040ab43627716667780ef1541a226dcf1da282c283825052cdfa23b4cb4eb9d; external-validation hash sha256:e7d0a682aef5796e20598862075168ba325d0bd0aba3b56914c69961ab09bc6c. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-17384a1d9c5f4b2e9aacbb6293204db
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-1478630680b84aee8dca359a52f20ba1
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-25dfe42a186e40c88717830e231034da
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-65ff4c19dea541b880d627eb4fbf8af3
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-8bbc1db0e28c4c60a61d554be47d9b92
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-c9f16c4a213a481daa225423696b55a4
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-8ba9957fa5604be4aeda7eff734f6ca5
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-e80af89368884e338a1956cbe7d0d943
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-6d9789e42f954cae8a05b6d471178de0
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-a9b062ada7ce4d2fa3f88d957d9f599a
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-0c3242203b3a40888edda75874c3b8d
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-48c6faedc14d49dfa5de2dd45ed6140d
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-d32732940c8e4867b80c81d9b01eb889
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-62b80d7f983a4bf9a9564d4afc30e02e
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-e7f8deb100824a408d3c2e0d1cb693a9

- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-94fbe6fbedf24c1cbfa7e834d4e4a7c9
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-8eff1b6e7bbb4496bc2632e3d3e31a71
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-4b5fae7b28894c28b650d185109427d2
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-7ad584d7eae4f169a17acb74b087530
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-bba06fb330a54307a9082b493a2453c3
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-c9cb9eedd61341a6b829cf445aa77b7c
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-6c8b4df5334a4f939f2d4f56b1edbf8
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-bcbb41bfe1ae435eac5bac8862f2f14d
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-ac18bbe2a7644934b24a8956f4b4a721
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-2e1cdb77c14c495e892c4336682b64a6
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-4be654f9e2a74d6cb58f62cf1baae0ec
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-b91423e5a7d349f78fefca0ed55980e8
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-1bc497e7e2f745f4b38b233207da6bc8

Comparison records:

- complete 48-file, 12,993,815-byte non-USPTO Open Reaction Database holdout retained
- all 131,209 conventional reaction rows inspected at the source-only boundary
- all 28 independently labeled azide-alkyne/click/cycloaddition identities selected before product opening
- 28 favorable, no adverse and no unresolved complete product vectors
- 112 exact element-preserving correspondences; each has two new cross-carrier adjacencies and three multiplicity relocations
- all V1--V8 absent, adverse, unresolved and failed-universal results preserved without awarding closure
- complete result vector sha256:ea07485d76d8023f8b055420a1bc68f2ada83477c7a5208885b6836e48d28689

Falsification boundary: The claim fails if any carrier, atom, support, new adjacency, multiplicity relocation, site class, prior attempt, payload or selected reaction is omitted; if more than one of 256 forms survives; if any selected ORD reaction lacks a complete element-preserving correspondence with exactly two generated cross-carrier adjacencies and positive finite multiplicity relocation; or if any adverse or unresolved row is filtered.

Measurement receipt: sha256:1b2d67267a0b328b4cd480ab8731788664b0b2f19ffaa4781b61509f6318e3c0.
Isolation certificate: sha256:7ca99a5475a7cfff7d1e64ea92481d665a4705e0acd9c674edd2e50102d409dd5.
Custody certificate: sha256:f991e32d633577823c9f1a81a9e53e20dff00331c3130acb9ce7aa1c6506a82.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The organic extension derives conjugation, resonance, aromatic recurrence and its measured stability excess, antiaromatic/nonaromatic distinction, conformer classes and populations, substitution, addition, elimination and composition-retaining rearrangement with complete adverse evidence. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no native numerical zero; structural absence is EmptyOne
- no negative irrational imaginary continuum fitted free random or imported native parameter
- no conventional addition name product database row atom map or measured outcome selects the survivor

- all V1--V8 absent adverse unresolved and failed-universal results remain preserved
- all 48 independent payloads and all 28 preselected reactions are required without post-outcome filtering

Immutable identities. Source manifest

sha256:cf29529d348e3afd8b335fba63b2659869778be7738ea111089139b14408ba7b; derivation seal sha256:4f0fcdfe3f92bf11281839185946e8d0f1160ac2f647681d6fb1acf392129667; engine receipt sha256:bcc6d8a50dce889570f775f89c2277b832be027ea697f31f1339535e1a42d118 at receipts/engine/model_admitted/SFT-CHEM-ADDITION-REACTION-FAMILY-009-bcc6d8a50dce8895.json; empirical hash sha256:2ddb1a7a2ae82118cf7bc09632d6925c7b7f2de09eb9360d02b6c7f8972009fa; measurement receipt sha256:1b2d67267a0b328b4cd480ab8731788664b0b2f19ffaa4781b61509f6318e3c0.

183. Exact complete inverse-addition elimination family law

Claim identity: SFT-CHEM-ELIMINATION-REACTION-FAMILY-010

Question and theorem. A complete elimination is the exact reversible boundary of complete addition: all atom and held-support occurrences remain, exactly two joining adjacencies are removed, and a positive finite multiplicity family returns to retained product incidences.

complete-single-source-and-multicarrier-products__every-source-atom-occurrence-retained__every-held-support-occurrence-retained__exact-two-source-adjacencies-removed-across-product-components__positive-finite-multiplicity-family-restored-with-base-retained__complete-productwise-same-adjacent-and-nonadjacent-site-family__value-free-elimination-product-vector-seal__fresh-unchanged-carrier-successor-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-QUANTUM-REVERSIBLE-MODEL-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001
- SFT-CHEM-ADDITION-REACTION-FAMILY-009

Generated grammar. Generate the literal product of carrier, atom, support, adjacency, multiplicity, site, observation and extension alternatives; decide every one of the 256 forms only from admitted Fold conservation, graph and reversible addition laws.

Boundary: Every positive finite complete source carrier and complete product-carrier family; every exact atom and held-support occurrence; every positive finite restored multiplicity family; every same, adjacent and non-adjacent site class; the complete IUPAC record; all 22 captured archive members; all 117 PDF pages; all 32 characterized products; all five unsuccessful rows; all seven optimization tables; and the complete 3a-prime/3a intermediate record.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-single-source-and-multicarrier-products__every-source-atom-occurrence-retained__every-held-support-occurrence-retained__exact-two-source-adjacencies-removed-across-product-components__positive-finite-multiplicity-family-restored-with-base-retained__complete-productwise-same-adjacent-and-nonadjacent-site-family__value-free-elimination-product-vector-seal__fresh-unchanged-carrier-successor-no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	<code>complete-single-source-and-multicarrier-products</code>	<code>selected-alkene-or-leaving-group-fragment-omitted</code>	A selected product cannot reconstruct the complete reaction carrier.
atoms	<code>every-source-atom-occurrence-retained</code>	<code>atom-erased-created-or-collapsed-to-formula</code>	An omitted leaving atom violates exact occurrence conservation.
supports	<code>every-held-support-occurrence-retained</code>	<code>support-erased-created-or-collapsed-to-order-number</code>	A scalar bond order cannot retain the held support identity.
adjacency	<code>exact-two-source-adjacencies-removed-across-product-components</code>	<code>named-leaving-group-or-arbitrary-cleavage-count</code>	A named group or free cleavage count imports a selector.
multiplicity	<code>positive-finite-multiplicity-family-restored-with-base-retained</code>	<code>fixed-one-layer-or-product-selected-order</code>	One chosen multiplicity omits the positive finite inverse family.
sites	<code>complete-productwise-same-adjacent-and-nonadjacent-site-family</code>	<code>beta-only-or-species-selected-site</code>	A beta-only convention omits same-site and positive finite non-adjacent forms.
observation	<code>value-free-elimination-product-vector-seal</code>	<code>product-scheme-or-yield-readable-before-seal</code>	An opened product scope could choose the survivor.
extension	<code>fresh-unchanged-carrier-successor-no-extra-rule</code>	<code>reaction-specific-exception-or-recomputed-prefix</code>	A named exception is not depth-independent.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 5963503343e638d671dc602e80e2648d5e05f6c156c8326a181a35507274584c.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: bd5194b50ca85729da8f0bfccfb05f33e77fb385d827d614830c879e8e77b71b.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 05cdf214e05f17273d400b12e7973dcf2a1b41d8bb125426f02668ea39425a73.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : e96d826113908428eb0a6650aefde015cc2bb037f4ebf84105552e6d41299899.

Independent reconstruction. Implementation hash

sha256 : a86009932a5ecf4e55e634da0d78ad8aabb54cfeabe2353aa6e11481645bcb24; independent certificate
sha256 : e132ada0e7c80df2eebbec5e951e70e64626add82f5a35e027938836b2f17509; external-validation
hash sha256 : 3739d33fb38ddac27a65d645fbbd6490ae24ccc02c44f1b8657db012fb87be3b. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were
preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC:E02038
- RSC:D4SC01905A:3a
- RSC:D4SC01905A:3b
- RSC:D4SC01905A:3c
- RSC:D4SC01905A:3d
- RSC:D4SC01905A:3e
- RSC:D4SC01905A:3f
- RSC:D4SC01905A:3g
- RSC:D4SC01905A:3h
- RSC:D4SC01905A:3i
- RSC:D4SC01905A:3j
- RSC:D4SC01905A:3k
- RSC:D4SC01905A:3l
- RSC:D4SC01905A:3m
- RSC:D4SC01905A:3n
- RSC:D4SC01905A:3o
- RSC:D4SC01905A:3p
- RSC:D4SC01905A:3q
- RSC:D4SC01905A:3r
- RSC:D4SC01905A:3s
- RSC:D4SC01905A:3t
- RSC:D4SC01905A:3u
- RSC:D4SC01905A:3v
- RSC:D4SC01905A:3w
- RSC:D4SC01905A:3x
- RSC:D4SC01905A:3y
- RSC:D4SC01905A:3z
- RSC:D4SC01905A:3aa
- RSC:D4SC01905A:3ab
- RSC:D4SC01905A:3ac

- RSC:D4SC01905A:3ad
- RSC:D4SC01905A:3ae
- RSC:D4SC01905A:3af
- RSC:D4SC01905A:UNSUCCESSFUL:1
- RSC:D4SC01905A:UNSUCCESSFUL:2
- RSC:D4SC01905A:UNSUCCESSFUL:3
- RSC:D4SC01905A:UNSUCCESSFUL:4
- RSC:D4SC01905A:UNSUCCESSFUL:5
- RSC:D4SC01905A:S1
- RSC:D4SC01905A:S2
- RSC:D4SC01905A:S3
- RSC:D4SC01905A:S4
- RSC:D4SC01905A:S5
- RSC:D4SC01905A:S6
- RSC:D4SC01905A:S7
- RSC:D4SC01905A:3A-PRIME-TIME-COURSE
- EUROPE-PMC:PMC11186341:SUPPLEMENT

Comparison records:

- complete 22-member Europe PMC archive and all 117 supplementary PDF pages retained
- all 32 characterized product blocks retained; every reported product displays unsaturation
- all 32 incomplete coproduct carriers remain explicitly unresolved for complete atom/support balance
- all five unsuccessful or low-elimination rows preserved without success filtering
- all seven optimization tables and 34 rows including the printed source gap retained
- the six-row 3a-prime/3a time course preserves decreasing intermediate and increasing product
- complete result vector sha256:94c7f50610c27ed788f933403f738fda709d17c1c89016b84110ffb81b097b7f

Falsification boundary: The claim fails if more than one of 256 generated forms survives; if any atom, support, adjacency, restored layer, site class or successor is omitted; if the IUPAC inverse-addition/unsaturation relation fails; if any characterized product lacks its reported unsaturation; if any archive member, PDF page, product, unsuccessful row, optimization row, alternative or intermediate record is omitted; or if an incomplete conventional product block is falsely awarded complete atom/support balance.

Measurement receipt: sha256:9e7b5aca88735d9db50dcb1d4127531b448b6f1bbfdbc7c217121ae38a99aa2f.

Isolation certificate: sha256:2c742b7696d119c4d84c2f909288a975f2c79ef2117660297c10f0cdfbd5eb7b.

Custody certificate: sha256:90ac99caa9f57a0cb812ab3f8684ed827e27f58ce691c306211171cc7441665c.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The organic extension derives conjugation, resonance, aromatic recurrence and its measured stability excess, antiaromatic/nonaromatic distinction, conformer classes and populations, substitution, addition, elimination and composition-retaining rearrangement with complete adverse evidence. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no native numerical zero; structural absence is EmptyOne
- no negative irrational imaginary continuum fitted free random or imported native parameter
- no conventional elimination mechanism, product or measured value used to select the survivor
- incomplete conventional coproduct carriers remain unresolved rather than awarded

Immutable identities. Source manifest

sha256:28866f18f753e8f771a43b9d1e6fbfe12957b1cd2442bb7d4415b9854683be4a; derivation seal sha256:584fe8da47fb06302c1b01246a4248f10fd180634da5545283f1758a78a98ba4; engine receipt sha256:7dc26f50dfa404272ea47a9426cbda3c27f069ddb54120a1b4dcf8a589be1f27 at receipts/engine/model_admitted/SFT-CHEM-ELIMINATION-REACTION-FAMILY-010-7dc26f50dfa40427.json; empirical hash sha256:7c0944cfd14c5a67046c77688dbb6aa0f74ab9f67e6dea0384b236ae59104962; measurement receipt sha256:9e7b5aca88735d9db50dcb1d4127531b448b6f1bbfbdc7c217121ae38a99aa2f.

184. Exact composition-retaining molecular rearrangement law

Claim identity: SFT-CHEM-REARRANGEMENT-REACTION-FAMILY-011

Question and theorem. A complete rearrangement retains every atom and held-support occurrence while a positive finite family of held supports changes exact incidence; every alternative, path, degenerate trace and unchanged successor remains generated.

complete-source-and-terminal-carrier__every-atom-occurrence-retained-in-source-order__every-held-support-occurrence-retained-once__positive-finite-held-support-incidence-change__complete-direct-or-opened-reclosure-path-family__all-nonoriginal-incidences-and-degenerate-traces-generated__value-free-rearrangement-product-vector-seal__fresh-unchanged-carrier-successor-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-QUANTUM-REVERSIBLE-MODEL-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001

- SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001
- SFT-CHEM-ELIMINATION-REACTION-FAMILY-010

Generated grammar. Generate the literal product of carrier, atom, support, adjacency, path, alternative, observation and extension coordinates; decide all 256 forms only from admitted Fold conservation, graph and reversible laws.

Boundary: Every positive finite connected source/terminal carrier pair; every exact atom and held support; every nonoriginal target incidence; both direct and opened-reclosure paths; constitutionally degenerate traces; three complete IUPAC records; the preserved first blind surface; all 38 pages and all eight exact Claisen pairs.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-source-and-terminal-carrier__every-atom-occurrence-retained-in-source-order__every-held-support-occurrence-retained-once__positive-finite-held-support-incidence-change__complete-direct-or-opened-reclosure-path-family__all-nonoriginal-incidences-and-degenerate-traces-generated__value-free-rearrangement-product-vector-seal__fresh-unchanged-carrier-successor-no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-source-and-terminal-carrier	selected-group-or-product-fragment-only	A fragment cannot prove that the same complete carrier rearranged.
atoms	every-atom-occurrence-retained-in-source-order	atom-created-erased-or-collapsed-to-formula	A changed atom inventory is a different composition, not a rearrangement.
supports	every-held-support-occurrence-retained-once	support-created-erased-or-renamed	A replaced bond label cannot be traced through the rearrangement.
adjacency	positive-finite-held-support-incidence-change	unchanged-graph-or-unregistered-edit	No change is not a rearrangement and an arbitrary edit is not forced.
path	complete-direct-or-opened-reclosure-path-family	named-mechanism-or-single-imported-order	A conventional mechanism name cannot select the native path.
alternatives	all-nonoriginal-incidences-and-degenerate-traces-generated	species-selected-target-or-degenerate-path-erased	A named product omits other lawful target incidences and constitutionally degenerate traces.
observation	value-free-rearrangement-product-vector-seal	product-scope-readable-before-seal	An opened rearrangement scope could choose the survivor.
extension	fresh-unchanged-carrier-successor-no-extra-rule	reaction-specific-exception-or-recomputed-prefix	A named exception is not depth-independent.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f4e8a463127ad996ea9657acc5c769dd178feca9004a539a3d4df7e19de5bee4`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b48a65426cc4682980f40d3de37ef27021454d1d19c60d82f07d1fd5c2cb1081`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5beba579e6ef4662899f49b12c129452e11bd03f456549b0fe3de7a4418483ff`.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : e400eaac9e7aba78ad6d6e3fdb39922544f650afd3159711bda0bd19c88aa427.

Independent reconstruction. Implementation hash

sha256 : 31915155c874a00707372c3ead7eb1b4dec70018a859f48a28552103ab9ec35e; independent certificate sha256 : 2d5189c21edd0ca50a1d7ca56ac88706d034784113d8bb2395a9519f6b8a5e37; external-validation hash sha256 : b338c57430bd37519f1f37df53b80574d3fec6d8e0ea28d8df81b4b54087a444. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC:M03997
- IUPAC:R05194
- IUPAC:R05196
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:1
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:2
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:3
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:4
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:5
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:6
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:7
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:8
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAGE:3
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAGE:4
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAGE:37
- EUROPE-PMC:ORG-011:V1
- ACS-FIGSHARE:10.1021/ja803370x.s001

Comparison records:

- complete official 38-page supporting-information PDF retained
- all eight source/product structures independently atom-enumerated
- eight exact endpoint inventories favorable; no adverse or unresolved endpoint inventory
- all eight pairs exhibit a positive constitutional-incidence change
- all optimization, non-detection, signed stereochemical and spectral surfaces retained
- first incomplete blind surface preserved as unresolved without relabelling
- complete result vector sha256:e74ca9dd577cb372aedd708a584285c7a292d93377dda7691b7f264ed4e2e14b

Falsification boundary: The claim fails if more than one generated form survives; any atom, held support, alternative incidence, path, degenerate trace or successor is omitted; any of eight endpoint inventories differs; any pair lacks a positive connectivity change; any page, optimization, non-detection, signed, spectral or first-surface row is omitted; or the external record is allowed to select the law.

Measurement receipt: sha256 : 9faa4e295cc942e60e57cc63493498e8a9969bb226e49b2d9e8bf63316c588a8.
Isolation certificate: sha256 : 32d4d8b11ea6fe3b33f5e2df1e9ec6b3950669bfcf7a738773f0f18fe8449bc5.
Custody certificate: sha256 : 097ca1ec766152a2160d843af86d7992a2c050a08a5c2827c0917fe58ede93da.

Exact evidence highlight. The eight independently enumerated source/product inventories are C8H1203, C9H1403, C12H2003, C15H1803, C9H1403, C11H1803, C16H2003 and C16H2603. The printed paired mass inscriptions 179.06787, 193.08352, 235.13047, 269.1, 193.08352, 221.1, 283.1 and 289.17742 [M+Na]⁺ independently cross-check those enumerations without selecting the rearrangement law.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The organic extension derives conjugation, resonance, aromatic recurrence and its measured stability excess, antiaromatic/nonaromatic distinction, conformer classes and populations, substitution, addition, elimination and composition-retaining rearrangement with complete adverse evidence. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no native numerical zero; structural absence is EmptyOne
- no negative irrational imaginary continuum fitted free random or imported native parameter
- no reaction name mechanism yield selectivity temperature time mass formula or measured product selects the survivor
- the first incomplete blind carrier surface remains unresolved without relabelling

Immutable identities. Source manifest

sha256:cb351844398484d654b82bb023ade6db72d4776fb5e7d3b44314ded67b4f7f4f; derivation seal
 sha256:296499f3dbb6ff45080a42248436d0ce1f98111f3fb7265ac45365cf41e558bd; engine receipt
 sha256:905bd55a18ffe29d8484ad273fe9a6245121b3bf0b56f0166dd45fda91b3bdca at receipts/engine/model_admitted/SFT-CHEM-REARRANGEMENT-REACTION-FAMILY-011-905bd55a18ffe29d.json;
 empirical hash sha256:60b3fbb38ec94a826956c89e68778cccebee595aed503b8cbdbce99beb4a3bcb;
 measurement receipt sha256:9faa4e295cc942e60e57cc63493498e8a9969bb226e49b2d9e8bf63316c588a8.

185. Exact cyclic-transition and relative-orientation law

Claim identity: SFT-CHEM-PERICYCLIC-REACTION-FAMILY-012

Question and theorem. A pericyclic Fold transition retains every carrier and support inside one generated closed cycle and exhausts four joint face assignments into exactly two relative-orientation classes without measured-product selection.

complete-source-and-terminal-carriers__complete-generated-transition-cycle__every-held-support-retained-once__positive-finite-cycle-confined-incidence-change__complete-two-fibre-face-assignment-product__two-relative-classes-under-global-complement__value-free-cycle-and-stereochemistry-seal__fresh-unchanged-successor-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-QUANTUM-STATE-COMPOSITION-001

- SFT-QUANTUM-PHASE-INTERFERENCE-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001
- SFT-CHEM-ADDITION-REACTION-FAMILY-009
- SFT-CHEM-REARRANGEMENT-REACTION-FAMILY-011

Generated grammar. Generate the literal product of carrier, cycle, support, transition, phase, stereochemistry, observation and extension coordinates; decide all 256 forms only from admitted Fold conservation and composition laws.

Boundary: Every positive finite retained source/terminal carrier; every generated closed transition cycle; every moved held support; all four two-fibre face assignments; both global-complement classes; two complete IUPAC records; all 32 primary rows, 44 archive members, 12 article pages, 203 supplementary pages and the complete CIF.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-source-and-terminal-carriers__complete-generated-transition-cycle__every-held-support-retained-once__positive-finite-cycle-confined-incidence-change__complete-two-fibre-face-assignment-product__two-relative-classes-under-global-complement__value-free-cycle-and-stereochemistry-seal__fresh-unchanged-successor-no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-source-and-terminal-carriers	selected-product-fragment-or-missing-reactant	A partial carrier cannot establish a closed transformation.
cycle	complete-generated-transition-cycle	open-path-or-named-reaction-assumed	An open or named path does not force cyclic support.
supports	every-held-support-retained-once	support-created-erased-or-renamed	Unretained support destroys the exact transition trace.
transition	positive-finite-cycle-confined-incidence-change	sequential-story-or-target-selected-edit	A story or observed product cannot select the native operation.
phase	complete-two-fibre-face-assignment-product	imported-orbital-sign-or-complex-scalar	An imported sign or complex value is outside native arithmetic.
stereochemistry	two-relative-classes-under-global-complement	single-preferred-product-or-erased-alternative	One preferred product omits lawful alternatives.
observation	value-free-cycle-and-stereochemistry-seal	external-stereochemistry-open-before-seal	An opened target could select the law.
extension	fresh-unchanged-successor-no-extra-rule	species-exception-or-recomputed-prefix	A reaction-specific exception is not a general law.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256: 5308525a126c949bfa13a7cddbfb12b20224c957aded26e11a0bb05eb1e153447.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 833880f45a4eb767726b00140868865be7fdb2e73e60d7b938d69e82b8805b7d.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 49fd0c09f1c9737e3eea6aa00c943ade7a57d37bebdb7632c3fed74451a0cc6d.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: 58f1227720b54b0616af126e410e7afd99a60a6aea95285ddeaaec1311b211df.

Independent reconstruction. Implementation hash

sha256: 3990402d75ee0893fcf930b027391b621879d98b0c1c573b79bc7006d9d780e6; independent certificate sha256: 04f3eacfcf828a44eca21c60fbda6f1506b02f91c66d9448beaf13e017d67c14; external-validation hash sha256: ce8fbb56fcc9e9bbc247d970e94635a9c11cc1b554cc173b41d23c02869e6678. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC:P04491
- IUPAC:C01496
- PMC8162770:TABLE-1:ROW:1
- PMC8162770:TABLE-1:ROW:2
- PMC8162770:TABLE-1:ROW:3
- PMC8162770:TABLE-1:ROW:4
- PMC8162770:TABLE-1:ROW:5
- PMC8162770:TABLE-1:ROW:6
- PMC8162770:TABLE-1:ROW:7
- PMC8162770:TABLE-1:ROW:8
- PMC8162770:TABLE-1:ROW:9
- PMC8162770:TABLE-1:ROW:10
- PMC8162770:TABLE-1:ROW:11
- PMC8162770:TABLE-1:ROW:12
- PMC8162770:TABLE-1:ROW:13
- PMC8162770:TABLE-1:ROW:14
- PMC8162770:TABLE-1:ROW:15
- PMC8162770:TABLE-1:ROW:16
- PMC8162770:TABLE-1:ROW:17
- PMC8162770:TABLE-1:ROW:18
- PMC8162770:TABLE-1:ROW:19
- PMC8162770:TABLE-1:ROW:20
- PMC8162770:TABLE-1:ROW:21
- PMC8162770:TABLE-1:ROW:22

- PMC8162770:TABLE-1:ROW:23
- PMC8162770:TABLE-1:ROW:24
- PMC8162770:TABLE-1:ROW:25
- PMC8162770:TABLE-1:ROW:26
- PMC8162770:TABLE-1:ROW:27
- PMC8162770:TABLE-1:ROW:28
- PMC8162770:TABLE-1:ROW:29
- PMC8162770:TABLE-1:ROW:30
- PMC8162770:TABLE-1:ROW:31
- PMC8162770:TABLE-1:ROW:32
- PMC8162770:ARTICLE
- PMC8162770:SUPPLEMENT:S001
- PMC8162770:SUPPLEMENT:S002
- NCBI-OA:PMC8162770

Comparison records:

- complete official 44-member open-access package and all 43 files retained
- all 32 primary reaction rows retained in source order
- 28 reported experimental ratios and four unresolved rows retained
- 22 first-class preferences, five second-class preferences and one equal row observed
- both relative orientation classes reported in all 28 measured rows
- complete 12-page article, 203-page supplement and crystallographic file retained
- complete result vector sha256:223a6b5670d682fb1eca677e3e239119f56d4783fe158b6e2b45f2b68c156605

Falsification boundary: The claim fails if more than one generated form survives; any carrier, cycle edge, held support, face assignment, relative class or unchanged successor is omitted; any of 32 primary rows, 44 archive members, 12 article pages, 203 supplementary pages, the CIF, favorable, adverse, equal, absent or unresolved row is omitted; the reported vector lacks representatives of both generated classes; or any external outcome selects the native law.

Measurement receipt: sha256:96ccb759ab343a1af9746b221aaeb8bead717053a77f3a0056e25ff10e087c07.
Isolation certificate: sha256:a182103caa19c8b8125ae3f6819f68d4fd424f161604deac66aa8328c320fc78.
Custody certificate: sha256:c6efbf548434397eae1c9ea182573a1d1f03b930cbd2f4ac4e5ad5b7ec0f11be.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: ()
- Chemical Science and PubMed Central: ()

Registered target identities:

- None.

Meaning. The organic extension derives conjugation, resonance, aromatic recurrence and its measured stability excess, antiaromatic/nonaromatic distinction, conformer classes and populations, substitution, addition, elimination and composition-retaining rearrangement with complete adverse evidence. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no native numerical zero; structural absence is EmptyOne and may be displayed as 0

- no negative irrational imaginary continuum fitted free random or imported native parameter
- no orbital rule reaction name energy yield ratio temperature condition or measured product selects the survivor
- all favorable adverse equal absent unresolved computational and experimental rows remain preserved

Immutable identities. Source manifest

sha256:63429a36a8dca0e01eca85c570b62c41f8dddacaf6c29cad7465eaa07fd81a99; derivation seal
 sha256:620a92e7035cb67944b028d4f70ffb1c64965cd1bbbda7a14f06269501e5415a; engine receipt
 sha256:608a48d3e1be90c6dbafdc6fd68fa731841c9c7b8a69cca351b05e54689a871f at receipts/engine/model_admitted/SFT-CHEM-PERICYCLIC-REACTION-FAMILY-012-608a48d3e1be90c6.json; empirical
 hash sha256:0d4a0514a10e54c9567197add992e73c3fc473df51243f46c392b3108dfc3f94; measurement
 receipt sha256:96ccb759ab343a1af9746b221aaeb8bead717053a77f3a0056e25ff10e087c07.

186. Exact initiation, propagation and termination network law

Claim identity: SFT-CHEM-RADICAL-REACTION-NETWORK-013

Question and theorem. A complete radical network retains two exact held support labels from initiation through positive finite propagation and closes the same labels at termination, with inactive support represented only by EmptyOne.

complete-retained-network-carrier__two-exact-held-radical-supports__closed-pair-opens-to-two-retained-active-sites__one-active-support-and-one-bond-layer-relocate__positive-finite-contiguous-propagation-family__two-active-supports-close-to-EmptyOne__value-free-initiation-propagation-termination-seal__fresh-unchanged-carrier-successor-no-extra-rule

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-RXN-INTERMEDIATE-001
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001
- SFT-CHEM-ADDITION-REACTION-FAMILY-009
- SFT-CHEM-PERICYCLIC-REACTION-FAMILY-012

Generated grammar. Generate the literal product of carrier, support, initiation, propagation, recurrence, termination, observation and extension coordinates; decide all 256 forms only from admitted Fold laws.

Boundary: Every complete finite carrier, two retained radical labels, exact pair opening, every positive finite propagation recurrence, exact pair closure, all three IUPAC records, all 67 article table rows, all 24 archive members and all 29 article/supplement pages.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-retained-network-carrier__two-exact-held-radical-supports__closed-pair-opens-to-two-retained-active-sites__one-active-support-and-one-bond-layer-relocate__positive-finite-contiguous-propagation-family__two-active-supports-close-to-EmptyOne__value-free-initiation-propagation-termination-seal__fresh-unchanged-carrier-successor-no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-retained-network-carrier	selected-chain-fragment-or-erased-coproduct	A fragment cannot close a radical network.
support	two-exact-held-radical-supports	created-destroyed-or-dot-only-radical	A typographic dot does not retain support identity.
initiation	closed-pair-opens-to-two-retained-active-sites	free-radical-assumed-or-randomly-created	An assumed active center is not generated.
propagation	one-active-support-and-one-bond-layer-relocate	mass-increase-story-or-measured-chain-selector	A name or observed chain length cannot select the step.
recurrence	positive-finite-contiguous-propagation-family	single-step-or-unbounded-assertion	One example does not force the chain family.
termination	two-active-supports-close-to-EmptyOne	active-support-erased-or-numerical-zero	Erasure or numerical zero destroys reversibility.
observation	value-free-initiation-propagation-termination-seal	external-chain-vector-open-before-seal	Opened outcomes could choose the law.
extension	fresh-unchanged-carrier-successor-no-extra-rule	species-exception-or-recomputed-prefix	A species exception is not a general chain law.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f331de062c1719bdc0fb4693b92b1e9b057d0e5f7bb6b9343c828c52a40d795f`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:a3e980b41bc75fa655ef5b77085e37cfb84d2c054d7f65770143761500b93196`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3ed11f5b4cd5b7edc0b894162d7438996f603b032a498c62326cf1a0db8eb5c7`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:04d3cfffed9cbeec3a0952196b6450fa2203fcf648efb12a053a48007b35500a1`.

Independent reconstruction. Implementation hash

`sha256:0f9ad3232463ca679b0d4576d7070eeb59e9e4a0de80b2ac5f68264e1185e585`; independent certificate `sha256:183dd66840f6b214df97a611d21c02cb84a8dc1678f0b2f8c95e8819cdd37f75`; external-validation hash `sha256:006c04fe386cb54184c83c2354d98174cbe68de47bca21fb23f2fd431f3ee7fb`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC:C00960
- IUPAC:I03042
- IUPAC:T06274
- PMC11598545:TABLE:1:ROW:1
- PMC11598545:TABLE:1:ROW:2
- PMC11598545:TABLE:1:ROW:3
- PMC11598545:TABLE:1:ROW:4
- PMC11598545:TABLE:1:ROW:5
- PMC11598545:TABLE:1:ROW:6
- PMC11598545:TABLE:1:ROW:7
- PMC11598545:TABLE:1:ROW:8
- PMC11598545:TABLE:1:ROW:9
- PMC11598545:TABLE:1:ROW:10
- PMC11598545:TABLE:1:ROW:11
- PMC11598545:TABLE:1:ROW:12
- PMC11598545:TABLE:1:ROW:13
- PMC11598545:TABLE:1:ROW:14
- PMC11598545:TABLE:1:ROW:15
- PMC11598545:TABLE:1:ROW:16
- PMC11598545:TABLE:1:ROW:17
- PMC11598545:TABLE:1:ROW:18
- PMC11598545:TABLE:1:ROW:19
- PMC11598545:TABLE:1:ROW:20
- PMC11598545:TABLE:1:ROW:21
- PMC11598545:TABLE:1:ROW:22
- PMC11598545:TABLE:1:ROW:23
- PMC11598545:TABLE:1:ROW:24
- PMC11598545:TABLE:1:ROW:25
- PMC11598545:TABLE:1:ROW:26
- PMC11598545:TABLE:1:ROW:27
- PMC11598545:TABLE:1:ROW:28
- PMC11598545:TABLE:1:ROW:29
- PMC11598545:TABLE:1:ROW:30
- PMC11598545:TABLE:1:ROW:31
- PMC11598545:TABLE:1:ROW:32
- PMC11598545:TABLE:1:ROW:33

- PMC11598545:TABLE:1:ROW:34
- PMC11598545:TABLE:1:ROW:35
- PMC11598545:TABLE:1:ROW:36
- PMC11598545:TABLE:1:ROW:37
- PMC11598545:TABLE:1:ROW:38
- PMC11598545:TABLE:1:ROW:39
- PMC11598545:TABLE:1:ROW:40
- PMC11598545:TABLE:1:ROW:41
- PMC11598545:TABLE:1:ROW:42
- PMC11598545:TABLE:1:ROW:43
- PMC11598545:TABLE:1:ROW:44
- PMC11598545:TABLE:1:ROW:45
- PMC11598545:TABLE:1:ROW:46
- PMC11598545:TABLE:1:ROW:47
- PMC11598545:TABLE:2:ROW:1
- PMC11598545:TABLE:2:ROW:2
- PMC11598545:TABLE:2:ROW:3
- PMC11598545:TABLE:2:ROW:4
- PMC11598545:TABLE:2:ROW:5
- PMC11598545:TABLE:2:ROW:6
- PMC11598545:TABLE:2:ROW:7
- PMC11598545:TABLE:2:ROW:8
- PMC11598545:TABLE:2:ROW:9
- PMC11598545:TABLE:2:ROW:10
- PMC11598545:TABLE:2:ROW:11
- PMC11598545:TABLE:2:ROW:12
- PMC11598545:TABLE:3:ROW:1
- PMC11598545:TABLE:3:ROW:2
- PMC11598545:TABLE:3:ROW:3
- PMC11598545:TABLE:3:ROW:4
- PMC11598545:TABLE:4:ROW:1
- PMC11598545:TABLE:4:ROW:2
- PMC11598545:TABLE:4:ROW:3
- PMC11598545:TABLE:4:ROW:4
- PMC11598545:ARTICLE
- PMC11598545:SUPPLEMENT:S001
- NCBI-OA:PMC11598545

Comparison records:

- complete official 24-member package and all 23 files retained
- all four primary tables and all 67 rows retained in source order
- initiation, positive propagation, two-active-support termination and chain-length dependence retained
- all 24 article pages and all five supplementary pages retained
- all external energy, rate, chain-length, temperature, concentration, uncertainty, sign, unit and absence inscriptions retained
- complete result vector sha256:d943e9683af16bb424d13b9def0d077a6cb02dff36a7b72e85f62be132712dd

Falsification boundary: The claim fails if more than one generated form survives; either held radical support is created, erased or renamed; initiation, positive propagation, termination or contiguous order is omitted; active absence is made numerical; any IUPAC record, archive member, one of 67 table rows, 29 pages, measured value, sign, unit, uncertainty, adverse or absent field is omitted; or an external outcome selects the native law.

Measurement receipt: sha256:d11894de785d359b7079e8ebf619e77798d61ca07a47b631c1ab5571535fb247.

Isolation certificate: sha256:4c69eb677ba00056a01cb6b97c3118b9a5cf9388acc7991a3579406ef2f48889.

Custody certificate: sha256:016f6df93b8fd0dd0b3445a7ac13056a2d9b88f841d53e8f644b2fbb8097fbb2.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: ()
- Polymers and PubMed Central: ()

Registered target identities:

- None.

Meaning. The organic extension derives conjugation, resonance, aromatic recurrence and its measured stability excess, antiaromatic/nonaromatic distinction, conformer classes and populations, substitution, addition, elimination and composition-retaining rearrangement with complete adverse evidence. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no native numerical zero; inactive support is structural EmptyOne
- no negative irrational imaginary continuum fitted free random or imported native parameter
- no radical dot reaction name rate energy temperature chain length or measured outcome selects the survivor
- all signs units uncertainties adverse absent and unresolved fields remain preserved

Immutable identities. Source manifest

sha256:a9d1a745f20556e014ec748fd09952e1f50df8459a4ccc9d441600c81d655b9d; derivation seal
 sha256:1ce3cb988bfcd9c6b2355f47de5a0eb06d13f1c685405b07b90d95f1648589d; engine receipt
 sha256:97ec902a0830f6578f2ca981be588a9906b4ff0d7eaa7a277e29c778f790860 at receipts/engine/model_admitted/SFT-CHEM-RADICAL-REACTION-NETWORK-013-97ec902a0830f657.json; empirical hash
 sha256:714cd7f10077dcfb017c65a2b8f3228c3ea8144caa24f2b25e5352d40e6e4f52; measurement receipt
 sha256:d11894de785d359b7079e8ebf619e77798d61ca07a47b631c1ab5571535fb247.

187. Exact selectivity partitions and complete product distributions

Claim identity: SFT-CHEM-SELECTIVITY-COMplete-DISTRIBUTION-014

Question and theorem. Chemo-, regio- and stereoselectivity are exact partitions of the complete positive-finite product support, with every reported amount attached only after structural sealing and no major-product filter.

complete-positive-finite-product-support__exact-chemo-site-partition__exact-regio-site-partition__exact-stereo-site-partition__postseal-held-amount-record-per-reported-product__complete-registered-product-distribution__exact-partition-and-EmptyOne-absence__fresh-product-successor-preserves-all-prior-classes

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-CAT-SELECTIVITY-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-CHEM-STEREO-DIASTEREOMER-001
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001
- SFT-CHEM-PERICYCLIC-REACTION-FAMILY-012
- SFT-CHEM-RADICAL-REACTION-NETWORK-013

Generated grammar. Generate the Cartesian product of the eight registered binary structural decisions exactly once.

Boundary: Eight dimensions exhaust carrier, three selectivity partitions, amount custody, complete observation, native arithmetic and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-positive-finite-product-support__exact-chemo-site-partition__exact-regio-site-partition__exact-stereo-site-partition__postseal-held-amount-record-per-reported-product__complete-registered-product-distribution__exact-partition-and-EmptyOne-absence__fresh-product-successor-preserves-all-prior-classes`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-positive-finite-product-support	major-product-only-carrier	A selected product erases the alternative support.
chemo	exact-chemo-site-partition	named-functional-group-preference	A reaction name cannot select a functional-group outcome.
regio	exact-regio-site-partition	imported-direction-rule	An imported orientation rule cannot select a site.
stereo	exact-stereo-site-partition	preferred-stereoisomer-only	A preferred stereoisomer erases lawful alternatives.
amount	postseal-held-amount-record-per-reported-product	amount-selects-product-support	A measured amount cannot choose which products are generated.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
observation	complete-registered-product-distribution	favourable-or-major-row-filter	Filtering destroys falsifiability.
arithmetic	exact-partition-and-EmptyOne-absence	signed-decimal-probability-native-law	External continuum arithmetic is outside native forcing.
extension	fresh-product-successor-preserves-all-prior-classes	reaction-specific-exception	An exception fitted to a reaction is an extra rule.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:ad2f8eb320c018af1097926cb335f45e6c719a8296e3aa4c776983e64f3d332d.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:8c9d48d3447227b1e221e47bee565cbdcc59f1babbf3e95b58fb0880134de6e2.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:4841844bc9e05a635550a5ece7d53e23d592f25cc085544e59ce5b6ded12a203.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:6d621bf8220b2414b50089e0601834d14fd42f5913c74430e7aa8bf63bf059c3.

Independent reconstruction. Implementation hash

sha256:0e96be579d7ad226c41883944c9ed2cd9f10bce03d57759606ddfdb6f01b01a7; independent certificate sha256:d232ff3253ab8aae5cdbc197deb7a79b6cc92a982fdbc90e552bdda8853d8; external-validation hash sha256:cda0c88afa1ea82a19033c22551c502f773369701c13b0d872363e7a94b4599b. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-C01051
- IUPAC-R05243
- IUPAC-S05991
- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC
- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC
- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC
- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC
- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC
- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC
- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC
- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC
- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC
- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC
- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC
- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC
- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC
- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC
- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC

- 1451

- 1452

- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC
- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC
- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC

Comparison records:

- all 130 preregistered ORD reaction rows and 130 outcomes retained in source order
- all 152 products, 302 product identifiers and 195 product measurements retained
- all 19 multi-product rows retained and no major-product filter applied
- all amount absences and 147 positive external numeric inscriptions retained; no observed signed-zero or negative inscription hidden
- complete result vector sha256:611cbab61acea638fe76edf8c44c4e2fe88ae11e4a26568369c8253eab7c31

Falsification boundary: The claim fails if more than one form survives; any generated product or selectivity partition is erased; a reported amount selects product support; any one of 130 preregistered reactions, 130 outcomes, 152 products, 302 identifiers, 195 measurement records, adverse or absent row is omitted; a major-product filter is used; or external outcomes enter native candidate generation or elimination.

Measurement receipt: sha256:a66b8c5f16fb059b4f8a24f573efc0d58d8e095af0286acb027cbce6eacda243.

Isolation certificate: sha256:8663edd73fb7f115fff82e97e7791b94751401e01398bab782d1a025bb167a41.

Custody certificate: sha256:5dc412ea203378f9dfd578719adf2c249dc8004cf45d0f2f39849ebe20352d74.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: ()
- Open Reaction Database: ()

Registered target identities:

- None.

Meaning. The organic extension derives conjugation, resonance, aromatic recurrence and its measured stability excess, antiaromatic/nonaromatic distinction, conformer classes and populations, substitution, addition, elimination and composition-retaining rearrangement with complete adverse evidence. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no native numerical zero, negative, irrational, imaginary, continuum probability, fitted, free, random or imported parameter
- no named reaction, measured yield, product amount or major-product designation selects the survivor
- all products, identifiers, measurements, signs, absences, adverse and unresolved rows remain preserved

Immutable identities. Source manifest

sha256:a2cc90e99300cabd510a67f45b268d58dd17e777ca72dd21540fd84fbc82533c; derivation seal

sha256:db20964edbdba021996149bf9cc646271d6bcd15d2e2337ddd9c8335fd1af97f6; engine receipt

sha256:765110af963aac06676adda2fffc9873d77d02848f7ae3c0c7c4095aa8d4ed000 at receipts/engine/model_admitted/SFT-CHEM-SELECTIVITY-COMPLETE-DISTRIBUTION-014-765110af963aac06.json;

empirical hash sha256:73058c9fb4118cebffe99ae8585e32303127d6edb88545f5289da4357d7a7144;

measurement receipt sha256:a66b8c5f16fb059b4f8a24f573efc0d58d8e095af0286acb027cbce6eacda243.

188. Exact protecting-group reversible state and restoration

Claim identity: SFT-CHEM-PROTECTING-GROUP-REVERSIBLE-STATE-015

Question and theorem. A complete carrier undergoes a temporary held functional transform, remains unchanged through a declared challenge and exactly restores its source function with endpoint absence represented by EmptyOne.

complete-retained-functional-carrier__held-target-function-identity__temporary-exact-functional-transform__one-retained-protecting-support__held-state-unchanged-under-declared-challenge__exact-source-function-restoration__endpoi

nt-protector-absence-is-EmptyOne__fresh-carrier-successor-preserves-cycle

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-QUANTUM-REVERSIBLE-MODEL-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001
- SFT-CHEM-SELECTIVITY-COMPLETE-DISTRIBUTION-014

Generated grammar. Generate the literal product of eight registered binary protection-cycle decisions.

Boundary: Eight dimensions exhaust carrier, target, transform, protector, challenge, restore, absence and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *complete-retained-functional-carrier__held-target-function-identity__temporary-exact-functional-transform__one-retained-protecting-support__held-state-unchanged-under-declared-challenge__exact-source-function-restoration__endpoint-protector-absence-is-EmptyOne__fresh-carrier-successor-preserves-cycle*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-retained-functional-carrier	reactive-fragment-only	A fragment cannot certify restoration.
target	held-target-function-identity	named-group-assumption	A name is not an exact target.
transform	temporary-exact-functional-transform	permanent-or-erasing-change	A permanent or erasing change is not protection.
protector	one-retained-protecting-support	free-or-discarded-protector	An untracked protector cannot reverse the step.
challenge	held-state-unchanged-under-declared-challenge	unverified-inertness-story	A label alone cannot establish a protected interval.
restore	exact-source-function-restoration	approximate-or-renamed-endpoint	Approximate restoration loses identity.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
absence	endpoint-protector-absence-is-EmptyOne	protector-equals-numerical-zero	Numerical zero is not structural absence.
extension	fresh-carrier-successor-preserves-cycle	molecule-specific-exception	An exception is an extra rule.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:310659c7997128b5ddd2eace1aab1ed2a9fb99ebdad938e75522098c3c922ae0.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:6bc1d2738fb9ef8e2d59ea441e8805144e3e9d807f19c44aea2dad262a740ea5.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:1b8140662fce16e5581e3ff8d2bb14bf7a102d6e3beb0a133e8c237fa8d2e4c7.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:48a90fca7fddefa3dbe979d3e0280e6670ced80f484356a07744dabdedcfb20f.

Independent reconstruction. Implementation hash

sha256:5e3dcab196e6f145de20061f47f745725d0801d06ee627fe3b661cbfe3a5c7e3; independent certificate sha256:712a42c624386e03df76e47f87270a1ad4f251ad8cb6bce13b6d186f61a53e9b; external-validation hash sha256:f1c3cfb37cff98d35ea74cd9dd4191e1d675ab1229de02a699ce1e3c856b9c41. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-09560
- IUPAC-09560
- IUPAC-09560
- IUPAC-09466
- IUPAC-09466

Comparison records:

- complete protecting-group and deprotection records retained
- temporary transform, protected interval and removal features reproduced
- narrow polymeric-resist scope retained without inflating it to exact restoration
- exact restoration remains the independently generated formal result

Falsification boundary: The claim fails if more than one form survives; carrier, target, protector, protected interval or exact endpoint is lost; endpoint absence is numerical; either complete IUPAC record or its narrow scope is omitted; or an external phrase selects the native law.

Measurement receipt: sha256:b392b3084272093f7942a9ae1b9edf0e35aa45f7568fc83d4c2c36eb70a021a9.

Isolation certificate: sha256:f14ca665ba07628ff96f95f449ad6509c62ca6c8b922a01965e528c35b8574f7.

Custody certificate: sha256:ac16816341d479e504a6745930c46b4914b9287bc7b624f9309b49129d67cd0d.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: ()

Registered target identities:

- None.

Meaning. The organic extension derives conjugation, resonance, aromatic recurrence and its measured stability excess, antiaromatic/nonaromatic distinction, conformer classes and populations, substitution, addition, elimination and composition-retaining rearrangement with complete adverse evidence. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no source phrase or named group selects the survivor
- narrow source scope is preserved without inflation

Immutable identities. Source manifest

sha256:4613433d03e0f65e47284665502846aef62f0e79c6ab42f47e48f1895eea8cab; derivation seal
 sha256:a45ed9f807ad730055c35608146e51be20cf3d0fadc88bd05ce272cd2d6ad1d4; engine receipt
 sha256:2f78de0d43da42ba65f04328fb2b46e04bda108b69952c52500c1736da1b0681 at receipts/engine/model_admitted/SFT-CHEM-PROTECTING-GROUP-REVERSIBLE-STATE-015-2f78de0d43da42ba.json;
 empirical hash sha256:0f771539e4b7b1c1b13b1a26b146d223f0bfc201e79a0ef9ceb956f928ead07c;
 measurement receipt sha256:b392b3084272093f7942a9ae1b9edf0e35aa45f7568fc83d4c2c36eb70a021a9.

189. Exhaustive retrosynthetic decomposition and exact forward reconstruction

Claim identity: SFT-CHEM-RETROSYNTHETIC-DECOMPOSITION-RECONSTRUCTION-016

Question and theorem. Every positive binary decomposition tree of a complete finite target is generated exactly once and every path forward-reconstructs the exact target.

*complete-held-target-carrier__every-positive-binary-split-generated__recursive-decomposition-to-One-leaves__finite-a
 cyclic-unique-tree-census__every-edge-has-exact-forward-inverse__all-generated-paths-forward-reconstruct-target__v
 alue-free-graph-sealed-before-comparison__fresh-occurrence-successor-generates-all-new-splits*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-ALG-TREES-GRAPHS-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007

- SFT-CHEM-PARALLEL-MECHANISM-COMPOSITION-008
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001
- SFT-CHEM-PROTECTING-GROUP-REVERSIBLE-STATE-015

Generated grammar. Generate the literal product of eight registered binary decomposition decisions.

Boundary: Eight dimensions exhaust target, cuts, recursion, graph, forward inverse, paths, observation and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-held-target-carrier__every-positive-binary-split-generated__recursive-decomposition-to-One-leaves__finite-acyclic-unique-tree-census__every-edge-has-exact-forward-inverse__all-generated-paths-forward-reconstruct-target__value-free-graph-sealed-before-comparison__fresh-occurrence-successor-generates-all-new-splits`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
target	complete-held-target-carrier	selected-target-fragment	A fragment cannot certify the synthesis target.
cuts	every-positive-binary-split-generated	chosen-disconnection-only	One favored disconnection is not exhaustive.
recursion	recursive-decomposition-to-One-leaves	one-step-retrosynthetic-story	One step does not close the graph.
graph	finite-acyclic-unique-tree-census	cyclic-or-duplicate-search	Cycles or duplicates corrupt the finite census.
forward	every-edge-has-exact-forward-inverse	unverified-plausible-route	Plausibility is not reconstruction.
paths	all-generated-paths-forward-reconstruct-target	successful-route-only	Keeping one successful route hides alternatives.
observation	value-free-graph-sealed-before-comparison	external-mechanism-selects-tree	External examples cannot select the graph.
extension	fresh-occurrence-successor-generates-all-new-splits	fixed-depth-exception	A fixed lookup is not a general law.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:0b72f405b358e02add2df4b70db9039d419032dea626816a305efebd23a5a1b1`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f11e5afa2d124f6b341da50e464139d1dcc1d5f36dafd64334fa5cab8a52231f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c441958ce77e7521e7c5006be201d30c30542c5a676fe175a5cfccd5a91ed057`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:76ae0f6df30327df9f68536a0529cc2256a80be00f88b5f9a901cd20dfae8fab`.

Independent reconstruction. Implementation hash

`sha256:c505502ee1044ae4513beda99fa7df4b031794ac957ecfc093682ba7e14ffb62`; independent certificate

sha256:845ca4787d9e8730f92804de357cf217ede94aaafc820b7dc64e145a41c769f9a; external-validation hash sha256:b4fd754800c7a55d50285133f1ecaabc29d72d503367ff316784c61f70b0ed73. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-M03804
- IUPAC-M03804
- IUPAC-M03804
- IUPAC-M03804
- IUPAC-R05174
- IUPAC-R05174
- IUPAC-R05174

Comparison records:

- both complete IUPAC records retained
- five four-leaf trees and fourteen five-leaf successor trees generated exactly
- all nineteen bounded trees forward-reconstruct their exact target
- alternative mechanisms and incomplete-evidence disclosures retained

Falsification boundary: The claim fails if more than one form survives; a split or tree is omitted or duplicated; a path does not exactly reconstruct the target; either complete IUPAC record, alternatives or incomplete-evidence disclosure is omitted; or an external route selects the graph.

Measurement receipt: sha256:fa7ab759bca0a1484a7676937e44db9fa076f0cd04d18e9324336f8e97d179ab.

Isolation certificate: sha256:d1a444cec6e98fd81b898a4c5a65f277333d93fb074b1510bc514c0ca5b7dcdb.

Custody certificate: sha256:b6360800cc16a14600a5ad66346b9339e720f8abc032f8b537750111b986348a.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: ()

Registered target identities:

- None.

Meaning. The organic extension derives conjugation, resonance, aromatic recurrence and its measured stability excess, antiaromatic/nonaromatic distinction, conformer classes and populations, substitution, addition, elimination and composition-retaining rearrangement with complete adverse evidence. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external route or measured success selects a tree
- no successful-path-only filtering

Immutable identities. Source manifest

sha256:a56aa3b5fee5e22920a6b31ce5989d9eebe727d22690bb342163ce4e5b12e3ef; derivation seal

sha256:d703b40b3725707312d9dc451a7fde6937343f8253e40f5678df48899f972bc3; engine receipt

sha256:2a460669175e8ea2cdb4c29de14025a2f470ad886c2534d9c71004c8ca9980ed at receipts/engine/model_admitted/SFT-CHEM-RETROSYNTHETIC-DECOMPOSITION-RECONSTRUCTION-016-2a460669175e8ea2.json; empirical hash

sha256:b1ab1ddea6485c0cb7d4e4bc1790146f52a01b747a2bd151ce69fc21213f982d; measurement receipt

sha256:fa7ab759bca0a1484a7676937e44db9fa076f0cd04d18e9324336f8e97d179ab.

24. Quantitative electrochemistry - 13 admitted claims

Half-reactions, cell composition, transport, polarization, interfaces and storage handoff retain one-owner ledgers. This entire registered subcategory is included; no proper subset is presented as field completion.

190. Fold half-reaction identity and held transfer orientation

Claim identity: SFT-CHEM-HALF-REACTION-IDENTITY-ORIENTATION-001

Question and theorem. A half-reaction retains complete species, phases, positive held electron occurrences, reversible source/terminal orientation and one explicit reference half-cell; no negative electron-count proof value is required.

exact-half-reaction-identity__complete-species-phase-carrier__positive-count-of-held-electron-carriers__held-source-terminal-orientation__exact-inverse-half-reaction-pair__held-reference-half-cell-identity__complete-half-reaction-record__fresh-species-successor-preserves-prior-record

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-INORGANIC-ACID-BASE-REDOX-NETWORK-017
- SFT-CHEM-RETROSYNTHETIC-DECOMPOSITION-RECONSTRUCTION-016

Generated grammar. Generate the literal product of eight registered binary half-reaction decisions.

Boundary: Eight dimensions exhaust identity, species, transfer, orientation, inverse pairing, reference, record and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *exact-half-reaction-identity__complete-species-phase-carrier__positive-count-of-held-electron-carriers__held-source-terminal-orientation__exact-inverse-half-reaction-pair__held-reference-half-cell-identity__complete-half-reaction-record__fresh-species-successor-preserves-prior-record*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
identity	exact-half-reaction-identity	equation-text-only	Text cannot retain a process identity.
species	complete-species-phase-carrier	selected-or-unphased-species	Missing species or phase changes the process.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
transfer	positive-count-of-held-electron-carriers	signed-electron-count	A negative number hides carrier identity and orientation.
orientation	held-source-terminal-orientation	sign-as-direction	A sign is not a reversible direction record.
pairing	exact-inverse-half-reaction-pair	open-oxidation-or-reduction-story	An open half cannot identify its inverse.
reference	held-reference-half-cell-identity	unbound-potential-answer	Potential is relational and needs its reference.
record	complete-half-reaction-record	favourable-species-only	Deleting species destroys stoichiometric closure.
extension	fresh-species-successor-preserves-prior-record	reaction-specific-exception	An exception is an extra rule.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:d06685a3b619eaf266d1d6e78f58ef6c21c4df93bb1a34b68a81c37fe4a34ddf.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:bd95fcc5c3bb51943f81504fca4ae5c867f858be873499fc1c7fa62bdf123ba.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:cf3594fe5bac90085b923b7a4c996d7c7f517df094cdd1f85553e02657d30a50.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:b5b46756384bfedfdfee5bae2a382c1e2d5932100a66b050ff879d8d9a175bc5.

Independent reconstruction. Implementation hash

sha256:b89b8b335fe14db5d189b71e59b04d98119ca923e2b875abdfa1331b97d56234; independent certificate sha256:5536cf4aa39131e16a2e309a968d4b65c4cb07ec40d6e4f211b1bfb1e9c8e150; external-validation hash sha256:311e69a59aba80acffa8611a641b0ceb2c696a646e013b7c85ca12d0c700299e. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-RT06783
- IUPAC-RT06783
- IUPAC-RT06783
- NIST-JPCRD-STANDARD-ELECTRODE-POTENTIALS-1989
- NIST-JPCRD-STANDARD-ELECTRODE-POTENTIALS-1989
- NIST-JPCRD-STANDARD-ELECTRODE-POTENTIALS-1989

Comparison records:

- complete IUPAC redox-potential record retained
- all 22 NIST pages and 99,794 extracted characters retained
- paired half-reactions, orientation, reference and standard-state conditions reproduced
- all conventional table values signs phases units uncertainty and absences retained downstream

Falsification boundary: The claim fails if more than one form survives; species, phase, transfer carrier, orientation, inverse or reference is omitted; the complete IUPAC record or any of 22 NIST pages is omitted; or a conventional sign or value selects the native law.

Measurement receipt: sha256:52d85a7c8887ad2a07b92dcce37d68daaab5788fbbc6e9e27ed435b6b2d4ba7d.
Isolation certificate: sha256:ea222d5b73969e56c2f291c9c5f88443191e53a685c2a943080df326198f4747.
Custody certificate: sha256:02527f1d032f76bb707954d1177912ee523e555d76d64e2e2dedf9fad4d468b2.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: ()
- National Institute of Standards and Technology: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no conventional sign potential value or external equation selects the survivor
- all 22 NIST pages and the complete IUPAC record remain downstream evidence

Immutable identities. Source manifest

sha256:649797855f25d2434c81651c60103f1d0546756a154fa99ffb1b182919a74066; derivation seal
sha256:e77411d610a906f9761d070b053507f1951bba22202a8d190479d978635d8613; engine receipt
sha256:b62e7c5b5b4ea98b691734a7099344965bc555c1b0f8e2d4f9ec5e6433a2cd1a at receipts/engine/model_admitted/SFT-CHEM-HALF-REACTION-IDENTITY-ORIENTATION-001-b62e7c5b5b4ea98b.json;
empirical hash sha256:ce1c40ec7b294dbf37b361f75659afa125ba76fe5d69b3dade138886b39777aa;
measurement receipt sha256:52d85a7c8887ad2a07b92dcce37d68daaab5788fbbc6e9e27ed435b6b2d4ba7d.

191. Fold electrode-potential chemical relation

Claim identity: SFT-CHEM-ELECTRODE-POTENTIAL-CHEMICAL-RELATION-002

Question and theorem. Electrode potential is the exact held above, below or coincident relation between complete half-cell work-per-carrier accounts at one reference and condition.

complete-half-cell-transfer-account__one-held-reference-electrode__same-held-standard-condition__exact-work-per-held-transfer-carrier__held-above-below-or-coincident-orientation__structural-EmptyOne-reference-coincidence__complete-standard-potential-reference-vector__common-work-successor-preserves-order

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-COMBINATORICS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001

- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-HALF-REACTION-IDENTITY-ORIENTATION-001

Generated grammar. Generate the literal product of eight registered electrode-potential decisions.

Boundary: Eight dimensions exhaust account, reference, condition, normalization, orientation, coincidence, record and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: complete-half-cell-transfer-account__one-held-reference-electrode__same-held-standard-condition__exact-work-per-held-transfer-carrier__held-above-below-or-coincident-orientation__structural-EmptyOne-reference-coincidence__complete-standard-potential-reference-vector__common-work-successor-preserves-order. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
account	complete-half-cell-transfer-account	voltage-answer-only	A displayed voltage does not retain the chemical transfer account.
reference	one-held-reference-electrode	unbound-electrode-number	An unbound number has no electrode reference.
condition	same-held-standard-condition	mixed-temperature-or-state	Changing conditions destroys comparability.
normalization	exact-work-per-held-transfer-carrier	unnormalized-total-work	Total work changes with carrier multiplicity.
orientation	held-above-below-or-coincident-orientation	signed-native-number	A sign imports a negative proof magnitude.
coincidence	structural-EmptyOne-reference-coincidence	numerical-zero-potential	Numerical zero is not a native magnitude.
record	complete-standard-potential-reference-vector	selected-standard-potential	A selected value can hide species, phase, uncertainty or estimates.
extension	common-work-successor-preserves-order	species-specific-offset	A fitted offset adds an unforced rule.

Adverse controls. All mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256 : 36a6db7a1050a8764cfef7753455c96761a2f487f5b5466d089c38461135e5e9.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 90883bf9557f0498b465620c0ec6ca33aee55e8bec6c6fb687e2e1343d5db55.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256 : 293bf53c14752afdcfbfb148fb2af22f70a967c27d4571da9a47ef906fedbb33.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256 : da7994a1f3cf7fe3a0a3384dbdeab713351b3e512b7ff368811b28e0de8d9d44.

Independent reconstruction. Implementation hash

sha256 : c9d2c4b54b46ff835928f72caffecbb252d7beac09184f40e8288d96e3edb15a; independent certificate sha256 : 7bd6f5e406d0fc7d19e9c72e582df2b6b3145ec4b3de05827f977c98b79dc0ce; external-validation

hash sha256:ca7f6f03018a5a5ff49fe49dfc0204f0c8202e6e811d1651eeb05d3bc6593775. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-JRES-AGCL-STANDARD-POTENTIAL-1954
- NIST-JRES-AGCL-STANDARD-POTENTIAL-1954
- NIST-JRES-AGCL-STANDARD-POTENTIAL-1954
- NIST-JRES-AGCL-STANDARD-POTENTIAL-1954
- NIST-JRES-AGCL-STANDARD-POTENTIAL-1954
- NIST-JRES-AGCL-STANDARD-POTENTIAL-1954
- NIST-JRES-AGCL-STANDARD-POTENTIAL-1954

Comparison records:

- complete eight-page NIST Ag/AgCl source retained
- all 7 separately registered claim targets retained
- all conventional values signs zero glyphs fitted equations smoothing corrections uncertainties adverse and absent records remain downstream provenance

Falsification boundary: The claim halts if the survivor is nonunique; any reference, condition, species, phase, potential, uncertainty, adverse discrepancy or NIST page is omitted; or a measured value or fitted source model selects the law.

Measurement receipt: sha256:b812574e95152879e919ab05508da067f65e4ff30d1669a94b208758f0f98de6.

Isolation certificate: sha256:3a50dcaddbeafe3f14894362011cfe6f5ebc33fe3882fa9282eb2b544d54f3df.

Custody certificate: sha256:35e5bf5c71eaa204566baedda5b432d505e1dbcf8bf6208c88ec52e60599e7b.

Registered source descriptions:

- National Bureau of Standards / National Institute of Standards and Technology: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured standard potential temperature or uncertainty selects the survivor
- least-squares smoothing and adjustable source parameters remain downstream provenance

Immutable identities. Source manifest

sha256:0e6aa0cac4711f2ceb9959a2fd2af6de8185072bc2287ac4573d03c6d1d21b02; derivation seal

sha256:fac649ade28bde5dbc6748a2e14dfaceb975af1888950b6ad88186ca1065ab5; engine receipt

sha256:8df16ac7856b666f091b5e72f6f00e7291e042d4043619f34d8a2b29b6e8b675 at receipts/engine/model_admitted/SFT-CHEM-ELECTRODE-POTENTIAL-CHEMICAL-RELATION-002-8df16ac7856b666f.json; empirical hash sha256:fe2c29f0d4ad5b855230e4d2d6b457ad40f5ed8385b022a74adac7361ade9369; measurement receipt sha256:b812574e95152879e919ab05508da067f65e4ff30d1669a94b208758f0f98de6.

192. Fold cell-potential composition law

Claim identity: SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003

Question and theorem. Two admitted half-cell coordinates at one reference and condition compose by exact positive sum or Take along a held cell path, with EmptyOne at coincidence and exact reversal.

two-distinct-held-half-cell-carriers__one-common-held-reference__one-common-held-condition__held-source-terminal-reaction-path__exact-positive-sum-or-Take-composition__structural-EmptyOne-cell-coincidence__complete-full-cell-potential-vector__exact-cell-reversal-preserves-positive-separation

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-QUANTUM-REVERSIBLE-MODEL-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-HALF-REACTION-IDENTITY-ORIENTATION-001
- SFT-CHEM-ELECTRODE-POTENTIAL-CHEMICAL-RELATION-002

Generated grammar. Generate the literal product of eight registered full-cell composition decisions.

Boundary: Eight dimensions exhaust carriers, reference, condition, path, composition, coincidence, record and reversal.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *two-distinct-held-half-cell-carriers__one-common-held-reference__one-common-held-condition__held-source-terminal-reaction-path__exact-positive-sum-or-Take-composition__structural-EmptyOne-cell-coincidence__complete-full-cell-potential-vector__exact-cell-reversal-preserves-positive-separation*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carriers	two-distinct-held-half-cell-carriers	anonymous-voltage-pair	Anonymous values lose the two chemical half-cells.
reference	one-common-held-reference	mixed-reference-subtraction	Different references cannot compose directly.
condition	one-common-held-condition	mixed-condition-subtraction	Different conditions destroy a cell comparison.
path	held-source-terminal-reaction-path	unordered-potential-difference	An unordered difference loses reaction direction.
composition	exact-positive-sum-or-Take-composition	signed-arithmetic-premise	Signed arithmetic imports negative proof values.
coincidence	structural-EmptyOne-cell-coincidence	numerical-zero-cell-potential	Numerical zero is not a native proof magnitude.
record	complete-full-cell-potential-vector	selected-cell-voltage	A selected voltage can conceal half-cell and condition rows.
reverse	exact-cell-reversal-preserves-positive-separation	irreversible-sign-flip	A sign flip does not reconstruct the chemical path.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:1e281e07e6da6377a6340ec59573c77ac680664859607b0f07bcfab84c6cb8bd.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: f8d98370ee32227302c1c25abf1e9bc4e8112c362db7beee0e279e2d318dc700.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: a455b27c4a18b53034bae58e023774538dd74fae869e5d614d391197980a0d40.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: c038614dd1164c684674ca9fc4391d6ab93802bb658f5abfad76a4c88f0ef7e8.

Independent reconstruction. Implementation hash

sha256: e9439bf1f3d4662665aa8fe082029327e7467fb86d111c341f2ec1f9c18f0f0f; independent certificate sha256: 55bdc72d9f2f140d03e9423c7490ddbc923236a1ce43ffd25eeef75b72abb4b; external-validation hash sha256: 8bd85e0715d633e9d42e086542cb3d463339b005ec2b476a4baf60b63aa65c49. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-JRES-AGCL-COMplete-CELL-1954
- NIST-JRES-AGCL-COMplete-CELL-1954
- NIST-JRES-AGCL-COMplete-CELL-1954
- NIST-JRES-AGCL-COMplete-CELL-1954
- NIST-JRES-AGCL-COMplete-CELL-1954
- NIST-JRES-AGCL-COMplete-CELL-1954
- NIST-JRES-AGCL-COMplete-CELL-1954

Comparison records:

- complete eight-page NIST Ag/AgCl source retained
- all 7 separately registered claim targets retained
- all conventional values signs zero glyphs fitted equations smoothing corrections uncertainties adverse and absent records remain downstream provenance

Falsification boundary: The claim halts if the survivor is nonunique; a half-cell, reference, condition, path, EMF row, correction, reverse convention or NIST page is omitted; or a measured value selects composition.

Measurement receipt: sha256: 57597fa43a5fe9e22bdfbec6a6dbc301ab24c77364aa4c2f18e7bf2ea1ce3734.

Isolation certificate: sha256: c872f10514cc98e8b0fa97276a992eb3f7e94ec8efaa84dc79b54a293e195788.

Custody certificate: sha256: e35cb4da68bfac8d9e061bed691d138b126ccb7d7a98ec664fe0eda260d1856a.

Registered source descriptions:

- National Bureau of Standards / National Institute of Standards and Technology: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured EMF molality or temperature selects composition
- all measurement and unapplied-correction records remain downstream provenance

Immutable identities. Source manifest

sha256:880e4199e13122e61820bb2b715abd9985c47baf21aea6870d26e4bc3dfb984f; derivation seal sha256:954fdb00f2642a7723fb6f69fa42c3b92885e811782d10b8f4f16293f9c38dc0; engine receipt sha256:35ddc2ae557f0174e49a4ef763c668e56d90dd93c6574514ee8f398a0a72fbae at receipts/engine/model_admitted/SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003-35ddc2ae557f0174.json; empirical hash sha256:fa6e9bdbb3004f8973c76cf9c4ae27d72c08bcb1a6ca0b6c3f3e7da2fe750047; measurement receipt sha256:57597fa43a5fe9e22bdfbec6a6dbc301ab24c77364aa4c2f18e7bf2ea1ce3734.

193. Fold concentration-dependent potential law

Claim identity: SFT-CHEM-CONCENTRATION-DEPENDENT-POTENTIAL-004

Question and theorem. A generated activity quotient changes potential by an exact held direction and positive generator-layer-per-carrier ratio; exact standard-state equality is EmptyOne.

complete-half-reaction-activity-state__held-product-reactant-or-coincident-orientation__complete-generated-doubling-support__layer-count-additive-product-composition__exact-layer-per-held-carrier-ratio__structural-EmptyOne-standard-state__complete-concentration-potential-series__common-generator-successor-preserves-separation

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-COMBINATORICS-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-SOLUTION-001
- SFT-CHEM-ACTIVITY-NONIDEAL-COMPOSITION-009
- SFT-CHEM-HALF-REACTION-IDENTITY-ORIENTATION-001
- SFT-CHEM-ELECTRODE-POTENTIAL-CHEMICAL-RELATION-002
- SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003

Generated grammar. Generate the literal product of eight registered concentration-potential decisions.

Boundary: Eight dimensions exhaust state, quotient, support, composition, transfer, coincidence, record and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-half-reaction-activity-state__held-product-reactant-or-coincident-orientation__complete-generated-doubling-support__layer-count-additive-product-composition__exact-layer-per-held-carrier-ratio__structural-EmptyOne-standard-state__complete-concentration-potential-series__common-generator-successor-preserves-separation`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
state	complete-half-reaction-activity-state	concentration-number-only	A concentration number does not retain chemical identity or condition.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
quotient	held-product-reactant-or-coincident-orientation	signed-or-continuum-activity-coordinate	A signed continuum coordinate imports an ungenerated domain.
support	complete-generated-doubling-support	arbitrary-real-activity-premise	An arbitrary real activity is outside exact Fold arithmetic.
composition	layer-count-additive-product-composition	imported-logarithm	A logarithm cannot select the native law.
transfer	exact-layer-per-held-carrier-ratio	unscaled-electron-count	Potential displacement must retain transferred-carrier multiplicity.
coincidence	structural-EmptyOne-standard-state	numerical-zero-standard-state	Numerical zero is not a native proof value.
record	complete-concentration-potential-series	selected-concentration-point	A selected point can hide adverse curvature and uncertainty.
extension	common-generator-successor-preserves-separation	refitted-concentration-coefficient	Refitting at each scale adds free choices.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:b37c66491e0dc9a259f0f5318d7614d4e1b323d8897d8cfc76b6d7639067db5f.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:80380616cdaefcb7d640e8ab7ff5e008b14022cd0cb504e72d110e80c397c3f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b6b3ad23f080e778da727ffbd1db583a5c83d52af7ffc9ade267d05920eb10e7.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:964c4457778dc183853b32d27a08200515c1c1143042ceb203a49842e0f6ad87.

Independent reconstruction. Implementation hash

sha256:5e60c84e438556369b3e364544aa381eacc678f7ee4a5ace84b2f8204dc4b2f0; independent certificate sha256:2c155fa3445e7b119d2ae7ba9410ae48313b6f7f57b445155dd8aca62c33e3a6; external-validation hash sha256:5279a54847ad22fbc0a8db9e1a3e602ff4cf6d2864c1a71b5628709ad171d519. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-JRES-AGCL-HCL-CONCENTRATION-SERIES-1954
- NIST-JRES-AGCL-HCL-CONCENTRATION-SERIES-1954
- NIST-JRES-AGCL-HCL-CONCENTRATION-SERIES-1954
- NIST-JRES-AGCL-HCL-CONCENTRATION-SERIES-1954
- NIST-JRES-AGCL-HCL-CONCENTRATION-SERIES-1954
- NIST-JRES-AGCL-HCL-CONCENTRATION-SERIES-1954
- NIST-JRES-AGCL-HCL-CONCENTRATION-SERIES-1954
- NIST-JRES-AGCL-HCL-CONCENTRATION-SERIES-1954

Comparison records:

- complete eight-page NIST Ag/AgCl source retained
- all 8 separately registered claim targets retained
- all conventional values signs zero glyphs fitted equations smoothing corrections uncertainties adverse and absent records remain downstream provenance

Falsification boundary: The claim halts if the survivor is nonunique; any molality, temperature, potential, activity, uncertainty, anomaly or NIST page is omitted; or a logarithm, fitted coefficient or measured trend selects the native law.

Measurement receipt: sha256:da4cb0b8511f6b0c983e5f21ec378d9a472c11c822516c9b6779e10bb3916c0d.

Isolation certificate: sha256:835b2b2306c423f14c2ea200be6ef87152893ba3de373cff5f86f3329eb00b51.

Custody certificate: sha256:741e45989656bfd653dbcfc5f08ec5347d7c821a09c1ccb2ffa0d733e15583c5.

Registered source descriptions:

- National Bureau of Standards / National Institute of Standards and Technology: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no imported logarithm Debye-Huckel coefficient smoothing law or measured series selects the survivor
- all conventional source equations fitted terms anomalies and values remain downstream provenance

Immutable identities. Source manifest

sha256:1759c7d05903bcacd2999cf589fabda6781211778b374e87966510139eafc750; derivation seal
 sha256:84787e0ee9366e0ce863d237a5b8a30e0bd55376b4f71e184dfeafeee9262da0; engine receipt
 sha256:1eea2b72f83d08bbffdc8291adba1c830764410ba065d8d056ca12a8bdf5bfce at receipts/engine/model_admitted/SFT-CHEM-CONCENTRATION-DEPENDENT-POTENTIAL-004-1eea2b72f83d08bb.json;
 empirical hash sha256:1bee4a05d79b1edb595299e0b762148132e14d9df9596afdc816adc6202ec7e2;
 measurement receipt sha256:da4cb0b8511f6b0c983e5f21ec378d9a472c11c822516c9b6779e10bb3916c0d.

194. Fold electrochemical work and reaction-direction correspondence

Claim identity: SFT-CHEM-ELECTROCHEMICAL-WORK-REACTION-DIRECTION-005

Question and theorem. Electrochemical work is the exact positive product of counted transferred carriers and held potential separation, with separate chemical/electrical path direction and structural EmptyOne equilibrium.

complete-cell-chemical-electrical-custody__positive-counted-transfer-carriers__positive-potential-separation-with-held-direction__exact-carrier-potential-product__held-work-reaction-correspondence__structural-EmptyOne-equilibrium__complete-cell-work-equilibrium-vector__exact-path-reversal-preserves-positive-work

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-PHYS-PLASMA-COLLECTIVE-001

- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003

Generated grammar. Generate the literal product of eight registered electrochemical-work decisions.

Boundary: Eight dimensions exhaust custody, carrier, potential, composition, direction, equilibrium, record and reversal.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-cell-chemical-electrical-custody__positive-counted-transfer-carriers__positive-potential-separation-with-held-direction__exact-carrier-potential-product__held-work-reaction-correspondence__structural-EmptyOne-equilibrium__complete-cell-work-equilibrium-vector__exact-path-reversal-preserves-positive-work`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
custody	complete-cell-chemical-electrical-custody	anonymous-energy-number	An anonymous energy loses its reaction and cell.
carrier	positive-counted-transfer-carriers	uncounted-charge	Uncounted charge cannot establish chemical work.
potential	positive-potential-separation-with-held-direction	signed-voltage-premise	A signed voltage imports negative proof magnitude.
composition	exact-carrier-potential-product	free-energy-postulate	A named free-energy formula does not derive work custody.
direction	held-work-reaction-correspondence	unoriented-work	Unoriented work cannot determine the reaction path.
equilibrium	structural-EmptyOne-equilibrium	numerical-zero-work	Numerical zero is not a native proof magnitude.
record	complete-cell-work-equilibrium-vector	selected-cell-result	A selected result can hide conditions and uncertainty.
reverse	exact-path-reversal-preserves-positive-work	irreversible-sign-change	A sign change cannot reconstruct both transfer paths.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:474e12e357e2df1543cf48b0f609c0626c4630aa6015dc05fe69feeab1d2259b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1fe8cbbd55c077bd7625d261403868b88ef6b45ab8a847fe64bc400ea998e92b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:25940f27801a08836482d11b88134254d49e469749d7745c2ecd8a714285eab6`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:f878dfef999b2acdc5cccfb0738996c4107385623909f99dcc8bd90563da81f9`.

Independent reconstruction. Implementation hash

`sha256:1816786c033a7f22ce5ba01a9939a4f09b00597ccdf31c1aa026f1b094c48116`; independent certificate

`sha256:c851200dc93fd26ea1154d20a283b30dd5d84c4bb74ce97bff0b07f4e934bc8d`; external-validation

hash sha256:7010b23f6a89c59adc55776aeb8234460b4323546a5e8017f3ac06cacdd83e1e. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS
- NIST-JRES-AGCL-STANDARD-POTENTIAL-1954
- NIST-CODATA-AGCL-WORK-CROSS-RECONSTRUCTION
- IUPAC-GREEN-BOOK-ELECTROCHEMISTRY-2007
- IUPAC-GREEN-BOOK-ELECTROCHEMISTRY-2007
- NIST-CODATA-AGCL-WORK-CROSS-RECONSTRUCTION
- NIST-CODATA-AGCL-WORK-CROSS-RECONSTRUCTION
- ECHEM-005-SOURCE-INVENTORY

Comparison records:

- complete 519-page shared post-seal source family retained
- all 8 separately registered claim targets retained
- all measured values units uncertainties corrections historical disagreements anomalous absent and unresolved records remain downstream provenance

Falsification boundary: The claim halts if the survivor is nonunique; any carrier, path, potential, work, equilibrium, uncertainty, disagreement or source page is omitted; or an external value selects the law.

Measurement receipt: sha256:0e676ccbcbe869154025c95455eaba484fdc04df44fadae517db1bdfe0c9a75.

Isolation certificate: sha256:3ea97cce1efca1d252454f9c7c617c907a36f85d7e97229723e98a1433b6b5d9.

Custody certificate: sha256:b4c56c21a1e448cd760df3eee60ef383d1a4a77b224cc251c4c32f1aa735faef.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no potential Faraday work or equilibrium outcome selects the survivor
- all conventional signs historical disagreements and uncertainties remain downstream provenance

Immutable identities. Source manifest

sha256:9cd4f44aa177d97037d127ca22335b9f12d4cba6dc2219bbb64326ae8394eb19; derivation seal

sha256:d68668a05104753f3147d360939387a11aa5bc5cb6f8e3bc662c735bb239bbb6; engine receipt

sha256:f50794cbe0530156b52fb4cbaacf18b7d9d570dd8d4ad254bd4a325cba757057 at receipts/engine/

model_admitted/SFT-CHEM-ELECTROCHEMICAL-WORK-REACTION-DIRECTION-005-f50794cbe0530156.

json; empirical hash sha256:58415873a377199ec648af9964f43aecc8db891c7838b9d6bfccf615f5097cb7;

measurement receipt sha256:0e676ccbcbe869154025c95455eaba484fdc04df44fadae517db1bdfe0c9a75.

195. Fold electrolysis and product-amount law

Claim identity: SFT-CHEM-ELECTROLYSIS-PRODUCT-AMOUNT-006

Question and theorem. Electrolysis product amount is the exact transferred-carrier to stoichiometric-carrier ratio with complete product, path and remainder custody.

complete-process-electrode-product-custody__positive-counted-transfer-occurrences__positive-carriers-per-product-occurrence__exact-carrier-to-product-ratio__complete-products-plus-held-remainder__structural-EmptyOne-no-complete-product__complete-charge-product-amount-vector__like-batches-compose-by-counted-addition

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-PLASMA-COLLECTIVE-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-HALF-REACTION-IDENTITY-ORIENTATION-001
- SFT-CHEM-ELECTROCHEMICAL-WORK-REACTION-DIRECTION-005

Generated grammar. Generate the literal product of eight registered electrolysis-product decisions.

Boundary: Eight dimensions exhaust custody, charge, stoichiometry, amount, remainder, absence, record and composition.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *complete-process-electrode-product-custody__positive-counted-transfer-occurrences__positive-carriers-per-product-occurrence__exact-carrier-to-product-ratio__complete-products-plus-held-remainder__structural-EmptyOne-no-complete-product__complete-charge-product-amount-vector__like-batches-compose-by-counted-addition*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
custody	complete-process-electrode-product-custody	mass-answer-only	A mass answer loses electrolysis identity and path.
charge	positive-counted-transfer-occurrences	continuous-charge-premise	A continuum charge does not expose transferred distinctions.
stoichiometry	positive-carriers-per-product-occurrence	empirical-equivalent-factor	A fitted equivalent factor does not retain carrier demand.
amount	exact-carrier-to-product-ratio	rounded-product-count	Rounding destroys exact carrier custody.
remainder	complete-products-plus-held-remainder	discarded-incomplete-product	Discarding a remainder loses transferred distinctions.
absence	structural-EmptyOne-no-complete-product	numerical-zero-product	Numerical zero is not a native product magnitude.
record	complete-charge-product-amount-vector	selected-coulometer-result	A selected mass-per-charge value can hide experimental corrections.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
composition	like-batches-compo se-by-counted-addi tion	batch-specific-conversion	A batch-specific rule adds an unforced parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:580e6173aee195cf17312c687318eac76ceac69964cf1fefad85e860c5d9edfb.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:e91b70c7c6b5c757e88743165004162fa8227fbaed6a084c1e3d504a46c01deb.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:cf8ebb8b3127be8105569a218758277841b5de521e9d07b863d8a1259113d966.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:ac25682a14f521968f5092399ec656a4c3d4d1c39a42bfaaca2882b8023d910e.

Independent reconstruction. Implementation hash

sha256:f12172bd792180e79f9e93ef0d3625ac6a2316e0d9ce542a4e19f48852fbaeff; independent certificate sha256:08ee5b6dc176d648deee6a7e6a29509de17c04ffc7c391837b516d7a23e2de3a; external-validation hash sha256:39fe478c7f9a16fdd001fcec7aa71d39d76fab3f404f7ba03eea46eb1c184b58. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-JRES-SILVER-ELECTROCHEMICAL-EQUIVALENT-1980
- NIST-JRES-SILVER-ELECTROCHEMICAL-EQUIVALENT-1980
- NIST-JRES-SILVER-ELECTROCHEMICAL-EQUIVALENT-1980
- NIST-JRES-SILVER-ELECTROCHEMICAL-EQUIVALENT-1980
- NIST-CODATA-2022-ALL-CONSTANTS
- NIST-JRES-SILVER-ELECTROCHEMICAL-EQUIVALENT-1980
- NIST-JRES-SILVER-ELECTROCHEMICAL-EQUIVALENT-1980
- ECHEM-006-SOURCE-INVENTORY

Comparison records:

- complete 519-page shared post-seal source family retained
- all 8 separately registered claim targets retained
- all measured values units uncertainties corrections historical disagreements anomalous absent and unresolved records remain downstream provenance

Falsification boundary: The claim halts if the survivor is nonunique; any run, charge, time, mass, correction, uncertainty, adverse comparison or source page is omitted; or an external equivalent selects the law.

Measurement receipt: sha256:376582e3d16aa14c36a5436521939d0648feb13269a773de44acdce199c1c85f.

Isolation certificate: sha256:3434a59f436b73dc2c52779d7da4c3567c1dd87264eda530406f9fa551bd5bec.

Custody certificate: sha256:fe9f36f6f8d6a849552bb7b2fbec5bc860555ccdfb7e86834de6caa61f18b682.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no coulometer mass current time Faraday or fitted correction selects the survivor
- every correction uncertainty adverse comparison and historical value remains downstream provenance

Immutable identities. Source manifest

sha256:3ce88bae3a81d7454bd3de7fcb4fc9fa93e947c223b6199ef1d12cf370d45e7; derivation seal
 sha256:985cc4028dbb36658311fddf01b22342237474d077972c52d1f82708fdcb47a5; engine receipt
 sha256:95fcb8396d3792c917e768e341f501e27ab8e3392a38f6fedf6e16e020eea2df at receipts/engine/model_admitted/SFT-CHEM-ELECTROLYSIS-PRODUCT-AMOUNT-006-95fcb8396d3792c9.json; empirical
 hash sha256:0576a5180862d3fd81545bf3ae9f9aecc0676d9b2e21baedef0115bcd6ec6db7; measurement
 receipt sha256:376582e3d16aa14c36a5436521939d0648feb13269a773de44acdce199c1c85f.

196. Fold ionic conductivity relation

Claim identity: SFT-CHEM-IONIC-CONDUCTIVITY-RELATION-007

Question and theorem. Ionic conductivity is the exact positive sum of every species-resolved carried distinction per finite held path resource under one composition and condition.

*complete-species-resolved-carrier-vector__held-species-transport-direction__complete-held-electrolyte-composition__
 one-common-held-condition__finite-held-path-resource-account__exact-sum-of-species-responses__complete-traceable
 -conductivity-vector__positive-species-successor-increases-response*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-CPLX-TIME-SPACE-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-COUPLED-MASS-HEAT-CHARGE-TRANSPORT-019
- SFT-CHEM-ELECTROLYSIS-PRODUCT-AMOUNT-006

Generated grammar. Generate the literal product of eight registered ionic-conductivity decisions.

Boundary: Eight dimensions exhaust carrier, direction, composition, condition, path, aggregation, record and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *complete-species-resolved-carrier-vector__held-species-transport-direction__complete-held-electrolyte-composition__one-common-held-condition__finite-held-path-resource-account__exact-sum-of-species-responses__complete-traceable-conductivity-vector__positive-species-successor-increases-response*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-species-resolved-carrier-vector	bulk-conductance-number	A bulk number loses the transporting ions.
direction	held-species-transport-direction	signed-flux-premise	Signed flux imports negative proof magnitude.
composition	complete-held-electrolyte-composition	composition-free-conductivity	Conductivity without composition is not chemically identified.
condition	one-common-held-condition	mixed-temperature-response	Mixed conditions cannot form one conductivity.
path	finite-held-path-resource-account	continuum-gradient-premise	A continuum gradient hides counted transport support.
aggregation	exact-sum-of-species-responses	fitted-mixture-coefficient	A fitted coefficient adds an unforced parameter.
record	complete-traceable-conductivity-vector	selected-conductivity-value	A selected value can hide composition, temperature and uncertainty.
extension	positive-species-successor-increases-response	fixed-species-count	A fixed count cannot extend to new admitted carriers.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 4d2470b93f39611dde724513f0f9eaa1108a8fa2fae6cac3dea3af630c2d8987.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: e929c80daf978dbf2a0242c3a3131b9550300cb18fb8d1c22d4e4b9a57b27ca3.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 226b12b5832909eb0fde012a53fb0497daf51b395db018496197acdc0f040b31.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 827c40ed7175b98ae2d8e17b30d03733af68dc4c63a9d1ab8e3d33f7d4318730.

Independent reconstruction. Implementation hash
sha256: 32db447dc8dfc187259edf87ac9e4e5fee883e066457428388352009a2a74461; independent certificate
sha256: 22d22649b757b4de78147c2b539ef014bdf831b63a29d280567156fde9739915; external-validation
hash sha256: a5e8c414b63b5ec8376420626753895ec67618793847c156f4abf40ba9335fcd. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-SRM-3190-CONDUCTIVITY-CERTIFICATE
- NIST-SRM-3190-CONDUCTIVITY-CERTIFICATE
- NIST-SRM-3190-CONDUCTIVITY-CERTIFICATE
- NIST-SRM-3190-CONDUCTIVITY-CERTIFICATE
- NIST-SRM-3190-CONDUCTIVITY-CERTIFICATE
- NIST-SP-260-142-PRIMARY-CONDUCTIVITY

- NIST-SP-260-176-SRM-CATALOG
- ECHEM-007-SOURCE-INVENTORY

Comparison records:

- complete 519-page shared post-seal source family retained
- all 8 separately registered claim targets retained
- all measured values units uncertainties corrections historical disagreements anomalous absent and unresolved records remain downstream provenance

Falsification boundary: The claim halts if the survivor is nonunique; any ion, composition, temperature, cell, value, unit, uncertainty, method, catalog row or source page is omitted; or a measured value selects the law.

Measurement receipt: sha256:d771b6819dcf97620014a5d66b4b96e6c61b3dfdb44e95daaa04a51ca462e12d.

Isolation certificate: sha256:425d6ab22836e78b68f539bcc05c40f0673dbf40d4cab0c63b761904458425.

Custody certificate: sha256:5d86246d354343738b577ff0161cecd832a7a2eead471d0c05f8f56bef94a1e.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no certified conductivity composition temperature or uncertainty selects the survivor
- continuum source equations remain downstream provenance

Immutable identities. Source manifest

sha256:f627d63ff40ad2cca2bdb4b00fe2efb0e072ddbf7234795a552662fe1cf66185; derivation seal

sha256:f5508e399957b376e3839df692009a4ac6cb0a2e1aa0cb1b4005380623f331be; engine receipt

sha256:e49c9f8ca3137d6b1618bc56d96e16e55354569f0a6dee3f404af2dec1b670bd at receipts/engine/model_admitted/SFT-CHEM-IONIC-CONDUCTIVITY-RELATION-007-e49c9f8ca3137d6b.json; empirical

hash sha256:8fe337bd96d7de1d3d22688e9aa8d9473b98887694a3272eeacef3479dad7894; measurement

receipt sha256:d771b6819dcf97620014a5d66b4b96e6c61b3dfdb44e95daaa04a51ca462e12d.

197. Fold ionic mobility and transference law

Claim identity: SFT-CHEM-IONIC-MOBILITY-TRANSFERENCE-008

Question and theorem. Each ionic mobility is an exact positive traversal-per-carrier-resource ratio and each transference part is that species contribution divided by the complete total, summing exactly to One.

complete-held-ionic-species-identity__held-species-mobility-direction__exact-traversal-per-carrier-resource-ratio__common-composition-condition-path__exact-species-contribution-partition__transference-parts-sum-exactly-to-One__structural-EmptyOne-absent-species__complete-mobility-transference-vector

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-ORDER-LATTICE-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-COUPLED-MASS-HEAT-CHARGE-TRANSPORT-019
- SFT-CHEM-IONIC-CONDUCTIVITY-RELATION-007

Generated grammar. Generate the literal product of eight registered mobility-transference decisions.

Boundary: Eight dimensions exhaust identity, direction, mobility, composition, partition, whole, absence and record.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-held-ionic-species-identity__held-species-mobility-direction__exact-traversal-per-carrier-resource-ratio__common-composition-condition-path__exact-species-contribution-partition__transference-parts-sum-exactly-to-One__structural-EmptyOne-absent-species__complete-mobility-transference-vector`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
identity	complete-held-ionic-species-identity	anonymous-mobile-charge	Anonymous charge loses chemical carrier identity.
direction	held-species-mobility-direction	signed-mobility	A sign imports negative proof magnitude.
mobility	exact-traversal-per-carrier-resource-ratio	continuum-velocity-field-quotient	A continuum derivative is not a counted transport proof.
composition	common-composition-condition-path	isolated-ion-answer	An isolated answer loses the complete electrolyte.
partition	exact-species-contribution-partition	independent-fitted-transport-numbers	Independent fits need not conserve the whole current.
whole	transference-parts-sum-exactly-to-One	approximately-normalized-sum	Approximate normalization can hide lost current.
absence	structural-EmptyOne-absent-species	numerical-zero-absent-ion	Numerical zero conflates absence with a measured response.
record	complete-mobility-transference-vector	selected-transport-number	A selected result can hide concentration change and anomalies.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:02f812d56ce00bd001540ab96306b436a1f31741a931b1414814944e6cf5b57a`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:c9d930c2c46b15e0545cdcbf4fb96919de3e32bc778ba7e9dd7e3d78f7e300fb`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:deb74a94328e51aaf9a496a2858384945dbc64f5eaf66e708810a21ed289f7a0`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:5f2812902ac512945e8756d59bb95fadc72194e572d2d3cc9de411d54f4a8b03`.

Independent reconstruction. Implementation hash

sha256:0396a975292bce5e6a8d67661d237f6be03bb627ee412452f9d76caaa2309166; independent certificate sha256:792481a243911abccfc36729104289fef0d407956adffb86fd9a06995b18b7c0; external-validation hash sha256:fac3bfae37f5092796f07f9c4abc2f064ddbd10912b65dcabdd0513d65d7d267. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NBS-JRES-TRANSFERENCE-CONCENTRATION-1931
- NBS-JRES-TRANSFERENCE-CONCENTRATION-1931
- NBS-JRES-TRANSFERENCE-CONCENTRATION-1931
- NBS-JRES-TRANSFERENCE-CONCENTRATION-1931
- NBS-JRES-TRANSFERENCE-CONCENTRATION-1931
- IUPAC-GREEN-BOOK-ELECTROCHEMISTRY-2007
- NBS-JRES-TRANSFERENCE-CONCENTRATION-1931
- ECHEM-008-SOURCE-INVENTORY

Comparison records:

- complete 519-page shared post-seal source family retained
- all 8 separately registered claim targets retained
- all measured values units uncertainties corrections historical disagreements anomalous absent and unresolved records remain downstream provenance

Falsification boundary: The claim halts if the survivor is nonunique; any species, direction, concentration, mobility, transference, anomaly, absent row or source page is omitted; or a measured value selects the law.

Measurement receipt: sha256:c9b6c35809bf4df5b10ee803e1c0660f3eef25da41f9a22b81d23432c0f6fffb1.

Isolation certificate: sha256:e5191331b190390d0da51f6175b8e2868e422e4a81d09ce86b6ff455718d90fa.

Custody certificate: sha256:e55b23bdcebe42de264318027ae6f3eff82d88389947323200457631e75a1f25.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no transference mobility concentration direction or anomaly selects the survivor
- all conventional derivatives fits and unresolved results remain downstream provenance

Immutable identities. Source manifest

sha256:da3653509853cb6cc13bc6c15fad8fa85a588022049e0dc18667528ed589b5c0; derivation seal

sha256:6438d5bc11a4e0ffc4bee72eeabcc8b80d5d296974e46460908b0a6644d8d797; engine receipt

sha256:df5f677264d69df93f29920c45789f8264a359214049c983d7a137b763ea7682 at receipts/engine/model_admitted/SFT-CHEM-IONIC-MOBILITY-TRANSFERENCE-008-df5f677264d69df9.json; empirical

hash sha256:1f91f789b9e89ffa413e2aec0f227c6397774d78bf0da2bc6a23c8829c0a4f7b; measurement

receipt sha256:c9b6c35809bf4df5b10ee803e1c0660f3eef25da41f9a22b81d23432c0f6fffb1.

198. Fold electrode reaction-rate law

Claim identity: SFT-CHEM-ELECTRODE-REACTION-RATE-009

Question and theorem. Electrode reaction rate is an exact positive count of forward and reverse reaction events per positive counted resource, retaining reaction, interface, condition, potential orientation and structural balance.

complete-reaction-interface-custody__positive-counted-forward-reverse-events__exact-events-per-counted-resource__held-potential-orientation__positive-Take-with-held-rate-direction__structural-EmptyOne-balanced-events__complete-current-potential-condition-vector__event-successor-recomputes-exact-rate

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-COMP-CPLX-TIME-SPACE-001
- SFT-CHEM-HALF-REACTION-IDENTITY-ORIENTATION-001
- SFT-CHEM-ELECTROCHEMICAL-WORK-REACTION-DIRECTION-005
- SFT-CHEM-IONIC-MOBILITY-TRANSFERENCE-008

Generated grammar. Generate the literal product of eight registered electrode-rate decisions.

Boundary: Eight dimensions exhaust reaction custody, events, resource, potential, net, balance, record and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *complete-reaction-interface-custody__positive-counted-forward-reverse-events__exact-events-per-counted-resource__held-potential-orientation__positive-Take-with-held-rate-direction__structural-EmptyOne-balanced-events__complete-current-potential-condition-vector__event-successor-recomputes-exact-rate*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
reaction	complete-reaction-interface-custody	anonymous-current	Current alone loses reaction identity.
events	positive-counted-forward-reverse-events	continuum-current-premise	Continuum flow hides event support.
resource	exact-events-per-counted-resource	unbounded-rate	Rate without resource is undefined.
potential	held-potential-orientation	signed-overpotential-premise	Signed potential imports negative magnitude.
net	positive-Take-with-held-rate-direction	subtracted-negative-rate	Signed subtraction can produce negative proof values.
balance	structural-EmptyOne-balanced-events	numerical-zero-net-current	Numerical zero is not a native magnitude.
record	complete-current-potential-condition-vector	selected-current-point	One point hides the curve and conditions.
extension	event-successor-recomputes-exact-rate	fitted-exchange-current-exponential	A fit imports answer-producing parameters.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: eb2afce6f92f19cfc6619eb9c3c2df4ae055d3446b47a783d2aa13faa5f20e62.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: a7d84ff06a7b7d7cab194293a998abfaf859fd562106196313b7b0b302a889ea.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 2a942639138e968dd0571b72849480a7fda32a4bebdb429ad49cfb5fea74f4d6.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 1a1bd155a3a629f4f3d1785d3da9972db3dcb0981a35b3498ad7a6e6a45aa589.

Independent reconstruction. Implementation hash

sha256: 9b4f3fcb4eda9a8711438df743c1dd17676dc01890d2b09dfc54b6842d9a3102; independent certificate sha256: 74f835dfc17aae9c3ff891b5f6b6edb9ee6969aafdf0b5ddb7257918a90db2ee; external-validation hash sha256: a854f826faa7ec2b8d8ac979c8d39d1dade2050bc47cd0a88a7ff5d43d9bf541. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NBS-JRES-GALVANIC-COUPLES-1950
- NBS-JRES-GALVANIC-COUPLES-1950
- NBS-JRES-GALVANIC-COUPLES-1950
- NBS-JRES-IRON-CORROSION-1957
- NBS-JRES-IRON-CORROSION-1957
- NBS-JRES-IRON-CORROSION-1957
- ECHEM-009-012-SHARED-SOURCES
- ECHEM-009-012-SOURCE-INVENTORY

Comparison records:

- complete 73-page shared post-seal source family retained
- all 8 separately registered claim targets retained
- all measured values units uncertainties fits discrepancies adverse absent and unresolved records remain downstream provenance

Falsification boundary: The claim halts if the survivor is nonunique; any reaction, interface, event direction, resource, current, potential, condition, adverse record or source page is omitted; or an external curve selects the law.

Measurement receipt: sha256: aecafb9be6c34a64e27d2afd816e28ecb9766c9fbd2ecb9d5a689a15928992e7.

Isolation certificate: sha256: d815082de99cbddd55a94a4b15d7b2d147eb0187dece7e942adaabb950a0f183.

Custody certificate: sha256: c824bc81a4f20f5d4f24042193881be6014bbbab238169d20d468f68e9e654e0.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external current potential curve or rate selects the survivor
- all external signs fits discontinuities and IR drops remain downstream provenance

Immutable identities. Source manifest

sha256:572d9974b3287ffcdd8c90584547cbf1fbf40226ec20be2495bfb583bc068b; derivation seal
 sha256:4cdc5950be5d4ad56615546f815aa8865ba759615088770d0ba3052f6150b804; engine receipt
 sha256:85b5ca8c3be8347f4175c93f9804dae6a843c619ddaaae3900d170b3b7c570c0 at receipts/engine/model_admitted/SFT-CHEM-ELECTRODE-REACTION-RATE-009-85b5ca8c3be8347f.json; empirical hash
 sha256:8e5e6e52bbfc1c42a5c3805ced46e8b9fddeab81ef537be9fc7311c6b2c92182; measurement receipt
 sha256:aecafb9be6c34a64e27d2afd816e28ecb9766c9fbd2ecb9d5a689a15928992e7.

199. Fold overpotential and polarization law

Claim identity: SFT-CHEM-OVERPOTENTIAL-POLARIZATION-010

Question and theorem. Polarization is an ordered finite current-potential response about one held equilibrium reference, with anodic, cathodic and equilibrium sides separated from positive exact magnitudes.

one-held-equilibrium-reference__held-anodic-cathodic-equilibrium-side__exact-positive-potential-distance__held-current-direction-positive-magnitude__structural-EmptyOne-equilibrium-point__complete-ordered-finite-polarization-curve__common-electrode-reference-condition__complete-raw-potential-current-vector

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-CHEM-ELECTRODE-POTENTIAL-CHEMICAL-RELATION-002
- SFT-CHEM-ELECTRODE-REACTION-RATE-009

Generated grammar. Generate the literal product of eight registered polarization decisions.

Boundary: Eight dimensions exhaust reference, orientation, magnitude, current, equilibrium, curve, condition and record.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *one-held-equilibrium-reference__held-anodic-cathodic-equilibrium-side__exact-positive-potential-distance__held-current-direction-positive-magnitude__structural-EmptyOne-equilibrium-point__complete-ordered-finite-polarization-curve__common-electrode-reference-condition__complete-raw-potential-current-vector*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
reference	one-held-equilibrium-reference	reference-free-voltage	Overpotential needs equilibrium reference.
orientation	held-anodic-cathodic-equilibrium-side	signed-potential-offset	A sign imports negative magnitude.
magnitude	exact-positive-potential-distance	continuum-difference-premise	A continuum difference is not an exact proof.
current	held-current-direction-positive-magnitude	anonymous-current	Current loses reaction direction.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
equilibrium	structural-EmptyOne-equilibrium-point	numerical-zero-origin	Numerical zero is not native equilibrium.
curve	complete-ordered-finite-polarization-curve	selected-polarization-point	One point hides response organization.
condition	common-electrode-reference-condition	mixed-condition-curve	Mixed conditions are incomparable.
record	complete-raw-potential-current-vector	fitted-tafel-line	A fitted line can hide deviations.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:1be55c21342f49ab2584c88a87d91f8930fa48e1aac31a663a803de84e3ede20.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:95e65cd8263fc22e81c6ecbcbca6ebe1f27dc0042989f4c94ec08101d3db515e.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:6687ccdcc5d0eb64dfa48ab8733f9afd5e11e89730357e3592ceed70be9a1fe7.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:831a0ee1764b6f28bf61810fd574f4a400669570dd669523eb7fa92f58cf0a0e.

Independent reconstruction. Implementation hash

sha256:b2ff6afd1ddfa901ed2e1fc4983334df8ef8fcf42606979ead04ef9311637f6f; independent certificate sha256:17777db881a2f94efdcf8115330dde96447feb93e57c6149362aba6cb91f48d2; external-validation hash sha256:d7c0fd602901814aa28cb3f6693c64fb7fc96540d5c2a50f29cdfafd9afa552d. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NBS-JRES-GALVANIC-COUPLES-1950
- NBS-JRES-GALVANIC-COUPLES-1950
- NBS-JRES-IRON-CORROSION-1957
- NBS-JRES-IRON-CORROSION-1957
- NBS-JRES-IRON-CORROSION-1957
- NBS-JRES-GALVANIC-COUPLES-1950
- NBS-JRES-IRON-CORROSION-1957
- ECHEM-009-012-SOURCE-INVENTORY

Comparison records:

- complete 73-page shared post-seal source family retained
- all 8 separately registered claim targets retained
- all measured values units uncertainties fits discrepancies adverse absent and unresolved records remain downstream provenance

Falsification boundary: The claim halts if the survivor is nonunique; any equilibrium, scan direction, point, reversal, break, IR drop, uncertainty or source page is omitted; or an external curve selects the law.

Measurement receipt: sha256:bb98396737ec419940c02d06c6e2228e459657068c930d7e880ea17392045510.
 Isolation certificate: sha256:172d6fe6dd6429b8cd3c7c6cadd68437d9b208fb663bdfc3d21fd6833deb4786.
 Custody certificate: sha256:ff3102259f0bb882ab61c4accc8b0c5fae045fa62833e263b9fe437f4ae43a9b.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external polarization value or fitted line selects the survivor
- all scan reversals breaks hysteresis IR drops and uncertainties remain downstream provenance

Immutable identities. Source manifest

sha256:77547cb9d107e2b000ac3e2b359dcd9ab155e905ac8f0757f6aa7ecf4a7b4832; derivation seal
 sha256:d32c4ad951f4da7d550ea6b336971d6e3e489e1fe8947d9dbb74692d8db55ff1; engine receipt
 sha256:5ffa4d639d74352f13b1a56b0464c7fe5ed2aa3f27c4f4091db16346e33fc1b4 at receipts/engine/model_admitted/SFT-CHEM-OVERPOTENTIAL-POLARIZATION-010-5ffa4d639d74352f.json; empirical
 hash sha256:748e3be2955c885ea8fc577ae91dbf245d1a1bffa5968261c4bb6c6c7313ae1; measurement
 receipt sha256:bb98396737ec419940c02d06c6e2228e459657068c930d7e880ea17392045510.

200. Fold double-layer interfacial-charge organization

Claim identity: SFT-CHEM-DOUBLE-LAYER-INTERFACIAL-CHARGE-011

Question and theorem. An electric double layer is a complete finite ordered interface support with held electrolyte, carrier identities and opposed sides, exact positive potential separation and counted carrier-per-potential organization.

one-held-electrode-electrolyte-interface__complete-finite-ordered-interface-layers__held-species-resolved-charge-carriers__held-opposed-interface-sides__exact-positive-interface-potential-separation__exact-carrier-per-potential-ratio__structural-EmptyOne-interface-coincidence__complete-interface-potential-capacitance-vector

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-CHEM-IONIC-CONDUCTIVITY-RELATION-007
- SFT-CHEM-OVERPOTENTIAL-POLARIZATION-010

Generated grammar. Generate the literal product of eight registered double-layer decisions.

Boundary: Eight dimensions exhaust interface, support, carrier, orientation, potential, capacitance, coincidence and record.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: one-held-electrode-electrolyte-interface__complete-finite-ordered-interface-layers__held-species-resolved-charge-carriers__held-opposed-interface-sides__exact-positive-interface-potential-separation__exact-carrier-per-potential-ratio__structural-EmptyOne-interface-coincidence__complete-interface-potential-capacitance-vector. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
interface	one-held-electrode-electrolyte-interface	bulk-charge-number	Bulk charge loses interface identity.
support	complete-finite-ordered-interface-layers	continuum-profile-premise	Continuum profile is not finite proof support.
carrier	held-species-resolved-charge-carriers	anonymous-charge-density	Anonymous density loses carrier identity.
orientation	held-opposed-interface-sides	signed-interfacial-charge	A sign imports negative magnitude.
potential	exact-positive-interface-potential-separation	unreferenced-potential	Potential needs interfacial separation.
capacitance	exact-carrier-per-potential-ratio	fitted-capacitance-parameter	A fit does not derive charge custody.
coincidence	structural-EmptyOne-interface-coincidence	numerical-zero-charge-state	Numerical zero is not native.
record	complete-interface-potential-capacitance-vector	selected-capacitance-point	One point hides spatial and composition response.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: aac23d457e5d6f9f622adfa8cc8175c7891b5502ff995809f189a28d54b85ac6.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 8fb06443e2804079846c590549aa00609f2d93380cfc030ab055f36ef2c79c7c.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: c812c14d3b8ca01c3a491ce728546bee48a4c3fb4682c32193af3100f880108c.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 7eabc843c68accec1c45f0a4e71b8af675055a3f039bf9ce9725ff1505d51b70d.

Independent reconstruction. Implementation hash

sha256: 93ca749245ccaf245e4d29d262e00b9be914a5072bbf3d8637cd4690ef58d3f6; independent certificate sha256: 5c3432c1a08b9261dd144e1146243ea5443e50590ce1682b6fbbe250e026ca27; external-validation hash sha256: 107db43167a54d5727d490f67a69162799034cb3a9613c428f28646703f4d349. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-GRAPHENE-DOUBLE-LAYER-2020
- NIST-GRAPHENE-DOUBLE-LAYER-2020
- NIST-GRAPHENE-DOUBLE-LAYER-2020
- NIST-GRAPHENE-DOUBLE-LAYER-2020
- NIST-GRAPHENE-DOUBLE-LAYER-2020

- NIST-GRAPHENE-DOUBLE-LAYER-2020
- NIST-GRAPHENE-DOUBLE-LAYER-2020
- ECHEM-011-SOURCE-INVENTORY

Comparison records:

- complete 73-page shared post-seal source family retained
- all 8 separately registered claim targets retained
- all measured values units uncertainties fits discrepancies adverse absent and unresolved records remain downstream provenance

Falsification boundary: The claim halts if the survivor is nonunique; any interface, composition, layer, carrier, potential, surface response, model provenance, limit or source page is omitted; or an external value selects the law.

Measurement receipt: sha256:b7faacc86c89a7a304681bdaa42ae3dd52e98a3249791f4c714924edf605d26a.

Isolation certificate: sha256:201898eb8c0a91fc2e691cb952565619ad201e81d1c1e6c08f7f6bb079e68d23.

Custody certificate: sha256:b2153fc71c61ddb3b34f140068db4b18ea77166cb5b1319a11e262555a585f72.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external potential capacitance or numerical model selects the survivor
- all fit parameters screening hysteresis and resolution limits remain downstream provenance

Immutable identities. Source manifest

sha256:aa4b7acb62eedbbcc15ecde9e60fae5b46cf827490d960a223a8b2a4110d481d; derivation seal

sha256:75c59db6833323405a0c4aa11ba40ebcb0566e952f73e4a0aed25e10fb4c2b1a; engine receipt

sha256:113855e15da76e59f3f7b907613c113c661f80c6d1fa5051a234e893998cda18 at receipts/engine/model_admitted/SFT-CHEM-DOUBLE-LAYER-INTERFACIAL-CHARGE-011-113855e15da76e59.json;

empirical hash sha256:ffa524ae12702dfab716ec001a14bfa11b11013021b0b381c430b376fdb805e5;

measurement receipt sha256:b7faacc86c89a7a304681bdaa42ae3dd52e98a3249791f4c714924edf605d26a.

201. Fold coupled corrosion reaction-network law

Claim identity: SFT-CHEM-CORROSION-REACTION-NETWORK-012

Question and theorem. Corrosion is a coupled anodic-cathodic network under one retained material and environment, with exact counted path rates, synchronized support, held excess orientation and structural balance.

held-material-environment-custody__complete-anodic-cathodic-network__positive-counted-path-events__exact-events-per-resource-ratio__exact-synchronized-path-rate__held-excess-path-orientation__structural-EmptyOne-balanced-excess__complete-potential-current-rate-mass-loss-vector

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-REDOX-COUPLING-001

- SFT-CHEM-ELECTRODE-REACTION-RATE-009
- SFT-CHEM-OVERPOTENTIAL-POLARIZATION-010

Generated grammar. Generate the literal product of eight registered corrosion-network decisions.

Boundary: Eight dimensions exhaust material, network, events, resource, coupling, orientation, balance and record.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-material-environment-custody__complete-anodic-cathodic-network__positive-counted-path-events__exact-event-s-per-resource-ratio__exact-synchronized-path-rate__held-excess-path-orientation__structural-EmptyOne-balanced-excess__complete-potential-current-rate-mass-loss-vector. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
material	held-material-environment-custody	material-free-rate	A rate without material is not corrosion identity.
network	complete-anodic-cathodic-network	single-reaction-answer	Corrosion requires coupled reactions.
events	positive-counted-path-events	continuum-loss-rate	A continuum rate hides events.
resource	exact-events-per-resource-ratio	unbounded-corrosion-rate	Rate needs resource support.
coupling	exact-synchronized-path-rate	independent-fitted-rates	Independent fits need not conserve current.
orientation	held-excess-path-orientation	signed-net-current	A sign imports negative magnitude.
balance	structural-EmptyOne-balanced-excess	numerical-zero-corrosion-current	Numerical zero is not native balance.
record	complete-potential-current-rate-mass-loss-vector	selected-corrosion-rate	One rate hides potential, mass loss and adverse rows.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256 : fe800b49c05a3645f1b7f5047ca92192122dbdc8828cc1d51657b33c1d9cd62e.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 3fa147072b0743c0cdbf2fec7ace41820a5d6fd4d0089cd795e492bc8a184a30.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256 : 09dfe81b79108e0a26d536de496b845d094470cba2494da670b8827edc60962c.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256 : 6bdb613188eebe1afe887883ba48a4a3a48ea4a9dd522a5a895b12c587c193fa.

Independent reconstruction. Implementation hash

sha256 : f578554c02bc9af575b3dd6eeb0a75ea1d7ab44b1e5e087cda415178454cb976; independent certificate sha256 : 2fac2ecde6d9ccd59a82cb304d2d2fc351b88d6d007ee67d28ab1e58316e0334; external-validation hash sha256 : da0aaf8ab0c70134930dee2da17a3b29e25f9b8a40f445cabdb97dfee433988b. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NBS-JRES-IRON-CORROSION-1957
- NBS-JRES-IRON-CORROSION-1957
- NBS-JRES-IRON-CORROSION-1957
- NBS-JRES-IRON-CORROSION-1957
- NBS-JRES-IRON-CORROSION-1957
- NBS-JRES-GALVANIC-COUPLES-1950
- NBS-JRES-IRON-CORROSION-1957
- ECHEM-009-012-SOURCE-INVENTORY

Comparison records:

- complete 73-page shared post-seal source family retained
- all 8 separately registered claim targets retained
- all measured values units uncertainties fits discrepancies adverse absent and unresolved records remain downstream provenance

Falsification boundary: The claim halts if the survivor is nonunique; any material, environment, reaction path, potential, current, rate, mass loss, discrepancy, control or source page is omitted; or an external result selects the law.

Measurement receipt: sha256:0e47987d5044ada97e0777c57bbc8f2bccd00e195a46790dfb5f9433a94fc3b0.

Isolation certificate: sha256:2228d790dd0cccdcbd72c01b38e75e22b483f1dc384778164e404745db3fa560.

Custody certificate: sha256:a027d7d59738f44447e14d776ee6f930b69f8968fddb2fe33891254463b05805.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external corrosion current mass loss or fitted ratio selects the survivor
- all discrepancies estimates controls and IR drops remain downstream provenance

Immutable identities. Source manifest

sha256:3510fc7f8f23ae560bf1477b259616e7b405405536e56a702aaf7268d6589c33; derivation seal

sha256:f490fbb3d4193489d5d4a518ed4a476f497f99c039d03c3cb767cd5a13f21f5f; engine receipt

sha256:aa576b702605524bb8518fc3148afd07d094598abddfe116f24c045459a0a7ea at receipts/engine

/model_admitted/SFT-CHEM-CORROSION-REACTION-NETWORK-012-aa576b702605524b.json; empirical

hash sha256:1cd5a4aa2fd13919f0ed955fc0fe364735d29a64b97ca86dd905595dca6a5ac4; measurement

receipt sha256:0e47987d5044ada97e0777c57bbc8f2bccd00e195a46790dfb5f9433a94fc3b0.

202. Fold electrochemical storage ownership and handoff law

Claim identity: SFT-CHEM-ELECTROCHEMICAL-STORAGE-HANDOFF-013

Question and theorem. Electrochemical storage retains one exact owner per coordinate: Chemistry owns species and reactions, Materials owns bulk response and degradation, and Engineering owns implementation, joined only by explicit directed claim handoffs.

complete-storage-coordinate-custody__chemistry-owns-species-reactions__materials-own-bulk-device-response__engineering-owns-implementation__exactly-one-owner-per-coordinate__explicit-directed-claim-handoff__complete-chemistry-material-record-pair__new-coordinate-requires-new-unique-owner

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003
- SFT-CHEM-CORROSION-REACTION-NETWORK-012
- SFT-MAT-DEGR-CORROSION-001
- SFT-ENG-REQUIREMENT-001

Generated grammar. Generate the literal product of eight registered storage-handoff decisions.

Boundary: Eight dimensions exhaust subject, Chemistry scope, Materials scope, Engineering scope, ownership, handoff, record and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-storage-coordinate-custody__chemistry-owns-species-reactions__materials-own-bulk-device-response__engineering-owns-implementation__exactly-one-owner-per-coordinate__explicit-directed-claim-handoff__complete-chemistry-material-record-pair__new-coordinate-requires-new-unique-owner`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
subject	complete-storage-coordinate-custody	anonymous-storage-number	A device number loses the chemical/material subject.
chemistry	chemistry-owns-species-reactions	chemistry-owns-device-performance	Bulk response exceeds Chemistry ownership.
materials	materials-own-bulk-device-response	materials-owns-reaction-law	Reaction identity exceeds Materials ownership.
engineering	engineering-owns-implementation	engineering-owns-natural-law	Implementation does not select natural law.
ownership	exactly-one-owner-per-coordinate	overlapping-branch-ownership	Duplicate ownership destroys exact provenance.
handoff	explicit-directed-claim-handoff	untraced-cross-branch-use	An untraced use loses dependency custody.
record	complete-chemistry-material-record-pair	selected-device-summary	One summary hides chemical and material records.
extension	new-coordinate-requires-new-unique-owner	application-redefines-ownership	An application cannot change branch law.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:5f5765463a92bc3ffc730b8ce79dbd7a3edb23052fa4bf1a057e6dd7ba1e34b1`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:a7703d796f8e230e6546277acf69481c652913cf70f7ee28b8de67f7e9e63d9e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:c477816e651f5fa0ec24bccb1a8db37230889f0a9c701e215587f1f43d7383b8.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:fd35e709db5ed26c546525d0b5dba295065844946ccb47853cfdca768e184dee.

Independent reconstruction. Implementation hash

sha256:0f3ea25bef5934f0178fc465ccb7406d33d529afb24bbfadd15cfbeffec47ace; independent certificate
sha256:522b3b68c0c1504d81d37dfef00374786f44c78b81a9bc483549891c7d77458; external-validation
hash sha256:0de9ce51b537382193b0e8741208f00ba6b92bc3d21013da432b492f1ceb7f23. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003
- SFT-MAT-DEGR-CORROSION-001
- SFT-ENG-REQUIREMENT-001
- SFT-V3-LIVE-CLAIM-CENSUS
- SFT-V3-LIVE-DEPENDENCY-GRAPH
- SFT-CHEM-MAT-PAIRED-RECEIPTS
- NIST-METAL-ADDITIVE-CORROSION
- SFT-V3-BRANCH-OWNERSHIP-BOUNDARY

Comparison records:

- three admitted branch-owner certificates retained
- two directed handoffs and complete 97,292-byte NIST Materials source retained
- development-observed source status disclosed and never relabelled blind

Falsification boundary: The claim halts if the survivor is nonunique; any owner, coordinate, admitted receipt, directed handoff, external material record or duplication control is omitted; or an application outcome selects the ownership law.

Measurement receipt: sha256:5d13e5471cd765f0166c60fa3fdbff0614c541dac9840646b3dd31b1287d046.

Isolation certificate: sha256:6c9eb344ca81f0c33debe0894219f0afc74310c50a0ce015a3b9a6d0d8286c93.

Custody certificate: sha256:4ae406e53d901fe3561c5e6f2667ebdef08b37045fa13a927bc905736f3b246e.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no device result source status or application selects ownership
- no duplicated branch ownership or untraced cross-branch use

Immutable identities. Source manifest

sha256:32cb87a0c32622be2adcfb699da7951aad92c4a0df0d52e86d075da160fa2d84; derivation seal

sha256:f5a82eaae94ab90d3d7ac7fcc3141643c412ce3fbc5991ac3783a76849e15295; engine receipt

sha256:ed5079835f482468f5b5dc7eab3d87ecc8412b18b99374216e300a1dcfb02692 at receipts/engin

e/model_admitted/SFT-CHEM-ELECTROCHEMICAL-STORAGE-HANDOFF-013-ed5079835f482468.json;
 empirical hash sha256:496543b458ab89590b620118e0adfe1ba8fa9a3e4307b2f65f7ec93b11f1f100;
 measurement receipt sha256:5d13e5471cd765f0166c60fa3fdbfff0614c541dac9840646b3dd31b1287d046.

25. Nuclear chemistry and radiochemistry - 12 admitted claims

Nuclear carrier changes and chemical consequences retain activity, branching, separation, isotope and radiation custody. This entire registered subcategory is included; no proper subset is presented as field completion.

203. Fold nuclide chemical-carrier law

Claim identity: SFT-CHEM-NUCLEAR-CHEMICAL-CARRIER-001

Question and theorem. A chemical nuclide carrier retains element, positive nucleon count, nuclear state, chemical species, phase and positive occurrence as one exact identity vector.

held-element-identity__positive-nucleon-count__held-nuclear-state__held-chemical-species__held-chemical-phase__positive-counted-occurrence__complete-nuclide-species-vector__fresh-occurrence-preserves-identity

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-ELEM-ISOTOPE-001
- SFT-PHYS-NUCLEAR-RADIOACTIVITY-001

Generated grammar. Generate the literal product of eight registered carrier decisions.

Boundary: Eight dimensions exhaust element, nuclide, state, species, phase, occurrence, record and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *held-element-identity__positive-nucleon-count__held-nuclear-state__held-chemical-species__held-chemical-phase__positive-counted-occurrence__complete-nuclide-species-vector__fresh-occurrence-preserves-identity*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
element	held-element-identity	anonymous-radiation	Radiation alone loses element.
nuclide	positive-nucleon-count	element-only-identity	Element alone collapses isotopes.
state	held-nuclear-state	ground-state-assumption	Nuclear state cannot be assumed.
species	held-chemical-species	bare-nuclide-answer	Chemistry requires species identity.
phase	held-chemical-phase	phase-free-species	Phase affects the chemical record.
occurrence	positive-counted-occurrence	continuum-amount-premise	Continuum amount hides occurrences.
record	complete-nuclide-species-vector	selected-nuclide-row	One row hides carrier diversity.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
extension	fresh-occurrence-p reserves-identity	lookup-table-exception	A lookup cannot define identity.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:0618f410bce1622e87fe16c0e1db32daca60bcd2cccd1de3a36e772977c70fc.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:3858c3e498b0e04ced7ac71c03f37d1093614d6d06aab64ea0d01255db558b10.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b97d8a968969baffe4cc1e59ac1c8e4d2505176acde3a818afef014308ea54c5.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:a71000c77dfa72a521a359747a1fb6be2fc9edd6ca24a38e3ca97bb58c785669.

Independent reconstruction. Implementation hash

sha256:f0eaea3928b1ef71daa8a5a42e6c7b812f2f2429e5ce567c65dbfdec47dbeff4; independent certificate sha256:137808b4440e6df61274e602619aa9579c07eeb2f2a555b7f6bb1c793a883e31; external-validation hash sha256:9a6f93134b854d0a877380a1f6cb2f86550c683a55663af1710f024139766377. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-NUCLEAR-PHYSICS-DATA-ISOTOPIC-COMPOSITION
- NIST-SRM-4239A-STRONTIUM-90-2022
- NIST-SRM-4239A-STRONTIUM-90-2022
- NIST-SRM-4239A-STRONTIUM-90-2022
- NIST-SRM-4239A-STRONTIUM-90-2022
- NIST-RADIOACTIVITY-SRMS-GENERAL-INFO-2025
- NIST-SRM-4239A-STRONTIUM-90-2022
- NUCHEM-001-004-SOURCE-INVENTORY

Comparison records:

- complete four-source post-seal family retained as 8 PDF pages, 2 HTML documents and 34,442 extracted characters
- all 8 separately registered claim targets retained
- all measured values units uncertainties assumptions discrepancies adverse absent unavailable and unresolved records remain downstream provenance

Falsification boundary: The claim halts if the survivor is nonunique; any element, nuclide, state, species, phase, occurrence, uncertainty, adverse or unavailable row, page or byte is omitted; or an external value selects the law.

Measurement receipt: sha256:f260ea1dd627185d71c68f2d28944de62179f55c60566fef1d18c3f5b95e1540.
Isolation certificate: sha256:78015c48801b6df327b45633264ec732d17bf1f9bd88e2054676bbf77361c8e1.
Custody certificate: sha256:bb1e763e1fbaaaa2ed4a66a016d1df52f5041acddb5f328d32a8bd3242a858a5.

Registered source descriptions:

- National Institute of Standards and Technology: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external nuclide table chemical matrix activity or uncertainty selects the survivor
- unreported nuclear state is retained as unreported rather than inferred

Immutable identities. Source manifest

sha256:3184b39379b82e6d7006c824357ddf2d7dc49c0c46664b84e94ffb7dbe0e4aef; derivation seal
 sha256:d55d44dfe8127d2f2b88e44118d49a46290ecb313fc76c5ff14d72c0227a2cd8; engine receipt
 sha256:86fdd019ae04bf7dc01fb957f5c136b658218aaecab526898530a2461965824f at receipts/engin
 e/model_admitted/SFT-CHEM-NUCLEAR-CHEMICAL-CARRIER-001-86fdd019ae04bf7d.json; empirical hash
 sha256:06785541016e56d548e22963997ec18afee51b2cce3166da593934bb07674d39; measurement receipt
 sha256:f260ea1dd627185d71c68f2d28944de62179f55c60566fef1d18c3f5b95e1540.

204. Fold radioactive chemical-transformation network

Claim identity: SFT-CHEM-RADIOACTIVE-CHEMICAL-TRANSFORMATION-002

Question and theorem. A radioactive chemical transformation is a complete directed network retaining parent, daughter, both chemical species, channel and positive counted events, with structural absence and reconstructive extension.

*held-parent-nuclide__held-daughter-nuclide__held-parent-daughter-species__held-decay-channel__positive-counted-tr
 ansformations__complete-directed-transformation-network__structural-EmptyOne-no-edge__successor-retains-prior-n
 etwork*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-PHYS-NUCLEAR-RADIOACTIVE-DECAY-TERMINAL-005
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-ELEM-ISOTOPE-001
- SFT-CHEM-NUCLEAR-CHEMICAL-CARRIER-001

Generated grammar. Generate the literal product of eight registered transformation decisions.

Boundary: Eight dimensions exhaust parent, daughter, chemistry, channel, events, network, absence and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-parent-nuclide__he
 ld-daughter-nuclide__held-parent-daughter-species__held-decay-channel__positive-counted
 -transformations__complete-directed-transformation-network__structural-EmptyOne-no-edge
 __successor-retains-prior-network. Closure is depth_independent; minimality and named-shape uniqueness
 are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
parent	held-parent-nuclid e	parent-number-only	Parent number loses identity.
daughter	held-daughter-nucl ide	anonymous-product	Product loses daughter identity.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
chemistry	held-parent-daughter-species	nuclear-edge-only	Chemical states would be lost.
channel	held-decay-channel	unlabelled-transition	Channel identity is required.
events	positive-counted-transformations	continuum-decay-flow	Flow hides events.
network	complete-directed-transformation-network	selected-decay-edge	One edge hides branching.
absence	structural-EmptyOne-no-edge	numerical-zero-daughter	Numerical zero is not native absence.
extension	successor-retains-prior-network	daughter-overwrites-parent	Overwriting destroys reconstruction.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: f31bdf01dd691a772697eedef0a97b164b847244ed82a18917b0c57e4d245f7f.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: b6ec6dcaed70f2ba339612f860b0abbf5becbdfae724c8c4d2a6ae7055da07a1.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: eaede9bb30d999b548e60d4f2e7c57a2df9c9103848c1a4ecade49240dfd58bb.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: ee4280e87ffdc93e3a5da0bf88833e4b34457f37f655c34878b70972761d3719.

Independent reconstruction. Implementation hash

sha256: 61e4f97b87f20a2c0c8a3ac8b95314f58c422a9273a337c2f8d044d800af3e5c; independent certificate
sha256: 2a3eab12145b880f1b54e0e58b061de653438c27b5ad8bf25ffb3188f54f40a8; external-validation
hash sha256: ae8fa8f740531f73251a81d0e2f9cc7107ce173b233fd09982599267d7a68a2c. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-SRM-4239A-STRONTIUM-90-2022
- NIST-SRM-4239A-STRONTIUM-90-2022
- NIST-SRM-4239A-STRONTIUM-90-2022
- NIST-SRM-4324C-URANIUM-232-2025
- NIST-SRM-4324C-URANIUM-232-2025
- NIST-SRM-4324C-URANIUM-232-2025
- NUCHEM-001-004-SHARED-SOURCES
- NUCHEM-001-004-SOURCE-INVENTORY

Comparison records:

- complete four-source post-seal family retained as 8 PDF pages, 2 HTML documents and 34,442 extracted characters
- all 8 separately registered claim targets retained
- all measured values units uncertainties assumptions discrepancies adverse absent unavailable and unresolved records remain downstream provenance

Falsification boundary: The claim halts if the survivor is nonunique; any parent, daughter, chemical state, channel, edge, equilibrium, assumption, disagreement, page or byte is omitted; or an external network selects the law.

Measurement receipt: sha256:8f7db7cffff46a871d6d3aec89ccbaa2788308873f47023b1cd7aab0bd5ca2256.

Isolation certificate: sha256:c21323d9e3224b49abc36e85af9f7fdd9150bf86441751797f14d1b1850a9bdd.

Custody certificate: sha256:bf72dfad8344f40c8ce4cadf04e3feff4e9a03712c794a3eb71156ce7368dff6.

Registered source descriptions:

- National Institute of Standards and Technology: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external decay chain or equilibrium record selects the survivor
- no source assumption or disagreement is suppressed

Immutable identities. Source manifest

sha256:9f3c2952565da087c75b36b8f88ad3579e0531114046bda103c4b91f9e5bcb6f; derivation seal

sha256:5f661b610007058ff6c8b290e1302c84b0a0250d464c633fd8c4fde0cf787770; engine receipt

sha256:d9f532123edcae0e686f4a4ccfbddacc43605f56dfddc3719b6f603c298d5687f at receipts/engine/model_admitted/SFT-CHEM-RADIOACTIVE-CHEMICAL-TRANSFORMATION-002-d9f532123edcae0e.json

; empirical hash sha256:96ed1ed2a1fac5d6559c2ecae9d843556d38fc411c13bffddfa95ef013a026df;

measurement receipt sha256:8f7db7cffff46a871d6d3aec89ccbaa2788308873f47023b1cd7aab0bd5ca2256.

205. Fold activity-amount-time law

Claim identity: SFT-CHEM-ACTIVITY-AMOUNT-TIME-003

Question and theorem. Activity is the exact positive ratio of counted transformations to counted resource intervals for a held nuclide species, while retained amount is positive Take or structural absence.

held-nuclide-species__positive-counted-initial-occurrences__positive-counted-transformations__positive-counted-resource-intervals__exact-transformations-per-resource__positive-Take-or-EmptyOne-retained-amount__complete-activity-amount-time-vector__ledger-successor-recomputes-relation

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-MEAS-AMOUNT-001
- SFT-PHYS-NUCLEAR-RADIOACTIVE-DECAY-TERMINAL-005
- SFT-CHEM-NUCLEAR-CHEMICAL-CARRIER-001
- SFT-CHEM-RADIOACTIVE-CHEMICAL-TRANSFORMATION-002

Generated grammar. Generate the literal product of eight registered activity decisions.

Boundary: Eight dimensions exhaust identity, amount, events, time, activity, remaining amount, record and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-nuclide-species__positive-counted-initial-occurrences__positive-counted-transformations__positive-counted-resource-intervals__exact-transformations-per-resource__positive-Take-or-EmptyOne-retained-amount__complete-activity-amount-time-vector__ledger-successor-recomputes-relation

. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
identity	held-nuclide-species	anonymous-activity-number	Activity needs nuclide/species custody.
amount	positive-counted-initial-occurrences	continuum-amount-premise	Continuum amount hides occurrences.
events	positive-counted-transformations	expected-decay-number	Expectation is not event custody.
time	positive-counted-resource-intervals	continuous-time-premise	Continuum time is not proof support.
activity	exact-transformations-per-resource	fitted-decay-constant	A fit cannot define activity.
remaining	positive-Take-or-EmptyOne-retained-amount	signed-amount-difference	Signed subtraction imports negative values.
record	complete-activity-amount-time-vector	selected-reference-time	One time hides evolution.
extension	ledger-successor-recomputes-relation	differential-equation-premise	A differential equation is not required.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:6a26dbace277c08586e8e760625c4a853d2bc6062d3f001bb5b28a4b48daa109.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:9ee121e0a8694d1f42bfe6bd27e949b8e66e46b7b46e1280da4e2c35f7f75cc7.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ef1be320201b51e6eacdab23d6ef0c5ef4ed2af259ef7d937a5cbf09bad403c4.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:beebca7074c8f8895bcde32950a7834031e813eb83f9b3f8d654f21169af8dab.

Independent reconstruction. Implementation hash

sha256:f918e10aabc29948051757bde51163db4c7fa5e2b743caed82f120b385792897; independent certificate sha256:0ea6d7e541ba26a1a333853f6dd418534546ac390d7fcac5fb251f3c4d908345; external-validation hash sha256:0c8d9c2a01931f67399adbf9a2fb4cc15715d6e64772d3d10845ae4020d96950. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- NIST-RADIOACTIVITY-SRMS-GENERAL-INFO-2025
- NIST-RADIOACTIVITY-SRMS-GENERAL-INFO-2025
- NIST-RADIOACTIVITY-SRMS-GENERAL-INFO-2025
- NIST-SRM-4239A-STRONTIUM-90-2022
- NIST-SRM-4239A-STRONTIUM-90-2022
- NIST-SRM-4239A-STRONTIUM-90-2022
- NIST-SRM-4324C-URANIUM-232-2025

- NUCHEM-001-004-SOURCE-INVENTORY

Comparison records:

- complete four-source post-seal family retained as 8 PDF pages, 2 HTML documents and 34,442 extracted characters
- all 8 separately registered claim targets retained
- all measured values units uncertainties assumptions discrepancies adverse absent unavailable and unresolved records remain downstream provenance

Falsification boundary: The claim halts if the survivor is nonunique; any identity, amount form, event, interval, activity, reference time, uncertainty, correction, adverse or unavailable row, page or byte is omitted; or a measurement selects the law.

Measurement receipt: sha256:e9a261aff4fd04b0c18e635f849ee22658b1f40d884c694edb3ee297eafef212.

Isolation certificate: sha256:d9ff1cd165893ba9164e2433f0072cadad903cdc2531983ce9e792432c5fe843.

Custody certificate: sha256:82483bcd4007ecf8a8ef5542f1cd1a66a68e33510a5605f383957d51ef0c19db.

Registered source descriptions:

- National Institute of Standards and Technology: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured activity half-life reference time or unit selects the survivor
- all decimal conventional continuum and signed inscriptions remain downstream provenance

Immutable identities. Source manifest

sha256:d81b99e17d48965bed9322eae6c741c33f4255b697e7861f64aa70fdf06b0732; derivation seal

sha256:5e59312566e993fda923a294af6a4402b26cc1b95d70009f0153cee32fdb57c3; engine receipt

sha256:e332f2d356c8111e51ad00d890645a4fa7b8be0e149d4f5f9d707e71d86d5e0b at

receipts/engine/model_admitted/SFT-CHEM-ACTIVITY-AMOUNT-TIME-003-e332f2d356c8111e.json;

empirical hash sha256:6b7892c44757763a3af01ba72211f04bf745e452123643e2ae3153e93e84f492;

measurement receipt sha256:e9a261aff4fd04b0c18e635f849ee22658b1f40d884c694edb3ee297eafef212.

206. Fold radioactive branching chemical-yield law

Claim identity: SFT-CHEM-RADIOACTIVE-BRANCHING-CHEMICAL-YIELD-004

Question and theorem. Radioactive chemical yield retains every held decay channel and daughter species, positive branch and recovered event counts, exact recovery ratios and a complete branch partition of One.

held-decay-channel__held-daughter-chemical-species__positive-counted-branch-events__positive-counted-recovered-events__exact-recovered-per-branch-ratio__complete-branch-partition-sums-to-One__complete-daughter-yield-vector__new-branch-repartitions-complete-total

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-PHYS-DECAY-WIDTH-BRANCHING-LIFETIME-TERMINAL-006
- SFT-CHEM-RADIOACTIVE-CHEMICAL-TRANSFORMATION-002

- SFT-CHEM-ACTIVITY-AMOUNT-TIME-003

Generated grammar. Generate the literal product of eight registered branching decisions.

Boundary: Eight dimensions exhaust channel, daughter, events, recovery, yield, partition, record and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-decay-channel__held-daughter-chemical-species__positive-counted-branch-events__positive-counted-recovered-events__exact-recovered-per-branch-ratio__complete-branch-partition-sums-to-One__complete-daughter-yield-vector__new-branch-repartitions-complete-total. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
channel	held-decay-channel	unlabelled-yield	Yield needs channel identity.
daughter	held-daughter-chemical-species	anonymous-product-amount	Product amount loses species.
events	positive-counted-branch-events	continuum-branch-probability	Probability hides event support.
recovery	positive-counted-recovered-events	fitted-chemical-recovery	A fitted factor is not custody.
yield	exact-recovered-per-branch-ratio	percentage-premise	A percentage is a display form.
partition	complete-branch-partition-sums-to-One	independent-branch-fractions	Independent values need not close.
record	complete-daughter-yield-vector	selected-favorable-daughter	Selection hides adverse/absent branches.
extension	new-branch-repartitions-complete-total	renormalize-after-omission	Renormalization hides missing branches.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:dc86736b14c945dffec5f7075509eff42bb53295d579562948a6b0a9bcfcc2c3.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:f61e6511c309d1791a76b857122d48c6b8285efbd1b88fc1f7c3056b927d98b5.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:04d81801f3d121383420d6ef12cbfb8437ecb90e3e1fa363385d160765795737.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:eb7df2f8b23b6a774f4bd7b4bce1dd6517cd0644a21c502fdad5971642f4a3f0.

Independent reconstruction. Implementation hash

sha256:8fc95440e85a971a71ecbdaca86695c60fcf80dc6a82446ac2e28eb3d66f045b; independent certificate sha256:1b19ba2a82d7a86b64ad9b2289935575c4a792c29710d40fc10c6b56ea3c1346; external-validation hash sha256:62281b6cb62f2a43ccf5a90e443a789c5816c0724a3a0b4604d5105b1122c976. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-SRM-4324C-URANIUM-232-2025

- NIST-SRM-4324C-URANIUM-232-2025
- NIST-RADIOACTIVITY-SRMS-GENERAL-INFO-2025
- NIST-SRM-4239A-STRONTIUM-90-2022
- NIST-SRM-4324C-URANIUM-232-2025
- NIST-SRM-4324C-URANIUM-232-2025
- NUCHEM-001-004-SHARED-SOURCES
- NUCHEM-001-004-SOURCE-INVENTORY

Comparison records:

- complete four-source post-seal family retained as 8 PDF pages, 2 HTML documents and 34,442 extracted characters
- all 8 separately registered claim targets retained
- all measured values units uncertainties assumptions discrepancies adverse absent unavailable and unresolved records remain downstream provenance

Falsification boundary: The claim halts if the survivor is nonunique; any channel, daughter, branch, recovery, method, assumption, discrepancy, uncertainty, unavailable row, page or byte is omitted; or external outcomes select the law.

Measurement receipt: sha256:7b19a8fe7cbf30015ae3f52dbe6154ddbe3058274d12d48c1d834c4a048c778d.

Isolation certificate: sha256:9b5c17e6c5cb1a3fef1e83a63f3bd52ec42c6129b3ff9ac726308c99fa88e98b.

Custody certificate: sha256:200e72583a4786a6d02e5afb5cc46ea2db8837387724c8de9f138f1e311ae230.

Registered source descriptions:

- National Institute of Standards and Technology: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external branch fraction daughter list or recovery result selects the survivor
- an unavailable numerical branch fraction remains unavailable and is never fabricated or renormalized

Immutable identities. Source manifest

sha256:4633a113bef65768756f17ff2b3d7074d0f23c8d30e9c29f79048e7ffe732ec6; derivation seal
 sha256:e54430222891de4335ab567ceff615b299d2b28924a4046ae1933f2e561d95de; engine receipt
 sha256:85705f51398e1561e5be35cf9f9806fa5878679dac6c5abaaf16435f2fe3ee6c at receipts/engine/
 model_admitted/SFT-CHEM-RADIOACTIVE-BRANCHING-CHEMICAL-YIELD-004-85705f51398e1561.json;
 empirical hash sha256:1b672443dcf1dc74d0cef5f5747bc768841c9df8211f8ecbd2811dd247b86eed;
 measurement receipt sha256:7b19a8fe7cbf30015ae3f52dbe6154ddbe3058274d12d48c1d834c4a048c778d.

207. Fold transient and secular radiochemical-equilibrium law

Claim identity: SFT-CHEM-RADIOCHEMICAL-EQUILIBRIUM-005

Question and theorem. Radiochemical equilibrium is an exact persistent parent/daughter activity-ratio class over complete positive ordered resource intervals: persistent non-One is transient, persistent One is secular, and nonpersistent recurrence is structural absence.

held-parent-daughter-identity__positive-ordered-resource-intervals__exact-counted-activity-pair__held-ratio-or-structural-absence__persistent-nonOne-ratio-forces-transient__persistent-One-ratio-forces-secular__complete-parent-daughter-time-vector__finite-successor-recomputes-regime

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-RADIOACTIVE-CHEMICAL-TRANSFORMATION-002
- SFT-CHEM-ACTIVITY-AMOUNT-TIME-003
- SFT-CHEM-RADIOACTIVE-BRANCHING-CHEMICAL-YIELD-004

Generated grammar. Generate the literal product of eight registered equilibrium decisions.

Boundary: Eight dimensions exhaust identity, support, activity, ratio, transient, secular, record and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-parent-daughter-identity__positive-ordered-resource-intervals__exact-counted-activity-pair__held-ratio-or-structural-absence__persistent-nonOne-ratio-forces-transient__persistent-One-ratio-forces-secular__complete-parent-daughter-time-vector__finite-successor-recomputes-regime. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
identity	held-parent-daughter-identity	anonymous-activity-pair	Equilibrium needs parent/daughter identity.
support	positive-ordered-resource-intervals	continuous-time-premise	Continuum time is not proof support.
activity	exact-counted-activity-pair	fitted-decay-curve	A fitted curve cannot define activity.
relation	held-ratio-or-structural-absence	signed-activity-difference	A signed scalar imports negative values.
transient	persistent-nonOne-ratio-forces-transient	named-transient-assumption	A name cannot establish a regime.
secular	persistent-One-ratio-forces-secular	approximate-equality-tolerance	A tolerance is a free parameter.
record	complete-parent-daughter-time-vector	selected-timepoint	One point cannot establish recurrence.
extension	finite-successor-recomputes-regime	differential-equation-premise	No imported differential equation is needed.

Adverse controls. All mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:a9c62cada930affa54c97c70a94c9ebdb9c938734720a656f4cb32492a53138f.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:720dbade1d46550917942824a02ed7dad670e645aac6c4af052941af32c2fa53.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:3b97b38a9367da1ed323e7b6ce32fe87729e1773924ac82b8d4557c92093c582.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:5dec36e137f7359e18d203c6f80db3ff481276f1213203a4a323e68d5c791c3f.

Independent reconstruction. Implementation hash

sha256:28671ab11135789aa900c4aef4737d04b3b48a3db21997192b805a21a4a8b74a; independent certificate sha256:9fa7d7f3fd2c5acf82c75ffab669bfedc9b46b990f6cfe5ace3f89ba252d3d87; external-validation hash sha256:073a808c9771f5c56d12d385b3c20b290f28fec9b83416302d700fba77dd578a. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-SRM-4239A-STRONTIUM-90-2022
- NUCHEM-005-NIST-SRM-PAIR
- NUCHEM-005-NIST-SRM-PAIR
- NUCHEM-005-NIST-SRM-PAIR
- NIST-SRM-4324C-URANIUM-232-2025
- NUCHEM-005-NIST-SRM-PAIR
- NUCHEM-005-008-SHARED-SOURCES
- NUCHEM-005-008-SOURCE-INVENTORY

Comparison records:

- complete five-source post-seal family retained as 266 PDF pages, 1 HTML document and 410,095 extracted characters
- all 8 separately registered claim targets retained
- all measured values units uncertainties assumptions corrections estimates discrepancies losses adverse absent unavailable and unresolved records remain downstream provenance

Falsification boundary: The claim halts if the survivor is nonunique; any parent, daughter, interval, activity, regime, assumption, uncertainty, unavailable row, page or byte is omitted; or an external equilibrium label selects the law.

Measurement receipt: sha256:032c121d31db68feedce1aeac533f82e443b5308501a30a821884222b96e466f.

Isolation certificate: sha256:501d7c358c6d91701fcbf98a991ec078a6dce92c5e6a17641183c1fd02959dbb.

Custody certificate: sha256:40d93b9e7d2343d68ccd3cef3f3b41092d03479a02173929e4c9573d6a56d899.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured activity time support or conventional decay model selects the survivor
- qualitative equilibrium records cannot fabricate an unavailable numerical time series

Immutable identities. Source manifest

sha256:1b34828bd6207da19dd9f2e12075239cead591b4bffc4cc7c2ffd4b588fc83ef; derivation seal sha256:7bcba1b3c66d3c00ff5a014a600327f70a1ece92d35cec92fb03b3d34878c88f; engine receipt sha256:9639309d02ccbeaa7794ad8e25ec0a28d0714ec5ce3e0a1a0a27c2c7a695cdb7 at receipts/engine/model_admitted/SFT-CHEM-RADIOCHEMICAL-EQUILIBRIUM-005-9639309d02ccbeaa.json; empirical hash sha256:b8ffeaab31ca33855714763b93ae424b7da72d60d064590804420b8185629ade; measurement receipt sha256:032c121d31db68feedce1aeac533f82e443b5308501a30a821884222b96e466f.

208. Fold isotope-exchange reaction law

Claim identity: SFT-CHEM-ISOTOPE-EXCHANGE-006

Question and theorem. Isotope exchange preserves exact element, isotope and carrier identities and all positive isotope/carrier totals, carries direction as a held positive excess, and forces its exchange quotient by cross-products.

held-element-light-heavy-isotopes__held-distinct-chemical-carriers__positive-complete-four-count-inventory__exact-isotope-and-carrier-conservation__held-direction-positive-Take__exact-cross-product-exchange-quotient__equal-forward-reverse-closes-EmptyOne__successor-preserves-identities-and-totals

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-ELEM-ISOTOPE-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-NUCLEAR-CHEMICAL-CARRIER-001
- SFT-CHEM-RADIOACTIVE-CHEMICAL-TRANSFORMATION-002

Generated grammar. Generate the literal product of eight registered exchange decisions.

Boundary: Eight dimensions exhaust identity, carriers, inventory, conservation, direction, quotient, equilibrium and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *held-element-light-heavy-isotopes__held-distinct-chemical-carriers__positive-complete-four-count-inventory__exact-isotope-and-carrier-conservation__held-direction-positive-Take__exact-cross-product-exchange-quotient__equal-forward-reverse-closes-EmptyOne__successor-preserves-identities-and-totals*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
identity	held-element-light-heavy-isotopes	mass-number-only	Mass number alone loses elemental and isotope identity.
carriers	held-distinct-chemical-carriers	phase-free-exchange	Exchange requires two chemical carriers.
inventory	positive-complete-four-count-inventory	continuum-concentration-remise	Continuum concentration hides occurrences.
conservation	exact-isotope-and-carrier-conservation	isotope-label-overwrite	Overwriting destroys custody.
direction	held-direction-positive-Take	signed-exchange-extent	A signed scalar imports negative values.
quotient	exact-cross-product-exchange-quotient	fitted-exchange-constant	A fit cannot define exchange.
equilibrium	equal-forward-reverse-closes-EmptyOne	numerical-zero-net	Numerical zero is not native balance.
extension	successor-preserve-identities-and-totals	lookup-fractionation-factor	A lookup cannot select a transition.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: be709f4d0ff936e6f5324ad8ae1fc1f8df0f34c1ea713da8dc2db10f48b11642.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 0cb8aefd385e8492eb546c9409c815b0b0f32523bf6d2b655d9cc2d92a782d9c.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 4bfc8ae5a4e42fafb538a940a0383f580f81c0b9ebddaa3d4d0a237e26934e21.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: a1d402e3611c1c292d1336e3c4d2a39345d66264eed6b89833396a94fe6e70d6.

Independent reconstruction. Implementation hash

sha256: e1bd466d6a0431d42a6eeb88db0fcf28438e5d9d90e5abcc18b758be97308730; independent certificate sha256: 9dfcd4759ab1e8a96bfab1a35a0e676956b6af5c3918fd7bd68caa0c917fd914; external-validation hash sha256: ec9a61a40262d62fa367d62d1a7a55e90a2c4a800198e42710a5bfd45b8905af. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- USGS-WRIR-02-4172-ISOTOPE-EQUILIBRIUM-CONSTANTS
- USGS-WRIR-02-4172-ISOTOPE-EQUILIBRIUM-CONSTANTS
- USGS-WRIR-02-4172-ISOTOPE-EQUILIBRIUM-CONSTANTS
- USGS-WRIR-02-4172-ISOTOPE-EQUILIBRIUM-CONSTANTS
- USGS-WRIR-02-4172-ISOTOPE-EQUILIBRIUM-CONSTANTS
- USGS-WRIR-02-4172-ISOTOPE-EQUILIBRIUM-CONSTANTS
- NUCHEM-005-008-SHARED-SOURCES
- NUCHEM-005-008-SOURCE-INVENTORY

Comparison records:

- complete five-source post-seal family retained as 266 PDF pages, 1 HTML document and 410,095 extracted characters
- all 8 separately registered claim targets retained
- all measured values units uncertainties assumptions corrections estimates discrepancies losses adverse absent unavailable and unresolved records remain downstream provenance

Falsification boundary: The claim halts if the survivor is nonunique; any isotope, carrier, conservation, quotient, example condition, nonideality, adverse row, page or byte is omitted; or an external quotient selects the law.

Measurement receipt: sha256: 0a508103a270400dfb42c2f34a62931c00106c1cf533a8286fec2ada926503f7.

Isolation certificate: sha256: ae758ae9b2ac13497da7548f4bbd3dc78750722f461357bdf135459becf28861.

Custody certificate: sha256: 5349ba90eeb9eebecc50f847733b6882ad5ece519079afdca965b771fbb0877b.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external alpha table equilibrium constant or species result selects the survivor
- every nonideality assumption and unavailable row remains

Immutable identities. Source manifest

sha256:b31e7848b0938fe51028e462925dad6492e1ae92f301e4fecbb8a1622d007e96; derivation seal sha256:7392731162add5b42c574e48d5ade338edc17960a8a8dcba7c7b32b2d109f965; engine receipt sha256:a94b6eb90aef2a3a5d04dd4c3135c1f43fd086d6c826078c91a09f2dcee42a5e at receipts/engine/model_admitted/SFT-CHEM-ISOTOPE-EXCHANGE-006-a94b6eb90aef2a3a.json; empirical hash sha256:0455df8f03e85f5bc41e9a06ff40c3fdc7b0426232d08a440bc732e983d0f60; measurement receipt sha256:0a508103a270400dfb42c2f34a62931c00106c1cf533a8286fec2ada926503f7.

209. Fold equilibrium isotope-fractionation law

Claim identity: SFT-CHEM-EQUILIBRIUM-ISOTOPE-FRACTIONATION-007

Question and theorem. Equilibrium isotope fractionation is the exact ratio of two positive heavy-per-light phase ratios, with held enrichment or structural coincidence, exchange balance and a stable factor.

held-light-heavy-isotopes__held-distinct-phase-pair__positive-complete-isotope-counts__exact-heavy-per-light-ratios__exact-ratio-of-ratios__held-enrichment-or-EmptyOne-coincidence__exchange-balance-plus-stable-factor__complete-vector-successor-recomputes

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-ELEM-ISOTOPE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-ISOTOPE-EXCHANGE-006

Generated grammar. Generate the literal product of eight registered fractionation decisions.

Boundary: Eight dimensions exhaust identity, phases, inventory, ratios, factor, orientation, equilibrium and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-light-heavy-isotopes__held-distinct-phase-pair__positive-complete-isotope-counts__exact-heavy-per-light-ratios__exact-ratio-of-ratios__held-enrichment-or-EmptyOne-coincidence__exchange-balance-plus-stable-factor__complete-vector-successor-recomputes. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
identity	held-light-heavy-isotopes	anonymous-heavy-light	Heavy/light labels need isotope identity.
phases	held-distinct-phase-pair	single-phase-ratio	Fractionation compares two carriers or phases.
inventory	positive-complete-isotope-counts	continuum-abundance-premise	Continuum abundance hides occurrences.
ratios	exact-heavy-per-light-ratios	decimal-isotope-ratio	A decimal is only a display form.
factor	exact-ratio-of-ratios	fitted-fractionation-factor	No fitted factor is admissible.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
orientation	held-enrichment-or-EmptyOne-coincidence	signed-delta-premise	A signed delta imports negative proof values.
equilibrium	exchange-balance-plus-stable-factor	named-equilibrium-assumption	A label cannot prove equilibrium.
extension	complete-vector-successor-recomputes	selected-isotope-pair	One selected pair hides the complete system.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256 : f4d8d9d4bf80a42ad50824829cf9f4c9e273d6d7cf01592eab738541b8196916.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : da19742a955339b0f22fba64be094ec1ffcbb6836a32d9eb1163f51c71d439af.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256 : d1c9116ef34052510837382ba228b3208a69410f7f6c9d469a7cbff35e6ab82b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256 : 2a45349c0b21fe6158df5ff911ce1eda7950726351e89b4505c169ba3b267d65.

Independent reconstruction. Implementation hash

sha256 : 81e8b833464efce2c34bdecafb1b8e6bbd3648ed9a021d97880b345391d68168; independent certificate sha256 : 3b01fd8dd1f09539f2fdeb39de1dda3911a1f17669512fff034467ebebe070f7; external-validation hash sha256 : bbd9c285793de734719d941c8e8674c526a40b2938d1960a7d1fe7c2e5497d75. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- USGS-PP-440-KK-STABLE-ISOTOPE-FRACTIONATION
- USGS-PP-440-KK-STABLE-ISOTOPE-FRACTIONATION
- USGS-PP-440-KK-STABLE-ISOTOPE-FRACTIONATION
- USGS-PP-440-KK-STABLE-ISOTOPE-FRACTIONATION
- USGS-PP-440-KK-STABLE-ISOTOPE-FRACTIONATION
- USGS-WRIR-02-4172-ISOTOPE-EQUILIBRIUM-CONSTANTS
- NUCHEM-005-008-SHARED-SOURCES
- NUCHEM-005-008-SOURCE-INVENTORY

Comparison records:

- complete five-source post-seal family retained as 266 PDF pages, 1 HTML document and 410,095 extracted characters
- all 8 separately registered claim targets retained
- all measured values units uncertainties assumptions corrections estimates discrepancies losses adverse absent unavailable and unresolved records remain downstream provenance

Falsification boundary: The claim halts if the survivor is nonunique; any isotope ratio, phase, factor, orientation, temperature, fit, estimate, discrepancy, unavailable row, page or byte is omitted; or an external factor selects the law.

Measurement receipt: sha256 : ae4dd750472a65e441131efac138f2fe6e71f01e9c5734d670fb5022e9489525.
Isolation certificate: sha256 : 529ba92f4690191819cf5acd6035d0b0dbb93e593d0acbb52ddce74271d6c7550.
Custody certificate: sha256 : 611665ef01059de46099437bd950198e92fe30f6dddceae0cbed101563a8d5b.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured fitted calculated inferred or estimated alpha value selects the survivor
- external signed deltas remain downstream measurement inscriptions

Immutable identities. Source manifest

sha256:d54c2454fbf6a3d9b93627d8d3b30b09ee056fb153cb80c797fccd081aa1bbb4; derivation seal sha256:31ee1108f5c6a0ac8fad3ef929d63d66d2d26116b70e315a7b72c6c30de055d8; engine receipt sha256:0ba2c2105037fa8f4bf1adba65b7bb8322c063726a0d379ebac9121a2ee2e44c at receipts/engine/model_admitted/SFT-CHEM-EQUILIBRIUM-ISOTOPE-FRACTIONATION-007-0ba2c2105037fa8f.json; empirical hash sha256:b0d40bf6b306548710d3273e30ee16b74bd633f05b6c426a8ead5a29a5578e54; measurement receipt sha256:ae4dd750472a65e441131efac138f2fe6e71f01e9c5734d670fb5022e9489525.

210. Fold kinetic isotope-fractionation law

Claim identity: SFT-CHEM-KINETIC-ISOTOPE-FRACTIONATION-008

Question and theorem. Kinetic isotope fractionation retains one reaction and two isotope identities over complete positive ordered resources, forces exact product rates and their factor, and records faster class and remainder without signed or negative native values.

held-reaction-light-heavy-identities__positive-ordered-resource-intervals__positive-counted-isotope-products__exact-products-per-resource-rates__exact-light-heavy-rate-ratio__held-faster-class-or-EmptyOne__positive-Take-or-EmptyOne-remainder__finite-time-series-successor-recomputes

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-ELEM-ISOTOPE-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-ACTIVITY-AMOUNT-TIME-003
- SFT-CHEM-ISOTOPE-EXCHANGE-006
- SFT-CHEM-EQUILIBRIUM-ISOTOPE-FRACTIONATION-007

Generated grammar. Generate the literal product of eight registered kinetic decisions.

Boundary: Eight dimensions exhaust identity, support, events, rates, factor, orientation, inventory and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-reaction-light-heavy-identities__positive-ordered-resource-intervals__positive-counted-isotope-products__exact-products-per-resource-rates__exact-light-heavy-rate-ratio__held-faster-class-or-EmptyOne__positive-Take-or-EmptyOne-remainder__finite-time-series-successor-recomputes. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
identity	held-reaction-light-heavy-identities	mass-only-rate	Rate needs reaction and isotope identity.
support	positive-ordered-resource-intervals	continuous-time-premise	Continuum time is not proof support.
events	positive-counted-isotope-products	expected-product-yield	Expectation is not event custody.
rates	exact-products-per-resource-rates	fitted-rate-constants	Fitted constants cannot define the native rates.
factor	exact-light-heavy-rate-ratio	imported-kinetic-isotope-effect	No imported factor is admissible.
orientation	held-faster-class-or-EmptyOne	signed-rate-difference	A signed difference imports negative values.
inventory	positive-Take-or-EmptyOne-remainder	negative-depleted-reactant	Depletion cannot produce a negative proof value.
extension	finite-time-series-successor-recomputes	Rayleigh-continuum-premise	A continuum law is not required.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: ca3da4f2e5948ca221a7584925ebb457d32346668103764f7fe74375b36f93e7.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 140e3043752cc502f5b7fd8e7c5fbacc45cea1b8aa79d56d57eeb47d5c85fbd5.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 68b49bca846b1c51534e01bdaf27406f288b6d7b18f5efefed1f1104ef07601a.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 48afdc2c3da3c25693447572add222ff06a0f1c9a9968f5d8e2d08f2c336570d.

Independent reconstruction. Implementation hash

sha256: fbe63740d034b3f3d4ba24fd6c2b9753e8f1bb292ab802726ed9c1589fe2d5a5; independent certificate sha256: 74d25ea8625052cf2834e2747df0d86e91acdb82000635a03c115338e65b0582; external-validation hash sha256: ff45f6968e6292286db034ba468210682e3df4de91fef502b92d11a40d4ac67f. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NBS-RP729-ELECTROLYTIC-HYDROGEN-OXYGEN-FRACTIONATION-1934
- NBS-RP729-ELECTROLYTIC-HYDROGEN-OXYGEN-FRACTIONATION-1934
- NBS-RP729-ELECTROLYTIC-HYDROGEN-OXYGEN-FRACTIONATION-1934
- NBS-RP729-ELECTROLYTIC-HYDROGEN-OXYGEN-FRACTIONATION-1934
- NBS-RP729-ELECTROLYTIC-HYDROGEN-OXYGEN-FRACTIONATION-1934
- NBS-RP729-ELECTROLYTIC-HYDROGEN-OXYGEN-FRACTIONATION-1934
- NUCHEM-005-008-SHARED-SOURCES
- NUCHEM-005-008-SOURCE-INVENTORY

Comparison records:

- complete five-source post-seal family retained as 266 PDF pages, 1 HTML document and 410,095 extracted characters
- all 8 separately registered claim targets retained
- all measured values units uncertainties assumptions corrections estimates discrepancies losses adverse absent unavailable and unresolved records remain downstream provenance

Falsification boundary: The claim halts if the survivor is nonunique; any reaction path, resource point, product, rate, factor, steady state, correction, loss, estimate, limitation, unreported row, page or byte is omitted; or an external response selects the law.

Measurement receipt: sha256:1353aa80d3100c33d67f1c292c4db731aae02f57be5ecef4388be8ab981178cb.

Isolation certificate: sha256:6877210ecd50c4b453df0f203447195114aa0d92b3f50df5df46034c13f99fe9.

Custody certificate: sha256:5c370fbae9cb933bb052c4609403f6557699296c17e4e98c6e4123d040de4bbb.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured kinetic response correction curve or conventional kinetic model selects the survivor
- every flow reversal loss estimate correction uncertainty limitation and unreported cell remains

Immutable identities. Source manifest

sha256:c850b7459642d5771d4f6148473ae128dec85a7fb29dc9dfd4bb10854b1d4d2d; derivation seal sha256:0e4a450e6d361a29a9ac7d0a53785f94c545a609b078bc7e26ccd35cc9c50908; engine receipt sha256:f9b52ef14fc8be4e08d25067f9632d8048bc99c10416aa2e38d980b637b851f6 at receipts/engine/model_admitted/SFT-CHEM-KINETIC-ISOTOPE-FRACTIONATION-008-f9b52ef14fc8be4e.json; empirical hash sha256:b0cd0755e5fba02e114e74c4facc6d8321df73c138b2fa007f4898581024f0cc; measurement receipt sha256:1353aa80d3100c33d67f1c292c4db731aae02f57be5ecef4388be8ab981178cb.

211. Fold radiotracer custody and inference law

Claim identity: SFT-CHEM-RADIOTRACER-CUSTODY-INFERENCE-009

Question and theorem. A radiotracer retains nuclide and chemical-carrier identity over complete positive ordered observation support; recovery is exact observed per administered custody, unobserved custody is positive or structural absence, and inference cannot exceed observations.

held-tracer-and-chemical-carrier__positive-ordered-observation-support__complete-location-time-vector__positive-administered-and-observed-counts__exact-observed-per-administered-ratio__positive-Take-or-EmptyOne-unobserved__inference-bounded-by-observed-support__successor-preserves-identity-and-complete-record

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-CHEM-NUCLEAR-CHEMICAL-CARRIER-001
- SFT-CHEM-ACTIVITY-AMOUNT-TIME-003

• SFT-CHEM-RADIOCHEMICAL-EQUILIBRIUM-005

Generated grammar. Generate the literal product of eight registered radiotracer decisions.

Boundary: Eight dimensions exhaust identity, support, locations, events, recovery, loss, inference and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-tracer-and-chemical-carrier__positive-ordered-observation-support__complete-location-time-vector__positive-administered-and-observed-counts__exact-observed-per-administered-ratio__positive-Take-or-EmptyOne-unobserved__inference-bounded-by-observed-support__successor-preserves-identity-and-complete-record. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
identity	held-tracer-and-chemical-carrier	activity-only-tracer	Activity alone loses nuclide and chemistry.
support	positive-ordered-observation-support	continuous-concentration-field	A continuum field hides observations.
locations	complete-location-time-vector	selected-detector-location	Selection cannot establish complete transport.
events	positive-administered-and-observed-counts	expected-count-premise	Expectation is not observed-event custody.
recovery	exact-observed-per-administered-ratio	fitted-recovery-coefficient	A fit cannot define recovery.
loss	positive-Take-or-EmptyOne-unobserved	negative-missing-activity	Missing tracer cannot be a negative proof value.
inference	inference-bounded-by-observed-support	model-selected-localization	A model cannot create an observation.
extension	successor-preserve-identity-and-complete-record	renormalized-selected-series	Renormalization erases custody.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:1412287ad3834bc04016ca32739e5b1777be764c43d103d1079a00d429519ee4.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:0648c1efedc71495922d6547263b9ce54219611c7c6210aaa03ef51edefa5db9.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:894a3d5950a9aa5145b681a161d6558d913085e0de38b863d9618f3b26c76c7f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:38240d4975673aa06c619eb09fda9c28c1b1256d6e249f330ca68ee4349f74ac.

Independent reconstruction. Implementation hash

sha256:7240cc9387387f5d8c0186348072974144bd245c63b8ff9620608c95ce4fa1b0; independent certificate sha256:7e7de3d4e6ccf9d4b595bbd73d1412098909e433a780a489f11124d9e5dbe5fd; external-validation hash sha256:5e1debeb74b3c80ebbcbafda7efb522d94579cd5e055675ddbffdada5c7712d6a3. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IAEA-TCS-31-RADIOTRACER-RTD-2008
- IAEA-TCS-31-RADIOTRACER-RTD-2008
- IAEA-TCS-31-RADIOTRACER-RTD-2008
- IAEA-TCS-31-RADIOTRACER-RTD-2008
- IAEA-TCS-31-RADIOTRACER-RTD-2008
- IAEA-TCS-31-RADIOTRACER-RTD-2008
- IAEA-TCS-31-RADIOTRACER-RTD-2008
- NUCHEM-009-012-SOURCE-INVENTORY

Comparison records:

- complete four-source post-seal family retained as 370 PDF pages and 837,013 extracted characters
- all 8 separately registered claim targets retained
- all measured values units uncertainties assumptions corrections fits estimates losses adverse absent unavailable and unresolved records remain downstream provenance; disclosed pre-seal snippets are never relabelled blind

Falsification boundary: The claim halts if the survivor is nonunique; any tracer, chemistry, detector, location, count, recovery, loss, correction, inference, adverse row, page or byte is omitted; or an external RTD outcome selects the law.

Measurement receipt: sha256:35cb2a47c1ac05325a53edfc061d816ff24bc6987a6af67adfd2fcfedc34124b.

Isolation certificate: sha256:c4acd5e7b74e8cf5ba41604a765038432ebf7aa447be9bdeaa2e2af7e08b3333.

Custody certificate: sha256:be090dd21821420b9eec220e894707ada8ae3a4c064861b71370899e35cf3e1b.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured recovery localization curve or fitted RTD model selects the survivor
- models and recommendations remain distinct from observations

Immutable identities. Source manifest

sha256:d42323406029af95d013041e4955078a5e5405dc4727446a9d15dd914312b659; derivation seal
 sha256:0487ef58304f40dad79e377f656254a561266b2804bd1c0c060debc2ed0c06ae; engine receipt
 sha256:d5a8c7ac0e12d9c4213e247e3cf32ecc25b7c62fa5205d7b634d2c8b3adc5fe9 at receipts/engine/
 model_admitted/SFT-CHEM-RADIOTRACER-CUSTODY-INFERENCE-009-d5a8c7ac0e12d9c4.json;
 empirical hash sha256:29811bab9979c98cab5d382ee63a834fd2a8bc8d035a282f0c044ef3fc01496f;
 measurement receipt sha256:35cb2a47c1ac05325a53edfc061d816ff24bc6987a6af67adfd2fcfedc34124b.

212. Fold radiochemical separation and decontamination law

Claim identity: SFT-CHEM-RADIOCHEMICAL-SEPARATION-DECONTAMINATION-010

Question and theorem. Radiochemical separation retains target and contaminant identities across complete feed/product/waste streams, conserves each species exactly, forces recovery and decontamination ratios by counts, and composes only balanced stages.

held-target-and-contaminant-species__complete-feed-product-waste-streams__positive-species-resolved-counts__exact-target-and-contaminant-conservation__exact-product-target-per-feed-target-ratio__exact-contaminant-fraction-ratio__positive-Take-or-EmptyOne-residual__successor-composes-balanced-stages

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-NUCLEAR-CHEMICAL-CARRIER-001
- SFT-CHEM-RADIOACTIVE-CHEMICAL-TRANSFORMATION-002
- SFT-CHEM-RADIOTRACER-CUSTODY-INFERENCE-009

Generated grammar. Generate the literal product of eight registered separation decisions.

Boundary: Eight dimensions exhaust identity, streams, inventory, balance, recovery, decontamination, absence and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *held-target-and-contaminant-species__complete-feed-product-waste-streams__positive-species-resolved-counts__exact-target-and-contaminant-conservation__exact-product-target-per-feed-target-ratio__exact-contaminant-fraction-ratio__positive-Take-or-EmptyOne-residual__successor-composes-balanced-stages*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
identity	held-target-and-contaminant-species	total-radioactivity-only	Total activity loses target and contaminant identity.
streams	complete-feed-product-waste-streams	product-only-record	A product alone hides feed and waste.
inventory	positive-species-resolved-counts	continuum-concentration-premise	Continuum concentration hides occurrences.
balance	exact-target-and-contaminant-conservation	unrecorded-process-loss	Unrecorded loss breaks custody.
recovery	exact-product-target-per-feed-target-ratio	percent-recovery-premise	A displayed percentage cannot define recovery.
decontamination	exact-contaminant-fraction-ratio	fitted-separation-factor	A fit cannot define decontamination.
absence	positive-Take-or-EmptyOne-residual	numerical-zero-contaminant	Numerical zero is not native absence.
extension	successor-composes-balanced-stages	selected-best-stage	Selecting a best stage erases the process.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256 : 525a89263a4c3d2f87ceb2619ac6c5b5155c2e6adeb09e3ade4ccc4985b42692.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : f8ae0b83dc96661b830969c7ce0c50e08fe3e013ae1dfdc732df6f216f96a3f4.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:41cf4b3e5b3e8d3cf2c3b320e02dd2f5b72c70a353eec15afab6cfeb95079dc0.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:b4e5a34bbe2cea192a6723dad3171e306d3e0010cd73bcfb49f9abf04090f4e2.

Independent reconstruction. Implementation hash

sha256:4a3a1952c78fbaff6822be9ca84f79f3f06863fff569b2f6099b77219e45aa3; independent certificate sha256:9130338d50a95d9c886bf254c5a41e85e9823761b42d1326634c387b143d8448; external-validation hash sha256:d3542d56dbfebebbca4c18be368e11b871c1344bdfa00c158a16d54d671d0ad0. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- DOE-OSTI-1580278-HFSLM-ISOTOPE-HARVESTING
- DOE-OSTI-1580278-HFSLM-ISOTOPE-HARVESTING
- DOE-OSTI-1580278-HFSLM-ISOTOPE-HARVESTING
- DOE-OSTI-1580278-HFSLM-ISOTOPE-HARVESTING
- DOE-OSTI-1580278-HFSLM-ISOTOPE-HARVESTING
- DOE-OSTI-1580278-HFSLM-ISOTOPE-HARVESTING
- DOE-OSTI-1580278-HFSLM-ISOTOPE-HARVESTING
- NUCHEM-009-012-SOURCE-INVENTORY

Comparison records:

- complete four-source post-seal family retained as 370 PDF pages and 837,013 extracted characters
- all 8 separately registered claim targets retained
- all measured values units uncertainties assumptions corrections fits estimates losses adverse absent unavailable and unresolved records remain downstream provenance; disclosed pre-seal snippets are never relabelled blind

Falsification boundary: The claim halts if the survivor is nonunique; any species, stream, balance, recovery, decontamination, stage, detection limit, uncertainty, adverse result, page or byte is omitted; or an external separation value selects the law.

Measurement receipt: sha256:01e91489b7b12fb07161d2b3841b056b759fd35203ba8b7326c184c9169d84d3.

Isolation certificate: sha256:433afabdfb8f66aa0255106e24c48b2f7e90326651609b7945253f8337514f31.

Custody certificate: sha256:3aad9828cbb7e7d8848baf09b106feaa93f40743c158d955f16af2ea97dddafa.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured recovery extraction factor or selected successful run chooses the survivor
- the adverse one-percent run and every limit remain

Immutable identities. Source manifest

sha256:5ea0e053bc9890a025923bd8dd1bbff54d6587deab4200de9df926673102c11f; derivation seal
 sha256:d8237fd59aa16099c1e38834dd5c0df6b4da3ac71bb3d276bf8d746d52dc3749; engine receipt
 sha256:eb846cb0c2554f3bd873aa45f902039b1b71aeb7d79c3de52825322b67d01873 at receipts/engine/model_admitted/SFT-CHEM-RADIOCHEMICAL-SEPARATION-DECONTAMINATION-010-eb846cb0c2554f3b.json; empirical hash
 sha256:85462cd52401aadf445f304a6a02e7025a8b537ac9c3e0a851e4bc27df15bbe4; measurement receipt
 sha256:01e91489b7b12fb07161d2b3841b056b759fd35203ba8b7326c184c9169d84d3.

213. Fold fission-product chemical-distribution law

Claim identity: SFT-CHEM-FISSION-PRODUCT-CHEMICAL-DISTRIBUTION-011

Question and theorem. Physics supplies fission-product identities; Chemistry retains every nuclide, species, phase and location with positive events, partitions the complete support exactly and permits only explicit identity-conserving chemical redistribution.

explicit-physics-product-handoff__held-nuclide-and-chemical-species__complete-phase-location-support__positive-species-resolved-events__exact-complete-partition-of-One__explicit-chemical-redistribution-network__nuclide-change-requires-physics-handoff__successor-retains-complete-time-sample-vector

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-PHYS-NUCLEAR-RADIOACTIVE-DECAY-TERMINAL-005
- SFT-PHYS-DECAY-WIDTH-BRANCHING-LIFETIME-TERMINAL-006
- SFT-CHEM-NUCLEAR-CHEMICAL-CARRIER-001
- SFT-CHEM-RADIOACTIVE-CHEMICAL-TRANSFORMATION-002
- SFT-CHEM-RADIOCHEMICAL-SEPARATION-DECONTAMINATION-010

Generated grammar. Generate the literal product of eight registered fission-product decisions.

Boundary: Eight dimensions exhaust handoff, identity, support, events, partition, chemistry, boundary and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *explicit-physics-product-handoff__held-nuclide-and-chemical-species__complete-phase-location-support__positive-species-resolved-events__exact-complete-partition-of-One__explicit-chemical-redistribution-network__nuclide-change-requires-physics-handoff__successor-retains-complete-time-sample-vector*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
handoff	explicit-physics-product-handoff	chemistry-selects-fission-products	Chemistry cannot select nuclear products.
identity	held-nuclide-and-chemical-species	mass-chain-only	Mass chain loses nuclide identity.
support	complete-phase-location-support	bulk-release-fraction	One bulk fraction loses phase and location.
events	positive-species-resolved-events	expected-yield-premise	Expectation is not counted custody.
partition	exact-complete-partition-of-One	renormalized-selected-products	Selection changes the distribution.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
chemistry	explicit-chemical-redistribution-network	fixed-chemical-form-assumption	A name cannot freeze chemistry.
boundary	nuclide-change-requires-physics-handoff	silent-decay-inside-chemistry	Chemistry cannot silently change nuclides.
extension	successor-retains-complete-time-sample-vector	selected-terminal-sample	One sample cannot establish the distribution.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:c2d085226134cc44674183a5be1b629b6593f5a37095916732af79cb49bc7383.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:8cae2552c9114c32e075be4827050de084735d171b4221e88d00d7112996176a.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:03fd71b465091a77977d94975623d31ff7c146c18a3174d85169704ea3a06615.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:3f4638693b9fde0036f03bd2a5fab36e7d7d31fdb40bb9e1fc30322f1480dfc2.

Independent reconstruction. Implementation hash

sha256:6b64f4e58c2b18c1d9951f9b223654f9c4c4a2a7a98472afe32ab06f88a56cfb; independent certificate sha256:1235d657019b86bfd3e3779eccf593a724eebc9c0c4f8a8435833c0a0d401933; external-validation hash sha256:2a22e0718f7010518966155aefd265ece737d999b3c138af919a23a6c4b8abe7. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- DOE-ORNL-4865-FISSION-PRODUCT-BEHAVIOR-MSRE
- DOE-ORNL-4865-FISSION-PRODUCT-BEHAVIOR-MSRE
- DOE-ORNL-4865-FISSION-PRODUCT-BEHAVIOR-MSRE
- DOE-ORNL-4865-FISSION-PRODUCT-BEHAVIOR-MSRE
- DOE-ORNL-4865-FISSION-PRODUCT-BEHAVIOR-MSRE
- DOE-ORNL-4865-FISSION-PRODUCT-BEHAVIOR-MSRE
- DOE-ORNL-4865-FISSION-PRODUCT-BEHAVIOR-MSRE
- NUCHEM-009-012-SOURCE-INVENTORY

Comparison records:

- complete four-source post-seal family retained as 370 PDF pages and 837,013 extracted characters
- all 8 separately registered claim targets retained
- all measured values units uncertainties assumptions corrections fits estimates losses adverse absent unavailable and unresolved records remain downstream provenance; disclosed pre-seal snippets are never relabelled blind

Falsification boundary: The claim halts if the survivor is nonunique; any supplied product, species, phase, location, sample, distribution, transfer, inventory basis, uncontrolled condition, discrepancy, page or byte is omitted; or an external chemical grouping selects the law.

Measurement receipt: sha256:1d2fecadaf135c13de60abed662ecf8681b4059f6a76e80ed009aab229c127aa.
 Isolation certificate: sha256:18bf04ce46dbc443ab49e5320b7475c9d181aaa9cf19356976c2262096084a79.
 Custody certificate: sha256:874a77aaebbb200682698b1580b6949026ced91719c843f4d7de28da80e64093.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- Chemistry cannot select fission-product existence or use an inventory model to choose the survivor
- all missing balances and conjectures remain unresolved evidence

Immutable identities. Source manifest

sha256:45991bc1aef4e6ffd01f66ff07725024dab8eb118bc42c5ac9caa0fb7373af5b; derivation seal
 sha256:2a8ef91b618836a043ac94f791b6049459bc8e195773fcc18799fcf06f6db5d7; engine receipt
 sha256:d3f311c2d0ccb2b91ce59d58f202b1c4dd1252a58315f66fad7fc7b0b51b1376 at receipts/engine/model_admitted/SFT-CHEM-FISSION-PRODUCT-CHEMICAL-DISTRIBUTION-011-d3f311c2d0ccb2b9.json;
 empirical hash sha256:1abb95ad8d1ded97e9d0de32c98c76b110b7c98fd67a4454a8ee58a391c9b011;
 measurement receipt sha256:1d2fecadaf135c13de60abed662ecf8681b4059f6a76e80ed009aab229c127aa.

214. Fold radiation-chemistry reaction-network law

Claim identity: SFT-CHEM-RADIATION-CHEMISTRY-REACTION-NETWORK-012

Question and theorem. Physics supplies positive deposited resource; Chemistry retains complete medium/reactant/product/channel identity, counted product events, exact product-per-resource yields, complete channel partition and explicit reaction or structural termination.

explicit-positive-deposited-resource-handoff__held-medium-reactant-product-identities__complete-directed-reaction-network__positive-product-event-counts__exact-product-per-deposited-resource-ratio__exact-complete-channel-partition-of-One__explicit-reaction-or-EmptyOne-termination__successor-retains-and-recomputes-complete-network

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-RADIOACTIVE-CHEMICAL-TRANSFORMATION-002
- SFT-CHEM-ACTIVITY-AMOUNT-TIME-003

Generated grammar. Generate the literal product of eight registered radiolysis decisions.

Boundary: Eight dimensions exhaust handoff, identity, network, events, yield, partition, closure and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *explicit-positive-deposited-resource-handoff__held-medium-reactant-product-identities__complete-directed-react*

ion-network__positive-product-event-counts__exact-product-per-deposited-resource-ratio__exact-complete-channel-partition-of-One__explicit-reaction-or-EmptyOne-termination__successor-retains-and-recomputes-complete-network. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
handoff	explicit-positive-deposited-resource-handoff	chemistry-defines-deposited-energy	Deposited resource belongs to Physics.
identity	held-medium-reactant-product-identities	anonymous-radical-yield	A yield needs medium and species identity.
network	complete-directed-reaction-network	single-net-equation	A net equation hides channels and intermediates.
events	positive-product-event-counts	continuum-concentration-premise	Continuum concentration hides occurrences.
yield	exact-product-per-deposited-resource-ratio	fitted-G-value	A fitted value cannot define native yield.
partition	exact-complete-channel-partition-of-One	selected-major-products	Major-product selection erases support.
closure	explicit-reaction-or-EmptyOne-termination	negative-consumption-ledger	Consumption cannot be a negative proof value.
extension	successor-retains-and-recomputes-complete-network	differential-kinetic-model-premise	A continuum kinetic model is not required.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:130ce93877c3ae7fa3454ba000aef4c86504140480d7f5162e947dd1f8fd87bc.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:4689cde66f237c91844b54e1169c1f7544a2ae3d084cb67fcf7223154cefea4f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:e454c640aee66dde6f37a2f9e7f057f18b6bea41b50a9382b766e8f38af8ca73.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:1e2888c5d7a1cf079ee20e385e35358750add754a12e86e737fd9f53178e8660.

Independent reconstruction. Implementation hash

sha256:f96e81d2bd7f4b9f43da6ff362e680123550399c7a1795736ec69f968ea83f43; independent certificate sha256:069f0ac3fa9fad569c975b73c1840a1c59bfff855e7058a515eda61f1cb09cbb1; external-validation hash sha256:0933b1fe0acf9e5928573effd16a0640503a23b191951acbab740c65ee9be25a. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NBS-NSRDS-45-RADIATION-CHEMISTRY-NITROUS-OXIDE
- NBS-NSRDS-45-RADIATION-CHEMISTRY-NITROUS-OXIDE

- NBS-NSRDS-45-RADIATION-CHEMISTRY-NITROUS-OXIDE
- NBS-NSRDS-45-RADIATION-CHEMISTRY-NITROUS-OXIDE
- NBS-NSRDS-45-RADIATION-CHEMISTRY-NITROUS-OXIDE
- NBS-NSRDS-45-RADIATION-CHEMISTRY-NITROUS-OXIDE
- NBS-NSRDS-45-RADIATION-CHEMISTRY-NITROUS-OXIDE
- NUCHEM-009-012-SOURCE-INVENTORY

Comparison records:

- complete four-source post-seal family retained as 370 PDF pages and 837,013 extracted characters
- all 8 separately registered claim targets retained
- all measured values units uncertainties assumptions corrections fits estimates losses adverse absent unavailable and unresolved records remain downstream provenance; disclosed pre-seal snippets are never relabelled blind

Falsification boundary: The claim halts if the survivor is nonunique; any resource, species, channel, event, yield, condition, reaction, correction, tentative or adverse row, page or byte is omitted; or an external G value selects the law.

Measurement receipt: sha256:f5f2d18f63fcd6e7f18be51f1a2e7dc1bf1133267947fded2fda215d2c56c91a.

Isolation certificate: sha256:ccfc0240798aba77e1254c6829c5d0dee9ebab1c3e0b0fe697515d80ab17872b.

Custody certificate: sha256:a8df85fd293c453be1de8f93d9da287bd575ff0493f9f7c2545a1817d444609e.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no preferred G value reaction table or conventional radiolysis model selects the survivor
- all corrections tentative classifications missing results nonlinearities and unreliable rows remain

Immutable identities. Source manifest

sha256:6c4779bd1eb99ae2f40fdb601eb0309022677f919ad31003270cc8709192d15; derivation seal

sha256:6e0e3cecd69f5fc9e04c3bea12846ff3bf3b4a70ae8ae0d53a6449d1007c7da6; engine receipt

sha256:ad00dad43f0a51d743cca24641be1ecab903298cc0e95290f78091432c2a8088 at receipts/engine/model_admitted/SFT-CHEM-RADIATION-CHEMISTRY-REACTION-NETWORK-012-ad00dad43f0a51d7.js

on; empirical hash sha256:0797e794b325b0452d90626d6082b0bbb05350bfc23060b214adfe4d601eefdd;

measurement receipt sha256:f5f2d18f63fcd6e7f18be51f1a2e7dc1bf1133267947fded2fda215d2c56c91a.

26. Analytical chemistry, spectroscopy and separation science - 22 admitted claims

Accuracy, precision, sensitivity, NMR, emission, diffraction, chromatography and multimodal identity retain all rows. This entire registered subcategory is included; no proper subset is presented as field completion.

215. Fold analytical accuracy and trueness law

Claim identity: SFT-CHEM-ANALYTICAL-ACCURACY-TRUENESS-001

Question and theorem. Traceable reference and held sample results force exact side/distance or structural coincidence with complete uncertainty and method boundaries.

traceable-reference-identity__held-sample-measurand-method__complete-ordered-result-support__held-side-positive-distance-or-EmptyOne__exact-observed-reference-relation__complete-exact-uncertainty-support__registered-material-method-boundary__successor-retains-and-recomputes-all-results

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-ANALYTICAL-SAMPLE-001
- SFT-CHEM-ANALYTICAL-CALIBRATION-001

Generated grammar. Generate the literal product of eight registered analytical decisions.

Boundary: Eight dimensions exhaust reference, measurand, support, comparison, trueness, uncertainty, boundary and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *traceable-reference-identity__held-sample-measurand-method__complete-ordered-result-support__held-side-positive-distance-or-EmptyOne__exact-observed-reference-relation__complete-exact-uncertainty-support__registered-material-method-boundary__successor-retains-and-recomputes-all-results*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
reference	traceable-reference-identity	nominal-value-premise	A name is not a reference.
measurand	held-sample-measurand-method	result-without-measurand	No comparison identity exists.
support	complete-ordered-result-support	selected-result	Selection hides variation.
comparison	held-side-positive-distance-or-EmptyOne	signed-error-scalar	A sign is not a native proof value.
trueness	exact-observed-reference-relation	declared-accurate-label	A label cannot force trueness.
uncertainty	complete-exact-uncertainty-support	single-certain-value	Alternatives are hidden.
boundary	registered-material-method-boundary	universal-method-accuracy	Accuracy is method and material bound.
extension	successor-retains-and-recomputes-all-results	renormalized-new-results	Renormalization erases custody.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256: a1ee1fab58bc32e84b2b6eb2fd9022e8818fe8847ad99dda785e86ee43284a4f.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 76e4fdb1b20824f0e35e31625bbfd17baf11b28609b03b578335eb6c261196f8.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:ecaca67607e25584b0fa3c9df17531fdeaa6667eab761ea39e1d3c8435a8bb3d.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:1024bcb38030bd216559589a0556a078d5419e5db3722c4cf17f75e35138ad3e.

Independent reconstruction. Implementation hash

sha256:91f6750d40671ed7044c13aa5368d48d966567ddf8b935669c3b3202983cd4bb; independent certificate sha256:6eadd2c99b10b0b493e1195703aa6d3a21793f3b5f2e92820cd03a62e825caf5; external-validation hash sha256:5ef8a6830ad7c1206f56414ec373b3fb7821b3489fb0003b0a3bd2c926fecf08. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-SRM-1959-CERTIFICATE-2011
- NIST-SRM-1959-CERTIFICATE-2011
- NIST-SRM-1959-CERTIFICATE-2011
- NIST-SRM-1959-CERTIFICATE-2011
- NIST-SRM-1959-CERTIFICATE-2011
- NIST-SRM-1959-CERTIFICATE-2011
- NIST-SRM-1959-CERTIFICATE-2011
- ANAL-001-005-SOURCE-INVENTORY

Comparison records:

- complete six-source post-seal family retained as 83 PDF pages, one HTML document and 180,366 extracted characters
- all 8 separately registered claim targets retained
- all values units uncertainties assumptions corrections fits estimates losses adverse absent unavailable unresolved historical and archived-status records remain downstream provenance; disclosed pre-seal snippets are never relabelled blind

Falsification boundary: The claim halts if the survivor is nonunique; any registered analytical row, uncertainty, adverse result, page or byte is omitted; or an external result selects the law. Eight dimensions exhaust reference, measurand, support, comparison, trueness, uncertainty, boundary and extension.

Measurement receipt: sha256:f972f3d1785d9e532ddf90d6d946daf828418f20a715775f5d89bc8b2dfc6b64.
Isolation certificate: sha256:9530559e15b4c29cc3cb10699a1af4facbbcf60be50cf4943f178d16e08296bc.
Custody certificate: sha256:ba9b8a86dc8f993d1a8bf44d67111eed2d664afea6210dd49a660378ce0b8ab1.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external value threshold model or selected successful result chooses the survivor
- all favorable adverse absent unavailable unresolved uncertainty assumption correction fit and loss rows remain

Immutable identities. Source manifest

sha256:9ede9259444779059157ee66587f248491baeebf18558d28b972601bac1a1dd3; derivation seal

sha256:053675455555661d77350a2353aa1fd0c1a1b3648174e31c576d3cbcefef6c7b; engine receipt sha256:e96550c2bc1a718b8bcffbac6e1f45c9a920e926701e2132cf1637123820ccf3 at receipts/engine/model_admitted/SFT-CHEM-ANALYTICAL-ACCURACY-TRUENESS-001-e96550c2bc1a718b.json; empirical hash sha256:4a0e7e788459eee44edd46c5bee93a4af46b17becdcb2decac5ccc2f48b2e5e4; measurement receipt sha256:f972f3d1785d9e532ddf90d6d946daf828418f20a715775f5d89bc8b2dfc6b64.

216. Fold analytical precision and repeatability law

Claim identity: SFT-CHEM-ANALYTICAL-PRECISION-REPEATABILITY-002

Question and theorem. Same-condition replicates force all exact pair distances and extreme spread while retaining every adverse replicate separately from trueness.

held-sample-measurand-method__same-held-repeatability-conditions__complete-positive-ordered-replicates__all-pair-positive-distance-or-EmptyOne__exact-extreme-distance-or-EmptyOne__repeatability-class-held-separately__all-adverse-replicates-retained__successor-retains-and-recomputes-pairs

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-ANALYTICAL-SAMPLE-001
- SFT-CHEM-ANALYTICAL-ACCURACY-TRUENESS-001

Generated grammar. Generate the literal product of eight registered analytical decisions.

Boundary: Eight dimensions exhaust identity, conditions, support, pair comparison, spread, class, adverse custody and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-sample-measurand-method__same-held-repeatability-conditions__complete-positive-ordered-replicates__all-pair-positive-distance-or-EmptyOne__exact-extreme-distance-or-EmptyOne__repeatability-class-held-separately__all-adverse-replicates-retained__successor-retains-and-recomputes-pairs. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
identity	held-sample-measurand-method	replicates-without-sample	Results cannot be paired.
conditions	same-held-repeatability-conditions	mixed-condition-pool	Repeatability is obscured.
support	complete-positive-ordered-replicates	summary-only	A summary erases rows.
comparison	all-pair-positive-distance-or-EmptyOne	signed-deviation-list	Signed values are not native proof distances.
spread	exact-extreme-distance-or-EmptyOne	fitted-dispersion-parameter	A fit cannot define observed spread.
class	repeatability-class-held-separately	precision-equals-trueness	Agreement with a reference is distinct.
adverse	all-adverse-replicates-retained	outlier-deleted	Deletion inflates precision.
extension	successor-retains-and-recomputes-pairs	summary-renormalized	Earlier distances disappear.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:56faba0b937d45f0d3aaffb2769c09cea5d09c2f46df06eadd098beffa9ed736.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:c93d4b15a7dd507325832a18a0f0002b9b155c7d17ca7735d51ae281e8d7c2a8.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:3a5f066926e2aa6841abe247d4351e7c7f74dad98eefc20291dc4269c96a0c99.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:6a7dc01e6666e0316fee45cc7bdd13529f5bfff7163248a3021d46c33b319d436.

Independent reconstruction. Implementation hash

sha256:701718defaeafb05117042d36cb8b784d76a8b8ed418210c7575a07f36e593a3; independent certificate sha256:b7d9225589653e9122c9080ebd0bd4435aa087ca800e7b09c0f0ebb2c8f7f5e2; external-validation hash sha256:7b04f1cda4ab1efdafabf73a7c036ab083ac5831dfad864bebf99ce9cc3c4b80. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-SRM-1959-CERTIFICATION-2010
- NIST-SRM-1959-CERTIFICATION-2010
- NIST-SRM-1959-CERTIFICATION-2010
- NIST-SRM-1959-CERTIFICATION-2010
- NIST-SRM-1959-CERTIFICATION-2010
- NIST-SRM-1959-CERTIFICATION-2010
- NIST-SRM-1959-CERTIFICATION-2010
- ANAL-001-005-SOURCE-INVENTORY

Comparison records:

- complete six-source post-seal family retained as 83 PDF pages, one HTML document and 180,366 extracted characters
- all 8 separately registered claim targets retained
- all values units uncertainties assumptions corrections fits estimates losses adverse absent unavailable unresolved historical and archived-status records remain downstream provenance; disclosed pre-seal snippets are never relabelled blind

Falsification boundary: The claim halts if the survivor is nonunique; any registered analytical row, uncertainty, adverse result, page or byte is omitted; or an external result selects the law. Eight dimensions exhaust identity, conditions, support, pair comparison, spread, class, adverse custody and extension.

Measurement receipt: sha256:8bfde17b4b3ea51b9e8b24d9f035b12bb6b19e5e1e764239079a0bc39cb6667f.

Isolation certificate: sha256:0cf02c7f69b097458420b35dc01cc4d437820c2195d4f793ca3fda11279ae9b3.

Custody certificate: sha256:86da70c37628449c5cb18243a01a86586c69ffdc6ac05992a2436b325e116946.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external value threshold model or selected successful result chooses the survivor
- all favorable adverse absent unavailable unresolved uncertainty assumption correction fit and loss rows remain

Immutable identities. Source manifest

sha256:a4c1cd4ee66b3cca2ce9dbd8b605134a9537b4251e2c359aa964167800ab9912; derivation seal
 sha256:a9eaea3afed71fdc7ec70366d3fa7eda337a63e4c3edd128a8074bd04863eb0e; engine receipt
 sha256:189762298384f7f436f22c3a9f6ef858a6333ae7bcd3d4c498ce01b5f713812b at receipts/engine/model_admitted/SFT-CHEM-ANALYTICAL-PRECISION-REPEATABILITY-002-189762298384f7f4.json;
 empirical hash sha256:907abab347336307786742c1cd846d61ae3c8f6257a2f270ffd61d2018f19be9;
 measurement receipt sha256:8bfde17b4b3ea51b9e8b24d9f035b12bb6b19e5e1e764239079a0bc39cb6667f.

217. Fold analytical sensitivity law

Claim identity: SFT-CHEM-ANALYTICAL-SENSITIVITY-003

Question and theorem. Ordered positive analyte inputs and paired responses force exact local response/input ratios, held direction and finite method-bounded segments with drift custody.

held-analyte-method-condition__positive-ordered-input-support__paired-complete-response-support__held-response-side-positive-change-or-EmptyOne__exact-local-response-per-input-ratio__finite-registered-segment-domain__complete-drift-noise-custody__successor-retains-and-adds-segment

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-ANALYTICAL-CALIBRATION-001
- SFT-CHEM-ANALYTICAL-PRECISION-REPEATABILITY-002

Generated grammar. Generate the literal product of eight registered analytical decisions.

Boundary: Eight dimensions exhaust identity, input, response, change, relation, domain, noise and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-analyte-method-condition__positive-ordered-input-support__paired-complete-response-support__held-response-side-positive-change-or-EmptyOne__exact-local-response-per-input-ratio__finite-registered-segment-domain__complete-drift-noise-custody__successor-retains-and-adds-segment. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
identity	held-analyte-method-condition	anonymous-response	No sensitivity is attributable.
input	positive-ordered-input-support	continuum-input-premise	Uncounted support cannot be audited.
response	paired-complete-response-support	response-fit-only	Observed rows are erased.
change	held-response-side-positive-change-or-EmptyOne	signed-response-change	A sign is not a proof magnitude.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
relation	exact-local-response-per-input-ratio	fitted-global-slope	A global fit can hide local behavior.
domain	finite-registered-segment-domain	universal-linearity	Sensitivity is condition-bound.
noise	complete-drift-noise-custody	drift-noise-erased	Sensitivity alone cannot establish resolution.
extension	successor-retains-and-adds-segment	refitted-history	A refit changes earlier relations.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 76dfd2e8542e35abee9ba647e93ceb34f14da9ed6e5ba13ae4e1a0a0687e7a9c.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: ae04ae2c2a6c7e4852366c9d9e8a15f729c75cb4e0494a9f91ac39a613ada9da.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 378e8c1753f9319a828e295b06c0077d4e47156d0c31261f2ba5e438c56ecc68.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 9877ca39e3e99a20c0b9a96f1b54bb2d4cc5939d60d5f04befdd98bb327bf3f9.

Independent reconstruction. Implementation hash

sha256: 84983f0846a3fecce17c4270a05d7c7ab2b379291ef15d469660df4e585840bc; independent certificate sha256: b6eb7b78863cba5376a9254d87fb1291e6e97992eea17dc89d31330a447a2395; external-validation hash sha256: 2c40fac8efc09106e76f39f157cdcca5eb91044879a7ce306ed2ee608147e165. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-SP-260-126-NTRM-GAS-STANDARDS
- NIST-SP-260-126-NTRM-GAS-STANDARDS
- NIST-SP-260-126-NTRM-GAS-STANDARDS
- NIST-SP-260-126-NTRM-GAS-STANDARDS
- NIST-SP-260-126-NTRM-GAS-STANDARDS
- NIST-SP-260-126-NTRM-GAS-STANDARDS
- NIST-SP-260-126-NTRM-GAS-STANDARDS
- ANAL-001-005-SOURCE-INVENTORY

Comparison records:

- complete six-source post-seal family retained as 83 PDF pages, one HTML document and 180,366 extracted characters
- all 8 separately registered claim targets retained
- all values units uncertainties assumptions corrections fits estimates losses adverse absent unavailable unresolved historical and archived-status records remain downstream provenance; disclosed pre-seal snippets are never relabelled blind

Falsification boundary: The claim halts if the survivor is nonunique; any registered analytical row, uncertainty, adverse result, page or byte is omitted; or an external result selects the law. Eight dimensions exhaust identity, input, response, change, relation, domain, noise and extension.

Measurement receipt: sha256:8c4fbe970cc36c2a23c75cf793eb52c509d93f707cc8b0510d917977b4d247a6.
 Isolation certificate: sha256:1f362b7aa2ad62aa491c61bf6dbdd12957d5f30567e88bbef9ec977796e63596.
 Custody certificate: sha256:34c6fb60ab137eb1c4398232b7c8e5cac5bb538a5f79222e9e8aae51c6bfede4.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external value threshold model or selected successful result chooses the survivor
- all favorable adverse absent unavailable unresolved uncertainty assumption correction fit and loss rows remain

Immutable identities. Source manifest

sha256:f971f26dc29d95111a601db8bf0f384f7d6b90cac7afa4423bf58fd2553f0f09; derivation seal
 sha256:f4051f6f7c1aa24db9d02974c377f40eab133499585d348e4588a0e8dbaec0e3; engine receipt
 sha256:157e7368c283e846a5dbfee9a656fdea0db746654f0037104ba9985a2647fca2 at receipts/engine/model_admitted/SFT-CHEM-ANALYTICAL-SENSITIVITY-003-157e7368c283e846.json; empirical hash
 sha256:3c19b9dde70c64e662b3855c36911290ab7044ae69e6e0394c287fe91f59d9f6; measurement receipt
 sha256:8c4fbe970cc36c2a23c75cf793eb52c509d93f707cc8b0510d917977b4d247a6.

218. Fold detection and quantification boundary law

Claim identity: SFT-CHEM-ANALYTICAL-DETECTION-QUANTIFICATION-004

Question and theorem. Complete blank and low-level supports force method-bounded distinguishability, explicit error custody, and separate detection and quantification classes or structural absence.

held-analyte-method-condition__complete-positive-blank-support__complete-positive-ordered-level-support__exact-blank-sample-distinguishability__false-positive-negative-custody__repeatability-and-recovery-required__detected-quantified-label-or-EmptyOne__successor-retains-and-recomputes-levels

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-ANALYTICAL-PRECISION-REPEATABILITY-002
- SFT-CHEM-ANALYTICAL-SENSITIVITY-003

Generated grammar. Generate the literal product of eight registered analytical decisions.

Boundary: Eight dimensions exhaust identity, blank, levels, detection, errors, quantification, absence and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-analyte-method-condition__complete-positive-blank-support__complete-positive-ordered-level-support__exact-blank-sample-distinguishability__false-positive-negative-custody__repeatability-and-recovery-required__detected-quantified-label-or-EmptyOne__successor-retains-and-recomputes-levels. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
identity	held-analyte-method-condition	unidentified-signal	No analyte boundary exists.
blank	complete-positive-blank-support	single-numerical-zero-blank	Numerical zero cannot represent background.
levels	complete-positive-ordered-level-support	selected-low-result	Selection hides failures.
detection	exact-blank-sample-distinguishability	declared-limit-number	A number alone cannot prove distinction.
errors	false-positive-negative-custody	false-results-erased	Error classes are hidden.
quantification	repeatability-and-recovery-required	detection-equals-quantification	Identity is weaker than measurement.
absence	detected-quantified-label-or-EmptyOne	negative-undetected-value	Undetected is not negative amount.
extension	successor-retains-and-recomputes-levels	threshold-refitted	A refit changes prior classifications.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : c2a1775008063e56638a3fe9d0e0d76e2d8c0a72ce282f82363bc788db75af37.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 864bb3ce20b14daea805f9f59bbcab620727d4fa2373eb5cb4a042b517f6fe1.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 42239c1e242c3ff8483a9cf6353d7b8b45f8054f7a9d8973eb1c3f30d2dacab3.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : acad53158937adfd628df92732c73980499cc0f8cf649d5efc4e03a09e7e409f.

Independent reconstruction. Implementation hash

sha256 : 4388f073dec320155e2ba339491e488aa4643a01d4260cd8cce01cb4a3dce304; independent certificate sha256 : 83d84a20bc395093589b98205ea9da783a51ddbb414e42f97bce88fc0ce1bf47; external-validation hash sha256 : d911ebb537efe2b1c28985ffc49164af63f961ae51a276f3998ad0c6de70f706. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-SP958-DETECTION-QUANTIFICATION-PAGES-164-166
- NIST-SP958-DETECTION-QUANTIFICATION-PAGES-164-166
- NIST-SP958-DETECTION-QUANTIFICATION-PAGES-164-166
- NIST-SP958-DETECTION-QUANTIFICATION-PAGES-164-166
- NIST-SP958-DETECTION-QUANTIFICATION-PAGES-164-166
- NIST-SP958-DETECTION-QUANTIFICATION-PAGES-164-166
- NIST-SP958-DETECTION-QUANTIFICATION-PAGES-164-166
- ANAL-001-005-SOURCE-INVENTORY

Comparison records:

- complete six-source post-seal family retained as 83 PDF pages, one HTML document and 180,366 extracted characters
- all 8 separately registered claim targets retained
- all values units uncertainties assumptions corrections fits estimates losses adverse absent unavailable unresolved historical and archived-status records remain downstream provenance; disclosed pre-seal snippets are never relabelled blind

Falsification boundary: The claim halts if the survivor is nonunique; any registered analytical row, uncertainty, adverse result, page or byte is omitted; or an external result selects the law. Eight dimensions exhaust identity, blank, levels, detection, errors, quantification, absence and extension.

Measurement receipt: sha256:87b4eefea8e5ac78624176cc177272592d8789e937c557f9d8a78706b202e6ab.
Isolation certificate: sha256:00cb3171a8736dafc26eec6325b225cd836976f141707525a7681c3e71b8fc27.
Custody certificate: sha256:d76c460a91571f061fca9fe70f5637c62b9fb1dfd68e398b8a15d2cf49155326.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external value threshold model or selected successful result chooses the survivor
- all favorable adverse absent unavailable unresolved uncertainty assumption correction fit and loss rows remain

Immutable identities. Source manifest

sha256:4a0b7a8e71902e6419f79aa0439e6a2fedd6e1cfdff7b242057132f23de84e07; derivation seal
sha256:8c78349068dc38b83e7a467f700093790e927394f714f54f38087c4ba63bcee7; engine receipt
sha256:d2d81bc678902abb4709f33583789ec28190133e7c3c5e1c6a91c87ff911ed68 at receipts/engine/model_admitted/SFT-CHEM-ANALYTICAL-DETECTION-QUANTIFICATION-004-d2d81bc678902abb.json
; empirical hash sha256:ebb731d5f7843145a01a800bc5401e1b9580e992d29e508d81cdf57a87a6cabd;
measurement receipt sha256:87b4eefea8e5ac78624176cc177272592d8789e937c557f9d8a78706b202e6ab.

219. Fold analytical selectivity and interference-matrix law

Claim identity: SFT-CHEM-ANALYTICAL-SELECTIVITY-INTERFERENCE-005

Question and theorem. Held analyte and complete interferent baselines and mixtures force exact method-bounded response distinction, interference side/distance or structural coincidence, retaining every error row.

held-analyte-method-condition__complete-declared-interferent-matrix__analyte-and-interferent-baselines-held__all-analyte-interferent-mixtures-held__held-side-positive-distance-or-EmptyOne__exact-response-distinguishability-class__favorable-adverse-error-rows-retained__successor-retains-and-adds-interferent

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-ANALYTICAL-SELECTIVITY-001
- SFT-CHEM-ANALYTICAL-SENSITIVITY-003
- SFT-CHEM-ANALYTICAL-DETECTION-QUANTIFICATION-004

Generated grammar. Generate the literal product of eight registered analytical decisions.

Boundary: Eight dimensions exhaust identity, matrix, baselines, mixtures, comparison, selectivity, adverse custody and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-analyte-method-condition__complete-declared-interferent-matrix__analyte-and-interferent-baselines-held__all-analyte-interferent-mixtures-held__held-side-positive-distance-or-EmptyOne__exact-response-distinguishability-class__favorable-adverse-error-rows-retained__successor-retains-and-adds-interferent. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
identity	held-analyte-method-condition	response-without-analyte	No target is attributable.
matrix	complete-declared-interferent-matrix	selected-interferent	Selection can hide interference.
baselines	analyte-and-interferent-baselines-held	mixture-only-result	Attribution is impossible.
mixtures	all-analyte-interferent-mixtures-held	accepted-mixtures-only	Adverse mixtures disappear.
comparison	held-side-positive-distance-or-EmptyOne	signed-bias-scalar	A sign alone loses response custody.
selectivity	exact-response-distinguishability-class	universal-specificity-label	Selectivity is method-bound.
adverse	favorable-adverse-error-rows-retained	false-positive-rows-erased	Erasure inflates selectivity.
extension	successor-retains-and-adds-interferent	matrix-renormalized	Prior comparisons disappear.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:6e374d1410e4a6acdc3517d9e4352cdb76a40e7ee9a232ba23dadb5b05f7b44d.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:b9c913d19eb9ca19709577a996dfb8fd9667f3c71823a4aa65f1d843cbe6868f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:1aa43a7644648064e058b8da1f0a6b8347d003ab49e640bfbe75f30a4a2773e1.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:e2383cd7b6a9dcef8585fb5d8fe929427962a36ebaea648f11ae7f6a766d3b1.

Independent reconstruction. Implementation hash

sha256:013da9849e036d04c30755b5e8c304d99a9e9c218722b5872d310b60d3cb0d58; independent certificate sha256:c43cdf0bc32fbb3fd730ef0c814104c14eeecd2bd88f312d4fa236558c902162; external-validation hash sha256:994c55ecf2952c8146755aaf038779bc35a5da985263871473edc3ed8d84bae0. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- NIST-IMS-FENTANYL-INTERFERENTS-2019
- NIST-IMS-FENTANYL-INTERFERENTS-2019
- NIST-IMS-FENTANYL-INTERFERENTS-2019
- NIST-IMS-FENTANYL-INTERFERENTS-2019
- NIST-IMS-FENTANYL-INTERFERENTS-2019
- NIST-IMS-FENTANYL-INTERFERENTS-2019
- NIST-IMS-FENTANYL-INTERFERENTS-2019
- ANAL-001-005-SOURCE-INVENTORY

Comparison records:

- complete six-source post-seal family retained as 83 PDF pages, one HTML document and 180,366 extracted characters
- all 8 separately registered claim targets retained
- all values units uncertainties assumptions corrections fits estimates losses adverse absent unavailable unresolved historical and archived-status records remain downstream provenance; disclosed pre-seal snippets are never relabelled blind

Falsification boundary: The claim halts if the survivor is nonunique; any registered analytical row, uncertainty, adverse result, page or byte is omitted; or an external result selects the law. Eight dimensions exhaust identity, matrix, baselines, mixtures, comparison, selectivity, adverse custody and extension.

Measurement receipt: sha256: faf29be119d0ce0b196b4eecbe3cbbec5a4e9fa7f9a8b3c6e84de38a028b3496.

Isolation certificate: sha256: c1d7befc73e7baf8c69beabd22dc68425a1dddb1be1bb8a24ffff241d62e19ab4.

Custody certificate: sha256: 4c4e50824a8daabcf4f2c70dc0e39e02613daee77db34dfc68119e58ed9346be.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external value threshold model or selected successful result chooses the survivor
- all favorable adverse absent unavailable unresolved uncertainty assumption correction fit and loss rows remain

Immutable identities. Source manifest

sha256: af961c7942ce4ede414885258e4db49aa4e47b2c1da57531d493be155c2c0488; derivation seal

sha256: 71f0fc3c5a41b44d4a9b40d0a9664c11fb3c090d47524fb43d6b410b33cc0942; engine receipt

sha256: 68ab2e8497202157e6cbd1ab8b8303cd46671384123312c65c4595230f1c28bf at receipts/engine/model_admitted/SFT-CHEM-ANALYTICAL-SELECTIVITY-INTERFERENCE-005-68ab2e8497202157.json

; empirical hash sha256: aaefa704ed69e3887be1538d13350975e3ab798b7b3057d6253c7a93eb7587ff;

measurement receipt sha256: faf29be119d0ce0b196b4eecbe3cbbec5a4e9fa7f9a8b3c6e84de38a028b3496.

220. Fold NMR chemical-shift relation

Claim identity: SFT-CHEM-NMR-CHEMICAL-SHIFT-006

Question and theorem. Held molecular, nucleus-site, reference and condition identities force an exact held-side chemical-shift relation with complete row custody.

held-molecular-carrier__held-nucleus-identity__complete-nucleus-site-map__held-reference-comparison__held-solvent-and-condition__held-side-positive-exact-ratio-or-EmptyOne__complete-shift-uncertainty-ambiguity-custody__success-or-retains-and-appends-complete-sites

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001
- SFT-CHEM-ANALYTICAL-SELECTIVITY-INTERFERENCE-005

Generated grammar. Generate the literal product of eight registered NMR-custody decisions.

Boundary: Eight dimensions exhaust carrier, nucleus, site, reference, environment, exact relation, custody and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *held-molecular-carrier__held-nucleus-identity__complete-nucleus-site-map__held-reference-comparison__held-solvent-and-condition__held-side-positive-exact-ratio-or-EmptyOne__complete-shift-uncertainty-ambiguity-custody__successor-retains-and-appends-complete-sites*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
identity	held-molecular-carrier	detached-shift-number	A scalar cannot identify a molecular observation.
nucleus	held-nucleus-identity	element-only-spectrum	Element name loses isotope and observed nucleus.
site	complete-nucleus-site-map	unordered-peak-list	An unordered list loses molecular attribution.
reference	held-reference-comparison	absolute-frequency-relabelled-shift	A shift exists only relative to a named reference.
environment	held-solvent-and-condition	unconditioned-universal-shift	Solvent and conditions can change the observation.
relation	held-side-positive-exact-ratio-or-EmptyOne	signed-or-floating-shift-premise	A sign or float is not a native exact relation.
custody	complete-shift-uncertainty-ambiguity-custody	selected-assigned-peaks	Selection hides ambiguity and adverse rows.
extension	successor-retains-and-appends-complete-sites	renormalized-added-spectrum	Renormalization erases prior custody.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1f7e6353517b5301bbcef740b57e7cca55c9724e2263d6742a3888acf4da4da4`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:61959db69affc65a52fc7a57f4936d131b36877a629fed3003fc0938bd4cf4eb`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 493660828e373ed7f32199e7b2edfb2df0cd95d4da562ccc5ee57641690d413a.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 9fe663665a1055c5cf29dba85b7d72c65570e568b51ef090884aa3c7d3017955.

Independent reconstruction. Implementation hash

sha256 : 9bc104094769803b2db3a8fd139495dec498447e560d4db0b2dc40c57a7e5c25; independent certificate
sha256 : c93252630999aeba724cda5c2482b27b2facb1e1d18fa993260b7ae924749b92; external-validation
hash sha256 : d803c577a81d7a58693bd1b39e522d09334a6452421f973f5930b93efe0dbf0f. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-NMR-NOMENCLATURE-2001
- BMRB-ENTRY-68-CHEMICAL-SHIFTS
- IUPAC-NMR-NOMENCLATURE-2001
- BMRB-ENTRY-68-CHEMICAL-SHIFTS
- IUPAC-NMR-NOMENCLATURE-2001
- BMRB-ENTRY-68-CHEMICAL-SHIFTS
- IUPAC-NMR-NOMENCLATURE-2001
- ANAL-006-008-NMR-SOURCE-INVENTORY

Comparison records:

- complete ten-source post-seal NMR family retained as 24 PDF pages, five HTML documents, four NMR-STAR records and 1,633 measured rows
- all 8 separately registered claim targets retained
- signed external coupling values are held orientations plus exact positive magnitudes; eleven external zero-rate inscriptions are unresolved structural EmptyOne
- all pre-seal identity exposure remains disclosed and is never relabelled blind

Falsification boundary: The claim halts if its survivor is nonunique; any registered NMR value, site, reference, condition, unit, uncertainty, ambiguity, absent field, adverse orientation, page or source byte is omitted; the independent reconstruction disagrees; or any external result selects the law. Eight dimensions exhaust carrier, nucleus, site, reference, environment, exact relation, custody and extension.

Measurement receipt: sha256 : 6cc7cec7ae161f2fa72698546cfd84fb75c49639545a7a3c6b5704a6d65aca3f.
Isolation certificate: sha256 : dfaac91274bbc529dec9e94bf9819ffaafdacbc90ab6af5626fe1c1c9aca7992.
Custody certificate: sha256 : eabbb1dfd0927e76b36322fcb624f2025f92ad35a1be33cd99e0ea906f5ca4ae.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter

- no external value reference equation conventional model threshold or selected successful row chooses the survivor
- all measured adverse absent unavailable unresolved error ambiguity bound condition unit and historical inscriptions remain

Immutable identities. Source manifest

sha256:cdc3db479dbe7db8ce61024dc6506ce43f3859141c40c3a5495f2364707928bb; derivation seal sha256:0bdbc3ff3175551ae2dde49e5506b2bfb75934691d4f9311ec58f72c525dba5a; engine receipt sha256:07525c04e8cfea4d1d1044356d602447ededeeef6335f2949216987d8c1482677 at receipts/engine/model_admitted/SFT-CHEM-NMR-CHEMICAL-SHIFT-006-07525c04e8cfea4d.json; empirical hash sha256:6875b7766e73a1c147f8ac4481543a67fa45b7a2e016a79daefb532553f38abd; measurement receipt sha256:6cc7cec7ae161f2fa72698546cfd84fb75c49639545a7a3c6b5704a6d65aca3f.

221. Fold NMR scalar spin-coupling relation

Claim identity: SFT-CHEM-NMR-SPIN-COUPLING-007

Question and theorem. Two held nucleus sites, their counted path and held spin orientation force an exact positive coupling magnitude or structural absence with complete custody.

held-two-nucleus-site-pair__held-preserving-alternating-or-unresolved-spin-relation__positive-counted-bonding-path__held-molecule-and-condition__exact-positive-magnitude-or-EmptyOne__site-swap-invariant-pair-relation__complete-value-error-bound-and-absence-custody__successor-retains-and-appends-complete-pairs

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-CHEM-ELECTRON-COUNT-SPIN-002
- SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001
- SFT-CHEM-NMR-CHEMICAL-SHIFT-006

Generated grammar. Generate the literal product of eight registered NMR-custody decisions.

Boundary: Eight dimensions exhaust pair, spin relation, counted path, condition, exact magnitude, symmetry, custody and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *held-two-nucleus-site-pair__held-preserving-alternating-or-unresolved-spin-relation__positive-counted-bonding-path__held-molecule-and-condition__exact-positive-magnitude-or-EmptyOne__site-swap-invariant-pair-relation__complete-value-error-bound-and-absence-custody__successor-retains-and-appends-complete-pairs*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
pair	held-two-nucleus-site-pair	single-peak-coupling-label	Coupling is a relation between two retained sites.
spin	held-preserving-alternating-or-unresolved-spin-relation	unsigned-magnitude-only	Magnitude alone loses the held spin hand.
path	positive-counted-bonding-path	coupling-without-bonding-path	A detached value cannot establish molecular connectivity.
environment	held-molecule-and-condition	unconditioned-universal-coupling	The reported value is condition-bound.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
magnitude	exact-positive-magnitude-or-EmptyOne	floating-or-fitted-coupling	A float or fitted equation cannot force the relation.
symmetry	site-swap-invariant-pair-relation	ordered-pair-changes-value	Relabelling the pair cannot alter a scalar coupling.
custody	complete-value-error-bound-and-absence-custody	selected-couplings-only	Selection hides limits and adverse rows.
extension	successor-retains-and-appends-complete-pairs	refitted-added-couplings	Refitting can change earlier records.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:b202328e4a64f41d0b00ed1bf8f394cc0ec701768a250de4f89b48e708d52d42.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:962e937cc6e441a3cbff466c3891ee473aef56d6cef0f4351301937d32f62694.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:cc2061b0df9f12cddeb4a3f3fac6f0091b885a2f12c711240223c107516de1db.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:abc09bd7494af6bf706ab8a2800c9b6ddf27076f2e76de69f41620339158250f.

Independent reconstruction. Implementation hash

sha256:e496e46ac1a463b3057193dcd3de1984e3b3a8b5eb907f0586382758160358f5; independent certificate sha256:ff4528f87e2a3d4cb39d473022dbf91485709302418c61067193a19841379747; external-validation hash sha256:cbabc74475e674fdec1dff21ad1b5c13c1d9f2bbf8e4026eb3a614e6133b590c. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-NMR-NOMENCLATURE-2001
- BMRB-ENTRY-16582-SCALAR-COUPPLINGS
- IUPAC-NMR-NOMENCLATURE-2001
- BMRB-ENTRY-16582-SCALAR-COUPPLINGS
- IUPAC-NMR-NOMENCLATURE-2001
- BMRB-ENTRY-16582-SCALAR-COUPPLINGS
- IUPAC-NMR-NOMENCLATURE-2001
- ANAL-006-008-NMR-SOURCE-INVENTORY

Comparison records:

- complete ten-source post-seal NMR family retained as 24 PDF pages, five HTML documents, four NMR-STAR records and 1,633 measured rows
- all 8 separately registered claim targets retained
- signed external coupling values are held orientations plus exact positive magnitudes; eleven external zero-rate inscriptions are unresolved structural EmptyOne
- all pre-seal identity exposure remains disclosed and is never relabelled blind

Falsification boundary: The claim halts if its survivor is nonunique; any registered NMR value, site, reference, condition, unit, uncertainty, ambiguity, absent field, adverse orientation, page or source byte is omitted; the independent reconstruction disagrees; or any external result selects the law. Eight dimensions exhaust pair, spin relation, counted path, condition, exact magnitude, symmetry, custody and extension.

Measurement receipt: sha256:f59520809b826cd2c1285f9954b7442b57fe4db43b0c2a936ad7af77a29b3372.

Isolation certificate: sha256:12c6e07d46e16d4301564f3cbf8b04810a5d6e7133a680f163e71919c2eba2d9.

Custody certificate: sha256:a48c33f60a22523348e0cea89365f3fcd95747e2c70fcac1470a726dbfef4126.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external value reference equation conventional model threshold or selected successful row chooses the survivor
- all measured adverse absent unavailable unresolved error ambiguity bound condition unit and historical inscriptions remain

Immutable identities. Source manifest

sha256:9d19fa490229807c17a41ac649cfd33e127812c1de41544aca089f10ff0c0469; derivation seal

sha256:08bc37bcd3d75ac5d22741da29c4d666979f81d3a915c0a273d213ecf2952813; engine receipt

sha256:e7f9cf5d8813ce2c7e445d785d8914f60e566db7e2c787649bb2cd94bcf7c366 at

receipts/engine/model_admitted/SFT-CHEM-NMR-SPIN-COUPLING-007-e7f9cf5d8813ce2c.json;

empirical hash sha256:2091f569b181b890471ccbefe04e6dd0599b4e8af240682e8f1d736627cccb08;

measurement receipt sha256:f59520809b826cd2c1285f9954b7442b57fe4db43b0c2a936ad7af77a29b3372.

222. Fold NMR relaxation and exchange law

Claim identity: SFT-CHEM-NMR-RELAXATION-EXCHANGE-008

Question and theorem. Held sites, states, processes and finite observation conditions force exact positive relaxation-time or exchange-rate records with complete absence custody.

held-molecular-carrier-and-site__held-source-and-destination-states__held-relaxation-or-exchange-process__finite-positive-observation-interval__exact-positive-time-or-counted-transition-rate__held-condition-unit-and-uncertainty__measured-bounded-unavailable-unresolved-custody__successor-retains-and-appends-complete-transitions

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001
- SFT-CHEM-NMR-SPIN-COUPLING-007

Generated grammar. Generate the literal product of eight registered NMR-custody decisions.

Boundary: Eight dimensions exhaust carrier/site, states, process, finite resource, exact relation, observation, adversity and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `held-molecular-carrier-and-site__held-source-and-destination-states__held-relaxation-or-exchange-process__finite-positive-observation-interval__exact-positive-time-or-counted-transition-rate__held-condition-unit-and-uncertainty__measured-bounded-unavailable-unresolved-custody__successor-retains-and-appends-complete-transitions`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	held-molecular-carrier-and-site	detached-relaxation-or-exchange-number	A number alone cannot identify the relaxing or exchanging molecule.
states	held-source-and-destination-states	single-state-decay-label	A transition requires source and destination states.
process	held-relaxation-or-exchange-process	generic-dynamics-label	Different NMR processes cannot be conflated.
resource	finite-positive-observation-interval	continuous-time-premise	An ungenerated continuum is not an observation ledger.
relation	exact-positive-time-or-counted-transition-rate	imported-exponential-or-fitted-rate	A fitted decay form cannot force the transition law.
observation	held-condition-unit-and-uncertainty	unconditioned-universal-dynamics	Relaxation and exchange are method and condition bound.
adversity	measured-bounded-unavailable-unresolved-custody	missing-or-unfavorable-rows-erased	Erasure manufactures a complete trend.
extension	successor-retains-and-appends-complete-transitions	refitted-added-timescale	Refitting changes prior evidence.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:9b2c0a6b12cfacdf1ce25693cfb5a79cecb8a764c2c5fe86f60b02015a8096be.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:8e188fa3429a52c4c2b147a11838d7d9ab97c6e39e2817da40a74f4f1118fc6e.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:1c1995ce3b28d41c5f48a4ee57b8a0647cb54b6bb6561c5130bc6be8e0a14e7e.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:68405aa12f05a613b585245d9fd32d5946564147055647e790f039060de1d7b9.

Independent reconstruction. Implementation hash

sha256:699be59c5b749122cdb65e24cdc7c49d7f5a47e9a1c6fefb4112c785c6722d3f; independent certificate sha256:a7ab14d0cf7f4c057122b618d38c858ba4df856108f4d5c35a093e901a3aef1c; external-validation hash sha256:85ac57c06daf598795eb1367a13c55f072bca12c6081099b22343faf8bdbb40b. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-NMR-NOMENCLATURE-2001
- BMRB-ENTRY-52365-RELAXATION

- BMRB-ENTRY-27257-H-EXCHANGE
- IUPAC-NMR-NOMENCLATURE-2001
- BMRB-ENTRY-52365-RELAXATION
- BMRB-ENTRY-27257-H-EXCHANGE
- IUPAC-NMR-NOMENCLATURE-2001
- ANAL-006-008-NMR-SOURCE-INVENTORY

Comparison records:

- complete ten-source post-seal NMR family retained as 24 PDF pages, five HTML documents, four NMR-STAR records and 1,633 measured rows
- all 8 separately registered claim targets retained
- signed external coupling values are held orientations plus exact positive magnitudes; eleven external zero-rate inscriptions are unresolved structural EmptyOne
- all pre-seal identity exposure remains disclosed and is never relabelled blind

Falsification boundary: The claim halts if its survivor is nonunique; any registered NMR value, site, reference, condition, unit, uncertainty, ambiguity, absent field, adverse orientation, page or source byte is omitted; the independent reconstruction disagrees; or any external result selects the law. Eight dimensions exhaust carrier/site, states, process, finite resource, exact relation, observation, adversity and extension.

Measurement receipt: sha256:b2488b0436a9ba02acba25ff45e33656bb7bc2c542eb78f6089f2f4e01bce669.
Isolation certificate: sha256:ac2e98f75196a5e00b9ffbe8184237afbca9d3605e9948e4db87ea809e55aedef.
Custody certificate: sha256:5a77c1feac1c24f33b78ec4e6a75a9d45421026745749201f9d6cbb34f3b368a.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external value reference equation conventional model threshold or selected successful row chooses the survivor
- all measured adverse absent unavailable unresolved error ambiguity bound condition unit and historical inscriptions remain

Immutable identities. Source manifest

sha256:e64f58e65cb775b14c1f18eb8aa68da2f0393b6519eba2b8fd5d3060c5ac05bd; derivation seal
sha256:f96c309ba64803c7734c29746ad0465f2ab78f90db23ed41f2593d5bd9f14938; engine receipt
sha256:cf3149e0d7e99648e36cdf8db7ee0d35e3acd938fbbc3cc9578bb015da018fe5 at receipts/engine/model_admitted/SFT-CHEM-NMR-RELAXATION-EXCHANGE-008-cf3149e0d7e99648.json; empirical hash
sha256:8971de89d827dd4d49696574042dfeaf13e0449d7321ceb11912786c65dbed95; measurement receipt
sha256:b2488b0436a9ba02acba25ff45e33656bb7bc2c542eb78f6089f2f4e01bce669.

223. Fold Raman transition and intensity law

Claim identity: SFT-CHEM-RAMAN-TRANSITION-INTENSITY-009

Question and theorem. Held molecular transitions force a held Raman side, exact positive shift/intensity support and complete conditioned line custody.

held-molecular-carrier__held-initial-and-final-state__held-polarizability-state-transformation__held-side-positive-exact-shift-or-EmptyOne__exact-positive-event-ratio-or-EmptyOne__held-excitation-condition-and-unit__complete-line-intensity-uncertainty-custody__successor-retains-and-appends-complete-lines

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001
- SFT-CHEM-MOLECULAR-POLARIZABILITY-006
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-SELECTION-RULE-STRUCTURE-010
- SFT-CHEM-VIBRATIONAL-FREQUENCY-009
- SFT-CHEM-NMR-RELAXATION-EXCHANGE-008

Generated grammar. Generate the literal product of eight registered photoluminescence-custody decisions.

Boundary: Eight dimensions exhaust carrier, states, polarizability, position, intensity, condition, custody and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *held-molecular-carrier__held-initial-and-final-state__held-polarizability-state-transformation__held-side-positive-exact-shift-or-EmptyOne__exact-positive-event-ratio-or-EmptyOne__held-excitation-condition-and-unit__complete-line-intensity-uncertainty-custody__successor-retains-and-appends-complete-lines*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	held-molecular-carrier	detached-raman-number	A number cannot identify the scattering molecule.
states	held-initial-and-final-state	peak-without-state-transition	A peak alone loses its molecular transition.
polarizability	held-polarizability-state-transformation	imported-raman-selection-rule	A named rule cannot replace the derived response change.
position	held-side-positive-exact-shift-or-EmptyOne	signed-floating-line-position	A signed float is not a native exact relation.
intensity	exact-positive-event-ratio-or-EmptyOne	arbitrary-normalized-height	An arbitrary height loses counted comparison support.
condition	held-excitation-condition-and-unit	unconditioned-universal-spectrum	Excitation, phase, temperature and instrument boundary affect observation.
custody	complete-line-intensity-uncertainty-custody	selected-prominent-lines	Selecting peaks hides weak, adverse, unresolved and absent rows.
extension	successor-retains-and-appends-complete-lines	refitted-added-spectrum	Renormalizing after extension changes prior evidence.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:1ecec4fb60abfa98bc3d9a12c6045ff51c9c5f0c755ef07912541df11aaa488a.

- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:60c8e64e39e4fc8ee817efce4fec5ebedee4f41bf9bb85c194ac7b9f98f67155.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:cad3e58163a937aea7c3ae62dc98514f897d11c2595add14ee2ee8312fe8ca49.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:51022cb8515e57e69981b297f3fc885a90b910656856af91e98fbeddcce59906.

Independent reconstruction. Implementation hash

sha256:e084b30bbb73fbc75b40a557377d87b96c4bcab7900bdbe674d2e17646680367; independent certificate sha256:6f560f702cc76e6e22bb75b0c89511facba3b060c2b42ff577df818b3c220e0c; external-validation hash sha256:81a7f82e703bf6ea9ca4da9d57930850c2c26683c1218a72330738ec6102e953. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-RAMAN-RELATIVE-INTENSITY-STANDARDS
- NIST-SRM-2241-RAMAN-785-NM
- NIST-SRM-2242A-RAMAN-532-NM
- NIST-RAMAN-RELATIVE-INTENSITY-STANDARDS
- NIST-SRM-2241-RAMAN-785-NM
- NIST-SRM-2242A-RAMAN-532-NM
- NIST-RAMAN-RELATIVE-INTENSITY-STANDARDS
- ANAL-009-011-COMPLETE-SOURCE-INVENTORY

Comparison records:

- complete sixteen-artifact post-seal surface retained: 73 PDF pages, six HTML documents, two XML records, two certified workbooks and 3,184,576 source bytes
- all 8 separately registered claim targets retained
- all coefficients, curves, yields, lifetimes, errors, outliers, adverse, absent, unavailable and unresolved observations retained
- external continuum spreadsheet replay remains downstream evidence and never selects the Fold-native law

Falsification boundary: The claim halts if its survivor is nonunique; any registered value, line, coefficient, curve, condition, unit, uncertainty, adverse, absent, unavailable, unresolved or source surface is omitted; the separate reconstruction disagrees; or an external result selects the native law. Eight dimensions exhaust carrier, states, polarizability, position, intensity, condition, custody and extension.

Measurement receipt: sha256:8320e6a923186157324e50fcc4499e30874d91d405cabea6a1ffee7038a6f560.

Isolation certificate: sha256:f8a708264342648984fcb3c2841f1d97ebf7725c01f540176689862df970c3ce.

Custody certificate: sha256:0a551594ab198a492f7cecf6c6ef20101f94f5a20c53ba77d5c105d861dcd03.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external coefficient fit value formula threshold prominent line or favorable row chooses the survivor
- all measured favorable adverse absent unavailable unresolved error bound condition unit and transport disclosures remain

Immutable identities. Source manifest

sha256:3ab0ea8f0ed4f09fdc4590f13db46db1559274482b4a93afe93a8b9b1012daf6; derivation seal
 sha256:1915a7606998972ceb1c1e9e15642835b43efe01a036c95c4bc41295adccb078; engine receipt
 sha256:3a31c40ef538e5177cd4f20be15f0ed5fccee352a787ae53854be9d84301568c at receipts/engine/model_admitted/SFT-CHEM-RAMAN-TRANSITION-INTENSITY-009-3a31c40ef538e517.json; empirical
 hash sha256:67bd545b781b51d7c1f7f6dbae5daf30b3437114b01def6ae596176d8789ce4a; measurement
 receipt sha256:8320e6a923186157324e50fcc4499e30874d91d405cabea6a1ffee7038a6f560.

224. Fold fluorescence yield and lifetime law

Claim identity: SFT-CHEM-FLUORESCENCE-YIELD-LIFETIME-010

Question and theorem. A held excitation forces a complete radiative/nonradiative partition, exact counted yield and finite lifetime custody.

held-molecular-carrier-and-excitation__held-initial-and-final-molecular-states__complete-radiative-nonradiative-partition__exact-positive-emission-support-or-EmptyOne__exact-outcome-over-excitation-ratio__exact-positive-observation-lifetime-or-EmptyOne__complete-emission-yield-lifetime-condition-custody__successor-retains-and-repartitions-complete-counts

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-SELECTION-RULE-STRUCTURE-010
- SFT-CHEM-PHOTOCHEM-001
- SFT-CHEM-COMPETING-CHANNEL-BRANCHING-006
- SFT-CHEM-NMR-RELAXATION-EXCHANGE-008
- SFT-CHEM-RAMAN-TRANSITION-INTENSITY-009

Generated grammar. Generate the literal product of eight registered photoluminescence-custody decisions.

Boundary: Eight dimensions exhaust carrier, transition, channels, emission, yield, lifetime, custody and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *held-molecular-carrier-and-excitation__held-initial-and-final-molecular-states__complete-radiative-nonradiative-partition__exact-positive-emission-support-or-EmptyOne__exact-outcome-over-excitation-ratio__exact-positive-observation-lifetime-or-EmptyOne__complete-emission-yield-lifetime-condition-custody__successor-retains-and-repartitions-complete-counts*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	held-molecular-carrier-and-excitation	detached-fluorescence-number	A number cannot identify the emitting molecule.
transition	held-initial-and-final-molecular-states	emission-peak-without-states	A peak alone loses its transition endpoints.
channels	complete-radiative-nonradiative-partition	radiative-channel-only	Ignoring nonradiative outcomes inflates yield.
emission	exact-positive-emission-support-or-EmptyOne	signed-floating-emission-value	A signed float is not a native emission relation.
yield	exact-outcome-over-excitation-ratio	fitted-quantum-yield	A fitted scalar can hide omitted outcomes.
lifetime	exact-positive-observation-lifetime-or-EmptyOne	imported-continuous-decay-law	A continuum decay form cannot force a finite observation.
custody	complete-emission-yield-lifetime-condition-custody	selected-bright-favorable-records	Selection erases quenching, uncertainty and missing rows.
extension	successor-retains-and-repartitions-complete-counts	refitted-added-channel	Refitting changes prior yields.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:0189ccedbb209f987b2d7bfff8435305d9d1d037fc5b82f54a9cbe2c3e068fc9c.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:66a98c7cc3ccf7022b90157bb1c6e317a472e074ed0034077f774f214d201d4d.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:8529c34d069626096e99df25525511dd0aca219267d0a84a579a8220623a6e2f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:27f81d85d80527af16220cdb1b19f1996a181a8368ef220e0d781939bbc2bb5a.

Independent reconstruction. Implementation hash

sha256:39548a9708c67fb391373752947396ce5a32827d2d280f6a6343c00fd7d25e24; independent certificate sha256:cc2accce72397717d63b2ebe890cecd5e3aa7ee843fbd122b3d4d373f00da1; external-validation hash sha256:bd571acb518835cd68205f6c4817ba2cc7792f72594dbcf9a5d7352e5c8a016f. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-FLUORESCENCE-STANDARDS-2010
- IUPAC-PHOTOLUMINESCENCE-QUANTUM-YIELD-2011
- NIST-SRM-2941A-FLUORESCENCE-GREEN
- NIST-IR-7458-FLUORESCENCE-GUIDE
- NIST-NLM-PASH-PHOSPHORESCENCE-2018
- IUPAC-FLUORESCENCE-STANDARDS-2010
- IUPAC-PHOTOLUMINESCENCE-QUANTUM-YIELD-2011

- ANAL-009-011-COMPLETE-SOURCE-INVENTORY

Comparison records:

- complete sixteen-artifact post-seal surface retained: 73 PDF pages, six HTML documents, two XML records, two certified workbooks and 3,184,576 source bytes
- all 8 separately registered claim targets retained
- all coefficients, curves, yields, lifetimes, errors, outliers, adverse, absent, unavailable and unresolved observations retained
- external continuum spreadsheet replay remains downstream evidence and never selects the Fold-native law

Falsification boundary: The claim halts if its survivor is nonunique; any registered value, line, coefficient, curve, condition, unit, uncertainty, adverse, absent, unavailable, unresolved or source surface is omitted; the separate reconstruction disagrees; or an external result selects the native law. Eight dimensions exhaust carrier, transition, channels, emission, yield, lifetime, custody and extension.

Measurement receipt: sha256:8742783851e961a0b76bb8d7ec30e892f7c60121bb90d6c1478339414709b15f.
Isolation certificate: sha256:ece838ab18af8bd2a82a7d813dea0b1a752d2fa81f9b9d0bed4af548df6ec8da.
Custody certificate: sha256:3ecaebec992785c2bf3bfa433b806efdb368910d0f60363b1a53c2303b7d1df2.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external coefficient fit value formula threshold prominent line or favorable row chooses the survivor
- all measured favorable adverse absent unavailable unresolved error bound condition unit and transport disclosures remain

Immutable identities. Source manifest

sha256:72dd73322b42e50c495ca281a1c85ef1ba5614c6fb80f35ea5953769dda69fc2; derivation seal
sha256:17362ba27fd87500541dc9bee10a35eebd0ec059ff5207c266e0d390d6e2cb7f; engine receipt
sha256:db992b28c2c1144acf643fcf9c1cbc7fea7c72b6ad92c99cb01c0f4430772c63 at receipts/engine/model_admitted/SFT-CHEM-FLUORESCENCE-YIELD-LIFETIME-010-db992b28c2c1144a.json; empirical
hash sha256:3a7912ba54349ecdcf8c3872f52372c82d71fe4a42d6f79dbdfdb684e5255d34; measurement
receipt sha256:8742783851e961a0b76bb8d7ec30e892f7c60121bb90d6c1478339414709b15f.

225. Fold phosphorescence intersystem law

Claim identity: SFT-CHEM-PHOSPHORESCENCE-INTERSYSTEM-011

Question and theorem. Held distinct spin states force a complete excitation-intersystem-emission path with exact observed or structural-absence custody.

held-molecular-carrier__held-distinct-source-and-intersystem-spin-states__held-excitation-intersystem-final-state-path__exact-positive-emission-support-or-EmptyOne__exact-outcome-over-excitation-ratio-or-EmptyOne__exact-positive-finite-lifetime-or-EmptyOne__complete-emission-lifetime-spin-condition-custody__successor-retains-and-appends-complete-paths

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-PHYS-QUANTUM-SPIN-001
- SFT-CHEM-ELECTRON-COUNT-SPIN-002
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-SELECTION-RULE-STRUCTURE-010
- SFT-CHEM-PHOTOCHEM-001
- SFT-CHEM-COMPETING-CHANNEL-BRANCHING-006
- SFT-CHEM-FLUORESCENCE-YIELD-LIFETIME-010

Generated grammar. Generate the literal product of eight registered photoluminescence-custody decisions.

Boundary: Eight dimensions exhaust carrier, spin, path, emission, yield, lifetime, custody and extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `held-molecular-carrier__held-distinct-source-and-intersystem-spin-states__held-excitation-intersystem-final-state-path__exact-positive-emission-support-or-EmptyOne__exact-outcome-over-excitation-ratio-or-EmptyOne__exact-positive-finite-lifetime-or-EmptyOne__complete-emission-lifetime-spin-condition-custody__successor-retains-and-appends-complete-paths`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	held-molecular-carrier	detached-phosphorescence-number	A number cannot identify the emitting molecule.
spin	held-distinct-source-and-intersystem-spin-states	phosphorescence-without-spin-path	The process loses its defining intersystem distinction.
path	held-excitation-intersystem-final-state-path	single-emission-arrow	A direct arrow erases the intermediate state.
emission	exact-positive-emission-support-or-EmptyOne	signed-floating-emission-value	A signed float is not native exact support.
yield	exact-outcome-over-excitation-ratio-or-EmptyOne	fitted-phosphorescence-yield	A fitted scalar can hide lost channels.
lifetime	exact-positive-finite-lifetime-or-EmptyOne	imported-continuous-tail	A continuum tail is not a finite observation ledger.
custody	complete-emission-lifetime-spin-condition-custody	selected-long-lived-emissions	Selection erases quenched, unavailable and unresolved rows.
extension	successor-retains-and-appends-complete-paths	refitted-added-intersystem-path	Refitting changes earlier paths.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9ffe80cbde5bba43610b194f37aa49d1b046419e6990560ff29a4c66ac283226`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:457b72b1f5386f3f2622113bd4c0505f8786946a8ce1e168c02ca2b94180328c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:b2545fbc6c9da19ad891c6e4777239981e12cf9bcd8fe9ed9ce02467c15ce5ff.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:951beb9cab004709c750ab77758a15cd756aa9405da931d8a11dc49748f090cd.

Independent reconstruction. Implementation hash

sha256:dcb3bcce543598fbf78ab8fcec4d63a91bac4695d513c63e97360bc806d8e846; independent certificate sha256:49bb980d74274dc88bf21b9eb9e8732b990f48deb37fe486d7c0e1491eca0b51; external-validation hash sha256:89d5e388562d0d36ed07f74b9f330cf7257de9a3ff0eebf6665fbdd2da6d5105. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-NLM-PASH-PHOSPHORESCENCE-2018
- IUPAC-PHOSPHORESCENCE-LIFETIME
- NIST-NLM-PASH-PHOSPHORESCENCE-2018
- IUPAC-PHOSPHORESCENCE-LIFETIME
- NIST-NLM-PASH-PHOSPHORESCENCE-2018
- IUPAC-PHOSPHORESCENCE-LIFETIME
- NIST-NLM-PASH-PHOSPHORESCENCE-2018
- ANAL-009-011-COMPLETE-SOURCE-INVENTORY

Comparison records:

- complete sixteen-artifact post-seal surface retained: 73 PDF pages, six HTML documents, two XML records, two certified workbooks and 3,184,576 source bytes
- all 8 separately registered claim targets retained
- all coefficients, curves, yields, lifetimes, errors, outliers, adverse, absent, unavailable and unresolved observations retained
- external continuum spreadsheet replay remains downstream evidence and never selects the Fold-native law

Falsification boundary: The claim halts if its survivor is nonunique; any registered value, line, coefficient, curve, condition, unit, uncertainty, adverse, absent, unavailable, unresolved or source surface is omitted; the separate reconstruction disagrees; or an external result selects the native law. Eight dimensions exhaust carrier, spin, path, emission, yield, lifetime, custody and extension.

Measurement receipt: sha256:d9f48b11513593e23d0274dfca84710c538f6b22f0580923c44b3dff7efc835b.

Isolation certificate: sha256:2382dec011b62027b9c16d4cf2e0463ddc3a0503ad17f1693e7dc8750f345add.

Custody certificate: sha256:7ba8cdcd899b00772457636cf5280fbb5f3fec687bc69778fafeb6900bfc25b3.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external coefficient fit value formula threshold prominent line or favorable row chooses the survivor
- all measured favorable adverse absent unavailable unresolved error bound condition unit and transport disclosures remain

Immutable identities. Source manifest

sha256:f6f654ce73b6e12d6075267162eea7f5dd0d2bb040863b964861d7694d923eda; derivation seal
 sha256:221ebcc21d6fc8dd773d20d49a20ef554963049ed77ae9481ea92b1c9ffdc147; engine receipt
 sha256:7402f1ff97095dd5342c8fec51864a26e957882bcb2ed377f3991353955af400 at receipts/engine/model_admitted/SFT-CHEM-PHOSPHORESCENCE-INTERSYSTEM-011-7402f1ff97095dd5.json; empirical
 hash sha256:438023a479115027a40a56286fa8dd90a221f6a5e73b660c04508c827719abac; measurement
 receipt sha256:d9f48b11513593e23d0274dfca84710c538f6b22f0580923c44b3dff7efc835b.

226. Fold infrared line-position and intensity law

Claim identity: SFT-CHEM-INFRARED-LINE-INTENSITY-012

Question and theorem. Held vibrational state pairs force exact positive infrared line coordinates and counted responses with complete condition, uncertainty and absent-line custody.

held-molecule-and-phase__held-initial-final-vibrational-states__exact-positive-line-coordinate-or-EmptyOne__exact-counted-transmission-response-or-EmptyOne__held-dipole-state-transformation__held-complete-measurement-condition__complete-line-error-bound-absence-custody__successor-appends-without-changing-prior-lines

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001
- SFT-CHEM-SPEC-INFRARED-001
- SFT-CHEM-VIBRATIONAL-FREQUENCY-009
- SFT-CHEM-RAMAN-TRANSITION-INTENSITY-009

Generated grammar. Generate the literal product of the eight frozen ANAL-012 custody decisions.

Boundary: Eight binary dimensions exhaust the registered ANAL-012 carrier, relation, exact-support, condition, custody and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *held-molecule-and-phase__held-initial-final-vibrational-states__exact-positive-line-coordinate-or-EmptyOne__exact-counted-transmission-response-or-EmptyOne__held-dipole-state-transformation__held-complete-measurement-condition__complete-line-error-bound-absence-custody__successor-appends-without-changing-prior-lines*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	held-molecule-and-phase	detached-ir-number	A number does not identify its molecule.
states	held-initial-final-vibrational-states	peak-without-vibrational-states	A peak alone loses both state endpoints.
coordinate	exact-positive-line-coordinate-or-EmptyOne	signed-floating-wavenumber	A signed float is not a native coordinate.
response	exact-counted-transmission-response-or-EmptyOne	arbitrary-absorbance-height	An arbitrary height loses counted comparison support.
selection	held-dipole-state-transformation	imported-ir-selection-rule	A named convention cannot replace the derived state distinction.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
condition	held-complete-measurement-condition	unconditioned-spectrum	Phase, temperature and instrument boundary affect observation.
custody	complete-line-error-bound-absence-custody	selected-strong-bands	Peak selection erases weak and adverse rows.
extension	successor-append-without-changing-prior-lines	refitted-added-spectrum	Refitting can change prior evidence.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:81b34bcc83219bf628be5ddcecb691cc0bea5c0a6f03bc1d8b22990934fa9834.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:2ab2c23734e660c98f19225ef83676acd2affa51316688caab8566080c21a04c.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:c9e98887892e89c4e1f17446372c4785602c05d8d229f6cf3d8b5547ab2d5eb9.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:7bf3dfb7d8ebaaa14502af96609930d118bda4b202d563982414d4fa5d0813aa.

Independent reconstruction. Implementation hash

sha256:f9cf5dae32a540c5026f21320e7e3b3dfc20f1494179b52de92eab682b076a3d; independent certificate sha256:50e6b3d46d212d930016692106d6d65817f4e2df59041c79219536fdc9948f9c; external-validation hash sha256:8e61666b732bbf0528f0fb8eef8cd1e62d8dfc9abe0e93babbeb21e917b54e4c. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-WEBBOOK-SRD69-GUIDE
- NIST-WEBBOOK-BENZENE-MULTIMODAL
- NIST-WEBBOOK-ACETONE-MULTIMODAL
- NIST-WEBBOOK-CYCLOHEXANE-MULTIMODAL
- NIST-WEBBOOK-SRD69-GUIDE
- NIST-WEBBOOK-BENZENE-MULTIMODAL
- NIST-WEBBOOK-ACETONE-MULTIMODAL
- NIST-WEBBOOK-CYCLOHEXANE-MULTIMODAL

Comparison records:

- complete 218-artifact post-seal surface retained: 22,221,914 source bytes, 173 PDF pages, seven HTML surfaces, 63 IR vectors, three UV-visible vectors, three mass vectors, 91 rotational lines, 619 chromatography rows and 385 neutron rows
- all 8 separately registered claim targets retained
- all favorable, adverse, absent, unavailable, unresolved, predicted, fitted, uncertain, superseded and transport records retained
- external values and conventional models remain downstream evidence and never select the Fold-native survivor

Falsification boundary: ANAL-012 halts if its survivor is nonunique; any registered source, target, value, line, page, condition, unit, uncertainty, favorable, adverse, absent, unavailable, unresolved, fitted, superseded or transport row is omitted; the implementation-distinct reconstruction disagrees; any dependency is missing; or external evidence selects the native law.

Measurement receipt: sha256:b7a0134374f12aec7af261af6810bee4ef95c12c7070e8c63ecc790a6ba1a50b.
 Isolation certificate: sha256:6ed5b836ece8bb70a21c1ded97427833b4da814268f8918599ef37313eb266b9.
 Custody certificate: sha256:0c08bbec13e14242bd7255fc18eac49fccf66ed19b91a0b53a6b860cb32cc284.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external value equation model fit sign zero outcome prominent row or favorable subset selects the survivor
- all favorable adverse absent unavailable unresolved predicted fitted uncertain superseded and transport rows remain

Immutable identities. Source manifest

sha256:be7bd240798d41c57f2b296cfb7145bc55d966c5f8db1848cc01284bb7cc37f8; derivation seal
 sha256:2dcd1ff8431b059c1f780b1b65d2009ca7ca7301db81c9e564a93c5b08314e3c; engine receipt
 sha256:f02f6937e976f9e1f538adeba977dd325215403f87020176180fbf5920f1b426 at receipts/engine/model_admitted/SFT-CHEM-INFRARED-LINE-INTENSITY-012-f02f6937e976f9e1.json; empirical hash
 sha256:4e7badebc93a2a2b1b58f3d54fe4497da8cc9598696c31941ce70c49affc6c4a; measurement receipt
 sha256:b7a0134374f12aec7af261af6810bee4ef95c12c7070e8c63ecc790a6ba1a50b.

227. Fold UV-visible line-position and intensity law

Claim identity: SFT-CHEM-UVVISIBLE-LINE-INTENSITY-013

Question and theorem. Held electronic state pairs force exact positive absorption coordinates and counted response support with complete solvent, condition and uncertainty custody.

held-molecule-and-electronic-carrier__held-initial-final-electronic-states__exact-positive-absorption-coordinate-or-EmptyOne__exact-counted-absorption-response-or-EmptyOne__complete-electronic-transition-support__held-solvent-phase-temperature-condition__complete-band-error-bound-absence-custody__successor-retains-and-appends-complete-bands

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001
- SFT-CHEM-SPEC-UVVIS-001
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-FLUORESCENCE-YIELD-LIFETIME-010

Generated grammar. Generate the literal product of the eight frozen ANAL-013 custody decisions.

Boundary: Eight binary dimensions exhaust the registered ANAL-013 carrier, relation, exact-support, condition, custody and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-molecule-and-electronic-carrier__held-initial-final-electronic-states__exact-positive-absorption-coordinate-or-EmptyOne__exact-counted-absorption-response-or-EmptyOne__complete-electronic-transition-support__held-solvent-phase-temperature-condition__complete-band-error-bound-abs

ence-custody__successor-retains-and-appends-complete-bands. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	held-molecule-and-electronic-carrier	detached-uvvis-number	A number does not identify its molecule.
states	held-initial-final-electronic-states	band-without-electronic-states	A band alone loses its state endpoints.
coordinate	exact-positive-absorption-coordinate-or-EmptyOne	signed-floating-wavelength	A signed float is not native support.
response	exact-counted-absorption-response-or-EmptyOne	fitted-extinction-scalar	A fitted scalar can hide source counts.
partition	complete-electronic-transition-support	bright-transition-only	Selecting bright transitions erases the remaining support.
condition	held-solvent-phase-temperature-condition	solvent-free-universal-band	Solvent and condition alter observation.
custody	complete-band-error-bound-absence-custody	selected-favorable-bands	Selection erases shoulders, errors and absent rows.
extension	successor-retains-and-appends-complete-bands	renormalized-added-band	Renormalization changes prior responses.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: bd27026ff9eeb3d250d6ccd58584798f28fa6e0ff65b85292ff17914dc18acd9.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: f5d8edc537c224fb1e6589dd3203040d8d91b175eb9c65e6e825801f5d28a587.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 5a6acf62bc8a47ff1c34dfb52a90a55d7dfe37a781b8522161d4a47e6663c4b1.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 31b2fba961a824808edf9443cfb357b5323a934b10e0a32563073bbfac29fdad.

Independent reconstruction. Implementation hash

sha256: df81f3d689392bd4ec49e26da2429f6f1f8aa2fe7e043f5b1e39729df2a74985; independent certificate sha256: 46f2e181035900649df883715fad2a944075e9d708bb9828924ec7ec6d13c6c0; external-validation hash sha256: a14379d85ca941f8f913512211d639aa67518ef98132dd97e1f202e680f945d0. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-WEBBOOK-SRD69-GUIDE
- NIST-WEBBOOK-BENZENE-MULTIMODAL
- NIST-WEBBOOK-ACETONE-MULTIMODAL
- NIST-WEBBOOK-SRD69-GUIDE

- NIST-WEBBOOK-BENZENE-MULTIMODAL
- NIST-WEBBOOK-ACETONE-MULTIMODAL
- NIST-WEBBOOK-SRD69-GUIDE
- NIST-WEBBOOK-BENZENE-MULTIMODAL

Comparison records:

- complete 218-artifact post-seal surface retained: 22,221,914 source bytes, 173 PDF pages, seven HTML surfaces, 63 IR vectors, three UV-visible vectors, three mass vectors, 91 rotational lines, 619 chromatography rows and 385 neutron rows
- all 8 separately registered claim targets retained
- all favorable, adverse, absent, unavailable, unresolved, predicted, fitted, uncertain, superseded and transport records retained
- external values and conventional models remain downstream evidence and never select the Fold-native survivor

Falsification boundary: ANAL-013 halts if its survivor is nonunique; any registered source, target, value, line, page, condition, unit, uncertainty, favorable, adverse, absent, unavailable, unresolved, fitted, superseded or transport row is omitted; the implementation-distinct reconstruction disagrees; any dependency is missing; or external evidence selects the native law.

Measurement receipt: sha256:db853c05581bb2b9042cab71a18818d459d3a29ce68e0a433a89c07427b7ebc3.

Isolation certificate: sha256:6bd6628b57ab0bc069de0fc18276b178856213e03fffedba07ad27611d7c81f8.

Custody certificate: sha256:fe8082ec3c90afc125f8d356eb7cac9fc8125faf1df91e8245854e408b57a35b.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external value equation model fit sign zero outcome prominent row or favorable subset selects the survivor
- all favorable adverse absent unavailable unresolved predicted fitted uncertain superseded and transport rows remain

Immutable identities. Source manifest

sha256:ba5d9bb7e697b80aaad4fdca2ca362811491c5afceec3233c0a46a72d7d6118c6; derivation seal
 sha256:bc26ddd829e5fd311e9d37da64f0c9d8239161df152f6f9695dcbc9557b98643; engine receipt
 sha256:bdba363b0775c4185cef2caa2f190c6b8ecd358a1bd7fbcc186fea28841dd724 at receipts/engine/model_admitted/SFT-CHEM-UVVISIBLE-LINE-INTENSITY-013-bdba363b0775c418.json; empirical hash
 sha256:eaf930d9facda11fbb37652a2410388fe5dcd02420877da929883d102295f498; measurement receipt
 sha256:db853c05581bb2b9042cab71a18818d459d3a29ce68e0a433a89c07427b7ebc3.

228. Fold mass, isotope and fragmentation spectrum law

Claim identity: SFT-CHEM-MASS-ISOTOPE-FRAGMENT-VECTOR-014

Question and theorem. Held molecular and ionic carriers force a complete exact mass-to-charge, isotope and fragmentation event vector without weak-peak selection.

held-molecular-parent-and-ion-carriers__held-isotope-and-charge-composition__exact-positive-mass-charge-ratio__held-parent-fragment-transition-path__exact-event-over-reference-event-ratio__held-ionization-and-acquisition-condition__complete-peak-isotope-error-absence-custody__successor-retains-and-appends-complete-fragments

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001
- SFT-CHEM-SPEC-MASS-001
- SFT-CHEM-ELEM-ISOTOPE-001
- SFT-CHEM-NUCLEAR-ELECTRONIC-COMPOSITION-012

Generated grammar. Generate the literal product of the eight frozen ANAL-014 custody decisions.

Boundary: Eight binary dimensions exhaust the registered ANAL-014 carrier, relation, exact-support, condition, custody and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-molecular-parent-and-ion-carriers__held-isotope-and-charge-composition__exact-positive-mass-charge-ratio__held-parent-fragment-transition-path__exact-event-over-reference-event-ratio__held-ionization-and-acquisition-condition__complete-peak-isotope-error-absence-custody__success-or-retains-and-appends-complete-fragments. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	held-molecular-parent-and-ion-carriers	mass-peaks-without-molecule	Peaks alone lose molecular identity.
composition	held-isotope-and-charge-composition	nominal-mass-only	Nominal mass loses isotope composition.
coordinate	exact-positive-mass-charge-ratio	floating-mass-to-charge	A float is not native exact support.
fragmentation	held-parent-fragment-transition-path	peak-without-fragment-path	A peak alone loses its parent path.
response	exact-event-over-reference-event-ratio	normalized-major-peak-only	Major-only normalization erases weak events.
condition	held-ionization-and-acquisition-condition	unconditioned-ionization	Ionization and instrument boundary affect fragmentation.
custody	complete-peak-isotope-error-absence-custody	selected-diagnostic-peaks	Selection erases weak, isotope and unresolved rows.
extension	successor-retains-and-appends-complete-fragments	refitted-added-fragment	Refitting changes prior event ratios.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256: 959513b8f8cb6f298990b9250ef8b2f65acab9cba6159896fbb6cdb9b50e294e.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: dd99abef90e47a2cd587688dc27c59560f2b4ffe6306d33cbfecb2bc5a1aaa1b.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 929b6219db255bc22d444d9835989bf020e21125b462e26dd969b2f9265060c1.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 4937a2d3a6aa9aaaa2e4636d83dc7e1ea5d588a578d996f3cb1baf84adf60ed9.

Independent reconstruction. Implementation hash

sha256 : 3ded677b43cf0439d5ca5779827b4ab9d1d8ad5ccb538667aa5fce0bfc0fe928; independent certificate
sha256 : 4694f5af3fcb1cb25a9a1169b0e99f5a809305f1b9a954e410f2875d4f8d117; external-validation
hash sha256 : c1607780f93f8c605b5696fab1b43bd5ae996dc6d1e69f35d90e39ebcacbc859. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were
preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-WEBBOOK-SRD69-GUIDE
- NIST-WEBBOOK-BENZENE-MULTIMODAL
- NIST-WEBBOOK-ACETONE-MULTIMODAL
- NIST-WEBBOOK-CYCLOHEXANE-MULTIMODAL
- NIST-WEBBOOK-SRD69-GUIDE
- NIST-WEBBOOK-BENZENE-MULTIMODAL
- NIST-WEBBOOK-ACETONE-MULTIMODAL
- NIST-WEBBOOK-CYCLOHEXANE-MULTIMODAL

Comparison records:

- complete 218-artifact post-seal surface retained: 22,221,914 source bytes, 173 PDF pages, seven HTML surfaces, 63 IR vectors,
three UV-visible vectors, three mass vectors, 91 rotational lines, 619 chromatography rows and 385 neutron rows
- all 8 separately registered claim targets retained
- all favorable, adverse, absent, unavailable, unresolved, predicted, fitted, uncertain, superseded and transport records retained
- external values and conventional models remain downstream evidence and never select the Fold-native survivor

Falsification boundary: ANAL-014 halts if its survivor is nonunique; any registered source, target, value, line, page, condition,
unit, uncertainty, favorable, adverse, absent, unavailable, unresolved, fitted, superseded or transport row is omitted; the
implementation-distinct reconstruction disagrees; any dependency is missing; or external evidence selects the native law.

Measurement receipt: sha256 : 63910a33e03fb0fd7e011bfb4bd755d1c45d385f12d132d39123f2bdc1ac3ed6.
Isolation certificate: sha256 : e4463d126ba7f28bbc47b78831fbbd73b154ce73c8da55487ae7fa8391f2f7cf.
Custody certificate: sha256 : 06b77d61db2bd44697b3e1d3bd6ca14e527b798ccb2004e8f49a3fdd39e23edf.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is
closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external value equation model fit sign zero outcome prominent row or favorable subset selects the survivor
- all favorable adverse absent unavailable unresolved predicted fitted uncertain superseded and transport rows remain

Immutable identities. Source manifest

sha256 : 34ce43b4d9b038cdeb9fc56ea29239112781d43b12663d1089068e85f00d8928; derivation seal

sha256:23f812da331d71765c593a6546dcc3cad5c29243e4f96bc0cb1d7487d42339bc; engine receipt sha256:ba5cd4595e4c3ceecfe6ea9a8c23330df8a8259df70711bc22e290122edc64e9 at receipts/engine/model_admitted/SFT-CHEM-MASS-ISOTOPE-FRAGMENT-VECTOR-014-ba5cd4595e4c3cee.json; empirical hash sha256:767dec03e713ca4c446182c08dec10ff063bb7b57d3cafa569304f9ddfae859b; measurement receipt sha256:63910a33e03fb0fd7e011bfb4bd755d1c45d385f12d132d39123f2bdc1ac3ed6.

229. Fold rotational spectrum line law

Claim identity: SFT-CHEM-ROTATIONAL-SPECTRUM-LINE-015

Question and theorem. Held geometry, axis, isotopologue and rotational-state pairs force a complete exact frequency/intensity/error line vector.

held-molecule-geometry-isotopologue__held-initial-final-rotational-states__held-molecular-axis-and-symmetry__exact-positive-frequency-or-EmptyOne__exact-positive-line-response-or-EmptyOne__held-energy-temperature-unit-condition__complete-frequency-error-intensity-state-custody__successor-appends-without-changing-prior-lines

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001
- SFT-CHEM-SPEC-ROT-VIB-001
- SFT-CHEM-ROTATIONAL-CONSTANT-010
- SFT-PHYS-MOLECULAR-SPECTROSCOPY-TERMINAL-005

Generated grammar. Generate the literal product of the eight frozen ANAL-015 custody decisions.

Boundary: Eight binary dimensions exhaust the registered ANAL-015 carrier, relation, exact-support, condition, custody and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-molecule-geometry-isotopologue__held-initial-final-rotational-states__held-molecular-axis-and-symmetry__exact-positive-frequency-or-EmptyOne__exact-positive-line-response-or-EmptyOne__held-energy-temperature-unit-condition__complete-frequency-error-intensity-state-custody__successor-appends-without-changing-prior-lines. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	held-molecule-geometry-isotopologue	detached-microwave-line	A line does not identify its molecule.
states	held-initial-final-rotational-states	line-without-rotational-states	A line alone loses quantum-state endpoints.
axis	held-molecular-axis-and-symmetry	axis-free-rotational-number	Geometry-dependent rotation needs a held axis.
coordinate	exact-positive-frequency-or-EmptyOne	floating-frequency	A float is not native exact support.
response	exact-positive-line-response-or-EmptyOne	arbitrary-line-strength	An arbitrary strength loses event support.
condition	held-energy-temperature-unit-condition	unconditioned-catalog-line	Temperature and state-energy boundary matter.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
custody	complete-frequency-error-intensity-state-custody	selected-detected-lines	Selection erases errors and predicted/unresolved rows.
extension	successor-append-without-changing-prior-lines	refitted-catalog-extension	Refitting can change earlier lines.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256: 8a40b24bf3dd37a97e0b00d332a91d6fd4c504f03ab426bbacae41647f4b6673.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 85f3620a933491cf6b1b7ac7d7279cbd5fe87efc668b5b9035339be7594a1ae4.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 7d809df1991b7b2e5cbf47940577b09568528b7efe63d93a86d754ba994c6f65.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: 523b42cfbaf915e6abc50e9f71bdab9e0d36e2b650c45b8b68a02dcd5e8aaa0.

Independent reconstruction. Implementation hash

sha256: a0a69daa173e45c802d917e30485e58c5cb7e3e5f415cf12078c219d151773ae; independent certificate sha256: a28a20c9a13e9eaddfe4cbde03ecf918bcdabbdb77601a325c2443314d355113; external-validation hash sha256: 5cc20bb7ad21c5ca2f376917f53236f1062f9ac3087d868bda437db12a097abd. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NASA-JPL-MOLECULAR-SPECTROSCOPY-CATALOG
- NASA-JPL-CO-ROTATIONAL-LINE-CATALOG-028001
- NASA-JPL-MOLECULAR-SPECTROSCOPY-CATALOG
- NASA-JPL-CO-ROTATIONAL-LINE-CATALOG-028001
- NASA-JPL-MOLECULAR-SPECTROSCOPY-CATALOG
- NASA-JPL-CO-ROTATIONAL-LINE-CATALOG-028001
- NASA-JPL-MOLECULAR-SPECTROSCOPY-CATALOG
- NASA-JPL-CO-ROTATIONAL-LINE-CATALOG-028001

Comparison records:

- complete 218-artifact post-seal surface retained: 22,221,914 source bytes, 173 PDF pages, seven HTML surfaces, 63 IR vectors, three UV-visible vectors, three mass vectors, 91 rotational lines, 619 chromatography rows and 385 neutron rows
- all 8 separately registered claim targets retained
- all favorable, adverse, absent, unavailable, unresolved, predicted, fitted, uncertain, superseded and transport records retained
- external values and conventional models remain downstream evidence and never select the Fold-native survivor

Falsification boundary: ANAL-015 halts if its survivor is nonunique; any registered source, target, value, line, page, condition, unit, uncertainty, favorable, adverse, absent, unavailable, unresolved, fitted, superseded or transport row is omitted; the implementation-distinct reconstruction disagrees; any dependency is missing; or external evidence selects the native law.

Measurement receipt: sha256: f04c29c88ec37822d12d54fac463e6d20855911a530989438c74fbff858ce72e.

Isolation certificate: sha256: 280a015c3edd38fc734ee973fea5470985fbb5d91d92adfe11cdcd37cb46f7a3.

Custody certificate: sha256:bdfce47978d0bc6b9c43f318da74011e170a2ed47523dd105ce80b67431a8d30.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external value equation model fit sign zero outcome prominent row or favorable subset selects the survivor
- all favorable adverse absent unavailable unresolved predicted fitted uncertain superseded and transport rows remain

Immutable identities. Source manifest

sha256:17cbb807cf6dbeac6bd5eb1ff54c4afd6aa343eaa2e61c39ec3512533bca0da2; derivation seal
 sha256:0b840d1dcdb2f0b1614526ba7d99117a9d11ad72dabaebeb6915faaaaced126a; engine receipt
 sha256:e3079d4d40dad58b944ef2b5b17ef06bff8285d7d2a102358fb2019b92ef4942 at receipts/engine/
 model_admitted/SFT-CHEM-ROTATIONAL-SPECTRUM-LINE-015-e3079d4d40dad58b.json; empirical hash
 sha256:17fe8e6c38b1d01352df18c1b626af16e7bd852d57f409f48bee986b0d65abbc; measurement receipt
 sha256:f04c29c88ec37822d12d54fac463e6d20855911a530989438c74fbff858ce72e.

230. Fold X-ray diffraction chemical-structure law

Claim identity: SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016

Question and theorem. A held crystal carrier and reciprocal support force exact diffraction coordinates, responses and a chemistry-owned structure correspondence.

*held-crystal-phase-and-composition__explicit-physics-xray-scattering-handoff__held-direct-and-reciprocal-support__e
 xact-positive-spacing-or-held-angle__exact-counted-relative-intensity-or-EmptyOne__complete-cell-symmetry-site-corr
 espondence__complete-reflection-error-condition-custody__successor-retains-prior-reflections-and-structure*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001
- SFT-PHYS-WAVE-DIFFRACTION-001
- SFT-MAT-CRYST-UNIT-CELL-001
- SFT-MAT-CRYST-RECIPROCAL-001

Generated grammar. Generate the literal product of the eight frozen ANAL-016 custody decisions.

Boundary: Eight binary dimensions exhaust the registered ANAL-016 carrier, relation, exact-support, condition, custody and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-crystal-phase-and-composition__explicit-physics-xray-scattering-handoff__held-direct-and-reciprocal-support__exact-positive-spacing-or-held-angle__exact-counted-relative-intensity-or-EmptyOne__complete-cell-symmetry-site-correspondence__complete-reflection-error-condition-custody__successor-retains-prior-reflections-and-structure. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	held-crystal-phase-and-composition	pattern-without-crystal	A pattern alone loses crystal identity.
probe	explicit-physics-x-ray-scattering-handoff	chemistry-owned-xray-probe	Probe production belongs to Physics.
support	held-direct-and-reciprocal-support	line-list-without-reciprocal-support	Lines alone lose the generated lattice relation.
coordinate	exact-positive-spacing-or-held-angle	floating-two-theta	A float is not native exact support.
response	exact-counted-relative-intensity-or-EmptyOne	arbitrary-diffraction-intensity	An arbitrary height loses comparison support.
structure	complete-cell-symmetry-site-correspondence	database-structure-label-only	A label cannot establish structure correspondence.
custody	complete-reflection-error-condition-custody	selected-major-reflections	Selection erases weak and adverse reflections.
extension	successor-retains-prior-reflections-and-structure	refined-added-pattern	Refinement can change prior evidence.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:3ee6584602617d278dcc9159ae31543ddbfe11eedf8f741cd029a6dc43fccc0.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:c1a69d8a51cd21499c17bbca82e4e7f0f6e4f9249bc6d8f33c690f789223f8b3.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:1c21a85f3e815a0f52abcb47ddd0c0a1de642bd4603bd87a01852ffb8bc37e63.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:a65f50c75c9d183a636a7fcda5dd4fa0369fc0ad2e5539df0038c03ccd860463.

Independent reconstruction. Implementation hash

sha256:b47f660681890f4fc2aafb5a9bfc56b0c4f19220058c188c7c07a8e3bb8ae5e; independent certificate sha256:b7dbb77e329da299e4f00abf9584b0ccddcb5b962915019d2a8517c01c286d50; external-validation hash sha256:99986586491d122e5f904762fbb32549372d9e9cc59d386b22459ab608211bc5. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-SRM-674-XRAY-INTENSITY-SET
- NIST-SRM-676A-DIFFRACTION-STANDARD
- NIST-SRM-674-XRAY-INTENSITY-SET
- NIST-SRM-676A-DIFFRACTION-STANDARD
- NIST-SRM-674-XRAY-INTENSITY-SET
- NIST-SRM-676A-DIFFRACTION-STANDARD

- NIST-SRM-674-XRAY-INTENSITY-SET
- NIST-SRM-676A-DIFFRACTION-STANDARD

Comparison records:

- complete 218-artifact post-seal surface retained: 22,221,914 source bytes, 173 PDF pages, seven HTML surfaces, 63 IR vectors, three UV-visible vectors, three mass vectors, 91 rotational lines, 619 chromatography rows and 385 neutron rows
- all 8 separately registered claim targets retained
- all favorable, adverse, absent, unavailable, unresolved, predicted, fitted, uncertain, superseded and transport records retained
- external values and conventional models remain downstream evidence and never select the Fold-native survivor

Falsification boundary: ANAL-016 halts if its survivor is nonunique; any registered source, target, value, line, page, condition, unit, uncertainty, favorable, adverse, absent, unavailable, unresolved, fitted, superseded or transport row is omitted; the implementation-distinct reconstruction disagrees; any dependency is missing; or external evidence selects the native law.

Measurement receipt: sha256:35aff32247bdfdf672f3ba1892d9d7c8ba6ec3434fc1471bd883f1533f9ee422.

Isolation certificate: sha256:dca2d92e335ba5defe7a96d1881109a91ff246d552c71613c1c6232d67d4ab9a.

Custody certificate: sha256:c24d9fee278ec2984f9454073c11fabda1d7454b58f1cc006864e4a1e24efea9.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external value equation model fit sign zero outcome prominent row or favorable subset selects the survivor
- all favorable adverse absent unavailable unresolved predicted fitted uncertain superseded and transport rows remain

Immutable identities. Source manifest

sha256:46e05e2405bf9919cf120e2305acd60dc48e96a043da761b3ef03005b2b59b46; derivation seal

sha256:1048f44c3e32ee7e04026dd31b04a8a28e36a703b4b2a5bf7d96281a51dd9004; engine receipt

sha256:5ee17101561f8ab2362d74ff43b67bbfd64664300a7e8c2723462537ce64b621 at receipts/engine/

model_admitted/SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016-5ee17101561f8ab2.json; empirical

hash sha256:cb6daf6bf2dab9073b49b02b217c344264e142dbe38480a9b42d52a7f9e0913d; measurement

receipt sha256:35aff32247bdfdf672f3ba1892d9d7c8ba6ec3434fc1471bd883f1533f9ee422.

231. Fold electron/neutron diffraction correspondence law

Claim identity: SFT-CHEM-ELECTRON-NEUTRON-DIFFRACTION-017

Question and theorem. Held probe identity and one held chemical structure force separately retained electron and neutron diffraction correspondences without probe conflation.

held-material-phase-and-composition__held-electron-or-neutron-probe__explicit-physics-scattering-handoff__held-reciprocal-and-site-support__exact-positive-diffraction-coordinate-or-EmptyOne__probe-specific-structure-response-correspondence__complete-probe-reflection-error-custody__successor-appends-probe-records-without-conflation

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-CHEM-ANALYTICAL-COMplete-RECORD-001
- SFT-PHYS-MATTER-SCATTERING-001
- SFT-MAT-CRYST-RECIPROCAL-001
- SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016

Generated grammar. Generate the literal product of the eight frozen ANAL-017 custody decisions.

Boundary: Eight binary dimensions exhaust the registered ANAL-017 carrier, relation, exact-support, condition, custody and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `held-material-phase-and-composition__held-electron-or-neutron-probe__explicit-physics-scattering-handoff__held-reciprocal-and-site-support__exact-positive-diffraction-coordinate-or-EmptyOne__probe-specific-structure-response-correspondence__complete-probe-reflection-error-custody__successor-appends-probe-records-without-conflation`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	held-material-phase-and-composition	diffraction-without-material	A pattern alone loses material identity.
probe	held-electron-or-neutron-probe	probe-agnostic-pattern	Electron and neutron interactions cannot be conflated.
handoff	explicit-physics-scattering-handoff	chemistry-invents-probe-law	Probe scattering belongs to Physics.
support	held-reciprocal-and-site-support	peaks-without-reciprocal-state	Peaks alone lose structural support.
coordinate	exact-positive-diffraction-coordinate-or-EmptyOne	floating-diffraction-coordinate	A float is not native exact support.
correspondence	probe-specific-structure-response-correspondence	same-response-for-both-probes	Different probes retain different response relations.
custody	complete-probe-reflection-error-custody	selected-agreeing-reflections	Agreement selection erases adverse rows.
extension	successor-appends-probe-records-without-conflation	refitted-mixed-probe-pattern	Refitting can change prior evidence.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:80d78a1f9275a6d8970a9bcf1b47fbb896d2171b023ec304b5391bd292a8be3f`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b76fcd8c0d58ddedadc2e69372481eaaafc837f8f8589e16557c9b1237ab0c`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e029a59996c433b72a9bb16a552947a0f78e447c1d34640d1d5ce16cd3696e1f`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:08ce6084ff38a3a96b8b4a70b5d453d2b7c602ca94059b0cd6c169dd906eaf98`.

Independent reconstruction. Implementation hash

sha256:af2fdf5b5d80c8a5f536a64ca3d56b9c4da43e34e693a3d3ff766c7fde59e2b2; independent certificate sha256:da5a9a3b0ef8010b86f67bf12d3faf9533d5b3522f68d9c32274221bc88f81ab; external-validation hash sha256:e070fe6cccc4f501f516752a2025d1b02842066dd0a163f043f38d668501e7c4. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-ELECTRON-DIFFRACTION-DATABASE-REPORT
- NIST-NEUTRON-SCATTERING-LENGTHS
- NIST-SRM-676A-DIFFRACTION-STANDARD
- NIST-ELECTRON-DIFFRACTION-DATABASE-REPORT
- NIST-NEUTRON-SCATTERING-LENGTHS
- NIST-SRM-676A-DIFFRACTION-STANDARD
- NIST-ELECTRON-DIFFRACTION-DATABASE-REPORT
- NIST-NEUTRON-SCATTERING-LENGTHS

Comparison records:

- complete 218-artifact post-seal surface retained: 22,221,914 source bytes, 173 PDF pages, seven HTML surfaces, 63 IR vectors, three UV-visible vectors, three mass vectors, 91 rotational lines, 619 chromatography rows and 385 neutron rows
- all 8 separately registered claim targets retained
- all favorable, adverse, absent, unavailable, unresolved, predicted, fitted, uncertain, superseded and transport records retained
- external values and conventional models remain downstream evidence and never select the Fold-native survivor

Falsification boundary: ANAL-017 halts if its survivor is nonunique; any registered source, target, value, line, page, condition, unit, uncertainty, favorable, adverse, absent, unavailable, unresolved, fitted, superseded or transport row is omitted; the implementation-distinct reconstruction disagrees; any dependency is missing; or external evidence selects the native law.

Measurement receipt: sha256:f17c503e1ed05d084f5ccb530b64866e94e738093b2845da37470faa84d5afc7.

Isolation certificate: sha256:d97858c277f48853dcfce5420aac1d94ec4319c32f975d48d397ec311f855737.

Custody certificate: sha256:1ab6bfc8f8c75b8347dd80814affe61c796aa3322f2834cdabba5e1ae23c09c9.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external value equation model fit sign zero outcome prominent row or favorable subset selects the survivor
- all favorable adverse absent unavailable unresolved predicted fitted uncertain superseded and transport rows remain

Immutable identities. Source manifest

sha256:eb9c0f6bfa42ce190655f604de9291a8280057f7134cadca49a13365db2fa4c98; derivation seal

sha256:aa8e0e18eee66141ad84d151f423e3ff2339e7e36e5cfe8a2cef0f1062b440b7; engine receipt

sha256:cf768f015cdb8251479ab5617c55d02a8637a5be053c313f15f9024fbd3f4553 at receipts/engine/model_admitted/SFT-CHEM-ELECTRON-NEUTRON-DIFFRACTION-017-cf768f015cdb8251.json; empirical hash sha256:3c4618c2fdd96719589ec7a8c4c9610e83dc3ce8af2d776c36942cde0df7f669; measurement

receipt sha256 : f17c503e1ed05d084f5ccb530b64866e94e738093b2845da37470faa84d5afc7.

232. Fold chromatographic retention and resolution law

Claim identity: SFT-CHEM-CHROMATOGRAPHIC-RETENTION-RESOLUTION-018

Question and theorem. Held analyte, matrix and mobile/stationary phases force complete exact retention, peak and pairwise resolution custody at fixed conditions.

held-analyte-and-matrix__held-mobile-and-stationary-phases__exact-positive-retention-coordinate-or-EmptyOne__complete-positive-peak-support__exact-pairwise-resolution-relation__held-temperature-flow-column-condition__complete-coelution-error-absence-custody__successor-retains-and-appends-complete-peaks

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001
- SFT-CHEM-ANALYTICAL-SAMPLE-001
- SFT-CHEM-ANALYTICAL-SELECTIVITY-INTERFERENCE-005
- SFT-CHEM-COUPLED-MASS-HEAT-CHARGE-TRANSPORT-019

Generated grammar. Generate the literal product of the eight frozen ANAL-018 custody decisions.

Boundary: Eight binary dimensions exhaust the registered ANAL-018 carrier, relation, exact-support, condition, custody and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-analyte-and-matrix__held-mobile-and-stationary-phases__exact-positive-retention-coordinate-or-EmptyOne__complete-positive-peak-support__exact-pairwise-resolution-relation__held-temperature-flow-column-condition__complete-coelution-error-absence-custody__successor-retains-and-appends-complete-peaks. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
analyte	held-analyte-and-matrix	retention-number-without-analyte	A time/index alone loses analyte identity.
phases	held-mobile-and-stationary-phases	phase-free-retention	Retention depends on both phase identities.
coordinate	exact-positive-retention-coordinate-or-EmptyOne	floating-retention-value	A float is not native exact support.
peak	complete-positive-peak-support	apex-only-peak	An apex alone loses complete peak support.
resolution	exact-pairwise-resolution-relation	named-separated-or-not	A category hides pairwise distances and widths.
condition	held-temperature-flow-column-condition	unconditioned-retention-index	Temperature, flow and phase conditions matter.
custody	complete-coelution-error-absence-custody	selected-resolved-analyses	Selection erases coelution and unresolved rows.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
extension	successor-retains-and-appends-complete-peaks	refitted-added-analyte	Refitting can change earlier peaks.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256 : b52b59d40799e087350adfcc0533500eda93ff29bd3ef9dfb383470d77fcf063.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 45728329dfa537ebf5e25bba7b593f34d5b9e8033f91964c632213490c57cdeb.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256 : 1b2b0f87c590fec31fde28afc8527140096a99cd726806c29b235f48c75b8082.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256 : 544bf02b1986d8e50aeb4dfd827d56b465eee1f7cd2e9b28e42f08dce685049a.

Independent reconstruction. Implementation hash

sha256 : 4954e185bd735a7dc73b9f6e9ece1fe3c639794981854c8ba231027d1dfb6cb8; independent certificate sha256 : 8a302abf72168db625dbd0fe45fcc92f5031e4a773a218432bd334feb3c5a362; external-validation hash sha256 : cdbf3aa2ce6eab2c04da9cfdc7d9ee9684a135c51042b388a374fd9931c6b09c. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-WEBBOOK-GAS-CHROMATOGRAPHY
- NIST-WEBBOOK-BENZENE-MULTIMODAL
- NIST-WEBBOOK-CYCLOHEXANE-MULTIMODAL
- NIST-WEBBOOK-GAS-CHROMATOGRAPHY
- NIST-WEBBOOK-BENZENE-MULTIMODAL
- NIST-WEBBOOK-CYCLOHEXANE-MULTIMODAL
- NIST-WEBBOOK-GAS-CHROMATOGRAPHY
- NIST-WEBBOOK-BENZENE-MULTIMODAL

Comparison records:

- complete 218-artifact post-seal surface retained: 22,221,914 source bytes, 173 PDF pages, seven HTML surfaces, 63 IR vectors, three UV-visible vectors, three mass vectors, 91 rotational lines, 619 chromatography rows and 385 neutron rows
- all 8 separately registered claim targets retained
- all favorable, adverse, absent, unavailable, unresolved, predicted, fitted, uncertain, superseded and transport records retained
- external values and conventional models remain downstream evidence and never select the Fold-native survivor

Falsification boundary: ANAL-018 halts if its survivor is nonunique; any registered source, target, value, line, page, condition, unit, uncertainty, favorable, adverse, absent, unavailable, unresolved, fitted, superseded or transport row is omitted; the implementation-distinct reconstruction disagrees; any dependency is missing; or external evidence selects the native law.

Measurement receipt: sha256 : 0e0572b10224466ef4b24a3f12db8512f7d4196174f6f76eace062a10f67def0.

Isolation certificate: sha256 : 2dd828a75a0b2872a2e4a52dc7f39227f82532f1d0b566495c6ac14c43f11a8b.

Custody certificate: sha256 : cd6306d9134756615b7a565aa1af00f3ff3749693a0b5187df0671c955f7abb1.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external value equation model fit sign zero outcome prominent row or favorable subset selects the survivor
- all favorable adverse absent unavailable unresolved predicted fitted uncertain superseded and transport rows remain

Immutable identities. Source manifest

sha256:1f23ee34accb810b4310cfa3975fe2af3b73df281371dbf236c89baf2c0a8579; derivation seal
 sha256:de7ea467f340a9bbe840cafc9ca93dfa4110b1e096700ba7816dacdf4f41b199; engine receipt
 sha256:288a23013dd25c34ddffb33fa425241df15bc5f904557f41d8d9de548a56a550 at receipts/engine/model_admitted/SFT-CHEM-CHROMATOGRAPHIC-RETENTION-RESOLUTION-018-288a23013dd25c34.json;
 empirical hash sha256:bbae759ed69df53dd3d4a0d40ed3b981beeb862a30ebf7c31ca146f14eac27e1;
 measurement receipt sha256:0e0572b10224466ef4b24a3f12db8512f7d4196174f6f76eace062a10f67def0.

233. Fold electrophoretic mobility and separation law

Claim identity: SFT-CHEM-ELECTROPHORETIC-MOBILITY-SEPARATION-019

Question and theorem. Held species, charge orientation and medium force exact positive mobility magnitude, trajectory and separation custody.

held-species-and-particle-identity__held-charge-motion-side__held-medium-pH-temperature-condition__held-field-direction-and-resource__exact-positive-mobility-magnitude-or-EmptyOne__exact-trajectory-and-pairwise-separation__complete-mobility-error-absence-custody__successor-retains-and-appends-species

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-ANALYTICAL-COMplete-RECORD-001
- SFT-CHEM-IONIC-MOBILITY-TRANSFERENCE-008
- SFT-CHEM-ANALYTICAL-SELECTIVITY-INTERFERENCE-005

Generated grammar. Generate the literal product of the eight frozen ANAL-019 custody decisions.

Boundary: Eight binary dimensions exhaust the registered ANAL-019 carrier, relation, exact-support, condition, custody and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-species-and-particle-identity__held-charge-motion-side__held-medium-pH-temperature-condition__held-field-direction-and-resource__exact-positive-mobility-magnitude-or-EmptyOne__exact-trajectory-and-pairwise-separation__complete-mobility-error-absence-custody__successor-retains-and-appends-species. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
species	held-species-and-particle-identity	mobility-without-species	A mobility value loses species identity.
charge	held-charge-motion-side	signed-negative-mobility	A signed number conflates orientation and magnitude.
medium	held-medium-pH-temperature-condition	medium-free-mobility	Medium and pH affect motion.
field	held-field-direction-and-resource	unrecorded-field	Motion needs a retained applied-field boundary.
mobility	exact-positive-mobility-magnitude-or-EmptyOne	floating-mobility	A float is not native exact support.
separation	exact-trajectory-and-pairwise-separation	named-separated-or-not	A category loses trajectories and distances.
custody	complete-mobility-error-absence-custody	selected-mobile-fraction	Selection erases immobile and adverse rows.
extension	successor-retains-and-appends-species	refitted-added-species	Refitting changes earlier mobilities.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : 842b306beb766a90bc8f0317f78223435de55dcfb1cb276f2db726bb7891bf34.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 70f7eecb0d75664434fac321a1eaf248a74f257e6b2edbe0a52c8e3bc4323068.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : d55ec7ddcf00b64e20d63b3c2537da64cde437795b40958382a0b9f0dcdd3248.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 62b632bc89fd4b24d0aa2a1d288b2f32fb428d8edee101d5ccfc0478dd4c80a7.

Independent reconstruction. Implementation hash

sha256 : 0333e2123ecc9a2bfdb15a045042832c9fdaf9d918cd51896ae625d268c5d50; independent certificate sha256 : 81e5c0aa3a184df04f2890c48eb68963ca81683c7b527be937e20f9ca11f1bac; external-validation hash sha256 : e4969285aae2217f16c43c057ae6338829b8828392108c0455a1a03aa046dc5f. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-SRM-1980-ELECTROPHORETIC-MOBILITY
- NIST-SP260-209-ZETA-MOBILITY
- NIST-SRM-1980-ELECTROPHORETIC-MOBILITY
- NIST-SP260-209-ZETA-MOBILITY
- NIST-SRM-1980-ELECTROPHORETIC-MOBILITY
- NIST-SP260-209-ZETA-MOBILITY
- NIST-SRM-1980-ELECTROPHORETIC-MOBILITY
- NIST-SP260-209-ZETA-MOBILITY

Comparison records:

- complete 218-artifact post-seal surface retained: 22,221,914 source bytes, 173 PDF pages, seven HTML surfaces, 63 IR vectors, three UV-visible vectors, three mass vectors, 91 rotational lines, 619 chromatography rows and 385 neutron rows
- all 8 separately registered claim targets retained
- all favorable, adverse, absent, unavailable, unresolved, predicted, fitted, uncertain, superseded and transport records retained
- external values and conventional models remain downstream evidence and never select the Fold-native survivor

Falsification boundary: ANAL-019 halts if its survivor is nonunique; any registered source, target, value, line, page, condition, unit, uncertainty, favorable, adverse, absent, unavailable, unresolved, fitted, superseded or transport row is omitted; the implementation-distinct reconstruction disagrees; any dependency is missing; or external evidence selects the native law.

Measurement receipt: sha256:af4973a6eead795a50580bb110a6c37538de0c3cb7d1a11a5ed9b17af3f96121.

Isolation certificate: sha256:704d88f7de48baf038d1fbcdd3315b99af46252a8010a5147a1a5648266671f5.

Custody certificate: sha256:cb5e8a7b10cf7c38a132cce920cabe06c79ecd9dbed20b0f134a791812b06ff9.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external value equation model fit sign zero outcome prominent row or favorable subset selects the survivor
- all favorable adverse absent unavailable unresolved predicted fitted uncertain superseded and transport rows remain

Immutable identities. Source manifest

sha256:d1b2cfd5c2c8fb4328898752eca49f865ac8f99c3615b374b72cabb4ffee4a56; derivation seal

sha256:fa5c7e9fc8d13748c45f32ceb512d3086a2df85e9ed59cc6c67f172e6cb61847; engine receipt

sha256:84321bf139917e3acddc89ace7c14511e4ab04364b65bc03c22a92b7b07c61e1 at receipts/engine/model_admitted/SFT-CHEM-ELECTROPHORETIC-MOBILITY-SEPARATION-019-84321bf139917e3a.json

; empirical hash sha256:4b2e87ad2c47bf0209e5247c2785955a25346c6f27e2e998d0d17ac5e24cb0b3;

measurement receipt sha256:af4973a6eead795a50580bb110a6c37538de0c3cb7d1a11a5ed9b17af3f96121.

234. Fold electroanalytical response law

Claim identity: SFT-CHEM-ELECTROANALYTICAL-RESPONSE-020

Question and theorem. Held analyte/electrode/cell identities and an ordered applied-potential path force a complete exact current-response trace with reaction and adverse custody.

held-analyte-electrode-cell-identity__held-ordered-applied-potential-path__held-oxidation-reduction-current-side__exact-positive-current-magnitude-or-EmptyOne__held-electrode-reaction-correspondence__held-scan-medium-reference-condition__complete-trace-error-background-custody__successor-appends-path-without-changing-prior-response

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-ANALYTICAL-COMplete-RECORD-001
- SFT-CHEM-ELECTRODE-POTENTIAL-CHEMICAL-RELATION-002

- SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003
- SFT-CHEM-ELECTRODE-REACTION-RATE-009

Generated grammar. Generate the literal product of the eight frozen ANAL-020 custody decisions.

Boundary: Eight binary dimensions exhaust the registered ANAL-020 carrier, relation, exact-support, condition, custody and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-analyte-electrode-cell-identity__held-ordered-applied-potential-path__held-oxidation-reduction-current-side__exact-positive-current-magnitude-or-EmptyOne__held-electrode-reaction-correspondence__held-scan-medium-reference-condition__complete-trace-error-background-custody__successor-appends-path-without-changing-prior-response. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
system	held-analyte-electrode-cell-identity	current-curve-without-system	A curve alone loses analyte and cell identity.
path	held-ordered-applied-potential-path	unordered-potential-points	An unordered set loses scan causality.
orientation	held-oxidation-reduction-current-side	signed-current-as-native-number	A signed number conflates direction and magnitude.
response	exact-positive-current-magnitude-or-EmptyOne	floating-current	A float is not native exact support.
reaction	held-electrode-reaction-correspondence	peak-without-electrode-reaction	A peak alone loses chemical transition identity.
condition	held-scan-medium-reference-condition	scan-rate-free-curve	Scan rate, medium and reference affect response.
custody	complete-trace-error-background-custody	selected-peak-values	Peak selection erases baseline and adverse trace rows.
extension	successor-appends-path-without-changing-prior-response	refitted-longer-scan	Refitting changes prior trace points.

Adverse controls. All mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256 : dae7ea26b9d5d951235b1d1a702e2e113b8b3610c3d3d88c67f31308bda52ea3.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 239d570855b887c1e990e463461b602b5db97304d48e0bda45ccd5149aede58b.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256 : 4809d55c6f34ed9ac0bb024ce91114c64f2d7598ee7621828112af6cb2f43b12.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256 : 1031f26eda8c4a7db8a2229fc666ca72fcb0c787d0c55b1ade01ff7ae65a3cb0.

Independent reconstruction. Implementation hash

sha256 : 587bf347adecb08d8573964ebcc101ae95b3d372893c55ded75cde3fdb2d39fb; independent certificate sha256 : c4bd49e641c7c800a24abb62dd6f1cdbcfd6d9b1f1bea5ca81a0440ffa26361ec; external-validation

hash sha256:c8e1e4c5da71f913eb599f94dc782a66dbb04ae0e7b6a252ca28a74f630e5017. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-ELECTROCHEMICAL-METHODS-2019
- NIST-VOLTAMMETRIC-LOD-STUDY
- IUPAC-ELECTROCHEMICAL-METHODS-2019
- NIST-VOLTAMMETRIC-LOD-STUDY
- IUPAC-ELECTROCHEMICAL-METHODS-2019
- NIST-VOLTAMMETRIC-LOD-STUDY
- IUPAC-ELECTROCHEMICAL-METHODS-2019
- NIST-VOLTAMMETRIC-LOD-STUDY

Comparison records:

- complete 218-artifact post-seal surface retained: 22,221,914 source bytes, 173 PDF pages, seven HTML surfaces, 63 IR vectors, three UV-visible vectors, three mass vectors, 91 rotational lines, 619 chromatography rows and 385 neutron rows
- all 8 separately registered claim targets retained
- all favorable, adverse, absent, unavailable, unresolved, predicted, fitted, uncertain, superseded and transport records retained
- external values and conventional models remain downstream evidence and never select the Fold-native survivor

Falsification boundary: ANAL-020 halts if its survivor is nonunique; any registered source, target, value, line, page, condition, unit, uncertainty, favorable, adverse, absent, unavailable, unresolved, fitted, superseded or transport row is omitted; the implementation-distinct reconstruction disagrees; any dependency is missing; or external evidence selects the native law.

Measurement receipt: sha256:1f42897071f17d5a5a3186207022aa532e6b6f5915d53e681644a97507a839ad.

Isolation certificate: sha256:ede9be31ef73f5b43380d4c09046acc95fe1838f9980926e48405c2b4b33b16f.

Custody certificate: sha256:28a5f7c57617fe6fda84c32e80c7ae1a9e5d26cad61cd0ad544d3068aa48cb9d.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external value equation model fit sign zero outcome prominent row or favorable subset selects the survivor
- all favorable adverse absent unavailable unresolved predicted fitted uncertain superseded and transport rows remain

Immutable identities. Source manifest

sha256:726ca57546b5c7888028fc07ef1195beceb31a761f6bc752f5d96798ba621d5c; derivation seal

sha256:628a718815f9e9ce723d477692e569c7121409692575bbf7c0da8b90b1896662; engine receipt

sha256:0a47e880278b7797759fd54efefe124803eebb0d4b2cdb69e752b4134f00f21f at receipts/engine/model_admitted/SFT-CHEM-ELECTROANALYTICAL-RESPONSE-020-0a47e880278b7797.json; empirical

hash sha256:10fcc66c9198a7a663a0761b375d128c1a8023ddd4f6405f7e6d9be59a70a120; measurement

receipt sha256:1f42897071f17d5a5a3186207022aa532e6b6f5915d53e681644a97507a839ad.

235. Fold multimodal molecular identity reconstruction law

Claim identity: SFT-CHEM-MULTIMODAL-MOLECULAR-IDENTITY-021

Question and theorem. Complete orthogonal analytical records force the unique common molecular carrier or halt without identity inflation.

one-held-molecular-carrier__complete-orthogonal-record-family__complete-generated-carrier-candidate-set__exact-record-candidate-incidence__exact-complete-support-intersection__one-unique-common-carrier-or-halt__complete-conflict-absence-unresolved-custody__successor-intersects-without-changing-prior-records

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-ANALYTICAL-COMplete-RECORD-001
- SFT-CHEM-NMR-CHEMICAL-SHIFT-006
- SFT-CHEM-RAMAN-TRANSITION-INTENSITY-009
- SFT-CHEM-INFRARED-LINE-INTENSITY-012
- SFT-CHEM-UVVISIBLE-LINE-INTENSITY-013
- SFT-CHEM-MASS-ISOTOPE-FRAGMENT-VECTOR-014
- SFT-CHEM-ROTATIONAL-SPECTRUM-LINE-015

Generated grammar. Generate the literal product of the eight frozen ANAL-021 custody decisions.

Boundary: Eight binary dimensions exhaust the registered ANAL-021 carrier, relation, exact-support, condition, custody and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *one-held-molecular-carrier__complete-orthogonal-record-family__complete-generated-carrier-candidate-set__exact-record-candidate-incidence__exact-complete-support-intersection__one-unique-common-carrier-or-halt__complete-conflict-absence-unresolved-custody__successor-intersects-without-changing-prior-records*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-held-molecular-carrier	independent-modality-identities	Separate names do not establish one carrier.
records	complete-orthogonal-record-family	selected-diagnostic-modalities	Selection can manufacture identity.
candidates	complete-generated-carrier-candidate-set	database-first-answer	A first answer is not exhaustive.
comparison	exact-record-candidate-incidence	fitted-similarity-score	A fitted score imports a threshold.
intersection	exact-complete-support-intersection	majority-vote-identity	A vote can ignore contradiction.
uniqueness	one-unique-common-carrier-or-halt	best-ranked-candidate	Ranking does not force uniqueness.
adversity	complete-conflict-absence-unresolved-custody	discarded-conflicting-modality	Discarding conflict invalidates reconstruction.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
extension	successor-intersects-without-changing-prior-records	refitted-new-modality	Refitting can change earlier evidence.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256 : a17c9cc0e4f4522ab2fd421ea79bbd889d9ba9091c46c1bcc7b62b1b71f46d4a.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 7313453d9fc2ca56681647c0ed1dfb4e7641e056576d5486c0122e64aca05f5c.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256 : bd9979e1416e7f6657cd2bf61374a8ff6b6b6dd3cc2a32baeb501ec2eef6484c41.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256 : 268dc27d82e3249a159273533ba9204f73234ba6d1a5b3cdfe855e648886cb1a.

Independent reconstruction. Implementation hash

sha256 : 9ba7f52168b703a83c2caf1fd93fcc5f1a625be6818d3dc7322e031a582304bc; independent certificate sha256 : c401b0dd2b0ee238b1b1ad32a3ad2cfc616bed82034de9de999b52edc85e31b1; external-validation hash sha256 : a62dcc2dd0b699169ca2f9fa58cc51329a08c79dfd6979af93f60f663ca876a8. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-WEBBOOK-BENZENE-MULTIMODAL
- NIST-WEBBOOK-ACETONE-MULTIMODAL
- NIST-WEBBOOK-CYCLOHEXANE-MULTIMODAL
- SFT-V3-ANAL-001-011-IMMUTABLE-EVIDENCE
- NIST-WEBBOOK-BENZENE-MULTIMODAL
- NIST-WEBBOOK-ACETONE-MULTIMODAL
- NIST-WEBBOOK-CYCLOHEXANE-MULTIMODAL
- SFT-V3-ANAL-001-011-IMMUTABLE-EVIDENCE

Comparison records:

- complete 218-artifact post-seal surface retained: 22,221,914 source bytes, 173 PDF pages, seven HTML surfaces, 63 IR vectors, three UV-visible vectors, three mass vectors, 91 rotational lines, 619 chromatography rows and 385 neutron rows
- all 8 separately registered claim targets retained
- all favorable, adverse, absent, unavailable, unresolved, predicted, fitted, uncertain, superseded and transport records retained
- external values and conventional models remain downstream evidence and never select the Fold-native survivor

Falsification boundary: ANAL-021 halts if its survivor is nonunique; any registered source, target, value, line, page, condition, unit, uncertainty, favorable, adverse, absent, unavailable, unresolved, fitted, superseded or transport row is omitted; the implementation-distinct reconstruction disagrees; any dependency is missing; or external evidence selects the native law.

Measurement receipt: sha256 : db2fac7c992effa8c0f87bfd5dd77ffd6bbae6a588d9d1408607a40e5d2c2720.

Isolation certificate: sha256 : 8f1578fa24942d46c0260ff972629a9a3952f06c0f0aa2cbe486def92515453e.

Custody certificate: sha256 : a5f1579668fd288fe6d218c0fb6efd362de449898d8568b1f0ee1a1210b395e1.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external value equation model fit sign zero outcome prominent row or favorable subset selects the survivor
- all favorable adverse absent unavailable unresolved predicted fitted uncertain superseded and transport rows remain

Immutable identities. Source manifest

sha256:f2c3360fde0733fb25d80eacc7f908cecc5696bd46f22fcb64220dc3a7a42bef; derivation seal
 sha256:fc0fac5ae8adf880aaebeaa43a7f288225c3f01d8f1ad2fae923b41cd2556bd6; engine receipt
 sha256:ce1108185e2b7ad7e02149909424887e4173c6c4c8068043a198a7ccdd87f0b2 at receipts/engine/model_admitted/SFT-CHEM-MULTIMODAL-MOLECULAR-IDENTITY-021-ce1108185e2b7ad7.json;
 empirical hash sha256:a003cce9f31c804f17e333e0602a87365e426036994b38b3a72da2cad16e3c66;
 measurement receipt sha256:db2fac7c992effa8c0f87bfd5dd77ffd6bbae6a588d9d1408607a40e5d2c2720.

236. Fold complete analytical performance and uncertainty budget

Claim identity: SFT-CHEM-ANALYTICAL-PERFORMANCE-BUDGET-022

Question and theorem. One analytical result is admissible only with jointly retained traceability, trueness, precision, sensitivity, selectivity, detection/quantification and uncertainty custody.

held-result-sample-measurand-method__complete-reference-traceability-chain__exact-trueness-relation-or-EmptyOne__complete-replicate-precision-relation__exact-sensitivity-selectivity-domain__separate-detection-quantification-custody__complete-component-dependency-uncertainty-budget__successor-retains-and-appends-validation-records

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001
- SFT-CHEM-ANALYTICAL-ACCURACY-TRUENESS-001
- SFT-CHEM-ANALYTICAL-PRECISION-REPEATABILITY-002
- SFT-CHEM-ANALYTICAL-SENSITIVITY-003
- SFT-CHEM-ANALYTICAL-DETECTION-QUANTIFICATION-004
- SFT-CHEM-ANALYTICAL-SELECTIVITY-INTERFERENCE-005
- SFT-CHEM-MULTIMODAL-MOLECULAR-IDENTITY-021

Generated grammar. Generate the literal product of the eight frozen ANAL-022 custody decisions.

Boundary: Eight binary dimensions exhaust the registered ANAL-022 carrier, relation, exact-support, condition, custody and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-result-sample-measurand-method__complete-reference-traceability-chain__exact-trueness-relation-or-EmptyOne__complete-replicate-precision-relation__exact-sensitivity-selectivity-domain__separate-detection-quantification-custody__complete-component-dependency-uncertainty-budget__successor-retains-and-appends-validation-records. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
result	held-result-sample-measurand-method	detached-final-number	A final number loses its analytical carrier.
traceability	complete-reference-traceability-chain	untraced-reference	An untraced result has no comparison chain.
trueness	exact-trueness-relation-or-EmptyOne	precision-as-trueness	Repeatability cannot establish reference agreement.
precision	complete-replicate-precision-relation	single-observation-precision	One observation cannot establish repeatability.
response	exact-sensitivity-selectivity-domain	sensitivity-without-domain	A slope without support/domain is incomplete.
decision	separate-detection-quantification-custody	single-detection-threshold	Detection cannot substitute for quantification.
uncertainty	complete-component-dependency-uncertainty-budget	single-combined-error-number	One scalar hides components and alternatives.
extension	successor-retains-and-appends-validation-records	replaced-validation-study	Replacement loses prior evidence.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:df30a809e8d90d9d5dae9318eec34195278aa3fb80977d4bb2dd4e2dcd48b3f.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:4513891e667d5c6d954ae932b7b618128eb136c832856a5130a7890219f3b68d.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:4ec1642ac139f161bc5398d5ebba244c21bdc158a053af2ae46b6ad2e3a39b3e.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:60ae218134bb7211075570e0931fba05e635e477a83c5a5f8a3b8e8251b893af.

Independent reconstruction. Implementation hash

sha256:7fb1f159d647305e96bd700dac57aab792fadf7540f392b17c82f0ee95524bc9; independent certificate sha256:2683bad0d7b5f641d96552051cfb1c29f07867dcfb47105f4b21015b484d5964; external-validation hash sha256:7434aa354cf36cb9d0bacee51e3713eaa6a1ca411224f3f8312f262ad326283. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- SFT-V3-ANAL-001-011-IMMUTABLE-EVIDENCE
- NIST-VOLTAMMETRIC-LOD-STUDY
- NIST-SRM-1980-ELECTROPHORETIC-MOBILITY
- SFT-V3-ANAL-001-011-IMMUTABLE-EVIDENCE
- NIST-VOLTAMMETRIC-LOD-STUDY
- NIST-SRM-1980-ELECTROPHORETIC-MOBILITY

- SFT-V3-ANAL-001-011-IMMUTABLE-EVIDENCE
- NIST-VOLTAMMETRIC-LOD-STUDY

Comparison records:

- complete 218-artifact post-seal surface retained: 22,221,914 source bytes, 173 PDF pages, seven HTML surfaces, 63 IR vectors, three UV-visible vectors, three mass vectors, 91 rotational lines, 619 chromatography rows and 385 neutron rows
- all 8 separately registered claim targets retained
- all favorable, adverse, absent, unavailable, unresolved, predicted, fitted, uncertain, superseded and transport records retained
- external values and conventional models remain downstream evidence and never select the Fold-native survivor

Falsification boundary: ANAL-022 halts if its survivor is nonunique; any registered source, target, value, line, page, condition, unit, uncertainty, favorable, adverse, absent, unavailable, unresolved, fitted, superseded or transport row is omitted; the implementation-distinct reconstruction disagrees; any dependency is missing; or external evidence selects the native law.

Measurement receipt: sha256:2a585abdbfedbd9ae39d60b680998f8728f9fd9bc44b37b2bdb59327e9d204c0.

Isolation certificate: sha256:719b9143ae69853d0b8e80eef2dc56274e23a684f0fdd412d1a426cf91006cf8.

Custody certificate: sha256:7d4d0b1b590531742acad71276f669037c428aab544c7fada899379a4221e0a0.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external value equation model fit sign zero outcome prominent row or favorable subset selects the survivor
- all favorable adverse absent unavailable unresolved predicted fitted uncertain superseded and transport rows remain

Immutable identities. Source manifest

sha256:148a72b42a5aecb3a153ceb8fd447a5dc178e2d12c92a9c05ad3c706594196ca; derivation seal

sha256:23e640fd07bb96e243d9bb4bccc5b16f744824dfd8278b1c70a4a5ce5f1b46bb; engine receipt

sha256:bdb6476cc6078def755b0a43e42de3ff7b6d18cf4c50783057ef2f9c9ad878de at receipts/engine/model_admitted/SFT-CHEM-ANALYTICAL-PERFORMANCE-BUDGET-022-bdb6476cc6078def.json;

empirical hash sha256:1acd3d8e2a8860e0fda55b1e620105f81f0097a55fa334e9d8c62b7f66d2b383;

measurement receipt sha256:2a585abdbfedbd9ae39d60b680998f8728f9fd9bc44b37b2bdb59327e9d204c0.

27. Computational chemistry and cheminformatics - 14 admitted claims

Chemical data structures are exact organizations of information with reversible provenance and complete finite enumerations. This entire registered subcategory is included; no proper subset is presented as field completion.

237. Fold canonical molecular-graph encoding law

Claim identity: SFT-CHEM-CANONICAL-MOLECULAR-GRAPH-ENCODING-001

Question and theorem. A held finite molecular carrier forces a reversible position-independent exact word over complete atom, bond, orientation and provenance distinctions.

held-finite-molecular-carrier__complete-held-atom-identities__complete-held-bond-identities__retained-isotope-charge-stereo-fibres__exhaustive-equivalence-class-canonical-word__exact-word-to-graph-reconstruction__held-source-and-translation-custody__append-only-new-carrier-registration

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001

Generated grammar. Generate the literal product of the eight frozen COMP-001 chemical-information custody decisions.

Boundary: Eight binary dimensions exhaust the registered COMP-001 identity, exact-support, equivalence, provenance, falsification and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-finite-molecular-carrier__complete-held-atom-identities__complete-held-bond-identities__retained-isotope-charge-stereo-fibres__exhaustive-equivalence-class-canonical-word__exact-word-to-graph-reconstruction__held-source-and-translation-custody__append-only-new-carrier-registration. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	held-finite-molecular-carrier	detached-symbol-string	detached-symbol-string closes a distinction required by carrier.
atoms	complete-held-atom-identities	unlabelled-vertex-count	unlabelled-vertex-count closes a distinction required by atoms.
bonds	complete-held-bond-identities	adjacency-without-bond-state	adjacency-without-bond-state closes a distinction required by bonds.
orientation	retained-isotope-charge-stereo-fibres	discarded-isotope-charge-stereo	discarded-isotope-charge-stereo closes a distinction required by orientation.
canonicity	exhaustive-equivalence-class-canonical-word	input-order-dependent-word	input-order-dependent-word closes a distinction required by canonicity.
decoding	exact-word-to-graph-reconstruction	one-way-text-token	one-way-text-token closes a distinction required by decoding.
custody	held-source-and-translation-custody	source-free-identifier	source-free-identifier closes a distinction required by custody.
extension	append-only-new-carrier-registration	renumbered-prior-graph	renumbered-prior-graph closes a distinction required by extension.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: cd749fe3e15db09d7dabe334a333cd5374ebe4142b87fb12f48b18d1c5152f90.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 842201e9234904380529d4846e74c2855d63142bdc0c51f9c765a000f8c9302a.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 603f2af3ceb9c048ccf8c3a2db5e7af0ce1cbaf28f95d215f6c17fb8a1e025b3.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: eac026c1d687599055b6b0d68c9c71de0289cfe308d1efd55bc2028c3d24fba3.

Independent reconstruction. Implementation hash

sha256:342ca1dee796514da7841231721c5664badbcd7267468575a749a117e1bb52fb; independent certificate sha256:438d7353243a95ec5baff2c21282989c45f8f7f3eedeb981ca73a8f7a3b215bc; external-validation hash sha256:e7624f3eab1c6fd9480b1eb0db4f6451c5aaaba31e5037eea794c4717ebdb7b0. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- PUBCHEM-PUG-REST-AUTHORITY
- PUBCHEM-C3H8O-FORMULA-CENSUS
- PUBCHEM-ASPIRIN-CONFORMERS
- PUBCHEM-ASPIRIN-SUBSTRUCTURE
- PUBCHEM-ASPIRIN-SIMILARITY
- CHEBI-REST-AUTHORITY
- RHEA-IMMUTABLE-REACTION-SMILES
- USPTO-IMMUTABLE-REACTION-SURFACE

Comparison records:

- complete 59-artifact 444,644,830-byte post-seal surface retained
- 12 PubChem records, four ChEBI cross-source records, 36,444 Rhea reactions, 50,016 USPTO reactions and 1,065,119 atom-mapped reactions retained
- all twelve registered invalid-property responses, low-confidence rows, conflicts, unavailable records and resource halts retained
- all 8 separately registered claim targets retained
- external formats, algorithms, scores and outcomes never select the Fold-native survivor

Falsification boundary: COMP-001 halts if its survivor is nonunique; a registered target, source route, graph, atom, bond, orientation, mapping, reaction row, provenance record, database conflict, adverse result or resource halt is omitted; the implementation-distinct reconstruction disagrees; a dependency is absent; or external data selects the native law.

Measurement receipt: sha256:58fd98a398fa285c4b4991fc559e868b01d7b8c2f65bd6058167ac6bec03db86.

Isolation certificate: sha256:b758e99588e49ad1444cd8de5ebf9b888bda2d3a94c24934f4993b13148c1ccf.

Custody certificate: sha256:9f4a97002d102ccc5fc0623a3e1006f7b9d199f7003151233447e6ca753ab423.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external database encoding algorithm score value equation model fit outcome or software library selects the survivor
- all favorable adverse absent unavailable unresolved low-confidence transport conflict and declared resource-halt rows remain

Immutable identities. Source manifest

sha256:99649812c1f946b32455c7188c9f9f46ca7bb9e57ed6bdcdc3b8f5ba74f8de58; derivation seal

sha256:07629d71514a248fe8f19841a78f79ea94be4580fd70c12a788a7d342d3fc9bd; engine receipt

sha256:9358e99b84fdf4738c170a8167e30bf44becc2ec010ed5ba04755eff83256b51 at receipts/engin

e/model_admitted/SFT-CHEM-CANONICAL-MOLECULAR-GRAPH-ENCODING-001-9358e99b84fdf473.json;
 empirical hash sha256:2848799eddf86d37311b5c34982ec093e28fd46df139f354db2d8691fc3f2988;
 measurement receipt sha256:58fd98a398fa285c4b4991fc559e868b01d7b8c2f65bd6058167ac6bec03db86.

238. Fold molecular-graph isomorphism and identity law

Claim identity: SFT-CHEM-MOLECULAR-GRAPH-ISOMORPHISM-002

Question and theorem. Two molecular records denote the same chemical carrier exactly when a generated bijection preserves every admitted atom, bond and held orientation distinction.

complete-graph-domain__complete-atom-bijection__all-atom-state-labels-preserved__all-bond-order-labels-preserved__all-held-fibres-preserved__complete-unresolved-cell-permutation-census__canonical-word-and-witness-equivalence__every-single-distinction-tamper-rejected

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-MULTIMODAL-MOLECULAR-IDENTITY-021

Generated grammar. Generate the literal product of the eight frozen COMP-002 chemical-information custody decisions.

Boundary: Eight binary dimensions exhaust the registered COMP-002 identity, exact-support, equivalence, provenance, falsification and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *complete-graph-domain__complete-atom-bijection__all-atom-state-labels-preserved__all-bond-order-labels-preserved__all-held-fibres-preserved__complete-unresolved-cell-permutation-census__canonical-word-and-witness-equivalence__every-single-distinction-tamper-rejected*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
domain	complete-graph-domain	formula-only-comparison	formula-only-comparison closes a distinction required by domain.
mapping	complete-atom-bijection	partial-or-many-to-one-map	partial-or-many-to-one-map closes a distinction required by mapping.
atoms	all-atom-state-labels-preserved	element-only-match	element-only-match closes a distinction required by atoms.
bonds	all-bond-order-labels-preserved	connectivity-only-match	connectivity-only-match closes a distinction required by bonds.
orientation	all-held-fibres-preserved	stereo-isotope-charge-erasure	stereo-isotope-charge-erasure closes a distinction required by orientation.
enumeration	complete-unresolved-cell-permutation-census	one-heuristic-labelling	one-heuristic-labelling closes a distinction required by enumeration.
decision	canonical-word-and-witness-equivalence	database-id-equality	database-id-equality closes a distinction required by decision.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
adverse	every-single-distinction-tamper-rejected	near-match-accepted	near-match-accepted closes a distinction required by adverse.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:1d5386fc4ab03a5cb0608baeebf3bd6ba557ac396bfa48aa9f4692b7fc97a827.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:45a3f90e6b87204e2d8cad2218fc2ce914aed5373dfc7442f7e45d3592f9b21.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:1fe896d9ea8d4a87315ee1e18a77f7fb8ee321fc620344d005032ccb21dda316.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:52a5f766c8d623d01d000d2bc4c9413cc0166caa00fc24c8255c36fca6732e.

Independent reconstruction. Implementation hash

sha256:0ce4c49762124c8f456cf2df29fb7ed0c2f3b49c63b1a846e2296cb225eedf28; independent certificate sha256:6849bb7a25bbd224f3b2c40ca83e8d178d9196b7134378111a08027ce443bd1d; external-validation hash sha256:0630f9a02251e097d88d4cdac0b3230d099a81b35c82c801071340b22721ec20. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- PUBCHEM-PUG-REST-AUTHORITY
- PUBCHEM-C3H8O-FORMULA-CENSUS
- PUBCHEM-ASPIRIN-CONFORMERS
- PUBCHEM-ASPIRIN-SUBSTRUCTURE
- PUBCHEM-ASPIRIN-SIMILARITY
- CHEBI-REST-AUTHORITY
- RHEA-IMMUTABLE-REACTION-SMILES
- USPTO-IMMUTABLE-REACTION-SURFACE

Comparison records:

- complete 59-artifact 444,644,830-byte post-seal surface retained
- 12 PubChem records, four ChEBI cross-source records, 36,444 Rhea reactions, 50,016 USPTO reactions and 1,065,119 atom-mapped reactions retained
- all twelve registered invalid-property responses, low-confidence rows, conflicts, unavailable records and resource halts retained
- all 8 separately registered claim targets retained
- external formats, algorithms, scores and outcomes never select the Fold-native survivor

Falsification boundary: COMP-002 halts if its survivor is nonunique; a registered target, source route, graph, atom, bond, orientation, mapping, reaction row, provenance record, database conflict, adverse result or resource halt is omitted; the implementation-distinct reconstruction disagrees; a dependency is absent; or external data selects the native law.

Measurement receipt: sha256:51da66628e8196b4a63fe2dee126a2e15c1c0b855f1d859a343ba1bafd873b2b. Isolation certificate: sha256:02886fe995caec9d6a41d4b6cb5c96ce8aca7cc82992bb8e3e385f6af6aae4d0. Custody certificate: sha256:cf30a439f3861f8df2dd566be0fa8e8e7aa8486e9dc59e3c2e9555687dc5f963.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external database encoding algorithm score value equation model fit outcome or software library selects the survivor
- all favorable adverse absent unavailable unresolved low-confidence transport conflict and declared resource-halt rows remain

Immutable identities. Source manifest

sha256:6e10609c99f999df9326971ca982c8f74e4ae88ae513bb2e6783ba9468e5d5d1; derivation seal sha256:9302815cf77a2facb4f5b758aa0303bcc7ee3c7e0d2dcfeb67a1ae33b6008457; engine receipt sha256:e1895597e606243629700724394c81c68ad21c8f4c28595236293cf3267c2fc3 at receipts/engine/model_admitted/SFT-CHEM-MOLECULAR-GRAPH-ISOMORPHISM-002-e1895597e6062436.json; empirical hash sha256:a41046addaa974b151a5aaea00a551492b0f32b041bb37c921e8ec572ae75168; measurement receipt sha256:51da66628e8196b4a63fe2dee126a2e15c1c0b855f1d859a343ba1bafd873b2b.

239. Fold chemical substructure law

Claim identity: SFT-CHEM-CHEMICAL-SUBSTRUCTURE-RELATION-003

Question and theorem. A chemical graph is a substructure of another exactly when at least one generated injective carrier map preserves every registered query atom, bond and held-state distinction.

held-query-graph__held-target-graph__injective-carrier-map__registered-atom-label-preservation__exact-bond-incidence-preservation__requested-held-fibres-preserved__all-embedding-census__structural-absence-with-falsification-trace

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-COMP-ALG-TREES-GRAPHS-001

Generated grammar. Generate the literal product of the eight frozen COMP-003 chemical-information custody decisions.

Boundary: Eight binary dimensions exhaust the registered COMP-003 identity, exact-support, equivalence, provenance, falsification and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-query-graph__held-target-graph__injective-carrier-map__registered-atom-label-preservation__exact-bond-incidence-preservation__requested-held-fibres-preserved__all-embedding-census__structural-absence-with-falsification-trace. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
query	held-query-graph	text-fragment-query	text-fragment-query closes a distinction required by query.
target	held-target-graph	database-hit-label	database-hit-label closes a distinction required by target.
mapping	injective-carrier-map	noninjective-match	noninjective-match closes a distinction required by mapping.
atoms	registered-atom-label-preservation	wildcard-unregistered-atoms	wildcard-unregistered-atoms closes a distinction required by atoms.
bonds	exact-bond-incidence-preservation	path-length-only-match	path-length-only-match closes a distinction required by bonds.
orientation	requested-held-fibres-preserved	stereo-charge-isotope-ignored	stereo-charge-isotope-ignored closes a distinction required by orientation.
enumeration	all-embedding-census	first-hit-search	first-hit-search closes a distinction required by enumeration.
adverse	structural-absence-with-falsification-trace	failed-query-silently-empty	failed-query-silently-empty closes a distinction required by adverse.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:0dbe9ec9af8a5636bad9ae850427f5440cdfd3addba7474227ffaf585974a082.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:f5eabbe43dca4bbdf7fa82e9ba7bee6b86db0ea7a37ab32b8259eae03c348b93.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:663017ea9d05617eacc47beffd9dbfc1b2d61c1c0b39cfead8bf7ab9ce61a91e.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:a97943b0749ef1561b8d1992800aefd2198f23ec3b5aeb68ce37b2a4c9e2b740.

Independent reconstruction. Implementation hash

sha256:5d26037a4c716ad745a18b20adb0abbfe618c5fe2149254434523bcf2e93a59d; independent certificate sha256:7d24516de3b1802ff30cd66a75c1d610eca3ae6d1582eecd5c90cc308b1c2f3e; external-validation hash sha256:777c2b395a79e5ff38305dfbd2d8ea549d8dbc997e5399b0bc4debe04f75e131. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- PUBCHEM-PUG-REST-AUTHORITY
- PUBCHEM-C3H8O-FORMULA-CENSUS
- PUBCHEM-ASPIRIN-CONFORMERS
- PUBCHEM-ASPIRIN-SUBSTRUCTURE
- PUBCHEM-ASPIRIN-SIMILARITY
- CHEBI-REST-AUTHORITY
- RHEA-IMMUTABLE-REACTION-SMILES
- USPTO-IMMUTABLE-REACTION-SURFACE

Comparison records:

- complete 59-artifact 444,644,830-byte post-seal surface retained

- 12 PubChem records, four ChEBI cross-source records, 36,444 Rhea reactions, 50,016 USPTO reactions and 1,065,119 atom-mapped reactions retained
- all twelve registered invalid-property responses, low-confidence rows, conflicts, unavailable records and resource halts retained
- all 8 separately registered claim targets retained
- external formats, algorithms, scores and outcomes never select the Fold-native survivor

Falsification boundary: COMP-003 halts if its survivor is nonunique; a registered target, source route, graph, atom, bond, orientation, mapping, reaction row, provenance record, database conflict, adverse result or resource halt is omitted; the implementation-distinct reconstruction disagrees; a dependency is absent; or external data selects the native law.

Measurement receipt: sha256:4f97025ce0b7b6526c789dcc70c2b260f9f8d24eff475d05a79f2208847bc2e8.

Isolation certificate: sha256:ed3816ee85763d5663964a2172bee80429258d6c99645853e95ac2727f161e8a.

Custody certificate: sha256:88c5ff1aeae3eda7e2a2a2f98bd37797fdec0d1d56c916b5cc2871c90f79bd44.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external database encoding algorithm score value equation model fit outcome or software library selects the survivor
- all favorable adverse absent unavailable unresolved low-confidence transport conflict and declared resource-halt rows remain

Immutable identities. Source manifest

sha256:2dfc17753df435772d7e513cf8e651d88a2f2ecf67ef58f9a91cb51974b01e2f; derivation seal sha256:02c39105c13b05357a35d8b7186c2091282ff0c59c55b95b57da2601aa06da0e; engine receipt sha256:81874d352c9c124722c3902bff20deb33032b47f643d7bb26065a2b23e6638f1 at receipts/engine/model_admitted/SFT-CHEM-CHEMICAL-SUBSTRUCTURE-RELATION-003-81874d352c9c1247.json; empirical hash sha256:80b4fbc320d0ecefaf0ed46ee9b16f3476437568e5ece0589995acbc7ac11cb36; measurement receipt sha256:4f97025ce0b7b6526c789dcc70c2b260f9f8d24eff475d05a79f2208847bc2e8.

240. Fold constitutional-isomer enumeration law

Claim identity: SFT-CHEM-CONSTITUTIONAL-ISOMER-ENUMERATION-004

Question and theorem. A declared finite elemental support forces every connected valence-admissible bond-incidence graph, quotiented only by the exact molecular identity law.

held-element-multiplicity-support__complete-generated-incidence-support__one-connected-molecular-carrier__admitted-element-capacity-bound__canonical-graph-equivalence-quotient__complete-bounded-candidate-census__every-member-and-rejection-witness__new-support-appends-new-census

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001

- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-BOND-ORDER-001

Generated grammar. Generate the literal product of the eight frozen COMP-004 chemical-information custody decisions.

Boundary: Eight binary dimensions exhaust the registered COMP-004 identity, exact-support, equivalence, provenance, falsification and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-element-multiplicity-support__complete-generated-incidence-support__one-connected-molecular-carrier__admitted-element-capacity-bound__canonical-graph-equivalence-quotient__complete-bounded-candidate-census__every-member-and-rejection-witness__new-support-appends-new-census. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
composition	held-element-multiplicity-support	unheld-formula-total	unheld-formula-total closes a distinction required by composition.
edges	complete-generated-incidence-support	catalogued-bond-patterns	catalogued-bond-patterns closes a distinction required by edges.
connectedness	one-connected-molecular-carrier	disconnected-fragment-admission	disconnected-fragment-admission closes a distinction required by connectedness.
valence	admitted-element-capacity-bound	unbounded-edge-placement	unbounded-edge-placement closes a distinction required by valence.
identity	canonical-graph-equivalence-quotient	drawing-name-identity	drawing-name-identity closes a distinction required by identity.
enumeration	complete-bounded-candidate-census	known-isomer-list	known-isomer-list closes a distinction required by enumeration.
certificate	every-member-and-rejection-witness	count-without-members	count-without-members closes a distinction required by certificate.
extension	new-support-appends-new-census	retroactive-recount	retroactive-recount closes a distinction required by extension.

Adverse controls. All mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:1ceab303df9f655eae637e59dcc1d74e1a20288c9e56c252d82353d907bd4b2c.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:0ce0a4e884bb6a0f088b0d132c89f10b06b53a451d2c69dada78e36feebee7c5.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:74980bd3360daf2a03ba8aab2950ec4b1a09dd134ca92eed4793b1a2dbf9bebd.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:500025714943f9b88387ab2ec31899a5b5227b002345a41b627d90e088c1cc9f.

Independent reconstruction. Implementation hash

sha256:8bc60ae6b321cc48913ead24996beb94becfe20e842cbaace023497821ee8865; independent certificate sha256:c15e1c5e477a36f2732743fa4acdbef6047ca5fc5bd7578fa8f35c6c4a1c6a91; external-validation hash sha256:e8366b0d1da07215f29e70372457ed96e9fd9d8c61818fef9961091632b035b6. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- PUBCHEM-PUG-REST-AUTHORITY
- PUBCHEM-C3H8O-FORMULA-CENSUS
- PUBCHEM-ASPIRIN-CONFORMERS
- PUBCHEM-ASPIRIN-SUBSTRUCTURE
- PUBCHEM-ASPIRIN-SIMILARITY
- CHEBI-REST-AUTHORITY
- RHEA-IMMUTABLE-REACTION-SMILES
- USPTO-IMMUTABLE-REACTION-SURFACE

Comparison records:

- complete 59-artifact 444,644,830-byte post-seal surface retained
- 12 PubChem records, four ChEBI cross-source records, 36,444 Rhea reactions, 50,016 USPTO reactions and 1,065,119 atom-mapped reactions retained
- all twelve registered invalid-property responses, low-confidence rows, conflicts, unavailable records and resource halts retained
- all 8 separately registered claim targets retained
- external formats, algorithms, scores and outcomes never select the Fold-native survivor

Falsification boundary: COMP-004 halts if its survivor is nonunique; a registered target, source route, graph, atom, bond, orientation, mapping, reaction row, provenance record, database conflict, adverse result or resource halt is omitted; the implementation-distinct reconstruction disagrees; a dependency is absent; or external data selects the native law.

Measurement receipt: sha256:825cac40ad2dd6075ce06056887c1c9ec09b6027d073130ce159ae46b6ebaaef.

Isolation certificate: sha256:d488d98fa8c8b5b8f0e26cc96d7d8e5fb1f53962128282a379b25cdee0d46f42.

Custody certificate: sha256:953c8ead94af28dc9622d40e613dd66e0b9c55902a4f2749ccf2e5158fddb701.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external database encoding algorithm score value equation model fit outcome or software library selects the survivor
- all favorable adverse absent unavailable unresolved low-confidence transport conflict and declared resource-halt rows remain

Immutable identities. Source manifest

sha256:53d6d7e0c7d541ec3172f52d3aacfd9bc1e3b682f0ee309aaa2a85bf339e1532; derivation seal

sha256:5f273f254fc2e647b21e98638b9427ebcd52f72eef643cd8bf36d93d5f3d8c00; engine receipt

sha256:9c7ef55ae880cd0da55070786eb517262dc6d84da61d57180eb92088e7395705 at receipts/engine/model_admitted/SFT-CHEM-CONSTITUTIONAL-ISOMER-ENUMERATION-004-9c7ef55ae880cd0d.json;

empirical hash sha256:61b1557dc90da0fb1b3b486afd4fa251603b7a24d883069124b43c70d68585f5;

measurement receipt sha256:825cac40ad2dd6075ce06056887c1c9ec09b6027d073130ce159ae46b6ebaaef.

241. Fold stereoisomer enumeration law

Claim identity: SFT-CHEM-STEREOMER-ENUMERATION-005

Question and theorem. A fixed constitutional graph and its forced held orientation sites generate every fibre assignment, with graph automorphisms removing only exact duplicates.

fixed-canonical-constitutional-graph__forced-held-orientation-sites__two-fibre-held-orientation-labels__complete-generated-assignment-support__exact-automorphism-quotient__orientation-preserving-graph-identity__member-assignment-and-equivalence-certificate__single-fibre-tamper-rejected

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-CHEM-STEREO-DIASTEREOMER-001

Generated grammar. Generate the literal product of the eight frozen COMP-005 chemical-information custody decisions.

Boundary: Eight binary dimensions exhaust the registered COMP-005 identity, exact-support, equivalence, provenance, falsification and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *fixed-canonical-constitutional-graph__forced-held-orientation-sites__two-fibre-held-orientation-labels__complete-generated-assignment-support__exact-automorphism-quotient__orientation-preserving-graph-identity__member-assignment-and-equivalence-certificate__single-fibre-tamper-rejected*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
constitution	fixed-canonical-constitutional-graph	formula-without-graph	formula-without-graph closes a distinction required by constitution.
sites	forced-held-orientation-sites	assumed-stereocentre-count	assumed-stereocentre-count closes a distinction required by sites.
labels	two-fibre-held-orientation-labels	signed-coordinate-stereo	signed-coordinate-stereo closes a distinction required by labels.
support	complete-generated-assignment-support	unconditional-power-of-two-count	unconditional-power-of-two-count closes a distinction required by support.
symmetry	exact-automorphism-quotient	manual-meso-exception	manual-meso-exception closes a distinction required by symmetry.
identity	orientation-preserving-graph-identity	name-or-drawing-comparison	name-or-drawing-comparison closes a distinction required by identity.
certificate	member-assignment-and-equivalence-certificate	count-only-output	count-only-output closes a distinction required by certificate.
adverse	single-fibre-tamper-rejected	orientation-erasure	orientation-erasure closes a distinction required by adverse.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:54507eaf98c2537285c5c907d5b2caa3465ebbb3fbedf51a3611b0117ce34193.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:7c7877f5c1d6ade56bdc39215e9a3fcfc3b6579aa9757a0a1c553a449ed2df1f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b414c399c9810e6ce7712b600d930cbdc83cad7eaaf17f0b5374fa1c84f948b3.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:71f1ee69b6a8aeaae82c3900ff3bdfb877bc1c263a24ac4c638bc247647abcfa.

Independent reconstruction. Implementation hash

sha256:fafedcdf57c297d8656cb8976b4f3ddfc21ceec7cfa231703656d1831f4cb20c; independent certificate sha256:da97297dfb7c34bbc4c5c8aefa193f1ff70977d0421167a06df5b787b18fde38; external-validation hash sha256:eb6b6cbe346c606585fbd9fa57c8a7b76645cb4d8448dbaffb0369c738fe529c. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- PUBCHEM-PUG-REST-AUTHORITY
- PUBCHEM-C3H8O-FORMULA-CENSUS
- PUBCHEM-ASPIRIN-CONFORMERS
- PUBCHEM-ASPIRIN-SUBSTRUCTURE
- PUBCHEM-ASPIRIN-SIMILARITY
- CHEBI-REST-AUTHORITY
- RHEA-IMMUTABLE-REACTION-SMILES
- USPTO-IMMUTABLE-REACTION-SURFACE

Comparison records:

- complete 59-artifact 444,644,830-byte post-seal surface retained
- 12 PubChem records, four ChEBI cross-source records, 36,444 Rhea reactions, 50,016 USPTO reactions and 1,065,119 atom-mapped reactions retained
- all twelve registered invalid-property responses, low-confidence rows, conflicts, unavailable records and resource halts retained
- all 8 separately registered claim targets retained
- external formats, algorithms, scores and outcomes never select the Fold-native survivor

Falsification boundary: COMP-005 halts if its survivor is nonunique; a registered target, source route, graph, atom, bond, orientation, mapping, reaction row, provenance record, database conflict, adverse result or resource halt is omitted; the implementation-distinct reconstruction disagrees; a dependency is absent; or external data selects the native law.

Measurement receipt: sha256:e95440c3406fb98779df519d8b3571a6fe70e8f6aed38769a859ea832c7bd1fc.

Isolation certificate: sha256:31fe663df5932e8635a3d1d915fa90a5d746dcde410aab804a87fe6a8d1455ac.

Custody certificate: sha256:0d2ff4df1bcd1e4c5419520fa0314411d32bd491b89a5eb3b0e5bb80799708ed.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external database encoding algorithm score value equation model fit outcome or software library selects the survivor
- all favorable adverse absent unavailable unresolved low-confidence transport conflict and declared resource-halt rows remain

Immutable identities. Source manifest

sha256:01f4be099a171a1be9067f80aa5f7e7a363f279a9285514fb428ff793ef35cdb; derivation seal
 sha256:d3fa5abc754a0db2d220ee1e48d11a610c251359bd6946f6e8690634dd3f5652; engine receipt
 sha256:749cbd4b1f5b0ef10a5af2ad36f7f4751ad217458b80c01ad6e2eda40f6340ab at receipts/engine/model_admitted/SFT-CHEM-STEREOMER-ENUMERATION-005-749cbd4b1f5b0ef1.json; empirical hash
 sha256:af8aba5d5e4952edc724546c7fc2a5efc11b4beed05373e261e7864455d2b38b; measurement receipt
 sha256:e95440c3406fb98779df519d8b3571a6fe70e8f6aed38769a859ea832c7bd1fc.

242. Fold conformer enumeration law

Claim identity: SFT-CHEM-CONFORMER-ENUMERATION-006

Question and theorem. A held molecular graph and its admitted rotatable bonds force complete finite torsion-fibre words at the declared resolution, quotiented by exact graph symmetry.

held-canonical-molecular-graph__forced-rotatable-bond-support__declared-finite-torsion-fibre-resolution__complete-torsion-word-support__exact-graph-symmetry-quotient__admitted-state-order-consumer-only__complete-member-and-equivalence-certificate__explicit-graph-and-resolution-boundary

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-CONFORMER-GENERATION-EQUIVALENCE-005
- SFT-CHEM-DIHEDRAL-TORSIONAL-STATE-004

Generated grammar. Generate the literal product of the eight frozen COMP-006 chemical-information custody decisions.

Boundary: Eight binary dimensions exhaust the registered COMP-006 identity, exact-support, equivalence, provenance, falsification and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-canonical-molecular-graph__forced-rotatable-bond-support__declared-finite-torsion-fibre-resolution__complete-torsion-word-support__exact-graph-symmetry-quotient__admitted-state-order-consumer-only__complete-member-and-equivalence-certificate__explicit-graph-and-resolution-boundary. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	held-canonical-molecular-graph	formula-only-conformer	formula-only-conformer closes a distinction required by carrier.
rotors	forced-rotatable-bond-support	all-single-bonds-assumed-rotatable	all-single-bonds-assumed-rotatable closes a distinction required by rotors.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
resolution	declared-finite-torsion-fibre-resolution	continuum-angle-premise	continuum-angle-premise closes a distinction required by resolution.
generation	complete-torsion-word-support	sampled-random-conformers	sampled-random-conformers closes a distinction required by generation.
symmetry	exact-graph-symmetry-quotient	duplicate-conformers-retained	duplicate-conformers-retained closes a distinction required by symmetry.
energy	admitted-state-order-consumer-only	fitted-force-field-selection	fitted-force-field-selection closes a distinction required by energy.
certificate	complete-member-and-equivalence-certificate	lowest-only-output	lowest-only-output closes a distinction required by certificate.
boundary	explicit-graph-and-resolution-boundary	unbounded-conformer-claim	unbounded-conformer-claim closes a distinction required by boundary.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 913738491d622822d226b5390e203d59652ac9f756aaee40a672c04c02eb88fd.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: e995fd6fc772129b28e29b2f1d167f4ee0cb252560cb6a3f86b603fd9ae576a3.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: da1c81dcfa96a5e3a6206b1385db3c8eb767f6e4d406c79276d4cad5fd59a758.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 88fbf55db67b4ca77705dc85b29fc405d47da37691c02ad4690bee3d5c881812.

Independent reconstruction. Implementation hash

sha256: 8997092f881a5f0113efb62fe3670dd9713ca20cc488782f9191ce68cc94887c; independent certificate sha256: daf8e291d1a2c7fe8973e0fd0ef3d133baff75b4a6cf4ff0e1aa876aa17565f0; external-validation hash sha256: d6ab6ebb10a24c7b39f82ac0e9961bdd270e80c3098b27ac4dac32731776b868. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- PUBCHEM-PUG-REST-AUTHORITY
- PUBCHEM-C3H8O-FORMULA-CENSUS
- PUBCHEM-ASPIRIN-CONFORMERS
- PUBCHEM-ASPIRIN-SUBSTRUCTURE
- PUBCHEM-ASPIRIN-SIMILARITY
- CHEBI-REST-AUTHORITY
- RHEA-IMMUTABLE-REACTION-SMILES
- USPTO-IMMUTABLE-REACTION-SURFACE

Comparison records:

- complete 59-artifact 444,644,830-byte post-seal surface retained
- 12 PubChem records, four ChEBI cross-source records, 36,444 Rhea reactions, 50,016 USPTO reactions and 1,065,119 atom-mapped reactions retained

- all twelve registered invalid-property responses, low-confidence rows, conflicts, unavailable records and resource halts retained
- all 8 separately registered claim targets retained
- external formats, algorithms, scores and outcomes never select the Fold-native survivor

Falsification boundary: COMP-006 halts if its survivor is nonunique; a registered target, source route, graph, atom, bond, orientation, mapping, reaction row, provenance record, database conflict, adverse result or resource halt is omitted; the implementation-distinct reconstruction disagrees; a dependency is absent; or external data selects the native law.

Measurement receipt: sha256:035a9c972eec69e2605055bd2903aeae2ead7b9981feb95dd8920f63d299339f.
Isolation certificate: sha256:7220a7a605b568ce6193867631c6c396b6c49ab3e5745897af91b9981a5fb6cc.
Custody certificate: sha256:8a9fd17d77117e632a49a2d5fe15762f12c0d434b8ead3941935f0fd37951941.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external database encoding algorithm score value equation model fit outcome or software library selects the survivor
- all favorable adverse absent unavailable unresolved low-confidence transport conflict and declared resource-halt rows remain

Immutable identities. Source manifest

sha256:9d2be8c2c5738afd36d0c1911d3cdcfcdfb9acbeceb9fa880e23a2a214ff746d; derivation seal
sha256:cf389026b1b26fe8f8e85eeb6056581a1f8f85f65b2933c317451deab28961ef; engine receipt
sha256:3a8960886113dbba8b92ba068a31a01faa620f7e1fa3f66b125dd30a0aa9644b at receipts/engine/model_admitted/SFT-CHEM-CONFORMER-ENUMERATION-006-3a8960886113dbba.json; empirical hash
sha256:dd5253ff7ffc90928ba0bb2e1629a7aa63e7f40736ca557fd71b75d64a73a75c; measurement receipt
sha256:035a9c972eec69e2605055bd2903aeae2ead7b9981feb95dd8920f63d299339f.

243. Fold reaction-graph generation law

Claim identity: SFT-CHEM-REACTION-GRAPH-GENERATION-007

Question and theorem. Held reactant graphs force the complete finite support of identity-conserving product graphs reachable by registered chemical transitions.

complete-held-reactant-graphs__registered-identity-conserving-graph-transitions__complete-carrier-conservation__complete-bond-change-trace__all-reachable-product-support__canonical-product-graph-quotient__all-products-absence-and-rejection-custody__declared-depth-and-transition-resource

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001

Generated grammar. Generate the literal product of the eight frozen COMP-007 chemical-information custody decisions.

Boundary: Eight binary dimensions exhaust the registered COMP-007 identity, exact-support, equivalence, provenance, falsification and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-held-reactant-graphs__registered-identity-conserving-graph-transitions__complete-carrier-conservation__complete-bond-change-trace__all-reachable-product-support__canonical-product-graph-quotient__all-products-absence-and-rejection-custody__declared-depth-and-transition-resource`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
source	complete-held-reactant-graphs	reaction-name-without-reactants	reaction-name-without-reactants closes a distinction required by source.
moves	registered-identity-conserving-graph-transitions	imported-template-library	imported-template-library closes a distinction required by moves.
carriers	complete-carrier-conservation	untracked-atom-creation-loss	untracked-atom-creation-loss closes a distinction required by carriers.
bonds	complete-bond-change-trace	product-only-connectivity	product-only-connectivity closes a distinction required by bonds.
generation	all-reachable-product-support	major-product-search	major-product-search closes a distinction required by generation.
identity	canonical-product-graph-quotient	string-product-deduplication	string-product-deduplication closes a distinction required by identity.
custody	all-products-absence-and-rejection-custody	favorable-products-only	favorable-products-only closes a distinction required by custody.
resource	declared-depth-and-transition-resource	unbounded-reaction-claim	unbounded-reaction-claim closes a distinction required by resource.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 826c9825fba8b8c43b529c11e40877b0212c95f0f7c264e512fbc49aa084cf5e.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 2de28837bafc977988cad6f66252ccdd869fe22b09803864eb54ba7d12ed534a.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 551f668ad28ac2d044cd190b8bf0c314864b1d03c6d329ec51fe87c3264a994f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: eedb345b06ca87d92828517c214769bc5f4c8322fb6b5369c66c7e6eaa9fd84.

Independent reconstruction. Implementation hash

sha256: da3e7caa9192344c4bb0ffefa3b31503b0c9ddb531623d3d80602d4b8f010ac3; independent certificate sha256: 5dab10cc3c6a834b6e336b9c3e074f9e465dca60b14b045f397b5886494f72d7; external-validation hash sha256: 4b039b37ca4dc3e014573c1f3a5000ce6245a5221e0405cdaaa040d4375671aa. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- PUBCHEM-PUG-REST-AUTHORITY
- PUBCHEM-C3H8O-FORMULA-CENSUS
- PUBCHEM-ASPIRIN-CONFORMERS
- PUBCHEM-ASPIRIN-SUBSTRUCTURE
- PUBCHEM-ASPIRIN-SIMILARITY
- CHEBI-REST-AUTHORITY
- RHEA-IMMUTABLE-REACTION-SMILES
- USPTO-IMMUTABLE-REACTION-SURFACE

Comparison records:

- complete 59-artifact 444,644,830-byte post-seal surface retained
- 12 PubChem records, four ChEBI cross-source records, 36,444 Rhea reactions, 50,016 USPTO reactions and 1,065,119 atom-mapped reactions retained
- all twelve registered invalid-property responses, low-confidence rows, conflicts, unavailable records and resource halts retained
- all 8 separately registered claim targets retained
- external formats, algorithms, scores and outcomes never select the Fold-native survivor

Falsification boundary: COMP-007 halts if its survivor is nonunique; a registered target, source route, graph, atom, bond, orientation, mapping, reaction row, provenance record, database conflict, adverse result or resource halt is omitted; the implementation-distinct reconstruction disagrees; a dependency is absent; or external data selects the native law.

Measurement receipt: sha256:e7ccffca3e9e45beb9fb28981fb54a431a7c631cf99b7f985703775adc776306.

Isolation certificate: sha256:ae383fe429569a4df1ba40eb1b26066745d891bde837fe8f1294faa55eb69fd0.

Custody certificate: sha256:fb7d6c5ec39d923ea1a186a644aa0d70753db50a352c5fcbecbf9fc2a7cf254.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external database encoding algorithm score value equation model fit outcome or software library selects the survivor
- all favorable adverse absent unavailable unresolved low-confidence transport conflict and declared resource-halt rows remain

Immutable identities. Source manifest

sha256:16079312b3a5238a3ecb0f5355fabbf07380d74ee194277dda11c99295b80dc6; derivation seal

sha256:a01b225ed56f4a89bc5d5286c8bc99408258caf57985aeb50274c61ebdbd4f5; engine receipt

sha256:0a51b0dedcd06539bd1e935aa395bc2a4443cc15a4c8af99d033c79c7be20b0b at receipts/engine/model_admitted/SFT-CHEM-REACTION-GRAPH-GENERATION-007-0a51b0dedcd06539.json; empirical

hash sha256:6e62930a5bb4dc13811917b6b7cad5730386de6433b5c34d379d61404debe01d; measurement

receipt sha256:e7ccffca3e9e45beb9fb28981fb54a431a7c631cf99b7f985703775adc776306.

244. Fold atom-mapping and reaction-balance law

Claim identity: SFT-CHEM-ATOM-MAPPING-REACTION-BALANCE-008

Question and theorem. A lawful reaction requires a complete source-to-product carrier bijection and exact elemental, isotope and charge-hand custody across every component.

held-reactant-atom-support__held-product-atom-support__complete-source-product-carrier-bijection__element-by-element-conservation__isotope-carrier-conservation__held-charge-hand-and-magnitude-custody__all-reactant-product-components-retained__mapping-inventory-and-change-certificate

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-RXN-MECHANISM-001

Generated grammar. Generate the literal product of the eight frozen COMP-008 chemical-information custody decisions.

Boundary: Eight binary dimensions exhaust the registered COMP-008 identity, exact-support, equivalence, provenance, falsification and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *held-reactant-atom-support__held-product-atom-support__complete-source-product-carrier-bijection__element-by-element-conservation__isotope-carrier-conservation__held-charge-hand-and-magnitude-custody__all-reactant-product-components-retained__mapping-inventory-and-change-certificate*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
sources	held-reactant-atom-support	unmapped-reactant-string	unmapped-reactant-string closes a distinction required by sources.
products	held-product-atom-support	unmapped-product-string	unmapped-product-string closes a distinction required by products.
mapping	complete-source-product-carrier-bijection	partial-atom-map	partial-atom-map closes a distinction required by mapping.
elements	element-by-element-conservation	mass-total-only-balance	mass-total-only-balance closes a distinction required by elements.
isotopes	isotope-carrier-conservation	isotope-label-erasure	isotope-label-erasure closes a distinction required by isotopes.
charge	held-charge-hand-and-magnitude-custody	signed-net-only-balance	signed-net-only-balance closes a distinction required by charge.
components	all-reactant-product-components-retained	selected-main-component	selected-main-component closes a distinction required by components.
certificate	mapping-inventory-and-change-certificate	balanced-boolean-only	balanced-boolean-only closes a distinction required by certificate.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256 : c9cb42c7f4392a16016e68bf91d5b25d78ce46d294bceeb1cf5ecfe36ad4f961.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 969b1dea18a47120ed78fe72f2c1ab3f2242cbb8c3eb4640d2d09f00da78aac1.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256 : 872ca02f333f0ca1de1a59a6dfdda760554d783c90d76fffacd7bdd4720d5031.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256 : 9087c1e30f7c986803642145fb2537be7c93680a7f1ee81db87d64efa96fc393.

Independent reconstruction. Implementation hash

sha256 : bd0c30a5d875fa5cf51d6e3c5574f6827c32f9e567c27abe62cf0bd94f21a458; independent certificate sha256 : 1d4c7f4fba58ee3f0cc5409feb048dc00bac59682f39f305e7ff82d8efa29fba; external-validation hash sha256 : 311aa26ff99275ddc3f03ffb560953811a663e073db4a6bc96aabea047898ae7. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- PUBCHEM-PUG-REST-AUTHORITY
- PUBCHEM-C3H8O-FORMULA-CENSUS
- PUBCHEM-ASPIRIN-CONFORMERS
- PUBCHEM-ASPIRIN-SUBSTRUCTURE
- PUBCHEM-ASPIRIN-SIMILARITY
- CHEBI-REST-AUTHORITY
- RHEA-IMMUTABLE-REACTION-SMILES
- USPTO-IMMUTABLE-REACTION-SURFACE

Comparison records:

- complete 59-artifact 444,644,830-byte post-seal surface retained
- 12 PubChem records, four ChEBI cross-source records, 36,444 Rhea reactions, 50,016 USPTO reactions and 1,065,119 atom-mapped reactions retained
- all twelve registered invalid-property responses, low-confidence rows, conflicts, unavailable records and resource halts retained
- all 8 separately registered claim targets retained
- external formats, algorithms, scores and outcomes never select the Fold-native survivor

Falsification boundary: COMP-008 halts if its survivor is nonunique; a registered target, source route, graph, atom, bond, orientation, mapping, reaction row, provenance record, database conflict, adverse result or resource halt is omitted; the implementation-distinct reconstruction disagrees; a dependency is absent; or external data selects the native law.

Measurement receipt: sha256 : d04bc95936f4ab6e55e991522c7f279f05516b3ca3f7e2325a59d3f7d438fe91.

Isolation certificate: sha256 : f146ef386e770ab74449e95c7ec72cc7ad1ef9ff580a27ff55e23807471b0233.

Custody certificate: sha256 : e5fc95beed8ad92ad64763637c8c4c442b54485888357865a5a3a2b1c76eb77f.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external database encoding algorithm score value equation model fit outcome or software library selects the survivor
- all favorable adverse absent unavailable unresolved low-confidence transport conflict and declared resource-halt rows remain

Immutable identities. Source manifest

sha256:36073696d93fa005248cebc26cd43c71a87c3df9afaa5ff22b117ab719992601; derivation seal sha256:895ced194111f069bb4ed855bdf8b5c2cc00f8aee03d4edea12d48bac93f8172; engine receipt sha256:377ac33f481e02a143540d265734e67549444095bb986f0577191b755fd46fc7 at receipts/engine/model_admitted/SFT-CHEM-ATOM-MAPPING-REACTION-BALANCE-008-377ac33f481e02a1.json; empirical hash sha256:d7efa29d32f40c35cd1de4039ffc8f859a8e226c48dcef51d4d898c0781fe9f0; measurement receipt sha256:d04bc95936f4ab6e55e991522c7f279f05516b3ca3f7e2325a59d3f7d438fe91.

245. Fold mechanism-search and proof-trace law

Claim identity: SFT-CHEM-MECHANISM-SEARCH-PROOF-TRACE-009

Question and theorem. Mechanism search enumerates every simple path in the admitted chemical transition graph; only paths whose every step has an admitted law and complete carrier trace may survive.

held-complete-chemical-states__admitted-chemical-transition-edge__complete-simple-path-enumeration__stepwise-carrier-custody__all-parallel-route-support__declared-cycle-and-resource-boundary__proof-kernel-stepwise-acceptance__premise-alternative-step-and-terminal-trace

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007
- SFT-CHEM-PARALLEL-MECHANISM-COMPOSITION-008

Generated grammar. Generate the literal product of the eight frozen COMP-009 chemical-information custody decisions.

Boundary: Eight binary dimensions exhaust the registered COMP-009 identity, exact-support, equivalence, provenance, falsification and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *held-complete-chemical-states__admitted-chemical-transition-edge__complete-simple-path-enumeration__stepwise-carrier-custody__all-parallel-route-support__declared-cycle-and-resource-boundary__proof-kernel-stepwise-acceptance__premise-alternative-step-and-terminal-trace*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
states	held-complete-chemical-states	unidentified-intermediate-nodes	unidentified-intermediate-nodes closes a distinction required by states.
transitions	admitted-chemical-transition-edge	learned-or-named-template-edge	learned-or-named-template-edge closes a distinction required by transitions.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
search	complete-simple-path-enumeration	first-successful-path	first-successful-path closes a distinction required by search.
carriers	stepwise-carrier-custody	net-reaction-only	net-reaction-only closes a distinction required by carriers.
branching	all-parallel-route-support	major-route-only	major-route-only closes a distinction required by branching.
cycles	declared-cycle-and-resource-boundary	silent-loop-truncation	silent-loop-truncation closes a distinction required by cycles.
acceptance	proof-kernel-stepwise-acceptance	score-selected-mechanism	score-selected-mechanism closes a distinction required by acceptance.
certificate	premise-alternative-step-and-terminal-trace	terminal-product-only	terminal-product-only closes a distinction required by certificate.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:dbc7af921d88dfa1c654046381fed4aa6f22f21b3ebf9d02cdc9722fd9fd27f9.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:bf862cdf4a266d22f6f44df39fb0c3842f4788ee29f6ccf1860f910485d697eb.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:6ea704b3155292e2cd7e82fc4340e920b788f1c5d3f806373e32b2eb1771ecdb.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:d65f675841bdb94d6d79ba62d8f7dc1e74bf20dca612b4c27bb22cf9c7308822.

Independent reconstruction. Implementation hash

sha256:354d6029d942e283ea06139b5be2ff9d861a2fd417671ed8468f5d1c0788410d; independent certificate sha256:8e2a651c5ff8c46ca5f0703d1fb4537f38c52f3cf7bef2f4cde068224538bb04; external-validation hash sha256:6d7edfdd50e3311190bf51a7f3da97d7881630a162f97bf2969eab76a41ceb0c. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- PUBCHEM-PUG-REST-AUTHORITY
- PUBCHEM-C3H8O-FORMULA-CENSUS
- PUBCHEM-ASPIRIN-CONFORMERS
- PUBCHEM-ASPIRIN-SUBSTRUCTURE
- PUBCHEM-ASPIRIN-SIMILARITY
- CHEBI-REST-AUTHORITY
- RHEA-IMMUTABLE-REACTION-SMILES
- USPTO-IMMUTABLE-REACTION-SURFACE

Comparison records:

- complete 59-artifact 444,644,830-byte post-seal surface retained
- 12 PubChem records, four ChEBI cross-source records, 36,444 Rhea reactions, 50,016 USPTO reactions and 1,065,119 atom-mapped reactions retained
- all twelve registered invalid-property responses, low-confidence rows, conflicts, unavailable records and resource halts retained

- all 8 separately registered claim targets retained
- external formats, algorithms, scores and outcomes never select the Fold-native survivor

Falsification boundary: COMP-009 halts if its survivor is nonunique; a registered target, source route, graph, atom, bond, orientation, mapping, reaction row, provenance record, database conflict, adverse result or resource halt is omitted; the implementation-distinct reconstruction disagrees; a dependency is absent; or external data selects the native law.

Measurement receipt: sha256:a1653ba13b6118da5f5db2c1e55ed253211c68e57c7246b8f972cc7985000352.

Isolation certificate: sha256:97f7d14c9172eea42099e58b8c18f42ec9371613e3082f59d74429455daae838.

Custody certificate: sha256:644974722fa70e3aecdc66fbc46381122c11a08426c1d0d60bf188693b1be561.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external database encoding algorithm score value equation model fit outcome or software library selects the survivor
- all favorable adverse absent unavailable unresolved low-confidence transport conflict and declared resource-halt rows remain

Immutable identities. Source manifest

sha256:ed8e19feb1da542e6b4bb55f329a84714df4e097f3c7df0b072a5c5587b34662; derivation seal

sha256:33f23870d833750ccce5963c4c058d1e2f5b41545a6f62f7686258d374900792; engine receipt

sha256:f9d9df0b6c21854dbda0889cd90926a3b88bf180d49dbf3677e3b3f202d75d9d at receipts/engine/model_admitted/SFT-CHEM-MECHANISM-SEARCH-PROOF-TRACE-009-f9d9df0b6c21854d.json; empirical

hash sha256:7a085110ec0f4ab573a896dbcca683262ea36c23b44a56dd5dd0dc8c73043a02; measurement

receipt sha256:a1653ba13b6118da5f5db2c1e55ed253211c68e57c7246b8f972cc7985000352.

246. Fold exact molecular similarity and distinction law

Claim identity: SFT-CHEM-EXACT-MOLECULAR-SIMILARITY-DISTINCTION-010

Question and theorem. Molecular comparison is the exact retained intersection and two held difference supports of admitted graph distinctions; no fitted weight or opaque scalar chooses similarity.

held-left-molecular-graph__held-right-molecular-graph__all-registered-exact-graph-distinctions__exact-shared-distinction-support__separate-left-and-right-difference-support__exact-isomorphism-retained-separately__componentwise-exact-partial-order__every-unshared-distinction-preserved

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-MOLECULAR-GRAPH-ISOMORPHISM-002
- SFT-CHEM-CHEMICAL-SUBSTRUCTURE-RELATION-003

Generated grammar. Generate the literal product of the eight frozen COMP-010 chemical-information custody decisions.

Boundary: Eight binary dimensions exhaust the registered COMP-010 identity, exact-support, equivalence, provenance, falsification and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `held-left-molecular-graph__held-right-molecular-graph__all-registered-exact-graph-distinctions__exact-shared-distinction-support__separate-left-and-right-difference-support__exact-isomorphism-retained-separately__componentwise-exact-partial-order__every-unshared-distinction-preserved`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
left	held-left-molecular-graph	identifier-only-left	identifier-only-left closes a distinction required by left.
right	held-right-molecular-graph	identifier-only-right	identifier-only-right closes a distinction required by right.
features	all-registered-exact-graph-distinctions	learned-descriptor-vector	learned-descriptor-vector closes a distinction required by features.
intersection	exact-shared-distinction-support	weighted-common-score	weighted-common-score closes a distinction required by intersection.
differences	separate-left-and-right-difference-support	symmetric-distance-only	symmetric-distance-only closes a distinction required by differences.
identity	exact-isomorphism-retained-separately	thresholded-same-molecule	thresholded-same-molecule closes a distinction required by identity.
ordering	componentwise-exact-partial-order	fitted-ranking-weight	fitted-ranking-weight closes a distinction required by ordering.
adverse	every-unshared-distinction-preserved	near-neighbour-conflation	near-neighbour-conflation closes a distinction required by adverse.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:ed4c5f8ad4dcaa31a646fc51aa081fc6e0ba20e35883d83365682e8804fedf77`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b931a8e327fdf3f90945ca06db713efa0c62a45fb136f78b36b0e6661bcb3864`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:54eb203b8b229aa396b79213a4f989165490fc411c3dc48f003aec7bf8fae43d`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:67b4332506c9ec011ccad8bd71b50f3f2b8af84097a9acf06adeb0ef92efc65e`.

Independent reconstruction. Implementation hash

`sha256:9ae8df792edca382aed4258b86f20b1c4053c70ca2c0497e17d337845817e23c`; independent certificate `sha256:cfb06b41f3a0b7d50da18a994829699c02cedc6388740b39c7ee5df066f72675`; external-validation hash `sha256:a64880e363f097e5fd7217917239742c6d52c4a73d9bc12f510dad5e6b7d7255`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- PUBCHEM-PUG-REST-AUTHORITY

- PUBCHEM-C3H8O-FORMULA-CENSUS
- PUBCHEM-ASPIRIN-CONFORMERS
- PUBCHEM-ASPIRIN-SUBSTRUCTURE
- PUBCHEM-ASPIRIN-SIMILARITY
- CHEBI-REST-AUTHORITY
- RHEA-IMMUTABLE-REACTION-SMILES
- USPTO-IMMUTABLE-REACTION-SURFACE

Comparison records:

- complete 59-artifact 444,644,830-byte post-seal surface retained
- 12 PubChem records, four ChEBI cross-source records, 36,444 Rhea reactions, 50,016 USPTO reactions and 1,065,119 atom-mapped reactions retained
- all twelve registered invalid-property responses, low-confidence rows, conflicts, unavailable records and resource halts retained
- all 8 separately registered claim targets retained
- external formats, algorithms, scores and outcomes never select the Fold-native survivor

Falsification boundary: COMP-010 halts if its survivor is nonunique; a registered target, source route, graph, atom, bond, orientation, mapping, reaction row, provenance record, database conflict, adverse result or resource halt is omitted; the implementation-distinct reconstruction disagrees; a dependency is absent; or external data selects the native law.

Measurement receipt: sha256:25c7ad4eac94c605788ea59f26303de9965f31e3b74ef32c25c192beef65effa.
Isolation certificate: sha256:3d6dfaa077c89bf5d3fe543398c3087bd9e2cc93d78d03f26497556f3c7fbad5.
Custody certificate: sha256:ac55d71dadd89818f6afd899277d1b5168df07075107cfd24c6097ee3f2696f2.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external database encoding algorithm score value equation model fit outcome or software library selects the survivor
- all favorable adverse absent unavailable unresolved low-confidence transport conflict and declared resource-halt rows remain

Immutable identities. Source manifest

sha256:002b6aff720e8e7f3ef81a81b14ce375e9b6d1fc87c583df5ecc2e91ea56b72b; derivation seal
sha256:3bce7844e90668556aabe698dc0d259697e180304613ddb006bb3b111e3fbaba; engine receipt
sha256:97898c7179d8b2b877d76ac80c08c81960fc0b23d454a616632cc1b67c1d3648 at receipts/engine/model_admitted/SFT-CHEM-EXACT-MOLECULAR-SIMILARITY-DISTINCTION-010-97898c7179d8b2b8.j
son; empirical hash sha256:8c34ffdafb1e10084c2e181f7455ca2d2059314bdbd4447065d41ee27475c165;
measurement receipt sha256:25c7ad4eac94c605788ea59f26303de9965f31e3b74ef32c25c192beef65effa.

247. Fold chemical database identity and provenance law

Claim identity: SFT-CHEM-DATABASE-IDENTITY-PROVENANCE-011

Question and theorem. Every external representation must bind reversibly to one held chemical carrier, its exact source identity, transformation trace, version and observation boundary.

held-canonical-chemical-carrier__exact-source-record-identity__captured-version-and-byte-identity__reversible-format-to-carrier-translation__carrier-preserving-cross-source-map__append-only-supersession-history__explicit-postseal-observation-boundary__all-conflicts-and-unavailable-records-retained

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-MULTIMODAL-MOLECULAR-IDENTITY-021

Generated grammar. Generate the literal product of the eight frozen COMP-011 chemical-information custody decisions.

Boundary: Eight binary dimensions exhaust the registered COMP-011 identity, exact-support, equivalence, provenance, falsification and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *held-canonical-chemical-carrier__exact-source-record-identity__captured-version-and-byte-identity__reversible-format-to-carrier-translation__carrier-preserving-cross-source-map__append-only-supersession-history__explicit-postseal-observation-boundary__all-conflicts-and-unavailable-records-retained*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	held-canonical-chemical-carrier	database-row-as-molecule	database-row-as-molecule closes a distinction required by carrier.
source	exact-source-record-identity	source-free-copied-value	source-free-copied-value closes a distinction required by source.
version	captured-version-and-byte-identity	mutable-current-record	mutable-current-record closes a distinction required by version.
representation	reversible-format-to-carrier-translation	format-specific-identity	format-specific-identity closes a distinction required by representation.
crosswalk	carrier-preserving-cross-source-map	name-only-cross-database-join	name-only-cross-database-join closes a distinction required by crosswalk.
history	append-only-supersession-history	latest-value-overwrite	latest-value-overwrite closes a distinction required by history.
observation	explicit-postseal-observation-boundary	derivation-database-conflation	derivation-database-conflation closes a distinction required by observation.
adverse	all-conflicts-and-unavailable-records-retained	source-disagreement-erasure	source-disagreement-erasure closes a distinction required by adverse.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9c37838e96b86d8fe2b189efe7d7ac6724410d3b5db200aa26218b0531e4e3aa`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f51e8d625230f4c6f65563e7b9b2fe81017473bf81bc16371a897cfb67555d2f`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 923a9650bc10a2c3841a1cac64b3401bfb78617b779fb59018a6d1fabb4138ab.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: ff9e9cfe889071dc16489146b2df89c990cee14414335988e76ac90e1e698b5a.

Independent reconstruction. Implementation hash

sha256: c27cbd16ca10e6cfabd3ba65791ad59310212e54b93e216892cc45886ca3ddad; independent certificate
sha256: a5ea792199f629797914fe68451998eda826573e2906c8e99707c0fb53f2b757; external-validation
hash sha256: e5d45dfaafc2667049a4c76e28e2647e5fc8a2084c302008493c6a1bd4402a61c. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- PUBCHEM-PUG-REST-AUTHORITY
- PUBCHEM-C3H8O-FORMULA-CENSUS
- PUBCHEM-ASPIRIN-CONFORMERS
- PUBCHEM-ASPIRIN-SUBSTRUCTURE
- PUBCHEM-ASPIRIN-SIMILARITY
- CHEBI-REST-AUTHORITY
- RHEA-IMMUTABLE-REACTION-SMILES
- USPTO-IMMUTABLE-REACTION-SURFACE

Comparison records:

- complete 59-artifact 444,644,830-byte post-seal surface retained
- 12 PubChem records, four ChEBI cross-source records, 36,444 Rhea reactions, 50,016 USPTO reactions and 1,065,119 atom-mapped reactions retained
- all twelve registered invalid-property responses, low-confidence rows, conflicts, unavailable records and resource halts retained
- all 8 separately registered claim targets retained
- external formats, algorithms, scores and outcomes never select the Fold-native survivor

Falsification boundary: COMP-011 halts if its survivor is nonunique; a registered target, source route, graph, atom, bond, orientation, mapping, reaction row, provenance record, database conflict, adverse result or resource halt is omitted; the implementation-distinct reconstruction disagrees; a dependency is absent; or external data selects the native law.

Measurement receipt: sha256: b5de3861d06006210c1f2eb3b7c1e5292b556e6f2845656b3c071f77659e69f7.

Isolation certificate: sha256: 9a89f336fa49126b72198101ffa18014b488f560785e7d139c2edc8cec2292c2.

Custody certificate: sha256: 3d6eaf52eb79f3179fd5fff1b84dc79acdfc57b1a022cc01e0a4260a68fa9aa5.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter

- no external database encoding algorithm score value equation model fit outcome or software library selects the survivor
- all favorable adverse absent unavailable unresolved low-confidence transport conflict and declared resource-halt rows remain

Immutable identities. Source manifest

sha256:b78d5662503fa8296a0739e9b6da4156c7cedff9a47588fbf073dd71c056c8ce; derivation seal sha256:e28c740c2db6d5a1b79c54c342a4f5b43e94e427403be9693120d03d1d1f6448; engine receipt sha256:287ab043a520733650c816cff1436f8890349b10fff721e8990a102a6c7cd40e3 at receipts/engine/model_admitted/SFT-CHEM-DATABASE-IDENTITY-PROVENANCE-011-287ab043a5207336.json; empirical hash sha256:396952a2bc5e3a5918f938b295ae8dd5950304b803d3c4800be2fc31558dd672; measurement receipt sha256:b5de3861d06006210c1f2eb3b7c1e5292b556e6f2845656b3c071f77659e69f7.

248. Fold symbolic chemical-property evaluation law

Claim identity: SFT-CHEM-SYMBOLIC-PROPERTY-EVALUATION-012

Question and theorem. A symbolic property evaluator may consume only admitted chemical carriers and laws, returning exact rational/count/held results with a complete dependency and transformation trace.

held-canonical-chemical-carrier__registered-admitted-property-law__exact-count-and-rational-arithmetic__complete-admitted-dependency-trace__held-unit-and-correspondence-boundary__exact-bound-and-status-custody__missing-distinction-halt__input-law-steps-result-certificate

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-COMP-ALG-SYMBOLIC-001
- SFT-CHEM-CROSS-PROPERTY-MOLECULAR-VECTOR-014

Generated grammar. Generate the literal product of the eight frozen COMP-012 chemical-information custody decisions.

Boundary: Eight binary dimensions exhaust the registered COMP-012 identity, exact-support, equivalence, provenance, falsification and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: held-canonical-chemical-carrier__registered-admitted-property-law__exact-count-and-rational-arithmetic__complete-admitted-dependency-trace__held-unit-and-correspondence-boundary__exact-bound-and-status-custody__missing-distinction-halt__input-law-steps-result-certificate. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
input	held-canonical-chemical-carrier	unparsed-chemical-name	unparsed-chemical-name closes a distinction required by input.
property	registered-admitted-property-law	opaque-prediction-target	opaque-prediction-target closes a distinction required by property.
arithmetic	exact-count-and-rational-arithmetic	floating-or-continuum-evaluator	floating-or-continuum-evaluator closes a distinction required by arithmetic.
dependencies	complete-admitted-dependency-trace	hidden-library-model	hidden-library-model closes a distinction required by dependencies.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
units	held-unit-and-corr espondence-boundar y	unitless-number	unitless-number closes a distinction required by units.
uncertainty	exact-bound-and-st atus-custody	discarded-source-bound	discarded-source-bound closes a distinction required by uncertainty.
unsupported	missing-distinctio n-halt	silent-extrapolation	silent-extrapolation closes a distinction required by unsupported.
certificate	input-law-steps-re sult-certificate	value-only-output	value-only-output closes a distinction required by certificate.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:52ac31597d73c5440e73e8b7b2f5dd0f2baf6768ea1b7bc22a8d70ef901d9b89.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:f32d73fbc17117f3b81b13d26df09b0ef2f83178e37e25d7aa123708a354d8a2.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:072480dc63115aa985c4b5759da00e0d6645914a9ba6927c8c480cc26c3681a3.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:59591710036d956bb7573dbc2d4da5baecfe840c2b4e5d339df33dddc57c3ce4.

Independent reconstruction. Implementation hash

sha256:4b10c5589f3491ccf9b8ad5708cd258d4cb74b1ce4734190c109c63b7b6c19bf; independent certificate
sha256:2231fef83ada7ac699eb2de6c9b86c5f7de8bbf69164834435de8bf9a3b64b3b; external-validation
hash sha256:6ea273c008c4b4ebb4caee0dc263494146e42b3d3b2df50693d37b7bd493f433. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- PUBCHEM-PUG-REST-AUTHORITY
- PUBCHEM-C3H8O-FORMULA-CENSUS
- PUBCHEM-ASPIRIN-CONFORMERS
- PUBCHEM-ASPIRIN-SUBSTRUCTURE
- PUBCHEM-ASPIRIN-SIMILARITY
- CHEBI-REST-AUTHORITY
- RHEA-IMMUTABLE-REACTION-SMILES
- USPTO-IMMUTABLE-REACTION-SURFACE

Comparison records:

- complete 59-artifact 444,644,830-byte post-seal surface retained
- 12 PubChem records, four ChEBI cross-source records, 36,444 Rhea reactions, 50,016 USPTO reactions and 1,065,119 atom-mapped reactions retained
- all twelve registered invalid-property responses, low-confidence rows, conflicts, unavailable records and resource halts retained
- all 8 separately registered claim targets retained
- external formats, algorithms, scores and outcomes never select the Fold-native survivor

Falsification boundary: COMP-012 halts if its survivor is nonunique; a registered target, source route, graph, atom, bond, orientation, mapping, reaction row, provenance record, database conflict, adverse result or resource halt is omitted; the implementation-distinct reconstruction disagrees; a dependency is absent; or external data selects the native law.

Measurement receipt: sha256 : a6675ebb641d971bc086844fedbd3ab83bd73b7b71cf518995b622c4cc5867f0.

Isolation certificate: sha256 : 714d14d514d1743bd950379e803e7d5df7e3ba11ef818cff65bf6659e9ecacd2.

Custody certificate: sha256 : eb0ad11d2299b1979ec1bb8bb0fc8b6ab5a3dfc1fdfe4bf2b27ea6ba79236dd0.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external database encoding algorithm score value equation model fit outcome or software library selects the survivor
- all favorable adverse absent unavailable unresolved low-confidence transport conflict and declared resource-halt rows remain

Immutable identities. Source manifest

sha256 : b864070ddffdc1d6d6efe7a4b78b62f9d70cf98124c5b26f7811338cae5c4c1; derivation seal

sha256 : e588e308c20e25c58236e44999c0219f63f2ee9cf131adecfaf073f8a65c312e; engine receipt

sha256 : 763febb44d5029ce32b130fce8f17669da1805dbbbeb533ff892b33e61234dd83 at receipts/engine/model_admitted/SFT-CHEM-SYMBOLIC-PROPERTY-EVALUATION-012-763febb44d5029ce.json; empirical

hash sha256 : ee5ff1c621412fda8326f3600819340e60c5747d200c9a83cc4838ddaabccbb4; measurement

receipt sha256 : a6675ebb641d971bc086844fedbd3ab83bd73b7b71cf518995b622c4cc5867f0.

249. Fold chemical prediction uncertainty and applicability law

Claim identity: SFT-CHEM-PREDICTION-UNCERTAINTY-APPLICABILITY-013

Question and theorem. A chemical prediction is admissible only on complete registered support; every missing distinction, unresolved equivalence and resource boundary is returned exactly and causes a halt outside scope.

registered-chemical-support-domain__complete-required-distinction-record__exact-missing-and-bounded-distinction-vector__support-inclusion-decision__mandatory-unsupported-state-halt__all-adverse-absent-and-unresolved-custody__new-evidence-appends-new-domain__prediction-scope-missing-set-and-trace

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-COMP-LEARN-LEARNING-LIMITS-001

Generated grammar. Generate the literal product of the eight frozen COMP-013 chemical-information custody decisions.

Boundary: Eight binary dimensions exhaust the registered COMP-013 identity, exact-support, equivalence, provenance, falsification and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `registered-chemical-support-domain__complete-required-distinction-record__exact-missing-and-bounded-distinction-vector__support-inclusion-decision__mandatory-unsupported-state-halt__all-adverse-absent-and-unresolved-custody__new-evidence-appends-new-domain__prediction-scope-missing-set-and-trace`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
domain	registered-chemical-support-domain	implicit-training-domain	implicit-training-domain closes a distinction required by domain.
input	complete-required-distinction-record	best-effort-partial-record	best-effort-partial-record closes a distinction required by input.
uncertainty	exact-missing-and-bounded-distinction-vector	single-confidence-score	single-confidence-score closes a distinction required by uncertainty.
applicability	support-inclusion-decision	similarity-threshold	similarity-threshold closes a distinction required by applicability.
extrapolation	mandatory-unsupported-state-halt	silent-out-of-domain-value	silent-out-of-domain-value closes a distinction required by extrapolation.
adverse	all-adverse-absent-and-unresolved-custody	discarded-failed-cases	discarded-failed-cases closes a distinction required by adverse.
extension	new-evidence-appends-new-domain	retroactive-domain-expansion	retroactive-domain-expansion closes a distinction required by extension.
certificate	prediction-scope-missing-set-and-trace	prediction-without-scope	prediction-without-scope closes a distinction required by certificate.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: ee88ab23209872f7d835b1f96a09de5bd35822cbbaa002fbe98c1a9c7c753b26.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 6df36f4d97e51ea8d3f4ff60ee326aeb08278d4ea5ff9d91bb6a55bb1be2132.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 2c6f8bd563b2b2ea87e78eeaf91e53743b498a333a6af9de4b021a0a17e197f.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: e35ee8ad6549a4708f4783a3adf71ed78456048acd8832cef448bf3998c07fb9.

Independent reconstruction. Implementation hash

sha256: 48e19fb7187c3749081372aa941338c9302991aa57db9a53b6feba85e8f01525; independent certificate sha256: 19c3c02e0f9e03d19366c88feb3780da57f967b694a868f4f49e2cad5f7ebbc; external-validation hash sha256: 8d38e2757112db9d05b6d20bace0013d79ce86b25e45bad748e3bc0c7cbf5c1a. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- PUBCHEM-PUG-REST-AUTHORITY
- PUBCHEM-C3H8O-FORMULA-CENSUS

- PUBCHEM-ASPIRIN-CONFORMERS
- PUBCHEM-ASPIRIN-SUBSTRUCTURE
- PUBCHEM-ASPIRIN-SIMILARITY
- CHEBI-REST-AUTHORITY
- RHEA-IMMUTABLE-REACTION-SMILES
- USPTO-IMMUTABLE-REACTION-SURFACE

Comparison records:

- complete 59-artifact 444,644,830-byte post-seal surface retained
- 12 PubChem records, four ChEBI cross-source records, 36,444 Rhea reactions, 50,016 USPTO reactions and 1,065,119 atom-mapped reactions retained
- all twelve registered invalid-property responses, low-confidence rows, conflicts, unavailable records and resource halts retained
- all 8 separately registered claim targets retained
- external formats, algorithms, scores and outcomes never select the Fold-native survivor

Falsification boundary: COMP-013 halts if its survivor is nonunique; a registered target, source route, graph, atom, bond, orientation, mapping, reaction row, provenance record, database conflict, adverse result or resource halt is omitted; the implementation-distinct reconstruction disagrees; a dependency is absent; or external data selects the native law.

Measurement receipt: sha256:624fc4678f7b1a9b6502f0ff81abe283ed468ff7c0a65921a90b91ebd6499284.

Isolation certificate: sha256:406a05c84cddd4a12e3adc853807be40f291fad5adf9f721a17fd9ab896c88be.

Custody certificate: sha256:942d7c4e92434f7e63924150d21f00271d7f4b77567171330162fafe02ddcfef.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external database encoding algorithm score value equation model fit outcome or software library selects the survivor
- all favorable adverse absent unavailable unresolved low-confidence transport conflict and declared resource-halt rows remain

Immutable identities. Source manifest

sha256:b6102de52e2b6cc574755dfe68c30d75bc2f0ba4e57afa0d8a717ccbb17b890b; derivation seal
 sha256:c13c875a897cb153c455f4061e22d5d23e02228afa996116315ffca004e88642; engine receipt
 sha256:188e0bf716fd6ca34b210699b19054bb71fddd051f7efd37e473d1b198f9c69e at receipts/engine/model_admitted/SFT-CHEM-PREDICTION-UNCERTAINTY-APPLICABILITY-013-188e0bf716fd6ca3.json;
 empirical hash sha256:be7a346f225a35ec3e836f1ed764d0d205f8210ab2e3a213339040a49314ca02;
 measurement receipt sha256:624fc4678f7b1a9b6502f0ff81abe283ed468ff7c0a65921a90b91ebd6499284.

250. Fold classical and quantum chemical algorithm correspondence law

Claim identity: SFT-CHEM-CLASSICAL-QUANTUM-ALGORITHM-CORRESPONDENCE-014

Question and theorem. The same admitted chemical transition executed by a classical exact machine and a reversible complete-support Fold machine has one branchwise state projection and the same observed terminal carrier.

one-shared-admitted-chemical-law__one-held-chemical-input-carrier__exact-classical-transition-trace__complete-reversible-branchwise-trace__exact-branch-state-projection__held-observation-and-retained-record__separate-time-space

-branch-record-cost__identical-canonical-terminal-carrier

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-OPERATIONAL-CLASSICAL-QUANTUM-CORRESPONDENCE-015
- SFT-COMP-FORM-UNIVERSALITY-001

Generated grammar. Generate the literal product of the eight frozen COMP-014 chemical-information custody decisions.

Boundary: Eight binary dimensions exhaust the registered COMP-014 identity, exact-support, equivalence, provenance, falsification and extension forms; $2^8 = 256$ generated candidates.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *one-shared-admitted-chemical-law__one-held-chemical-input-carrier__exact-classical-transition-trace__complete-reversible-branchwise-trace__exact-branch-state-projection__held-observation-and-retained-record__separate-time-space-branch-record-cost__identical-canonical-terminal-carrier*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
chemistry	one-shared-admitted-chemical-law	different-classical-quantum-chemical-laws	different-classical-quantum-chemical-laws closes a distinction required by chemistry.
input	one-held-chemical-input-carrier	different-mode-specific-inputs	different-mode-specific-inputs closes a distinction required by input.
classical	exact-classical-transition-trace	opaque-classical-library-output	opaque-classical-library-output closes a distinction required by classical.
quantum	complete-reversible-branchwise-trace	amplitude-only-output	amplitude-only-output closes a distinction required by quantum.
mapping	exact-branch-state-projection	statistical-output-similarity	statistical-output-similarity closes a distinction required by mapping.
observation	held-observation-and-retained-record	collapse-erases-record	collapse-erases-record closes a distinction required by observation.
resource	separate-time-space-branch-record-cost	unreported-mode-cost	unreported-mode-cost closes a distinction required by resource.
terminal	identical-canonical-terminal-carrier	approximately-same-result	approximately-same-result closes a distinction required by terminal.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:10abd274c198f08090d0ba6de2198ae4b1f48359ab503356e8a7315b2ee2fa31`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b564527ad98fd07f4389441039984c2c16da68b0139366b791d622fa1d6c0baa`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:a158c41c3adff957f9edb545cbf8213392ff8a428584d92e161d0ffac863bda7.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:4430c59531288fb0d6c652a9ac3102986b3d196b898ec4741d904e238171de44.

Independent reconstruction. Implementation hash

sha256:004ec90aaf42f441b6e2bceec983a936847f1af94294f9d550b4d7d67d142d7b; independent certificate sha256:4d39c268a670cdeda3b0591efa173854a52bbe47a89b423f06218fc57e5ebe05; external-validation hash sha256:e50b64270ad516f2e3ec71892ada5a91f21ffbc02fd481fb724df787ea3ef058. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- PUBCHEM-PUG-REST-AUTHORITY
- PUBCHEM-C3H8O-FORMULA-CENSUS
- PUBCHEM-ASPIRIN-CONFORMERS
- PUBCHEM-ASPIRIN-SUBSTRUCTURE
- PUBCHEM-ASPIRIN-SIMILARITY
- CHEBI-REST-AUTHORITY
- RHEA-IMMUTABLE-REACTION-SMILES
- USPTO-IMMUTABLE-REACTION-SURFACE

Comparison records:

- complete 59-artifact 444,644,830-byte post-seal surface retained
- 12 PubChem records, four ChEBI cross-source records, 36,444 Rhea reactions, 50,016 USPTO reactions and 1,065,119 atom-mapped reactions retained
- all twelve registered invalid-property responses, low-confidence rows, conflicts, unavailable records and resource halts retained
- all 8 separately registered claim targets retained
- external formats, algorithms, scores and outcomes never select the Fold-native survivor

Falsification boundary: COMP-014 halts if its survivor is nonunique; a registered target, source route, graph, atom, bond, orientation, mapping, reaction row, provenance record, database conflict, adverse result or resource halt is omitted; the implementation-distinct reconstruction disagrees; a dependency is absent; or external data selects the native law.

Measurement receipt: sha256:facb9543c124496152767897345a4afe6e54fd506784c5dcc1a85b43675d610c.

Isolation certificate: sha256:950da84d4b0408ba2c47189513b0137b7812e27a2dafafcfa6d7b15a9a0805c1.

Custody certificate: sha256:2c0dba9444fd5cc8485f9c18607c10fc8924ca34706bd9be0fedec9fff6f96e3f.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no external database encoding algorithm score value equation model fit outcome or software library selects the survivor
- all favorable adverse absent unavailable unresolved low-confidence transport conflict and declared resource-halt rows remain

Immutable identities. Source manifest

sha256:18b81dee9c313a7e1b48d7e9876057e268db60e563544dc4d4e7549dc404ddea; derivation seal
 sha256:e534d776324594aa6bcf2fa3c9d381cbbba67e4aec169b33b63da1de43e67f3c0; engine receipt
 sha256:05f6e8c3de78a3172c1a3cd27412b0d93a9849171572c1a5a633dc7ec3f4dd19 at receipts/engine/model_admitted/SFT-CHEM-CLASSICAL-QUANTUM-ALGORITHM-CORRESPONDENCE-014-05f6e8c3de78a317.json; empirical hash
 sha256:3649f465e4048da9db584ddb58e8ea863ae55b1f064eac0e4adb2df8847878b3; measurement receipt
 sha256:facb9543c124496152767897345a4afe6e54fd506784c5dcc1a85b43675d610c.

28. Quantitative polymer chemistry - 13 admitted claims

Polymer size, dispersity, networks, sequence, architecture, gelation, conformation, phase and degradation are separately forced. This entire registered subcategory is included; no proper subset is presented as field completion.

251. Fold degree-of-polymerization law

Claim identity: SFT-CHEM-DEGREE-OF-POLYMERIZATION-001

Question and theorem. A complete finite chain forces its degree of polymerization as the exact positive repeat-unit count, equivalently the exact repeat-supported chain mass after separately retained end-group support is removed.

one-complete-finite-chain-carrier__held-repeat-unit-identity__positive-exact-repeat-count__repeat-supported-mass-separated-from-ends__exact-positive-rational-relation__terminal-support-retained__per-chain-result-before-population-summary__chain-repeat-end-and-ratio-trace

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-CHEM-POLYMER-DISTRIBUTION-001

Generated grammar. Generate the literal product of the eight frozen POLY-001 decisions.

Boundary: The eight POLY-001 dimensions exhaust the registered carrier, operation, boundary and certificate distinctions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *one-complete-finite-chain-carrier__held-repeat-unit-identity__positive-exact-repeat-count__repeat-supported-mass-separated-from-ends__exact-positive-rational-relation__terminal-support-retained__per-chain-result-before-population-summary__chain-repeat-end-and-ratio-trace*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-complete-finite-chain-carrier	sample-average-without-chain	sample-average-without-chain closes a distinction required by carrier.
repeat	held-repeat-unit-identity	anonymous-backbone-mass	anonymous-backbone-mass closes a distinction required by repeat.
extent	positive-exact-repeat-count	continuum-chain-length	continuum-chain-length closes a distinction required by extent.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
mass	repeat-supported-mass-separated-from-ends	total-mass-with-ends-conflated	total-mass-with-ends-conflated closes a distinction required by mass.
ratio	exact-positive-rational-relation	rounded-decimal-quotient	rounded-decimal-quotient closes a distinction required by ratio.
ends	terminal-support-retained	terminal-groups-erased	terminal-groups-erased closes a distinction required by ends.
population	per-chain-result-before-population-summary	single-value-replaces-population	single-value-replaces-population closes a distinction required by population.
certificate	chain-repeat-end-and-ratio-trace	degree-only-output	degree-only-output closes a distinction required by certificate.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:decad8eedfd52f9cb46767cea371ec90f951e17973f6b8d00e94d4dceec53de46.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:8ba2d4615d37a60a9378f993d38deaf390b3e57de59939acbb4492e9e50c276b.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:899c2b888e0e590f1e1ea65c4ac8df4a48a4597d54b7c9dacd12a407a610f937.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:abfb3f2a0843dd41b6048e9ac5dcd89efa5e10376701cf761fb2c30ad139a451.

Independent reconstruction. Implementation hash

sha256:e503805064a59ec4f1dd1b4049fe44538fe3e647fbe1f0f85483a7d1703831da; independent certificate sha256:2a6ef8f04ba6fd7f55fcbdba4c3df9f7280e1dbc0e0aaee47f506adf677670f1; external-validation hash sha256:226dd1c73453a61e0a5ad6d4d239d2dd2b381cf4d4704a7ee76ce151e6f54c8c. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-SRM-2888-POLYSTYRENE-CERTIFICATION
- NIST-PRECISION-DEUTERATED-POLYETHYLENE

Comparison records:

- complete 21-artifact, 28,928,563-byte and 279-page Polymer Chemistry source surface retained
- all eight separately registered claim targets reconstructed
- first extraction failure retried by full rendered-page OCR; the original failure remains in the evidence trail
- the one irreconcilable printed source-table arithmetic cell is retained as an external-source defect and never treated as an SFT failure
- all measured values units uncertainties corrections method disagreements favorable adverse absent unavailable inconsistent and unresolved rows remain downstream provenance

Falsification boundary: POLY-001 halts if its survivor is nonunique; a registered target, source byte, measured value, unit, uncertainty, adverse, absent, unavailable, inconsistent or unresolved row is omitted; a source outcome selects the native law; or the reconstruction retry is erased.

Measurement receipt: sha256:df670a6b838d3978c622ae11eb1667c4f602b4595779eede418a4e3fb05814c4.

Isolation certificate: sha256:2eece50c5df497b25905bd66ffa20bf9d53e867a6cd10191515557cdb47eb4f4.

Custody certificate: sha256:6d1e33162bd76a9c749b92aae2de53fb745d0a1496345e51e31dfdbf86aaa844.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured polymer value equation distribution trend or favorable row selects the survivor
- no first extraction failure source defect adverse row or unresolved record retires an obligation

Immutable identities. Source manifest

sha256:36549aef8987838c25ab05877088c9af8f952e33321e3eb9380ea3494f3b9500; derivation seal
 sha256:ce44bb9fc51af6710542638a3cdec514956f7694c83f47ca20e10111ce36d198; engine receipt
 sha256:065acfb40cb3eb8e7fb3598c77311df64f5c191fab1c24194e1bf1e960225041 at receipts/engine/model_admitted/SFT-CHEM-DEGREE-OF-POLYMERIZATION-001-065acfb40cb3eb8e.json; empirical hash
 sha256:2f8884bdfb4a7cdd99e3e74a03adabf4939d059e41e03ce71115b1d00ff1ba4b; measurement receipt
 sha256:df670a6b838d3978c622ae11eb1667c4f602b4595779eede418a4e3fb05814c4.

252. Fold number-average molecular-size law

Claim identity: SFT-CHEM-NUMBER-AVERAGE-MOLECULAR-SIZE-002

Question and theorem. For a complete finite polymer population, the number-average molecular size is the exact sum of every chain size repeated by its positive multiplicity divided by the complete positive chain count.

complete-resolved-chain-population__exact-positive-chain-sizes__positive-multiplicity-per-size__multiplicity-weighted-size-total__complete-chain-count-denominator__exact-positive-rational-average__sample-and-method-boundary-retained__all-rows-sums-ratio-and-boundary-trace

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-CHEM-POLYMER-DISTRIBUTION-001
- SFT-CHEM-DEGREE-OF-POLYMERIZATION-001

Generated grammar. Generate the literal product of the eight frozen POLY-002 decisions.

Boundary: The eight POLY-002 dimensions exhaust the registered carrier, operation, boundary and certificate distinctions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-resolved-chain-population__exact-positive-chain-sizes__positive-multiplicity-per-size__multiplicity-weighted-size-total__complete-chain-count-denominator__exact-positive-rational-average__sample-and-method-boundary-retained__all-rows-sums-ratio-and-boundary-trace`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
population	complete-resolved-chain-population	selected-representative-chains	selected-representative-chains closes a distinction required by population.
sizes	exact-positive-chain-sizes	binned-or-rounded-sizes	binned-or-rounded-sizes closes a distinction required by sizes.
multiplicity	positive-multiplicity-per-size	duplicate-carriers-collapsed	duplicate-carriers-collapsed closes a distinction required by multiplicity.
numerator	multiplicity-weighted-size-total	unweighted-size-total	unweighted-size-total closes a distinction required by numerator.
denominator	complete-chain-count-denominator	mass-total-denominator	mass-total-denominator closes a distinction required by denominator.
arithmetic	exact-positive-rational-average	floating-average	floating-average closes a distinction required by arithmetic.
scope	sample-and-method-boundary-retained	unbounded-population-claim	unbounded-population-claim closes a distinction required by scope.
certificate	all-rows-sums-ratio-and-boundary-trace	mean-only-output	mean-only-output closes a distinction required by certificate.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:b04ecc681a263de26c705bc7e82fcb3e6c5312a134e05f842e2cad2728c743c4.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:f9f23e7616337e47dae794508bef81eb03338b23faa1590a22c17aa185297191.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:be368c2163adb4f7fbf24123db474df23476a6c84ad744f37adeca96099789cb.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:9ca6370d38ab23aa13799b7062d31c64c64afe9b3baa2f034ac266297bb7862b.

Independent reconstruction. Implementation hash

sha256:802763e1b5614d24d4a36f52702c4f1a917650b61b44b8478ea269fc64ccd8c2; independent certificate sha256:f3da9194ea3379a8ce041a7c27bbad4f51682e7e9dc3d287fdbcdb7044b59427f; external-validation hash sha256:11d99071ae0d6ab913591233a3a673db1b5b4d2d358a07ba0c69b0188cc343fd. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-SRM-2888-POLYSTYRENE-CERTIFICATION
- NIST-QUINTUPLE-DETECTOR-COPOLYMER

Comparison records:

- complete 21-artifact, 28,928,563-byte and 279-page Polymer Chemistry source surface retained
- all eight separately registered claim targets reconstructed
- first extraction failure retried by full rendered-page OCR; the original failure remains in the evidence trail
- the one irreconcilable printed source-table arithmetic cell is retained as an external-source defect and never treated as an SFT failure

- all measured values units uncertainties corrections method disagreements favorable adverse absent unavailable inconsistent and unresolved rows remain downstream provenance

Falsification boundary: POLY-002 halts if its survivor is nonunique; a registered target, source byte, measured value, unit, uncertainty, adverse, absent, unavailable, inconsistent or unresolved row is omitted; a source outcome selects the native law; or the reconstruction retry is erased.

Measurement receipt: sha256:3fcea4dcf46245bbfc9ea9572e9eb7f83833ebd587a01544980e903dc06cfc55.

Isolation certificate: sha256:296c9aeaca46fe3a1fcf0c2715f627a41a0b46f656d3c40442c112cb63318c64.

Custody certificate: sha256:6d8a0720bd84125307a0a9b5ec76cdd0868f648b3f59ce573b72ad440dcb02ae.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured polymer value equation distribution trend or favorable row selects the survivor
- no first extraction failure source defect adverse row or unresolved record retires an obligation

Immutable identities. Source manifest

sha256:529352cd75318fee640fe927b1108679f06ff424c02101f9ee723147c74a8c92; derivation seal

sha256:49fef9141a85a464f0d47fbdaae861b33faec2cdd975135c4ad199172d1a21ec; engine receipt

sha256:2074e8e1207661559765c1d88e51a4665a4b76b8536020332a437b08988ac534 at receipts/engin

e/model_admitted/SFT-CHEM-NUMBER-AVERAGE-MOLECULAR-SIZE-002-2074e8e120766155.json;

empirical hash sha256:743bd3d564d216281555200da98520de7d4497f5adcc185740f1748998401e1f;

measurement receipt sha256:3fcea4dcf46245bbfc9ea9572e9eb7f83833ebd587a01544980e903dc06cfc55.

253. Fold mass-weighted molecular-size law

Claim identity: SFT-CHEM-MASS-WEIGHTED-MOLECULAR-SIZE-003

Question and theorem. For a complete finite polymer population, the mass-weighted molecular size is the exact second size moment divided by the exact first size moment, with every multiplicity retained.

complete-resolved-chain-population__exact-positive-chain-sizes__positive-multiplicity-per-size__one-held-mass-copy-per-carrier-mass__complete-second-size-moment__complete-first-size-moment__exact-positive-rational-moment-ratio__all-rows-moments-ratio-and-boundary-trace

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-CHEM-POLYMER-DISTRIBUTION-001
- SFT-CHEM-NUMBER-AVERAGE-MOLECULAR-SIZE-002

Generated grammar. Generate the literal product of the eight frozen POLY-003 decisions.

Boundary: The eight POLY-003 dimensions exhaust the registered carrier, operation, boundary and certificate distinctions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-resolved-chain-population__exact-positive-chain-sizes__positive-multiplicity-per-size__one-held-mass-copy-per-carrier-mass__complete-second-size-moment__complete-first-size-moment__exact-positive-rational-moment-ratio__all-rows-moments-ratio-and-boundary-trace`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
population	complete-resolved-chain-population	selected-heavy-chains	selected-heavy-chains closes a distinction required by population.
sizes	exact-positive-chain-sizes	nominal-class-labels	nominal-class-labels closes a distinction required by sizes.
multiplicity	positive-multiplicity-per-size	frequency-erased	frequency-erased closes a distinction required by multiplicity.
weight	one-held-mass-copy-per-carrier-mass	free-statistical-weight	free-statistical-weight closes a distinction required by weight.
numerator	complete-second-size-moment	first-moment-only	first-moment-only closes a distinction required by numerator.
denominator	complete-first-size-moment	chain-count-denominator	chain-count-denominator closes a distinction required by denominator.
arithmetic	exact-positive-rational-moment-ratio	floating-moment-ratio	floating-moment-ratio closes a distinction required by arithmetic.
certificate	all-rows-moments-ratio-and-boundary-trace	weighted-mean-only-output	weighted-mean-only-output closes a distinction required by certificate.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:76eb7cac6ca7a28149a5a4862118c84542b4c1a004e110f8ec9a31ade300bb4d`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:93c10d64347ccb3bc45bfefa9aa69a8a0900e65f38b855bb647e0c3ef8bc8b79`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:08cbd895f5a4b016104fa62349cc0890fd2da093ed2240b95785a730c87dac4a`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:31675aa3a515f3bea3cc33fc42ec878ce6e197e275fa7efab5cff39e7826e763`.

Independent reconstruction. Implementation hash

`sha256:edf01d1b5cc3d48de15bce322bec38fad36621e16f016634f98455eed33a13c8`; independent certificate `sha256:4f2b623208c4ef2eb44b9afbc4877ee4fac60c2229ad4f0a6e2ae46311a600c2`; external-validation hash `sha256:41170c8b2c3a36529e5b3362cae64dfac896017f4b1273364ef2abb536af3fd7`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- NIST-SRM-2888-POLYSTYRENE-CERTIFICATION
- NIST-SRM-2886-POLYETHYLENE-CERTIFICATION

Comparison records:

- complete 21-artifact, 28,928,563-byte and 279-page Polymer Chemistry source surface retained
- all eight separately registered claim targets reconstructed
- first extraction failure retried by full rendered-page OCR; the original failure remains in the evidence trail
- the one irreconcilable printed source-table arithmetic cell is retained as an external-source defect and never treated as an SFT failure
- all measured values units uncertainties corrections method disagreements favorable adverse absent unavailable inconsistent and unresolved rows remain downstream provenance

Falsification boundary: POLY-003 halts if its survivor is nonunique; a registered target, source byte, measured value, unit, uncertainty, adverse, absent, unavailable, inconsistent or unresolved row is omitted; a source outcome selects the native law; or the reconstruction retry is erased.

Measurement receipt: sha256:6c677ddafd64b1a4df23a0db2d32f75110894d6e7da83370592eef8d4c145328.

Isolation certificate: sha256:3b2a99d0c2f517e3a7ccb88386dc286c98bb4704e406422bb9ba83dcbf0cc596.

Custody certificate: sha256:bd6193a879b680e8b788926bb9ef18a0c90f2937181703b99dfa743f29f3b78d.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured polymer value equation distribution trend or favorable row selects the survivor
- no first extraction failure source defect adverse row or unresolved record retires an obligation

Immutable identities. Source manifest

sha256:b10cc6c3cf2bf43a0d3517b858ca812d072dc3c6826cb18011938ae5154a1788; derivation seal
 sha256:b79248a3616b9d1630d46f9b350c44eeef5f3813067bf78c11da8abcb5c68318; engine receipt
 sha256:f176593b59c0cf7b53b5ef492515908a9ac47ac2b26f712874bd3c7d7b0c6151 at receipts/engine/model_admitted/SFT-CHEM-MASS-WEIGHTED-MOLECULAR-SIZE-003-f176593b59c0cf7b.json; empirical
 hash sha256:acaaf9f70d21be09bb6275175b4671b5d3b29d1f9a947ba7c2105a433dcd076d; measurement
 receipt sha256:6c677ddafd64b1a4df23a0db2d32f75110894d6e7da83370592eef8d4c145328.

254. Fold polymer dispersity law

Claim identity: SFT-CHEM-POLYMER-DISPERSITY-004

Question and theorem. Polymer dispersity is the exact positive ratio of the separately forced mass-weighted and number-average molecular sizes for the same complete source-bound population.

one-shared-complete-population__forced-number-average-with-trace__forced-mass-average-with-trace__mass-average-over-number-average__exact-positive-rational-result__no-distribution-shape-assumed__all-source-bound-rows-retained__population-both-moments-ratio-and-boundary-trace

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001

- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-CHEM-POLYMER-DISTRIBUTION-001
- SFT-CHEM-MASS-WEIGHTED-MOLECULAR-SIZE-003

Generated grammar. Generate the literal product of the eight frozen POLY-004 decisions.

Boundary: The eight POLY-004 dimensions exhaust the registered carrier, operation, boundary and certificate distinctions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `one-shared-complete-population__forced-number-average-with-trace__forced-mass-average-with-trace__mass-average-over-number-average__exact-positive-rational-result__no-distribution-shape-assumed__all-source-bound-rows-retained__population-both-moments-ratio-and-boundary-trace`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
population	one-shared-complete-population	different-populations-for-moments	different-populations-for-moments closes a distinction required by population.
number	forced-number-average-with-trace	number-average-untraced	number-average-untraced closes a distinction required by number.
mass	forced-mass-average-with-trace	mass-average-untraced	mass-average-untraced closes a distinction required by mass.
ratio	mass-average-over-number-average	subtracted-spread	subtracted-spread closes a distinction required by ratio.
arithmetic	exact-positive-rational-result	floating-fit-statistic	floating-fit-statistic closes a distinction required by arithmetic.
distribution	no-distribution-shape-assumed	named-distribution-assumption	named-distribution-assumption closes a distinction required by distribution.
adverse	all-source-bound-rows-retained	unresolved-rows-discarded	unresolved-rows-discarded closes a distinction required by adverse.
certificate	population-both-moments-ratio-and-boundary-trace	scalar-only-output	scalar-only-output closes a distinction required by certificate.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:7b1d3c0ce553fd9200ba87bf0c79cdebc2d761c5afc872c5250ee3c9cfbba821`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:67ea8165bdd01751628f550a17429857872f760b8336e1fd56aec26c35d8fae5`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:2f36bd25142053ddf3ac211e2024c7676a2e589808db47b63a064ac4a24b713d`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:dab73aeaf58627c9e3555d85ef6487ede885e3271f98bf5caf9bfb34e6cbe92e`.

Independent reconstruction. Implementation hash

`sha256:948d351a3dab855014b281a20a7d1876637100b490c1ec8cdf5dbc594c76398c`; independent certificate `sha256:34fd7d179ebb653e03895dfc20c51d6ddf8f0ffe937111a944040826b5451ec7`; external-validation

hash sha256:57f174c2fc590d5dc3982f798ff4b2ed37179f7fc42a15d7a8781fa9222e7920. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-SRM-2886-POLYETHYLENE-CERTIFICATION
- NIST-MONODISPERSE-PAMS-KINETIC-NETWORK

Comparison records:

- complete 21-artifact, 28,928,563-byte and 279-page Polymer Chemistry source surface retained
- all eight separately registered claim targets reconstructed
- first extraction failure retried by full rendered-page OCR; the original failure remains in the evidence trail
- the one irreconcilable printed source-table arithmetic cell is retained as an external-source defect and never treated as an SFT failure
- all measured values units uncertainties corrections method disagreements favorable adverse absent unavailable inconsistent and unresolved rows remain downstream provenance

Falsification boundary: POLY-004 halts if its survivor is nonunique; a registered target, source byte, measured value, unit, uncertainty, adverse, absent, unavailable, inconsistent or unresolved row is omitted; a source outcome selects the native law; or the reconstruction retry is erased.

Measurement receipt: sha256:ac1d5a6dfc38b2d2a70be4cd3439e17b1360270cb2d6b5ab617b6a305c5eedbb.

Isolation certificate: sha256:aebbf7048329bcf166d00737f736fa4e886b417989d85fa1e6b8127f1025c003.

Custody certificate: sha256:84119466eb3e140291353528e41793bce2bb96c376bf0b907ed046743dba1626.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured polymer value equation distribution trend or favorable row selects the survivor
- no first extraction failure source defect adverse row or unresolved record retires an obligation

Immutable identities. Source manifest

sha256:23d8f9d4b9b1d46a8f9e3c247d6504f94fe7fe0482fe393976a1b08ca5aa876f; derivation seal

sha256:c84b75a48604f285a4c13d26fcd1c56e9a76700870846f57aeb1c7a07e9ae13a; engine receipt

sha256:d32e9721f4366db655e96e401760ab71b4c49e6ce0bba8717c7073b6a7424203 at

receipts/engine/model_admitted/SFT-CHEM-POLYMER-DISPERSITY-004-d32e9721f4366db6.json;

empirical hash sha256:acb2ac2e260bac573701c9c4801d1d3649d620e29e66c8726b5c6c478f94a6d8;

measurement receipt sha256:ac1d5a6dfc38b2d2a70be4cd3439e17b1360270cb2d6b5ab617b6a305c5eedbb.

255. Fold chain-growth polymerization-network law

Claim identity: SFT-CHEM-CHAIN-GROWTH-POLYMERIZATION-NETWORK-005

Question and theorem. A chain-growth polymerization is the complete finite transition network in which initiation creates a held active chain end, propagation appends one retained monomer, transfer moves the active end, and termination closes it without losing carrier custody.

complete-held-chain-and-active-end-states__explicit-active-end-creation-transition__one-monomer-append-transition__active-end-transfer-transition-retained__explicit-terminal-chain-transition__all-simple-registered-network-paths__complete-carrier-and-active-end-custody__states-transitions-paths-resources-and-terminal-trace

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-CHEM-POLYMER-DISTRIBUTION-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-REACTION-GRAPH-GENERATION-007
- SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007

Generated grammar. Generate the literal product of the eight frozen POLY-005 decisions.

Boundary: The eight POLY-005 dimensions exhaust the registered carrier, operation, boundary and certificate distinctions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *complete-held-chain-and-active-end-states__explicit-active-end-creation-transition__one-monomer-append-transition__active-end-transfer-transition-retained__explicit-terminal-chain-transition__all-simple-registered-network-paths__complete-carrier-and-active-end-custody__states-transitions-paths-resources-and-terminal-trace*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
states	complete-held-chain-and-active-end-states	conversion-scalar-only	conversion-scalar-only closes a distinction required by states.
initiation	explicit-active-end-creation-transition	implicit-chain-start	implicit-chain-start closes a distinction required by initiation.
propagation	one-monomer-append-transition	continuous-growth-rate	continuous-growth-rate closes a distinction required by propagation.
transfer	active-end-transfer-transition-retained	transfer-path-erased	transfer-path-erased closes a distinction required by transfer.
termination	explicit-terminal-chain-transition	silent-process-stop	silent-process-stop closes a distinction required by termination.
paths	all-simple-registered-network-paths	major-path-only	major-path-only closes a distinction required by paths.
custody	complete-carrier-and-active-end-custody	unbalanced-monomer-loss	unbalanced-monomer-loss closes a distinction required by custody.
certificate	states-transitions-paths-resources-and-terminal-trace	conversion-only-output	conversion-only-output closes a distinction required by certificate.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 7c99a87e4ea6d47af80458c0c28793a4cc4546fce768e98163d10249fddb69d9.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: d160ce3ae221e2e489d5701bf80716143ae30ee0e9a2baf55e7f05dbb6ed2c2b.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 5812aa50d70763bf38eee13e9ef454658f93aa66912bf55012a97d952aa3138e.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 62c9abdcdf41093f0f0308f0964d0e7f961c9ac8952979b269caa1700add2a83.

Independent reconstruction. Implementation hash

sha256: 477bdc3cac3da8cc5b97363c63373b00116b23b40c54cac8e0e81b45ceb822f2; independent certificate sha256: ad0e4c5b1be7dd203973ffa0b0db8924f5812226f9bddadae33d57f70b28b806; external-validation hash sha256: 37c4998bc4bb412ff87f153d16bf4d837e2d77680526a6f9ea300c89c76995a9. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-STOCHASTIC-PHOTOPOLYMER-NETWORK-GROWTH
- NIST-MONODISPERSE-PAMS-KINETIC-NETWORK

Comparison records:

- complete 21-artifact, 28,928,563-byte and 279-page Polymer Chemistry source surface retained
- all eight separately registered claim targets reconstructed
- first extraction failure retried by full rendered-page OCR; the original failure remains in the evidence trail
- the one irreconcilable printed source-table arithmetic cell is retained as an external-source defect and never treated as an SFT failure
- all measured values units uncertainties corrections method disagreements favorable adverse absent unavailable inconsistent and unresolved rows remain downstream provenance

Falsification boundary: POLY-005 halts if its survivor is nonunique; a registered target, source byte, measured value, unit, uncertainty, adverse, absent, unavailable, inconsistent or unresolved row is omitted; a source outcome selects the native law; or the reconstruction retry is erased.

Measurement receipt: sha256: 19f0e143ae69317ffe063837ed03dc25a47abdbd3a844d09ffd3fe229dd26a7c.

Isolation certificate: sha256: 8c7189714569d06ab2fb35a39b406a16febdcc0cc4f03c634ecbca324f8421067.

Custody certificate: sha256: 225418f7c6dfdfda7672351b972c2a617e1490480911d1c3ac0afb66887372b0.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured polymer value equation distribution trend or favorable row selects the survivor

- no first extraction failure source defect adverse row or unresolved record retires an obligation

Immutable identities. Source manifest

sha256:c1127aff9a7eefec437c6bbaaad35d6500f4b045e8347404b39c249081aff770; derivation seal sha256:4d807915ac5c892d24316ec59b9d0a388894a7b194fb7684fc4a039706487144; engine receipt sha256:b480d01f119976bedf205450bca2e715f2dfffb93d68068dd43ac228f451a8c9d at receipts/engine/model_admitted/SFT-CHEM-CHAIN-GROWTH-POLYMERIZATION-NETWORK-005-b480d01f119976be.json; empirical hash sha256:eee73764abec9d6be2b30b869ca9c53fac9dc99552da20478edc74622cd262cf; measurement receipt sha256:19f0e143ae69317ffe063837ed03dc25a47abdbd3a844d09ffd3fe229dd26a7c.

256. Fold step-growth polymerization-network law

Claim identity: SFT-CHEM-STEP-GROWTH-POLYMERIZATION-NETWORK-006

Question and theorem. In a finite step-growth population every admitted intermolecular bond merges exactly two connected components into one; the retained molecule count and complete reactive-group transition network are therefore forced at every step.

complete-molecule-and-reactive-group-population__one-explicit-intermolecular-bond-transition__each-bond-merges-two-components__every-reactive-group-identity-retained__all-connected-component-products__cycle-and-intermolecular-bonds-distinguished__exact-initial-minus-merge-count__groups-bonds-components-products-and-boundary-trace

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-CHEM-POLYMER-DISTRIBUTION-001
- SFT-CHEM-CHAIN-GROWTH-POLYMERIZATION-NETWORK-005

Generated grammar. Generate the literal product of the eight frozen POLY-006 decisions.

Boundary: The eight POLY-006 dimensions exhaust the registered carrier, operation, boundary and certificate distinctions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *complete-molecule-and-reactive-group-population__one-explicit-intermolecular-bond-transition__each-bond-merges-two-components__every-reactive-group-identity-retained__all-connected-component-products__cycle-and-intermolecular-bonds-distinguished__exact-initial-minus-merge-count__groups-bonds-components-products-and-boundary-trace*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
population	complete-molecule-and-reactive-group-population	mean-chain-only	mean-chain-only closes a distinction required by population.
reaction	one-explicit-intermolecular-bond-transition	conversion-without-bond	conversion-without-bond closes a distinction required by reaction.
components	each-bond-merges-two-components	component-count-untracked	component-count-untracked closes a distinction required by components.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
groups	every-reactive-group-identity-retained	reactive-groups-erased	reactive-groups-erased closes a distinction required by groups.
products	all-connected-component-products	selected-chain-products	selected-chain-products closes a distinction required by products.
cycles	cycle-and-intermolecular-bonds-distinguished	cycle-bond-counts-as-merge	cycle-bond-counts-as-merge closes a distinction required by cycles.
arithmetic	exact-initial-minus-merge-count	imported-conversion-equation	imported-conversion-equation closes a distinction required by arithmetic.
certificate	groups-bonds-components-products-and-boundary-trace	degree-only-output	degree-only-output closes a distinction required by certificate.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : c19746d41c7d7ee69e72948736c308c2d4f6075153eada9651dcc191b0357799.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 7658533bfe5fb7ced224226f9e2598f78fdfff67e701b38b6a8f27a050863073.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 6477965c004e6c2a001d8ce9e6a928ec0456563d30e453a84ae9875a9dc983bc.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : fed64efdf6583bfd788a98c548dddfa39fe7e228c44e7ed8a7441d59bcb142a.

Independent reconstruction. Implementation hash

sha256 : e5054e4e09c26d4386f87893eba73e9151bc7ce2818f4e1772cd8a434beab4a5; independent certificate
sha256 : de06f66e6b7fadab41bb13440164e8e81acf866e4852af843c2d0c6927655cf3; external-validation
hash sha256 : 36a8e50fddc11ca2b34f8ef7570eaf1c10d1cf0bd4a9e539c97ebe5ebfa1cd2c. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-STOCHASTIC-PHOTOPOLYMER-NETWORK-GROWTH
- NIST-CROSSLINKED-PHOTOPOLYMER-CONVERSION

Comparison records:

- complete 21-artifact, 28,928,563-byte and 279-page Polymer Chemistry source surface retained
- all eight separately registered claim targets reconstructed
- first extraction failure retried by full rendered-page OCR; the original failure remains in the evidence trail
- the one irreconcilable printed source-table arithmetic cell is retained as an external-source defect and never treated as an SFT failure
- all measured values units uncertainties corrections method disagreements favorable adverse absent unavailable inconsistent and unresolved rows remain downstream provenance

Falsification boundary: POLY-006 halts if its survivor is nonunique; a registered target, source byte, measured value, unit, uncertainty, adverse, absent, unavailable, inconsistent or unresolved row is omitted; a source outcome selects the native law; or the reconstruction retry is erased.

Measurement receipt: sha256:7629e0807dd40b5e8fe4c1818dd8a194c03e18d807fe7dd7ec85f85764e90888.
 Isolation certificate: sha256:7783dfba01cf8cb1af506b4b7b667c429585b33133bed89dd88f4627c3570e4b.
 Custody certificate: sha256:a1d50ce8b093a4fcc7a6ae68411fe4c00e2654b7776ae48be94566d2f3c1a987.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured polymer value equation distribution trend or favorable row selects the survivor
- no first extraction failure source defect adverse row or unresolved record retires an obligation

Immutable identities. Source manifest

sha256:64c4991a4effd3650728d7265c53c923da735077f10e566d179074228f7cf3e0; derivation seal
 sha256:23753c915ce884226f37a5a2ec638f29d7cabd8d94e3ca142373fd54e76ff126; engine receipt
 sha256:0278e5e0e9f6a83640a7863751d0fd0f205a8ed657ca3e526fafdb2b5a2d2d66 at receipts/engine/model_admitted/SFT-CHEM-STEP-GROWTH-POLYMERIZATION-NETWORK-006-0278e5e0e9f6a836.json;
 empirical hash sha256:11fad6018269bc923f6cf28d93ccd992d35649601deab89991ff09d9f89d49c5;
 measurement receipt sha256:7629e0807dd40b5e8fe4c1818dd8a194c03e18d807fe7dd7ec85f85764e90888.

257. Fold copolymer sequence-composition law

Claim identity: SFT-CHEM-COPOLYMER-SEQUENCE-COMPOSITION-007

Question and theorem. A finite copolymer is an exact generated word of held monomer labels; composition is the complete positive label-count vector divided by word length while sequence order remains separately reconstructible.

complete-ordered-monomer-word__held-monomer-identities__complete-positive-label-count-vector__exact-label-count-over-word-length__every-position-and-neighbour-relation-retained__word-and-sample-boundaries-separated__append-one-held-label-extension__word-counts-ratios-sequence-and-boundary-trace

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-CHEM-POLYMER-DISTRIBUTION-001
- SFT-CHEM-STEP-GROWTH-POLYMERIZATION-NETWORK-006

Generated grammar. Generate the literal product of the eight frozen POLY-007 decisions.

Boundary: The eight POLY-007 dimensions exhaust the registered carrier, operation, boundary and certificate distinctions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: complete-ordered-monomer-word__held-monomer-identities__complete-positive-label-count-vector__exact-label-count-over-word-length__every-position-and-neighbour-relation-retained__word-and-sample-boundaries-separated__append-one-held-label-extension__word-counts-ratios-sequence-and-boundary-trace

ndaries-separated__append-one-held-label-extension__word-counts-ratios-sequence-and-boundary-trace. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
word	complete-ordered-monomer-word	composition-without-sequence	composition-without-sequence closes a distinction required by word.
labels	held-monomer-identities	anonymous-repeat-count	anonymous-repeat-count closes a distinction required by labels.
counts	complete-positive-label-count-vector	selected-monomer-fractions	selected-monomer-fractions closes a distinction required by counts.
composition	exact-label-count-over-word-length	rounded-percentages	rounded-percentages closes a distinction required by composition.
sequence	every-position-and-neighbour-relation-retained	random-sequence-premise	random-sequence-premise closes a distinction required by sequence.
population	word-and-sample-boundaries-separated	one-word-replaces-sample	one-word-replaces-sample closes a distinction required by population.
extension	append-one-held-label-extension	retroactive-sequence-fit	retroactive-sequence-fit closes a distinction required by extension.
certificate	word-counts-ratios-sequence-and-boundary-trace	composition-only-output	composition-only-output closes a distinction required by certificate.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : e377be9db3ad20255b6108262618191e8f787a673e3b80ce86672893456e3e62.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : ea7cc1d9100d5f418a73886b2f7228fcdc91e8935dc494c382b1b2318c1e3504.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : a0b336a148d374d2387e650de0e17de967f049bdc65dd66d19409c3e9080a500.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 4d1188693bb160cf5aa1eb8c1f70130f1ef9e482978a60fb085f37f3d6fa948f.

Independent reconstruction. Implementation hash

sha256 : c8caa7bd9136870e4a37a1724b7b8d9c62fc564643e11ce1a9e7e56e1872f467; independent certificate sha256 : 50bc95a40ee494317ab46a8999246426d1eedf2b330b6ce4cce131b11d57907a; external-validation hash sha256 : 946d513aaea698833517dd4474efeb15f260b60b1541a8ca4b7a6b904474346f. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-RING-OPENING-COPOLYMER-SEQUENCE
- NIST-PRECISION-DEUTERATED-POLYETHYLENE

Comparison records:

- complete 21-artifact, 28,928,563-byte and 279-page Polymer Chemistry source surface retained
- all eight separately registered claim targets reconstructed

- first extraction failure retried by full rendered-page OCR; the original failure remains in the evidence trail
- the one irreconcilable printed source-table arithmetic cell is retained as an external-source defect and never treated as an SFT failure
- all measured values units uncertainties corrections method disagreements favorable adverse absent unavailable inconsistent and unresolved rows remain downstream provenance

Falsification boundary: POLY-007 halts if its survivor is nonunique; a registered target, source byte, measured value, unit, uncertainty, adverse, absent, unavailable, inconsistent or unresolved row is omitted; a source outcome selects the native law; or the reconstruction retry is erased.

Measurement receipt: sha256:7052291dead69eccbeaa14e605b9ff190d1f135ecfcbae627dbcbfb804583992.
Isolation certificate: sha256:dab0464e531a5f84cfcae7921beeeffd09b9d63947b1abec6c569d8299614a9b.
Custody certificate: sha256:9603feb272aa2dd8d8930af858298ff0f02d14c45158315d4b73b8659c263822.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured polymer value equation distribution trend or favorable row selects the survivor
- no first extraction failure source defect adverse row or unresolved record retires an obligation

Immutable identities. Source manifest

sha256:711e329fae2347bdc9707b46ef9b41eae9817d53bbd655af8f6adf0f5710609c; derivation seal
sha256:92245416850713864393cb460b4f1a5d470531aee3ccce8de32eca499a9e10b0; engine receipt
sha256:d6bb8a00838a12fccec6bbe408e5ff8d71ff540a6548aa2ad478c7cc405e9d08c at receipts/engine/model_admitted/SFT-CHEM-COPOLYMER-SEQUENCE-COMPOSITION-007-d6bb8a00838a12fc.json;
empirical hash sha256:4da80f3903a1d77982ce7bb47d3646d604c5e2441995a970f5a24bdd3c304368;
measurement receipt sha256:7052291dead69eccbeaa14e605b9ff190d1f135ecfcbae627dbcbfb804583992.

258. Fold branched, star and network architecture law

Claim identity: SFT-CHEM-POLYMER-ARCHITECTURE-008

Question and theorem. Polymer architecture is the exact isomorphism class of its finite held monomer-incidence graph; linear, branched, star and crosslinked forms are forced by complete degree, component, cycle and branch-centre records.

complete-finite-polymer-incidence-graph__held-repeat-and-junction-identities__complete-covalent-incidence-support__degree-of-every-held-vertex__complete-connected-component-census__cycle-rank-retained__exact-labelled-graph-isomorphism-class__vertices-edges-degrees-components-cycles-and-map-trace

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-POLYMER-CHAIN-001

- SFT-CHEM-POLYMER-DISTRIBUTION-001
- SFT-CHEM-COPOLYMER-SEQUENCE-COMPOSITION-007
- SFT-CHEM-CANONICAL-MOLECULAR-GRAPH-ENCODING-001

Generated grammar. Generate the literal product of the eight frozen POLY-008 decisions.

Boundary: The eight POLY-008 dimensions exhaust the registered carrier, operation, boundary and certificate distinctions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: complete-finite-polymer-incidence-graph__held-repeat-and-junction-identities__complete-covalent-incidence-support__degree-of-every-held-vertex__complete-connected-component-census__cycle-rank-retained__exact-labelled-graph-isomorphism-class__vertices-edges-degrees-components-cycles-and-map-trace. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-finite-polymer-incidence-graph	architecture-name-only	architecture-name-only closes a distinction required by carrier.
vertices	held-repeat-and-junction-identities	unlabelled-node-total	unlabelled-node-total closes a distinction required by vertices.
edges	complete-covalent-incidence-support	connectivity-summary	connectivity-summary closes a distinction required by edges.
degree	degree-of-every-held-vertex	branch-count-only	branch-count-only closes a distinction required by degree.
components	complete-connected-component-census	disconnected-pieces-conflated	disconnected-pieces-conflated closes a distinction required by components.
cycles	cycle-rank-retained	tree-assumed	tree-assumed closes a distinction required by cycles.
identity	exact-labelled-graph-isomorphism-class	drawing-layout-identity	drawing-layout-identity closes a distinction required by identity.
certificate	vertices-edges-degrees-components-cycles-and-map-trace	architecture-label-only	architecture-label-only closes a distinction required by certificate.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256 : a2f39b4845d4d01dac1c3a846e31c10d86f5c21b6a967849f19cf4d3375cb6b7.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 600585c35486a0b20f276230d1a7b9b8f6ba207aae63507ce61b5ebdaf0e37fa.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256 : 26fa08cec5f8dfa3c33b0e779872cf9d30a8520423d8d5eb83d7a754d2eb134a.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256 : 2f14aa0ea61318d551ae70ff2304c01507cccd9368a7c75f0d134d3a8ce3cff6.

Independent reconstruction. Implementation hash

sha256 : b5725187525fd8762bedc3ca1a57243db161e5fde0a45d889d0ffe4dea7ff03f; independent certificate sha256 : af97fe5fa988e2937eeec78b88c1837a909fe28dc3c6d635fc4d2e79f1428f96; external-validation hash sha256 : 4d73b952c739864be7e2a05e2c8b2d89b92616e0bd9e4286464d96d4efee67de. The independent

program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- NIST-BRANCH-PLACEMENT-DILUTE-SOLUTION
- NIST-MACROMOLECULAR-ARCHITECTURES

Comparison records:

- complete 21-artifact, 28,928,563-byte and 279-page Polymer Chemistry source surface retained
- all eight separately registered claim targets reconstructed
- first extraction failure retried by full rendered-page OCR; the original failure remains in the evidence trail
- the one irreconcilable printed source-table arithmetic cell is retained as an external-source defect and never treated as an SFT failure
- all measured values units uncertainties corrections method disagreements favorable adverse absent unavailable inconsistent and unresolved rows remain downstream provenance

Falsification boundary: POLY-008 halts if its survivor is nonunique; a registered target, source byte, measured value, unit, uncertainty, adverse, absent, unavailable, inconsistent or unresolved row is omitted; a source outcome selects the native law; or the reconstruction retry is erased.

Measurement receipt: sha256:e82484c2869390844c2dffea8cfeab651c8b8b123d4dcf14bde5b575093514bb.

Isolation certificate: sha256:083900459fb075b28f80a6de28a65e54c9e54e3b078bd5a879e5efc2b9585b66.

Custody certificate: sha256:68078a9744675bf3a9be93d66ace65a3d53f0b25c1d148a070a8aececd2d4312.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured polymer value equation distribution trend or favorable row selects the survivor
- no first extraction failure source defect adverse row or unresolved record retires an obligation

Immutable identities. Source manifest

sha256:7fa5fe53237331227e08252685a0b36464a1ac7a462ff2748c212e8cac567791; derivation seal
sha256:d92cbb55c304b1ff78a2d85cc239a9482c70f48e7732d73b4ae8eaf320aa7d03; engine receipt
sha256:4047581cfcc8c92646c81323ff2ae72697c673a3a745b681fc38a18b4e821318 at
receipts/engine/model_admitted/SFT-CHEM-POLYMER-ARCHITECTURE-008-4047581cfcc8c926.json;
empirical hash sha256:f9ca28e23090003f745d8e4819c255114f8c6453964ee5538c1c0caf4636e79e;
measurement receipt sha256:e82484c2869390844c2dffea8cfeab651c8b8b123d4dcf14bde5b575093514bb.

259. Fold crosslink and finite-gelation boundary law

Claim identity: SFT-CHEM-CROSSLINK-GELATION-BOUNDARY-009

Question and theorem. At a declared finite observation scale, gelation is certified exactly when one retained covalent component joins the registered disjoint inlet and outlet boundaries; no completed infinite network is asserted.

complete-finite-crosslink-graph__disjoint-held-inlet-and-outlet-support__one-component-connects-both-boundaries__complete-component-and-path-census__declared-finite-observation-scale__first-registered-boundary-connecting-transit ion__every-unconnected-boundary-state-rejected__graph-boundaries-component-witness-and-scale-trace

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-CHEM-POLYMER-DISTRIBUTION-001
- SFT-CHEM-POLYMER-ARCHITECTURE-008

Generated grammar. Generate the literal product of the eight frozen POLY-009 decisions.

Boundary: The eight POLY-009 dimensions exhaust the registered carrier, operation, boundary and certificate distinctions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-finite-crosslink-graph__disjoint-held-inlet-and-outlet-support__one-component-connects-both-boundaries__complete-component-and-path-census__declared-finite-observation-scale__first-registered-boundary-connecting-transition__every-unconnected-boundary-state-rejected__graph-boundaries-component-witness-and-scale-trace`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
network	complete-finite-crosslink-graph	bulk-gel-label	bulk-gel-label closes a distinction required by network.
boundaries	disjoint-held-inlet-and-outlet-support	unregistered-system-extent	unregistered-system-extent closes a distinction required by boundaries.
connectivity	one-component-connects-both-boundaries	crosslink-count-threshold	crosslink-count-threshold closes a distinction required by connectivity.
enumeration	complete-component-and-path-census	sampled-path-test	sampled-path-test closes a distinction required by enumeration.
infinity	declared-finite-observation-scale	completed-infinite-network	completed-infinite-network closes a distinction required by infinity.
transition	first-registered-boundary-connecting-transition	assumed-universal-gel-point	assumed-universal-gel-point closes a distinction required by transition.
adverse	every-unconnected-boundary-state-rejected	near-spanning-network-accepted	near-spanning-network-accepted closes a distinction required by adverse.
certificate	graph-boundaries-component-witness-and-scale-trace	gel-boolean-only	gel-boolean-only closes a distinction required by certificate.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:21e1019fc48844611c0f4f8cc00303d18e3a46914914ed37b85165aa15b3f3f9`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:74cd4eeb536825309332e73509bc99eb0772dce65491f7e41543d3600943e28d`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:d76e0b7813046441ffe08942e1c10f1c1fb29355f407805a182412108c57ab0b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:7105f6dc44f0b54066bbf15ecf406906a9b05bc7415611cd675daad65168bc3e.

Independent reconstruction. Implementation hash

sha256:7792a583afd43df7260e96b6ab33f1d6c1a8224bdc70e17b89e6480cd50ef1cb; independent certificate sha256:a0be12b920cd23e0d9259dc703cab491f7b5174d08ffad9688e9e089b7dd809d; external-validation hash sha256:4e6bed2818de0c30b3437e539b11fb0d3f344bbb98daa3f9fa2d75199e513002. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-THERMOREVERSIBLE-GELATION-PERCOLATION
- NIST-BRIDGING-GELATION

Comparison records:

- complete 21-artifact, 28,928,563-byte and 279-page Polymer Chemistry source surface retained
- all eight separately registered claim targets reconstructed
- first extraction failure retried by full rendered-page OCR; the original failure remains in the evidence trail
- the one irreconcilable printed source-table arithmetic cell is retained as an external-source defect and never treated as an SFT failure
- all measured values units uncertainties corrections method disagreements favorable adverse absent unavailable inconsistent and unresolved rows remain downstream provenance

Falsification boundary: POLY-009 halts if its survivor is nonunique; a registered target, source byte, measured value, unit, uncertainty, adverse, absent, unavailable, inconsistent or unresolved row is omitted; a source outcome selects the native law; or the reconstruction retry is erased.

Measurement receipt: sha256:5c17be5f6925c62764432d97949ed3caada64710f678ac8e5579003c26f1b6c7.

Isolation certificate: sha256:bd3e17aa585b5979d107a2b2017f916e89444fae505c888a787e65f3693f2110.

Custody certificate: sha256:0f34880cb6d619619be2789ac4f79f043de7f730518f0b8eb0271bf20debc830.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured polymer value equation distribution trend or favorable row selects the survivor
- no first extraction failure source defect adverse row or unresolved record retires an obligation

Immutable identities. Source manifest

sha256:e81342bc1268e5ae3d24563e4795c2b8b3c3847ef31065bf2e281afadd553450; derivation seal

sha256:43ae0b496ba523d6a2e174f4b1934a13b0f1d88bb3d1fdac771bbc23737b2f33; engine receipt

sha256:f18e59a809621d8da060d00487821d16b807aacd236d6282cfdcab7da5c933b7 at receipts/engine/model_admitted/SFT-CHEM-CROSSLINK-GELATION-BOUNDARY-009-f18e59a809621d8d.json; empirical

hash sha256:7e61bb556b1ad2d0e429cfc1e565fa929eb5fdab502402c9fa7b477ed37799b1; measurement receipt sha256:5c17be5f6925c62764432d97949ed3caada64710f678ac8e5579003c26f1b6c7.

260. Fold polymer conformation and size law

Claim identity: SFT-CHEM-POLYMER-CONFORMATION-SIZE-010

Question and theorem. A finite chain conformation is a generated ordered position word and its squared size is the exact mean squared displacement from its exact centroid; no continuum random walk or irrational root enters the native result.

complete-ordered-chain-carrier__complete-generated-finite-conformation-support__exact-rational-position-word__exact-finite-centroid__exact-mean-squared-centroid-distance__explicit-state-and-weight-census__declared-chain-state-and-resolution-boundary__positions-centroid-squares-weights-and-boundary-trace

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-CHEM-POLYMER-DISTRIBUTION-001
- SFT-CHEM-POLYMER-ARCHITECTURE-008
- SFT-CHEM-CONFORMER-ENUMERATION-006

Generated grammar. Generate the literal product of the eight frozen POLY-010 decisions.

Boundary: The eight POLY-010 dimensions exhaust the registered carrier, operation, boundary and certificate distinctions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *complete-ordered-chain-carrier__complete-generated-finite-conformation-support__exact-rational-position-word__exact-finite-centroid__exact-mean-squared-centroid-distance__explicit-state-and-weight-census__declared-chain-state-and-resolution-boundary__positions-centroid-squares-weights-and-boundary-trace*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
chain	complete-ordered-chain-carrier	mass-only-chain	mass-only-chain closes a distinction required by chain.
states	complete-generated-finite-conformation-support	one-equilibrium-shape	one-equilibrium-shape closes a distinction required by states.
positions	exact-rational-position-word	continuum-coordinate-field	continuum-coordinate-field closes a distinction required by positions.
centre	exact-finite-centroid	assumed-origin	assumed-origin closes a distinction required by centre.
size	exact-mean-squared-centroid-distance	root-mean-radius	root-mean-radius closes a distinction required by size.
ensemble	explicit-state-and-weight-census	random-walk-premise	random-walk-premise closes a distinction required by ensemble.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
scope	declared-chain-state-and-resolution-boundary	asymptotic-universal-law	asymptotic-universal-law closes a distinction required by scope.
certificate	positions-centroid-squares-weights-and-boundary-trace	size-scalar-only	size-scalar-only closes a distinction required by certificate.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:6531e4161a061d4e24e4d397925239e04919502b0f3f6d261f4674a2c7aead69.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:e5e773f46a3dbcd18f66b763456748715a510999db6628f2058ee4d292dfb6b0.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:6613ee483760616cbff1134e58618d1e1da5aa8155edec9ac2d8e98cc36eb36f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:dfc2b2a2360fdc1e8966de8a3408373b2a15ed208d2dc41e0a5227a351f4a9be.

Independent reconstruction. Implementation hash

sha256:2595aa4fc6da936bfad3bd2940dae5bd0b6e0a554f8cba6d3c7b48b22142486d; independent certificate sha256:84b69d69f446e7d271710edc76d2d359cdc407517162ba519bb09d3c1261d1b9; external-validation hash sha256:7127c12ccb4f91be7816ce67696fce6674715f906415208af5b402cba3906ded. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-QUINTUPLE-DETECTOR-COPOLYMER
- NIST-BRANCH-PLACEMENT-DILUTE-SOLUTION

Comparison records:

- complete 21-artifact, 28,928,563-byte and 279-page Polymer Chemistry source surface retained
- all eight separately registered claim targets reconstructed
- first extraction failure retried by full rendered-page OCR; the original failure remains in the evidence trail
- the one irreconcilable printed source-table arithmetic cell is retained as an external-source defect and never treated as an SFT failure
- all measured values units uncertainties corrections method disagreements favorable adverse absent unavailable inconsistent and unresolved rows remain downstream provenance

Falsification boundary: POLY-010 halts if its survivor is nonunique; a registered target, source byte, measured value, unit, uncertainty, adverse, absent, unavailable, inconsistent or unresolved row is omitted; a source outcome selects the native law; or the reconstruction retry is erased.

Measurement receipt: sha256:d07a27598145a6042e5dd91d83be6ea0691e57f9f1080b065988f7015025b8f2.

Isolation certificate: sha256:cc02f4b58b2fee2fe612ea7c633c63d7da7c4d78628ed7a8a786797a292a284d.

Custody certificate: sha256:5c526a35169254046e42f922550d01717a642aefcfc07a9e14175f26c5658b57.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured polymer value equation distribution trend or favorable row selects the survivor
- no first extraction failure source defect adverse row or unresolved record retires an obligation

Immutable identities. Source manifest

sha256:698586872751d77d95551c5afac840410906afd553913acb444a735a633244da; derivation seal
 sha256:d6b2e4dd541ef3c2d77cd0cdeec96e5e25527a3c5eaba9e45321d0fd780ab4d4; engine receipt
 sha256:ebdb8895ec6882a4812462c711c1a4c33df8badafa4a201ae0046188f3552bbe at receipts/engine/model_admitted/SFT-CHEM-POLYMER-CONFORMATION-SIZE-010-ebdb8895ec6882a4.json; empirical
 hash sha256:6616ca2023df9fde6fa608abd267a1555d3279bb78e723f4b78e32d20f03a1b8; measurement
 receipt sha256:d07a27598145a6042e5dd91d83be6ea0691e57f9f1080b065988f7015025b8f2.

261. Fold polymer phase and transition law

Claim identity: SFT-CHEM-POLYMER-PHASE-TRANSITION-011

Question and theorem. A polymer molecular transition is the first ordered registered state at which a held molecular phase label changes, with composition, architecture, condition and hysteresis direction retained; bulk performance remains Materials-owned.

complete-polymer-molecular-carrier__ordered-complete-condition-state-record__held-molecular-phase-label-change__composition-and-architecture-retained__transition-direction-and-hysteresis-held__chemistry-transition-materials-performance-separated__all-no-change-and-multiple-change-rows-retained__carrier-states-label-change-direction-and-ownership-trace

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-CHEM-POLYMER-DISTRIBUTION-001
- SFT-CHEM-POLYMER-CONFORMATION-SIZE-010
- SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013

Generated grammar. Generate the literal product of the eight frozen POLY-011 decisions.

Boundary: The eight POLY-011 dimensions exhaust the registered carrier, operation, boundary and certificate distinctions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *complete-polymer-molecular-carrier__ordered-complete-condition-state-record__held-molecular-phase-label-change__composition-and-architecture-retained__transition-direction-and-hysteresis-held__chemistry-transition-materials-performance-separated__all-no-change-and-multiple-change-rows-retained__carrier-states-label-change-direction-and-ownership-trace*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-polymer-molecular-carrier	material-name-only	material-name-only closes a distinction required by carrier.
states	ordered-complete-condition-state-record	single-transition-temperature	single-transition-temperature closes a distinction required by states.
phase	held-molecular-phase-label-change	scalar-property-change-only	scalar-property-change-only closes a distinction required by phase.
composition	composition-and-architecture-retained	composition-erased	composition-erased closes a distinction required by composition.
direction	transition-direction-and-hysteresis-held	heating-cooling-conflated	heating-cooling-conflated closes a distinction required by direction.
boundary	chemistry-transition-materials-performance-separated	bulk-performance-imported	bulk-performance-imported closes a distinction required by boundary.
adverse	all-no-change-and-multiple-change-rows-retained	no-change-row-discarded	no-change-row-discarded closes a distinction required by adverse.
certificate	carrier-states-label-change-direction-and-ownership-trace	temperature-only-output	temperature-only-output closes a distinction required by certificate.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : bfd408f183e2beb5273964a35f0acc4ada659c5d1555cca31ad8782e0c72d91b.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : f02734b220e4e88a6e374b08ec110e05d3a07d997774b2790df75a122cfa818f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 3185e1490e8d5b2112a460d29920dd244e2f9cf2521e8f0053b2f219a338a091.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 0528a48572747baa8cd84660cd4b6c79f656ec3bf5998a3776acc7d5bb07e9a8.

Independent reconstruction. Implementation hash

sha256 : e0383d619e1a0b905be3d5b3d2ffafcf010c3b8a45880408adbbab57ade06cdd08; independent certificate sha256 : 3fa99e25cb6851beabd4ada9765bfd694af49f9dd48d6ddacdbeb98b4fe23276; external-validation hash sha256 : 0634d5210b6a6d32717a248af0bc0dafd3268d44745b24f920772ac0e652650c. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-PVOH-IONIC-LIQUID-GELATION
- NIST-CROSSLINKED-PHOTOPOLYMER-CONVERSION

Comparison records:

- complete 21-artifact, 28,928,563-byte and 279-page Polymer Chemistry source surface retained
- all eight separately registered claim targets reconstructed

- first extraction failure retried by full rendered-page OCR; the original failure remains in the evidence trail
- the one irreconcilable printed source-table arithmetic cell is retained as an external-source defect and never treated as an SFT failure
- all measured values units uncertainties corrections method disagreements favorable adverse absent unavailable inconsistent and unresolved rows remain downstream provenance

Falsification boundary: POLY-011 halts if its survivor is nonunique; a registered target, source byte, measured value, unit, uncertainty, adverse, absent, unavailable, inconsistent or unresolved row is omitted; a source outcome selects the native law; or the reconstruction retry is erased.

Measurement receipt: sha256:554602583cee2e8f57f89c9d6e5f513a28d8c6dc4f610e823ab05c7d8d5c2979.

Isolation certificate: sha256:d2d584a451b5011150ad1d26b828149115f00e786aed05022c282a8f9bde4c92.

Custody certificate: sha256:269cc449fcbfb53038fed37442774edeb6933d640308a7d84fbbe08c6f92c629.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured polymer value equation distribution trend or favorable row selects the survivor
- no first extraction failure source defect adverse row or unresolved record retires an obligation

Immutable identities. Source manifest

sha256:3d03b605cc7d1fb35095c0048dd51c353527575388681810ddb9f48a0c32bb4c; derivation seal

sha256:07602f4b96716c7fe13def030e91392a32d24150591b96b9919ec5da45fa040f; engine receipt

sha256:ee2dba715ca2a3b640a1e2c95156ff25817ad663a1a5f5e243760e2882f5f701 at receipts/engine/model_admitted/SFT-CHEM-POLYMER-PHASE-TRANSITION-011-ee2dba715ca2a3b6.json; empirical hash

sha256:19ec6346312069f517aaa3687b952f006efdc4337e32a0302eb97ce1fa02ba58; measurement receipt

sha256:554602583cee2e8f57f89c9d6e5f513a28d8c6dc4f610e823ab05c7d8d5c2979.

262. Fold polymer degradation and depolymerization-network law

Claim identity: SFT-CHEM-POLYMER-DEGRADATION-NETWORK-012

Question and theorem. Polymer degradation is the complete finite network of held scission, transfer, unzipping, crosslink and product transitions in which every repeat-unit carrier is retained in a terminal fragment or an explicitly released product.

complete-held-polymer-source-carrier__registered-scission-transfer-unzip-and-crosslink-transitions__all-simple-generated-product-paths__complete-terminal-fragment-identities__explicit-released-product-carriers__exact-repeat-unit-carrier-conservation__time-temperature-environment-boundary-retained__source-moves-paths-products-balance-and-condition-trace

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001

- SFT-CHEM-POLYMER-CHAIN-001
- SFT-CHEM-POLYMER-DISTRIBUTION-001
- SFT-CHEM-POLYMER-PHASE-TRANSITION-011
- SFT-CHEM-REACTION-GRAPH-GENERATION-007

Generated grammar. Generate the literal product of the eight frozen POLY-012 decisions.

Boundary: The eight POLY-012 dimensions exhaust the registered carrier, operation, boundary and certificate distinctions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-held-polymer-source-carrier__registered-scission-transfer-unzip-and-crosslink-transitions__all-simple-generated-product-paths__complete-terminal-fragment-identities__explicit-released-product-carriers__exact-repeat-unit-carrier-conservation__time-temperature-environment-boundary-retained__source-moves-paths-products-balance-and-condition-trace`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
source	complete-held-polymer-source-carrier	mass-loss-curve-only	mass-loss-curve-only closes a distinction required by source.
moves	registered-scission-transfer-unzip-and-crosslink-transitions	one-global-degradation-rate	one-global-degradation-rate closes a distinction required by moves.
paths	all-simple-generated-product-paths	dominant-product-path	dominant-product-path closes a distinction required by paths.
fragments	complete-terminal-fragment-identities	unidentified-residue	unidentified-residue closes a distinction required by fragments.
released	explicit-released-product-carriers	lost-mass-placeholder	lost-mass-placeholder closes a distinction required by released.
balance	exact-repeat-unit-carrier-conservation	approximate-mass-closure	approximate-mass-closure closes a distinction required by balance.
conditions	time-temperature-environment-boundary-retained	condition-free-mechanism	condition-free-mechanism closes a distinction required by conditions.
certificate	source-moves-paths-products-balance-and-condition-trace	remaining-mass-only	remaining-mass-only closes a distinction required by certificate.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1526a6c8a0fb9db8068079d6ea54578211c0b3f364307af38c00071124fd498b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:5d81b04510a775d1616b5419003f40b76a2d1d91917b9ddb543deb59bbcb3db`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:91e98902a405116f6fb06693c10de9d9561fedcc313607f33e50ac769a0e8745`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:54f937729db5f9ed62f949526e089ed387ad8f7f8856e93edbab59031158b479`.

Independent reconstruction. Implementation hash

sha256:9e9e528b381670b1efb975425ef54a558218217bf9ee9d1c05ced38276b52dbf; independent certificate sha256:3ed9a7e3742ee727d7a953e0d63eea79730aa605aad40a836c945439475b202d; external-validation hash sha256:e20327bcb3eb91419ddf3157423f1d7a76968b40a71ca3169068d0d3bed298f. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-POLYMER-PYROLYSIS-NETWORK
- NIST-POLYMER-DEPOLYMERIZATION-KINETICS

Comparison records:

- complete 21-artifact, 28,928,563-byte and 279-page Polymer Chemistry source surface retained
- all eight separately registered claim targets reconstructed
- first extraction failure retried by full rendered-page OCR; the original failure remains in the evidence trail
- the one irreconcilable printed source-table arithmetic cell is retained as an external-source defect and never treated as an SFT failure
- all measured values units uncertainties corrections method disagreements favorable adverse absent unavailable inconsistent and unresolved rows remain downstream provenance

Falsification boundary: POLY-012 halts if its survivor is nonunique; a registered target, source byte, measured value, unit, uncertainty, adverse, absent, unavailable, inconsistent or unresolved row is omitted; a source outcome selects the native law; or the reconstruction retry is erased.

Measurement receipt: sha256:69266dc0225989f31751f67573a674e256afb0f954be2a5ffbefbfcd3211b77a.

Isolation certificate: sha256:57df9fa47d46b476d4e1cc5285cc69dfb2a7c408b59228e91698c36f3cc4b6c7.

Custody certificate: sha256:455565e0748159c87b4e6726c30453c93bc6017403440b6c5e5fdd7baa94c548.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured polymer value equation distribution trend or favorable row selects the survivor
- no first extraction failure source defect adverse row or unresolved record retires an obligation

Immutable identities. Source manifest

sha256:f5880957589f224f8bda55a7b460688cc32a41c9efe362275a4d33db53acfb80; derivation seal

sha256:a32d8d6eaa063fc2124627b06e2d5d358621e2dffee916991b6d0c03f7a74ad0; engine receipt

sha256:bcc3b5f0f762be039c57e36ca19b4b76640c911843e2217d0b54ff303ca31f09 at receipts/engine/model_admitted/SFT-CHEM-POLYMER-DEGRADATION-NETWORK-012-bcc3b5f0f762be03.json; empirical

hash sha256:f5681e92edf69f8968747f505180756d6aac8f74e61fe940b6e2310c8bca8ee0; measurement

receipt sha256:69266dc0225989f31751f67573a674e256afb0f954be2a5ffbefbfcd3211b77a.

263. Fold polymer Chemistry-to-Materials handoff law

Claim identity: SFT-CHEM-POLYMER-MATERIALS-HANDOFF-013

Question and theorem. The Chemistry-to-Materials handoff pairs one complete molecular architecture and state record with one separately owned bulk-property record; neither branch duplicates nor selects the other's law.

complete-chemistry-architecture-and-state-record__complete-material-sample-and-property-record__one-exact-source-bound-paired-record__one-owner-per-law-and-result__chemistry-carrier-precedes-material-response__new-evidence-opens-separate-lawful-extension__missing-side-mandatory-halt__both-records-owners-dependencies-and-boundary-trace

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-CHEM-POLYMER-DISTRIBUTION-001
- SFT-CHEM-POLYMER-DEGRADATION-NETWORK-012
- SFT-MAT-CLASS-POLYMER-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MECH-STRESS-STRAIN-001
- SFT-MAT-PHASE-TRANSITION-001

Generated grammar. Generate the literal product of the eight frozen POLY-013 decisions.

Boundary: The eight POLY-013 dimensions exhaust the registered carrier, operation, boundary and certificate distinctions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *complete-chemistry-architecture-and-state-record__complete-material-sample-and-property-record__one-exact-source-bound-paired-record__one-owner-per-law-and-result__chemistry-carrier-precedes-material-response__new-evidence-opens-separate-lawful-extension__missing-side-mandatory-halt__both-records-owners-dependencies-and-boundary-trace*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
chemistry	complete-chemistry-architecture-and-state-record	polymer-name-only	polymer-name-only closes a distinction required by chemistry.
materials	complete-material-sample-and-property-record	bulk-value-without-carrier	bulk-value-without-carrier closes a distinction required by materials.
pairing	one-exact-source-bound-paired-record	unlinked-branch-records	unlinked-branch-records closes a distinction required by pairing.
ownership	one-owner-per-law-and-result	duplicated-property-law	duplicated-property-law closes a distinction required by ownership.
direction	chemistry-carrier-precedes-material-response	materials-selects-chemistry	materials-selects-chemistry closes a distinction required by direction.
feedback	new-evidence-opens-separate-lawful-extension	performance-retrofits-molecular-law	performance-retrofits-molecular-law closes a distinction required by feedback.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
adverse	missing-side-mandatory-halt	unpaired-record-silently-used	unpaired-record-silently-used closes a distinction required by adverse.
certificate	both-records-owner-s-dependencies-and-boundary-trace	cross-branch-link-only	cross-branch-link-only closes a distinction required by certificate.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : ad8c77bed20a0bb7f72bff8ea4b9929bf964d2e1e22f2caa29cc1526ced30eed.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 1a10a18c5c0e39253d3bef8b4b5801a89a8b944f34525674aec645dd5af4e7e5.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : f033ef8de77fda18425a557f5092be2de1b00fe2afbafa884915c8bce1d7f343.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : cc43f5f91e9b620ec78d1655fd2ccd58a54159c08d4d86506278ffad83aebb4c.

Independent reconstruction. Implementation hash

sha256 : 16b91b8f207ef0578d4038d6ba641ad01cb08c3969558953dff619cc1347897b; independent certificate
sha256 : 1a35581923eb147e8005813e6147bf6b4db8f22eccec567464b74ddce2cb5ba49; external-validation
hash sha256 : 319e0ad504fafefbfca2999c750276b4e682d3acd1364e8bc8de2815471294b31. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-CROSSLINKED-PHOTOPOLYMER-CONVERSION
- NIST-POLYMER-PROCESSING-METROLOGY

Comparison records:

- complete 21-artifact, 28,928,563-byte and 279-page Polymer Chemistry source surface retained
- all eight separately registered claim targets reconstructed
- first extraction failure retried by full rendered-page OCR; the original failure remains in the evidence trail
- the one irreconcilable printed source-table arithmetic cell is retained as an external-source defect and never treated as an SFT failure
- all measured values units uncertainties corrections method disagreements favorable adverse absent unavailable inconsistent and unresolved rows remain downstream provenance

Falsification boundary: POLY-013 halts if its survivor is nonunique; a registered target, source byte, measured value, unit, uncertainty, adverse, absent, unavailable, inconsistent or unresolved row is omitted; a source outcome selects the native law; or the reconstruction retry is erased.

Measurement receipt: sha256 : 12bbc5ecfab8f29b66662e346394a8d4f237636b628e6f34cec64797120750f1.
Isolation certificate: sha256 : dc9827a6e9ae54fe092d927b192ab137b4de6b98fdd2e51651dbc318f9dce1a2.
Custody certificate: sha256 : febfb963bc13687b5512ca609f0d6f2e2486bd05e939349908d6be53a6d30e340.

Registered source descriptions:

- None: ()

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no numerical zero negative irrational imaginary continuum fitted free random or imported native parameter
- no measured polymer value equation distribution trend or favorable row selects the survivor
- no first extraction failure source defect adverse row or unresolved record retires an obligation

Immutable identities. Source manifest

sha256:c3d996b05fc843ac202502f292f40a0e50c3444b3a753f172a142819fa79e14b; derivation seal
 sha256:b71633fcb9034e84f294f1e926ed6bbab313674d4c80720b6bd0d2a47afc7180; engine receipt
 sha256:a69103e6715074cfdb9843ef831a9ad775fc5f19039bb7b3a761e052361a1b07 at receipts/engine/model_admitted/SFT-CHEM-POLYMER-MATERIALS-HANDOFF-013-a69103e6715074cf.json; empirical
 hash sha256:0ebb11a036b550dc14fda49654478d465a678b55ae43cfa58c80a68a87e7a42e; measurement
 receipt sha256:12bbc5ecfab8f29b66662e346394a8d4f237636b628e6f34cec64797120750f1.

29. Complete external-validation vectors and Chemistry Grand Lock - 12 admitted claims

The complete dated evidence surface preserves every receipt, closure class, source, result class and formal boundary without aggregate scoring. This entire registered subcategory is included; no proper subset is presented as field completion.

264. Substantial molecular geometry validation vector

Claim identity: SFT-CHEM-VALIDATION-MOLECULAR-GEOMETRY-VECTOR-001

Question and theorem. VALID-001 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 14 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-001 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 14 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-BOND-LENGTH-STRENGTH-001
- SFT-CHEM-COORDINATION-GEOMETRY-HELD-ORIENTATION-004
- SFT-CHEM-COORDINATION-ISOMERISM-EQUIVALENCE-005
- SFT-CHEM-COORDINATION-NUMBER-INCIDENCE-COUNT-002
- SFT-CHEM-DEFECT-NONSTOICHIOMETRY-016
- SFT-CHEM-DIHEDRAL-TORSIONAL-STATE-004
- SFT-CHEM-ELECTRON-NEUTRON-DIFFRACTION-017
- SFT-CHEM-EQUILIBRIUM-BOND-LENGTH-001
- SFT-CHEM-LIGAND-DENTICITY-CHELATION-TOPOLOGY-003
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOLECULAR-BOND-ANGLE-003
- SFT-CHEM-MULTIMODAL-MOLECULAR-IDENTITY-021
- SFT-CHEM-SOLID-STATE-LOCAL-COORDINATION-015
- SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016

Generated grammar. Generate the complete eight-axis VALID-001 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 14 value-free claim identities registered for Chemistry VALID-001, their current receipt-bound evidence and all result classes.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-receipt-bound-membership__complete-valid-001-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
authority	complete-receipt-bound-membership	selected-headline	A headline omits the evidence surface.
relation	complete-valid-001-typed-evidence-vector	aggregate-score-or-fit	A score or fit can hide distinct failures.
custody	sealed-law-then-capability-closed-release	outcomes-in-forcing	Outcome access may select the law.
results	favourable-adverse-absent-unavailable-unresolved	favourable-only	Selective evidence invalidates the vector.
closure	each-declared-closure-scope-held	single-universal-scope	That overwrites finite boundaries.
formal	explicit-formal-standing-boundary	invented-empirical-successes	That fabricates data.
controls	omission-tamper-and-boundary-halt	omission-tolerated	A missing row cannot close a vector.
extension	dated-complete-extension-open	permanent-lock	A dated vector cannot bar discovery.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256: f3c1e021125c5f0769bde21e4e4682bcbe768ee42c1bd1b20e44ba5e63cda28d.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256: 783740ba6603fb309ea0a4efa49547b2def9c96cbd22c60ea5c7b6a8248f80e0.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 5fdbab30ecc3d91161ce4f628af43731e143872f1beb796a403330dedc434a73.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256: b97a5ce76480aff38b9a75a8ee9f63bfd86bbd0b7c0d4e9fb06270a6a3e73489.

Independent reconstruction. Implementation hash

sha256: ebb766680add8ad494e3d447edc51e205fdb57736b98d14ebd76d6737d57ea34; independent certificate
sha256: e8ae411787bb257e9ee1bb27af36a14a3fd53e75370fd69e95bb3eeca2de7790; external-validation
hash sha256: cf86f00c20bb7ca4a1d02c577071cc7d693be767457638c9dc2cf74c46276560. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-B00702-2026
- IUPAC-GOLD-BOOK-C01332
- NIST-CCCBDB-COMPLETE-EXPERIMENTAL-GEOMETRY
- IUPAC-GOLD-BOOK-I03294
- IUPAC-GOLD-BOOK-C01093
- IUPAC-GOLD-BOOK-E02069
- IUPAC-RED-BOOK-2005
- IUPAC-GOLD-BOOK-C01331
- NIST-CCCBDB-SRD101-SCF3-TICL4-FECO5
- IUPAC-S05735
- IUPAC-13807
- IUPAC-09464
- IUPAC-14011
- IUPAC-14026
- NIST-CCCBDB-SRD101-ETHANOL-EXPERIMENTAL-ROTATIONAL-BARRIER
- NIST-ELECTRON-DIFFRACTION-DATABASE-REPORT
- NIST-NEUTRON-SCATTERING-LENGTHS
- NIST-SRM-676A-DIFFRACTION-STANDARD
- NIST-CODATA-2022-ALL-CONSTANTS
- NIST-WEBBOOK-SRD69-H2-DIATOMIC-CONSTANTS
- NIST-WEBBOOK-SRD69-D2-DIATOMIC-CONSTANTS
- IUPAC-GOLD-BOOK-B00649
- IUPAC-GOLD-BOOK-C01012
- IUPAC-GOLD-BOOK-D01594
- IUPAC-GOLD-BOOK-H01881
- IUPAC-GOLD-BOOK-K03366
- IUPAC-GOLD-BOOK-L03518
- IUPAC-GOLD-BOOK-S06061-2026
- NIST-CCCBDB-SRD101-BF3-EXPERIMENTAL-GEOMETRY
- NIST-CCCBDB-SRD101-XEF2-EXPERIMENTAL-GEOMETRY
- NIST-CCCBDB-SRD101-XEF4-EXPERIMENTAL-GEOMETRY
- NIST-WEBBOOK-BENZENE-MULTIMODAL
- NIST-WEBBOOK-ACETONE-MULTIMODAL
- NIST-WEBBOOK-CYCLOHEXANE-MULTIMODAL
- SFT-V3-ANAL-001-011-IMMUTABLE-EVIDENCE
- IUPAC-12823

- NIST-SRM-674-XRAY-INTENSITY-SET

Comparison records:

- VALID-001: complete registered membership count 14
- complete Chemistry base: 263 claims; 255 empirical; 8 formal-only
- retained evidence: 26486 measurement/evidence lines and 1177 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-CHEM-BOND-LENGTH-STRENGTH-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-COORDINATION-GEOMETRY-HELD-ORIENTATION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=10; explicit-status-lines=3
- SFT-CHEM-COORDINATION-ISOMERISM-EQUIVALENCE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=14; explicit-status-lines=1
- SFT-CHEM-COORDINATION-NUMBER-INCIDENCE-COUNT-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=29; explicit-status-lines=1
- SFT-CHEM-DEFECT-NONSTOICHIOMETRY-016: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=13; explicit-status-lines=2
- SFT-CHEM-DIHEDRAL-TORSIONAL-STATE-004: source-custodied-post-seal-comparison; closure=finite_complete; sources=1; evidence-lines=76; explicit-status-lines=14
- SFT-CHEM-ELECTRON-NEUTRON-DIFFRACTION-017: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-EQUILIBRIUM-BOND-LENGTH-001: source-custodied-post-seal-comparison; closure=finite_complete; sources=3; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-LIGAND-DENTICITY-CHELATION-TOPOLOGY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=6; evidence-lines=30; explicit-status-lines=1
- SFT-CHEM-MOL-GEOMETRY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MOLECULAR-BOND-ANGLE-003: source-custodied-post-seal-comparison; closure=finite_complete; sources=3; evidence-lines=12; explicit-status-lines=8
- SFT-CHEM-MULTIMODAL-MOLECULAR-IDENTITY-021: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-SOLID-STATE-LOCAL-COORDINATION-015: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=11; explicit-status-lines=2
- SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2

Falsification boundary: Reject if any registered member, receipt, control, evidence file, result class, closure scope, formal-only boundary or source-custody row is missing, changed, duplicated or selected after outcome access.

Measurement receipt: sha256: ca76452f3008133651efc69a9d2a62fb1215a00d7b6f419efd07e103a263c6ba.

Isolation certificate: sha256: 8b99fd1bc221a493bc558e196880c2ae2a590761874c2f88ed4190497bdc8842.

Custody certificate: sha256: bad44b7bd856c74d48baca211529bd0daa9d5abcb1312077785e251835588893.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favorable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

Immutable identities. Source manifest

sha256:616538f4b3ef3ac8aa04545736de9066e7405823346ab2f47be25a1fee02297a; derivation seal sha256:4ef6f495cc88346c5ebe75270e6a3fa2b114c0e54404f10317377fae305722ad; engine receipt sha256:3fe068e9578065309505f4bd2b6b10c86331611573781be3c8fa75650ffa6f32 at receipts/engine/model_admitted/SFT-CHEM-VALIDATION-MOLECULAR-GEOMETRY-VECTOR-001-3fe068e957806530.json; empirical hash sha256:aa950281cd4d3962edfa70c4a4cc92ac7300ddfbbed13041e40cfe3b07dcca3c; measurement receipt sha256:ca76452f3008133651efc69a9d2a62fb1215a00d7b6f419efd07e103a263c6ba.

265. Substantial thermochemical validation vector

Claim identity: SFT-CHEM-VALIDATION-THERMOCHEMICAL-VECTOR-002

Question and theorem. VALID-002 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 13 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-002 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 13 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-BOND-DISSOCIATION-ENERGY-002
- SFT-CHEM-CHEMICAL-POTENTIAL-EQUIVALENT-COMPONENT-008
- SFT-CHEM-CROSS-PROPERTY-MOLECULAR-VECTOR-014
- SFT-CHEM-ENTHALPY-EQUIVALENT-STATE-006
- SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007
- SFT-CHEM-HEAT-WORK-TRANSFER-PARTITION-004
- SFT-CHEM-INTERNAL-ENERGY-COMPOSITION-003
- SFT-CHEM-MOLECULAR-FORMATION-ENERGY-013
- SFT-CHEM-MOLECULAR-IONIZATION-ENERGY-007
- SFT-CHEM-SOLVATION-DISSOLUTION-FREE-ORDER-015
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-VALIDATION-MOLECULAR-GEOMETRY-VECTOR-001

Generated grammar. Generate the complete eight-axis VALID-002 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 13 value-free claim identities registered for Chemistry VALID-002, their current receipt-bound evidence and all result classes.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-receipt-bound-membership__complete-valid-002-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
authority	complete-receipt-bound-membership	selected-headline	A headline omits the evidence surface.
relation	complete-valid-002-typed-evidence-vector	aggregate-score-or-fit	A score or fit can hide distinct failures.
custody	sealed-law-then-capability-closed-release	outcomes-in-forcing	Outcome access may select the law.
results	favourable-adverse-absent-unavailable-unresolved	favourable-only	Selective evidence invalidates the vector.
closure	each-declared-closure-scope-held	single-universal-scope	That overwrites finite boundaries.
formal	explicit-formal-standing-boundary	invented-empirical-successes	That fabricates data.
controls	omission-tamper-and-boundary-halt	omission-tolerated	A missing row cannot close a vector.
extension	dated-complete-extension-open	permanent-lock	A dated vector cannot bar discovery.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256: 065256c9b7b4bbd465a467e4196d419e7e732fbb2af782471e94b4d6a7ee2ac0.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256: 2e9ae016709d5e626303c00e6ec7b5a059ce9682ec3bf93e8a96f08e6a2f804a.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 8f886e569e4942deb86716f3cd4ee6153a4ef3999f3f671914897de52670dd89.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256: 6a4c4898592e4d89620397600708b461aa9cd4ca6094f3ca092fd8c858ca62be.

Independent reconstruction. Implementation hash

sha256: ebb766680add8ad494e3d447edc51e205fdb57736b98d14ebd76d6737d57ea34; independent certificate
sha256: daf856d606b560788c249eaa6cc747817d6b1306cac44a6329e47a179aa8bfb6; external-validation
hash sha256: 7586149b603a7f2dec36a2e33f0e3f47c49daa939879c241c721ea08844b5288. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- APS-PRA-49-2460-1994

- PRL-UDEM-H-1S2S-1997
- PRL-PARTHEY-HD-1S2S-SHIFT-2010
- SI-DEFINING-LIGHT-SPEED
- JCP-LIU-H2-DISSOCIATION-2009
- PRA-HUSSELS-D2-DISSOCIATION-2022
- NIST-TRC-THERMOML-JCED-2019-9B00414
- SFT-V3-ADMITTED-PROP-001
- SFT-V3-ADMITTED-PROP-002
- SFT-V3-ADMITTED-PROP-003
- SFT-V3-ADMITTED-PROP-004
- SFT-V3-ADMITTED-PROP-005
- SFT-V3-ADMITTED-PROP-006
- SFT-V3-ADMITTED-PROP-007
- SFT-V3-ADMITTED-PROP-008
- SFT-V3-ADMITTED-PROP-009
- SFT-V3-ADMITTED-PROP-010
- SFT-V3-ADMITTED-PROP-011
- SFT-V3-ADMITTED-PROP-012
- SFT-V3-ADMITTED-PROP-013
- NIST-CHEMISTRY-WEBBOOK-SRD69-WATER-FLUID-PROPERTIES
- NIST-JANAF-N-007-NO2-GAS
- NIST-JANAF-N-032-N2O4-GAS
- NIST-CCCBDB-SRD-101-COMPLETE-EXPERIMENTAL-FORMATION-ENERGY
- NIST-CCCBDB-THERMODYNAMIC-REFERENCE-STATES
- NIST-WEBBOOK-SRD69-GAS-PHASE-ION-THERMOCHEMISTRY
- NIST-CCCBDB-SRD101-EXPERIMENTAL-IONIZATION-ENERGY
- FREESOLV-0.52-COMMIT-6C7D19B
- NIST-TRC-THERMOML-JCED-2016-61-1470-1476
- NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- IUPAC-GOLD-BOOK-REACTION-DIRECTION-COMPOSITE-2026
- IUPAC-GOLD-BOOK-REACTION-THERMOCHEMISTRY-COMPOSITE-2026

Comparison records:

- VALID-002: complete registered membership count 13
- complete Chemistry base: 263 claims; 255 empirical; 8 formal-only
- retained evidence: 26486 measurement/evidence lines and 1177 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-CHEM-BOND-DISSOCIATION-ENERGY-002: source-custodied-post-seal-comparison; closure=finite_complete; sources=6; evidence-lines=11; explicit-status-lines=7

- SFT-CHEM-CHEMICAL-POTENTIAL-EQUIVALENT-COMPONENT-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=83; explicit-status-lines=0
- SFT-CHEM-CROSS-PROPERTY-MOLECULAR-VECTOR-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=13; evidence-lines=9033; explicit-status-lines=0
- SFT-CHEM-ENTHALPY-EQUIVALENT-STATE-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=22; explicit-status-lines=0
- SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=72; explicit-status-lines=0
- SFT-CHEM-HEAT-WORK-TRANSFER-PARTITION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=22; explicit-status-lines=1
- SFT-CHEM-INTERNAL-ENERGY-COMPOSITION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=21; explicit-status-lines=0
- SFT-CHEM-MOLECULAR-FORMATION-ENERGY-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=2106; explicit-status-lines=0
- SFT-CHEM-MOLECULAR-IONIZATION-ENERGY-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=13; explicit-status-lines=1
- SFT-CHEM-SOLVATION-DISSOLUTION-FREE-ORDER-015: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=810; explicit-status-lines=1
- SFT-CHEM-STATE-ENERGY-ORDER-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=313; explicit-status-lines=6
- SFT-CHEM-THERMO-DIRECTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-THERMO-REACTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1

Falsification boundary: Reject if any registered member, receipt, control, evidence file, result class, closure scope, formal-only boundary or source-custody row is missing, changed, duplicated or selected after outcome access.

Measurement receipt: sha256:6c495c2158b9d083e5de963b1d95f1b1ef06ee73e08f7dd4299dafbb1f03b3c1.

Isolation certificate: sha256:91be8b38a57b73b1d5fde0f744eec5ac34c588d92ce95369017b6c4580c4c35c.

Custody certificate: sha256:162f38248feef2c80c5973a9f41b39630f9cd562d4245e7cf50094d6c47667a1.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favorable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

Immutable identities. Source manifest

sha256:c6b84b886662b550c1dbc54eeab8178d8069a4eebd6ecc97c17d00c40fe61865; derivation seal
 sha256:4513f834d1eb8eaeabab8da177a58ebe2a03c7d8ae8e900c174cdb3f9956494f4; engine receipt
 sha256:1b3d75c0246ea717bcd61969f05ed282f94f8eeb0d24e8b3de03092c828693ad at receipts/engine/model_admitted/SFT-CHEM-VALIDATION-THERMOCHEMICAL-VECTOR-002-1b3d75c0246ea717.json;
 empirical hash sha256:338d5d1105de99afce3d9b4865a7f9d22fa38f31417c8a859debf5cdd7fa23fb;
 measurement receipt sha256:6c495c2158b9d083e5de963b1d95f1b1ef06ee73e08f7dd4299dafbb1f03b3c1.

266. Substantial equilibrium validation vector

Claim identity: SFT-CHEM-VALIDATION-EQUILIBRIUM-VECTOR-003

Question and theorem. VALID-003 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 18 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-003 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 18 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-ACTIVITY-NONIDEAL-COMPOSITION-009
- SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003
- SFT-CHEM-CHEMICAL-POTENTIAL-EQUIVALENT-COMPONENT-008
- SFT-CHEM-COLLIGATIVE-COMPOSITION-RESPONSE-014
- SFT-CHEM-CONCENTRATION-DEPENDENT-POTENTIAL-004
- SFT-CHEM-ELECTRODE-POTENTIAL-CHEMICAL-RELATION-002
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-EQUILIBRIUM-BOND-LENGTH-001
- SFT-CHEM-EQUILIBRIUM-ISOTOPE-FRACTIONATION-007
- SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007
- SFT-CHEM-FUGACITY-EQUIVALENT-GAS-MIXTURE-010
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013
- SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011
- SFT-CHEM-RADIOCHEMICAL-EQUILIBRIUM-005
- SFT-CHEM-REVERSIBLE-KINETIC-EQUILIBRIUM-CORRESPONDENCE-009
- SFT-CHEM-SOLUTION-EQUILIBRIUM-001
- SFT-CHEM-SOLVATION-DISSOLUTION-FREE-ORDER-015
- SFT-CHEM-VALIDATION-THERMOCHEMICAL-VECTOR-002

Generated grammar. Generate the complete eight-axis VALID-003 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 18 value-free claim identities registered for Chemistry VALID-003, their current receipt-bound evidence and all result classes.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: complete-receipt-bound-membership__complete-valid-003-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-comp

lete-extension-open. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
authority	complete-receipt-bound-membership	selected-headline	A headline omits the evidence surface.
relation	complete-valid-003-typed-evidence-vector	aggregate-score-or-fit	A score or fit can hide distinct failures.
custody	sealed-law-then-capability-closed-release	outcomes-in-forcing	Outcome access may select the law.
results	favourable-adverse-absent-unavailable-unresolved	favourable-only	Selective evidence invalidates the vector.
closure	each-declared-closure-scope-held	single-universal-scope	That overwrites finite boundaries.
formal	explicit-formal-standing-boundary	invented-empirical-successes	That fabricates data.
controls	omission-tamper-and-boundary-halt	omission-tolerated	A missing row cannot close a vector.
extension	dated-complete-extension-open	permanent-lock	A dated vector cannot bar discovery.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:8ddf495893b48fef1b9e1240f513b549226ca3baa51c4354fab20bbb3f885e00.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:e3d7be7ba4f4a36be1d76a8919f4257c53536a731abf183454f18fe0b7a87390.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:8a370e80217d5de84303375bddcda9e48e7a22e0b55230f3aee6864c01e39c8e.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:7696b2246d4b3c2b579ecc65141fa107ea19573cc459c0f920d2cdb8a5867b13.

Independent reconstruction. Implementation hash

sha256:ebb766680add8ad494e3d447edc51e205fdb57736b98d14ebd76d6737d57ea34; independent certificate
sha256:b96d07cd8468df04307f50d27c5ac2ef013ce98220a840d4524fbaef5909becb; external-validation
hash sha256:987956f36e0a28e7d405d7959696b6f02e678f82ca50cee43f4d2c4bf7daf9d8. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-TRC-THERMOML-JCED-2019-9B00694
- NIST-JRES-AGCL-COMplete-CELL-1954
- NIST-TRC-THERMOML-JCED-2019-9B00414
- NIST-TRC-THERMOML-FPE-2013-337-60-66
- NIST-TRC-THERMOML-FPE-2016-420-14-19

- NIST-TRC-THERMOML-JCT-2009-41-1439-1445
- NIST-JRES-AGCL-HCL-CONCENTRATION-SERIES-1954
- NIST-JRES-AGCL-STANDARD-POTENTIAL-1954
- IUPAC-GOLD-BOOK-CHEMICAL-EQUILIBRIUM-COMPOSITE-2026
- NIST-CODATA-2022-ALL-CONSTANTS
- NIST-WEBBOOK-SRD69-H2-DIATOMIC-CONSTANTS
- NIST-WEBBOOK-SRD69-D2-DIATOMIC-CONSTANTS
- USGS-PP-440-KK-STABLE-ISOTOPE-FRACTIONATION
- USGS-WRIR-02-4172-ISOTOPE-EQUILIBRIUM-CONSTANTS
- NUCHEM-005-008-SHARED-SOURCES
- NUCHEM-005-008-SOURCE-INVENTORY
- NIST-JANAF-N-007-NO2-GAS
- NIST-JANAF-N-032-N2O4-GAS
- NIST-TRC-THERMOML-FPE-2019-485-145-152
- NIST-TRC-THERMOML-JCT-2012-47-260-266
- IUPAC-GOLDBOOK-P04533-PHASE-RULE
- NIST-PHASE-DIAGRAM-GENERAL-DISCUSSION-GLOSSARY
- NIST-SRM-4239A-STRONTIUM-90-2022
- NUCHEM-005-NIST-SRM-PAIR
- NIST-SRM-4324C-URANIUM-232-2025
- NATURE-COMMUNICATIONS-S41467-023-40190-4-COMplete
- IUPAC-GOLD-BOOK-SOLUTION-EQUILIBRIUM-COMPOSITE-2026
- FREESOLV-0.52-COMMIT-6C7D19B
- NIST-TRC-THERMOML-JCED-2016-61-1470-1476

Comparison records:

- VALID-003: complete registered membership count 18
- complete Chemistry base: 263 claims; 255 empirical; 8 formal-only
- retained evidence: 26486 measurement/evidence lines and 1177 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-CHEM-ACTIVITY-NONIDEAL-COMPOSITION-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=213; explicit-status-lines=1
- SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=7; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-CHEMICAL-POTENTIAL-EQUIVALENT-COMPONENT-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=83; explicit-status-lines=0
- SFT-CHEM-COLLIGATIVE-COMPOSITION-RESPONSE-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=285; explicit-status-lines=0
- SFT-CHEM-CONCENTRATION-DEPENDENT-POTENTIAL-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2

- SFT-CHEM-ELECTRODE-POTENTIAL-CHEMICAL-RELATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=7; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-EQ-CHEMICAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-EQUILIBRIUM-BOND-LENGTH-001: source-custodied-post-seal-comparison; closure=finite_complete; sources=3; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-EQUILIBRIUM-ISOTOPE-FRACTIONATION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=72; explicit-status-lines=0
- SFT-CHEM-FUGACITY-EQUIVALENT-GAS-MIXTURE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=105; explicit-status-lines=0
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=128; explicit-status-lines=0
- SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=25; explicit-status-lines=0
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=27; explicit-status-lines=0
- SFT-CHEM-RADIOCHEMICAL-EQUILIBRIUM-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-REVERSIBLE-KINETIC-EQUILIBRIUM-CORRESPONDENCE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=176; explicit-status-lines=2
- SFT-CHEM-SOLUTION-EQUILIBRIUM-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-SOLVATION-DISSOLUTION-FREE-ORDER-015: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=810; explicit-status-lines=1

Falsification boundary: Reject if any registered member, receipt, control, evidence file, result class, closure scope, formal-only boundary or source-custody row is missing, changed, duplicated or selected after outcome access.

Measurement receipt: sha256:50530cb821f5777a82c363e70211afabbf3d5efcb0abd58658d21b61a20d2862.

Isolation certificate: sha256:cce981abb189c0331737f6320c12a37cb05193dafeab48e0bdc6f673d6239526.

Custody certificate: sha256:2b02d84a17788fc440bf9440c57b885180d8468db111f9003262481da3ffff10.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favorable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude

- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

Immutable identities. Source manifest

sha256:446cae1c547c41ae506bcd5278d1e82f72ebaaf46cc47678836da1baffaeaa5e; derivation seal sha256:aba9550f9e75ce9e3e8faebc8b5e5a2b7fe55790c3d07f9ee133444da45851d5; engine receipt sha256:b9759323b6410152ed69b482f2a2687dd0b4c3835f2075ddcc1ab2a4cd379af7 at receipts/engine/model_admitted/SFT-CHEM-VALIDATION-EQUILIBRIUM-VECTOR-003-b9759323b6410152.json; empirical hash sha256:7780dfc8540e82ef38006b3398c469d9bc90437550c0a855d3d4a2412b07b12b; measurement receipt sha256:50530cb821f5777a82c363e70211afabbf3d5efcb0abd58658d21b61a20d2862.

267. Substantial kinetic validation vector

Claim identity: SFT-CHEM-VALIDATION-KINETIC-VECTOR-004

Question and theorem. VALID-004 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 21 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-004 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 21 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-ACTIVATION-BARRIER-VALUE-RELATION-004
- SFT-CHEM-CATALYTIC-TURNOVER-CYCLE-FREQUENCY-010
- SFT-CHEM-COMPETING-CHANNEL-BRANCHING-006
- SFT-CHEM-CONCENTRATION-DEPENDENCE-RELATION-002
- SFT-CHEM-DIFFUSION-LIMITED-REACTION-BOUNDARY-011
- SFT-CHEM-ELECTRODE-REACTION-RATE-009
- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-ORDER-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-KINETIC-ISOTOPE-EFFECT-RELATION-012
- SFT-CHEM-KINETIC-ISOTOPE-FRACTIONATION-008
- SFT-CHEM-MECHANISM-SEARCH-PROOF-TRACE-009
- SFT-CHEM-PARALLEL-MECHANISM-COMPOSITION-008
- SFT-CHEM-REACTION-DYNAMICS-SCATTERING-PRODUCT-STATE-013
- SFT-CHEM-REVERSIBLE-KINETIC-EQUILIBRIUM-CORRESPONDENCE-009
- SFT-CHEM-RXN-INTERMEDIATE-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007
- SFT-CHEM-TEMPERATURE-DEPENDENCE-RELATION-003
- SFT-CHEM-TRANSITION-STATE-EQUIVALENT-BOUNDARY-005
- SFT-CHEM-VALIDATION-EQUILIBRIUM-VECTOR-003

Generated grammar. Generate the complete eight-axis VALID-004 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 21 value-free claim identities registered for Chemistry VALID-004, their current receipt-bound evidence and all result classes.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-receipt-bound-membership__complete-valid-004-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
authority	complete-receipt-bound-membership	selected-headline	A headline omits the evidence surface.
relation	complete-valid-004-typed-evidence-vector	aggregate-score-or-fit	A score or fit can hide distinct failures.
custody	sealed-law-then-capability-closed-release	outcomes-in-forcing	Outcome access may select the law.
results	favourable-adverse-absent-unavailable-unresolved	favourable-only	Selective evidence invalidates the vector.
closure	each-declared-closure-scope-held	single-universal-scope	That overwrites finite boundaries.
formal	explicit-formal-standing-boundary	invented-empirical-successes	That fabricates data.
controls	omission-tamper-and-boundary-halt	omission-tolerated	A missing row cannot close a vector.
extension	dated-complete-extension-open	permanent-lock	A dated vector cannot bar discovery.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256: 98d67066b44cd4f3dfc8c5e07cb1e51df2036b768d6aeee9e1f737802efe9ec3.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256: a852a2875d67bd44d070d2e298ca5f693a52ef1c472d0e2e691160b7b45ef490.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: f224c629a4ab6d503a9e533038b81d52d444752f3ff8c5f757a43a265bbfd6b1.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256: 44478a499687c412e84e88aae2fd183de533432ec27e6fa7e27d66982be10d9d.

Independent reconstruction. Implementation hash

sha256: ebb766680add8ad494e3d447edc51e205fdb57736b98d14ebd76d6737d57ea34; independent certificate
sha256: ddf1a7a02e1558a4cf00391ff2c610f940ef04786beeeeb4b5bcf7ce10f4c0e5; external-validation
hash sha256: 2c92d6779bb2ae839d46b88f7f735b40650949fca94ea186a105e32b77780b0a. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- NIST-SRD-101-CCCBDB-EXPERIMENTAL-INTERNAL-ROTATION-BARRIERS-COMPLETE

- NATURE-NANOTECHNOLOGY-S41565-021-00959-4-COMPLETE
- COMMUNICATIONS-CHEMISTRY-S42004-024-01239-7-COMPLETE
- RSC-PCCP-2014-C3CP54664K-TABLE-2
- NATURE-COMMUNICATIONS-S41467-025-68008-5-COMPLETE
- NBS-JRES-GALVANIC-COUPLES-1950
- NBS-JRES-IRON-CORROSION-1957
- ECHEM-009-012-SHARED-SOURCES
- ECHEM-009-012-SOURCE-INVENTORY
- NIST-SRD17-1972TIN-KOO3-15
- NIST-SRD17-1967WIL2017-1
- NIST-SRD17-1967AVR-KRA501-503-1
- NIST-SRD17-1962ASH-BUR685-691-1
- IUPAC-GOLD-BOOK-ACTIVATION-COMPOSITE-2026
- IUPAC-GOLD-BOOK-O04322-2026
- IUPAC-GOLD-BOOK-REACTION-RATE-COMPOSITE-2026
- NATURE-COMMUNICATIONS-S41467-024-44753-X-COMPLETE
- NBS-RP729-ELECTROLYTIC-HYDROGEN-OXYGEN-FRACTIONATION-1934
- NUCHEM-005-008-SHARED-SOURCES
- NUCHEM-005-008-SOURCE-INVENTORY
- PUBCHEM-PUG-REST-AUTHORITY
- PUBCHEM-C3H8O-FORMULA-CENSUS
- PUBCHEM-ASPIRIN-CONFORMERS
- PUBCHEM-ASPIRIN-SUBSTRUCTURE
- PUBCHEM-ASPIRIN-SIMILARITY
- CHEBI-REST-AUTHORITY
- RHEA-IMMUTABLE-REACTION-SMILES
- USPTO-IMMUTABLE-REACTION-SURFACE
- NATURE-COMMUNICATIONS-S41467-026-70199-4-SOURCE-DATA
- NATURE-COMMUNICATIONS-S41467-025-66587-X-COMPLETE
- NATURE-COMMUNICATIONS-S41467-023-40190-4-COMPLETE
- IUPAC-GOLD-BOOK-I03096-2026
- IUPAC-GOLD-BOOK-REACTION-MECHANISM-COMPOSITE-2026
- NATURE-COMMUNICATIONS-S41467-024-49814-9-COMPLETE
- RCSB-PDB-8WZF-8WZV
- CXIDB-221
- ACS-JPCA-2015-JP505790M-COMPLETE-TEMPERATURE-RATE-SERIES
- KYRIAKOU-ET-AL-ACS-NANO-NN500703K-COMPLETE

Comparison records:

- VALID-004: complete registered membership count 21
- complete Chemistry base: 263 claims; 255 empirical; 8 formal-only
- retained evidence: 26486 measurement/evidence lines and 1177 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-CHEM-ACTIVATION-BARRIER-VALUE-RELATION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=52; explicit-status-lines=3
- SFT-CHEM-CATALYTIC-TURNOVER-CYCLE-FREQUENCY-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=510; explicit-status-lines=2
- SFT-CHEM-COMPETING-CHANNEL-BRANCHING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=16; explicit-status-lines=2
- SFT-CHEM-CONCENTRATION-DEPENDENCE-RELATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=17; explicit-status-lines=1
- SFT-CHEM-DIFFUSION-LIMITED-REACTION-BOUNDARY-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=264; explicit-status-lines=3
- SFT-CHEM-ELECTRODE-REACTION-RATE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=55; explicit-status-lines=0
- SFT-CHEM-KIN-ACTIVATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-KIN-ORDER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-KIN-RATE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-KINETIC-ISOTOPE-EFFECT-RELATION-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=84; explicit-status-lines=3
- SFT-CHEM-KINETIC-ISOTOPE-FRACTIONATION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-MECHANISM-SEARCH-PROOF-TRACE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-PARALLEL-MECHANISM-COMPOSITION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=39; explicit-status-lines=3
- SFT-CHEM-REACTION-DYNAMICS-SCATTERING-PRODUCT-STATE-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=65; explicit-status-lines=2
- SFT-CHEM-REVERSIBLE-KINETIC-EQUILIBRIUM-CORRESPONDENCE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=176; explicit-status-lines=2
- SFT-CHEM-RXN-INTERMEDIATE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=2
- SFT-CHEM-RXN-MECHANISM-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=28; explicit-status-lines=5
- SFT-CHEM-TEMPERATURE-DEPENDENCE-RELATION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=27; explicit-status-lines=1

- SFT-CHEM-TRANSITION-STATE-EQUIVALENT-BOUNDARY-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=10; explicit-status-lines=1

Falsification boundary: Reject if any registered member, receipt, control, evidence file, result class, closure scope, formal-only boundary or source-custody row is missing, changed, duplicated or selected after outcome access.

Measurement receipt: sha256:d266c3cc932882909d0a272d92e510a3050ae645198dbb673921bfc05b030b61.

Isolation certificate: sha256:6e578573561599aa22339991f6606895d482d519894a3b0fb37eb4053367bbbc.

Custody certificate: sha256:cc3fcc59f9d9cc9daced2070b2efbd14cf2664be2b0bf960586e9f1341b49a31.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favorable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

Immutable identities. Source manifest

sha256:0acce24c269cc2f932cdcf82f6d0348b17f7c5324e0b1764da6c32f636eaf3b1; derivation seal

sha256:1fd9f0c1cdd4347e343a78174c816d4750ace8ba5d801d13edacdfccbe39168e; engine receipt

sha256:662842be7a53f0a99166310a507a1b9da3a2928248ac355ef5f20e89947a6d39 at receipts/engine

/model_admitted/SFT-CHEM-VALIDATION-KINETIC-VECTOR-004-662842be7a53f0a9.json; empirical

hash sha256:5c1ea4beecf02bd079a49565585e7c8a2e70aa94ca4194fd35eb190f27ec022b; measurement

receipt sha256:d266c3cc932882909d0a272d92e510a3050ae645198dbb673921bfc05b030b61.

268. Substantial spectroscopy validation vector

Claim identity: SFT-CHEM-VALIDATION-SPECTROSCOPY-VECTOR-005

Question and theorem. VALID-005 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 26 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-005 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 26 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-ELECTRON-NEUTRON-DIFFRACTION-017
- SFT-CHEM-FLUORESCENCE-YIELD-LIFETIME-010
- SFT-CHEM-INFRARED-LINE-INTENSITY-012
- SFT-CHEM-MASS-ISOTOPE-FRAGMENT-VECTOR-014
- SFT-CHEM-MOLECULAR-MAGNETIC-RESPONSE-012
- SFT-CHEM-MOLECULAR-QUANTUM-MEASUREMENT-REDUCTION-014

- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-NMR-CHEMICAL-SHIFT-006
- SFT-CHEM-NMR-RELAXATION-EXCHANGE-008
- SFT-CHEM-NMR-SPIN-COUPLING-007
- SFT-CHEM-PHOSPHORESCENCE-INTERSYSTEM-011
- SFT-CHEM-RAMAN-TRANSITION-INTENSITY-009
- SFT-CHEM-RESOLVED-ROVIBRONIC-SPIN-COMPOSITION-013
- SFT-CHEM-ROTATIONAL-CONSTANT-010
- SFT-CHEM-ROTATIONAL-SPECTRUM-LINE-015
- SFT-CHEM-SELECTION-RULE-STRUCTURE-010
- SFT-CHEM-SMITHIUM-JOINT-DETECTION-001
- SFT-CHEM-SMITHIUM-SPECTROSCOPIC-CLASSES-001
- SFT-CHEM-SPEC-INFRARED-001
- SFT-CHEM-SPEC-MASS-001
- SFT-CHEM-SPEC-ROT-VIB-001
- SFT-CHEM-SPEC-UVVIS-001
- SFT-CHEM-UVVISIBLE-LINE-INTENSITY-013
- SFT-CHEM-VALIDATION-SMITHIUM-COMPLETE-FAMILY-001
- SFT-CHEM-VIBRATIONAL-FREQUENCY-009
- SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016
- SFT-CHEM-VALIDATION-KINETIC-VECTOR-004

Generated grammar. Generate the complete eight-axis VALID-005 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 26 value-free claim identities registered for Chemistry VALID-005, their current receipt-bound evidence and all result classes.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-receipt-bound-membership__complete-valid-005-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
authority	complete-receipt-bound-membership	selected-headline	A headline omits the evidence surface.
relation	complete-valid-005-typed-evidence-vector	aggregate-score-or-fit	A score or fit can hide distinct failures.
custody	sealed-law-then-capability-closed-release	outcomes-in-forcing	Outcome access may select the law.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
results	favourable-adverse -absent-unavailable -unresolved	favourable-only	Selective evidence invalidates the vector.
closure	each-declared-closure-scope-held	single-universal-scope	That overwrites finite boundaries.
formal	explicit-formal-standing-boundary	invented-empirical-successes	That fabricates data.
controls	omission-tamper-and-boundary-halt	omission-tolerated	A missing row cannot close a vector.
extension	dated-complete-extension-open	permanent-lock	A dated vector cannot bar discovery.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:a3bdc79f5c87d1d83dfd03912b2d5ea560d97d1ac39ee6b44607b02a88a334fd.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:ce817215abcc0335d5990b506775d0a8e9b3820ed565bc4e05fb5239e339e96a.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:0857141ceba146569a75d8b665a8139356857c82fa72aea6aea4ed32e4a1f015.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:0fb95e0658a5f274eb4f4585eea3818f52583ef1e0fc3321d69ded86e35924f6.

Independent reconstruction. Implementation hash

sha256:ebb766680add8ad494e3d447edc51e205fdb57736b98d14ebd76d6737d57ea34; independent certificate sha256:415ae63cbfe79ddc792d78c9bca1142b7209d7adc0656e5d953f69dde6226033; external-validation hash sha256:ce0986b0ef2ff22a9eb1308d13b455939eaf39a63d6af0e3003b3cd329201c2f. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-ELECTRON-DIFFRACTION-DATABASE-REPORT
- NIST-NEUTRON-SCATTERING-LENGTHS
- NIST-SRM-676A-DIFFRACTION-STANDARD
- IUPAC-FLUORESCENCE-STANDARDS-2010
- IUPAC-PHOTOLUMINESCENCE-QUANTUM-YIELD-2011
- NIST-SRM-2941A-FLUORESCENCE-GREEN
- NIST-IR-7458-FLUORESCENCE-GUIDE
- NIST-NLM-PASH-PHOSPHORESCENCE-2018
- ANAL-009-011-COMPLETE-SOURCE-INVENTORY
- NIST-WEBBOOK-SRD69-GUIDE
- NIST-WEBBOOK-BENZENE-MULTIMODAL
- NIST-WEBBOOK-ACETONE-MULTIMODAL
- NIST-WEBBOOK-CYCLOHEXANE-MULTIMODAL
- NIST-MOLECULAR-MICROWAVE-SPECTRAL-DATABASES-SRD-114-115-117

- NIST-JPCRD-3-609-1974-DIATOMIC-MICROWAVE-SPECTRAL-TABLES
- NIST-MDS2-3389-CAH-QUANTUM-STATE-TRACKING-CONTROL-V1.0.1
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- IUPAC-NMR-NOMENCLATURE-2001
- BMRB-ENTRY-68-CHEMICAL-SHIFTS
- ANAL-006-008-NMR-SOURCE-INVENTORY
- BMRB-ENTRY-52365-RELAXATION
- BMRB-ENTRY-27257-H-EXCHANGE
- BMRB-ENTRY-16582-SCALAR-COUPPLINGS
- IUPAC-PHOSPHORESCENCE-LIFETIME
- NIST-RAMAN-RELATIVE-INTENSITY-STANDARDS
- NIST-SRM-2241-RAMAN-785-NM
- NIST-SRM-2242A-RAMAN-532-NM
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-HD-D2-RESOLVED-DIATOMIC-SURFACE-2026
- NIST-CCCBDB-COMPLETE-ROTATIONAL-CONSTANT-SURFACE
- NASA-JPL-MOLECULAR-SPECTROSCOPY-CATALOG
- NASA-JPL-CO-ROTATIONAL-LINE-CATALOG-028001
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- IUPAC-GOLD-BOOK-INFRARED-SPECTROSCOPY-COMPOSITE-2026
- IUPAC-GOLD-BOOK-MASS-SPECTROSCOPY-COMPOSITE-2026
- IUPAC-GOLD-BOOK-ROTATION-VIBRATION-COMPOSITE-2026
- IUPAC-GOLD-BOOK-UVVISIBLE-SPECTROSCOPY-COMPOSITE-2026
- IUPAC-SUPERHEAVY-2018
- IUPAC-PERIODIC-TABLE-2022
- NIST-ATOMIC-SPECTROSCOPY
- NIST-CCCBDB-COMPLETE-DISPLAYED-FUNDAMENTAL-FREQUENCY-SURFACE
- NIST-SRM-674-XRAY-INTENSITY-SET

Comparison records:

- VALID-005: complete registered membership count 26
- complete Chemistry base: 263 claims; 255 empirical; 8 formal-only
- retained evidence: 26486 measurement/evidence lines and 1177 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-CHEM-ELECTRON-NEUTRON-DIFFRACTION-017: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-FLUORESCENCE-YIELD-LIFETIME-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-INFRARED-LINE-INTENSITY-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2

- SFT-CHEM-MASS-ISOTOPE-FRAGMENT-VECTOR-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-MOLECULAR-MAGNETIC-RESPONSE-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=181; explicit-status-lines=1
- SFT-CHEM-MOLECULAR-QUANTUM-MEASUREMENT-REDUCTION-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=345; explicit-status-lines=0
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=80; explicit-status-lines=16
- SFT-CHEM-NMR-CHEMICAL-SHIFT-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-NMR-RELAXATION-EXCHANGE-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-NMR-SPIN-COUPPLING-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-PHOSPHORESCENCE-INTERSYSTEM-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-RAMAN-TRANSITION-INTENSITY-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-RESOLVED-ROVIBRONIC-SPIN-COMPOSITION-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=107; explicit-status-lines=2
- SFT-CHEM-ROTATIONAL-CONSTANT-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3024; explicit-status-lines=1
- SFT-CHEM-ROTATIONAL-SPECTRUM-LINE-015: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-SELECTION-RULE-STRUCTURE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=75; explicit-status-lines=11
- SFT-CHEM-SMITHIUM-JOINT-DETECTION-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SMITHIUM-SPECTROSCOPIC-CLASSES-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SPEC-INFRARED-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-SPEC-MASS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-SPEC-ROT-VIB-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-SPEC-UVVIS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-UVVISIBLE-LINE-INTENSITY-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-VALIDATION-SMITHIUM-COMPLETE-FAMILY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=6; explicit-status-lines=4
- SFT-CHEM-VIBRATIONAL-FREQUENCY-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2016; explicit-status-lines=1

- SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2

Falsification boundary: Reject if any registered member, receipt, control, evidence file, result class, closure scope, formal-only boundary or source-custody row is missing, changed, duplicated or selected after outcome access.

Measurement receipt: sha256:a727ad77935babb5d35289a29d9e87b3dd88eaa117e9a63ed645d23714f3c999.

Isolation certificate: sha256:4d1405fbfda58b8c70e6ad05eb503a1c8fd56035f1f870f623c2c2df158fef4d.

Custody certificate: sha256:25055437085e0dc1164ad2179b87790e73938b717012ebe702d8d2f8e7d3f91f.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favorable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

Immutable identities. Source manifest

sha256:da75ec2f6827f8635e13a446edd0993255d4dfcddf8ac02ba413d040d21166b7; derivation seal

sha256:257523ff8b716f12bfc927d651d4fa3aa1075b667462c9a9b15b00af269a42c1; engine receipt

sha256:9cc81bb257a521fb272f5e4fe832f9cb8beb471edfdabf8303d523b84ab3a270 at receipts/engine/model_admitted/SFT-CHEM-VALIDATION-SPECTROSCOPY-VECTOR-005-9cc81bb257a521fb.json;

empirical hash sha256:754005883b33b82a0fdb7b9b44c4576ee976c8d42074926818fc7f1fc52a1ddd;

measurement receipt sha256:a727ad77935babb5d35289a29d9e87b3dd88eaa117e9a63ed645d23714f3c999.

269. Substantial electrochemical validation vector

Claim identity: SFT-CHEM-VALIDATION-ELECTROCHEMICAL-VECTOR-006

Question and theorem. VALID-006 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 17 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-006 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 17 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003
- SFT-CHEM-CONCENTRATION-DEPENDENT-POTENTIAL-004
- SFT-CHEM-CORROSION-REACTION-NETWORK-012
- SFT-CHEM-DOUBLE-LAYER-INTERFACIAL-CHARGE-011
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-ELECTROCHEMICAL-STORAGE-HANDOFF-013

- SFT-CHEM-ELECTROCHEMICAL-WORK-REACTION-DIRECTION-005
- SFT-CHEM-ELECTRODE-POTENTIAL-CHEMICAL-RELATION-002
- SFT-CHEM-ELECTRODE-REACTION-RATE-009
- SFT-CHEM-ELECTROLYSIS-PRODUCT-AMOUNT-006
- SFT-CHEM-HALF-REACTION-IDENTITY-ORIENTATION-001
- SFT-CHEM-INORGANIC-ACID-BASE-REDOX-NETWORK-017
- SFT-CHEM-IONIC-CONDUCTIVITY-RELATION-007
- SFT-CHEM-IONIC-MOBILITY-TRANSFERENCE-008
- SFT-CHEM-OVERPOTENTIAL-POLARIZATION-010
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-VALIDATION-SPECTROSCOPY-VECTOR-005

Generated grammar. Generate the complete eight-axis VALID-006 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 17 value-free claim identities registered for Chemistry VALID-006, their current receipt-bound evidence and all result classes.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-receipt-bound-membership__complete-valid-006-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open`. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
authority	complete-receipt-bound-membership	selected-headline	A headline omits the evidence surface.
relation	complete-valid-006-typed-evidence-vector	aggregate-score-or-fit	A score or fit can hide distinct failures.
custody	sealed-law-then-capability-closed-release	outcomes-in-forcing	Outcome access may select the law.
results	favourable-adverse-absent-unavailable-unresolved	favourable-only	Selective evidence invalidates the vector.
closure	each-declared-closure-scope-held	single-universal-scope	That overwrites finite boundaries.
formal	explicit-formal-standing-boundary	invented-empirical-successes	That fabricates data.
controls	omission-tamper-and-boundary-halt	omission-tolerated	A missing row cannot close a vector.
extension	dated-complete-extension-open	permanent-lock	A dated vector cannot bar discovery.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:1bada631a5f2c1998344379bc6f66bdc3effadcb15cd02d24f1e50cf9db607d9.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:447793820c04a60f75353fbc5a8f3248954a95837df009d0f9474b0338638599.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ac443e029c8a49bde8f3061acce7c75f1de1dc71a4f7ba0bd0e7e23031d7ee6e.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:90a895443a417c3fdebb62a44c4ecea78b1da9cd21aa6209620dd2d2cc84ef67.

Independent reconstruction. Implementation hash

sha256:ebb766680add8ad494e3d447edc51e205fdb57736b98d14ebd76d6737d57ea34; independent certificate sha256:42dcf322f6cf78b422e3adaee2782676e0b3c329dd8cc4ca87658b8a85315ee1; external-validation hash sha256:908c931ab90ffa9770744d39005c13e0cc7ef229dab6e457fd87a956e67fce9e. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- NIST-JRES-AGCL-COMplete-CELL-1954
- NIST-JRES-AGCL-HCL-CONCENTRATION-SERIES-1954
- NBS-JRES-IRON-CORROSION-1957
- NBS-JRES-GALVANIC-COUPLES-1950
- ECHEM-009-012-SOURCE-INVENTORY
- NIST-GRAPHENE-DOUBLE-LAYER-2020
- ECHEM-011-SOURCE-INVENTORY
- IUPAC-GOLD-BOOK-ELECTROCHEMICAL-CELL-COMPOSITE-2026
- SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003
- SFT-MAT-DEGR-CORROSION-001
- SFT-ENG-REQUIREMENT-001
- SFT-V3-LIVE-CLAIM-CENSUS
- SFT-V3-LIVE-DEPENDENCY-GRAPH
- SFT-CHEM-MAT-PAIRED-RECEIPTS
- NIST-METAL-ADDITIVE-CORROSION
- SFT-V3-BRANCH-OWNERSHIP-BOUNDARY
- NIST-CODATA-2022-ALL-CONSTANTS
- NIST-JRES-AGCL-STANDARD-POTENTIAL-1954
- NIST-CODATA-AGCL-WORK-CROSS-RECONSTRUCTION
- IUPAC-GREEN-BOOK-ELECTROCHEMISTRY-2007
- ECHEM-005-SOURCE-INVENTORY
- ECHEM-009-012-SHARED-SOURCES
- NIST-JRES-SILVER-ELECTROCHEMICAL-EQUIVALENT-1980
- ECHEM-006-SOURCE-INVENTORY

- IUPAC-RT06783
- NIST-JPCRD-STANDARD-ELECTRODE-POTENTIALS-1989
- IUPAC-L03508
- IUPAC-L03511
- IUPAC-L03510
- IUPAC-O04362
- IUPAC-R05222
- NIST-SRM-3190-CONDUCTIVITY-CERTIFICATE
- NIST-SP-260-142-PRIMARY-CONDUCTIVITY
- NIST-SP-260-176-SRM-CATALOG
- ECHEM-007-SOURCE-INVENTORY
- NBS-JRES-TRANSFERENCE-CONCENTRATION-1931
- ECHEM-008-SOURCE-INVENTORY
- IUPAC-GOLD-BOOK-REDOX-COMPOSITE-2026
- IUPAC-GOLD-BOOK-O04365-2026

Comparison records:

- VALID-006: complete registered membership count 17
- complete Chemistry base: 263 claims; 255 empirical; 8 formal-only
- retained evidence: 26486 measurement/evidence lines and 1177 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=7; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-CONCENTRATION-DEPENDENT-POTENTIAL-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-CORROSION-REACTION-NETWORK-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-DOUBLE-LAYER-INTERFACIAL-CHARGE-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ELECTROCHEM-CELL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELECTROCHEMICAL-STORAGE-HANDOFF-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=1
- SFT-CHEM-ELECTROCHEMICAL-WORK-REACTION-DIRECTION-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ELECTRODE-POTENTIAL-CHEMICAL-RELATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=7; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ELECTRODE-REACTION-RATE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ELECTROLYSIS-PRODUCT-AMOUNT-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-HALF-REACTION-IDENTITY-ORIENTATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=6; evidence-lines=4; explicit-status-lines=0

- SFT-CHEM-INORGANIC-ACID-BASE-REDOX-NETWORK-017: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=14; explicit-status-lines=1
- SFT-CHEM-IONIC-CONDUCTIVITY-RELATION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-IONIC-MOBILITY-TRANSFERENCE-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-OVERPOTENTIAL-POLARIZATION-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-REDOX-COUPPLING-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-REDOX-OXIDATION-STATE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1

Falsification boundary: Reject if any registered member, receipt, control, evidence file, result class, closure scope, formal-only boundary or source-custody row is missing, changed, duplicated or selected after outcome access.

Measurement receipt: sha256:b9829ad790bccd9792ba7765a43711d566f970b1c295566e951194ad8003d1ec.

Isolation certificate: sha256:3d2081c68d9be9d7fb8afdcba1a7533e7f892a8ef1e21e16a19ec4ecefcc5a.

Custody certificate: sha256:3adc9356f1d83014cb4563a0e7e01b39f5e1f873bbb6bc36ca2e9a3310c186ec.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favorable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

Immutable identities. Source manifest

sha256:caab3a2049564a7a74032d0131e1bf43ad0278492082d83cc92575cad72ebc13; derivation seal

sha256:2e6f0e3f9126ba35fe178b31790b66c4013620f3c4fd25a973b5bd1a337374e5; engine receipt

sha256:4b0221d819d6fc23faa0cbd3c9c2948f441546b4344f8bdce8b39789dbadb897 at receipts/engine/model_admitted/SFT-CHEM-VALIDATION-ELECTROCHEMICAL-VECTOR-006-4b0221d819d6fc23.json;

empirical hash sha256:cee5c84e633725b6976e0603af9571c1de2f91bbf7b76abdd48d865093930803;

measurement receipt sha256:b9829ad790bccd9792ba7765a43711d566f970b1c295566e951194ad8003d1ec.

270. Substantial inorganic and coordination validation vector

Claim identity: SFT-CHEM-VALIDATION-INORGANIC-COORDINATION-VECTOR-007

Question and theorem. VALID-007 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 39 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-007 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 39 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-COMPLEX-SPIN-STATE-ORDER-007
- SFT-CHEM-CONFIGURATION-ORDER-PATH-011
- SFT-CHEM-COORDINATION-ENTITY-RETAINED-IDENTITY-001
- SFT-CHEM-COORDINATION-GEOMETRY-HELD-ORIENTATION-004
- SFT-CHEM-COORDINATION-ISOMERISM-EQUIVALENCE-005
- SFT-CHEM-COORDINATION-NUMBER-INCIDENCE-COUNT-002
- SFT-CHEM-DEFECT-NONSTOICHIOMETRY-016
- SFT-CHEM-ELECTRON-COUNT-SPIN-002
- SFT-CHEM-INORGANIC-ACID-BASE-REDOX-NETWORK-017
- SFT-CHEM-INORGANIC-COLOUR-ELECTRONIC-TRANSITION-008
- SFT-CHEM-INORGANIC-MAGNETIC-STATE-009
- SFT-CHEM-INSERTION-ELIMINATION-PATHWAY-013
- SFT-CHEM-JOINT-CORRELATION-DISSOCIATION-007
- SFT-CHEM-LIGAND-DENTICITY-CHELATION-TOPOLOGY-003
- SFT-CHEM-LIGAND-STATE-SPLITTING-006
- SFT-CHEM-METAL-CLUSTER-BONDING-014
- SFT-CHEM-MOLECULAR-EXCLUSION-EXCHANGE-006
- SFT-CHEM-MOLECULAR-MAGNETIC-RESPONSE-012
- SFT-CHEM-MOLECULAR-QUANTUM-MEASUREMENT-REDUCTION-014
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-MULTICENTRE-DELOCALIZED-SUPPORT-008
- SFT-CHEM-NUCLEAR-ELECTRONIC-COMPOSITION-012
- SFT-CHEM-ORBITAL-SUPPORT-OCCUPANCY-003
- SFT-CHEM-ORGANOMETALLIC-ELECTRON-ACCOUNTING-011
- SFT-CHEM-ORGANOMETALLIC-METAL-CARBON-BOND-010
- SFT-CHEM-OXIDATIVE-ADDITION-REDUCTIVE-ELIMINATION-012
- SFT-CHEM-RESOLVED-ROVIBRONIC-SPIN-COMPOSITION-013
- SFT-CHEM-SELECTION-RULE-STRUCTURE-010
- SFT-CHEM-SMITHIUM-CHEMICAL-SEPARATION-001
- SFT-CHEM-SMITHIUM-DECAY-CHANNEL-LEDGER-001
- SFT-CHEM-SMITHIUM-ION-OXIDATION-LADDER-001
- SFT-CHEM-SMITHIUM-JOINT-DETECTION-001
- SFT-CHEM-SMITHIUM-LIFETIME-BOUNDARY-001
- SFT-CHEM-SMITHIUM-SPECTROSCOPIC-CLASSES-001
- SFT-CHEM-SMITHIUM-SYNTHESIS-CONSERVATION-001

- SFT-CHEM-SOLID-STATE-LOCAL-COORDINATION-015
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-STATE-SYMMETRY-DEGENERACY-005
- SFT-CHEM-VALIDATION-SMITHIUM-COMPLETE-FAMILY-001
- SFT-CHEM-VALIDATION-ELECTROCHEMICAL-VECTOR-006

Generated grammar. Generate the complete eight-axis VALID-007 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 39 value-free claim identities registered for Chemistry VALID-007, their current receipt-bound evidence and all result classes.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-receipt-bound-membership__complete-valid-007-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
authority	complete-receipt-bound-membership	selected-headline	A headline omits the evidence surface.
relation	complete-valid-007-typed-evidence-vector	aggregate-score-or-fit	A score or fit can hide distinct failures.
custody	sealed-law-then-capability-closed-release	outcomes-in-forcing	Outcome access may select the law.
results	favourable-adverse-absent-unavailable-unresolved	favourable-only	Selective evidence invalidates the vector.
closure	each-declared-closure-scope-held	single-universal-scope	That overwrites finite boundaries.
formal	explicit-formal-standing-boundary	invented-empirical-successes	That fabricates data.
controls	omission-tamper-and-boundary-halt	omission-tolerated	A missing row cannot close a vector.
extension	dated-complete-extension-open	permanent-lock	A dated vector cannot bar discovery.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256: 301279d1f8578919eca5fa3a503a7c945cde3cb6788e03d4c7725267e9b2f131.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256: b35290ed85cfd01f1395288c8d96ab2b762e5870b666a6af69277a4e9d22da61.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: ca6f44c27eeced18e43b9376404ceb8d18d7d9a645ef03bf631beb8289ee8b20.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt

sha256:a9bcfd00e9f1efe42fdbcf72c3a260892df5a93378317340d23c883dacb128f2.

Independent reconstruction. Implementation hash

sha256:ebb766680add8ad494e3d447edc51e205fdb57736b98d14ebd76d6737d57ea34; independent certificate sha256:c588e102c5f53a5274bd84e6049ce2ee3337026a1c6704600da78fc9bf1350c5; external-validation hash sha256:8e2d5901b017904c644c16559b4758b2f30785670f9dd3f51c2a84ca12baa66b. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-HT06789
- IUPAC-LT07067
- IUPAC-ST07104
- NIST-CCCBDB-SRD101-ETHANOL-EXPERIMENTAL-ROTATIONAL-BARRIER
- IUPAC-GOLD-BOOK-C01330-C00930-L03518
- NIST-CCCBDB-SRD101-FECO5-FERROCENE
- IUPAC-GOLD-BOOK-C01332
- NIST-CCCBDB-COMplete-EXPERIMENTAL-GEOMETRY
- IUPAC-GOLD-BOOK-I03294
- IUPAC-GOLD-BOOK-C01093
- IUPAC-GOLD-BOOK-E02069
- IUPAC-RED-BOOK-2005
- IUPAC-GOLD-BOOK-C01331
- NIST-CCCBDB-SRD101-SCF3-TICL4-FECO5
- IUPAC-S05735
- IUPAC-13807
- IUPAC-09464
- IUPAC-14011
- IUPAC-14026
- NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- IUPAC-L03508
- IUPAC-L03511
- IUPAC-L03510
- IUPAC-O04362
- IUPAC-R05222
- IUPAC-08245
- IUPAC-L03519
- IUPAC-L03520
- IUPAC-M03868
- NIST-WEBBOOK-C13007926-UVVIS-COMplete
- NIST-WEBBOOK-C12082085-UVVIS-COMplete
- NIST-WEBBOOK-C12108111-UVVIS-COMplete

- NIST-WEBBOOK-C12146360-UVVIS-COMPLETE
- IUPAC-M03689
- IUPAC-P04404
- IUPAC-D01668
- NIST-MOLECULAR-MICROWAVE-SPECTRAL-DATABASES-SRD-114-115-117
- NIST-JPCRD-3-609-1974-DIATOMIC-MICROWAVE-SPECTRAL-TABLES
- IUPAC-M03924
- IUPAC-I03058
- IUPAC-E02038
- APS-PRA-49-2460-1994
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-DISSOCIATION
- IUPAC-GOLD-BOOK-B00649
- IUPAC-GOLD-BOOK-C01012
- IUPAC-GOLD-BOOK-D01594
- IUPAC-GOLD-BOOK-H01881
- IUPAC-GOLD-BOOK-K03366
- IUPAC-GOLD-BOOK-L03518
- IUPAC-LT06764
- IUPAC-L03517
- NIST-WEBBOOK-C13939065-UVVIS-COMPLETE
- NIST-WEBBOOK-C14040110-UVVIS-COMPLETE
- IUPAC-CT06769
- IUPAC-IT06779
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- NIST-MDS2-3389-CAH-QUANTUM-STATE-TRACKING-CONTROL-V1.0.1
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- IUPAC-GOLD-BOOK-08789-2026
- NIST-CCCBDB-SRD101-DIBORANE
- NIST-CCCBDB-SRD101-BENZENE
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-HD-D2-DIATOMIC-CONSTANTS-2026
- IUPAC-E01913
- IUPAC-O04328
- IUPAC-O04367
- IUPAC-R05223
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-HD-D2-RESOLVED-DIATOMIC-SURFACE-2026
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- IUPAC-12823
- IUPAC-SUPERHEAVY-2018

- IUPAC-PERIODIC-TABLE-2022
- NIST-ATOMIC-SPECTROSCOPY

Comparison records:

- VALID-007: complete registered membership count 39
- complete Chemistry base: 263 claims; 255 empirical; 8 formal-only
- retained evidence: 26486 measurement/evidence lines and 1177 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-CHEM-COMPLEX-SPIN-STATE-ORDER-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=12; explicit-status-lines=2
- SFT-CHEM-CONFIGURATION-ORDER-PATH-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=56; explicit-status-lines=0
- SFT-CHEM-COORDINATION-ENTITY-RETAINED-IDENTITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=26; explicit-status-lines=1
- SFT-CHEM-COORDINATION-GEOMETRY-HELD-ORIENTATION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=10; explicit-status-lines=3
- SFT-CHEM-COORDINATION-ISOMERISM-EQUIVALENCE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=14; explicit-status-lines=1
- SFT-CHEM-COORDINATION-NUMBER-INCIDENCE-COUNT-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=29; explicit-status-lines=1
- SFT-CHEM-DEFECT-NONSTOICHIOMETRY-016: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=13; explicit-status-lines=2
- SFT-CHEM-ELECTRON-COUNT-SPIN-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=27; explicit-status-lines=6
- SFT-CHEM-INORGANIC-ACID-BASE-REDOX-NETWORK-017: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=14; explicit-status-lines=1
- SFT-CHEM-INORGANIC-COLOUR-ELECTRONIC-TRANSITION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=12; explicit-status-lines=1
- SFT-CHEM-INORGANIC-MAGNETIC-STATE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=13; explicit-status-lines=0
- SFT-CHEM-INSERTION-ELIMINATION-PATHWAY-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=10; explicit-status-lines=3
- SFT-CHEM-JOINT-CORRELATION-DISSOCIATION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=30; explicit-status-lines=14
- SFT-CHEM-LIGAND-DENTICITY-CHELATION-TOPOLOGY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=6; evidence-lines=30; explicit-status-lines=1
- SFT-CHEM-LIGAND-STATE-SPLITTING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=13; explicit-status-lines=3
- SFT-CHEM-METAL-CLUSTER-BONDING-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=10; explicit-status-lines=1
- SFT-CHEM-MOLECULAR-EXCLUSION-EXCHANGE-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=80; explicit-status-lines=12
- SFT-CHEM-MOLECULAR-MAGNETIC-RESPONSE-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=181; explicit-status-lines=1

- SFT-CHEM-MOLECULAR-QUANTUM-MEASUREMENT-REDUCTION-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=345; explicit-status-lines=0
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=80; explicit-status-lines=16
- SFT-CHEM-MULTICENTRE-DELOCALIZED-SUPPORT-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=41; explicit-status-lines=16
- SFT-CHEM-NUCLEAR-ELECTRONIC-COMPOSITION-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=433; explicit-status-lines=0
- SFT-CHEM-ORBITAL-SUPPORT-OCCUPANCY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=366; explicit-status-lines=7
- SFT-CHEM-ORGANOMETALLIC-ELECTRON-ACCOUNTING-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=10; explicit-status-lines=2
- SFT-CHEM-ORGANOMETALLIC-METAL-CARBON-BOND-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=11; explicit-status-lines=0
- SFT-CHEM-OXIDATIVE-ADDITION-REDUCTIVE-ELIMINATION-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=10; explicit-status-lines=0
- SFT-CHEM-RESOLVED-ROVIBRONIC-SPIN-COMPOSITION-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=107; explicit-status-lines=2
- SFT-CHEM-SELECTION-RULE-STRUCTURE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=75; explicit-status-lines=11
- SFT-CHEM-SMITHIUM-CHEMICAL-SEPARATION-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SMITHIUM-DECAY-CHANNEL-LEDGER-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SMITHIUM-ION-OXIDATION-LADDER-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SMITHIUM-JOINT-DETECTION-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SMITHIUM-LIFETIME-BOUNDARY-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SMITHIUM-SPECTROSCOPIC-CLASSES-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SMITHIUM-SYNTHESIS-CONSERVATION-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SOLID-STATE-LOCAL-COORDINATION-015: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=11; explicit-status-lines=2
- SFT-CHEM-STATE-ENERGY-ORDER-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=313; explicit-status-lines=6
- SFT-CHEM-STATE-SYMMETRY-DEGENERACY-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=379; explicit-status-lines=11
- SFT-CHEM-VALIDATION-SMITHIUM-COMPLETE-FAMILY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=6; explicit-status-lines=4

Falsification boundary: Reject if any registered member, receipt, control, evidence file, result class, closure scope, formal-only boundary or source-custody row is missing, changed, duplicated or selected after outcome access.

Measurement receipt: sha256:0db8d7194a06f7e0d8b28ca5f97a3ae9c63970c4b19e0787b236f058b7177366.
 Isolation certificate: sha256:70b4587c9af389f82893212a2a1ecfbcdcb9d4650d9cb9aee643ded2c12dc7da.
 Custody certificate: sha256:25cbc21ce0e8f65f84397aaf4c0e50d39dd7d38975f3d9e9c73e355e79acfaf3.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favorable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

Immutable identities. Source manifest

sha256:bed92fc4822d2ede6a1ce90027f23337361dec3804d7a98635a68fc9c3dc8788; derivation seal
 sha256:93e4c1a14d492ebe5cc39245d93c2014d46006012a628f8c8ff5aad3eab63a8b; engine receipt
 sha256:98548c90b133401da439ea2860dd337c23e05f24dffcba2896742aa4f179d870 at receipts/engine/model_admitted/SFT-CHEM-VALIDATION-INORGANIC-COORDINATION-VECTOR-007-98548c90b133401d.json; empirical hash
 sha256:37742f387bae8f778786daea5cb685abba86a0309636884f5f288ccf449b5d1c; measurement receipt
 sha256:0db8d7194a06f7e0d8b28ca5f97a3ae9c63970c4b19e0787b236f058b7177366.

271. Substantial organic-reaction validation vector

Claim identity: SFT-CHEM-VALIDATION-ORGANIC-REACTION-VECTOR-008

Question and theorem. VALID-008 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 33 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-008 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 33 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-ADDITION-REACTION-FAMILY-009
- SFT-CHEM-ANTIAROMATIC-NONAROMATIC-DISTINCTION-004
- SFT-CHEM-AROMATIC-RECURRENCE-STABILITY-003
- SFT-CHEM-CATALYTIC-TURNOVER-CYCLE-FREQUENCY-010
- SFT-CHEM-COMPETING-CHANNEL-BRANCHING-006
- SFT-CHEM-CONFORMER-GENERATION-EQUIVALENCE-005
- SFT-CHEM-CONFORMER-POPULATION-ORDERING-006
- SFT-CHEM-CONJUGATED-SUPPORT-001

- SFT-CHEM-DIFFUSION-LIMITED-REACTION-BOUNDARY-011
- SFT-CHEM-ELECTROPHILIC-SUBSTITUTION-FAMILY-008
- SFT-CHEM-ELIMINATION-REACTION-FAMILY-010
- SFT-CHEM-INORGANIC-ACID-BASE-REDOX-NETWORK-017
- SFT-CHEM-INORGANIC-COLOUR-ELECTRONIC-TRANSITION-008
- SFT-CHEM-INORGANIC-MAGNETIC-STATE-009
- SFT-CHEM-KINETIC-ISOTOPE-EFFECT-RELATION-012
- SFT-CHEM-NUCLEOPHILIC-SUBSTITUTION-FAMILY-007
- SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001
- SFT-CHEM-PARALLEL-MECHANISM-COMPOSITION-008
- SFT-CHEM-PERICYCLIC-REACTION-FAMILY-012
- SFT-CHEM-PROTECTING-GROUP-REVERSIBLE-STATE-015
- SFT-CHEM-RADICAL-REACTION-NETWORK-013
- SFT-CHEM-REACTION-DYNAMICS-SCATTERING-PRODUCT-STATE-013
- SFT-CHEM-REARRANGEMENT-REACTION-FAMILY-011
- SFT-CHEM-RESONANCE-EQUIVALENT-REPRESENTATION-002
- SFT-CHEM-RETROSYNTHETIC-DECOMPOSITION-RECONSTRUCTION-016
- SFT-CHEM-REVERSIBLE-KINETIC-EQUILIBRIUM-CORRESPONDENCE-009
- SFT-CHEM-SELECTIVITY-COMPLETE-DISTRIBUTION-014
- SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-CHEM-STEREO-DIASTEREOMER-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-CHEM-STEREoisomer-ENUMERATION-005
- SFT-CHEM-VALIDATION-INORGANIC-COORDINATION-VECTOR-007

Generated grammar. Generate the complete eight-axis VALID-008 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 33 value-free claim identities registered for Chemistry VALID-008, their current receipt-bound evidence and all result classes.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: complete-receipt-bound-membership__complete-valid-008-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
authority	complete-receipt-bound-membership	selected-headline	A headline omits the evidence surface.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
relation	complete-valid-008-typed-evidence-vector	aggregate-score-or-fit	A score or fit can hide distinct failures.
custody	sealed-law-then-capability-closed-release	outcomes-in-forcing	Outcome access may select the law.
results	favourable-adverse-absent-unavailable-unresolved	favourable-only	Selective evidence invalidates the vector.
closure	each-declared-closure-scope-held	single-universal-scope	That overwrites finite boundaries.
formal	explicit-formal-standing-boundary	invented-empirical-successes	That fabricates data.
controls	omission-tamper-and-boundary-halt	omission-tolerated	A missing row cannot close a vector.
extension	dated-complete-extension-open	permanent-lock	A dated vector cannot bar discovery.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:4dd2a95d4ff92565af3eba7f8bf0d069fbd645d2f8ac900bb9464e7c0ff7a2a9.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:ec562a0671d5c00c5e9c8ae33ff1035f25f9834dd2ff59df3145817806b4c8b4.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ff319304464bc47f183f83593be444ffd9ed3287356f7985c8f3cb3f5a4e9ea9.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:355489339e3cda45dec2dfe909b11c3c34d042056387c9c4a97b865d05eff3d8.

Independent reconstruction. Implementation hash

sha256:ebb766680add8ad494e3d447edc51e205fdb57736b98d14ebd76d6737d57ea34; independent certificate
sha256:ba790a540e613f7099c81208e94d240cd896fc807ec9477b154d5dd61a620768; external-validation
hash sha256:037be004cadca817e14f43753ed38088a96ed80337a1c1e6e421302bcab84b66. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-17384a1d9c5f4b2e9aaccbb6293204db
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-1478630680b84aee8dca359a52f20ba1
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-25dfe42a186e40c88717830e231034da
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-65ff4c19dea541b880d627eb4fbf8af3
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-8bbc1db0e28c4c60a61d554be47d9b92
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-c9f16c4a213a481daa225423696b55a4
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-8ba9957fa5604be4aeda7eff734f6ca5
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-e80af89368884e338a1956cbe7d0d943
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-6d9789e42f954cae8a05b6d471178de0
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-a9b062ada7ce4d2fa3f88d957d9f599a

- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-0c3242203b3a408882edda75874c3b8d
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-48c6faedc14d49dfa5de2dd45ed6140d
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-d32732940c8e4867b80c81d9b01eb889
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-62b80d7f983a4bf9a9564d4afc30e02e
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-e7f8deb100824a408d3c2e0d1cb693a9
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-94fbe6fbedf24c1cbfa7e834d4e4a7c9
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-8eff1b6e7bbb4496bc2632e3d3e31a71
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-4b5fae7b28894c28b650d185109427d2
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-7ad584d7eae4f169a17acb74b087530
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-bba06fb330a54307a9082b493a2453c3
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-c9cb9eedd61341a6b829cf445aa77b7c
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-6c8b4df5334a4f939f2d4f56b1edbf8
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-bcbb41bfe1ae435eac5bac8862f2f14d
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-ac18bbe2a7644934b24a8956f4b4a721
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-2e1cdb77c14c495e892c4336682b64a6
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-4be654f9e2a74d6cb58f62cf1baae0ec
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-b91423e5a7d349f78fefca0ed55980e8
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-1bc497e7e2f745f4b38b233207da6bc8
- IUPAC-A00441
- IUPAC-AT06987
- NIST-WEBBOOK-C71432-THERMOCHEMISTRY
- NIST-CCCBDB-1120532-NEUTRAL-EXPERIMENTAL
- NIST-CCCBDB-629209-NEUTRAL-EXPERIMENTAL
- IUPAC-A00442
- IUPAC-R05333
- NIST-WEBBOOK-C110838-THERMOCHEMISTRY
- NIST-WEBBOOK-C110827-THERMOCHEMISTRY
- NIST-CCCBDB-SRD101-C71432-COMPLETE-EXPERIMENTAL-DATA
- NIST-CCCBDB-SRD101-C110838-COMPLETE-EXPERIMENTAL-DATA
- NIST-CCCBDB-SRD101-C110827-COMPLETE-EXPERIMENTAL-DATA
- NATURE-NANOTECHNOLOGY-S41565-021-00959-4-COMPLETE
- COMMUNICATIONS-CHEMISTRY-S42004-024-01239-7-COMPLETE
- IUPAC-C01262
- IUPAC-C01258
- IUPAC-C01259
- NIST-CCCBDB-106978-NEUTRAL-EXPERIMENTAL
- IUPAC-CT01038
- NIST-CCCBDB-106978-INTERNAL-ROTATION

- NIST-CCCBDB-109660-INTERNAL-ROTATION-1
- OSTI-22415852-PRIMARY-RECORD
- OSTI-22415852-PUBLIC-FULLTEXT-ENDPOINT
- PUBMED-25591384-COMPLETE-FULLTEXT-PROVIDER-SURFACE
- ACS-JP404315T-COMPLETE-PRIMARY-AND-SI-SURFACE
- ACS-FIGSHARE-EXACT-RESOURCE-DOI-JP404315T
- ACS-FIGSHARE-4012420-JP404315T-SI-001
- CORE-43583184-DOWNLOAD-PDF-LEGACY
- CORE-43583184-DOWNLOAD-CURRENT
- IUPAC-C01267
- IUPAC-08786
- IUPAC-D01583
- NIST-CCCBDB-106990-NEUTRAL-EXPERIMENTAL
- NIST-CCCBDB-591935-NEUTRAL-GEOMETRY
- NIST-WEBBOOK-C106990-SPECTRA
- NIST-WEBBOOK-C106990-UVVIS-JCAMP-INDEX-0
- NATURE-COMMUNICATIONS-S41467-025-68008-5-COMPLETE
- IUPAC-E02020
- IUPAC-08167
- NATURE-CHEMISTRY-S41557-022-01101-0-ARTICLE
- NATURE-CHEMISTRY-41557-2022-1101-MOESM1
- IUPAC:E02038
- RSC:D4SC01905A:3a
- RSC:D4SC01905A:3b
- RSC:D4SC01905A:3c
- RSC:D4SC01905A:3d
- RSC:D4SC01905A:3e
- RSC:D4SC01905A:3f
- RSC:D4SC01905A:3g
- RSC:D4SC01905A:3h
- RSC:D4SC01905A:3i
- RSC:D4SC01905A:3j
- RSC:D4SC01905A:3k
- RSC:D4SC01905A:3l
- RSC:D4SC01905A:3m
- RSC:D4SC01905A:3n
- RSC:D4SC01905A:3o
- RSC:D4SC01905A:3p

- RSC:D4SC01905A:3q
- RSC:D4SC01905A:3r
- RSC:D4SC01905A:3s
- RSC:D4SC01905A:3t
- RSC:D4SC01905A:3u
- RSC:D4SC01905A:3v
- RSC:D4SC01905A:3w
- RSC:D4SC01905A:3x
- RSC:D4SC01905A:3y
- RSC:D4SC01905A:3z
- RSC:D4SC01905A:3aa
- RSC:D4SC01905A:3ab
- RSC:D4SC01905A:3ac
- RSC:D4SC01905A:3ad
- RSC:D4SC01905A:3ae
- RSC:D4SC01905A:3af
- RSC:D4SC01905A:UNSUCCESSFUL:1
- RSC:D4SC01905A:UNSUCCESSFUL:2
- RSC:D4SC01905A:UNSUCCESSFUL:3
- RSC:D4SC01905A:UNSUCCESSFUL:4
- RSC:D4SC01905A:UNSUCCESSFUL:5
- RSC:D4SC01905A:S1
- RSC:D4SC01905A:S2
- RSC:D4SC01905A:S3
- RSC:D4SC01905A:S4
- RSC:D4SC01905A:S5
- RSC:D4SC01905A:S6
- RSC:D4SC01905A:S7
- RSC:D4SC01905A:3A-PRIME-TIME-COURSE
- EUROPE-PMC:PMC11186341:SUPPLEMENT
- IUPAC-L03508
- IUPAC-L03511
- IUPAC-L03510
- IUPAC-O04362
- IUPAC-R05222
- IUPAC-08245
- IUPAC-L03519
- IUPAC-L03520

- IUPAC-M03868
- NIST-WEBBOOK-C13007926-UVVIS-COMPLETE
- NIST-WEBBOOK-C12082085-UVVIS-COMPLETE
- NIST-WEBBOOK-C12108111-UVVIS-COMPLETE
- NIST-WEBBOOK-C12146360-UVVIS-COMPLETE
- IUPAC-M03689
- IUPAC-P04404
- IUPAC-D01668
- NIST-MOLECULAR-MICROWAVE-SPECTRAL-DATABASES-SRD-114-115-117
- NIST-JPCRD-3-609-1974-DIATOMIC-MICROWAVE-SPECTRAL-TABLES
- NATURE-COMMUNICATIONS-S41467-024-44753-X-COMPLETE
- IUPAC-N04249
- IUPAC-08191
- IUPAC-E02130
- IUPAC-S06078
- IUPAC-H02809
- PUBCHEM-PUG-BROMOMETHANE-STRUCTURE
- PUBCHEM-PUG-HYDROXIDE-STRUCTURE
- PUBCHEM-PUG-METHANOL-STRUCTURE
- PUBCHEM-PUG-BROMIDE-STRUCTURE
- IUPAC-GOLD-BOOK-FUNCTIONAL-GROUP-COMPOSITE-2026
- IUPAC-GOLD-BOOK-ORGANIC-REACTION-FAMILY-COMPOSITE-2026
- NATURE-COMMUNICATIONS-S41467-026-70199-4-SOURCE-DATA
- IUPAC:P04491
- IUPAC:C01496
- PMC8162770:TABLE-1:ROW:1
- PMC8162770:TABLE-1:ROW:2
- PMC8162770:TABLE-1:ROW:3
- PMC8162770:TABLE-1:ROW:4
- PMC8162770:TABLE-1:ROW:5
- PMC8162770:TABLE-1:ROW:6
- PMC8162770:TABLE-1:ROW:7
- PMC8162770:TABLE-1:ROW:8
- PMC8162770:TABLE-1:ROW:9
- PMC8162770:TABLE-1:ROW:10
- PMC8162770:TABLE-1:ROW:11
- PMC8162770:TABLE-1:ROW:12
- PMC8162770:TABLE-1:ROW:13

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- PMC8162770:TABLE-1:ROW:31
- PMC8162770:TABLE-1:ROW:32
- PMC8162770:ARTICLE
- PMC8162770:SUPPLEMENT:S001
- PMC8162770:SUPPLEMENT:S002
- NCBI-OA:PMC8162770
- IUPAC-09560
- IUPAC-09466
- IUPAC:C00960
- IUPAC:I03042
- IUPAC:T06274
- PMC11598545:TABLE:1:ROW:1
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- PMC11598545:TABLE:1:ROW:3
- PMC11598545:TABLE:1:ROW:4
- PMC11598545:TABLE:1:ROW:5
- PMC11598545:TABLE:1:ROW:6
- PMC11598545:TABLE:1:ROW:7
- PMC11598545:TABLE:1:ROW:8
- PMC11598545:TABLE:1:ROW:9
- PMC11598545:TABLE:1:ROW:10

- PMC11598545:TABLE:1:ROW:11
- PMC11598545:TABLE:1:ROW:12
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- PMC11598545:TABLE:1:ROW:45
- PMC11598545:TABLE:1:ROW:46
- PMC11598545:TABLE:1:ROW:47
- PMC11598545:TABLE:2:ROW:1

- PMC11598545:TABLE:2:ROW:2
- PMC11598545:TABLE:2:ROW:3
- PMC11598545:TABLE:2:ROW:4
- PMC11598545:TABLE:2:ROW:5
- PMC11598545:TABLE:2:ROW:6
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- PMC11598545:TABLE:2:ROW:12
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- PMC11598545:TABLE:3:ROW:2
- PMC11598545:TABLE:3:ROW:3
- PMC11598545:TABLE:3:ROW:4
- PMC11598545:TABLE:4:ROW:1
- PMC11598545:TABLE:4:ROW:2
- PMC11598545:TABLE:4:ROW:3
- PMC11598545:TABLE:4:ROW:4
- PMC11598545:ARTICLE
- PMC11598545:SUPPLEMENT:S001
- NCBI-OA:PMC11598545
- NATURE-COMMUNICATIONS-S41467-025-66587-X-COMplete
- IUPAC:M03997
- IUPAC:R05194
- IUPAC:R05196
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:1
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:2
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:3
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:4
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:5
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:6
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:7
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:8
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAGE:3
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAGE:4
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAGE:37
- EUROPE-PMC:ORG-011:V1

- ACS-FIGSHARE:10.1021/ja803370x.s001
- IUPAC-R05326
- IUPAC-08205
- IUPAC-C01309
- IUPAC-M03804
- IUPAC-R05174
- NATURE-COMMUNICATIONS-S41467-023-40190-4-COMplete
- IUPAC-C01051
- IUPAC-R05243
- IUPAC-S05991
- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC
- NATURE-COMMUNICATIONS-S41467-024-49814-9-COMplete
- RCSB-PDB-8WZF-8WZV
- CXIDB-221
- IUPAC-GOLD-BOOK-CHIRALITY-COMPOSITE-2026
- IUPAC-GOLD-BOOK-DIASTEREOMER-COMPOSITE-2026
- IUPAC-GOLD-BOOK-ENANTIOMER-COMPOSITE-2026
- PUBCHEM-PUG-REST-AUTHORITY
- PUBCHEM-C3H8O-FORMULA-CENSUS
- PUBCHEM-ASPIRIN-CONFORMERS
- PUBCHEM-ASPIRIN-SUBSTRUCTURE
- PUBCHEM-ASPIRIN-SIMILARITY
- CHEBI-REST-AUTHORITY
- RHEA-IMMUTABLE-REACTION-SMILES
- USPTO-IMMUTABLE-REACTION-SURFACE

Comparison records:

- VALID-008: complete registered membership count 33
- complete Chemistry base: 263 claims; 255 empirical; 8 formal-only
- retained evidence: 26486 measurement/evidence lines and 1177 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-CHEM-ADDITION-REACTION-FAMILY-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=28; evidence-lines=7; explicit-status-lines=3
- SFT-CHEM-ANTIAROMATIC-NONAROMATIC-DISTINCTION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=21; explicit-status-lines=5
- SFT-CHEM-AROMATIC-RECURRENCE-STABILITY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=9; evidence-lines=20; explicit-status-lines=2
- SFT-CHEM-CATALYTIC-TURNOVER-CYCLE-FREQUENCY-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=510; explicit-status-lines=2
- SFT-CHEM-COMPETING-CHANNEL-BRANCHING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=16; explicit-status-lines=2

- SFT-CHEM-CONFORMER-GENERATION-EQUIVALENCE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=16; explicit-status-lines=3
- SFT-CHEM-CONFORMER-POPULATION-ORDERING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=14; evidence-lines=20; explicit-status-lines=3
- SFT-CHEM-CONJUGATED-SUPPORT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=7; evidence-lines=18; explicit-status-lines=0
- SFT-CHEM-DIFFUSION-LIMITED-REACTION-BOUNDARY-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=264; explicit-status-lines=3
- SFT-CHEM-ELECTROPHILIC-SUBSTITUTION-FAMILY-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=6; explicit-status-lines=1
- SFT-CHEM-ELIMINATION-REACTION-FAMILY-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=47; evidence-lines=7; explicit-status-lines=1
- SFT-CHEM-INORGANIC-ACID-BASE-REDOX-NETWORK-017: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=14; explicit-status-lines=1
- SFT-CHEM-INORGANIC-COLOUR-ELECTRONIC-TRANSITION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=12; explicit-status-lines=1
- SFT-CHEM-INORGANIC-MAGNETIC-STATE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=13; explicit-status-lines=0
- SFT-CHEM-KINETIC-ISOTOPE-EFFECT-RELATION-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=84; explicit-status-lines=3
- SFT-CHEM-NUCLEOPHILIC-SUBSTITUTION-FAMILY-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=9; evidence-lines=7; explicit-status-lines=1
- SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-PARALLEL-MECHANISM-COMPOSITION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=39; explicit-status-lines=3
- SFT-CHEM-PERICYCLIC-REACTION-FAMILY-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=38; evidence-lines=7; explicit-status-lines=2
- SFT-CHEM-PROTECTING-GROUP-REVERSIBLE-STATE-015: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=4; explicit-status-lines=0
- SFT-CHEM-RADICAL-REACTION-NETWORK-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=73; evidence-lines=6; explicit-status-lines=1
- SFT-CHEM-REACTION-DYNAMICS-SCATTERING-PRODUCT-STATE-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=65; explicit-status-lines=2
- SFT-CHEM-REARRANGEMENT-REACTION-FAMILY-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=16; evidence-lines=7; explicit-status-lines=2
- SFT-CHEM-RESONANCE-EQUIVALENT-REPRESENTATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=21; explicit-status-lines=0
- SFT-CHEM-RETROSYNTHETIC-DECOMPOSITION-RECONSTRUCTION-016: source-custodied-post-seal-comparison; closure=depth_independent; sources=7; evidence-lines=4; explicit-status-lines=0
- SFT-CHEM-REVERSIBLE-KINETIC-EQUILIBRIUM-CORRESPONDENCE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=176; explicit-status-lines=2

- SFT-CHEM-SELECTIVITY-COMPLETE-DISTRIBUTION-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=133; evidence-lines=5; explicit-status-lines=1
- SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=28; explicit-status-lines=5
- SFT-CHEM-STEREO-CHIRALITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STEREO-DIASTEREOMER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STEREO-ENANTIOMER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STEREOISOMER-ENUMERATION-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2

Falsification boundary: Reject if any registered member, receipt, control, evidence file, result class, closure scope, formal-only boundary or source-custody row is missing, changed, duplicated or selected after outcome access.

Measurement receipt: sha256:f97f4d849770b3ff4f5ff729e9643fb301937210a45a00d8790f6216b555be41.

Isolation certificate: sha256:1297e0b967b0277424b9d6e3908fa1abf0cfec1519ad00f5aee55df1edf1025f.

Custody certificate: sha256:d83aac44b494318762fd96475eed9e0c72fb633803ebfe2c70f95e9a7c5141bc.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favorable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

Immutable identities. Source manifest

sha256:8ebb8e922c0e77f82789fea7c307e984766519a5572fe26db76548d1dc5d24d; derivation seal

sha256:3cb80844555b998c052c2117bd83e4f3286805f6c8a0515e7db0d1df30394c4f; engine receipt

sha256:0d3727e75b924eb416297d3822bcba72c734c24ac08ae6514bcbeb9c39a61995 at receipts/engine/model_admitted/SFT-CHEM-VALIDATION-ORGANIC-REACTION-VECTOR-008-0d3727e75b924eb4.json;

empirical hash sha256:93976f10eafadb4caf83a9b551173be5436cd29a7bd24a83d99a25556e5e325a;

measurement receipt sha256:f97f4d849770b3ff4f5ff729e9643fb301937210a45a00d8790f6216b555be41.

272. Substantial polymer validation vector

Claim identity: SFT-CHEM-VALIDATION-POLYMER-VECTOR-009

Question and theorem. VALID-009 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 15 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-009 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 15 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-CHAIN-GROWTH-POLYMERIZATION-NETWORK-005
- SFT-CHEM-COPOLYMER-SEQUENCE-COMPOSITION-007
- SFT-CHEM-CROSSLINK-GELATION-BOUNDARY-009
- SFT-CHEM-DEGREE-OF-POLYMERIZATION-001
- SFT-CHEM-MASS-WEIGHTED-MOLECULAR-SIZE-003
- SFT-CHEM-NUMBER-AVERAGE-MOLECULAR-SIZE-002
- SFT-CHEM-POLYMER-ARCHITECTURE-008
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-CHEM-POLYMER-CONFORMATION-SIZE-010
- SFT-CHEM-POLYMER-DEGRADATION-NETWORK-012
- SFT-CHEM-POLYMER-DISPERSITY-004
- SFT-CHEM-POLYMER-DISTRIBUTION-001
- SFT-CHEM-POLYMER-MATERIALS-HANDOFF-013
- SFT-CHEM-POLYMER-PHASE-TRANSITION-011
- SFT-CHEM-STEP-GROWTH-POLYMERIZATION-NETWORK-006
- SFT-CHEM-VALIDATION-ORGANIC-REACTION-VECTOR-008

Generated grammar. Generate the complete eight-axis VALID-009 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 15 value-free claim identities registered for Chemistry VALID-009, their current receipt-bound evidence and all result classes.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-receipt-bound-membership__complete-valid-009-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
authority	complete-receipt-bound-membership	selected-headline	A headline omits the evidence surface.
relation	complete-valid-009-typed-evidence-vector	aggregate-score-or-fit	A score or fit can hide distinct failures.
custody	sealed-law-then-capability-closed-release	outcomes-in-forcing	Outcome access may select the law.
results	favourable-adverse-absent-unavailable-unresolved	favourable-only	Selective evidence invalidates the vector.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
closure	each-declared-closure-scope-held	single-universal-scope	That overwrites finite boundaries.
formal	explicit-formal-standing-boundary	invented-empirical-successes	That fabricates data.
controls	omission-tamper-and-boundary-halt	omission-tolerated	A missing row cannot close a vector.
extension	dated-complete-extension-open	permanent-lock	A dated vector cannot bar discovery.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:5b7ca2ee698b44de9e2a5b68e1baa96074f15821315d9968c250fee29acc0b6d.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:44d143ae4620b556d8d04effc8358101636c811d96eb32811fa07cefdd8115a6.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:39ebbb58d2377b4fef765b584a4728070ad02bc699e3832264c87019d3e9cfd4.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:aabf93e3a768890194c34cd5b8164be31159269e45d8e18a05e2b12b6d342890.

Independent reconstruction. Implementation hash

sha256:ebb766680add8ad494e3d447edc51e205fdb57736b98d14ebd76d6737d57ea34; independent certificate sha256:4c717c594e32940a3de1b991239332a447ac61d1f1b54cbd420993fec2d00b27; external-validation hash sha256:547b14c7e62cc7c2973cbb56d77d50808573c9e4116dc014294dacd7e738fcdb. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-STOCHASTIC-PHOTOPOLYMER-NETWORK-GROWTH
- NIST-MONODISPERSE-PAMS-KINETIC-NETWORK
- NIST-RING-OPENING-COPOLYMER-SEQUENCE
- NIST-PRECISION-DEUTERATED-POLYETHYLENE
- NIST-THERMOREVERSIBLE-GELATION-PERCOLATION
- NIST-BRIDGING-GELATION
- NIST-SRM-2888-POLYSTYRENE-CERTIFICATION
- NIST-SRM-2886-POLYETHYLENE-CERTIFICATION
- NIST-QUINTUPLE-DETECTOR-COPOLYMER
- NIST-BRANCH-PLACEMENT-DILUTE-SOLUTION
- NIST-MACROMOLECULAR-ARCHITECTURES
- IUPAC-GOLD-BOOK-POLYMER-CHAIN-COMPOSITE-2026
- NIST-POLYMER-PYROLYSIS-NETWORK
- NIST-POLYMER-DEPOLYMERIZATION-KINETICS
- IUPAC-GOLD-BOOK-POLYMER-DISTRIBUTION-COMPOSITE-2026
- NIST-CROSSLINKED-PHOTOPOLYMER-CONVERSION

- NIST-POLYMER-PROCESSING-METROLOGY
- NIST-PVOH-IONIC-LIQUID-GELATION

Comparison records:

- VALID-009: complete registered membership count 15
- complete Chemistry base: 263 claims; 255 empirical; 8 formal-only
- retained evidence: 26486 measurement/evidence lines and 1177 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-CHEM-CHAIN-GROWTH-POLYMERIZATION-NETWORK-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-COPOLYMER-SEQUENCE-COMPOSITION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-CROSSLINK-GELATION-BOUNDARY-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-DEGREE-OF-POLYMERIZATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-MASS-WEIGHTED-MOLECULAR-SIZE-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-NUMBER-AVERAGE-MOLECULAR-SIZE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-ARCHITECTURE-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-CHAIN-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-POLYMER-CONFORMATION-SIZE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-DEGRADATION-NETWORK-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-DISPERSITY-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-DISTRIBUTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-POLYMER-MATERIALS-HANDOFF-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-PHASE-TRANSITION-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-STEP-GROWTH-POLYMERIZATION-NETWORK-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2

Falsification boundary: Reject if any registered member, receipt, control, evidence file, result class, closure scope, formal-only boundary or source-custody row is missing, changed, duplicated or selected after outcome access.

Measurement receipt: sha256:33f9c2851050fca48e7482255dc1f6246144c3fbe0b049fcf817affc06d30037.

Isolation certificate: sha256:5113964312592368b2053de149fc4d6bae15d67d79098d68fdb228d2343eb88.

Custody certificate: sha256:e93cfc68795859a0f7f9532e89cedfec8c7547c32598cc61628f141fa8a94aad.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favorable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

Immutable identities. Source manifest

sha256:90afe7ca00825494baf6604d6d514e21752f48a75828b2440ce634a4eb8c00d4; derivation seal sha256:3be24e173954e74d2418f700bc90842f2cf555e5bc1632c08919a5124feedfe3; engine receipt sha256:03f3a14da345577d0ee5823ac7cc232f682ca86eaa271d9003229c895112cfc9 at receipts/engine/model_admitted/SFT-CHEM-VALIDATION-POLYMER-VECTOR-009-03f3a14da345577d.json; empirical hash sha256:247f034ad0be1ea3aa4f994b0050a5e72acb0a9129864fbd9ca83f17e906d27; measurement receipt sha256:33f9c2851050fca48e7482255dc1f6246144c3f5e0b049fcf817affc06d30037.

273. Cross-source Chemistry reproducibility vector

Claim identity: SFT-CHEM-VALIDATION-CROSS-SOURCE-REPRODUCIBILITY-VECTOR-010

Question and theorem. VALID-010 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 129 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-010 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 129 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-ACTIVITY-AMOUNT-TIME-003
- SFT-CHEM-ADDITION-REACTION-FAMILY-009
- SFT-CHEM-ANALYTICAL-ACCURACY-TRUENESS-001
- SFT-CHEM-ANALYTICAL-DETECTION-QUANTIFICATION-004
- SFT-CHEM-ANALYTICAL-PERFORMANCE-BUDGET-022
- SFT-CHEM-ANALYTICAL-PRECISION-REPEATABILITY-002
- SFT-CHEM-ANALYTICAL-SELECTIVITY-INTERFERENCE-005
- SFT-CHEM-ANALYTICAL-SENSITIVITY-003
- SFT-CHEM-ANTIAROMATIC-NONAROMATIC-DISTINCTION-004
- SFT-CHEM-AROMATIC-RECURRENCE-STABILITY-003
- SFT-CHEM-ATOM-MAPPING-REACTION-BALANCE-008
- SFT-CHEM-BOND-DISSOCIATION-ENERGY-002
- SFT-CHEM-CANONICAL-MOLECULAR-GRAPH-ENCODING-001
- SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003

- SFT-CHEM-CHAIN-GROWTH-POLYMERIZATION-NETWORK-005
- SFT-CHEM-CHEMICAL-SUBSTRUCTURE-RELATION-003
- SFT-CHEM-CHROMATOGRAPHIC-RETENTION-RESOLUTION-018
- SFT-CHEM-CLASSICAL-QUANTUM-ALGORITHM-CORRESPONDENCE-014
- SFT-CHEM-COLLIGATIVE-COMPOSITION-RESPONSE-014
- SFT-CHEM-COMPLEX-SPIN-STATE-ORDER-007
- SFT-CHEM-CONCENTRATION-DEPENDENT-POTENTIAL-004
- SFT-CHEM-CONFORMER-ENUMERATION-006
- SFT-CHEM-CONFORMER-GENERATION-EQUIVALENCE-005
- SFT-CHEM-CONFORMER-POPULATION-ORDERING-006
- SFT-CHEM-CONJUGATED-SUPPORT-001
- SFT-CHEM-CONSTITUTIONAL-ISOMER-ENUMERATION-004
- SFT-CHEM-COORDINATION-ENTITY-RETAINED-IDENTITY-001
- SFT-CHEM-COORDINATION-GEOMETRY-HELD-ORIENTATION-004
- SFT-CHEM-COORDINATION-ISOMERISM-EQUIVALENCE-005
- SFT-CHEM-COORDINATION-NUMBER-INCIDENCE-COUNT-002
- SFT-CHEM-COPOLYMER-SEQUENCE-COMPOSITION-007
- SFT-CHEM-CORROSION-REACTION-NETWORK-012
- SFT-CHEM-COUPLED-MASS-HEAT-CHARGE-TRANSPORT-019
- SFT-CHEM-CROSS-PROPERTY-MOLECULAR-VECTOR-014
- SFT-CHEM-CROSSLINK-GELATION-BOUNDARY-009
- SFT-CHEM-DATABASE-IDENTITY-PROVENANCE-011
- SFT-CHEM-DEFECT-NONSTOICHIOMETRY-016
- SFT-CHEM-DEGREE-OF-POLYMERIZATION-001
- SFT-CHEM-DOUBLE-LAYER-INTERFACIAL-CHARGE-011
- SFT-CHEM-ELECTROANALYTICAL-RESPONSE-020
- SFT-CHEM-ELECTROCHEMICAL-STORAGE-HANDOFF-013
- SFT-CHEM-ELECTROCHEMICAL-WORK-REACTION-DIRECTION-005
- SFT-CHEM-ELECTRODE-POTENTIAL-CHEMICAL-RELATION-002
- SFT-CHEM-ELECTRODE-REACTION-RATE-009
- SFT-CHEM-ELECTROLYSIS-PRODUCT-AMOUNT-006
- SFT-CHEM-ELECTRON-NEUTRON-DIFFRACTION-017
- SFT-CHEM-ELECTROPHILIC-SUBSTITUTION-FAMILY-008
- SFT-CHEM-ELECTROPHORETIC-MOBILITY-SEPARATION-019
- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001
- SFT-CHEM-ELIMINATION-REACTION-FAMILY-010
- SFT-CHEM-EQUILIBRIUM-BOND-LENGTH-001
- SFT-CHEM-EQUILIBRIUM-ISOTOPE-FRACTIONATION-007

- SFT-CHEM-EXACT-MOLECULAR-SIMILARITY-DISTINCTION-010
- SFT-CHEM-FINITE-MICROSTATE-SUPPORT-001
- SFT-CHEM-FISSION-PRODUCT-CHEMICAL-DISTRIBUTION-011
- SFT-CHEM-FLUORESCENCE-YIELD-LIFETIME-010
- SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007
- SFT-CHEM-HALF-REACTION-IDENTITY-ORIENTATION-001
- SFT-CHEM-INFRARED-LINE-INTENSITY-012
- SFT-CHEM-INORGANIC-ACID-BASE-REDOX-NETWORK-017
- SFT-CHEM-INORGANIC-COLOUR-ELECTRONIC-TRANSITION-008
- SFT-CHEM-INORGANIC-MAGNETIC-STATE-009
- SFT-CHEM-INSERTION-ELIMINATION-PATHWAY-013
- SFT-CHEM-INTERMOLECULAR-BINDING-011
- SFT-CHEM-IONIC-CONDUCTIVITY-RELATION-007
- SFT-CHEM-IONIC-MOBILITY-TRANSFERENCE-008
- SFT-CHEM-ISOTOPE-EXCHANGE-006
- SFT-CHEM-JOINT-CORRELATION-DISSOCIATION-007
- SFT-CHEM-KINETIC-ISOTOPE-FRACTIONATION-008
- SFT-CHEM-LIGAND-DENTICITY-CHELATION-TOPOLOGY-003
- SFT-CHEM-LIGAND-STATE-SPLITTING-006
- SFT-CHEM-MASS-ISOTOPE-FRAGMENT-VECTOR-014
- SFT-CHEM-MASS-WEIGHTED-MOLECULAR-SIZE-003
- SFT-CHEM-MECHANISM-SEARCH-PROOF-TRACE-009
- SFT-CHEM-METAL-CLUSTER-BONDING-014
- SFT-CHEM-MOLECULAR-BOND-ANGLE-003
- SFT-CHEM-MOLECULAR-DIFFUSION-RELATION-016
- SFT-CHEM-MOLECULAR-DIPOLE-MAGNITUDE-005
- SFT-CHEM-MOLECULAR-ELECTRON-AFFINITY-008
- SFT-CHEM-MOLECULAR-FORMATION-ENERGY-013
- SFT-CHEM-MOLECULAR-GRAPH-ISOMORPHISM-002
- SFT-CHEM-MOLECULAR-IONIZATION-ENERGY-007
- SFT-CHEM-MOLECULAR-MAGNETIC-RESPONSE-012
- SFT-CHEM-MULTICENTRE-DELOCALIZED-SUPPORT-008
- SFT-CHEM-MULTIMODAL-MOLECULAR-IDENTITY-021
- SFT-CHEM-NMR-CHEMICAL-SHIFT-006
- SFT-CHEM-NMR-RELAXATION-EXCHANGE-008
- SFT-CHEM-NMR-SPIN-COUPLING-007
- SFT-CHEM-NUCLEAR-CHEMICAL-CARRIER-001
- SFT-CHEM-NUCLEOPHILIC-SUBSTITUTION-FAMILY-007

- SFT-CHEM-NUMBER-AVERAGE-MOLECULAR-SIZE-002
- SFT-CHEM-OVERPOTENTIAL-POLARIZATION-010
- SFT-CHEM-OXIDATIVE-ADDITION-REDUCTIVE-ELIMINATION-012
- SFT-CHEM-PERICYCLIC-REACTION-FAMILY-012
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011
- SFT-CHEM-PHOSPHORESCENCE-INTERSYSTEM-011
- SFT-CHEM-POLYMER-ARCHITECTURE-008
- SFT-CHEM-POLYMER-CONFORMATION-SIZE-010
- SFT-CHEM-POLYMER-DEGRADATION-NETWORK-012
- SFT-CHEM-POLYMER-DISPERSITY-004
- SFT-CHEM-POLYMER-MATERIALS-HANDOFF-013
- SFT-CHEM-POLYMER-PHASE-TRANSITION-011
- SFT-CHEM-PREDICTION-UNCERTAINTY-APPLICABILITY-013
- SFT-CHEM-PROTECTING-GROUP-REVERSIBLE-STATE-015
- SFT-CHEM-RADIATION-CHEMISTRY-REACTION-NETWORK-012
- SFT-CHEM-RADICAL-REACTION-NETWORK-013
- SFT-CHEM-RADIOACTIVE-BRANCHING-CHEMICAL-YIELD-004
- SFT-CHEM-RADIOACTIVE-CHEMICAL-TRANSFORMATION-002
- SFT-CHEM-RADIOCHEMICAL-EQUILIBRIUM-005
- SFT-CHEM-RADIOCHEMICAL-SEPARATION-DECONTAMINATION-010
- SFT-CHEM-RADIOTRACER-CUSTODY-INFERENCE-009
- SFT-CHEM-RAMAN-TRANSITION-INTENSITY-009
- SFT-CHEM-REACTION-GRAPH-GENERATION-007
- SFT-CHEM-REARRANGEMENT-REACTION-FAMILY-011
- SFT-CHEM-RESONANCE-EQUIVALENT-REPRESENTATION-002
- SFT-CHEM-RETROSYNTHETIC-DECOMPOSITION-RECONSTRUCTION-016
- SFT-CHEM-ROTATIONAL-SPECTRUM-LINE-015
- SFT-CHEM-SELECTIVITY-COMPLETE-DISTRIBUTION-014
- SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007
- SFT-CHEM-SOLID-STATE-LOCAL-COORDINATION-015
- SFT-CHEM-SOLVATION-DISSOLUTION-FREE-ORDER-015
- SFT-CHEM-STEP-GROWTH-POLYMERIZATION-NETWORK-006
- SFT-CHEM-STEREoisomer-ENUMERATION-005
- SFT-CHEM-SYMBOLIC-PROPERTY-EVALUATION-012
- SFT-CHEM-TEMPERATURE-CORRESPONDENCE-002
- SFT-CHEM-THERMAL-CONDUCTIVITY-RELATION-018
- SFT-CHEM-UVVISIBLE-LINE-INTENSITY-013
- SFT-CHEM-VISCOUS-TRANSPORT-RELATION-017

- SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016
- SFT-CHEM-VALIDATION-POLYMER-VECTOR-009

Generated grammar. Generate the complete eight-axis VALID-010 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 129 value-free claim identities registered for Chemistry VALID-010, their current receipt-bound evidence and all result classes.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-receipt-bound-membership__complete-valid-010-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
authority	complete-receipt-bound-membership	selected-headline	A headline omits the evidence surface.
relation	complete-valid-010-typed-evidence-vector	aggregate-score-or-fit	A score or fit can hide distinct failures.
custody	sealed-law-then-capability-closed-release	outcomes-in-forcing	Outcome access may select the law.
results	favourable-adverse-absent-unavailable-unresolved	favourable-only	Selective evidence invalidates the vector.
closure	each-declared-closure-scope-held	single-universal-scope	That overwrites finite boundaries.
formal	explicit-formal-standing-boundary	invented-empirical-successes	That fabricates data.
controls	omission-tamper-and-boundary-halt	omission-tolerated	A missing row cannot close a vector.
extension	dated-complete-extension-open	permanent-lock	A dated vector cannot bar discovery.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:1f04883c78e4f1e6d359fd63d5c3d07ec0827a6dc2a62588eb5d14e3fa768d14.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:fd41e97d8d45629319aaba40ccc80f277994c34780b72010b7af57ee9884f2a5.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ff164087e46d78f0eb737147c81930e885002ed18099ceb7529eded0e5469a53.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:0d9f045e7d4507f8046f275b88976b820704967acb4513d529b45d04126cdc42.

Independent reconstruction. Implementation hash

sha256:ebb766680add8ad494e3d447edc51e205fdb57736b98d14ebd76d6737d57ea34; independent certificate
sha256:b946672a90c9e45737910fb00a3f1c3914bb62c202361f07435bb59293e07a56; external-validation

hash sha256: 761fb8157a183f59607b53b6e25b323a39f0217e0e779d902adfa163cf31b998. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-RADIOACTIVITY-SRMS-GENERAL-INFO-2025
- NIST-SRM-4239A-STRONTIUM-90-2022
- NIST-SRM-4324C-URANIUM-232-2025
- NUCHEM-001-004-SOURCE-INVENTORY
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-17384a1d9c5f4b2e9aaccbb6293204db
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-1478630680b84aee8dca359a52f20ba1
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-25dfe42a186e40c88717830e231034da
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-65ff4c19dea541b880d627eb4fbf8af3
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-8bbc1db0e28c4c60a61d554be47d9b92
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-c9f16c4a213a481daa225423696b55a4
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-8ba9957fa5604be4aeda7eff734f6ca5
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-e80af89368884e338a1956cbe7d0d943
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-6d9789e42f954cae8a05b6d471178de0
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-a9b062ada7ce4d2fa3f88d957d9f599a
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-0c3242203b3a408882edda75874c3b8d
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-48c6faedc14d49dfa5de2dd45ed6140d
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-d32732940c8e4867b80c81d9b01eb889
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-62b80d7f983a4bf9a9564d4afc30e02e
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-e7f8deb100824a408d3c2e0d1cb693a9
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-94fbe6fbedf24c1cbfa7e834d4e4a7c9
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-8eff1b6e7bbb4496bc2632e3d3e31a71
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-4b5fae7b28894c28b650d185109427d2
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-7ad584d7eae4f169a17acb74b087530
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-bba06fb330a54307a9082b493a2453c3
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-c9cb9eedd61341a6b829cf445aa77b7c
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-6c8b4df5334a4f939f2d4f56b1edbf8
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-bcbb41bfe1ae435eac5bac8862f2f14d
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-ac18bbe2a7644934b24a8956f4b4a721
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-2e1cdb77c14c495e892c4336682b64a6
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-4be654f9e2a74d6cb58f62cf1baae0ec
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-b91423e5a7d349f78fefca0ed55980e8
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-1bc497e7e2f745f4b38b233207da6bc8
- NIST-SRM-1959-CERTIFICATE-2011
- ANAL-001-005-SOURCE-INVENTORY
- NIST-SP958-DETECTION-QUANTIFICATION-PAGES-164-166

- SFT-V3-ANAL-001-011-IMMUTABLE-EVIDENCE
- NIST-VOLTAMMETRIC-LOD-STUDY
- NIST-SRM-1980-ELECTROPHORETIC-MOBILITY
- NIST-SRM-1959-CERTIFICATION-2010
- NIST-IMS-FENTANYL-INTERFERENTS-2019
- NIST-SP-260-126-NTRM-GAS-STANDARDS
- IUPAC-A00441
- IUPAC-AT06987
- NIST-WEBBOOK-C71432-THERMOCHEMISTRY
- NIST-CCCBDB-1120532-NEUTRAL-EXPERIMENTAL
- NIST-CCCBDB-629209-NEUTRAL-EXPERIMENTAL
- IUPAC-A00442
- IUPAC-R05333
- NIST-WEBBOOK-C110838-THERMOCHEMISTRY
- NIST-WEBBOOK-C110827-THERMOCHEMISTRY
- NIST-CCCBDB-SRD101-C71432-COMPLETE-EXPERIMENTAL-DATA
- NIST-CCCBDB-SRD101-C110838-COMPLETE-EXPERIMENTAL-DATA
- NIST-CCCBDB-SRD101-C110827-COMPLETE-EXPERIMENTAL-DATA
- PUBCHEM-PUG-REST-AUTHORITY
- PUBCHEM-C3H8O-FORMULA-CENSUS
- PUBCHEM-ASPIRIN-CONFORMERS
- PUBCHEM-ASPIRIN-SUBSTRUCTURE
- PUBCHEM-ASPIRIN-SIMILARITY
- CHEBI-REST-AUTHORITY
- RHEA-IMMUTABLE-REACTION-SMILES
- USPTO-IMMUTABLE-REACTION-SURFACE
- APS-PRA-49-2460-1994
- PRL-UDEM-H-1S2S-1997
- PRL-PARTHEY-HD-1S2S-SHIFT-2010
- SI-DEFINING-LIGHT-SPEED
- JCP-LIU-H2-DISSOCIATION-2009
- PRA-HUSSELS-D2-DISSOCIATION-2022
- NIST-JRES-AGCL-COMPLETE-CELL-1954
- NIST-STOCHASTIC-PHOTOPOLYMER-NETWORK-GROWTH
- NIST-MONODISPERSE-PAMS-KINETIC-NETWORK
- NIST-WEBBOOK-GAS-CHROMATOGRAPHY
- NIST-WEBBOOK-BENZENE-MULTIMODAL
- NIST-WEBBOOK-CYCLOHEXANE-MULTIMODAL

- NIST-TRC-THERMOML-FPE-2013-337-60-66
- NIST-TRC-THERMOML-FPE-2016-420-14-19
- NIST-TRC-THERMOML-JCT-2009-41-1439-1445
- IUPAC-HT06789
- IUPAC-LT07067
- IUPAC-ST07104
- NIST-JRES-AGCL-HCL-CONCENTRATION-SERIES-1954
- IUPAC-C01262
- IUPAC-C01258
- IUPAC-C01259
- NIST-CCCBDB-106978-NEUTRAL-EXPERIMENTAL
- IUPAC-CT01038
- NIST-CCCBDB-106978-INTERNAL-ROTATION
- NIST-CCCBDB-109660-INTERNAL-ROTATION-1
- OSTI-22415852-PRIMARY-RECORD
- OSTI-22415852-PUBLIC-FULLTEXT-ENDPOINT
- PUBMED-25591384-COMPLETE-FULLTEXT-PROVIDER-SURFACE
- ACS-JP404315T-COMPLETE-PRIMARY-AND-SI-SURFACE
- ACS-FIGSHARE-EXACT-RESOURCE-DOI-JP404315T
- ACS-FIGSHARE-4012420-JP404315T-SI-001
- CORE-43583184-DOWNLOAD-PDF-LEGACY
- CORE-43583184-DOWNLOAD-CURRENT
- IUPAC-C01267
- IUPAC-08786
- IUPAC-D01583
- NIST-CCCBDB-106990-NEUTRAL-EXPERIMENTAL
- NIST-CCCBDB-591935-NEUTRAL-GEOMETRY
- NIST-WEBBOOK-C106990-SPECTRA
- NIST-WEBBOOK-C106990-UVVIS-JCAMP-INDEX-0
- IUPAC-GOLD-BOOK-C01330-C00930-L03518
- NIST-CCCBDB-SRD101-FECO5-FERROCENE
- IUPAC-GOLD-BOOK-C01332
- NIST-CCCBDB-COMPLETE-EXPERIMENTAL-GEOMETRY
- IUPAC-GOLD-BOOK-I03294
- IUPAC-GOLD-BOOK-C01093
- IUPAC-GOLD-BOOK-E02069
- IUPAC-RED-BOOK-2005
- IUPAC-GOLD-BOOK-C01331

- NIST-CCCBDB-SRD101-SCF3-TICL4-FECO5
- NIST-RING-OPENING-COPOLYMER-SEQUENCE
- NIST-PRECISION-DEUTERATED-POLYETHYLENE
- NBS-JRES-IRON-CORROSION-1957
- NBS-JRES-GALVANIC-COUPLES-1950
- ECHEM-009-012-SOURCE-INVENTORY
- NIST-TRC-THERMOML-JCED-2015-60-3621-3630
- NIST-TRC-THERMOML-JCED-2006-51-680-685
- NIST-TRC-THERMOML-JCED-2014-59-757-763
- SFT-V3-ADMITTED-PROP-001
- SFT-V3-ADMITTED-PROP-002
- SFT-V3-ADMITTED-PROP-003
- SFT-V3-ADMITTED-PROP-004
- SFT-V3-ADMITTED-PROP-005
- SFT-V3-ADMITTED-PROP-006
- SFT-V3-ADMITTED-PROP-007
- SFT-V3-ADMITTED-PROP-008
- SFT-V3-ADMITTED-PROP-009
- SFT-V3-ADMITTED-PROP-010
- SFT-V3-ADMITTED-PROP-011
- SFT-V3-ADMITTED-PROP-012
- SFT-V3-ADMITTED-PROP-013
- NIST-THERMOREVERSIBLE-GELATION-PERCOLATION
- NIST-BRIDGING-GELATION
- IUPAC-S05735
- IUPAC-13807
- IUPAC-09464
- IUPAC-14011
- IUPAC-14026
- NIST-SRM-2888-POLYSTYRENE-CERTIFICATION
- NIST-GRAPHENE-DOUBLE-LAYER-2020
- ECHEM-011-SOURCE-INVENTORY
- IUPAC-ELECTROCHEMICAL-METHODS-2019
- SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003
- SFT-MAT-DEGR-CORROSION-001
- SFT-ENG-REQUIREMENT-001
- SFT-V3-LIVE-CLAIM-CENSUS
- SFT-V3-LIVE-DEPENDENCY-GRAPH

- SFT-CHEM-MAT-PAIRED-RECEIPTS
- NIST-METAL-ADDITIVE-CORROSION
- SFT-V3-BRANCH-OWNERSHIP-BOUNDARY
- NIST-CODATA-2022-ALL-CONSTANTS
- NIST-JRES-AGCL-STANDARD-POTENTIAL-1954
- NIST-CODATA-AGCL-WORK-CROSS-RECONSTRUCTION
- IUPAC-GREEN-BOOK-ELECTROCHEMISTRY-2007
- ECHEM-005-SOURCE-INVENTORY
- ECHEM-009-012-SHARED-SOURCES
- NIST-JRES-SILVER-ELECTROCHEMICAL-EQUIVALENT-1980
- ECHEM-006-SOURCE-INVENTORY
- NIST-ELECTRON-DIFFRACTION-DATABASE-REPORT
- NIST-NEUTRON-SCATTERING-LENGTHS
- NIST-SRM-676A-DIFFRACTION-STANDARD
- IUPAC-E02020
- IUPAC-08167
- NATURE-CHEMISTRY-S41557-022-01101-0-ARTICLE
- NATURE-CHEMISTRY-41557-2022-1101-MOESM1
- NIST-SP260-209-ZETA-MOBILITY
- NIST-SRD17-1972TIN-KOO3-15
- NIST-SRD17-1967WIL2017-1
- NIST-SRD17-1967AVR-KRA501-503-1
- NIST-SRD17-1962ASH-BUR685-691-1
- IUPAC:E02038
- RSC:D4SC01905A:3a
- RSC:D4SC01905A:3b
- RSC:D4SC01905A:3c
- RSC:D4SC01905A:3d
- RSC:D4SC01905A:3e
- RSC:D4SC01905A:3f
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- RSC:D4SC01905A:3ac
- RSC:D4SC01905A:3ad
- RSC:D4SC01905A:3ae
- RSC:D4SC01905A:3af
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- RSC:D4SC01905A:UNSUCCESSFUL:2
- RSC:D4SC01905A:UNSUCCESSFUL:3
- RSC:D4SC01905A:UNSUCCESSFUL:4
- RSC:D4SC01905A:UNSUCCESSFUL:5
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- RSC:D4SC01905A:S5
- RSC:D4SC01905A:S6
- RSC:D4SC01905A:S7
- RSC:D4SC01905A:3A-PRIME-TIME-COURSE
- EUROPE-PMC:PMC11186341:SUPPLEMENT
- NIST-WEBBOOK-SRD69-H2-DIATOMIC-CONSTANTS
- NIST-WEBBOOK-SRD69-D2-DIATOMIC-CONSTANTS
- USGS-PP-440-KK-STABLE-ISOTOPE-FRACTIONATION
- USGS-WRIR-02-4172-ISOTOPE-EQUILIBRIUM-CONSTANTS
- NUCHEM-005-008-SHARED-SOURCES
- NUCHEM-005-008-SOURCE-INVENTORY

- NIST-MDS2-3389-CAH-PLUS-QUANTUM-JUMP-THERMOMETRY
- NIST-CHEMISTRY-WEBBOOK-SRD69-WATER-GAS-CALORIMETRIC-TABLE
- DOE-ORNL-4865-FISSION-PRODUCT-BEHAVIOR-MSRE
- NUCHEM-009-012-SOURCE-INVENTORY
- IUPAC-FLUORESCENCE-STANDARDS-2010
- IUPAC-PHOTOLUMINESCENCE-QUANTUM-YIELD-2011
- NIST-SRM-2941A-FLUORESCENCE-GREEN
- NIST-IR-7458-FLUORESCENCE-GUIDE
- NIST-NLM-PASH-PHOSPHORESCENCE-2018
- ANAL-009-011-COMPLETE-SOURCE-INVENTORY
- NIST-JANAF-N-007-NO2-GAS
- NIST-JANAF-N-032-N2O4-GAS
- IUPAC-RT06783
- NIST-JPCRD-STANDARD-ELECTRODE-POTENTIALS-1989
- NIST-WEBBOOK-SRD69-GUIDE
- NIST-WEBBOOK-ACETONE-MULTIMODAL
- IUPAC-L03508
- IUPAC-L03511
- IUPAC-L03510
- IUPAC-O04362
- IUPAC-R05222
- IUPAC-08245
- IUPAC-L03519
- IUPAC-L03520
- IUPAC-M03868
- NIST-WEBBOOK-C13007926-UVVIS-COMPLETE
- NIST-WEBBOOK-C12082085-UVVIS-COMPLETE
- NIST-WEBBOOK-C12108111-UVVIS-COMPLETE
- NIST-WEBBOOK-C12146360-UVVIS-COMPLETE
- IUPAC-M03689
- IUPAC-P04404
- IUPAC-D01668
- NIST-MOLECULAR-MICROWAVE-SPECTRAL-DATABASES-SRD-114-115-117
- NIST-JPCRD-3-609-1974-DIATOMIC-MICROWAVE-SPECTRAL-TABLES
- IUPAC-M03924
- IUPAC-I03058
- IUPAC-E02038
- NIST-CCCBDB-SRD-101-DIMER-BINDING

- NIST-FARADAY-C8FD00092A-WATER-CLUSTER-DISSOCIATION
- NIST-JPCRD-1.555757-ION-CLUSTER-SCOPE
- NIST-SRM-3190-CONDUCTIVITY-CERTIFICATE
- NIST-SP-260-142-PRIMARY-CONDUCTIVITY
- NIST-SP-260-176-SRM-CATALOG
- ECHEM-007-SOURCE-INVENTORY
- NBS-JRES-TRANSFERENCE-CONCENTRATION-1931
- ECHEM-008-SOURCE-INVENTORY
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-DISSOCIATION
- NBS-RP729-ELECTROLYTIC-HYDROGEN-OXYGEN-FRACTIONATION-1934
- IUPAC-GOLD-BOOK-B00649
- IUPAC-GOLD-BOOK-C01012
- IUPAC-GOLD-BOOK-D01594
- IUPAC-GOLD-BOOK-H01881
- IUPAC-GOLD-BOOK-K03366
- IUPAC-GOLD-BOOK-L03518
- IUPAC-LT06764
- IUPAC-L03517
- NIST-WEBBOOK-C13939065-UVVIS-COMPLETE
- NIST-WEBBOOK-C14040110-UVVIS-COMPLETE
- NIST-SRM-2886-POLYETHYLENE-CERTIFICATION
- IUPAC-CT06769
- IUPAC-IT06779
- NIST-CCCBDB-SRD101-BF3-EXPERIMENTAL-GEOMETRY
- NIST-CCCBDB-SRD101-XEF2-EXPERIMENTAL-GEOMETRY
- NIST-CCCBDB-SRD101-XEF4-EXPERIMENTAL-GEOMETRY
- NIST-TRC-THERMOML-JCED-2011-56-4840-4848
- NIST-TRC-THERMOML-FPE-2017-437-34-42
- NIST-TRC-THERMOML-FPE-2008-271-43-52
- NIST-NBS-JCP-59-2254-1973
- NIST-CCCBDB-SRD101-EXPERIMENTAL-DIPOLES
- NIST-CCCBDB-COMPLETE-ELECTRON-AFFINITY-CATALOG
- NIST-CCCBDB-EXPERIMENTAL-MOLECULAR-ELECTRON-AFFINITY
- NIST-WEBBOOK-SRD69-GAS-PHASE-ION-THERMOCHEMISTRY
- NIST-CCCBDB-SRD-101-COMPLETE-EXPERIMENTAL-FORMATION-ENERGY
- NIST-CCCBDB-THERMODYNAMIC-REFERENCE-STATES
- NIST-CCCBDB-SRD101-EXPERIMENTAL-IONIZATION-ENERGY
- IUPAC-GOLD-BOOK-08789-2026

- NIST-CCCBDB-SRD101-DIBORANE
- NIST-CCCBDB-SRD101-BENZENE
- IUPAC-NMR-NOMENCLATURE-2001
- BMRB-ENTRY-68-CHEMICAL-SHIFTS
- ANAL-006-008-NMR-SOURCE-INVENTORY
- BMRB-ENTRY-52365-RELAXATION
- BMRB-ENTRY-27257-H-EXCHANGE
- BMRB-ENTRY-16582-SCALAR-COUPPLINGS
- NIST-NUCLEAR-PHYSICS-DATA-ISOTOPIC-COMPOSITION
- IUPAC-N04249
- IUPAC-08191
- IUPAC-E02130
- IUPAC-S06078
- IUPAC-H02809
- PUBCHEM-PUG-BROMOMETHANE-STRUCTURE
- PUBCHEM-PUG-HYDROXIDE-STRUCTURE
- PUBCHEM-PUG-METHANOL-STRUCTURE
- PUBCHEM-PUG-BROMIDE-STRUCTURE
- NIST-QUINTUPLE-DETECTOR-COPOLYMER
- IUPAC-O04367
- IUPAC-R05223
- IUPAC:P04491
- IUPAC:C01496
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- PMC8162770:SUPPLEMENT:S002
- NCBI-OA:PMC8162770
- IUPAC-GOLDBOOK-P04533-PHASE-RULE
- NIST-PHASE-DIAGRAM-GENERAL-DISCUSSION-GLOSSARY
- IUPAC-PHOSPHORESCENCE-LIFETIME
- NIST-BRANCH-PLACEMENT-DILUTE-SOLUTION
- NIST-MACROMOLECULAR-ARCHITECTURES
- NIST-POLYMER-PYROLYSIS-NETWORK
- NIST-POLYMER-DEPOLYMERIZATION-KINETICS
- NIST-CROSSLINKED-PHOTOPOLYMER-CONVERSION
- NIST-POLYMER-PROCESSING-METROLOGY
- NIST-PVOH-IONIC-LIQUID-GELATION
- IUPAC-09560
- IUPAC-09466
- NBS-NSRDS-45-RADIATION-CHEMISTRY-NITROUS-OXIDE
- IUPAC:C00960
- IUPAC:I03042
- IUPAC:T06274
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- NCBI-OA:PMC11598545
- NUCHEM-001-004-SHARED-SOURCES
- NUCHEM-005-NIST-SRM-PAIR
- DOE-OSTI-1580278-HFSLM-ISOTOPE-HARVESTING
- IAEA-TCS-31-RADIOTRACER-RTD-2008
- NIST-RAMAN-RELATIVE-INTENSITY-STANDARDS
- NIST-SRM-2241-RAMAN-785-NM
- NIST-SRM-2242A-RAMAN-532-NM

- IUPAC:M03997
- IUPAC:R05194
- IUPAC:R05196
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:1
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:2
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:3
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:4
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- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:8
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- ACS-FIGSHARE:10.1021/ja803370x.s001:PAGE:4
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAGE:37
- EUROPE-PMC:ORG-011:V1
- ACS-FIGSHARE:10.1021/ja803370x.s001
- IUPAC-R05326
- IUPAC-08205
- IUPAC-C01309
- IUPAC-M03804
- IUPAC-R05174
- NASA-JPL-MOLECULAR-SPECTROSCOPY-CATALOG
- NASA-JPL-CO-ROTATIONAL-LINE-CATALOG-028001
- IUPAC-C01051
- IUPAC-R05243
- IUPAC-S05991
- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC
- NATURE-COMMUNICATIONS-S41467-024-49814-9-COMplete
- RCSB-PDB-8WZF-8WZV
- CXIDB-221
- IUPAC-12823
- FREESOLV-0.52-COMMIT-6C7D19B
- NIST-TRC-THERMOML-JCED-2016-61-1470-1476
- NIST-CODATA-2022-BOLTZMANN
- NIM-NIST-ACOUSTIC-BOLTZMANN-2017
- NIST-JOHNSON-NOISE-BOLTZMANN-2011
- NIST-TRC-THERMOML-JCED-2013-58-663-670
- NIST-TRC-THERMOML-JCT-2019-133-135-142

- NIST-TRC-THERMOML-FPE-2018-477-78-86
- NIST-TRC-THERMOML-FPE-2018-474-6-13
- NIST-TRC-THERMOML-JCED-2005-50-1038-1042
- NIST-TRC-THERMOML-FPE-2017-453-13-23
- NIST-SRM-674-XRAY-INTENSITY-SET

Comparison records:

- VALID-010: complete registered membership count 129
- complete Chemistry base: 263 claims; 255 empirical; 8 formal-only
- retained evidence: 26486 measurement/evidence lines and 1177 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-CHEM-ACTIVITY-AMOUNT-TIME-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ADDITION-REACTION-FAMILY-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=28; evidence-lines=7; explicit-status-lines=3
- SFT-CHEM-ANALYTICAL-ACCURACY-TRUENESS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ANALYTICAL-DETECTION-QUANTIFICATION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ANALYTICAL-PERFORMANCE-BUDGET-022: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-ANALYTICAL-PRECISION-REPEATABILITY-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ANALYTICAL-SELECTIVITY-INTERFERENCE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ANALYTICAL-SENSITIVITY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ANTIAROMATIC-NONAROMATIC-DISTINCTION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=21; explicit-status-lines=5
- SFT-CHEM-AROMATIC-RECURRENCE-STABILITY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=9; evidence-lines=20; explicit-status-lines=2
- SFT-CHEM-ATOM-MAPPING-REACTION-BALANCE-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-BOND-DISSOCIATION-ENERGY-002: source-custodied-post-seal-comparison; closure=finite_complete; sources=6; evidence-lines=11; explicit-status-lines=7
- SFT-CHEM-CANONICAL-MOLECULAR-GRAPH-ENCODING-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=7; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-CHAIN-GROWTH-POLYMERIZATION-NETWORK-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-CHEMICAL-SUBSTRUCTURE-RELATION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2

- SFT-CHEM-CHROMATOGRAPHIC-RETENTION-RESOLUTION-018: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-CLASSICAL-QUANTUM-ALGORITHM-CORRESPONDENCE-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-COLLIGATIVE-COMPOSITION-RESPONSE-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=285; explicit-status-lines=0
- SFT-CHEM-COMPLEX-SPIN-STATE-ORDER-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=12; explicit-status-lines=2
- SFT-CHEM-CONCENTRATION-DEPENDENT-POTENTIAL-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-CONFORMER-ENUMERATION-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-CONFORMER-GENERATION-EQUIVALENCE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=16; explicit-status-lines=3
- SFT-CHEM-CONFORMER-POPULATION-ORDERING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=14; evidence-lines=20; explicit-status-lines=3
- SFT-CHEM-CONJUGATED-SUPPORT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=7; evidence-lines=18; explicit-status-lines=0
- SFT-CHEM-CONSTITUTIONAL-ISOMER-ENUMERATION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-COORDINATION-ENTITY-RETAINED-IDENTITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=26; explicit-status-lines=1
- SFT-CHEM-COORDINATION-GEOMETRY-HELD-ORIENTATION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=10; explicit-status-lines=3
- SFT-CHEM-COORDINATION-ISOMERISM-EQUIVALENCE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=14; explicit-status-lines=1
- SFT-CHEM-COORDINATION-NUMBER-INCIDENCE-COUNT-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=29; explicit-status-lines=1
- SFT-CHEM-COPOLYMER-SEQUENCE-COMPOSITION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-CORROSION-REACTION-NETWORK-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-COUPLED-MASS-HEAT-CHARGE-TRANSPORT-019: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=240; explicit-status-lines=0
- SFT-CHEM-CROSS-PROPERTY-MOLECULAR-VECTOR-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=13; evidence-lines=9033; explicit-status-lines=0
- SFT-CHEM-CROSSLINK-GELATION-BOUNDARY-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-DATABASE-IDENTITY-PROVENANCE-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-DEFECT-NONSTOICHIOMETRY-016: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=13; explicit-status-lines=2
- SFT-CHEM-DEGREE-OF-POLYMERIZATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2

- SFT-CHEM-DOUBLE-LAYER-INTERFACIAL-CHARGE-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ELECTROANALYTICAL-RESPONSE-020: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-ELECTROCHEMICAL-STORAGE-HANDOFF-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=1
- SFT-CHEM-ELECTROCHEMICAL-WORK-REACTION-DIRECTION-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ELECTRODE-POTENTIAL-CHEMICAL-RELATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=7; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ELECTRODE-REACTION-RATE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ELECTROLYSIS-PRODUCT-AMOUNT-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ELECTRON-NEUTRON-DIFFRACTION-017: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-ELECTROPHILIC-SUBSTITUTION-FAMILY-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=6; explicit-status-lines=1
- SFT-CHEM-ELECTROPHORETIC-MOBILITY-SEPARATION-019: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=55; explicit-status-lines=0
- SFT-CHEM-ELIMINATION-REACTION-FAMILY-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=47; evidence-lines=7; explicit-status-lines=1
- SFT-CHEM-EQUILIBRIUM-BOND-LENGTH-001: source-custodied-post-seal-comparison; closure=finite_complete; sources=3; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-EQUILIBRIUM-ISOTOPE-FRACTIONATION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-EXACT-MOLECULAR-SIMILARITY-DISTINCTION-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-FINITE-MICROSTATE-SUPPORT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=395; explicit-status-lines=0
- SFT-CHEM-FISSION-PRODUCT-CHEMICAL-DISTRIBUTION-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-FLUORESCENCE-YIELD-LIFETIME-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=72; explicit-status-lines=0
- SFT-CHEM-HALF-REACTION-IDENTITY-ORIENTATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=6; evidence-lines=4; explicit-status-lines=0
- SFT-CHEM-INFRARED-LINE-INTENSITY-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-INORGANIC-ACID-BASE-REDOX-NETWORK-017: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=14; explicit-status-lines=1

- SFT-CHEM-INORGANIC-COLOUR-ELECTRONIC-TRANSITION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=12; explicit-status-lines=1
- SFT-CHEM-INORGANIC-MAGNETIC-STATE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=13; explicit-status-lines=0
- SFT-CHEM-INSERTION-ELIMINATION-PATHWAY-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=10; explicit-status-lines=3
- SFT-CHEM-INTERMOLECULAR-BINDING-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=1305; explicit-status-lines=99
- SFT-CHEM-IONIC-CONDUCTIVITY-RELATION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-IONIC-MOBILITY-TRANSFERENCE-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ISOTOPE-EXCHANGE-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-JOINT-CORRELATION-DISSOCIATION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=30; explicit-status-lines=14
- SFT-CHEM-KINETIC-ISOTOPE-FRACTIONATION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-LIGAND-DENTICITY-CHELATION-TOPOLOGY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=6; evidence-lines=30; explicit-status-lines=1
- SFT-CHEM-LIGAND-STATE-SPLITTING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=13; explicit-status-lines=3
- SFT-CHEM-MASS-ISOTOPE-FRAGMENT-VECTOR-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-MASS-WEIGHTED-MOLECULAR-SIZE-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-MECHANISM-SEARCH-PROOF-TRACE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-METAL-CLUSTER-BONDING-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=10; explicit-status-lines=1
- SFT-CHEM-MOLECULAR-BOND-ANGLE-003: source-custodied-post-seal-comparison; closure=finite_complete; sources=3; evidence-lines=12; explicit-status-lines=8
- SFT-CHEM-MOLECULAR-DIFFUSION-RELATION-016: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=174; explicit-status-lines=1
- SFT-CHEM-MOLECULAR-DIPOLE-MAGNITUDE-005: source-custodied-post-seal-comparison; closure=finite_complete; sources=2; evidence-lines=15; explicit-status-lines=10
- SFT-CHEM-MOLECULAR-ELECTRON-AFFINITY-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=102; explicit-status-lines=0
- SFT-CHEM-MOLECULAR-FORMATION-ENERGY-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=2106; explicit-status-lines=0
- SFT-CHEM-MOLECULAR-GRAPH-ISOMORPHISM-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-MOLECULAR-IONIZATION-ENERGY-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=13; explicit-status-lines=1

- SFT-CHEM-MOLECULAR-MAGNETIC-RESPONSE-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=181; explicit-status-lines=1
- SFT-CHEM-MULTICENTRE-DELOCALIZED-SUPPORT-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=41; explicit-status-lines=16
- SFT-CHEM-MULTIMODAL-MOLECULAR-IDENTITY-021: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-NMR-CHEMICAL-SHIFT-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-NMR-RELAXATION-EXCHANGE-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-NMR-SPIN-COUPPLING-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-NUCLEAR-CHEMICAL-CARRIER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-NUCLEOPHILIC-SUBSTITUTION-FAMILY-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=9; evidence-lines=7; explicit-status-lines=1
- SFT-CHEM-NUMBER-AVERAGE-MOLECULAR-SIZE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-OVERPOTENTIAL-POLARIZATION-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-OXIDATIVE-ADDITION-REDUCTIVE-ELIMINATION-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=10; explicit-status-lines=0
- SFT-CHEM-PERICYCLIC-REACTION-FAMILY-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=38; evidence-lines=7; explicit-status-lines=2
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=27; explicit-status-lines=0
- SFT-CHEM-PHOSPHORESCENCE-INTERSYSTEM-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-POLYMER-ARCHITECTURE-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-CONFORMATION-SIZE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-DEGRADATION-NETWORK-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-DISPERSITY-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-MATERIALS-HANDOFF-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-PHASE-TRANSITION-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-PREDICTION-UNCERTAINTY-APPLICABILITY-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-PROTECTING-GROUP-REVERSIBLE-STATE-015: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=4; explicit-status-lines=0

- SFT-CHEM-RADIATION-CHEMISTRY-REACTION-NETWORK-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-RADICAL-REACTION-NETWORK-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=73; evidence-lines=6; explicit-status-lines=1
- SFT-CHEM-RADIOACTIVE-BRANCHING-CHEMICAL-YIELD-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-RADIOACTIVE-CHEMICAL-TRANSFORMATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-RADIOCHEMICAL-EQUILIBRIUM-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-RADIOCHEMICAL-SEPARATION-DECONTAMINATION-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-RADIOTRACER-CUSTODY-INFERENCE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-RAMAN-TRANSITION-INTENSITY-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-REACTION-GRAPH-GENERATION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-REARRANGEMENT-REACTION-FAMILY-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=16; evidence-lines=7; explicit-status-lines=2
- SFT-CHEM-RESONANCE-EQUIVALENT-REPRESENTATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=21; explicit-status-lines=0
- SFT-CHEM-RETROSYNTHETIC-DECOMPOSITION-RECONSTRUCTION-016: source-custodied-post-seal-comparison; closure=depth_independent; sources=7; evidence-lines=4; explicit-status-lines=0
- SFT-CHEM-ROTATIONAL-SPECTRUM-LINE-015: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-SELECTIVITY-COMPLETE-DISTRIBUTION-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=133; evidence-lines=5; explicit-status-lines=1
- SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=28; explicit-status-lines=5
- SFT-CHEM-SOLID-STATE-LOCAL-COORDINATION-015: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=11; explicit-status-lines=2
- SFT-CHEM-SOLVATION-DISSOLUTION-FREE-ORDER-015: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=810; explicit-status-lines=1
- SFT-CHEM-STEP-GROWTH-POLYMERIZATION-NETWORK-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-STEREOISOMER-ENUMERATION-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-SYMBOLIC-PROPERTY-EVALUATION-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-TEMPERATURE-CORRESPONDENCE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=10; explicit-status-lines=0
- SFT-CHEM-THERMAL-CONDUCTIVITY-RELATION-018: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=665; explicit-status-lines=0

- SFT-CHEM-UVVISIBLE-LINE-INTENSITY-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-VISCOUS-TRANSPORT-RELATION-017: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=434; explicit-status-lines=1
- SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2

Falsification boundary: Reject if any registered member, receipt, control, evidence file, result class, closure scope, formal-only boundary or source-custody row is missing, changed, duplicated or selected after outcome access.

Measurement receipt: sha256: ae508b51c9deeebc215808af2c3097fc80448689be1d697383ddc2b0deb37650.

Isolation certificate: sha256: 6284c5bcff4d381c884680ded03ef127a86ae20690f21498e7e92696837aabb8.

Custody certificate: sha256: 1918c235588d86df5ef9806c8003ade2e193ec35d670eb90dfda2519892575ea.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favorable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

Immutable identities. Source manifest

sha256: 5da8f549131d3aa1a17d0cbd2013993c8e3afa53b1d885781b3546abe11af1d0; derivation seal

sha256: 9456411c77b4a98e79d445ed808e0cb862e145bca107412dc2c31a591ceb3ab2; engine receipt

sha256: 043d3795d9b725736b67badcaf95545999e67162868c74f7a0fa4b3a066dff5e at receipts/engine/model_admitted/SFT-CHEM-VALIDATION-CROSS-SOURCE-REPRODUCIBILITY-VECTOR-010-043d3795d9b72573.json; empirical hash

sha256: b4d411650993e447c240d1d956e649d197301c458a9f6f41dc7d3d50ed2de689; measurement receipt

sha256: ae508b51c9deeebc215808af2c3097fc80448689be1d697383ddc2b0deb37650.

274. Complete adverse and out-of-bound Chemistry vector

Claim identity: SFT-CHEM-VALIDATION-ADVERSE-OUT-OF-BOUND-VECTOR-011

Question and theorem. VALID-011 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 263 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-011 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 263 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-AMPHOTERIC-001

- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-ACTIVATION-BARRIER-VALUE-RELATION-004
- SFT-CHEM-ACTIVITY-AMOUNT-TIME-003
- SFT-CHEM-ACTIVITY-NONIDEAL-COMPOSITION-009
- SFT-CHEM-ADDITION-REACTION-FAMILY-009
- SFT-CHEM-ANALYTICAL-ACCURACY-TRUENESS-001
- SFT-CHEM-ANALYTICAL-CALIBRATION-001
- SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001
- SFT-CHEM-ANALYTICAL-DETECTION-QUANTIFICATION-004
- SFT-CHEM-ANALYTICAL-PERFORMANCE-BUDGET-022
- SFT-CHEM-ANALYTICAL-PRECISION-REPEATABILITY-002
- SFT-CHEM-ANALYTICAL-SAMPLE-001
- SFT-CHEM-ANALYTICAL-SELECTIVITY-001
- SFT-CHEM-ANALYTICAL-SELECTIVITY-INTERFERENCE-005
- SFT-CHEM-ANALYTICAL-SENSITIVITY-003
- SFT-CHEM-ANTIAROMATIC-NONAROMATIC-DISTINCTION-004
- SFT-CHEM-AROMATIC-RECURRENCE-STABILITY-003
- SFT-CHEM-ATOM-MAPPING-REACTION-BALANCE-008
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-DISSOCIATION-ENERGY-002
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-LENGTH-STRENGTH-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-CANONICAL-MOLECULAR-GRAPH-ENCODING-001
- SFT-CHEM-CAT-CATALYST-001
- SFT-CHEM-CAT-PATHWAY-001
- SFT-CHEM-CAT-SELECTIVITY-001
- SFT-CHEM-CATALYTIC-TURNOVER-CYCLE-FREQUENCY-010
- SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003
- SFT-CHEM-CHAIN-GROWTH-POLYMERIZATION-NETWORK-005
- SFT-CHEM-CHEMICAL-POTENTIAL-EQUIVALENT-COMPONENT-008
- SFT-CHEM-CHEMICAL-SUBSTRUCTURE-RELATION-003

- SFT-CHEM-CHROMATOGRAPHIC-RETENTION-RESOLUTION-018
- SFT-CHEM-CLASSICAL-QUANTUM-ALGORITHM-CORRESPONDENCE-014
- SFT-CHEM-COLLIGATIVE-COMPOSITION-RESPONSE-014
- SFT-CHEM-COLLOID-DISPERSION-001
- SFT-CHEM-COMPETING-CHANNEL-BRANCHING-006
- SFT-CHEM-COMPLEX-SPIN-STATE-ORDER-007
- SFT-CHEM-CONCENTRATION-DEPENDENCE-RELATION-002
- SFT-CHEM-CONCENTRATION-DEPENDENT-POTENTIAL-004
- SFT-CHEM-CONFIGURATION-ORDER-PATH-011
- SFT-CHEM-CONFORMER-ENUMERATION-006
- SFT-CHEM-CONFORMER-GENERATION-EQUIVALENCE-005
- SFT-CHEM-CONFORMER-POPULATION-ORDERING-006
- SFT-CHEM-CONJUGATED-SUPPORT-001
- SFT-CHEM-CONSTITUTIONAL-ISOMER-ENUMERATION-004
- SFT-CHEM-COORDINATION-ENTITY-RETAINED-IDENTITY-001
- SFT-CHEM-COORDINATION-GEOMETRY-HELD-ORIENTATION-004
- SFT-CHEM-COORDINATION-ISOMERISM-EQUIVALENCE-005
- SFT-CHEM-COORDINATION-NUMBER-INCIDENCE-COUNT-002
- SFT-CHEM-COPOLYMER-SEQUENCE-COMPOSITION-007
- SFT-CHEM-CORROSION-REACTION-NETWORK-012
- SFT-CHEM-COUPLED-MASS-HEAT-CHARGE-TRANSPORT-019
- SFT-CHEM-CROSS-PROPERTY-MOLECULAR-VECTOR-014
- SFT-CHEM-CROSSLINK-GELATION-BOUNDARY-009
- SFT-CHEM-DATABASE-IDENTITY-PROVENANCE-011
- SFT-CHEM-DEFECT-NONSTOICHIOMETRY-016
- SFT-CHEM-DEGREE-OF-POLYMERIZATION-001
- SFT-CHEM-DIFFUSION-LIMITED-REACTION-BOUNDARY-011
- SFT-CHEM-DIHEDRAL-TORSIONAL-STATE-004
- SFT-CHEM-DOUBLE-LAYER-INTERFACIAL-CHARGE-011
- SFT-CHEM-ELECTROANALYTICAL-RESPONSE-020
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-ELECTROCHEMICAL-STORAGE-HANDOFF-013
- SFT-CHEM-ELECTROCHEMICAL-WORK-REACTION-DIRECTION-005
- SFT-CHEM-ELECTRODE-POTENTIAL-CHEMICAL-RELATION-002
- SFT-CHEM-ELECTRODE-REACTION-RATE-009
- SFT-CHEM-ELECTROLYSIS-PRODUCT-AMOUNT-006
- SFT-CHEM-ELECTRON-COUNT-SPIN-002
- SFT-CHEM-ELECTRON-NEUTRON-DIFFRACTION-017

- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-ELECTRONIC-STATE-IDENTITY-001
- SFT-CHEM-ELECTROPHILIC-SUBSTITUTION-FAMILY-008
- SFT-CHEM-ELECTROPHORETIC-MOBILITY-SEPARATION-019
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-ATOMIC-WEIGHT-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-GROUP-PERIOD-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-ELEM-ISOTOPE-001
- SFT-CHEM-ELEM-PERIODIC-BOUNDARY-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001
- SFT-CHEM-ELEM-VALENCE-001
- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001
- SFT-CHEM-ELIMINATION-REACTION-FAMILY-010
- SFT-CHEM-ENTHALPY-EQUIVALENT-STATE-006
- SFT-CHEM-ENTROPY-MULTIPLICITY-CORRESPONDENCE-005
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-EQUILIBRIUM-BOND-LENGTH-001
- SFT-CHEM-EQUILIBRIUM-ISOTOPE-FRACTIONATION-007
- SFT-CHEM-EXACT-MOLECULAR-SIMILARITY-DISTINCTION-010
- SFT-CHEM-FINITE-MICROSTATE-SUPPORT-001
- SFT-CHEM-FISSION-PRODUCT-CHEMICAL-DISTRIBUTION-011
- SFT-CHEM-FLUORESCENCE-YIELD-LIFETIME-010
- SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007
- SFT-CHEM-FUGACITY-EQUIVALENT-GAS-MIXTURE-010
- SFT-CHEM-HALF-REACTION-IDENTITY-ORIENTATION-001
- SFT-CHEM-HEAT-WORK-TRANSFER-PARTITION-004
- SFT-CHEM-INFRARED-LINE-INTENSITY-012
- SFT-CHEM-INORGANIC-ACID-BASE-REDOX-NETWORK-017
- SFT-CHEM-INORGANIC-COLOUR-ELECTRONIC-TRANSITION-008
- SFT-CHEM-INORGANIC-MAGNETIC-STATE-009
- SFT-CHEM-INSERTION-ELIMINATION-PATHWAY-013
- SFT-CHEM-INTERFACE-TRANSFER-001
- SFT-CHEM-INTERMOLECULAR-BINDING-011
- SFT-CHEM-INTERNAL-ENERGY-COMPOSITION-003
- SFT-CHEM-IONIC-CONDUCTIVITY-RELATION-007

- SFT-CHEM- IONIC-MOBILITY-TRANSFERENCE-008
- SFT-CHEM-ISOTOPE-EXCHANGE-006
- SFT-CHEM-JOINT-CORRELATION-DISSOCIATION-007
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-ORDER-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-KINETIC-ISOTOPE-EFFECT-RELATION-012
- SFT-CHEM-KINETIC-ISOTOPE-FRACTIONATION-008
- SFT-CHEM-LIGAND-DENTICITY-CHELATION-TOPOLOGY-003
- SFT-CHEM-LIGAND-STATE-SPLITTING-006
- SFT-CHEM-MASS-ISOTOPE-FRAGMENT-VECTOR-014
- SFT-CHEM-MASS-WEIGHTED-MOLECULAR-SIZE-003
- SFT-CHEM-MEAS-AMOUNT-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-MEAS-NOMENCLATURE-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MECHANISM-SEARCH-PROOF-TRACE-009
- SFT-CHEM-METAL-CLUSTER-BONDING-014
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-NETWORK-001
- SFT-CHEM-MOL-SUPRAMOLECULAR-001
- SFT-CHEM-MOLECULAR-BOND-ANGLE-003
- SFT-CHEM-MOLECULAR-DIFFUSION-RELATION-016
- SFT-CHEM-MOLECULAR-DIPOLE-MAGNITUDE-005
- SFT-CHEM-MOLECULAR-ELECTRON-AFFINITY-008
- SFT-CHEM-MOLECULAR-EXCLUSION-EXCHANGE-006
- SFT-CHEM-MOLECULAR-FORMATION-ENERGY-013
- SFT-CHEM-MOLECULAR-GRAPH-ISOMORPHISM-002
- SFT-CHEM-MOLECULAR-IONIZATION-ENERGY-007
- SFT-CHEM-MOLECULAR-MAGNETIC-RESPONSE-012
- SFT-CHEM-MOLECULAR-POLARIZABILITY-006

- SFT-CHEM-MOLECULAR-QUANTUM-MEASUREMENT-REDUCTION-014
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-MULTICENTRE-DELOCALIZED-SUPPORT-008
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013
- SFT-CHEM-MULTIMODAL-MOLECULAR-IDENTITY-021
- SFT-CHEM-NET-AUTOCATALYSIS-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-NMR-CHEMICAL-SHIFT-006
- SFT-CHEM-NMR-RELAXATION-EXCHANGE-008
- SFT-CHEM-NMR-SPIN-COUPLING-007
- SFT-CHEM-NUCLEAR-CHEMICAL-CARRIER-001
- SFT-CHEM-NUCLEAR-ELECTRONIC-COMPOSITION-012
- SFT-CHEM-NUCLEOPHILIC-SUBSTITUTION-FAMILY-007
- SFT-CHEM-NUMBER-AVERAGE-MOLECULAR-SIZE-002
- SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012
- SFT-CHEM-OPERATIONAL-CLASSICAL-QUANTUM-CORRESPONDENCE-015
- SFT-CHEM-ORBITAL-SUPPORT-OCCUPANCY-003
- SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001
- SFT-CHEM-ORGANOMETALLIC-ELECTRON-ACCOUNTING-011
- SFT-CHEM-ORGANOMETALLIC-METAL-CARBON-BOND-010
- SFT-CHEM-OVERPOTENTIAL-POLARIZATION-010
- SFT-CHEM-OXIDATIVE-ADDITION-REDUCTIVE-ELIMINATION-012
- SFT-CHEM-PARALLEL-MECHANISM-COMPOSITION-008
- SFT-CHEM-PERICYCLIC-REACTION-FAMILY-012
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011
- SFT-CHEM-PHOSPHORESCENCE-INTERSYSTEM-011
- SFT-CHEM-PHOTOCHEM-001
- SFT-CHEM-POLYMER-ARCHITECTURE-008
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-CHEM-POLYMER-CONFORMATION-SIZE-010
- SFT-CHEM-POLYMER-DEGRADATION-NETWORK-012
- SFT-CHEM-POLYMER-DISPERSITY-004
- SFT-CHEM-POLYMER-DISTRIBUTION-001
- SFT-CHEM-POLYMER-MATERIALS-HANDOFF-013
- SFT-CHEM-POLYMER-PHASE-TRANSITION-011
- SFT-CHEM-PRED-G-BLOCK-001

- SFT-CHEM-PRED-PERIODIC-ENDPOINT-001
- SFT-CHEM-PRED-SMITHIUM-001
- SFT-CHEM-PREDICTION-UNCERTAINTY-APPLICABILITY-013
- SFT-CHEM-PROTECTING-GROUP-REVERSIBLE-STATE-015
- SFT-CHEM-RADIATION-CHEMISTRY-REACTION-NETWORK-012
- SFT-CHEM-RADICAL-REACTION-NETWORK-013
- SFT-CHEM-RADIOACTIVE-BRANCHING-CHEMICAL-YIELD-004
- SFT-CHEM-RADIOACTIVE-CHEMICAL-TRANSFORMATION-002
- SFT-CHEM-RADIOCHEMICAL-EQUILIBRIUM-005
- SFT-CHEM-RADIOCHEMICAL-SEPARATION-DECONTAMINATION-010
- SFT-CHEM-RADIOTRACER-CUSTODY-INFERENCE-009
- SFT-CHEM-RAMAN-TRANSITION-INTENSITY-009
- SFT-CHEM-REACTION-DYNAMICS-SCATTERING-PRODUCT-STATE-013
- SFT-CHEM-REACTION-GRAPH-GENERATION-007
- SFT-CHEM-REARRANGEMENT-REACTION-FAMILY-011
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-RESOLVED-ROVIBRONIC-SPIN-COMPOSITION-013
- SFT-CHEM-RESONANCE-EQUIVALENT-REPRESENTATION-002
- SFT-CHEM-RETROSYNTHETIC-DECOMPOSITION-RECONSTRUCTION-016
- SFT-CHEM-REVERSIBLE-KINETIC-EQUILIBRIUM-CORRESPONDENCE-009
- SFT-CHEM-ROTATIONAL-CONSTANT-010
- SFT-CHEM-ROTATIONAL-SPECTRUM-LINE-015
- SFT-CHEM-ROVIBRONIC-COMPOSITION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-INTERMEDIATE-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-SELECTION-RULE-STRUCTURE-010
- SFT-CHEM-SELECTIVITY-COMPLETE-DISTRIBUTION-014
- SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007
- SFT-CHEM-SMITHIUM-CHEMICAL-SEPARATION-001
- SFT-CHEM-SMITHIUM-DECAY-CHANNEL-LEDGER-001
- SFT-CHEM-SMITHIUM-ION-OXIDATION-LADDER-001
- SFT-CHEM-SMITHIUM-JOINT-DETECTION-001
- SFT-CHEM-SMITHIUM-LIFETIME-BOUNDARY-001
- SFT-CHEM-SMITHIUM-SPECTROSCOPIC-CLASSES-001
- SFT-CHEM-SMITHIUM-SYNTHESIS-CONSERVATION-001
- SFT-CHEM-SOLID-STATE-LOCAL-COORDINATION-015

- SFT-CHEM-SOLUTION-EQUILIBRIUM-001
- SFT-CHEM-SOLVATION-DISSOLUTION-FREE-ORDER-015
- SFT-CHEM-SPEC-INFRARED-001
- SFT-CHEM-SPEC-MASS-001
- SFT-CHEM-SPEC-ROT-VIB-001
- SFT-CHEM-SPEC-UVVIS-001
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-STATE-SYMMETRY-DEGENERACY-005
- SFT-CHEM-STEP-GROWTH-POLYMERIZATION-NETWORK-006
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-CHEM-STEREO-DIASTEREOMER-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-CHEM-STEREOISOMER-ENUMERATION-005
- SFT-CHEM-STOICH-COEFFICIENT-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STOICH-LIMITING-001
- SFT-CHEM-STOICH-MIXTURE-001
- SFT-CHEM-STOICH-SOLUTION-001
- SFT-CHEM-STOICH-YIELD-001
- SFT-CHEM-SURFACE-ADSORPTION-001
- SFT-CHEM-SYMBOLIC-PROPERTY-EVALUATION-012
- SFT-CHEM-TEMPERATURE-CORRESPONDENCE-002
- SFT-CHEM-TEMPERATURE-DEPENDENCE-RELATION-003
- SFT-CHEM-THERMAL-CONDUCTIVITY-RELATION-018
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-TRANSITION-STATE-EQUIVALENT-BOUNDARY-005
- SFT-CHEM-UVVISIBLE-LINE-INTENSITY-013
- SFT-CHEM-VALIDATION-SMITHIUM-COMPLETE-FAMILY-001
- SFT-CHEM-VIBRATIONAL-FREQUENCY-009
- SFT-CHEM-VISCOUS-TRANSPORT-RELATION-017
- SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016
- SFT-CHEM-VALIDATION-CROSS-SOURCE-REPRODUCIBILITY-VECTOR-010

Generated grammar. Generate the complete eight-axis VALID-011 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 263 value-free claim identities registered for Chemistry VALID-011, their current receipt-bound evidence and all result classes.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-receipt-bound-membership__complete-valid-011-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
authority	complete-receipt-bound-membership	selected-headline	A headline omits the evidence surface.
relation	complete-valid-011-typed-evidence-vector	aggregate-score-or-fit	A score or fit can hide distinct failures.
custody	sealed-law-then-capability-closed-release	outcomes-in-forcing	Outcome access may select the law.
results	favourable-adverse-absent-unavailable-unresolved	favourable-only	Selective evidence invalidates the vector.
closure	each-declared-closure-scope-held	single-universal-scope	That overwrites finite boundaries.
formal	explicit-formal-standing-boundary	invented-empirical-successes	That fabricates data.
controls	omission-tamper-and-boundary-halt	omission-tolerated	A missing row cannot close a vector.
extension	dated-complete-extension-open	permanent-lock	A dated vector cannot bar discovery.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256: fc9fc6df1b04a5ce136c403bc1436b546963a4a32b3460effa69aecb8a299285.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256: 21221e73c5fbf39717d08befcd183423ae7bf43522087958782474bf6decea30.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: c3c285d25933ef831d30f8071fdcc65841329389a710405bb55c1f5772134932.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256: f0e0e4e5796429ca345efb1dd9d8be6175e1200dda2ed0ae034a77eacf58542d.

Independent reconstruction. Implementation hash

sha256: ebb766680add8ad494e3d447edc51e205fdb57736b98d14ebd76d6737d57ea34; independent certificate
sha256: 705b6ea49a34173bdbf88602f077fd87d268a1f3b9e211d71b281a45bd10bf97; external-validation
hash sha256: 98811d8f4411a00115ec97e4809f9cf18c5f9a83dc6444acb647af80a9e9f262. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ACID-BASE-COMPOSITE-2026
- IUPAC-GOLD-BOOK-A00306-2026

- IUPAC-DIDAC-E15-2026
- IUPAC-GOLD-BOOK-LEWIS-COMPOSITE-2026
- IUPAC-GOLD-BOOK-P04915-2026
- NIST-SRD-101-CCCBDB-EXPERIMENTAL-INTERNAL-ROTATION-BARRIERS-COMPLETE
- NIST-RADIOACTIVITY-SRMS-GENERAL-INFO-2025
- NIST-SRM-4239A-STRONTIUM-90-2022
- NIST-SRM-4324C-URANIUM-232-2025
- NUCHEM-001-004-SOURCE-INVENTORY
- NIST-TRC-THERMOML-JCED-2019-9B00694
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-17384a1d9c5f4b2e9aacbb6293204db
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-1478630680b84aee8dca359a52f20ba1
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- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-65ff4c19dea541b880d627eb4fbf8af3
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-8bbc1db0e28c4c60a61d554be47d9b92
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- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-c9cb9eedd61341a6b829cf445aa77b7c
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-6c8b4df5334a4f939f2d4f56b1edbf8
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- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-1bc497e7e2f745f4b38b233207da6bc8
- NIST-SRM-1959-CERTIFICATE-2011

- ANAL-001-005-SOURCE-INVENTORY
- IUPAC-GOLD-BOOK-ANALYTICAL-CALIBRATION-COMPOSITE-2026
- IUPAC-NIST-COMplete-ANALYTICAL-RECORD-COMPOSITE-2026
- NIST-SP958-DETECTION-QUANTIFICATION-PAGES-164-166
- SFT-V3-ANAL-001-011-IMMUTABLE-EVIDENCE
- NIST-VOLTAMMETRIC-LOD-STUDY
- NIST-SRM-1980-ELECTROPHORETIC-MOBILITY
- NIST-SRM-1959-CERTIFICATION-2010
- IUPAC-GOLD-BOOK-ANALYTICAL-SAMPLE-COMPOSITE-2026
- IUPAC-GOLD-BOOK-ANALYTICAL-SELECTIVITY-COMPOSITE-2026
- NIST-IMS-FENTANYL-INTERFERENTS-2019
- NIST-SP-260-126-NTRM-GAS-STANDARDS
- IUPAC-A00441
- IUPAC-AT06987
- NIST-WEBBOOK-C71432-THERMOCHEMISTRY
- NIST-CCCBDB-1120532-NEUTRAL-EXPERIMENTAL
- NIST-CCCBDB-629209-NEUTRAL-EXPERIMENTAL
- IUPAC-A00442
- IUPAC-R05333
- NIST-WEBBOOK-C110838-THERMOCHEMISTRY
- NIST-WEBBOOK-C110827-THERMOCHEMISTRY
- NIST-CCCBDB-SRD101-C71432-COMplete-EXPERIMENTAL-DATA
- NIST-CCCBDB-SRD101-C110838-COMplete-EXPERIMENTAL-DATA
- NIST-CCCBDB-SRD101-C110827-COMplete-EXPERIMENTAL-DATA
- PUBCHEM-PUG-REST-AUTHORITY
- PUBCHEM-C3H8O-FORMULA-CENSUS
- PUBCHEM-ASPIRIN-CONFORMERS
- PUBCHEM-ASPIRIN-SUBSTRUCTURE
- PUBCHEM-ASPIRIN-SIMILARITY
- CHEBI-REST-AUTHORITY
- RHEA-IMMUTABLE-REACTION-SMILES
- USPTO-IMMUTABLE-REACTION-SURFACE
- EMBL-EBI-CHEBI-BIOMOLECULAR-BOUNDARY-2026
- IUPAC-GOLD-BOOK-CT07009-2026
- IUPAC-GOLD-BOOK-C01384-2026
- APS-PRA-49-2460-1994
- PRL-UDEM-H-1S2S-1997
- PRL-PARTHEY-HD-1S2S-SHIFT-2010

- SI-DEFINING-LIGHT-SPEED
- JCP-LIU-H2-DISSOCIATION-2009
- PRA-HUSSELS-D2-DISSOCIATION-2022
- IUPAC-GOLD-BOOK-IT07058-2026
- IUPAC-GOLD-BOOK-B00702-2026
- IUPAC-GOLD-BOOK-08789-2026
- IUPAC-GOLD-BOOK-B00707-2026
- IUPAC-GOLD-BOOK-BOND-POLARITY-COMPOSITE-2026
- CHEM-AUTHORITY-CATALYST-COMPOSITE-2026
- CHEM-AUTHORITY-CATALYTIC-PATHWAY-COMPOSITE-2026
- IUPAC-GOLD-BOOK-CATALYTIC-SELECTIVITY-COMPOSITE-2026
- NATURE-NANOTECHNOLOGY-S41565-021-00959-4-COMPLETE
- NIST-JRES-AGCL-COMPLETE-CELL-1954
- NIST-STOCHASTIC-PHOTOPOLYMER-NETWORK-GROWTH
- NIST-MONODISPERSE-PAMS-KINETIC-NETWORK
- NIST-TRC-THERMOML-JCED-2019-9B00414
- NIST-WEBBOOK-GAS-CHROMATOGRAPHY
- NIST-WEBBOOK-BENZENE-MULTIMODAL
- NIST-WEBBOOK-CYCLOHEXANE-MULTIMODAL
- NIST-TRC-THERMOML-FPE-2013-337-60-66
- NIST-TRC-THERMOML-FPE-2016-420-14-19
- NIST-TRC-THERMOML-JCT-2009-41-1439-1445
- IUPAC-GOLD-BOOK-COLLOIDAL-DISPERSION-COMPOSITE-2026
- COMMUNICATIONS-CHEMISTRY-S42004-024-01239-7-COMPLETE
- IUPAC-HT06789
- IUPAC-LT07067
- IUPAC-ST07104
- RSC-PCCP-2014-C3CP54664K-TABLE-2
- NIST-JRES-AGCL-HCL-CONCENTRATION-SERIES-1954
- NIST-CCCBDB-SRD101-ETHANOL-EXPERIMENTAL-ROTATIONAL-BARRIER
- IUPAC-C01262
- IUPAC-C01258
- IUPAC-C01259
- NIST-CCCBDB-106978-NEUTRAL-EXPERIMENTAL
- IUPAC-CT01038
- NIST-CCCBDB-106978-INTERNAL-ROTATION
- NIST-CCCBDB-109660-INTERNAL-ROTATION-1
- OSTI-22415852-PRIMARY-RECORD

- OSTI-22415852-PUBLIC-FULLTEXT-ENDPOINT
- PUBMED-25591384-COMPLETE-FULLTEXT-PROVIDER-SURFACE
- ACS-JP404315T-COMPLETE-PRIMARY-AND-SI-SURFACE
- ACS-FIGSHARE-EXACT-RESOURCE-DOI-JP404315T
- ACS-FIGSHARE-4012420-JP404315T-SI-001
- CORE-43583184-DOWNLOAD-PDF-LEGACY
- CORE-43583184-DOWNLOAD-CURRENT
- IUPAC-C01267
- IUPAC-08786
- IUPAC-D01583
- NIST-CCCBDB-106990-NEUTRAL-EXPERIMENTAL
- NIST-CCCBDB-591935-NEUTRAL-GEOMETRY
- NIST-WEBBOOK-C106990-SPECTRA
- NIST-WEBBOOK-C106990-UVVIS-JCAMP-INDEX-0
- IUPAC-GOLD-BOOK-C01330-C00930-L03518
- NIST-CCCBDB-SRD101-FECO5-FERROCENE
- IUPAC-GOLD-BOOK-C01332
- NIST-CCCBDB-COMPLETE-EXPERIMENTAL-GEOMETRY
- IUPAC-GOLD-BOOK-I03294
- IUPAC-GOLD-BOOK-C01093
- IUPAC-GOLD-BOOK-E02069
- IUPAC-RED-BOOK-2005
- IUPAC-GOLD-BOOK-C01331
- NIST-CCCBDB-SRD101-SCF3-TICL4-FECO5
- NIST-RING-OPENING-COPOLYMER-SEQUENCE
- NIST-PRECISION-DEUTERATED-POLYETHYLENE
- NBS-JRES-IRON-CORROSION-1957
- NBS-JRES-GALVANIC-COUPLES-1950
- ECHEM-009-012-SOURCE-INVENTORY
- NIST-TRC-THERMOML-JCED-2015-60-3621-3630
- NIST-TRC-THERMOML-JCED-2006-51-680-685
- NIST-TRC-THERMOML-JCED-2014-59-757-763
- SFT-V3-ADMITTED-PROP-001
- SFT-V3-ADMITTED-PROP-002
- SFT-V3-ADMITTED-PROP-003
- SFT-V3-ADMITTED-PROP-004
- SFT-V3-ADMITTED-PROP-005
- SFT-V3-ADMITTED-PROP-006

- SFT-V3-ADMITTED-PROP-007
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- SFT-V3-ADMITTED-PROP-013
- NIST-THERMOREVERSIBLE-GELATION-PERCOLATION
- NIST-BRIDGING-GELATION
- IUPAC-S05735
- IUPAC-13807
- IUPAC-09464
- IUPAC-14011
- IUPAC-14026
- NIST-SRM-2888-POLYSTYRENE-CERTIFICATION
- NATURE-COMMUNICATIONS-S41467-025-68008-5-COMPLETE
- NIST-GRAPHENE-DOUBLE-LAYER-2020
- ECHEM-011-SOURCE-INVENTORY
- IUPAC-ELECTROCHEMICAL-METHODS-2019
- IUPAC-GOLD-BOOK-ELECTROCHEMICAL-CELL-COMPOSITE-2026
- SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003
- SFT-MAT-DEGR-CORROSION-001
- SFT-ENG-REQUIREMENT-001
- SFT-V3-LIVE-CLAIM-CENSUS
- SFT-V3-LIVE-DEPENDENCY-GRAPH
- SFT-CHEM-MAT-PAIRED-RECEIPTS
- NIST-METAL-ADDITIVE-CORROSION
- SFT-V3-BRANCH-OWNERSHIP-BOUNDARY
- NIST-CODATA-2022-ALL-CONSTANTS
- NIST-JRES-AGCL-STANDARD-POTENTIAL-1954
- NIST-CODATA-AGCL-WORK-CROSS-RECONSTRUCTION
- IUPAC-GREEN-BOOK-ELECTROCHEMISTRY-2007
- ECHEM-005-SOURCE-INVENTORY
- ECHEM-009-012-SHARED-SOURCES
- NIST-JRES-SILVER-ELECTROCHEMICAL-EQUIVALENT-1980
- ECHEM-006-SOURCE-INVENTORY
- NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- NIST-ELECTRON-DIFFRACTION-DATABASE-REPORT

- NIST-NEUTRON-SCATTERING-LENGTHS
- NIST-SRM-676A-DIFFRACTION-STANDARD
- IUPAC-GOLD-BOOK-ELECTRONEGATIVITY-COMPOSITE-2026
- IUPAC-GOLD-BOOK-ET07026-2026
- IUPAC-E02020
- IUPAC-08167
- NATURE-CHEMISTRY-S41557-022-01101-0-ARTICLE
- NATURE-CHEMISTRY-41557-2022-1101-MOESM1
- NIST-SP260-209-ZETA-MOBILITY
- IUPAC-GOLD-BOOK-A00499-2026
- IUPAC-GOLD-BOOK-R05258-2026
- IUPAC-GOLD-BOOK-C01022-2026
- IUPAC-PERIODIC-TABLE-2022
- IUPAC-GOLD-BOOK-I03158-2026
- IUPAC-GOLD-BOOK-I03331-2026
- IUPAC-GOLD-BOOK-V06588-2026
- NIST-SRD17-1972TIN-KOO3-15
- NIST-SRD17-1967WIL2017-1
- NIST-SRD17-1967AVR-KRA501-503-1
- NIST-SRD17-1962ASH-BUR685-691-1
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- RSC:D4SC01905A:3e
- RSC:D4SC01905A:3f
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- RSC:D4SC01905A:3A-PRIME-TIME-COURSE
- EUROPE-PMC:PMC11186341:SUPPLEMENT
- NIST-CHEMISTRY-WEBBOOK-SRD69-WATER-FLUID-PROPERTIES
- IUPAC-GOLD-BOOK-CHEMICAL-EQUILIBRIUM-COMPOSITE-2026
- NIST-WEBBOOK-SRD69-H2-DIATOMIC-CONSTANTS
- NIST-WEBBOOK-SRD69-D2-DIATOMIC-CONSTANTS
- USGS-PP-440-KK-STABLE-ISOTOPE-FRACTIONATION
- USGS-WRIR-02-4172-ISOTOPE-EQUILIBRIUM-CONSTANTS
- NUCHEM-005-008-SHARED-SOURCES
- NUCHEM-005-008-SOURCE-INVENTORY
- NIST-MDS2-3389-CAH-PLUS-QUANTUM-JUMP-THERMOMETRY

- NIST-CHEMISTRY-WEBBOOK-SRD69-WATER-GAS-CALORIMETRIC-TABLE
- DOE-ORNL-4865-FISSION-PRODUCT-BEHAVIOR-MSRE
- NUCHEM-009-012-SOURCE-INVENTORY
- IUPAC-FLUORESCENCE-STANDARDS-2010
- IUPAC-PHOTOLUMINESCENCE-QUANTUM-YIELD-2011
- NIST-SRM-2941A-FLUORESCENCE-GREEN
- NIST-IR-7458-FLUORESCENCE-GUIDE
- NIST-NLM-PASH-PHOSPHORESCENCE-2018
- ANAL-009-011-COMPLETE-SOURCE-INVENTORY
- NIST-JANAF-N-007-NO2-GAS
- NIST-JANAF-N-032-N2O4-GAS
- NIST-TRC-THERMOML-FPE-2019-485-145-152
- IUPAC-RT06783
- NIST-JPCRD-STANDARD-ELECTRODE-POTENTIALS-1989
- NIST-WEBBOOK-SRD69-GUIDE
- NIST-WEBBOOK-ACETONE-MULTIMODAL
- IUPAC-L03508
- IUPAC-L03511
- IUPAC-L03510
- IUPAC-O04362
- IUPAC-R05222
- IUPAC-08245
- IUPAC-L03519
- IUPAC-L03520
- IUPAC-M03868
- NIST-WEBBOOK-C13007926-UVVIS-COMPLETE
- NIST-WEBBOOK-C12082085-UVVIS-COMPLETE
- NIST-WEBBOOK-C12108111-UVVIS-COMPLETE
- NIST-WEBBOOK-C12146360-UVVIS-COMPLETE
- IUPAC-M03689
- IUPAC-P04404
- IUPAC-D01668
- NIST-MOLECULAR-MICROWAVE-SPECTRAL-DATABASES-SRD-114-115-117
- NIST-JPCRD-3-609-1974-DIATOMIC-MICROWAVE-SPECTRAL-TABLES
- IUPAC-M03924
- IUPAC-I03058
- IUPAC-E02038
- IUPAC-GOLD-BOOK-INTERFACE-TRANSFER-COMPOSITE-2026

- NIST-CCCBDB-SRD-101-DIMER-BINDING
- NIST-FARADAY-C8FD00092A-WATER-CLUSTER-DISSOCIATION
- NIST-JPCRD-1.555757-ION-CLUSTER-SCOPE
- NIST-SRM-3190-CONDUCTIVITY-CERTIFICATE
- NIST-SP-260-142-PRIMARY-CONDUCTIVITY
- NIST-SP-260-176-SRM-CATALOG
- ECHEM-007-SOURCE-INVENTORY
- NBS-JRES-TRANSFERENCE-CONCENTRATION-1931
- ECHEM-008-SOURCE-INVENTORY
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-DISSOCIATION
- IUPAC-GOLD-BOOK-ACTIVATION-COMPOSITE-2026
- IUPAC-GOLD-BOOK-O04322-2026
- IUPAC-GOLD-BOOK-REACTION-RATE-COMPOSITE-2026
- NATURE-COMMUNICATIONS-S41467-024-44753-X-COMPLETE
- NBS-RP729-ELECTROLYTIC-HYDROGEN-OXYGEN-FRACTIONATION-1934
- IUPAC-GOLD-BOOK-B00649
- IUPAC-GOLD-BOOK-C01012
- IUPAC-GOLD-BOOK-D01594
- IUPAC-GOLD-BOOK-H01881
- IUPAC-GOLD-BOOK-K03366
- IUPAC-GOLD-BOOK-L03518
- IUPAC-LT06764
- IUPAC-L03517
- NIST-WEBBOOK-C13939065-UVVIS-COMPLETE
- NIST-WEBBOOK-C14040110-UVVIS-COMPLETE
- NIST-SRM-2886-POLYETHYLENE-CERTIFICATION
- IUPAC-GOLD-BOOK-A00297-2026
- IUPAC-GOLD-BOOK-M03986-2026
- IUPAC-GOLD-BOOK-CT01038-2026
- IUPAC-GOLD-BOOK-M03987-2026
- IUPAC-GOLD-BOOK-S06236-2026
- IUPAC-GOLD-BOOK-C01039-2026
- IUPAC-GOLD-BOOK-08013-2026
- IUPAC-GOLD-BOOK-08133-2026
- IUPAC-CT06769
- IUPAC-IT06779
- IUPAC-GOLD-BOOK-S06061-2026
- IUPAC-GOLD-BOOK-I03098-2026

- IUPAC-GOLD-BOOK-I03289-2026
- IUPAC-GOLD-BOOK-M04002-2026
- IUPAC-GOLD-BOOK-N04112-2026
- IUPAC-GOLD-BOOK-13867-2026
- NIST-CCCBDB-SRD101-BF3-EXPERIMENTAL-GEOMETRY
- NIST-CCCBDB-SRD101-XEF2-EXPERIMENTAL-GEOMETRY
- NIST-CCCBDB-SRD101-XEF4-EXPERIMENTAL-GEOMETRY
- NIST-TRC-THERMOML-JCED-2011-56-4840-4848
- NIST-TRC-THERMOML-FPE-2017-437-34-42
- NIST-TRC-THERMOML-FPE-2008-271-43-52
- NIST-NBS-JCP-59-2254-1973
- NIST-CCCBDB-SRD101-EXPERIMENTAL-DIPOLES
- NIST-CCCBDB-COMPLETE-ELECTRON-AFFINITY-CATALOG
- NIST-CCCBDB-EXPERIMENTAL-MOLECULAR-ELECTRON-AFFINITY
- NIST-WEBBOOK-SRD69-GAS-PHASE-ION-THERMOCHEMISTRY
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- NIST-CCCBDB-SRD-101-COMPLETE-EXPERIMENTAL-FORMATION-ENERGY
- NIST-CCCBDB-THERMODYNAMIC-REFERENCE-STATES
- NIST-CCCBDB-SRD101-EXPERIMENTAL-IONIZATION-ENERGY
- NIST-CCCBDB-SRD101-EXPERIMENTAL-POLARIZABILITIES
- NIST-MDS2-3389-CAH-QUANTUM-STATE-TRACKING-CONTROL-V1.0.1
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- NIST-CCCBDB-SRD101-DIBORANE
- NIST-CCCBDB-SRD101-BENZENE
- NIST-TRC-THERMOML-JCT-2012-47-260-266
- IUPAC-GOLD-BOOK-AUTOCATALYSIS-COMPOSITE-2026
- IUPAC-GOLD-BOOK-REACTION-NETWORK-COMPOSITE-2026
- IUPAC-NMR-NOMENCLATURE-2001
- BMRB-ENTRY-68-CHEMICAL-SHIFTS
- ANAL-006-008-NMR-SOURCE-INVENTORY
- BMRB-ENTRY-52365-RELAXATION
- BMRB-ENTRY-27257-H-EXCHANGE
- BMRB-ENTRY-16582-SCALAR-COUPPLINGS
- NIST-NUCLEAR-PHYSICS-DATA-ISOTOPIC-COMPOSITION
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-HD-D2-DIATOMIC-CONSTANTS-2026
- IUPAC-N04249
- IUPAC-08191
- IUPAC-E02130

- IUPAC-S06078
- IUPAC-H02809
- PUBCHEM-PUG-BROMOMETHANE-STRUCTURE
- PUBCHEM-PUG-HYDROXIDE-STRUCTURE
- PUBCHEM-PUG-METHANOL-STRUCTURE
- PUBCHEM-PUG-BROMIDE-STRUCTURE
- NIST-QUINTUPLE-DETECTOR-COPOLYMER
- IUPAC-GOLD-BOOK-FUNCTIONAL-GROUP-COMPOSITE-2026
- IUPAC-GOLD-BOOK-ORGANIC-REACTION-FAMILY-COMPOSITE-2026
- IUPAC-E01913
- IUPAC-O04328
- IUPAC-O04367
- IUPAC-R05223
- NATURE-COMMUNICATIONS-S41467-026-70199-4-SOURCE-DATA
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- IUPAC-GOLD-BOOK-CHEMICAL-PHASE-COMPOSITE-2026
- IUPAC-GOLDBOOK-P04533-PHASE-RULE
- NIST-PHASE-DIAGRAM-GENERAL-DISCUSSION-GLOSSARY
- IUPAC-PHOSPHORESCENCE-LIFETIME
- IUPAC-GOLD-BOOK-PHOTOCHEMISTRY-COMPOSITE-2026
- NIST-BRANCH-PLACEMENT-DILUTE-SOLUTION
- NIST-MACROMOLECULAR-ARCHITECTURES
- IUPAC-GOLD-BOOK-POLYMER-CHAIN-COMPOSITE-2026
- NIST-POLYMER-PYROLYSIS-NETWORK
- NIST-POLYMER-DEPOLYMERIZATION-KINETICS
- IUPAC-GOLD-BOOK-POLYMER-DISTRIBUTION-COMPOSITE-2026
- NIST-CROSSLINKED-PHOTOPOLYMER-CONVERSION
- NIST-POLYMER-PROCESSING-METROLOGY
- NIST-PVOH-IONIC-LIQUID-GELATION
- IUPAC-09560
- IUPAC-09466
- NBS-NSRDS-45-RADIATION-CHEMISTRY-NITROUS-OXIDE
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- PMC11598545:TABLE:2:ROW:11
- PMC11598545:TABLE:2:ROW:12
- PMC11598545:TABLE:3:ROW:1
- PMC11598545:TABLE:3:ROW:2
- PMC11598545:TABLE:3:ROW:3
- PMC11598545:TABLE:3:ROW:4
- PMC11598545:TABLE:4:ROW:1
- PMC11598545:TABLE:4:ROW:2
- PMC11598545:TABLE:4:ROW:3
- PMC11598545:TABLE:4:ROW:4
- PMC11598545:ARTICLE
- PMC11598545:SUPPLEMENT:S001
- NCBI-OA:PMC11598545
- NUCHEM-001-004-SHARED-SOURCES
- NUCHEM-005-NIST-SRM-PAIR
- DOE-OSTI-1580278-HFSLM-ISOTOPE-HARVESTING
- IAEA-TCS-31-RADIOTRACER-RTD-2008
- NIST-RAMAN-RELATIVE-INTENSITY-STANDARDS
- NIST-SRM-2241-RAMAN-785-NM
- NIST-SRM-2242A-RAMAN-532-NM
- NATURE-COMMUNICATIONS-S41467-025-66587-X-COMPLETE
- IUPAC:M03997
- IUPAC:R05194

- IUPAC:R05196
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:1
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:2
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:3
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:4
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:5
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:6
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:7
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:8
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAGE:3
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAGE:4
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAGE:37
- EUROPE-PMC:ORG-011:V1
- ACS-FIGSHARE:10.1021/ja803370x.s001
- IUPAC-GOLD-BOOK-REDOX-COMPOSITE-2026
- IUPAC-GOLD-BOOK-O04365-2026
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-HD-D2-RESOLVED-DIATOMIC-SURFACE-2026
- IUPAC-R05326
- IUPAC-08205
- IUPAC-C01309
- IUPAC-M03804
- IUPAC-R05174
- NATURE-COMMUNICATIONS-S41467-023-40190-4-COMplete
- NIST-CCCBDB-COMplete-ROTATIONAL-CONSTANT-SURFACE
- NASA-JPL-MOLECULAR-SPECTROSCOPY-CATALOG
- NASA-JPL-CO-ROTATIONAL-LINE-CATALOG-028001
- NIST-MOLECULAR-SPECTROSCOPY-H2-D2-SUCCESSOR
- IUPAC-GOLD-BOOK-CHEMICAL-REACTION-COMPOSITE-2026
- IUPAC-GOLD-BOOK-I03096-2026
- IUPAC-GOLD-BOOK-REACTION-MECHANISM-COMPOSITE-2026
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- IUPAC-C01051
- IUPAC-R05243
- IUPAC-S05991
- ORD-FEAF1B793C6D408AAEC1CAC7CC3CEADC
- NATURE-COMMUNICATIONS-S41467-024-49814-9-COMplete
- RCSB-PDB-8WZF-8WZV
- CXIDB-221

- IUPAC-12823
- IUPAC-GOLD-BOOK-SOLUTION-EQUILIBRIUM-COMPOSITE-2026
- FREESOLV-0.52-COMMIT-6C7D19B
- NIST-TRC-THERMOML-JCED-2016-61-1470-1476
- IUPAC-GOLD-BOOK-INFRARED-SPECTROSCOPY-COMPOSITE-2026
- IUPAC-GOLD-BOOK-MASS-SPECTROSCOPY-COMPOSITE-2026
- IUPAC-GOLD-BOOK-ROTATION-VIBRATION-COMPOSITE-2026
- IUPAC-GOLD-BOOK-UVVISIBLE-SPECTROSCOPY-COMPOSITE-2026
- IUPAC-GOLD-BOOK-CHIRALITY-COMPOSITE-2026
- IUPAC-GOLD-BOOK-DIASTEREOMER-COMPOSITE-2026
- IUPAC-GOLD-BOOK-ENANTIOMER-COMPOSITE-2026
- IUPAC-GOLD-BOOK-C01034-2026
- IUPAC-GOLD-BOOK-M03722-2026
- IUPAC-GOLD-BOOK-S06026-2026
- IUPAC-GOLD-BOOK-D01771-2026
- IUPAC-GOLD-BOOK-M03949-2026
- IUPAC-GOLD-BOOK-S05746-2026
- IUPAC-GOLD-BOOK-C01041-2026
- IUPAC-GOLD-BOOK-ADSORPTION-COMPOSITE-2026
- NIST-CODATA-2022-BOLTZMANN
- NIM-NIST-ACOUSTIC-BOLTZMANN-2017
- NIST-JOHNSON-NOISE-BOLTZMANN-2011
- ACS-JPCA-2015-JP505790M-COMplete-TEMPERATURE-RATE-SERIES
- NIST-TRC-THERMOML-JCED-2013-58-663-670
- NIST-TRC-THERMOML-JCT-2019-133-135-142
- NIST-TRC-THERMOML-FPE-2018-477-78-86
- IUPAC-GOLD-BOOK-REACTION-DIRECTION-COMPOSITE-2026
- IUPAC-GOLD-BOOK-REACTION-THERMOCHEMISTRY-COMPOSITE-2026
- KYRIAKOU-ET-AL-ACS-NANO-NN500703K-COMplete
- IUPAC-SUPERHEAVY-2018
- NIST-ATOMIC-SPECTROSCOPY
- NIST-CCCBDB-COMplete-DISPLAYED-FUNDAMENTAL-FREQUENCY-SURFACE
- NIST-TRC-THERMOML-FPE-2018-474-6-13
- NIST-TRC-THERMOML-JCED-2005-50-1038-1042
- NIST-TRC-THERMOML-FPE-2017-453-13-23
- NIST-SRM-674-XRAY-INTENSITY-SET

Comparison records:

- VALID-011: complete registered membership count 263

- complete Chemistry base: 263 claims; 255 empirical; 8 formal-only
- retained evidence: 26486 measurement/evidence lines and 1177 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-CHEM-AB-ACID-BASE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-AB-AMPHOTERIC-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-AB-BUFFER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-AB-LEWIS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-AB-PROTON-TRANSFER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ACTIVATION-BARRIER-VALUE-RELATION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=52; explicit-status-lines=3
- SFT-CHEM-ACTIVITY-AMOUNT-TIME-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ACTIVITY-NONIDEAL-COMPOSITION-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=213; explicit-status-lines=1
- SFT-CHEM-ADDITION-REACTION-FAMILY-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=28; evidence-lines=7; explicit-status-lines=3
- SFT-CHEM-ANALYTICAL-ACCURACY-TRUENESS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ANALYTICAL-CALIBRATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ANALYTICAL-DETECTION-QUANTIFICATION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ANALYTICAL-PERFORMANCE-BUDGET-022: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-ANALYTICAL-PRECISION-REPEATABILITY-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ANALYTICAL-SAMPLE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ANALYTICAL-SELECTIVITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ANALYTICAL-SELECTIVITY-INTERFERENCE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ANALYTICAL-SENSITIVITY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ANTIAROMATIC-NONAROMATIC-DISTINCTION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=21; explicit-status-lines=5
- SFT-CHEM-AROMATIC-RECURRENCE-STABILITY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=9; evidence-lines=20; explicit-status-lines=2

- SFT-CHEM-ATOM-MAPPING-REACTION-BALANCE-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-BOND-CHEMICAL-BOND-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-BOND-COVALENT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-BOND-DISSOCIATION-ENERGY-002: source-custodied-post-seal-comparison; closure=finite_complete; sources=6; evidence-lines=11; explicit-status-lines=7
- SFT-CHEM-BOND-IONIC-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-BOND-LENGTH-STRENGTH-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-BOND-METALLIC-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-BOND-ORDER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-BOND-POLARITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-CANONICAL-MOLECULAR-GRAPH-ENCODING-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-CAT-CATALYST-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-CAT-PATHWAY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-CAT-SELECTIVITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-CATALYTIC-TURNOVER-CYCLE-FREQUENCY-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=510; explicit-status-lines=2
- SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=7; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-CHAIN-GROWTH-POLYMERIZATION-NETWORK-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-CHEMICAL-POTENTIAL-EQUIVALENT-COMPONENT-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=83; explicit-status-lines=0
- SFT-CHEM-CHEMICAL-SUBSTRUCTURE-RELATION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-CHROMATOGRAPHIC-RETENTION-RESOLUTION-018: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-CLASSICAL-QUANTUM-ALGORITHM-CORRESPONDENCE-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-COLLIGATIVE-COMPOSITION-RESPONSE-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=285; explicit-status-lines=0

- SFT-CHEM-COLLOID-DISPERSION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-COMPETING-CHANNEL-BRANCHING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=16; explicit-status-lines=2
- SFT-CHEM-COMPLEX-SPIN-STATE-ORDER-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=12; explicit-status-lines=2
- SFT-CHEM-CONCENTRATION-DEPENDENCE-RELATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=17; explicit-status-lines=1
- SFT-CHEM-CONCENTRATION-DEPENDENT-POTENTIAL-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-CONFIGURATION-ORDER-PATH-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=56; explicit-status-lines=0
- SFT-CHEM-CONFORMER-ENUMERATION-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-CONFORMER-GENERATION-EQUIVALENCE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=16; explicit-status-lines=3
- SFT-CHEM-CONFORMER-POPULATION-ORDERING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=14; evidence-lines=20; explicit-status-lines=3
- SFT-CHEM-CONJUGATED-SUPPORT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=7; evidence-lines=18; explicit-status-lines=0
- SFT-CHEM-CONSTITUTIONAL-ISOMER-ENUMERATION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-COORDINATION-ENTITY-RETAINED-IDENTITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=26; explicit-status-lines=1
- SFT-CHEM-COORDINATION-GEOMETRY-HELD-ORIENTATION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=10; explicit-status-lines=3
- SFT-CHEM-COORDINATION-ISOMERISM-EQUIVALENCE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=14; explicit-status-lines=1
- SFT-CHEM-COORDINATION-NUMBER-INCIDENCE-COUNT-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=29; explicit-status-lines=1
- SFT-CHEM-COPOLYMER-SEQUENCE-COMPOSITION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-CORROSION-REACTION-NETWORK-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-COUPLED-MASS-HEAT-CHARGE-TRANSPORT-019: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=240; explicit-status-lines=0
- SFT-CHEM-CROSS-PROPERTY-MOLECULAR-VECTOR-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=13; evidence-lines=9033; explicit-status-lines=0
- SFT-CHEM-CROSSLINK-GELATION-BOUNDARY-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-DATABASE-IDENTITY-PROVENANCE-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-DEFECT-NONSTOICHIOMETRY-016: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=13; explicit-status-lines=2

- SFT-CHEM-DEGREE-OF-POLYMERIZATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-DIFFUSION-LIMITED-REACTION-BOUNDARY-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=264; explicit-status-lines=3
- SFT-CHEM-DIHEDRAL-TORSIONAL-STATE-004: source-custodied-post-seal-comparison; closure=finite_complete; sources=1; evidence-lines=76; explicit-status-lines=14
- SFT-CHEM-DOUBLE-LAYER-INTERFACIAL-CHARGE-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ELECTROANALYTICAL-RESPONSE-020: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-ELECTROCHEM-CELL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELECTROCHEMICAL-STORAGE-HANDOFF-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=1
- SFT-CHEM-ELECTROCHEMICAL-WORK-REACTION-DIRECTION-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ELECTRODE-POTENTIAL-CHEMICAL-RELATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=7; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ELECTRODE-REACTION-RATE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ELECTROLYSIS-PRODUCT-AMOUNT-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ELECTRON-COUNT-SPIN-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=27; explicit-status-lines=6
- SFT-CHEM-ELECTRON-NEUTRON-DIFFRACTION-017: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-ELECTRONEGATIVITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELECTRONIC-STATE-IDENTITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=2
- SFT-CHEM-ELECTROPHILIC-SUBSTITUTION-FAMILY-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=6; explicit-status-lines=1
- SFT-CHEM-ELECTROPHORETIC-MOBILITY-SEPARATION-019: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELEM-ATOMIC-WEIGHT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELEM-ELEMENT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELEM-GROUP-PERIOD-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELEM-ION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1

- SFT-CHEM-ELEM-ISOTOPE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELEM-PERIODIC-BOUNDARY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELEM-PERIODIC-ORDER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELEM-VALENCE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=55; explicit-status-lines=0
- SFT-CHEM-ELIMINATION-REACTION-FAMILY-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=47; evidence-lines=7; explicit-status-lines=1
- SFT-CHEM-ENTHALPY-EQUIVALENT-STATE-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=22; explicit-status-lines=0
- SFT-CHEM-ENTROPY-MULTIPLICITY-CORRESPONDENCE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=22; explicit-status-lines=1
- SFT-CHEM-EQ-CHEMICAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-EQUILIBRIUM-BOND-LENGTH-001: source-custodied-post-seal-comparison; closure=finite_complete; sources=3; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-EQUILIBRIUM-ISOTOPE-FRACTIONATION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-EXACT-MOLECULAR-SIMILARITY-DISTINCTION-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-FINITE-MICROSTATE-SUPPORT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=395; explicit-status-lines=0
- SFT-CHEM-FISSION-PRODUCT-CHEMICAL-DISTRIBUTION-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-FLUORESCENCE-YIELD-LIFETIME-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=72; explicit-status-lines=0
- SFT-CHEM-FUGACITY-EQUIVALENT-GAS-MIXTURE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=105; explicit-status-lines=0
- SFT-CHEM-HALF-REACTION-IDENTITY-ORIENTATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=6; evidence-lines=4; explicit-status-lines=0
- SFT-CHEM-HEAT-WORK-TRANSFER-PARTITION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=22; explicit-status-lines=1
- SFT-CHEM-INFRARED-LINE-INTENSITY-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-INORGANIC-ACID-BASE-REDOX-NETWORK-017: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=14; explicit-status-lines=1

- SFT-CHEM-INORGANIC-COLOUR-ELECTRONIC-TRANSITION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=12; explicit-status-lines=1
- SFT-CHEM-INORGANIC-MAGNETIC-STATE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=13; explicit-status-lines=0
- SFT-CHEM-INSERTION-ELIMINATION-PATHWAY-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=10; explicit-status-lines=3
- SFT-CHEM-INTERFACE-TRANSFER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-INTERMOLECULAR-BINDING-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=1305; explicit-status-lines=99
- SFT-CHEM-INTERNAL-ENERGY-COMPOSITION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=21; explicit-status-lines=0
- SFT-CHEM-IONIC-CONDUCTIVITY-RELATION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-IONIC-MOBILITY-TRANSFERENCE-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ISOTOPE-EXCHANGE-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-JOINT-CORRELATION-DISSOCIATION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=30; explicit-status-lines=14
- SFT-CHEM-KIN-ACTIVATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-KIN-ORDER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-KIN-RATE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-KINETIC-ISOTOPE-EFFECT-RELATION-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=84; explicit-status-lines=3
- SFT-CHEM-KINETIC-ISOTOPE-FRACTIONATION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-LIGAND-DENTICITY-CHELATION-TOPOLOGY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=6; evidence-lines=30; explicit-status-lines=1
- SFT-CHEM-LIGAND-STATE-SPLITTING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=13; explicit-status-lines=3
- SFT-CHEM-MASS-ISOTOPE-FRAGMENT-VECTOR-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-MASS-WEIGHTED-MOLECULAR-SIZE-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-MEAS-AMOUNT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1

- SFT-CHEM-MEAS-FORMULA-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MEAS-NOMENCLATURE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MEAS-SUBSTANCE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MEAS-TRACEABILITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MEAS-UNCERTAINTY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MECHANISM-SEARCH-PROOF-TRACE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-METAL-CLUSTER-BONDING-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=10; explicit-status-lines=1
- SFT-CHEM-MOL-GEOMETRY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MOL-INTERMOLECULAR-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MOL-ISOMER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MOL-MOLECULE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MOL-NETWORK-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MOL-SUPRAMOLECULAR-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MOLECULAR-BOND-ANGLE-003: source-custodied-post-seal-comparison; closure=finite_complete; sources=3; evidence-lines=12; explicit-status-lines=8
- SFT-CHEM-MOLECULAR-DIFFUSION-RELATION-016: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=174; explicit-status-lines=1
- SFT-CHEM-MOLECULAR-DIPOLE-MAGNITUDE-005: source-custodied-post-seal-comparison; closure=finite_complete; sources=2; evidence-lines=15; explicit-status-lines=10
- SFT-CHEM-MOLECULAR-ELECTRON-AFFINITY-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=102; explicit-status-lines=0
- SFT-CHEM-MOLECULAR-EXCLUSION-EXCHANGE-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=80; explicit-status-lines=12
- SFT-CHEM-MOLECULAR-FORMATION-ENERGY-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=2106; explicit-status-lines=0
- SFT-CHEM-MOLECULAR-GRAPH-ISOMORPHISM-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-MOLECULAR-IONIZATION-ENERGY-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=13; explicit-status-lines=1
- SFT-CHEM-MOLECULAR-MAGNETIC-RESPONSE-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=181; explicit-status-lines=1

- SFT-CHEM-MOLECULAR-POLARIZABILITY-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=256; explicit-status-lines=1
- SFT-CHEM-MOLECULAR-QUANTUM-MEASUREMENT-REDUCTION-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=345; explicit-status-lines=0
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=80; explicit-status-lines=16
- SFT-CHEM-MULTICENTRE-DELOCALIZED-SUPPORT-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=41; explicit-status-lines=16
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=128; explicit-status-lines=0
- SFT-CHEM-MULTIMODAL-MOLECULAR-IDENTITY-021: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-NET-AUTOCATALYSIS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-NET-REACTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-NMR-CHEMICAL-SHIFT-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-NMR-RELAXATION-EXCHANGE-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-NMR-SPIN-COUPPLING-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-NUCLEAR-CHEMICAL-CARRIER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-NUCLEAR-ELECTRONIC-COMPOSITION-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=433; explicit-status-lines=0
- SFT-CHEM-NUCLEOPHILIC-SUBSTITUTION-FAMILY-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=9; evidence-lines=7; explicit-status-lines=1
- SFT-CHEM-NUMBER-AVERAGE-MOLECULAR-SIZE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=25; explicit-status-lines=0
- SFT-CHEM-OPERATIONAL-CLASSICAL-QUANTUM-CORRESPONDENCE-015: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-ORBITAL-SUPPORT-OCCUPANCY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=366; explicit-status-lines=7
- SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ORGANOMETALLIC-ELECTRON-ACCOUNTING-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=10; explicit-status-lines=2
- SFT-CHEM-ORGANOMETALLIC-METAL-CARBON-BOND-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=11; explicit-status-lines=0

- SFT-CHEM-OVERPOTENTIAL-POLARIZATION-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-OXIDATIVE-ADDITION-REDUCTIVE-ELIMINATION-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=10; explicit-status-lines=0
- SFT-CHEM-PARALLEL-MECHANISM-COMPOSITION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=39; explicit-status-lines=3
- SFT-CHEM-PERICYCLIC-REACTION-FAMILY-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=38; evidence-lines=7; explicit-status-lines=2
- SFT-CHEM-PHASE-CHEMICAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=27; explicit-status-lines=0
- SFT-CHEM-PHOSPHORESCENCE-INTERSYSTEM-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-PHOTOCHEM-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-POLYMER-ARCHITECTURE-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-CHAIN-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-POLYMER-CONFORMATION-SIZE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-DEGRADATION-NETWORK-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-DISPERSITY-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-DISTRIBUTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-POLYMER-MATERIALS-HANDOFF-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-PHASE-TRANSITION-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-PRED-G-BLOCK-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-PRED-PERIODIC-ENDPOINT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-PRED-SMITHIUM-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-PREDICTION-UNCERTAINTY-APPLICABILITY-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-PROTECTING-GROUP-REVERSIBLE-STATE-015: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=4; explicit-status-lines=0
- SFT-CHEM-RADIATION-CHEMISTRY-REACTION-NETWORK-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2

- SFT-CHEM-RADICAL-REACTION-NETWORK-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=73; evidence-lines=6; explicit-status-lines=1
- SFT-CHEM-RADIOACTIVE-BRANCHING-CHEMICAL-YIELD-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-RADIOACTIVE-CHEMICAL-TRANSFORMATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-RADIOCHEMICAL-EQUILIBRIUM-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-RADIOCHEMICAL-SEPARATION-DECONTAMINATION-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-RADIOTRACER-CUSTODY-INFERENCE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-RAMAN-TRANSITION-INTENSITY-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-REACTION-DYNAMICS-SCATTERING-PRODUCT-STATE-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=65; explicit-status-lines=2
- SFT-CHEM-REACTION-GRAPH-GENERATION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-REARRANGEMENT-REACTION-FAMILY-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=16; evidence-lines=7; explicit-status-lines=2
- SFT-CHEM-REDOX-COUPPLING-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-REDOX-OXIDATION-STATE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-RESOLVED-ROVIBRONIC-SPIN-COMPOSITION-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=107; explicit-status-lines=2
- SFT-CHEM-RESONANCE-EQUIVALENT-REPRESENTATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=21; explicit-status-lines=0
- SFT-CHEM-RETROSYNTHETIC-DECOMPOSITION-RECONSTRUCTION-016: source-custodied-post-seal-comparison; closure=depth_independent; sources=7; evidence-lines=4; explicit-status-lines=0
- SFT-CHEM-REVERSIBLE-KINETIC-EQUILIBRIUM-CORRESPONDENCE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=176; explicit-status-lines=2
- SFT-CHEM-ROTATIONAL-CONSTANT-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3024; explicit-status-lines=1
- SFT-CHEM-ROTATIONAL-SPECTRUM-LINE-015: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-ROVIBRONIC-COMPOSITION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=6; explicit-status-lines=2
- SFT-CHEM-RXN-IDENTITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-RXN-INTERMEDIATE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=2
- SFT-CHEM-RXN-MECHANISM-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1

- SFT-CHEM-SELECTION-RULE-STRUCTURE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=75; explicit-status-lines=11
- SFT-CHEM-SELECTIVITY-COMPLETE-DISTRIBUTION-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=133; evidence-lines=5; explicit-status-lines=1
- SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=28; explicit-status-lines=5
- SFT-CHEM-SMITHIUM-CHEMICAL-SEPARATION-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SMITHIUM-DECAY-CHANNEL-LEDGER-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SMITHIUM-ION-OXIDATION-LADDER-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SMITHIUM-JOINT-DETECTION-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SMITHIUM-LIFETIME-BOUNDARY-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SMITHIUM-SPECTROSCOPIC-CLASSES-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SMITHIUM-SYNTHESIS-CONSERVATION-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SOLID-STATE-LOCAL-COORDINATION-015: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=11; explicit-status-lines=2
- SFT-CHEM-SOLUTION-EQUILIBRIUM-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-SOLVATION-DISSOLUTION-FREE-ORDER-015: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=810; explicit-status-lines=1
- SFT-CHEM-SPEC-INFRARED-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-SPEC-MASS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-SPEC-ROT-VIB-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-SPEC-UVVIS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STATE-ENERGY-ORDER-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=313; explicit-status-lines=6
- SFT-CHEM-STATE-SYMMETRY-DEGENERACY-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=379; explicit-status-lines=11
- SFT-CHEM-STEP-GROWTH-POLYMERIZATION-NETWORK-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-STEREO-CHIRALITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STEREO-DIASTEREOMER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1

- SFT-CHEM-STEREO-ENANTIOMER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STEREOISOMER-ENUMERATION-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-STOICH-COEFFICIENT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STOICH-COMPOSITION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STOICH-CONSERVATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STOICH-LIMITING-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STOICH-MIXTURE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STOICH-SOLUTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STOICH-YIELD-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-SURFACE-ADSORPTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-SYMBOLIC-PROPERTY-EVALUATION-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-TEMPERATURE-CORRESPONDENCE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=10; explicit-status-lines=0
- SFT-CHEM-TEMPERATURE-DEPENDENCE-RELATION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=27; explicit-status-lines=1
- SFT-CHEM-THERMAL-CONDUCTIVITY-RELATION-018: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=665; explicit-status-lines=0
- SFT-CHEM-THERMO-DIRECTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-THERMO-REACTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-TRANSITION-STATE-EQUIVALENT-BOUNDARY-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=10; explicit-status-lines=1
- SFT-CHEM-UVVISIBLE-LINE-INTENSITY-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-VALIDATION-SMITHIUM-COMPLETE-FAMILY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=6; explicit-status-lines=4
- SFT-CHEM-VIBRATIONAL-FREQUENCY-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2016; explicit-status-lines=1
- SFT-CHEM-VISCOUS-TRANSPORT-RELATION-017: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=434; explicit-status-lines=1
- SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2

Falsification boundary: Reject if any registered member, receipt, control, evidence file, result class, closure scope, formal-only boundary or source-custody row is missing, changed, duplicated or selected after outcome access.

Measurement receipt: sha256:36e5658979976b641c3108291b7609ed4fa3ce3617fde72f4164a35f6bb2ac9d.

Isolation certificate: sha256:29c484e7c980c85e1c79cbd0fd286864d066e765c32bcd0cbf5a29233a2e9b3.

Custody certificate: sha256:5363c0974ef027fdd2f536e53b967df72bbe5a3718e65443d48732c2a5b66a13.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favorable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

Immutable identities. Source manifest

sha256:c48a529cae15fdc0c9c4612473f07c5a17c130c08b2ca9e28ec8c3b8d9ac2e2d; derivation seal sha256:75df6445649f85bd1ee8e5a963338ebade01ae668c496a2c210e4d822d9ecf92; engine receipt sha256:68919fc8d349fd2cd6b1b9bfbfc3792289a792ecc12800e660c7c1db3d67b4853 at receipts/engine/model_admitted/SFT-CHEM-VALIDATION-ADVERSE-OUT-OF-BOUND-VECTOR-011-68919fc8d349fd2c.json; empirical hash sha256:0c0e6157b9e4bd44917520f28946fea996db65a7339d1cbf5d03ffe09caf5d3e; measurement receipt sha256:36e5658979976b641c3108291b7609ed4fa3ce3617fde72f4164a35f6bb2ac9d.

275. Chemistry empirical Grand Lock

Claim identity: SFT-CHEM-VALIDATION-EMPIRICAL-GRAND-LOCK-012

Question and theorem. VALID-012 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 263 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

VALID-012 uniquely requires a complete member-by-member, receipt-bound, post-seal evidence vector across all 263 registered claims, retaining actual closure scopes, all outcomes and every explicit unmeasured boundary.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-ACTIVATION-BARRIER-VALUE-RELATION-004
- SFT-CHEM-ACTIVITY-AMOUNT-TIME-003

- SFT-CHEM-ACTIVITY-NONIDEAL-COMPOSITION-009
- SFT-CHEM-ADDITION-REACTION-FAMILY-009
- SFT-CHEM-ANALYTICAL-ACCURACY-TRUENESS-001
- SFT-CHEM-ANALYTICAL-CALIBRATION-001
- SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001
- SFT-CHEM-ANALYTICAL-DETECTION-QUANTIFICATION-004
- SFT-CHEM-ANALYTICAL-PERFORMANCE-BUDGET-022
- SFT-CHEM-ANALYTICAL-PRECISION-REPEATABILITY-002
- SFT-CHEM-ANALYTICAL-SAMPLE-001
- SFT-CHEM-ANALYTICAL-SELECTIVITY-001
- SFT-CHEM-ANALYTICAL-SELECTIVITY-INTERFERENCE-005
- SFT-CHEM-ANALYTICAL-SENSITIVITY-003
- SFT-CHEM-ANTIAROMATIC-NONAROMATIC-DISTINCTION-004
- SFT-CHEM-AROMATIC-RECURRENCE-STABILITY-003
- SFT-CHEM-ATOM-MAPPING-REACTION-BALANCE-008
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-DISSOCIATION-ENERGY-002
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-LENGTH-STRENGTH-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-CANONICAL-MOLECULAR-GRAPH-ENCODING-001
- SFT-CHEM-CAT-CATALYST-001
- SFT-CHEM-CAT-PATHWAY-001
- SFT-CHEM-CAT-SELECTIVITY-001
- SFT-CHEM-CATALYTIC-TURNOVER-CYCLE-FREQUENCY-010
- SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003
- SFT-CHEM-CHAIN-GROWTH-POLYMERIZATION-NETWORK-005
- SFT-CHEM-CHEMICAL-POTENTIAL-EQUIVALENT-COMPONENT-008
- SFT-CHEM-CHEMICAL-SUBSTRUCTURE-RELATION-003
- SFT-CHEM-CHROMATOGRAPHIC-RETENTION-RESOLUTION-018
- SFT-CHEM-CLASSICAL-QUANTUM-ALGORITHM-CORRESPONDENCE-014
- SFT-CHEM-COLLIGATIVE-COMPOSITION-RESPONSE-014
- SFT-CHEM-COLLOID-DISPERSION-001
- SFT-CHEM-COMPETING-CHANNEL-BRANCHING-006

- SFT-CHEM-COMPLEX-SPIN-STATE-ORDER-007
- SFT-CHEM-CONCENTRATION-DEPENDENCE-RELATION-002
- SFT-CHEM-CONCENTRATION-DEPENDENT-POTENTIAL-004
- SFT-CHEM-CONFIGURATION-ORDER-PATH-011
- SFT-CHEM-CONFORMER-ENUMERATION-006
- SFT-CHEM-CONFORMER-GENERATION-EQUIVALENCE-005
- SFT-CHEM-CONFORMER-POPULATION-ORDERING-006
- SFT-CHEM-CONJUGATED-SUPPORT-001
- SFT-CHEM-CONSTITUTIONAL-ISOMER-ENUMERATION-004
- SFT-CHEM-COORDINATION-ENTITY-RETAINED-IDENTITY-001
- SFT-CHEM-COORDINATION-GEOMETRY-HELD-ORIENTATION-004
- SFT-CHEM-COORDINATION-ISOMERISM-EQUIVALENCE-005
- SFT-CHEM-COORDINATION-NUMBER-INCIDENCE-COUNT-002
- SFT-CHEM-COPOLYMER-SEQUENCE-COMPOSITION-007
- SFT-CHEM-CORROSION-REACTION-NETWORK-012
- SFT-CHEM-COUPLED-MASS-HEAT-CHARGE-TRANSPORT-019
- SFT-CHEM-CROSS-PROPERTY-MOLECULAR-VECTOR-014
- SFT-CHEM-CROSSLINK-GELATION-BOUNDARY-009
- SFT-CHEM-DATABASE-IDENTITY-PROVENANCE-011
- SFT-CHEM-DEFECT-NONSTOICHIOMETRY-016
- SFT-CHEM-DEGREE-OF-POLYMERIZATION-001
- SFT-CHEM-DIFFUSION-LIMITED-REACTION-BOUNDARY-011
- SFT-CHEM-DIHEDRAL-TORSIONAL-STATE-004
- SFT-CHEM-DOUBLE-LAYER-INTERFACIAL-CHARGE-011
- SFT-CHEM-ELECTROANALYTICAL-RESPONSE-020
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-ELECTROCHEMICAL-STORAGE-HANDOFF-013
- SFT-CHEM-ELECTROCHEMICAL-WORK-REACTION-DIRECTION-005
- SFT-CHEM-ELECTRODE-POTENTIAL-CHEMICAL-RELATION-002
- SFT-CHEM-ELECTRODE-REACTION-RATE-009
- SFT-CHEM-ELECTROLYSIS-PRODUCT-AMOUNT-006
- SFT-CHEM-ELECTRON-COUNT-SPIN-002
- SFT-CHEM-ELECTRON-NEUTRON-DIFFRACTION-017
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-ELECTRONIC-STATE-IDENTITY-001
- SFT-CHEM-ELECTROPHILIC-SUBSTITUTION-FAMILY-008
- SFT-CHEM-ELECTROPHORETIC-MOBILITY-SEPARATION-019
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001

- SFT-CHEM-ELEM-ATOMIC-WEIGHT-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-GROUP-PERIOD-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-ELEM-ISOTOPE-001
- SFT-CHEM-ELEM-PERIODIC-BOUNDARY-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001
- SFT-CHEM-ELEM-VALENCE-001
- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001
- SFT-CHEM-ELIMINATION-REACTION-FAMILY-010
- SFT-CHEM-ENTHALPY-EQUIVALENT-STATE-006
- SFT-CHEM-ENTROPY-MULTIPLICITY-CORRESPONDENCE-005
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-EQUILIBRIUM-BOND-LENGTH-001
- SFT-CHEM-EQUILIBRIUM-ISOTOPE-FRACTIONATION-007
- SFT-CHEM-EXACT-MOLECULAR-SIMILARITY-DISTINCTION-010
- SFT-CHEM-FINITE-MICROSTATE-SUPPORT-001
- SFT-CHEM-FISSION-PRODUCT-CHEMICAL-DISTRIBUTION-011
- SFT-CHEM-FLUORESCENCE-YIELD-LIFETIME-010
- SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007
- SFT-CHEM-FUGACITY-EQUIVALENT-GAS-MIXTURE-010
- SFT-CHEM-HALF-REACTION-IDENTITY-ORIENTATION-001
- SFT-CHEM-HEAT-WORK-TRANSFER-PARTITION-004
- SFT-CHEM-INFRARED-LINE-INTENSITY-012
- SFT-CHEM-INORGANIC-ACID-BASE-REDOX-NETWORK-017
- SFT-CHEM-INORGANIC-COLOUR-ELECTRONIC-TRANSITION-008
- SFT-CHEM-INORGANIC-MAGNETIC-STATE-009
- SFT-CHEM-INSERTION-ELIMINATION-PATHWAY-013
- SFT-CHEM-INTERFACE-TRANSFER-001
- SFT-CHEM-INTERMOLECULAR-BINDING-011
- SFT-CHEM-INTERNAL-ENERGY-COMPOSITION-003
- SFT-CHEM-IONIC-CONDUCTIVITY-RELATION-007
- SFT-CHEM-IONIC-MOBILITY-TRANSFERENCE-008
- SFT-CHEM-ISOTOPE-EXCHANGE-006
- SFT-CHEM-JOINT-CORRELATION-DISSOCIATION-007
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-ORDER-001

- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-KINETIC-ISOTOPE-EFFECT-RELATION-012
- SFT-CHEM-KINETIC-ISOTOPE-FRACTIONATION-008
- SFT-CHEM-LIGAND-DENTICITY-CHELATION-TOPOLOGY-003
- SFT-CHEM-LIGAND-STATE-SPLITTING-006
- SFT-CHEM-MASS-ISOTOPE-FRAGMENT-VECTOR-014
- SFT-CHEM-MASS-WEIGHTED-MOLECULAR-SIZE-003
- SFT-CHEM-MEAS-AMOUNT-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-MEAS-NOMENCLATURE-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MECHANISM-SEARCH-PROOF-TRACE-009
- SFT-CHEM-METAL-CLUSTER-BONDING-014
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-NETWORK-001
- SFT-CHEM-MOL-SUPRAMOLECULAR-001
- SFT-CHEM-MOLECULAR-BOND-ANGLE-003
- SFT-CHEM-MOLECULAR-DIFFUSION-RELATION-016
- SFT-CHEM-MOLECULAR-DIPOLE-MAGNITUDE-005
- SFT-CHEM-MOLECULAR-ELECTRON-AFFINITY-008
- SFT-CHEM-MOLECULAR-EXCLUSION-EXCHANGE-006
- SFT-CHEM-MOLECULAR-FORMATION-ENERGY-013
- SFT-CHEM-MOLECULAR-GRAPH-ISOMORPHISM-002
- SFT-CHEM-MOLECULAR-IONIZATION-ENERGY-007
- SFT-CHEM-MOLECULAR-MAGNETIC-RESPONSE-012
- SFT-CHEM-MOLECULAR-POLARIZABILITY-006
- SFT-CHEM-MOLECULAR-QUANTUM-MEASUREMENT-REDUCTION-014
- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009
- SFT-CHEM-MULTICENTRE-DELOCALIZED-SUPPORT-008
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013
- SFT-CHEM-MULTIMODAL-MOLECULAR-IDENTITY-021

- SFT-CHEM-NET-AUTOCATALYSIS-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-NMR-CHEMICAL-SHIFT-006
- SFT-CHEM-NMR-RELAXATION-EXCHANGE-008
- SFT-CHEM-NMR-SPIN-COUPLING-007
- SFT-CHEM-NUCLEAR-CHEMICAL-CARRIER-001
- SFT-CHEM-NUCLEAR-ELECTRONIC-COMPOSITION-012
- SFT-CHEM-NUCLEOPHILIC-SUBSTITUTION-FAMILY-007
- SFT-CHEM-NUMBER-AVERAGE-MOLECULAR-SIZE-002
- SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012
- SFT-CHEM-OPERATIONAL-CLASSICAL-QUANTUM-CORRESPONDENCE-015
- SFT-CHEM-ORBITAL-SUPPORT-OCCUPANCY-003
- SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001
- SFT-CHEM-ORGANOMETALLIC-ELECTRON-ACCOUNTING-011
- SFT-CHEM-ORGANOMETALLIC-METAL-CARBON-BOND-010
- SFT-CHEM-OVERPOTENTIAL-POLARIZATION-010
- SFT-CHEM-OXIDATIVE-ADDITION-REDUCTIVE-ELIMINATION-012
- SFT-CHEM-PARALLEL-MECHANISM-COMPOSITION-008
- SFT-CHEM-PERICYCLIC-REACTION-FAMILY-012
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011
- SFT-CHEM-PHOSPHORESCENCE-INTERSYSTEM-011
- SFT-CHEM-PHOTOCHEM-001
- SFT-CHEM-POLYMER-ARCHITECTURE-008
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-CHEM-POLYMER-CONFORMATION-SIZE-010
- SFT-CHEM-POLYMER-DEGRADATION-NETWORK-012
- SFT-CHEM-POLYMER-DISPERSITY-004
- SFT-CHEM-POLYMER-DISTRIBUTION-001
- SFT-CHEM-POLYMER-MATERIALS-HANDOFF-013
- SFT-CHEM-POLYMER-PHASE-TRANSITION-011
- SFT-CHEM-PRED-G-BLOCK-001
- SFT-CHEM-PRED-PERIODIC-ENDPOINT-001
- SFT-CHEM-PRED-SMITHIUM-001
- SFT-CHEM-PREDICTION-UNCERTAINTY-APPLICABILITY-013
- SFT-CHEM-PROTECTING-GROUP-REVERSIBLE-STATE-015
- SFT-CHEM-RADIATION-CHEMISTRY-REACTION-NETWORK-012

- SFT-CHEM-RADICAL-REACTION-NETWORK-013
- SFT-CHEM-RADIOACTIVE-BRANCHING-CHEMICAL-YIELD-004
- SFT-CHEM-RADIOACTIVE-CHEMICAL-TRANSFORMATION-002
- SFT-CHEM-RADIOCHEMICAL-EQUILIBRIUM-005
- SFT-CHEM-RADIOCHEMICAL-SEPARATION-DECONTAMINATION-010
- SFT-CHEM-RADIOTRACER-CUSTODY-INFERENCE-009
- SFT-CHEM-RAMAN-TRANSITION-INTENSITY-009
- SFT-CHEM-REACTION-DYNAMICS-SCATTERING-PRODUCT-STATE-013
- SFT-CHEM-REACTION-GRAPH-GENERATION-007
- SFT-CHEM-REARRANGEMENT-REACTION-FAMILY-011
- SFT-CHEM-REDOX-COUPPING-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-RESOLVED-ROVIBRONIC-SPIN-COMPOSITION-013
- SFT-CHEM-RESONANCE-EQUIVALENT-REPRESENTATION-002
- SFT-CHEM-RETROSYNTHETIC-DECOMPOSITION-RECONSTRUCTION-016
- SFT-CHEM-REVERSIBLE-KINETIC-EQUILIBRIUM-CORRESPONDENCE-009
- SFT-CHEM-ROTATIONAL-CONSTANT-010
- SFT-CHEM-ROTATIONAL-SPECTRUM-LINE-015
- SFT-CHEM-ROVIBRONIC-COMPOSITION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-INTERMEDIATE-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-SELECTION-RULE-STRUCTURE-010
- SFT-CHEM-SELECTIVITY-COMPLETE-DISTRIBUTION-014
- SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007
- SFT-CHEM-SMITHIUM-CHEMICAL-SEPARATION-001
- SFT-CHEM-SMITHIUM-DECAY-CHANNEL-LEDGER-001
- SFT-CHEM-SMITHIUM-ION-OXIDATION-LADDER-001
- SFT-CHEM-SMITHIUM-JOINT-DETECTION-001
- SFT-CHEM-SMITHIUM-LIFETIME-BOUNDARY-001
- SFT-CHEM-SMITHIUM-SPECTROSCOPIC-CLASSES-001
- SFT-CHEM-SMITHIUM-SYNTHESIS-CONSERVATION-001
- SFT-CHEM-SOLID-STATE-LOCAL-COORDINATION-015
- SFT-CHEM-SOLUTION-EQUILIBRIUM-001
- SFT-CHEM-SOLVATION-DISSOLUTION-FREE-ORDER-015
- SFT-CHEM-SPEC-INFRARED-001
- SFT-CHEM-SPEC-MASS-001
- SFT-CHEM-SPEC-ROT-VIB-001

- SFT-CHEM-SPEC-UVVIS-001
- SFT-CHEM-STATE-ENERGY-ORDER-004
- SFT-CHEM-STATE-SYMMETRY-DEGENERACY-005
- SFT-CHEM-STEP-GROWTH-POLYMERIZATION-NETWORK-006
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-CHEM-STEREO-DIASTEREOMER-001
- SFT-CHEM-STEREO-ENANTIOMER-001
- SFT-CHEM-STEREOISOMER-ENUMERATION-005
- SFT-CHEM-STOICH-COEFFICIENT-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STOICH-LIMITING-001
- SFT-CHEM-STOICH-MIXTURE-001
- SFT-CHEM-STOICH-SOLUTION-001
- SFT-CHEM-STOICH-YIELD-001
- SFT-CHEM-SURFACE-ADSORPTION-001
- SFT-CHEM-SYMBOLIC-PROPERTY-EVALUATION-012
- SFT-CHEM-TEMPERATURE-CORRESPONDENCE-002
- SFT-CHEM-TEMPERATURE-DEPENDENCE-RELATION-003
- SFT-CHEM-THERMAL-CONDUCTIVITY-RELATION-018
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-TRANSITION-STATE-EQUIVALENT-BOUNDARY-005
- SFT-CHEM-UVVISIBLE-LINE-INTENSITY-013
- SFT-CHEM-VALIDATION-SMITHIUM-COMPLETE-FAMILY-001
- SFT-CHEM-VIBRATIONAL-FREQUENCY-009
- SFT-CHEM-VISCOUS-TRANSPORT-RELATION-017
- SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016
- SFT-CHEM-VALIDATION-ADVERSE-OUT-OF-BOUND-VECTOR-011

Generated grammar. Generate the complete eight-axis VALID-012 evidence-preservation product before capability-closed release of its frozen vector.

Boundary: Exactly the 263 value-free claim identities registered for Chemistry VALID-012, their current receipt-bound evidence and all result classes.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: complete-receipt-bound-membership__complete-valid-012-typed-evidence-vector__sealed-law-then-capability-closed-release__favourable-adverse-absent-unavailable-unresolved__each-declared-closure-scope-held__explicit-formal-standing-boundary__omission-tamper-and-boundary-halt__dated-complete-extension-open. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
authority	complete-receipt-bound-membership	selected-headline	A headline omits the evidence surface.
relation	complete-valid-012-typed-evidence-vector	aggregate-score-or-fit	A score or fit can hide distinct failures.
custody	sealed-law-then-capability-closed-release	outcomes-in-forcing	Outcome access may select the law.
results	favourable-adverse-absent-unavailable-unresolved	favourable-only	Selective evidence invalidates the vector.
closure	each-declared-closure-scope-held	single-universal-scope	That overwrites finite boundaries.
formal	explicit-formal-standing-boundary	invented-empirical-successes	That fabricates data.
controls	omission-tamper-and-boundary-halt	omission-tolerated	A missing row cannot close a vector.
extension	dated-complete-extension-open	permanent-lock	A dated vector cannot bar discovery.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:646f6a94bd422d165b415901cf9313d05bdd6b57384a90053cfb1425fa9352af.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:15fbdc8fe7973e262113a4f490917d3682889a7d4a014507458151688816f579.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:08a63170c99cb7a98464b390a43d5f5dc9750708eb099d9ff51b6c7e5a9056ac.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:84e9fec68eea28474574ee64e58e4056ab2059a0cd15de9e9d0f895dbfc296bc.

Independent reconstruction. Implementation hash

sha256:ebb766680add8ad494e3d447edc51e205fdb57736b98d14ebd76d6737d57ea34; independent certificate sha256:705b6ea49a34173bdbf88602f077fd87d268a1f3b9e211d71b281a45bd10bf97; external-validation hash sha256:1d007e0bc51b55fc59704370bb0b9cce03ed048dfe5e043a213519416f58f04d. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ACID-BASE-COMPOSITE-2026
- IUPAC-GOLD-BOOK-A00306-2026
- IUPAC-DIDAC-E15-2026
- IUPAC-GOLD-BOOK-LEWIS-COMPOSITE-2026
- IUPAC-GOLD-BOOK-P04915-2026
- NIST-SRD-101-CCCBDB-EXPERIMENTAL-INTERNAL-ROTATION-BARRIERS-COMplete
- NIST-RADIOACTIVITY-SRMS-GENERAL-INFO-2025

- NIST-SRM-4239A-STRONTIUM-90-2022
- NIST-SRM-4324C-URANIUM-232-2025
- NUCHEM-001-004-SOURCE-INVENTORY
- NIST-TRC-THERMOML-JCED-2019-9B00694
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-17384a1d9c5f4b2e9aaccbb6293204db
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-1478630680b84aee8dca359a52f20ba1
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-25dfe42a186e40c88717830e231034da
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-65ff4c19dea541b880d627eb4fbf8af3
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-8bbc1db0e28c4c60a61d554be47d9b92
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-c9f16c4a213a481daa225423696b55a4
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-8ba9957fa5604be4aeda7eff734f6ca5
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-e80af89368884e338a1956cbe7d0d943
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-6d9789e42f954cae8a05b6d471178de0
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-a9b062ada7ce4d2fa3f88d957d9f599a
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-0c3242203b3a408882edda75874c3b8d
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-48c6faedc14d49dfa5de2dd45ed6140d
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-d32732940c8e4867b80c81d9b01eb889
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-62b80d7f983a4bf9a9564d4afc30e02e
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-e7f8deb100824a408d3c2e0d1cb693a9
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-94fbe6fbedf24c1cbfa7e834d4e4a7c9
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-8eff1b6e7bbb4496bc2632e3d3e31a71
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-4b5fae7b28894c28b650d185109427d2
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-7ad584d7eae4f169a17acb74b087530
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-bba06fb330a54307a9082b493a2453c3
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-c9cb9eedd61341a6b829cf445aa77b7c
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-6c8b4df5334a4f939f2d4f56b1edbf8
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-bcbb41bfe1ae435eac5bac8862f2f14d
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-ac18bbe2a7644934b24a8956f4b4a721
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-2e1cdb77c14c495e892c4336682b64a6
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-4be654f9e2a74d6cb58f62cf1baae0ec
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-b91423e5a7d349f78fefca0ed55980e8
- ORD:ord_dataset-d92976309c3a48a3a64a4cf5e7048086:ord-1bc497e7e2f745f4b38b233207da6bc8
- NIST-SRM-1959-CERTIFICATE-2011
- ANAL-001-005-SOURCE-INVENTORY
- IUPAC-GOLD-BOOK-ANALYTICAL-CALIBRATION-COMPOSITE-2026
- IUPAC-NIST-COMPLETE-ANALYTICAL-RECORD-COMPOSITE-2026
- NIST-SP958-DETECTION-QUANTIFICATION-PAGES-164-166
- SFT-V3-ANAL-001-011-IMMUTABLE-EVIDENCE

- NIST-VOLTAMMETRIC-LOD-STUDY
- NIST-SRM-1980-ELECTROPHORETIC-MOBILITY
- NIST-SRM-1959-CERTIFICATION-2010
- IUPAC-GOLD-BOOK-ANALYTICAL-SAMPLE-COMPOSITE-2026
- IUPAC-GOLD-BOOK-ANALYTICAL-SELECTIVITY-COMPOSITE-2026
- NIST-IMS-FENTANYL-INTERFERENTS-2019
- NIST-SP-260-126-NTRM-GAS-STANDARDS
- IUPAC-A00441
- IUPAC-AT06987
- NIST-WEBBOOK-C71432-THERMOCHEMISTRY
- NIST-CCCBDB-1120532-NEUTRAL-EXPERIMENTAL
- NIST-CCCBDB-629209-NEUTRAL-EXPERIMENTAL
- IUPAC-A00442
- IUPAC-R05333
- NIST-WEBBOOK-C110838-THERMOCHEMISTRY
- NIST-WEBBOOK-C110827-THERMOCHEMISTRY
- NIST-CCCBDB-SRD101-C71432-COMPLETE-EXPERIMENTAL-DATA
- NIST-CCCBDB-SRD101-C110838-COMPLETE-EXPERIMENTAL-DATA
- NIST-CCCBDB-SRD101-C110827-COMPLETE-EXPERIMENTAL-DATA
- PUBCHEM-PUG-REST-AUTHORITY
- PUBCHEM-C3H8O-FORMULA-CENSUS
- PUBCHEM-ASPIRIN-CONFORMERS
- PUBCHEM-ASPIRIN-SUBSTRUCTURE
- PUBCHEM-ASPIRIN-SIMILARITY
- CHEBI-REST-AUTHORITY
- RHEA-IMMUTABLE-REACTION-SMILES
- USPTO-IMMUTABLE-REACTION-SURFACE
- EMBL-EBI-CHEBI-BIOMOLECULAR-BOUNDARY-2026
- IUPAC-GOLD-BOOK-CT07009-2026
- IUPAC-GOLD-BOOK-C01384-2026
- APS-PRA-49-2460-1994
- PRL-UDEM-H-1S2S-1997
- PRL-PARTHEY-HD-1S2S-SHIFT-2010
- SI-DEFINING-LIGHT-SPEED
- JCP-LIU-H2-DISSOCIATION-2009
- PRA-HUSSELS-D2-DISSOCIATION-2022
- IUPAC-GOLD-BOOK-IT07058-2026
- IUPAC-GOLD-BOOK-B00702-2026

- IUPAC-GOLD-BOOK-08789-2026
- IUPAC-GOLD-BOOK-B00707-2026
- IUPAC-GOLD-BOOK-BOND-POLARITY-COMPOSITE-2026
- CHEM-AUTHORITY-CATALYST-COMPOSITE-2026
- CHEM-AUTHORITY-CATALYTIC-PATHWAY-COMPOSITE-2026
- IUPAC-GOLD-BOOK-CATALYTIC-SELECTIVITY-COMPOSITE-2026
- NATURE-NANOTECHNOLOGY-S41565-021-00959-4-COMPLETE
- NIST-JRES-AGCL-COMPLETE-CELL-1954
- NIST-STOCHASTIC-PHOTOPOLYMER-NETWORK-GROWTH
- NIST-MONODISPERSE-PAMS-KINETIC-NETWORK
- NIST-TRC-THERMOML-JCED-2019-9B00414
- NIST-WEBBOOK-GAS-CHROMATOGRAPHY
- NIST-WEBBOOK-BENZENE-MULTIMODAL
- NIST-WEBBOOK-CYCLOHEXANE-MULTIMODAL
- NIST-TRC-THERMOML-FPE-2013-337-60-66
- NIST-TRC-THERMOML-FPE-2016-420-14-19
- NIST-TRC-THERMOML-JCT-2009-41-1439-1445
- IUPAC-GOLD-BOOK-COLLOIDAL-DISPERSION-COMPOSITE-2026
- COMMUNICATIONS-CHEMISTRY-S42004-024-01239-7-COMPLETE
- IUPAC-HT06789
- IUPAC-LT07067
- IUPAC-ST07104
- RSC-PCCP-2014-C3CP54664K-TABLE-2
- NIST-JRES-AGCL-HCL-CONCENTRATION-SERIES-1954
- NIST-CCCBDB-SRD101-ETHANOL-EXPERIMENTAL-ROTATIONAL-BARRIER
- IUPAC-C01262
- IUPAC-C01258
- IUPAC-C01259
- NIST-CCCBDB-106978-NEUTRAL-EXPERIMENTAL
- IUPAC-CT01038
- NIST-CCCBDB-106978-INTERNAL-ROTATION
- NIST-CCCBDB-109660-INTERNAL-ROTATION-1
- OSTI-22415852-PRIMARY-RECORD
- OSTI-22415852-PUBLIC-FULLTEXT-ENDPOINT
- PUBMED-25591384-COMPLETE-FULLTEXT-PROVIDER-SURFACE
- ACS-JP404315T-COMPLETE-PRIMARY-AND-SI-SURFACE
- ACS-FIGSHARE-EXACT-RESOURCE-DOI-JP404315T
- ACS-FIGSHARE-4012420-JP404315T-SI-001

- CORE-43583184-DOWNLOAD-PDF-LEGACY
- CORE-43583184-DOWNLOAD-CURRENT
- IUPAC-C01267
- IUPAC-08786
- IUPAC-D01583
- NIST-CCCBDB-106990-NEUTRAL-EXPERIMENTAL
- NIST-CCCBDB-591935-NEUTRAL-GEOMETRY
- NIST-WEBBOOK-C106990-SPECTRA
- NIST-WEBBOOK-C106990-UVVIS-JCAMP-INDEX-0
- IUPAC-GOLD-BOOK-C01330-C00930-L03518
- NIST-CCCBDB-SRD101-FECO5-FERROCENE
- IUPAC-GOLD-BOOK-C01332
- NIST-CCCBDB-COMplete-EXPERIMENTAL-GEOMETRY
- IUPAC-GOLD-BOOK-I03294
- IUPAC-GOLD-BOOK-C01093
- IUPAC-GOLD-BOOK-E02069
- IUPAC-RED-BOOK-2005
- IUPAC-GOLD-BOOK-C01331
- NIST-CCCBDB-SRD101-SCF3-TICL4-FECO5
- NIST-RING-OPENING-COPOLYMER-SEQUENCE
- NIST-PRECISION-DEUTERATED-POLYETHYLENE
- NBS-JRES-IRON-CORROSION-1957
- NBS-JRES-GALVANIC-COUPLES-1950
- ECHEM-009-012-SOURCE-INVENTORY
- NIST-TRC-THERMOML-JCED-2015-60-3621-3630
- NIST-TRC-THERMOML-JCED-2006-51-680-685
- NIST-TRC-THERMOML-JCED-2014-59-757-763
- SFT-V3-ADMITTED-PROP-001
- SFT-V3-ADMITTED-PROP-002
- SFT-V3-ADMITTED-PROP-003
- SFT-V3-ADMITTED-PROP-004
- SFT-V3-ADMITTED-PROP-005
- SFT-V3-ADMITTED-PROP-006
- SFT-V3-ADMITTED-PROP-007
- SFT-V3-ADMITTED-PROP-008
- SFT-V3-ADMITTED-PROP-009
- SFT-V3-ADMITTED-PROP-010
- SFT-V3-ADMITTED-PROP-011

- SFT-V3-ADMITTED-PROP-012
- SFT-V3-ADMITTED-PROP-013
- NIST-THERMOREVERSIBLE-GELATION-PERCOLATION
- NIST-BRIDGING-GELATION
- IUPAC-S05735
- IUPAC-13807
- IUPAC-09464
- IUPAC-14011
- IUPAC-14026
- NIST-SRM-2888-POLYSTYRENE-CERTIFICATION
- NATURE-COMMUNICATIONS-S41467-025-68008-5-COMPLETE
- NIST-GRAPHENE-DOUBLE-LAYER-2020
- ECHEM-011-SOURCE-INVENTORY
- IUPAC-ELECTROCHEMICAL-METHODS-2019
- IUPAC-GOLD-BOOK-ELECTROCHEMICAL-CELL-COMPOSITE-2026
- SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003
- SFT-MAT-DEGR-CORROSION-001
- SFT-ENG-REQUIREMENT-001
- SFT-V3-LIVE-CLAIM-CENSUS
- SFT-V3-LIVE-DEPENDENCY-GRAPH
- SFT-CHEM-MAT-PAIRED-RECEIPTS
- NIST-METAL-ADDITIVE-CORROSION
- SFT-V3-BRANCH-OWNERSHIP-BOUNDARY
- NIST-CODATA-2022-ALL-CONSTANTS
- NIST-JRES-AGCL-STANDARD-POTENTIAL-1954
- NIST-CODATA-AGCL-WORK-CROSS-RECONSTRUCTION
- IUPAC-GREEN-BOOK-ELECTROCHEMISTRY-2007
- ECHEM-005-SOURCE-INVENTORY
- ECHEM-009-012-SHARED-SOURCES
- NIST-JRES-SILVER-ELECTROCHEMICAL-EQUIVALENT-1980
- ECHEM-006-SOURCE-INVENTORY
- NIST-CHEMISTRY-WEBBOOK-SRD69-DIATOMIC-CONSTANTS-2025
- NIST-ELECTRON-DIFFRACTION-DATABASE-REPORT
- NIST-NEUTRON-SCATTERING-LENGTHS
- NIST-SRM-676A-DIFFRACTION-STANDARD
- IUPAC-GOLD-BOOK-ELECTRONEGATIVITY-COMPOSITE-2026
- IUPAC-GOLD-BOOK-ET07026-2026
- IUPAC-E02020

- IUPAC-08167
- NATURE-CHEMISTRY-S41557-022-01101-0-ARTICLE
- NATURE-CHEMISTRY-41557-2022-1101-MOESM1
- NIST-SP260-209-ZETA-MOBILITY
- IUPAC-GOLD-BOOK-A00499-2026
- IUPAC-GOLD-BOOK-R05258-2026
- IUPAC-GOLD-BOOK-C01022-2026
- IUPAC-PERIODIC-TABLE-2022
- IUPAC-GOLD-BOOK-I03158-2026
- IUPAC-GOLD-BOOK-I03331-2026
- IUPAC-GOLD-BOOK-V06588-2026
- NIST-SRD17-1972TIN-KOO3-15
- NIST-SRD17-1967WIL2017-1
- NIST-SRD17-1967AVR-KRA501-503-1
- NIST-SRD17-1962ASH-BUR685-691-1
- IUPAC:E02038
- RSC:D4SC01905A:3a
- RSC:D4SC01905A:3b
- RSC:D4SC01905A:3c
- RSC:D4SC01905A:3d
- RSC:D4SC01905A:3e
- RSC:D4SC01905A:3f
- RSC:D4SC01905A:3g
- RSC:D4SC01905A:3h
- RSC:D4SC01905A:3i
- RSC:D4SC01905A:3j
- RSC:D4SC01905A:3k
- RSC:D4SC01905A:3l
- RSC:D4SC01905A:3m
- RSC:D4SC01905A:3n
- RSC:D4SC01905A:3o
- RSC:D4SC01905A:3p
- RSC:D4SC01905A:3q
- RSC:D4SC01905A:3r
- RSC:D4SC01905A:3s
- RSC:D4SC01905A:3t
- RSC:D4SC01905A:3u
- RSC:D4SC01905A:3v

- RSC:D4SC01905A:3w
- RSC:D4SC01905A:3x
- RSC:D4SC01905A:3y
- RSC:D4SC01905A:3z
- RSC:D4SC01905A:3aa
- RSC:D4SC01905A:3ab
- RSC:D4SC01905A:3ac
- RSC:D4SC01905A:3ad
- RSC:D4SC01905A:3ae
- RSC:D4SC01905A:3af
- RSC:D4SC01905A:UNSUCCESSFUL:1
- RSC:D4SC01905A:UNSUCCESSFUL:2
- RSC:D4SC01905A:UNSUCCESSFUL:3
- RSC:D4SC01905A:UNSUCCESSFUL:4
- RSC:D4SC01905A:UNSUCCESSFUL:5
- RSC:D4SC01905A:S1
- RSC:D4SC01905A:S2
- RSC:D4SC01905A:S3
- RSC:D4SC01905A:S4
- RSC:D4SC01905A:S5
- RSC:D4SC01905A:S6
- RSC:D4SC01905A:S7
- RSC:D4SC01905A:3A-PRIME-TIME-COURSE
- EUROPE-PMC:PMC11186341:SUPPLEMENT
- NIST-CHEMISTRY-WEBBOOK-SRD69-WATER-FLUID-PROPERTIES
- IUPAC-GOLD-BOOK-CHEMICAL-EQUILIBRIUM-COMPOSITE-2026
- NIST-WEBBOOK-SRD69-H2-DIATOMIC-CONSTANTS
- NIST-WEBBOOK-SRD69-D2-DIATOMIC-CONSTANTS
- USGS-PP-440-KK-STABLE-ISOTOPE-FRACTIONATION
- USGS-WRIR-02-4172-ISOTOPE-EQUILIBRIUM-CONSTANTS
- NUCHEM-005-008-SHARED-SOURCES
- NUCHEM-005-008-SOURCE-INVENTORY
- NIST-MDS2-3389-CAH-PLUS-QUANTUM-JUMP-THERMOMETRY
- NIST-CHEMISTRY-WEBBOOK-SRD69-WATER-GAS-CALORIMETRIC-TABLE
- DOE-ORNL-4865-FISSION-PRODUCT-BEHAVIOR-MSRE
- NUCHEM-009-012-SOURCE-INVENTORY
- IUPAC-FLUORESCENCE-STANDARDS-2010
- IUPAC-PHOTOLUMINESCENCE-QUANTUM-YIELD-2011

- NIST-SRM-2941A-FLUORESCENCE-GREEN
- NIST-IR-7458-FLUORESCENCE-GUIDE
- NIST-NLM-PASH-PHOSPHORESCENCE-2018
- ANAL-009-011-COMPLETE-SOURCE-INVENTORY
- NIST-JANAF-N-007-NO2-GAS
- NIST-JANAF-N-032-N2O4-GAS
- NIST-TRC-THERMOML-FPE-2019-485-145-152
- IUPAC-RT06783
- NIST-JPCRD-STANDARD-ELECTRODE-POTENTIALS-1989
- NIST-WEBBOOK-SRD69-GUIDE
- NIST-WEBBOOK-ACETONE-MULTIMODAL
- IUPAC-L03508
- IUPAC-L03511
- IUPAC-L03510
- IUPAC-O04362
- IUPAC-R05222
- IUPAC-08245
- IUPAC-L03519
- IUPAC-L03520
- IUPAC-M03868
- NIST-WEBBOOK-C13007926-UVVIS-COMPLETE
- NIST-WEBBOOK-C12082085-UVVIS-COMPLETE
- NIST-WEBBOOK-C12108111-UVVIS-COMPLETE
- NIST-WEBBOOK-C12146360-UVVIS-COMPLETE
- IUPAC-M03689
- IUPAC-P04404
- IUPAC-D01668
- NIST-MOLECULAR-MICROWAVE-SPECTRAL-DATABASES-SRD-114-115-117
- NIST-JPCRD-3-609-1974-DIATOMIC-MICROWAVE-SPECTRAL-TABLES
- IUPAC-M03924
- IUPAC-I03058
- IUPAC-E02038
- IUPAC-GOLD-BOOK-INTERFACE-TRANSFER-COMPOSITE-2026
- NIST-CCCBDB-SRD-101-DIMER-BINDING
- NIST-FARADAY-C8FD00092A-WATER-CLUSTER-DISSOCIATION
- NIST-JPCRD-1.555757-ION-CLUSTER-SCOPE
- NIST-SRM-3190-CONDUCTIVITY-CERTIFICATE
- NIST-SP-260-142-PRIMARY-CONDUCTIVITY

- NIST-SP-260-176-SRM-CATALOG
- ECHEM-007-SOURCE-INVENTORY
- NBS-JRES-TRANSFERENCE-CONCENTRATION-1931
- ECHEM-008-SOURCE-INVENTORY
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-DISSOCIATION
- IUPAC-GOLD-BOOK-ACTIVATION-COMPOSITE-2026
- IUPAC-GOLD-BOOK-O04322-2026
- IUPAC-GOLD-BOOK-REACTION-RATE-COMPOSITE-2026
- NATURE-COMMUNICATIONS-S41467-024-44753-X-COMPLETE
- NBS-RP729-ELECTROLYTIC-HYDROGEN-OXYGEN-FRACTIONATION-1934
- IUPAC-GOLD-BOOK-B00649
- IUPAC-GOLD-BOOK-C01012
- IUPAC-GOLD-BOOK-D01594
- IUPAC-GOLD-BOOK-H01881
- IUPAC-GOLD-BOOK-K03366
- IUPAC-GOLD-BOOK-L03518
- IUPAC-LT06764
- IUPAC-L03517
- NIST-WEBBOOK-C13939065-UVVIS-COMPLETE
- NIST-WEBBOOK-C14040110-UVVIS-COMPLETE
- NIST-SRM-2886-POLYETHYLENE-CERTIFICATION
- IUPAC-GOLD-BOOK-A00297-2026
- IUPAC-GOLD-BOOK-M03986-2026
- IUPAC-GOLD-BOOK-CT01038-2026
- IUPAC-GOLD-BOOK-M03987-2026
- IUPAC-GOLD-BOOK-S06236-2026
- IUPAC-GOLD-BOOK-C01039-2026
- IUPAC-GOLD-BOOK-08013-2026
- IUPAC-GOLD-BOOK-08133-2026
- IUPAC-CT06769
- IUPAC-IT06779
- IUPAC-GOLD-BOOK-S06061-2026
- IUPAC-GOLD-BOOK-I03098-2026
- IUPAC-GOLD-BOOK-I03289-2026
- IUPAC-GOLD-BOOK-M04002-2026
- IUPAC-GOLD-BOOK-N04112-2026
- IUPAC-GOLD-BOOK-13867-2026
- NIST-CCCBDB-SRD101-BF3-EXPERIMENTAL-GEOMETRY

- NIST-CCCBDB-SRD101-XEF2-EXPERIMENTAL-GEOMETRY
- NIST-CCCBDB-SRD101-XEF4-EXPERIMENTAL-GEOMETRY
- NIST-TRC-THERMOML-JCED-2011-56-4840-4848
- NIST-TRC-THERMOML-FPE-2017-437-34-42
- NIST-TRC-THERMOML-FPE-2008-271-43-52
- NIST-NBS-JCP-59-2254-1973
- NIST-CCCBDB-SRD101-EXPERIMENTAL-DIPOLES
- NIST-CCCBDB-COMPLETE-ELECTRON-AFFINITY-CATALOG
- NIST-CCCBDB-EXPERIMENTAL-MOLECULAR-ELECTRON-AFFINITY
- NIST-WEBBOOK-SRD69-GAS-PHASE-ION-THERMOCHEMISTRY
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-EXCHANGE-2025
- NIST-CCCBDB-SRD-101-COMPLETE-EXPERIMENTAL-FORMATION-ENERGY
- NIST-CCCBDB-THERMODYNAMIC-REFERENCE-STATES
- NIST-CCCBDB-SRD101-EXPERIMENTAL-IONIZATION-ENERGY
- NIST-CCCBDB-SRD101-EXPERIMENTAL-POLARIZABILITIES
- NIST-MDS2-3389-CAH-QUANTUM-STATE-TRACKING-CONTROL-V1.0.1
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-TRANSITIONS
- NIST-CCCBDB-SRD101-DIBORANE
- NIST-CCCBDB-SRD101-BENZENE
- NIST-TRC-THERMOML-JCT-2012-47-260-266
- IUPAC-GOLD-BOOK-AUTOCATALYSIS-COMPOSITE-2026
- IUPAC-GOLD-BOOK-REACTION-NETWORK-COMPOSITE-2026
- IUPAC-NMR-NOMENCLATURE-2001
- BMRB-ENTRY-68-CHEMICAL-SHIFTS
- ANAL-006-008-NMR-SOURCE-INVENTORY
- BMRB-ENTRY-52365-RELAXATION
- BMRB-ENTRY-27257-H-EXCHANGE
- BMRB-ENTRY-16582-SCALAR-COUPPLINGS
- NIST-NUCLEAR-PHYSICS-DATA-ISOTOPIIC-COMPOSITION
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-HD-D2-DIATOMIC-CONSTANTS-2026
- IUPAC-N04249
- IUPAC-08191
- IUPAC-E02130
- IUPAC-S06078
- IUPAC-H02809
- PUBCHEM-PUG-BROMOMETHANE-STRUCTURE
- PUBCHEM-PUG-HYDROXIDE-STRUCTURE
- PUBCHEM-PUG-METHANOL-STRUCTURE

- PUBCHEM-PUG-BROMIDE-STRUCTURE
- NIST-QUINTUPLE-DETECTOR-COPOLYMER
- IUPAC-GOLD-BOOK-FUNCTIONAL-GROUP-COMPOSITE-2026
- IUPAC-GOLD-BOOK-ORGANIC-REACTION-FAMILY-COMPOSITE-2026
- IUPAC-E01913
- IUPAC-O04328
- IUPAC-O04367
- IUPAC-R05223
- NATURE-COMMUNICATIONS-S41467-026-70199-4-SOURCE-DATA
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- IUPAC:C01496
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- IUPAC-GOLD-BOOK-CHEMICAL-PHASE-COMPOSITE-2026
- IUPAC-GOLDBOOK-P04533-PHASE-RULE
- NIST-PHASE-DIAGRAM-GENERAL-DISCUSSION-GLOSSARY
- IUPAC-PHOSPHORESCENCE-LIFETIME
- IUPAC-GOLD-BOOK-PHOTOCHEMISTRY-COMPOSITE-2026
- NIST-BRANCH-PLACEMENT-DILUTE-SOLUTION
- NIST-MACROMOLECULAR-ARCHITECTURES
- IUPAC-GOLD-BOOK-POLYMER-CHAIN-COMPOSITE-2026
- NIST-POLYMER-PYROLYSIS-NETWORK
- NIST-POLYMER-DEPOLYMERIZATION-KINETICS
- IUPAC-GOLD-BOOK-POLYMER-DISTRIBUTION-COMPOSITE-2026
- NIST-CROSSLINKED-PHOTOPOLYMER-CONVERSION
- NIST-POLYMER-PROCESSING-METROLOGY
- NIST-PVOH-IONIC-LIQUID-GELATION
- IUPAC-09560
- IUPAC-09466
- NBS-NSRDS-45-RADIATION-CHEMISTRY-NITROUS-OXIDE
- IUPAC:C00960
- IUPAC:I03042
- IUPAC:T06274
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- NUCHEM-001-004-SHARED-SOURCES
- NUCHEM-005-NIST-SRM-PAIR
- DOE-OSTI-1580278-HFSLM-ISOTOPE-HARVESTING
- IAEA-TCS-31-RADIOTRACER-RTD-2008
- NIST-RAMAN-RELATIVE-INTENSITY-STANDARDS
- NIST-SRM-2241-RAMAN-785-NM
- NIST-SRM-2242A-RAMAN-532-NM
- NATURE-COMMUNICATIONS-S41467-025-66587-X-COMPLETE
- IUPAC:M03997
- IUPAC:R05194
- IUPAC:R05196
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:1
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:2
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAIR:3
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- ACS-FIGSHARE:10.1021/ja803370x.s001:PAGE:4
- ACS-FIGSHARE:10.1021/ja803370x.s001:PAGE:37
- EUROPE-PMC:ORG-011:V1
- ACS-FIGSHARE:10.1021/ja803370x.s001
- IUPAC-GOLD-BOOK-REDOX-COMPOSITE-2026
- IUPAC-GOLD-BOOK-O04365-2026
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-HD-D2-RESOLVED-DIATOMIC-SURFACE-2026
- IUPAC-R05326
- IUPAC-08205
- IUPAC-C01309
- IUPAC-M03804
- IUPAC-R05174
- NATURE-COMMUNICATIONS-S41467-023-40190-4-COMplete
- NIST-CCCBDB-COMplete-ROTATIONAL-CONSTANT-SURFACE
- NASA-JPL-MOLECULAR-SPECTROSCOPY-CATALOG
- NASA-JPL-CO-ROTATIONAL-LINE-CATALOG-028001
- NIST-MOLECULAR-SPECTROSCOPY-H2-D2-SUCCESSOR
- IUPAC-GOLD-BOOK-CHEMICAL-REACTION-COMPOSITE-2026
- IUPAC-GOLD-BOOK-I03096-2026
- IUPAC-GOLD-BOOK-REACTION-MECHANISM-COMPOSITE-2026
- NIST-CHEMISTRY-WEBBOOK-SRD69-H2-SELECTION-SURFACE
- IUPAC-C01051
- IUPAC-R05243
- IUPAC-S05991
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- NATURE-COMMUNICATIONS-S41467-024-49814-9-COMplete
- RCSB-PDB-8WZF-8WZV
- CXIDB-221
- IUPAC-12823
- IUPAC-GOLD-BOOK-SOLUTION-EQUILIBRIUM-COMPOSITE-2026
- FREESOLV-0.52-COMMIT-6C7D19B
- NIST-TRC-THERMOML-JCED-2016-61-1470-1476
- IUPAC-GOLD-BOOK-INFRARED-SPECTROSCOPY-COMPOSITE-2026

- IUPAC-GOLD-BOOK-MASS-SPECTROSCOPY-COMPOSITE-2026
- IUPAC-GOLD-BOOK-ROTATION-VIBRATION-COMPOSITE-2026
- IUPAC-GOLD-BOOK-UVVISIBLE-SPECTROSCOPY-COMPOSITE-2026
- IUPAC-GOLD-BOOK-CHIRALITY-COMPOSITE-2026
- IUPAC-GOLD-BOOK-DIASTEREOMER-COMPOSITE-2026
- IUPAC-GOLD-BOOK-ENANTIOMER-COMPOSITE-2026
- IUPAC-GOLD-BOOK-C01034-2026
- IUPAC-GOLD-BOOK-M03722-2026
- IUPAC-GOLD-BOOK-S06026-2026
- IUPAC-GOLD-BOOK-D01771-2026
- IUPAC-GOLD-BOOK-M03949-2026
- IUPAC-GOLD-BOOK-S05746-2026
- IUPAC-GOLD-BOOK-C01041-2026
- IUPAC-GOLD-BOOK-ADSORPTION-COMPOSITE-2026
- NIST-CODATA-2022-BOLTZMANN
- NIM-NIST-ACOUSTIC-BOLTZMANN-2017
- NIST-JOHNSON-NOISE-BOLTZMANN-2011
- ACS-JPCA-2015-JP505790M-COMPLETE-TEMPERATURE-RATE-SERIES
- NIST-TRC-THERMOML-JCED-2013-58-663-670
- NIST-TRC-THERMOML-JCT-2019-133-135-142
- NIST-TRC-THERMOML-FPE-2018-477-78-86
- IUPAC-GOLD-BOOK-REACTION-DIRECTION-COMPOSITE-2026
- IUPAC-GOLD-BOOK-REACTION-THERMOCHEMISTRY-COMPOSITE-2026
- KYRIAKOU-ET-AL-ACS-NANO-NN500703K-COMPLETE
- IUPAC-SUPERHEAVY-2018
- NIST-ATOMIC-SPECTROSCOPY
- NIST-CCCBDB-COMPLETE-DISPLAYED-FUNDAMENTAL-FREQUENCY-SURFACE
- NIST-TRC-THERMOML-FPE-2018-474-6-13
- NIST-TRC-THERMOML-JCED-2005-50-1038-1042
- NIST-TRC-THERMOML-FPE-2017-453-13-23
- NIST-SRM-674-XRAY-INTENSITY-SET

Comparison records:

- VALID-012: complete registered membership count 263
- complete Chemistry base: 263 claims; 255 empirical; 8 formal-only
- retained evidence: 26486 measurement/evidence lines and 1177 source-identity occurrences
- all favorable adverse absent unavailable unresolved tampered and scope-boundary rows retained; omission control rejected True
- SFT-CHEM-AB-ACID-BASE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1

- SFT-CHEM-AB-AMPHOTERIC-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-AB-BUFFER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-AB-LEWIS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-AB-PROTON-TRANSFER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ACTIVATION-BARRIER-VALUE-RELATION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=52; explicit-status-lines=3
- SFT-CHEM-ACTIVITY-AMOUNT-TIME-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ACTIVITY-NONIDEAL-COMPOSITION-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=213; explicit-status-lines=1
- SFT-CHEM-ADDITION-REACTION-FAMILY-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=28; evidence-lines=7; explicit-status-lines=3
- SFT-CHEM-ANALYTICAL-ACCURACY-TRUENESS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ANALYTICAL-CALIBRATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ANALYTICAL-DETECTION-QUANTIFICATION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ANALYTICAL-PERFORMANCE-BUDGET-022: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-ANALYTICAL-PRECISION-REPEATABILITY-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ANALYTICAL-SAMPLE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ANALYTICAL-SELECTIVITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ANALYTICAL-SELECTIVITY-INTERFERENCE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ANALYTICAL-SENSITIVITY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ANTIAROMATIC-NONAROMATIC-DISTINCTION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=21; explicit-status-lines=5
- SFT-CHEM-AROMATIC-RECURRENCE-STABILITY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=9; evidence-lines=20; explicit-status-lines=2
- SFT-CHEM-ATOM-MAPPING-REACTION-BALANCE-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1

- SFT-CHEM-BOND-CHEMICAL-BOND-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-BOND-COVALENT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-BOND-DISSOCIATION-ENERGY-002: source-custodied-post-seal-comparison; closure=finite_complete; sources=6; evidence-lines=11; explicit-status-lines=7
- SFT-CHEM-BOND-IONIC-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-BOND-LENGTH-STRENGTH-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-BOND-METALLIC-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-BOND-ORDER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-BOND-POLARITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-CANONICAL-MOLECULAR-GRAPH-ENCODING-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-CAT-CATALYST-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-CAT-PATHWAY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-CAT-SELECTIVITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-CATALYTIC-TURNOVER-CYCLE-FREQUENCY-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=510; explicit-status-lines=2
- SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=7; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-CHAIN-GROWTH-POLYMERIZATION-NETWORK-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-CHEMICAL-POTENTIAL-EQUIVALENT-COMPONENT-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=83; explicit-status-lines=0
- SFT-CHEM-CHEMICAL-SUBSTRUCTURE-RELATION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-CHROMATOGRAPHIC-RETENTION-RESOLUTION-018: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-CLASSICAL-QUANTUM-ALGORITHM-CORRESPONDENCE-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-COLLIGATIVE-COMPOSITION-RESPONSE-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=285; explicit-status-lines=0
- SFT-CHEM-COLLOID-DISPERSION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-COMPETING-CHANNEL-BRANCHING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=16; explicit-status-lines=2

- SFT-CHEM-COMPLEX-SPIN-STATE-ORDER-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=12; explicit-status-lines=2
- SFT-CHEM-CONCENTRATION-DEPENDENCE-RELATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=17; explicit-status-lines=1
- SFT-CHEM-CONCENTRATION-DEPENDENT-POTENTIAL-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-CONFIGURATION-ORDER-PATH-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=56; explicit-status-lines=0
- SFT-CHEM-CONFORMER-ENUMERATION-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-CONFORMER-GENERATION-EQUIVALENCE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=16; explicit-status-lines=3
- SFT-CHEM-CONFORMER-POPULATION-ORDERING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=14; evidence-lines=20; explicit-status-lines=3
- SFT-CHEM-CONJUGATED-SUPPORT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=7; evidence-lines=18; explicit-status-lines=0
- SFT-CHEM-CONSTITUTIONAL-ISOMER-ENUMERATION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-COORDINATION-ENTITY-RETAINED-IDENTITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=26; explicit-status-lines=1
- SFT-CHEM-COORDINATION-GEOMETRY-HELD-ORIENTATION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=10; explicit-status-lines=3
- SFT-CHEM-COORDINATION-ISOMERISM-EQUIVALENCE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=14; explicit-status-lines=1
- SFT-CHEM-COORDINATION-NUMBER-INCIDENCE-COUNT-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=29; explicit-status-lines=1
- SFT-CHEM-COPOLYMER-SEQUENCE-COMPOSITION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-CORROSION-REACTION-NETWORK-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-COUPLED-MASS-HEAT-CHARGE-TRANSPORT-019: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=240; explicit-status-lines=0
- SFT-CHEM-CROSS-PROPERTY-MOLECULAR-VECTOR-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=13; evidence-lines=9033; explicit-status-lines=0
- SFT-CHEM-CROSSLINK-GELATION-BOUNDARY-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-DATABASE-IDENTITY-PROVENANCE-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-DEFECT-NONSTOICHIOMETRY-016: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=13; explicit-status-lines=2
- SFT-CHEM-DEGREE-OF-POLYMERIZATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-DIFFUSION-LIMITED-REACTION-BOUNDARY-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=264; explicit-status-lines=3

- SFT-CHEM-DIHEDRAL-TORSIONAL-STATE-004: source-custodied-post-seal-comparison; closure=finite_complete; sources=1; evidence-lines=76; explicit-status-lines=14
- SFT-CHEM-DOUBLE-LAYER-INTERFACIAL-CHARGE-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ELECTROANALYTICAL-RESPONSE-020: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-ELECTROCHEM-CELL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELECTROCHEMICAL-STORAGE-HANDOFF-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=1
- SFT-CHEM-ELECTROCHEMICAL-WORK-REACTION-DIRECTION-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ELECTRODE-POTENTIAL-CHEMICAL-RELATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=7; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ELECTRODE-REACTION-RATE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ELECTROLYSIS-PRODUCT-AMOUNT-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ELECTRON-COUNT-SPIN-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=27; explicit-status-lines=6
- SFT-CHEM-ELECTRON-NEUTRON-DIFFRACTION-017: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-ELECTRONEGATIVITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELECTRONIC-STATE-IDENTITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=2
- SFT-CHEM-ELECTROPHILIC-SUBSTITUTION-FAMILY-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=6; explicit-status-lines=1
- SFT-CHEM-ELECTROPHORETIC-MOBILITY-SEPARATION-019: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELEM-ATOMIC-WEIGHT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELEM-ELEMENT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELEM-GROUP-PERIOD-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELEM-ION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELEM-ISOTOPE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELEM-PERIODIC-BOUNDARY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1

- SFT-CHEM-ELEM-PERIODIC-ORDER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELEM-VALENCE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=55; explicit-status-lines=0
- SFT-CHEM-ELIMINATION-REACTION-FAMILY-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=47; evidence-lines=7; explicit-status-lines=1
- SFT-CHEM-ENTHALPY-EQUIVALENT-STATE-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=22; explicit-status-lines=0
- SFT-CHEM-ENTROPY-MULTIPLICITY-CORRESPONDENCE-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=22; explicit-status-lines=1
- SFT-CHEM-EQ-CHEMICAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-EQUILIBRIUM-BOND-LENGTH-001: source-custodied-post-seal-comparison; closure=finite_complete; sources=3; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-EQUILIBRIUM-ISOTOPE-FRACTIONATION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-EXACT-MOLECULAR-SIMILARITY-DISTINCTION-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-FINITE-MICROSTATE-SUPPORT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=395; explicit-status-lines=0
- SFT-CHEM-FISSION-PRODUCT-CHEMICAL-DISTRIBUTION-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-FLUORESCENCE-YIELD-LIFETIME-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=72; explicit-status-lines=0
- SFT-CHEM-FUGACITY-EQUIVALENT-GAS-MIXTURE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=105; explicit-status-lines=0
- SFT-CHEM-HALF-REACTION-IDENTITY-ORIENTATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=6; evidence-lines=4; explicit-status-lines=0
- SFT-CHEM-HEAT-WORK-TRANSFER-PARTITION-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=22; explicit-status-lines=1
- SFT-CHEM-INFRARED-LINE-INTENSITY-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-INORGANIC-ACID-BASE-REDOX-NETWORK-017: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=14; explicit-status-lines=1
- SFT-CHEM-INORGANIC-COLOUR-ELECTRONIC-TRANSITION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=12; explicit-status-lines=1
- SFT-CHEM-INORGANIC-MAGNETIC-STATE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=13; explicit-status-lines=0

- SFT-CHEM-INSERTION-ELIMINATION-PATHWAY-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=10; explicit-status-lines=3
- SFT-CHEM-INTERFACE-TRANSFER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-INTERMOLECULAR-BINDING-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=1305; explicit-status-lines=99
- SFT-CHEM-INTERNAL-ENERGY-COMPOSITION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=21; explicit-status-lines=0
- SFT-CHEM-IONIC-CONDUCTIVITY-RELATION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-IONIC-MOBILITY-TRANSFERENCE-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-ISOTOPE-EXCHANGE-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-JOINT-CORRELATION-DISSOCIATION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=30; explicit-status-lines=14
- SFT-CHEM-KIN-ACTIVATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-KIN-ORDER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-KIN-RATE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-KINETIC-ISOTOPE-EFFECT-RELATION-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=84; explicit-status-lines=3
- SFT-CHEM-KINETIC-ISOTOPE-FRACTIONATION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-LIGAND-DENTICITY-CHELATION-TOPOLOGY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=6; evidence-lines=30; explicit-status-lines=1
- SFT-CHEM-LIGAND-STATE-SPLITTING-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=13; explicit-status-lines=3
- SFT-CHEM-MASS-ISOTOPE-FRAGMENT-VECTOR-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-MASS-WEIGHTED-MOLECULAR-SIZE-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-MEAS-AMOUNT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MEAS-FORMULA-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MEAS-NOMENCLATURE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1

- SFT-CHEM-MEAS-SUBSTANCE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MEAS-TRACEABILITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MEAS-UNCERTAINTY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MECHANISM-SEARCH-PROOF-TRACE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-METAL-CLUSTER-BONDING-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=10; explicit-status-lines=1
- SFT-CHEM-MOL-GEOMETRY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MOL-INTERMOLECULAR-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MOL-ISOMER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MOL-MOLECULE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MOL-NETWORK-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MOL-SUPRAMOLECULAR-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-MOLECULAR-BOND-ANGLE-003: source-custodied-post-seal-comparison; closure=finite_complete; sources=3; evidence-lines=12; explicit-status-lines=8
- SFT-CHEM-MOLECULAR-DIFFUSION-RELATION-016: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=174; explicit-status-lines=1
- SFT-CHEM-MOLECULAR-DIPOLE-MAGNITUDE-005: source-custodied-post-seal-comparison; closure=finite_complete; sources=2; evidence-lines=15; explicit-status-lines=10
- SFT-CHEM-MOLECULAR-ELECTRON-AFFINITY-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=102; explicit-status-lines=0
- SFT-CHEM-MOLECULAR-EXCLUSION-EXCHANGE-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=80; explicit-status-lines=12
- SFT-CHEM-MOLECULAR-FORMATION-ENERGY-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=2106; explicit-status-lines=0
- SFT-CHEM-MOLECULAR-GRAPH-ISOMORPHISM-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-MOLECULAR-IONIZATION-ENERGY-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=13; explicit-status-lines=1
- SFT-CHEM-MOLECULAR-MAGNETIC-RESPONSE-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=181; explicit-status-lines=1
- SFT-CHEM-MOLECULAR-POLARIZABILITY-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=256; explicit-status-lines=1
- SFT-CHEM-MOLECULAR-QUANTUM-MEASUREMENT-REDUCTION-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=345; explicit-status-lines=0

- SFT-CHEM-MOLECULAR-STATE-TRANSITION-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=80; explicit-status-lines=16
- SFT-CHEM-MULTICENTRE-DELOCALIZED-SUPPORT-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=41; explicit-status-lines=16
- SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=128; explicit-status-lines=0
- SFT-CHEM-MULTIMODAL-MOLECULAR-IDENTITY-021: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-NET-AUTOCATALYSIS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-NET-REACTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-NMR-CHEMICAL-SHIFT-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-NMR-RELAXATION-EXCHANGE-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-NMR-SPIN-COUPPLING-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-NUCLEAR-CHEMICAL-CARRIER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-NUCLEAR-ELECTRONIC-COMPOSITION-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=433; explicit-status-lines=0
- SFT-CHEM-NUCLEOPHILIC-SUBSTITUTION-FAMILY-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=9; evidence-lines=7; explicit-status-lines=1
- SFT-CHEM-NUMBER-AVERAGE-MOLECULAR-SIZE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=25; explicit-status-lines=0
- SFT-CHEM-OPERATIONAL-CLASSICAL-QUANTUM-CORRESPONDENCE-015: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-ORBITAL-SUPPORT-OCCUPANCY-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=366; explicit-status-lines=7
- SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-ORGANOMETALLIC-ELECTRON-ACCOUNTING-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=10; explicit-status-lines=2
- SFT-CHEM-ORGANOMETALLIC-METAL-CARBON-BOND-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=11; explicit-status-lines=0
- SFT-CHEM-OVERPOTENTIAL-POLARIZATION-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-OXIDATIVE-ADDITION-REDUCTIVE-ELIMINATION-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=10; explicit-status-lines=0

- SFT-CHEM-PARALLEL-MECHANISM-COMPOSITION-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=39; explicit-status-lines=3
- SFT-CHEM-PERICYCLIC-REACTION-FAMILY-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=38; evidence-lines=7; explicit-status-lines=2
- SFT-CHEM-PHASE-CHEMICAL-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-PHASE-RULE-STRUCTURAL-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=27; explicit-status-lines=0
- SFT-CHEM-PHOSPHORESCENCE-INTERSYSTEM-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-PHOTOCHEM-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-POLYMER-ARCHITECTURE-008: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-CHAIN-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-POLYMER-CONFORMATION-SIZE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-DEGRADATION-NETWORK-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-DISPERSITY-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-DISTRIBUTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-POLYMER-MATERIALS-HANDOFF-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-POLYMER-PHASE-TRANSITION-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-PRED-G-BLOCK-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-PRED-PERIODIC-ENDPOINT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-PRED-SMITHIUM-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-PREDICTION-UNCERTAINTY-APPLICABILITY-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-PROTECTING-GROUP-REVERSIBLE-STATE-015: source-custodied-post-seal-comparison; closure=depth_independent; sources=5; evidence-lines=4; explicit-status-lines=0
- SFT-CHEM-RADIATION-CHEMISTRY-REACTION-NETWORK-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-RADICAL-REACTION-NETWORK-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=73; evidence-lines=6; explicit-status-lines=1
- SFT-CHEM-RADIOACTIVE-BRANCHING-CHEMICAL-YIELD-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2

- SFT-CHEM-RADIOACTIVE-CHEMICAL-TRANSFORMATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-RADIOCHEMICAL-EQUILIBRIUM-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-RADIOCHEMICAL-SEPARATION-DECONTAMINATION-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-RADIOTRACER-CUSTODY-INFERENCE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=3; explicit-status-lines=2
- SFT-CHEM-RAMAN-TRANSITION-INTENSITY-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-REACTION-DYNAMICS-SCATTERING-PRODUCT-STATE-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=65; explicit-status-lines=2
- SFT-CHEM-REACTION-GRAPH-GENERATION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-REARRANGEMENT-REACTION-FAMILY-011: source-custodied-post-seal-comparison; closure=depth_independent; sources=16; evidence-lines=7; explicit-status-lines=2
- SFT-CHEM-REDOX-COUPPLING-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-REDOX-OXIDATION-STATE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-RESOLVED-ROVIBRONIC-SPIN-COMPOSITION-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=107; explicit-status-lines=2
- SFT-CHEM-RESONANCE-EQUIVALENT-REPRESENTATION-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=4; evidence-lines=21; explicit-status-lines=0
- SFT-CHEM-RETROSYNTHETIC-DECOMPOSITION-RECONSTRUCTION-016: source-custodied-post-seal-comparison; closure=depth_independent; sources=7; evidence-lines=4; explicit-status-lines=0
- SFT-CHEM-REVERSIBLE-KINETIC-EQUILIBRIUM-CORRESPONDENCE-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=176; explicit-status-lines=2
- SFT-CHEM-ROTATIONAL-CONSTANT-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=3024; explicit-status-lines=1
- SFT-CHEM-ROTATIONAL-SPECTRUM-LINE-015: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-ROVIBRONIC-COMPOSITION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=6; explicit-status-lines=2
- SFT-CHEM-RXN-IDENTITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-RXN-INTERMEDIATE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=2
- SFT-CHEM-RXN-MECHANISM-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-SELECTION-RULE-STRUCTURE-010: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=75; explicit-status-lines=11
- SFT-CHEM-SELECTIVITY-COMPLETE-DISTRIBUTION-014: source-custodied-post-seal-comparison; closure=depth_independent; sources=133; evidence-lines=5; explicit-status-lines=1

- SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=28; explicit-status-lines=5
- SFT-CHEM-SMITHIUM-CHEMICAL-SEPARATION-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SMITHIUM-DECAY-CHANNEL-LEDGER-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SMITHIUM-ION-OXIDATION-LADDER-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SMITHIUM-JOINT-DETECTION-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SMITHIUM-LIFETIME-BOUNDARY-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SMITHIUM-SPECTROSCOPIC-CLASSES-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SMITHIUM-SYNTHESIS-CONSERVATION-001: formal-only-standing-or-operational-boundary; closure=depth_independent; sources=0; evidence-lines=0; explicit-status-lines=1
- SFT-CHEM-SOLID-STATE-LOCAL-COORDINATION-015: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=11; explicit-status-lines=2
- SFT-CHEM-SOLUTION-EQUILIBRIUM-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-SOLVATION-DISSOLUTION-FREE-ORDER-015: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=810; explicit-status-lines=1
- SFT-CHEM-SPEC-INFRARED-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-SPEC-MASS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-SPEC-ROT-VIB-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-SPEC-UVVIS-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STATE-ENERGY-ORDER-004: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=313; explicit-status-lines=6
- SFT-CHEM-STATE-SYMMETRY-DEGENERACY-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=379; explicit-status-lines=11
- SFT-CHEM-STEP-GROWTH-POLYMERIZATION-NETWORK-006: source-custodied-post-seal-comparison; closure=depth_independent; sources=2; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-STEREO-CHIRALITY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STEREO-DIASTEREOMER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STEREO-ENANTIOMER-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STEREOISOMER-ENUMERATION-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2

- SFT-CHEM-STOICH-COEFFICIENT-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STOICH-COMPOSITION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STOICH-CONSERVATION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STOICH-LIMITING-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STOICH-MIXTURE-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STOICH-SOLUTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-STOICH-YIELD-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-SURFACE-ADSORPTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-SYMBOLIC-PROPERTY-EVALUATION-012: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=5; explicit-status-lines=2
- SFT-CHEM-TEMPERATURE-CORRESPONDENCE-002: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=10; explicit-status-lines=0
- SFT-CHEM-TEMPERATURE-DEPENDENCE-RELATION-003: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=27; explicit-status-lines=1
- SFT-CHEM-THERMAL-CONDUCTIVITY-RELATION-018: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=665; explicit-status-lines=0
- SFT-CHEM-THERMO-DIRECTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-THERMO-REACTION-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2; explicit-status-lines=1
- SFT-CHEM-TRANSITION-STATE-EQUIVALENT-BOUNDARY-005: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=10; explicit-status-lines=1
- SFT-CHEM-UVVISIBLE-LINE-INTENSITY-013: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2
- SFT-CHEM-VALIDATION-SMITHIUM-COMPLETE-FAMILY-001: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=6; explicit-status-lines=4
- SFT-CHEM-VIBRATIONAL-FREQUENCY-009: source-custodied-post-seal-comparison; closure=depth_independent; sources=1; evidence-lines=2016; explicit-status-lines=1
- SFT-CHEM-VISCOUS-TRANSPORT-RELATION-017: source-custodied-post-seal-comparison; closure=depth_independent; sources=3; evidence-lines=434; explicit-status-lines=1
- SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016: source-custodied-post-seal-comparison; closure=depth_independent; sources=8; evidence-lines=4; explicit-status-lines=2

Falsification boundary: Reject if any registered member, receipt, control, evidence file, result class, closure scope, formal-only boundary or source-custody row is missing, changed, duplicated or selected after outcome access.

Measurement receipt: sha256:56d014a6865286a1f4d7ac97a8463f0b9e20ccaf92fb87c3fc84ebf89a2908c5.

Isolation certificate: sha256:0649cbe978b7cf4bc753f722508f9a16a5e32c22f3a291e9f31e3b132bfd8beb.

Custody certificate: sha256:02a99ce611705f989c3101ced0542381bddbf0cb670f093c6583fc56e89a6609.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no target value, outcome, match rate or measurement selects the validation law
- no fitted threshold, favorable-only sample or omitted member
- no formal-only standing boundary relabelled as a measured outcome
- no finite-complete claim relabelled depth-independent or conversely
- no numerical absence, negative, irrational, imaginary, floating, fitted or free proof magnitude
- no first failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

Immutable identities. Source manifest

sha256:6d5adf68aa88afd038839456cdc5bdb7aa939ee20e4f8fc3f69dc0df54ab564e; derivation seal
 sha256:7db7b5d1c9aa1e012e8021faeebb3001222a71e57d0fecde98f0e9656f46ac52; engine receipt
 sha256:be4b72344aeaefd342ef43b7660e19fe73fcd6cda30541ca198debcc72f994a9 at receipts/engine/model_admitted/SFT-CHEM-VALIDATION-EMPIRICAL-GRAND-LOCK-012-be4b72344aeaefd3.json;
 empirical hash sha256:529f3a1f2f4ca5796cf90ed1d01f900ecbfa0ab1ce5060296f2a120006701d56;
 measurement receipt sha256:56d014a6865286a1f4d7ac97a8463f0b9e20ccaf92fb87c3fc84ebf89a2908c5.

30. Cross-branch ownership and handoff laws - 6 admitted claims

Chemistry exports exact laws through directed dependencies while downstream branches retain their own distinct ownership. This entire registered subcategory is included; no proper subset is presented as field completion.

276. Chemistry-to-Materials one-owner handoff

Claim identity: SFT-CHEM-MATERIALS-OWNERSHIP-HANDOFF-001

Question and theorem. The Chemistry handoff to materials preserves complete paired admitted records and exactly one owner per coordinate: chemistry owns species, bonds, reactions and local structure while materials owns bulk response.

HAND-001 uniquely requires the complete receipt-bound pair in which chemistry owns species, bonds, reactions and local structure while materials owns bulk response; consumption is a directed dependency and never duplicate ownership.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-VALIDATION-EMPIRICAL-GRAND-LOCK-012
- SFT-CHEM-POLYMER-MATERIALS-HANDOFF-013
- SFT-MAT-MEAS-COMPOSITION-001
- SFT-MAT-MEAS-PROPERTY-001

Generated grammar. Generate the complete eight-axis HAND-001 ownership product before opening the registered paired evidence.

Boundary: Exactly the registered HAND-001 owner identities, directed relations, paired receipt records, controls and extension boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-owned-coordinate__exactly-one-owner__chemistry-to-materials-directed-one-owner-handoff__directed-consumer-edge__complete-paired-receipt-records__value-free-registry-before-outcomes__duplicate-missing-or-tampered-halts__dated-complete-extension-open`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
subject	complete-owned-coordinate	anonymous-cross-branch-result	An anonymous result loses the owned coordinate.
ownership	exactly-one-owner	overlapping-or-ownerless	Duplicate or absent ownership destroys exact provenance.
relation	chemistry-to-materials-directed-one-owner-handoff	branch-name-association	A label alone is not a handoff.
dependency	directed-consumer-edge	copied-law	Copying a law creates duplicate ownership.
records	complete-paired-receipt-records	selected-summary	A summary can hide one side of the boundary.
custody	value-free-registry-before-outcomes	outcome-before-registration	Outcome access may choose the boundary.
falsification	duplicate-missing-or-tampered-halts	omission-tolerated	A missing owner, record or edge cannot close.
extension	dated-complete-extension-open	permanent-branch-lock	A dated graph cannot bar discovery.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256: 6772d88a5ad91028cbca9c07aa90676eb42b7e37e4c2324da1656568fb198359.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256: 87519e6c64024b2783a95a6972790304b9617d9f2f8a0bb35c48b3b085518e6d.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: b92857632a4d29ded2afd903db515711f1f67b159d3b2f71fe00a209fc32125c.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256: 1314b03906c8d63630baedb0abe6681c53b1b437ab68a18e24c3406b39d55b01.

Independent reconstruction. Implementation hash

sha256: 36c71852157bec55c2eb61cdff11c2c3b52af1887d6242e8b4f1bcdfd66dc595; independent certificate sha256: 537aa47d846df42b791d16f6a9a02f73bb853a82d51e52e94e39140a52c8de0f; external-validation hash sha256: ba5f87c5d0d0e304b116080f6a31c5a8abb9e29d79034ad9183fa0237c5cfc35. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- NIST-CROSSLINKED-PHOTOPOLYMER-CONVERSION
- NIST-POLYMER-PROCESSING-METROLOGY
- NIST-MATERIALS-DIVISION-2026-07-24

- NIST-REFERENCE-MATERIALS-2026-07-24

Comparison records:

- HAND-001: value-free owner registry preceded outcome access
- frozen graph: 1484 claims, 25013 dependency edges, 18553 cross-branch uses
- paired external surface: 17 records, 31 source identities, 53 evidence lines
- all selected evidence rows preserved; omission control rejected True
- chemistry owns SFT-CHEM-POLYMER-MATERIALS-HANDOFF-013; sources=2; evidence-lines=5; receipt=sha256:a69103e6715074cfdb9843ef831a9ad775fc5f19039bb7b3a761e052361a1b07
- materials owns SFT-MAT-MEAS-COMPOSITION-001; sources=1; evidence-lines=3; receipt=sha256:12f436cc0234aa79c73e3e13c80607ee40d2765efedab4566e0e1d3af6e789e3
- materials owns SFT-MAT-MEAS-PROPERTY-001; sources=1; evidence-lines=3; receipt=sha256:bf49f2ad1f00be8833e35eae3aad141cd5213b11094b232ae4fb632a3bad6535

Falsification boundary: Reject if any registered owner, paired receipt, external record, registration, dependency edge, root trace, control or extension boundary is missing, duplicated, changed or opened before the value-free owner registry.

Measurement receipt: sha256 : f62e0d7a244ed054e95fd6630916da4c0d7f227482e3390022bf385b5f33a323.

Isolation certificate: sha256 : 9eaa89a94164f54a0159155b4e6ccd16af5ebd22e6bb029ce00b2301a5796c94.

Custody certificate: sha256 : cfeb3531773a8c5de8e81eea245327428dabf9b68d912202e80649bd455238a0.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no downstream outcome, preferred owner, measurement value or match result selects the law
- no branch may duplicate ownership of a coordinate it consumes
- no application record may redefine a natural-law owner
- no missing dependency or source record is silently ignored
- no fitted threshold, aggregate score, numerical zero, negative, irrational, imaginary or floating proof magnitude
- no failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

Immutable identities. Source manifest

sha256 : 84149933b84994da8167cb5464ccc448e6b91769e043d2e64ccfbff9973ef183; derivation seal

sha256 : fc9c1e2812ef1f26942a6b2f93173bd9e01d1c0f749b5fd848a1fa243bf5743d; engine receipt

sha256 : da67f6404c234aeb9db221306817032248d20f581db03732828bafed8d718fdbe at receipts/engine/model_admitted/SFT-CHEM-MATERIALS-OWNERSHIP-HANDOFF-001-da67f6404c234aeb.json; empirical

hash sha256 : fbb10e25912fd5f16f6d01e996a21895a90048eb523535271701ed3e9fda6a61; measurement

receipt sha256 : f62e0d7a244ed054e95fd6630916da4c0d7f227482e3390022bf385b5f33a323.

277. Chemistry-to-Biology one-owner handoff

Claim identity: SFT-CHEM-BIOLOGY-OWNERSHIP-HANDOFF-002

Question and theorem. The Chemistry handoff to biology preserves complete paired admitted records and exactly one owner per coordinate: chemistry owns molecular identity and reaction while biology owns living organization and function.

HAND-002 uniquely requires the complete receipt-bound pair in which chemistry owns molecular identity and reaction while biology owns living organization and function; consumption is a directed dependency and never duplicate ownership.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-VALIDATION-EMPIRICAL-GRAND-LOCK-012
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001
- SFT-BIO-PROTEIN-FUNCTION-001
- SFT-BIO-METABOLIC-TRANSFORMATION-001
- SFT-CHEM-MATERIALS-OWNERSHIP-HANDOFF-001

Generated grammar. Generate the complete eight-axis HAND-002 ownership product before opening the registered paired evidence.

Boundary: Exactly the registered HAND-002 owner identities, directed relations, paired receipt records, controls and extension boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-owned-coordinate__exactly-one-owner__chemistry-to-biology-directed-one-owner-handoff__directed-consumer-edge__complete-paired-receipt-records__value-free-registry-before-outcomes__duplicate-missing-or-tampered-halts__dated-complete-extension-open`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
subject	complete-owned-coordinate	anonymous-cross-branch-result	An anonymous result loses the owned coordinate.
ownership	exactly-one-owner	overlapping-or-ownerless	Duplicate or absent ownership destroys exact provenance.
relation	chemistry-to-biology-directed-one-owner-handoff	branch-name-association	A label alone is not a handoff.
dependency	directed-consumer-edge	copied-law	Copying a law creates duplicate ownership.
records	complete-paired-receipt-records	selected-summary	A summary can hide one side of the boundary.
custody	value-free-registry-before-outcomes	outcome-before-registration	Outcome access may choose the boundary.
falsification	duplicate-missing-or-tampered-halts	omission-tolerated	A missing owner, record or edge cannot close.
extension	dated-complete-extension-open	permanent-branch-lock	A dated graph cannot bar discovery.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:06d98c345a5ce96f3dbb01584fc6a6b1fe551a5887e48cafea28dbe60139daa9.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:3e1c66f7aed300103a5d42668838d91b3411fc1882a243a1591321213a4fad56.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:c301c1dda90912425441daa4369722a71087a659c701da0f955a0577de22d704.

- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256 : c6377be36ebf6f18b2cc5334a72ef7ac197bf551f69b1a300df5f526ce25161b.

Independent reconstruction. Implementation hash

sha256 : 36c71852157bec55c2eb61cdff11c2c3b52af1887d6242e8b4f1bcdfd66dc595; independent certificate
sha256 : 537aa47d846df42b791d16f6a9a02f73bb853a82d51e52e94e39140a52c8de0f; external-validation
hash sha256 : 45253e5cf046d0570b0e7d057fcca80774adec0e299d2fd50c5d71f1f89a359a. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- EMBL-EBI-CHEBI-BIOMOLECULAR-BOUNDARY-2026
- SFT-BIO-SRC-NCBI-PROTEIN-FOLDING-V1
- SFT-BIO-SRC-SEQUENCE-ONTOLOGY-OBO-RELEASE-V1
- SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM
- SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION

Comparison records:

- HAND-002: value-free owner registry preceded outcome access
- frozen graph: 1484 claims, 25013 dependency edges, 18553 cross-branch uses
- paired external surface: 17 records, 31 source identities, 53 evidence lines
- all selected evidence rows preserved; omission control rejected **True**
- chemistry owns SFT-CHEM-BIOMOLECULAR-BOUNDARY-001; sources=1; evidence-lines=2;
receipt=sha256:dfed7cd5d124ff390399d5adf8b06a01a7edcc102a9e6d0ff59beb201bfd547e
- biology owns SFT-BIO-PROTEIN-FUNCTION-001; sources=2; evidence-lines=4;
receipt=sha256:946afec8074bb304b554d02c2b91056966d13d930868d873a389d21ac5435252
- biology owns SFT-BIO-METABOLIC-TRANSFORMATION-001; sources=2; evidence-lines=4;
receipt=sha256:0f1ee13bab8ff013eb1e629c8ea7f95d710817fb54744617342881f9e019ffa3

Falsification boundary: Reject if any registered owner, paired receipt, external record, registration, dependency edge, root trace, control or extension boundary is missing, duplicated, changed or opened before the value-free owner registry.

Measurement receipt: sha256 : ff9f25e95ba6420d0694ed12fc8eb9bcd875a72cadb6e7b36542d0c342b31597.

Isolation certificate: sha256 : c6480aa16e1574137897713f0f9e73df45514d649af367b11ce4ba61ec936a72.

Custody certificate: sha256 : 3c14cbee459037a716b52173dbfb1464606e46bf98ed6f92fe544fd90c90d058.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no downstream outcome, preferred owner, measurement value or match result selects the law
- no branch may duplicate ownership of a coordinate it consumes
- no application record may redefine a natural-law owner
- no missing dependency or source record is silently ignored

- no fitted threshold, aggregate score, numerical zero, negative, irrational, imaginary or floating proof magnitude
- no failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

Immutable identities. Source manifest

sha256:6fb1ca127997ac4d47465039d99a34d679be44a88fd9a8a748f52248a41883d4; derivation seal sha256:b20704b4e35a560b84ce020e477f4a5f9605000b2233dc1402c1d83fb0a1b05c; engine receipt sha256:37d9b00972e3dd38e15e09ed6de31b355990e32b6d1044d1b4f7e83487c56ade at receipts/engine/model_admitted/SFT-CHEM-BIOLOGY-OWNERSHIP-HANDOFF-002-37d9b00972e3dd38.json; empirical hash sha256:e02b98a8e92625b560206e88090ed0ef461828388f3a0560701b6180b57d4f64; measurement receipt sha256:ff9f25e95ba6420d0694ed12fc8eb9bcd875a72cadb6e7b36542d0c342b31597.

278. Chemistry-to-Medicine one-owner handoff

Claim identity: SFT-CHEM-MEDICINE-OWNERSHIP-HANDOFF-003

Question and theorem. The Chemistry handoff to medicine preserves complete paired admitted records and exactly one owner per coordinate: chemistry owns substances and transformations while medicine owns intervention and health outcome.

HAND-003 uniquely requires the complete receipt-bound pair in which chemistry owns substances and transformations while medicine owns intervention and health outcome; consumption is a directed dependency and never duplicate ownership.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-VALIDATION-EMPIRICAL-GRAND-LOCK-012
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-MED-INTERVENTION-001
- SFT-MED-DOSE-001
- SFT-MED-RESPONSE-001
- SFT-CHEM-BIOLOGY-OWNERSHIP-HANDOFF-002

Generated grammar. Generate the complete eight-axis HAND-003 ownership product before opening the registered paired evidence.

Boundary: Exactly the registered HAND-003 owner identities, directed relations, paired receipt records, controls and extension boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: complete-owned-coordinate__exactly-one-owner__chemistry-to-medicine-directed-one-owner-handoff__directed-consumer-edge__complete-paired-receipt-records__value-free-registry-before-outcomes__duplicate-missing-or-tampered-halts__dated-complete-extension-open. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
subject	complete-owned-coordinate	anonymous-cross-branch-result	An anonymous result loses the owned coordinate.
ownership	exactly-one-owner	overlapping-or-ownerless	Duplicate or absent ownership destroys exact provenance.
relation	chemistry-to-medicine-directed-one-owner-handoff	branch-name-association	A label alone is not a handoff.
dependency	directed-consumer-edge	copied-law	Copying a law creates duplicate ownership.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
records	complete-paired-receipt-records	selected-summary	A summary can hide one side of the boundary.
custody	value-free-registry-before-outcomes	outcome-before-registration	Outcome access may choose the boundary.
falsification	duplicate-missing-or-tampered-halts	omission-tolerated	A missing owner, record or edge cannot close.
extension	dated-complete-extension-open	permanent-branch-lock	A dated graph cannot bar discovery.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:5ffd5f86c11fcdacdeea82b4c6f1fae387b7878baa1d0c4661ac2d371ce7890e.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:7fc80e9598c00adc081b9bd098fcb8998409b60869918fb1f32b647168ac9307.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:4ee75155bc025dd6b2217bd111772c3882e37d4d77a9a429043cdfba394c2728.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:9e3f03ac7450e5c76ca494515236db263a39c86db6a93d1f4ee98d046d917d39.

Independent reconstruction. Implementation hash

sha256:36c71852157bec55c2eb61cdff11c2c3b52af1887d6242e8b4f1bcdfd66dc595; independent certificate
sha256:537aa47d846df42b791d16f6a9a02f73bb853a82d51e52e94e39140a52c8de0f; external-validation
hash sha256:8c2e41653ae4a931aea27cc45560664a6fbb4d5516b3a90329c8fe4759ee69b9. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-C01039-2026
- MED-ICH-E6R3-001
- MED-ICH-E9R1-001
- MED-FDA-EXPOSURE-RESPONSE-001
- MED-FDA-BIOMARKER-001

Comparison records:

- HAND-003: value-free owner registry preceded outcome access
- frozen graph: 1484 claims, 25013 dependency edges, 18553 cross-branch uses
- paired external surface: 17 records, 31 source identities, 53 evidence lines
- all selected evidence rows preserved; omission control rejected True
- chemistry owns SFT-CHEM-MEAS-SUBSTANCE-001; sources=1; evidence-lines=2;
receipt=sha256:c24408e53e46d5873b5c2f77d638045df99bf6250d5cbac3895cc9a092df15e7
- medicine owns SFT-MED-INTERVENTION-001; sources=2; evidence-lines=4;
receipt=sha256:ef389261cfe2919237eefe5a95ed42263554fa6271eb87e13685b1d43ed1ddcc
- medicine owns SFT-MED-DOSE-001; sources=3; evidence-lines=4;
receipt=sha256:ac1d83eb8670e428a755b801006e386ef850299d00310595b34b68fdbbac76a5

- medicine owns SFT-MED-RESPONSE-001; sources=3; evidence-lines=4;
receipt=sha256:3a051896a773bd06407d83654f7e55a5f64aca695299eace57d1e33edef69367

Falsification boundary: Reject if any registered owner, paired receipt, external record, registration, dependency edge, root trace, control or extension boundary is missing, duplicated, changed or opened before the value-free owner registry.

Measurement receipt: sha256:ff7e97759412487d56013182f08a1d3ee50f01ccac87271cf8751fb4f9c45f69.

Isolation certificate: sha256:53516fe173e169bbacd2120e3f3a3a9e338f5b9aa1bf4c7ff705cb0c1ca7876f.

Custody certificate: sha256:3ba8bb3b36932b4b2ef0a76297f2853740cecefc5b828f636bcdf4a1652abe19.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no downstream outcome, preferred owner, measurement value or match result selects the law
- no branch may duplicate ownership of a coordinate it consumes
- no application record may redefine a natural-law owner
- no missing dependency or source record is silently ignored
- no fitted threshold, aggregate score, numerical zero, negative, irrational, imaginary or floating proof magnitude
- no failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

Immutable identities. Source manifest

sha256:695527d64b6972f19ac35c93531266d2eec314982198ec2dbc6c2f470f12f500; derivation seal
sha256:a777576641f0303f9649abd48a66a79c3ca91950ea2659bcebb4344257ae9823; engine receipt
sha256:4b264c200dae449b3bebcf5d8af248097a539a72e2c411204f133b0215bfc35 at receipts/engine/model_admitted/SFT-CHEM-MEDICINE-OWNERSHIP-HANDOFF-003-4b264c200dae449b.json; empirical
hash sha256:2168e5b99b28b00eeb3daf4a2eb954f3b8e138f2dde9b87827b5ffcf45e95cc6; measurement
receipt sha256:ff7e97759412487d56013182f08a1d3ee50f01ccac87271cf8751fb4f9c45f69.

279. Chemistry-to-Earth and Environmental Science one-owner handoff

Claim identity: SFT-CHEM-EARTH-ENVIRONMENT-OWNERSHIP-HANDOFF-004

Question and theorem. The Chemistry handoff to earth_environment preserves complete paired admitted records and exactly one owner per coordinate: chemistry owns species and reactions while Earth and Environmental Science owns history, transport and planetary context.

HAND-004 uniquely requires the complete receipt-bound pair in which chemistry owns species and reactions while Earth and Environmental Science owns history, transport and planetary context; consumption is a directed dependency and never duplicate ownership.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-VALIDATION-EMPIRICAL-GRAND-LOCK-012
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-EARTH-ENVIRONMENTAL-TRANSFORMATION-001

- SFT-EARTH-ENVIRONMENTAL-TRANSPORT-001
- SFT-CHEM-MEDICINE-OWNERSHIP-HANDOFF-003

Generated grammar. Generate the complete eight-axis HAND-004 ownership product before opening the registered paired evidence.

Boundary: Exactly the registered HAND-004 owner identities, directed relations, paired receipt records, controls and extension boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-owned-coordinate__exactly-one-owner__chemistry-to-earth_environment-directed-one-owner-handoff__directed-consumer-edge__complete-paired-receipt-records__value-free-registry-before-outcomes__duplicate-missing-or-tampered-halts__dated-complete-extension-open`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
subject	complete-owned-coordinate	anonymous-cross-branch-result	An anonymous result loses the owned coordinate.
ownership	exactly-one-owner	overlapping-or-ownerless	Duplicate or absent ownership destroys exact provenance.
relation	chemistry-to-earth_environment-directed-one-owner-handoff	branch-name-association	A label alone is not a handoff.
dependency	directed-consumer-edge	copied-law	Copying a law creates duplicate ownership.
records	complete-paired-receipt-records	selected-summary	A summary can hide one side of the boundary.
custody	value-free-registry-before-outcomes	outcome-before-registration	Outcome access may choose the boundary.
falsification	duplicate-missing-or-tampered-halts	omission-tolerated	A missing owner, record or edge cannot close.
extension	dated-complete-extension-open	permanent-branch-lock	A dated graph cannot bar discovery.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256: e917c84dcef57bf4bf9796c4435f6e89d07ec7aca7bd7fd5e8e073f61531199f.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256: 47baf1d877be819d8b08c2be664cff4edbed0071e81e0a25155d26c4b4b8398f.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: a81d74338a00b32c756de8987d2304c0bfd339bce87b44062236f2c35515532e.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256: e78acb0e0d487b399f0baa895c8dd018cf91472fac019ba7959e25eb2b10e8e4.

Independent reconstruction. Implementation hash

sha256: 36c71852157bec55c2eb61cdf11c2c3b52af1887d6242e8b4f1bcdfd66dc595; independent certificate
sha256: 537aa47d846df42b791d16f6a9a02f73bb853a82d51e52e94e39140a52c8de0f; external-validation
hash sha256: 4019caf1cefa0c81ddd82163ede62729baf87be8374449f7ba807a6589de6835. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-C01039-2026
- IUPAC-GOLD-BOOK-CHEMICAL-REACTION-COMPOSITE-2026
- EPA-AIRDATA-001
- WQP-WEB-SERVICES-001
- USGS-LANDSAT-ACCESS-001
- NEON-OPEN-DATA-001

Comparison records:

- HAND-004: value-free owner registry preceded outcome access
- frozen graph: 1484 claims, 25013 dependency edges, 18553 cross-branch uses
- paired external surface: 17 records, 31 source identities, 53 evidence lines
- all selected evidence rows preserved; omission control rejected **True**
- chemistry owns SFT-CHEM-MEAS-SUBSTANCE-001; sources=1; evidence-lines=2;
receipt=sha256:c24408e53e46d5873b5c2f77d638045df99bf6250d5cbae3895cc9a092df15e7
- chemistry owns SFT-CHEM-RXN-IDENTITY-001; sources=1; evidence-lines=2;
receipt=sha256:ec1b06b42e323744b8166f5498671868cd4f9ce5ef68c1f43d718a894712e42d
- earth_environment owns SFT-EARTH-ENVIRONMENTAL-TRANSFORMATION-001; sources=4; evidence-lines=4;
receipt=sha256:90f90906ca26785c8256f3e093512560e27bfb09ca386250231eb0f66bc24fb8
- earth_environment owns SFT-EARTH-ENVIRONMENTAL-TRANSPORT-001; sources=4; evidence-lines=4;
receipt=sha256:4406c9886170dbcd2c3b7a8348b855d249b42b605272afc2ba4a02ae4cde94c0

Falsification boundary: Reject if any registered owner, paired receipt, external record, registration, dependency edge, root trace, control or extension boundary is missing, duplicated, changed or opened before the value-free owner registry.

Measurement receipt: sha256:3663472269afd8f9beb171052324bdf7ad927e39a2b20d8bd4e6dd264bb5ec8f.

Isolation certificate: sha256:d2ff74804fae9335fd16bff15438c7cde7fe7ddae8fbd4cb5d7a47353fa69267.

Custody certificate: sha256:878688d194651d911c45cc6cd3a80d70b63d16d172af55fec0092b82d9cca384.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no downstream outcome, preferred owner, measurement value or match result selects the law
- no branch may duplicate ownership of a coordinate it consumes
- no application record may redefine a natural-law owner
- no missing dependency or source record is silently ignored
- no fitted threshold, aggregate score, numerical zero, negative, irrational, imaginary or floating proof magnitude
- no failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

Immutable identities. Source manifest

sha256:c556bb94401562c5d98af17f95e8090ae40a2ba58b80c3a1b0e04870acd0226e; derivation seal sha256:06a854cd62a5597232687f79813d2833ec21ffd8988c18dcd926262cce9ecf35; engine receipt sha256:1b37f27441880372331632b8bbf9e7856610b2a8155b08f6ae9a18aedd70cd2a at receipts/engine/model_admitted/SFT-CHEM-EARTH-ENVIRONMENT-OWNERSHIP-HANDOFF-004-1b37f27441880372.json; empirical hash sha256:0c87a510abb3c44ceb560ed0b2bf8ec231b5c3edb833b1ccf5e9d9eeaf350a51; measurement receipt sha256:3663472269afd8f9beb171052324bdf7ad927e39a2b20d8bd4e6dd264bb5ec8f.

280. Chemistry-to-Astronomy one-owner handoff

Claim identity: SFT-CHEM-ASTRONOMY-OWNERSHIP-HANDOFF-005

Question and theorem. The Chemistry handoff to astronomy_cosmology preserves complete paired admitted records and exactly one owner per coordinate: chemistry owns molecular structure and spectra while astronomy owns source population and cosmic context.

HAND-005 uniquely requires the complete receipt-bound pair in which chemistry owns molecular structure and spectra while astronomy owns source population and cosmic context; consumption is a directed dependency and never duplicate ownership.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-VALIDATION-EMPIRICAL-GRAND-LOCK-012
- SFT-CHEM-SPEC-ROT-VIB-001
- SFT-ASTRO-SPECTRUM-001
- SFT-ASTRO-SOURCE-001
- SFT-CHEM-EARTH-ENVIRONMENT-OWNERSHIP-HANDOFF-004

Generated grammar. Generate the complete eight-axis HAND-005 ownership product before opening the registered paired evidence.

Boundary: Exactly the registered HAND-005 owner identities, directed relations, paired receipt records, controls and extension boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: complete-owned-coordinate__exactly-one-owner__chemistry-to-astronomy_cosmology-directed-one-owner-handoff__directed-consumer-edge__complete-paired-receipt-records__value-free-registry-before-outcomes__duplicate-missing-or-tampered-halts__dated-complete-extension-open. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
subject	complete-owned-coordinate	anonymous-cross-branch-result	An anonymous result loses the owned coordinate.
ownership	exactly-one-owner	overlapping-or-ownerless	Duplicate or absent ownership destroys exact provenance.
relation	chemistry-to-astronomy_cosmology-directed-one-owner-handoff	branch-name-association	A label alone is not a handoff.
dependency	directed-consumer-edge	copied-law	Copying a law creates duplicate ownership.
records	complete-paired-receipt-records	selected-summary	A summary can hide one side of the boundary.
custody	value-free-registry-before-outcomes	outcome-before-registration	Outcome access may choose the boundary.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
falsification	duplicate-missing-or-tampered-halts	omission-tolerated	A missing owner, record or edge cannot close.
extension	dated-complete-extension-open	permanent-branch-lock	A dated graph cannot bar discovery.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256 : a849c0e767ab9ca3eda3b8428986944bc641c4c96f31b5d14f357e8fb5ce4288.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256 : 61173f32649b5b624d7faa78ad4fbf7d0153e43e668eaecf34f1cbe118e75521.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 4cf7574d295467b4b1f32698138d863d7d63c4925f1b4db31c2f20353d865ddd.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256 : d906957c465ce6b3ddc384cf7a49bb989683260e79a0b8b06a82ab5f12847e22.

Independent reconstruction. Implementation hash

sha256 : 36c71852157bec55c2eb61cdff11c2c3b52af1887d6242e8b4f1bcdfd66dc595; independent certificate
sha256 : 537aa47d846df42b791d16f6a9a02f73bb853a82d51e52e94e39140a52c8de0f; external-validation
hash sha256 : 5e9e51e8a4f6d65aaf1ce4fd6ce4006decffd28d5ef9520496fdf59eb7426613. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ROTATION-VIBRATION-COMPOSITE-2026
- ESA-GAIA-ARCHIVE-001

Comparison records:

- HAND-005: value-free owner registry preceded outcome access
- frozen graph: 1484 claims, 25013 dependency edges, 18553 cross-branch uses
- paired external surface: 17 records, 31 source identities, 53 evidence lines
- all selected evidence rows preserved; omission control rejected **True**
- chemistry owns SFT-CHEM-SPEC-ROT-VIB-001; sources=1; evidence-lines=2;
receipt=sha256:20367bb1fe68cb5795f63f6d62da369e7cfc5295098f621cc6736c7fd3075e3d
- astronomy_cosmology owns SFT-ASTRO-SPECTRUM-001; sources=1; evidence-lines=2;
receipt=sha256:3c2ae757898fd4ca1063639281403d361117113843e1a4bd69144ffa55655594
- astronomy_cosmology owns SFT-ASTRO-SOURCE-001; sources=1; evidence-lines=2;
receipt=sha256:5473ed30f43a3ee1099bf327c08e196a9abcd888adb958b68d7c85d51681e958

Falsification boundary: Reject if any registered owner, paired receipt, external record, registration, dependency edge, root trace, control or extension boundary is missing, duplicated, changed or opened before the value-free owner registry.

Measurement receipt: sha256 : 707c47d702aed4f2d435c944b113b3f27433ab5a61043fa350c89929acb2e86f.

Isolation certificate: sha256 : 8682d64213ce1f17bd8485c65805798b96dcddc7c1073b84691453a045920e5f.

Custody certificate: sha256 : 221327f122ac2ea8cad631377de5b853879cc829d3e69009db687a5b10a73687.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no downstream outcome, preferred owner, measurement value or match result selects the law
- no branch may duplicate ownership of a coordinate it consumes
- no application record may redefine a natural-law owner
- no missing dependency or source record is silently ignored
- no fitted threshold, aggregate score, numerical zero, negative, irrational, imaginary or floating proof magnitude
- no failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

Immutable identities. Source manifest

sha256:177675ea5e0b2f12f1265f53b966eb74502ffffde433439c79617c82d03212276; derivation seal
sha256:afa485cd8eedd301606afb300ff203e222c5ba821965a8f3ec0bc09d4b0586fb; engine receipt
sha256:355716812aceedef1e07165e27f12f265628a8be4993e37976cb743068c1d1a9 at receipts/engine/model_admitted/SFT-CHEM-ASTRONOMY-OWNERSHIP-HANDOFF-005-355716812aceedef.json; empirical
hash sha256:2c68bf7477c90df6a6288373e5f9f3e075514891b0a0a9f21380ac9b16e9d71e; measurement
receipt sha256:707c47d702aed4f2d435c944b113b3f27433ab5a61043fa350c89929acb2e86f.

281. Chemistry cross-branch one-owner completeness certificate

Claim identity: SFT-CHEM-CROSS-BRANCH-ONE-OWNER-COMPLETENESS-006

Question and theorem. Across the complete frozen graph of 1484 admitted claims and 25013 dependency edges, every claim has exactly one branch owner, every claim reaches the root theorem, and cross-branch use remains a directed dependency rather than duplicate ownership.

HAND-006 uniquely requires one owner for each of 1484 frozen claims, complete root reachability, and all 25013 directed dependency edges, with lawful extension adding rather than rewriting ownership.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-CHEM-VALIDATION-EMPIRICAL-GRAND-LOCK-012
- SFT-CHEM-MATERIALS-OWNERSHIP-HANDOFF-001
- SFT-CHEM-BIOLOGY-OWNERSHIP-HANDOFF-002
- SFT-CHEM-MEDICINE-OWNERSHIP-HANDOFF-003
- SFT-CHEM-EARTH-ENVIRONMENT-OWNERSHIP-HANDOFF-004
- SFT-CHEM-ASTRONOMY-OWNERSHIP-HANDOFF-005

Generated grammar. Generate the complete eight-axis HAND-006 ownership product before opening the registered paired evidence.

Boundary: Exactly the registered HAND-006 owner identities, directed relations, paired receipt records, controls and extension boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: complete-owned-coordinate__exactly-one-owner__complete-one-owner-root-traced-consumer-graph__directed-consumer-edge__complete-paired-receipt-records__value-free-registry-before-outcomes__duplicate-missing-or-tampered-halts__dated-complete-extension-open. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
subject	complete-owned-coordinate	anonymous-cross-branch-result	An anonymous result loses the owned coordinate.
ownership	exactly-one-owner	overlapping-or-ownerless	Duplicate or absent ownership destroys exact provenance.
relation	complete-one-owner-root-traced-consumer-graph	branch-name-association	A label alone is not a handoff.
dependency	directed-consumer-edge	copied-law	Copying a law creates duplicate ownership.
records	complete-paired-receipt-records	selected-summary	A summary can hide one side of the boundary.
custody	value-free-registry-before-outcomes	outcome-before-registration	Outcome access may choose the boundary.
falsification	duplicate-missing-or-tampered-halts	omission-tolerated	A missing owner, record or edge cannot close.
extension	dated-complete-extension-open	permanent-branch-lock	A dated graph cannot bar discovery.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:8e2a512f3aaa9b3b3ad6573330fa92578c4643f22a53128e51a25184eb8e3937.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:adf3e2e55f835f60b3926e42826818ac3404bbc6ee13c614e135e57f10df82ac.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:85096c41ea7f9d678097dfaa5a4bc9f52f75a167c428017517e0913a69b06ff4.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:9ba3b7e525195342aa90c9fa6274fc24e28f6c501e82608aba2681ce4870509b.

Independent reconstruction. Implementation hash

sha256:36c71852157bec55c2eb61cdff11c2c3b52af1887d6242e8b4f1bcdfd66dc595; independent certificate
sha256:537aa47d846df42b791d16f6a9a02f73bb853a82d51e52e94e39140a52c8de0f; external-validation
hash sha256:26dcd1db42a28a1c375368f9ae565cf7ed6de36b26de2f310121e6b9b7f55bc4. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- NIST-CROSSLINKED-PHOTOPOLYMER-CONVERSION
- NIST-POLYMER-PROCESSING-METROLOGY
- NIST-MATERIALS-DIVISION-2026-07-24
- NIST-REFERENCE-MATERIALS-2026-07-24
- EMBL-EBI-CHEBI-BIOMOLECULAR-BOUNDARY-2026
- SFT-BIO-SRC-NCBI-PROTEIN-FOLDING-V1
- SFT-BIO-SRC-SEQUENCE-ONTOLOGY-OBO-RELEASE-V1
- SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM

- SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION
- IUPAC-GOLD-BOOK-C01039-2026
- MED-ICH-E6R3-001
- MED-ICH-E9R1-001
- MED-FDA-EXPOSURE-RESPONSE-001
- MED-FDA-BIOMARKER-001
- IUPAC-GOLD-BOOK-CHEMICAL-REACTION-COMPOSITE-2026
- EPA-AIRDATA-001
- WQP-WEB-SERVICES-001
- USGS-LANDSAT-ACCESS-001
- NEON-OPEN-DATA-001
- IUPAC-GOLD-BOOK-ROTATION-VIBRATION-COMPOSITE-2026
- ESA-GAIA-ARCHIVE-001

Comparison records:

- HAND-006: value-free owner registry preceded outcome access
- frozen graph: 1484 claims, 25013 dependency edges, 18553 cross-branch uses
- paired external surface: 17 records, 31 source identities, 53 evidence lines
- all selected evidence rows preserved; omission control rejected True
- complete owner graph: 1484 unique owners; 25013 dependency edges; 18553 cross-branch edges; all root-traced

Falsification boundary: Reject if any registered owner, paired receipt, external record, registration, dependency edge, root trace, control or extension boundary is missing, duplicated, changed or opened before the value-free owner registry.

Measurement receipt: sha256 : 383c763e06484143273081b7f16eadd2412d5ba357824fef24159dbdf3ab8efb.

Isolation certificate: sha256 : b4ae59f7185bc3683c0c6a39c342dd13e0879eda0c9c55e9342610089c9cf9d5.

Custody certificate: sha256 : ee0f01091f44f8b97e7d7ce2090a2616bd9da086f58fb97dda350085adeddedf.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no downstream outcome, preferred owner, measurement value or match result selects the law
- no branch may duplicate ownership of a coordinate it consumes
- no application record may redefine a natural-law owner
- no missing dependency or source record is silently ignored
- no fitted threshold, aggregate score, numerical zero, negative, irrational, imaginary or floating proof magnitude
- no failed route retires an obligation
- no engine, verifier, prior receipt, certificate or admitted claim change

Immutable identities. Source manifest

sha256:ae800fd5a6813dfbc6447318d13230f25db6ea94fe24150d3b8dc78af60ee69b; derivation seal
 sha256:3df97c2c59859c1634ade3095943cc636e78f6f49c8a91fd3c40ff71b3a3b90d; engine receipt
 sha256:19a909f22474460812b3821628c27dbe0e7c24f8cf56919a0012cd618cde839e at receipts/engine/
 model_admitted/SFT-CHEM-CROSS-BRANCH-ONE-OWNER-COMPLETENESS-006-19a909f224744608.json
 ; empirical hash sha256:4e9bae10393e5103e069849653345c1132efa3e2fb6a9835e8713d7ea4971f99;
 measurement receipt sha256:383c763e06484143273081b7f16eadd2412d5ba357824fef24159dbdf3ab8efb.

31. Integrated empirical synthesis

The complete branch contains 273 empirically compared claims and eight explicitly formal-only boundaries. The frozen VALID vector reopens no law and awards no aggregate score: it verifies that all 263 pre-VALID Chemistry claims retain their own current receipts, controls, closure types and evidence classes. Across that surface it preserves 26,486 evidence lines and 1,177 source-identity occurrences. The final six handoff laws then test seventeen paired empirical records and the complete owner/dependency graph. Every unfavorable, absent, unavailable and unresolved row remains available for criticism without substituting for the separately admitted law.

32. Standing predictions and direct falsifiers

- Direct observation inconsistent with g-block opening at $Z=121$ falsifies that registered extrapolation.
- A confirmed element-126 carrier inconsistent with $N=184$, $A=310$ or $[Og] \ 8s2 \ 5g6$ falsifies the corresponding Smithium claims.
- A positive measured Smithium lifetime, spectrum, decay or chemistry outside the registered generated consequence classes falsifies the relevant successor claim; present non-observation is retained as non-observation.
- A verified atomic carrier beyond the registered $Z=137$ structural endpoint falsifies that exact boundary.
- Any omitted source row, changed source identity, replay mismatch, accepted tampered control, duplicate owner or target access before seal invalidates the affected claim.
- A later lawful discovery is not prohibited by “completion”; it must be added through a new claim, complete grammar and successor evidence vector.

33. Reproducibility and current verification state

The complete HAND family reproduced six of six receipt hashes exactly. The combined VALID/HAND focused suite passed eight tests; lightweight repository validation passed at 1,490 total V3 claims; the engine and verification-authority seals remained valid. The full heavy all-branch command is intentionally reserved for the final global Grand Lock and is not misrepresented as rerun for this paper draft. The release evidence map binds every Chemistry package and current receipt by hash.

34. Conclusion

Chemistry is complete to the registered current standard: 272 of 272 obligations, 281 live admitted claims, 71,936 exhaustively enumerated candidates, 281 unique survivors, 1,124 mandatory controls, 281 independent reconstructions, 273 post-seal empirical packages and eight formal-only boundaries. This is not a consensus summary or an opaque prediction system. It is an open, claim-by-claim chain from the root theorem through exact structure, explicit falsification and observation. Its authority is reproducibility. Its completion remains open to every lawful correction and discovery.

References and authoritative records

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- NIST. *Artificial Intelligence Risk Management Framework 1.0*. <https://airc.nist.gov/airmf-resources/airmf/3-sec-characteristics/>.